Microlocal Analysis and Index Theory

A streamlined edition of two sets of lecture notes by Richard Melrose

After R. B. Melrose

Introduction to Microlocal Analysis (MIT, 2003) and
Lectures on Pseudodifferential Operators (MIT 18.157, 2005–2006)

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Preface

This book combines two sets of lecture notes by Richard Melrose: *Introduction to Microlocal Analysis* (MIT, version 0.5B, 2003), a revised version of the notes from several courses he taught at MIT, and *Lectures on Pseudodifferential Operators* (MIT course 18.157, Fall 2005, revised 2006).

Microlocal analysis is a geometric theory of distributions, or equivalently a theory of geometric distributions. Rather than general distributions, which are like general continuous functions only worse, it studies the distributions that actually arise from differential and integral equations. Conormal distributions and, more generally, Lagrangian distributions are smooth away from a well-defined set and have a precisely described singularity there. Their singularities are the main subject.

Prerequisites and suggested reading

The reader should already know some distribution theory and some basic analysis, and some familiarity with analysis on manifolds will help. If one or two of these are missing they can be picked up along the way. A reader who has none of them should first do some preliminary reading. Friedlander's book [3] is a good introduction to distributions, and Volume I of Hörmander's treatise [7] gives a much fuller treatment that also serves as a general reference. For analysis on manifolds, most of what we need can be found in Munkres' book [9] or in Spivak's short book [12]. Chapter 1 reviews tempered distributions and the Fourier transform, so that the book is reasonably self-contained.

About this edition

This is a streamlined edition. The material from the two sources has been merged and reorganized into five parts.

- Part I builds the calculus: tempered distributions and compactifications, pseudodifferential operators on $\mathbb{R}^n$, microlocalization, conormal distributions and operators on manifolds.
- Part II covers the global calculi on $\mathbb{R}^n$: the isotropic calculus, and traces and determinants.
- Part III covers boundary problems and propagation: boundary problems, the Hamilton–Jacobi and transport equations, the wave equation and Weyl asymptotics.
- Part IV is index theory, taken from the 18.157 notes: homotopy invariance of the index, Toeplitz operators, Bott periodicity, product-type operators, the families index theorem and eta forms.
- Part V covers the algebraic structure of the calculus: Hochschild homology.

Notation has been simplified and made uniform throughout. Proofs that were only sketched, or left as exercises but needed later, have been completed, and errors have been corrected. The "addenda" to the 18.157 lectures have been moved into the main text or the exercises. Draft material has been removed, including chapters that existed in the originals only as

headings. Each chapter ends with exercises and a short note on its main departures from the original notes.

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Part I

The calculus

Chapter 1

Tempered distributions and the Fourier transform

This chapter sets out the analytic framework for the rest of the book. Microlocal analysis studies distributions with special singularities, and we work in the space $\mathcal{S}'(\mathbb{R}^n)$ of tempered distributions, the dual of the Schwartz space $\mathcal{S}(\mathbb{R}^n)$. The Fourier transform is an isomorphism of both spaces, and it is the main tool for everything that follows.

We review the Schwartz space and its topology (Section 1.1), define tempered distributions and the basic operations on them (Section 1.2), and prove two elementary lemmas (Section 1.3). These lemmas give a proof of the Fourier inversion formula that only uses the structure of distributions (Section 1.4). Section 1.5 applies the Fourier transform to differential operators. With the Fourier transform available, we prove the two main structural results of the theory in Section 1.6: the representation of tempered distributions as derivatives of bounded continuous functions, and the Schwartz kernel theorem. The last two sections cover compactifications of a vector space. The radial compactification $\overline{\mathbb{R}^n}$ is a compact manifold with boundary. Schwartz functions on $\mathbb{R}^n$ are exactly the smooth functions on $\overline{\mathbb{R}^n}$ that vanish to infinite order at the boundary, and smooth functions on $\overline{\mathbb{R}^n}$ are classical symbols of order zero. This is the geometric picture behind the symbol spaces of Chapter 2.

Notation. Multi-indices are $\alpha \in \mathbb{N}_0^n$, with $|\alpha| = \alpha_1 + \cdots + \alpha_n$, $\alpha! = \alpha_1! \cdots \alpha_n!$, $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. We write $\partial_j = \partial/\partial x_j$, $D_j = -i\partial_j$, $D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}$ and $\langle x \rangle = (1 + |x|^2)^{1/2}$. Note that

$$\max(1, |x|) \le \langle x \rangle \le 1 + |x|, \qquad |x^\alpha| \le \langle x \rangle^{|\alpha|}, \qquad \langle x \rangle^{2N} = \sum_{|\gamma| \le N} \frac{N!}{(N - |\gamma|)!\, \gamma!}\, x^{2\gamma}. \tag{1.0.1}$$

1.1 Schwartz test functions

1.1.1 Smoothness

A function $\phi \colon \mathbb{R}^n \to \mathbb{C}$ is differentiable at $\bar{x}$ if there is an affine function $\ell_{\bar{x}}(x) = c + \sum_j d_j (x_j - \bar{x}_j)$ with $|\phi(x) - \ell_{\bar{x}}(x)| = o(|x - \bar{x}|)$. Then $c = \phi(\bar{x})$ and $d_j = \partial_j \phi(\bar{x})$. Iterating, ϕ is k times differentiable if its partial derivatives are $(k-1)$ times differentiable. In that case Taylor's theorem gives, for each $\bar{x}$,

$$\phi(x) = \sum_{|\alpha| \le k} \frac{i^{|\alpha|} (D^\alpha \phi)(\bar{x})}{\alpha!} (x - \bar{x})^\alpha + o(|x - \bar{x}|^k), \qquad x \to \bar{x},$$

because $\partial^\alpha = i^{|\alpha|}D^\alpha$. Having such polynomial approximations of all orders at every point does *not* by itself imply smoothness (Exercise 1.1). We write $\mathcal{C}^\infty(\mathbb{R}^n)$ for the infinitely differentiable functions. Its subspace

$$\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n) = \{u \in \mathcal{C}^\infty(\mathbb{R}^n);\ \sup_x |D^\alpha u(x)| < \infty\ \forall\, \alpha\}$$

consists of the smooth functions all of whose derivatives are bounded.

1.1.2 The Schwartz space

Definition 1.1.1. The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ of smooth functions of rapid decrease consists of those $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$ such that

$$\sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta \phi(x)| < \infty \quad \text{for all } \alpha, \beta \in \mathbb{N}_0^n.$$

For $k \in \mathbb{N}_0$ we set

$$\|\phi\|_k = \max_{|\alpha|+|\beta|\le k} \sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta \phi(x)|, \tag{1.1.1}$$

and define the distance

$$d(\phi,\psi) = \sum_{k=0}^{\infty} 2^{-k} \frac{\|\phi-\psi\|_k}{1+\|\phi-\psi\|_k}. \tag{1.1.2}$$

Each $\|\cdot\|_k$ is a norm: it is clearly a seminorm, and $\|\phi\|_k \ge \sup|\phi|$. The norms increase with k, that is $\|\phi\|_k \le \|\phi\|_{k+1}$. By (1.0.1) an equivalent family of norms is $\max_{|\beta|\le k} \sup\langle x\rangle^k |D^\beta\phi|$. More precisely, each $\sup\langle x\rangle^p|x^\alpha D^\beta\phi|$ is bounded by a constant times $\|\phi\|_{p+|\alpha|+|\beta|}$, and we use this freely. That d is a metric is Exercise 1.3. A sequence converges, or is Cauchy, for d if and only if it converges, or is Cauchy, for every norm $\|\cdot\|_k$ (Exercise 1.4). Two elementary inclusions relate the balls of the metric and the norms. If $d(\phi,0) < 2^{-k-1}$ then $\|\phi\|_k < 1$. Conversely, for every p,

$$d(\phi,0) \le \sum_{k\le p} 2^{-k}\|\phi\|_k + \sum_{k>p} 2^{-k} \le 2\|\phi\|_p + 2^{-p}. \tag{1.1.3}$$

Proposition 1.1.2. *With the metric d, $\mathcal{S}(\mathbb{R}^n)$ is a complete metric space; it is a Fréchet space.*

Proof. Let (ϕ_j) be Cauchy for d, so it is Cauchy in each norm $\|\cdot\|_k$. Then for all α,β the functions $x^\alpha D^\beta\phi_j$ converge uniformly on $\mathbb{R}^n$ to bounded continuous functions $g_{\alpha\beta}$. If a sequence of C^1 functions converges uniformly and its first derivatives also converge uniformly, then the limit is C^1 and its derivatives are the limits of the derivatives. Applying this repeatedly shows that $\phi = g_{00}$ is smooth and $D^\beta\phi = g_{0\beta}$. Hence $g_{\alpha\beta} = x^\alpha D^\beta\phi$, which is bounded, so $\phi \in \mathcal{S}(\mathbb{R}^n)$. Finally $\|\phi_j - \phi\|_k \to 0$ for each k, so $d(\phi_j,\phi)\to 0$. The metric is translation invariant and given by a countable family of seminorms, so $\mathcal{S}(\mathbb{R}^n)$ is a Fréchet space. □

The space is not trivial. For example $\exp(-|x|^2) \in \mathcal{S}(\mathbb{R}^n)$, since every derivative is a polynomial times $\exp(-|x|^2)$. Every $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ is in $\mathcal{S}(\mathbb{R}^n)$.

Lemma 1.1.3. *Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\chi = 1$ on $|x| \le 1$, and set $\chi_R(x) = \chi(x/R)$. Then for every k there is C_k such that*

$$\|(1-\chi_R)\phi\|_k \le C_k R^{-1}\|\phi\|_{k+1}, \qquad \phi \in \mathcal{S}(\mathbb{R}^n),\ R \ge 1.$$

In particular $\chi_R\phi \to \phi$ in $\mathcal{S}(\mathbb{R}^n)$ as $R\to\infty$, so $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$.

Proof. By Leibniz' formula, $x^\alpha D^\beta((1-\chi_R)\phi)$ is a sum of terms $x^\alpha D^\gamma(1-\chi_R)\, D^{\beta-\gamma}\phi$. Each $D^\gamma(1-\chi_R)$ is bounded uniformly in $R \geq 1$, because $D^\gamma \chi_R = R^{-|\gamma|}(D^\gamma\chi)(x/R)$. All of these terms vanish in $|x| \leq R$. There $1 \leq |x|/R \leq \langle x\rangle/R$, so each term is bounded by $CR^{-1}\sup\langle x\rangle|x^\alpha D^{\beta-\gamma}\phi| \leq C'R^{-1}\|\phi\|_{k+1}$. □

1.1.3 Continuous linear maps

Since the topology is given by the increasing family of norms $\|\cdot\|_k$, continuity of linear maps reduces to estimates.

Lemma 1.1.4. *A linear map $T\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^m)$ is continuous if and only if for every k there are N and C with $\|T\phi\|_k \leq C\|\phi\|_N$ for all $\phi \in \mathcal{S}(\mathbb{R}^n)$. A linear functional $u\colon \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ is continuous if and only if there are N and C with $|u(\phi)| \leq C\|\phi\|_N$ for all ϕ.*

Proof. If the estimates hold, then $\phi_j \to \phi$ implies $\|T\phi_j - T\phi\|_k \to 0$ for all k, hence $T\phi_j \to T\phi$. Conversely, suppose T is continuous at 0 and fix k. There is $r > 0$ such that $d(\phi, 0) < r$ implies $d(T\phi, 0) < 2^{-k-1}$, and hence $\|T\phi\|_k < 1$. Choose p with $2^{-p} < r/2$ and $\epsilon = r/4$. By (1.1.3), $\|\phi\|_p < \epsilon$ implies $d(\phi, 0) < r$. By homogeneity it follows that $\|T\phi\|_k \leq \epsilon^{-1}\|\phi\|_p$ for all ϕ. The functional case is the same argument with $|\cdot|$ in place of $\|\cdot\|_k$. □

Using this criterion it is straightforward to check that the following maps are continuous.

- *Multiplication and differentiation.* The maps $x_j, D_j\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ satisfy $\|x_j\phi\|_k + \|D_j\phi\|_k \leq C\|\phi\|_{k+1}$. Indeed, $x^\alpha D^\beta(x_j\phi) = x^\alpha x_j D^\beta\phi - i\beta_j x^\alpha D^{\beta-e_j}\phi$.
- *Multiplication by $\mathcal{C}^\infty_{\mathrm{b}}$ functions.* If $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ then, by Leibniz' formula

$$D^\alpha(a\phi) = \sum_{\beta\leq\alpha} \binom{\alpha}{\beta} D^\beta a\, D^{\alpha-\beta}\phi, \tag{1.1.4}$$

  we have $\|a\phi\|_k \leq C_k\|\phi\|_k$, with C_k depending on finitely many derivatives of a. So $\mathcal{S}(\mathbb{R}^n)$ is a module over $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$. In particular $\langle x\rangle^{-s} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ for $s \geq 0$ acts on $\mathcal{S}(\mathbb{R}^n)$.
- *Products.* Pointwise multiplication $\mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ makes $\mathcal{S}(\mathbb{R}^n)$ into a commutative algebra without identity.
- *Exterior product.* The bilinear map

$$\mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^m) \ni (\phi, \psi) \longmapsto \phi \boxtimes \psi \in \mathcal{S}(\mathbb{R}^{n+m}), \qquad \phi \boxtimes \psi(x, y) = \phi(x)\psi(y),$$

  satisfies $\|\phi\boxtimes\psi\|_k \leq \|\phi\|_k\|\psi\|_k$, since $(x, y)^{(\alpha,\alpha')}D^{(\beta,\beta')}(\phi\boxtimes\psi) = (x^\alpha D^\beta\phi)\boxtimes(y^{\alpha'}D^{\beta'}\psi)$.
- *Restriction.* If $\mathbb{R}^k \subset \mathbb{R}^n$ is the subspace $x_j = 0$, $j > k$, then $\pi_k^*\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^k)$, $\pi_k^* f(y) = f(y_1, \ldots, y_k, 0, \ldots, 0)$, is continuous and surjective (Exercise 1.6).

1.2 Tempered distributions

1.2.1 Definition

Integration $\int\colon \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ is continuous, since $|\int \phi\, dx| \leq \int\langle x\rangle^{-n-1}dx \cdot \sup\langle x\rangle^{n+1}|\phi|$. Combined with multiplication it gives the bilinear pairing $(\phi, \psi) \mapsto \int \phi\psi\, dx$. For fixed $\phi \in \mathcal{S}(\mathbb{R}^n)$ the functional

$$T_\phi\colon \mathcal{S}(\mathbb{R}^n) \ni \psi \longmapsto \int_{\mathbb{R}^n} \phi(x)\psi(x)\, dx \tag{1.2.1}$$

satisfies $|T_\phi(\psi)| \le C\|\psi\|_{n+1}$.

Definition 1.2.1. The space $\mathcal{S}'(\mathbb{R}^n)$ of *tempered distributions* is the space of continuous linear functionals $u\colon \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$. By Lemma 1.1.4 a linear functional u is in $\mathcal{S}'(\mathbb{R}^n)$ if and only if there are k and C with

$$|u(\phi)| \le C\|\phi\|_k \quad \forall\, \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.2.2}$$

We also write $\langle u, \phi\rangle = u(\phi)$.

Lemma 1.2.2. *The map $\mathcal{S}(\mathbb{R}^n) \ni \phi \mapsto T_\phi \in \mathcal{S}'(\mathbb{R}^n)$ is injective.*

Proof. $T_\phi(\bar\phi) = \int |\phi|^2 dx$, so $T_\phi = 0$ implies $\phi = 0$. □

More generally, if u is a measurable function with $\langle x\rangle^{-N} u \in L^1(\mathbb{R}^n)$ for some N, then $T_u(\psi) = \int u\psi\, dx$ satisfies $|T_u(\psi)| \le \|\langle x\rangle^{-N} u\|_{L^1} \sup\langle x\rangle^N |\psi|$, so $T_u \in \mathcal{S}'(\mathbb{R}^n)$. This applies to bounded continuous functions, to $L^2(\mathbb{R}^n)$ and to polynomials. If u is continuous and $T_u = 0$ then $u = 0$: otherwise $\operatorname{Re} u$ or $\operatorname{Im} u$ has a strict sign near some point, and pairing with a non-negative bump function supported there gives a contradiction. For $u \in L^2(\mathbb{R}^n)$ see Exercise 1.8. We therefore identify such functions with the distributions they define, and simply write u for T_u.

1.2.2 The weak topology

We give $\mathcal{S}'(\mathbb{R}^n)$ the *weak topology*. This is the weakest topology in which all the maps $\mathcal{S}'(\mathbb{R}^n) \ni u \mapsto u(\phi) \in \mathbb{C}$, $\phi \in \mathcal{S}(\mathbb{R}^n)$, are continuous. A basis of neighbourhoods of u_0 is given by the sets

$$\{u \in \mathcal{S}'(\mathbb{R}^n);\ |u(\phi_j) - u_0(\phi_j)| < \epsilon_j,\ j = 1, \dots, p\}, \tag{1.2.3}$$

where $\{\phi_1, \dots, \phi_p\} \subset \mathcal{S}(\mathbb{R}^n)$ is finite and $\epsilon_j > 0$. A map into $\mathcal{S}'(\mathbb{R}^n)$ is continuous if and only if its composites with all the evaluations $u \mapsto u(\phi)$ are continuous.

Proposition 1.2.3. *$\mathcal{S}(\mathbb{R}^n)$, embedded by $\phi \mapsto T_\phi$, is dense in $\mathcal{S}'(\mathbb{R}^n)$.*

Proof. It suffices to show the following. For $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\phi_1, \dots, \phi_p \in \mathcal{S}(\mathbb{R}^n)$ there is $\psi \in \mathcal{S}(\mathbb{R}^n)$ with $T_\psi(\phi_j) = u(\phi_j)$ for all j. Then T_ψ lies in every neighbourhood (1.2.3) of u. Let $\phi'_1, \dots, \phi'_q$ be a basis of $\operatorname{span}\{\phi_j\}$ that is orthonormal for the L^2 inner product, and set

$$\psi = \sum_{l=1}^{q} u(\phi'_l)\, \overline{\phi'_l} \in \mathcal{S}(\mathbb{R}^n).$$

Then $T_\psi(\phi'_m) = \sum_l u(\phi'_l) \int \overline{\phi'_l}\phi'_m\, dx = u(\phi'_m)$ for each m. By linearity, $T_\psi = u$ on the span of the ϕ_j. □

1.2.3 Operations by duality

If $T\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous and linear and has a continuous *transpose* T^t, meaning $\int (T\phi)\psi = \int \phi\, (T^t\psi)$, then $(Tu)(\psi) = u(T^t\psi)$ defines a weakly continuous extension $T\colon \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$. It is consistent with the inclusion $\mathcal{S} \subset \mathcal{S}'$, and it is the unique continuous extension by Proposition 1.2.3. Integration by parts shows that $D_j^t = -D_j$, and clearly $x_j^t = x_j$ and $a^t = a$ for multiplication by $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$. Thus for $u \in \mathcal{S}'(\mathbb{R}^n)$, $\psi \in \mathcal{S}(\mathbb{R}^n)$ and $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ we define

$$D_j u(\psi) = u(-D_j\psi), \qquad x_j u(\psi) = u(x_j\psi), \qquad au(\psi) = u(a\psi). \tag{1.2.4}$$

These are continuous maps $\mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ which extend the corresponding operations on $\mathcal{S}(\mathbb{R}^n)$. Consequently, if u is a bounded continuous function then $x^\beta D^\alpha u \in \mathcal{S}'(\mathbb{R}^n)$ for all α, β.

Similarly, the exterior product extends to distributions. For $u \in \mathcal{S}'(\mathbb{R}^n)$ and $v \in \mathcal{S}'(\mathbb{R}^m)$ there is a unique $u \boxtimes v \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $(u \boxtimes v)(\phi \boxtimes \psi) = u(\phi)v(\psi)$. We only need the special case $1 \boxtimes v$, which is constructed directly in Section 1.3. The general case follows from the kernel theorem in Section 1.6 (Exercise 1.13).

1.2.4 Linear changes of variables

A linear map $L\colon \mathbb{R}^n \to \mathbb{R}^n$, $(Lx)_j = \sum_k L_{jk}x_k$, is invertible if and only if $\det L \neq 0$, if and only if there is $c > 0$ with $c|x| \leq |Lx| \leq c^{-1}|x|$ for all x. For $L \in \mathrm{GL}(n, \mathbb{R})$ the pull-back is $L^*\phi = \phi \circ L$. By the chain rule

$$D_j(L^*\phi) = \sum_{k=1}^{n} L_{kj}\, L^*(D_k\phi), \tag{1.2.5}$$

so $x^\alpha D^\beta(L^*\phi)$ is a linear combination of the functions $x^\alpha L^*(D^\gamma\phi)$, $|\gamma| = |\beta|$. Since $\langle x\rangle \leq C\langle Lx\rangle$, $L^*\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous, and it is an isomorphism with inverse $(L^{-1})^*$. Changing variables $y = Lx$ in $\int \phi(Lx)\psi(x)\,dx$ gives

$$T_{L^*\phi}(\psi) = T_\phi(|\det L|^{-1}(L^{-1})^*\psi), \tag{1.2.6}$$

and so we define, for $u \in \mathcal{S}'(\mathbb{R}^n)$,

$$L^*u(\psi) = u\big(|\det L|^{-1}(L^{-1})^*\psi\big), \qquad \psi \in \mathcal{S}(\mathbb{R}^n).$$

This is an isomorphism $L^*\colon \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ extending L^* on $\mathcal{S}(\mathbb{R}^n)$. By density, (1.2.5) and the identity $L^*(x_j u) = (L^*x_j)(L^*u)$ continue to hold for $u \in \mathcal{S}'(\mathbb{R}^n)$.

1.2.5 Support

Definition 1.2.4. A distribution $u \in \mathcal{S}'(\mathbb{R}^n)$ *vanishes* on an open set $O \subset \mathbb{R}^n$ if $u(\phi) = 0$ for all $\phi \in \mathcal{C}_c^\infty(O)$. The *support* $\operatorname{supp} u$ is the complement of the union of all open sets on which u vanishes.

If u vanishes on each O_i, then it vanishes on $\bigcup_i O_i$. Indeed, a $\phi \in \mathcal{C}_c^\infty(\bigcup_i O_i)$ has compact support, which is covered by finitely many O_i, and a finite partition of unity splits ϕ into pieces supported in these sets. So u vanishes on $\mathbb{R}^n \setminus \operatorname{supp} u$. For continuous functions this agrees with the usual support (Exercise 1.10). If $\operatorname{supp} u \subset F$, F closed, and $\phi \in \mathcal{S}(\mathbb{R}^n)$ vanishes in a neighbourhood of F, then $u(\phi) = 0$. To see this, note that $\chi_R\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n \setminus F)$ in the notation of Lemma 1.1.3, and $\chi_R\phi \to \phi$ in $\mathcal{S}(\mathbb{R}^n)$.

1.3 Two little lemmas

The two lemmas of this section solve the simplest distributional equations, $D_j u = 0$ and $x_j u = 0$. They are good exercises in the definitions. More importantly, they are exactly what we need for the proof of the Fourier inversion formula in Section 1.4. Each is proved by a decomposition of test functions that is dual to the equation.

Write $x = (x', x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$ (by relabelling, any variable x_j can be treated in the same way). Define the integration map

$$I\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^{n-1}), \qquad I\phi(x') = \int_{\mathbb{R}} \phi(x', x_n)\,dx_n. \tag{1.3.1}$$

Differentiating under the integral and using $\langle x'\rangle^p \le \langle x\rangle^p$ gives $\|I\phi\|_k \le C\|\phi\|_{k+2}$, so I is continuous. For $v \in \mathcal{S}'(\mathbb{R}^{n-1})$ we may therefore define

$$(1 \boxtimes v)(\phi) = v(I\phi), \qquad \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.3.2}$$

Then $1 \boxtimes v \in \mathcal{S}'(\mathbb{R}^n)$, and $(1 \boxtimes v)(\phi' \boxtimes \phi_n) = v(\phi')\int \phi_n$. This is the "distribution $v(x')$ regarded as a function of x independent of x_n". (In the original notes it is written $\pi^* v$, with $\pi(x) = x'$.) Since $I(\theta \boxtimes \rho) = \theta$ whenever $\int \rho = 1$, the map I is surjective and $v \mapsto 1 \boxtimes v$ is injective.

Lemma 1.3.1. *Let $u \in \mathcal{S}'(\mathbb{R}^n)$. Then $D_n u = 0$ if and only if $u = 1 \boxtimes v$ for some $v \in \mathcal{S}'(\mathbb{R}^{n-1})$, and then v is unique.*

Proof. If $\psi \in \mathcal{S}(\mathbb{R}^n)$ then $I(D_n\psi) = 0$, since $\psi \to 0$ as $|x_n| \to \infty$. Hence $D_n(1 \boxtimes v)(\psi) = v(I(-D_n\psi)) = 0$.

For the converse, fix $\rho \in \mathcal{S}(\mathbb{R})$ with $\int \rho = 1$. We claim that every $\phi \in \mathcal{S}(\mathbb{R}^n)$ can be written

$$\phi = (I\phi) \boxtimes \rho + D_n\psi, \qquad \psi \in \mathcal{S}(\mathbb{R}^n). \tag{1.3.3}$$

Set $\zeta = \phi - (I\phi) \boxtimes \rho \in \mathcal{S}(\mathbb{R}^n)$, so that $I\zeta = 0$. Define

$$\psi(x', x_n) = i\int_{-\infty}^{x_n} \zeta(x', t)\, dt = -i\int_{x_n}^{\infty} \zeta(x', t)\, dt,$$

where the two expressions agree because $I\zeta = 0$. Then $D_n\psi = \zeta$, and ψ is smooth. Every derivative of ψ is either a derivative of ζ (if it involves x_n) or of the form $i\int_{-\infty}^{x_n} D_{x'}^{\beta'}\zeta\, dt$, where $D_{x'}^{\beta'}\zeta$ again has integral zero in x_n. For $g \in \mathcal{S}(\mathbb{R}^n)$ with $\int g(x', t)\, dt = 0$ and every N,

$$\Big|\int_{-\infty}^{x_n} g(x', t)\, dt\Big| \le \sup\big(\langle x'\rangle^N \langle t\rangle^{N+2}|g|\big)\, \langle x'\rangle^{-N} \int_{|t|\ge|x_n|} \langle t\rangle^{-N-2} dt \le C\langle x'\rangle^{-N}\langle x_n\rangle^{-N-1}.$$

Here we use the first integral for $x_n \le 0$ and the second for $x_n \ge 0$. Since $\langle x\rangle \le \langle x'\rangle\langle x_n\rangle$, all $\langle x\rangle^N D^\beta\psi$ are bounded, so $\psi \in \mathcal{S}(\mathbb{R}^n)$. This proves (1.3.3).

Now suppose $D_n u = 0$. Then $u(D_n\psi) = -(D_n u)(\psi) = 0$, so by (1.3.3) $u(\phi) = u((I\phi)\boxtimes\rho)$. The map $\theta \mapsto \theta\boxtimes\rho$ is continuous $\mathcal{S}(\mathbb{R}^{n-1}) \to \mathcal{S}(\mathbb{R}^n)$, so $v(\theta) = u(\theta\boxtimes\rho)$ defines $v \in \mathcal{S}'(\mathbb{R}^{n-1})$, and $u(\phi) = v(I\phi)$, that is $u = 1\boxtimes v$. Uniqueness follows from the injectivity of $v \mapsto 1\boxtimes v$. □

Applying the lemma repeatedly gives the version we need. For $x \in \mathbb{R}^k$ and $z \in \mathbb{R}^m$, define $(1 \boxtimes v)(\Phi) = v\big(\int_{\mathbb{R}^k} \Phi(x, \cdot)\, dx\big)$ for $v \in \mathcal{S}'(\mathbb{R}^m)$ and $\Phi \in \mathcal{S}(\mathbb{R}^{k+m})$.

Corollary 1.3.2. *If $u \in \mathcal{S}'(\mathbb{R}^{k+m}_{x,z})$ satisfies $D_{x_j} u = 0$ for $j = 1, \dots, k$, then $u = 1 \boxtimes v$ for a unique $v \in \mathcal{S}'(\mathbb{R}^m)$. Moreover $z_j(1 \boxtimes v) = 1 \boxtimes (z_j v)$.*

Proof. Induct on k. By Lemma 1.3.1 (in the variable x_1), $u = 1 \boxtimes v_1$ with $v_1 \in \mathcal{S}'(\mathbb{R}^{k-1+m})$. Integration in x_1 commutes with D_{x_j} for $j \ge 2$, so $0 = D_{x_j}(1 \boxtimes v_1) = 1 \boxtimes (D_{x_j} v_1)$. By injectivity, $D_{x_j} v_1 = 0$ for $j \ge 2$, and the inductive hypothesis applies to v_1. The last identity holds because integration in x commutes with multiplication by z_j. □

The second lemma is a special case of Schwartz' description of distributions supported at a point (Theorem 1.5.1).

Lemma 1.3.3. *If $u \in \mathcal{S}'(\mathbb{R}^n)$ and $x_j u = 0$ for $j = 1, \dots, n$, then $u = c\delta$ for some $c \in \mathbb{C}$. Here $\delta(\phi) = \phi(0)$.*

Proof. Choose $\rho \in \mathcal{C}^\infty_{\rm c}(\mathbb{R}^n)$ with $\rho = 1$ on $|x| \le \frac12$ and $\rho = 0$ on $|x| \ge \frac34$. We claim that every $\phi \in \mathcal{S}(\mathbb{R}^n)$ can be written

$$\phi = \phi(0)\rho + \sum_{j=1}^n x_j \psi_j, \qquad \psi_j \in \mathcal{S}(\mathbb{R}^n). \tag{1.3.4}$$

By Taylor's formula $\phi(x) = \phi(0) + \sum_j x_j \zeta_j(x)$ with $\zeta_j(x) = \int_0^1 (\partial_j\phi)(tx)\,dt \in \mathcal{C}^\infty(\mathbb{R}^n)$, so $\rho\phi = \phi(0)\rho + \sum_j x_j \rho\zeta_j$ with $\rho\zeta_j \in \mathcal{C}^\infty_{\rm c}(\mathbb{R}^n)$. The function $(1-\rho)\phi$ vanishes for $|x| \le \frac12$. The function $|x|^{-2}(1-\rho)$ is smooth with all derivatives bounded, so $\zeta = |x|^{-2}(1-\rho)\phi \in \mathcal{S}(\mathbb{R}^n)$ and $(1-\rho)\phi = \sum_j x_j (x_j\zeta)$. Thus (1.3.4) holds with $\psi_j = \rho\zeta_j + x_j\zeta$. If $x_j u = 0$ for all j, then $u(x_j\psi_j) = (x_j u)(\psi_j) = 0$, so $u(\phi) = \phi(0)u(\rho) = c\,\delta(\phi)$ with $c = u(\rho)$. □

1.4 The Fourier transform

1.4.1 On Schwartz functions

For $\phi \in \mathcal{S}(\mathbb{R}^n)$ we set

$$\mathcal{F}\phi(\xi) = \hat\phi(\xi) = \int_{\mathbb{R}^n} e^{-ix\cdot\xi}\phi(x)\,dx, \qquad \mathcal{G}\psi(x) = \int_{\mathbb{R}^n} e^{ix\cdot\xi}\psi(\xi)đ\xi, \tag{1.4.1}$$

where $đ\xi = (2\pi)^{-n}d\xi$. We will show that $\mathcal{G} = \mathcal{F}^{-1}$. The integrals converge absolutely, with $|\hat\phi(\xi)| \le \int \langle x\rangle^{-n-1}dx\,\sup\langle x\rangle^{n+1}|\phi|$. Differentiation under the integral sign is justified because $x^\beta\phi$ is again integrable, and gives $D^\beta_\xi\hat\phi = \mathcal{F}((-x)^\beta\phi)$. Integration by parts gives $\xi_j\hat\phi = \mathcal{F}(D_j\phi)$. Together:

$$\mathcal{F}(D_j\phi) = \xi_j\mathcal{F}\phi, \qquad \mathcal{F}(x_j\phi) = -D_{\xi_j}\mathcal{F}\phi, \qquad \xi^\alpha D^\beta_\xi\hat\phi = \mathcal{F}\big(D^\alpha((-x)^\beta\phi)\big). \tag{1.4.2}$$

Proposition 1.4.1. *$\mathcal{F}$ and $\mathcal{G}$ are continuous linear maps $\mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$, with $\|\mathcal{F}\phi\|_k \le C_k\|\phi\|_{k+n+1}$.*

Proof. For $|\alpha| + |\beta| \le k$, by (1.4.2),

$$|\xi^\alpha D^\beta_\xi\hat\phi(\xi)| \le \int \langle x\rangle^{-n-1}dx\ \sup_x \langle x\rangle^{n+1}|D^\alpha(x^\beta\phi)| \le C\|\phi\|_{k+n+1}.$$

The function $D^\alpha(x^\beta\phi)$ is a combination of terms $x^{\beta'}D^{\alpha'}\phi$ with $\beta' \le \beta$, $\alpha' \le \alpha$. Since $\mathcal{G}\psi(x) = (2\pi)^{-n}\mathcal{F}\psi(-x)$, the same holds for $\mathcal{G}$. □

The same argument applies to partial Fourier transforms. For example $\mathcal{F}_2\Phi(x,\xi) = \int e^{-iy\cdot\xi}\Phi(x,y)\,dy$ is continuous on $\mathcal{S}(\mathbb{R}^{2n})$: the weights in x simply pass through the integral.

Lemma 1.4.2. *$\mathcal{F}(e^{-|x|^2})(\xi) = \pi^{n/2}e^{-|\xi|^2/4}$ and $\mathcal{G}(e^{-|\xi|^2/4})(x) = \pi^{-n/2}e^{-|x|^2}$.*

Proof. Since $e^{-|x|^2} = \prod_j e^{-x_j^2}$, the integral factorizes and it suffices to take $n = 1$. Let $g(\xi) = \int e^{-ix\xi}e^{-x^2}dx$. Differentiating and integrating by parts,

$$g'(\xi) = \frac{i}{2}\int e^{-ix\xi}\,\frac{d}{dx}e^{-x^2}dx = -\frac{i}{2}\int\Big(\frac{d}{dx}e^{-ix\xi}\Big)e^{-x^2}dx = -\frac{\xi}{2}g(\xi).$$

So $g(\xi) = g(0)e^{-\xi^2/4}$, and $g(0) = \int e^{-x^2}dx = \pi^{1/2}$. For the second formula, change variables $\xi = 2\eta$: $\int e^{ix\xi}e^{-|\xi|^2/4}d\xi = 2^n\int e^{-i(-2x)\cdot\eta}e^{-|\eta|^2}d\eta = 2^n\pi^{n/2}e^{-|x|^2}$. Multiplying by $(2\pi)^{-n}$ gives the result. □

Theorem 1.4.3 (Fourier inversion)**.** *$\mathcal{F}\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is an isomorphism with inverse $\mathcal{G}$.*

Proof. We use the *idea* of the Schwartz kernel theorem, not the theorem itself, since the Fourier transform is used to prove it. Define $K\colon \mathcal{S}(\mathbb{R}^{2n}) \to \mathbb{C}$ by

$$K(\Phi) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{ix\cdot\xi}\, \mathcal{F}_2\Phi(x,\xi) đ\xi\, dx.$$

The integrand is in $\mathcal{S}(\mathbb{R}^{2n})$, and $\mathcal{F}_2$ is continuous, so $K \in \mathcal{S}'(\mathbb{R}^{2n})$. For $\zeta, \psi \in \mathcal{S}(\mathbb{R}^n)$ we have $\mathcal{F}_2(\zeta \boxtimes \psi) = \zeta \boxtimes \hat{\psi}$, so

$$K(\zeta \boxtimes \psi) = \int \zeta(x)\, (\mathcal{G}\mathcal{F}\psi)(x)\, dx. \tag{1.4.3}$$

That is, K is the Schwartz kernel of $\mathcal{G} \circ \mathcal{F}$.

Step 1: the equations satisfied by K. We claim that

$$(x_j - y_j)K = 0, \qquad (D_{x_j} + D_{y_j})K = 0, \qquad j = 1, \dots, n. \tag{1.4.4}$$

By (1.4.2) applied in y, $\mathcal{F}_2((x_j - y_j)\Phi) = (x_j + D_{\xi_j})\mathcal{F}_2\Phi$. All integrands below are Schwartz functions of (x,ξ), so integrating by parts in ξ gives

$$K((x_j - y_j)\Phi) = \iint e^{ix\cdot\xi}(x_j + D_{\xi_j})\mathcal{F}_2\Phi đ\xi\, dx = \iint (x_j e^{ix\cdot\xi} - D_{\xi_j} e^{ix\cdot\xi})\mathcal{F}_2\Phi đ\xi\, dx = 0.$$

Similarly $\mathcal{F}_2((D_{x_j} + D_{y_j})\Phi) = (D_{x_j} + \xi_j)\mathcal{F}_2\Phi$, and integrating by parts in x shows that $K((D_{x_j} + D_{y_j})\Phi) = 0$. This gives (1.4.4).

Step 2: change of variables. Let $\Lambda(x,z) = (x, x-z)$. This is a linear involution of $\mathbb{R}^{2n}$ with $|\det \Lambda| = 1$. Set $K' = \Lambda^* K$, so that $K'(\Phi) = K(\Phi\circ\Lambda)$. Since $(D_{x_j} + D_{y_j})(\Phi\circ\Lambda) = (D_{x_j}\Phi)\circ\Lambda$ and $(x_j - y_j)(\Phi \circ \Lambda) = (z_j\Phi) \circ \Lambda$, the equations (1.4.4) become

$$D_{x_j} K' = 0, \qquad z_j K' = 0, \qquad j = 1, \dots, n.$$

Step 3: the two lemmas. By Corollary 1.3.2, $K' = 1 \boxtimes f$ with $f \in \mathcal{S}'(\mathbb{R}^n_z)$, and $1 \boxtimes (z_j f) = z_j K' = 0$, so $z_j f = 0$ for all j. By Lemma 1.3.3, $f = c\delta$. Hence $K'(\Phi) = c\int \Phi(x,0)\, dx$, and since Λ is an involution, $K(\Phi) = K'(\Phi \circ \Lambda) = c\int \Phi(x,x)\, dx$. By (1.4.3), $\int \zeta\, \mathcal{G}\mathcal{F}\psi = c\int \zeta\psi$ for all ζ, so $\mathcal{G}\mathcal{F}\psi = c\psi$ for every $\psi \in \mathcal{S}(\mathbb{R}^n)$.

Step 4: the constant. By Lemma 1.4.2, $\mathcal{G}\mathcal{F}(e^{-|x|^2}) = \pi^{n/2}\mathcal{G}(e^{-|\xi|^2/4}) = e^{-|x|^2}$, so $c = 1$.

Thus $\mathcal{G}\mathcal{F} = \mathrm{Id}$. With $R\phi(x) = \phi(-x)$ we have $\mathcal{G} = (2\pi)^{-n} R\mathcal{F} = (2\pi)^{-n}\mathcal{F}R$, and therefore $\mathcal{F}\mathcal{G} = (2\pi)^{-n}\mathcal{F}R\mathcal{F} = R\mathcal{G}\mathcal{F}R = \mathrm{Id}$ as well. □

1.4.2 On tempered distributions

Fubini's theorem gives the symmetry

$$\int \hat{\phi}\, \psi\, dx = \int \phi\, \hat{\psi}\, dx, \qquad \phi, \psi \in \mathcal{S}(\mathbb{R}^n), \tag{1.4.5}$$

so $\mathcal{F}^t = \mathcal{F}$. As in Section 1.2.3 we define

$$\mathcal{F}u(\phi) = \hat{u}(\phi) = u(\hat{\phi}), \qquad \mathcal{G}u(\phi) = u(\mathcal{G}\phi), \qquad u \in \mathcal{S}'(\mathbb{R}^n),\ \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.4.6}$$

Corollary 1.4.4. *The map $\mathcal{F}$ of (1.4.6) is the unique continuous extension of the Fourier transform to $\mathcal{S}'(\mathbb{R}^n)$. It is an isomorphism with inverse $\mathcal{G}$, both maps are continuous for the weak topology, and for $u \in \mathcal{S}'(\mathbb{R}^n)$*

$$\mathcal{F}(D_j u) = \xi_j \mathcal{F}u, \qquad \mathcal{F}(x_j u) = -D_{\xi_j}\mathcal{F}u.$$

Proof. Weak continuity is immediate from (1.4.6). The definition is consistent with (1.4.5), and $\mathcal{G}\mathcal{F}u(\phi) = u(\mathcal{F}\mathcal{G}\phi) = u(\phi)$, and similarly $\mathcal{F}\mathcal{G} = \mathrm{Id}$. For the identities, $\mathcal{F}(D_j u)(\phi) = u(-D_j\hat{\phi}) = u(\mathcal{F}(x_j\phi)) = (\xi_j\hat{u})(\phi)$. Likewise $\mathcal{F}(x_j u)(\phi) = u(x_j\hat{\phi}) = u(\mathcal{F}(D_j\phi)) = \hat{u}(D_j\phi) = (-D_j\hat{u})(\phi)$. □

1.4.3 Plancherel's theorem

Complex conjugation of (1.4.1) gives $\overline{\hat{\psi}} = (2\pi)^n \mathcal{G}\bar{\psi}$. Combining this with (1.4.5) gives the adjoint identity

$$\int \phi\, \overline{\hat{\psi}}\, d\xi = (2\pi)^n \int (\mathcal{G}\phi)\, \bar{\psi}\, dx, \qquad \phi, \psi \in \mathcal{S}(\mathbb{R}^n). \tag{1.4.7}$$

Proposition 1.4.5 (Plancherel)**.** *The Fourier transform extends by continuity from $\mathcal{S}(\mathbb{R}^n)$ to an isomorphism $\mathcal{F}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$, consistent with its action on $\mathcal{S}'(\mathbb{R}^n)$, and*

$$\int \hat{u}\, \overline{\hat{v}}\, d\xi = (2\pi)^n \int u\bar{v}\, dx, \qquad \|\hat{u}\|_{L^2} = (2\pi)^{n/2} \|u\|_{L^2}.$$

Proof. Taking $\phi = \hat{u}$ in (1.4.7) with $u, v \in \mathcal{S}(\mathbb{R}^n)$ gives Parseval's formula, and $v = u$ gives the norm identity on $\mathcal{S}(\mathbb{R}^n)$. Now $\mathcal{C}_c^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ (a standard fact of measure theory), so $\mathcal{F}$ extends uniquely to a map $L^2 \to L^2$ that is $(2\pi)^{n/2}$ times an isometry. Its range is closed and contains $\mathcal{S}(\mathbb{R}^n)$, so it is all of $L^2(\mathbb{R}^n)$. The extension agrees with (1.4.6). Indeed, if $u_j \to u$ in L^2 with $u_j \in \mathcal{S}$, then $u_j \to u$ and $\hat{u}_j \to \hat{u}$ weakly in $\mathcal{S}'$, by the Cauchy–Schwarz inequality. □

1.5 Differential operators

The simplest Fourier transforms of distributions follow directly from the definitions. Since $\hat{\delta}(\phi) = \delta(\hat{\phi}) = \int \phi\, dx$, Corollary 1.4.4 gives

$$\mathcal{F}\delta = 1, \qquad \mathcal{F}(D^\alpha \delta) = \xi^\alpha, \quad \alpha \in \mathbb{N}_0^n. \tag{1.5.1}$$

Here $D^\alpha\delta(\phi) = \delta((-D)^\alpha\phi) = (-1)^{|\alpha|} D^\alpha\phi(0)$.

Theorem 1.5.1. *If $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\operatorname{supp} u \subset \{0\}$, then $u = \sum_{|\alpha|\le k} c_\alpha D^\alpha\delta$ for some k and constants c_α. Consequently the Fourier transform is an isomorphism*

$$\mathcal{F}\colon \{u \in \mathcal{S}'(\mathbb{R}^n);\ \operatorname{supp} u \subset \{0\}\} \longrightarrow \mathbb{C}[\xi]$$

onto the polynomials in ξ.

Proof. Let $|u(\phi)| \le C\|\phi\|_k$. First suppose that $\phi \in \mathcal{S}(\mathbb{R}^n)$ vanishes to order k at 0, meaning $D^\mu\phi(0) = 0$ for $|\mu| \le k$. Applying Taylor's formula to $D^\mu\phi$ gives $|D^\mu\phi(x)| \le C|x|^{k+1-|\mu|}$ for $|x| \le 1$ and $|\mu| \le k$. Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\chi = 1$ near 0 and $\operatorname{supp}\chi \subset \{|x| \le 1\}$, and let $\chi_\epsilon(x) = \chi(x/\epsilon)$ for $0 < \epsilon \le 1$. Then $(1-\chi_\epsilon)\phi$ vanishes near 0, so $u((1-\chi_\epsilon)\phi) = 0$ and $u(\phi) = u(\chi_\epsilon\phi)$. For $|\alpha| + |\beta| \le k$, the function $x^\alpha D^\beta(\chi_\epsilon\phi)$ is supported in $|x| \le \epsilon$, and it is a sum of terms $\epsilon^{-|\gamma|}(D^\gamma\chi)(x/\epsilon)\, x^\alpha D^{\beta-\gamma}\phi$, each bounded by $C\epsilon^{-|\gamma|}\epsilon^{k+1-|\beta|+|\gamma|} \le C\epsilon$. Hence $|u(\phi)| \le C\|\chi_\epsilon\phi\|_k \le C'\epsilon \to 0$, and $u(\phi) = 0$.

For general ϕ, the function $\phi - \chi\sum_{|\alpha|\le k} \partial^\alpha\phi(0)x^\alpha/\alpha!$ vanishes to order k at 0. So $u(\phi) = \sum_{|\alpha|\le k} a_\alpha \partial^\alpha\phi(0)$ with $a_\alpha = u(\chi x^\alpha)/\alpha!$. This is of the stated form, since $\partial^\alpha\phi(0) = i^{|\alpha|}D^\alpha\phi(0) = (-i)^{|\alpha|}(D^\alpha\delta)(\phi)$. The final statement follows from (1.5.1), since $\mathcal{F}$ is injective and $D^\alpha\delta$ has support $\{0\}$. □

Lemma 1.3.3 is a special case. If $x_j u = 0$ for all j, then u vanishes on $\mathbb{R}^n \setminus \{0\}$, because there $\phi = \sum_j x_j(x_j|x|^{-2}\phi)$ for $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n \setminus 0)$. So $\operatorname{supp} u \subset \{0\}$, and then $x_j \sum c_\alpha D^\alpha\delta = 0$ forces $c_\alpha = 0$ for $\alpha \ne 0$. Indeed, applying $\mathcal{F}$ gives $D_{\xi_j} \sum c_\alpha \xi^\alpha = 0$ for all j, so the polynomial is constant.

1.5.1 Constant coefficients

A differential operator with constant coefficients $P(D) = \sum_{|\alpha|\le m} c_\alpha D^\alpha$ acts on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$. By Corollary 1.4.4,

$$\widehat{P(D)u}(\xi) = P(\xi)\hat{u}(\xi), \qquad P(\xi) = \sum_{|\alpha|\le m} c_\alpha \xi^\alpha, \qquad u \in \mathcal{S}'(\mathbb{R}^n). \tag{1.5.2}$$

The polynomial $P(\xi)$ is the *(full) characteristic polynomial*, or symbol, of $P(D)$, and it determines $P(D)$. By the inversion formula, for $u \in \mathcal{S}(\mathbb{R}^n)$,

$$P(D)u(x) = \int e^{ix\cdot\xi} P(\xi)\hat{u}(\xi)đ\xi = \int \Big(\int e^{i(x-y)\cdot\xi} P(\xi)u(y)\,dy\Big)đ\xi. \tag{1.5.3}$$

The last integral is *iterated*: the y-integral is done first. The case $P \equiv 1$ is the inversion formula.

1.5.2 Variable coefficients

By the module property (1.1.4), a differential operator

$$P(x,D) = \sum_{|\alpha|\le m} p_\alpha(x) D^\alpha, \qquad p_\alpha \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n),$$

is continuous on $\mathcal{S}(\mathbb{R}^n)$ and, by (1.2.4), on $\mathcal{S}'(\mathbb{R}^n)$. Inserting the inversion formula for $D^\alpha\phi$ gives

$$P(x,D)\phi(x) = \int e^{ix\cdot\xi} P(x,\xi)\hat{\phi}(\xi)đ\xi = \int \Big(\int e^{i(x-y)\cdot\xi} P(x,\xi)\phi(y)\,dy\Big)đ\xi, \tag{1.5.4}$$

again as an iterated integral, where $P(x,\xi) = \sum_{|\alpha|\le m} p_\alpha(x)\xi^\alpha$ is the *(full) symbol* of $P(x,D)$. In the notation of Chapter 2, $P(x,D) = \mathrm{Op}(P)$. Pseudodifferential operators are obtained by replacing the polynomial $P(x,\xi)$ in (1.5.4) by more general symbols.

1.6 Two big theorems

We now prove two structure theorems. The first says that tempered distributions are "not too bad": they are finite sums of derivatives of bounded continuous functions multiplied by polynomials. The second, the Schwartz kernel theorem, says that every continuous linear operator $\mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$ is given by a distributional kernel. The kernel theorem was long considered hard to prove, but the Fourier transform gives a fairly direct proof. Both proofs rest on the same estimate.

1.6.1 A basic estimate

For $N, M \in \mathbb{N}_0$ consider the differential operator with polynomial coefficients

$$A_{N,M} = (1 + |D|^2)^M \langle x\rangle^{2N} \colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n), \qquad |D|^2 = D_1^2 + \cdots + D_n^2.$$

It is an isomorphism, with continuous inverse $A_{N,M}^{-1}\psi = \langle x\rangle^{-2N}\mathcal{G}\big(\langle\xi\rangle^{-2M}\hat{\psi}\big)$. This uses (1.5.2) and the fact that $\langle\cdot\rangle^{-2N}$ and $\langle\cdot\rangle^{-2M}$ are in $\mathcal{C}^\infty_{\mathrm{b}}$. Its transpose is $A_{N,M}^t = \langle x\rangle^{2N}(1 + |D|^2)^M$. For $s \in \mathbb{R}$ we use the weighted L^2 norm of the Fourier transform,

$$\|\psi\|_{(-s)} = \big\|\langle\xi\rangle^{-s}\hat{\psi}\big\|_{L^2}.$$

Lemma 1.6.1. *Let $k \in \mathbb{N}_0$ and choose $2N \geq k$ and $2M \geq k + 2n$. Then there is C such that*

$$\|A_{N,M}^{-1}\psi\|_k \leq C\|\psi\|_{(-n)}, \qquad \psi \in \mathcal{S}(\mathbb{R}^n).$$

Proof. Let $f = \mathcal{G}(\langle\xi\rangle^{-2M}\hat{\psi})$. For $|\alpha|+|\beta| \leq k$, the function $x^\alpha D^\beta(\langle x\rangle^{-2N} f)$ is a sum of terms $x^\alpha D^{\beta-\gamma}(\langle x\rangle^{-2N})D^\gamma f$, and $|x^\alpha D^\mu \langle x\rangle^{-2N}| \leq C\langle x\rangle^{|\alpha|-2N} \leq C$. Also, by the Cauchy–Schwarz inequality,

$$|D^\gamma f(x)| = \Big|\int e^{ix\cdot\xi}\xi^\gamma \langle\xi\rangle^{-2M}\hat{\psi}\,d\!\!\!^{-}\xi\Big| \leq (2\pi)^{-n}\big\|\langle\xi\rangle^{k+n-2M}\big\|_{L^2}\,\|\psi\|_{(-n)},$$

and $\langle\xi\rangle^{k+n-2M} \leq \langle\xi\rangle^{-n}$ is square integrable. □

Lemma 1.6.2. *Let $s > n/2$ and let $\mathcal{H}_s$ be the Hilbert space of measurable f on $\mathbb{R}^n$ with $\langle\xi\rangle^{-s} f \in L^2$, with inner product $(f,g)_s = \int\langle\xi\rangle^{-2s} f\bar{g}\,d\xi$. Then:*

(i) $\mathcal{S}(\mathbb{R}^n)$ is dense in $\mathcal{H}_s$. For each $x \in \mathbb{R}^n$, $e_x(\xi) = e^{-ix\cdot\xi}$ is in $\mathcal{H}_s$, and $x \mapsto e_x \in \mathcal{H}_s$ is bounded and continuous.

(ii) If $\lambda\colon \mathcal{H}_s \to \mathbb{C}$ is a bounded linear functional, then $w(x) = \lambda(e_x)$ is a bounded continuous function and $\lambda(\hat{\psi}) = \int w(x)\psi(x)\,dx$ for all $\psi \in \mathcal{S}(\mathbb{R}^n)$.

Proof. Multiplication by $\langle\xi\rangle^{-s}$ is a unitary map $\mathcal{H}_s \to L^2$ which maps $\mathcal{S}$ onto $\mathcal{S}$, so density follows from density of $\mathcal{S}$ in L^2. Next, $\|e_x\|_s^2 = \int\langle\xi\rangle^{-2s}d\xi < \infty$, since $2s > n$. Also $\|e_x - e_{x'}\|_s^2 = \int\langle\xi\rangle^{-2s}|e^{-ix\cdot\xi} - e^{-ix'\cdot\xi}|^2 d\xi \to 0$ as $x' \to x$, by dominated convergence. For (ii), the Riesz representation theorem gives $h \in \mathcal{H}_s$ with $\lambda(f) = \int\langle\xi\rangle^{-2s} f\bar{h}\,d\xi$. Then $g = \langle\xi\rangle^{-2s}\bar{h} = \langle\xi\rangle^{-s}\cdot\langle\xi\rangle^{-s}\bar{h}$ is a product of two L^2 functions, so $g \in L^1$. By Fubini,

$$\lambda(\hat{\psi}) = \int g(\xi)\int e^{-ix\cdot\xi}\psi(x)\,dx\,d\xi = \int\Big(\int g(\xi)e^{-ix\cdot\xi}d\xi\Big)\psi(x)\,dx,$$

and $\int g(\xi)e^{-ix\cdot\xi}d\xi = \lambda(e_x) = w(x)$. This is bounded by $\|g\|_{L^1}$, and continuous by (i). □

1.6.2 The structure theorem

Theorem 1.6.3 (Schwartz representation theorem)**.** *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfy $|u(\phi)| \leq C\|\phi\|_k$. Then there are finitely many bounded continuous functions $u_{\alpha\beta}$ such that*

$$u = \sum_{|\alpha|+|\beta|\leq 2k+2n+2} x^\beta D^\alpha u_{\alpha\beta}.$$

More precisely, $u = \langle x\rangle^{2N}(1+|D|^2)^M w$ with w bounded and continuous and N, M as in Lemma 1.6.1. Conversely, every such expression is in $\mathcal{S}'(\mathbb{R}^n)$.

Proof. Take $N = \lceil k/2\rceil$ and $M = \lceil k/2\rceil + n$, so that $2N + 2M \leq 2k + 2n + 2$, and write $A = A_{N,M}$. By Lemma 1.6.1, $|u(A^{-1}\psi)| \leq C'\|\psi\|_{(-n)}$. So $\lambda(\hat{\psi}) = u(A^{-1}\psi)$ is bounded on the dense subspace $\mathcal{S} \subset \mathcal{H}_n$ and extends to a bounded functional on $\mathcal{H}_n$. By Lemma 1.6.2 there is a bounded continuous w with $u(A^{-1}\psi) = \int w\psi\,dx$. Putting $\psi = A\phi$,

$$u(\phi) = \int w\,A\phi\,dx = (A^t w)(\phi) = \big(\langle x\rangle^{2N}(1+|D|^2)^M w\big)(\phi).$$

Expanding $\langle x\rangle^{2N}(1+|D|^2)^M$ by (1.0.1) gives the sum, with $u_{\alpha\beta}$ constant multiples of w. The converse was noted after (1.2.4). □

1.6.3 The Schwartz kernel theorem

If $K \in \mathcal{S}'(\mathbb{R}^{n+m})$, define $O_K \colon \mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$ by

$$O_K(\psi)(\phi) = K(\phi \boxtimes \psi), \qquad \phi \in \mathcal{S}(\mathbb{R}^n),\ \psi \in \mathcal{S}(\mathbb{R}^m). \tag{1.6.1}$$

Formally $O_K\psi(x) = \int K(x,y)\psi(y)\,dy$. If $|K(\Phi)| \le C\|\Phi\|_k$ then $|O_K(\psi)(\phi)| \le C\|\phi\|_k\|\psi\|_k$. Hence $O_K\psi \in \mathcal{S}'(\mathbb{R}^n)$, and O_K is continuous from $\mathcal{S}(\mathbb{R}^m)$ to $\mathcal{S}'(\mathbb{R}^n)$ with the weak topology. The kernel theorem is the converse. Two lemmas are needed first.

Lemma 1.6.4. *Let $B \colon \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^m) \to \mathbb{C}$ be bilinear and separately continuous. Then there are C and p with $|B(\phi,\psi)| \le C\|\phi\|_p\|\psi\|_p$.*

Proof. For $j \in \mathbb{N}$ let $E_j = \{\phi;\ |B(\phi,\psi)| \le j\|\psi\|_j\ \forall \psi \in \mathcal{S}(\mathbb{R}^m)\}$. Each E_j is closed, being an intersection of closed sets, since $B(\cdot,\psi)$ is continuous. The E_j cover $\mathcal{S}(\mathbb{R}^n)$, because each $B(\phi,\cdot)$ is a continuous functional and so satisfies an estimate (1.2.2). By the Baire category theorem and Proposition 1.1.2, some E_j contains a d-ball around some ϕ_0. By (1.1.3) it then contains $\{\phi_0 + \phi;\ \|\phi\|_q < \epsilon\}$ for some q and $\epsilon > 0$. For $\|\phi\|_q < \epsilon$, $|B(\phi,\psi)| \le |B(\phi_0+\phi,\psi)| + |B(\phi_0,\psi)| \le 2j\|\psi\|_j$. By homogeneity $|B(\phi,\psi)| \le 2j\epsilon^{-1}\|\phi\|_q\|\psi\|_j$, and we can take $p = \max(q,j)$. □

Lemma 1.6.5. *Finite sums of exterior products $\phi \boxtimes \psi$, $\phi \in \mathcal{S}(\mathbb{R}^n)$, $\psi \in \mathcal{S}(\mathbb{R}^m)$, are dense in $\mathcal{S}(\mathbb{R}^{n+m})$.*

Proof. By Lemma 1.1.3 it suffices to approximate $\Phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{n+m})$, say with support in the open cube $(-L,L)^{n+m}$. Extend Φ to a smooth function Φ_{per} that is $2L$-periodic in each variable. Its Fourier coefficients c_κ, $\kappa \in \mathbb{Z}^{n+m}$, decay rapidly (integrate by parts), so the Fourier series $\sum_\kappa c_\kappa e^{i\pi\kappa\cdot(x,y)/L}$ converges to Φ_{per} uniformly with all derivatives. Choose $\chi \in \mathcal{C}^\infty_{\mathrm{c}}((-L,L)^n)$ and $\chi' \in \mathcal{C}^\infty_{\mathrm{c}}((-L,L)^m)$ with $\chi\boxtimes\chi' = 1$ on $\operatorname{supp}\Phi$. Then $\chi\boxtimes\chi'\cdot\Phi_{\mathrm{per}} = \Phi$. Each term $\chi \boxtimes \chi' \cdot e^{i\pi\kappa\cdot(x,y)/L}$ is the exterior product of $\chi e^{i\pi\kappa'\cdot x/L}$ and $\chi' e^{i\pi\kappa''\cdot y/L}$, where $\kappa = (\kappa',\kappa'')$. The partial sums, multiplied by $\chi \boxtimes \chi'$, converge to Φ in every $\mathcal{C}^k$ norm with supports in a fixed compact set, and hence in $\mathcal{S}(\mathbb{R}^{n+m})$. □

Theorem 1.6.6 (Schwartz kernel theorem)**.** *For every linear map $A \colon \mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$ that is continuous for the weak topology on $\mathcal{S}'(\mathbb{R}^n)$, there is a unique $K \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $A = O_K$. Thus $K \mapsto O_K$ is a bijection between $\mathcal{S}'(\mathbb{R}^{n+m})$ and the continuous linear maps $\mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$. Moreover $K = \langle x\rangle^{2N}\langle y\rangle^{2N}(1+|D_x|^2)^M(1+|D_y|^2)^M k$ for some bounded continuous function k on $\mathbb{R}^{n+m}$ and some N, M.*

We call $K = K_A$ the *Schwartz kernel* of A, and write formally $A\psi(x) = \int K_A(x,y)\psi(y)\,dy$.

Proof. Uniqueness follows from Lemma 1.6.5. For existence, let $B(\phi,\psi) = (A\psi)(\phi)$. This is separately continuous: in ϕ because $A\psi \in \mathcal{S}'$, and in ψ by the weak continuity of A. By Lemma 1.6.4, $|B(\phi,\psi)| \le C\|\phi\|_p\|\psi\|_p$. Choose $2N \ge p$ and $2M \ge p + 2\max(n,m)$, and let $A_x = A_{N,M}$ on $\mathbb{R}^n$ and $A_y = A_{N,M}$ on $\mathbb{R}^m$. By Lemma 1.6.1, applied in each set of variables, $B'(\phi,\psi) = B(A_x^{-1}\phi, A_y^{-1}\psi)$ satisfies $|B'(\phi,\psi)| \le C'\|\phi\|_{(-n)}\|\psi\|_{(-m)}$. So $b(\hat\phi,\hat\psi) = B'(\phi,\psi)$ extends to a bounded bilinear form b on $\mathcal{H}_n(\mathbb{R}^n) \times \mathcal{H}_m(\mathbb{R}^m)$. Set

$$k(x,y) = b(e_x, e_y).$$

By Lemma 1.6.2(i) and the bilinear bound, k is bounded and continuous. For fixed ϕ, Lemma 1.6.2(ii) applied to $b(\hat\phi,\cdot)$ gives $B'(\phi,\psi) = \int b(\hat\phi,e_y)\psi(y)\,dy$. For fixed y, applied to

$b(\cdot, e_y)$, it gives $b(\hat{\phi}, e_y) = \int k(x,y)\phi(x)\,dx$. Hence $B'(\phi,\psi) = \iint k(x,y)\phi(x)\psi(y)\,dx\,dy$, and so

$$(A\psi)(\phi) = B'(A_x\phi, A_y\psi) = \iint k\; A_xA_y(\phi\boxtimes\psi)\,dx\,dy.$$

The operator A_xA_y is a differential operator with polynomial coefficients on $\mathbb{R}^{n+m}$, hence continuous on $\mathcal{S}(\mathbb{R}^{n+m})$, and k is bounded. So $K(\Phi) = \iint k\, A_xA_y\Phi$ defines $K \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $O_K = A$, and $K = (A_xA_y)^t k$ is of the stated form. □

1.6.4 Examples of kernels

Amongst tempered distributions, $\mathcal{S}(\mathbb{R}^n)$ consists of the "trivial" examples, and studying the singularities of tempered distributions amounts to studying the quotient $\mathcal{S}'(\mathbb{R}^n)/\mathcal{S}(\mathbb{R}^n)$. This could reasonably be called the space of *tempered microfunctions*. A typical singular distribution is δ. Its definition says that δ is the Schwartz kernel of the map $\mathcal{S}(\mathbb{R}^n) \ni \phi \mapsto \phi(0) \in \mathbb{C} = \mathcal{S}(\mathbb{R}^0)$. More generally, $\phi \mapsto D^\alpha\phi(0)$ has kernel $(-D)^\alpha\delta$, since $(-D)^\alpha\delta(\phi) = \delta(D^\alpha\phi)$. For the identity operator on $\mathcal{S}(\mathbb{R}^n)$, $K_{\mathrm{Id}}(\phi\boxtimes\psi) = \int\phi\psi\,dx$, so $K_{\mathrm{Id}} = \delta(x-y)$. This notation means the pull-back of $1\boxtimes\delta$ under the change of variables of Step 2 of Theorem 1.4.3; the proof of that theorem identified $K_{\mathcal{GF}} = \delta(x-y)$. If K_A is the kernel of A, then the kernel of $D^\alpha\circ A$ is $D_x^\alpha K_A$ and the kernel of $A\circ D^\alpha$ is $(-D_y)^\alpha K_A$. In particular the operator $P(x,D)$ of (1.5.4) has kernel

$$K_{P(x,D)}(x,y) = \sum_{|\alpha|\le m} p_\alpha(x)D_x^\alpha\delta(x-y) = \int e^{i(x-y)\cdot\xi}P(x,\xi)đ\xi.$$

The second expression is an oscillatory integral, interpreted via (1.5.4). The kernels of pseudodifferential operators in Chapter 2 are of the same form with more general symbols $P(x,\xi)$.

1.7 Radial compactification

A real vector space W of dimension n is a smooth manifold, but $\mathcal{C}^\infty(W)$ puts no restriction on behaviour at infinity. A good way to describe functions with controlled behaviour at infinity is to embed W as a dense open subset of a compact manifold X, possibly with boundary or corners, by a diffeomorphism onto its image. Restriction then identifies $\mathcal{C}^\infty(X)$ with a subspace of $\mathcal{C}^\infty(W)$, since a smooth function is determined by its values on a dense set. Several such compactifications exist. We construct them first for $\mathbb{R}^n$ and then check how they behave under linear (and affine) changes of coordinates. A compactification that is invariant under $\mathrm{GL}(n,\mathbb{R})$ is well defined for every vector space. For us the most important one is the *radial compactification*. Smooth functions on it are the classical symbols of order 0, and Schwartz functions are the smooth functions on it that vanish to infinite order at the boundary.

1.7.1 The one-point compactification and its defect

Identify $\mathbb{R}^n$ with the hyperplane $\{1\}\times\mathbb{R}^n \subset \mathbb{R}^{n+1}$, and let $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ be the unit sphere. The line through the south pole $(-1,0)$ and $(1,z)$ is $t \mapsto (2t-1, tz)$. It meets $\mathbb{S}^n$ again at $t = 4/(4+|z|^2)$, which gives the stereographic projection

$$P(z) = \Big(\frac{4-|z|^2}{4+|z|^2}, \frac{4z}{4+|z|^2}\Big), \qquad P^{-1}(Z_0,Z) = \frac{2Z}{1+Z_0}, \quad Z_0 > -1.$$

P is a diffeomorphism of $\mathbb{R}^n$ onto $\mathbb{S}^n$ minus the south pole. The *one-point compactification* ${}^1\overline{\mathbb{R}^n} = \mathbb{R}^n \cup \{\infty\}$ is $\mathbb{S}^n$ with this identification. The reflection $Z_0 \mapsto -Z_0$ corresponds to the inversion $z \mapsto \iota(z) = 4z/|z|^2$. Hence a function $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ extends smoothly to ${}^1\overline{\mathbb{R}^n}$ if and only if $f(\iota(w))$ extends smoothly across $w = 0$. Orthogonal transformations $O \in \mathrm{O}(n)$ lift to the diffeomorphisms $(Z_0, Z) \mapsto (Z_0, OZ)$. General linear transformations do not.

Proposition 1.7.1. *$G \in \mathrm{GL}(n, \mathbb{R})$ extends to a $\mathcal{C}^1$ map of ${}^1\overline{\mathbb{R}^n}$ fixing ∞ if and only if G is conformal, that is $G = cO$ with $c > 0$ and $O \in \mathrm{O}(n)$. In particular, for $n \geq 2$ the scaling $(z_1, z') \mapsto (sz_1, z')$, $s \neq 1$, does not extend.*

Proof. In the coordinate $w = \iota(z)$ near ∞, G becomes $h(w) = \iota(G\iota(w)) = |w|^2 Gw/|Gw|^2$ for $w \neq 0$, with $h(0) = 0$. If $G = cO$ then $h(w) = c^{-1}Ow$, which is smooth. Conversely, suppose h is differentiable at 0. Since $h(tw) = th(w)$ for $t > 0$, we get $h(w) = \lim_{t \downarrow 0} h(tw)/t = h'(0)w$, so h is linear, say $h(w) = Hw$. Then $G^{-1}Hw = q(w)w$ with $q(w) = |w|^2/|Gw|^2$. So every vector is an eigenvector of $G^{-1}H$, which is therefore a multiple cI of the identity. Hence $q \equiv c$, that is $|Gw|^2 = c^{-1}|w|^2$, and G is conformal. □

Thus the one-point compactification depends on a conformal structure on W, and not only on its linear structure (Exercise 1.14). It reappears in the proof of the index theorem, but it is not the right compactification for symbols.

1.7.2 The radial compactification

Instead of the south pole, project from the origin. The ray from 0 through $(1, z)$ meets the upper half of the unit sphere at the point

$$\mathrm{RC}\colon \mathbb{R}^n \ni z \longmapsto \Big(\frac{1}{\langle z\rangle}, \frac{z}{\langle z\rangle}\Big) \in \mathbb{S}^n_+ = \{X = (X_0, X') \in \mathbb{R}^{n+1};\ X_0 \geq 0,\ |X| = 1\}. \quad (1.7.1)$$

This is a diffeomorphism onto the open hemisphere $\{X_0 > 0\}$, with inverse $X \mapsto X'/X_0$. The closed hemisphere $\mathbb{S}^n_+$ is a compact manifold with boundary $\{X_0 = 0\} \cong \mathbb{S}^{n-1}$, diffeomorphic to a closed ball.

Definition 1.7.2. The *radial compactification* $\overline{\mathbb{R}^n}$ is $\mathbb{S}^n_+$ with $\mathbb{R}^n$ identified with its interior by RC. Its boundary is the *sphere at infinity*. $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$ denotes the functions f on $\mathbb{R}^n$ such that $f \circ \mathrm{RC}^{-1}$ extends to a smooth function on $\mathbb{S}^n_+$, smooth up to the boundary. We write $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n})$ for the subspace of those that vanish to infinite order at the boundary, with all derivatives.

The function $\rho = X_0$, pulled back to $\langle z\rangle^{-1}$, is a *boundary defining function*: it is smooth, positive in the interior, and vanishes simply on the boundary. Near the boundary there is a convenient product structure.

Lemma 1.7.3. *The map $z \mapsto (s, \omega) = (|z|^{-1}, z/|z|)$, $|z| > 1$, extends to a diffeomorphism of a neighbourhood $\{X_0 < |X'|\}$ of the boundary of $\overline{\mathbb{R}^n}$ onto $[0, 1)_s \times \mathbb{S}^{n-1}$. Hence $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ is in $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$ if and only if $f(z) = F(|z|^{-1}, z/|z|)$ in $|z| > 1$ for some $F \in \mathcal{C}^\infty([0, 1) \times \mathbb{S}^{n-1})$. Moreover*

$$\langle z\rangle^{-1} = s(1 + s^2)^{-1/2}, \qquad \frac{z}{\langle z\rangle} = (1 + s^2)^{-1/2}\omega$$

are in $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$.

Proof. In terms of X, $s = X_0/|X'|$ and $\omega = X'/|X'|$, which are smooth where $X' \neq 0$. The inverse is $X = (s, \omega)/(1 + s^2)^{1/2}$, which is smooth on $[0, \infty) \times \mathbb{S}^{n-1}$. The region $|z| > 1$ corresponds to $X_0 < |X'|$, that is $s < 1$. □

Remark 1.7.4. As a set, $\overline{\mathbb{R}^n} = \mathbb{R}^n \sqcup \mathbb{S}^{n-1}$, where the boundary point $(0,\omega)$ is the limit of $\mathrm{RC}(t\omega)$ as $t \to \infty$, the "end" of the ray in direction ω. Lemma 1.7.3 shows that the smooth structure is generated by the functions on $\mathbb{R}^n$ which, outside a compact set, are homogeneous of degree 0 (such as $z/|z|$), or homogeneous of degree -1 (such as $|z|^{-1}$). A function homogeneous of degree 0 for $|z| > 1$ extends to $\overline{\mathbb{R}^n}$ with boundary value its restriction to the unit sphere. A function homogeneous of negative integral degree $-k$ is s^k times such a function, and so vanishes at the boundary. For an abstract vector space W we set $\mathbb{S}W = (W \setminus 0)/\mathbb{R}^+$ and $\overline{W} = W \sqcup \mathbb{S}W$. The smooth structure is defined through any linear isomorphism $W \to \mathbb{R}^n$, and it does not depend on the choice by Proposition 1.7.5 below.

1.7.3 Invariance

Proposition 1.7.5. *Every affine map $T(z) = Gz + a$, $G \in \mathrm{GL}(n,\mathbb{R})$, $a \in \mathbb{R}^n$, extends to a diffeomorphism $\tilde{T}$ of $\overline{\mathbb{R}^n}$. Explicitly,*

$$\tilde{T}(X) = \frac{MX}{|MX|}, \qquad M(X_0, X') = (X_0, GX' + aX_0).$$

In particular the radial compactification $\overline{W}$ of any real vector space is well defined, and translations act on it smoothly.

Proof. M is an invertible linear map of $\mathbb{R}^{n+1}$ that preserves $\{X_0 > 0\}$ and $\{X_0 = 0\}$. If $X \in \mathbb{S}^n_+$ then $MX \neq 0$, so $\tilde{T}$ is a smooth map of a neighbourhood of $\mathbb{S}^n_+$ into $\mathbb{S}^n_+$. Its inverse is the same construction for M^{-1}, which corresponds to T^{-1}. Since $\mathrm{RC}(z)$ is a positive multiple of $(1,z)$ and $M(1,z) = (1, Gz + a)$, we get $\tilde{T}(\mathrm{RC}(z)) = \mathrm{RC}(Tz)$. □

The infinitesimal version is also useful. It shows which vector fields on $\mathbb{R}^n$ extend smoothly, and with what vanishing at infinity.

Lemma 1.7.6. *In the coordinates (s,ω) of Lemma 1.7.3,*

$$\partial_{z_j} = s\big(-\omega_j\, s\partial_s + W_j\big), \qquad z_i\partial_{z_j} = -\omega_i\omega_j\, s\partial_s + \omega_i W_j,$$

where W_j is the smooth vector field on $\mathbb{S}^{n-1}$ given by orthogonal projection of e_j onto the tangent space, $W_j = \sum_k (\delta_{jk} - \omega_j\omega_k)\partial_{\omega_k}$. Thus the linear vector fields $z_i\partial_{z_j}$ extend to smooth vector fields on $\overline{\mathbb{R}^n}$ tangent to the boundary. The constant vector fields ∂_{z_j} extend to ρ times smooth vector fields tangent to the boundary. Conversely, the radial field $s\partial_s = -\sum_j z_j\partial_{z_j}$ and the rotation fields $\Omega_{jk} = z_j\partial_{z_k} - z_k\partial_{z_j}$ span, over $\mathcal{C}^\infty([0,1)\times\mathbb{S}^{n-1})$, the smooth vector fields tangent to the boundary near the boundary.

Proof. From $s = |z|^{-1}$ and $\omega = z/|z|$: $\partial_{z_j} s = -z_j|z|^{-3} = -s^2\omega_j$ and $\partial_{z_j}\omega_k = s(\delta_{jk} - \omega_j\omega_k)$. This gives the first formula, and the second follows on multiplying by $z_i = \omega_i/s$. The fields Ω_{jk} are tangent to the spheres $|z| = \text{const}$, are homogeneous of degree 0, and span the tangent space of $\mathbb{S}^{n-1}$ at every point. □

1.7.4 Schwartz functions

Proposition 1.7.7. *Pull-back by* RC *identifies $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n})$ with $\mathcal{S}(\mathbb{R}^n)$, as sets and as Fréchet spaces.*

Proof. Both conditions are about behaviour in $|z| > 1$, so we work in the collar $[0,1)\times\mathbb{S}^{n-1}$ and write $f(z) = F(s,\omega)$.

Suppose $F \in \dot{\mathcal{C}}^\infty$. By Lemma 1.7.6, applying the first formula repeatedly and using $V(s^p g) = s^p(V - p\omega_j)g$ for $V = -\omega_j s\partial_s + W_j$, we get $\partial_z^\beta f = s^{|\beta|} Q_\beta F$, where Q_β is a differential operator with smooth coefficients on $[0,1) \times \mathbb{S}^{n-1}$. Also $z^\alpha = s^{-|\alpha|}\omega^\alpha$. Since F and all its derivatives vanish to infinite order at $s = 0$, $z^\alpha \partial^\beta f = s^{|\beta|-|\alpha|}\omega^\alpha Q_\beta F$ is bounded. So $f \in \mathcal{S}(\mathbb{R}^n)$.

Conversely, let $f \in \mathcal{S}(\mathbb{R}^n)$. Every differential operator with smooth coefficients on $[0,1) \times \mathbb{S}^{n-1}$ is a combination, with smooth coefficients, of products of ∂_s and the Ω_{jk}. On $s > 0$ we have $\partial_s = -s^{-1}\sum_j z_j \partial_{z_j} = -|z|\sum_j z_j \partial_{z_j}$, and Ω_{jk} has polynomial coefficients. Hence every derivative of F is a finite sum of smooth bounded coefficients times $|z|^p\, z^\alpha \partial_z^\beta f$, which is $O(|z|^{-N}) = O(s^N)$ for every N. A function on $(0,1) \times \mathbb{S}^{n-1}$ all of whose derivatives are $O(s^N)$ for all N extends by zero to an element of $\dot{\mathcal{C}}^\infty([0,1)\times\mathbb{S}^{n-1})$. Indeed, each derivative extends continuously by 0, and the difference quotient in s at $s = 0$ is $O(s^{N-1}) \to 0$, so by induction the extension is $\mathcal{C}^k$ for all k with vanishing derivatives at $s = 0$. The estimates in both directions are uniform, so the identification is a topological isomorphism. □

Hence the dual space $\mathcal{S}'(\mathbb{R}^n)$ is the dual of $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n})$. The dual of $\dot{\mathcal{C}}^\infty$ on a compact manifold with boundary is the space of *extendible distributions*, so tempered distributions on $\mathbb{R}^n$ are exactly the extendible distributions on $\overline{\mathbb{R}^n}$. This is one way of "bringing infinity in".

1.7.5 Symbols as functions on the radial compactification

Proposition 1.7.8. *Let $a \in \mathcal{C}^\infty(\overline{\mathbb{R}^n})$. Then:*

(i) $a \in \mathcal{C}^\infty(\mathbb{R}^n)$ and $|D^\alpha a(z)| \le C_\alpha \langle z\rangle^{-|\alpha|}$ for all α, so a is a symbol of order 0 in the sense of Chapter 2;

(ii) there are $a_j \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$ (the Taylor coefficients of a at $s = 0$) such that for every N

$$r_N(z) = a(z) - \sum_{j<N} |z|^{-j} a_j(z/|z|) \quad \text{satisfies} \quad |D^\alpha r_N(z)| \le C_{N,\alpha}|z|^{-N-|\alpha|}, \quad |z| > 1;$$

(iii) for each polynomial p of degree m, $\langle z\rangle^{-m} p \in \mathcal{C}^\infty(\overline{\mathbb{R}^n})$, with boundary value $p_m(\omega)$, where p_m is the homogeneous part of degree m.

Proof. For (i), the proof of Proposition 1.7.7 gives $\partial^\alpha a = s^{|\alpha|} Q_\alpha F$ in $|z| > 1$, with $Q_\alpha F$ bounded. Since $s \le 2\langle z\rangle^{-1}$ there, this gives the estimate. In $|z| \le 1$ it is trivial. For (ii), Taylor's formula in s gives $F(s,\omega) = \sum_{j<N} s^j a_j(\omega) + s^N R_N(s,\omega)$ with R_N smooth, and $\partial^\alpha(s^N R_N) = s^{N+|\alpha|} Q'_\alpha R_N$ as before. For (iii), $\langle z\rangle^{-m} z^\alpha = (z/\langle z\rangle)^\alpha \langle z\rangle^{|\alpha|-m}$ is a product of components of X' and the power $X_0^{m-|\alpha|}$, so it is smooth. At $X_0 = 0$ only the terms with $|\alpha| = m$ survive, and there $X' = \omega$. □

More generally, if $\rho \in \mathcal{C}^\infty(\overline{W})$ is any boundary defining function, the space

$$S^z_{\mathrm{ph}}(W) = \rho^{-z}\mathcal{C}^\infty(\overline{W}), \qquad z \in \mathbb{C},$$

is the space of *classical* (or *polyhomogeneous*) symbols of order z on W. It does not depend on the choice of ρ, since two boundary defining functions differ by a positive smooth factor. By Proposition 1.7.5 it is invariant under affine changes of variables. Part (ii) of Proposition 1.7.8 is the asymptotic expansion in homogeneous terms of decreasing degree that defines polyhomogeneous symbols in Chapter 2. Using Borel's lemma one shows there that, conversely, every symbol of order 0 with such an expansion is in $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$.

1.8 Further compactifications

1.8.1 The quadratic compactification

Projecting the hemisphere $\mathbb{S}^n_+$ onto its last n coordinates gives a homeomorphism onto the closed unit ball $\mathbb{B}^n$. Composing with RC gives the *quadratic compactification*

$$\mathrm{Q}\colon \mathbb{R}^n \ni z \longmapsto \frac{z}{\langle z\rangle} \in \mathbb{B}^n = \{y \in \mathbb{R}^n;\ |y| \le 1\}, \tag{1.8.1}$$

a diffeomorphism onto the open ball. We write ${}^{q}\overline{\mathbb{R}^n}$ for $\mathbb{B}^n$ with this identification. The projection $\mathbb{S}^n_+ \to \mathbb{B}^n$ is smooth, but its inverse $y \mapsto ((1-|y|^2)^{1/2}, y)$ is not smooth at $|y| = 1$. So the identity of $\mathbb{R}^n$ extends to a smooth map $\overline{\mathbb{R}^n} \to {}^{q}\overline{\mathbb{R}^n}$ but not conversely, and $\mathcal{C}^\infty({}^{q}\overline{\mathbb{R}^n}) \subsetneq \mathcal{C}^\infty(\overline{\mathbb{R}^n})$. Concretely, $\langle z\rangle^{-1} = (1-|y|^2)^{1/2}$ is not smooth on ${}^{q}\overline{\mathbb{R}^n}$. The function $1-|y|^2 = \langle z\rangle^{-2}$ is a boundary defining function there, and near the boundary $(t,\omega) = (|z|^{-2}, z/|z|)$ are smooth coordinates, which explains the name.

Proposition 1.8.1. *(i) Pull-back by* Q *identifies* $\dot{\mathcal{C}}^\infty({}^{q}\overline{\mathbb{R}^n})$*, the smooth functions on* $\mathbb{R}^n$ *supported in* $\mathbb{B}^n$*, with* $\mathcal{S}(\mathbb{R}^n)$*.*

(ii) Every $G \in \mathrm{GL}(n,\mathbb{R})$ *extends to a diffeomorphism of* ${}^{q}\overline{\mathbb{R}^n}$*, given by* $y \mapsto Gy/(1-|y|^2 + |Gy|^2)^{1/2}$*.*

(iii) For $n \ge 2$ *and* $a \ne 0$*, the translation* $z \mapsto z + a$ *does not extend to a* $\mathcal{C}^1$ *map of* ${}^{q}\overline{\mathbb{R}^n}$*.*

Proof. (i) Let $\pi(X) = X'$. If $g \in \dot{\mathcal{C}}^\infty(\mathbb{B}^n)$ then $g\circ\pi \in \dot{\mathcal{C}}^\infty(\mathbb{S}^n_+)$, because π is smooth and maps the boundary to the boundary. Conversely, let $f \in \dot{\mathcal{C}}^\infty(\mathbb{S}^n_+)$ and $g = f\circ\pi^{-1}$. A derivative of order k of g is a sum of products of derivatives of f at $\pi^{-1}(y)$ with derivatives of $X_0 = (1-|y|^2)^{1/2}$ of order at most k, which are $O(X_0^{1-2k})$. The derivatives of f are $O(X_0^N)$ for all N. So all derivatives of g tend to 0 at $|y| = 1$, and g extended by 0 is smooth. Now apply Proposition 1.7.7.

(ii) With $y = z/\langle z\rangle$ we have $\langle z\rangle^2 = (1-|y|^2)^{-1}$, so $Gz/\langle Gz\rangle = Gy/(1-|y|^2+|Gy|^2)^{1/2}$. The denominator is positive on $\mathbb{B}^n$, since $Gy \ne 0$ when $|y| = 1$, and the inverse is given by G^{-1}.

(iii) Let $\rho = (1-|y|^2)^{1/2} = \langle z\rangle^{-1}$. Since $\langle z+a\rangle^2 = \langle z\rangle^2 + 2z\cdot a + |a|^2$, the translation becomes

$$y \longmapsto \frac{y+\rho a}{(1+2\rho\, y\cdot a + \rho^2|a|^2)^{1/2}} = y + \rho\big(a - (y\cdot a)y\big) + \rho^2 h(y,\rho),$$

with h smooth in (y,ρ). Choose a unit vector $y_0 \perp a$, which is possible as $n \ge 2$, and set $e = a/|a|$. Along $y = ty_0$, $t \uparrow 1$, the e-component of the image is $|a|\rho + \rho^2 h_e$ with $\rho = (1-t^2)^{1/2}$. Its derivative in t is unbounded as $t \to 1$, so the map is not $\mathcal{C}^1$ up to the boundary. □

Thus the quadratic compactification is well defined for a vector space, but not for an affine space. In terms of vector fields, near the boundary $z_i\partial_{z_j} = -2\omega_i\omega_j\, t\partial_t + \omega_i W_j$ is smooth, while $\partial_{z_j} = t^{1/2}(-2\omega_j\, t\partial_t + W_j)$ is not. This is one reason to prefer the radial compactification, on which translations act smoothly. The other is Proposition 1.7.8: classical symbols with terms of odd degree, such as $\langle z\rangle^{-1}$, are smooth on $\overline{\mathbb{R}^n}$ but not on ${}^{q}\overline{\mathbb{R}^n}$.

1.8.2 Relation to the one-point compactification

In the coordinate $w = \iota(z) = 4z/|z|^2$ near $\infty \in {}^1\overline{\mathbb{R}^n}$ we have $w = 4s\omega$ in terms of the coordinates of Lemma 1.7.3, and this is smooth on $[0,1) \times \mathbb{S}^{n-1}$. So the identity extends to a smooth map

$$\beta\colon \overline{\mathbb{R}^n} \longrightarrow {}^1\overline{\mathbb{R}^n},$$

which collapses the sphere at infinity to the point ∞ and is a diffeomorphism elsewhere. In particular $\mathcal{C}^\infty({}^1\overline{\mathbb{R}^n}) \subset \mathcal{C}^\infty(\overline{\mathbb{R}^n})$. Near the boundary, β is the introduction of polar coordinates $(s,\omega) \mapsto s\omega$ around ∞. In the language of later chapters, $\overline{\mathbb{R}^n}$ is the *blow-up* $[{}^1\overline{\mathbb{R}^n}, \{\infty\}]$. In general, a manifold with boundary X together with a smooth map $\beta\colon X \to M$ is the blow-up of M at p if β is a diffeomorphism of $X \setminus \partial X$ onto $M \setminus \{p\}$, maps ∂X to p, and polar coordinates around p (in some local coordinates) lift to a diffeomorphism of a neighbourhood of ∂X onto $[0,\epsilon) \times \mathbb{S}^{n-1}$. This does not depend on the choice of local coordinates (Exercise 1.19).

1.8.3 Subspaces and relative compactifications

Proposition 1.8.2. *If $V \subset W$ is a linear subspace, then the inclusion extends to a smooth embedding $\overline{V} \hookrightarrow \overline{W}$, and likewise ${}^q\overline{V} \hookrightarrow {}^q\overline{W}$. Every injective linear map $V \to W$ extends to a smooth map $\overline{V} \to \overline{W}$.*

Proof. By Proposition 1.7.5 we may choose coordinates with $V = \mathbb{R}^k \times \{0\}$. Then $\mathrm{RC}_n(v,0) = (\langle v\rangle^{-1}, v/\langle v\rangle, 0)$, which is the standard embedding $\mathbb{S}^k_+ \hookrightarrow \mathbb{S}^n_+$ composed with RC_k. The quadratic case is the same. An injective linear map is an isomorphism onto its image followed by an inclusion. □

Linear maps that are not injective do not extend in general (Exercise 1.16): the points at infinity "do not know where to go". The remedy is to compactify relative to a subspace. Let $V \subset W$ and choose a complement U, so $W = V \times U$, together with Euclidean metrics on V and U. Consider

$$R_V\colon W \ni (v,u) \longmapsto \big(\mathrm{RC}_U(u),\ \mathrm{RC}_V(v/\langle u\rangle)\big) \in \overline{U} \times \overline{V}. \tag{1.8.2}$$

Explicitly, $R_V(v,u) = (t,u',s,v')$ with

$$t = \frac{1}{\langle u\rangle}, \quad u' = \frac{u}{\langle u\rangle}, \quad s = \frac{\langle u\rangle}{(1+|u|^2+|v|^2)^{1/2}}, \quad v' = \frac{v}{(1+|u|^2+|v|^2)^{1/2}},$$

satisfying $t^2+|u'|^2 = 1 = s^2+|v'|^2$. Here we used $\langle v/\langle u\rangle\rangle = (1+|u|^2+|v|^2)^{1/2}/\langle u\rangle$. The map $(v,u) \mapsto (v/\langle u\rangle, u)$ is a diffeomorphism of W, so R_V is a diffeomorphism onto the interior of the compact manifold with corners $\overline{U} \times \overline{V}$, with inverse $(t,u',s,v') \mapsto (v'/(st), u'/t)$. We write ${}^V\overline{W}$ for $\overline{U} \times \overline{V}$ with this identification, and call it the *radial compactification of W relative to V*.

Proposition 1.8.3. *Every affine map of W whose linear part maps V into itself extends to a diffeomorphism of ${}^V\overline{W}$. In particular ${}^V\overline{W}$ does not depend on the choice of complement U or of the metrics. The short exact sequence $0 \to V \to W \to W/V \to 0$ extends to smooth maps*

$$\overline{V} \longrightarrow {}^V\overline{W} \longrightarrow \overline{W/V}.$$

If $A\colon V \to W$ is any linear map with null space N, then A extends to a smooth map ${}^N\overline{V} \to \overline{W}$.

Proof. An affine map preserving the direction V has the form $T(v,u) = (Gv+Su+a,\ Hu+b)$, with $G \in \mathrm{GL}(V)$, $H \in \mathrm{GL}(U)$, $S\colon U \to V$ linear, $a \in V$ and $b \in U$. In the coordinates $(\nu, u) = (v/\langle u\rangle, u)$ it becomes

$$(\nu,u) \longmapsto \Big(c(u)(G\nu + S\tfrac{u}{\langle u\rangle} + a\langle u\rangle^{-1}),\ Hu+b\Big), \qquad c(u) = \frac{\langle u\rangle}{\langle Hu+b\rangle}.$$

By Proposition 1.7.5, $u \mapsto Hu+b$ extends to a diffeomorphism of $\overline{U}$, and $\langle Hu+b\rangle^{-1}$ is the pull-back of the boundary defining function $\langle u\rangle^{-1}$ under this map. Hence c is a positive function in $\mathcal{C}^\infty(\overline{U})$. The functions $u/\langle u\rangle$ and $\langle u\rangle^{-1}$ are also in $\mathcal{C}^\infty(\overline{U})$. Thus the first component is an affine map of V, with invertible linear part, depending smoothly on the parameter $p \in \overline{U}$. The formula of Proposition 1.7.5, with M depending smoothly on p, shows that it extends smoothly to $\overline{U} \times \overline{V}$. The same holds for the inverse. Changing the complement or the metrics is such a linear map, so ${}^V\overline{W}$ does not depend on these choices.

The inclusion $v \mapsto (v,0)$ becomes $v \mapsto (\mathrm{RC}_U(0), \mathrm{RC}_V(v))$, which extends to $\overline{V} \cong \{\mathrm{RC}_U(0)\} \times \overline{V}$. The projection $(v,u) \mapsto u$, with $U \cong W/V$, extends as the projection $\overline{U} \times \overline{V} \to \overline{U}$. Finally, write A as the projection $V \to V/N$ followed by the injective map $V/N \to W$ induced by A, and use the first part of this proposition together with Proposition 1.8.2. □

Remark 1.8.4. In the trivial cases $V = \{0\}$ and $V = W$ the relative compactification reduces to $\overline{W}$ (Exercise 1.17). For a non-trivial subspace, ${}^V\overline{W}$ is canonically the blow-up $[\overline{W}, \mathbb{S}V]$ of $\overline{W}$ along the sphere at infinity of V, which is a submanifold of the boundary. Similarly, for a product $W = V \times U$ the identity map extends to neither $\overline{V} \times \overline{U} \to \overline{W}$ nor its inverse. However, the doubly-relative compactification (Exercise 1.18) is canonically both $[\overline{V \times U}; \mathbb{S}U, \mathbb{S}V]$ and $[\overline{V} \times \overline{U}; \mathbb{S}V \times \mathbb{S}U]$. We do not prove these identifications here. Blow-up of submanifolds is taken up when it is needed later in the book.

1.9 Exercises

Exercise 1.1. Suppose $\phi\colon \mathbb{R}^n \to \mathbb{C}$ is such that for each $\bar{x} \in \mathbb{R}^n$ and each $k \in \mathbb{N}_0$ there are $\epsilon_k > 0$ and a polynomial $p_k(x;\bar{x})$ with $|\phi(x) - p_k(x;\bar{x})| \le \epsilon_k^{-1}|x-\bar{x}|^{k+1}$ for $|x-\bar{x}| \le \epsilon_k$. Must ϕ be infinitely differentiable? (Consider $\phi(x) = e^{-1/x^2}\sin(e^{1/x^2})$ on $\mathbb{R}$.)

Exercise 1.2. Show that $u(x) = e^x\cos(e^x)$ is a tempered distribution on $\mathbb{R}$, in the sense that there is $T \in \mathcal{S}'(\mathbb{R})$ with $T(\phi) = \int u\phi\,dx$ for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$. Show that T is unique. (Write u as the derivative of a bounded function.) Note that $\int u\phi\,dx$ need not converge absolutely for $\phi \in \mathcal{S}(\mathbb{R})$.

Exercise 1.3. Show that d in (1.1.2) is a metric on $\mathcal{S}(\mathbb{R}^n)$. (If $\|\cdot\|$ is a norm, show that $\frac{\|u+v\|}{1+\|u+v\|} \le \frac{\|u\|}{1+\|u\|} + \frac{\|v\|}{1+\|v\|}$.)

Exercise 1.4. Show that a sequence $\phi_j \in \mathcal{S}(\mathbb{R}^n)$ is Cauchy, respectively converges to ϕ, for d if and only if it is Cauchy, respectively converges to ϕ, for each norm $\|\cdot\|_k$.

Exercise 1.5. Show that a bilinear map $B\colon \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^p) \to \mathbb{C}$ is (jointly) continuous if and only if $|B(\phi,\psi)| \le C\|\phi\|_N\|\psi\|_N$ for some C and N. Combined with Lemma 1.6.4, this shows that separate continuity implies joint continuity.

Exercise 1.6. Show that the restriction map $\pi_k^*\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^k)$ is continuous and surjective.

Exercise 1.7. Give a second proof of Lemma 1.4.2. For $n = 1$ write $\int e^{-ix\xi}e^{-x^2}dx = e^{-\xi^2/4}\int e^{-(x+\frac{i}{2}\xi)^2}dx$, interpret the integral as a contour integral, and shift the contour to $\operatorname{Im} x = -\xi/2$.

Exercise 1.8. Show that (1.2.1) makes sense for $\phi \in L^2(\mathbb{R}^n)$, and that the resulting map $L^2(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is injective.

Exercise 1.9. (A weak Sobolev embedding.)

(a) Show that if $f \in L^1(\mathbb{R}^n)$ then $v(x) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(t)\,dt$ is bounded and continuous and $D_1 \cdots D_n v = (-i)^n f$ in $\mathcal{S}'(\mathbb{R}^n)$. Show that $(1+|x|)^{-n-1}L^2(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$.

(b) Show that if $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\langle\xi\rangle^s\hat{u} \in L^2(\mathbb{R}^n)$ for some $s > n/2 + k$, then $u \in \mathcal{C}^k(\mathbb{R}^n)$ and $D^\alpha u$ is bounded and continuous for $|\alpha| \le k$.

Exercise 1.10. Show that for $\phi \in \mathcal{S}(\mathbb{R}^n)$ (indeed for ϕ continuous and polynomially bounded), the support of T_ϕ in the sense of Definition 1.2.4 is the closure of $\{x;\ \phi(x) \neq 0\}$. Show that $\operatorname{supp}\delta = \{0\}$.

Exercise 1.11. Show that a differential operator P with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is *local*: $\operatorname{supp}(Pu) \subset \operatorname{supp} u$ for all $u \in \mathcal{S}'(\mathbb{R}^n)$. The converse, for a linear operator $P\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ with $\operatorname{supp}(Pu) \subset \operatorname{supp} u$ and (for simplicity) $\operatorname{supp}(Pu) \subset K$ for a fixed compact K, is Peetre's theorem. How would you try to prove it? (No full proof required.)

Exercise 1.12. Find a bounded continuous function w on $\mathbb{R}$ with $\delta = (1 + D^2)w$, as predicted by Theorem 1.6.3. (Try $w = \frac{1}{2}e^{-|x|}$.) Can δ be written as Dw with w bounded and continuous?

Exercise 1.13. Use Theorem 1.6.6 to show that for $u \in \mathcal{S}'(\mathbb{R}^n)$ and $v \in \mathcal{S}'(\mathbb{R}^m)$ there is a unique $u \boxtimes v \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $(u \boxtimes v)(\phi \boxtimes \psi) = u(\phi)v(\psi)$.

Exercise 1.14. Formulate precisely the notion of a conformal structure on a real vector space W (a Euclidean metric up to positive multiples). Show that the one-point compactification of W depends only on such a structure. If $V \subset W$ carries the induced conformal structure, show that the inclusion extends to a smooth map ${}^1\overline{V} \to {}^1\overline{W}$.

Exercise 1.15. Show that the quadratic compactification can be defined for any real vector space, and that as a set it is again $W \sqcup \mathbb{S}W$, with a different smooth structure. (Use Proposition 1.8.1(ii).)

Exercise 1.16. Show that the projection $\mathbb{R}^2 \ni (z_1, z_2) \mapsto z_1 \in \mathbb{R}$ does not extend to a continuous map $\overline{\mathbb{R}^2} \to \overline{\mathbb{R}}$. (Consider the curves $z = (c, t)$, $t \to \infty$.) Show more generally that a linear map between vector spaces of positive dimension extends continuously to the radial (or quadratic) compactifications if and only if it is injective.

Exercise 1.17. Show from (1.8.2) that ${}^{\{0\}}\overline{W}$ and ${}^{W}\overline{W}$ are both canonically $\overline{W}$.

Exercise 1.18. (Doubly-relative compactification.) For $W = V \times U$ with metrics on V and U, consider

$$W \ni (v,u) \longmapsto \Big(\frac{1}{\langle u\rangle}, \frac{1}{\langle v\rangle}, \frac{\langle u\rangle}{\langle (u,v)\rangle}, \frac{\langle v\rangle}{\langle (u,v)\rangle}, \frac{v}{\langle v\rangle}, \frac{(v,u)}{\langle (u,v)\rangle}, \frac{u}{\langle u\rangle}\Big) \in \mathbb{R}^4 \times V \times W \times U.$$

Show that this is a diffeomorphism onto its image, and that the closure of the image is a compact manifold with corners. It has three boundary hypersurfaces if $\dim U, \dim V \ge 2$. Denote it by ${}^{U,V}\overline{U \times V}$. Show that translations, and linear maps preserving both U and V, lift to diffeomorphisms. Show that the inclusions of $\overline{U}$ and $\overline{V}$ extend smoothly, and that the identity of W extends to smooth maps from ${}^{U,V}\overline{U \times V}$ to both $\overline{U \times V}$ and $\overline{U} \times \overline{V}$. Draw a picture for $\dim U = \dim V = 1$.

Exercise 1.19. Show that a change of local coordinates fixing $0 \in \mathbb{R}^n$ induces a diffeomorphism of $[0, \epsilon) \times \mathbb{S}^{n-1}$, for some $\epsilon > 0$, under polar coordinates $(r, \omega) \mapsto r\omega$. (Treat linear changes first, using Proposition 1.7.5. Then reduce to the case where the Jacobian at 0 is the identity.) Conclude that the blow-up of a point is well defined.

Notes

This chapter merges Chapter 1 of *Introduction to Microlocal Analysis* with Lecture 1 (and its addenda) of *Lectures on Pseudodifferential Operators*. The "two big theorems" were only stated in the original, with outlines of proofs in the problems. Here they are proved in full, using a single estimate (Lemma 1.6.1) and a Riesz representation argument in place of the Hilbert–Schmidt argument suggested originally. Separate continuity is upgraded to joint continuity by a Baire category argument, and uniqueness of the kernel uses the density of tensor products. The classification of distributions supported at a point is proved in all dimensions. Both compactification discussions have been unified around the explicit formulae (1.7.1) and Proposition 1.7.5, which replace the Lie algebra arguments of the original. The relative compactification is written as a product $\overline{U} \times \overline{V}$ by the change of variables $v \mapsto v/\langle u \rangle$. For the kernel theorem and the structure of distributions see [7, Chapter 5 and §7.1] and [3]. Compactifications, blow-ups and manifolds with corners are treated systematically in Melrose's work on the *b*-calculus, see [16].

Chapter 2

Pseudodifferential operators on Euclidean space

This chapter constructs the basic algebra of pseudodifferential operators on $\mathbb{R}^n$, the "traditional" calculus with symbols of type $(1,0)$ whose coefficients are uniformly smooth in the base variables. The starting point is the Fourier representation of a differential operator with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$,

$$P(x,D)u(x) = \int e^{i(x-y)\cdot\xi} P(x,\xi)u(y)\,dy đ\xi = \int e^{ix\cdot\xi} P(x,\xi)\hat{u}(\xi) đ\xi, \qquad u \in \mathcal{S}(\mathbb{R}^n),$$

where $P(x,\xi) = \sum_{|\alpha|\le m} p_\alpha(x)\xi^\alpha$ is the characteristic polynomial. We generalize this formula by replacing the polynomial $P(x,\xi)$ by a *symbol*, a smooth function satisfying uniform estimates at infinity, and we even allow it to depend on the "incoming" variable y, so that operators are given by

$$Au(x) = \int e^{i(x-y)\cdot\xi} a(x,y,\xi)u(y)\,dy đ\xi, \qquad u \in \mathcal{S}(\mathbb{R}^n). \tag{2.0.1}$$

Such integrals are in general not absolutely convergent. Section 2.1 develops the symbol spaces, including asymptotic summation. In Section 2.2 the integral (2.0.1) is interpreted as an oscillatory integral and shown to define a continuous operator on $\mathcal{S}(\mathbb{R}^n)$. Section 2.3 shows that every such operator can be written with an amplitude independent of y (reduction to the left symbol), and Section 2.4 deduces from this that the operators form an order-filtered $*$-algebra. The remaining sections develop the consequences: principal symbols, ellipticity and parametrices, elliptic regularity, boundedness on L^2 and on (weighted) Sobolev spaces, polyhomogeneous (classical) operators and invariance under affine changes of variables. Coordinate invariance, and hence operators on manifolds, is the subject of Chapters 3 and 5.

Throughout, $\langle\xi\rangle = (1+|\xi|^2)^{1/2}$, $D = -i\partial$, $đ\xi = (2\pi)^{-n}d\xi$, and the Schwartz space carries the norms $\|\phi\|_k = \max_{|\alpha|+|\beta|\le k} \sup_x |x^\alpha D^\beta \phi(x)|$ of Chapter 1. The bilinear pairing of a distribution u with a test function ϕ is written $u(\phi)$ or $\langle u,\phi\rangle$, and $(u,v) = \int u\bar{v}\,dx$ is the L^2 inner product.

2.1 Symbols

A polynomial $p(\xi)$ of degree at most m satisfies $|p(\xi)| \le C\langle\xi\rangle^m$, and since $D^\alpha_\xi p$ has degree at most $m-|\alpha|$, also $|D^\alpha_\xi p(\xi)| \le C_\alpha\langle\xi\rangle^{m-|\alpha|}$. If $P(x,\xi)$ is the characteristic polynomial of a differential operator of order m with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ (all derivatives bounded) then

$$|D^\alpha_x D^\beta_\xi P(x,\xi)| \le C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|} \quad \text{on } \mathbb{R}^n\times\mathbb{R}^n,\ \alpha,\beta\in\mathbb{N}_0^n.$$

There is no reason to have the same number of base and fibre variables, so in general we define:

Definition 2.1.1. For $m \in \mathbb{R}$ the space $S^m(\mathbb{R}^p;\mathbb{R}^n)$ of *symbols of order* m consists of the functions $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^n)$ satisfying

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| \le C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|} \quad \text{on } \mathbb{R}^p\times\mathbb{R}^n, \text{ for all } \alpha\in\mathbb{N}_0^p,\ \beta\in\mathbb{N}_0^n. \tag{2.1.1}$$

It is given the topology of the norms

$$\|a\|_{N,m} = \max_{|\alpha|+|\beta|\le N}\ \sup_{(z,\xi)\in\mathbb{R}^p\times\mathbb{R}^n} \langle\xi\rangle^{-m+|\beta|}|D_z^\alpha D_\xi^\beta a(z,\xi)|, \qquad N\in\mathbb{N}_0.$$

We set $S^\infty(\mathbb{R}^p;\mathbb{R}^n) = \bigcup_m S^m(\mathbb{R}^p;\mathbb{R}^n)$ and $S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n) = \bigcap_m S^m(\mathbb{R}^p;\mathbb{R}^n)$, the latter with the topology given by all the norms $\|\cdot\|_{N,m}$, $N\in\mathbb{N}_0$, $m\in\mathbb{R}$; its elements are called *residual* symbols.

Remark 2.1.2. The estimates are uniform in the "base" variables z; this is the analogue of the condition that the coefficients be in $\mathcal{C}_{\mathrm{b}}^\infty$. For an open set $\Omega\subset\mathbb{R}^p$ one defines $S^m(\Omega;\mathbb{R}^n)$ in the same way, requiring (2.1.1) on $\Omega\times\mathbb{R}^n$. The class of symbols is not the only reasonable one (see Exercise 2.9), but it is large enough to contain everything we need at first and small enough for the calculus to work easily.

Since $\langle\xi\rangle^m \le \langle\xi\rangle^{m'}$ for $m\le m'$ we have $\|a\|_{N,m'} \le \|a\|_{N,m}$, so there are continuous inclusions $S^m(\mathbb{R}^p;\mathbb{R}^n)\hookrightarrow S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ for $m\le m'$; the symbol spaces form a filtration of $S^\infty(\mathbb{R}^p;\mathbb{R}^n)$. A residual symbol is a smooth function all of whose derivatives are bounded by $C_{\alpha\beta N}\langle\xi\rangle^{-N}$ for every N.

We shall use repeatedly the following elementary inequality.

Lemma 2.1.3 (Peetre's inequality)**.** *For all $x,y\in\mathbb{R}^n$ and $s\in\mathbb{R}$,*

$$\langle x+y\rangle^s \le 2^{|s|/2}\langle x\rangle^s\langle y\rangle^{|s|}.$$

Proof. Since $|x+y|^2\le 2|x|^2+2|y|^2$, $1+|x+y|^2\le 2(1+|x|^2)(1+|y|^2)$, i.e. $\langle x+y\rangle\le\sqrt{2}\langle x\rangle\langle y\rangle$. This gives the case $s\ge 0$. Applied to $x=(x+y)+(-y)$ it gives $\langle x\rangle\le\sqrt{2}\langle x+y\rangle\langle y\rangle$, i.e. $\langle x+y\rangle^{-1}\le\sqrt{2}\langle x\rangle^{-1}\langle y\rangle$, and raising this to the power $|s|$ gives the case $s<0$. □

Lemma 2.1.4. *Let $m,m'\in\mathbb{R}$.*

(i) $\langle\xi\rangle^m \in S^m(\mathbb{R}^p;\mathbb{R}^n)$.

(ii) *Multiplication defines a continuous bilinear map*

$$S^m(\mathbb{R}^p;\mathbb{R}^n)\times S^{m'}(\mathbb{R}^p;\mathbb{R}^n)\to S^{m+m'}(\mathbb{R}^p;\mathbb{R}^n).$$

(iii) *D_z^α maps $S^m(\mathbb{R}^p;\mathbb{R}^n)$ continuously into itself and D_ξ^β maps it continuously into $S^{m-|\beta|}(\mathbb{R}^p;\mathbb{R}^n)$.*

(iv) *$S^m(\mathbb{R}^p;\mathbb{R}^n)$ is a Fréchet space.*

(v) *If $a\in\mathcal{C}^\infty(\mathbb{R}^p\times\mathbb{R}^n)$ has all derivatives bounded and $a(z,\xi)=0$ for $|\xi|\ge R$, then $a\in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$.*

Proof. (i) By induction on $|\beta|$, $D_\xi^\beta \langle\xi\rangle^m = \langle\xi\rangle^{m-2|\beta|} P_\beta(\xi)$ with P_β a polynomial of degree at most $|\beta|$: indeed $D_{\xi_j}(\langle\xi\rangle^s P) = \langle\xi\rangle^{s-2}(-is\xi_j P + \langle\xi\rangle^2 D_{\xi_j} P)$ and the polynomial in brackets has degree at most $\deg P + 1$. Hence $|D_\xi^\beta \langle\xi\rangle^m| \le C_\beta \langle\xi\rangle^{m-|\beta|}$.

(ii) By Leibniz' formula,

$$\langle\xi\rangle^{-m-m'+|\beta|} |D_z^\alpha D_\xi^\beta (ab)| \le \sum_{\mu\le\alpha,\ \gamma\le\beta} \binom{\alpha}{\mu}\binom{\beta}{\gamma} \langle\xi\rangle^{-m+|\gamma|} |D_z^\mu D_\xi^\gamma a|\, \langle\xi\rangle^{-m'+|\beta-\gamma|} |D_z^{\alpha-\mu} D_\xi^{\beta-\gamma} b|,$$

so $\|ab\|_{N,m+m'} \le C_N \|a\|_{N,m} \|b\|_{N,m'}$.

(iii) is immediate from the definition: $\|D_z^\alpha a\|_{N,m} \le \|a\|_{N+|\alpha|,m}$ and $\|D_\xi^\beta a\|_{N,m-|\beta|} \le \|a\|_{N+|\beta|,m}$.

(iv) If (a_k) is Cauchy in S^m then, for each α, β, $\langle\xi\rangle^{-m+|\beta|} D_z^\alpha D_\xi^\beta a_k$ converges uniformly. Hence $a_k \to a$ in $\mathcal{C}^\infty$ with all derivatives converging locally uniformly, and the uniform limits are $\langle\xi\rangle^{-m+|\beta|} D_z^\alpha D_\xi^\beta a$; it follows that $a \in S^m$ and $a_k \to a$ in S^m.

(v) $\langle\xi\rangle^N |D_z^\alpha D_\xi^\beta a| \le \langle R\rangle^N \sup |D_z^\alpha D_\xi^\beta a|$. □

In particular $S^m \cdot S^{m'} = S^{m+m'}$, since $a = \langle\xi\rangle^m \cdot (\langle\xi\rangle^{-m} a)$. The inclusion $S^m \hookrightarrow S^{m'}$, $m' > m$, is never dense. Instead we have the following technical but very useful result.

Lemma 2.1.5. *Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $0 \le \chi \le 1$, $\chi(\xi) = 1$ for $|\xi| \le 1$ and $\chi(\xi) = 0$ for $|\xi| \ge 2$. For $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ and $0 < \epsilon \le 1$ set $a_\epsilon(z,\xi) = \chi(\epsilon\xi) a(z,\xi)$. Then $a_\epsilon \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$, the family (a_ϵ) is bounded in $S^m(\mathbb{R}^p;\mathbb{R}^n)$, and $a_\epsilon \to a$ in $S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ as $\epsilon \to 0$ for every $m' > m$. In particular $S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$ is dense in $S^m(\mathbb{R}^p;\mathbb{R}^n)$ in the topology of $S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ for every $m' > m$.*

Proof. The function $\chi(\epsilon\xi)$ is bounded in $S^0(\mathbb{R}^n)$ uniformly in $\epsilon \in (0,1]$: for $\beta \ne 0$, $D_\xi^\beta(\chi(\epsilon\xi)) = \epsilon^{|\beta|}(D^\beta\chi)(\epsilon\xi)$ is supported in $1/\epsilon \le |\xi| \le 2/\epsilon$, where $\langle\xi\rangle \le 1 + |\xi| \le 3/\epsilon$, so it is bounded by $3^{|\beta|} \sup|D^\beta\chi|\, \langle\xi\rangle^{-|\beta|}$. By Lemma 2.1.4(ii) the family a_ϵ is bounded in S^m, and $a_\epsilon \in S^{-\infty}$ by Lemma 2.1.4(v). Now $D_z^\alpha D_\xi^\beta (a - a_\epsilon)$ is bounded by $C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|}$ uniformly in ϵ and vanishes for $|\xi| < 1/\epsilon$. Hence, for $m' > m$,

$$\sup \langle\xi\rangle^{-m'+|\beta|} |D_z^\alpha D_\xi^\beta (a - a_\epsilon)| \le C_{\alpha\beta} \sup_{|\xi|\ge 1/\epsilon} \langle\xi\rangle^{m-m'} \le C_{\alpha\beta}\, \epsilon^{m'-m} \to 0. \qquad \square$$

One cannot take $m' = m$ here, for essentially the same reason that $\mathcal{S}(\mathbb{R}^n)$ is not dense in $\mathcal{C}_b^\infty(\mathbb{R}^n)$: a limit in S^m of residual symbols satisfies $\langle\xi\rangle^{-m} a \to 0$ as $|\xi| \to \infty$, so for instance $1 \in S^0(\mathbb{R}^p;\mathbb{R}^n)$ is not in the closure of $S^{-\infty}$ if $n > 0$.

2.1.1 Ellipticity of symbols

Lemma 2.1.6. *Let $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ and let $U \subset \mathbb{R}^p \times \mathbb{R}^n$ be an open set on which $|a(z,\xi)| \ge \epsilon\langle\xi\rangle^m$ for some $\epsilon > 0$. Then on U*

$$|D_z^\alpha D_\xi^\beta (a^{-1})| \le C_{\alpha\beta} \langle\xi\rangle^{-m-|\beta|},$$

and if moreover $a > 0$ on U then, for every $s \in \mathbb{R}$, $|D_z^\alpha D_\xi^\beta (a^s)| \le C_{\alpha\beta s}\langle\xi\rangle^{sm-|\beta|}$ on U. The constants depend only on ϵ, s and the norms of a. In particular, if $U = \mathbb{R}^p \times \mathbb{R}^n$, then $a^{-1} \in S^{-m}(\mathbb{R}^p;\mathbb{R}^n)$ and, when $a > 0$, $a^s \in S^{sm}(\mathbb{R}^p;\mathbb{R}^n)$.

Proof. Let $s \in \mathbb{R}$ (with $s = -1$ if a is not positive). We claim that on U

$$D_z^\alpha D_\xi^\beta(a^s) = \sum_{j=0}^{|\alpha|+|\beta|} a^{s-j} G_j, \qquad G_j = G_j^{\alpha\beta} \in S^{jm-|\beta|}(\mathbb{R}^p;\mathbb{R}^n),$$

where the G_j are universal polynomials in the derivatives of a (hence defined on all of $\mathbb{R}^p \times \mathbb{R}^n$). This holds for $\alpha = \beta = 0$ with $G_0 = 1$. If it holds for (α, β) then

$$D_{z_l}(a^{s-j}G_j) = a^{s-j-1}((s-j)(D_{z_l}a)G_j) + a^{s-j}D_{z_l}G_j,$$

with $(D_{z_l}a)G_j \in S^{(j+1)m-|\beta|}$ and $D_{z_l}G_j \in S^{jm-|\beta|}$; similarly for D_{ξ_l}, where $(D_{\xi_l}a)G_j \in S^{(j+1)m-|\beta|-1}$ and $D_{\xi_l}G_j \in S^{jm-|\beta|-1}$. This proves the claim. On U, $|a|^{s-j} \le C\langle\xi\rangle^{(s-j)m}$: for $s-j \ge 0$ by the upper bound $|a| \le C\langle\xi\rangle^m$, for $s-j < 0$ by the lower bound. Hence $|a^{s-j}G_j| \le C\langle\xi\rangle^{(s-j)m+jm-|\beta|} = C\langle\xi\rangle^{sm-|\beta|}$. □

Definition 2.1.7. A symbol $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ is *elliptic* if there are $\epsilon > 0$ and C such that

$$|a(z,\xi)| \ge \epsilon\langle\xi\rangle^m \quad \text{for all } z \in \mathbb{R}^p,\ |\xi| \ge C. \tag{2.1.2}$$

Note that ϵ and C are independent of z, so this is a notion of *uniform* ellipticity. Condition (2.1.2) is equivalent to the existence of $\epsilon > 0$ and C' with $|a(z,\xi)| \ge \epsilon\langle\xi\rangle^m - C'\langle\xi\rangle^{m-1}$ everywhere. Indeed, the latter gives $|a| \ge \frac{\epsilon}{2}\langle\xi\rangle^m$ once $\langle\xi\rangle \ge 2C'/\epsilon$; conversely, if (2.1.2) holds then for $|\xi| \le C$ we have $\epsilon\langle\xi\rangle^m \le \epsilon\langle C\rangle\langle\xi\rangle^{m-1}$, so $|a| \ge \epsilon\langle\xi\rangle^m - \epsilon\langle C\rangle\langle\xi\rangle^{m-1}$ everywhere.

Lemma 2.1.8. *If $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ is elliptic there exists $b \in S^{-m}(\mathbb{R}^p;\mathbb{R}^n)$ such that $ab - 1 \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$. Conversely, if $ab - 1 \in S^{-\infty}$ for some $b \in S^{-m}$, then a is elliptic.*

Proof. Let C be as in (2.1.2) and choose $\psi \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $\psi(\xi) = 0$ for $|\xi| \le C+1$ and $\psi(\xi) = 1$ for $|\xi| \ge C+2$; then $\psi \in S^0(\mathbb{R}^n)$. Set $b = \psi/a$ where $\psi \ne 0$ and $b = 0$ elsewhere; b is smooth since it vanishes on a neighbourhood of $\{|\xi| \le C+1\}$ and $a \ne 0$ on $|\xi| > C$. Applying Lemma 2.1.6 on $U = \{|\xi| > C\}$ and Leibniz' formula shows $b \in S^{-m}$, and $ab - 1 = \psi - 1$ is smooth, independent of z and supported in $|\xi| \le C+2$, hence residual.

Conversely, if $ab = 1 + e$ with $e \in S^{-\infty}$, choose C with $|e| \le \frac{1}{2}$ for $|\xi| \ge C$. Since $|b| \le C'\langle\xi\rangle^{-m}$, for $|\xi| \ge C$ we get $\frac{1}{2} \le |a||b| \le C'\langle\xi\rangle^{-m}|a|$, which is (2.1.2) with $\epsilon = 1/(2C')$. □

2.1.2 Asymptotic summation

Suppose $a_j \in S^{m-j}(\mathbb{R}^p;\mathbb{R}^n)$, $j \in \mathbb{N}_0$. The decreasing orders mean that these symbols become smaller and smaller as $|\xi| \to \infty$, but the series $\sum_j a_j$ need not converge. Nevertheless it can be "summed" asymptotically.

Definition 2.1.9. Let $m_j \in \mathbb{R}$ be a sequence decreasing to $-\infty$ and $a_j \in S^{m_j}(\mathbb{R}^p;\mathbb{R}^n)$. We write

$$a \sim \sum_{j=0}^{\infty} a_j$$

if $a \in S^{m_0}(\mathbb{R}^p;\mathbb{R}^n)$ and $a - \sum_{j<N} a_j \in S^{m_N}(\mathbb{R}^p;\mathbb{R}^n)$ for every N; a is then called an *asymptotic sum* of the series. Most often $m_j = m - j$.

Proposition 2.1.10 (Borel's lemma for symbols)**.** *Every series $\sum_j a_j$ with $a_j \in S^{m_j}(\mathbb{R}^p;\mathbb{R}^n)$, $m_j \downarrow -\infty$, has an asymptotic sum, and the asymptotic sum is unique up to an additive term in $S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$.*

Proof. Uniqueness: if a and a' are both asymptotic sums then $a - a' = (a - \sum_{j<N} a_j) - (a' - \sum_{j<N} a_j) \in S^{m_N}$ for every N, so $a - a' \in S^{-\infty}$.

Existence: we cut off each term near the origin in ξ, more and more drastically as j increases. Choose $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $\phi(\xi) = 0$ for $|\xi| \leq 1$ and $\phi(\xi) = 1$ for $|\xi| \geq 2$ (e.g. $\phi = 1 - \chi$ with χ as in Lemma 2.1.5). As in the proof of Lemma 2.1.5, $\phi(\epsilon\xi)$ is bounded in $S^0(\mathbb{R}^n)$ for $\epsilon \in (0, 1]$, so $\|\phi(\epsilon\xi)a_j\|_{N,m_j} \leq C_{N,j}$ with $C_{N,j}$ independent of ϵ. Moreover $\phi(\epsilon\xi)a_j$ vanishes for $|\xi| < 1/\epsilon$, where $\langle\xi\rangle^{-1} \leq \epsilon$; replacing m_j by $m_j + 1$ in the norm therefore gains a factor ϵ:

$$\|\phi(\epsilon\xi)a_j\|_{N,m_j+1} \leq C_{N,j}\,\epsilon. \tag{2.1.3}$$

Choose $\epsilon_j \in (0, 1]$ with $C_{N,j}\epsilon_j \leq 2^{-j}$ for all $N \leq j$, and set

$$a(z, \xi) = \sum_{j=0}^{\infty} \phi(\epsilon_j\xi)a_j(z, \xi).$$

The ϵ_j may be taken decreasing to 0, so on each set $\{|\xi| \leq R\}$ only finitely many terms are non-zero and a is a well-defined smooth function. Fix $M \in \mathbb{N}_0$. Since $m_j \downarrow -\infty$ there is J_M with $m_j + 1 \leq m_M$ for $j \geq J_M$; for such j and $j \geq N$, $\|\phi(\epsilon_j\xi)a_j\|_{N,m_M} \leq \|\phi(\epsilon_j\xi)a_j\|_{N,m_j+1} \leq 2^{-j}$. Hence $\sum_{j\geq M} \phi(\epsilon_j\xi)a_j$ converges absolutely in each norm of the Fréchet space S^{m_M} and so converges there; its limit is the pointwise sum. Thus

$$a - \sum_{j<M} a_j = \sum_{j<M} (\phi(\epsilon_j\xi) - 1)a_j + \sum_{j\geq M} \phi(\epsilon_j\xi)a_j \in S^{-\infty} + S^{m_M},$$

where the first sum is residual by Lemma 2.1.4(v). □

The following simple observations allow asymptotic sums to be manipulated formally.

Lemma 2.1.11. (i) *If T is a linear map with $T(S^k(\mathbb{R}^p; \mathbb{R}^n)) \subset S^{k+\mu}(\mathbb{R}^{p'}; \mathbb{R}^n)$ for all k (e.g. multiplication by an element of S^μ, differentiation, complex conjugation, restriction to a linear subspace of $\mathbb{R}^p$) and $a \sim \sum_j a_j$, then $Ta \sim \sum_j Ta_j$.*

(ii) *If $a \sim \sum_j a_j$ and $b \sim \sum_k b_k$, with $a_j \in S^{m-j}$, $b_k \in S^{m'-k}$, then $ab \sim \sum_l \sum_{j+k=l} a_j b_k$.*

(iii) (Rearrangement) *Let $a_{j,k} \in S^{m-j-k}(\mathbb{R}^p; \mathbb{R}^n)$ for $j, k \in \mathbb{N}_0$. If $c_j \sim \sum_k a_{j,k}$ for each j and $c \sim \sum_j c_j$, then $c \sim \sum_l (\sum_{j+k=l} a_{j,k})$.*

Proof. (i) $Ta - \sum_{j<N} Ta_j = T(a - \sum_{j<N} a_j) \in S^{m_N+\mu}$. (ii) $ab - \sum_{j+k<N} a_j b_k = (a - \sum_{j<N} a_j)b + \sum_{j<N} a_j(b - \sum_{k<N-j} b_k) \in S^{m+m'-N}$. (iii) $c - \sum_{j+k<N} a_{j,k} = (c - \sum_{j<N} c_j) + \sum_{j<N}(c_j - \sum_{k<N-j} a_{j,k}) \in S^{m-N}$. □

2.2 Amplitudes and pseudodifferential operators

We now interpret (2.0.1). It turns out to be convenient to allow the amplitude $a(x, y, \xi)$ to grow polynomially in the direction of $x - y$; this extra freedom makes no difference to the class of operators (Theorem 2.3.1) but is very useful, for instance for weighted estimates.

Definition 2.2.1. For $m, w \in \mathbb{R}$ let $\mathcal{A}^m_w(\mathbb{R}^n)$ be the space of $a \in \mathcal{C}^\infty(\mathbb{R}^n_x \times \mathbb{R}^n_y \times \mathbb{R}^n_\xi)$ such that

$$|D^\alpha_x D^\beta_y D^\gamma_\xi a(x, y, \xi)| \leq C_{\alpha\beta\gamma}\langle x - y\rangle^w \langle\xi\rangle^{m-|\gamma|} \quad \text{for all } \alpha, \beta, \gamma \in \mathbb{N}^n_0, \tag{2.2.1}$$

with the norms $\|a\|_{N,m,w}$, the best constants in (2.2.1) for $|\alpha| + |\beta| + |\gamma| \leq N$. Elements of $\mathcal{A}^m_w(\mathbb{R}^n)$ are called *amplitudes* of order m and weight w. We set $\mathcal{A}^{-\infty}_w(\mathbb{R}^n) = \bigcap_m \mathcal{A}^m_w(\mathbb{R}^n)$.

Lemma 2.2.2. *Multiplication by $\langle x-y\rangle^w$ is an isomorphism of Fréchet spaces from $S^m(\mathbb{R}^{2n}_{(x,y)};\mathbb{R}^n_\xi)$ onto $\mathcal{A}^m_w(\mathbb{R}^n)$. More generally, call $f\in\mathcal{C}^\infty(\mathbb{R}^{2n})$* of weight w' *if $|D_x^\alpha D_y^\beta f|\le C_{\alpha\beta}\langle x-y\rangle^{w'}$ for all α,β; then multiplication by such f maps $\mathcal{A}^m_w(\mathbb{R}^n)$ continuously into $\mathcal{A}^m_{w+w'}(\mathbb{R}^n)$.*

Proof. The second statement is immediate from Leibniz' formula. The function $\langle z\rangle^w$ is a symbol of order w in $z\in\mathbb{R}^n$ (with no base variables) by Lemma 2.1.4(i), so $|D_z^\alpha\langle z\rangle^w|\le C\langle z\rangle^{w-|\alpha|}\le C\langle z\rangle^w$. Hence $\langle x-y\rangle^{\pm w}$ is of weight $\pm w$, and multiplication by $\langle x-y\rangle^w$ and by $\langle x-y\rangle^{-w}$ are mutually inverse continuous maps between $S^m(\mathbb{R}^{2n};\mathbb{R}^n)=\mathcal{A}^m_0(\mathbb{R}^n)$ and $\mathcal{A}^m_w(\mathbb{R}^n)$. □

Thus $\mathcal{A}^m_w(\mathbb{R}^n)=\langle x-y\rangle^w S^m(\mathbb{R}^{2n};\mathbb{R}^n)$, and the results of Section 2.1 transfer to amplitudes. In particular, by Lemma 2.1.5, if $a\in\mathcal{A}^m_w(\mathbb{R}^n)$ and χ is as there, then $\chi(\epsilon\xi)a\in\mathcal{A}^{-\infty}_w(\mathbb{R}^n)$ is bounded in $\mathcal{A}^m_w(\mathbb{R}^n)$ and converges to a in $\mathcal{A}^{m'}_w(\mathbb{R}^n)$ as $\epsilon\to 0$, for every $m'>m$. Note also the following elementary facts:

- $\mathcal{A}^m_w\subset\mathcal{A}^{m'}_{w'}$ if $m\le m'$ and $w\le w'$; $\mathcal{A}^m_w\cdot\mathcal{A}^{m'}_{w'}\subset\mathcal{A}^{m+m'}_{w+w'}$;
- D_{x_j}, D_{y_j} preserve $\mathcal{A}^m_w$, D_{ξ_j} maps it to $\mathcal{A}^{m-1}_w$ and multiplication by ξ_j maps it to $\mathcal{A}^{m+1}_w$;
- $(x-y)^\alpha$ is of weight $|\alpha|$, so $(x-y)^\alpha\mathcal{A}^m_w\subset\mathcal{A}^m_{w+|\alpha|}$;
- if $a\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ then both $a(x,\xi)$ and $a(y,\xi)$ lie in $\mathcal{A}^m_0(\mathbb{R}^n)$; and if $a\in\mathcal{A}^m_w$ then $a(x,x,\xi)\in S^m(\mathbb{R}^n;\mathbb{R}^n)$.

2.2.1 The kernel as an oscillatory integral

If $a\in\mathcal{A}^m_w(\mathbb{R}^n)$ with $m<-n$ then

$$K(x,y)=\int e^{i(x-y)\cdot\xi}a(x,y,\xi)đ\xi$$

converges absolutely and locally uniformly, so defines a continuous function with $|K(x,y)|\le C\langle x-y\rangle^w$. For general m we integrate by parts. Let $\Delta_x=D_{x_1}^2+\cdots+D_{x_n}^2$ be the (positive) Laplacian. Then

$$(1+\Delta_x)^q e^{i(x-y)\cdot\xi}=\langle\xi\rangle^{2q}e^{i(x-y)\cdot\xi},\qquad (1+\Delta_y)^q e^{i(x-y)\cdot\xi}=\langle\xi\rangle^{2q}e^{i(x-y)\cdot\xi}. \tag{2.2.2}$$

Proposition 2.2.3. *Let $a\in\mathcal{A}^m_w(\mathbb{R}^n)$ and let $q\in\mathbb{N}_0$ satisfy $m-2q<-n$. For $\phi\in\mathcal{S}(\mathbb{R}^{2n})$ the integral*

$$I(a)(\phi)=\int_{\mathbb{R}^{3n}} e^{i(x-y)\cdot\xi}\langle\xi\rangle^{-2q}(1+\Delta_x)^q\big[a(x,y,\xi)\phi(x,y)\big]\,dx\,dyđ\xi \tag{2.2.3}$$

converges absolutely, is independent of the choice of such q, and satisfies

$$|I(a)(\phi)|\le C\|a\|_{2q,m,w}\|\phi\|_k \tag{2.2.4}$$

for some $k=k(q,w,n)$ and $C=C(q,w,n,m)$. Thus $I(a)\in\mathcal{S}'(\mathbb{R}^{2n})$ and $a\mapsto I(a)$ is continuous in the sense of (2.2.4). Moreover:

(i) *if $m<-n$, $I(a)$ is the continuous function $K(x,y)=\int e^{i(x-y)\cdot\xi}a(x,y,\xi)đ\xi$;*

(ii) *if a_k is bounded in $\mathcal{A}^m_w(\mathbb{R}^n)$ and $a_k \to a$ in $\mathcal{A}^{m'}_w(\mathbb{R}^n)$ for some $m' > m$, then $I(a_k)(\phi) \to I(a)(\phi)$ for every $\phi \in \mathcal{S}(\mathbb{R}^{2n})$; in particular, for χ as in Lemma 2.1.5,*

$$I(a)(\phi) = \lim_{\epsilon\to 0} \int e^{i(x-y)\cdot\xi}\chi(\epsilon\xi)a(x,y,\xi)\phi(x,y)\,dx\,dy đ\xi;$$

(iii) *the same value is obtained if $(1+\Delta_x)^q$ is replaced by $(1+\Delta_y)^q$ in (2.2.3).*

Part (ii) justifies calling $I(a)$ an *oscillatory integral*: it is the limit of the absolutely convergent integrals obtained by cutting off the amplitude at large frequencies, and it is the unique extension of the absolutely convergent case that is continuous in the sense of (ii).

Proof. By Leibniz' formula $(1+\Delta_x)^q[a\phi]$ is a linear combination of terms $D_x^\mu a\, D_x^\nu\phi$ with $|\mu|+|\nu| \le 2q$, and $|D_x^\mu a| \le \|a\|_{2q,m,w}\langle x-y\rangle^w\langle\xi\rangle^m$. Since $\langle x-y\rangle \le \sqrt{2}\langle(x,y)\rangle$ we have $\langle x-y\rangle^w \le 2^{|w|/2}\langle(x,y)\rangle^{|w|}$. Hence the integrand in (2.2.3) is bounded by

$$C\|a\|_{2q,m,w}\langle\xi\rangle^{m-2q}\langle(x,y)\rangle^{|w|}\sum_{|\nu|\le 2q}|D^\nu\phi(x,y)| \le C'\|a\|_{2q,m,w}\|\phi\|_k\langle\xi\rangle^{m-2q}\langle(x,y)\rangle^{-2n-1}$$

for $k \ge 2q+|w|+2n+1$, which is integrable since $m-2q<-n$. This proves absolute convergence and (2.2.4).

If q and $q+1$ are both admissible, put $F = \langle\xi\rangle^{-2q-2}(1+\Delta_x)^q[a\phi]$. For fixed (y,ξ), F and all its x-derivatives are rapidly decreasing in x, so integration by parts and (2.2.2) give

$$\int e^{i(x-y)\cdot\xi}(1+\Delta_x)F\,dx = \int((1+\Delta_x)e^{i(x-y)\cdot\xi})F\,dx = \langle\xi\rangle^2\int e^{i(x-y)\cdot\xi}F\,dx.$$

Integrating in (y,ξ) (both sides being absolutely integrable) shows that the integrals for $q+1$ and q agree. So $I(a)$ is independent of q.

(i) If $m<-n$ we may take $q=0$, and then Fubini's theorem gives $I(a)(\phi) = \int K\phi$.

(ii) Choose q with $m'-2q<-n$ and apply (2.2.4) with m' in place of m to $a-a_k$. The particular case follows from the remarks after Lemma 2.2.2 and (i).

(iii) Denote by $I'(a)(\phi)$ the integral with Δ_y in place of Δ_x. Exactly the same arguments show that it converges absolutely, is independent of q and satisfies (2.2.4), hence (ii). For $a \in \mathcal{A}^{-\infty}_w$ both $I(a)(\phi)$ and $I'(a)(\phi)$ equal $\int e^{i(x-y)\cdot\xi}a\phi\,dx\,dy đ\xi$ (take $q=0$). By (ii), applied to $\chi(\epsilon\xi)a$, they agree for all a. □

Distributional operations on $I(a)$ correspond to simple operations on the amplitude.

Lemma 2.2.4. *For $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ and $j=1,\dots,n$, the following identities hold in $\mathcal{S}'(\mathbb{R}^{2n})$:*

$$D_{x_j}I(a) = I((\xi_j + D_{x_j})a), \qquad D_{y_j}I(a) = I((D_{y_j}-\xi_j)a),$$
$$(x_j-y_j)I(a) = I((x_j-y_j)a) = I(-D_{\xi_j}a).$$

Moreover $fI(a) = I(fa)$ for every f of weight w' in the sense of Lemma 2.2.2. In particular, for every $\alpha \in \mathbb{N}_0^n$, $(x-y)^\alpha I(a) = I((-D_\xi)^\alpha a)$.

Proof. Such an f and all its derivatives are polynomially bounded, so $f\phi \in \mathcal{S}(\mathbb{R}^{2n})$ for $\phi \in \mathcal{S}(\mathbb{R}^{2n})$, and the integrals defining $I(a)(f\phi)$ and $I(fa)(\phi)$ by (2.2.3) (with the same q) are identical. This gives $fI(a) = I(fa)$, and in particular the first equality in the second line.

For the remaining identities, suppose first that $a \in \mathcal{A}^{-\infty}_w$. Then all integrals below converge absolutely and

$$D_{x_j}I(a)(\phi) = -\int e^{i(x-y)\cdot\xi}a\,D_{x_j}\phi = \int D_{x_j}(e^{i(x-y)\cdot\xi}a)\phi = \int e^{i(x-y)\cdot\xi}(\xi_j a + D_{x_j}a)\phi,$$

integrating by parts in x. The identity for D_{y_j} is the same. Since $(x_j - y_j)e^{i(x-y)\cdot\xi} = D_{\xi_j}e^{i(x-y)\cdot\xi}$, integration by parts in ξ (the boundary terms vanish because a is rapidly decreasing in ξ) gives $I((x_j - y_j)a) = I(-D_{\xi_j}a)$. For general a, apply these identities to $\chi(\epsilon\xi)a$ and let $\epsilon \to 0$, using Proposition 2.2.3(ii) for both sides: the amplitudes involved are bounded in $\mathcal{A}^{m+1}_{w+1}$ and converge in $\mathcal{A}^{m'+1}_{w+1}$ for $m' > m$. The last statement follows by induction. □

2.2.2 Pseudodifferential operators

Proposition 2.2.5. *Let $a \in \mathcal{A}^m_w(\mathbb{R}^n)$. For $u \in \mathcal{S}(\mathbb{R}^n)$ define $\operatorname{Op}(a)u \in \mathcal{S}'(\mathbb{R}^n)$ by*

$$\langle \operatorname{Op}(a)u, v\rangle = I(a)(v \otimes u), \qquad v \in \mathcal{S}(\mathbb{R}^n), \tag{2.2.5}$$

where $(v \otimes u)(x, y) = v(x)u(y)$. Then $\operatorname{Op}(a)u \in \mathcal{S}(\mathbb{R}^n)$ and $\operatorname{Op}(a) : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous. For $q \in \mathbb{N}_0$ with $m - 2q < -n$,

$$\operatorname{Op}(a)u(x) = \int e^{i(x-y)\cdot\xi}\langle\xi\rangle^{-2q}(1 + \Delta_y)^q\big[a(x, y, \xi)u(y)\big]\, dy\text{đ}\xi, \tag{2.2.6}$$

an absolutely convergent integral; if $m < -n$ this is just $\int e^{i(x-y)\cdot\xi}a(x, y, \xi)u(y)\, dy\text{đ}\xi$. In general

$$\operatorname{Op}(a)u(x) = \lim_{\epsilon\to 0}\int e^{i(x-y)\cdot\xi}\chi(\epsilon\xi)a(x, y, \xi)u(y)\, dy\text{đ}\xi.$$

Proof. Let $F(x)$ denote the right side of (2.2.6). As in the proof of Proposition 2.2.3, but now using Peetre's inequality $\langle x - y\rangle^w \le 2^{|w|/2}\langle x\rangle^{|w|}\langle y\rangle^{|w|}$, its integrand is bounded by $C\|a\|_{2q,m,w}\langle x\rangle^{|w|}\langle\xi\rangle^{m-2q}\langle y\rangle^{|w|}\sum_{|\nu|\le 2q}|D^\nu u(y)|$. Hence the integral converges absolutely, F is continuous by dominated convergence, and

$$|F(x)| \le C\|a\|_{2q,m,w}\|u\|_k\langle x\rangle^{|w|} \tag{2.2.7}$$

for some k. By Proposition 2.2.3(iii) and Fubini's theorem, $I(a)(v \otimes u) = \int v(x)F(x)\, dx$, so $\operatorname{Op}(a)u = F$. The limit formula follows in the same way from Proposition 2.2.3(ii), the convergence being locally uniform in x by dominated convergence.

To see that $\operatorname{Op}(a)u \in \mathcal{S}$, note that (2.2.5) and Lemma 2.2.4 give

$$D_{x_j}\operatorname{Op}(a)u = \operatorname{Op}((\xi_j + D_{x_j})a)u, \qquad x_j\operatorname{Op}(a)u = \operatorname{Op}(a)(x_ju) + \operatorname{Op}(-D_{\xi_j}a)u.$$

For instance $\langle D_{x_j}\operatorname{Op}(a)u, v\rangle = -I(a)(D_{x_j}v \otimes u) = (D_{x_j}I(a))(v \otimes u)$, and $(x_j - y_j)(v \otimes u) = (x_jv) \otimes u - v \otimes (x_ju)$. By induction, for all α, β, $x^\alpha D^\beta\operatorname{Op}(a)u$ is a finite sum of terms $\operatorname{Op}(a')(x^{\alpha'}u)$ with $a' \in \mathcal{A}^{m+|\beta|}_w$, $|\alpha'| \le |\alpha|$, the a' depending continuously on a. By (2.2.7) each such term is a continuous function bounded by $C\|a\|_{N,m,w}\|u\|_{k'}\langle x\rangle^{|w|}$. Since all distributional derivatives of $\operatorname{Op}(a)u$ are continuous, $\operatorname{Op}(a)u \in \mathcal{C}^\infty(\mathbb{R}^n)$; and since $\langle x\rangle^{2L}$ is a polynomial in x, we get $\sup\langle x\rangle^{2L-|w|}|D^\beta\operatorname{Op}(a)u| \le C\|a\|_{N,m,w}\|u\|_{k''}$ for all L and β. Thus $\operatorname{Op}(a)u \in \mathcal{S}(\mathbb{R}^n)$ and $\operatorname{Op}(a)$ is continuous on $\mathcal{S}(\mathbb{R}^n)$. □

Definition 2.2.6. For $m \in \mathbb{R}$, $\Psi^m(\mathbb{R}^n)$ is the space of operators $\operatorname{Op}(a) : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ with $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ for some $w \in \mathbb{R}$. Its elements are the *pseudodifferential operators of order m* (of type $(1, 0)$, with uniformly bounded coefficients). We set $\Psi^\infty(\mathbb{R}^n) = \bigcup_m \Psi^m(\mathbb{R}^n)$ and $\Psi^{-\infty}(\mathbb{R}^n) = \bigcap_m \Psi^m(\mathbb{R}^n)$. The distribution $K_A = I(a)$ is the *Schwartz kernel* of $A = \operatorname{Op}(a)$.

Since $\mathcal{A}^m_w \subset \mathcal{A}^{m'}_{w'}$ for $m \le m'$, $w \le w'$, each $\Psi^m(\mathbb{R}^n)$ is a linear space and $\Psi^m(\mathbb{R}^n) \subset \Psi^{m'}(\mathbb{R}^n)$ for $m \le m'$. By (2.2.5), K_A is determined by A on the products $v \otimes u$, and these

span a dense subspace of $\mathcal{S}(\mathbb{R}^{2n})$ (see Chapter 1), so K_A is determined by A even though the amplitude is not.

The two extreme cases are amplitudes independent of y and amplitudes independent of x. For $a, b \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ we write

$$\operatorname{Op}(a) = \operatorname{Op}(a(x,\xi)), \qquad \operatorname{Op}_R(b) = \operatorname{Op}(b(y,\xi)),$$

the *left* and *right quantizations*; they lie in $\Psi^m(\mathbb{R}^n)$.

Proposition 2.2.7. *Let $a, b \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ and $u \in \mathcal{S}(\mathbb{R}^n)$. Then*

$$\operatorname{Op}(a)u(x) = \int e^{ix\cdot\xi} a(x,\xi)\hat{u}(\xi)\text{đ}\xi, \tag{2.2.8}$$

$$\widehat{\operatorname{Op}_R(b)u}(\xi) = \int e^{-iy\cdot\xi} b(y,\xi)u(y)\,dy, \tag{2.2.9}$$

Moreover, if $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ and $b \in S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$ then $a(x,\xi)b(y,\xi) \in \mathcal{A}_0^{m+m'}(\mathbb{R}^n)$ and $\operatorname{Op}(a)\operatorname{Op}_R(b) = \operatorname{Op}(a(x,\xi)b(y,\xi)) \in \Psi^{m+m'}(\mathbb{R}^n)$.

Proof. If $a \in S^{-\infty}$, (2.2.8) follows from the absolutely convergent formula of Proposition 2.2.5 and Fubini. In general, let $a_\epsilon = \chi(\epsilon\xi)a$. Then $\operatorname{Op}(a_\epsilon)u(x) \to \operatorname{Op}(a)u(x)$ by Proposition 2.2.5, while $\int e^{ix\cdot\xi}a_\epsilon\hat{u}\text{đ}\xi \to \int e^{ix\cdot\xi}a\hat{u}\text{đ}\xi$ by dominated convergence, since $|a_\epsilon| \le C\langle\xi\rangle^m$ uniformly and $\hat{u} \in \mathcal{S}$.

For (2.2.9) let $g(\xi)$ be the right side and $B = \operatorname{Op}_R(b)$. If $b \in S^{-\infty}$, g is integrable and Fubini gives $Bu(x) = \int e^{ix\cdot\xi}g(\xi)\text{đ}\xi$, so $\widehat{Bu} = g$ by Fourier inversion. In general, with $b_\epsilon = \chi(\epsilon\xi)b$ and $B_\epsilon = \operatorname{Op}_R(b_\epsilon)$, we have $\widehat{B_\epsilon u} = g_\epsilon \to g$ pointwise by dominated convergence; on the other hand $B_\epsilon u \to Bu$ in $\mathcal{S}'(\mathbb{R}^n)$, by Proposition 2.2.3(ii), so $\widehat{B_\epsilon u} \to \widehat{Bu}$ in $\mathcal{S}'$. Since g_ϵ is bounded by $C\langle\xi\rangle^m\|u\|_{L^1}$ uniformly in ϵ, also $g_\epsilon \to g$ in $\mathcal{S}'$, and so $\widehat{Bu} = g$.

Finally let $c(x,y,\xi) = a(x,\xi)b(y,\xi) \in \mathcal{A}_0^{m+m'}$ and choose q with $m+m'-2q < -n$. By (2.2.6) and Fubini,

$$\operatorname{Op}(c)u(x) = \int e^{ix\cdot\xi}a(x,\xi)\langle\xi\rangle^{-2q}\Big[\int e^{-iy\cdot\xi}(1+\Delta_y)^q\big(b(y,\xi)u(y)\big)dy\Big]\text{đ}\xi.$$

Integrating by parts in y (for fixed ξ, bu is rapidly decreasing with all derivatives) the inner integral is $\langle\xi\rangle^{2q}g(\xi)$, so by (2.2.8) and (2.2.9), $\operatorname{Op}(c)u(x) = \int e^{ix\cdot\xi}a(x,\xi)\widehat{Bu}(\xi)\text{đ}\xi = \operatorname{Op}(a)Bu(x)$. □

Example 2.2.8. Let $P = \sum_{|\alpha|\le m} p_\alpha(x)D^\alpha$ be a differential operator with coefficients $p_\alpha \in \mathcal{C}_{\mathrm{b}}^\infty(\mathbb{R}^n)$; we denote the space of these by $\operatorname{Diff}_{\mathrm{b}}^m(\mathbb{R}^n)$. Its characteristic polynomial $p(x,\xi) = \sum_\alpha p_\alpha(x)\xi^\alpha$ lies in $S^m(\mathbb{R}^n;\mathbb{R}^n)$ and, since $\widehat{D^\alpha u} = \xi^\alpha\hat{u}$, (2.2.8) and Fourier inversion give $P = \operatorname{Op}(p)$. Thus $\operatorname{Diff}_{\mathrm{b}}^m(\mathbb{R}^n) \subset \Psi^m(\mathbb{R}^n)$. Similarly, multiplication by $f \in \mathcal{C}_{\mathrm{b}}^\infty(\mathbb{R}^n)$ is $\operatorname{Op}(f) \in \Psi^0(\mathbb{R}^n)$, and the Fourier multiplier $\langle D\rangle^s = \operatorname{Op}(\langle\xi\rangle^s)$, $\widehat{\langle D\rangle^s u} = \langle\xi\rangle^s\hat{u}$, lies in $\Psi^s(\mathbb{R}^n)$ for every $s \in \mathbb{R}$.

2.2.3 Adjoints and the action on tempered distributions

Proposition 2.2.9. *If $A = \operatorname{Op}(a) \in \Psi^m(\mathbb{R}^n)$ with $a \in \mathcal{A}_w^m(\mathbb{R}^n)$, then $A^* = \operatorname{Op}(a^*)$ with $a^*(x,y,\xi) = \overline{a(y,x,\xi)} \in \mathcal{A}_w^m(\mathbb{R}^n)$ satisfies*

$$(Au, v) = (u, A^*v) \quad \textit{for all } u, v \in \mathcal{S}(\mathbb{R}^n).$$

In particular $A^* \in \Psi^m(\mathbb{R}^n)$, and the formal adjoint is determined by this identity. Consequently, the transpose $A^t\phi = \overline{A^*\bar{\phi}}$, which satisfies $\int(Au)\phi = \int u(A^t\phi)$ for $u, \phi \in \mathcal{S}$, is also continuous on $\mathcal{S}(\mathbb{R}^n)$.

Proof. Clearly $a^* \in \mathcal{A}^m_w$, depending continuously on a in each $\mathcal{A}^{m'}_w$. We must show $I(a)(\bar v \otimes u) = \overline{I(a^*)(\bar u \otimes v)}$, since the left side is (Au, v) and the right side is $\overline{(A^*v, u)} = (u, A^*v)$. For $a \in \mathcal{A}^{-\infty}_w$ this follows from the absolutely convergent expressions:

$$\overline{I(a^*)(\bar u \otimes v)} = \int e^{-i(x-y)\cdot\xi} a(y,x,\xi) u(x) \overline{v(y)}\, dx\, dy đ\xi = \int e^{i(x-y)\cdot\xi} a(x,y,\xi) \overline{v(x)} u(y)\, dx\, dy đ\xi,$$

after exchanging the names of x and y. The general case follows by replacing a by $\chi(\epsilon\xi)a$ and using Proposition 2.2.3(ii). Uniqueness of A^* is clear since $\mathcal{S}$ is dense in L^2. □

This shows one advantage of allowing general amplitudes: closure under adjoints is immediate. It also allows every $A \in \Psi^m(\mathbb{R}^n)$ to be extended to tempered distributions by duality:

$$(Au)(\phi) = u(A^t\phi), \qquad u \in \mathcal{S}'(\mathbb{R}^n),\ \phi \in \mathcal{S}(\mathbb{R}^n). \tag{2.2.10}$$

Since A^t is continuous on $\mathcal{S}$, this defines $Au \in \mathcal{S}'(\mathbb{R}^n)$, the map $A : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is continuous (for the weak topology), and it extends A on $\mathcal{S}(\mathbb{R}^n)$; as $\mathcal{S}(\mathbb{R}^n)$ is dense in $\mathcal{S}'(\mathbb{R}^n)$ this extension is unique. Since $(AB)^t = B^tA^t$, the extensions satisfy $(AB)u = A(Bu)$ for $u \in \mathcal{S}'$. For example $\widehat{\langle D\rangle^s u} = \langle\xi\rangle^s \hat u$ continues to hold for $u \in \mathcal{S}'(\mathbb{R}^n)$.

2.3 Reduction to the left symbol

The main step towards the composition theorem is to show that every pseudodifferential operator can be written, uniquely, as the left quantization of a symbol independent of y:

Theorem 2.3.1 (Reduction)**.** *For each $A \in \Psi^m(\mathbb{R}^n)$ there is a unique $\sigma_L(A) \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ with $A = \mathrm{Op}(\sigma_L(A))$. If $A = \mathrm{Op}(a)$ with $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ then*

$$\sigma_L(A)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D^\alpha_\xi D^\alpha_y a(x,y,\xi)\Big|_{y=x}. \tag{2.3.1}$$

Here the terms with $|\alpha| = j$ are grouped together; they have order $m - j$. The symbol $\sigma_L(A)$ is the *left* (or *full*) *symbol* of A. The proof occupies the rest of this section.

2.3.1 Taylor expansion of the amplitude

The key observation is the last identity of Lemma 2.2.4: multiplying the amplitude by $(x-y)^\alpha$ appears to raise its weight, but actually lowers the order of the operator by $|\alpha|$.

Lemma 2.3.2. *Let $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ and $N \in \mathbb{N}$. Then*

$$\mathrm{Op}(a) = \sum_{j=0}^{N-1} \mathrm{Op}(a_j) + R_N, \qquad a_j(x,\xi) = \sum_{|\alpha|=j} \frac{i^{|\alpha|}}{\alpha!} D^\alpha_\xi D^\alpha_y a(x,y,\xi)\Big|_{y=x} \in S^{m-j}(\mathbb{R}^n;\mathbb{R}^n),$$

with $R_N \in \Psi^{m-N}(\mathbb{R}^n)$; in fact $R_N = \mathrm{Op}(r_N)$ with $r_N \in \mathcal{A}^{m-N}_{w_+}(\mathbb{R}^n)$, $w_+ = \max(w,0)$.

Proof. Taylor's formula with integral remainder, applied to a in the variable y about $y = x$, gives

$$a(x,y,\xi) = \sum_{|\alpha|<N} \frac{(y-x)^\alpha}{\alpha!} (\partial^\alpha_y a)(x,x,\xi) + \sum_{|\alpha|=N} (y-x)^\alpha \rho_\alpha(x,y,\xi),$$

$$\rho_\alpha(x,y,\xi) = \frac{N}{\alpha!} \int_0^1 (1-t)^{N-1} (\partial^\alpha_y a)(x, x + t(y-x), \xi)\, dt.$$

We claim $\rho_\alpha \in \mathcal{A}^m_{w_+}$. Differentiating under the integral, $D_x^\mu D_y^\nu D_\xi^\gamma$ of the integrand is a linear combination, with coefficients that are polynomials in t bounded on $[0,1]$, of derivatives of $\partial_y^\alpha a$ evaluated at $(x, x+t(y-x), \xi)$, and these are bounded by $C\langle t(x-y)\rangle^w \langle\xi\rangle^{m-|\gamma|} \le C\langle x-y\rangle^{w_+}\langle\xi\rangle^{m-|\gamma|}$, since $\langle t(x-y)\rangle \le \langle x-y\rangle$ and $\langle t(x-y)\rangle \ge 1$. Now apply Lemma 2.2.4: for any $b \in \mathcal{A}^m_{w'}$,

$$I((y-x)^\alpha b) = (-1)^{|\alpha|}(x-y)^\alpha I(b) = I(D_\xi^\alpha b).$$

With $b = (\partial_y^\alpha a)(x,x,\xi)/\alpha!$, and $\partial_y^\alpha = i^{|\alpha|}D_y^\alpha$, this shows that the term of order α in the Taylor sum contributes $\operatorname{Op}(\frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_y^\alpha a(x,y,\xi)|_{y=x})$. With $b = \rho_\alpha$ it shows that the remainder term contributes $\operatorname{Op}(D_\xi^\alpha \rho_\alpha)$, with $D_\xi^\alpha \rho_\alpha \in \mathcal{A}^{m-N}_{w_+}$. □

2.3.2 Residual operators

Applying Borel's lemma to the series in Lemma 2.3.2 we will find a symbol b with $\operatorname{Op}(a) - \operatorname{Op}(b) \in \Psi^{m-N}$ for all N. To finish the proof of Theorem 2.3.1 we need to identify the residual operators. Note that *a priori* an element of $\Psi^{-\infty}(\mathbb{R}^n)$ is only known to be of the form $\operatorname{Op}(a_M)$ with $a_M \in \mathcal{A}^{-M}_{w_M}$ for each M, where the weight w_M may depend on M.

Proposition 2.3.3. *For a continuous linear operator $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ with Schwartz kernel K the following are equivalent:*

(a) $A \in \Psi^{-\infty}(\mathbb{R}^n)$;

(b) $K \in \mathcal{C}^\infty(\mathbb{R}^{2n})$ *and for all* $\alpha, \beta \in \mathbb{N}_0^n$ *and* $N \in \mathbb{N}_0$

$$|D_x^\alpha D_y^\beta K(x,y)| \le C_{\alpha\beta N}\langle x-y\rangle^{-N}; \tag{2.3.2}$$

(c) $A = \operatorname{Op}(c)$ *for some* $c \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$.

Proof. (c)⇒(a) is immediate, since $S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n) \subset \mathcal{A}^M_0(\mathbb{R}^n)$ for every M.

(a)⇒(b). For each $M > n$ write $A = \operatorname{Op}(a_M)$ with $a_M \in \mathcal{A}^{-M}_{w_M}$. By Proposition 2.2.3(i), $I(a_M)$ is a continuous, polynomially bounded function; two such functions which give the same integral against all $v \otimes u$ coincide, so $K = I(a_M)$ is a continuous function independent of M. By Lemma 2.2.4, for all α, β, γ,

$$(x-y)^\gamma D_x^\alpha D_y^\beta K = I(b), \qquad b = (-D_\xi)^\gamma(\xi + D_x)^\alpha (D_y - \xi)^\beta a_M \in \mathcal{A}^{-M+|\alpha|+|\beta|-|\gamma|}_{w_M},$$

where the powers of $\xi + D_x$ and $D_y - \xi$ are understood componentwise. Taking $\gamma = 0$ and $M > n + |\alpha| + |\beta|$ shows that all distributional derivatives of K are continuous functions, so $K \in \mathcal{C}^\infty(\mathbb{R}^{2n})$; then $|(x-y)^\gamma D_x^\alpha D_y^\beta K| \le C\langle x-y\rangle^{w_M}$ for all γ. Since $\langle x-y\rangle^{2L}$ is a polynomial in $x-y$, this gives (2.3.2).

(b)⇒(c). Set $g(x,z) = K(x, x-z)$. Since $D_x g = (D_x K + D_y K)(x, x-z)$ and $D_z g = -(D_y K)(x, x-z)$, (2.3.2) gives $|D_x^\alpha D_z^\beta g(x,z)| \le C_{\alpha\beta N}\langle z\rangle^{-N}$, so $\sup_{x,z}|z^\gamma D_x^\alpha D_z^\beta g| < \infty$ for all α, β, γ. Let

$$c(x,\xi) = \int e^{-iz\cdot\xi} g(x,z)\, dz.$$

Then $\xi^\beta D_\xi^\gamma D_x^\alpha c$ is the Fourier transform in z of $D_z^\beta((-z)^\gamma D_x^\alpha g)$, which is bounded in L^1_z uniformly in x; hence $\sup_{x,\xi}|\xi^\beta D_\xi^\gamma D_x^\alpha c| < \infty$ for all α, β, γ, i.e. $c \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$. By Fourier inversion $g(x,z) = \int e^{iz\cdot\xi} c(x,\xi) d\!\!\!^-\xi$, so by Proposition 2.2.3(i) the kernel of $\operatorname{Op}(c)$ is $\int e^{i(x-y)\cdot\xi} c(x,\xi) d\!\!\!^-\xi = g(x, x-y) = K(x,y)$. Hence $A = \operatorname{Op}(c)$. □

Proof of Theorem 2.3.1. Let $A = \mathrm{Op}(a)$, $a \in \mathcal{A}^m_w$, and let $a_j \in S^{m-j}(\mathbb{R}^n;\mathbb{R}^n)$ be as in Lemma 2.3.2. By Proposition 2.1.10 choose $b \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ with $b \sim \sum_j a_j$. Then for every N

$$A - \mathrm{Op}(b) = R_N - \mathrm{Op}\Big(b - \sum_{j<N} a_j\Big) \in \Psi^{m-N}(\mathbb{R}^n),$$

so $A - \mathrm{Op}(b) \in \Psi^{-\infty}(\mathbb{R}^n)$ and, by Proposition 2.3.3, $A - \mathrm{Op}(b) = \mathrm{Op}(c)$ with $c \in S^{-\infty}$. Thus $A = \mathrm{Op}(b+c)$ and $\sigma_L(A) = b + c$ satisfies (2.3.1).

For uniqueness, suppose $\mathrm{Op}(e) = 0$ with $e \in S^m(\mathbb{R}^n;\mathbb{R}^n)$. By (2.2.8), $\int e^{ix\cdot\xi} e(x,\xi)\hat{u}(\xi)đ\xi = 0$ for all $u \in \mathcal{S}$. For fixed x, $\hat{u}$ ranges over all of $\mathcal{S}(\mathbb{R}^n)$, so the continuous, polynomially bounded function $\xi \mapsto e^{ix\cdot\xi}e(x,\xi)$ vanishes identically. Hence $e = 0$. □

Corollary 2.3.4. (i) $\mathrm{Op} : S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ *is a bijection with inverse* σ_L*; in particular every* $A \in \Psi^m(\mathbb{R}^n)$ *has an amplitude in* $\mathcal{A}^m_0(\mathbb{R}^n)$*, so the weights* w *play no role in the definition of* $\Psi^m(\mathbb{R}^n)$*.*

(ii) $\Psi^{-\infty}(\mathbb{R}^n) = \mathrm{Op}\big(S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)\big)$*, and for* $A \in \Psi^m(\mathbb{R}^n)$ *and any* $k \in \mathbb{R}$*,* $A \in \Psi^k(\mathbb{R}^n)$ *if and only if* $\sigma_L(A) \in S^k(\mathbb{R}^n;\mathbb{R}^n)$*.*

(iii) (Asymptotic summation of operators) *If* $A_j \in \Psi^{m-j}(\mathbb{R}^n)$*, there exists* $A \in \Psi^m(\mathbb{R}^n)$*, unique up to* $\Psi^{-\infty}(\mathbb{R}^n)$*, with* $A - \sum_{j<N} A_j \in \Psi^{m-N}(\mathbb{R}^n)$ *for all* N*; namely any* A *with* $\sigma_L(A) \sim \sum_j \sigma_L(A_j)$*.*

Proof. (i) and (ii) are restatements of Theorem 2.3.1 and Proposition 2.3.3, using that σ_L is linear. (iii) follows from (i), (ii) and Proposition 2.1.10. □

2.4 Composition and adjoints

We can now prove the fundamental result of this chapter.

Theorem 2.4.1 (Composition)**.** *The space* $\Psi^\infty(\mathbb{R}^n)$ *is an order-filtered* $*$*-algebra of continuous operators on* $\mathcal{S}(\mathbb{R}^n)$ *(and on* $\mathcal{S}'(\mathbb{R}^n)$*):*

$$A \in \Psi^m(\mathbb{R}^n) \Longrightarrow A^* \in \Psi^m(\mathbb{R}^n), \qquad A \in \Psi^m(\mathbb{R}^n),\ B \in \Psi^{m'}(\mathbb{R}^n) \Longrightarrow AB \in \Psi^{m+m'}(\mathbb{R}^n).$$

Moreover the left symbols satisfy

$$\sigma_L(A^*)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha \overline{\sigma_L(A)(x,\xi)}, \tag{2.4.1}$$

$$\sigma_L(AB)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha \sigma_L(A)(x,\xi)\, D_x^\alpha \sigma_L(B)(x,\xi). \tag{2.4.2}$$

Formally, (2.4.2) reads $\sigma_L(AB) = e^{iD_\eta\cdot D_y}\sigma_L(A)(x,\eta)\sigma_L(B)(y,\xi)\big|_{y=x,\eta=\xi}$.

Proof. The statement about adjoints is Proposition 2.2.9. By that proposition, the adjoint of $\mathrm{Op}(a)$, $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, has amplitude $\overline{a(y,\xi)}$, so Theorem 2.3.1 (with $D_y^\alpha \overline{a(y,\xi)}|_{y=x} = D_x^\alpha \overline{a(x,\xi)}$) gives (2.4.1).

Right symbols. Every $B \in \Psi^{m'}(\mathbb{R}^n)$ can be written as $B = \mathrm{Op}_R(\sigma_R(B))$ with $\sigma_R(B) = \overline{\sigma_L(B^*)} \in S^{m'}$: indeed $B = (B^*)^* = \mathrm{Op}(\sigma_L(B^*))^*$, whose amplitude is $\overline{\sigma_L(B^*)(y,\xi)}$. Taking complex conjugates in (2.4.1) applied to B, and using $\overline{D^\alpha f} = (-1)^{|\alpha|}D^\alpha \bar{f}$ together with Lemma 2.1.11(i),

$$\sigma_R(B) \sim \sum_\beta \frac{(-i)^{|\beta|}}{\beta!} D_\xi^\beta D_x^\beta \sigma_L(B). \tag{2.4.3}$$

Composition. Let $a = \sigma_L(A)$ and $b = \sigma_R(B)$. By Proposition 2.2.7, $AB = \mathrm{Op}(a)\,\mathrm{Op}_R(b) = \mathrm{Op}(a(x,\xi)b(y,\xi))$ with $a(x,\xi)b(y,\xi) \in \mathcal{A}_0^{m+m'}$, so $AB \in \Psi^{m+m'}(\mathbb{R}^n)$, and Theorem 2.3.1 gives

$$\sigma_L(AB) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha [a\, D_x^\alpha b].$$

The map $f \mapsto D_\xi^\alpha[a\, D_x^\alpha f]$ lowers orders by $|\alpha|$ and shifts them by m, so by (2.4.3) and Lemma 2.1.11(i),(iii), with $c = \sigma_L(B)$,

$$\sigma_L(AB) \sim \sum_\gamma \sum_{\alpha+\beta=\gamma} \frac{i^{|\alpha|}(-i)^{|\beta|}}{\alpha!\,\beta!} D_\xi^\alpha [a\, D_\xi^\beta D_x^\gamma c],$$

the terms with $|\gamma| = l$ having order $m + m' - l$. By Leibniz' formula $D_\xi^\alpha[aD_\xi^\beta f] = \sum_{\mu\le\alpha} \binom{\alpha}{\mu} D_\xi^\mu a\, D_\xi^{\gamma-\mu} f$ when $\alpha+\beta=\gamma$, and using $\frac{\gamma!}{\alpha!\beta!}\binom{\alpha}{\mu} = \binom{\gamma}{\mu}\binom{\gamma-\mu}{\alpha-\mu}$, the coefficient of $\frac{1}{\gamma!} D_\xi^\mu a\, D_\xi^{\gamma-\mu} D_x^\gamma c$ in the inner sum is

$$\binom{\gamma}{\mu} i^{|\mu|} \sum_{\delta\le\gamma-\mu} \binom{\gamma-\mu}{\delta} i^{|\delta|}(-i)^{|\gamma-\mu-\delta|} = \binom{\gamma}{\mu} i^{|\mu|} \prod_{l=1}^n (i-i)^{\gamma_l-\mu_l},$$

which vanishes unless $\mu = \gamma$, when it is $i^{|\gamma|}$. Hence the inner sum equals $\frac{i^{|\gamma|}}{\gamma!} D_\xi^\gamma a\, D_x^\gamma c$, which is (2.4.2).

The operators extend continuously to $\mathcal{S}'(\mathbb{R}^n)$ by (2.2.10). □

Applying Theorem 2.3.1 to the amplitude $b(y,\xi)$ shows that left and right symbols are related by

$$\sigma_L(\mathrm{Op}_R(b)) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha b, \tag{2.4.4}$$

so that formally $\sigma_L = e^{iD_x\cdot D_\xi}\sigma_R$ and, by (2.4.3), $\sigma_R = e^{-iD_x\cdot D_\xi}\sigma_L$. The map $\mathrm{Op}_R : S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ is a bijection with inverse σ_R; see Exercise 2.6 for a "centrist" alternative.

2.5 The principal symbol

By (2.4.4) and (2.4.3), $\sigma_L(A) - \sigma_R(A) \in S^{m-1}(\mathbb{R}^n;\mathbb{R}^n)$ for $A \in \Psi^m(\mathbb{R}^n)$: the full symbol depends on the choice of quantization, but its leading part does not. For this reason we consider the quotient spaces

$$S^m(\mathbb{R}^p;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^p;\mathbb{R}^n),$$

and write $[a]$ for the class of $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$.

Definition 2.5.1. The *principal symbol map* $\sigma_m : \Psi^m(\mathbb{R}^n) \to S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n)$ is $\sigma_m(A) = [\sigma_L(A)] = [\sigma_R(A)]$.

Unlike σ_L or σ_R, σ_m depends on m: one needs to know that the order is at most m before it is defined. If $A = \mathrm{Op}(a)$ with $a \in \mathcal{A}_w^m$, then $\sigma_m(A) = [a(x,x,\xi)]$ by Theorem 2.3.1. For a differential operator $P = \sum_{|\alpha|\le m} p_\alpha D^\alpha \in \mathrm{Diff}_{\mathrm{b}}^m(\mathbb{R}^n)$, $\sigma_m(P)$ is represented by $\sum_{|\alpha|=m} p_\alpha(x)\xi^\alpha$.

Proposition 2.5.2. *For every $m \in \mathbb{R}$ the sequence*

$$0 \longrightarrow \Psi^{m-1}(\mathbb{R}^n) \hookrightarrow \Psi^m(\mathbb{R}^n) \xrightarrow{\sigma_m} S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n) \longrightarrow 0$$

is exact. Moreover, for $A \in \Psi^m(\mathbb{R}^n)$ and $B \in \Psi^{m'}(\mathbb{R}^n)$,

$$\sigma_{m+m'}(AB) = \sigma_m(A)\,\sigma_{m'}(B), \qquad \sigma_m(A^*) = \overline{\sigma_m(A)}, \tag{2.5.1}$$

and the commutator $[A,B] = AB - BA$ lies in $\Psi^{m+m'-1}(\mathbb{R}^n)$ with

$$\sigma_{m+m'-1}([A,B]) = \frac{1}{i}\{\sigma_m(A),\sigma_{m'}(B)\}, \qquad \{a,b\} = \sum_{j=1}^{n}\Big(\frac{\partial a}{\partial \xi_j}\frac{\partial b}{\partial x_j} - \frac{\partial a}{\partial x_j}\frac{\partial b}{\partial \xi_j}\Big). \tag{2.5.2}$$

Proof. Surjectivity of σ_m holds because $\sigma_m(\mathrm{Op}(a)) = [a]$, and its null space is Ψ^{m-1} by Corollary 2.3.4(ii). All terms with $|\alpha| \geq 1$ in (2.4.2) and (2.4.1) have order at most $m+m'-1$, respectively $m-1$, which gives (2.5.1). In particular $\sigma_{m+m'}([A,B]) = 0$, so $[A,B] \in \Psi^{m+m'-1}$; the Poisson bracket is well defined on the classes (changing a by an element of S^{m-1} changes $\{a,b\}$ by an element of $S^{m+m'-2}$). With $a = \sigma_L(A)$, $b = \sigma_L(B)$, the terms with $|\alpha| = 1$ in (2.4.2) give

$$\sigma_L([A,B]) - i\sum_j (D_{\xi_j} a\, D_{x_j} b - D_{\xi_j} b\, D_{x_j} a) \in S^{m+m'-2},$$

and $iD_{\xi_j} a\, D_{x_j} b = -i\,\partial_{\xi_j} a\,\partial_{x_j} b$, which gives (2.5.2). □

The simplicity of (2.5.1) compared with (2.4.2) is achieved at the expense of a great loss of information. Still, many problems can be solved using it; it says that the principal symbol gives a homomorphism from the filtered algebra $\Psi^0(\mathbb{R}^n)$ onto the *commutative* algebra $S^0(\mathbb{R}^n;\mathbb{R}^n)/S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$.

2.6 Ellipticity and parametrices

Definition 2.6.1. An operator $A \in \Psi^m(\mathbb{R}^n)$ is *elliptic* if it is invertible modulo $\Psi^{-\infty}(\mathbb{R}^n)$ with an approximate inverse of order $-m$, i.e. if there exists $B \in \Psi^{-m}(\mathbb{R}^n)$ with

$$AB - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n). \tag{2.6.1}$$

Such a B is called a (right) *parametrix* of A.

Thus ellipticity is invertibility in the quotient algebra $\Psi^\infty(\mathbb{R}^n)/\Psi^{-\infty}(\mathbb{R}^n)$, with the inverse in $\Psi^{-m}(\mathbb{R}^n)/\Psi^{-\infty}(\mathbb{R}^n)$. The point is that it is a property of the principal symbol alone. Both at the level of symbols and of operators we use the following Neumann series argument.

Lemma 2.6.2. *If $e \in S^{-1}(\mathbb{R}^p;\mathbb{R}^n)$ there is $f \in S^{-1}(\mathbb{R}^p;\mathbb{R}^n)$ with $(1+e)(1+f) - 1 \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$. If $E \in \Psi^{-1}(\mathbb{R}^n)$ there is $F \in \Psi^{-1}(\mathbb{R}^n)$ with $(\mathrm{Id}+E)(\mathrm{Id}+F) - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$ and $(\mathrm{Id}+F)(\mathrm{Id}+E) - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$.*

Proof. We give the operator version; the symbol version is identical, with Proposition 2.1.10 in place of Corollary 2.3.4(iii). By Theorem 2.4.1, $(-E)^j \in \Psi^{-j}$, so by Corollary 2.3.4(iii) there is $F \in \Psi^{-1}$ with $F - \sum_{j=1}^N (-E)^j \in \Psi^{-N-1}$ for all N. Since $(\mathrm{Id}+E)\sum_{j=0}^N (-E)^j = \mathrm{Id} - (-E)^{N+1} = \sum_{j=0}^N (-E)^j(\mathrm{Id}+E)$,

$$(\mathrm{Id}+E)(\mathrm{Id}+F) - \mathrm{Id} = -(-E)^{N+1} + (\mathrm{Id}+E)\Big(F - \sum_{j=1}^{N}(-E)^j\Big) \in \Psi^{-N-1}(\mathbb{R}^n)$$

for every N, and similarly for $(\mathrm{Id}+F)(\mathrm{Id}+E) - \mathrm{Id}$. □

Theorem 2.6.3. *For $A \in \Psi^m(\mathbb{R}^n)$ the following conditions are equivalent:*

(a) *A is elliptic;*

(b) *there exists $b \in S^{-m}(\mathbb{R}^n;\mathbb{R}^n)$ such that $\sigma_m(A)\cdot[b] = 1$ in $S^0(\mathbb{R}^n;\mathbb{R}^n)/S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$;*

(c) *there exists $b \in S^{-m}(\mathbb{R}^n;\mathbb{R}^n)$ such that $\sigma_L(A)\,b - 1 \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$;*

(d) *$\sigma_L(A)$ is an elliptic symbol: there is $\epsilon > 0$ such that $|\sigma_L(A)(x,\xi)| \geq \epsilon\langle\xi\rangle^m$ for $|\xi| \geq 1/\epsilon$.*

Condition (d) *holds for one representative of $\sigma_m(A)$ if and only if it holds for all of them (with different ϵ).*

Proof. The last statement follows from the equivalent form $|a| \geq \epsilon\langle\xi\rangle^m - C\langle\xi\rangle^{m-1}$ of ellipticity noted after Definition 2.1.7. We prove (a)⇒(b)⇒(c)⇔(d) and (c)⇒(a).

(a)⇒(b): since $\mathrm{Id} = \mathrm{Op}(1)$, $\sigma_0(\mathrm{Id}) = 1$; applying (2.5.1) to (2.6.1) gives $\sigma_m(A)\sigma_{-m}(B) = 1$.

(b)⇒(c): (b) means that $\sigma_L(A)b = 1 + e$ with $e \in S^{-1}$. With f as in Lemma 2.6.2, $\sigma_L(A)\,b(1+f) = (1+e)(1+f) \in 1 + S^{-\infty}$, and $b(1+f) \in S^{-m}$.

(c)⇔(d) is Lemma 2.1.8.

(c)⇒(a): let $B_1 = \mathrm{Op}(b)$. By (2.4.2), $\sigma_L(AB_1) - \sigma_L(A)b \in S^{-1}$, so $E = AB_1 - \mathrm{Id} \in \Psi^{-1}(\mathbb{R}^n)$. With F as in Lemma 2.6.2, $B = B_1(\mathrm{Id}+F) \in \Psi^{-m}(\mathbb{R}^n)$ satisfies $AB - \mathrm{Id} = (\mathrm{Id}+E)(\mathrm{Id}+F) - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$. □

In Definition 2.6.1 we took B to be a right parametrix. We could equally well have taken a left parametrix:

Lemma 2.6.4. *$A \in \Psi^m(\mathbb{R}^n)$ is elliptic if and only if there exists $B' \in \Psi^{-m}(\mathbb{R}^n)$ with $B'A - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$. If B is a right and B' a left parametrix then $B - B' \in \Psi^{-\infty}(\mathbb{R}^n)$.*

Proof. If $B'A - \mathrm{Id} \in \Psi^{-\infty}$ then $\sigma_{-m}(B')\sigma_m(A) = 1$, and since multiplication of symbols is commutative, (b) of Theorem 2.6.3 holds, so A is elliptic. Conversely, if A is elliptic, let b be as in Theorem 2.6.3(c) and $B_1 = \mathrm{Op}(b)$. As in the proof above, $B_1A = \mathrm{Id}+E'$ with $E' \in \Psi^{-1}$, and with F' as in Lemma 2.6.2 (for E'), $B' = (\mathrm{Id}+F')B_1$ satisfies $B'A - \mathrm{Id} \in \Psi^{-\infty}$. Finally, if $AB = \mathrm{Id}+E$ and $B'A = \mathrm{Id}+E'$ with $E, E' \in \Psi^{-\infty}$, then $B' + B'E = B'(AB) = (B'A)B = B + E'B$, so $B - B' = B'E - E'B \in \Psi^{-\infty}$. □

Thus a parametrix of an elliptic operator is automatically two-sided, and it is unique modulo $\Psi^{-\infty}(\mathbb{R}^n)$.

2.6.1 Elliptic regularity

One of the main reasons that residual operators are "residual" is that they are smoothing.

Lemma 2.6.5. *Let $E \in \Psi^{-\infty}(\mathbb{R}^n)$ have kernel K. For $u \in \mathcal{S}'(\mathbb{R}^n)$, Eu is the smooth function $x \mapsto u(K(x,\cdot))$, and all its derivatives are polynomially bounded. In particular $E : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n) \cap \mathcal{C}^\infty(\mathbb{R}^n)$.*

Proof. Write $k_x = K(x,\cdot)$. By (2.3.2) and Peetre's inequality $\langle y\rangle^N \leq 2^{N/2}\langle x\rangle^N\langle x-y\rangle^N$,

$$\sup_y |y^\gamma D_y^\beta D_x^\alpha K(x,y)| \leq C_{\alpha\beta\gamma}\langle x\rangle^{|\gamma|}.$$

Hence $x \mapsto k_x$ is a smooth map into $\mathcal{S}(\mathbb{R}^n)$ with $D_x^\alpha k_x = (D_x^\alpha K)(x,\cdot)$ and $\|D_x^\alpha k_x\|_j \leq C\langle x\rangle^j$: for instance $t^{-1}(k_{x+te_l} - k_x) - (\partial_{x_l}K)(x,\cdot) = \int_0^1 ((\partial_{x_l}K)(x+ste_l,\cdot) - (\partial_{x_l}K)(x,\cdot))ds$, whose $\mathcal{S}$-norms are $O(|t|)$ by the bounds on the second derivatives. Since $|u(\phi)| \leq C\|\phi\|_j$ for some

j, the function $f(x) = u(k_x)$ is smooth with $D^\alpha f(x) = u(D_x^\alpha k_x)$ and $|D^\alpha f(x)| \le C_\alpha \langle x\rangle^j$. Finally, by (2.2.10) and Proposition 2.2.9, $(Eu)(\phi) = u(E^t\phi)$ with $E^t\phi = \int \phi(x) k_x\, dx$; this integral converges in $\mathcal{S}(\mathbb{R}^n)$ (the integrand is continuous with norms bounded by $C|\phi(x)|\langle x\rangle^j$), so $u(E^t\phi) = \int \phi(x) u(k_x)\, dx$, i.e. $Eu = f$. □

Directly from the existence of parametrices we deduce the regularity of solutions of elliptic equations.

Proposition 2.6.6. *If $A \in \Psi^m(\mathbb{R}^n)$ is elliptic and $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfies $Au \in \mathcal{S}(\mathbb{R}^n)$ (e.g. $Au = 0$), then $u \in \mathcal{C}^\infty(\mathbb{R}^n)$.*

Proof. Let $B \in \Psi^{-m}(\mathbb{R}^n)$ be a left parametrix, $BA = \mathrm{Id} + E$ with $E \in \Psi^{-\infty}$. Then $u = B(Au) - Eu$, where $B(Au) \in \mathcal{S}(\mathbb{R}^n)$ and $Eu \in \mathcal{C}^\infty(\mathbb{R}^n)$ by Lemma 2.6.5. □

A more precise, quantitative version is given in Proposition 2.8.4; a local version requires the microlocal methods of Chapter 3.

2.6.2 The Laplacian

Let $g_{ij} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, $i, j = 1, \dots, n$, be the components of a *uniformly positive* Riemannian metric on $\mathbb{R}^n$: $g_{ij} = g_{ji}$ and, for some $\epsilon > 0$,

$$\sum_{i,j=1}^n g_{ij}(x)\xi_i\xi_j \ge \epsilon|\xi|^2 \quad \text{for all } x, \xi \in \mathbb{R}^n. \tag{2.6.2}$$

Let $g(x) = \det(g_{ij}(x))$ and let $(g^{ij}(x))$ be the inverse matrix. Since the g_{ij} are bounded, the eigenvalues of (g_{ij}) lie in $[\epsilon, C]$; hence $\epsilon^n \le g \le C^n$, and by Cramer's rule $g^{ij}, \sqrt{g} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, with $\sum g^{ij}\xi_i\xi_j \ge C^{-1}|\xi|^2$. The *Laplacian* of the metric is the differential operator

$$\Delta_g = \sum_{i,j=1}^n \frac{1}{\sqrt{g}} D_{x_i} g^{ij} \sqrt{g}\, D_{x_j} = \sum_{i,j} g^{ij} D_{x_i} D_{x_j} + \sum_{i,j} \frac{1}{\sqrt{g}} (D_{x_i}(g^{ij}\sqrt{g})) D_{x_j}.$$

It is characterized by the integration by parts formula

$$\int_{\mathbb{R}^n} \sum_{i,j} g^{ij}(x)\, D_{x_j}\phi\, \overline{D_{x_i}\psi}\, dg = \int_{\mathbb{R}^n} \Delta_g \phi\, \overline{\psi}\, dg \quad \text{for all } \phi, \psi \in \mathcal{S}(\mathbb{R}^n), \qquad dg = \sqrt{g}\, dx,$$

which follows from $\int D_{x_i} f\, \bar{h}\, dx = \int f\, \overline{D_{x_i} h}\, dx$. In particular Δ_g is formally self-adjoint and non-negative with respect to the measure dg.

Proposition 2.6.7. *$\Delta_g \in \mathrm{Diff}^2_{\mathrm{b}}(\mathbb{R}^n) \subset \Psi^2(\mathbb{R}^n)$ is elliptic, with $\sigma_2(\Delta_g) = [\sum_{i,j} g^{ij}(x)\xi_i\xi_j]$. Hence there is $B \in \Psi^{-2}(\mathbb{R}^n)$ with $\Delta_g B - \mathrm{Id}$, $B\Delta_g - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$, and if $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfies $\Delta_g u = 0$ then $u \in \mathcal{C}^\infty(\mathbb{R}^n)$.*

Proof. The coefficients are in $\mathcal{C}^\infty_{\mathrm{b}}$ by the remarks above, and the principal symbol satisfies $\sum g^{ij}\xi_i\xi_j \ge C^{-1}|\xi|^2 \ge (2C)^{-1}\langle\xi\rangle^2$ for $|\xi| \ge 1$. The rest follows from Theorem 2.6.3, Lemma 2.6.4 and Proposition 2.6.6. □

Note that the upper bound on the g_{ij}, which comes from the assumption $g_{ij} \in \mathcal{C}^\infty_{\mathrm{b}}$, is needed to obtain the lower bound on the principal symbol $\sum g^{ij}\xi_i\xi_j$.

2.7 Boundedness on L^2

So far pseudodifferential operators have been considered as continuous operators on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$. Using the composition theorem we can now prove "finite order" regularity results, the basic one being:

Theorem 2.7.1. *Every $A \in \Psi^0(\mathbb{R}^n)$ extends by continuity from $\mathcal{S}(\mathbb{R}^n)$ to a bounded linear operator $A : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$, which coincides with the restriction to $L^2(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$ of the action (2.2.10) on $\mathcal{S}'(\mathbb{R}^n)$.*

The proof is in two stages: residual operators are handled by direct estimation, and the general case is reduced to that one by Hörmander's trick of using an approximate square root.

Lemma 2.7.2 (Schur)**.** *If K is a measurable function on $\mathbb{R}^{2n}$ such that*

$$M_1 = \sup_{x\in\mathbb{R}^n} \int_{\mathbb{R}^n} |K(x,y)|\,dy < \infty, \qquad M_2 = \sup_{y\in\mathbb{R}^n} \int_{\mathbb{R}^n} |K(x,y)|\,dx < \infty,$$

then $T\phi(x) = \int K(x,y)\phi(y)\,dy$ defines a bounded operator on $L^2(\mathbb{R}^n)$ with $\|T\| \le \sqrt{M_1 M_2}$.

Proof. By the Cauchy–Schwarz inequality applied to $|K|^{1/2} \cdot |K|^{1/2}|\phi|$,

$$|T\phi(x)|^2 \le \int |K(x,y)|\,dy \int |K(x,y)||\phi(y)|^2\,dy \le M_1 \int |K(x,y)||\phi(y)|^2\,dy,$$

and integrating in x (Tonelli) gives $\|T\phi\|^2 \le M_1 M_2 \|\phi\|^2$. □

By Proposition 2.3.3, the kernel of $G \in \Psi^{-\infty}(\mathbb{R}^n)$ satisfies $|K(x,y)| \le C\langle x-y\rangle^{-n-1}$, and $\int \langle z\rangle^{-n-1} dz < \infty$. Hence every $G \in \Psi^{-\infty}(\mathbb{R}^n)$ is bounded on $L^2(\mathbb{R}^n)$.

We say that a real symbol $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ is *positive elliptic* if $a \ge \epsilon\langle\xi\rangle^m$ for some $\epsilon > 0$. By Lemma 2.1.6, $a^s \in S^{sm}$ for all $s \in \mathbb{R}$.

Proposition 2.7.3. *Let $A \in \Psi^m(\mathbb{R}^n)$ be self-adjoint, $A = A^*$, and suppose that $\sigma_m(A)$ has a positive elliptic representative. Then there exists $B \in \Psi^{m/2}(\mathbb{R}^n)$ with $B = B^*$ and*

$$A = B^2 + G, \qquad G \in \Psi^{-\infty}(\mathbb{R}^n).$$

Proof. Let a be a positive elliptic representative of $\sigma_m(A)$ and $b_0 = a^{1/2} \in S^{m/2}(\mathbb{R}^n;\mathbb{R}^n)$. Let $B_0 = \frac{1}{2}(\mathrm{Op}(b_0) + \mathrm{Op}(b_0)^*)$; then $B_0 = B_0^*$ and $\sigma_{m/2}(B_0) = [b_0]$ since b_0 is real. By (2.5.1), $\sigma_m(B_0^2) = [b_0^2] = [a]$, so $E_1 = A - B_0^2 \in \Psi^{m-1}(\mathbb{R}^n)$.

We proceed inductively. Suppose $B_j \in \Psi^{m/2-j}(\mathbb{R}^n)$, $B_j = B_j^*$, have been chosen for $j \le N$ such that, with $S_N = \sum_{j=0}^N B_j$,

$$E_{N+1} = A - S_N^2 \in \Psi^{m-N-1}(\mathbb{R}^n);$$

this has been done for $N = 0$. Note $E_{N+1}^* = E_{N+1}$. For $B_{N+1} \in \Psi^{m/2-N-1}(\mathbb{R}^n)$,

$$A - (S_N + B_{N+1})^2 = E_{N+1} - S_N B_{N+1} - B_{N+1} S_N - B_{N+1}^2.$$

Since $S_N - B_0 \in \Psi^{m/2-1}$, all terms on the right are of order $m-N-2$ except $E_{N+1} - B_0 B_{N+1} - B_{N+1} B_0 \in \Psi^{m-N-1}$, whose principal symbol is

$$\sigma_{m-N-1}(E_{N+1}) - 2b_0\,\sigma_{m/2-N-1}(B_{N+1}).$$

Let e represent $\sigma_{m-N-1}(E_{N+1})$; by Lemma 2.1.6, $e/(2b_0) \in S^{m/2-N-1}$. Let $B_{N+1} = \frac{1}{2}(\mathrm{Op}(e/2b_0) + \mathrm{Op}(e/2b_0)^*)$. Since E_{N+1} is self-adjoint, $[e] = [\bar{e}]$ and so $\sigma_{m/2-N-1}(B_{N+1}) = [e/(2b_0)]$. Then $A - S_{N+1}^2 \in \Psi^{m-N-2}$, completing the induction.

By Corollary 2.3.4(iii) choose $B' \in \Psi^{m/2}$ with $B' - S_N \in \Psi^{m/2-N-1}$ for all N, and set $B = \frac{1}{2}(B' + B'^*)$. Since $S_N = S_N^*$, also $B - S_N = \frac{1}{2}((B' - S_N) + (B' - S_N)^*) \in \Psi^{m/2-N-1}$. Writing $B = S_N + R_N$,

$$A - B^2 = E_{N+1} - S_N R_N - R_N S_N - R_N^2 \in \Psi^{m-N-1}(\mathbb{R}^n)$$

for every N. Thus $G = A - B^2 \in \Psi^{-\infty}(\mathbb{R}^n)$. □

Proof of Theorem 2.7.1. Let $a = \sigma_L(A) \in S^0$ and $C = 1 + \sup|a|^2$. Then $C - A^*A \in \Psi^0$ is self-adjoint and by (2.5.1) its principal symbol is represented by $C - |a|^2 \geq 1$, which is positive elliptic of order 0. By Proposition 2.7.3, $C - A^*A = B^2 + G$ with $B = B^* \in \Psi^0$ and $G \in \Psi^{-\infty}$; G is bounded on L^2 by Schur's lemma. For $\phi \in \mathcal{S}(\mathbb{R}^n)$,

$$\|A\phi\|^2 = (A^*A\phi, \phi) = C\|\phi\|^2 - \|B\phi\|^2 - (G\phi, \phi) \leq (C + \|G\|_{L^2 \to L^2})\|\phi\|^2.$$

Since $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ (Exercise 2.14), A extends uniquely to a bounded operator on L^2. If $\phi_k \to u$ in L^2 then $\phi_k \to u$ in $\mathcal{S}'$, so $A\phi_k \to Au$ in $\mathcal{S}'$ by the continuity of (2.2.10); hence the two extensions agree. □

2.8 Sobolev spaces

Recall that for $k \in \mathbb{N}_0$ the Sobolev space $H^k(\mathbb{R}^n)$ consists of the $u \in L^2(\mathbb{R}^n)$ with $D^\alpha u \in L^2(\mathbb{R}^n)$ for all $|\alpha| \leq k$. By Plancherel's theorem this means $\xi^\alpha \hat{u} \in L^2$ for $|\alpha| \leq k$, and since

$$\langle \xi \rangle^{2k} = (1 + \xi_1^2 + \cdots + \xi_n^2)^k = \sum_{|\beta| \leq k} c_\beta \, \xi^{2\beta}, \qquad c_\beta > 0, \tag{2.8.1}$$

these finitely many conditions are equivalent to the single condition $\langle \xi \rangle^k \hat{u} \in L^2$, i.e. $\langle D \rangle^k u \in L^2(\mathbb{R}^n)$, where $\langle D \rangle^s = \mathrm{Op}(\langle \xi \rangle^s) \in \Psi^s(\mathbb{R}^n)$ as in Example 2.2.8. For negative integers $-k$, one defines

$$H^{-k}(\mathbb{R}^n) = \Big\{ u \in \mathcal{S}'(\mathbb{R}^n);\ u = \sum_{|\beta| \leq k} D^\beta u_\beta,\ u_\beta \in L^2(\mathbb{R}^n) \Big\},$$

and again $u \in H^{-k}(\mathbb{R}^n)$ if and only if $\langle D \rangle^{-k} u \in L^2$. Indeed, if $u = \sum D^\beta u_\beta$ then $\langle \xi \rangle^{-k} \hat{u} = \sum \xi^\beta \langle \xi \rangle^{-k} \hat{u}_\beta \in L^2$ since $|\xi^\beta| \leq \langle \xi \rangle^k$. Conversely, if $\langle \xi \rangle^{-k} \hat{u} \in L^2$, (2.8.1) gives $u = \sum_{|\beta| \leq k} D^\beta u_\beta$ with $\hat{u}_\beta = c_\beta \xi^\beta \langle \xi \rangle^{-2k} \hat{u}$, and $|\hat{u}_\beta| \leq c_\beta \langle \xi \rangle^{-k} |\hat{u}|$, so $u_\beta \in L^2$. In view of this we define:

Definition 2.8.1. For $s \in \mathbb{R}$, the Sobolev space of order s is

$$H^s(\mathbb{R}^n) = \{ u \in \mathcal{S}'(\mathbb{R}^n);\ \langle D \rangle^s u \in L^2(\mathbb{R}^n) \}, \qquad \|u\|_{H^s}^2 = \|\langle D \rangle^s u\|_{L^2}^2 = \int \langle \xi \rangle^{2s} |\hat{u}(\xi)|^2 d\!\!\bar{}\xi.$$

For $q \in \mathbb{R}$, the weighted Sobolev space is $\langle x \rangle^q H^s(\mathbb{R}^n) = \{ u \in \mathcal{S}'(\mathbb{R}^n);\ \langle x \rangle^{-q} u \in H^s(\mathbb{R}^n) \}$, with norm $\|u\|_{\langle x \rangle^q H^s} = \|\langle x \rangle^{-q} u\|_{H^s}$.

$H^s(\mathbb{R}^n)$ is a Hilbert space, isometric via $\langle D \rangle^s$ to $L^2(\mathbb{R}^n)$, and $H^{s'}(\mathbb{R}^n) \subset H^s(\mathbb{R}^n)$ for $s' \geq s$. Unfortunately these spaces get *smaller* as s increases, whereas $\Psi^m(\mathbb{R}^n)$ gets bigger with m. We set

$$H^\infty(\mathbb{R}^n) = \bigcap_s H^s(\mathbb{R}^n), \qquad H^{-\infty}(\mathbb{R}^n) = \bigcup_s H^s(\mathbb{R}^n).$$

Then $\mathcal{S} \subsetneq H^\infty \subsetneq H^{-\infty} \subsetneq \mathcal{S}'$: for instance $\langle x\rangle^{-n} \in H^\infty \setminus \mathcal{S}$ (all its derivatives are $O(\langle x\rangle^{-n})$, hence square integrable), $\delta_0 \in H^{-\infty} \setminus H^\infty$ (as $\hat\delta_0 = 1$), and $1 \notin H^{-\infty}$ (as $\hat 1 = (2\pi)^n\delta_0$). In particular $H^{-\infty}$ does not capture all tempered distributions; the weighted spaces remedy this (Lemma 2.8.3). Note also the Sobolev embedding: if $s > n/2$ then $H^s(\mathbb{R}^n)$ is contained in the bounded continuous functions, since $\int|\hat u|\,\text{đ}\xi \le \big(\int\langle\xi\rangle^{-2s}\text{đ}\xi\big)^{1/2}\|u\|_{H^s}$.

Theorem 2.8.2. *For all $q, s, M \in \mathbb{R}$, each $A \in \Psi^M(\mathbb{R}^n)$ defines a continuous linear map*

$$A : \langle x\rangle^q H^s(\mathbb{R}^n) \longrightarrow \langle x\rangle^q H^{s-M}(\mathbb{R}^n).$$

Proof. First let $q = 0$. The operator $\langle D\rangle^{s'}$ is an isometric isomorphism $H^t \to H^{t-s'}$ for all t, with inverse $\langle D\rangle^{-s'}$. By the composition theorem $B = \langle D\rangle^{s-M}A\langle D\rangle^{-s} \in \Psi^0(\mathbb{R}^n)$, which is bounded on L^2 by Theorem 2.7.1. For $u \in H^s$, $v = \langle D\rangle^s u \in L^2$ and $Au = \langle D\rangle^{M-s}Bv$ (as operators on $\mathcal{S}'$), so $\|Au\|_{H^{s-M}} = \|Bv\|_{L^2} \le \|B\|\|u\|_{H^s}$.

$$\begin{array}{ccc} H^s(\mathbb{R}^n) & \xrightarrow{\ A\ } & H^{s-M}(\mathbb{R}^n) \\ {\scriptstyle\langle D\rangle^s}\big\downarrow & & \big\downarrow{\scriptstyle\langle D\rangle^{s-M}} \\ L^2(\mathbb{R}^n) & \xrightarrow{\ B\ } & L^2(\mathbb{R}^n) \end{array}$$

For general q, multiplication by $\langle x\rangle^q$ is by definition an isometry of H^s onto $\langle x\rangle^q H^s$, so it suffices to show that $Q = \langle x\rangle^{-q}A\langle x\rangle^q \in \Psi^M(\mathbb{R}^n)$. Write $A = \mathrm{Op}(a)$ with $a \in S^M(\mathbb{R}^n;\mathbb{R}^n)$. Then the kernel of Q is $fI(a)$ with $f(x,y) = \langle x\rangle^{-q}\langle y\rangle^q$. Now $D_x^\alpha D_y^\beta f = (D^\alpha\langle x\rangle^{-q})(D^\beta\langle y\rangle^q)$ is bounded by $C\langle x\rangle^{-q}\langle y\rangle^q \le C2^{|q|/2}\langle x-y\rangle^{|q|}$, by Lemma 2.1.4(i) and Peetre's inequality; so f has weight $|q|$, and by Lemma 2.2.4, $fI(a) = I(fa)$ with $fa \in \mathcal{A}^M_{|q|}(\mathbb{R}^n)$. Hence $Q = \mathrm{Op}(fa) \in \Psi^M(\mathbb{R}^n)$. □

This is a typical instance where amplitudes of non-zero weight are useful; the left symbol of Q is of course given by Theorem 2.3.1.

Lemma 2.8.3. $\mathcal{S}'(\mathbb{R}^n) = \bigcup_{M\in\mathbb{N}_0}\langle x\rangle^M H^{-M}(\mathbb{R}^n)$.

Proof. Let $u \in \mathcal{S}'(\mathbb{R}^n)$. There are C and k with $|u(\phi)| \le C\max_{|\alpha|,|\beta|\le k}\sup|x^\alpha D^\beta\phi|$. For $|\alpha|, |\beta| \le k$ the operator $x^\alpha D^\beta\langle x\rangle^{-k}$ is a differential operator of order $\le k$ with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ (by Leibniz' formula, as $x^\alpha D^\gamma\langle x\rangle^{-k}$ is bounded with all derivatives for $|\alpha| \le k$), so it maps H^{n+k} to H^n boundedly by Theorem 2.8.2. With the Sobolev embedding this gives $\sup|x^\alpha D^\beta\phi| \le C\|\langle x\rangle^k\phi\|_{H^N}$, $N = n + k$. Hence $\psi \mapsto u(\langle x\rangle^{-k}\langle D\rangle^{-N}\psi)$, $\psi \in \mathcal{S}(\mathbb{R}^n)$, is bounded in the L^2 norm, and by the Riesz representation theorem there is $h \in L^2(\mathbb{R}^n)$ with $u(\langle x\rangle^{-k}\langle D\rangle^{-N}\psi) = \int h\psi\,dx$. Since $\langle x\rangle^{-k}\langle D\rangle^{-N}$ is an isomorphism of $\mathcal{S}(\mathbb{R}^n)$ with inverse $\langle D\rangle^N\langle x\rangle^k$, and $\langle D\rangle^N$ is its own transpose,

$$u(\phi) = \int h\,\langle D\rangle^N(\langle x\rangle^k\phi)\,dx = (\langle x\rangle^k\langle D\rangle^N h)(\phi),$$

i.e. $u = \langle x\rangle^k\langle D\rangle^N h \in \langle x\rangle^k H^{-N} \subset \langle x\rangle^M H^{-M}$ with $M = N$. Here we used that $\langle x\rangle^k H^{-N} \subset \langle x\rangle^M H^{-M}$ for $k \le M$, $N \le M$, which holds because multiplication by $\langle x\rangle^{k-M} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ preserves H^{-N} by Theorem 2.8.2. □

Alternatively this follows from the structure theorem for tempered distributions: every $u \in \mathcal{S}'(\mathbb{R}^n)$ is a finite sum $\sum x^\alpha D^\beta u_{\alpha\beta}$ with $u_{\alpha\beta}$ bounded and continuous, and bounded continuous functions lie in $\langle x\rangle^n L^2(\mathbb{R}^n)$.

The elliptic regularity result can now be refined.

Proposition 2.8.4. *Let $A \in \Psi^m(\mathbb{R}^n)$ be elliptic. If $u \in \langle x\rangle^{p'} H^{s'}(\mathbb{R}^n)$ and $Au \in \langle x\rangle^{p} H^{s}(\mathbb{R}^n)$, then $u \in \langle x\rangle^{p''} H^{s+m}(\mathbb{R}^n)$ with $p'' = \max(p,p')$. In particular, if $Au \in \langle x\rangle^{p} H^{\infty}$ then $u \in \langle x\rangle^{p''} H^{\infty}$.*

Proof. Let $B \in \Psi^{-m}$ be a parametrix, $BA = \mathrm{Id} + G$ with $G \in \Psi^{-\infty}$. Then by Theorem 2.8.2

$$u = B(Au) - Gu \in \langle x\rangle^{p} H^{s+m}(\mathbb{R}^n) + \langle x\rangle^{p'} H^{\infty}(\mathbb{R}^n) \subset \langle x\rangle^{p''} H^{s+m}(\mathbb{R}^n),$$

using that $\langle x\rangle^{p} H^{t} \subset \langle x\rangle^{p''} H^{t}$ for $p \le p''$ (as in the proof of Lemma 2.8.3). □

2.9 Polyhomogeneous operators

So far we have considered operators whose amplitudes satisfy the symbol estimates. Some weakenings of these estimates are indicated in Exercise 2.9. Here instead we consider a restriction, to symbols with asymptotic expansions in homogeneous functions; this is the most commonly used class of pseudodifferential operators.

A function $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^n)$ is *homogeneous of degree $\mu \in \mathbb{R}$ in $|\xi| \ge 1$* if

$$a(z,t\xi) = t^\mu a(z,\xi) \quad \text{for all } z,\ t \ge 1,\ |\xi| \ge 1.$$

If such a function also satisfies the uniform regularity estimates

$$\sup_{z\in\mathbb{R}^p,\ |\xi|\le 2} |D_z^\alpha D_\xi^\beta a(z,\xi)| < \infty \quad \text{for all } \alpha,\beta, \tag{2.9.1}$$

then $a \in S^\mu(\mathbb{R}^p;\mathbb{R}^n)$. Indeed, differentiating the homogeneity relation shows that $D_z^\alpha D_\xi^\beta a$ is homogeneous of degree $\mu - |\beta|$ in $|\xi| > 1$, hence by continuity in $|\xi| \ge 1$, so for $|\xi| \ge 1$

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| = |\xi|^{\mu-|\beta|} |D_z^\alpha D_\xi^\beta a(z,\xi/|\xi|)| \le C\langle\xi\rangle^{\mu-|\beta|},$$

while for $|\xi| \le 1$ the estimates are given by (2.9.1). Conversely (2.9.1) holds for any symbol. A function homogeneous of degree μ in $|\xi| \ge 1$ which lies in $S^{\mu-1}$ vanishes in $|\xi| \ge 1$, since $|a(z,\xi)| = t^{-\mu}|a(z,t\xi)| \le Ct^{-1}\langle\xi\rangle^{\mu-1}$ for all $t \ge 1$.

Definition 2.9.1. For $m \in \mathbb{R}$, the space $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n) \subset S^m(\mathbb{R}^p;\mathbb{R}^n)$ of (*one-step*) *polyhomogeneous* or *classical* symbols consists of those a for which there exist $a_{m-j} \in S^{m-j}(\mathbb{R}^p;\mathbb{R}^n)$, homogeneous of degree $m-j$ in $|\xi| \ge 1$, $j \in \mathbb{N}_0$, such that

$$a \sim \sum_{j=0}^{\infty} a_{m-j}.$$

By the remark preceding the definition, the terms a_{m-j} are uniquely determined by a in $|\xi| \ge 1$ (if $\sum_{j<N}(a_{m-j} - a'_{m-j}) \in S^{m-N}$ then inductively each difference is homogeneous and of lower order, so vanishes in $|\xi| \ge 1$). Clearly $S^{-\infty} \subset S^m_{\mathrm{ph}}$ for every m (take all $a_{m-j} = 0$), and $S^m_{\mathrm{ph}} \cap S^{m-1} = S^{m-1}_{\mathrm{ph}}$: if $a \in S^m_{\mathrm{ph}} \cap S^{m-1}$ then $a_m \in S^{m-1}$ vanishes in $|\xi| \ge 1$, so $a_m \in S^{-\infty}$ and $a \sim a'_{m-1} + \sum_{j\ge 2} a_{m-j}$ with $a'_{m-1} = a_{m-1} + a_m$, which is homogeneous of degree $m-1$ in $|\xi| \ge 1$. The choice of 1 in "$|\xi| \ge 1$" is immaterial: if h is homogeneous of degree μ in $|\xi| \ge R$, then $h'(z,\xi) = \rho(\xi)|\xi|^\mu R^{-\mu} h(z, R\xi/|\xi|)$, with $\rho \in \mathcal{C}^\infty$ vanishing near 0 and equal to 1 in $|\xi| \ge 1$, is homogeneous of degree μ in $|\xi| \ge 1$ and $h - h'$ has compact support in ξ, uniformly in z, so is residual.

Lemma 2.9.2. (i) *$S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n) \cdot S^{m'}_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n) \subset S^{m+m'}_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$; D_{z_j} preserves S^m_{ph}, D_{ξ_j} maps S^m_{ph} into S^{m-1}_{ph}, and complex conjugation and restriction to linear subspaces of $\mathbb{R}^p$ preserve polyhomogeneity.*

(ii) *If* $b_j \in S^{m-j}_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ *and* $b \sim \sum_j b_j$ *then* $b \in S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$.

(iii) *Polynomials in* ξ *with* $\mathcal{C}^\infty_{\mathrm{b}}$ *coefficients are polyhomogeneous, and* $\langle\xi\rangle^s \in S^s_{\mathrm{ph}}(\mathbb{R}^n)$ *for all* $s \in \mathbb{R}$*; multiplication by* $\langle\xi\rangle^s$ *is an isomorphism* $S^m_{\mathrm{ph}} \to S^{m+s}_{\mathrm{ph}}$.

Proof. (i) Homogeneity is preserved by products, D_z, conjugation and restriction, and D_ξ lowers the degree of homogeneity by one (in $|\xi| > 1$, hence by continuity in $|\xi| \geq 1$). The asymptotic expansions are then given by Lemma 2.1.11(i),(ii).

(ii) Write $b_j \sim \sum_k b_{j,k}$ with $b_{j,k}$ homogeneous of degree $m-j-k$. Then $c_l = \sum_{j+k=l} b_{j,k}$ is homogeneous of degree $m-l$ in $|\xi| \geq 1$, and $b \sim \sum_l c_l$ by Lemma 2.1.11(iii).

(iii) A polynomial $\sum_{|\alpha|\leq k} p_\alpha(z)\xi^\alpha$ is the finite sum of its homogeneous parts. For $\langle\xi\rangle^s$, let ρ be as above. For $|\xi| \geq 1$, $\langle\xi\rangle^s = |\xi|^s(1+|\xi|^{-2})^{s/2}$, and by the binomial series, for $K \in \mathbb{N}$,

$$\langle\xi\rangle^s - \sum_{k<K} \binom{s/2}{k} \rho(\xi)|\xi|^{s-2k} = |\xi|^{s-2K} h_K(|\xi|^{-2}) \quad \text{in } |\xi| \geq 2,$$

where h_K is real-analytic on $[0, \frac{1}{4}]$. Since $|\xi|^{-2}$ is homogeneous of degree -2 in $|\xi| \geq 1$, hence a symbol of order -2 there, the chain rule shows that $h_K(|\xi|^{-2})$ satisfies symbol estimates of order 0 in $|\xi| \geq 2$; so the left side is in S^{s-2K}. Thus $\langle\xi\rangle^s \sim \sum_k \binom{s/2}{k}\rho(\xi)|\xi|^{s-2k}$. The last statement follows from (i), with inverse multiplication by $\langle\xi\rangle^{-s}$. □

Definition 2.9.3. For $m \in \mathbb{R}$, $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ is the space of operators $\mathrm{Op}(a)$ with $a = \langle x-y\rangle^w \tilde{a}$, $\tilde{a} \in S^m_{\mathrm{ph}}(\mathbb{R}^{2n};\mathbb{R}^n)$, for some $w \in \mathbb{R}$.

Proposition 2.9.4. *An operator* $A \in \Psi^m(\mathbb{R}^n)$ *lies in* $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ *if and only if* $\sigma_L(A) \in S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$. *The spaces* $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ *form an order-filtered* $*$*-subalgebra of* $\Psi^\infty(\mathbb{R}^n)$ *containing* $\Psi^{-\infty}(\mathbb{R}^n)$ *and* $\mathrm{Diff}^m_{\mathrm{b}}(\mathbb{R}^n)$, *and for every* $A \in \Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ *also* $\sigma_R(A) \in S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$.

Proof. If $\sigma_L(A) \in S^m_{\mathrm{ph}}$ then $A = \mathrm{Op}(\sigma_L(A))$ is of the required form with $w = 0$. Conversely let $A = \mathrm{Op}(a)$, $a = \langle x-y\rangle^w\tilde{a}$. By Leibniz' formula

$$D_y^\alpha a(x,y,\xi)\big|_{y=x} = \sum_{\gamma\leq\alpha} \binom{\alpha}{\gamma} c_\gamma\, D_y^{\alpha-\gamma}\tilde{a}(x,y,\xi)\big|_{y=x}, \qquad c_\gamma = D_y^\gamma\langle x-y\rangle^w\big|_{y=x} \in \mathbb{C},$$

so by Lemma 2.9.2(i) each term in (2.3.1) is polyhomogeneous, of order $m-|\alpha|$, and $\sigma_L(A) \in S^m_{\mathrm{ph}}$ by Lemma 2.9.2(ii). The formulas (2.4.1), (2.4.2) and (2.4.3), together with Lemma 2.9.2, show that A^*, AB and $\sigma_R(A)$ are polyhomogeneous when A and B are. The remaining statements follow from $S^{-\infty} \subset S^m_{\mathrm{ph}}$ and Lemma 2.9.2(iii). □

This subalgebra is often denoted simply $\Psi^m(\mathbb{R}^n)$, its elements being called *classical* pseudodifferential operators. For $A \in \Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ the principal symbol has a preferred representative, the leading term a_m in the expansion of $\sigma_L(A)$, and it is natural to identify $\sigma_m(A)$ with the function a_m, homogeneous of degree m on $|\xi| \geq 1$ (or by extension on $\mathbb{R}^n \times (\mathbb{R}^n \setminus 0)$). Conversely, if $A \in \Psi^m(\mathbb{R}^n)$ and $\sigma_m(A)$ has a representative a_m homogeneous of degree m in $|\xi| \geq 1$, then $A - \mathrm{Op}(a_m) \in \Psi^{m-1}(\mathbb{R}^n)$; that is,

$$\sigma_m(A) \text{ has a homogeneous representative} \iff A \in \Psi^m_{\mathrm{ph}}(\mathbb{R}^n) + \Psi^{m-1}(\mathbb{R}^n).$$

It is usually the case that a construction in $\Psi^\infty(\mathbb{R}^n)$ applied to elements of $\Psi^\infty_{\mathrm{ph}}(\mathbb{R}^n)$ yields elements of $\Psi^\infty_{\mathrm{ph}}(\mathbb{R}^n)$. As an illustration:

Proposition 2.9.5. *If* $A \in \Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ *is elliptic, then every parametrix of* A *lies in* $\Psi^{-m}_{\mathrm{ph}}(\mathbb{R}^n)$.

Proof. Since parametrices are unique modulo $\Psi^{-\infty}(\mathbb{R}^n) \subset \Psi^{-m}_{\mathrm{ph}}(\mathbb{R}^n)$, it suffices to construct one in Ψ^{-m}_{ph}. Let a_m be the leading term of $\sigma_L(A)$. By Theorem 2.6.3(d), $|\sigma_L(A)| \geq \epsilon\langle\xi\rangle^m$ for $|\xi| \geq R$, hence $|a_m| \geq \frac{\epsilon}{2}\langle\xi\rangle^m$ for $|\xi| \geq R' \geq 1$; by homogeneity $|a_m(x,\theta)| = R'^{-m}|a_m(x,R'\theta)| \geq \frac{\epsilon}{2}(\langle R'\rangle/R')^m \geq c > 0$ for $|\theta| = 1$, so $|a_m| \geq c|\xi|^m$ in $|\xi| \geq 1$. As $\nabla_\xi a_m$ is bounded on $|\xi| \leq 2$, uniformly in x, there is $\delta \in (0, \frac{1}{2})$ with $|a_m| \geq \frac{c}{2}$ on $|\xi| \geq 1-\delta$. Choose $\rho \in \mathcal{C}^\infty(\mathbb{R}^n)$ vanishing in $|\xi| \leq 1-\delta$ and equal to 1 in $|\xi| \geq 1$, and set $b = \rho/a_m$. By Lemma 2.1.6 (applied on $\{|\xi| > 1-\delta\}$) $b \in S^{-m}$, and b is homogeneous of degree $-m$ in $|\xi| \geq 1$, so $b \in S^{-m}_{\mathrm{ph}}$. Let $B_0 = \mathrm{Op}(b) \in \Psi^{-m}_{\mathrm{ph}}$. Then $E = AB_0 - \mathrm{Id} \in \Psi^0_{\mathrm{ph}}$ and $\sigma_L(E) \equiv a_m b - 1 = \rho - 1 \equiv 0$ modulo S^{-1}, so $\sigma_L(E) \in S^0_{\mathrm{ph}} \cap S^{-1} = S^{-1}_{\mathrm{ph}}$. The operator F of Lemma 2.6.2 has $\sigma_L(F) \sim \sum_{j\geq 1}\sigma_L((-E)^j)$, which is polyhomogeneous by Lemma 2.9.2(ii) and Proposition 2.9.4. Hence $B = B_0(\mathrm{Id}+F) \in \Psi^{-m}_{\mathrm{ph}}(\mathbb{R}^n)$ is a parametrix. □

As a subspace, $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ is not closed in $S^m(\mathbb{R}^p;\mathbb{R}^n)$: since it contains $S^{-\infty}$, its closure contains $S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ for every $m' < m$, by Lemma 2.1.5. To capture its properties one strengthens the topology: in addition to the symbol norms $\|a\|_{N,m}$ one uses the norms

$$\sup_{z\in\mathbb{R}^p,\ 1\leq|\xi|\leq 2} |D_z^\alpha D_\xi^\beta a_{m-j}(z,\xi)| \quad \text{and} \quad \Big\|a - \sum_{j=0}^{k} a_{m-j}\Big\|_{N,m-k-1}, \qquad j,k,N \in \mathbb{N}_0,\ \alpha,\beta,$$

on the terms of the expansion. With the corresponding metric topology, $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ is complete (Exercise 2.8). When $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ needs to be topologized, it is this topology which is transferred by the identification $\sigma_L : \Psi^m_{\mathrm{ph}}(\mathbb{R}^n) \to S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$.

2.10 Affine invariance

The algebra $\Psi^\infty(\mathbb{R}^n)$ is invariant under affine changes of coordinates. Let $T : \mathbb{R}^n \to \mathbb{R}^n$, $Tx = Lx + c$, with $L \in \mathrm{GL}(n,\mathbb{R})$ and $c \in \mathbb{R}^n$, and let $T^*f = f \circ T$, an isomorphism of $\mathcal{S}(\mathbb{R}^n)$ and of $\mathcal{S}'(\mathbb{R}^n)$.

Proposition 2.10.1. *If $A \in \Psi^m(\mathbb{R}^n)$ then $A_T = T^*A(T^*)^{-1} \in \Psi^m(\mathbb{R}^n)$, and*

$$\sigma_L(A_T)(x,\xi) = \sigma_L(A)(Tx, (L^t)^{-1}\xi), \tag{2.10.1}$$

where L^t is the transpose of L. The same holds for $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$.

Proof. Let $a = \sigma_L(A)$, $f \in \mathcal{S}(\mathbb{R}^n)$ and $g = (T^*)^{-1}f = f \circ T^{-1}$. Substituting $y = Ly' + c$, $\hat{g}(\xi) = |\det L| e^{-ic\cdot\xi}\hat{f}(L^t\xi)$. Hence, by (2.2.8),

$$\begin{aligned} A_T f(x) = Ag(Tx) &= \int e^{i(Lx+c)\cdot\xi} a(Tx,\xi)|\det L| e^{-ic\cdot\xi}\hat{f}(L^t\xi)đ\xi \\ &= \int e^{ix\cdot\eta} a(Tx,(L^t)^{-1}\eta)\hat{f}(\eta)đ\eta, \end{aligned}$$

with $\eta = L^t\xi$. The function $a(Tx,(L^t)^{-1}\eta)$ is in $S^m(\mathbb{R}^n;\mathbb{R}^n)$, since $C^{-1}\langle\eta\rangle \leq \langle(L^t)^{-1}\eta\rangle \leq C\langle\eta\rangle$ and derivatives transform linearly; so $A_T = \mathrm{Op}\big(a(Tx,(L^t)^{-1}\eta)\big)$. If a is polyhomogeneous, so is $a(Tx,(L^t)^{-1}\eta)$: its terms are homogeneous in $|(L^t)^{-1}\eta| \geq 1$, which contains $|\eta| \geq R$ for some R, and by the remark after Definition 2.9.1 this suffices. □

In particular, for a translation $Tx = x + c$ we get $\sigma_L(A_T)(x,\xi) = \sigma_L(A)(x+c,\xi)$, and for a linear map there is no Jacobian factor in (2.10.1). This invariance means that the spaces $\Psi^m(V)$ and $\Psi^m_{\mathrm{ph}}(V)$ of operators on $\mathcal{S}(V)$ are well defined for any finite-dimensional real vector space (or affine space) V, with symbols defined on $V \times V^*$. Invariance under general changes of coordinates, which is needed to define pseudodifferential operators on manifolds, is considerably more subtle and is discussed in Chapters 3 and 5.

2.11 Exercises

Exercise 2.1. Show in detail that $\langle\xi\rangle^m \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ for every $m \in \mathbb{R}$ and every p, and deduce that $S^m(\mathbb{R}^p;\mathbb{R}^n)\cdot S^{m'}(\mathbb{R}^p;\mathbb{R}^n) = S^{m+m'}(\mathbb{R}^p;\mathbb{R}^n)$.

Exercise 2.2. Let $n=2$ and identify $\mathbb{R}^2_\xi$ with $\mathbb{C}$ via $\zeta = \xi_1 + i\xi_2$. Let $a \in S^1(\mathbb{R}^2)$ (a symbol in ξ alone) be elliptic. Show that for $r>0$ sufficiently large

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{1}{a(re^{i\theta})}\frac{d}{d\theta}a(re^{i\theta})\,d\theta$$

exists, is an integer, and is independent of r. Conclude that there is an elliptic symbol $a \in S^1(\mathbb{R}^2)$ such that there is no $b \in S^1(\mathbb{R}^2)$ with $b \neq 0$ on all of $\mathbb{R}^2$ and $b(\xi) = a(\xi)$ for $|\xi| > r$, for any r. *Hint:* consider a equal to $\xi_1 + i\xi_2$ for $|\xi| \geq 1$.

Exercise 2.3. Show that a symbol $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ which satisfies $|a(z,\xi)| \leq C\langle\xi\rangle^{m'}$ for some $m' < m$ is necessarily in $S^{m'+\epsilon}(\mathbb{R}^p;\mathbb{R}^n)$ for every $\epsilon > 0$. *Hint:* use the interpolation inequality $\sup_I |f'|^2 \leq C\sup_I|f|(\sup_I|f''| + \delta^{-2}\sup_I|f|)$ for functions on an interval I of length δ, on balls of radius comparable to $\langle\xi\rangle$ in ξ and of radius 1 in z, and iterate.

Exercise 2.4. Let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\psi(\xi) = 1$ in $|\xi| < 1$. Show that if $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p\times\mathbb{R}^n)$ then $c_\phi(z,\xi) = \phi(z,\xi/|\xi|)(1-\psi(\xi)) \in S^0(\mathbb{R}^p;\mathbb{R}^n)$. For $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ define the *cone support* of a by its complement: $(\bar z,\bar\xi) \in \mathbb{R}^p\times(\mathbb{R}^n\setminus 0)$ is *not* in $\operatorname{cone\,supp}(a)$ if there exists $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p\times\mathbb{R}^n)$ with $\phi(\bar z,\bar\xi/|\bar\xi|) \neq 0$ such that $c_\phi a \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$. Show that for $a \in S^m$ and $b \in S^{m'}$

$$\operatorname{cone\,supp}(ab) \subset \operatorname{cone\,supp}(a) \cap \operatorname{cone\,supp}(b).$$

If $\operatorname{cone\,supp}(a) = \emptyset$, does it follow that $a \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$? (Consider $p=n=1$ and $a(z,\xi) = \chi(\xi/(1+z^2))$ with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ supported in $1 \leq |t| \leq 2$.)

Exercise 2.5. Show that the Schwartz kernel of any $A \in \Psi^\infty(\mathbb{R}^n)$ is smooth away from the diagonal $\{x=y\}$, and that it is rapidly decreasing, with all derivatives, as $|x-y| \to \infty$. *Hint:* by Lemma 2.2.4, $(x-y)^\alpha K_A$ becomes increasingly smooth as $|\alpha|$ increases.

Exercise 2.6. For $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ let $\mathrm{Op}_W(a) = \mathrm{Op}(a(\frac{x+y}{2},\xi))$ be its *Weyl quantization*. Show that $\mathrm{Op}_W(a) \in \Psi^m(\mathbb{R}^n)$, that $\mathrm{Op}_W(a)^* = \mathrm{Op}_W(\bar a)$, and that $\sigma_L(\mathrm{Op}_W(a)) \sim \sum_\alpha \frac{(i/2)^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha a$. Deduce that $\mathrm{Op}_W : S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ is a bijection.

Exercise 2.7. (Polyhomogeneous symbols as smooth functions.) Show that $a \in \mathcal{C}^\infty(\mathbb{R}^p\times\mathbb{R}^n)$ is in $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ if and only if a satisfies (2.9.1) and the function $(z,\theta,r) \mapsto r^m a(z,\theta/r)$ on $\mathbb{R}^p\times\mathbb{S}^{n-1}\times(0,1]$ extends to a smooth function on $\mathbb{R}^p\times\mathbb{S}^{n-1}\times[0,1]$ all of whose derivatives are bounded. Relate the Taylor series at $r=0$ to the terms a_{m-j}.

Exercise 2.8. Show that $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ is complete with respect to the norms introduced at the end of Section 2.9, that it is not closed in $S^m(\mathbb{R}^p;\mathbb{R}^n)$, and that its closure in $S^m(\mathbb{R}^p;\mathbb{R}^n)$ contains $S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ for every $m' < m$.

Exercise 2.9. For $0 \leq \delta < \rho \leq 1$ let $S^m_{\rho,\delta}(\mathbb{R}^p;\mathbb{R}^n)$ be the space of $a \in \mathcal{C}^\infty(\mathbb{R}^p\times\mathbb{R}^n)$ with $|D_z^\alpha D_\xi^\beta a| \leq C_{\alpha\beta}\langle\xi\rangle^{m-\rho|\beta|+\delta|\alpha|}$. Show that these spaces are closed under multiplication (orders adding), that $D_\xi^\beta D_z^\alpha$ maps $S^m_{\rho,\delta}$ into $S^{m-\rho|\beta|+\delta|\alpha|}_{\rho,\delta}$, and that series $\sum_j a_j$ with $a_j \in S^{m-j(\rho-\delta)}_{\rho,\delta}$ have asymptotic sums. The symbols of this chapter are those of type $(1,0)$. Which of the steps in the proof of Theorem 2.3.1 use $\rho=1$, $\delta=0$, and how must they be modified? (See [7, Section 7.8] and [8, Section 18.1].)

Exercise 2.10. Show that for any $s \geq 0$ the unit ball in $H^s(\mathbb{R}^n)$ is *not* precompact in $L^2(\mathbb{R}^n)$, i.e. there is a sequence $f_j \in H^s(\mathbb{R}^n)$ with $\|f_j\|_{H^s} \leq 1$ and no subsequence convergent in $L^2(\mathbb{R}^n)$.

Exercise 2.11. Show that for any $R > 0$ and $N > 0$ the subspace $\{u \in H^N(\mathbb{R}^n);\ u(x) = 0 \text{ in } |x| > R\}$ of $H^N(\mathbb{R}^n)$ is compactly included in $L^2(\mathbb{R}^n)$, i.e. the intersection of the unit ball of $H^N(\mathbb{R}^n)$ with this subspace is precompact in $L^2(\mathbb{R}^n)$. *Hint:* for $N > n/2$ one can use the Sobolev embedding and the Arzelà–Ascoli theorem; in general, with $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 on $|x| \leq R$, write $\hat{u} = \widehat{\phi u} = (2\pi)^{-n}\hat{\phi} * \hat{u}$ to see that the Fourier transforms are equicontinuous.

Exercise 2.12. Using Exercise 2.11, or otherwise, show that for any $\epsilon > 0$ the inclusion $\langle x\rangle^{-\epsilon} H^\epsilon(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n)$ is compact, i.e. any sequence f_j such that $\langle x\rangle^\epsilon f_j$ is bounded in $H^\epsilon(\mathbb{R}^n)$ has a subsequence convergent in $L^2(\mathbb{R}^n)$. *Hint:* choose $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi(x) = 1$ in $|x| < 1$ and, for each k, consider the sequence $\phi(x/k)f_j$. Show that its Fourier transforms are equicontinuous and bounded in $\langle \xi\rangle^{-\epsilon} L^2$, deduce that it has a subsequence convergent in $L^2(\mathbb{R}^n)$, and estimate $\|(1 - \phi(x/k))f_j\|_{L^2} \leq C\langle k\rangle^{-\epsilon}$. Conclude by diagonalization.

Exercise 2.13. Prove (2.10.1) directly from the definition of $\mathrm{Op}(a)$ via amplitudes, and show that the right symbol transforms in the same way: $\sigma_R(A_T)(y,\xi) = \sigma_R(A)(Ty, (L^t)^{-1}\xi)$. *Hint:* show that $(A_T)^* = (A^*)_T$.

Exercise 2.14. Show that $\mathcal{S}(\mathbb{R}^n)$, and even $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, is dense in $L^2(\mathbb{R}^n)$.

Exercise 2.15. Show that any continuous linear operator $\mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ has Schwartz kernel in $\mathcal{S}(\mathbb{R}^{2n})$.

Notes

The main departures from Melrose's original text are as follows. The oscillatory integral $I(a)$ is defined directly by the integration by parts formula (2.2.3) rather than by extension by continuity, and all identities between kernels are derived from Lemma 2.2.4. The composition theorem is proved right after the reduction theorem, with the passage from right to left symbols in (2.4.2) carried out explicitly. The residual characterization (Proposition 2.3.3) takes into account that the weight of the amplitudes may depend on the order. The weighted Sobolev estimates use the weight $\langle x\rangle^{-q}\langle y\rangle^q$ as a multiplier of amplitudes, avoiding a case distinction. We have also added the commutator formula (2.5.2). The treatment of L^2 boundedness via an approximate square root is due to Hörmander; for alternative treatments of the calculus see [8], [15] and [11].

Chapter 3

Microlocalization

The pseudodifferential operators of Chapter 2 are not local: the support of Au is in general not contained in the support of u. They are, however, *pseudolocal*: they do not create new singularities. This chapter refines that statement in two directions. First, singularities of a distribution are not only located at points of $\mathbb{R}^n$ but also have directions: the *wavefront set* $\mathrm{WF}(u)$ is a closed conic subset of $T^*\mathbb{R}^n \setminus 0 = \mathbb{R}^n \times (\mathbb{R}^n \setminus 0)$ which projects onto the singular support. Secondly, an operator $A \in \Psi^m(\mathbb{R}^n)$ has an *essential support* $\mathrm{WF}'(A) \subset T^*\mathbb{R}^n \setminus 0$, the set where its symbol is not rapidly decreasing, and A is *microlocal*: $\mathrm{WF}(Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u)$. Conversely, where A is elliptic it has a microlocal parametrix, so $\mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{Char}_m(A)$.

Sections 3.1 and 3.2 deal with supports and singular supports, Sections 3.3 to 3.5 with the essential support, ellipticity at a point and microlocal parametrices, and Section 3.6 with the invariance of the calculus under changes of coordinates; this is what makes it possible to transfer everything to manifolds in Chapter 5. The wavefront set is introduced in Section 3.7 and characterized in terms of the Fourier transform in Section 3.8. The remaining sections develop the calculus of wavefront sets: invariance under diffeomorphisms, the wavefront set of the kernel of a pseudodifferential operator, and the behaviour of wavefront sets under pairing, exterior products, restriction, projection, multiplication, pull-back and push-forward, culminating in the wavefront relation of an operator with a distributional kernel.

Conventions

We use freely the results of Chapters 1 and 2. We write $u(\phi)$ for the (bilinear) pairing of a distribution with a test function and $\hat{u} = \mathcal{F}u$ for the Fourier transform. A function f on a set $U \subset \mathbb{R}^p \times \mathbb{R}^n$ is *rapidly decreasing on* U if $\sup_{(x,\xi)\in U} \langle \xi \rangle^N |f(x,\xi)| < \infty$ for every N; similarly for functions of ξ alone. For $\bar{\xi} \in \mathbb{R}^n \setminus 0$, $\bar{x} \in \mathbb{R}^p$ and $\epsilon > 0$ we use the open cones

$$V_\epsilon(\bar{\xi}) = \Big\{ \xi \in \mathbb{R}^n \setminus 0;\ \Big| \frac{\xi}{|\xi|} - \frac{\bar{\xi}}{|\bar{\xi}|} \Big| < \epsilon \Big\}, \qquad \Gamma_\epsilon(\bar{x},\bar{\xi}) = \{x; |x-\bar{x}| < \epsilon\} \times V_\epsilon(\bar{\xi}).$$

A set $W \subset \mathbb{R}^p \times (\mathbb{R}^n \setminus 0)$ is *conic* if $(x,\xi) \in W \Rightarrow (x, t\xi) \in W$ for all $t > 0$. Every conic neighbourhood of $(\bar{x},\bar{\xi})$ contains some $\Gamma_\epsilon(\bar{x},\bar{\xi})$. For a conic set W we put $W^- = \{(x,-\xi); (x,\xi) \in W\}$. We shall often use the elementary inequality

$$|se - tf| \geq \tfrac{1}{2}|e-f|\,(s+t), \qquad s,t \geq 0,\ |e| = |f| = 1, \tag{3.0.1}$$

which follows from $|se-tf|^2 = (s-t)^2 + st|e-f|^2 \geq \frac{1}{2}|e-f|^2(s^2+t^2)$ (note $|e-f| \leq 2$), and the inequality $\big|\frac{v}{|v|} - \frac{w}{|w|}\big| \leq \frac{2|v-w|}{|w|}$ for $v, w \neq 0$.

From Chapter 2 we recall in particular: every $A \in \Psi^m(\mathbb{R}^n)$ is $A = \mathrm{Op}(\sigma_L(A)) = \mathrm{Op}_R(\sigma_R(A))$ with unique left and right symbols in $S^m(\mathbb{R}^n;\mathbb{R}^n)$; if $A = \mathrm{Op}(a)$ for an amplitude $a \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$ (an amplitude of weight 0), then its kernel is the oscillatory integral $K_A = I(a) = \int e^{i(x-y)\cdot\xi} a(x,y,\xi)\,đ\xi$ and

$$\sigma_L(A)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_y^\alpha a(x,y,\xi)\big|_{y=x}; \tag{3.0.2}$$

$\mathrm{Op}(a)\,\mathrm{Op}_R(b) = \mathrm{Op}(a(x,\xi)b(y,\xi))$ for symbols a, b; $\Psi^{-\infty}(\mathbb{R}^n)$ consists of the operators whose kernels are smooth and satisfy $|D_x^\alpha D_y^\beta K(x,y)| \le C_{\alpha\beta N}\langle x-y\rangle^{-N}$ for all α,β,N; such operators map $\mathcal{S}'(\mathbb{R}^n)$ into $\mathcal{C}^\infty(\mathbb{R}^n)$, with $Eu(x) = u(K_E(x,\cdot))$; every $A \in \Psi^m(\mathbb{R}^n)$ acts continuously on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, the latter by $(Au)(\phi) = u(A^t\phi)$; and an operator $A \in \Psi^m(\mathbb{R}^n)$ which is *(globally) elliptic*, in the sense that $|\sigma_L(A)(x,\xi)| \ge \epsilon\langle\xi\rangle^m$ for $|\xi| \ge R$ and some $\epsilon, R > 0$, has a parametrix $B \in \Psi^{-m}(\mathbb{R}^n)$ with $AB - \mathrm{Id},\ BA - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$.

3.1 Supports and singular supports

Recall from Chapter 1 that the support $\operatorname{supp} u$ of $u \in \mathcal{S}'(\mathbb{R}^n)$ is the complement of the largest open set on which u vanishes; equivalently, $\bar x \notin \operatorname{supp} u$ if and only if there exists $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(\bar x) \ne 0$ and $\phi u = 0$. We write

$$\mathcal{C}_c^{-\infty}(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ \operatorname{supp} u \text{ compact}\}.$$

An element $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ acts on all of $\mathcal{C}^\infty(\mathbb{R}^n)$ by $u(\phi) = u(\psi\phi)$, where $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equals 1 near $\operatorname{supp} u$; this does not depend on ψ. Since u is tempered, $|u(\phi)| \le C\|\psi\phi\|_k$, so u has *finite order*: for a compact neighbourhood L of $\operatorname{supp} u$,

$$|u(\phi)| \le C \sum_{|\beta|\le k} \sup_L |D^\beta\phi|, \qquad \phi \in \mathcal{C}^\infty(\mathbb{R}^n). \tag{3.1.1}$$

3.1.1 The calculus of supports

If $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is continuous and linear, its Schwartz kernel $K_A \in \mathcal{S}'(\mathbb{R}^{2n})$ has a support, and we set $\operatorname{supp}(A) = \operatorname{supp}(K_A)$. A subset $G \subset \mathbb{R}^n \times \mathbb{R}^n$ defines a *relation*, that is a map on subsets,

$$G \circ S = \{x \in \mathbb{R}^n;\ \exists\, y \in S \text{ with } (x,y) \in G\}, \qquad S \subset \mathbb{R}^n.$$

In terms of the projections $\pi_L(x,y) = x$, $\pi_R(x,y) = y$,

$$\begin{array}{ccc} & \mathbb{R}^{2n} & \\ {\scriptstyle\pi_L}\swarrow & & \searrow{\scriptstyle\pi_R} \\ \mathbb{R}^n & & \mathbb{R}^n \end{array} \qquad G \circ S = \pi_L\big(\pi_R^{-1}(S) \cap G\big).$$

If S is compact and G is closed then $G \circ S$ is closed: the restriction of π_L to the closed set $\pi_R^{-1}(S) \cap G$ is proper (the preimage of a compact set K is a closed subset of $K \times S$), and proper continuous maps into $\mathbb{R}^n$ are closed.

Proposition 3.1.1 (Calculus of supports)**.** *If $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is continuous and linear then*

$$\operatorname{supp}(A\phi) \subset \operatorname{supp}(A) \circ \operatorname{supp}(\phi) \quad \textit{for all } \phi \in \mathcal{C}_c^\infty(\mathbb{R}^n).$$

Proof. Since $\operatorname{supp}(A) \circ \operatorname{supp}(\phi)$ is closed, a point $\bar{x}$ outside it has a neighbourhood disjoint from it; choose $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ supported in this neighbourhood with $\psi(\bar{x}) \neq 0$. If $(x, y) \in \operatorname{supp}(A)$ with $y \in \operatorname{supp}\phi$ then $x \in \operatorname{supp}(A) \circ \operatorname{supp}(\phi)$, so $x \notin \operatorname{supp}\psi$. Thus $\operatorname{supp}(A) \cap (\operatorname{supp}\psi \times \operatorname{supp}\phi) = \emptyset$. For every $\chi \in \mathcal{S}(\mathbb{R}^n)$ the function $(\chi\psi) \boxtimes \phi$ vanishes near $\operatorname{supp} K_A$, hence $(\psi A\phi)(\chi) = K_A((\chi\psi) \boxtimes \phi) = 0$. So $\psi A\phi = 0$ and $\bar{x} \notin \operatorname{supp}(A\phi)$. □

For instance, the kernel of a differential operator is supported in the diagonal $\mathrm{Diag} = \{(x, x)\}$, so differential operators are *local*: $\operatorname{supp}(P\phi) \subset \operatorname{supp}\phi$. A pseudodifferential operator, on the other hand, generally has $\operatorname{supp}(A) = \mathbb{R}^{2n}$; the kernel of $\langle D\rangle^{-2}$ on $\mathbb{R}$, for example, is $\frac{1}{2}e^{-|x-y|}$.

3.1.2 Singular support

Definition 3.1.2. The *singular support* of $u \in \mathcal{S}'(\mathbb{R}^n)$ is the set $\operatorname{sing\,supp} u$ of points $\bar{x}$ such that there is no $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ and $\phi u \in \mathcal{C}^\infty(\mathbb{R}^n)$.

If $\phi u \in \mathcal{C}^\infty$ and $\phi(\bar{x}) \neq 0$ then $\phi \neq 0$ on a ball B around $\bar{x}$, and for every $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(B)$, $\psi u = (\psi/\phi)\phi u \in \mathcal{C}^\infty$. Hence the complement of $\operatorname{sing\,supp} u$ is open, $\operatorname{sing\,supp} u$ is closed, and $\bar{x} \notin \operatorname{sing\,supp} u$ if and only if $\phi u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ for some $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi = 1$ near $\bar{x}$. Directly from the definitions,

$$\operatorname{sing\,supp} u \subset \operatorname{supp} u, \qquad \operatorname{sing\,supp} u = \emptyset \iff u \in \mathcal{C}^\infty(\mathbb{R}^n),$$

the second because u is smooth if it is smooth near each point.

3.2 Pseudolocality

The first observation is that the kernel of a pseudodifferential operator is smooth, and rapidly decreasing, away from the diagonal.

Lemma 3.2.1. *Let $a \in S^m(\mathbb{R}^{2n}; \mathbb{R}^n)$ be an amplitude and let $\rho \in \mathcal{C}^\infty(\mathbb{R}^n)$ have all derivatives bounded and vanish near 0. Then $\rho(x-y)I(a)(x, y)$ is the kernel of an element of $\Psi^{-\infty}(\mathbb{R}^n)$: it is smooth and*

$$|D_x^\beta D_y^\gamma(\rho(x-y)I(a))| \leq C_{\beta\gamma N}\langle x-y\rangle^{-N} \quad \text{for all } \beta, \gamma, N.$$

Proof. By the kernel identities of Chapter 2,

$$D_x^\beta D_y^\gamma[(x-y)^\alpha I(a)] = I((\xi + D_x)^\beta (D_y - \xi)^\gamma (-D_\xi)^\alpha a), \tag{3.2.1}$$

and the amplitude on the right has order $m + |\beta| + |\gamma| - |\alpha|$. If this is less than $-n$ the oscillatory integral converges absolutely and uniformly, so for $|\alpha| = N > m + n + k$ the function $k_\alpha = (x-y)^\alpha I(a)$ is of class $\mathcal{C}^k$ with all derivatives of order at most k bounded.

Suppose $\rho(z) = 0$ for $|z| < \delta$. Using $|z|^{2N} = \sum_{|\alpha|=N} \frac{N!}{\alpha!} z^{2\alpha}$ we can write

$$\rho(z) = \sum_{|\alpha|=N} \frac{N!}{\alpha!} h_\alpha(z) z^\alpha, \qquad h_\alpha(z) = \frac{\rho(z) z^\alpha}{|z|^{2N}},$$

where $h_\alpha \in \mathcal{C}^\infty(\mathbb{R}^n)$ and, since $z^\alpha |z|^{-2N}$ is homogeneous of degree $-N$ on $|z| \geq \delta$, $|D^\mu h_\alpha(z)| \leq C_\mu \langle z\rangle^{-N}$. Then $\rho(x-y)I(a) = \sum_{|\alpha|=N} \frac{N!}{\alpha!} h_\alpha(x-y) k_\alpha$ as distributions, and by Leibniz' formula this is a $\mathcal{C}^k$ function whose derivatives of order at most k are bounded by $C\langle x-y\rangle^{-N}$. Since k and $N > m + n + k$ are arbitrary, the lemma follows. □

Proposition 3.2.2. *If $A \in \Psi^m(\mathbb{R}^n)$ then* $\operatorname{sing\,supp}(K_A) \subset \mathrm{Diag}$. *Moreover, if $\phi, \psi \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ have supports at positive distance from each other, then $\phi A \psi \in \Psi^{-\infty}(\mathbb{R}^n)$.*

Proof. Write $K_A = I(a)$ with $a = \sigma_L(A)(x,\xi)$. If $\bar{x} \neq \bar{y}$, choose ρ as in Lemma 3.2.1 with $\rho = 1$ near $\bar{x} - \bar{y}$; then $K_A = \rho(x-y)K_A$ near $(\bar{x},\bar{y})$, which is smooth. For the second statement, if $|x-y| \geq \delta > 0$ whenever $x \in \operatorname{supp}\phi$, $y \in \operatorname{supp}\psi$, choose ρ with $\rho(z) = 1$ for $|z| \geq \delta$ and $\rho(z) = 0$ for $|z| \leq \delta/2$. The kernel of $\phi A \psi$ is $\phi(x)\psi(y)K_A = \phi(x)\psi(y)\rho(x-y)K_A$, and multiplication by $\phi(x)\psi(y)$, a function with bounded derivatives, preserves the estimates of Lemma 3.2.1. □

We also note how residual operators act on distributions of compact support.

Lemma 3.2.3. *If $E \in \Psi^{-\infty}(\mathbb{R}^n)$ and $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ then $Eu \in \mathcal{S}(\mathbb{R}^n)$.*

Proof. $Eu(x) = u(K_E(x,\cdot))$, and differentiating under u, $x^\alpha D^\gamma Eu(x) = u\big(x^\alpha D_x^\gamma K_E(x,\cdot)\big)$. By (3.1.1) this is bounded by $C \sup_{y\in L, |\beta|\leq k} |x^\alpha D_x^\gamma D_y^\beta K_E(x,y)|$, which is finite since $\langle x\rangle \leq C_L\langle x-y\rangle$ for y in the compact set L and $|D_x^\gamma D_y^\beta K_E| \leq C_N\langle x-y\rangle^{-N}$. □

Definition 3.2.4. A continuous linear operator $A : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is *pseudolocal* if $\operatorname{sing\,supp}(Au) \subset \operatorname{sing\,supp}(u)$ for all $u \in \mathcal{S}'(\mathbb{R}^n)$.

Proposition 3.2.5. *Pseudodifferential operators are pseudolocal.*

Proof. Let $A \in \Psi^m(\mathbb{R}^n)$, $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\bar{x} \notin \operatorname{sing\,supp} u$. Choose $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi = 1$ on a neighbourhood U of $\bar{x}$ and $\phi u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ with $\psi = 1$ near $\bar{x}$. Then

$$\psi A u = \psi A(\phi u) + \psi A(1-\phi)u.$$

The first term is in $\mathcal{S}(\mathbb{R}^n)$, since A preserves $\mathcal{S}$. Since $\operatorname{supp}\psi$ is a compact subset of the open set U and $1-\phi = 0$ on U, the supports of ψ and $1-\phi$ are at positive distance, so $\psi A(1-\phi) \in \Psi^{-\infty}(\mathbb{R}^n)$ by Proposition 3.2.2, and the second term is smooth. Hence $\psi A u \in \mathcal{C}^\infty$ and $\bar{x} \notin \operatorname{sing\,supp}(Au)$. □

3.3 Cone support and essential support

Definition 3.3.1. The *cone support* of $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ is the set $\operatorname{cone\,supp}(a) \subset \mathbb{R}^p \times (\mathbb{R}^n \setminus 0)$ of points $(\bar{x},\bar{\xi})$ such that a is *not* rapidly decreasing on any set $\Gamma_\epsilon(\bar{x},\bar{\xi})$, $\epsilon > 0$.

The complement of $\operatorname{cone\,supp}(a)$ is open and conic, so $\operatorname{cone\,supp}(a)$ is a closed conic subset of $\mathbb{R}^p \times (\mathbb{R}^n \setminus 0)$. Only the behaviour of a for large $|\xi|$ matters: changing a on a set where $|\xi|$ is bounded does not change the cone support. A priori the definition only involves a itself, but in fact all derivatives decay as well.

Lemma 3.3.2. *Let $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ be rapidly decreasing on $\Gamma_\epsilon(\bar{x},\bar{\xi})$. Then for every $\epsilon' < \epsilon$ and all α, β, $D_x^\alpha D_\xi^\beta a$ is rapidly decreasing on $\Gamma_{\epsilon'}(\bar{x},\bar{\xi})$. In particular* $\operatorname{cone\,supp}(D_x^\alpha D_\xi^\beta a) \subset \operatorname{cone\,supp}(a)$.

Proof. Since $D_{x_j} a \in S^m$ and $D_{\xi_j} a \in S^{m-1}$, by induction (with finitely many intermediate values of ϵ between ϵ' and ϵ) it suffices to show that $\partial_{x_j} a$ and $\partial_{\xi_j} a$ are rapidly decreasing on $\Gamma_{\epsilon''} = \Gamma_{\epsilon''}(\bar{x},\bar{\xi})$ for any $\epsilon'' < \epsilon$. Fix $M \geq |m|$; we show that both derivatives are $O(\langle\xi\rangle^{(m-M)/2})$ on $\Gamma_{\epsilon''}$, which suffices as M is arbitrary. Since all derivatives of a are $O(\langle\xi\rangle^m)$ we may restrict to $|\xi| \geq R$, for an R to be chosen.

Let $(x,\xi)\in\Gamma_{\epsilon''}$ and $0<t\le\epsilon-\epsilon''$. Then $(x+se_j,\xi)\in\Gamma_\epsilon$ for $|s|\le t$, and Taylor's formula gives

$$|t\,\partial_{x_j}a(x,\xi)|\le|a(x+te_j,\xi)|+|a(x,\xi)|+\tfrac12t^2\sup_{|s|\le t}|\partial_{x_j}^2a(x+se_j,\xi)|\le 2C_M\langle\xi\rangle^{-M}+Ct^2\langle\xi\rangle^m.$$

Taking $t=(\epsilon-\epsilon'')\langle\xi\rangle^{-(M+m)/2}$ gives $|\partial_{x_j}a(x,\xi)|\le C'\langle\xi\rangle^{(m-M)/2}$.

For the ξ-derivative let $0<t\le1$ and $|s|\le t$. Then $\big|\frac{\xi+se_j}{|\xi+se_j|}-\frac{\xi}{|\xi|}\big|\le\frac{2}{|\xi|}$, so $(x,\xi+se_j)\in\Gamma_\epsilon$ if $|\xi|\ge R=2/(\epsilon-\epsilon'')+1$; also $\langle\xi+se_j\rangle\ge\frac12\langle\xi\rangle$. As before

$$|t\,\partial_{\xi_j}a(x,\xi)|\le C\langle\xi\rangle^{-M}+Ct^2\langle\xi\rangle^m,$$

and $t=\langle\xi\rangle^{-(M+m)/2}$ gives $|\partial_{\xi_j}a|\le C'\langle\xi\rangle^{(m-M)/2}$. □

Corollary 3.3.3. *Let $a\in S^m(\mathbb{R}^p;\mathbb{R}^n)$ and let $K\subset\mathbb{R}^p$ be compact with* $\operatorname{cone\,supp}(a)\cap(K\times(\mathbb{R}^n\setminus0))=\emptyset$. *Then $\phi a\in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$ for every $\phi\in\mathcal{C}^\infty_{\mathrm c}(\mathbb{R}^p)$ with* $\operatorname{supp}\phi\subset K$. *In particular, if a has compact support in x and* $\operatorname{cone\,supp}(a)=\emptyset$, *then $a\in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$.*

Proof. Each point of the compact set $K\times\mathbb{S}^{n-1}$ has a neighbourhood $\Gamma_\epsilon(\bar x,\bar\xi)$ on which a is rapidly decreasing, hence (by Lemma 3.3.2) with all derivatives on $\Gamma_{\epsilon/2}(\bar x,\bar\xi)$. Finitely many of the latter cover $K\times(\mathbb{R}^n\setminus0)$, so every $D_x^\alpha D_\xi^\beta a$ is rapidly decreasing on $K\times\mathbb{R}^n$, and by Leibniz' formula $\phi a\in S^{-\infty}$. □

Remark 3.3.4. The compactness in x cannot be dropped: a symbol may have empty cone support without being in $S^{-\infty}$, because the definition only controls it locally uniformly in x; see Exercise 3.6.

Definition 3.3.5. The *essential support* (or *operator wavefront set*) of $A\in\Psi^m(\mathbb{R}^n)$ is the closed conic set

$$\mathrm{WF}'(A)=\operatorname{cone\,supp}\big(\sigma_L(A)\big)\subset T^*\mathbb{R}^n\setminus0.$$

The essential support can be computed from any amplitude.

Lemma 3.3.6. *If $A=\mathrm{Op}(a)$ with $a\in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$ then*

$$\mathrm{WF}'(A)\subset\{(x,\xi);\ (x,x,\xi)\in\operatorname{cone\,supp}(a)\}.$$

Proof. Suppose $(\bar x,\bar x,\bar\xi)\notin\operatorname{cone\,supp}(a)$, so a is rapidly decreasing on $\{|x-\bar x|<\epsilon,|y-\bar x|<\epsilon,\xi\in V_\epsilon(\bar\xi)\}$ for some $\epsilon>0$. By Lemma 3.3.2 (applied with (x,y) as base variables) all derivatives of a are rapidly decreasing on the corresponding set with $\epsilon/2$ in place of ϵ, so each term $D_\xi^\alpha D_y^\alpha a(x,y,\xi)|_{y=x}$ in (3.0.2) is rapidly decreasing on $\Gamma_{\epsilon/2}(\bar x,\bar\xi)$. By (3.0.2), $\sigma_L(A)-\sum_{|\alpha|<N}\frac{i^{|\alpha|}}{\alpha!}D_\xi^\alpha D_y^\alpha a|_{y=x}\in S^{m-N}$, so $|\sigma_L(A)|\le C_N\langle\xi\rangle^{m-N}$ on $\Gamma_{\epsilon/2}(\bar x,\bar\xi)$ for every N, that is $(\bar x,\bar\xi)\notin\mathrm{WF}'(A)$. □

Proposition 3.3.7. *For $A\in\Psi^m(\mathbb{R}^n)$ and $B\in\Psi^{m'}(\mathbb{R}^n)$:*

(i) $\mathrm{WF}'(A)=\operatorname{cone\,supp}(\sigma_R(A))$;

(ii) $\mathrm{WF}'(A+B)\subset\mathrm{WF}'(A)\cup\mathrm{WF}'(B)$ *(if $m=m'$) and* $\mathrm{WF}'(AB)\subset\mathrm{WF}'(A)\cap\mathrm{WF}'(B)$;

(iii) $\mathrm{WF}'(A^*)=\mathrm{WF}'(A)$ *and* $\mathrm{WF}'(A^t)=\mathrm{WF}'(A)^-$;

(iv) $\mathrm{WF}'(\mathrm{Id})=T^*\mathbb{R}^n\setminus0$, $\mathrm{WF}'(E)=\emptyset$ *for $E\in\Psi^{-\infty}(\mathbb{R}^n)$, and for multiplication by $\phi\in\mathcal{C}^\infty_{\mathrm b}(\mathbb{R}^n)$,* $\mathrm{WF}'(\phi)=\operatorname{supp}\phi\times(\mathbb{R}^n\setminus0)$.

Proof. (i) $A = \mathrm{Op}(b(y,\xi))$ with $b = \sigma_R(A)$, and $(x,x,\xi) \in \operatorname{cone}\operatorname{supp}(b(y,\xi))$ if and only if $(x,\xi) \in \operatorname{cone}\operatorname{supp}(b)$, so $\mathrm{WF}'(A) \subset \operatorname{cone}\operatorname{supp}(\sigma_R(A))$ by Lemma 3.3.6. Conversely $\sigma_R(A) \sim \sum_\beta \frac{(-i)^{|\beta|}}{\beta!} D_\xi^\beta D_x^\beta \sigma_L(A)$, and the argument in the proof of Lemma 3.3.6 shows $\operatorname{cone}\operatorname{supp}(\sigma_R(A)) \subset \operatorname{cone}\operatorname{supp}(\sigma_L(A))$.

(ii) The first statement is clear. For the second, let $a = \sigma_L(A)$ and $b = \sigma_R(B)$, so that $AB = \mathrm{Op}(a(x,\xi)b(y,\xi))$. If $(\bar{x},\bar{\xi}) \notin \mathrm{WF}'(A)$ then $a(x,\xi)b(y,\xi)$ is rapidly decreasing on $\{|x-\bar{x}| < \epsilon, \xi \in V_\epsilon(\bar{\xi})\}$ for some ϵ, since b is polynomially bounded; if $(\bar{x},\bar{\xi}) \notin \mathrm{WF}'(B) = \operatorname{cone}\operatorname{supp}(b)$, the same holds with the roles of x and y exchanged. In either case $(\bar{x},\bar{x},\bar{\xi}) \notin \operatorname{cone}\operatorname{supp}(a(x,\xi)b(y,\xi))$, and Lemma 3.3.6 applies.

(iii) The adjoint of $\mathrm{Op}(a)$, $a = \sigma_L(A)$, is $\mathrm{Op}(\overline{a(y,\xi)})$, and its transpose is $\mathrm{Op}(a(y,-\xi))$ (the kernel of A^t is $K_A(y,x) = \int e^{i(x-y)\cdot\xi} a(y,-\xi)\,d\!\!\bar{}\,\xi$). By Lemma 3.3.6, $\mathrm{WF}'(A^*) \subset \mathrm{WF}'(A)$ and $\mathrm{WF}'(A^t) \subset \mathrm{WF}'(A)^-$; applying this to A^* and A^t gives equality.

(iv) is immediate from the definition, since $\sigma_L(\mathrm{Id}) = 1$, $\sigma_L(E) \in S^{-\infty}$ and $\sigma_L(\phi) = \phi(x)$. □

Lemma 3.3.8. *Let $A \in \Psi^m(\mathbb{R}^n)$ and $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\mathrm{WF}'(A) \cap (\operatorname{supp}\phi \times (\mathbb{R}^n \setminus 0)) = \emptyset$. Then ϕA and $A\phi$ lie in $\Psi^{-\infty}(\mathbb{R}^n)$. In particular, if $\mathrm{WF}'(A) = \emptyset$ then $\phi A, A\phi \in \Psi^{-\infty}(\mathbb{R}^n)$ for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$.*

Proof. $\sigma_L(\phi A) = \phi\sigma_L(A) \in S^{-\infty}$ by Corollary 3.3.3. Similarly $A\phi = \mathrm{Op}(\sigma_R(A)(y,\xi)\phi(y)) = \mathrm{Op}_R(\phi\sigma_R(A))$ and $\phi\sigma_R(A) \in S^{-\infty}$, by Proposition 3.3.7(i) and Corollary 3.3.3. □

Note that $\mathrm{WF}'(A) = \emptyset$ does *not* imply $A \in \Psi^{-\infty}(\mathbb{R}^n)$ (Remark 3.3.4); this is why compactly supported cut-offs appear in many arguments below.

3.3.1 Microlocal cut-offs

We shall constantly use operators which are "microlocally the identity" near a given closed cone and vanish microlocally away from it. Fix $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $0 \le \chi \le 1$, $\chi(t) = 1$ for $|t| \le \frac{1}{2}$ and $\chi(t) = 0$ for $|t| \ge 1$. For $z = (\bar{x},\bar{\xi}) \in T^*\mathbb{R}^n \setminus 0$ and $0 < \epsilon < 1$ set

$$\gamma_{z,\epsilon}(x,\xi) = \chi\Big(\frac{x-\bar{x}}{\epsilon}\Big)\chi\Big(\frac{1}{\epsilon}\Big(\frac{\xi}{|\xi|} - \frac{\bar{\xi}}{|\bar{\xi}|}\Big)\Big)\big(1-\chi(\epsilon\xi)\big). \tag{3.3.1}$$

The last factor vanishes for $|\xi| \le 1/(2\epsilon)$, and functions which are smooth and homogeneous of degree 0 in $|\xi| \ge c > 0$ are symbols of order 0 there, so $\gamma_{z,\epsilon} \in S^0(\mathbb{R}^n;\mathbb{R}^n)$. It takes values in $[0,1]$, is supported in $\overline{\Gamma_\epsilon(z)}$ and equals 1 on $\Gamma_{\epsilon/2}(z) \cap \{|\xi| \ge 1/\epsilon\}$. Hence $G_{z,\epsilon} = \mathrm{Op}(\gamma_{z,\epsilon}) \in \Psi^0(\mathbb{R}^n)$ satisfies

$$\mathrm{WF}'(G_{z,\epsilon}) \subset \overline{\Gamma_\epsilon(z)}, \qquad \mathrm{WF}'(\mathrm{Id} - G_{z,\epsilon}) \cap \Gamma_{\epsilon/2}(z) = \emptyset, \qquad G_{z,\epsilon} = \psi G_{z,\epsilon} \tag{3.3.2}$$

for any $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 on $\{|x-\bar{x}| \le \epsilon\}$.

Lemma 3.3.9 (Microlocal cut-off)**.** *Let $W_1, W_2 \subset T^*\mathbb{R}^n \setminus 0$ be disjoint closed conic sets with $\pi(W_1)$ bounded, where $\pi(x,\xi) = x$. Then there is $a \in S^0(\mathbb{R}^n;\mathbb{R}^n)$, real-valued with compact support in x, such that $C = \mathrm{Op}(a)$ satisfies*

$$\mathrm{WF}'(C) \cap W_2 = \emptyset, \qquad \mathrm{WF}'(\mathrm{Id} - C) \cap W_1 = \emptyset.$$

Proof. The set $W_1 \cap (\mathbb{R}^n \times \mathbb{S}^{n-1})$ is compact. For each of its points z there is $\epsilon_z > 0$ with $\overline{\Gamma_{2\epsilon_z}(z)} \cap W_2 = \emptyset$, since W_2 is closed and conic. Choose finitely many $z_1, \dots, z_N$ such that the cones $\Gamma_{\epsilon_j/2}(z_j)$, $\epsilon_j = 2\epsilon_{z_j}$, cover W_1, and put $g = \sum_j \gamma_{z_j,\epsilon_j} \in S^0$. Then $g \ge 0$, $g \ge 1$ on $\bigcup_j \Gamma_{\epsilon_j/2}(z_j) \cap \{|\xi| \ge R\}$ with $R = \max_j 1/\epsilon_j$, and $\operatorname{supp} g \subset \bigcup_j \overline{\Gamma_{\epsilon_j}(z_j)}$. Let $\Phi \in \mathcal{C}^\infty(\mathbb{R})$

with $\Phi(t) = 0$ for $t \leq 0$ and $\Phi(t) = 1$ for $t \geq 1$, and set $a = \Phi \circ g$. By the chain rule $a \in S^0$, and a has compact support in x. Since a vanishes outside $\bigcup_j \overline{\Gamma_{\epsilon_j}(z_j)}$, which is a closed cone disjoint from W_2, $\mathrm{WF}'(C) \cap W_2 = \emptyset$. Since $1 - a$ vanishes on the open cone $\bigcup_j \Gamma_{\epsilon_j/2}(z_j)$ for $|\xi| \geq R$, and this cone contains W_1, $\mathrm{WF}'(\mathrm{Id} - C) \cap W_1 = \emptyset$. □

3.4 The characteristic variety

Definition 3.4.1. $A \in \Psi^m(\mathbb{R}^n)$ is *elliptic at* $(\bar{x}, \bar{\xi}) \in T^*\mathbb{R}^n \setminus 0$ if there is $\epsilon > 0$ such that

$$|\sigma_L(A)(x, \xi)| \geq \epsilon |\xi|^m \quad \text{on } \Gamma_\epsilon(\bar{x}, \bar{\xi}) \cap \{|\xi| \geq 1/\epsilon\}.$$

The set of such points is the *elliptic set* $\mathrm{Ell}_m(A)$; its complement $\mathrm{Char}_m(A) = T^*\mathbb{R}^n \setminus 0 \setminus \mathrm{Ell}_m(A)$ is the *characteristic variety* of A.

If $a' - \sigma_L(A) \in S^{m-1}$ then $|a' - \sigma_L(A)| \leq C\langle\xi\rangle^{m-1} \leq \frac{\epsilon}{2}|\xi|^m$ for $|\xi|$ large, so the condition holds for a' with $\epsilon/2$ in place of ϵ (after shrinking the cone). Thus ellipticity at a point depends only on the principal symbol $\sigma_m(A)$; in particular $\sigma_R(A)$ may be used instead of $\sigma_L(A)$. Clearly $\mathrm{Ell}_m(A)$ is open and conic, so $\mathrm{Char}_m(A)$ is a closed conic subset of $T^*\mathbb{R}^n \setminus 0$. If A is elliptic at z then $\sigma_L(A)$ is not rapidly decreasing near z, so

$$\mathrm{Ell}_m(A) \subset \mathrm{WF}'(A).$$

For example, multiplication by $\phi \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is elliptic (of order 0) at $(\bar{x}, \bar{\xi})$ if and only if $\phi(\bar{x}) \neq 0$, and $G_{z,\epsilon}$ in (3.3.2) is elliptic at z. The operator A is globally elliptic in the sense of Chapter 2 if and only if the estimate in Definition 3.4.1 holds uniformly, with one ϵ for all $(\bar{x}, \bar{\xi})$.

Proposition 3.4.2. *For $A \in \Psi^m(\mathbb{R}^n)$ and $B \in \Psi^{m'}(\mathbb{R}^n)$,*

$$\mathrm{Char}_{m+m'}(AB) = \mathrm{Char}_m(A) \cup \mathrm{Char}_{m'}(B).$$

Proof. We may use the representatives $a = \sigma_L(A)$, $b = \sigma_L(B)$ and ab of the principal symbols, since $\sigma_L(AB) - ab \in S^{m+m'-1}$. If A and B are elliptic at z then $|ab| \geq \epsilon^2|\xi|^{m+m'}$ on a suitable $\Gamma_\epsilon(z) \cap \{|\xi| \geq 1/\epsilon\}$, so AB is elliptic at z. Conversely if $|ab| \geq \epsilon|\xi|^{m+m'}$ there, then since $|b| \leq C\langle\xi\rangle^{m'} \leq 2^{|m'|}C|\xi|^{m'}$ for $|\xi| \geq 1$, we get $|a| \geq \epsilon'|\xi|^m$, so A is elliptic at z, and similarly B is. □

3.5 Microlocal parametrices

The essential support allows us to refine the notion of a parametrix of an elliptic operator to that of a parametrix at a point.

Proposition 3.5.1. *If $A \in \Psi^m(\mathbb{R}^n)$ and $z \notin \mathrm{Char}_m(A)$, there exists a* microlocal parametrix *for A at z: an operator $B \in \Psi^{-m}(\mathbb{R}^n)$ with*

$$z \notin \mathrm{WF}'(AB - \mathrm{Id}) \quad \textit{and} \quad z \notin \mathrm{WF}'(BA - \mathrm{Id}).$$

Moreover there is then an open conic neighbourhood Γ of z with $\Gamma \cap (\mathrm{WF}'(AB - \mathrm{Id}) \cup \mathrm{WF}'(BA - \mathrm{Id})) = \emptyset$.

Proof. The last statement holds for any B with the first property, as essential supports are closed. Let $a = \sigma_L(A)$ and choose $\epsilon \in (0, \frac{1}{2})$ and $c_0 > 0$ such that $|a| \geq c_0|\xi|^m$ on $\Gamma_{2\epsilon}(z) \cap \{|\xi| \geq 1/(2\epsilon)\}$. Let $\gamma = \gamma_{z,\epsilon}$ and $L = \mathrm{Op}(\gamma)$. Then $z \notin \mathrm{WF}'(\mathrm{Id} - L)$ by (3.3.2), and $\operatorname{supp}\gamma \subset \overline{\Gamma_\epsilon(z)} \cap \{|\xi| \geq 1/(2\epsilon)\}$, so that $|a| \geq c_0|\xi|^m$ on $\operatorname{supp}\gamma$.

Let $\Lambda = \mathrm{Op}(\langle\xi\rangle^{2m}) \in \Psi^{2m}(\mathbb{R}^n)$ and consider

$$J = (\mathrm{Id} - L)\Lambda + A^*A \in \Psi^{2m}(\mathbb{R}^n).$$

Its principal symbol is represented by $(1-\gamma)\langle\xi\rangle^{2m} + |a|^2$. Since $0 \leq \gamma \leq 1$, at a point with $|\xi| \geq R$ either $\gamma \leq \frac{1}{2}$, and then this is at least $\frac{1}{2}\langle\xi\rangle^{2m}$, or $\gamma > \frac{1}{2}$, and then the point is in $\operatorname{supp}\gamma$ and the expression is at least $c_0^2|\xi|^{2m} \geq c\langle\xi\rangle^{2m}$. Since $\sigma_L(J)$ differs from this representative by a symbol of order $2m-1$, J is globally elliptic and has a parametrix $H \in \Psi^{-2m}(\mathbb{R}^n)$, $HJ - \mathrm{Id} = E_1 \in \Psi^{-\infty}$. Set $B_1 = HA^* \in \Psi^{-m}(\mathbb{R}^n)$. Then

$$B_1A - \mathrm{Id} = H\big(J - (\mathrm{Id} - L)\Lambda\big) - \mathrm{Id} = E_1 - H(\mathrm{Id} - L)\Lambda,$$

and $\mathrm{WF}'(H(\mathrm{Id} - L)\Lambda) \subset \mathrm{WF}'(\mathrm{Id} - L) \not\ni z$ by Proposition 3.3.7. So $z \notin \mathrm{WF}'(B_1A - \mathrm{Id})$: B_1 is a left microlocal parametrix.

In the same way $J' = (\mathrm{Id} - L)\Lambda + AA^*$ is globally elliptic; if H' is a parametrix, $JH' - \mathrm{Id} = E_2 \in \Psi^{-\infty}$, then $B_2 = A^*H'$ satisfies $AB_2 - \mathrm{Id} = E_2 - (\mathrm{Id} - L)\Lambda H'$, so $z \notin \mathrm{WF}'(AB_2 - \mathrm{Id})$. Write $R_1 = B_1A - \mathrm{Id}$ and $R_2 = AB_2 - \mathrm{Id}$. Then

$$B_2 = (B_1A - R_1)B_2 = B_1(\mathrm{Id} + R_2) - R_1B_2 = B_1 + B_1R_2 - R_1B_2,$$

so $z \notin \mathrm{WF}'(B_2 - B_1)$, and $AB_1 - \mathrm{Id} = R_2 + A(B_1 - B_2)$ shows $z \notin \mathrm{WF}'(AB_1 - \mathrm{Id})$. Thus $B = B_1$ is a two-sided microlocal parametrix. □

The same argument shows that a microlocal parametrix at z is unique up to an operator whose essential support does not contain z (Exercise 3.11).

3.6 Coordinate invariance

For an open set $\Omega \subset \mathbb{R}^n$ let $\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ be the smooth functions with compact support in Ω and

$$\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ \operatorname{supp} u \text{ is a compact subset of } \Omega\}.$$

An element $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$ acts on $\mathcal{C}^\infty(\Omega)$ by $u(\phi) = u(\psi\phi)$, $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equal to 1 near $\operatorname{supp} u$.

Let $F : \Omega \to \Omega'$ be a diffeomorphism between open subsets of $\mathbb{R}^n$, with inverse G, and let $F^*\phi = \phi \circ F$, so $F^* : \mathcal{C}^\infty_{\mathrm{c}}(\Omega') \to \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$. If $u \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ and $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, the change of variables $y = F(x)$ gives

$$\int u(F(x))\phi(x)\,dx = \int u(y)\,\phi(G(y))\,|\det DG(y)|\,dy. \tag{3.6.1}$$

Lemma 3.6.1. *For $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega')$ the formula*

$$(F^*u)(\phi) = u\big(|\det DG|\,G^*\phi\big), \qquad \phi \in \mathcal{S}(\mathbb{R}^n),$$

(where $G^*\phi = \phi \circ G \in \mathcal{C}^\infty(\Omega')$) defines $F^*u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$ with $\operatorname{supp} F^*u = G(\operatorname{supp} u)$. It extends F^* on $\mathcal{C}^\infty_{\mathrm{c}}(\Omega')$, is continuous in the sense that $F^*u_j \to F^*u$ weakly if $u_j \to u$ weakly and the supports of the u_j lie in a fixed compact subset of Ω', and satisfies $(F_1 \circ F_2)^* = F_2^*F_1^*$; in particular $G^*F^* = \mathrm{Id}$.

Proof. u acts on $\mathcal{C}^\infty(\Omega')$, and $\phi \mapsto |\det DG| G^*\phi$ is continuous from $\mathcal{S}(\mathbb{R}^n)$ to $\mathcal{C}^\infty(\Omega')$, so $F^*u \in \mathcal{S}'(\mathbb{R}^n)$. If ϕ vanishes near $G(\operatorname{supp} u)$ then $|\det DG| G^*\phi$ vanishes near $\operatorname{supp} u$, so $\operatorname{supp} F^*u \subset G(\operatorname{supp} u)$, and equality follows from $G^*F^*u = u$. The consistency with smooth functions is (3.6.1), continuity is clear, and the composition law follows from the chain rule for Jacobians. □

If $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega')$ and $\rho_\epsilon(x) = \epsilon^{-n}\rho(x/\epsilon)$ with $\rho \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^n)$, $\int \rho = 1$, then $\rho_\epsilon * u \in \mathcal{C}_{\mathrm{c}}^\infty(\Omega')$ for small ϵ, with supports in a fixed compact set, and $\rho_\epsilon * u \to u$ weakly. So $\mathcal{C}_{\mathrm{c}}^\infty(\Omega')$ is weakly dense in $\mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega')$ in this strong sense (see also Exercise 3.5), and F^* is the unique such continuous extension.

Now let $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ have kernel with compact support in $\Omega \times \Omega$. By Proposition 3.1.1, A maps $\mathcal{C}_{\mathrm{c}}^\infty(\Omega)$, indeed $\mathcal{S}(\mathbb{R}^n)$, into $\mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega)$, and we define

$$A_F = G^* \circ A \circ F^* : \mathcal{C}_{\mathrm{c}}^\infty(\Omega') \to \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega'). \tag{3.6.2}$$

By (3.6.1) its kernel is

$$K_{A_F}(x', y') = K_A(G(x'), G(y'))\, |\det DG(y')|,$$

that is, $K_{A_F} = |\det DG(y')|\, (G \times G)^* K_A$, where $(G \times G)^*$ is the pull-back of Lemma 3.6.1 for the diffeomorphism $G \times G : \Omega' \times \Omega' \to \Omega \times \Omega$; it has compact support in $\Omega' \times \Omega'$, and we extend it by zero to $\mathbb{R}^{2n}$, so $A_F : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$. The diffeomorphism F induces the map

$$\chi_F : \Omega \times \mathbb{R}^n \to \Omega' \times \mathbb{R}^n, \qquad \chi_F(x, \xi) = (F(x), (DF(x)^t)^{-1}\xi),$$

which is a diffeomorphism, linear and invertible on each fibre. (In the language of Chapter 5, χ_F is the inverse of the cotangent map $F^* : T^*_{F(x)}\Omega' \to T^*_x\Omega$, $\eta \mapsto DF(x)^t\eta$.)

Theorem 3.6.2 (Coordinate invariance)**.** *Let $F : \Omega \to \Omega'$ be a diffeomorphism of open subsets of $\mathbb{R}^n$ and $A \in \Psi^m(\mathbb{R}^n)$ an operator whose kernel has compact support in $\Omega \times \Omega$. Then $A_F \in \Psi^m(\mathbb{R}^n)$, its kernel has compact support in $\Omega' \times \Omega'$, and*

$$\sigma_m(A_F) \circ \chi_F = \sigma_m(A), \qquad \mathrm{WF}'(A_F) = \chi_F(\mathrm{WF}'(A)),$$
$$\mathrm{Char}_m(A_F) \cap \pi^{-1}(\Omega') = \chi_F(\mathrm{Char}_m(A) \cap \pi^{-1}(\Omega)),$$

where $\pi(x, \xi) = x$ and the first identity is understood as $\sigma_L(A_F)(F(x), (DF(x)^t)^{-1}\xi) - \sigma_L(A)(x, \xi) \in S^{m-1}$ for x in a neighbourhood of the support, both symbols vanishing for x outside compact subsets of Ω, Ω' respectively.

Proof. Let $K \subset \Omega$ be compact with $\operatorname{supp} K_A \subset K \times K$, and let $\psi \in \mathcal{C}_{\mathrm{c}}^\infty(\Omega)$ equal 1 near K. Then $\psi A = A$, so $a = \sigma_L(A)$ satisfies $a = \psi a$ by uniqueness of the left symbol, and $A = \psi A \psi = \mathrm{Op}(b_0)$ with $b_0(x, y, \xi) = \psi(x)\psi(y)a(x, \xi)$. Similarly $\sigma_L(A_F)$ is supported in $F(K) \times \mathbb{R}^n$.

Let $L = \operatorname{supp} \psi$. Choose $\delta' > 0$ so small that the closed $2\delta'$-neighbourhood of $F(L)$ lies in Ω' and that, whenever $\operatorname{dist}(x', F(L)) \le \delta'$ and $|x' - y'| \le \delta'$, the matrix

$$T(x', y') = \int_0^1 DG(y' + t(x' - y'))\, dt$$

is invertible with $|\det T| \ge c > 0$ and $\|T^{-1}\| \le C$; this is possible since $T(x', x') = DG(x')$ and $F(L)$ is compact. Then $G(x') - G(y') = T(x', y')(x' - y')$. By uniform continuity of F on a compact neighbourhood of L, choose $\delta > 0$ such that $x, y \in L$, $|x - y| \le 2\delta$ imply $|F(x) - F(y)| < \delta'$.

Let $\mu \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal 1 on $|z| \le \delta$ and vanish for $|z| \ge 2\delta$, and split $A = A_1 + A_2$ with $K_{A_1} = \mu(x-y)K_A$, $K_{A_2} = (1-\mu(x-y))K_A$. By Lemma 3.2.1, $A_2 \in \Psi^{-\infty}(\mathbb{R}^n)$, so K_{A_2} is smooth, and it has compact support in $\Omega \times \Omega$. Hence $K_{(A_2)_F}$ is smooth with compact support in $\Omega' \times \Omega'$, in particular it satisfies the estimates characterizing $\Psi^{-\infty}(\mathbb{R}^n)$, and $(A_2)_F \in \Psi^{-\infty}(\mathbb{R}^n)$.

The kernel of A_1 is $I(b)$ with $b(x,y,\xi) = \mu(x-y)\psi(x)\psi(y)a(x,\xi) \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$, which vanishes unless $x, y \in L$ and $|x-y| \le 2\delta$. If $b \in S^{-\infty}$ the integral converges absolutely and the substitution $\xi = S(x',y')\eta$, $S = (T^t)^{-1}$, gives

$$\begin{aligned} K_{(A_1)_F}(x',y') &= \int e^{i(x'-y')\cdot\eta} c(x',y',\eta)\, đ\eta, \\ c(x',y',\eta) &= b(G(x'),G(y'),S(x',y')\eta)\,\frac{|\det DG(y')|}{|\det T(x',y')|}, \end{aligned} \tag{3.6.3}$$

where c is extended by zero outside the compact set $\{(x',y');\ G(x'),G(y') \in L,\ |G(x') - G(y')| \le 2\delta\}$; on this set $|x'-y'| < \delta'$, so c is smooth. For general b we apply this to $\chi(\epsilon\xi)b$ and let $\epsilon \to 0$: both sides of (3.6.3) converge in $\mathcal{S}'(\mathbb{R}^{2n})$, by the continuity of I on amplitudes and of the pull-back on compactly supported distributions. Thus $(A_1)_F = \mathrm{Op}(c)$, provided we show $c \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$.

Write $f(x',y')$ for the smooth factor in c. By the chain rule, $D_{x'}^\alpha D_{y'}^\beta D_\eta^\gamma [b(G(x'),G(y'),S\eta)]$ is a finite sum of terms

$$M(x',y')\,\eta^\nu \left(D_x^{\alpha'} D_y^{\beta'} D_\xi^\mu b\right)(G(x'),G(y'),S\eta), \qquad |\nu| \le |\mu| - |\gamma|,$$

with smooth coefficients M: each η-derivative produces one ξ-derivative of b and no power of η, while an x'- or y'-derivative falling on $S(x',y')\eta$ produces one ξ-derivative and at most one factor of η. On the support, $\langle S\eta\rangle$ is comparable to $\langle\eta\rangle$, so each term is bounded by $C\langle\eta\rangle^{|\nu|}\langle\eta\rangle^{m-|\mu|} \le C\langle\eta\rangle^{m-|\gamma|}$. With Leibniz' formula for the factor f, this shows $c \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$.

Thus $A_F = (A_1)_F + (A_2)_F \in \Psi^m(\mathbb{R}^n)$. By Chapter 2, $\sigma_m(A_F)$ is represented by $c(x',x',\eta)$. Since $T(x',x') = DG(x')$,

$$c(x',x',\eta) = \mu(0)\psi(G(x'))^2 a\big(G(x'),(DG(x')^t)^{-1}\eta\big) = a\big(x, DF(x)^t\eta\big), \quad x = G(x'),$$

using $DG(x')^{-1} = DF(x)$ and $\psi a = a$. With $(x',\eta) = \chi_F(x,\xi)$ this is $a(x,\xi)$, proving the statement about principal symbols. The same chain rule estimates show that composition with χ_F^{-1} maps symbols of order $m-1$ supported over L to symbols of order $m-1$, so the class is well defined.

For the essential support we use Lemma 3.3.6. Let $(\bar x',\bar\eta) \in \Omega' \times (\mathbb{R}^n \setminus 0)$ with $\chi_F^{-1}(\bar x',\bar\eta) = (\bar x,\bar\xi) \notin \mathrm{WF}'(A)$, so a, and hence $b(x,y,\xi)$ for y arbitrary, is rapidly decreasing for $(x,\xi) \in \Gamma_\epsilon(\bar x,\bar\xi)$. Since $S(\bar x',\bar x')\bar\eta = \bar\xi$ and S, G are continuous, for (x',y') close to $(\bar x',\bar x')$ and $\eta \in V_{\epsilon'}(\bar\eta)$ with ϵ' small we have $(G(x'),S(x',y')\eta) \in \Gamma_\epsilon(\bar x,\bar\xi)$; as $\langle S\eta\rangle \ge c\langle\eta\rangle$, c is rapidly decreasing there. Hence $(\bar x',\bar x',\bar\eta) \notin \operatorname{cone}\operatorname{supp}(c)$ and $(\bar x',\bar\eta) \notin \mathrm{WF}'(A_F)$. Points over $\mathbb{R}^n \setminus F(K)$ are not in $\mathrm{WF}'(A_F)$, since $\sigma_L(A_F)$ vanishes there. Thus $\mathrm{WF}'(A_F) \subset \chi_F(\mathrm{WF}'(A))$. Since $A = (A_F)_G$, the reverse inclusion follows by symmetry. The statement about characteristic varieties follows from that about principal symbols, since χ_F is homogeneous of degree one in the fibres and satisfies $c^{-1}|\xi| \le |(DF(x)^t)^{-1}\xi| \le c|\xi|$ locally uniformly in x; so it maps the sets $\Gamma_\epsilon(z) \cap \{|\xi| \ge 1/\epsilon\}$ into and onto sets of the same type. □

This invariance shows that the principal symbol, essential support and characteristic variety of a pseudodifferential operator are invariantly defined on the cotangent bundle; this is the basis of the extension of the calculus to manifolds in Chapter 5. It also gives us the freedom to choose local coordinates adapted to a particular problem.

3.7 The wavefront set

We first define the wavefront set for distributions of compact support and then localize.

Definition 3.7.1. For $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$,

$$\mathrm{WF}(u) = \bigcap\{\mathrm{Char}_0(A);\ A \in \Psi^0(\mathbb{R}^n),\ Au \in \mathcal{C}^\infty(\mathbb{R}^n)\}.$$

Thus $z \notin \mathrm{WF}(u)$ if and only if there is $A \in \Psi^0(\mathbb{R}^n)$, elliptic at z, with $Au \in \mathcal{C}^\infty(\mathbb{R}^n)$. As an intersection of closed conic sets, $\mathrm{WF}(u) \subset T^*\mathbb{R}^n \setminus 0$ is closed and conic. The wavefront set refines the singular support. Let $\pi : T^*\mathbb{R}^n \setminus 0 \to \mathbb{R}^n$ be the projection.

Proposition 3.7.2. *For $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$, $\pi(\mathrm{WF}(u)) = \operatorname{sing\,supp}(u)$.*

Proof. If $\bar{x} \notin \operatorname{sing\,supp} u$ there is $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ and $\phi u \in \mathcal{C}^\infty$. As a multiplication operator $\phi \in \Psi^0(\mathbb{R}^n)$ is elliptic at $(\bar{x}, \bar{\xi})$ for every $\bar{\xi} \neq 0$, so $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$. Thus $\pi(\mathrm{WF}(u)) \subset \operatorname{sing\,supp}(u)$.

Conversely suppose $(\bar{x}, \omega) \notin \mathrm{WF}(u)$ for every $\omega \in \mathbb{S}^{n-1}$. For each ω there is $A_\omega \in \Psi^0$ with $A_\omega u \in \mathcal{C}^\infty$ and $|\sigma_L(A_\omega)| \geq \epsilon_\omega$ on $\Gamma_{\epsilon_\omega}(\bar{x}, \omega) \cap \{|\xi| \geq 1/\epsilon_\omega\}$. By compactness of the sphere there are finitely many ω_j such that the $V_{\epsilon_j}(\omega_j)$, $\epsilon_j = \epsilon_{\omega_j}$, cover $\mathbb{R}^n \setminus 0$. Let $A_j = A_{\omega_j}$, $\epsilon_0 = \min_j \epsilon_j$, $R_0 = \max_j 1/\epsilon_j$ and

$$B = \sum_j A_j^* A_j \in \Psi^0(\mathbb{R}^n).$$

Since $A_j u \in \mathcal{C}^\infty \cap \mathcal{S}'$ and A_j^* is pseudolocal (Proposition 3.2.5), $Bu \in \mathcal{C}^\infty(\mathbb{R}^n)$. Moreover $\sigma_L(B) = \sum_j |\sigma_L(A_j)|^2 + r$ with $r \in S^{-1}$, and $\sum_j |\sigma_L(A_j)|^2 \geq \epsilon_0^2$ when $|x - \bar{x}| < \epsilon_0$ and $|\xi| \geq R_0$. Let $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, $0 \leq \phi \leq 1$, with support in $\{|x - \bar{x}| < \epsilon_0\}$ and $\phi = 1$ near $\bar{x}$. Then $\sigma_L(B + (1 - \phi)) = \sum_j |\sigma_L(A_j)|^2 + (1 - \phi) + r$ and, for $|\xi| \geq R_0$, the sum of the first two terms is at least $\min(\epsilon_0^2, 1)$ (consider separately $|x - \bar{x}| < \epsilon_0$ and $|x - \bar{x}| \geq \epsilon_0$). As $|r| \leq C\langle\xi\rangle^{-1}$, the operator $B + (1 - \phi)$ is globally elliptic, so has a parametrix $Q \in \Psi^0(\mathbb{R}^n)$:

$$\mathrm{Id} = QB + Q(1 - \phi) - E, \qquad E \in \Psi^{-\infty}(\mathbb{R}^n).$$

Let $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\psi(\bar{x}) \neq 0$ and $\operatorname{supp}\psi$ contained in the interior of $\{\phi = 1\}$. Then $\psi u = \psi QBu + \psi Q(1 - \phi)u - \psi Eu$. The first term is smooth by pseudolocality, the second because $\psi Q(1 - \phi) \in \Psi^{-\infty}$ by Proposition 3.2.2, and the third because $E \in \Psi^{-\infty}$. Hence $\bar{x} \notin \operatorname{sing\,supp} u$. □

Lemma 3.7.3. *Let $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ and $z \notin \mathrm{WF}(u)$. Then there is an open conic neighbourhood Γ of z such that $Cu \in \mathcal{C}^\infty(\mathbb{R}^n)$ for every $C \in \Psi^m(\mathbb{R}^n)$, $m \in \mathbb{R}$, with $\mathrm{WF}'(C) \subset \Gamma$.*

Proof. Let $A \in \Psi^0$ be elliptic at z with $Au \in \mathcal{C}^\infty$ and let Q be a microlocal parametrix for A at z (Proposition 3.5.1), so $QA = \mathrm{Id} + R$ with $z \notin \mathrm{WF}'(R)$. Let Γ be the complement of $\mathrm{WF}'(R)$, an open conic neighbourhood of z, and let $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal 1 near $\operatorname{supp} u$. If $\mathrm{WF}'(C) \subset \Gamma$ then

$$Cu = CQ(Au) - CR\psi u.$$

The first term is smooth by pseudolocality. Since $\mathrm{WF}'(CR) \subset \mathrm{WF}'(C) \cap \mathrm{WF}'(R) = \emptyset$, $CR\psi \in \Psi^{-\infty}$ by Lemma 3.3.8, so the second term is smooth too. □

In particular, $\mathrm{WF}(u + v) \subset \mathrm{WF}(u) \cup \mathrm{WF}(v)$ for $u, v \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$: if z is in neither set, take $C = G_{z,\epsilon}$ with $\overline{\Gamma_\epsilon(z)}$ inside both of the cones given by the lemma.

Proposition 3.7.4. *If $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$, $A \in \Psi^m(\mathbb{R}^n)$ and $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ then*

$$\mathrm{WF}(\phi A u) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u).$$

Proof. Let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal 1 near $\operatorname{supp} u$.

Suppose first $z \notin \mathrm{WF}'(A)$. Since $\mathrm{WF}'(A)$ is closed and conic, there is $\epsilon > 0$ with $\overline{\Gamma_\epsilon(z)} \cap \mathrm{WF}'(A) = \emptyset$; let $C = G_{z,\epsilon}$, which is elliptic at z. Then $\mathrm{WF}'(C\phi A) \subset \mathrm{WF}'(C) \cap \mathrm{WF}'(A) = \emptyset$, and $C\phi A = \psi' C \phi A$ for a suitable $\psi' \in \mathcal{C}^\infty_{\mathrm{c}}$ by (3.3.2), so $C\phi A \in \Psi^{-\infty}$ by Lemma 3.3.8. Hence $C(\phi A u) \in \mathcal{C}^\infty$ and $z \notin \mathrm{WF}(\phi A u)$.

Suppose next $z \notin \mathrm{WF}(u)$ and let Γ be as in Lemma 3.7.3. Choose ϵ with $\overline{\Gamma_\epsilon(z)} \subset \Gamma$ and let $D = G_{z,\epsilon}$, so $\mathrm{WF}'(D) \subset \Gamma$ and $Du \in \mathcal{C}^\infty$; and let $C = G_{z,\epsilon/4}$, which is elliptic at z with $\mathrm{WF}'(C) \subset \overline{\Gamma_{\epsilon/4}(z)}$, a set disjoint from $\mathrm{WF}'(\mathrm{Id} - D)$ by (3.3.2). Then

$$C\phi A u = C\phi A(Du) + C\phi A(\mathrm{Id} - D)\psi u.$$

The first term is smooth by pseudolocality, as $Du \in \mathcal{C}^\infty \cap \mathcal{S}'$. In the second term, $\mathrm{WF}'(C\phi A(\mathrm{Id} - D)) \subset \mathrm{WF}'(C) \cap \mathrm{WF}'(\mathrm{Id} - D) = \emptyset$, so $C\phi A(\mathrm{Id} - D)\psi \in \Psi^{-\infty}$ by Lemma 3.3.8. Hence $C(\phi A u) \in \mathcal{C}^\infty$ and $z \notin \mathrm{WF}(\phi A u)$. □

Taking $A = \mathrm{Id}$ we see that $\mathrm{WF}(\phi u) \subset \mathrm{WF}(u)$ for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$ and $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}$, and $\mathrm{WF}(\phi u) = \mathrm{WF}(u)$ if $\phi = 1$ near $\operatorname{supp} u$. This makes the following extension consistent.

Definition 3.7.5. For $u \in \mathcal{S}'(\mathbb{R}^n)$,

$$\mathrm{WF}(u) = \bigcup_{\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)} \mathrm{WF}(\phi u) \subset T^*\mathbb{R}^n \setminus 0.$$

Proposition 3.7.6. *Let $u, v \in \mathcal{S}'(\mathbb{R}^n)$.*

(i) *If $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and $\phi(\bar{x}) \neq 0$ then $\mathrm{WF}(u) \cap \pi^{-1}(\bar{x}) = \mathrm{WF}(\phi u) \cap \pi^{-1}(\bar{x})$.*

(ii) *$\mathrm{WF}(u)$ is closed and conic, and $\pi(\mathrm{WF}(u)) = \operatorname{sing\,supp}(u)$.*

(iii) *$\mathrm{WF}(u + v) \subset \mathrm{WF}(u) \cup \mathrm{WF}(v)$.*

Proof. (i) One inclusion is the definition. Conversely let $z = (\bar{x}, \bar{\xi}) \notin \mathrm{WF}(\phi u)$ and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$. Choose $\phi' \in \mathcal{C}^\infty_{\mathrm{c}}$ supported in $\{\phi \neq 0\}$ with $\phi' = 1$ near $\bar{x}$, and write $\psi u = \theta(\phi u) + \psi(1 - \phi')u$ with $\theta = \psi\phi'/\phi \in \mathcal{C}^\infty_{\mathrm{c}}$. Then $z \notin \mathrm{WF}(\theta\phi u)$ by Proposition 3.7.4, while $\psi(1 - \phi')u$ vanishes near $\bar{x}$, so its wavefront set does not meet $\pi^{-1}(\bar{x})$ by Proposition 3.7.2. Hence $z \notin \mathrm{WF}(\psi u)$.

(ii) By (i), $\mathrm{WF}(u) \cap \pi^{-1}(U) = \mathrm{WF}(\phi u) \cap \pi^{-1}(U)$ on the open set $U = \{\phi \neq 0\}$, so $\mathrm{WF}(u)$ is closed, being locally closed; it is clearly conic. Also $\bar{x} \notin \operatorname{sing\,supp} u$ if and only if $\bar{x} \notin \operatorname{sing\,supp}(\phi u)$ for some (any) ϕ as in (i), so the second statement follows from Proposition 3.7.2.

(iii) follows from (i) and the corresponding statement for $\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$. □

Theorem 3.7.7 (Microlocality)**.** *If $A \in \Psi^m(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$ then*

$$\mathrm{WF}(Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u).$$

Proof. Let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and choose $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} \phi$. Then $\phi A u = \phi A(\psi u) + \phi A(1 - \psi)u$, and the second term is smooth by Proposition 3.2.2. By Proposition 3.7.4, $\mathrm{WF}(\phi A u) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(\psi u) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u)$. Taking the union over ϕ gives the result. □

Combined with microlocal parametrices this gives the basic regularity result.

Theorem 3.7.8 (Microlocal elliptic regularity)**.** *If $A \in \Psi^m(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$ then*

$$\mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{Char}_m(A).$$

Proof. Let $z \notin \mathrm{Char}_m(A)$ and let B be a microlocal parametrix at z, so $BA = \mathrm{Id} + R$ with $z \notin \mathrm{WF}'(R)$. Then $u = B(Au) - Ru$, and by Theorem 3.7.7 and Proposition 3.7.6(iii), $\mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{WF}'(R)$, which does not contain z if $z \notin \mathrm{WF}(Au)$. □

In particular, if A is elliptic at every point (that is, $\mathrm{Char}_m(A) = \emptyset$), then $\mathrm{WF}(u) = \mathrm{WF}(Au)$. For instance $\mathrm{WF}(u)$ is unchanged by the Fourier multipliers $\langle D\rangle^s$.

3.8 The wavefront set and the Fourier transform

We now give the more classical characterization of the wavefront set, in terms of the Fourier transform. We use the following standard facts about $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$, which follow from (3.1.1) and the Fourier inversion formula by approximating the integrals below by Riemann sums in $\mathcal{S}(\mathbb{R}^n)$: $\hat{u}(\xi) = u(e^{-ix\cdot\xi})$ is a smooth function with $|\hat{u}(\xi)| \leq C\langle\xi\rangle^k$; for $\phi \in \mathcal{S}(\mathbb{R}^n)$,

$$u(\phi) = \int \hat{u}(\xi)\,\hat{\phi}(-\xi)đ\xi, \qquad \widehat{\phi u}(\xi) = \int \hat{\phi}(\xi-\eta)\,\hat{u}(\eta)đ\eta. \tag{3.8.1}$$

Conversely, a continuous polynomially bounded function is the Fourier transform of a unique element of $\mathcal{S}'(\mathbb{R}^n)$.

For $v \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$ let

$$\Sigma(v) = \{\xi \in \mathbb{R}^n \setminus 0;\ (x,\xi) \in \mathrm{WF}(v) \text{ for some } x\}.$$

Since $\mathrm{WF}(v)$ is closed, conic and lies over the compact set $\operatorname{supp} v$, $\mathrm{WF}(v) \cap (\mathbb{R}^n \times \mathbb{S}^{n-1})$ is compact, so $\Sigma(v)$ is a closed cone in $\mathbb{R}^n \setminus 0$, and $\mathrm{WF}(v) \subset \operatorname{supp} v \times \Sigma(v)$.

Lemma 3.8.1. *Let $v \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$.*

(i) *If $\Gamma \subset \mathbb{R}^n \setminus 0$ is a closed cone with $\Gamma \cap \Sigma(v) = \emptyset$, then $\hat{v}$ is rapidly decreasing on Γ.*

(ii) *If $\hat{v}$ is rapidly decreasing on an open cone V then $\mathrm{WF}(v) \cap (\mathbb{R}^n \times V) = \emptyset$.*

(iii) *If $\bar{x} \in \mathbb{R}^n$ and V is an open cone containing $\{\xi;\ (\bar{x},\xi) \in \mathrm{WF}(v)\}$, there is $r > 0$ such that $\Sigma(\phi v) \subset V$ for all $\phi \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)$ supported in $\{|x - \bar{x}| < r\}$.*

Proof. (i) By compactness there is an open cone $\Gamma' \supset \Gamma$ with $\overline{\Gamma'} \cap \Sigma(v) = \emptyset$ (the distance between $\Gamma \cap \mathbb{S}^{n-1}$ and the compact set $\Sigma(v) \cap \mathbb{S}^{n-1}$ is positive). Choose $g \in S^0(\mathbb{R}^n)$, a function of ξ only, with $g = 1$ on $\Gamma \cap \{|\xi| \geq 1\}$ and $\operatorname{supp} g \subset \Gamma'$; for instance $g = h(\xi/|\xi|)(1 - \chi(\xi))$ with $h \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$. Then $g(D) = \mathrm{Op}(g)$ has $\mathrm{WF}'(g(D)) \subset \mathbb{R}^n \times \overline{\Gamma'}$, which is disjoint from $\mathrm{WF}(v)$, so $g(D)v \in \mathcal{C}^\infty$ by Theorem 3.7.7. Let $\psi_1, \psi_2 \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)$ with $\psi_1 = 1$ near $\operatorname{supp} v$ and $\psi_2 = 1$ near $\operatorname{supp} \psi_1$. Then

$$g(D)v = \psi_2 g(D)v + (1 - \psi_2)g(D)\psi_1 v,$$

where the first term is in $\mathcal{C}_{\mathrm{c}}^{\infty}$ and the second is in $\mathcal{S}$ by Proposition 3.2.2 and Lemma 3.2.3. Thus $g\hat{v} = \widehat{g(D)v} \in \mathcal{S}$, and $\hat{v}$ is rapidly decreasing on Γ.

(ii) Let $z = (\bar{x},\bar{\xi})$ with $\bar{\xi} \in V$, and choose $\epsilon > 0$ with $\overline{V_\epsilon(\bar{\xi})} \subset V$ and $g \in S^0(\mathbb{R}^n)$ a function of ξ with $g = 1$ on $V_{\epsilon/2}(\bar{\xi}) \cap \{|\xi| \geq 1\}$, supported in $V_\epsilon(\bar{\xi})$. Then $g(D)$ is elliptic at

z, and $\widehat{g(D)v} = g\hat{v}$ is rapidly decreasing, hence integrable together with $\xi^\alpha g\hat{v}$ for all α, so $g(D)v \in \mathcal{C}^\infty$ by Fourier inversion. Thus $z \notin \mathrm{WF}(v)$.

(iii) Otherwise there are $x_j \to \bar{x}$ and unit vectors $\xi_j \notin V$ with $(x_j, \xi_j) \in \mathrm{WF}(v)$; a subsequence has $\xi_j \to \xi$, and $(\bar{x}, \xi) \in \mathrm{WF}(v)$ by closedness, $\xi \notin V$ since the complement of V is closed, a contradiction. Since $\mathrm{WF}(\phi v) \subset \mathrm{WF}(v) \cap (\operatorname{supp}\phi \times \mathbb{R}^n)$, the claim follows. □

Proposition 3.8.2 (Fourier characterization)**.** *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ and $(\bar{x}, \bar{\xi}) \in T^*\mathbb{R}^n \setminus 0$. Then $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$ if and only if there exist $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ and $\epsilon > 0$ such that $\widehat{\phi u}$ is rapidly decreasing on $V_\epsilon(\bar{\xi})$:*

$$|\widehat{\phi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \quad \text{for } \xi \in V_\epsilon(\bar{\xi}),\ N \in \mathbb{N}.$$

More precisely, if $K \subset \mathbb{R}^n$ is compact, $\Gamma \subset \mathbb{R}^n \setminus 0$ is a closed cone and $\mathrm{WF}(u) \cap (K \times \Gamma) = \emptyset$, then $\widehat{\phi u}$ is rapidly decreasing on Γ for every $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\operatorname{supp}\phi \subset K$.

Proof. If $\widehat{\phi u}$ is rapidly decreasing on $V_\epsilon(\bar{\xi})$ then $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(\phi u)$ by Lemma 3.8.1(ii), and so $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$ by Proposition 3.7.6(i).

For the last statement, $\mathrm{WF}(\phi u) \subset \mathrm{WF}(u) \cap (K \times \mathbb{R}^n)$, so $\Sigma(\phi u) \cap \Gamma = \emptyset$ and Lemma 3.8.1(i) applies. Finally, if $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$, closedness of $\mathrm{WF}(u)$ and a compactness argument as in Lemma 3.8.1(iii) give $r, \epsilon > 0$ with $\mathrm{WF}(u) \cap (K \times \overline{V_\epsilon(\bar{\xi})}) = \emptyset$ for $K = \{|x - \bar{x}| \leq r\}$, and any ϕ supported in K with $\phi(\bar{x}) \neq 0$ will do. □

Example 3.8.3. Since $\hat{\delta} = 1$ and $\operatorname{sing\,supp}\delta = \{0\}$, $\mathrm{WF}(\delta) = \{0\} \times (\mathbb{R}^n \setminus 0)$: indeed $\widehat{\phi\delta} = \phi(0)$ is not rapidly decreasing in any direction if $\phi(0) \neq 0$. More generally, if $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ is not smooth then $\Sigma(u) \neq \emptyset$, since otherwise $\hat{u} \in \mathcal{S}$ by Lemma 3.8.1(i).

Corollary 3.8.4. *For $u \in \mathcal{S}'(\mathbb{R}^n)$, $\mathrm{WF}(\bar{u}) = \mathrm{WF}(u)^-$.*

Proof. $\widehat{\phi\bar{u}}(\xi) = \overline{\widehat{\bar{\phi}u}(-\xi)}$, so this follows from Proposition 3.8.2 applied with $\bar{\phi}$. □

3.9 Invariance under diffeomorphisms

For a diffeomorphism $F : \Omega \to \Omega'$ between open subsets of $\mathbb{R}^n$ we use the map $\chi_F(x, \xi) = (F(x), (DF(x)^t)^{-1}\xi)$ of Section 3.6. Its inverse is

$$\chi_F^{-1}(y, \eta) = (G(y), DF(G(y))^t\eta), \qquad G = F^{-1},$$

the cotangent map $\eta \mapsto DF(x)^t\eta$, $x = G(y)$, which we denote by F^* when applied to subsets: for $W \subset T^*\Omega' \setminus 0$,

$$F^*W = \{(x, DF(x)^t\eta);\ (F(x), \eta) \in W\}.$$

Theorem 3.9.1. *If $F : \Omega \to \Omega'$ is a diffeomorphism between open subsets of $\mathbb{R}^n$ and $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega')$, then*

$$\mathrm{WF}(F^*u) = F^* \mathrm{WF}(u) = \{(x, DF(x)^t\eta);\ (F(x), \eta) \in \mathrm{WF}(u)\}.$$

Proof. Let $(x, \xi) \in T^*\Omega \setminus 0$ with $w = \chi_F(x, \xi) \notin \mathrm{WF}(u)$; we show $(x, \xi) \notin \mathrm{WF}(F^*u)$. Let Γ be an open conic neighbourhood of w as in Lemma 3.7.3, choose $\epsilon > 0$ with $\overline{\Gamma_\epsilon(w)} \subset \Gamma$ and $\{|y - F(x)| \leq \epsilon\} \subset \Omega'$, and let $\theta \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ equal 1 on $\{|y - F(x)| \leq \epsilon\}$. The operator $B = \theta G_{w,\epsilon}\theta \in \Psi^0(\mathbb{R}^n)$ has kernel with compact support in $\Omega' \times \Omega'$, $\mathrm{WF}'(B) \subset \overline{\Gamma_\epsilon(w)} \subset \Gamma$ and is elliptic at w. So $Bu \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$.

By Theorem 3.6.2, applied to the diffeomorphism $G : \Omega' \to \Omega$, the operator $\tilde{B} = B_G = F^* B G^*$ is in $\Psi^0(\mathbb{R}^n)$ and is elliptic at $\chi_G(w) = (x, \xi)$. Both $\tilde{B}$ (as an element of $\Psi^0(\mathbb{R}^n)$ acting on $\mathcal{S}'$) and $F^* \circ B \circ G^*$ are weakly continuous on $\mathcal{C}_c^{-\infty}(\Omega)$ and agree on $\mathcal{C}_c^\infty(\Omega)$, so by density they agree on $\mathcal{C}_c^{-\infty}(\Omega)$. Hence $\tilde{B}(F^*u) = F^* B G^* F^* u = F^*(Bu) \in \mathcal{C}^\infty$, and $(x, \xi) \notin \mathrm{WF}(F^*u)$.

We have shown $\mathrm{WF}(F^*u) \subset \chi_F^{-1}(\mathrm{WF}(u)) = F^* \mathrm{WF}(u)$. Applying this to G and F^*u, and using $G^* F^* u = u$, gives the reverse inclusion. □

If L is an invertible linear map of $\mathbb{R}^n$ and $u \in \mathcal{S}'(\mathbb{R}^n)$, then by localization (Proposition 3.7.6) the theorem also applies to L^*u: $\mathrm{WF}(L^*u) = \{(x, L^t\eta);\ (Lx, \eta) \in \mathrm{WF}(u)\}$.

3.10 The wavefront set of a kernel

The kernels of pseudodifferential operators form an interesting class of distributions. Their wavefront sets are determined by, and determine, the essential supports of the operators.

Proposition 3.10.1. *If $A \in \Psi^m(\mathbb{R}^n)$ then*

$$\mathrm{WF}(K_A) = \{(x, x, \xi, -\xi) \in T^*\mathbb{R}^{2n} \setminus 0;\ (x, \xi) \in \mathrm{WF}'(A)\}.$$

Proof. By Proposition 3.2.2 and Proposition 3.7.6(ii), $\mathrm{WF}(K_A)$ lies over the diagonal. Fix $\bar{x} \in \mathbb{R}^n$, let $\mu \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal 1 near 0 and let A_μ have kernel $\mu(x-y)K_A$; by Lemma 3.2.1, $A - A_\mu \in \Psi^{-\infty}$, so $a_\mu = \sigma_L(A_\mu)$ differs from $a = \sigma_L(A)$ by an element of $S^{-\infty}$. For $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ consider

$$k(x, y) = \phi(x)\mu(x-y)K_A(x, y) \in \mathcal{C}_c^{-\infty}(\mathbb{R}^{2n}), \qquad w = L^*k, \quad L(x, z) = (x, x - z),$$

so $w(x, z) = \phi(x)\mu(z)K_A(x, x-z)$, and $k = K_A$ near $(\bar{x}, \bar{x})$. Since $K_{A_\mu} = \int e^{i(x-y)\cdot\xi} a_\mu(x, \xi) đ\xi$, the distribution $L^*K_{A_\mu}$ is the inverse Fourier transform of $a_\mu(x, \xi)$ in ξ, and so

$$\hat{w}(\zeta, \xi) = \int e^{-ix\cdot\zeta} \phi(x) a_\mu(x, \xi)\, dx,$$

the Fourier transform being taken in $(x, z) \mapsto (\zeta, \xi)$. Since ϕa_μ is a symbol with compact support in x, integration by parts in x gives

$$|\hat{w}(\zeta, \xi)| \leq C_N \langle\zeta\rangle^{-N} \langle\xi\rangle^m \quad \text{for all } N. \tag{3.10.1}$$

Points of the wavefront set and the covectors transform under L by Theorem 3.9.1: since $DL^t(\xi, \eta) = (\xi + \eta, -\eta)$,

$$(x, 0; \zeta, \eta') \in \mathrm{WF}(w) \iff (x, x; \zeta + \eta', -\eta') \in \mathrm{WF}(k). \tag{3.10.2}$$

Inclusion $\subset$. By (3.10.1), $\hat{w}$ is rapidly decreasing on every closed cone $\{|\xi| \leq c|\zeta|\}$, so $\Sigma(w) \subset \{\zeta = 0\}$ by Lemma 3.8.1(ii). Since w is smooth for $z \neq 0$, $\mathrm{WF}(w) \subset \{(x, 0; 0, \xi)\}$, and (3.10.2) gives $\mathrm{WF}(k) \subset \{(x, x; \xi, -\xi)\}$.

Now suppose $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}'(A)$. Then a_μ and all its derivatives are rapidly decreasing on $\Gamma_\epsilon(\bar{x}, \bar{\xi})$ for some ϵ (Lemma 3.3.2); take $\operatorname{supp}\phi \subset \{|x - \bar{x}| < \epsilon\}$. Then integration by parts gives $|\hat{w}(\zeta, \xi)| \leq C_N \langle\zeta\rangle^{-N} \langle\xi\rangle^{-N}$ for $\xi \in V_\epsilon(\bar{\xi})$, which with (3.10.1) shows that $\hat{w}$ is rapidly decreasing on the cone $\{(\zeta, \xi);\ \xi \in V_\epsilon(\bar{\xi}),\ |\zeta| < |\xi|\}$, a conic neighbourhood of $(0, \bar{\xi})$. Hence $(\bar{x}, 0; 0, \bar{\xi}) \notin \mathrm{WF}(w)$, and by (3.10.2), $(\bar{x}, \bar{x}; \bar{\xi}, -\bar{\xi}) \notin \mathrm{WF}(k)$, so not in $\mathrm{WF}(K_A)$.

Inclusion $\supset$. Suppose $(\bar{x}, \bar{x}, \bar{\xi}, -\bar{\xi}) \notin \mathrm{WF}(K_A)$. By closedness there are $r, \theta > 0$ such that $(x, x, \xi, -\xi) \notin \mathrm{WF}(K_A)$ for $|x - \bar{x}| < r$, $\xi \in V_\theta(\bar{\xi})$. Take $\operatorname{supp}\phi \subset \{|x - \bar{x}| < r\}$ and

$\phi = 1$ near $\bar{x}$. Then, by (3.10.2) and the first part, $\Sigma(w) \subset \{(0,\xi);\ \xi \notin V_\theta(\bar{\xi})\}$, which is disjoint from the closed cone $C = \{(\zeta,\xi) \neq 0;\ |\zeta| \leq |\xi|,\ \xi \in \overline{V_{\theta/2}(\bar{\xi})}\}$. By Lemma 3.8.1(i), $\hat{w}$ is rapidly decreasing on C. By Fourier inversion in ζ, for $\xi \in V_{\theta/2}(\bar{\xi})$,

$$|\phi(x)a_\mu(x,\xi)| \leq \int_{|\zeta|\leq|\xi|} |\hat{w}(\zeta,\xi)|đ\zeta + \int_{|\zeta|>|\xi|} |\hat{w}(\zeta,\xi)|đ\zeta \leq C_N\langle\xi\rangle^{n-N} + C_N\langle\xi\rangle^{m+n-N},$$

using (3.10.1) in the second integral. So a_μ, and hence a, is rapidly decreasing on $\Gamma_{r'}(\bar{x},\bar{\xi})$ for small r', that is $(\bar{x},\bar{\xi}) \notin \mathrm{WF}'(A)$. □

In particular $\mathrm{WF}'(A)$ and $\mathrm{WF}(K_A)$ determine each other, and $\mathrm{WF}'(A) = \emptyset$ if and only if $K_A \in \mathcal{C}^\infty(\mathbb{R}^{2n})$.

3.11 The calculus of wavefront sets: pairing

We now wish to understand, in terms of wavefront sets, three fundamental operations on distributions: pull-back F^*u by a smooth map, push-forward F_*u, and multiplication u_1u_2. None of them is defined on all distributions. To analyse them we first discuss some more elementary operations:

pairing:	$(u,v) \longmapsto \int u(x)v(x)\,dx,$
exterior product:	$(u,v) \longmapsto (u \boxtimes v)(x,y) = u(x)v(y),$
restriction:	$u(x,y) \longmapsto u(x,0),$
projection:	$u(x,y) \longmapsto \int u(x,y)\,dy,$
invariance:	$u \longmapsto F^*u,\ F$ a diffeomorphism.

The last was treated in Section 3.9. Restriction and diffeomorphisms are special cases of pull-back and projection is a special case of push-forward; conversely, we shall express the three fundamental operations in terms of the elementary ones. Throughout, all distributions have compact support; by the local nature of the wavefront set this is not a serious restriction.

The pairing $\int u_1u_2\,dx$ of two distributions is the first example of an operation which requires a condition on wavefront sets. We give Melrose's operator-theoretic definition; in Section 3.13 it is recovered as the integral of the product (Exercise 3.15). We need a transposition identity.

Lemma 3.11.1. *Let $D \in \Psi^m(\mathbb{R}^n)$ and $u,v \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ with $\mathrm{WF}'(D) \cap \mathrm{WF}(u) = \emptyset$ and $\mathrm{WF}'(D^t) \cap \mathrm{WF}(v) = \emptyset$. Then Du and D^tv are smooth and, for any $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} u \cup \operatorname{supp} v$,*

$$v(\psi Du) = u(\psi D^tv).$$

Proof. Smoothness follows from Theorem 3.7.7. Both sides are independent of the choice of ψ. Let $\psi_1 \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal 1 near $\operatorname{supp} u \cup \operatorname{supp} v$ and let $\psi = \psi_2$ equal 1 near $\operatorname{supp}\psi_1$. The closed conic sets $W_1 = \mathrm{WF}'(D) \cap (\operatorname{supp}\psi_1 \times \mathbb{R}^n)$ and $W_2 = \mathrm{WF}(u)$ are disjoint and $\pi(W_1)$ is bounded, so Lemma 3.3.9 gives $B \in \Psi^0(\mathbb{R}^n)$ with $\mathrm{WF}'(B) \cap \mathrm{WF}(u) = \emptyset$ and $\mathrm{WF}'(\mathrm{Id}-B) \cap W_1 = \emptyset$. Write $u = u_s + u_r$ with $u_s = \psi_1Bu$ and $u_r = \psi_1(\mathrm{Id}-B)u$. By microlocality Bu is smooth, so $u_s \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, and since D^tv is smooth, the definition of D on $\mathcal{S}'$ gives

$$v(\psi Du_s) = v(Du_s) = (D^tv)(u_s) = \int (D^tv)u_s\,dx = u_s(\psi D^tv).$$

For u_r, let $E = \psi_2 D\psi_1(\mathrm{Id}-B)\psi_1$. Since $\mathrm{WF}'(D\psi_1) \subset W_1$, $\mathrm{WF}'(E) \subset W_1 \cap \mathrm{WF}'(\mathrm{Id}-B) = \emptyset$, and by Lemma 3.3.8 $E \in \Psi^{-\infty}(\mathbb{R}^n)$ with kernel $K_E \in \mathcal{C}_c^\infty(\mathbb{R}^{2n})$. As $u = \psi_1 u$, $\psi_2 D u_r = Eu$, while

$$u_r(\psi_2 D^t v) = u((\mathrm{Id}-B^t)\psi_1\psi_2 D^t v) = u(\psi_1(\mathrm{Id}-B^t)\psi_1 D^t \psi_2 v) = u(E^t v),$$

using $u = \psi_1 u$, $v = \psi_2 v$ and $\psi_1\psi_2 = \psi_1$. Finally $v(Eu) = u(E^t v)$ for E with kernel in $\mathcal{C}_c^\infty(\mathbb{R}^{2n})$: both sides equal $(v \boxtimes u)(K_E)$, since $Eu(x) = u(K_E(x,\cdot))$, $E^t v(y) = v(K_E(\cdot, y))$, and the identity $v_x(u_y(\Phi(x,y))) = u_y(v_x(\Phi(x,y)))$ holds for $\Phi = \phi \boxtimes \phi'$, hence by continuity and density for all $\Phi \in \mathcal{S}(\mathbb{R}^{2n})$. Adding the two identities proves the lemma. □

Proposition 3.11.2 (Pairing)**.** *Let $u_1, u_2 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ satisfy*

$$(x,\xi) \in \mathrm{WF}(u_1) \Longrightarrow (x,-\xi) \notin \mathrm{WF}(u_2), \quad \textit{i.e.} \quad \mathrm{WF}(u_1) \cap \mathrm{WF}(u_2)^- = \emptyset. \tag{3.11.1}$$

Then there exists $A \in \Psi^0(\mathbb{R}^n)$ with

$$\mathrm{WF}'(A) \cap \mathrm{WF}(u_1) = \emptyset, \qquad \mathrm{WF}'(\mathrm{Id}-A) \cap \mathrm{WF}(u_2)^- = \emptyset, \tag{3.11.2}$$

and for any such A and any $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} u_1 \cup \operatorname{supp} u_2$ the number

$$P(u_1,u_2) = u_2(\psi A u_1) + u_1(\psi(\mathrm{Id}-A)^t u_2)$$

is independent of A and ψ. The pairing P is bilinear, symmetric, and $P(u_1,u_2) = u_2(u_1) = \int u_1 u_2\, dx$ if $u_1 \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ (and similarly if $u_2 \in \mathcal{C}_c^\infty$).

Proof. Existence of A follows from Lemma 3.3.9 with $W_1 = \mathrm{WF}(u_2)^-$, $W_2 = \mathrm{WF}(u_1)$. By microlocality and $\mathrm{WF}'((\mathrm{Id}-A)^t) = \mathrm{WF}'(\mathrm{Id}-A)^-$ (Proposition 3.3.7), Au_1 and $(\mathrm{Id}-A)^t u_2$ are smooth, so $P(u_1,u_2)$ is defined; it is clearly independent of ψ. If A' is another choice, $D = A - A'$ satisfies $\mathrm{WF}'(D) \cap \mathrm{WF}(u_1) = \emptyset$, and since $D = (\mathrm{Id}-A') - (\mathrm{Id}-A)$ also $\mathrm{WF}'(D) \cap \mathrm{WF}(u_2)^- = \emptyset$, that is $\mathrm{WF}'(D^t) \cap \mathrm{WF}(u_2) = \emptyset$. The difference of the two expressions is $u_2(\psi D u_1) - u_1(\psi D^t u_2)$, which vanishes by Lemma 3.11.1.

For linearity in the first argument, suppose (u_1,u_2) and (u_1',u_2) both satisfy (3.11.1). Then $(\mathrm{WF}(u_1) \cup \mathrm{WF}(u_1')) \cap \mathrm{WF}(u_2)^- = \emptyset$, and an A obtained from Lemma 3.3.9 with $W_2 = \mathrm{WF}(u_1) \cup \mathrm{WF}(u_1')$ satisfies (3.11.2) for the three pairs (u_1,u_2), (u_1',u_2) and (u_1+u_1',u_2), since $\mathrm{WF}(u_1+u_1') \subset \mathrm{WF}(u_1) \cup \mathrm{WF}(u_1')$; with this A the expressions are visibly additive. Linearity in the second argument is the same. For symmetry note that $A' = \mathrm{Id} - A^t$ satisfies (3.11.2) for the pair (u_2,u_1), and the corresponding expression is $u_1(\psi(\mathrm{Id}-A^t)u_2) + u_2(\psi A u_1)$. If $u_1 \in \mathcal{C}_c^\infty$ we may take $A = \mathrm{Id}$, and if $u_2 \in \mathcal{C}_c^\infty$, $A = 0$. □

The *sesquilinear* pairing $\langle u_1,u_2\rangle = \int u_1 \overline{u_2}\, dx$ is then defined as $P(u_1,\overline{u_2})$. Since $\mathrm{WF}(\overline{u_2}) = \mathrm{WF}(u_2)^-$ (Corollary 3.8.4), the condition becomes simply $\mathrm{WF}(u_1) \cap \mathrm{WF}(u_2) = \emptyset$, and $\langle u_1,u_2\rangle = \langle Au_1,u_2\rangle + \langle u_1,(\mathrm{Id}-A^*)u_2\rangle$ for any A with $\mathrm{WF}'(A) \cap \mathrm{WF}(u_1) = \emptyset = \mathrm{WF}'(\mathrm{Id}-A) \cap \mathrm{WF}(u_2)$.

3.12 Exterior products, restriction and projection

3.12.1 Exterior product

The exterior product $u_1 \boxtimes u_2$ of $u_1 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^p)$ and $u_2 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^k)$ is always defined (Chapter 1); it lies in $\mathcal{C}_c^{-\infty}(\mathbb{R}^{p+k})$, $(u_1 \boxtimes u_2)(\Phi) = u_1(u_2(\Phi(x,\cdot)))$, and its Fourier transform is

$$\widehat{u_1 \boxtimes u_2}(\xi,\eta) = \hat{u}_1(\xi)\,\hat{u}_2(\eta).$$

Proposition 3.12.1. *For $u_1 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^p)$ and $u_2 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^k)$,*

$$\mathrm{WF}(u_1 \boxtimes u_2) \subset \big(\mathrm{WF}(u_1) \times \mathrm{WF}(u_2)\big) \cup \{(x,y,0,\eta);\ x \in \operatorname{supp} u_1,\ (y,\eta) \in \mathrm{WF}(u_2)\} \cup \{(x,y,\xi,0);\ (x,\xi) \in \mathrm{WF}(u_1),\ y \in \operatorname{supp} u_2\},$$

where $\mathrm{WF}(u_1) \times \mathrm{WF}(u_2)$ *means* $\{(x,y,\xi,\eta); (x,\xi) \in \mathrm{WF}(u_1), (y,\eta) \in \mathrm{WF}(u_2)\}$.

Proof. Let $(\bar x, \bar y, \bar\xi, \bar\eta)$ be outside the right side. We localize with $\phi_1(x)\phi_2(y)$, $\phi_1(\bar x) \neq 0 \neq \phi_2(\bar y)$, using $\phi_1\phi_2(u_1 \boxtimes u_2) = (\phi_1 u_1) \boxtimes (\phi_2 u_2)$, whose Fourier transform is $\widehat{\phi_1 u_1}(\xi)\widehat{\phi_2 u_2}(\eta)$. If $\bar x \notin \operatorname{supp} u_1$ or $\bar y \notin \operatorname{supp} u_2$ we can make this zero. Otherwise one of the following holds.

(a) $\bar\xi \neq 0$ and $(\bar x, \bar\xi) \notin \mathrm{WF}(u_1)$. Choose ϕ_1 by Proposition 3.8.2 with $\widehat{\phi_1 u_1}$ rapidly decreasing on $V_\epsilon(\bar\xi)$. On the conic neighbourhood $\{(\xi,\eta);\ \xi \in V_\epsilon(\bar\xi),\ |\eta| < C|\xi|\}$ of $(\bar\xi, \bar\eta)$, where $C > |\bar\eta|/|\bar\xi|$, we have $\langle(\xi,\eta)\rangle \le (1+C)\langle\xi\rangle$ and $|\widehat{\phi_2 u_2}(\eta)| \le C'\langle\xi\rangle^M$, so the product is rapidly decreasing.

(b) $\bar\eta \neq 0$ and $(\bar y, \bar\eta) \notin \mathrm{WF}(u_2)$: the same with the factors exchanged.

These cases exhaust the possibilities: if both $\bar\xi \neq 0$ and $\bar\eta \neq 0$, the point is not in $\mathrm{WF}(u_1) \times \mathrm{WF}(u_2)$, so (a) or (b) holds; if $\bar\xi = 0$ then $\bar\eta \neq 0$ and, $\bar x$ being in $\operatorname{supp} u_1$, $(\bar y, \bar\eta) \notin \mathrm{WF}(u_2)$, so (b) holds; similarly if $\bar\eta = 0$. □

3.12.2 Restriction

Write $\mathbb{R}^{p+k} \ni (x,y)$, let $Y = \{y = 0\}$ and $\iota : \mathbb{R}^p \to \mathbb{R}^{p+k}$, $\iota(x) = (x,0)$. We want to define $\iota^* u = u \restriction Y$. No reasonable such map exists on all distributions (think of $\delta(y)$), but it does under a condition on the wavefront set: $\mathrm{WF}(u)$ should not meet the *conormal bundle*

$$N^*Y = \{(x,0;0,\eta);\ x \in \mathbb{R}^p,\ \eta \in \mathbb{R}^k \setminus 0\}.$$

For $\delta > 0$ consider the closed cone $C_\delta = \{(\xi,\eta) \in \mathbb{R}^{p+k} \setminus 0;\ |\xi| \le \delta|\eta|\}$ and let $\mathcal{R}_\delta$ be the set of $w \in \mathcal{C}_c^{-\infty}(\mathbb{R}^{p+k})$ with $\hat w$ rapidly decreasing on C_δ. If $w \in \mathcal{R}_\delta$ then, since $|\hat w| \le C\langle(\xi,\eta)\rangle^M$,

$$\int |\hat w(\xi,\eta)|\, d\eta \le \int_{|\eta| \le |\xi|/\delta} C\langle(\xi,\eta)\rangle^M\, d\eta + \int_{|\eta| > |\xi|/\delta} C_N \langle\eta\rangle^{-N}\, d\eta \le C'\langle\xi\rangle^{M+k}, \tag{3.12.1}$$

locally uniformly in ξ, so

$$g_w(\xi) = \int \hat w(\xi,\eta)đ\eta \qquad (đ\eta = (2\pi)^{-k}d\eta)$$

is continuous and polynomially bounded, and we define $\iota^* w = \mathcal{F}^{-1} g_w \in \mathcal{S}'(\mathbb{R}^p)$.

Lemma 3.12.2. *Let $w \in \mathcal{R}_\delta$, $\rho \in \mathcal{C}_c^\infty(\mathbb{R}^k)$ with $\int \rho = 1$, and $\rho_\epsilon(y) = \epsilon^{-k}\rho(y/\epsilon)$.*

(i) *For $\phi \in \mathcal{S}(\mathbb{R}^p)$, $\iota^* w(\phi) = \lim_{\epsilon\to 0} w(\phi \boxtimes \rho_\epsilon)$.*

(ii) *If $w \in \mathcal{C}_c^\infty(\mathbb{R}^{p+k})$ then $\iota^* w = w(\cdot, 0)$.*

(iii) $\operatorname{supp} \iota^* w \subset \{x;\ (x,0) \in \operatorname{supp} w\}$*; in particular $\iota^* w \in \mathcal{C}_c^{-\infty}(\mathbb{R}^p)$, and $\iota^* w = 0$ if $\operatorname{supp} w \cap Y = \emptyset$.*

(iv) *For $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^p)$, $\phi(x)w \in \mathcal{R}_{\delta/2}$ and $\iota^*(\phi w) = \phi\, \iota^* w$.*

Proof. (i) By (3.8.1) in $\mathbb{R}^{p+k}$, $w(\phi \boxtimes \rho_\epsilon) = \iint \hat w(\xi,\eta)\hat\phi(-\xi)\hat\rho(-\epsilon\eta)đ\xi đ\eta$. As $\epsilon \to 0$, $\hat\rho(-\epsilon\eta) \to \hat\rho(0) = 1$ with $|\hat\rho| \le \|\rho\|_{L^1}$, and $|\hat w(\xi,\eta)\hat\phi(-\xi)|$ is integrable by (3.12.1); by dominated convergence the limit is $\int g_w(\xi)\hat\phi(-\xi)đ\xi = \iota^* w(\phi)$.

(ii) is clear from (i), and (iii) follows from (i) too: if $(\bar{x},0)\notin\operatorname{supp} w$ and ϕ is supported near $\bar{x}$, then $\phi\boxtimes\rho_\epsilon$ vanishes near $\operatorname{supp} w$ for small ϵ.

(iv) By Lemma 3.8.1(ii), $\Sigma(w)$ does not meet the interior of C_δ, which contains $C_{\delta/2}$; as $\Sigma(\phi w)\subset\Sigma(w)$, Lemma 3.8.1(i) shows $\phi w\in\mathcal{R}_{\delta/2}$. The identity follows from (i), since $(\phi w)(f\boxtimes\rho_\epsilon)=w(\phi f\boxtimes\rho_\epsilon)$. □

Theorem 3.12.3 (Restriction)**.** *Let $u\in\mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^{p+k})$ satisfy $\mathrm{WF}(u)\cap N^*Y=\emptyset$. There are $r,\delta>0$ such that $\chi u\in\mathcal{R}_\delta$ for every $\chi=\chi(y)\in\mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^k)$ supported in $\{|y|<r\}$, and*

$$\iota^*u=\iota^*(\chi u)\in\mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^p),\qquad \chi=1\ \textit{near}\ 0,$$

is independent of the choice of χ. The map $u\mapsto\iota^*u$ is linear, coincides with $u\mapsto u(\cdot,0)$ on $\mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^{p+k})$, satisfies $\iota^*u(\phi)=\lim_{\epsilon\to0}u(\phi\boxtimes\rho_\epsilon)$, $\iota^*(\phi u)=\phi\,\iota^*u$ for $\phi\in\mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^p)$, and

$$\mathrm{WF}(\iota^*u)\subset\iota^*\,\mathrm{WF}(u)=\{(x,\xi)\in T^*\mathbb{R}^p\setminus0;\ (x,0,\xi,\eta)\in\mathrm{WF}(u)\ \textit{for some}\ \eta\in\mathbb{R}^k\}.$$

Proof. The compact sets $Z_\delta=\mathrm{WF}(u)\cap\{|y|\le\delta\}\cap(\mathbb{R}^{p+k}\times C_\delta)\cap\{|(\xi,\eta)|=1\}$ decrease as $\delta\downarrow0$, and their intersection is contained in $\mathrm{WF}(u)\cap N^*Y=\emptyset$. Hence $Z_\delta=\emptyset$ for some $\delta>0$; take $r=\delta$. If $\operatorname{supp}\chi\subset\{|y|<r\}$ then $\mathrm{WF}(\chi u)\subset\mathrm{WF}(u)\cap\{|y|<r\}$, so $\Sigma(\chi u)\cap C_\delta=\emptyset$ and $\chi u\in\mathcal{R}_\delta$ by Lemma 3.8.1(i). If χ,χ' are two choices, $(\chi-\chi')u\in\mathcal{R}_\delta$ vanishes near Y, so $\iota^*(\chi u)=\iota^*(\chi' u)$ by Lemma 3.12.2(iii). Linearity is clear, and the other properties, except the last, follow from Lemma 3.12.2, since $\chi(y)\rho_\epsilon(y)=\rho_\epsilon(y)$ for small ϵ. If $u\in\mathcal{R}_{\delta'}$ for some $\delta'>0$ then ι^*u as defined before Lemma 3.12.2 agrees with $\iota^*(\chi u)$, by the limit formula.

For the wavefront set, let $(\bar{x},\bar{\xi})\in T^*\mathbb{R}^p\setminus0$ with $(\bar{x},0,\bar{\xi},\eta)\notin\mathrm{WF}(u)$ for all η. For $\theta\in(0,1)$ let

$$C'_\theta=C_\theta\cup\{(\xi,\eta);\ \xi\in\overline{V_\theta(\bar{\xi})}\},$$

a closed cone in $\mathbb{R}^{p+k}\setminus0$. As $\theta\downarrow0$ the compact sets $C'_\theta\cap\mathbb{S}^{p+k-1}$ decrease to the set of unit vectors (ξ,η) with $\xi=0$ or $\xi\in\mathbb{R}_+\bar{\xi}$, which is disjoint from the closed cone $\{(\xi,\eta);\ (\bar{x},0,\xi,\eta)\in\mathrm{WF}(u)\}$ by hypothesis and since $\mathrm{WF}(u)\cap N^*Y=\emptyset$. Hence for some θ the latter cone lies in the open cone $\mathbb{R}^{p+k}\setminus(C'_\theta\cup\{0\})$, and by Lemma 3.8.1(iii) there are $\phi\in\mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^p)$, $\phi(\bar{x})\neq0$, and $\chi(y)$ as above (with small supports) such that $\Sigma(\phi\chi u)\cap C'_\theta=\emptyset$. Then $\widehat{\phi\chi u}$ is rapidly decreasing on C'_θ; in particular $\phi\chi u\in\mathcal{R}_\theta$, and for $\xi\in V_\theta(\bar{\xi})$, where $(\xi,\eta)\in C'_\theta$ for all η,

$$|\widehat{\phi\,\iota^*u}(\xi)|=\Big|\int\widehat{\phi\chi u}(\xi,\eta)đ\eta\Big|\le C_N\int\langle(\xi,\eta)\rangle^{-N-k-1}\,d\eta\le C'_N\langle\xi\rangle^{-N}.$$

Here we used $\phi\iota^*u=\iota^*(\phi\chi u)$, by the independence of χ and Lemma 3.12.2(iv). By Proposition 3.8.2, $(\bar{x},\bar{\xi})\notin\mathrm{WF}(\iota^*u)$. □

One can picture this as follows: of $\mathrm{WF}(u)$ only the part over Y survives, and the covector (ξ,η) is projected to its component ξ along Y; the hypothesis guarantees that no non-zero covector is projected to zero.

3.12.3 Projection

Let $\pi_1:\mathbb{R}^{p+k}\to\mathbb{R}^p$, $\pi_1(x,y)=x$. For $u\in\mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^{p+k})$ define the push-forward, or "integral over the fibres", $(\pi_1)_*u\in\mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^p)$ by

$$\Big(\int u(x,y)\,dy\Big)(\phi)=(\pi_1)_*u(\phi)=u(\pi_1^*\phi),\qquad \pi_1^*\phi(x,y)=\phi(x),$$

which makes sense because u acts on $\mathcal{C}^\infty(\mathbb{R}^{p+k})$. Thus projection is dual to pull-back of functions. By duality, $\operatorname{supp}(\pi_1)_* u \subset \pi_1(\operatorname{supp} u)$, since $\operatorname{supp} \pi_1^* \phi = \pi_1^{-1}(\operatorname{supp} \phi)$. Moreover $(\pi_1)_*$ maps $\mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^{p+k})$ to $\mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^p)$, and

$$\widehat{(\pi_1)_* u}(\xi) = u(e^{-ix\cdot\xi}) = \hat{u}(\xi, 0), \qquad \phi\,(\pi_1)_* u = (\pi_1)_*((\pi_1^*\phi)u). \tag{3.12.2}$$

Proposition 3.12.4 (Projection)**.** *For $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^{p+k})$,*

$$\mathrm{WF}\big((\pi_1)_* u\big) \subset \{(x,\xi) \in T^*\mathbb{R}^p \setminus 0;\ (x, y, \xi, 0) \in \mathrm{WF}(u) \text{ for some } y \in \mathbb{R}^k\}.$$

Proof. Let $(\bar{x}, \bar{\xi})$ be outside the right side. We claim there are $r, \theta > 0$ such that $\mathrm{WF}(u)$ does not meet $\{|x - \bar{x}| < r\} \times \mathbb{R}^k \times C''_\theta$, where $C''_\theta = \{(\xi, \eta);\ \xi \in \overline{V_\theta(\bar{\xi})},\ |\eta| \le \theta|\xi|\}$. Otherwise there are $(x_j, y_j, \xi_j, \eta_j) \in \mathrm{WF}(u)$ with $|(\xi_j, \eta_j)| = 1$, $x_j \to \bar{x}$ and $(\xi_j, \eta_j) \in C''_{1/j}$; since y_j lies in the compact set $\pi_2(\operatorname{supp} u)$, a subsequence converges to a point $(\bar{x}, y, \bar{\xi}/|\bar{\xi}|, 0)$ of the closed set $\mathrm{WF}(u)$, a contradiction. Now let $\phi \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^p)$ be supported in $\{|x - \bar{x}| < r\}$ with $\phi(\bar{x}) \neq 0$. Then $\Sigma((\pi_1^*\phi)u) \cap C''_\theta = \emptyset$, so by Lemma 3.8.1(i) and (3.12.2), $\widehat{\phi(\pi_1)_* u}(\xi) = \widehat{(\pi_1^*\phi)u}(\xi, 0)$ is rapidly decreasing on $V_\theta(\bar{\xi})$. By Proposition 3.8.2, $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}((\pi_1)_* u)$. □

Corollary 3.12.5. *If $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^{p+k})$ and $\mathrm{WF}(u) \cap \{(x, y, \xi, 0);\ \xi \neq 0\} = \emptyset$ then $(\pi_1)_* u \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^p)$.*

The covectors $(x, y, \xi, 0)$ are exactly those which are conormal to the fibres $\{x\} \times \mathbb{R}^k$ over which we integrate. Thus the only singularities which survive integration are those conormal to the fibres; those which are even partially in the direction of integration are wiped out. This fits with the fact that integration in *all* variables leaves no singularities at all.

3.13 Products of distributions

We express the product as "$u_1(x)u_2(x + y)$ restricted to $y = 0$". Let $L : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be the linear map $L(x, y) = (x, x + y)$, with $\det L = 1$ and transpose $DL^t(\xi, \eta) = (\xi + \eta, \eta)$.

Theorem 3.13.1 (Products)**.** *Let $u_1, u_2 \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$ satisfy*

$$(x, \xi) \in \mathrm{WF}(u_1) \Longrightarrow (x, -\xi) \notin \mathrm{WF}(u_2).$$

Then $w = L^*(u_1 \boxtimes u_2) \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^{2n})$ satisfies $\mathrm{WF}(w) \cap N^*\{y = 0\} = \emptyset$, and the product

$$u_1 u_2 = \iota^* L^*(u_1 \boxtimes u_2) \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$$

has the following properties.

(i) $\operatorname{supp}(u_1 u_2) \subset \operatorname{supp} u_1 \cap \operatorname{supp} u_2$, *and for $\phi \in \mathcal{S}(\mathbb{R}^n)$ and ρ, ρ_ϵ as in Lemma 3.12.2 (now on $\mathbb{R}^n$), with $\check{\rho}_\epsilon(t) = \rho_\epsilon(-t)$,*

$$(u_1 u_2)(\phi) = \lim_{\epsilon \to 0} u_1\big(\phi \cdot (u_2 * \check{\rho}_\epsilon)\big).$$

(ii) *If u_1 or u_2 is smooth, $u_1 u_2$ is the usual product.*

(iii) $(\phi u_1)u_2 = \phi(u_1 u_2)$ *for $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$.*

(iv) *The wavefront set satisfies*

$$\begin{aligned}\mathrm{WF}(u_1u_2) \subset\ & \{(x,\xi);\ x\in\operatorname{supp}u_1,\ (x,\xi)\in\mathrm{WF}(u_2)\}\\ & \cup\{(x,\xi);\ x\in\operatorname{supp}u_2,\ (x,\xi)\in\mathrm{WF}(u_1)\}\\ & \cup\{(x,\xi_1+\xi_2);\ (x,\xi_1)\in\mathrm{WF}(u_1),\ (x,\xi_2)\in\mathrm{WF}(u_2)\}.\end{aligned}$$

Note that $\xi_1+\xi_2\neq 0$ in the last set, by the hypothesis.

Proof. By Theorem 3.9.1 (for the linear map L),

$$\mathrm{WF}(w)=\{(x,y;\xi+\eta,\eta);\ (x,x+y;\xi,\eta)\in\mathrm{WF}(u_1\boxtimes u_2)\}.$$

A point of $N^*\{y=0\}$ has $y=0$ and $\xi+\eta=0$, $\eta\neq 0$; by Proposition 3.12.1, $(x,x;-\eta,\eta)$ with both components non-zero can only lie in $\mathrm{WF}(u_1\boxtimes u_2)$ if $(x,-\eta)\in\mathrm{WF}(u_1)$ and $(x,\eta)\in\mathrm{WF}(u_2)$, which is excluded. So Theorem 3.12.3 applies. The support statement follows from that theorem, since $(x,0)\in\operatorname{supp}w$ means $(x,x)\in\operatorname{supp}u_1\times\operatorname{supp}u_2$. For the formula in (i), by Theorem 3.12.3 and the definition of L^*,

$$(u_1u_2)(\phi)=\lim_{\epsilon\to 0}(u_1\boxtimes u_2)(\phi(x)\rho_\epsilon(z-x))=\lim_{\epsilon\to 0}u_1(\phi(x)\,u_2(\rho_\epsilon(\cdot-x))),$$

and $u_2(\rho_\epsilon(\cdot-x))=(u_2*\check\rho_\epsilon)(x)$.

(ii) If u_2 is smooth, $u_2*\check\rho_\epsilon\to u_2$ in $\mathcal{C}^\infty$, so (i) gives $u_1(\phi u_2)$. If u_1 is smooth, $u_1(\phi\cdot(u_2*\check\rho_\epsilon))=u_2((u_1\phi)*\rho_\epsilon)\to u_2(u_1\phi)$.

(iii) is immediate from (i).

(iv) By Theorem 3.12.3 and the formula for $\mathrm{WF}(w)$,

$$\mathrm{WF}(u_1u_2)\subset\{(x,\xi+\eta);\ (x,x;\xi,\eta)\in\mathrm{WF}(u_1\boxtimes u_2)\},$$

and Proposition 3.12.1 gives the three sets. □

Remark 3.13.2. If the stronger condition $\Sigma(u_1)\cap(-\Sigma(u_2))=\emptyset$ holds, for example if u_1 and u_2 have small supports near a point $\bar x$ at which the hypothesis of the theorem holds (Lemma 3.8.1(iii)), then the familiar formula

$$\widehat{u_1u_2}(\xi)=\int\hat u_1(\xi-\eta)\hat u_2(\eta)đ\eta$$

holds, with an absolutely convergent integral. Indeed $\hat w(\xi,\eta)=\hat u_1(\xi-\eta)\hat u_2(\eta)$, and $w\in\mathcal{R}_\delta$ for small δ: choose open cones $V_i\supset\Sigma(u_i)$ with $\overline{V_1}\cap(-\overline{V_2})=\emptyset$; if $|\xi|\le\delta|\eta|$ then either $\eta\notin V_2$, where $\hat u_2(\eta)$ is rapidly decreasing and $|\hat u_1(\xi-\eta)|\le C\langle\eta\rangle^M$, or $\eta\in V_2$, and then $\xi-\eta$, whose direction differs from that of $-\eta$ by at most 2δ, lies in a closed cone disjoint from $\Sigma(u_1)$ if δ is small, so $\hat u_1(\xi-\eta)$ is rapidly decreasing in $\langle\eta\rangle$. The formula is then the definition of ι^*w.

3.14 Pull-back

Now let $F:\Omega_1\to\Omega_2$ be a smooth map between open sets $\Omega_1\subset\mathbb{R}^n$ and $\Omega_2\subset\mathbb{R}^m$; the dimensions may differ. When can we define F^*u for $u\in\mathcal{C}_c^{-\infty}(\Omega_2)$, and what is its wavefront set? For a general F there is no consistent definition for all distributions (consider F constant and $u=\delta$). We assume that F is *proper*, i.e. $F^{-1}(K)$ is compact for every compact

$K \subset \Omega_2$, mostly for convenience: then $F^*\phi = \phi \circ F \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)$. The differential of F defines the cotangent maps

$$F_x^* : \mathbb{R}^m = T^*_{F(x)}\Omega_2 \to T^*_x\Omega_1 = \mathbb{R}^n, \qquad \eta \mapsto DF(x)^t\eta, \quad \text{i.e. } \xi_i = \sum_j \frac{\partial F_j}{\partial x_i}(x)\eta_j,$$

and we set

$$N_F = \{(F(x),\eta) \in \Omega_2 \times (\mathbb{R}^m \setminus 0);\ x \in \Omega_1,\ DF(x)^t\eta = 0\},$$
$$F^*W = \{(x, DF(x)^t\eta);\ (F(x),\eta) \in W\} \quad \text{for } W \subset T^*\Omega_2 \setminus 0.$$

Note that F^* is a relation rather than a map between cotangent spaces. N_F is the set of covectors that are mapped to zero; it is empty if F is a submersion, and for ι of the previous section it is N^*Y.

The construction decomposes F into elementary maps. Let $U = \{(x,y) \in \Omega_1 \times \mathbb{R}^m;\ y + F(x) \in \Omega_2\}$ and

$$H : U \to \Omega_1 \times \Omega_2, \qquad H(x,y) = (x, y + F(x)),$$

a diffeomorphism with inverse $(x,z) \mapsto (x, z - F(x))$ and $\det DH = 1$; it maps $U \cap \{y = 0\}$ onto the graph of F. For a function, $F^*\phi = \iota^* H^*(1 \boxtimes \phi)$, $\iota(x) = (x,0)$.

Theorem 3.14.1 (Pull-back)**.** *Let $F : \Omega_1 \to \Omega_2$ be a proper smooth map and let $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_2)$ satisfy $\mathrm{WF}(u) \cap N_F = \emptyset$. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ equal 1 near the compact set $F^{-1}(\operatorname{supp} u)$. Then $H^*(\chi \boxtimes u) \in \mathcal{C}^{-\infty}_{\mathrm{c}}(U)$ satisfies the hypothesis of Theorem 3.12.3, and*

$$F^*u = \iota^* H^*(\chi \boxtimes u) \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1)$$

is independent of χ, with $\operatorname{supp} F^*u \subset F^{-1}(\operatorname{supp} u)$. This defines a linear map which extends $\phi \mapsto \phi \circ F$ on $\mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)$, satisfies

$$F^*u(\phi) = \lim_{\epsilon \to 0} u\Big(\int \chi(x)\phi(x)\rho_\epsilon(\cdot - F(x))\,dx\Big), \qquad \phi \in \mathcal{S}(\mathbb{R}^n),$$

for $\rho \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^m)$, $\int \rho = 1$, agrees with the pull-back of Lemma 3.6.1 if F is a diffeomorphism, and

$$\mathrm{WF}(F^*u) \subset F^* \mathrm{WF}(u) = \bigcup_{x \in \Omega_1} \{x\} \times F_x^*(\mathrm{WF}(u) \cap T^*_{F(x)}\Omega_2).$$

Proof. By Proposition 3.12.1, since $\mathrm{WF}(\chi) = \emptyset$, $\mathrm{WF}(\chi \boxtimes u) \subset \{(x,z;0,\eta);\ x \in \operatorname{supp}\chi,\ (z,\eta) \in \mathrm{WF}(u)\}$. Since $DH(x,y)^t(\xi,\eta) = (\xi + DF(x)^t\eta, \eta)$, Theorem 3.9.1 gives

$$\mathrm{WF}\big(H^*(\chi \boxtimes u)\big) \subset \{(x,y; DF(x)^t\eta, \eta);\ x \in \operatorname{supp}\chi,\ (y + F(x),\eta) \in \mathrm{WF}(u)\}.$$

A point of this set over $y = 0$ lies in $N^*\{y = 0\}$ only if $DF(x)^t\eta = 0$ and $(F(x),\eta) \in \mathrm{WF}(u)$, i.e. if $(F(x),\eta) \in N_F \cap \mathrm{WF}(u) = \emptyset$. Thus Theorem 3.12.3 applies, and gives $\operatorname{supp} F^*u \subset \{x \in \operatorname{supp}\chi;\ F(x) \in \operatorname{supp} u\}$ and $\mathrm{WF}(F^*u) \subset \{(x, DF(x)^t\eta);\ (F(x),\eta) \in \mathrm{WF}(u)\}$.

If χ' is another choice, $H^*((\chi - \chi') \boxtimes u)$ is supported in $\{(x,y);\ x \in \operatorname{supp}(\chi - \chi'),\ y + F(x) \in \operatorname{supp} u\}$, which does not meet $\{y = 0\}$ because $\chi = \chi'$ near $F^{-1}(\operatorname{supp} u)$; hence its restriction vanishes (Lemma 3.12.2(iii)). By Theorem 3.12.3, the formula for H^* in Lemma 3.6.1 (with Jacobian 1) and the definition of the exterior product,

$$F^*u(\phi) = \lim_{\epsilon \to 0} H^*(\chi \boxtimes u)(\phi \boxtimes \rho_\epsilon) = \lim_{\epsilon \to 0} (\chi \boxtimes u)\big(\phi(x)\rho_\epsilon(z - F(x))\big),$$

which is the stated formula. If u is smooth, the expression under the limit is $\int \chi(x)\phi(x)(u * \check{\rho}_\epsilon)(F(x))\,dx$, with $\check{\rho}_\epsilon(t) = \rho_\epsilon(-t)$, which converges to $\int \chi\phi\,(u \circ F)\,dx = \int \phi\,(u \circ F)\,dx$. If

F is a diffeomorphism with inverse G, substituting $x = G(z')$ shows that the function $\int \chi\phi(x)\rho_\epsilon(z - F(x))\,dx$ is $\rho_\epsilon * f$ with $f = |\det DG|\,(\chi\phi)\circ G \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)$, which converges to f in $\mathcal{C}^\infty$; so $F^*u(\phi) = u(f)$, which is the definition in Lemma 3.6.1 since $\chi \circ G = 1$ near $\operatorname{supp} u$. □

The hypothesis $\mathrm{WF}(u) \cap N_F = \emptyset$ is natural: without it the right side of the wavefront set bound would contain zero covectors. Restriction to a submanifold, products ($u_1u_2 = \Delta^*(u_1 \boxtimes u_2)$ with $\Delta(x) = (x, x)$, for which $N_\Delta = \{(x, x; \xi, -\xi)\}$) and diffeomorphisms are all special cases; see Exercise 3.16.

3.15 Push-forward

The dual operation is push-forward. Pull-back maps $\mathcal{C}^\infty_{\mathrm{c}}$ to $\mathcal{C}^\infty_{\mathrm{c}}$ (for proper F) but is not always defined on distributions. Dually:

Theorem 3.15.1 (Push-forward)**.** *Let $F : \Omega_1 \to \Omega_2$ be a smooth map from an open subset of $\mathbb{R}^n$ to an open subset of $\mathbb{R}^m$. For $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1)$,*

$$F_*u(\phi) = u(F^*\phi), \qquad \phi \in \mathcal{C}^\infty(\Omega_2),$$

defines $F_*u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_2)$, with $\operatorname{supp} F_*u \subset F(\operatorname{supp} u)$, and

$$\mathrm{WF}(F_*u) \subset \{(F(x), \eta);\ x \in \operatorname{supp} u,\ \eta \neq 0,\ DF(x)^t\eta = 0\} \cup \{(F(x), \eta);\ \eta \neq 0,\ (x, DF(x)^t\eta) \in \mathrm{WF}(u)\}.$$

Thus F_* is always defined, but even if $u \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ in general $F_*u \notin \mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)$; if F is a submersion the first set is empty and F_* maps $\mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ into $\mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)$. The two sets in the bound are really the same, if one regards $(x, 0)$ as belonging to the "wavefront set" of u at every point of $\operatorname{supp} u$.

Proof. F_*u is a distribution since $\phi \mapsto \psi F^*\phi$, $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ equal to 1 near $\operatorname{supp} u$, is continuous; the support statement is clear. Formally,

$$F_*u(\phi) = \int u(x)\phi(F(x))\,dx = \iint u(x)\delta(y - F(x))\phi(y)\,dy\,dx,$$

so we write $F_*u = (\pi_2)_*\tilde{H}^*(u \boxtimes \delta)$, where $\delta \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^m)$, $\tilde{H} : \Omega_1 \times \mathbb{R}^m \to \Omega_1 \times \mathbb{R}^m$, $\tilde{H}(x, y) = (x, y - F(x))$, is a diffeomorphism with $\det D\tilde{H} = 1$, and $\pi_2(x, y) = y$. To check this identity, note that $\tilde{H}^*(u \boxtimes \delta) \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1 \times \mathbb{R}^m)$ is supported in the compact graph $\{(x, F(x));\ x \in \operatorname{supp} u\}$ and, by Lemma 3.6.1,

$$(\pi_2)_*\tilde{H}^*(u \boxtimes \delta)(\phi) = (u \boxtimes \delta)(\psi(x)\phi(z + F(x))) = u\big(\psi\,\phi \circ F\big) = F_*u(\phi).$$

Now we estimate the wavefront set using the elementary operations. Since $\mathrm{WF}(\delta) = \{0\} \times (\mathbb{R}^m \setminus 0)$ (Example 3.8.3), Proposition 3.12.1 gives

$$\mathrm{WF}(u \boxtimes \delta) \subset \{(x, 0; \xi, \eta);\ (\xi, \eta) \neq 0,\ (x, \xi) \in \mathrm{WF}(u) \text{ or } (\xi = 0,\ x \in \operatorname{supp} u)\}.$$

Since $D\tilde{H}(x, y)^t(\xi, \eta) = (\xi - DF(x)^t\eta, \eta)$, Theorem 3.9.1 gives

$$\mathrm{WF}\big(\tilde{H}^*(u \boxtimes \delta)\big) \subset \{(x, F(x); \xi - DF(x)^t\eta, \eta);\ (x, \xi) \in \mathrm{WF}(u) \text{ or } (\xi = 0,\ x \in \operatorname{supp} u)\}.$$

Finally, by Proposition 3.12.4 (with the roles of the factors exchanged), $(y, \eta) \in \mathrm{WF}(F_*u)$ only if $\eta \neq 0$ and $(x, y; 0, \eta)$ lies in the last set for some x, i.e. $y = F(x)$ and $\xi = DF(x)^t\eta$ with either $(x, DF(x)^t\eta) \in \mathrm{WF}(u)$, or $DF(x)^t\eta = 0$ and $x \in \operatorname{supp} u$. □

The operator F_* has Schwartz kernel $K(y,x)=\delta(y-F(x))$: $F_*u(y)=\int K(y,x)u(x)\,dx$. Locally K is the pull-back of $\delta\boxtimes\chi$, $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$, under the diffeomorphism $(y,x)\mapsto(y-F(x),x)$, so by Proposition 3.12.1 and Theorem 3.9.1,

$$\mathrm{WF}(K)\subset\{(F(x),x;\eta,-DF(x)^t\eta);\ \eta\neq0\}.$$

In the notation of the next section (with the roles of the two factors interchanged), $\mathrm{WF}'(K)$ is contained in the relation $\{((F(x),\eta),(x,DF(x)^t\eta))\}$, and the bound of Theorem 3.15.1 reads $\mathrm{WF}(F_*u)\subset\mathrm{WF}'(K)\circ\mathrm{WF}(u)\cup\{(y,\eta);\ (y,x;\eta,0)\in\mathrm{WF}(K),\ x\in\operatorname{supp}u\}$.

3.16 The wavefront relation

One serious application of these results is to general operators with distributional kernels. Let $\Omega_1\subset\mathbb{R}^n$, $\Omega_2\subset\mathbb{R}^m$ be open and $K\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1\times\Omega_2)$. Then K defines an operator

$$A:\mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)\to\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1),\qquad Au(\phi)=K(\phi\boxtimes u),$$

formally $Au(x)=\int K(x,y)u(y)\,dy$. We set

$$\begin{aligned}\mathrm{WF}'(K)&=\{((x,\xi),(y,\eta));\ (x,y;\xi,-\eta)\in\mathrm{WF}(K)\}\subset(\Omega_1\times\mathbb{R}^n)\times(\Omega_2\times\mathbb{R}^m),\\ \mathrm{WF}_{\Omega_1}(K)&=\{(x,\xi)\in T^*\Omega_1\setminus0;\ (x,y;\xi,0)\in\mathrm{WF}(K)\text{ for some }y\},\\ \mathrm{WF}'_{\Omega_2}(K)&=\{(y,\eta)\in T^*\Omega_2\setminus0;\ (x,y;0,-\eta)\in\mathrm{WF}(K)\text{ for some }x\}.\end{aligned}$$

We regard $\mathrm{WF}'(K)$ as a relation: for $W\subset T^*\Omega_2\setminus0$,

$$\mathrm{WF}'(K)\circ W=\{(x,\xi)\in T^*\Omega_1\setminus0;\ ((x,\xi),(y,\eta))\in\mathrm{WF}'(K)\text{ for some }(y,\eta)\in W\}.$$

Theorem 3.16.1 (Wavefront relation)**.** *Let $K\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1\times\Omega_2)$ and let $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ equal 1 near the projection of* $\operatorname{supp}K$*. For every $u\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_2)$ with*

$$\mathrm{WF}(u)\cap\mathrm{WF}'_{\Omega_2}(K)=\emptyset\tag{3.16.1}$$

the product $K\cdot(\chi\boxtimes u)$ is defined, and

$$Au=(\pi_1)_*(K\cdot(\chi\boxtimes u))\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1)$$

extends the operator A above. It satisfies

$$\mathrm{WF}(Au)\subset\mathrm{WF}'(K)\circ\mathrm{WF}(u)\ \cup\ \{(x,\xi);\ (x,y;\xi,0)\in\mathrm{WF}(K)\text{ for some }y\in\operatorname{supp}u\}.$$

Proof. By Proposition 3.12.1, $\mathrm{WF}(\chi\boxtimes u)\subset\{(x,y;0,\eta);\ (y,\eta)\in\mathrm{WF}(u)\}$. The product condition of Theorem 3.13.1 fails only at points $(x,y;0,-\eta)\in\mathrm{WF}(K)$ with $(y,\eta)\in\mathrm{WF}(u)$, which are excluded by (3.16.1). If u is smooth the product is $K\cdot(\chi\otimes u)$ and $(\pi_1)_*(K\cdot(\chi\otimes u))(\phi)=K(\chi\phi\boxtimes u)=K(\phi\boxtimes u)$, so Au agrees with the previous definition. By Theorem 3.13.1(iv), a point of $\mathrm{WF}(K\cdot(\chi\boxtimes u))$ has the form $(x,y;\zeta_1+\zeta_2)$ where either $\zeta_1=(\xi_1,\eta_1)$ with $(x,y;\zeta_1)\in\mathrm{WF}(K)$, or $\zeta_1=0$ and $(x,y)\in\operatorname{supp}K$; and either $\zeta_2=(0,\eta_2)$ with $(y,\eta_2)\in\mathrm{WF}(u)$, or $\zeta_2=0$ and $y\in\operatorname{supp}u$. By Proposition 3.12.4, $(x,\xi)\in\mathrm{WF}(Au)$ requires such a point with $\zeta_1+\zeta_2=(\xi,0)$, $\xi\neq0$. If $\zeta_1=0$ the first component of $\zeta_1+\zeta_2$ vanishes, which is excluded. If $\zeta_2=0$ then $(x,y;\xi,0)\in\mathrm{WF}(K)$ with $y\in\operatorname{supp}u$. Otherwise $\xi_1=\xi$ and $\eta_1=-\eta_2$, so $(x,y;\xi,-\eta_2)\in\mathrm{WF}(K)$ with $(y,\eta_2)\in\mathrm{WF}(u)$, that is $(x,\xi)\in\mathrm{WF}'(K)\circ\mathrm{WF}(u)$. □

Remark 3.16.2. If $K = K_B$ for some $B \in \Psi^m(\mathbb{R}^n)$ with kernel of compact support in $\Omega \times \Omega$, then $Au = Bu$ for every $u \in \mathcal{C}_c^{-\infty}(\Omega)$. Indeed, by Theorem 3.13.1(i) with a product mollifier $\rho = \rho^{(1)} \boxtimes \rho^{(2)}$ on $\mathbb{R}^{2n}$, and $\phi \in \mathcal{C}_c^\infty$, $\psi \in \mathcal{C}_c^\infty$ equal to 1 near $\pi_2(\operatorname{supp} K)$,

$$Au(\phi) = \lim_{\epsilon\to 0} K\big((\phi\,(\chi * \check\rho_\epsilon^{(1)})) \boxtimes (\psi\,(u * \check\rho_\epsilon^{(2)}))\big) = \lim_{\epsilon\to 0}(Bu_\epsilon)(\phi) = \lim_{\epsilon\to 0} u_\epsilon(B^t\phi) = u(B^t\phi),$$

where $u_\epsilon = u * \check\rho_\epsilon^{(2)} \to u$ in $\mathcal{S}'$, since $\chi * \check\rho_\epsilon^{(1)} = 1$ near the projection of $\operatorname{supp} K$ for small ϵ.

Corollary 3.16.3. *Let $K \in \mathcal{C}_c^{-\infty}(\Omega_1 \times \Omega_2)$.*

(i) *If $\mathrm{WF}(K) \cap \{(x, y; \xi, 0)\} = \emptyset$ then A maps $\mathcal{C}_c^\infty(\Omega_2)$ into $\mathcal{C}_c^\infty(\Omega_1)$.*

(ii) *If $\mathrm{WF}(K) \cap \{(x, y; 0, \eta)\} = \emptyset$ then A extends to a linear map $A : \mathcal{C}_c^{-\infty}(\Omega_2) \to \mathcal{C}_c^{-\infty}(\Omega_1)$ with $\mathrm{WF}(Au) \subset \mathrm{WF}'(K) \circ \mathrm{WF}(u) \cup \mathrm{WF}_{\Omega_1}(K)$.*

Proof. Immediate from Theorem 3.16.1: in (i), $\mathrm{WF}(u) = \emptyset$ for smooth u, and in (ii), condition (3.16.1) is void. □

3.16.1 Applications

For a pseudodifferential operator $B \in \Psi^m(\mathbb{R}^n)$ with kernel of compact support, Proposition 3.10.1 gives

$$\mathrm{WF}'(K_B) = \{((x, \xi), (x, \xi));\ (x, \xi) \in \mathrm{WF}'(B)\},$$

a subset of the identity relation. In particular $\mathrm{WF}(K_B)$ contains no points of the form $(x, y; \xi, 0)$ or $(x, y; 0, \eta)$, so both parts of Corollary 3.16.3 apply, and by Remark 3.16.2 Theorem 3.16.1 reduces to $\mathrm{WF}(Bu) \subset \mathrm{WF}'(B) \cap \mathrm{WF}(u)$; this recovers microlocality (Theorem 3.7.7) for such operators. The next chapters will show that many other natural operators, such as the solution operators of hyperbolic equations, have kernels whose wavefront sets are well-behaved relations, and Theorem 3.16.1 then describes how they propagate singularities.

Remark 3.16.4. The restriction to compactly supported kernels is only for simplicity. If K is a distribution on $\Omega_1 \times \Omega_2$ with *proper* support, i.e. both projections $\operatorname{supp} K \to \Omega_i$ are proper, then A maps $\mathcal{C}_c^\infty(\Omega_2)$ to distributions of compact support in Ω_1, and the results of this section hold for $u \in \mathcal{C}_c^{-\infty}(\Omega_2)$, since only K on the compact set $\operatorname{supp} K \cap (\Omega_1 \times \operatorname{supp} u)$ is involved.

3.17 Exercises

Exercise 3.1. Is it true that every $E \in \Psi^{-\infty}(\mathbb{R}^n)$ defines a map from $\mathcal{S}'(\mathbb{R}^n)$ to $\mathcal{S}(\mathbb{R}^n)$? (Consider $E = \mathrm{Op}(e)$ with $e \in \mathcal{S}(\mathbb{R}^n_\xi)$ and $e(0) \neq 0$, acting on the constant function 1.) Compare with Lemma 3.2.3.

Exercise 3.2. Show that $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ by proving that if $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ is identically equal to 1 near the origin then

$$u_k(x) = \phi\Big(\frac{x}{k}\Big) \int e^{ix\cdot\xi} \phi\Big(\frac{\xi}{k}\Big) \hat u(\xi) đ\xi \in \mathcal{S}(\mathbb{R}^n) \quad \text{for } u \in L^2(\mathbb{R}^n),$$

and $u_k \to u$ in $L^2(\mathbb{R}^n)$. Can you see any relation to pseudodifferential operators here?

Exercise 3.3. For $k \in \mathbb{N}$ show that $u \in \mathcal{S}'(\mathbb{R}^n)$ can be written $u = \sum_{|\alpha|\le k} D^\alpha u_\alpha$ with $u_\alpha \in L^2(\mathbb{R}^n)$ if and only if $\langle D\rangle^{-k} u \in L^2(\mathbb{R}^n)$, that is, if and only if $u \in H^{-k}(\mathbb{R}^n)$.

Exercise 3.4. Let $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ be real with $a \geq 0$, and $A = \sum_{j=1}^n D_{x_j}^2 + a(x)$. Show that $Au = 0$ has no non-trivial solution $u \in L^2(\mathbb{R}^n)$. (Show first that $u \in H^2(\mathbb{R}^n)$, then compute (Au, u).)

Exercise 3.5. Show that for any open set $\Omega \subset \mathbb{R}^n$, $\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ is dense in $\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$ in the weak topology, i.e. every $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$ is the limit of a sequence in $\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ in $\mathcal{S}'(\mathbb{R}^n)$.

Exercise 3.6. Let $h \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ be supported in $\{1 < |\eta| < 2\}$ and not identically zero, and let $a(x,\xi) = h(\xi/\langle x\rangle)$. Show that $a \in S^0(\mathbb{R}^n;\mathbb{R}^n)$, that $\operatorname{conesupp}(a) = \emptyset$, so $\mathrm{WF}'(\mathrm{Op}(a)) = \emptyset$, but that $a \notin S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$, so $\mathrm{Op}(a) \notin \Psi^{-\infty}(\mathbb{R}^n)$.

Exercise 3.7. Show that for $u \in \mathcal{S}'(\mathbb{R}^n)$

$$\mathrm{WF}(u) = \bigcap\{\mathrm{Char}_0(A);\ A \in \Psi^0(\mathbb{R}^n),\ Au \in \mathcal{C}^\infty(\mathbb{R}^n)\},$$

so that Definition 3.7.1 extends verbatim.

Exercise 3.8. Compute the wavefront sets of the following distributions: $\delta \in \mathcal{S}'(\mathbb{R}^n)$; $|x| \in \mathcal{S}'(\mathbb{R}^n)$; the characteristic function $\chi_{\mathbb{B}}$ of the unit ball, $\chi_{\mathbb{B}}(x) = 1$ for $|x| \leq 1$ and 0 for $|x| > 1$. (For the last, use Theorem 3.9.1 to reduce to the Heaviside function of one variable, regarded as a function on $\mathbb{R}^n$.)

Exercise 3.9. Let $\Gamma \subset T^*\mathbb{R}^n \setminus 0$ be a closed cone and define

$$\mathcal{C}^{-\infty}_{c,\Gamma}(\mathbb{R}^n) = \{u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n);\ Au \in \mathcal{C}^\infty(\mathbb{R}^n)\ \forall\, A \in \Psi^0(\mathbb{R}^n) \text{ with } \mathrm{WF}'(A) \cap \Gamma = \emptyset\}.$$

Show that this is the space of $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ with $\mathrm{WF}(u) \subset \Gamma$. Describe a complete topology on this space with respect to which $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ is a dense subspace.

Exercise 3.10. Give an alternative proof of Proposition 3.5.1, not using global parametrices, along the following lines. Let $z \notin \mathrm{Char}_m(A)$, $a = \sigma_L(A)$, and choose $\epsilon > 0$ so small that $\gamma_\epsilon = \gamma_{z,\epsilon}$ is supported where $|a| \geq c|\xi|^m$. Then $b_0 = \gamma_\epsilon/a \in S^{-m}(\mathbb{R}^n;\mathbb{R}^n)$. With $B_0 = \mathrm{Op}(b_0)$, show that $G = \mathrm{Id} - AB_0 \in \Psi^0$ has $\sigma_0(G) = [1 - \gamma_\epsilon]$, and that with $C = \mathrm{Op}(\gamma_{z,\epsilon/4})$, $CG \in \Psi^{-1}$ and $z \notin \mathrm{WF}'((\mathrm{Id} - C)G)$. Writing $\mathrm{Id} - AB_0 = CG + R_1$, let $B_1 \in \Psi^{-1}$ be an asymptotic sum of $\sum_{j\geq 1}(CG)^j$ and $B = B_0(\mathrm{Id} + B_1)$. Show that $AB = \mathrm{Id} + R_2$ with $z \notin \mathrm{WF}'(R_2)$. Construct a left parametrix in the same way, or by applying the construction to A^* and taking adjoints.

Exercise 3.11. Show that microlocal parametrices are essentially unique: if B and B' satisfy $z \notin \mathrm{WF}'(AB' - \mathrm{Id})$ and $z \notin \mathrm{WF}'(BA - \mathrm{Id})$ then $z \notin \mathrm{WF}'(B - B')$.

Exercise 3.12. Given $(\bar{x}, \bar{\xi}) \in T^*\mathbb{R}^n \setminus 0$, construct $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ with

$$\mathrm{WF}(u) = \{(\bar{x}, t\bar{\xi});\ t > 0\}.$$

(Hint: take $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\int \phi \neq 0$ and $u(x) = \sum_{k\geq 1} k^{-2}\phi(k(x - \bar{x}))e^{ik^3 x\cdot\bar{\xi}}$, and use the Fourier transform. See [7, Section 8.1] for the construction of a distribution with any prescribed closed conic wavefront set.)

Exercise 3.13. Suppose $A \in \Psi^m(\mathbb{R}^n)$ has Schwartz kernel of compact support. If $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ use the elementary operations (and Proposition 3.10.1) to investigate under what conditions $\kappa(x,y) = K_A(x,y)u(y)$, and then $\gamma = (\pi_1)_*\kappa$, make sense. What can you say about $\mathrm{WF}(\gamma)$?

Exercise 3.14. Consider the projection $\pi_1 : \mathbb{R}^p \times \mathbb{R}^k \to \mathbb{R}^p$. Show that $(\pi_1)_*$ can be extended to some distributions which do not have compact support, for example to $\{u \in \mathcal{S}'(\mathbb{R}^{p+k});\ \operatorname{supp} u \cap (K \times \mathbb{R}^k) \text{ is compact for each compact } K \subset \mathbb{R}^p\}$, with values in distributions on $\mathbb{R}^p$ (that is, continuous linear functionals on $\mathcal{C}_c^\infty(\mathbb{R}^p)$), and that Proposition 3.12.4 remains valid.

Exercise 3.15. Let $u_1, u_2 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ satisfy (3.11.1). Show that $P(u_1, u_2) = (u_1 u_2)(\psi)$ for any $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} u_1 \cup \operatorname{supp} u_2$. (Use Theorem 3.13.1(i): replace u_2 by $u_2 * \check{\rho}_\epsilon$ in P, and use Lemma 3.11.1.) Deduce that $u_1 u_2 = u_2 u_1$.

Exercise 3.16. Check that Theorem 3.14.1 contains Theorem 3.12.3 (with $F = \iota$) and Theorem 3.13.1 (with $F = \Delta$, $\Delta(x) = (x, x)$, applied to $u_1 \boxtimes u_2$): show that the definitions agree, using the limit formulas, and that the wavefront set bounds coincide.

Notes

The definition of the wavefront set via pseudodifferential operators follows Melrose's notes, while the Fourier transform characterization of Section 3.8 is Hörmander's original definition; see [7, Chapter 8] for the classical treatment of the calculus of wavefront sets (pull-back, products, push-forward and the wavefront relation of kernels), including continuity statements in the spaces $\mathcal{D}'_\Gamma$ which we have not discussed, and [8, Chapter 18] for the essential support of pseudodifferential operators.

Compared with the original notes, the main changes are as follows. Essential supports are used together with compactly supported cut-offs throughout, because with uniformly bounded symbols $\mathrm{WF}'(A) = \emptyset$ does not imply $A \in \Psi^{-\infty}(\mathbb{R}^n)$. The operations on wavefront sets are all proved via the Fourier transform characterization and a single restriction theorem; products are defined as restrictions of $u_1(x)u_2(x+y)$ to $y = 0$ (rather than through the pairing), and the pull-back by a general map is defined through restriction to a graph, with a limit formula showing consistency with all the special cases. The wavefront relation theorem includes the term $\mathrm{WF}_{\Omega_1}(K)$, which is needed in general.

Chapter 4

Conormal distributions

The Dirac delta function, the Schwartz kernels of pseudodifferential operators and the travelling-wave solutions of hyperbolic equations are all examples of distributions that are smooth outside a submanifold Y and have a very particular, "symbolic", singularity at Y. In this chapter we introduce the corresponding spaces $I^m(X,Y)$ of *conormal distributions*. We start with the simplest case, Y a point in $\mathbb{R}^n$, where conormal distributions are simply inverse Fourier transforms of symbols; pass to a linear subspace $Y = \{0\} \times \mathbb{R}^d \subset \mathbb{R}^n$ (equivalently the zero section of a trivial vector bundle), where the Fourier transform is taken only in the normal variables; prove the invariance of these spaces under diffeomorphisms preserving Y; and then define $I^m(X,Y;\Omega^{\frac{1}{2}})$ for a closed embedded submanifold of a manifold, either through coordinates or through a normal fibration. The principal symbol is an invariantly defined half-density on the conormal bundle N^*Y, and gives a short exact sequence. Finally we show that the conormal distributions are characterized, up to an arbitrarily small loss in the order, by *iterated regularity*: stability of a fixed Sobolev regularity under repeated application of vector fields tangent to Y.

We use the symbol spaces of Chapter 2, the radial compactification of Chapter 1 and, for wavefront sets, Chapter 3. Pseudodifferential operators on manifolds are discussed only in Chapter 5; their action on conormal distributions is derived here in local coordinates and globalized in Chapter 8.

4.1 Symbols

4.1.1 Symbols with bounds

Recall from Chapter 2 that for $m \in \mathbb{R}$, $S^m(\mathbb{R}^p;\mathbb{R}^k)$ is the space of $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^k)$ such that

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| \leq C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|} \quad \forall\ \alpha,\beta, \tag{4.1.1}$$

with the Fréchet topology given by the best constants; $S^m(\mathbb{R}^k)$ denotes the case without parameters, $S^{-\infty} = \bigcap_m S^m$, and every sequence $a_j \in S^{m-j}$ has an asymptotic sum $a \sim \sum_j a_j$, unique modulo $S^{-\infty}$. We shall also write $S^m_c(\mathbb{R}^p;\mathbb{R}^k)$ for the subspace of symbols vanishing for z outside a compact subset of $\mathbb{R}^p$. Clearly $\partial_{\xi_j} : S^m \to S^{m-1}$, $\xi_j : S^m \to S^{m+1}$, and $S^m \cdot S^{m'} \subset S^{m+m'}$. Note that $S^{-\infty}(\mathbb{R}^k) = \mathcal{S}(\mathbb{R}^k)$.

The symbol spaces are invariant under linear changes of the variable: if $A(z) \in \mathrm{GL}(k,\mathbb{R})$ depends smoothly on z, with A, A^{-1} and all their derivatives bounded, then $a(z,A(z)\xi) \in S^m(\mathbb{R}^p;\mathbb{R}^k)$ whenever $a \in S^m(\mathbb{R}^p;\mathbb{R}^k)$, since $\langle A(z)\xi\rangle$ and $\langle\xi\rangle$ are uniformly comparable and each z-derivative falling on $A(z)\xi$ produces a factor of order one together with a ξ-derivative

of a. In particular the symbol spaces $S^m(W)$ are well defined for any finite-dimensional real vector space W.

The following density statement allows many identities to be proved first for Schwartz functions.

Lemma 4.1.1. *If $a \in S^m(\mathbb{R}^p;\mathbb{R}^k)$, $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$ equals 1 near 0 and $a_\epsilon(z,\xi) = \phi(\epsilon\xi)a(z,\xi)$, then a_ϵ is bounded in S^m for $0 < \epsilon \le 1$ and $a_\epsilon \to a$ in $S^{m'}$ for every $m' > m$ as $\epsilon \to 0$.*

Proof. The derivative $D^\gamma_\xi[\phi(\epsilon\xi)] = \epsilon^{|\gamma|}(D^\gamma\phi)(\epsilon\xi)$ is supported where $|\xi| \le C/\epsilon$, so it is bounded by $C_\gamma\langle\xi\rangle^{-|\gamma|}$ uniformly in ϵ; hence a_ϵ is bounded in S^m by Leibniz' formula. Moreover $a - a_\epsilon = (1-\phi(\epsilon\xi))a$ and its derivatives are supported in $|\xi| \ge c/\epsilon$, where $\langle\xi\rangle^{m-|\beta|} \le C\epsilon^{m'-m}\langle\xi\rangle^{m'-|\beta|}$; so $a - a_\epsilon \to 0$ in $S^{m'}$. □

4.1.2 Classical symbols

Let W be a real vector space of dimension k and $\overline{W}$ its radial compactification (Chapter 1), a compact manifold with boundary whose interior is W; the action of $\mathrm{GL}(W)$ and of translations extends smoothly to $\overline{W}$. We call the boundary $\mathbb{S}W = \partial\overline{W}$ the *sphere at infinity*; it is canonically $(W\setminus 0)/\mathbb{R}^+$. Let $\rho \in \mathcal{C}^\infty(\overline{W})$ be a boundary defining function, for instance $\rho = \langle\xi\rangle^{-1}$ in linear coordinates. Any two are related by $\rho' = f\rho$ with $0 < f \in \mathcal{C}^\infty(\overline{W})$, so

$$S^m_{\mathrm{ph}}(W) = \rho^{-m}\mathcal{C}^\infty(\overline{W}) = \{a \in \mathcal{C}^\infty(W);\ \rho^m a \in \mathcal{C}^\infty(\overline{W})\}$$

is independent of the choice; this is the space of *classical* (or polyhomogeneous) symbols of order m. (One may regard it as the space of sections of a trivial line bundle N_{-m} over $\overline{W}$ of which ρ^{-m} is a global trivializing section.)

Lemma 4.1.2. *For all $m, m' \in \mathbb{R}$:*

(i) $S^m_{\mathrm{ph}}(W)\cdot S^{m'}_{\mathrm{ph}}(W) = S^{m+m'}_{\mathrm{ph}}(W)$ *and* $S^m_{\mathrm{ph}}(W)$ *is invariant under* $\mathrm{GL}(W)$ *and translations;*

(ii) $S^m_{\mathrm{ph}}(W) \subset S^m(W)$*; in linear coordinates* $\xi^\alpha\partial^\beta_\xi : S^m_{\mathrm{ph}} \to S^{m+|\alpha|-|\beta|}_{\mathrm{ph}}$*;*

(iii) $\bigcap_N S^{m-N}_{\mathrm{ph}}(W) = \mathcal{S}(W)$*;*

(iv) $a \in S^m_{\mathrm{ph}}(W)$ *if and only if there are functions* $a_{m-j} \in \mathcal{C}^\infty(W\setminus 0)$*, homogeneous of degree* $m-j$*, such that for one (hence any)* $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ *equal to one near* 0*,*

$$a - \sum_{j<N}(1-\chi)a_{m-j} \in S^{m-N}(W) \quad \forall\ N. \tag{4.1.2}$$

Proof. (i) is immediate from the definition and the invariance of $\overline{W}$. For (ii) and (iv) use linear coordinates and, near the boundary, the coordinates $r = |\xi|^{-1} \in [0,1)$, $\omega = \xi/|\xi| \in \mathbb{S}^{k-1}$ on $\overline{W}$; then $\rho = r\,g$ with $g > 0$ smooth up to $r = 0$, so $a \in S^m_{\mathrm{ph}}$ means $a = r^{-m}f(r,\omega)$ with f smooth up to $r = 0$. In these coordinates $\xi_j\partial_{\xi_l}$ is a combination, with coefficients smooth up to $r = 0$, of $r\partial_r$ and vector fields on $\mathbb{S}^{k-1}$, while ∂_{ξ_l} is r times such a combination, as one sees from homogeneity ($\xi_j\partial_{\xi_l}$ is homogeneous of degree 0 and ∂_{ξ_l} of degree -1). Since $r\partial_r(r^{-m}f) = r^{-m}(r\partial_r - m)f$, repeated application of such vector fields to $r^{-m}f$ yields r^{-m} times bounded functions, so the estimates (4.1.1) hold in $|\xi| \ge 1$ (where $r \sim \langle\xi\rangle^{-1}$), and trivially in $|\xi| \le 1$; the mapping property in (ii) follows as well. Taylor's formula in r at $r = 0$, $f = \sum_{j<N} r^j f_j(\omega) + r^N f'_N(r,\omega)$, gives (4.1.2) with $a_{m-j}(\xi) = |\xi|^{m-j}f_j(\xi/|\xi|)$, the remainder $r^{N-m}f'_N$ being in S^{m-N} by the argument just given. Conversely, if (4.1.2)

holds, then near $r = 0$ the function $f = r^m a$ satisfies $f - \sum_{j<N} r^j f_j = O(r^N)$ together with all derivatives in $r\partial_r$ and ∂_ω, and $\partial_r = r^{-1}(r\partial_r)$; so f is $\mathcal{C}^{N-1}$ up to $r = 0$ for every N, i.e. smooth. Finally (iii): the intersection is contained in $S^{-\infty}(W) = \mathcal{S}(W)$ by (ii), and conversely a Schwartz function is r^∞ times a smooth function of (r, ω), by the same change of variables. □

The leading term in (4.1.2) is the *principal part* $\sigma_m(a) = a_m$; it is a smooth function on $W \setminus 0$ homogeneous of degree m, equivalently (multiplying by ρ^m and restricting) a smooth section of N_{-m} over $\mathbb{S}W$. Denote by $\mathcal{C}^\infty_{(m)}(W \setminus 0)$ the space of smooth functions homogeneous of degree m.

Proposition 4.1.3. *For every m the sequence*

$$0 \longrightarrow S^{m-1}_{\mathrm{ph}}(W) \longrightarrow S^m_{\mathrm{ph}}(W) \xrightarrow{\sigma_m} \mathcal{C}^\infty_{(m)}(W \setminus 0) \longrightarrow 0$$

is exact. Equivalently $\rho\mathcal{C}^\infty(\overline{W}) \to \mathcal{C}^\infty(\overline{W}) \to \mathcal{C}^\infty(\mathbb{S}W)$ is exact: restriction to the boundary is surjective and a smooth function vanishing at the boundary is ρ times a smooth function.

Proof. Multiplication by ρ^m reduces the statement to the case $m = 0$, where it is Taylor's theorem at the boundary (in the coordinates (r, ω) above) and the existence of extensions: $f(\omega) \mapsto (1 - \chi)f(\xi/|\xi|)$. □

Proposition 4.1.4 (Asymptotic completeness)**.** *If $a_j \in S^{m-j}_{\mathrm{ph}}(W)$, $j \in \mathbb{N}_0$, there exists $a \in S^m_{\mathrm{ph}}(W)$ such that $a - \sum_{j<N} a_j \in S^{m-N}_{\mathrm{ph}}(W)$ for all N; a is unique modulo $\mathcal{S}(W)$. The same holds, with the same proof, for symbols depending smoothly on parameters in a compact set.*

Proof. Uniqueness follows from Lemma 4.1.2(iii). Multiplying by ρ^m, it suffices to show: if X is a compact manifold with boundary, ρ a boundary defining function and $b_j \in \rho^j\mathcal{C}^\infty(X)$, there is $b \in \mathcal{C}^\infty(X)$ with $b - \sum_{j<N} b_j \in \rho^N\mathcal{C}^\infty(X)$ for all N. Using a collar $[0, 1)_r \times \partial X$ and a partition of unity this reduces to the following form of Borel's lemma: given $u_j \in \mathcal{C}^\infty(\partial X)$ there is $u \in \mathcal{C}^\infty_{\mathrm{c}}([0, 1) \times \partial X)$ with $\partial^j_r u(0, \cdot) = u_j$ for all j. (Indeed, write $b_j = \sum_l r^{j+l} c_{jl}(\omega)/(j+l)! + O(r^\infty)$ near the boundary and take the u with $u_p = \sum_{j\le p} c_{j,p-j}$; then $b - \sum_{j<N} b_j$ vanishes to order N at $r = 0$.) Choose $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ with $\chi = 1$ on $|r| < \frac{1}{2}$, $\chi = 0$ on $|r| > 1$, and set

$$u(r, \omega) = \sum_j \frac{r^j}{j!} u_j(\omega)\chi(r/\epsilon_j), \qquad 0 < \epsilon_j \le 1.$$

Each term vanishes for $r > \epsilon_j$, and for $l \le j$ and any differential operator Q of order q on ∂X, $|\partial^l_r Q(r^j u_j \chi(r/\epsilon_j)/j!)| \le C_{j,l,Q}\, \epsilon_j^{j-l}$, with $C_{j,l,Q}$ independent of ϵ_j, since $\partial^p_r[\chi(r/\epsilon_j)] = \epsilon_j^{-p}\chi^{(p)}(r/\epsilon_j)$ and $r \le \epsilon_j$ on the support. Fix an exhausting sequence $Q_1, Q_2, \dots$ of differential operators on ∂X. Choosing $\epsilon_j \le 2^{-j}/(1 + \max_{l+q\le j-1} C_{j,l,Q_q})$ imposes only finitely many conditions on each ϵ_j, and makes every differentiated series converge uniformly. Hence u is smooth and, since only the term j contributes to $\partial^j_r u(0, \cdot)$, $\partial^j_r u(0, \cdot) = u_j$. Parameters are handled by including the parameter derivatives among the Q_q. □

The symbol estimates (4.1.1) do not imply that a is classical: $S^m_{\mathrm{ph}}(W)$ is a closed subspace of $S^m(W)$ in the topology of $\rho^{-m}\mathcal{C}^\infty(\overline{W})$, and $\mathcal{S}(W)$ is not dense in it in that topology, though by Lemma 4.1.1 it is dense in the weaker topology of $S^{m'}$, $m' > m$. The relation between the two classes is explored in Exercises 4.1 and 4.2. For symbols depending on parameters $z \in \mathbb{R}^p$ we write $S^m_{\mathrm{ph}}(\mathbb{R}^p; \mathbb{R}^k)$ for those $a \in S^m(\mathbb{R}^p; \mathbb{R}^k)$ having an expansion

(4.1.2) with $a_{m-j} \in \mathcal{C}^\infty(\mathbb{R}^p \times (\mathbb{R}^k \setminus 0))$ homogeneous in ξ, uniformly bounded with all derivatives on $\mathbb{R}^p \times \mathbb{S}^{k-1}$, the remainders being in $S^{m-N}(\mathbb{R}^p; \mathbb{R}^k)$. All statements in this chapter hold for the classical spaces as well as for the spaces of symbols with bounds; we indicate the classical versions by a subscript "ph".

4.2 Conormal distributions at a point

Definition 4.2.1. For $m \in \mathbb{R}$ set

$$I^m_{\mathcal{S}}(\mathbb{R}^n, \{0\}) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ \hat{u} \in S^{m-\frac{n}{4}}(\mathbb{R}^n)\}, \qquad I^m_{\mathcal{S},\mathrm{ph}}(\mathbb{R}^n, \{0\}) = \mathcal{F}^{-1}(S^{m-\frac{n}{4}}_{\mathrm{ph}}(\mathbb{R}^n)).$$

The shift $n/4$ in the order is a normalization, which is explained by the behaviour of the order under the operations considered later (it makes the order of the Schwartz kernel of a pseudodifferential operator of order m, as a conormal distribution at the diagonal, equal to m). For example $\hat{\delta}_0 = 1$, so

$$\delta_0 \in I^{\frac{n}{4}}_{\mathcal{S},\mathrm{ph}}(\mathbb{R}^n, \{0\}).$$

Since $S^{m-1} \subset S^m$, $I^{m-1}_{\mathcal{S}} \subset I^m_{\mathcal{S}}$, and $\bigcap_m I^m_{\mathcal{S}}(\mathbb{R}^n, \{0\}) = \mathcal{F}^{-1}\mathcal{S}(\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^n)$.

Lemma 4.2.2. *Let $u \in I^m_{\mathcal{S}}(\mathbb{R}^n, \{0\})$ (resp. the classical space). Then*

(i) $z^\alpha D_z^\beta u \in I^{m+|\beta|-|\alpha|}_{\mathcal{S}}(\mathbb{R}^n, \{0\})$ *(resp. classical);*

(ii) if $m < -\frac{3n}{4} - j$, $j \in \mathbb{N}_0$, *then* $u \in \mathcal{C}^j(\mathbb{R}^n)$ *and* $D^\beta u \in \mathcal{C}^0_0(\mathbb{R}^n)$ *for* $|\beta| \leq j$;

(iii) if $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ *equals one near* 0, *then* $(1-\chi)u \in \mathcal{S}(\mathbb{R}^n)$; *in particular* $\operatorname{sing\,supp} u \subset \{0\}$;

(iv) the integral $\int u\,dz$, *defined as* $\int \chi u + \int (1-\chi)u$ *(independent of* χ*), equals* $\hat{u}(0)$;

(v) if $v \in I^{m'}_{\mathcal{S}}(\mathbb{R}^n, \{0\})$ *then* $u * v \in I^{m+m'-\frac{n}{4}}_{\mathcal{S}}(\mathbb{R}^n, \{0\})$.

Proof. (i) follows from $\mathcal{F}(z^\alpha D^\beta u) = (-D_\zeta)^\alpha(\zeta^\beta \hat{u})$ and Lemma 4.1.2(ii). (ii): if $m < -3n/4 - j$ then $\zeta^\beta \hat{u} \in S^{m-\frac{n}{4}+|\beta|} \subset L^1$ for $|\beta| \leq j$, so $D^\beta u$ is the inverse Fourier transform of an integrable function, hence continuous and vanishing at infinity by the Riemann–Lebesgue lemma. (iii): by (i) and (ii), $|z|^{2N} D^\beta u = \sum_{|\gamma|=N} c_\gamma z^{2\gamma} D^\beta u$ is bounded if $2N > m + |\beta| + 3n/4$; since $1-\chi$ vanishes near 0, $|D^\beta((1-\chi)u)| \leq C_{N,\beta}\langle z\rangle^{-2N}$ for all N and β. (iv): the definition is independent of χ since the two definitions agree on $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$; and $\int \chi u = \widehat{\chi u}(0)$, $\int (1-\chi)u = \widehat{(1-\chi)u}(0)$ add up to $\hat{u}(0)$. (v): $\widehat{u * v} = \hat{u}\hat{v} \in S^{m-\frac{n}{4}+m'-\frac{n}{4}} = S^{(m+m'-\frac{n}{4})-\frac{n}{4}}$. □

4.2.1 Densities and invariance

To see that these spaces are naturally associated to the origin of an abstract vector space we need densities. Let W be a real vector space of dimension n. For $t \in \mathbb{R}$, the space $\Omega^t W$ of *t-densities* consists of the maps $\nu : \Lambda^n W \setminus 0 \to \mathbb{R}$ with $\nu(cv) = |c|^t \nu(v)$ for $c \in \mathbb{R} \setminus 0$; it is a line, and $\Omega W = \Omega^1 W$. If $G \in \mathrm{GL}(W)$ then $G^*\nu = |\det G|^t \nu$; for $W = \mathbb{R}^n$ we write $|dz|^t$ for the t-density taking the value 1 on $e_1 \wedge \cdots \wedge e_n$. Densities are what one integrates invariantly: $\int G^*\mu = \int \mu$ for $\mu \in \mathcal{S}(W; \Omega W)$. The pairing $\Omega W \otimes \Omega W' \to \mathbb{R}$, $|dz| \otimes |d\zeta| \mapsto 1$ for dual bases, is independent of the basis, so $\Omega W' = (\Omega W)'$.

For $u \in \mathcal{S}(\mathbb{R}^n)$ and $G \in \mathrm{GL}(n, \mathbb{R})$ the change of variables $y = Gz$ gives

$$\widehat{G^*u}(\zeta) = \int e^{-iz\cdot\zeta} u(Gz)\,dz = |\det G|^{-1} \hat{u}((G^{-1})^t \zeta).$$

Thus if we define the Fourier transform to take values in densities, $\mathcal{F}u = \hat{u}(\zeta)|d\zeta|$, then $\mathcal{F}(G^*u) = ((G^{-1})^t)^*\mathcal{F}u$: the Fourier transform is a linear-invariant isomorphism

$$\mathcal{F} : \mathcal{S}(W) \longrightarrow \mathcal{S}(W'; \Omega W'), \qquad \mathcal{F}^{-1}\mu(w) = (2\pi)^{-n} \int_{W'} e^{i\langle w,\zeta\rangle}\mu(\zeta), \tag{4.2.1}$$

extending to $\mathcal{S}'(W) \to \mathcal{S}'(W'; \Omega W')$. Consequently

$$I^m(W, \{0\}) = \mathcal{F}^{-1}(S^{m-\frac{n}{4}}(W'; \Omega W')),$$

where $S^r(W'; \Omega W') = S^r(W') \cdot |d\zeta|$, is well defined independently of any choice of basis, and similarly for the classical space; the integral of Lemma 4.2.2(iv) becomes the invariant map $I^m(W, \{0\}; \Omega W) \ni u \mapsto (\mathcal{F}u)(0)$ if u is density-valued. The *principal symbol* of u is the class of $\mathcal{F}u$ in $S^{m-\frac{n}{4}}(W'; \Omega W')/S^{m-\frac{n}{4}-1}(W'; \Omega W')$ and, in the classical case, its principal part, a density on $W' \setminus 0$ homogeneous of degree $m - \frac{n}{4} + n = m + \frac{3n}{4}$ (the density factor $|d\zeta|$ being homogeneous of degree n). By Proposition 4.1.3 there are short exact sequences

$$I^{m-1}(W, \{0\}) \hookrightarrow I^m(W, \{0\}) \xrightarrow{\sigma_m} S^{m-\frac{n}{4}}(W'; \Omega W')/S^{m-\frac{n}{4}-1}(W'; \Omega W'),$$

and, in the classical case, onto the homogeneous densities of degree $m + \frac{3n}{4}$ on $W' \setminus 0$.

Invariance under non-linear changes of coordinates fixing 0 is a special case of Theorem 4.4.4 below.

4.2.2 Homogeneity and conormality

For classical conormal distributions the singularity at the origin can be described quite explicitly. We use two facts about homogeneous distributions [7, Theorems 3.2.3, 7.1.16, 7.1.18]: if $f \in \mathcal{C}^\infty(\mathbb{R}^n \setminus 0)$ is homogeneous of degree μ and $\mu \notin \{-n, -n-1, \dots\}$, then f has a unique extension to a distribution $\tilde{f} \in \mathcal{S}'(\mathbb{R}^n)$ homogeneous of degree μ; and the Fourier transform of a homogeneous distribution of degree μ which is smooth outside 0 is homogeneous of degree $-n-\mu$ and smooth outside 0.

Proposition 4.2.3. *Let $u \in I^m_{\mathcal{S},\mathrm{ph}}(\mathbb{R}^n, \{0\})$, so $\hat{u} \in S^\mu_{\mathrm{ph}}(\mathbb{R}^n)$ with $\mu = m - \frac{n}{4}$, and suppose $\mu \notin \mathbb{Z}$. Then there are functions $U_j \in \mathcal{C}^\infty(\mathbb{R}^n \setminus 0)$, homogeneous of degree $-n-\mu+j$ and determined by the homogeneous terms $\hat{u}_{\mu-j}$ in the expansion of $\hat{u}$, such that*

$$u - \sum_{j<N} U_j \in \mathcal{C}^r(\mathbb{R}^n) \quad \text{for } 0 \le r < N - \mu - n.$$

Conversely the U_j determine the terms $\hat{u}_{\mu-j}$, so u is determined modulo $\mathcal{S}(\mathbb{R}^n)$ by its expansion.

Proof. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal one near 0. Since $\mu - j$ is not an integer, $\hat{u}_{\mu-j}$ has a homogeneous extension $\tilde{a}_j$, and $U_j := \mathcal{F}^{-1}\tilde{a}_j$ is homogeneous of degree $-n-\mu+j$ and smooth outside 0. The difference $(1-\chi)\hat{u}_{\mu-j} - \tilde{a}_j = -\chi\tilde{a}_j$ has compact support, so its inverse Fourier transform is smooth. The remainder $\hat{u} - \sum_{j<N}(1-\chi)\hat{u}_{\mu-j} \in S^{\mu-N}$ has inverse Fourier transform in $\mathcal{C}^r$ if $\mu - N + r < -n$. The converse follows by applying the Fourier transform to the homogeneous terms, which are distinguished by their degrees. □

When μ is an integer the expansion of $\hat{u}$ is unchanged, but the expansion of u may contain terms $p(z)\log|z|$ with p a homogeneous polynomial (for $\mu \le -n$), coming from the homogeneous terms of degree $-n-l$, $l \in \mathbb{N}_0$, which have no homogeneous extension; and for $\mu \in \mathbb{N}_0$ the polynomial part of $\hat{u}$ corresponds to derivatives of the delta function at 0, invisible in $z \neq 0$.

4.2.3 Blow-up of the origin

A construction dual to radial compactification is the blow-up of the origin, which resolves the singularity of such homogeneous functions. Polar coordinates give a smooth surjective map

$$\beta : [0,\infty) \times \mathbb{S}^{n-1} \ni (r,\omega) \longmapsto r\omega \in \mathbb{R}^n,$$

a diffeomorphism of $(0,\infty)\times\mathbb{S}^{n-1}$ onto $\mathbb{R}^n \setminus 0$ which collapses $\{0\}\times\mathbb{S}^{n-1}$ to 0. Each U_j in Proposition 4.2.3 is of the form $r^{-n-\mu+j}$ times a smooth function on the sphere.

Lemma 4.2.4. *The action of* $\mathrm{GL}(n,\mathbb{R})$ *on* $\mathbb{R}^n$ *lifts under* β *to a smooth action on* $[0,\infty)\times\mathbb{S}^{n-1}$.

Proof. For $G \in \mathrm{GL}(n,\mathbb{R})$ the lift is $\tilde{G}(r,\omega) = (r|G\omega|,\, G\omega/|G\omega|)$, which is smooth up to $r=0$, satisfies $\beta\circ\tilde{G} = G\circ\beta$, and $\widetilde{GH} = \tilde{G}\tilde{H}$. (Infinitesimally: the linear vector fields $z_i\partial_{z_j}$ are homogeneous of degree 0, so on $(0,\infty)\times\mathbb{S}^{n-1}$ they are of the form $a(\omega)r\partial_r + V(\omega)$ with V a vector field on the sphere, which extends smoothly to $r=0$, tangent to the boundary.) □

Thus for a vector space W the blown-up space $[W,\{0\}]$, a manifold with boundary $\mathbb{S}W$ with a blow-down map onto W, is well defined; see Exercise 4.3.

4.3 The local model

4.3.1 Notation

Throughout the rest of the chapter n is the dimension of the ambient space, k the codimension of the submanifold and $d = n-k$ its dimension. In the local model

$$\mathbb{R}^n = \mathbb{R}^k_x \times \mathbb{R}^d_y, \qquad Y = \{x=0\}, \qquad z=(x,y),\ \zeta=(\xi,\eta),$$

and the order of a conormal distribution is related to the order of its symbol by

$$M = m + \frac{n}{4} - \frac{k}{2}. \tag{4.3.1}$$

For a point ($k=n$) this is $M = m - \frac{n}{4}$ as in Definition 4.2.1; for the diagonal in $\mathbb{R}^n\times\mathbb{R}^n$ ($n\to 2n$, $k=n$) it is $M=m$.

For an amplitude $c \in S^M(\mathbb{R}^n;\mathbb{R}^k)$, i.e. $c = c(x,y,\xi)$ satisfying the estimates (4.1.1) uniformly in (x,y), we define the distribution

$$\mathcal{I}[c](x,y) = \int_{\mathbb{R}^k} e^{ix\cdot\xi} c(x,y,\xi)\,đ\xi, \qquad \langle \mathcal{I}[c],\phi\rangle = \int \Big(\int e^{ix\cdot\xi} c(x,y,\xi)\phi(x,y)\,dx\,dy\Big) đ\xi, \tag{4.3.2}$$

for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$; here $đ\xi = (2\pi)^{-k}d\xi$. Writing $e^{ix\cdot\xi} = \langle\xi\rangle^{-2N}(1-\Delta_x)^N e^{ix\cdot\xi}$ and integrating by parts shows that the inner integral is $O(\langle\xi\rangle^{M-2N})$, so the outer integral converges for $2N > M+k$ and $\mathcal{I}[c]\in\mathcal{C}^{-\infty}(\mathbb{R}^n)$. When $M<-k$ the integral in (4.3.2) converges absolutely and $\mathcal{I}[c]$ is continuous. By Lemma 4.1.1, $\mathcal{I}[c_\epsilon]\to\mathcal{I}[c]$ in $\mathcal{C}^{-\infty}$ if $c_\epsilon\to c$ in $S^{M'}$, which justifies the formal manipulations below.

Lemma 4.3.1. *For* $c\in S^M(\mathbb{R}^n;\mathbb{R}^k)$*:*

(i) $D_{y_l}\mathcal{I}[c] = \mathcal{I}[D_{y_l}c]$, $D_{x_j}\mathcal{I}[c] = \mathcal{I}[\xi_j c + D_{x_j}c]$, $x_j\mathcal{I}[c] = \mathcal{I}[-D_{\xi_j}c]$ *and* $\psi\mathcal{I}[c] = \mathcal{I}[\psi c]$ *for* $\psi\in\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$*;*

(ii) *if* $M < -k - r$ *then* $\mathcal{I}[c] \in \mathcal{C}^r(\mathbb{R}^n)$ *with bounded derivatives of order* $\le r$*;*

(iii) $\mathcal{I}[c]$ *is smooth in* $x \neq 0$ *and, for every* $\epsilon > 0$, $\sup_{|x|\ge\epsilon}\langle x\rangle^N |D_x^\alpha D_y^\gamma \mathcal{I}[c]| < \infty$ *for all* N, α, γ.

Proof. (i) holds for c of order $< -k$ by differentiation under the integral and, for x_j, by $x_j e^{ix\cdot\xi} = D_{\xi_j} e^{ix\cdot\xi}$ and integration by parts; it extends by continuity. (ii) holds since then $D_x^\alpha D_y^\gamma \mathcal{I}[c]$, $|\alpha+\gamma| \le r$, is $\mathcal{I}[c']$ with c' of order $< -k$. (iii): by (i), $|x|^{2N} D_x^\alpha D_y^\gamma \mathcal{I}[c] = \mathcal{I}[c']$ with c' of order $M + |\alpha| - 2N$, which is bounded with all derivatives up to order r once $2N > M + |\alpha| + k + r$. □

Lemma 4.3.2 (Reduction)**.** *If* $c \in S^M(\mathbb{R}^n;\mathbb{R}^k)$ *and* $a \in S^M(\mathbb{R}^d;\mathbb{R}^k)$ *satisfies*

$$a(y,\xi) \sim \sum_\alpha \frac{1}{\alpha!}(-D_\xi)^\alpha \partial_x^\alpha c(0,y,\xi), \tag{4.3.3}$$

then $\mathcal{I}[c] - \mathcal{I}[a] \in \mathcal{C}^\infty(\mathbb{R}^n)$*. If* c *is classical, so is* a*.*

Proof. By Taylor's formula in x, $c(x,y,\xi) = \sum_{|\alpha|<N} \frac{x^\alpha}{\alpha!}\partial_x^\alpha c(0,y,\xi) + \sum_{|\alpha|=N} x^\alpha r_\alpha(x,y,\xi)$ with $r_\alpha = \frac{N}{\alpha!}\int_0^1 (1-t)^{N-1}\partial_x^\alpha c(tx,y,\xi)\,dt \in S^M(\mathbb{R}^n;\mathbb{R}^k)$. By Lemma 4.3.1(i), $\mathcal{I}[x^\alpha f] = \mathcal{I}[(-D_\xi)^\alpha f]$, so

$$\mathcal{I}[c] - \mathcal{I}[a] = \mathcal{I}[a_N - a] + \sum_{|\alpha|=N} \mathcal{I}[(-D_\xi)^\alpha r_\alpha], \quad a_N = \sum_{|\alpha|<N} \frac{1}{\alpha!}(-D_\xi)^\alpha \partial_x^\alpha c(0,y,\xi).$$

Both terms on the right have amplitudes of order $M - N$, so are in $\mathcal{C}^r$ if $N > M + k + r$. As N is arbitrary, $\mathcal{I}[c] - \mathcal{I}[a]$ is smooth. □

Definition 4.3.3. Let $\Omega \subset \mathbb{R}^n$ be open. The space $I_{\mathrm{c}}^m(\Omega, Y)$ consists of the $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega)$ of the form $u = \mathcal{I}[c] + f$ with $c \in S^M(\mathbb{R}^n;\mathbb{R}^k)$, M given by (4.3.1), and $f \in \mathcal{C}^\infty(\mathbb{R}^n)$; $I_{\mathrm{c,ph}}^m(\Omega, Y)$ is defined with $c \in S_{\mathrm{ph}}^M(\mathbb{R}^n;\mathbb{R}^k)$. Further, $I^m(\Omega, Y)$ is the space of $u \in \mathcal{C}^{-\infty}(\Omega)$ such that $\phi u \in I_{\mathrm{c}}^m(\Omega, Y)$ for all $\phi \in \mathcal{C}_{\mathrm{c}}^\infty(\Omega)$, and similarly in the classical case.

If $u = \mathcal{I}[c] + f \in I_{\mathrm{c}}^m(\Omega, Y)$ and $\chi \in \mathcal{C}_{\mathrm{c}}^\infty(\Omega)$ equals one near $\operatorname{supp} u$, then $u = \mathcal{I}[\chi c] + \chi f$, so the amplitude can be taken with compact support in (x,y). By Lemma 4.3.2 it can also be taken independent of x: $u = \mathcal{I}[a] + f'$ with $a \in S_{\mathrm{c}}^M(\mathbb{R}^d;\mathbb{R}^k)$ and f' smooth (not compactly supported in general). We call such an a a *full symbol* of u.

Proposition 4.3.4. *Let* $\Omega \subset \mathbb{R}^n$ *be open. Then (and likewise for the classical spaces):*

(i) $I^m(\Omega, Y)$ *is a* $\mathcal{C}^\infty(\Omega)$*-module,* $I^{m-1} \subset I^m$*,* $x_j : I^m \to I^{m-1}$*,* $D_{x_j} : I^m \to I^{m+1}$*, and every smooth vector field tangent to* Y *maps* I^m *to itself;*

(ii) *each* $u \in I^m(\Omega, Y)$ *is smooth in* $\Omega \setminus Y$*, and* $\bigcap_m I^m(\Omega, Y) = \mathcal{C}^\infty(\Omega)$*;*

(iii) *a full symbol of* $u \in I_{\mathrm{c}}^m(\Omega, Y)$ *is determined modulo* $S^{-\infty}$*; and* $u \in I_{\mathrm{c}}^{m-1}(\Omega, Y)$ *if and only if its full symbols lie in* S^{M-1}*;*

(iv) *if* $u_j \in I_{\mathrm{c}}^{m-j}(\Omega, Y)$ *have supports in a fixed compact set, there is* $u \in I_{\mathrm{c}}^m(\Omega, Y)$ *with* $u - \sum_{j<N} u_j \in I_{\mathrm{c}}^{m-N}(\Omega, Y)$ *for all* N*;*

(v) *if* $P = \sum_{|\alpha|\le q} p_\alpha(z) D_z^\alpha$ *with* $p_\alpha \in \mathcal{C}^\infty(\Omega)$*, then* $P : I^m(\Omega, Y) \to I^{m+q}(\Omega, Y)$ *and, if* $u \in I_{\mathrm{c}}^m$ *has full symbol* a*, then* Pu *has full symbol congruent to* $p_q(0,y;\xi,0)a(y,\xi)$ *modulo* S^{M+q-1}*, where* $p_q(z,\zeta) = \sum_{|\alpha|=q} p_\alpha(z)\zeta^\alpha$*.*

Proof. (i): for $\psi \in \mathcal{C}^\infty(\Omega)$ and $u \in I^m_{\mathrm{c}}$, $\psi u = \psi\chi u = \mathcal{I}[\psi\chi c] + \psi\chi f$ with χ as above. The rest follows from Lemma 4.3.1(i), since the vector fields tangent to Y are spanned over $\mathcal{C}^\infty$ by $x_i\partial_{x_j}$ and ∂_{y_l} (see Lemma 4.7.1 below) and $x_i\partial_{x_j}$ maps an amplitude of order M to $(-D_{\xi_i})(i\xi_j c + \partial_{x_j} c)$, of order M. (ii): smoothness off Y is Lemma 4.3.1(iii). If u lies in every I^m, then for each ϕ and r, ϕu is the sum of a smooth function and $\mathcal{I}[c]$ with c of order $< -k - r$, so $\phi u \in \mathcal{C}^r$. Conversely $\mathcal{C}^\infty_{\mathrm{c}}(\Omega) \subset I^m_{\mathrm{c}}$ trivially. (iii): suppose $\mathcal{I}[a] \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $a \in S^M_{\mathrm{c}}(\mathbb{R}^d;\mathbb{R}^k)$. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$ equal one near 0. For each y, $\mathcal{I}[a](\cdot,y)$ is the inverse Fourier transform of $a(y,\cdot)$, and it is the sum of $\chi(x)\mathcal{I}[a](x,y)$, smooth with compact support, and $(1-\chi(x))\mathcal{I}[a](x,y)$, which is in $\mathcal{S}(\mathbb{R}^k)$ by Lemma 4.3.1(iii), both with seminorms bounded with all y-derivatives. Hence $a \in S^{-\infty}$. If now $u = \mathcal{I}[a] + f = \mathcal{I}[b] + g$ then $\mathcal{I}[a-b]$ is smooth, so $a - b \in S^{-\infty}$. The second statement follows: if $u \in I^{m-1}_{\mathrm{c}}$ it has a full symbol $b \in S^{M-1}$, and any other full symbol differs from b by an element of $S^{-\infty}$. (iv): with full symbols $a_j \in S^{M-j}_{\mathrm{c}}$ supported in a fixed compact set of y, take $u = \chi\mathcal{I}[a]$ with $a \sim \sum a_j$ (with support in the same set) and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equal one near all the supports. (v): by (i) and Lemma 4.3.1(i), $D^\alpha\mathcal{I}[a] = \mathcal{I}[(\xi,0)^\alpha a] + \mathcal{I}[c']$ with c' of order $M + |\alpha| - 1$ (the y-derivatives do not raise the order), and $p_\alpha\mathcal{I}[b] = \mathcal{I}[p_\alpha b]$, whose reduction is $p_\alpha(0,y)b$ modulo order $M' - 1$ by (4.3.3). □

Remark 4.3.5. For $k = n$ (so $Y = \{0\}$) we recover the spaces of Section 4.2: $I^m_{\mathcal{S}}(\mathbb{R}^n,\{0\}) = I^m_{\mathrm{c}}(\mathbb{R}^n,\{0\}) + \mathcal{S}(\mathbb{R}^n)$. Indeed if $\hat{u} \in S^{m-\frac{n}{4}}$ then $u = \chi u + (1-\chi)u$ with $(1-\chi)u \in \mathcal{S}$ by Lemma 4.2.2(iii) and $\chi u = \mathcal{I}[\chi\hat{u}]$. Conversely $\mathcal{I}[c]$ with c compactly supported in z has Fourier transform $\int \hat{c}(\zeta-\xi;\xi)đ\xi$, where $\hat{c}(\cdot;\xi)$ is the Fourier transform of $c(\cdot,\xi)$; this is in $S^{m-\frac{n}{4}}$ by Peetre's inequality $\langle\xi\rangle^M \le 2^{|M|}\langle\zeta\rangle^M\langle\zeta-\xi\rangle^{|M|}$ (moving derivatives onto c after the substitution $\xi \mapsto \zeta - \xi$). In particular $I^m_{\mathcal{S}}(\mathbb{R}^n,\{0\})$ is a $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$-module, and in particular an $\mathcal{S}(\mathbb{R}^n)$-module.

Example 4.3.6 (An elliptic problem at a point)**.** Let $P = \sum_{|\alpha|\le q} p_\alpha(z)D^\alpha$ with $p_\alpha \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, and suppose P is *elliptic at* 0: $p_q(0,\zeta) = \sum_{|\alpha|=q} p_\alpha(0)\zeta^\alpha \neq 0$ for $\zeta \neq 0$. Then for every $f \in I^m_{\mathcal{S}}(\mathbb{R}^n,\{0\})$ there is $u \in I^{m-q}_{\mathcal{S}}(\mathbb{R}^n,\{0\})$ with $Pu - f \in \mathcal{S}(\mathbb{R}^n)$. Indeed, by Proposition 4.3.4(v) and Remark 4.3.5, $P : I^r_{\mathcal{S}} \to I^{r+q}_{\mathcal{S}}$ and $Pv - p_q(0,D)v \in I^{r+q-1}_{\mathcal{S}}$ for $v \in I^r_{\mathcal{S}}$. Choose $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near 0 and set $\hat{u}_0 = (1-\chi)\hat{f}/p_q(0,\cdot) \in S^{m-q-\frac{n}{4}}$, using that $(1-\chi)/p_q(0,\cdot) \in S^{-q}_{\mathrm{ph}}$. Then $p_q(0,D)u_0 - f \in \mathcal{S}$, so $Pu_0 = f - f_1$ with $f_1 \in I^{m-1}_{\mathcal{S}}$. Repeating with f replaced by f_1, we find $u_j \in I^{m-q-j}_{\mathcal{S}}$ with $P(u_0+\cdots+u_j) - f \in I^{m-j-1}_{\mathcal{S}}$; an asymptotic sum $u \sim \sum u_j$ (Proposition 4.3.4(iv)) then satisfies $Pu - f \in \bigcap_j I^{m-j}_{\mathcal{S}} = \mathcal{S}(\mathbb{R}^n)$. The same works in the classical spaces.

4.3.2 Pseudodifferential operators

We now let pseudodifferential operators on $\mathbb{R}^n$, as in Chapter 2, act on conormal distributions. For $a \in S^M_{\mathrm{c}}(\mathbb{R}^d;\mathbb{R}^k)$ the distribution $\mathcal{I}[a]$ is tempered, with Fourier transform

$$\widehat{\mathcal{I}[a]}(\xi,\eta) = \tilde{a}(\xi,\eta) = \int e^{-iy\cdot\eta}a(y,\xi)\,dy, \qquad |D^\beta_\xi D^\gamma_\eta \tilde{a}(\xi,\eta)| \le C_{N\beta\gamma}\langle\eta\rangle^{-N}\langle\xi\rangle^{M-|\beta|}, \tag{4.3.4}$$

the estimate following by integration by parts in y.

Proposition 4.3.7. *Let $A = \mathrm{Op}(b)$, $b \in S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$, and $u \in I^m_{\mathrm{c}}(\mathbb{R}^n,Y)$ with full symbol $a \in S^M_{\mathrm{c}}(\mathbb{R}^d;\mathbb{R}^k)$. Then $Au \in I^{m+m'}(\mathbb{R}^n,Y)$; more precisely $Au - \mathcal{I}[c] \in \mathcal{S}(\mathbb{R}^n)$ where $c \in S^{M+m'}(\mathbb{R}^n;\mathbb{R}^k)$ is given by*

$$c(x,y,\xi) = \int e^{iy\cdot\eta}b(x,y,\xi,\eta)\tilde{a}(\xi,\eta)đ\eta \sim \sum_\gamma \frac{1}{\gamma!}\partial^\gamma_\eta b(x,y,\xi,0)\,D^\gamma_y a(y,\xi). \tag{4.3.5}$$

In particular Au has full symbol (if b has compact support in z)

$$a'(y,\xi) \sim \sum_{\alpha,\gamma} \frac{1}{\alpha!\gamma!}(-D_\xi)^\alpha \partial_x^\alpha \Big(\partial_\eta^\gamma b(x,y,\xi,0) D_y^\gamma a(y,\xi)\Big)\Big|_{x=0}, \tag{4.3.6}$$

and $a' - b(0,y,\xi,0)a \in S^{M+m'-1}$. If b and a are classical, so is c.

Proof. Write $u = \mathcal{I}[a] - g$. Then g is smooth, and outside a compact set equals $\mathcal{I}[a]$, which vanishes for y outside the support of a and is rapidly decreasing in x by Lemma 4.3.1(iii); so $g \in \mathcal{S}(\mathbb{R}^n)$ and $Ag \in \mathcal{S}(\mathbb{R}^n)$. By (4.3.4), $A\mathcal{I}[a](x,y) = \int e^{ix\cdot\xi + iy\cdot\eta} b(x,y,\xi,\eta)\tilde{a}(\xi,\eta)đ\xi đ\eta$, the η-integral being absolutely convergent; this is $\mathcal{I}[c]$ with c as in (4.3.5). To estimate c, note that derivatives in (x,y) and ξ may be taken under the integral, and by Peetre's inequality

$$|\partial_z^\delta \partial_\xi^\beta b(z,\xi,\eta)| \le C\langle(\xi,\eta)\rangle^{m'-|\beta|} \le C'\langle\xi\rangle^{m'-|\beta|}\langle\eta\rangle^{|m'|+|\beta|};$$

combined with (4.3.4) (and y-derivatives falling on $e^{iy\cdot\eta}$ producing powers of η) this gives $c \in S^{M+m'}$. For the expansion, Taylor's formula in η gives $b(z,\xi,\eta) = \sum_{|\gamma|<L} \frac{\eta^\gamma}{\gamma!}\partial_\eta^\gamma b(z,\xi,0) + \sum_{|\gamma|=L} \eta^\gamma r_\gamma(z,\xi,\eta)$ with

$$|\partial_z^\delta \partial_\xi^\beta r_\gamma| \le C \sup_{0\le t\le 1} \langle(\xi,t\eta)\rangle^{m'-L-|\beta|} \le C\langle\xi\rangle^{m'-L-|\beta|}\langle\eta\rangle^{|m'-L|+|\beta|}.$$

Since $\int e^{iy\cdot\eta}\eta^\gamma \tilde{a} đ\eta = D_y^\gamma a$, the terms with $|\gamma| < L$ give the expansion and the remainder is of order $M + m' - L$ by the same estimate as before. Finally (4.3.6) follows from Lemma 4.3.2. In the classical case each term of (4.3.5) is classical, so the asymptotic sum is. □

The microlocal content of this computation is that u is singular only in the conormal directions to Y. Recall from Chapter 3 that $(\bar{z},\bar{\zeta}) \notin \mathrm{WF}(u)$ if and only if there is $A \in \Psi^0(\mathbb{R}^n)$ elliptic at $(\bar{z},\bar{\zeta})$ with $Au \in \mathcal{C}^\infty$. The conormal bundle of Y is $N^*Y = \{(0,y;\xi,0)\} \subset T^*\mathbb{R}^n$.

Proposition 4.3.8. *If $u \in I^m(\Omega, Y)$ then $\mathrm{WF}(u) \subset N^*Y \backslash 0$. More precisely, if $u \in I^m_{\mathrm{c}}(\Omega, Y)$ has full symbol a, then $(0,\bar{y};\bar{\xi},0) \notin \mathrm{WF}(u)$ if and only if a satisfies the estimates of $S^{-\infty}$ in $U \times \Gamma$, for some neighbourhood U of $\bar{y}$ and open cone $\Gamma \ni \bar{\xi}$.*

Proof. We may assume $u \in I^m_{\mathrm{c}}$. Over $\Omega \setminus Y$, u is smooth. Let $\bar{z} = (0,\bar{y})$ and $\bar{\zeta} = (\bar{\xi},\bar{\eta})$ with $\bar{\eta} \neq 0$. Choose $\delta > 0$ with $|\bar{\eta}| > 2\delta|\bar{\xi}|$, and $b = \phi(z)\psi(\zeta)$ with $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$, $\phi(\bar{z}) \neq 0$, $\psi \in S^0_{\mathrm{ph}}(\mathbb{R}^n)$ homogeneous of degree 0 in $|\zeta| \ge 1$, equal to one near $\bar{\zeta}/|\bar{\zeta}|$ there, and supported in $\{|\eta| \ge \delta|\xi|,\ |\zeta| \ge \frac{1}{2}\}$. On the support of ψ, $\langle\eta\rangle \ge c\langle\xi\rangle$, so by (4.3.4) the amplitude c of (4.3.5) satisfies $|c| \le C_N \int_{\langle\eta\rangle\ge c\langle\xi\rangle} \langle\eta\rangle^{-N}\langle\xi\rangle^M d\eta \le C'_N\langle\xi\rangle^{M-N+d}$ for all N, and similarly for derivatives; thus $c \in S^{-\infty}$ and $\mathrm{Op}(b)u \in \mathcal{C}^\infty$. Hence $\mathrm{WF}(u) \subset N^*Y$.

For the refined statement let $\bar{\zeta} = (\bar{\xi},0)$. If a is of order $-\infty$ in $U \times \Gamma$, take $b = \phi(z)\psi(\xi,\eta)$ as before but with ϕ supported in $\mathbb{R}^k \times U'$, $U' \Subset U$, and ψ supported in a small conic neighbourhood of $(\bar{\xi},0)$, so that $(y,\xi) \in U' \times \Gamma$ on the support of $\phi(x,y)\psi(\xi,0)$ for $|\xi| \ge 1$. Each term in (4.3.6) then vanishes to infinite order (the terms involve a only at points of $U' \times \Gamma$), so $Au \in \mathcal{C}^\infty$ and A is elliptic at $(\bar{z},\bar{\zeta})$. Conversely, if $(\bar{z},\bar{\zeta}) \notin \mathrm{WF}(u)$, choose $B = \mathrm{Op}(b)$ as just described with $b = 1$ on a conic neighbourhood of $(\bar{z},\bar{\zeta})$ and $\mathrm{WF}'(B) \cap \mathrm{WF}(u) = \emptyset$; then $Bu \in \mathcal{C}^\infty$ by Chapter 3, so its full symbol (4.3.6) is in $S^{-\infty}$, and near $(\bar{y},\bar{\xi})$ all terms with $(\alpha,\gamma) \neq 0$ vanish because b is constant there, leaving $a \in S^{-\infty}$ near $(\bar{y},\bar{\xi})$. □

4.4 Invariance

We now show that the spaces $I^m_{\rm c}(\Omega, Y)$ are invariant under diffeomorphisms preserving Y, and compute the effect on the full symbol modulo lower order terms. Let $F : \Omega' \to \Omega$ be a diffeomorphism between open subsets of $\mathbb{R}^n$ with $F(\Omega' \cap Y) = \Omega \cap Y$. Write $F(x,y) = (F'(x,y), F''(x,y))$; then $F'(0,y) = 0$, so $\partial_y F'(0,y) = 0$, and

$$g(y) = F''(0,y), \qquad A(y) = \partial_x F'(0,y) \in \mathrm{GL}(k,\mathbb{R}) \tag{4.4.1}$$

define a diffeomorphism g of $\Omega' \cap Y$ onto $\Omega \cap Y$ and the linear maps $A(y)$ induced by F_* on the normal spaces.

Lemma 4.4.1. *Let $V \subset \mathbb{R}^d$ be open, let g be a diffeomorphism of V onto $g(V)$ and let $A \in \mathcal{C}^\infty(V; \mathrm{GL}(k,\mathbb{R}))$. Set $F_0(x,y) = (A(y)x, g(y))$, a diffeomorphism of $\mathbb{R}^k \times V$ onto $\mathbb{R}^k \times g(V)$. If $a \in S^M_{\rm c}(\mathbb{R}^d; \mathbb{R}^k)$ has support in $g(V) \times \mathbb{R}^k$ then*

$$F_0^* \mathcal{I}[a] = \mathcal{I}[a_F], \qquad a_F(y,\eta) = |\det A(y)|^{-1} a(g(y), A(y)^{-t}\eta) \in S^M_{\rm c}(\mathbb{R}^d; \mathbb{R}^k),$$

where $A^{-t} = (A^t)^{-1}$; a_F is classical if a is.

Proof. For fixed y this is the linear change of variables $\xi = A(y)^{-t}\eta$ in $\int e^{iA(y)x\cdot\xi} a(g(y),\xi) đ\xi$. That a_F is a symbol follows from the invariance of symbol estimates under smooth families of linear maps, the support of a_F in y being compact. □

Lemma 4.4.2. *A vector field of the form*

$$V = \sum_{i,j,l} x_i x_j \alpha_{ijl}(z) \partial_{x_l} + \sum_{i,l} x_i \beta_{il}(z) \partial_{y_l}, \qquad \alpha_{ijl}, \beta_{il} \in \mathcal{C}^\infty(\Omega), \tag{4.4.2}$$

maps $I^m(\Omega, Y)$ to $I^{m-1}(\Omega, Y)$.

Proof. By Proposition 4.3.4(i), $x_i x_j \partial_{x_l}$ maps I^m into I^{m+1-2} and $x_i \partial_{y_l}$ maps I^m into I^{m-1}, and the coefficients are harmless. □

We also need conormal distributions depending smoothly on a parameter $s \in [0,1]$. A *smooth family* $u(s) \in I^m_{\rm c}(\Omega, Y)$ is one of the form $u(s) = \mathcal{I}[c(s)] + f(s)$ with $c \in S^M(\mathbb{R}_s \times \mathbb{R}^n; \mathbb{R}^k)$ and $f \in \mathcal{C}^\infty(\mathbb{R} \times \mathbb{R}^n)$, the supports of $u(s)$ lying in a fixed compact subset of Ω; that is, u is an element of $I^m_{\rm c}$ on $\mathbb{R}_s \times \mathbb{R}^n$ with respect to $\mathbb{R}_s \times Y$, restricted to $[0,1]$. By Proposition 4.3.4 applied in $n+1$ variables, $\partial_s u(s)$ and $\int_s^1 u(s')\, ds'$ are smooth families of order m, $V_s u(s)$ is a smooth family of order $m-1$ if V_s is of the form (4.4.2) with coefficients smooth in s, and families of orders $m - j$ have asymptotic sums.

Proposition 4.4.3. *Let H be a diffeomorphism from a neighbourhood O of $\bar z = (0, \bar y)$ onto its image such that $H(0,y) = (0,y)$ and $\partial_x H'(0,y) = \mathrm{Id}$ for $(0,y) \in O$. Then there is a neighbourhood O' of $\bar z$ such that $H^* u - u \in I^{m-1}_{\rm c}(\mathbb{R}^n, Y)$ for all $u \in I^m_{\rm c}(\mathbb{R}^n, Y)$ with support in O'; similarly in the classical case.*

Proof. Set $H_s(z) = z + s(H(z) - z)$, $s \in [0,1]$, so $H_0 = \mathrm{Id}$, $H_1 = H$ and $H_s(0,y) = (0,y)$. At points of Y, $dH_s = \left(\begin{smallmatrix} \mathrm{Id} & 0 \\ sB & \mathrm{Id} \end{smallmatrix}\right)$ with $B = \partial_x H''(0,y)$, which is invertible. By the inverse function theorem, applied uniformly for s in the compact set $[0,1]$, there is a ball $O_1 \ni \bar z$ such that each H_s is a diffeomorphism of O_1 onto its image, and a smaller ball O' contained in $H_s(O_1)$ for all s. For $w \in O'$ let

$$V_s(w) = (H - \mathrm{Id})(H_s^{-1}(w)),$$

a vector field depending smoothly on s. For smooth v the chain rule gives

$$\frac{d}{ds}H_s^*v = H_s^*(V_sv), \tag{4.4.3}$$

and the same holds for a smooth family of distributions $v(s)$ in the form $\frac{d}{ds}H_s^*v(s) = H_s^*(\partial_s v + V_s v)$: pairing with a test function ϕ, $\langle H_s^*v(s),\phi\rangle = \langle v(s),\phi_s\rangle$ with $\phi_s = |\det dH_s^{-1}|\,(H_s^{-1})^*\phi$ smooth in s, and the identity $\langle w, \partial_s\phi_s\rangle = \langle H_s^*(V_sw),\phi\rangle$, valid for smooth w by (4.4.3), extends to distributions by density.

Since $H'(x,y)-x = O(|x|^2)$, $H''(x,y)-y = O(|x|)$ and the x-component of $H_s^{-1}(w)$ vanishes on Y (so is a combination $\sum x_i(\dots)$), V_s is of the form (4.4.2) with coefficients smooth in s.

Now let $u \in I^m_{\mathrm{c}}$ have support in O'. We construct a smooth family $u(s)$, with supports in a fixed compact subset of O', such that

$$u(1) = u, \qquad \partial_s u(s) + V_s u(s) = r(s) \in \mathcal{C}^\infty_{\mathrm{c}}([0,1]\times O'), \qquad u(0)-u \in I^{m-1}_{\mathrm{c}}. \tag{4.4.4}$$

Set $u_0(s) = u$ and $f_1(s) = V_s u \in I^{m-1}$. Inductively, given smooth families $u_0,\dots,u_N$ with u_j of order $m-j$, $u_j(1)=0$ for $j\ge 1$, and $\partial_s(u_0+\dots+u_N)+V_s(u_0+\dots+u_N) = f_{N+1}$ of order $m-N-1$, put $u_{N+1}(s) = \int_s^1 f_{N+1}(s')\,ds'$. Then u_{N+1} has order $m-N-1$, vanishes at $s=1$, and the new error is $f_{N+2} = V_s u_{N+1}$, of order $m-N-2$. Let $\tilde u(s) \sim \sum_j u_j(s)$ be an asymptotic sum, multiplied by a cut-off equal to one near the supports. Then $\partial_s\tilde u + V_s\tilde u$ is a smooth family of every order, hence a smooth function of (s,z), and $\tilde u(1)-u = \tilde u(1) - \sum_{j<N} u_j(1)$ has order $m-N$ for every N, so is smooth. Thus $u(s) = \tilde u(s) - (\tilde u(1)-u)$ satisfies (4.4.4); note that $u(0)-u \equiv \sum_{j\ge1} u_j(0)$ modulo $\mathcal{C}^\infty$ has order $m-1$. Then

$$H^*u = H_1^*u(1) = H_0^*u(0) + \int_0^1 \frac{d}{ds}H_s^*u(s)\,ds = u(0) + \int_0^1 H_s^*r(s)\,ds,$$

and the last integral is smooth. Hence $H^*u - u = (u(0)-u) + \mathcal{C}^\infty \subset I^{m-1}_{\mathrm{c}}$. □

Theorem 4.4.4. *Let $F:\Omega'\to\Omega$ be a diffeomorphism of open subsets of $\mathbb{R}^n$ with $F(\Omega'\cap Y) = \Omega\cap Y$, and g, A as in (4.4.1). Then $F^*: I^m_{\mathrm{c}}(\Omega,Y)\to I^m_{\mathrm{c}}(\Omega',Y)$, and if $u\in I^m_{\mathrm{c}}(\Omega,Y)$ has full symbol a, then F^*u has full symbol congruent modulo S^{M-1} to*

$$|\det A(y)|^{-1} a(g(y), A(y)^{-t}\eta). \tag{4.4.5}$$

The same holds for the classical spaces.

Proof. Using a partition of unity and the module property we may assume that u has support in a small neighbourhood of $F(\bar z)$, $\bar z\in Y\cap\Omega'$ (away from Y, u and F^*u are smooth). Let $F_0(x,y) = (A(y)x, g(y))$, a diffeomorphism of $\mathbb{R}^k\times(Y\cap\Omega')$ onto $\mathbb{R}^k\times(Y\cap\Omega)$ (identifying Y with $\mathbb{R}^d$), and $H = F_0^{-1}\circ F$, defined near $\bar z$. Then $H(0,y) = (0,y)$ and $\partial_x H'(0,y) = A(y)^{-1}\partial_x F'(0,y) = \mathrm{Id}$, so Proposition 4.4.3 applies to H. With $u = \mathcal{I}[a]+f$ we have, by Lemma 4.4.1, $F_0^*u = \mathcal{I}[a_F] + F_0^*f$ (after cutting off), and $F^*u = H^*(F_0^*u) \in F_0^*u + I^{m-1}_{\mathrm{c}}$. □

4.5 Conormality at the zero section of a vector bundle

Let $\pi: W\to Y$ be a real vector bundle of rank k over a manifold Y of dimension d, so W has dimension $n = d+k$, and let $0_W\subset W$ be the zero section and W' the dual bundle. The linear invariance of the radial compactification allows it to be performed fibrewise: $\overline{W}$ is

a manifold with boundary, fibred over Y with fibres $\overline{W_y}$ (formally, the bundle associated to the frame bundle of W with fibre $\overline{\mathbb{R}^k}$), and $\rho^{-M}\mathcal{C}^\infty(\overline{W})$ is well defined. Symbols with bounds $S^M(W)$ on a bundle over a compact base, or with compact support in the base, are defined via local trivializations, by the linear invariance of the estimates.

Let $\Omega_{\mathrm{fib}} = \Omega_{\mathrm{fib}}(W')$ be the line bundle over W' whose fibre at $\zeta \in W'_y$ is $\Omega(W'_y)$, the densities on the fibre. The fibrewise version of (4.2.1) is an isomorphism

$$\mathcal{F}_{\mathrm{fib}} : \mathcal{S}(W) \longrightarrow \mathcal{S}(W';\Omega_{\mathrm{fib}}),$$

defined for (fibrewise) Schwartz functions; in a local trivialization $W|_U \cong U \times \mathbb{R}^k$ it is the partial Fourier transform in the fibre variables followed by multiplication by $|d\xi|$, and it does not depend on the trivialization. Its inverse is defined on symbols as an oscillatory integral.

Definition 4.5.1. For Y compact, $I^m(W, 0_W)$ is the space of distributions $u = \mathcal{F}_{\mathrm{fib}}^{-1}b + f$ with $b \in S^M(W';\Omega_{\mathrm{fib}})$ (that is, $b = \beta|d\xi|$ with $\beta \in S^M$ in local trivializations), $M = m + \frac{n}{4} - \frac{k}{2}$, and $f \in \mathcal{C}^\infty(W)$; similarly $I^m_{\mathrm{ph}}(W, 0_W)$ with b classical.

In a local trivialization, with fibre coordinates x and base coordinates y, $\mathcal{F}_{\mathrm{fib}}^{-1}(\beta|d\xi|) = \mathcal{I}[\beta]$, so $I^m(W, 0_W)$ consists of the distributions which are smooth outside 0_W and lie in the spaces of Definition 4.3.3 in every such local trivialization. Changes of trivialization are of the fibre-linear form of Lemma 4.4.1 (with g a change of coordinates on Y), under which $\beta|d\xi|$ transforms as a fibre density, so we obtain a well-defined symbol map with values in $S^M(W';\Omega_{\mathrm{fib}})/S^{M-1}(W';\Omega_{\mathrm{fib}})$, and, by Proposition 4.3.4(iii) and the surjectivity of the inverse Fourier transform, a short exact sequence

$$I^{m-1}(W, 0_W) \hookrightarrow I^m(W, 0_W) \xrightarrow{\sigma_m} S^M(W';\Omega_{\mathrm{fib}})/S^{M-1}(W';\Omega_{\mathrm{fib}}), \tag{4.5.1}$$

and in the classical case the corresponding sequence onto the smooth fibre densities on $W' \setminus 0$ homogeneous of degree $M + k$. For a product $W = Y \times \mathbb{R}^k$ the definition says that $I^m(Y \times \mathbb{R}^k, Y \times \{0\})$ consists of smooth functions of $y \in Y$ with values in $I^{m-\frac{d}{4}}(\mathbb{R}^k, \{0\})$ (up to smooth terms), the topology of $\mathcal{C}^\infty(Y; \rho^{-M}\mathcal{C}^\infty(\overline{\mathbb{R}^k})) = \rho^{-M}\mathcal{C}^\infty(Y \times \overline{\mathbb{R}^k})$ making precise what it means for a conormal distribution to depend smoothly on parameters.

Bundle isomorphisms, possibly covering a diffeomorphism of the base, act on $I^m(W, 0_W)$ by Lemma 4.4.1. For the extension to submanifolds, the essential point is invariance under diffeomorphisms which do not preserve the linear structure. If h is a diffeomorphism between neighbourhoods of 0_W which maps 0_W to itself, then at each point of 0_W the differential h_* preserves $T0_W$ and so induces a map on $T_yW/T_y0_W = W_y$.

Lemma 4.5.2. *Any diffeomorphism h of a neighbourhood of 0_W onto a neighbourhood of 0_W with $h(0_W) = 0_W$ factorizes, near 0_W, as $h = h_0 \circ h_1$ where h_0 is a bundle isomorphism (covering $h|_{0_W}$) and h_1 is the identity on 0_W with $(h_1)_* = \mathrm{Id}$ on $W_y = T_yW/T_y0_W$ for all y.*

Proof. Take h_0 to be the bundle map induced by h on the normal spaces, i.e. $h_0(w) = h_*(w)$ for $w \in W_y = N_y0_W$, covering $h|_{0_W}$; then $h_1 = h_0^{-1} \circ h$ has the stated properties. In local trivializations this is the factorization $F = F_0 \circ H$ used in the proof of Theorem 4.4.4. □

Proposition 4.5.3. *Let Y be compact and $h : N \to N'$ a diffeomorphism between open neighbourhoods of 0_W which is the identity on 0_W and induces the identity on W_y for all $y \in Y$. If $u \in I^m(W, 0_W)$ has support in N' then $h^*u - u \in I^{m-1}(W, 0_W)$; in particular $\sigma_m(h^*u) = \sigma_m(u)$. Combined with Lemma 4.5.2, every diffeomorphism h between neighbourhoods of 0_W mapping 0_W to itself acts on conormal distributions supported in its range, with $\sigma_m(h^*u) = h_0^*\sigma_m(u)$.*

Proof. Cover Y by finitely many open sets over which W is trivial, take a partition of unity ϕ_j of a neighbourhood of 0_W subordinate to their preimages and apply Proposition 4.4.3 (in the form of the proof of Theorem 4.4.4) to each $\phi_j u$, noting that $u - \sum \phi_j u$ is smooth. □

4.6 Conormal distributions on manifolds

4.6.1 Preliminaries

Manifolds are discussed systematically in Chapter 5; here we need only a few notions. Let X be a smooth manifold of dimension n. The density bundles $\Omega^t(X)$, $t \in \mathbb{R}$, are the line bundles with fibres $\Omega^t(T_pX)$; in local coordinates their sections are $v|dz|^t$, and under a change of coordinates $z = F(z')$, $v|dz|^t$ pulls back to $(v \circ F)|\det F'|^t|dz'|^t$. Smooth compactly supported densities can be integrated, and $\mathcal{C}^{-\infty}(X;\Omega^{\frac12})$ is the dual of $\mathcal{C}^\infty_{\mathrm{c}}(X;\Omega^{\frac12})$, the pairing of two half-densities being a density; locally an element of $\mathcal{C}^{-\infty}(X;\Omega^{\frac12})$ is $v|dz|^{1/2}$ with v a distribution in coordinates, transforming as above. A smooth vector field V acts on half-densities by the Lie derivative: in coordinates, with $V = \sum v_j\partial_{z_j}$,

$$\mathcal{L}_V(\phi|dz|^t) = (V\phi + t(\textstyle\sum_j \partial_{z_j} v_j)\phi)|dz|^t. \tag{4.6.1}$$

Let $Y \subset X$ be a closed embedded submanifold of codimension k. An *adapted chart* is a chart $\psi : U \to \psi(U) \subset \mathbb{R}^n = \mathbb{R}^k_x \times \mathbb{R}^d_y$ with $\psi(U \cap Y) = \psi(U) \cap \{x = 0\}$; such charts cover Y. The *normal bundle* NY has fibres $N_yY = T_yX/T_yY$, and the *conormal bundle* $N^*Y \subset T^*X$, with fibres $N^*_yY = \{\zeta \in T^*_yX;\ \zeta(T_yY) = 0\}$, is its dual. In an adapted chart, $N^*Y = \{(0, y; \xi, 0)\}$, and (y, ξ) are coordinates on N^*Y. If $F : \Omega' \to \Omega$ is a change of adapted charts as in Section 4.4, the pull-back $F^*(\xi \cdot dx) = (A(y)^t\xi) \cdot dx'$ at $(0, y)$, since $\partial_y F'(0, y) = 0$; so the induced identification of conormal bundles is

$$F^\sharp : N^*Y \ni (y, \eta) \longmapsto (g(y), A(y)^{-t}\eta) \in N^*Y, \tag{4.6.2}$$

(primed coordinates on the left, unprimed on the right) which is linear on the fibres.

Definition 4.6.1. The space $I^m(X, Y; \Omega^{\frac12})$ consists of those $u \in \mathcal{C}^{-\infty}(X;\Omega^{\frac12})$ which are smooth on $X \setminus Y$ and such that for every adapted chart $\psi : U \to \psi(U)$ and every $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$, $(\psi^{-1})^*(\phi u) = v|dz|^{1/2}$ with $v \in I^m_{\mathrm{c}}(\psi(U), \{x = 0\})$. The classical subspace $I^m_{\mathrm{ph}}(X, Y; \Omega^{\frac12})$ is defined in the same way. For a vector bundle E over X, $I^m(X, Y; E \otimes \Omega^{\frac12})$ is defined by requiring the coefficients in local trivializations to be in $I^m(X, Y; \Omega^{\frac12})$; similarly with $\Omega^{\frac12}$ omitted.

By Theorem 4.4.4 (the factor $|\det F'|^{1/2}$ being smooth) it suffices to check the condition for a family of adapted charts covering Y, and the definition agrees with Definition 4.3.3 for $X = \mathbb{R}^n$ and with Definition 4.5.1 for the zero section of a vector bundle; it is independent of the choice of transition functions for E by the module property.

4.6.2 Normal fibrations

Alternatively, $I^m(X, Y)$ can be defined by transferring Definition 4.5.1 from the normal bundle to X.

Theorem 4.6.2 (Tubular neighbourhood theorem)**.** *If $Y \subset X$ is a closed embedded submanifold, there are an open neighbourhood D of Y in X, an open neighbourhood D' of the zero section of NY, and a diffeomorphism $f : D \to D'$ (a* normal fibration*) such that*

$f|_Y$ is the natural identification of Y with 0_{NY} and, for each $y \in Y$, the map induced by f_* from $N_yY = T_yX/T_yY$ to $T_y(NY)/T_y0_{NY} = N_yY$ is the identity. If f_1, f_2 are two normal fibrations then $f_2 \circ f_1^{-1}$ is a diffeomorphism between neighbourhoods of 0_{NY} satisfying the hypotheses of Proposition 4.5.3.

Sketch of proof. Choose a Riemannian metric on X; it identifies N_yY with the orthogonal complement of T_yY in T_yX. The exponential map $NY \ni (y,v) \mapsto \exp_y(v)$ is smooth, is the identity on the zero section and has differential the identity at points of the zero section (with respect to the splitting $T_y(NY) = T_yY \oplus N_yY$), so it is a local diffeomorphism near 0_{NY}; a standard argument (using that it is injective on 0_{NY} and Y is closed) shows it is a diffeomorphism of a neighbourhood D' of 0_{NY} onto a neighbourhood D of Y, and f is its inverse. The last statement is immediate from the definition. □

Proposition 4.6.3. *Let $f : D \to D'$ be a normal fibration and Y compact. Then $u \in \mathcal{C}^{-\infty}(X)$ lies in $I^m(X,Y)$ if and only if u is smooth outside Y and, for some (hence any) $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(D)$ equal to one near Y, $(f^{-1})^*(\phi u) \in I^m(NY, 0_{NY})$.*

Proof. Local trivializations of NY composed with f are adapted charts for Y, and local trivializations of NY are adapted charts for 0_{NY}, so both conditions amount to the condition of Definition 4.6.1 in a particular covering family of adapted charts. Independence of f also follows from Proposition 4.5.3 and Theorem 4.6.2. □

4.6.3 The principal symbol

For $r \in \mathbb{R}$ let $S^r(N^*Y; \Omega^{\frac12})$ be the space of smooth half-densities on N^*Y which in each adapted chart are of the form $\beta(y,\xi)|dy\,d\xi|^{1/2}$ with $\phi\beta \in S^{r-\frac{k}{2}}(\mathbb{R}^d;\mathbb{R}^k)$ for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$ of the chart. This is invariant under the fibre-linear maps (4.6.2), so is well defined; the shift by $k/2$ reflects the fact that $|d\xi|^{1/2}$ is homogeneous of degree $k/2$ under fibre dilations, so that the order r is the total homogeneity. Write $S^{r-[1]}(N^*Y;\Omega^{\frac12}) = S^r(N^*Y;\Omega^{\frac12})/S^{r-1}(N^*Y;\Omega^{\frac12})$. In the classical case the principal parts are smooth half-densities on $N^*Y \setminus 0$ homogeneous of degree r; we denote this space $\mathcal{C}^\infty_{(r)}(N^*Y \setminus 0; \Omega^{\frac12})$.

Theorem 4.6.4. *There is a unique map*

$$\sigma_m : I^m(X,Y;\Omega^{\frac12}) \longrightarrow S^{m+\frac{n}{4}-[1]}(N^*Y;\Omega^{\frac12})$$

such that if $u = (\mathcal{I}[a]+f)|dx\,dy|^{1/2}$ in an adapted chart then $\sigma_m(u) = a(y,\xi)|dy\,d\xi|^{1/2}$ there. It gives a short exact sequence

$$0 \longrightarrow I^{m-1}(X,Y;\Omega^{\frac12}) \longrightarrow I^m(X,Y;\Omega^{\frac12}) \xrightarrow{\sigma_m} S^{m+\frac{n}{4}-[1]}(N^*Y;\Omega^{\frac12}) \longrightarrow 0, \tag{4.6.3}$$

and, in the classical case, a short exact sequence

$$0 \to I^{m-1}_{\mathrm{ph}}(X,Y;\Omega^{\frac12}) \to I^m_{\mathrm{ph}}(X,Y;\Omega^{\frac12}) \xrightarrow{\sigma_m} \mathcal{C}^\infty_{(m+\frac{n}{4})}(N^*Y \setminus 0;\Omega^{\frac12}) \to 0.$$

Moreover $\sigma_m(\psi u) = (\psi|_Y)\sigma_m(u)$ for $\psi \in \mathcal{C}^\infty(X)$, where $\psi|_Y$ is lifted to N^*Y.

Proof. We must check that the local definition is consistent. Let F be a change of adapted coordinates. If $u = v|dz|^{1/2}$ then $F^*u = (F^*v)|\det F'|^{1/2}|dz'|^{1/2}$. At $x = 0$, F' is block triangular with diagonal blocks $A(y)$ and $g'(y)$, so $|\det F'(0,y)|^{1/2} = |\det A|^{1/2}|\det g'|^{1/2}$; by Theorem 4.4.4 and the multiplication rule in Proposition 4.3.4(v), the full symbol of F^*u is congruent modulo S^{M-1} to

$$|\det A(y)|^{-\frac12}|\det g'(y)|^{\frac12} a(g(y), A(y)^{-t}\eta).$$

On the other hand the Jacobian determinant of $F^\sharp$ in (4.6.2) is $\det g'(y) \det A(y)^{-1}$, so $(F^\sharp)^*(a|dy\,d\xi|^{1/2})$ is exactly this function times $|dy\,d\eta|^{1/2}$. Hence σ_m is well defined, with $\sigma_m(F^*u) = (F^\sharp)^*\sigma_m(u)$; the classical case is the same. The multiplicative property holds because $\psi(x,y)-\psi(0,y) = \sum_j x_j\psi_j$ and x_j lowers the order. Exactness in the middle is local and is Proposition 4.3.4(iii); injectivity is trivial. For surjectivity, given $b \in S^{m+\frac{n}{4}}(N^*Y;\Omega^{\frac12})$, cover Y by adapted charts U_j (a locally finite cover) with a partition of unity ϕ_j on Y, write $\phi_j b = \beta_j|dy\,d\xi|^{1/2}$ and let $u = \sum_j \chi_j \mathcal{I}[\beta_j]|dz|^{1/2}$, with $\chi_j \in \mathcal{C}^\infty_{\mathrm{c}}(U_j)$ equal to one near $\operatorname{supp}\phi_j \times \{0\}$; then $\sigma_m(u) = \sum_j \phi_j b = b$. □

4.6.4 Properties

We collect the main properties of the spaces $I^m(X,Y;\Omega^{\frac12})$; all hold equally for the classical spaces.

Theorem 4.6.5. *Let $Y \subset X$ be a closed embedded submanifold of codimension k, $\dim X = n$.*

(i) Elements of $I^m(X,Y;\Omega^{\frac12})$ are smooth outside Y, $I^{m-1} \subset I^m$ and $\bigcap_m I^m(X,Y;\Omega^{\frac12}) = \mathcal{C}^\infty(X;\Omega^{\frac12})$.

(ii) (Asymptotic completeness) *If $u_j \in I^{m-j}(X,Y;\Omega^{\frac12})$, there exists $u \in I^m(X,Y;\Omega^{\frac12})$ with $u - \sum_{j<N} u_j \in I^{m-N}$ for all N.*

(iii) (Invariance) *If $F : X' \to X$ is a diffeomorphism and $Y' = F^{-1}(Y)$, then $F^* : I^m(X,Y;\Omega^{\frac12}) \to I^m(X',Y';\Omega^{\frac12})$ and $\sigma_m(F^*u) = (F^\sharp)^*\sigma_m(u)$, where $F^\sharp : N^*Y' \to N^*Y$ is the inverse of the restriction of $F^* : T^*X \to T^*X'$. The same holds for diffeomorphisms between open subsets, for distributions supported in the range.*

(iv) (Differential operators) *If $P \in \operatorname{Diff}^q(X;\Omega^{\frac12})$ then $P : I^m(X,Y;\Omega^{\frac12}) \to I^{m+q}(X,Y;\Omega^{\frac12})$ and $\sigma_{m+q}(Pu) = \sigma_q(P)|_{N^*Y}\,\sigma_m(u)$, where $\sigma_q(P)$ is the principal symbol, characterized by $P(e^{itf}v) = e^{itf}(t^q\sigma_q(P)(df)v + O(t^{q-1}))$ as $t \to \infty$ for real $f \in \mathcal{C}^\infty(X)$ and $v \in \mathcal{C}^\infty_{\mathrm{c}}(X;\Omega^{\frac12})$.*

(v) (Wavefront set) $\operatorname{WF}(u) \subset N^*Y \setminus 0$ *for $u \in I^m(X,Y;\Omega^{\frac12})$.*

Proof. All statements are local and invariant, so follow from the local results: (i) from Proposition 4.3.4(ii); (ii) from Proposition 4.3.4(iv) and a partition of unity; (iii) from Theorems 4.4.4 and 4.6.4; (iv) from Proposition 4.3.4(v), since in coordinates $\sigma_q(P)(z,\zeta) = p_q(z,\zeta)$ (the half-density factor and the lower order terms only contribute terms of order $q-1$); (v) from Proposition 4.3.8. □

Remark 4.6.6. If one works with functions rather than half-densities, the computation in the proof of Theorem 4.6.4 shows that the full symbol transforms by (4.4.5), i.e. $a|d\xi|$ transforms as a fibre density on N^*Y: the principal symbol of $u \in I^m(X,Y)$ is then a section of $\Omega_{\mathrm{fib}}(N^*Y)$ with coefficient of order M, as in (4.5.1). Half-densities make the symbol a half-density on the symplectic manifold N^*Y, which is the form used in Chapter 8.

4.7 Iterated regularity

We now show that conormal distributions are characterized by a regularity property which is manifestly invariant and does not involve the Fourier transform. Let $\mathcal{V}(X,Y)$ denote the space of smooth vector fields on X tangent to Y; it is a Lie algebra and a $\mathcal{C}^\infty(X)$-module.

Lemma 4.7.1. *In the local model, $\mathcal{V}(\Omega, Y)$ is spanned over $\mathcal{C}^\infty(\Omega)$ by the vector fields $x_i\partial_{x_j}$, $1 \le i, j \le k$, and ∂_{y_l}, $1 \le l \le d$.*

Proof. A vector field $V = \sum_j a_j\partial_{x_j} + \sum_l b_l\partial_{y_l}$ is tangent to $\{x = 0\}$ if and only if $a_j(0, y) = 0$ for all j, that is, $a_j = \sum_i x_i a_{ij}$ with $a_{ij} \in \mathcal{C}^\infty$ (Taylor's formula with integral remainder in x, valid in a neighbourhood of each point of Y; away from Y use $\partial_{x_j} = \sum_i (x_i/|x|^2)x_i\partial_{x_j}$ and a partition of unity). □

Definition 4.7.2. For $s \in \mathbb{R}$ let

$$IH^s(X, Y; \Omega^{\frac12}) = \{u \in H^s_{\mathrm{loc}}(X; \Omega^{\frac12});\ \mathcal{L}_{V_1} \cdots \mathcal{L}_{V_j} u \in H^s_{\mathrm{loc}}(X; \Omega^{\frac12})\ \forall\ j,\ V_i \in \mathcal{V}(X, Y)\},$$

and $I^*(X, Y; \Omega^{\frac12}) = \bigcup_s IH^s(X, Y; \Omega^{\frac12})$.

Here H^s_{loc} is defined via coordinates and is invariant under diffeomorphisms (Chapter 3); the zeroth order terms in (4.6.1) are irrelevant since H^s_{loc} is a $\mathcal{C}^\infty$-module, so in coordinates we may replace $\mathcal{L}_V$ by V acting on coefficients. Thus IH^s is manifestly invariant under diffeomorphisms preserving Y, and $\bigcap_s IH^s(X, Y; \Omega^{\frac12}) = \mathcal{C}^\infty(X; \Omega^{\frac12})$ by Sobolev embedding.

Lemma 4.7.3. *Let $u \in H^s(\mathbb{R}^n)$ have compact support. Then $u \in IH^s(\mathbb{R}^n, Y)$ if and only if*

$$x^\alpha D_x^\beta D_y^\gamma u \in H^s(\mathbb{R}^n) \quad \text{for all } \alpha, \beta, \gamma \text{ with } |\beta| \le |\alpha|, \tag{4.7.1}$$

or equivalently if and only if $D_x^\beta D_y^\gamma x^\alpha u \in H^s$ for all $|\beta| \le |\alpha|$.

Proof. If $V_i \in \mathcal{V}$, then by Lemma 4.7.1, moving coefficients to the left using $[W, \psi] = W\psi$, $V_1 \cdots V_j u$ is a combination with smooth coefficients of products of at most j of the generators $x_i\partial_{x_j}$, ∂_{y_l} applied to u; and such products are linear combinations with constant coefficients of $x^\alpha D_x^\beta D_y^\gamma$ with $|\alpha| = |\beta|$. Conversely, by induction on $|\alpha|$, $x^\alpha D_x^\beta$ with $|\alpha| = |\beta|$ is a combination of products of the $x_i D_{x_j}$, since $x_i D_{x_j} x^{\alpha'} D_x^{\beta'} = x_i x^{\alpha'} D_{x_j} D_x^{\beta'} + x_i (D_{x_j} x^{\alpha'}) D_x^{\beta'}$ and the last term is shorter. The cases $|\beta| < |\alpha|$ follow by multiplying by $\chi x^{\alpha''}$, with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to one on $\operatorname{supp} u$; and $D^\beta x^\alpha$ is a combination of $x^{\alpha-\delta} D^{\beta-\delta}$, which preserves the condition $|\beta| \le |\alpha|$. □

The key analytic step converts weighted L^2 bounds on all derivatives into symbol estimates.

Lemma 4.7.4. *Let $t \in \mathbb{R}$, $j \in \mathbb{N}_0$, $L > j + \frac{k}{2}$, and let $b \in \mathcal{C}^\infty(\mathbb{R}^k)$ satisfy $\langle\xi\rangle^{t+|\alpha|} D^\alpha b \in L^2(\mathbb{R}^k)$ for $|\alpha| \le L$. Then*

$$|D^\beta b(\xi)| \le C\langle\xi\rangle^{-t-\frac{k}{2}-|\beta|}, \quad |\beta| \le j, \qquad C = C_{k,j,t,L} \sum_{|\alpha|\le L} \|\langle\xi\rangle^{t+|\alpha|} D^\alpha b\|_{L^2}.$$

Conversely, if $b \in S^\mu(\mathbb{R}^k)$ then $\langle\xi\rangle^{t+|\alpha|} D^\alpha b \in L^2$ for all α provided $t < -\mu - \frac{k}{2}$.

Proof. The converse is immediate since $\langle\xi\rangle^{t+\mu} \in L^2$ if $2(t + \mu) < -k$. For the direct statement, let $A = \{\frac12 < |\zeta| < 2\}$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(A)$ equal to one on $|\zeta| = 1$. For $R \ge 1$ set $b_R(\zeta) = b(R\zeta)$. On RA we have $R/2 \le \langle\xi\rangle \le 3R$, so

$$\|D^\alpha b_R\|^2_{L^2(A)} = R^{2|\alpha|-k} \int_{RA} |D^\alpha b|^2 d\xi \le CR^{2|\alpha|-k} R^{-2(t+|\alpha|)} \|\langle\xi\rangle^{t+|\alpha|} D^\alpha b\|^2_{L^2}.$$

Hence $\|\chi b_R\|_{H^L} \le CR^{-t-\frac{k}{2}}$, and by Sobolev's embedding $|D^\beta b_R(\zeta)| \le CR^{-t-\frac{k}{2}}$ for $|\zeta| = 1$, $|\beta| \le j$. Given $|\xi| \ge 1$ take $R = |\xi|$, $\zeta = \xi/|\xi|$: then $D^\beta b(\xi) = R^{-|\beta|}(D^\beta b_R)(\zeta)$, which gives the estimate. For $|\xi| \le 1$ apply Sobolev's embedding to $\chi_0 b$ with $\chi_0 \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to one on the unit ball. □

Theorem 4.7.5. *In the local model, for compactly supported distributions,*

$$I^m_{\mathrm{c}}(\mathbb{R}^n,Y)\subset IH^s(\mathbb{R}^n,Y)\quad \text{if } s<-m-\frac{n}{4},\qquad IH^s(\mathbb{R}^n,Y)\cap\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)\subset I_{\mathrm{c}}^{-s-\frac{n}{4}}(\mathbb{R}^n,Y). \tag{4.7.2}$$

Consequently, for any closed embedded submanifold $Y\subset X$,

$$IH^s(X,Y;\Omega^{\frac12})\subset I^{-s-\frac{n}{4}}(X,Y;\Omega^{\frac12})\subset IH^{s'}(X,Y;\Omega^{\frac12})\quad\forall\ s'<s,$$

so $I^*(X,Y;\Omega^{\frac12})=\bigcup_m I^m(X,Y;\Omega^{\frac12})$.

Proof. First inclusion. Let $u=\mathcal{I}[c]+f$ with $c\in S^M$ compactly supported in (x,y) and $f\in\mathcal{C}^\infty_{\mathrm{c}}$. By Lemma 4.3.1(i), each $x^\alpha D_x^\beta D_y^\gamma$ with $|\alpha|=|\beta|$ maps $\mathcal{I}[c]$ to some $\mathcal{I}[c']$ with c' of order M and compact support, so by Lemma 4.7.3 it suffices to show $\mathcal{I}[c]\in H^s$. Let $\hat c(\zeta',\eta;\xi)$ be the Fourier transform of $c(\cdot,\cdot,\xi)$ in (x,y); since c has compact support in (x,y), $|\hat c(\zeta',\eta;\xi)|\le C_N\langle\zeta'\rangle^{-N}\langle\eta\rangle^{-N}\langle\xi\rangle^M$. Then (first for $M<-k$, by Fubini, then in general by Lemma 4.1.1)

$$\begin{aligned}\widehat{\mathcal{I}[c]}(\zeta,\eta)&=\int\hat c(\zeta-\xi,\eta;\xi)\,đ\xi,\\ |\widehat{\mathcal{I}[c]}(\zeta,\eta)|&\le C_N\langle\eta\rangle^{-N}\int\langle\zeta-\xi\rangle^{-N}\langle\xi\rangle^M\,d\xi\le C'_N\langle\eta\rangle^{-N}\langle\zeta\rangle^M\end{aligned}$$

by Peetre's inequality, for $N>k+|M|$. Since $\langle(\zeta,\eta)\rangle^s\le\langle\zeta\rangle^s\langle\eta\rangle^{|s|}$, $\langle(\zeta,\eta)\rangle^s\widehat{\mathcal{I}[c]}\in L^2$ provided $2(s+M)<-k$, i.e. $s<-M-\frac{k}{2}=-m-\frac{n}{4}$.

Second inclusion. Let $u\in IH^s$ have compact support, so $\hat u$ is smooth. By Lemma 4.7.3 and $\mathcal{F}(x^\alpha u)=(-D_\zeta)^\alpha\hat u$, $\zeta^\beta\eta^\gamma D_\zeta^\alpha\hat u\in\langle(\zeta,\eta)\rangle^{-s}L^2$ for $|\beta|\le|\alpha|$ and all γ. Since $\langle\zeta\rangle^{|\alpha|}\le C\sum_{|\beta|\le|\alpha|}|\zeta^\beta|$, similarly for $\langle\eta\rangle^N$, and $\langle(\zeta,\eta)\rangle^s\ge c\langle\zeta\rangle^s\langle\eta\rangle^{-|s|}$, we get

$$\langle\zeta\rangle^{s+|\alpha|}\langle\eta\rangle^N D_\zeta^\alpha\hat u(\zeta,\eta)\in L^2(\mathbb{R}^n)\quad\forall\ \alpha,N. \tag{4.7.3}$$

Let $a(y,\zeta)=\int e^{iy\cdot\eta}\hat u(\zeta,\eta)đ\eta$ be the partial Fourier transform of u in x; it is smooth, has compact support in y, and $u=\mathcal{I}[a]$. By Minkowski's and the Cauchy–Schwarz inequalities, for $N>|\gamma|+\frac{d}{2}$,

$$\|\langle\zeta\rangle^{s+|\alpha|}D_\zeta^\alpha D_y^\gamma a(y,\cdot)\|_{L^2}\le\int|\eta|^{|\gamma|}\|\langle\zeta\rangle^{s+|\alpha|}D_\zeta^\alpha\hat u(\cdot,\eta)\|_{L^2}đ\eta\le C\|\langle\eta\rangle^N\langle\zeta\rangle^{s+|\alpha|}D_\zeta^\alpha\hat u\|_{L^2(\mathbb{R}^n)},$$

uniformly in y. Applying Lemma 4.7.4 with $t=s$ to $D_y^\gamma a(y,\cdot)$ gives $|D_\zeta^\beta D_y^\gamma a|\le C\langle\zeta\rangle^{-s-\frac{k}{2}-|\beta|}$, so $a\in S^{-s-\frac{k}{2}}_{\mathrm{c}}$, which by (4.3.1) means $u\in I_{\mathrm{c}}^{-s-\frac{n}{4}}$.

The global statement follows by localization in adapted charts, using the invariance of IH^s and of I^m. Note that the middle space in the global chain is $I^{-s-\frac{n}{4}}$, and $I^{-s-\frac{n}{4}}\subset IH^{s'}$ for $s'<s$ by the first inclusion. □

Remark 4.7.6. Theorem 4.7.5 gives a second proof that $I^*(X,Y)$ is invariant under diffeomorphisms, and that the order is invariant up to an arbitrarily small loss; the sharp invariance of the order and of the principal symbol requires Theorem 4.4.4. For $Y=\{0\}$ the theorem is the statement that $D_\zeta^\alpha\hat u\in\langle\zeta\rangle^{-s-|\alpha|}L^2$ for all α implies $\hat u\in S^{-s-\frac{n}{2}}$, and conversely $S^\mu\subset\langle\zeta\rangle^{-s-|\alpha|}L^2$-type spaces for $s<-\mu-\frac{n}{2}$.

4.8 Examples

Example 4.8.1 (Delta sections)**.** In the local model, $\delta(x) = \delta(x_1)\cdots\delta(x_k) = \mathcal{I}[1]$, the full symbol being $1 \in S^0$; so $\delta(x) \in I^m_{\mathrm{ph}}(\mathbb{R}^n, Y)$ with $M = 0$, i.e. $m = \frac{k}{2} - \frac{n}{4} = \frac{1}{4}(\operatorname{codim} Y - \dim Y)$. Invariantly: the restriction to Y of the density bundle of X splits as $\Omega_Y X = \Omega Y \otimes \Omega(NY)$, where $\Omega(NY)$ is the bundle of densities on the fibres of the normal bundle. If $0 < \mu \in \mathcal{C}^\infty(Y; \Omega(NY))$ is a positive normal density, then

$$u_\mu : \mathcal{C}^\infty_{\mathrm{c}}(X; \Omega X) \ni \nu \longmapsto \int_Y \nu|_Y/\mu$$

is a well-defined distribution, and in adapted coordinates in which $\mu = \mu(y)|dx|$ it is $\mu(y)^{-1}\delta(x)$. So $u_\mu \in I^{\frac{1}{4}(k-d)}_{\mathrm{ph}}(X, Y)$ with full symbol $\mu(y)^{-1}$; with the normalization of Remark 4.6.6, its principal symbol is $\mu(y)^{-1}|d\xi|$, the fibre density on N^*Y dual to μ.

Example 4.8.2 (Poincaré dual forms)**.** Conormal sections of vector bundles are defined in Definition 4.6.1. Let X be a compact oriented manifold of dimension n and $Y \subset X$ a closed embedded oriented submanifold of dimension d. Near any point of Y choose oriented coordinates in which $Y = \{z_{d+1} = \cdots = z_n = 0\}$ and $(z_1, \ldots, z_d)$ are oriented coordinates on Y, and set

$$p_Y = \delta(z_{d+1})\cdots\delta(z_n)\, dz_{d+1} \wedge \cdots \wedge dz_n \in I^{\frac{1}{4}(n-2d)}_{\mathrm{ph}}(X, Y; \Lambda^{n-d}).$$

This is independent of the choices, because it is characterized by

$$\int_X \phi \wedge p_Y = \int_Y \phi \qquad \forall\ \phi \in \mathcal{C}^\infty(X; \Lambda^d),$$

as one checks in the given coordinates ($\phi \wedge p_Y$ only involves the coefficient of $dz_1 \wedge \cdots \wedge dz_d$ in ϕ). It follows that $dp_Y = 0$ in the distributional sense: for $\psi \in \mathcal{C}^\infty(X; \Lambda^{d-1})$, $\int_X \psi \wedge dp_Y = \pm \int_X d\psi \wedge p_Y = \pm \int_Y d\psi = 0$ by Stokes' theorem. The distributional de Rham cohomology, defined by the complex $d : \mathcal{C}^{-\infty}(X; \Lambda^j) \to \mathcal{C}^{-\infty}(X; \Lambda^{j+1})$, is canonically isomorphic to the smooth de Rham cohomology (this follows, for instance, from the Hodge theorem), and the class of p_Y in $H^{n-d}_{\mathrm{dR}}(X)$ is, by the displayed identity, the Poincaré dual of Y.

4.9 Exercises

Exercise 4.1. Show by induction that $a \in S^0(\mathbb{R}^n)$ if and only if $a \in \mathcal{C}^\infty(\mathbb{R}^n)$ and $V_1 \cdots V_N a \in L^\infty(\mathbb{R}^n)$ for all N and all choices of $V_j \in \{\xi_k \partial_{\xi_l}\}$ (including $N = 0$). (Hint: in $|\xi| \geq 1$, $\partial_{\xi_l} = \sum_k \xi_k |\xi|^{-2}\, \xi_k \partial_{\xi_l}$.)

Exercise 4.2. For a compact manifold with boundary X let $\mathcal{V}_{\mathrm{b}}(X)$ be the Lie algebra of smooth vector fields tangent to ∂X and

$$\mathcal{A}(X) = \{a \in \mathcal{C}^\infty(X \setminus \partial X) \cap L^\infty;\ V_1 \cdots V_N a \in L^\infty\ \forall\ N,\ V_i \in \mathcal{V}_{\mathrm{b}}(X)\}.$$

Using that the linear vector fields $\xi_k \partial_{\xi_l}$ span $\mathcal{V}_{\mathrm{b}}(\overline{\mathbb{R}^n})$ near the boundary, show that $S^m(\mathbb{R}^n) = \rho^{-m}\mathcal{A}(\overline{\mathbb{R}^n})$.

Exercise 4.3. Let W be a real vector space of dimension n. Define the sphere of W as $\mathbb{S}W = (W \setminus 0)/\mathbb{R}^+$. Show that $[W, \{0\}] = (W \setminus 0) \sqcup \mathbb{S}W$ has a unique structure of manifold with boundary, diffeomorphic to $[0, \infty) \times \mathbb{S}^{n-1}$, such that the blow-down map $\beta : [W, \{0\}] \to W$ (the identity on $W \setminus 0$, mapping $\mathbb{S}W$ to 0) is smooth and the action of $\mathrm{GL}(W)$ lifts to a smooth action (use Lemma 4.2.4).

Exercise 4.4. Let U be a finite-dimensional complex vector space and set $I^m(W,\{0\};U) = I^m(W,\{0\})\otimes U$. Show that the Fourier transform gives an isomorphism $I^m(W,\{0\};U)\to S^{m-\frac{n}{4}}(W';U\otimes\Omega W')$ (with $n=\dim W$), that $\Omega U'\otimes\Omega U$ is canonically trivial for a real vector space U, and that the integral of Lemma 4.2.2(iv) defines a linearly invariant map $I^m(W,\{0\};\Omega W)\to\mathbb{C}$.

Exercise 4.5. Let $u\in\mathcal{C}^{-\infty}(\mathbb{R}^n)$ be smooth outside 0 and homogeneous modulo $\mathcal{C}^\infty$ of degree h: $u(tz)=t^hu(z)+F(t,z)$ for $t>0$, $z\neq 0$, with $F\in\mathcal{C}^\infty((0,\infty)\times\mathbb{R}^n)$. Show that χu is a classical conormal distribution at 0 for $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, and determine its order.

Exercise 4.6. Show that every $u\in I^m(X,Y)$ with $\operatorname{supp}u\subset Y$ is of the form Pu_μ for some $P\in\mathrm{Diff}^q(X)$, with u_μ as in Example 4.8.1, at least locally. Which orders m can occur?

Exercise 4.7. Let $u\in I^m(X,Y;\Omega^{\frac12})$ be classical, with principal symbol nonvanishing on $N^*Y\setminus 0$. Show that $\mathrm{WF}(u)=N^*Y\setminus 0$, and that if $A\in\Psi^0(\mathbb{R}^n)$ satisfies $Au\in\mathcal{C}^\infty$ then the principal symbol of A vanishes on N^*Y (locally).

Notes

This chapter merges the treatment of conormal distributions in the 1998 lecture notes (where they are introduced through iterated regularity) with that of the 2005 lectures (where they are defined by Fourier transform of classical symbols, first at a point, then at the zero section of a vector bundle, then at a submanifold via a normal fibration). The opening argument of the earlier notes, relating iterated regularity to symbol estimates, has been completed (Lemma 4.7.4 and Theorem 4.7.5), and the invariance under diffeomorphisms is proved by Moser's homotopy method in the sharp form $H^*u-u\in I^{m-1}$, which yields the invariance of the principal symbol without the loss ϵ inherent in the Sobolev approach. We use Hörmander's normalization of the order, for which $\delta_0\in I^{n/4}(\mathbb{R}^n,\{0\})$ and the kernels of pseudodifferential operators of order m have order m; some formulas for the order in the 2005 lectures contain sign slips which have been corrected. The standard reference is [8, Section 18.2]; see also [6] for the more general Lagrangian distributions, of which conormal distributions are the special case associated to the Lagrangian submanifold N^*Y.

Chapter 5

Pseudodifferential operators on manifolds

In this chapter pseudodifferential operators are transferred from $\mathbb{R}^n$ to a compact manifold X, acting between sections of vector bundles. Two definitions are given and shown to agree: the invariant one, in which an operator of order m is identified with its Schwartz kernel, a distribution on X^2 conormal of order m to the diagonal (using the conormal distributions of Chapter 4); and the local one, in which an operator is required to be, in every coordinate patch, an element of the calculus on $\mathbb{R}^n$ of Chapter 2, and to have a kernel smooth away from the diagonal. With the coordinate invariance established in Chapters 2 and 3, the whole symbol calculus carries over. We then construct parametrices for elliptic operators, introduce Sobolev spaces on X, and prove that elliptic operators are Fredholm, with generalized inverses which are again pseudodifferential. Smoothing operators, which carry all the "errors" of the calculus, are discussed in some detail: they form an ideal of "infinite matrices" with a trace and a group of invertible elements, $G^{-\infty}$, which is the starting point for the index theory of later chapters. The main applications here are the Hodge theorem for elliptic complexes, in particular for the de Rham complex, and the existence of pseudodifferential projections onto prescribed subspaces. The facts about bounded operators on Hilbert space which are used are collected, with proofs, in Appendix A.

Throughout, unless explicitly stated otherwise, X is a *compact* manifold of dimension n; several definitions extend to the non-compact case with minor changes, which are indicated in remarks.

5.1 Smooth manifolds

5.1.1 $\mathcal{C}^\infty$ structures

Let X be a paracompact Hausdorff topological space. We describe the smooth structure of X through its algebra of smooth functions.

Definition 5.1.1. A $\mathcal{C}^\infty$ *structure* on X is a subspace $\mathcal{F} \subset \mathcal{C}^0(X;\mathbb{R})$ with the following property: for each $\bar{x} \in X$ there exist $f_1, \dots, f_n \in \mathcal{F}$ and an open neighbourhood $\Omega \ni \bar{x}$ such that

$$\kappa : \Omega \ni x \longmapsto (f_1(x), \dots, f_n(x)) \in \mathbb{R}^n \tag{5.1.1}$$

is a homeomorphism onto an open subset of $\mathbb{R}^n$ and every $f \in \mathcal{F}$ satisfies $f|_\Omega = g \circ \kappa$ for some $g \in \mathcal{C}^\infty(\kappa(\Omega))$. The map κ is then called a *coordinate system* (or chart) near $\bar{x}$. Two $\mathcal{C}^\infty$ structures $\mathcal{F}_1$, $\mathcal{F}_2$ are *compatible* if $\mathcal{F}_1 \cup \mathcal{F}_2$ is also a $\mathcal{C}^\infty$ structure.

Compatibility is an equivalence relation, and each equivalence class contains a unique maximal element, the union of the class (Exercise 5.1).

Definition 5.1.2. A $\mathcal{C}^\infty$ *manifold* is a paracompact Hausdorff space X with a maximal $\mathcal{C}^\infty$ structure, which is denoted $\mathcal{C}^\infty(X) \subset \mathcal{C}^0(X)$.

The maximal structure is an algebra (Exercise 5.2), and the number n of coordinate functions is locally constant; we always assume it is constant, $n = \dim X$. If $\kappa : \Omega \to \kappa(\Omega)$ is a coordinate system in X and $\mathcal{C}^\infty_{\mathrm{c}}(\Omega) \subset \mathcal{C}^\infty(X)$ denotes the space of smooth functions vanishing outside a compact subset of Ω, then

$$\kappa^* : \mathcal{C}^\infty_{\mathrm{c}}(\kappa(\Omega)) \longrightarrow \mathcal{C}^\infty_{\mathrm{c}}(\Omega) = \{u \in \mathcal{C}^\infty(X);\ u = 0 \text{ on } X \setminus K,\ K \Subset \Omega\}$$

is an isomorphism. Moreover $\mathcal{C}^\infty(X)$ is *local*: if $u : X \to \mathbb{R}$ and each $\bar{x} \in X$ has an open neighbourhood $\Omega_{\bar{x}}$ on which $u = f_{\bar{x}}$ for some $f_{\bar{x}} \in \mathcal{C}^\infty(X)$, then $u \in \mathcal{C}^\infty(X)$. The transition maps $\kappa' \circ \kappa^{-1}$ between two coordinate systems are diffeomorphisms between open subsets of $\mathbb{R}^n$, so this definition agrees with the usual one by atlases. A map $G : X \to Y$ between manifolds is $\mathcal{C}^\infty$ if $G^*\mathcal{C}^\infty(Y) \subset \mathcal{C}^\infty(X)$, that is, $u \circ G \in \mathcal{C}^\infty(X)$ for all $u \in \mathcal{C}^\infty(Y)$. Complex-valued smooth functions are those with real and imaginary parts in $\mathcal{C}^\infty(X)$.

5.1.2 Partitions of unity

Any $\mathcal{C}^\infty$ manifold admits partitions of unity subordinate to any open cover (Exercise 5.3). For compact manifolds this is elementary: for each point p choose a coordinate system near p inside an element of the cover and pull back a non-negative bump function on $\mathbb{R}^n$ which is positive at the image of p; by locality this gives an element of $\mathcal{C}^\infty(X)$. Finitely many of these are positive on a set covering X, and dividing each of them by their (positive) sum gives a finite partition of unity subordinate to the cover. For localizing operators a slightly stronger property is convenient.

Lemma 5.1.3. *If $\{U_i\}$ is an open cover of a compact manifold X there is a finite partition of unity $\{\phi_a\}$, $0 \le \phi_a \le 1$, $\sum_a \phi_a = 1$, each $\operatorname{supp}\phi_a$ contained in some U_i, such that*

$$\operatorname{supp}\phi_a \cap \operatorname{supp}\phi_b \neq \emptyset \Longrightarrow \operatorname{supp}\phi_a \cup \operatorname{supp}\phi_b \subset U_i \quad \textit{for some } i. \tag{5.1.2}$$

Proof. Choose a finite partition of unity $\{\chi_i\}$ with $\operatorname{supp}\chi_i \subset U_i$ (passing to a finite subcover). For $p \in X$ set

$$V_p = \bigcap_{i:\, p \in \operatorname{supp}\chi_i} U_i \ \setminus \bigcup_{i:\, p \notin \operatorname{supp}\chi_i} \operatorname{supp}\chi_i,$$

an open neighbourhood of p with the property that $V_p \cap \operatorname{supp}\chi_i \neq \emptyset$ implies $p \in \operatorname{supp}\chi_i$ and hence $V_p \subset U_i$. Choose finitely many p_j such that the $V_j = V_{p_j}$ cover X, a partition of unity $\{\psi_j\}$ with $\operatorname{supp}\psi_j \subset V_j$, and set $\phi_{ij} = \chi_i\psi_j$. If $\operatorname{supp}\phi_{ij} \cap \operatorname{supp}\phi_{i'j'} \neq \emptyset$ then $\operatorname{supp}\psi_j$ meets $\operatorname{supp}\chi_{i'}$, so $V_j \subset U_{i'}$, and therefore $\operatorname{supp}\phi_{ij} \subset U_{i'}$; since also $\operatorname{supp}\phi_{i'j'} \subset \operatorname{supp}\chi_{i'} \subset U_{i'}$, the union of the two supports lies in $U_{i'}$. □

5.2 Vector bundles, densities and distributions

5.2.1 Vector bundles

A (complex) *vector bundle* of rank N over X is a manifold E with a smooth surjection $\pi : E \to X$, each fibre $E_x = \pi^{-1}(x)$ carrying the structure of an N-dimensional complex vector space, such that each point of X has an open neighbourhood U and a diffeomorphism

$\pi^{-1}(U) \to U \times \mathbb{C}^N$ which is fibrewise linear and covers the identity (a *local trivialization*). Two trivializations differ by a smooth map $U \cap U' \to \mathrm{GL}(N, \mathbb{C})$. Real vector bundles are defined in the same way. The space of smooth sections is

$$\mathcal{C}^\infty(X; E) = \{u : X \to E \text{ smooth};\ \pi \circ u = \mathrm{Id}\};$$

it is a module over $\mathcal{C}^\infty(X)$, and over a trivializing open set a section is just an N-tuple of smooth functions. We shall often use charts $\kappa : W \to \kappa(W)$ over which the bundles under consideration are trivial; we call these *trivializing charts*. Since X is compact, it is covered by finitely many of them.

The cotangent bundle

For $\bar{x} \in X$ let $\mathcal{I}_{\bar{x}} = \{u \in \mathcal{C}^\infty(X); u(\bar{x}) = 0\}$ and let $\mathcal{I}^2_{\bar{x}}$ be the space of finite sums of products of pairs of elements of $\mathcal{I}_{\bar{x}}$. The cotangent space is

$$T^*_{\bar{x}} X = \mathcal{I}_{\bar{x}} / \mathcal{I}^2_{\bar{x}}, \qquad T^*X = \bigcup_{\bar{x} \in X} T^*_{\bar{x}} X,$$

and the differential of $f \in \mathcal{C}^\infty(X)$ is the map $df : X \to T^*X$, $df(\bar{x}) = [f - f(\bar{x})] \in T^*_{\bar{x}} X$. In a coordinate system $x_1, \dots, x_n$ near $\bar{x}$, Taylor's formula with integral remainder shows that $f - f(\bar{x}) - \sum_j \partial_j f(\bar{x})(x_j - \bar{x}_j) \in \mathcal{I}^2_{\bar{x}}$ (after multiplication by a cutoff equal to one near $\bar{x}$), so $dx_1, \dots, dx_n$ is a basis of $T^*_{\bar{x}} X$ and $df = \sum_j \partial_j f \, dx_j$. The coordinates $(x, \xi) \mapsto \sum_j \xi_j \, dx_j$ make T^*X a real vector bundle; if $y = y(x)$ is another coordinate system, the corresponding fibre coordinates are related by

$$\xi = \Big(\frac{\partial y}{\partial x}\Big)^T \eta, \quad \text{i.e. } \sum_j \xi_j \, dx_j = \sum_k \eta_k \, dy_k, \tag{5.2.1}$$

which is linear in the fibre variables with smooth invertible coefficients. Each df is a smooth section of T^*X.

Operations on bundles

Any functorial operation on finite-dimensional vector spaces produces new bundles from old, the transition functions being transformed accordingly. In particular we have the tangent bundle $T_x X = (T^*_x X)^*$, the dual bundle E', the bundles $E \otimes F$, $\mathrm{Hom}(E, F) = F \otimes E'$, $\mathrm{End}(E) = \mathrm{Hom}(E, E)$, and the exterior powers

$$\Lambda^k_x X = \{u : (T_x X)^k \to \mathbb{C} \text{ multilinear and antisymmetric}\}, \qquad \Lambda^1 X = T^*X \otimes \mathbb{C},$$

whose sections are the k-forms. If $E \to Y$ is a vector bundle and $G : X \to Y$ is smooth, the *pull-back bundle* $G^*E = \bigcup_{x \in X} E_{G(x)}$ is a smooth vector bundle over X; if $\phi \in \mathcal{C}^\infty(Y; E)$ then $G^*\phi \in \mathcal{C}^\infty(X; G^*E)$ and $\mathcal{C}^\infty(X; G^*E)$ is spanned over $\mathcal{C}^\infty(X)$ by $G^*\mathcal{C}^\infty(Y; E)$. If $E \to X$ and $F \to Y$ are bundles, the *exterior tensor product* $F \boxtimes E \to X \times Y$ has fibre $(F \boxtimes E)_{(x,y)} = F_x \otimes E_y$. On X^2 we use the projections $\pi_L(x, y) = x$ and $\pi_R(x, y) = y$, and the "big" homomorphism bundle

$$\mathrm{Hom}(E, F) = \pi_L^* F \otimes \pi_R^* E' = F \boxtimes E', \qquad \mathrm{Hom}(E, F)_{(x,y)} = \hom(E_y, F_x), \tag{5.2.2}$$

whose restriction to the diagonal is $\hom(E, F)$. A *hermitian metric* on E is a smoothly varying inner product on the fibres; one exists by a partition of unity argument.

5.2.2 Densities and integration

For $\alpha \in \mathbb{R}$ the bundle of α-densities is

$$\Omega^\alpha_x X = \{u : \Lambda^n T_x X \setminus 0 \to \mathbb{C};\ u(t\,v) = |t|^\alpha u(v)\ \forall t \in \mathbb{R} \setminus 0\},$$

a line bundle, and $\Omega = \Omega^1$. In local coordinates an α-density is $u = g(x)|dx|^\alpha$, and under a change of coordinates $y = y(x)$ the coefficient transforms by $g \mapsto g\,|\det(\partial y/\partial x)|^\alpha$. Thus, by the change of variables formula, if $u \in \mathcal{C}^\infty_{\mathrm{c}}(W;\Omega)$ is supported in a coordinate patch then $\int u = \int_{\mathbb{R}^n} g(x)\,dx$ is independent of the coordinates, and using a partition of unity the integral

$$\int_X : \mathcal{C}^\infty(X;\Omega) \longrightarrow \mathbb{C}$$

is well defined. Densities, unlike forms, require no orientation. A *smooth positive density* ν (one which is a positive multiple of $|dx|$ in every coordinate system) exists, again by a partition of unity, and any other density is $f\nu$ with $f \in \mathcal{C}^\infty(X)$. Given ν and a hermitian metric on E, the space $L^2(X;E)$ is the completion of $\mathcal{C}^\infty(X;E)$ with respect to

$$(u,v) = \int_X \langle u(x), v(x)\rangle_{E_x}\,\nu(x). \tag{5.2.3}$$

Different choices give the same space with equivalent norms. On X^2 we use the *right density bundle* $\Omega_R = \pi_R^*\Omega$; a section of $\mathrm{Hom}(E,F)\otimes\Omega_R$ is in local coordinates and trivializations a matrix $k(x,y)|dy|$.

5.2.3 Distributional sections

The topology on $\mathcal{C}^\infty(X;E)$ is given by the seminorms $\sup|D^\alpha(\kappa_i^{-1})^*(\phi_i u)|$, $\alpha \in \mathbb{N}_0^n$, for a finite cover by trivializing charts κ_i and a subordinate partition of unity ϕ_i (the N-tuples of functions being estimated componentwise); this makes $\mathcal{C}^\infty(X;E)$ a Fréchet space and is independent of choices. The space of distributional sections is the dual space

$$\mathcal{C}^{-\infty}(X;E) = \big(\mathcal{C}^\infty(X;E'\otimes\Omega)\big)', \tag{5.2.4}$$

with the weak topology. Integration of the pairing gives the inclusion $\mathcal{C}^\infty(X;E) \hookrightarrow \mathcal{C}^{-\infty}(X;E)$, $u \mapsto (\phi \mapsto \int_X \phi(u))$, and $\mathcal{C}^\infty(X;E)$ is dense. Multiplication by $\mathcal{C}^\infty(X)$ and the restriction to open sets are defined by duality; if $u \in \mathcal{C}^{-\infty}(X;E)$ and $\kappa : W \to \kappa(W)$ is a trivializing chart, then for $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ the localization ψu is represented by an N-tuple of compactly supported distributions on $\kappa(W)$, which we write $\kappa_*(\psi u) = (\kappa^{-1})^*(\psi u)$.

Distributional sections can equally be defined "algebraically".

Proposition 5.2.1. *For any vector bundle U over the compact manifold Y there is a natural isomorphism*

$$\mathcal{C}^{-\infty}(Y) \otimes_{\mathcal{C}^\infty(Y)} \mathcal{C}^\infty(Y;U) \cong \mathcal{C}^{-\infty}(Y;U).$$

Proof. The map $\sum_i u_i \otimes g_i \mapsto \sum_i u_i g_i$, where $(ug)(\psi) = u(g\cdot\psi)$ for $\psi \in \mathcal{C}^\infty(Y;U'\otimes\Omega)$ (so that $g\cdot\psi \in \mathcal{C}^\infty(Y;\Omega)$), is well defined on the tensor product since it is $\mathcal{C}^\infty(Y)$-balanced. To invert it, choose finitely many sections $g_i \in \mathcal{C}^\infty(Y;U)$ and $q_i \in \mathcal{C}^\infty(Y;U')$ with

$$\sum_i g_i \otimes q_i = \mathrm{Id}_U \in \mathcal{C}^\infty(Y;\mathrm{End}(U)). \tag{5.2.5}$$

These exist: take a finite partition of unity ρ_k subordinate to a cover by trivializing open sets U_k, local frames $e_{k,l}$ over U_k with dual frames $e^{k,l}$, functions $\psi_k \in \mathcal{C}^\infty_{\mathrm{c}}(U_k)$

equal to one on $\operatorname{supp}\rho_k$, and use the sections $\rho_k e_{k,l}$ and $\psi_k e^{k,l}$, extended by zero. Then $\sum_k\sum_l \rho_k e_{k,l}\otimes\psi_k e^{k,l} = \sum_k \rho_k\,\mathrm{Id} = \mathrm{Id}$. Define $u\mapsto \sum_i (u\cdot q_i)\otimes g_i$ for $u\in\mathcal{C}^{-\infty}(Y;U)$, where $u\cdot q_i\in\mathcal{C}^{-\infty}(Y)$ is the pairing. By (5.2.5), $\sum_i (u\cdot q_i)g_i = u$, while for $u=\sum_j u_j\otimes h_j$, $\sum_i\sum_j u_j(h_j\cdot q_i)\otimes g_i = \sum_j u_j\otimes\sum_i (h_j\cdot q_i)g_i = \sum_j u_j\otimes h_j$, using the $\mathcal{C}^\infty(Y)$-balancing. So the two maps are mutually inverse. □

The same argument shows that any $\mathcal{C}^\infty(Y)$-module of "distributions" which is local and invariant under multiplication by smooth functions, such as the conormal spaces below, can be extended to sections of bundles by tensoring.

5.3 Euclidean and conormal preliminaries

5.3.1 The calculus on $\mathbb{R}^n$

We collect the results from Chapters 2 and 3 which are used below. Here $\Psi^m(\mathbb{R}^n)$ is the space of operators $\mathrm{Op}(a)$, $a\in S^m(\mathbb{R}^n;\mathbb{R}^n)$, with kernels $K(x,y)=\int e^{i(x-y)\cdot\xi}a(x,\xi)đ\xi$.

(E1) $\Psi^m(\mathbb{R}^n)\subset\Psi^{m'}(\mathbb{R}^n)$ for $m\le m'$; $\Psi^{-\infty}(\mathbb{R}^n)=\bigcap_m\Psi^m(\mathbb{R}^n)$ consists of operators with smooth kernels, and every operator with kernel in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$ lies in $\Psi^{-\infty}(\mathbb{R}^n)$. Kernels of elements of $\Psi^m(\mathbb{R}^n)$ are smooth off the diagonal.

(E2) (Amplitudes) If $a(x,y,\xi)$ satisfies $|D_x^\alpha D_y^\beta D_\xi^\gamma a|\le C\langle\xi\rangle^{m-|\gamma|}$ then the operator with kernel $\int e^{i(x-y)\cdot\xi}a(x,y,\xi)đ\xi$ lies in $\Psi^m(\mathbb{R}^n)$, with principal symbol $a(x,x,\xi)$. Equivalently, every $A\in\Psi^m(\mathbb{R}^n)$ has a *left* symbol $a_L(x,\xi)$ and a *right* symbol $a_R(y,\xi)$, both in S^m, with $\sigma_m(A)=[a_L]=[a_R]$. In particular $\phi A\psi\in\Psi^m(\mathbb{R}^n)$ for $\phi,\psi\in\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, and the kernel of A may be multiplied by $\chi(x-y)$, $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near 0, modulo $\Psi^{-\infty}(\mathbb{R}^n)$.

(E3) (Composition and adjoints) $\Psi^m(\mathbb{R}^n)\circ\Psi^{m'}(\mathbb{R}^n)\subset\Psi^{m+m'}(\mathbb{R}^n)$ with $\sigma_{m+m'}(AB)=\sigma_m(A)\sigma_{m'}(B)$; the L^2 adjoint satisfies $A^*\in\Psi^m(\mathbb{R}^n)$, $\sigma_m(A^*)=\overline{\sigma_m(A)}$.

(E4) (Principal symbol) The sequence $0\to\Psi^{m-1}(\mathbb{R}^n)\to\Psi^m(\mathbb{R}^n)\xrightarrow{\sigma_m}S^m/S^{m-1}\to 0$ is exact, with $\sigma_m(\mathrm{Op}(a))=[a]$.

(E5) (Asymptotic completeness) If $A_j\in\Psi^{m-j}(\mathbb{R}^n)$ there exists $A\in\Psi^m(\mathbb{R}^n)$ with $A-\sum_{j<N}A_j\in\Psi^{m-N}(\mathbb{R}^n)$ for all N. If the kernels of the A_j have support in a fixed compact set K, A may be chosen with kernel supported in any given neighbourhood of K (replace A by $\chi A\chi$ with $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near both projections of K).

(E6) (Sobolev boundedness) Each $A\in\Psi^m(\mathbb{R}^n)$ is bounded $H^s(\mathbb{R}^n)\to H^{s-m}(\mathbb{R}^n)$ for every $s\in\mathbb{R}$; in particular $\Psi^0(\mathbb{R}^n)$ acts boundedly on $L^2(\mathbb{R}^n)$, and multiplication by $\phi\in\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is bounded on every $H^s(\mathbb{R}^n)$.

(E7) (Coordinate invariance) Let $\Phi:\Omega\to\Omega'$ be a diffeomorphism between open subsets of $\mathbb{R}^n$ and $A\in\Psi^m(\mathbb{R}^n)$ have kernel with compact support in $\Omega\times\Omega$. Then $\Phi_*A=(\Phi^{-1})^*\circ A\circ\Phi^*$, extended by zero, lies in $\Psi^m(\mathbb{R}^n)$, and

$$\sigma_m(\Phi_*A)(\Phi(x),\eta)=\sigma_m(A)(x,(d\Phi_x)^T\eta). \tag{5.3.1}$$

The symbol formula (5.3.1) follows from the proof of coordinate invariance: writing $G=\Phi^{-1}$ and $G(x)-G(y)=T(x,y)(x-y)$, the kernel of Φ_*A near the diagonal is given by the amplitude $c(x,y,\eta)=a(G(x),T(x,y)^{-T}\eta)\,|\det dG(y)|\,|\det T(x,y)|^{-1}$, and since $T(x,x)=dG(x)$ its value on the diagonal is $a(G(x),(dG_x)^{-T}\eta)$, which is (5.3.1) because $(dG_{\Phi(x')})^{-T}=(d\Phi_{x'})^T$. Formula (5.3.1) says precisely that the principal symbol is invariantly a function on the cotangent bundle: it is the pull-back under the induced map $T^*\Omega'\to T^*\Omega$.

5.3.2 Conormal distributions

We recall the definition from Chapter 4. Let M be a manifold of dimension N and $Y \subset M$ a closed embedded submanifold of codimension k. The space $I^m(M,Y)$ of distributions (or distributional densities) conormal to Y of order m consists of those u which are smooth on $M \setminus Y$ and such that, near each point of Y, in local coordinates $(w,z) \in \mathbb{R}^{N-k} \times \mathbb{R}^k$ in which $Y = \{z = 0\}$,

$$u = \int_{\mathbb{R}^k} e^{iz\cdot\zeta} a(w,\zeta)\,\text{đ}\zeta + v, \quad a \in S^{m+\frac{N}{4}-\frac{k}{2}}(\mathbb{R}^{N-k};\mathbb{R}^k),\ v \text{ smooth}. \tag{5.3.2}$$

(For distributional densities the formula is multiplied by the coordinate density $|dw\,dz|$.) The subspace in which the amplitudes a are classical symbols is denoted $I^m_{\mathrm{ph}}(M,Y)$. We use the following properties.

(C1) $I^m(M,Y)$ is a $\mathcal{C}^\infty(M)$-module, is local, and is invariant under diffeomorphisms: if $\Phi : M \to M'$ is a diffeomorphism with $\Phi(Y) = Y'$ then $\Phi^* I^m(M',Y') = I^m(M,Y)$. Moreover $I^m \subset I^{m'}$ for $m \le m'$ and $\bigcap_m I^m(M,Y) = \mathcal{C}^\infty(M)$.

(C2) The amplitude a in (5.3.2) may be allowed to depend on z as well, with symbol estimates uniform on compact sets in (w,z), without changing the space.

(C3) Differential operators of order p map $I^m(M,Y)$ into $I^{m+p}(M,Y)$, and multiplication by a smooth function vanishing on Y maps $I^m(M,Y)$ into $I^{m-1}(M,Y)$.

The order normalization is such that the Dirac delta at a point of $\mathbb{R}^N$ has order $N/4$. For a vector bundle E over M we set

$$I^m(M,Y;E) = I^m(M,Y) \otimes_{\mathcal{C}^\infty(M)} \mathcal{C}^\infty(M;E),$$

the space of finite sums $\sum_i u_i e_i$, $u_i \in I^m(M,Y)$, $e_i \in \mathcal{C}^\infty(M;E)$, modulo the relations generated by $\mathcal{C}^\infty(M)$-linearity. As in Proposition 5.2.1, if u has support in an open set over which E is trivial, with local frame e_i, then $u = \sum_i u_i e_i$ with $u_i \in I^m(M,Y)$, and there are natural inclusions

$$\mathcal{C}^\infty(M;E) \hookrightarrow I^m(M,Y;E) \hookrightarrow \mathcal{C}^{-\infty}(M;E).$$

Integration, restriction and push-forward

If M is compact and V is a fixed vector space, the integral of a distributional density, i.e. its pairing with the constant function 1, gives a linear map $\int_M : I^m(M,Y;V\otimes\Omega_M) \to V$.

Proposition 5.3.1. *Let $Z \subset M$ be an embedded submanifold transversal to Y, i.e. $T_pY + T_pZ = T_pM$ for all $p \in Y \cap Z$. Then restriction $\mathcal{C}^\infty(M;E) \to \mathcal{C}^\infty(Z;E|_Z)$ extends to a map*

$$I^m(M,Y;E) \longrightarrow I^{m+\frac{1}{4}\operatorname{codim} Z}(Z, Y\cap Z; E|_Z).$$

Proof. Elements supported away from Y are smooth, so by a partition of unity it suffices to consider u supported near a point $p \in Y \cap Z$, where E may be taken trivial. Let $z_1,\dots,z_k$ be local defining functions for Y and $t_1,\dots,t_q$ for Z, $q = \operatorname{codim} Z$. Transversality means precisely that the differentials of the z_j and t_l are linearly independent at p, so they can be completed by functions w' to a coordinate system (w',t,z) near p, in which $Y = \{z=0\}$ and $Z = \{t = 0\}$. In these coordinates $u = \int e^{iz\cdot\zeta} a(w',t,\zeta)\,\text{đ}\zeta + v$ with $a \in S^{m+\frac{N}{4}-\frac{k}{2}}$, and u depends smoothly on t as a distribution in (w',z). We set $u|_Z = \int e^{iz\cdot\zeta} a(w',0,\zeta)\,\text{đ}\zeta + v|_{t=0}$; this agrees with the restriction if $a \in S^{-\infty}$, and in general is the limit of the restrictions of the smooth functions obtained by replacing a by $a(w',t,\zeta)\chi(\zeta/R)$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$ equal to one near 0, as $R \to \infty$; hence it is independent of choices. Since $\dim Z = N - q$, the order m'' of $u|_Z$ is fixed by $m + \frac{N}{4} - \frac{k}{2} = m'' + \frac{N-q}{4} - \frac{k}{2}$, i.e. $m'' = m + \frac{q}{4}$. □

Now let $\phi : M \to B$ be a fibration, locally the projection of a product, transversal to Y in the sense that each fibre is transversal to Y: $T_pY + T_p\phi^{-1}(\phi(p)) = T_pM$ for $p \in Y$. Let Ω_ϕ be the bundle of densities on the fibres, so that $\Omega_M = \Omega_\phi \otimes \phi^*\Omega_B$.

Proposition 5.3.2. *If $\phi : M \to B$ is a fibration with compact fibres transversal to Y, and E is a vector bundle over B, then fibre integration extends from smooth sections to a linear map*

$$\phi_* : I^m(M, Y; \phi^*E \otimes \Omega_\phi) \longrightarrow \mathcal{C}^\infty(B; E).$$

Proof. For smooth sections, locally in B where $M = U \times Z$, $\phi_* u(b) = \int_Z u(b, z)$, which is smooth in b. By a partition of unity it suffices to consider u supported near a point $p \in Y$ and E trivial. By transversality, the defining functions of Y restrict to each fibre near p to functions with independent differentials, so we may choose coordinates (b, z', z'') near p in which ϕ is $(b, z', z'') \mapsto b$ and $Y = \{z' = 0\}$. Then, modulo a smooth term, $u = \int e^{iz'\cdot\zeta} a(b, z'', \zeta) đ\zeta \, |dz'\, dz''|$ with compact support in (b, z''). Choosing $\chi \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to one on the support of u in z',

$$\phi_* u(b) = \iint \chi(z') \int e^{iz'\cdot\zeta} a(b, z'', \zeta) đ\zeta \, dz'\, dz'' = \iint a(b, z'', \zeta) \hat{\chi}(-\zeta) đ\zeta \, dz'',$$

which is smooth in b since $\hat{\chi} \in \mathcal{S}$ and a is a symbol. (Formally, $\int e^{iz'\cdot\zeta} dz' = (2\pi)^k \delta(\zeta)$ and $\phi_* u(b) = \int a(b, z'', 0)\, dz''$.) □

5.4 Pseudodifferential operators on a compact manifold

5.4.1 Definition

Let E and F be complex vector bundles over the compact manifold X. The diagonal $\mathrm{Diag} = \{(x, x)\} \subset X^2$ is an embedded submanifold of codimension n, and the fibres $\{x\} \times X$ of π_L are transversal to it.

Definition 5.4.1. For $m \in \mathbb{R}$, the space of pseudodifferential operators of order m from sections of E to sections of F is

$$\Psi^m(X; E, F) = I^m(X^2, \mathrm{Diag}; \mathrm{Hom}(E, F) \otimes \Omega_R), \tag{5.4.1}$$

with $\mathrm{Hom}(E, F)$ as in (5.2.2). The operator defined by $A \in \Psi^m(X; E, F)$ is

$$Au(x) = \int_X A(x, y) u(y), \quad \text{i.e.} \quad Au = (\pi_L)_*(A \cdot \pi_R^* u), \qquad u \in \mathcal{C}^\infty(X; E). \tag{5.4.2}$$

We write $\Psi^m(X; E) = \Psi^m(X; E, E)$, $\Psi^m(X)$ for trivial line bundles, and Ψ^m_{ph} for the classical operators, defined using I^m_{ph}.

In (5.4.2), $\pi_R^* u \in \mathcal{C}^\infty(X^2; \pi_R^* E)$, the product with the kernel contracts E' with E and so gives an element of $I^m(X^2, \mathrm{Diag}; \pi_L^* F \otimes \Omega_R)$; since Ω_R is the density bundle of the fibres of π_L, which are transversal to Diag, Proposition 5.3.2 shows that $Au \in \mathcal{C}^\infty(X; F)$. Thus $A : \mathcal{C}^\infty(X; E) \to \mathcal{C}^\infty(X; F)$, and the operator determines its kernel, by the Schwartz kernel theorem; we identify the two, and also write K_A for the kernel. For $\phi, \psi \in \mathcal{C}^\infty(X)$, the kernel of $\phi A \psi$ is $\phi(x) A(x, y) \psi(y)$.

The space of *smoothing operators* is

$$\Psi^{-\infty}(X; E, F) = \bigcap_m \Psi^m(X; E, F) = \mathcal{C}^\infty(X^2; \mathrm{Hom}(E, F) \otimes \Omega_R), \tag{5.4.3}$$

by (C1). Such an operator, $Au(x) = \int_X A(x, y) u(y)$, extends to a continuous map $\mathcal{C}^{-\infty}(X; E) \to \mathcal{C}^\infty(X; F)$, by letting the distribution u act in the variable y.

5.4.2 Local description

We now connect Definition 5.4.1 with the calculus on $\mathbb{R}^n$. If $\kappa : W \to \kappa(W)$ is a chart over which E and F are trivialized, and A has kernel with compact support in $W \times W$, its *local representative* is the matrix of operators

$$\kappa_* A = (\kappa^{-1})^* \circ A \circ \kappa^* : \mathcal{C}^\infty_{\mathrm{c}}(\kappa(W))^{N_E} \longrightarrow \mathcal{C}^\infty_{\mathrm{c}}(\kappa(W))^{N_F},$$

where the right density factor is trivialized by the coordinate density.

Theorem 5.4.2. *A linear operator $A : \mathcal{C}^\infty(X;E) \to \mathcal{C}^\infty(X;F)$ is given by an element of $\Psi^m(X;E,F)$ if and only if*

(i) *if $\phi, \psi \in \mathcal{C}^\infty(X)$ have disjoint supports then $\phi A \psi$ has kernel in $\mathcal{C}^\infty(X^2; \operatorname{Hom}(E,F) \otimes \Omega_R)$, and*

(ii) *for every trivializing chart $\kappa : W \to \kappa(W)$ and every $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$, the entries of $\kappa_*(\psi A \psi)$ are elements of $\Psi^m(\mathbb{R}^n)$ (with kernels supported in $\kappa(\operatorname{supp}\psi)^2$).*

Moreover it suffices to require (ii) *for the charts of one finite cover of X by trivializing charts.*

Proof. Suppose $A \in \Psi^m(X;E,F)$. Its kernel is smooth outside the diagonal, which gives (i). For (ii), $\psi(x)\psi(y)A(x,y)$ is conormal of order m with compact support in $W \times W$, so by (C1) the kernel of $\kappa_*(\psi A \psi)$ is a matrix of compactly supported elements of $I^m(\mathbb{R}^{2n}, \mathrm{Diag})$. Use the linear coordinates $(x,z) = (x, x-y)$ on $\mathbb{R}^{2n}$, in which $\mathrm{Diag} = \{z = 0\}$; here $N = 2n$, $k = n$ and the symbol order in (5.3.2) is $m + \frac{2n}{4} - \frac{n}{2} = m$. By (C1) the local representation (5.3.2) holds in these coordinates, so with a partition of unity in x and a cutoff $\chi(z)$ equal to one near $z = 0$, each entry of the kernel is a finite sum of terms $\chi_j(x)\chi(x-y)\int e^{i(x-y)\cdot\xi} a_j(x,\xi)\,d\!\!\!^-\xi$, $a_j \in S^m$, plus a function in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$. By (E2) and (E1) these are kernels of elements of $\Psi^m(\mathbb{R}^n)$.

Conversely suppose (i) holds and (ii) holds for the charts $\kappa_i : W_i \to \kappa_i(W_i)$ of a finite cover. By (i), near any point $(\bar{x}, \bar{y})$ with $\bar{x} \neq \bar{y}$ the kernel of A coincides with that of $\phi A \psi$, where $\phi = 1$ near $\bar{x}$, $\psi = 1$ near $\bar{y}$ and $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$; so K_A is smooth off the diagonal. Near $(\bar{x}, \bar{x})$, with $\bar{x} \in W_i$ and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W_i)$ equal to one near $\bar{x}$, K_A coincides with the kernel of $\psi A \psi$, whose local representative is $\int e^{i(x-y)\cdot\xi} a(x,\xi)\,d\!\!\!^-\xi$ with $a \in S^m$: this is of the form (5.3.2) in the coordinates $(x, x-y)$. Since conormality is a local property, $K_A \in I^m(X^2, \mathrm{Diag}; \operatorname{Hom}(E,F) \otimes \Omega_R)$. In particular (ii) then holds for all trivializing charts. □

The second half of the proof, combined with the first, shows again that the compactly supported elements of $\Psi^m(\mathbb{R}^n)$ are invariant under diffeomorphisms, as in (E7). It also gives the following, which we use repeatedly to build operators on X out of operators on $\mathbb{R}^n$.

Corollary 5.4.3. *If $\kappa : W \to \kappa(W)$ is a trivializing chart and B is an $N_F \times N_E$ matrix of elements of $\Psi^m(\mathbb{R}^n)$ with kernels of compact support in $\kappa(W) \times \kappa(W)$, then $\kappa^* \circ B \circ (\kappa^{-1})^*$, extended by zero, is an element of $\Psi^m(X;E,F)$.*

5.4.3 Normal fibrations and reduced forms

To write the kernel of an operator supported in a coordinate patch as an inverse Fourier transform, one identifies a neighbourhood of the diagonal with its normal bundle. The normal bundle of $\mathrm{Diag} \subset \mathbb{R}^{2n}$ is identified with $T\mathbb{R}^n = \mathbb{R}^n \times \mathbb{R}^n$ via the left tangent

vectors $(w,0)$; there remain several natural choices of a *normal fibration*, i.e. a linear map $\mathbb{R}^{2n}\to T\mathbb{R}^n$ inducing this identification. The *left*, *right* and *Weyl* fibrations are

$$f_L(x,y)=(x,x-y),\quad f_R(x,y)=(y,x-y),\quad f_W(x,y)=(\tfrac{x+y}{2},x-y).$$

In each case the fibre variable is $x-y$ and one of x, y or $\frac{x+y}{2}$ is held fixed. Correspondingly an operator $A\in\Psi^m(\mathbb{R}^n)$ with compactly supported kernel has the three representations

$$K_A=\int e^{i(x-y)\cdot\xi}a_L(x,\xi)đ\xi=\int e^{i(x-y)\cdot\xi}a_R(y,\xi)đ\xi=\int e^{i(x-y)\cdot\xi}a_W(\tfrac{x+y}{2},\xi)đ\xi,\qquad(5.4.4)$$

with $a_L,a_R,a_W\in S^m$ (classical if A is): the first two exist by (E2), and the Weyl symbol is obtained in the same way by Taylor expansion of an amplitude about $x=y$ (see [8, Section 18.5]). All three have the same principal symbol, and they may be taken to have compact support in the base variable.

Lemma 5.4.4. *If $A\in\Psi^m(\mathbb{R}^n)$ has compactly supported kernel, then for $f\in\mathcal{S}(\mathbb{R}^n)$*

$$Af(x)=\int e^{ix\cdot\xi}a_L(x,\xi)\hat f(\xi)đ\xi,\qquad \widehat{Af}(\xi)=\int e^{-iy\cdot\xi}a_R(y,\xi)f(y)\,dy.\qquad(5.4.5)$$

Proof. The first formula is the definition of $\mathrm{Op}(a_L)$ after inserting $\hat f(\xi)=\int e^{-iy\cdot\xi}f(y)dy$. For the second, let $g(\xi)$ denote the right side, with a_R compactly supported in y; integration by parts in y shows that $|\xi^\alpha g(\xi)|\le C_\alpha\langle\xi\rangle^m$, so g is smooth and polynomially bounded, and by Fubini (first for a_R replaced by $a_R(y,\xi)\chi(\xi/R)$, then letting $R\to\infty$) $Af(x)=\int e^{ix\cdot\xi}g(\xi)đ\xi$, i.e. $\widehat{Af}=g$. □

5.4.4 Differential operators

A differential operator $P\in\mathrm{Diff}^k(X;E,F)$ is a linear map $\mathcal{C}^\infty(X;E)\to\mathcal{C}^\infty(X;F)$ which in any local coordinates and trivializations takes the form

$$P=\sum_{|\alpha|\le k}p_\alpha(x)D_x^\alpha,\quad p_\alpha\text{ smooth }N_F\times N_E\text{ matrices.}\qquad(5.4.6)$$

By Theorem 5.4.2, $\mathrm{Diff}^k(X;E,F)\subset\Psi^k_{\mathrm{ph}}(X;E,F)$: condition (i) holds trivially since $\phi P\psi=0$ if $\operatorname{supp}\phi\cap\operatorname{supp}\psi=\emptyset$, and $\psi P\psi=\sum\psi p_\alpha D^\alpha\psi$ in local coordinates is a composite of elements of $\Psi(\mathbb{R}^n)$. The kernel of P is $\sum_\alpha p_\alpha(x)D_x^\alpha\delta(x-y)|dy|$, conormal to the diagonal.

The leading part of (5.4.6) has an invariant meaning. Let $\bar x\in X$, $\Xi\in T^*_{\bar x}X$, choose a real-valued $f\in\mathcal{C}^\infty(X)$ with $df(\bar x)=\Xi$ and $u\in\mathcal{C}^\infty(X;E)$. Then by Leibniz' formula

$$e^{-itf}P(e^{itf}u)=\sum_{j=0}^k t^jU_j,\qquad U_k(\bar x)=\sum_{|\alpha|=k}p_\alpha(\bar x)\,\Xi^\alpha\,u(\bar x),\qquad(5.4.7)$$

where $\Xi^\alpha=(\partial f(\bar x))^\alpha$ in local coordinates; the left side is defined invariantly, so the coefficient of t^k is too. Hence the *principal symbol*

$$\sigma_k(P)(x,\xi)=\sum_{|\alpha|=k}p_\alpha(x)\xi^\alpha\in\mathcal{C}^\infty(T^*X;\hom(E,F))$$

is a well-defined section, homogeneous polynomial of degree k on the fibres; P is called *elliptic* if $\sigma_k(P)(x,\xi)$ is invertible for all $\xi\neq0$. For example, for the exterior derivative $d(e^{itf}u)=e^{itf}(it\,df\wedge u+du)$, so $\sigma_1(d)(x,\xi)=i\,\xi\wedge$.

5.5 The calculus

5.5.1 Localization

Lemma 5.5.1. *Let $\{U_i\}$ be a finite open cover of X and $A \in \Psi^m(X;E,F)$. Then*

$$A = \sum_i A_i + A', \qquad A_i \in \Psi^m(X;E,F),\ \operatorname{supp} K_{A_i} \Subset U_i \times U_i,\ A' \in \Psi^{-\infty}(X;E,F). \tag{5.5.1}$$

Proof. Let $\{\phi_a\}$ be a partition of unity as in Lemma 5.1.3. Then $A = \sum_{a,b} \phi_a A \phi_b$, the kernel of $\phi_a A \phi_b$ being $\phi_a(x)A(x,y)\phi_b(y)$. If $\operatorname{supp}\phi_a \cap \operatorname{supp}\phi_b = \emptyset$ this kernel vanishes near the diagonal, so is smooth. Otherwise $\operatorname{supp}\phi_a \cup \operatorname{supp}\phi_b \subset U_i$ for some i, and the term has kernel supported in $U_i \times U_i$. Collecting terms gives (5.5.1). □

Taking the U_i to be trivializing charts, each A_i in (5.5.1) is given by a matrix of compactly supported elements of $\Psi^m(\mathbb{R}^n)$, and conversely by Corollary 5.4.3.

Proposition 5.5.2. *Each $A \in \Psi^m(X;E,F)$ is continuous $\mathcal{C}^\infty(X;E) \to \mathcal{C}^\infty(X;F)$. If $R \in \Psi^{-\infty}(X;G,E)$ then $AR \in \Psi^{-\infty}(X;G,F)$.*

Proof. In (5.5.1), A' has a smooth kernel, so is continuous, and each A_i is, in local coordinates, a matrix of elements of $\Psi^m(\mathbb{R}^n)$ with compactly supported kernels; these are continuous on $\mathcal{S}(\mathbb{R}^n)$ and hence on smooth functions supported in a fixed compact set. The kernel of AR is $(x,z) \mapsto A(K_R(\cdot,z))(x)$; since $z \mapsto K_R(\cdot,z)$ is a smooth map into $\mathcal{C}^\infty(X;E)$ (after trivializing the bundles in z locally) and A is continuous and linear, this is smooth in (x,z). □

5.5.2 Adjoints, transposes and action on distributions

Fix a smooth positive density ν on X and hermitian metrics on E and F, so that $L^2(X;E)$ and $L^2(X;F)$ are defined by (5.2.3).

Proposition 5.5.3. *If $A \in \Psi^m(X;E,F)$ there is a unique $A^* \in \Psi^m(X;F,E)$ such that $(Au,v) = (u,A^*v)$ for all $u \in \mathcal{C}^\infty(X;E)$, $v \in \mathcal{C}^\infty(X;F)$, and $\sigma_m(A^*) = \sigma_m(A)^*$ (pointwise adjoint with respect to the metrics).*

Proof. Write $A(x,y) = k(x,y)\nu(y)$ and set $A^*(x,y) = k(y,x)^*\nu(y)$, where $k(y,x)^*$ is the adjoint of $k(y,x) \in \hom(E_x,F_y)$. The reflection $(x,y) \mapsto (y,x)$ preserves the diagonal, and I^m is invariant under complex conjugation (replace $a(w,\zeta)$ by $\overline{a(w,-\zeta)}$ in (5.3.2)), so by (C1) $A^* \in \Psi^m(X;F,E)$. The identity $(Au,v) = (u,A^*v)$ is the statement that both sides equal the pairing of the kernel with $\overline{v}(x) \otimes u(y)$. Uniqueness is clear. In local coordinates where $\nu = \nu_0(x)|dx|$ and the metrics are given by positive matrices h_E, h_F, the local representative of A^* is $\nu_0^{-1} h_E^{-1} B^* h_F \nu_0$ where B^* is the Euclidean adjoint of that of A, so (E3) gives $\sigma_m(A^*) = h_E^{-1}\sigma_m(A)^\dagger h_F$, which is the adjoint with respect to the metrics. □

Corollary 5.5.4. *If $A \in \Psi^m(X;F,G)$ and $R \in \Psi^{-\infty}(X;E,F)$ then AR and $R'A$ are smoothing, for any $R' \in \Psi^{-\infty}(X;G,E)$.*

Proof. AR is smoothing by Proposition 5.5.2, and $R'A = (A^*R'^*)^*$, the adjoint of a smoothing operator being smoothing. □

Similarly the *transpose* of A, with kernel $A^t(x,y) = A(y,x)^t$, is an element of $\Psi^m(X;F' \otimes \Omega, E' \otimes \Omega)$ (the density factors match, since $A(y,x) \in F_y \otimes E'_x \otimes \Omega_x$), and $\int_X (Au)\cdot\phi = \int_X u \cdot A^t\phi$ for smooth u and ϕ. We may therefore extend A to distributions by

$$A : \mathcal{C}^{-\infty}(X;E) \longrightarrow \mathcal{C}^{-\infty}(X;F), \qquad (Au)(\phi) = u(A^t\phi), \tag{5.5.2}$$

which is continuous and, by density, the unique continuous extension of A on $\mathcal{C}^\infty(X;E)$. All the identities below between operators on smooth sections therefore hold on distributions.

5.5.3 Composition

Theorem 5.5.5. *If E, F, G are vector bundles over the compact manifold X then*

$$\Psi^m(X;F,G)\circ\Psi^{m'}(X;E,F)\subset\Psi^{m+m'}(X;E,G).$$

Proof. Let $A\in\Psi^m(X;F,G)$, $B\in\Psi^{m'}(X;E,F)$. Choose a finite cover by charts U_c over which E, F and G are trivial, a subordinate partition of unity ϕ_c, and $\chi_c,\chi_c',\tilde\chi_c\in\mathcal{C}^\infty_{\mathrm{c}}(U_c)$ with $\chi_c=\chi_c'=1$ near $\operatorname{supp}\phi_c$ and $\tilde\chi_c=1$ near $\operatorname{supp}\chi_c\cup\operatorname{supp}\chi_c'$. Then

$$AB=\sum_c A\phi_cB=\sum_c\left(\chi_cA\phi_cB\chi_c'+(1-\chi_c)A\phi_cB+\chi_cA\phi_cB(1-\chi_c')\right).$$

Since $(1-\chi_c)$ and ϕ_c have disjoint supports, $(1-\chi_c)A\phi_c$ is smoothing, and so is its product with B by Corollary 5.5.4; similarly for the last term. Finally $\chi_cA\phi_cB\chi_c'=\chi_c(\tilde\chi_cA\tilde\chi_c)\phi_c(\tilde\chi_cB\tilde\chi_c)\chi_c'$ is, in the chart U_c, a product of matrices of elements of $\Psi^m(\mathbb{R}^n)$ and $\Psi^{m'}(\mathbb{R}^n)$ with compactly supported kernels, so lies in $\Psi^{m+m'}(\mathbb{R}^n)$ by (E3), and the result follows from Corollary 5.4.3. □

The local step can be seen directly from Lemma 5.4.4: writing the first factor in left-reduced and the second in right-reduced form, for $f\in\mathcal{S}(\mathbb{R}^n)$,

$$\begin{aligned}ABf(x)&=\int e^{ix\cdot\xi}a_L(x,\xi)\Big(\int e^{-iy\cdot\xi}b_R(y,\xi)f(y)\,dy\Big)đ\xi,\\ K_{AB}&=\int e^{i(x-y)\cdot\xi}a_L(x,\xi)b_R(y,\xi)đ\xi.\end{aligned}\tag{5.5.3}$$

The amplitude $a_L(x,\xi)b_R(y,\xi)$ satisfies the estimates of (E2) with order $m+m'$, so $AB\in\Psi^{m+m'}(\mathbb{R}^n)$ with principal symbol $a_L(x,\xi)b_R(x,\xi)$. Equivalently, K_{AB} is the restriction to $z=y$ of $\int e^{i(x-y)\cdot\xi}a_L(x,\xi)b_R(z,\xi)đ\xi$, which is conormal in $\mathbb{R}^{3n}$ to $\{x=y\}$; the submanifold $\{z=y\}$ is transversal to it, and Proposition 5.3.1 shows again that the result is conormal.

Remark 5.5.6. When one factor is smoothing, the composite can also be treated globally. With $\pi_S(x,y,z)=(x,y)$, $\pi_F(x,y,z)=(y,z)$ and $\pi_C(x,y,z)=(x,z)$ on X^3, the kernel of AR, R smoothing, is $(\pi_C)_*(\pi_S^*A\cdot\pi_F^*R)$, where the density factor of A in y is used for the integration. The pull-back π_S^*A is conormal to $\pi_S^{-1}(\mathrm{Diag})=\{x=y\}$, of order $m-\frac{n}{4}$ by the behaviour of conormal distributions under pull-back by submersions (Chapter 4), the product with the smooth section π_F^*R preserves this, and the fibres of π_C (on which y varies) are transversal to $\{x=y\}$, so the push-forward is smooth by Proposition 5.3.2. For general A and B, however, the corresponding product $\pi_S^*A\cdot\pi_F^*B$ is not defined in this simple way, since both factors are singular at the triple diagonal; this is why the proof above localizes.

5.5.4 Symbols and the principal symbol

Let $V\to X$ be a real vector bundle (we only need $V=T^*X$) and W a complex vector bundle over X, pulled back to V. The space $S^m(V;W)$ consists of the smooth sections a of W over V such that, in any local trivializations of V and W over a coordinate patch, and on any compact subset K of the patch,

$$|D_x^\alpha D_\xi^\beta a(x,\xi)|\le C_{K,\alpha,\beta}\langle\xi\rangle^{m-|\beta|}.$$

This is independent of the trivializations: under $(x,\xi)\mapsto(x,L(x)\xi)$, with L smooth and invertible, and a change $a\mapsto g(x)a$ of trivialization of W, the chain rule shows that each ξ-derivative of the new expression is a combination of ξ-derivatives of the old one with smooth coefficients, and each x-derivative produces terms $(\partial_x L)\xi\cdot\partial_\xi a$ which are of the same order; since $\langle L\xi\rangle$ and $\langle\xi\rangle$ are comparable on compact sets the estimates follow by induction. Since X is compact, $S^m(V;W)$ is a Fréchet space. Set $S^{m-[1]}(V;W)=S^m(V;W)/S^{m-1}(V;W)$. The classical symbols S^m_{ph} are defined in the same way; for these the leading part is a smooth section over $V\setminus 0$, homogeneous of degree m, i.e. a section over the sphere bundle $SV=(V\setminus 0)/\mathbb{R}^+$ of $N_m\otimes W$, where N_m is the line bundle whose sections are the functions homogeneous of degree m on $V\setminus 0$.

Proposition 5.5.7. *There is a unique linear map*

$$\sigma_m:\Psi^m(X;E,F)\longrightarrow S^{m-[1]}(T^*X;\hom(E,F))$$

such that for every trivializing chart $\kappa:W\to\kappa(W)$ and every $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(W)$, $\sigma_m(\kappa_*(\psi A\psi))$ is the local expression of $\psi^2\sigma_m(A)$, using the coordinates (5.2.1) on T^*W. It has the following properties.

(a) *The sequence $0\to\Psi^{m-1}(X;E,F)\hookrightarrow\Psi^m(X;E,F)\xrightarrow{\sigma_m}S^{m-[1]}(T^*X;\hom(E,F))\to 0$ is exact.*

(b) *$\sigma_{m+m'}(AB)=\sigma_m(A)\sigma_{m'}(B)$ and $\sigma_m(A^*)=\sigma_m(A)^*$.*

(c) *For $P\in\mathrm{Diff}^k(X;E,F)$, $\sigma_k(P)$ is the class of the polynomial symbol defined by (5.4.7).*

(d) *For classical operators $\sigma_m(A)$ is a homogeneous section and there is a short exact sequence*

$$\Psi^{m-1}_{\mathrm{ph}}(X;E,F)\hookrightarrow\Psi^m_{\mathrm{ph}}(X;E,F)\xrightarrow{\sigma_m}\mathcal{C}^\infty(S^*X;N_m\otimes\hom(E,F))\to 0.$$

Proof. Choose a finite cover by trivializing charts $\kappa_i:W_i\to\kappa_i(W_i)$ and $\psi_i\in\mathcal{C}^\infty_{\mathrm{c}}(W_i)$ with $\sum_i\psi_i^2=1$ (e.g. $\psi_i=\rho_i(\sum_k\rho_k^2)^{-1/2}$ for a partition of unity ρ_i). The principal symbol of $\kappa_{i*}(\psi_iA\psi_i)\in\Psi^m(\mathbb{R}^n)$ has a representative supported, in x, in a compact subset of $\kappa_i(W_i)$, well defined modulo S^{m-1} symbols with the same property; transferring it to T^*W_i by (5.2.1) and extending by zero gives $a_i\in S^{m-[1]}(T^*X;\hom(E,F))$. Set $\sigma_m(A)=\sum_i a_i$.

To check the required property, let $\kappa:W\to\kappa(W)$ and $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(W)$ be arbitrary. Then $\psi A\psi=\sum_i\psi_i^2\psi A\psi$ and $\psi_i^2(\psi A\psi)-\psi_i(\psi A\psi)\psi_i=\psi_i[\psi_i,\psi A\psi]$, which in the chart κ has principal symbol zero by (E3). The operator $\psi_i\psi A\psi\psi_i=\psi(\psi_iA\psi_i)\psi$ has kernel supported in $(W\cap W_i)^2$; in the chart κ_i its principal symbol is ψ^2 times that of $\kappa_{i*}(\psi_iA\psi_i)$, and by (E7) its principal symbol in the chart κ is the same invariant function on T^*X, namely ψ^2a_i. Summing over i gives $\sigma_m(\kappa_*(\psi A\psi))=\psi^2\sigma_m(A)$. Uniqueness follows by applying the property to the ψ_i: two such maps differ by b with $\psi_i^2b\in S^{m-1}$ for all i, so $b=\sum_i\psi_i^2b\in S^{m-1}$.

(a) Clearly σ_m vanishes on Ψ^{m-1}. If $\sigma_m(A)=0$ then for every trivializing chart and $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(W)$, $\kappa_*(\psi A\psi)$ has vanishing principal symbol, so lies in $\Psi^{m-1}(\mathbb{R}^n)$ by (E4); by Theorem 5.4.2, $A\in\Psi^{m-1}(X;E,F)$. For surjectivity define, with the notation above, the *quantization map*

$$q(a)=\sum_i\kappa_i^*\circ(\tilde\chi_i\,\mathrm{Op}(a_i)\tilde\chi_i)\circ(\kappa_i^{-1})^*,\tag{5.5.4}$$

where a_i is the local expression of ψ_i^2a in the chart κ_i (extended by zero outside $\kappa_i(W_i)$) and $\tilde\chi_i\in\mathcal{C}^\infty_{\mathrm{c}}(\kappa_i(W_i))$ is equal to one near the support of $\psi_i\circ\kappa_i^{-1}$. By Corollary 5.4.3,

$q(a) \in \Psi^m(X;E,F)$, and applying the defining property in an arbitrary chart, using (E7) as above, gives $\sigma_m(q(a)) = \sum_i \psi_i^2 a = a$. Note that q is linear and maps $S^{-\infty}$ into $\Psi^{-\infty}$.

(b) Let $\psi \in \mathcal{C}_c^\infty(W)$ and $\chi \in \mathcal{C}_c^\infty(W)$ equal to one near $\operatorname{supp}\psi$. Then $\psi AB\psi - (\psi A\chi)(\chi B\psi) = \psi A(1-\chi^2)B\psi$ is smoothing, by the argument in the proof of Theorem 5.5.5, and the local representatives of $\psi A\chi$ and $\chi B\psi$ have principal symbols $\psi\chi\sigma_m(A)$ and $\chi\psi\sigma_{m'}(B)$ (compare $\psi A\chi$ with $\tilde\psi A\tilde\psi$ for $\tilde\psi = 1$ on $\operatorname{supp}\chi$, using (E2)). So (E3) gives the multiplicativity. The statement for adjoints was proved in Proposition 5.5.3.

(c) is immediate from (5.4.6), and (d) follows from (a), since the quantization of a classical symbol is classical. □

Remark 5.5.8. Since $\Psi^m(X) = I^m(X^2, \mathrm{Diag}; \Omega_R)$ is a $\mathcal{C}^\infty(X^2)$-module, (5.4.1) may be written $\Psi^m(X;E,F) = \Psi^m(X) \otimes_{\mathcal{C}^\infty(X^2)} \mathcal{C}^\infty(X^2; \mathrm{Hom}(E,F))$. Thus an operator between sections of bundles is a finite sum $\sum_i A_i G_i$ with $A_i \in \Psi^m(X)$ and $G_i \in \mathcal{C}^\infty(X^2; \mathrm{Hom}(E,F))$, and in local trivializations it is simply a matrix of scalar operators.

Proposition 5.5.9. *If $A_j \in \Psi^{m-j}(X;E,F)$ for $j \in \mathbb{N}_0$, there exists $A \in \Psi^m(X;E,F)$, unique modulo $\Psi^{-\infty}(X;E,F)$, such that $A - \sum_{j<N} A_j \in \Psi^{m-N}(X;E,F)$ for all N; we write $A \sim \sum_j A_j$.*

Proof. Choose a finite cover by trivializing charts U_c, decompose each A_j as in Lemma 5.5.1 with respect to it, $A_j = \sum_c A_{j,c} + A_j'$, and note that the supports of the kernels of the $A_{j,c}$ may be taken to lie in a fixed compact subset K_c of $U_c \times U_c$ independent of j (use the same partition of unity for all j). By (E5) there are $B_c \in \Psi^m(X;E,F)$ with kernels supported in $U_c \times U_c$ and $B_c - \sum_{j<N} A_{j,c} \in \Psi^{m-N}$ for all N. Then $A = \sum_c B_c$ satisfies $A - \sum_{j<N} A_j = \sum_c (B_c - \sum_{j<N} A_{j,c}) - \sum_{j<N} A_j' \in \Psi^{m-N}$. If A and $\tilde A$ both have the property, $A - \tilde A \in \bigcap_N \Psi^{m-N} = \Psi^{-\infty}$. □

Thus $\Psi(X;E) = \bigcup_m \Psi^m(X;E)$ is a filtered $*$-algebra, the principal symbol map is a homomorphism onto the symbol algebra, and $\Psi^{-\infty}(X;E)$ is an ideal. Moreover, from (C3) or directly from Theorem 5.5.5,

$$\mathrm{Diff}^k(X;F,G) \circ \Psi^m(X;E,F) \subset \Psi^{k+m}(X;E,G), \qquad \sigma_{k+m}(PA) = \sigma_k(P)\sigma_m(A).$$

5.6 Ellipticity and parametrices

Definition 5.6.1. An operator $A \in \Psi^m(X;E,F)$ is *elliptic* if there is a symbol $b \in S^{-m}(T^*X; \mathrm{hom}(F,E))$ such that, for a representative a of $\sigma_m(A)$,

$$ba - \mathrm{Id}_E \in S^{-1}(T^*X; \mathrm{hom}(E)), \qquad ab - \mathrm{Id}_F \in S^{-1}(T^*X; \mathrm{hom}(F)).$$

Lemma 5.6.2. *Fix a metric on T^*X. Then $A \in \Psi^m(X;E,F)$ is elliptic if and only if a (any) representative a of $\sigma_m(A)$ is invertible for $|\xi| \geq R$, for some R, with $|a(x,\xi)^{-1}| \leq C\langle\xi\rangle^{-m}$ there. For classical operators this holds if and only if the homogeneous principal symbol is invertible on $T^*X \setminus 0$, and for differential operators it is the condition given in Section 5.4.*

Proof. If $ba = \mathrm{Id} + r$, $ab = \mathrm{Id} + r'$ with $r, r' \in S^{-1}$ then for $|\xi|$ large $|r|, |r'| < \frac{1}{2}$, so a has a left and a right inverse, hence is invertible (in particular the ranks of E and F coincide), and $a^{-1} = (\mathrm{Id} + r)^{-1} b$ satisfies the bound. Conversely, if the bound holds, let $\chi \in \mathcal{C}^\infty(T^*X)$ be equal to one for $|\xi| \leq R+1$ and vanish for $|\xi| \geq R+2$, and set $b = (1-\chi)a^{-1}$. Differentiating $a\,a^{-1} = \mathrm{Id}$ repeatedly, each derivative of a^{-1} is a sum of products of a^{-1} with derivatives of a, and the symbol estimates for $b \in S^{-m}$ follow by induction; $ab - \mathrm{Id} = -\chi\,\mathrm{Id}$ and $ba - \mathrm{Id}$

have compact support in ξ, so lie in $S^{-\infty}$. For a homogeneous symbol invertible on the sphere bundle the bound follows from compactness of S^*X and homogeneity. □

Theorem 5.6.3. *If $A \in \Psi^m(X;E,F)$ is elliptic, there exists $B \in \Psi^{-m}(X;F,E)$, unique modulo $\Psi^{-\infty}$, such that*

$$BA = \mathrm{Id}_E - R_E, \quad AB = \mathrm{Id}_F - R_F, \qquad R_E \in \Psi^{-\infty}(X;E),\ R_F \in \Psi^{-\infty}(X;F). \tag{5.6.1}$$

If A is classical, so is B.

Proof. With b as in Definition 5.6.1 let $B_0 = q(b) \in \Psi^{-m}(X;F,E)$. By Proposition 5.5.7, $AB_0 = \mathrm{Id}_F - S$ with $S \in \Psi^{-1}(X;F)$. By Proposition 5.5.9 choose $B_r \in \Psi^{-m}(X;F,E)$ with $B_r \sim \sum_{j\geq 0} B_0 S^j$. Then for each N,

$$AB_r \equiv AB_0 \sum_{j<N} S^j = (\mathrm{Id} - S) \sum_{j<N} S^j = \mathrm{Id} - S^N \mod \Psi^{-N},$$

so $AB_r - \mathrm{Id} \in \Psi^{-N}$ for all N, i.e. $AB_r = \mathrm{Id} - R_F$ with R_F smoothing. In the same way, starting from $B_0 A = \mathrm{Id} - S'$, there is B_l with $B_l A = \mathrm{Id} - R_E$. Then

$$B_l = B_l(AB_r + R_F) = (\mathrm{Id} - R_E)B_r + B_l R_F = B_r - R_E B_r + B_l R_F,$$

so $B_l - B_r \in \Psi^{-\infty}$ by Corollary 5.5.4, and $B = B_r$ satisfies both identities in (5.6.1). The same computation shows that any left parametrix and any right parametrix differ by a smoothing operator, which gives uniqueness. In the classical case each step stays in the classical calculus. □

Corollary 5.6.4 (Elliptic regularity)**.** *If $A \in \Psi^m(X;E,F)$ is elliptic, $u \in \mathcal{C}^{-\infty}(X;E)$ and $Au \in \mathcal{C}^\infty(X;F)$, then $u \in \mathcal{C}^\infty(X;E)$. In particular the null space of A on distributions,* $\mathrm{Nul}(A) = \{u \in \mathcal{C}^{-\infty}(X;E);\, Au = 0\}$*, consists of smooth sections.*

Proof. By (5.6.1), $u = BAu + R_E u$; here $BAu \in \mathcal{C}^\infty$ by Proposition 5.5.2 and $R_E u \in \mathcal{C}^\infty$ since R_E is smoothing. □

The parametrix reduces the study of A to that of operators $\mathrm{Id} + R$ with R smoothing: $\mathrm{Nul}(A) \subset \mathrm{Nul}(\mathrm{Id} - R_E)$ and $\mathrm{Ran}(A) \supset \mathrm{Ran}(\mathrm{Id} - R_F)$. In Section 5.8 we use this to show that elliptic operators are Fredholm, on smooth sections as well as on Sobolev spaces. The *index* of an elliptic operator is then

$$\mathrm{ind}(A) = \dim \mathrm{Nul}(A) - \dim\left(\mathcal{C}^\infty(X;F)/A\,\mathcal{C}^\infty(X;E)\right). \tag{5.6.2}$$

For a linear map between finite-dimensional spaces the corresponding integer is just the difference of the dimensions of the spaces; the index is a genuinely infinite-dimensional invariant. Its computation in terms of the principal symbol is the content of the Atiyah–Singer index theorem, the subject of Chapter 10 and the following chapters.

5.7 Sobolev spaces

5.7.1 Local facts

Lemma 5.7.1. *Let $K \subset \mathbb{R}^n$ be compact and $H^s_K = \{u \in H^s(\mathbb{R}^n); \operatorname{supp} u \subset K\}$.*

(a) *If $\Phi : \Omega \to \Omega'$ is a diffeomorphism of open sets and $K \subset \Omega$, then $\Phi_* = (\Phi^{-1})^* : H^s_K \to H^s_{\Phi(K)}$ is bounded, for every $s \in \mathbb{R}$.*

(b) *An operator with kernel in* $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$ *is bounded* $H^t(\mathbb{R}^n) \to H^{t'}(\mathbb{R}^n)$ *for all* t, t'.

(c) (Rellich) *If* $t < s$ *the inclusion* $H^s_K \hookrightarrow H^t(\mathbb{R}^n)$ *is compact.*

(d) $\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n) = \bigcup_s H^s_{\mathrm{c}}(\mathbb{R}^n)$, *and* $H^s(\mathbb{R}^n) \subset \mathcal{C}^k(\mathbb{R}^n)$, *with bounded inclusion into bounded* $\mathcal{C}^k$ *functions, if* $s > k + \frac{n}{2}$.

Proof. (a) Let $\Lambda_t = \mathrm{Op}(\langle\xi\rangle^t)$ and choose $\chi, \chi' \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ with $\chi = 1$ near K and $\chi' = 1$ near $\operatorname{supp}\chi$. For $u \in H^s_K$ put $f = \Lambda_s u \in L^2$; then

$$u = \chi\Lambda_{-s} f = \chi\Lambda_{-s}\chi' f + \chi\Lambda_{-s}(1-\chi')f.$$

The kernel of Λ_{-s} is $k(x-y)$ where k is smooth outside 0 and, together with all its derivatives, rapidly decreasing at infinity (since $|z|^{2N} D^\alpha k = \mathcal{F}^{-1}(\Delta^N_\xi(\xi^\alpha\langle\xi\rangle^{-s}))$ is bounded when $2N > n - s + |\alpha|$). So the second term is given by a smooth kernel $\chi(x)k(x-y)(1-\chi'(y))$, supported in $\operatorname{supp}\chi \times \mathbb{R}^n$ and with x-derivatives bounded by $C_N\langle y\rangle^{-N}$; it maps L^2 boundedly into smooth functions supported in $\operatorname{supp}\chi$, with each $\mathcal{C}^k$ norm bounded, and these are preserved by Φ_*. For the first term, $\Phi_*(\chi\Lambda_{-s}\chi' f) = \Phi_*(\chi\Lambda_{-s}\chi')\,\Phi_*(\chi' f)$, where $\Phi_*(\chi\Lambda_{-s}\chi') \in \Psi^{-s}(\mathbb{R}^n)$ by (E7), which is bounded $L^2 \to H^s$ by (E6), and $\|\Phi_*(\chi' f)\|_{L^2} \le C\|f\|_{L^2}$ since the Jacobian of Φ^{-1} is bounded on $\Phi(\operatorname{supp}\chi')$.

(b) Let T have kernel $k \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$. Then for each multi-index α the derivative $D^\alpha Tu(x) = u(D^\alpha_x k(x,\cdot))$ is bounded by the product $\|u\|_{H^t}\sup_x \|D^\alpha_x k(x,\cdot)\|_{H^{-t}}$ and Tu has support in a fixed compact set.

(c) Let $u_j \in H^s_K$, $\|u_j\|_{H^s} \le 1$, and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near K. Then $\hat{u}_j(\xi) = u_j(\chi e^{-ix\cdot\xi})$, so $\hat{u}_j$ and all its derivatives are bounded on compact sets uniformly in j. By the Arzelà–Ascoli theorem and a diagonal argument a subsequence (still denoted u_j) has $\hat{u}_j$ converging uniformly on compact sets. Then for any R,

$$\|u_j - u_k\|^2_{H^t} \le \int_{|\xi|\le R} \langle\xi\rangle^{2t}|\hat{u}_j - \hat{u}_k|^2 đ\xi + \langle R\rangle^{2(t-s)} \int \langle\xi\rangle^{2s}|\hat{u}_j - \hat{u}_k|^2 đ\xi,$$

where the last term is at most $4\langle R\rangle^{2(t-s)}$; choosing R large and then j, k large shows that (u_j) is Cauchy in H^t.

(d) A compactly supported distribution has finite order, so $|\hat{u}(\xi)| \le C\langle\xi\rangle^N$ for some N and $u \in H^{-N-n}$. If $s > k + \frac{n}{2}$ and $|\alpha| \le k$ then $\int |\xi^\alpha \hat{u}| đ\xi \le \|\langle\xi\rangle^{|\alpha|-s}\|_{L^2}\|u\|_{H^s}$, so $D^\alpha u$ is continuous and bounded. □

5.7.2 Sobolev spaces on a compact manifold

Fix a finite cover of X by charts $\kappa_i : W_i \to \kappa_i(W_i)$ over which E is trivial, and a subordinate partition of unity ρ_i.

Definition 5.7.2. For $s \in \mathbb{R}$,

$$H^s(X;E) = \{u \in \mathcal{C}^{-\infty}(X;E);\ \kappa_{i*}(\rho_i u) \in H^s(\mathbb{R}^n)^{N_E}\ \forall i\}, \qquad \|u\|^2_s = \sum_i \|\kappa_{i*}(\rho_i u)\|^2_{H^s}.$$

Proposition 5.7.3. (a) $H^s(X;E)$ *is independent of the choices made, with equivalent norms. A distribution* u *lies in* $H^s(X;E)$ *if and only if* $\kappa_*(\psi u) \in H^s(\mathbb{R}^n)^{N_E}$ *for every trivializing chart* κ *and* $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$; *and if* w *has support in a compact subset of* W *and* $\kappa_* w \in H^s$, *then* $w \in H^s(X;E)$, *with* $\|w\|_s \le C\|\kappa_* w\|_{H^s}$, C *depending only on the support.*

(b) $H^s(X;E)$ *is a Hilbert space in which* $\mathcal{C}^\infty(X;E)$ *is dense, and* $H^0(X;E) = L^2(X;E)$ *with equivalent norms.*

(c) $\bigcap_s H^s(X;E) = \mathcal{C}^\infty(X;E)$ *and* $\bigcup_s H^s(X;E) = \mathcal{C}^{-\infty}(X;E)$.

(d) *For* $t < s$ *the inclusion* $H^s(X;E) \hookrightarrow H^t(X;E)$ *is compact.*

(e) *The* L^2 *pairing* (5.2.3) *extends by continuity to a sesquilinear pairing* $H^s(X;E) \times H^{-s}(X;E) \to \mathbb{C}$, *and every continuous linear functional on* $H^s(X;E)$ *is of the form* $u \mapsto (u,w)$ *for a unique* $w \in H^{-s}(X;E)$.

Proof. (a) Let κ be a trivializing chart and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$. Then $\psi u = \sum_i \psi\rho_i u$, and $\kappa_{i*}(\psi\rho_i u) = (\kappa_{i*}\psi)\kappa_{i*}(\rho_i u) \in H^s$ (by (E6), after changing trivializations, which is multiplication by a smooth matrix) is supported in a compact subset of $\kappa_i(W \cap W_i)$, so by Lemma 5.7.1(a) applied to $\kappa \circ \kappa_i^{-1}$, $\kappa_*(\psi\rho_i u) \in H^s$, with norm estimates. This proves that membership for one choice of cover and partition implies the condition for all charts, which in turn implies membership for any other choice; the norm equivalences come with the estimates. The last statement follows in the same way.

(b) The map $u \mapsto (\kappa_{i*}(\rho_i u))_i$ is an isometry onto a subspace of $\bigoplus_i H^s(\mathbb{R}^n)^{N_E}$. Its range is closed: if $v_i^{(k)} = \kappa_{i*}(\rho_i u^{(k)}) \to v_i$ in H^s, then $u^{(k)} = \sum_i \kappa_i^* v_i^{(k)}$ converges in $\mathcal{C}^{-\infty}(X;E)$ to $u = \sum_i \kappa_i^* v_i$ (the v_i being supported in $\kappa_i(\operatorname{supp}\rho_i)$), and so $v_i = \kappa_{i*}(\rho_i u)$. For density, approximate each $\kappa_{i*}(\rho_i u)$ in H^s by smooth functions supported in a fixed compact subset of $\kappa_i(W_i)$ (by convolution with an approximate identity) and use (a). The identification of H^0 with L^2 follows since the density ν and the metric on E are given by smooth, positive, locally bounded factors.

(c) If $u \in H^s$ for all s, each $\kappa_{i*}(\rho_i u)$ is smooth by Lemma 5.7.1(d), so $u = \sum_i \rho_i u$ is smooth. Conversely, each $\kappa_{i*}(\rho_i u)$ is a compactly supported distribution, so lies in some H^{s_i}.

(d) Under the isometry of (b), $H^s(X;E)$ maps into $\bigoplus_i (H^s_{K_i})^{N_E}$ with $K_i = \kappa_i(\operatorname{supp}\rho_i)$, and the inclusion of this space into $\bigoplus_i H^t(\mathbb{R}^n)^{N_E}$ is compact by Lemma 5.7.1(c).

(e) Choose $\psi_i \in \mathcal{C}^\infty_{\mathrm{c}}(W_i)$ equal to one on $\operatorname{supp}\rho_i$. In local coordinates $\nu = \nu_i|dx|$ and the metric is h_i, so $(u,w) = \sum_i \int \langle h_i\nu_i\kappa_{i*}(\rho_i u), \kappa_{i*}(\psi_i w)\rangle dx$, which is bounded by $C\|u\|_s\|w\|_{-s}$ using $|\int f\bar g| \le \|f\|_{H^s}\|g\|_{H^{-s}}$ on $\mathbb{R}^n$ and (a). Conversely, let ℓ be a continuous linear functional on $H^s(X;E)$. By the Riesz representation theorem, $\ell(u)$ is the H^s inner product of u with some $w' \in H^s$, i.e.

$$\ell(u) = \sum_i \int \langle \kappa_{i*}(\rho_i u), \Lambda_{2s} v_i\rangle dx = \sum_i \int \langle \kappa_{i*}(\rho_i u), \chi_i \Lambda_{2s} v_i\rangle dx,$$

where $v_i = \kappa_{i*}(\rho_i w')$, $\Lambda_{2s} = \operatorname{Op}(\langle\xi\rangle^{2s})$ and $\chi_i \in \mathcal{C}^\infty_{\mathrm{c}}(\kappa_i(W_i))$ equals one on $\kappa_i(\operatorname{supp}\rho_i)$. Here $\chi_i\Lambda_{2s}v_i \in H^{-s}$ has compact support in $\kappa_i(W_i)$, so $w = \sum_i \rho_i\kappa_i^*(\nu_i^{-1}h_i^{-1}\chi_i\Lambda_{2s}v_i) \in H^{-s}(X;E)$ by (a), and $\ell(u) = (u,w)$. Uniqueness follows from density of $\mathcal{C}^\infty(X;E)$ in $H^s(X;E)$. □

Theorem 5.7.4. *Each* $A \in \Psi^m(X;E,F)$ *extends to a bounded operator* $A : H^s(X;E) \to H^{s-m}(X;F)$ *for every* $s \in \mathbb{R}$. *In particular smoothing operators map* H^s *into* H^t *for all* s,t, *and are compact on each* H^s; *operators of negative order are compact on each* H^s.

Proof. Use the decomposition (5.5.1) with respect to a cover by charts trivializing E and F. The smoothing part A' maps $H^s(X;E)$ to $H^t(X;F)$ for all t: decomposing $A' = \sum_{j,k} \rho_j A' \rho_k$, each term is, in the charts κ_j and κ_k, an operator with kernel in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$, so Lemma 5.7.1(b) and Proposition 5.7.3(a) apply. For A_i, with kernel supported in $K' \times K'$, $K' \Subset W_i$, choose $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W_i)$ equal to one near K'. Then $A_i u = A_i(\psi u)$, $\kappa_{i*}(\psi u) \in H^s$, and the local representative of A_i is a matrix of elements of $\Psi^m(\mathbb{R}^n)$, so maps it into H^{s-m}, with support in $\kappa_i(K')$; by Proposition 5.7.3(a), $A_i u \in H^{s-m}(X;F)$ with the appropriate bound. The

compactness statements follow from Proposition 5.7.3(d), since an operator of order $m < 0$ maps H^s boundedly into H^{s-m}, which includes compactly into H^s. □

Proposition 5.7.5. *Let $A \in \Psi^m(X;E,F)$ be elliptic. If $u \in \mathcal{C}^{-\infty}(X;E)$ and $Au \in H^{s-m}(X;F)$ then $u \in H^s(X;E)$, and for any $t < s$*

$$\|u\|_s \leq C(\|Au\|_{s-m} + \|u\|_t).$$

Consequently, for any elliptic $\Lambda \in \Psi^s(X;E)$,

$$H^s(X;E) = \{u \in \mathcal{C}^{-\infty}(X;E);\ \Lambda u \in L^2\} = \{u \in \mathcal{C}^{-\infty}(X;E);\ Pu \in L^2\ \forall P \in \Psi^s(X;E)\}.$$

Proof. By (5.6.1), $u = BAu + R_E u$, with $BAu \in H^s$ by Theorem 5.7.4 and $R_E u \in \mathcal{C}^\infty$ (note $u \in H^t$ for some t by Proposition 5.7.3(c)); the estimate follows from the bounds on B and R_E. For the last statement, an elliptic $\Lambda \in \Psi^s(X;E)$ exists (e.g. $q(\langle\xi\rangle^s \operatorname{Id}_E)$ for a metric on T^*X), and the inclusions follow from Theorem 5.7.4 and the first part. □

5.8 Fredholm theory of elliptic operators

Theorem 5.8.1. *Let $A \in \Psi^m(X;E,F)$ be elliptic, and let $A^* \in \Psi^m(X;F,E)$ be its adjoint.*

(a) *$N = \operatorname{Nul}(A)$ and $N^* = \operatorname{Nul}(A^*)$ (on distributions) are finite-dimensional subspaces of $\mathcal{C}^\infty(X;E)$ and $\mathcal{C}^\infty(X;F)$ respectively.*

(b) *For every $s \in \mathbb{R}$, $A : H^s(X;E) \to H^{s-m}(X;F)$ is Fredholm, with null space N and range $\{f \in H^{s-m}(X;F);\ (f,v) = 0\ \forall v \in N^*\}$.*

(c) *$A : \mathcal{C}^\infty(X;E) \to \mathcal{C}^\infty(X;F)$ has null space N and closed range, with complement N^*: $\mathcal{C}^\infty(X;F) = A\mathcal{C}^\infty(X;E) \oplus N^*$, an L^2-orthogonal sum.*

(d) *$\operatorname{ind}(A) = \dim N - \dim N^*$ is the index on $\mathcal{C}^\infty$ and on every H^s. It depends only on $\sigma_m(A)$; moreover $\operatorname{ind}(A^*) = -\operatorname{ind}(A)$, $\operatorname{ind}(AB) = \operatorname{ind}(A) + \operatorname{ind}(B)$ for elliptic $B \in \Psi^{m'}(X;G,E)$, and the index is constant on any family of elliptic operators which is continuous in the norm of $\mathcal{B}(H^s(X;E), H^{s-m}(X;F))$ for some s.*

Proof. Let B be a parametrix as in (5.6.1). By Theorem 5.7.4, $A : H^s \to H^{s-m}$ and $B : H^{s-m} \to H^s$ are bounded and R_E, R_F are compact on H^s and H^{s-m}, so A is Fredholm on these spaces by the parametrix characterization of Fredholm operators (Theorem A.3.3). Its null space on H^s is contained in N, and $N \subset \mathcal{C}^\infty \subset H^s$ by Corollary 5.6.4; so it equals N, which is therefore finite-dimensional. The adjoint is elliptic, since $\sigma_m(A^*) = \sigma_m(A)^*$ is inverted by b^*, so the same applies to N^*. This proves (a).

For (b), let $R_s = AH^s(X;E)$, which is closed, and $Y_s = \{f \in H^{s-m}; (f,v) = 0\ \forall v \in N^*\}$. If $u \in H^s$ and $v \in N^*$, then $(Au,v) = (u,A^*v) = 0$, by continuity from smooth u; so $R_s \subset Y_s$. If $R_s \neq Y_s$ there is a continuous linear functional on H^{s-m} which vanishes on R_s but not on Y_s. By Proposition 5.7.3(e) it is $f \mapsto (f,w)$ for some $w \in H^{m-s}(X;F)$; then $(u,A^*w) = (Au,w) = 0$ for all $u \in \mathcal{C}^\infty(X;E)$, so $A^*w = 0$, $w \in N^*$, and the functional vanishes on Y_s, a contradiction.

(c) If $f \in \mathcal{C}^\infty(X;F)$ is orthogonal to N^* then by (b) $f = Au$ with $u \in H^s$ for any s, and u is smooth by Corollary 5.6.4. The range on $\mathcal{C}^\infty$ is thus the orthocomplement of N^* in $\mathcal{C}^\infty(X;F)$, which is closed, and since $N^* \subset \mathcal{C}^\infty$, $f = (f - \pi_{N^*} f) + \pi_{N^*} f$ gives the decomposition, where π_{N^*} is orthogonal projection onto N^*.

(d) By (b), the codimension of the range on H^{s-m} is $\dim N^*$, the elements of N^* being linearly independent functionals on H^{s-m}; similarly on $\mathcal{C}^\infty$ by (c). If $B' \in \Psi^{m-1}(X;E,F)$,

then $B' : H^s \to H^{s-m}$ is compact (it maps into H^{s-m+1}), so $\operatorname{ind}(A+B') = \operatorname{ind}(A)$ by the stability of the index under compact perturbations (Theorem A.3.6). The formula for A^* is clear from (a), the product formula follows from Proposition A.3.5 applied to $H^s \to H^{s-m'} \to H^{s-m'-m}$, and the constancy along continuous families is Theorem A.3.6 again. □

In particular, for $R \in \Psi^{-\infty}(X;E)$ the operator $\mathrm{Id}+R$ is Fredholm of index zero on $L^2(X;E)$ and on $\mathcal{C}^\infty(X;E)$: it is elliptic of order 0, and $\mathrm{Id}+tR$, $t \in [0,1]$, is a continuous family connecting it to the identity. Note also that the range in (c) is closed in $\mathcal{C}^\infty(X;F)$ and has a finite-dimensional complement; closedness of the range is part of the Fredholm property, and is not implied by the finite codimension of its closure.

5.8.1 Generalized inverses

We need a lemma on composites with smoothing operators.

Lemma 5.8.2. *Let $R_1 \in \Psi^{-\infty}(X;G',G)$, $R_2 \in \Psi^{-\infty}(X;E,E')$, and let $T : H^t(X;E') \to H^{t'}(X;G')$ be bounded, for some $t,t' \in \mathbb{R}$. Then $R_1 T R_2 \in \Psi^{-\infty}(X;E,G)$, and its kernel depends continuously, in $\mathcal{C}^\infty(X^2; \operatorname{Hom}(E,G)\otimes\Omega_R)$, on T in the operator norm.*

Proof. The kernel of R_1TR_2 is $K(x,z) = (R_1\, T\, k_z)(x)$, where $k_z = K_{R_2}(\cdot,z)$ (with the fibre of $E'_z \otimes \Omega_z$ trivialized locally in z). The map $z \mapsto k_z$ is smooth into $H^t(X;E')$, so $z \mapsto Tk_z$ is smooth into $H^{t'}(X;G')$ with derivatives bounded by $\|T\|$ times those of k_z, and $R_1 : H^{t'}(X;G') \to \mathcal{C}^\infty(X;G)$ is continuous, with $D_x^\alpha(R_1 g)(x)$ the pairing of g with the smooth family $D_x^\alpha K_{R_1}(x,\cdot) \in H^{-t'}$. Hence K is smooth, and each $\mathcal{C}^k$ norm of K is bounded by $C_k\|T\|$. □

Proposition 5.8.3. *Let $A \in \Psi^m(X;E,F)$ be elliptic and let π_N, π_{N^*} be the orthogonal projections onto $N = \operatorname{Nul}(A)$ and $N^* = \operatorname{Nul}(A^*)$. These are smoothing operators of finite rank, and there is a unique $G \in \Psi^{-m}(X;F,E)$ such that*

$$GA = \mathrm{Id}_E - \pi_N, \qquad AG = \mathrm{Id}_F - \pi_{N^*}, \qquad G\pi_{N^*} = 0. \tag{5.8.1}$$

Moreover $G - B \in \Psi^{-\infty}$ for any parametrix B.

Proof. If $e_1,\dots,e_k$ is an orthonormal basis of $N \subset \mathcal{C}^\infty(X;E)$, then $\pi_N u = \sum_j (u,e_j)e_j$ has kernel $\sum_j e_j(x)\otimes\langle\cdot,e_j(y)\rangle\nu(y)$, which is smooth; similarly for π_{N^*}. For $f \in H^{s-m}(X;F)$ let Gf be the unique $u \in H^s(X;E)$ with $u \perp N$ and $Au = (\mathrm{Id}-\pi_{N^*})f$; it exists by Theorem 5.8.1(b), and does not depend on s. The map A restricted to $N^\perp \cap H^s$ is a continuous bijection onto the closed range, so G is bounded $H^{s-m} \to H^s$ by the open mapping theorem. The identities (5.8.1) follow directly from the definition, and they determine G. Now

$$\begin{aligned} G &= G(AB+R_F) = (\mathrm{Id}-\pi_N)B + GR_F, \\ GR_F &= (BA+R_E)GR_F = B(\mathrm{Id}-\pi_{N^*})R_F + R_E G R_F. \end{aligned}$$

Here $R_E G R_F$ is smoothing by Lemma 5.8.2, and the remaining terms other than B are smoothing by Corollary 5.5.4. Thus $G - B \in \Psi^{-\infty}$. □

The generalized inverse is the unique linear map from $\mathcal{C}^\infty(X;F)$ to $\mathcal{C}^\infty(X;E)$ which vanishes on N^*, has range $N^\perp$ (the orthocomplement in $\mathcal{C}^\infty(X;E)$) and inverts A as a map from $N^\perp$ to the range of A.

Corollary 5.8.4. *For each $s \in \mathbb{R}$ and each bundle E there exists an invertible, formally self-adjoint, elliptic $\Lambda_s \in \Psi^s(X;E)$ with $\Lambda_s^{-1} \in \Psi^{-s}(X;E)$ and $(\Lambda_s u, u) \geq 0$. In particular $\|\Lambda_s u\|_{L^2}$ is a norm on $H^s(X;E)$ equivalent to $\|u\|_s$, and $\Lambda_s : H^t \to H^{t-s}$ is an isomorphism for every t.*

Proof. Let $Q \in \Psi^{s/2}(X;E)$ be elliptic, and $\Lambda_s = Q^*Q + \pi$, where π is the orthogonal projection onto the finite-dimensional space $\mathrm{Nul}(Q) \subset \mathcal{C}^\infty$. Then Λ_s is self-adjoint and elliptic, with principal symbol $\sigma(Q)^*\sigma(Q)$. If $\Lambda_s u = 0$ then u is smooth and $0 = (\Lambda_s u, u) = \|Qu\|^2 + \|\pi u\|^2$, so $u \in \mathrm{Nul}(Q)$ and $u = \pi u = 0$. Thus $\mathrm{Nul}(\Lambda_s) = \mathrm{Nul}(\Lambda_s^*) = 0$, and Λ_s is invertible on $\mathcal{C}^\infty$ and on every H^t with inverse its generalized inverse, an element of Ψ^{-s}. □

Proposition 5.8.5. *Let $A \in \Psi^m(X;E,F)$ be elliptic. There exists a finite-rank $R \in \Psi^{-\infty}(X;E,F)$ such that $A + \epsilon R$ is invertible for all $\epsilon \neq 0$ if and only if $\mathrm{ind}(A) = 0$. If $A + R$ is invertible, its inverse lies in $\Psi^{-m}(X;F,E)$.*

Proof. If $A+\epsilon R$ is invertible for some ϵ then $\mathrm{ind}(A) = \mathrm{ind}(A+\epsilon R) = 0$, by Theorem 5.8.1(d). Conversely, if $\mathrm{ind}(A) = 0$, choose orthonormal bases e_j of N and f_j of N^* (of the same dimension) and set $Ru = \sum_j (u, e_j) f_j$. If $(A+\epsilon R)u = 0$ then $Au = -\epsilon Ru \in \mathrm{Ran}(A) \cap N^* = \{0\}$, so $u \in N$ and $Ru = 0$, hence $u = 0$. Thus $A + \epsilon R$ is injective, of index zero, hence surjective, and its inverse is its generalized inverse, an element of Ψ^{-m}. □

Theorem 5.8.6. *Let $A \in \Psi^m(X;E)$, $m > 0$, be elliptic and formally self-adjoint. Then $L^2(X;E)$ has an orthonormal basis of eigenfunctions $Ae_j = \lambda_j e_j$ with $e_j \in \mathcal{C}^\infty(X;E)$, $\lambda_j \in \mathbb{R}$, each eigenvalue of finite multiplicity, and $|\lambda_j| \to \infty$.*

Proof. The generalized inverse $G \in \Psi^{-m}(X;E)$ of A is self-adjoint (the characterizing identities (5.8.1) are preserved by taking adjoints, since $N = N^*$) and compact on L^2 by Theorem 5.7.4. By the spectral theorem for compact self-adjoint operators (Theorem A.2.3), L^2 is the orthogonal sum of $\mathrm{Nul}(G) = N$ and the eigenspaces of G for non-zero eigenvalues $\mu_j \to 0$, each finite-dimensional. If $Gu = \mu u$ with $\mu \neq 0$ then $u \perp N$ and $u = AGu/\mu = Au/\mu$, so $Au = \mu^{-1}u$ and u is smooth by elliptic regularity. Taking orthonormal bases of N (eigenvalue 0) and of these eigenspaces gives the result, with $\lambda_j = \mu_j^{-1}$. □

5.9 Smoothing operators

Most results about elliptic operators are ultimately reduced, via the parametrix, to properties of smoothing operators. We fix a smooth positive density ν and hermitian metrics, and write the kernel of $A \in \Psi^{-\infty}(X;E,F)$ as $A(x,y) = k_A(x,y)\nu(y)$ with $k_A \in \mathcal{C}^\infty(X^2; \mathrm{Hom}(E,F))$. By the Cauchy–Schwarz inequality,

$$\|Au\|_{L^2} \leq \|k_A\|_{L^2(X^2)} \|u\|_{L^2}, \tag{5.9.1}$$

and indeed A is Hilbert–Schmidt with $\|A\|_2 = \|k_A\|_{L^2(X^2)}$ (Proposition A.4.2); in particular it is compact on L^2.

Lemma 5.9.1. *If $A \in \Psi^m(X;E,F)$ with $m < -n/2$ then A is Hilbert–Schmidt on L^2.*

Proof. By Lemma 5.5.1 it suffices to consider an operator given in local coordinates by $\mathrm{Op}(a)$, $a \in S^m$, with kernel of compact support. Its kernel is $K(x, x-z) = \mathcal{F}^{-1}_{\xi \to z} a(x,\cdot)(z)$, so by Plancherel $\iint |K(x,y)|^2 dy\, dx = \iint |a(x,\xi)|^2 đ\xi\, dx$ over the compact x-support, which is finite since $\langle \xi \rangle^{2m}$ is integrable. □

5.9.1 The group $G^{-\infty}$

Lemma 5.9.2. *If $A \in \Psi^{-\infty}(X;E)$ and $\mathrm{Id}+A$ is invertible on $L^2(X;E)$, then $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+B$ with $B \in \Psi^{-\infty}(X;E)$. In particular this holds if $\|A\|_{\mathcal{B}(L^2)} < 1$, when $B = \sum_{l\geq 1}(-A)^l$.*

Proof. Let $T = (\mathrm{Id}+A)^{-1}$. From $T = \mathrm{Id} - AT = \mathrm{Id} - TA$ we get $T = \mathrm{Id} - A(\mathrm{Id} - TA) = \mathrm{Id} - A + ATA$, so $B = -A + ATA$, which is smoothing by Lemma 5.8.2. □

Definition 5.9.3. For a compact manifold X and a vector bundle E over it,

$$\begin{aligned} G^{-\infty}(X;E) &= \{A \in \Psi^{-\infty}(X;E);\ \mathrm{Id}+A \text{ is invertible on } L^2(X;E)\} \\ &= \{A;\ (\mathrm{Id}+A)^{-1} = \mathrm{Id}+B,\ B \in \Psi^{-\infty}(X;E)\}, \end{aligned}$$

topologized as a subset of the Fréchet space $\Psi^{-\infty}(X;E) = \mathcal{C}^\infty(X^2; \mathrm{Hom}(E) \otimes \Omega_R)$, with the group law $(A,B) \mapsto A + B + AB$, i.e. composition of $\mathrm{Id}+A$ and $\mathrm{Id}+B$.

By Theorem 5.8.1, $\mathrm{Id}+A$ is invertible on L^2 if and only if it is invertible on $\mathcal{C}^\infty(X;E)$, i.e. is injective there, since the index is zero.

Proposition 5.9.4. *$G^{-\infty}(X;E)$ is an open subset of $\Psi^{-\infty}(X;E)$ and a topological group.*

Proof. Let $A \in G^{-\infty}$ and $\delta = \|(\mathrm{Id}+A)^{-1}\|^{-1}$. If $\|B\|_{\mathcal{B}(L^2)} < \delta$ then $\mathrm{Id}+A+B = (\mathrm{Id}+A)(\mathrm{Id}+(\mathrm{Id}+A)^{-1}B)$ is invertible; by (5.9.1) this holds for B in a neighbourhood of zero in $\Psi^{-\infty}$, so $G^{-\infty}$ is open. Composition, with kernel $k_A + k_B + \int k_A(x,y)k_B(y,z)\nu(y)$, is continuous; the inverse $B = -A + A(\mathrm{Id}+A)^{-1}A$ depends continuously on A by Lemma 5.8.2, since $A \mapsto (\mathrm{Id}+A)^{-1}$ is continuous into $\mathcal{B}(L^2)$. □

One should think of $G^{-\infty}(X;E)$ as the group of invertible operators on $L^2(X;E)$ which differ from the identity by an operator with smooth kernel; it plays the role of an infinite-dimensional general linear group, as the following results show.

5.9.2 Finite-rank approximation

Lemma 5.9.5. *A smoothing operator $A \in \Psi^{-\infty}(X;E,F)$ has finite rank if and only if*

$$A(x,y) = \sum_{i=1}^{N} f_i(x)e_i(y)\nu(y), \qquad f_i \in \mathcal{C}^\infty(X;F),\ e_i \in \mathcal{C}^\infty(X;E'). \tag{5.9.2}$$

Proof. Such an operator has range in the span of the f_i. Conversely, if the range is finite-dimensional, it lies in $\mathcal{C}^\infty(X;F)$; let $f_1,\dots,f_N$ be an orthonormal basis of it. Then $Au = \sum_i (Au,f_i)f_i$ and $(Au,f_i) = (u,A^*f_i) = \int_X (A^*f_i)^\dagger(y)u(y)\nu(y)$, where $(A^*f_i)^\dagger \in \mathcal{C}^\infty(X;E')$ is the dual of A^*f_i under the metric; this is (5.9.2) with $e_i = (A^*f_i)^\dagger$. □

If the e_i and f_j in (5.9.2) are taken to be orthonormal, and $E = F$, one may orthonormalize all of them together, obtaining a finite orthonormal set $v_1,\dots,v_M \subset \mathcal{C}^\infty(X;E)$ and $Au = \sum_{j,k} a_{jk}(u,v_k)v_j$; the finite-rank smoothing operators with range and coefficients in a fixed such span form a subalgebra isomorphic, as an algebra, to $M(M,\mathbb{C})$.

Lemma 5.9.6. *The finite-rank operators are dense in $\Psi^{-\infty}(X;E,F)$. More precisely, for every A there are finite-rank A_N with $k_{A_N} \to k_A$ in $\mathcal{C}^\infty(X^2;\mathrm{Hom}(E,F))$, and for a smooth family $A(p)$ depending on parameters p in a compact manifold P the $A_N(p)$ may be chosen as in (5.9.2) with f_i, e_i independent of p and coefficients depending smoothly on p, converging uniformly with all derivatives.*

Proof. Consider first the torus $\mathbb{T}^n = \mathbb{R}^n/2\pi\mathbb{Z}^n$ with density $|d\theta|$ and trivial bundles. A kernel $a \in \mathcal{C}^\infty(\mathbb{T}^{2n})$ has a rapidly convergent Fourier expansion

$$a(\theta,\theta') = \sum_{I,J\in\mathbb{Z}^n} a_{IJ} e^{iI\cdot\theta} e^{-iJ\cdot\theta'}, \quad a_{IJ} = (2\pi)^{-2n} \int_{\mathbb{T}^{2n}} e^{-iI\cdot\theta+iJ\cdot\theta'} a(\theta,\theta')\, d\theta\, d\theta',$$

the coefficients being rapidly decreasing in (I,J) since a is smooth. Hence the truncations $a_N = \sum_{|I|,|J|\le N} a_{IJ} e^{iI\cdot\theta} e^{-iJ\cdot\theta'}$, which are finite-rank kernels, converge to a in $\mathcal{C}^\infty$, and continuously in parameters.

In general choose a finite cover of X by charts $\kappa_p : U_p \to \kappa_p(U_p)$ with bounded images, over which E and F are trivial, and a subordinate partition of unity ρ_p. Then $A = \sum_{p,q} \rho_p A \rho_q$. In the coordinates $\kappa_p(x)$, $\kappa_q(y)$ (rescaled and translated so that $\kappa_p(U_p)$ and $\kappa_q(U_q)$ lie in $(0,2\pi)^n$) the kernel of $\rho_p A \rho_q$, with respect to the coordinate density, is a matrix of functions $a \in \mathcal{C}^\infty_{\mathrm{c}}((0,2\pi)^{2n})$, which we regard as functions on $\mathbb{T}^{2n}$. Choose $\chi_p \in \mathcal{C}^\infty_{\mathrm{c}}(\kappa_p(U_p))$ equal to one near $\kappa_p(\operatorname{supp}\rho_p)$, similarly χ_q. Then $\chi_p(\theta) a_N(\theta,\theta')\chi_q(\theta')$ converges to a in $\mathcal{C}^\infty$, and is a finite sum of products of functions of θ and of θ', compactly supported in the coordinate patches. Transferring back to X and summing over p,q gives finite-rank smoothing operators converging to A; the functions involved depend only on N, not on parameters. □

Proposition 5.9.7. *$G^{-\infty}(X;E)$ is connected (by smooth paths).*

Proof. Let $A \in G^{-\infty}$ and $\delta = \|(\mathrm{Id}+A)^{-1}\|^{-1}_{\mathcal{B}(L^2)}$. As in the proof of Proposition 5.9.4, the convex set $\{A+B;\ \|B\|_{\mathcal{B}(L^2)} < \delta\} \cap \Psi^{-\infty}$ lies in $G^{-\infty}$. By Lemma 5.9.6 and (5.9.1) there is a finite-rank A_N in this set, and the segment from A to A_N lies in $G^{-\infty}$. Let $V \subset \mathcal{C}^\infty(X;E)$ be a finite-dimensional space containing the range of A_N and the range of A_N^*, with orthonormal basis $v_1,\dots,v_M$. Then A_N vanishes on $V^\perp$ and maps V to itself, with matrix a, so $\mathrm{Id}+A_N = (\mathrm{Id}_V + a) \oplus \mathrm{Id}_{V^\perp}$ and $\mathrm{Id}+a \in \mathrm{GL}(M,\mathbb{C})$. Since $\mathrm{GL}(M,\mathbb{C})$ is connected, there is a smooth path $g(t)$ in it from $\mathrm{Id}+a$ to Id, and $\sum_{j,k}(g(t)-\mathrm{Id})_{jk}(\cdot,v_k)v_j$ is a smooth path in $G^{-\infty}$ from A_N to 0. □

5.9.3 Odd K-theory

The homotopy classes of maps into $G^{-\infty}$ give a direct definition of odd K-theory: for a compact manifold X set

$$K^{-1}(X) = [X; G^{-\infty}(Y;E)], \tag{5.9.3}$$

the set of smooth homotopy classes of smooth maps $X \to G^{-\infty}(Y;E)$, where Y is any compact manifold of positive dimension and E any vector bundle over Y. Here a smooth map is an element

$$K \in \mathcal{C}^\infty(X \times Y^2; \mathrm{Hom}(E) \otimes \Omega_R) \ \text{ with } \ \mathrm{Id}+K(x,\cdot) \text{ invertible for every } x \in X,$$

and two such are homotopic if they are the ends of a smooth map $X \times [0,1] \to G^{-\infty}(Y;E)$. Implicit in (5.9.3) is the following fact; the group structure is induced by that of $G^{-\infty}$, and the relation with the usual definition by stable homotopy classes of maps into $\mathrm{GL}(N,\mathbb{C})$ is visible in the proof.

Proposition 5.9.8. *For any two choices (Y,E), (Z,F) with $\dim Y, \dim Z > 0$ there is a natural bijection $[X; G^{-\infty}(Y;E)] \cong [X; G^{-\infty}(Z;F)]$.*

Sketch of proof. It suffices to compare with the model $Y = \mathbb{S}$, $E = \mathbb{C}$. Embed $\mathrm{GL}(N,\mathbb{C})$ into $G^{-\infty}(\mathbb{S})$ by

$$(a_{kl}) \longmapsto \sum_{k,l=1}^{N} (a_{kl} - \delta_{kl})\, e^{ik\theta} e^{-il\theta'} \frac{|d\theta'|}{2\pi};$$

the identity is subtracted so that this is a group homomorphism. Given a smooth map $A : X \to G^{-\infty}(Y;E)$, Lemma 5.9.6 gives finite-rank approximations $A_{(N)}(x)$, with fixed functions f_i, e_i, converging uniformly with all derivatives; for N large the straight line from A to $A_{(N)}$ stays in $G^{-\infty}$, by the openness estimate, which is uniform since X is compact. Choosing an orthonormal basis of the span of the f_i and e_i, as in Proposition 5.9.7, identifies $\mathrm{Id} + A_{(N)}(x)$ with a smooth map into some $\mathrm{GL}(M,\mathbb{C})$, and hence, via the embedding, a map $\tilde{A} : X \to G^{-\infty}(\mathbb{S})$. A different basis conjugates $\tilde{A}$ by a fixed element of $G^{-\infty}(\mathbb{S})$, which can be connected to the identity by Proposition 5.9.7; adding redundant elements to the basis does not change $\tilde{A}$; and increasing N changes $A_{(N)}$ only within the convex neighbourhoods used above. So the class of $\tilde{A}$ is well defined, and applying the construction to $X \times [0,1]$ shows it depends only on the homotopy class of A. Reversing the roles, using an orthonormal set of smooth sections of E over Y (which exists in any finite number since $\dim Y > 0$) gives the inverse map. □

5.9.4 The trace

For $A \in \Psi^{-\infty}(X;E)$ the restriction of the kernel to the diagonal is a section of $\hom(E) \otimes \Omega$, and we set

$$\mathrm{Tr}(A) = \int_X \mathrm{tr}\,(A|_{\mathrm{Diag}}) = \int_X \mathrm{tr}\, k_A(x,x)\, \nu(x). \tag{5.9.4}$$

For a finite-rank operator $Au = \sum_{j,k} a_{jk}(u, v_k) v_j$ with orthonormal v_j as above, $k_A(x,x) = \sum_{j,k} a_{jk} v_j(x) \otimes v_k(x)^\dagger$, so $\mathrm{Tr}(A) = \sum_{j,k} a_{jk}(v_j, v_k) = \sum_j a_{jj}$, the trace of its matrix; since the matrix trace is invariant under change of basis, $\mathrm{tr}(b^{-1}ab) = \sum_{i,j,k} b^{-1}_{ij} a_{jk} b_{ki} = \mathrm{tr}\, a$, this is consistent.

Proposition 5.9.9. *The trace* (5.9.4) *is a continuous linear functional on* $\Psi^{-\infty}(X;E)$, *and*

(a) $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ *for* $A \in \Psi^{-\infty}(X;E,F)$, $B \in \Psi^{-\infty}(X;F,E)$;

(b) *each* $A \in \Psi^{-\infty}(X;E)$ *is of trace class on* $L^2(X;E)$, *with* $\|A\|_1 \le C\|k_A\|_{\mathcal{C}^k}$ *for some* k, *and* (5.9.4) *is its Hilbert space trace; in particular* $\mathrm{Tr}(A) = \lim_N \mathrm{Tr}(A_N)$ *for any finite-rank approximations as in Lemma 5.9.6;*

(c) $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ *for* $A \in \Psi^{-\infty}(X;E)$ *and* $B \in \Psi^m(X;E)$, *any* m.

Proof. Continuity is clear. (a) By Fubini,

$$\begin{aligned}\mathrm{Tr}(AB) &= \int_X \int_X \mathrm{tr}\,(k_A(x,y) k_B(y,x))\nu(y)\nu(x) \\ &= \int_X \int_X \mathrm{tr}\,(k_B(y,x) k_A(x,y))\nu(x)\nu(y) = \mathrm{Tr}(BA),\end{aligned}$$

using $\mathrm{tr}(ab) = \mathrm{tr}(ba)$ for $a \in \hom(E_y, F_x)$, $b \in \hom(F_x, E_y)$.

(b) Let s be an even integer with $s > n/2$ and Λ_s as in Corollary 5.8.4. Then $A = \Lambda_s^{-1}(\Lambda_s A)$, where Λ_s^{-1} is Hilbert–Schmidt by Lemma 5.9.1, and $\Lambda_s A$ is smoothing with $\|\Lambda_s A\|_2^2 = \int \|\Lambda_s k_A(\cdot,y)\|_{L^2}^2 \nu(y) \le C \int \|k_A(\cdot,y)\|_s^2 \nu(y) \le C' \|k_A\|_{\mathcal{C}^s}^2$. So A is of trace class with $\|A\|_1 \le \|\Lambda_s^{-1}\|_2 \|\Lambda_s A\|_2 \le C\|k_A\|_{\mathcal{C}^s}$ (Definition A.4.3). For finite-rank A, (5.9.2) gives

Hilbert space trace $\sum_i (f_i, e_i^\dagger) = \int \sum_i e_i(x)(f_i(x))\nu(x)$, which is (5.9.4). Both traces are continuous on $\Psi^{-\infty}$ (the first by the estimate just proved), so they agree by Lemma 5.9.6.

(c) Write $B = \Lambda_m(\Lambda_m^{-1}B)$ with $\Lambda_m^{-1}B \in \Psi^0$ bounded. The operators $A\Lambda_m$ and BA are smoothing, hence of trace class, and using that $\operatorname{Tr}(ST) = \operatorname{Tr}(TS)$ for S of trace class and T bounded (Proposition A.4.4), $\operatorname{Tr}(AB) = \operatorname{Tr}\big((A\Lambda_m)(\Lambda_m^{-1}B)\big) = \operatorname{Tr}(\Lambda_m^{-1}BA\Lambda_m) = \operatorname{Tr}\big((BA\Lambda_m)\Lambda_m^{-1}\big) = \operatorname{Tr}(BA)$. □

Remark 5.9.10. Using the trace one can define the Fredholm determinant: there is a unique entire function $\det(\mathrm{Id}+A)$ on $\Psi^{-\infty}(X;E)$ such that

$$\det\big((\mathrm{Id}+A)(\mathrm{Id}+B)\big) = \det(\mathrm{Id}+A)\det(\mathrm{Id}+B), \qquad \tfrac{d}{ds}\det(\mathrm{Id}+sA)|_{s=0} = \operatorname{Tr}(A),$$

and $A \in G^{-\infty}(X;E)$ if and only if $\det(\mathrm{Id}+A) \neq 0$. It follows that $G^{-\infty}(X;E)$ is dense in $\Psi^{-\infty}(X;E)$. Since $G^{-\infty}$ is connected, the determinant can be defined on $G^{-\infty}$ by

$$\det(\mathrm{Id}+A) = \exp\Big(\int_0^1 \operatorname{Tr}\big((\mathrm{Id}+\gamma(t))^{-1}\tfrac{d\gamma}{dt}\big)dt\Big)$$

for a smooth curve γ in $G^{-\infty}$ from 0 to A; the main points are independence of the curve and the extension to all of $\Psi^{-\infty}$. This is carried out in Chapter 10; compare also Chapter 7.

Proposition 5.9.11. *Let $A \in \Psi^{-\infty}(X;E)$. The set*

$$\operatorname{Spec}(A)\setminus\{0\} = \{z \in \mathbb{C}\setminus\{0\};\ z\,\mathrm{Id}-A \text{ is not invertible on } L^2(X;E)\}$$

is contained in $\{|z| \leq \|A\|\}$ and is discrete in $\mathbb{C}\setminus\{0\}$; each of its points is an eigenvalue, and the generalized eigenspace

$$E(z) = \{u \in L^2(X;E);\ (z\,\mathrm{Id}-A)^N u = 0 \text{ for some } N \in \mathbb{N}\}$$

is finite-dimensional and contained in $\mathcal{C}^\infty(X;E)$.

Proof. For $|z| > \|A\|$, $z - A = z(\mathrm{Id} - A/z)$ is invertible by the Neumann series. Since A is compact, the other statements, except smoothness, are the Riesz theory of compact operators (Theorem A.3.7). If $(z-A)^N u = 0$ with $z \neq 0$, expanding by the binomial theorem gives $z^N u = Ap(A)u$ for a polynomial p, so $u \in \operatorname{Ran}(A) \subset \mathcal{C}^\infty(X;E)$. □

Alternatively, $\operatorname{Spec}(A)\setminus\{0\}$ is the zero set of the entire function $z \mapsto \det(\mathrm{Id} - A/z)$ on $\mathbb{C}\setminus\{0\}$, which tends to one as $|z| \to \infty$.

5.10 Elliptic complexes and the Hodge theorem

5.10.1 Elliptic complexes

Let $E_0, E_1, \dots, E_N$ be vector bundles over X and $P_i \in \operatorname{Diff}^k(X; E_i, E_{i+1})$ differential operators with $P_{i+1}P_i = 0$, so that

$$0 \longrightarrow \mathcal{C}^\infty(X;E_0) \xrightarrow{P_0} \mathcal{C}^\infty(X;E_1) \xrightarrow{P_1} \cdots \xrightarrow{P_{N-1}} \mathcal{C}^\infty(X;E_N) \longrightarrow 0 \tag{5.10.1}$$

is a complex (we set $P_{-1} = 0$, $P_N = 0$). Then $\sigma_k(P_{i+1})\sigma_k(P_i) = \sigma_{2k}(P_{i+1}P_i) = 0$, and the complex is called *elliptic* if the symbol sequence

$$0 \to E_{0,x} \xrightarrow{\sigma_k(P_0)(x,\xi)} E_{1,x} \to \cdots \to E_{N,x} \to 0$$

is exact for every $(x,\xi) \in T^*X \setminus 0$. Its cohomology is $H^i = \operatorname{Nul}(P_i)/P_{i-1}\mathcal{C}^\infty(X;E_{i-1})$, where P_i acts on smooth sections.

Choose a positive density and hermitian metrics, and let P_i^* be the formal adjoints; they form the adjoint complex, $P_{i-1}^*P_i^* = (P_iP_{i-1})^* = 0$. Set

$$\Delta_i = P_i^*P_i + P_{i-1}P_{i-1}^* \in \operatorname{Diff}^{2k}(X;E_i).$$

Theorem 5.10.1 (Hodge)**.** *For an elliptic complex* (5.10.1) *on a compact manifold:*

(a) *each Δ_i is elliptic and formally self-adjoint, and its null space*

$$\mathcal{H}^i = \{u \in \mathcal{C}^\infty(X;E_i);\ P_iu = 0,\ P_{i-1}^*u = 0\}$$

is finite-dimensional;

(b) *there is an L^2-orthogonal decomposition*

$$\mathcal{C}^\infty(X;E_i) = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^\infty(X;E_{i-1}) \oplus P_i^*\mathcal{C}^\infty(X;E_{i+1}); \tag{5.10.2}$$

(c) *the ranges of the P_i on smooth sections are closed, and we have $\operatorname{Nul}(P_i) = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^\infty(X;E_{i-1})$, so that the natural map $\mathcal{H}^i \to H^i$ is an isomorphism; in particular the cohomology of the complex is finite-dimensional.*

Proof. (a) The principal symbol is $\sigma_{2k}(\Delta_i) = \sigma_k(P_i)^*\sigma_k(P_i) + \sigma_k(P_{i-1})\sigma_k(P_{i-1})^*$, a self-adjoint endomorphism. If it annihilates $v \in E_{i,x}$, $\xi \neq 0$, then $0 = \langle \sigma_{2k}(\Delta_i)v, v\rangle = |\sigma_k(P_i)v|^2 + |\sigma_k(P_{i-1})^*v|^2$, so $v \in \operatorname{Nul}\sigma_k(P_i) = \operatorname{Ran}\sigma_k(P_{i-1})$ by exactness, and also $v \perp \operatorname{Ran}\sigma_k(P_{i-1})$; hence $v = 0$. Being homogeneous of degree $2k$ and invertible on $T^*X \setminus 0$, the symbol is elliptic (Lemma 5.6.2). For $u \in \mathcal{C}^\infty$, $(\Delta_i u, u) = \|P_iu\|^2 + \|P_{i-1}^*u\|^2$, which identifies the null space, and it is finite-dimensional by Theorem 5.8.1.

(b) Since Δ_i is self-adjoint, $\operatorname{Nul}(\Delta_i^*) = \mathcal{H}^i$, and the generalized inverse G_i of Proposition 5.8.3 satisfies $\Delta_iG_i = \operatorname{Id} - \pi_i$, where π_i is orthogonal projection onto $\mathcal{H}^i$. Thus for $u \in \mathcal{C}^\infty(X;E_i)$

$$u = \pi_iu + P_{i-1}(P_{i-1}^*G_iu) + P_i^*(P_iG_iu).$$

The three spaces are mutually orthogonal: $(h, P_{i-1}v) = (P_{i-1}^*h, v) = 0$ and $(h, P_i^*w) = (P_ih, w) = 0$ for $h \in \mathcal{H}^i$, and $(P_{i-1}v, P_i^*w) = (P_iP_{i-1}v, w) = 0$.

(c) If $P_iu = 0$, then $w = P_i^*P_iG_iu$ satisfies $P_iw = P_i(u - \pi_iu - P_{i-1}P_{i-1}^*G_iu) = 0$, so $\|w\|^2 = (P_iG_iu, P_iw) = 0$. Hence $\operatorname{Nul}(P_i) = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^\infty(X;E_{i-1})$, and $P_{i-1}\mathcal{C}^\infty(X;E_{i-1}) = \operatorname{Nul}(P_i) \cap (\mathcal{H}^i)^\perp$ is closed in $\mathcal{C}^\infty$ (and its closure in L^2 meets $\mathcal{C}^\infty$ in itself). The map $\mathcal{H}^i \to H^i$ is therefore surjective, and it is injective since $\mathcal{H}^i \perp P_{i-1}\mathcal{C}^\infty(X;E_{i-1})$. □

5.10.2 The de Rham complex and the Laplacian on forms

The exterior derivative $d : \mathcal{C}^\infty(X;\Lambda^p) \to \mathcal{C}^\infty(X;\Lambda^{p+1})$ extends the differential of functions, satisfies $d^2 = 0$, and in local coordinates is $d = \sum_j dx_j \wedge \partial_{x_j} = i\sum_j dx_j \wedge D_{x_j}$. As computed in Section 5.4, $\sigma_1(d)(x,\xi) = i\,\xi\wedge$. The de Rham complex

$$0 \to \mathcal{C}^\infty(X;\Lambda^0) \xrightarrow{d} \mathcal{C}^\infty(X;\Lambda^1) \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{C}^\infty(X;\Lambda^n) \to 0 \tag{5.10.3}$$

is elliptic: for $\xi \neq 0$, the forms $\alpha \in \Lambda_x^p$ with $\xi \wedge \alpha = 0$ are exactly those of the form $\xi \wedge \beta$, $\beta \in \Lambda_x^{p-1}$, as one sees by choosing a basis $e_1 = \xi, e_2, \dots, e_n$ of T_x^*X and expanding α in the induced basis of Λ_x^p. The cohomology of (5.10.3) is the de Rham cohomology $H^p_{\mathrm{dR}}(X)$.

Choose a Riemannian metric g on X, which induces hermitian metrics on the Λ^p and the Riemannian density. Let d^* be the formal adjoint of d and

$$\Delta = dd^* + d^*d \in \mathrm{Diff}^2(X;\Lambda^p), \qquad \sigma_2(\Delta)(x,\xi) = |\xi|_g^2 \,\mathrm{Id},$$

since $\sigma_1(d^*) = -i\,\iota_\xi$ (contraction with the dual vector of ξ) and $\xi\wedge\iota_\xi + \iota_\xi\,\xi\wedge = |\xi|^2$. On functions, Δ is the Laplace–Beltrami operator, non-negative with null space the locally constant functions.

Corollary 5.10.2 (Hodge theorem)**.** *On a compact Riemannian manifold, the space $\mathcal{H}^p$ of harmonic p-forms, $\Delta u = 0$ (equivalently $du = 0$ and $d^*u = 0$), is finite-dimensional, every de Rham cohomology class contains a unique harmonic representative, and*

$$\mathcal{C}^\infty(X;\Lambda^p) = \mathcal{H}^p \oplus d\,\mathcal{C}^\infty(X;\Lambda^{p-1}) \oplus d^*\mathcal{C}^\infty(X;\Lambda^{p+1}).$$

Thus the Betti numbers $b_p = \dim H^p_{\mathrm{dR}}(X)$ are finite. On the full form bundle $\Lambda^* = \bigoplus_p \Lambda^p$, the operator $d + d^*$ satisfies $(d+d^*)^2 = \Delta$, so is elliptic, and it maps even forms to odd forms.

Corollary 5.10.3. *The index of $d + d^* : \mathcal{C}^\infty(X;\Lambda^{\mathrm{ev}}) \to \mathcal{C}^\infty(X;\Lambda^{\mathrm{odd}})$ is the Euler characteristic $\chi(X) = \sum_p (-1)^p b_p$.*

Proof. If $(d+d^*)u = 0$ then $\Delta u = 0$; conversely if $\Delta u = 0$ then each component of u satisfies $du_p = 0$, $d^*u_p = 0$. So the null space of $d + d^*$ on even forms is $\bigoplus_p \mathcal{H}^{2p}$; its adjoint is $d + d^*$ on odd forms, with null space $\bigoplus_p \mathcal{H}^{2p+1}$. Now apply Theorem 5.8.1(d) and Corollary 5.10.2. □

In particular, the index of this operator does not depend on the Riemannian metric, as also follows from the homotopy invariance of the index.

5.11 Pseudodifferential projections

In this section E is a hermitian bundle over the compact manifold X, with L^2 defined by a positive density.

Proposition 5.11.1. *If $P \in \Psi^0(X;E)$ satisfies $P^2 - P \in \Psi^{-\infty}(X;E)$ then there exists $\Pi \in \Psi^0(X;E)$ with $\Pi^2 = \Pi$ and $\Pi - P \in \Psi^{-\infty}(X;E)$.*

Proof. Let $R = P^2 - P$, which commutes with P. For $\lambda \notin \{0,1\}$ a direct computation gives

$$(\lambda - P)(P + \lambda - 1) = \lambda(\lambda-1) - R = \lambda(\lambda-1)\big(\mathrm{Id} - S_\lambda\big), \qquad S_\lambda = \frac{R}{\lambda(\lambda-1)}, \tag{5.11.1}$$

and all these operators commute. Hence if $\lambda(\lambda-1) \notin \mathrm{Spec}(R)$, then $\lambda - P$ is invertible, with

$$(\lambda - P)^{-1} = \frac{P+\lambda-1}{\lambda(\lambda-1)}(\mathrm{Id} - S_\lambda)^{-1} = \frac{P+\lambda-1}{\lambda(\lambda-1)} + \frac{P+\lambda-1}{\lambda(\lambda-1)}\,S_\lambda(\mathrm{Id} - S_\lambda)^{-1}.$$

Since R is compact, $\mathrm{Spec}(R)\setminus\{0\}$ is countable (Theorem A.3.7), so the set of $r \in (\frac14, \frac34)$ such that the circle $\gamma_r = \{|\lambda - 1| = r\}$ meets $\{\lambda; \lambda(\lambda-1) \in \mathrm{Spec}(R)\}$ is countable, and we may fix r with $\gamma = \gamma_r$ disjoint from $\mathrm{Spec}(P)$. Let

$$\Pi = \frac{1}{2\pi i}\oint_\gamma (\lambda - P)^{-1} d\lambda.$$

The first term in the expansion of the resolvent contributes, by the residue theorem (the only pole inside γ is at $\lambda = 1$), exactly P. In the second term, $S_\lambda(\mathrm{Id} - S_\lambda)^{-1} = S_\lambda + S_\lambda(\mathrm{Id} - S_\lambda)^{-1}S_\lambda$ is smoothing, with kernel depending continuously on $\lambda \in \gamma$ by Lemma 5.8.2, and remains so after composition with $P + \lambda - 1$; so its integral is smoothing. Hence $\Pi - P \in \Psi^{-\infty}$.

That $\Pi^2 = \Pi$ is the standard property of Riesz projections: with $\gamma' = \gamma_{r'}$, $r' < r$ close to r and also disjoint from $\mathrm{Spec}(P)$, the resolvent identity $(\lambda - P)^{-1}(\mu - P)^{-1} = ((\lambda - P)^{-1} - (\mu - P)^{-1})/(\mu - \lambda)$ gives

$$\Pi^2 = \frac{1}{(2\pi i)^2} \oint_{\gamma'} \oint_{\gamma} \frac{(\lambda - P)^{-1} - (\mu - P)^{-1}}{\mu - \lambda}\, d\mu\, d\lambda = \frac{1}{2\pi i} \oint_{\gamma'} (\lambda - P)^{-1} d\lambda = \Pi,$$

since $\oint_\gamma (\mu - \lambda)^{-1} d\mu = 2\pi i$ for λ on γ' (inside γ) and $\oint_{\gamma'} (\mu - \lambda)^{-1} d\lambda = 0$ for μ on γ; the integral over γ' equals that over γ by Cauchy's theorem, the resolvent being holomorphic between the circles. □

Proposition 5.11.2. *Let $P \in \Psi^0(X; E)$ satisfy $P^2 - P \in \Psi^{-\infty}(X; E)$, and let $F \subset H^s(X; E)$ be a closed subspace for which there are $A, B \in \Psi^{-\infty}(X; E)$ with*

$$(\mathrm{Id} - P)u = Au \quad \forall u \in F \quad \text{and} \quad (P + B)H^s(X; E) \subset F.$$

Then there exists $B' \in \Psi^{-\infty}(X; E)$ such that $(P + B')^2 = P + B'$ and $F = (P + B')H^s(X; E)$.

Proof. Assume first that $s = 0$. All modifications below change P by smoothing operators, and the hypotheses are preserved, with new A and B, under such changes: if $P' = P + C$ with C smoothing, then $\mathrm{Id} - P' = A - C$ on F and $P' + (B - C) = P + B$.

Step 1. By Proposition 5.11.1 we may assume that P is a projection, $P^2 = P$. Then $\mathrm{Ran}(\mathrm{Id} - P) = \mathrm{Nul}(P)$.

Step 2. Let $E_0 = F \cap \mathrm{Nul}(P)$. On E_0, $u = (\mathrm{Id} - P)u = Au$, so $E_0 \subset \mathrm{Nul}(\mathrm{Id} - A)$, which is finite-dimensional and consists of smooth sections (Theorem 5.8.1). Let $e_1, \dots, e_k$ be an orthonormal basis of E_0 and $f_j = (\mathrm{Id} - P^*)e_j \in \mathcal{C}^\infty(X; E)$; then $(e_l, f_j) = ((\mathrm{Id} - P)e_l, e_j) = \delta_{lj}$, so $Qu = \sum_j (u, f_j)e_j$ is a smoothing projection onto E_0. Moreover $QP = 0$, since $P^* f_j = 0$, and $PQ = 0$, since $e_j \in \mathrm{Nul}(P)$; hence $P_1 = P + Q$ is a projection. If $u \in F$ and $P_1 u = 0$, then $Pu = -Qu \in E_0 \subset \mathrm{Nul}(P)$, so $Pu = P^2 u = 0$, then $Qu = 0$, $u \in E_0$ and $u = Qu = 0$. Thus, replacing P by P_1, we may assume $F \cap \mathrm{Nul}(P) = \{0\}$.

Step 3. Let $G = F + \mathrm{Nul}(P)$. Since $\mathrm{Id} + B = (P + B) + (\mathrm{Id} - P)$, the range of $\mathrm{Id} + B$ lies in G. By Theorem 5.8.1, $\mathrm{Ran}(\mathrm{Id} + B)$ is closed, with L^2-orthogonal complement $N' = \mathrm{Nul}(\mathrm{Id} + B^*) \subset \mathcal{C}^\infty(X; E)$ of finite dimension. Hence $G = \mathrm{Ran}(\mathrm{Id} + B) \oplus (G \cap N')$ is closed, and if C' is the orthocomplement of $G \cap N'$ in N' then $L^2(X; E) = G \oplus C'$ with $C' \subset \mathcal{C}^\infty$ finite-dimensional. Let $c_1, \dots, c_l$ be a basis of C'. Since $C' \cap \mathrm{Nul}(P) = \{0\}$, the $c'_j = Pc_j$ are linearly independent; let h_j be the dual basis of their span with respect to the L^2 inner product and $g_j = P^* h_j$. Then $Q'u = \sum_j (u, g_j)c'_j$ is a smoothing operator with $Q'P = Q'$, $PQ' = Q'$, $Q'c'_j = c'_j$, so $Q'^2 = Q'$ and $P_2 = P - Q'$ is a projection. One checks that $\mathrm{Nul}(P_2) = \mathrm{Nul}(P) \oplus C'$: indeed $P_2 c_j = c'_j - Q'Pc_j = 0$, $\mathrm{Nul}(P) \subset \mathrm{Nul}(P_2)$, and if $P_2 u = 0$ then $Pu = Q'u = \sum a_j Pc_j$, so $u - \sum a_j c_j \in \mathrm{Nul}(P)$. Consequently $L^2 = F \oplus \mathrm{Nul}(P_2)$: the sum is $G \oplus C'$, and if $f = v + c$ with $f \in F$, $v \in \mathrm{Nul}(P)$, $c \in C'$, then $c \in G \cap C' = \{0\}$ and $f \in F \cap \mathrm{Nul}(P) = \{0\}$. Replacing P by P_2, we may therefore assume

$$L^2(X; E) = F \oplus \mathrm{Nul}(P), \tag{5.11.2}$$

a direct, not necessarily orthogonal, sum of closed subspaces.

Step 4. Let P' be the projection with range F and null space $\mathrm{Nul}(P)$, which is bounded by (5.11.2) and the closed graph theorem. Since $\mathrm{Nul}(P') = \mathrm{Nul}(P) = \mathrm{Ran}(\mathrm{Id}-P)$, we have $P'(\mathrm{Id}-P) = 0$ and $P(\mathrm{Id}-P') = 0$, so $P' = P'P$ and $P = PP'$. Hence

$$P' - P = (\mathrm{Id}-P)P' = AP',$$

since $P'u \in F$ and $\mathrm{Id}-P = A$ on F. Also $P'(P+B) = P+B$ since $(P+B)L^2 \subset F$, so $P'-P = P'P-P = B-P'B = (\mathrm{Id}-P')B$. Combining,

$$P'-P = AP' = A(P+(\mathrm{Id}-P')B) = AP + A(\mathrm{Id}-P')B,$$

which is smoothing by Corollary 5.5.4 and Lemma 5.8.2. Thus $B' = P'-P$ (relative to the original P, plus the smoothing changes made in Steps 1–3) has the required properties.

For general s, let Λ_s be as in Corollary 5.8.4. Then $\tilde{P} = \Lambda_s P \Lambda_s^{-1} \in \Psi^0$ satisfies $\tilde{P}^2 - \tilde{P} \in \Psi^{-\infty}$, $\tilde{F} = \Lambda_s F \subset L^2$ is closed, and the hypotheses hold with $\tilde{A} = \Lambda_s A \Lambda_s^{-1}$, $\tilde{B} = \Lambda_s B \Lambda_s^{-1}$. The case $s = 0$ gives $\tilde{B}'$, and $B' = \Lambda_s^{-1}\tilde{B}'\Lambda_s$ has the required properties. □

Remark 5.11.3. The functional calculus of elliptic operators, through the heat kernel e^{-tA}, the resolvent $(A-\lambda)^{-1}$ with parameter, and the complex powers A^z of a positive elliptic operator, is not developed here; for these see [11] and [15].

5.12 Exercises

Exercise 5.1. Show that compatibility, as defined in Definition 5.1.1, is an equivalence relation on $\mathcal{C}^\infty$ structures. Conclude that there is a unique maximal $\mathcal{C}^\infty$ structure containing any given $\mathcal{C}^\infty$ structure.

Exercise 5.2. Let $\mathcal{F}$ be a $\mathcal{C}^\infty$ structure on X and let O_a, $a \in \mathcal{A}$, be a covering of X by coordinate neighbourhoods as in (5.1.1). Show that the maximal $\mathcal{C}^\infty$ structure containing $\mathcal{F}$ consists of all functions on X which, on each O_a, are of the form $g \circ \kappa_a$ with g smooth. Conclude that the maximal $\mathcal{C}^\infty$ structure is an algebra.

Exercise 5.3 (Partitions of unity)**.** Show that any $\mathcal{C}^\infty$ manifold admits partitions of unity. That is, if O_a, $a \in \mathcal{A}$, is an open cover of X, then there exist $\rho_{a,i} \in \mathcal{C}^\infty(X)$, $a \in \mathcal{A}$, $i \in \mathbb{N}$, with $0 \le \rho_{a,i} \le 1$, each $\rho_{a,i}$ vanishing outside a compact subset $K_{a,i} \subset O_a$, such that each point has a neighbourhood meeting only finitely many of the $K_{a,i}$, and $\sum_{a,i} \rho_{a,i} = 1$.

Exercise 5.4. Fill in the details in the proof of Proposition 5.3.1: with u_R obtained by replacing a by $a(w',t,\zeta)\chi(\zeta/R)$, show that $u_R \to u$ in $\mathcal{C}^{-\infty}(M)$ and that $u_R|_Z$ converges in $\mathcal{C}^{-\infty}(Z)$ as $R \to \infty$. Deduce that $u|_Z$ does not depend on the coordinates used, and that $(\phi u)|_Z = \phi|_Z\, u|_Z$ for $\phi \in \mathcal{C}^\infty(M)$.

Exercise 5.5. Show that a continuous linear map $A : \mathcal{C}^\infty(X;E) \to \mathcal{C}^\infty(X;F)$ is smoothing if and only if it extends to a continuous linear map $\mathcal{C}^{-\infty}(X;E) \to \mathcal{C}^\infty(X;F)$. Show that smoothing operators are *not* characterized as the bounded operators $L^2(X) \to \mathcal{C}^\infty(X)$, by considering $Au = (u,g)\cdot 1$ with $g \in L^2(X) \setminus \mathcal{C}^\infty(X)$.

Exercise 5.6. Let $A \in \Psi^m_{\mathrm{ph}}(X;E,F)$ and let $f \in \mathcal{C}^\infty(X)$ be real-valued with $df(\bar{x}) \neq 0$. Show that for $u \in \mathcal{C}^\infty(X;E)$, as $t \to +\infty$, $e^{-itf}A(e^{itf}u)(\bar{x}) = t^m \sigma_m(A)(\bar{x}, df(\bar{x}))u(\bar{x}) + O(t^{m-1})$, where $\sigma_m(A)$ is the homogeneous principal symbol. (Use the left-reduced form in local coordinates and Taylor's formula for f.) Conclude that this gives an invariant definition of the principal symbol of classical operators extending (5.4.7).

Exercise 5.7. Show that an elliptic $A \in \Psi^m(X;E)$ which is formally self-adjoint has index zero, and that the same is true of any elliptic operator in $\Psi^m(X;E)$ whose principal symbol can be connected to that of a self-adjoint elliptic operator through elliptic symbols.

Exercise 5.8. Let X be compact, oriented and Riemannian, of dimension n, and let $* : \Lambda^p \to \Lambda^{n-p}$ be the Hodge star. Show that $d^* = \pm * d *$ on p-forms and that $*$ maps harmonic p-forms isomorphically onto harmonic $(n-p)$-forms. Deduce Poincaré duality $b_p = b_{n-p}$, and that $\chi(X) = 0$ if n is odd.

Exercise 5.9. Assuming the properties of the Fredholm determinant in Remark 5.9.10, show that $G^{-\infty}(X;E)$ is dense in $\Psi^{-\infty}(X;E)$. Prove this directly using Lemma 5.9.6 and the density of $\mathrm{GL}(M,\mathbb{C})$ in $M(M,\mathbb{C})$.

Notes

This chapter combines Chapter 5 of the 2003 notes with Lectures 5–8 of the 2005 lectures. The invariant definition of pseudodifferential operators through conormal kernels comes from the latter, and the local definition from the former; Theorem 5.4.2 shows that they agree. The source text gave only a sketch of the Fredholm property, carried out on $\mathcal{C}^\infty$ with L^2 estimates; here it is proved on Sobolev spaces, which are defined in Section 5.7, and the range is characterized by duality. The Hodge theorem, only announced in the 2003 notes, is proved for general elliptic complexes as in the 2005 lectures. The existence of the projection Π in Proposition 5.11.1 was stated without proof; we use a contour integral. The functional calculus (heat kernel, resolvent, complex powers) announced in the original is not included; see [11, 15]. For a full treatment of pseudodifferential operators on manifolds see [8, Chapter 18] and [15].

Part II

Global calculi, traces and determinants

Chapter 6

Isotropic and scattering calculi

In this chapter two *global* calculi of pseudodifferential operators on $\mathbb{R}^n$ are constructed and analysed. In the *isotropic* calculus the symbols are required to satisfy estimates in which the space variables x and the dual variables ξ play completely symmetric roles: each derivative, in x or in ξ, improves the decay by a factor of $(1+|x|+|\xi|)^{-1}$. In the *scattering* calculus derivatives in x gain powers of $\langle x\rangle^{-1}$ and derivatives in ξ gain powers of $\langle\xi\rangle^{-1}$, separately. Both calculi are subalgebras of (weighted versions of) the traditional calculus $\Psi^m(\mathbb{R}^n)$ of Chapter 2, and we use the results proved there freely. The isotropic calculus is especially simple because its residual algebra $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ consists of the operators with kernels in $\mathcal{S}(\mathbb{R}^{2n})$; these operators are compact (indeed Hilbert–Schmidt) on $L^2(\mathbb{R}^n)$. This residual algebra, the associated group $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ of invertible perturbations of the identity, and the approximation of residual operators by finite rank operators built from the eigenfunctions of the harmonic oscillator, are the basic objects for the discussion of traces and determinants in Chapter 7. Thus this chapter is preparatory: the constructions (adjoints, composition, parametrices, generalized inverses, Sobolev spaces, L^2 estimates, complex powers) reappear later for operators on manifolds, and here they can be carried out without the complications of coordinate changes.

Unlike the original notes, we work with general (not necessarily polyhomogeneous) symbols by default and indicate the polyhomogeneous subclasses separately.

Notation. Points of $\mathbb{R}^{2n} = \mathbb{R}^n_x \times \mathbb{R}^n_\xi$ are written $w = (x,\xi)$ and $\langle w\rangle = \langle(x,\xi)\rangle = (1+|x|^2+|\xi|^2)^{1/2}$. We repeatedly use the following elementary inequalities. For $v, w \in \mathbb{R}^N$ and $s \in \mathbb{R}$ (Peetre's inequality)

$$\langle w+v\rangle^s \le 2^{|s|/2}\langle w\rangle^s\langle v\rangle^{|s|}, \tag{6.0.1}$$

which follows from $1+|w+v|^2 \le 2(1+|w|^2)(1+|v|^2)$, applied to (w,v) if $s \ge 0$ and to $(w+v,-v)$ if $s<0$. For $z \in \mathbb{R}^p$, $\xi \in \mathbb{R}^n$ and $\theta \in [0,1]$,

$$\max(\langle z\rangle,\langle\xi\rangle) \le \langle(z,\xi)\rangle \le \langle z\rangle\langle\xi\rangle, \qquad \langle(z,\xi)\rangle \ge \langle z\rangle^\theta\langle\xi\rangle^{1-\theta}, \tag{6.0.2}$$

the last by the arithmetic–geometric mean inequality, $\langle z\rangle^{2\theta}\langle\xi\rangle^{2(1-\theta)} \le \theta\langle z\rangle^2 + (1-\theta)\langle\xi\rangle^2 \le \langle(z,\xi)\rangle^2$. We write $m_+ = \max(m,0)$.

6.1 Symbol classes

6.1.1 Isotropic symbols

Definition 6.1.1. For $m \in \mathbb{R}$ and $N \in \mathbb{N}$ the space of *isotropic symbols* of order m is

$$\Gamma^m(\mathbb{R}^N) = \{a \in \mathcal{C}^\infty(\mathbb{R}^N);\ |\partial^\alpha a(w)| \le C_\alpha\langle w\rangle^{m-|\alpha|}\ \forall\,\alpha \in \mathbb{N}_0^N\}.$$

It is a Fréchet space with the norms $\|a\|_{m,k} = \max_{|\alpha|\le k} \sup_w \langle w\rangle^{-m+|\alpha|}|\partial^\alpha a(w)|$. We set $\Gamma^{-\infty}(\mathbb{R}^N) = \bigcap_m \Gamma^m(\mathbb{R}^N) = \mathcal{S}(\mathbb{R}^N)$ and $\Gamma^\infty(\mathbb{R}^N) = \bigcup_m \Gamma^m(\mathbb{R}^N)$. Vector- and matrix-valued symbols are defined in the same way.

Lemma 6.1.2. (i) $\Gamma^m \cdot \Gamma^{m'} \subset \Gamma^{m+m'}$, $\partial^\alpha : \Gamma^m \to \Gamma^{m-|\alpha|}$, $\Gamma^m \subset \Gamma^{m'}$ *for* $m \le m'$, *and* $\langle w\rangle^s \in \Gamma^s$ *for all* $s \in \mathbb{R}$.

(ii) *If* $a \in \Gamma^0(\mathbb{R}^N;\mathbb{C}^M)$ *and* F *is smooth on a neighbourhood of the (compact) closure of* $a(\mathbb{R}^N)$, *then* $F \circ a \in \Gamma^0(\mathbb{R}^N)$.

(iii) *If* L *is a linear isomorphism of* $\mathbb{R}^N$ *then* $a \mapsto a \circ L$ *is an isomorphism of* $\Gamma^m(\mathbb{R}^N)$.

(iv) $\mathcal{S}(\mathbb{R}^N)$ *is dense in* $\Gamma^m(\mathbb{R}^N)$ *in the topology of* $\Gamma^{m'}(\mathbb{R}^N)$ *for any* $m' > m$.

Proof. (i) follows from Leibniz' formula and, for $\langle w\rangle^s$, from the fact that $\partial^\alpha\langle w\rangle^s$ is a sum of terms $p(w)\langle w\rangle^{s-2k}$ with p a polynomial of degree at most $2k-|\alpha|$. (ii) By the chain rule $\partial^\alpha(F\circ a)$ is a sum of terms $(\partial^k F)(a)\,\partial^{\alpha_1}a\cdots\partial^{\alpha_k}a$ with $\alpha_1+\cdots+\alpha_k=\alpha$, $|\alpha_i|\ge 1$, each bounded by $C\langle w\rangle^{-|\alpha|}$. (iii) holds since $c\langle w\rangle \le \langle Lw\rangle \le C\langle w\rangle$. (iv) Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^N)$ equal 1 near 0 and $a_k = \chi(w/k)a$. The functions $\chi(w/k)$ are bounded in Γ^0 uniformly in $k\ge 1$ and $1-\chi(w/k)$ vanishes for $|w|\le ck$, where $\langle w\rangle^{m-m'} \le Ck^{m-m'}$; hence $\|a-a_k\|_{m',j} \le C_j k^{m-m'} \to 0$. □

Lemma 6.1.3 (Asymptotic completeness)**.** *If* $a_j \in \Gamma^{m-j}(\mathbb{R}^N)$ *for* $j \in \mathbb{N}_0$, *there exists* $a \in \Gamma^m(\mathbb{R}^N)$ *such that*

$$a - \sum_{j<k} a_j \in \Gamma^{m-k}(\mathbb{R}^N) \quad \forall\, k \in \mathbb{N}_0;$$

a *is unique modulo* $\mathcal{S}(\mathbb{R}^N)$ *and we write* $a \sim \sum_j a_j$.

Proof. Choose $\psi \in \mathcal{C}^\infty(\mathbb{R}^N)$ with $\psi(w)=0$ for $|w|\le 1$, $\psi(w)=1$ for $|w|\ge 2$, and set $\psi_\epsilon(w)=\psi(\epsilon w)$ for $0<\epsilon\le 1$. Then $\partial^\alpha\psi_\epsilon = \epsilon^{|\alpha|}(\partial^\alpha\psi)(\epsilon w)$ is, for $\alpha\ne 0$, supported in $1/\epsilon \le |w| \le 2/\epsilon$, so $|\partial^\alpha\psi_\epsilon| \le C_\alpha\langle w\rangle^{-|\alpha|}$ uniformly in ϵ; moreover $\langle w\rangle \ge 1/\epsilon$ on $\operatorname{supp}\psi_\epsilon$. Hence for $k<j$ and $\ell\in\mathbb{N}_0$

$$\|\psi_\epsilon a_j\|_{m-k,\ell} \le C_{j,\ell} \sup_{\operatorname{supp}\psi_\epsilon} \langle w\rangle^{k-j} \le C_{j,\ell}\,\epsilon.$$

Choose $\epsilon_j \downarrow 0$ with $C_{j,j}\epsilon_j \le 2^{-j}$ (we may take $C_{j,\ell}$ increasing in ℓ and valid for all $k<j$). Then for each k and ℓ the series $\sum_{j>k}\psi_{\epsilon_j}a_j$ converges in the norm $\|\cdot\|_{m-k,\ell}$, since all but finitely many terms are bounded by 2^{-j}. Thus $a=\sum_j \psi_{\epsilon_j}a_j$ converges in Γ^m and

$$a - \sum_{j<k} a_j = \sum_{j<k}(\psi_{\epsilon_j}-1)a_j + \psi_{\epsilon_k}a_k + \sum_{j>k}\psi_{\epsilon_j}a_j \in \mathcal{C}^\infty_{\mathrm{c}} + \Gamma^{m-k} + \Gamma^{m-k}.$$

Uniqueness holds because $\bigcap_k \Gamma^{m-k} = \mathcal{S}(\mathbb{R}^N)$. □

Sums with steps of two, $a\sim\sum_j a_j$ with $a_j\in\Gamma^{m-2j}$, are covered by inserting zero terms.

6.1.2 Scattering symbols

Definition 6.1.4. For $m,l\in\mathbb{R}$ the space $S^{m,l}(\mathbb{R}^p;\mathbb{R}^n)$ of *scattering symbols* consists of the $a\in\mathcal{C}^\infty(\mathbb{R}^p_z\times\mathbb{R}^n_\xi)$ such that

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| \le C_{\alpha\beta}\langle z\rangle^{l-|\alpha|}\langle\xi\rangle^{m-|\beta|} \quad \forall\,\alpha,\beta.$$

Here m is the order in the fibre variables and l the order in the base variables.

The following properties are clear: $S^{m,l}\cdot S^{m',l'}\subset S^{m+m',l+l'}$, $D_z: S^{m,l}\to S^{m,l-1}$, $D_\xi: S^{m,l}\to S^{m-1,l}$, $S^{m,l}(\mathbb{R}^p;\mathbb{R}^n)\subset\langle z\rangle^l S^m(\mathbb{R}^p;\mathbb{R}^n)$, and $\bigcap_{m,l} S^{m,l}(\mathbb{R}^n;\mathbb{R}^n)=\mathcal{S}(\mathbb{R}^{2n})$.

Lemma 6.1.5. *If $a_j\in S^{m-j,l-j}(\mathbb{R}^p;\mathbb{R}^n)$, $j\in\mathbb{N}_0$, there exists $a\in S^{m,l}(\mathbb{R}^p;\mathbb{R}^n)$ with $a-\sum_{j<k}a_j\in S^{m-k,l-k}(\mathbb{R}^p;\mathbb{R}^n)$ for all k; a is unique modulo $\mathcal{S}(\mathbb{R}^{p+n})$.*

Proof. Repeat the proof of Lemma 6.1.3 with $\psi_\epsilon(z,\xi)=1-(1-\psi(\epsilon z))(1-\psi(\epsilon\xi))$, which is bounded in $S^{0,0}$ uniformly in ϵ, vanishes where $|z|\le 1/\epsilon$ and $|\xi|\le 1/\epsilon$ (so $1-\psi_\epsilon$ has compact support), and satisfies $\langle z\rangle^{-1}\langle\xi\rangle^{-1}\le\epsilon$ on its support. Thus for $k<j$ the $S^{m-k,l-k}$ norms of $\psi_\epsilon a_j$ are bounded by $C\sup_{\operatorname{supp}\psi_\epsilon}(\langle z\rangle\langle\xi\rangle)^{k-j}\le C\epsilon$. □

The two classes are related as follows; this replaces the corresponding lemma of the original notes, where the power of $1+|x|^2$ is misprinted.

Lemma 6.1.6. *For $z\in\mathbb{R}^p$, $\xi\in\mathbb{R}^n$,*

$$\Gamma^m(\mathbb{R}^{p+n})\subset\begin{cases}S^{m,m}(\mathbb{R}^p;\mathbb{R}^n), & m\ge 0,\\ \bigcap_{m\le t\le 0}S^{m-t,t}(\mathbb{R}^p;\mathbb{R}^n), & m\le 0.\end{cases}$$

In particular $\Gamma^m(\mathbb{R}^{p+n})\subset\langle z\rangle^{m_+}S^m(\mathbb{R}^p;\mathbb{R}^n)$.

Proof. By (6.0.2), $|\partial_z^\alpha\partial_\xi^\beta a|\le C\langle(z,\xi)\rangle^m\langle z\rangle^{-|\alpha|}\langle\xi\rangle^{-|\beta|}$. If $m\ge 0$, $\langle(z,\xi)\rangle^m\le\langle z\rangle^m\langle\xi\rangle^m$. If $m<0$ and $m\le t\le 0$, take $\theta=t/m\in[0,1]$ in (6.0.2) to get $\langle(z,\xi)\rangle^m\le\langle z\rangle^t\langle\xi\rangle^{m-t}$. □

6.1.3 Polyhomogeneous symbols and radial compactification

A function $a\in\mathcal{C}^\infty(\mathbb{R}^N)$ is *homogeneous of degree μ outside the unit ball* if $a(tw)=t^\mu a(w)$ for $|w|\ge 1$, $t\ge 1$. Such a function lies in $\Gamma^\mu(\mathbb{R}^N)$, since on $|w|\ge 1$ its derivatives of order k are homogeneous of degree $\mu-k$.

Definition 6.1.7. The space $\Gamma^m_{\mathrm{ph}}(\mathbb{R}^N)$ of *polyhomogeneous* (classical) isotropic symbols consists of those $a\in\Gamma^m(\mathbb{R}^N)$ for which there are a_{m-j}, homogeneous of degree $m-j$ outside the unit ball, with $a\sim\sum_j a_{m-j}$. The *principal symbol* of a is the restriction of a_m to the sphere $\mathbb{S}^{N-1}$.

The polyhomogeneous classes are closed under multiplication and differentiation ($\partial_{w_j}:\Gamma^m_{\mathrm{ph}}\to\Gamma^{m-1}_{\mathrm{ph}}$), contain $\langle w\rangle^s\in\Gamma^s_{\mathrm{ph}}$ (expand $\langle w\rangle^s=|w|^s(1+|w|^{-2})^{s/2}$ binomially for $|w|\ge 2$ and cut the terms off near 0), and if $a_j\in\Gamma^{m-j}_{\mathrm{ph}}$ and $a\sim\sum_j a_j$ then $a\in\Gamma^m_{\mathrm{ph}}$, as one sees by collecting the finitely many terms of each degree of homogeneity in the expansions of the a_j.

The *radial compactification* is the map

$$\mathrm{RC}:\mathbb{R}^N\ni w\longmapsto(\langle w\rangle^{-1},w\langle w\rangle^{-1})\in\mathbb{S}^N_+=\{X=(X_0,X')\in\mathbb{R}^{1+N};\ |X|=1,\ X_0\ge 0\},$$

a diffeomorphism of $\mathbb{R}^N$ onto the interior of the closed half-sphere $\mathbb{S}^N_+$, a compact manifold with boundary $\partial\mathbb{S}^N_+=\{X_0=0\}\cong\mathbb{S}^{N-1}$. The function $\rho=X_0$ is a boundary defining function and $\rho\circ\mathrm{RC}=\langle w\rangle^{-1}$. We denote by $\dot{\mathcal{C}}^\infty(\mathbb{S}^N_+)$ the smooth functions vanishing to infinite order at the boundary. The space $\mathcal{L}(\mathbb{S}^N_+)=\bigcup_{k\in\mathbb{N}_0}\rho^{-k}\mathcal{C}^\infty(\mathbb{S}^N_+)$ is the space of *Laurent functions* on the half-sphere; more generally $\rho^\mu\mathcal{C}^\infty(\mathbb{S}^N_+)$, $\mu\in\mathbb{C}$, is the space of products $\rho^\mu v$ with v smooth, regarded as functions on the interior.

Proposition 6.1.8. (i) *For every $m\in\mathbb{R}$, $\mathrm{RC}^*(\rho^{-m}\mathcal{C}^\infty(\mathbb{S}^N_+))=\Gamma^m_{\mathrm{ph}}(\mathbb{R}^N)$, and the principal symbol of $\mathrm{RC}^*(\rho^{-m}F)$ is $F|_{\partial\mathbb{S}^N_+}$.*

(ii) $\mathrm{RC}^*\,\dot{\mathcal{C}}^\infty(\mathbb{S}^N_+) = \mathcal{S}(\mathbb{R}^N)$.

(iii) *If $f \in \Gamma^{-k}(\mathbb{R}^N)$, $k \in \mathbb{N}$, then $f \circ \mathrm{RC}^{-1}$ extends to a $\mathcal{C}^{k-1}$ function on $\mathbb{S}^N_+$ vanishing at the boundary, and $\Gamma^{-k}(\mathbb{R}^N) \ni f \mapsto f \circ \mathrm{RC}^{-1} \in \mathcal{C}^{k-1}(\mathbb{S}^N_+)$ is continuous.*

Sketch of proof. Near the boundary use the coordinates $\rho \in [0,1)$ and $\theta = X'/|X'| \in \mathbb{S}^{N-1}$; in terms of w, $\rho = \langle w\rangle^{-1}$ and $\theta = w/|w|$, so $|w| = (1-\rho^2)^{1/2}/\rho$. In polar coordinates $\partial_{w_j} = \theta_j\partial_r + r^{-1}\Omega_j$, where Ω_j is the vector field on $\mathbb{S}^{N-1}$ obtained by projecting e_j tangentially, and $\partial_r = -r\rho^3\partial_\rho$. Hence

$$V_j := \langle w\rangle\partial_{w_j} = -\theta_j(1-\rho^2)^{1/2}\rho\partial_\rho + (1-\rho^2)^{-1/2}\Omega_j. \tag{6.1.1}$$

These are smooth vector fields on $\mathbb{S}^N_+$ tangent to the boundary, and since $\sum_j \theta_j\Omega_j = 0$ they span, over $\mathcal{C}^\infty$, the fields $\rho\partial_\rho$ and Ω_j, that is all smooth vector fields tangent to the boundary. An induction shows that $\langle w\rangle^{|\alpha|}\partial^\alpha$ is a combination, with coefficients in Γ^0, of products of at most $|\alpha|$ of the V_j. Consequently $\mathrm{RC}^*\,\mathcal{C}^\infty(\mathbb{S}^N_+) \subset \Gamma^0$. Conversely, a derivative of order j of $f \circ \mathrm{RC}^{-1}$ in (ρ,θ) is ρ^{-j} times a combination of at most j-fold products of the V_j applied to f; this gives (iii), since such products applied to $f \in \Gamma^{-k}$ are $O(\langle w\rangle^{-k}) = O(\rho^k)$, and also (ii), since for $f \in \mathcal{S}$ these are $O(\rho^K)$ for every K (conversely, $F \in \dot{\mathcal{C}}^\infty$ can be written $F = \rho^K F_K$ with F_K smooth for every K, so $\mathrm{RC}^*\,F \in \bigcap_K \Gamma^{-K} = \mathcal{S}$ by the first argument).

For (i), if $F \in \mathcal{C}^\infty(\mathbb{S}^N_+)$, Taylor's formula in ρ gives $F = \sum_{j<K}\rho^j F_j(\theta) + \rho^K R_K$ near the boundary. Since $\rho = |w|^{-1}(1+|w|^{-2})^{-1/2}$ for $|w| \geq 1$, each $\rho^{j-m}F_j(\theta)$ pulls back to an asymptotic sum of functions homogeneous of degrees $m-j-2i$, $i \in \mathbb{N}_0$, while $\rho^{K-m}R_K$ pulls back into Γ^{m-K}. Thus $\mathrm{RC}^*(\rho^{-m}F) \in \Gamma^m_{\mathrm{ph}}$, with leading term $|w|^m F(0,\theta)$. Conversely, a function homogeneous of degree $m-j$ outside the unit ball is $|w|^{m-j}f(\theta) = \rho^{j-m}(1-\rho^2)^{(m-j)/2}f(\theta)$, which lies in $\rho^{-m}\rho^j\mathcal{C}^\infty$ near the boundary. Given $a \sim \sum_j a_{m-j}$, Borel's lemma gives $F \in \mathcal{C}^\infty(\mathbb{S}^N_+)$ with $\rho^{-m}F$ having the same expansion, and then $a - \mathrm{RC}^*(\rho^{-m}F) \in \mathcal{S}(\mathbb{R}^N) = \mathrm{RC}^*\,\dot{\mathcal{C}}^\infty$ by (ii). □

Thus $\Gamma^{\mathbb{Z}}_{\mathrm{ph}}(\mathbb{R}^N) = \bigcup_{k\in\mathbb{Z}}\Gamma^k_{\mathrm{ph}}(\mathbb{R}^N)$ is identified with the Laurent functions $\mathcal{L}(\mathbb{S}^N_+)$ and $\Gamma^0_{\mathrm{ph}}(\mathbb{R}^N)$ with $\mathcal{C}^\infty(\mathbb{S}^N_+)$.

6.2 Isotropic operators

Recall the left quantization $\mathrm{Op}(a)u(x) = \int e^{i(x-y)\cdot\xi}a(x,\xi)u(y)\,dy\text{đ}\xi$ and, for an amplitude $a(x,y,\xi)$, the same formula with $a(x,y,\xi)$.

Definition 6.2.1. For $m \in \mathbb{R}$ the space of *isotropic pseudodifferential operators* of order m on $\mathbb{R}^n$ is

$$\Psi^m_{\mathrm{iso}}(\mathbb{R}^n) = \{\mathrm{Op}(a);\ a \in \Gamma^m(\mathbb{R}^{2n})\}, \qquad \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n) = \{\mathrm{Op}(a);\ a \in \Gamma^m_{\mathrm{ph}}(\mathbb{R}^{2n})\}.$$

We set $\Psi^{-\infty}_{\mathrm{iso}} = \bigcap_m \Psi^m_{\mathrm{iso}}$ and $\Psi^\infty_{\mathrm{iso}} = \bigcup_m \Psi^m_{\mathrm{iso}}$.

By Lemma 6.1.6, if $a \in \Gamma^m(\mathbb{R}^{2n})$ then $a = \langle x\rangle^{m+}b$ with $b \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, so $\mathrm{Op}(a) = \langle x\rangle^{m+}\mathrm{Op}(b) \in \langle x\rangle^{m+}\Psi^m(\mathbb{R}^n)$. In particular isotropic operators act continuously on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, by the results of Chapter 2. The kernel of $\mathrm{Op}(a)$ is $K(x,y) = \int e^{i(x-y)\cdot\xi}a(x,\xi)\text{đ}\xi$, i.e. the partial inverse Fourier transform of a in ξ evaluated at $(x, x-y)$; hence a is determined by $\mathrm{Op}(a)$ and we call $\sigma_L(\mathrm{Op}(a)) = a$ the *(full) left symbol*. Since partial Fourier transformation and linear changes of variables preserve $\mathcal{S}$:

Lemma 6.2.2. *An operator A is in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ if and only if it has Schwartz kernel in $\mathcal{S}(\mathbb{R}^{2n})$, and $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) \ni A \mapsto K_A \in \mathcal{S}(\mathbb{R}^{2n})$ is a bijection.*

We topologize $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ by this identification.

Example 6.2.3. Since $\mathrm{Op}(x^\beta\xi^\alpha) = x^\beta D^\alpha$, every differential operator with polynomial coefficients, $P = \sum_{|\alpha|+|\beta|\le m} c_{\alpha\beta}x^\beta D^\alpha$, lies in $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$. Conversely, a differential operator lies in $\Psi^\infty_{\mathrm{iso}}(\mathbb{R}^n)$ only if its coefficients are polynomials (a smooth function with all derivatives of order k bounded by $C\langle x\rangle^{m-k}$ for all k is a polynomial). The basic example is the *harmonic oscillator* $H = \sum_j (D_j^2 + x_j^2)$, with symbol $|x|^2 + |\xi|^2$, studied in Section 6.8.

6.2.1 Amplitudes and reduction

As in Chapter 2, adjoints and products are obtained by reducing operators defined by amplitudes to left-reduced form. Note that a left symbol $a(x,\xi) \in \Gamma^m(\mathbb{R}^{2n})$, regarded as a function of (x,y,ξ), is *not* in general in $\Gamma^m(\mathbb{R}^{3n})$, so we use a hybrid class of amplitudes.

Definition 6.2.4. For $m \in \mathbb{R}$, $\mathcal{A}^m(\mathbb{R}^{3n})$ is the space of $a \in \mathcal{C}^\infty(\mathbb{R}^{3n}_{x,y,\xi})$ for which there is $w \ge 0$ such that for all multi-indices $\alpha' \le \alpha$, $\beta' \le \beta$, $\gamma' \le \gamma$

$$|D_x^\alpha D_y^\beta D_\xi^\gamma a| \le C\langle(x,\xi)\rangle^{m-k'}\langle x-y\rangle^{w+k'}\langle\xi\rangle^{-|\gamma-\gamma'|}, \qquad k' = |\alpha'| + |\beta'| + |\gamma'|. \tag{6.2.1}$$

Thus any subset of the derivatives may be used to improve the decay in (x,ξ), at the cost of a corresponding growth in $|x-y|$, while the remaining ξ-derivatives still give decay in ξ.

Lemma 6.2.5. (i) *If $a \in \Gamma^m(\mathbb{R}^{2n})$ and $b \in \Gamma^{m'}(\mathbb{R}^{2n})$ then $a(x,\xi)b(y,\xi) \in \mathcal{A}^{m+m'}(\mathbb{R}^{3n})$.*

(ii) *If $a \in \Gamma^m(\mathbb{R}^{2n})$ and $\lambda \in [0,1]$ then $a(x+\lambda(y-x),\xi) \in \mathcal{A}^m(\mathbb{R}^{3n})$.*

(iii) *If $a \in \mathcal{A}^m(\mathbb{R}^{3n})$ then $|D_x^\alpha D_y^\beta D_\xi^\gamma a| \le C\langle x\rangle^{m_+}\langle x-y\rangle^w\langle\xi\rangle^{m-|\gamma|}$.*

Proof. (i) By Leibniz' formula $D_x^\alpha D_y^\beta D_\xi^\gamma(ab)$ is a sum of terms $(D_x^\alpha D_\xi^{\gamma_1}a)(x,\xi)(D_y^\beta D_\xi^{\gamma_2}b)(y,\xi)$ with $\gamma_1+\gamma_2 = \gamma$. Given $\gamma' \le \gamma$, choose $\gamma_i' \le \gamma_i$ with $|\gamma_1'|+|\gamma_2'| = |\gamma'|$. Using $\langle(x,\xi)\rangle^{-1} \le \langle\xi\rangle^{-1}$ we get $|D_x^\alpha D_\xi^{\gamma_1}a| \le C\langle(x,\xi)\rangle^{m-|\alpha'|-|\gamma_1'|}\langle\xi\rangle^{-|\gamma_1-\gamma_1'|}$, and similarly, followed by (6.0.1) with $v = (y-x,0)$,

$$\begin{aligned}|D_y^\beta D_\xi^{\gamma_2}b(y,\xi)| &\le C\langle(y,\xi)\rangle^{m'-|\beta'|-|\gamma_2'|}\langle\xi\rangle^{-|\gamma_2-\gamma_2'|}\\ &\le C'\langle(x,\xi)\rangle^{m'-|\beta'|-|\gamma_2'|}\langle x-y\rangle^{|m'|+|\beta'|+|\gamma_2'|}\langle\xi\rangle^{-|\gamma_2-\gamma_2'|}.\end{aligned}$$

Multiplying gives (6.2.1) with $w = |m'|$. (ii) With $\bar x = x+\lambda(y-x)$ the derivative is a constant times $(\partial_x^{\alpha+\beta}D_\xi^\gamma a)(\bar x,\xi)$, bounded by $C\langle(\bar x,\xi)\rangle^{m-k'}\langle\xi\rangle^{-|\gamma-\gamma'|}$, and (6.0.1) gives $\langle(\bar x,\xi)\rangle^{m-k'} \le C\langle(x,\xi)\rangle^{m-k'}\langle x-y\rangle^{|m|+k'}$. (iii) Take $\alpha' = \beta' = \gamma' = 0$ and use $\langle(x,\xi)\rangle^m \le \langle x\rangle^{m_+}\langle\xi\rangle^m$. □

By (iii), $\langle x\rangle^{-m_+}a$ is an amplitude of the traditional calculus of Chapter 2 (of order m with weight $\langle x-y\rangle^w$), so $\mathrm{Op}(a) = \langle x\rangle^{m_+}\mathrm{Op}(\langle x\rangle^{-m_+}a) \in \langle x\rangle^{m_+}\Psi^m(\mathbb{R}^n)$ is a well-defined continuous operator on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$.

Proposition 6.2.6 (Reduction)**.** *If $a \in \mathcal{A}^m(\mathbb{R}^{3n})$ then $\mathrm{Op}(a) \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ and*

$$\sigma_L(\mathrm{Op}(a)) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!}(D_\xi^\alpha D_y^\alpha a)(x,x,\xi), \tag{6.2.2}$$

where the term of index α lies in $\Gamma^{m-2|\alpha|}(\mathbb{R}^{2n})$.

Proof. Let $c_\alpha(x,\xi) = \frac{i^{|\alpha|}}{\alpha!}(D_\xi^\alpha D_y^\alpha a)(x,x,\xi)$. A derivative $D_x^\mu D_\xi^\gamma c_\alpha$ is a sum of values at $y = x$ of derivatives of a containing $|\alpha|$ derivatives in y, $|\alpha|+|\gamma|$ in ξ and $|\mu|$ in x or y; applying (6.2.1) with all derivatives used for decay, and $\langle x-y\rangle = 1$ at $y = x$, shows $c_\alpha \in \Gamma^{m-2|\alpha|}(\mathbb{R}^{2n})$.

Taylor's formula in y around x gives

$$a(x,y,\xi) = \sum_{|\alpha|<N} \frac{(y-x)^\alpha}{\alpha!}(\partial_y^\alpha a)(x,x,\xi) + \sum_{|\alpha|=N} \frac{N(y-x)^\alpha}{\alpha!}\int_0^1 (1-t)^{N-1}(\partial_y^\alpha a)(x,x+t(y-x),\xi)\,dt.$$

Since $(y-x)^\alpha e^{i(x-y)\cdot\xi} = (-D_\xi)^\alpha e^{i(x-y)\cdot\xi}$, integration by parts in ξ (legitimate for these oscillatory integrals, as in Chapter 2) gives

$$\begin{aligned} \operatorname{Op}(a) &= \sum_{|\alpha|<N} \operatorname{Op}(c_\alpha) + \operatorname{Op}(r_N), \\ r_N(x,y,\xi) &= \sum_{|\alpha|=N} \frac{N i^N}{\alpha!}\int_0^1 (1-t)^{N-1}(D_\xi^\alpha D_y^\alpha a)(x,x+t(y-x),\xi)\,dt. \end{aligned} \tag{6.2.3}$$

We estimate r_N. A derivative $D_x^\mu D_y^\nu D_\xi^\gamma$ of the integrand is, by the chain rule, a combination with bounded coefficients of derivatives of a evaluated at $(x, x+t(y-x),\xi)$ which contain the N derivatives D_y^α, the $N+|\gamma|$ derivatives $D_\xi^{\alpha+\gamma}$ and further x- and y-derivatives. Applying (6.2.1) using exactly the $2N$ derivatives $D_y^\alpha D_\xi^\alpha$ for decay, and noting that the first argument is still x and $|x-(x+t(y-x))| \le |x-y|$, we find

$$|D_x^\mu D_y^\nu D_\xi^\gamma r_N| \le C\langle(x,\xi)\rangle^{m-2N}\langle x-y\rangle^{w+2N}\langle\xi\rangle^{-|\gamma|}.$$

Let $2N > m$ and $s = N - m/2$. By (6.0.2), $\langle(x,\xi)\rangle^{-2s} \le \langle x\rangle^{-s}\langle\xi\rangle^{-s}$, so $\langle x\rangle^s r_N$ is an amplitude of the traditional calculus of order $-s$ with weight $\langle x-y\rangle^{w+2N}$ (derivatives of $\langle x\rangle^s$ are bounded by $C\langle x\rangle^s$). By Chapter 2, $\operatorname{Op}(r_N) = \langle x\rangle^{-s}Q_N$ with $Q_N \in \Psi^{-s}(\mathbb{R}^n)$, so the left symbol of $\operatorname{Op}(r_N)$ lies in $\langle x\rangle^{-s}S^{-s}(\mathbb{R}^n;\mathbb{R}^n)$: all its derivatives are bounded by $C\langle x\rangle^{-s}\langle\xi\rangle^{-s}$.

Now $\operatorname{Op}(a)$ has a left symbol $\sigma \in \langle x\rangle^{m+}S^m(\mathbb{R}^n;\mathbb{R}^n)$ and, by (6.2.3) and uniqueness of left symbols, $\sigma - \sum_{|\alpha|<N} c_\alpha \in \langle x\rangle^{-s}S^{-s}$. Choose $b \in \Gamma^m(\mathbb{R}^{2n})$ with $b \sim \sum_\alpha c_\alpha$ (Lemma 6.1.3). Then for every N

$$\sigma - b = \Big(\sigma - \sum_{|\alpha|<N} c_\alpha\Big) - \Big(b - \sum_{|\alpha|<N} c_\alpha\Big) \in \langle x\rangle^{-s}S^{-s} + \Gamma^{m-2N},$$

and all derivatives of elements of both spaces are bounded by $C\langle x\rangle^{-s}\langle\xi\rangle^{-s}$, with $s = N - m/2$ arbitrarily large. Hence $\sigma - b \in \mathcal{S}(\mathbb{R}^{2n})$, so $\sigma \in \Gamma^m(\mathbb{R}^{2n})$ and $\sigma \sim \sum_\alpha c_\alpha$. □

6.2.2 The algebra of isotropic operators

Theorem 6.2.7. *The spaces $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ form an order-filtered $*$-algebra with residual algebra $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, the operators with kernels in $\mathcal{S}(\mathbb{R}^{2n})$. Precisely, if $A = \operatorname{Op}(a) \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ and $B = \operatorname{Op}(b) \in \Psi^{m'}_{\mathrm{iso}}(\mathbb{R}^n)$ then*

(i) *the formal adjoint A^*, defined by $(Au,v) = (u,A^*v)$ for $u,v \in \mathcal{S}(\mathbb{R}^n)$, lies in $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ and $\sigma_L(A^*) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!}D_\xi^\alpha D_x^\alpha \bar a$;*

(ii) *$AB \in \Psi^{m+m'}_{\mathrm{iso}}(\mathbb{R}^n)$ and*

$$\sigma_L(AB) = a\#b \sim \sum_\alpha \frac{1}{\alpha!}\partial_\xi^\alpha a\, D_x^\alpha b, \qquad a\#b - ab \in \Gamma^{m+m'-2}(\mathbb{R}^{2n}); \tag{6.2.4}$$

(iii) *the subspaces $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ form a $*$-subalgebra;*

(iv) *the bilinear maps $\Gamma^m \times \Gamma^{m'} \ni (a,b) \mapsto a\#b \in \Gamma^{m+m'}$ and*

$$\Gamma^m \times \Gamma^{m'} \ni (a,b) \mapsto a\#b - \sum_{|\alpha|<N} \frac{1}{\alpha!}\partial_\xi^\alpha a\, D_x^\alpha b \in \Gamma^{m+m'-2N}$$

are continuous.

Proof. (i) The kernel of A^* is $\overline{K_A(y,x)} = \int e^{i(x-y)\cdot\xi}\bar a(y,\xi)đ\xi$, so $A^* = \mathrm{Op}(\bar a(y,\xi))$, and $\bar a(y,\xi) \in \mathcal{A}^m$ by Lemma 6.2.5(ii) with $\lambda = 1$; apply Proposition 6.2.6.

(ii) Applying (i) to $B^* = \mathrm{Op}(c)$ we find $B = \mathrm{Op}(c)^* = \mathrm{Op}(b_R(y,\xi))$ with $b_R = \bar c \in \Gamma^{m'}$; taking conjugates in the expansion of (i), $b_R \sim \sum_\gamma \frac{(-i)^{|\gamma|}}{\gamma!} D_\xi^\gamma D_x^\gamma b$. For $u \in \mathcal{S}$, $\widehat{Bu}(\xi) = \int e^{-iy\cdot\xi} b_R(y,\xi)u(y)\,dy$, hence $ABu(x) = \int e^{ix\cdot\xi} a(x,\xi)\widehat{Bu}(\xi)đ\xi$, i.e. $AB = \mathrm{Op}(a(x,\xi)b_R(y,\xi))$. By Lemma 6.2.5(i) and Proposition 6.2.6, $AB \in \Psi^{m+m'}_{\mathrm{iso}}$ with $\sigma_L(AB) \sim \sum_\beta \frac{i^{|\beta|}}{\beta!} D_\xi^\beta (a\, D_x^\beta b_R)$. Inserting the expansion of b_R (with errors in $\Gamma^{m'-2N}$, which contribute errors in $\Gamma^{m+m'-2N}$) we obtain a double series whose terms of index (β,γ) lie in $\Gamma^{m+m'-2|\beta|-2|\gamma|}$. Grouping the terms with $|\beta|+|\gamma| = j$ we get a finite sum which, by the identity of formal power series in commuting variables

$$e^{iD_y\cdot(D_\eta+D_\xi)} e^{-iD_y\cdot D_\xi} = e^{iD_y\cdot D_\eta}$$

applied to $a(x,\eta)b(y,\xi)$ and evaluated at $y = x$, $\eta = \xi$, equals $\sum_{|\alpha|=j} \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha a\, D_x^\alpha b$. Since $i^{|\alpha|}D_\xi^\alpha = \partial_\xi^\alpha$ this is (6.2.4); the terms with $\alpha \neq 0$ lie in $\Gamma^{m+m'-2}$.

(iii) All terms in the expansions are polyhomogeneous when a, b are, and asymptotic sums of polyhomogeneous symbols are polyhomogeneous.

(iv) By the closed graph theorem for Fréchet spaces it suffices to check that $a \mapsto a\#b$ (for fixed b) and $b \mapsto a\#b$ have closed graphs (separately continuous bilinear maps of Fréchet spaces are continuous). If $a_k \to a$ in Γ^m and $a_k\#b \to c$ in $\Gamma^{m+m'}$, then for $u, v \in \mathcal{S}(\mathbb{R}^n)$, $\langle \mathrm{Op}(a_k\#b)u, v\rangle = \langle \mathrm{Op}(a_k)\,\mathrm{Op}(b)u, v\rangle$; the left side tends to $\langle \mathrm{Op}(c)u, v\rangle$ and the right side to $\langle \mathrm{Op}(a\#b)u, v\rangle$, since $\langle \mathrm{Op}(a)f, v\rangle = \iint e^{ix\cdot\xi} a(x,\xi)\hat f(\xi)v(x)\,dx đ\xi$ depends continuously on $a \in \Gamma^m$ for $f, v \in \mathcal{S}$. Thus $c = a\#b$. The second map is handled in the same way. □

Remark 6.2.8. The *principal symbol* $\sigma_m(A)$ of $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is the class of $\sigma_L(A)$ in Γ^m/Γ^{m-1}; there is a short exact sequence $0 \to \Psi^{m-1}_{\mathrm{iso}} \to \Psi^m_{\mathrm{iso}} \to \Gamma^m/\Gamma^{m-1} \to 0$, and $\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B)$, $\sigma_m(A^*) = \overline{\sigma_m(A)}$. For $A \in \Psi^m_{\mathrm{iso,ph}}$ we identify $\sigma_m(A)$ with the principal symbol of $\sigma_L(A)$, a function in $\mathcal{C}^\infty(\mathbb{S}^{2n-1})$. A feature of the isotropic calculus is that the expansions above proceed in steps of two: $AB - \mathrm{Op}(ab) \in \Psi^{m+m'-2}_{\mathrm{iso}}$ and, by (6.2.4), $\sigma_L([A,B]) - (-i)\{a,b\} \in \Gamma^{m+m'-4}$, where $\{a,b\} = \sum_j (\partial_{\xi_j} a\,\partial_{x_j} b - \partial_{x_j} a\,\partial_{\xi_j} b)$. Under the identification of Proposition 6.1.8, the theorem shows that the Laurent functions $\mathcal{L}(\mathbb{S}^{2n}_+)$ carry a non-commutative product, for which $\mathcal{C}^\infty(\mathbb{S}^{2n}_+)$ is a subalgebra.

Remark 6.2.9. If E and F are finite-dimensional vector spaces, $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n; E, F)$ denotes the operators from $\mathcal{S}(\mathbb{R}^n; E)$ to $\mathcal{S}(\mathbb{R}^n; F)$ given, after a choice of bases, by matrices of elements of $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$; equivalently their left symbols are in $\Gamma^m(\mathbb{R}^{2n}; \mathrm{Hom}(E,F))$. All results of this chapter extend to this setting with the obvious changes, and we use them freely; in Section 6.7 we work with $\mathbb{C}^N$-valued operators explicitly.

6.2.3 Invariance under the Fourier transform

A property which distinguishes the isotropic calculus sharply from the traditional one is its invariance under the Fourier transform.

Proposition 6.2.10. *If $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ then $\hat{A} = \mathcal{F}A\mathcal{F}^{-1} \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, i.e. $\widehat{Au} = \hat{A}\hat{u}$; if $A = \mathrm{Op}(b(y,\xi))$ is written in right-reduced form, $b \in \Gamma^m(\mathbb{R}^{2n})$, then $\sigma_L(\hat{A})(\xi, z) = b(-z, \xi)$. The same holds for $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$.*

Proof. The right-reduced form exists by the proof of Theorem 6.2.7(ii). Suppose first $b \in \mathcal{S}(\mathbb{R}^{2n})$. Then $\widehat{Au}(\xi) = \int e^{-iy\cdot\xi} b(y,\xi)u(y)\,dy$, and substituting $u(y) = \int e^{iy\cdot\eta}\hat{u}(\eta)đ\eta$ and then $y = -z$,

$$\widehat{Au}(\xi) = \iint e^{i(\xi-\eta)\cdot z} b(-z,\xi)\hat{u}(\eta)\,d\eta đz,$$

which is $\mathrm{Op}(c)\hat{u}$ with $c(\xi, z) = b(-z, \xi)$, ξ now playing the role of the base variable and z that of the dual variable. Clearly $c \in \Gamma^m$ if $b \in \Gamma^m$, by Lemma 6.1.2(iii). For general $b \in \Gamma^m$ choose $b_k \in \mathcal{S}$ with $b_k \to b$ in Γ^{m+1} (Lemma 6.1.2(iv)); for $u, v \in \mathcal{S}$ both $\langle \mathcal{F}\,\mathrm{Op}(b_k(y,\xi))\mathcal{F}^{-1}u, v\rangle$ and $\langle \mathrm{Op}(c_k)u, v\rangle$ depend continuously on $b_k \in \Gamma^{m+1}$, so the identity persists in the limit. □

6.3 Scattering operators

Definition 6.3.1. For $m, l \in \mathbb{R}$ set $\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n) = \{\mathrm{Op}(a);\ a \in S^{m,l}(\mathbb{R}^n;\mathbb{R}^n)\}$.

Thus $\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n) \subset \langle x\rangle^l \Psi^m(\mathbb{R}^n)$, and by Lemma 6.1.6, $\Psi^m_{\mathrm{iso}} \subset \Psi^{m,m}_{\mathrm{sc}}$ if $m \geq 0$ and $\Psi^m_{\mathrm{iso}} \subset \Psi^{m-t,t}_{\mathrm{sc}}$ for $m \leq t \leq 0$. Since $\bigcap_{m,l} S^{m,l}(\mathbb{R}^n;\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^{2n})$,

$$\bigcap_{m,l} \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n) = \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n). \tag{6.3.1}$$

Typical elements are the differential operators $\sum_{|\alpha|\leq m} c_\alpha(x)D^\alpha$ with $|\partial^\beta c_\alpha| \leq C_\beta\langle x\rangle^{l-|\beta|}$; for instance the Laplacian $\Delta = \sum_j D_j^2$ is in $\Psi^{2,0}_{\mathrm{sc}}(\mathbb{R}^n)$, and $\Delta + 1$ is elliptic in the scattering sense below, although it is not an isotropic operator.

For scattering operators one uses the amplitudes $\mathcal{A}^{m,l}_{\mathrm{sc}}$: smooth functions $a(x, y, \xi)$ such that for some $w \geq 0$

$$|D_x^\alpha D_y^\beta D_\xi^\gamma a| \leq C\langle x\rangle^{l-|\alpha'|-|\beta'|}\langle x-y\rangle^{w+|\alpha'|+|\beta'|}\langle\xi\rangle^{m-|\gamma|} \qquad \forall\, \alpha' \leq \alpha,\ \beta' \leq \beta.$$

As in Lemma 6.2.5, $a(x,\xi)b(y,\xi) \in \mathcal{A}^{m+m',l+l'}_{\mathrm{sc}}$ if $a \in S^{m,l}$, $b \in S^{m',l'}$ (use (6.0.1) for $\langle y\rangle^{l'-|\beta'|}$), and $a(x + \lambda(y-x), \xi) \in \mathcal{A}^{m,l}_{\mathrm{sc}}$.

Theorem 6.3.2. *The spaces $\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$ form a bi-filtered $*$-algebra, $\Psi^{m,l}_{\mathrm{sc}} \circ \Psi^{m',l'}_{\mathrm{sc}} \subset \Psi^{m+m',l+l'}_{\mathrm{sc}}$, with residual space (6.3.1). The left symbols of adjoints and products are given by the formulæ of Theorem 6.2.7, the expansions now being asymptotic in the sense of Lemma 6.1.5; in particular $AB - \mathrm{Op}(ab) \in \Psi^{m+m'-1,l+l'-1}_{\mathrm{sc}}(\mathbb{R}^n)$.*

Proof. The proof of Proposition 6.2.6 applies with the following changes. For $a \in \mathcal{A}^{m,l}_{\mathrm{sc}}$ the terms $c_\alpha = \frac{i^{|\alpha|}}{\alpha!}(D_\xi^\alpha D_y^\alpha a)(x,x,\xi)$ lie in $S^{m-|\alpha|,l-|\alpha|}$, and the remainder in (6.2.3) satisfies $|D_x^\mu D_y^\nu D_\xi^\gamma r_N| \leq C\langle x\rangle^{l-N}\langle x-y\rangle^{w+N}\langle\xi\rangle^{m-N-|\gamma|}$. Hence $\mathrm{Op}(r_N) \in \langle x\rangle^{l-N}\Psi^{m-N}(\mathbb{R}^n)$ and the left symbol σ of $\mathrm{Op}(a)$ satisfies $\sigma - \sum_{|\alpha|<N} c_\alpha \in \langle x\rangle^{l-N}S^{m-N}(\mathbb{R}^n;\mathbb{R}^n)$. If $b \in S^{m,l}$ with $b \sim \sum c_\alpha$, then $\sigma - b \in \bigcap_N \langle x\rangle^{l-N}S^{m-N}(\mathbb{R}^n;\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^{2n})$. Adjoints and products are then treated exactly as in Theorem 6.2.7. □

Remark 6.3.3. Polyhomogeneous scattering symbols are defined using the radial compactification in each factor: $S^{m,l}_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n) = \langle z\rangle^l\langle\xi\rangle^m(\mathrm{RC}\times\mathrm{RC})^*\mathcal{C}^\infty(\mathbb{S}^p_+\times\mathbb{S}^n_+)$. By Proposition 6.1.8 applied in each factor these are scattering symbols, and they are preserved by D_z, D_ξ (which lower the respective orders) and by products. If $\Psi^{m,l}_{\mathrm{sc,ph}}(\mathbb{R}^n) = \mathrm{Op}(S^{m,l}_{\mathrm{ph}})$ then $\Psi^{m,l}_{\mathrm{sc,ph}}\circ\Psi^{m',l'}_{\mathrm{sc,ph}}\subset\Psi^{m+m',l+l'}_{\mathrm{sc,ph}}$: the terms of the composition formula are polyhomogeneous, an asymptotic sum $\sum_j c_j$ with $c_j\in S^{m-j,l-j}_{\mathrm{ph}}$ can be realized by a polyhomogeneous symbol using Borel's lemma for the series $\sum_j(\rho_1\rho_2)^jG_j$ on the manifold with corners $\mathbb{S}^p_+\times\mathbb{S}^n_+$ (ρ_1,ρ_2 the two boundary defining functions), and $\mathcal{S}(\mathbb{R}^{p+n})$ is contained in every $S^{m,l}_{\mathrm{ph}}$. Repeating the computation in the proof of Proposition 6.2.10 shows that $\mathcal{F}\,\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)\,\mathcal{F}^{-1} = \Psi^{l,m}_{\mathrm{sc}}(\mathbb{R}^n)$: the Fourier transform interchanges the two orders.

6.4 The residual isotropic algebra

The residual algebra $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, a subalgebra of the residual algebra $\Psi^{-\infty}(\mathbb{R}^n)$ of the traditional calculus, has two important properties not shared by the latter: its elements are compact on $L^2(\mathbb{R}^n)$, and it is a 'corner' in the bounded operators. We use the strong (dual) topology on $\mathcal{S}'(\mathbb{R}^n)$, in which a set is bounded if and only if it is weakly bounded (Banach–Steinhaus, $\mathcal{S}$ being Fréchet), and convergence means uniform convergence on bounded subsets of $\mathcal{S}$.

Lemma 6.4.1. *If $u\in\mathcal{S}'(\mathbb{R}^q)$ and $F\in\mathcal{S}(\mathbb{R}^p\times\mathbb{R}^q)$ then $f(z)=\langle u,F(z,\cdot)\rangle$ lies in $\mathcal{S}(\mathbb{R}^p)$, with $\partial_z^\alpha f=\langle u,\partial_z^\alpha F(z,\cdot)\rangle$, and $(u,F)\mapsto f$ is continuous (hypocontinuous) and bounded on products of bounded sets.*

Proof. The map $z\mapsto F(z,\cdot)\in\mathcal{S}(\mathbb{R}^q)$ is smooth, by Taylor's formula with remainder and the uniform Schwartz bounds on the derivatives of F. The distribution u has finite order: $|\langle u,\phi\rangle|\le C\sum_{|\beta|+|\gamma|\le k}\sup|y^\beta D^\gamma\phi|$. Hence

$$|z^\delta\partial_z^\alpha f(z)|\le C\sum_{|\beta|+|\gamma|\le k}\sup_y|z^\delta y^\beta D_y^\gamma\partial_z^\alpha F(z,y)|,$$

which is bounded. □

Proposition 6.4.2. *A continuous linear operator $A:\mathcal{S}(\mathbb{R}^n)\to\mathcal{S}'(\mathbb{R}^n)$ lies in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ if and only if it extends to a continuous linear map*

$$A:\mathcal{S}'(\mathbb{R}^n)\to\mathcal{S}(\mathbb{R}^n).$$

In particular each $A\in\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ maps bounded subsets of $L^2(\mathbb{R}^n)$ into bounded subsets of $\mathcal{S}(\mathbb{R}^n)$.

Proof. If $K_A\in\mathcal{S}(\mathbb{R}^{2n})$ then $Au(x)=\langle u,K_A(x,\cdot)\rangle$ for $u\in\mathcal{S}'$ by Lemma 6.4.1, and $\sup_x|x^\gamma D^\alpha Au(x)|\le\sup_{\phi\in B}|\langle u,\phi\rangle|$ where $B=\{x^\gamma D_x^\alpha K_A(x,\cdot);\ x\in\mathbb{R}^n\}$ is bounded in $\mathcal{S}$, which is continuity for the strong topology.

Conversely suppose $A:\mathcal{S}'\to\mathcal{S}$ is continuous. The map $\mathbb{R}^n\ni y\mapsto\delta_y\in\mathcal{S}'$ is smooth, with $\partial_{y_j}\delta_y=-\partial_j\delta_y$, since difference quotients of $\phi\in\mathcal{S}$ converge uniformly for ϕ in bounded sets. For each N the set $\{\langle y\rangle^N\partial^\beta\delta_y;\ y\in\mathbb{R}^n,\ |\beta|\le N\}$ is weakly, hence strongly, bounded in $\mathcal{S}'$, because $|\langle\langle y\rangle^N\partial^\beta\delta_y,\phi\rangle|=\langle y\rangle^N|\partial^\beta\phi(y)|$ is bounded for each $\phi\in\mathcal{S}$. Its image under A is bounded in $\mathcal{S}$. Thus $K(x,y)=(A\delta_y)(x)$ is smooth and $\langle y\rangle^N|x^\gamma D_x^\alpha D_y^\beta K(x,y)|$ is bounded for all N,γ,α and $|\beta|\le N$, i.e. $K\in\mathcal{S}(\mathbb{R}^{2n})$. Finally, for $\phi\in\mathcal{S}$ the Riemann sums of $\int\phi(y)\delta_y\,dy$ converge to ϕ in $\mathcal{S}'$, so $A\phi=\int\phi(y)A\delta_y\,dy=\int K(\cdot,y)\phi(y)\,dy$ and K is the kernel of A. □

Proposition 6.4.3. *Each $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ is a Hilbert–Schmidt, hence compact, operator on $L^2(\mathbb{R}^n)$, and $\|A\|_{\mathcal{B}(L^2)} \leq \|K_A\|_{L^2(\mathbb{R}^{2n})}$; in particular the inclusion $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) \to \mathcal{B}(L^2(\mathbb{R}^n))$ is continuous.*

Proof. $K_A \in \mathcal{S}(\mathbb{R}^{2n}) \subset L^2(\mathbb{R}^{2n})$. (Alternatively: by Proposition 6.4.2, A maps the unit ball of L^2 into a bounded set in $\mathcal{S}$, and bounded subsets of $\mathcal{S}$ are precompact in L^2 by the Arzelà–Ascoli theorem applied to $\langle x\rangle^{n+1}u$.) □

Lemma 6.4.4. *If $A_1, A_2 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and B is a bounded operator on $L^2(\mathbb{R}^n)$, or more generally any continuous linear operator $B : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$, then $A_1BA_2 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. Moreover for each k there are k' and C such that, for bounded B,*

$$p_k(K_{A_1BA_2}) \leq C\, p_{k'}(K_{A_1})\, p_{k'}(K_{A_2})\, \|B\|_{\mathcal{B}(L^2)}, \tag{6.4.1}$$

where $p_k(K) = \sum_{|\alpha|+|\beta|+|\gamma|+|\delta|\leq k} \sup |x^\alpha y^\beta D_x^\gamma D_y^\delta K(x,y)|$.

Proof. For bounded B the kernel of $C = A_1BA_2$ is

$$K_C(x,y) = \big(B[K_{A_2}(\cdot,y)], \overline{K_{A_1}(x,\cdot)}\big)_{L^2},$$

and differentiating under the inner product (the maps $y \mapsto K_{A_2}(\cdot,y)$, $x \mapsto K_{A_1}(x,\cdot)$ are smooth into L^2)

$$|x^\alpha y^\beta D_x^\gamma D_y^\delta K_C(x,y)| \leq \|B\| \sup_y \|y^\beta D_y^\delta K_{A_2}(\cdot,y)\|_{L^2} \sup_x \|x^\alpha D_x^\gamma K_{A_1}(x,\cdot)\|_{L^2},$$

and $\|f(x,\cdot)\|_{L^2} \leq C \sup_{x'} \langle x'\rangle^n |f(x,x')|$. For general B with kernel $K_B \in \mathcal{S}'(\mathbb{R}^{2n})$ (Schwartz kernel theorem), $K_C(x,y) = \langle K_B, K_{A_1}(x,\cdot) \otimes K_{A_2}(\cdot,y)\rangle$, the pairing being in the variables (x',y') of $K_{A_1}(x,x')K_{A_2}(y',y) \in \mathcal{S}(\mathbb{R}^{4n})$; this is in $\mathcal{S}(\mathbb{R}^{2n})$ by Lemma 6.4.1. □

Note that $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ is an ideal in $\Psi^{\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, but not in $\mathcal{B}(L^2(\mathbb{R}^n))$.

6.4.1 Finite rank operators and perturbations of the identity

If B is a bounded operator on $L^2(\mathbb{R}^n)$ of finite rank, let $f_1, \ldots, f_N$ be an orthonormal basis of its range; then $u \mapsto (Bu, f_j)$ are continuous functionals, given by $g_j \in L^2$, and $K_B = \sum_j f_j(x)\overline{g_j(y)}$. If $B \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ the range of B and of B^* lie in $\mathcal{S}$, so $f_j, g_j \in \mathcal{S}(\mathbb{R}^n)$; conversely such kernels are in $\mathcal{S}(\mathbb{R}^{2n})$. The finite rank elements of $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ form an ideal in $\Psi^{\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Lemma 6.4.5. *If K is a compact operator on a Hilbert space $\mathcal{H}$ then for each $t \in \mathbb{C}$, $\mathrm{Id} + tK$ has finite-dimensional null space and closed range of finite codimension, and* $\operatorname{ind}(\mathrm{Id}+tK) = \dim \mathrm{Nul}(\mathrm{Id}+tK) - \dim \mathrm{Nul}(\mathrm{Id}+\bar{t}K^*) = 0$.

Proof. Let $T = \mathrm{Id} + K$ (replace K by tK). On $N_0 = \mathrm{Nul}(T)$ we have $u = -Ku$, so the unit ball of N_0 is contained in K(unit ball), which is precompact; hence $\dim N_0 < \infty$. Let $R_0 = N_0^\perp$ and $R_1 = \mathrm{Ran}(T)$. If $f_k = Tu_k \to f$ with $u_k \in R_0$, then (u_k) is bounded: otherwise along a subsequence $\|u_k\| \to \infty$, and $v_k = u_k/\|u_k\|$ satisfies $v_k = -Kv_k + f_k/\|u_k\|$; by compactness a subsequence converges to some $v \in R_0$ with $\|v\| = 1$ and $Tv = 0$, a contradiction. Then $u_k = -Ku_k + f_k$ has a convergent subsequence, with limit u satisfying $Tu = f$; so R_1 is closed, and $N_1 = R_1^\perp = \mathrm{Nul}(T^*)$ is finite-dimensional by the first argument applied to K^*.

To see that the index is locally constant in t, fix t_0, let $T = \mathrm{Id} + t_0 K$ and write $\mathrm{Id} + tK = T + (t - t_0)K$ as a 2×2 matrix with respect to the decompositions $\mathcal{H} = N_0 \oplus R_0$ (domain) and $\mathcal{H} = N_1 \oplus R_1$ (range):

$$\mathrm{Id} + tK = \begin{pmatrix} (t-t_0)K_{00} & (t-t_0)K_{01} \\ (t-t_0)K_{10} & D(t) \end{pmatrix}, \qquad D(t) = \tilde{T} + (t - t_0)K_{11},$$

where $\tilde{T} : R_0 \to R_1$ is the restriction of T, invertible by the closed graph theorem. For $|t - t_0|$ small $D(t)$ is invertible and

$$\begin{pmatrix} P & Q \\ R & D \end{pmatrix} = \begin{pmatrix} \mathrm{Id} & QD^{-1} \\ 0 & \mathrm{Id} \end{pmatrix} \begin{pmatrix} P - QD^{-1}R & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} \mathrm{Id} & 0 \\ D^{-1}R & \mathrm{Id} \end{pmatrix}.$$

The outer factors are invertible, so null space and cokernel of $\mathrm{Id} + tK$ are isomorphic to those of the linear map $G(t) = P - QD^{-1}R : N_0 \to N_1$ between finite-dimensional spaces, whose index is $\dim N_0 - \dim N_1 = \mathrm{ind}(\mathrm{Id} + t_0 K)$. Thus $t \mapsto \mathrm{ind}(\mathrm{Id} + tK)$ is locally constant on $\mathbb{C}$, hence constant, and it vanishes at $t = 0$. □

Proposition 6.4.6. *Let $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. Then $\mathrm{Id} + A$ is Fredholm of index zero on $L^2(\mathbb{R}^n)$: its null space N_0 is finite-dimensional and contained in $\mathcal{S}(\mathbb{R}^n)$, and its range is closed and is the orthocomplement of the null space $N_1 \subset \mathcal{S}(\mathbb{R}^n)$ of $\mathrm{Id} + A^*$. If Π_0, Π_1 denote the orthogonal projections onto N_0, N_1, then $\Pi_0, \Pi_1 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and there is $B \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ such that*

$$(\mathrm{Id} + A)(\mathrm{Id} + B) = \mathrm{Id} - \Pi_1, \qquad (\mathrm{Id} + B)(\mathrm{Id} + A) = \mathrm{Id} - \Pi_0. \tag{6.4.2}$$

Moreover there is a finite rank $A' \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, of rank $\dim N_0$, such that $\mathrm{Id} + A + sA'$ is invertible on $L^2(\mathbb{R}^n)$ for all $s \neq 0$.

Proof. By Proposition 6.4.3 and Lemma 6.4.5, $\mathrm{Id} + A$ is Fredholm of index zero. If $u \in N_0$ then $u = -Au \in \mathcal{S}$, and similarly $N_1 \subset \mathcal{S}$, so $\Pi_0 = \sum_j f_j \otimes \bar{f}_j$ and $\Pi_1 = \sum_j g_j \otimes \bar{g}_j$, for orthonormal bases $f_j \in N_0$, $g_j \in N_1$, are in $\Psi^{-\infty}_{\mathrm{iso}}$. The restriction of $\mathrm{Id} + A$ is a bijection $N_0^\perp \to N_1^\perp = \mathrm{Ran}(\mathrm{Id} + A)$, with bounded inverse; extending this inverse by 0 on N_1 gives a bounded operator $\mathrm{Id} + B$ satisfying (6.4.2). From the first identity $B = -A - AB - \Pi_1$, and from the second $AB = -A^2 - ABA - A\Pi_0$; hence

$$B = -A + A^2 + ABA + A\Pi_0 - \Pi_1 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$$

by Lemma 6.4.4. Finally, since $\dim N_0 = \dim N_1 = N$, the operator $A' = \sum_{j=1}^N g_j \otimes \bar{f}_j$ maps N_0 isomorphically onto N_1 and vanishes on $N_0^\perp$; with respect to $L^2 = N_0 \oplus N_0^\perp \to N_1 \oplus N_1^\perp$, $\mathrm{Id} + A + sA'$ is the direct sum of $sA'|_{N_0}$ and $(\mathrm{Id} + A)|_{N_0^\perp}$, invertible for $s \neq 0$. □

Remark 6.4.7. Once the trace on $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ has been introduced (Chapter 7), the vanishing of the index also follows from (6.4.2): $\mathrm{ind}(\mathrm{Id} + A) = \mathrm{Tr}\,\Pi_0 - \mathrm{Tr}\,\Pi_1 = \mathrm{Tr}([A, B]) = 0$. The argument above is independent of the trace.

Corollary 6.4.8. *If $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and $\mathrm{Id} + A$ is injective, or surjective, on $L^2(\mathbb{R}^n)$, then $\mathrm{Id} + A$ is invertible and its inverse lies in $\mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.*

Proof. By index zero both N_0 and N_1 vanish, and $(\mathrm{Id} + A)^{-1} = \mathrm{Id} + B$. □

Corollary 6.4.9. *For $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $\mathrm{Id} + A$ is Fredholm of index zero as an operator on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$: in both cases its null space is N_0 and its range is $\{f;\ \Pi_1 f = 0\}$, a closed subspace of codimension $\dim N_1$.*

Proof. The identities (6.4.2) hold on $\mathcal{S}$ and, by continuity and density, on $\mathcal{S}'$. If $u \in \mathcal{S}'$ and $(\mathrm{Id} + A)u = 0$ then $u = -Au \in \mathcal{S}$, so $u \in N_0$. The range contains $\mathrm{Ran}(\mathrm{Id} - \Pi_1) = \mathrm{Nul}(\Pi_1)$ and is contained in it, since $((\mathrm{Id} + A)u, v) = (u, (\mathrm{Id} + A^*)v) = 0$ for $v \in N_1$. □

6.5 Exponential and logarithm

Proposition 6.5.1. *For $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ the series*

$$\exp(A) = \sum_{j\geq 0} \frac{1}{j!} A^j$$

converges in $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, and $\exp : \Psi^{-\infty}_{\mathrm{iso}} \to \mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$ is continuous and entire (holomorphic along every complex line, locally uniformly). Each $\exp(A)$ is invertible with inverse $\exp(-A)$. If $\|A\|_{\mathcal{B}(L^2)} < 1$ the series

$$\log(\mathrm{Id}+A) = \sum_{j\geq 1} \frac{(-1)^{j+1}}{j} A^j$$

converges in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $\exp(\log(\mathrm{Id}+A)) = \mathrm{Id}+A$, and $\log(\exp(A)) = A$ if $\|A\|_{\mathcal{B}(L^2)} < \log 2$. In particular the range of $\exp$ contains the neighbourhood $\{\mathrm{Id}+A;\ \|A\|_{\mathcal{B}(L^2)} < 1\}$ of Id in $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Proof. For $j \geq 2$ we write $A^j = A\,A^{j-2}A$ (with $A^0 = \mathrm{Id}$); then by (6.4.1) $p_k(K_{A^j}) \leq C\,p_{k'}(K_A)^2\,\|A\|^{j-2}_{\mathcal{B}(L^2)}$. Thus both series converge absolutely in every seminorm, uniformly for A in sets on which $p_{k'}(K_A)$ and $\|A\|$ are bounded (respectively $\|A\| \leq r < 1$), which by Proposition 6.4.3 include neighbourhoods of each point. The partial sums are continuous polynomial maps, so the limits are continuous and holomorphic along complex lines. The identities $\exp(A)\exp(-A) = \mathrm{Id}$, $\exp(\log(\mathrm{Id}+A)) = \mathrm{Id}+A$ and $\log(\exp A) = A$ (where $\|\exp A - \mathrm{Id}\| \leq e^{\|A\|} - 1 < 1$) hold for the norm-convergent series in the Banach algebra $\mathcal{B}(L^2)$, and convergence in $\Psi^{-\infty}_{\mathrm{iso}}$ implies convergence in $\mathcal{B}(L^2)$. □

6.6 Elliptic operators

Definition 6.6.1. An operator $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ with left symbol a is *elliptic* if there are $c, R > 0$ such that $a(w)$ is invertible and $|a(w)^{-1}| \leq c^{-1}\langle w\rangle^{-m}$ for $|w| \geq R$ (for $N = 1$: $|a(w)| \geq c\langle w\rangle^m$). For $A \in \Psi^m_{\mathrm{iso,ph}}$ this holds if and only if the principal symbol $\sigma_m(A)$ is invertible at every point of $\mathbb{S}^{2n-1}$. Similarly $A \in \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$ is elliptic if $|a(x,\xi)| \geq c\langle x\rangle^l\langle\xi\rangle^m$ for $|x|+|\xi| \geq R$.

Theorem 6.6.2. *If $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is elliptic there is $B \in \Psi^{-m}_{\mathrm{iso}}(\mathbb{R}^n)$, unique modulo $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, with*

$$AB - \mathrm{Id},\ BA - \mathrm{Id} \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n); \tag{6.6.1}$$

if $A \in \Psi^m_{\mathrm{iso,ph}}$ then $B \in \Psi^{-m}_{\mathrm{iso,ph}}$. Consequently, if $u \in \mathcal{S}'(\mathbb{R}^n)$ and $Au \in \mathcal{S}(\mathbb{R}^n)$ then $u \in \mathcal{S}(\mathbb{R}^n)$. The same holds for elliptic $A \in \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$, with $B \in \Psi^{-m,-l}_{\mathrm{sc}}(\mathbb{R}^n)$.

Proof. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$ equal 1 on $|w| \leq R$. Choose $F \in \mathcal{C}^\infty(\mathbb{C})$ with $F(z) = 1/z$ for $|z| \geq c$; then $F(\langle w\rangle^{-m}a) \in \Gamma^0$ by Lemma 6.1.2(ii), and $b_0 = (1-\chi)\langle w\rangle^{-m}F(\langle w\rangle^{-m}a) \in \Gamma^{-m}$ satisfies $ab_0 = 1-\chi$. (For matrix symbols use a smooth function equal to the inverse on the closed set of matrices M with $|M^{-1}| \leq c^{-1}$.) If a is polyhomogeneous so is b_0, since $1/a = a_m^{-1}\sum_k (-r)^k$ asymptotically, with $r = (a-a_m)/a_m \in \Gamma^{-1}_{\mathrm{ph}}$ for large $|w|$. By (6.2.4), $B_0 = \mathrm{Op}(b_0)$ satisfies $AB_0 = \mathrm{Id}-E$ with $\sigma_L(E) = \chi + \Gamma^{-2}$, so $E \in \Psi^{-2}_{\mathrm{iso}}$. Choose $B_R \in \Psi^{-m}_{\mathrm{iso}}$ with $\sigma_L(B_R) \sim \sum_j \sigma_L(B_0E^j)$, the terms having orders $-m-2j$. Then for each N, $B_R - \sum_{j<N} B_0E^j \in \Psi^{-m-2N}_{\mathrm{iso}}$ and

$$AB_R = \sum_{j<N} (\mathrm{Id}-E)E^j + \Psi^{-2N}_{\mathrm{iso}} = \mathrm{Id}-E^N + \Psi^{-2N}_{\mathrm{iso}} = \mathrm{Id}+\Psi^{-2N}_{\mathrm{iso}}.$$

Similarly $B_0 A = \mathrm{Id} - E'$ with $E' \in \Psi^{-2}_{\mathrm{iso}}$, and $B_L \sim \sum_j E'^j B_0$ satisfies $B_L A - \mathrm{Id} \in \Psi^{-\infty}_{\mathrm{iso}}$. Then $B_L - B_R = B_L(\mathrm{Id} - AB_R) + (B_L A - \mathrm{Id})B_R \in \Psi^{-\infty}_{\mathrm{iso}}$, so $B = B_R$ satisfies (6.6.1); uniqueness follows in the same way. If $Au \in \mathcal{S}$ then $u = B(Au) - (BA - \mathrm{Id})u \in \mathcal{S}$, since residual operators map $\mathcal{S}'$ into $\mathcal{S}$ (Proposition 6.4.2). In the scattering case the same construction works with $E \in \Psi^{-1,-1}_{\mathrm{sc}}$ and asymptotic sums in the sense of Lemma 6.1.5, the residual space being (6.3.1). □

Combining the symbolic construction with the analysis of Section 6.4 we see that elliptic operators have generalized inverses with finite rank errors.

Proposition 6.6.3. *Let $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ be elliptic. Then $N_0 = \mathrm{Nul}(A)$ and $N_1 = \mathrm{Nul}(A^*)$ (on $\mathcal{S}'(\mathbb{R}^n)$) are finite-dimensional subspaces of $\mathcal{S}(\mathbb{R}^n)$, and if Π_0, Π_1 are the L^2-orthogonal projections onto them, there is $G \in \Psi^{-m}_{\mathrm{iso}}(\mathbb{R}^n)$ such that*

$$AG = \mathrm{Id} - \Pi_1, \qquad GA = \mathrm{Id} - \Pi_0. \tag{6.6.2}$$

Hence A is Fredholm on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, with range $\{f;\ \Pi_1 f = 0\}$ and index $\dim N_0 - \dim N_1$. If A is polyhomogeneous then so is G. The same holds for elliptic $A \in \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$, with $G \in \Psi^{-m,-l}_{\mathrm{sc}}(\mathbb{R}^n)$.

Proof. Let B be a parametrix, $AB = \mathrm{Id} - E_R$, $BA = \mathrm{Id} - E_L$ with $E_R, E_L \in \Psi^{-\infty}_{\mathrm{iso}}$. If $Au = 0$, $u \in \mathcal{S}'$, then $u = E_L u$, so $N_0 \subset \mathrm{Nul}(\mathrm{Id} - E_L)$, which is finite-dimensional and contained in $\mathcal{S}$ by Corollary 6.4.9; the same applies to A^*, which is elliptic.

We show that $\mathrm{Ran}_{\mathcal{S}}(A) = \{f \in \mathcal{S};\ f \perp N_1\}$. Applying Proposition 6.4.6 to $-E_R$ gives $B' \in \Psi^{-\infty}_{\mathrm{iso}}$ and a finite rank orthogonal projection Π, with range $W \subset \mathcal{S}$, such that $AB(\mathrm{Id} + B') = (\mathrm{Id} - E_R)(\mathrm{Id} + B') = \mathrm{Id} - \Pi$. Hence $V = A(\mathcal{S}) \supset \{f \in \mathcal{S};\ f \perp W\}$. Since $W \subset \mathcal{S}$ is finite-dimensional, every $v \in V$ decomposes as $v = (v - P_W v) + P_W v$ with $v - P_W v \in V$, so $V = (W^\perp \cap \mathcal{S}) \oplus (V \cap W)$, and with $W_1 = W \ominus (V \cap W)$ we get $V = \{f \in \mathcal{S};\ f \perp W_1\}$. On the other hand $W_1 = \{g \in \mathcal{S};\ g \perp V\}$ (using the density of $W^\perp \cap \mathcal{S}$ in $W^\perp$), and $g \in \mathcal{S}$ is orthogonal to $A(\mathcal{S})$ if and only if $A^* g = 0$. Thus $W_1 = N_1$, as claimed.

Next, if $f \in \mathcal{S}'$ and $\Pi_1 f = 0$, then $f = AB(\mathrm{Id} + B')f + \Pi f$ with $\Pi f \in \mathcal{S}$ and $\Pi f \perp N_1$ (as $(Au, v) = (u, A^* v) = 0$ for $v \in N_1$), so $\Pi f \in A(\mathcal{S})$ and $f \in A(\mathcal{S}')$. Conversely $A(\mathcal{S}') \perp N_1$. Hence we may define $G : \mathcal{S}' \to \mathcal{S}'$ by $G = 0$ on N_1 and, for $\Pi_1 f = 0$, $Gf = u$ where u is the unique solution of $Au = f$ with $\Pi_0 u = 0$. Then (6.6.2) holds on $\mathcal{S}'$, and G maps $\mathcal{S}$ to $\mathcal{S}$ (elliptic regularity). The restriction $G : \mathcal{S} \to \mathcal{S}$ has closed graph, hence is continuous: if $f_k \to f$ and $Gf_k \to u$ in $\mathcal{S}$ then $Au = (\mathrm{Id} - \Pi_1)f$ and $\Pi_0 u = 0$, so $u = Gf$. Multiplying (6.6.2) by B gives $(\mathrm{Id} - E_L)G = B - B\Pi_1$ and $G(\mathrm{Id} - E_R) = B - \Pi_0 B$, so

$$G = B - B\Pi_1 + E_L G = B - B\Pi_1 + E_L B - E_L \Pi_0 B + E_L G E_R.$$

Here $E_L G E_R$ is continuous from $\mathcal{S}'$ to $\mathcal{S}$, since $E_R : \mathcal{S}' \to \mathcal{S}$, $G : \mathcal{S} \to \mathcal{S}$ and $E_L : \mathcal{S} \to \mathcal{S}$ are; so it is in $\Psi^{-\infty}_{\mathrm{iso}}$ by Proposition 6.4.2. All other terms except B are residual, so $G \in B + \Psi^{-\infty}_{\mathrm{iso}} \subset \Psi^{-m}_{\mathrm{iso}}$. The statements about $\mathcal{S}$ follow in the same way, and the scattering case is identical. □

6.7 L^2 boundedness and compactness

Proposition 6.7.1. *Every $A \in \Psi^0_{\mathrm{iso}}(\mathbb{R}^n; \mathbb{C}^N)$ is bounded on $L^2(\mathbb{R}^n; \mathbb{C}^N)$, and every $A \in \Psi^{-\epsilon}_{\mathrm{iso}}(\mathbb{R}^n; \mathbb{C}^N)$, $\epsilon > 0$, is compact. Elements of Ψ^m_{iso} with $m < -3n/2$ are Hilbert–Schmidt.*

Proof. By Lemma 6.1.6, $\Psi^0_{\mathrm{iso}}(\mathbb{R}^n) \subset \Psi^0(\mathbb{R}^n)$, which is bounded on L^2 by Chapter 2. If $a \in \Gamma^{-M}$, $M > n$, the kernel of $\operatorname{Op}(a)$ satisfies $(x-y)^\beta K(x,y) = \int e^{i(x-y)\cdot\xi}(-D_\xi)^\beta a(x,\xi)đ\xi$, and $\int \langle (x,\xi)\rangle^{-M} d\xi = C\langle x\rangle^{n-M}$ (substitute $\xi = \langle x\rangle\eta$), so $|K(x,y)| \le C_N\langle x\rangle^{n-M}\langle x-y\rangle^{-N}$ for all N; this is in $L^2(\mathbb{R}^{2n})$ if $M > 3n/2$. Now let $A \in \Psi^{-\epsilon}_{\mathrm{iso}}$ and $T = A^*A \in \Psi^{-2\epsilon}_{\mathrm{iso}}$, a bounded self-adjoint operator. For j large, $T^{2^j} \in \Psi^{-2^{j+1}\epsilon}_{\mathrm{iso}}$ is Hilbert–Schmidt, hence compact. If S is bounded and self-adjoint with S^2 compact, then S is compact: for $u_k \rightharpoonup u$ weakly, $\|S(u_k-u)\|^2 = (S^2(u_k-u), u_k-u) \le \|S^2(u_k-u)\|\|u_k-u\| \to 0$. By induction T is compact, and then $\|A(u_k-u)\|^2 = (T(u_k-u),u_k-u) \to 0$, so A is compact. □

We next identify the norm of $A \in \Psi^0_{\mathrm{iso}}$ modulo compact operators. The lower bound uses *coherent states*. For $w = (x_0,\xi_0) \in \mathbb{R}^{2n}$ let $(T_w u)(x) = e^{ix\cdot\xi_0}u(x-x_0)$, a unitary operator on $L^2(\mathbb{R}^n)$ preserving $\mathcal{S}(\mathbb{R}^n)$; a direct change of variables shows

$$T_w^{-1}\operatorname{Op}(a)T_w = \operatorname{Op}(a(\cdot + w)). \tag{6.7.1}$$

Let $u_0 = \pi^{-n/4}e^{-|x|^2/2}$, so $\|u_0\| = 1$.

Lemma 6.7.2. *Let $w_k \in \mathbb{R}^{2n}$ with $|w_k| \to \infty$ and $\phi_k = T_{w_k}u_0$. Then $\|\phi_k\| = 1$, $\phi_k \rightharpoonup 0$ weakly in $L^2(\mathbb{R}^n)$, and for every $a \in \Gamma^0(\mathbb{R}^{2n}; M(N,\mathbb{C}))$*

$$\sup_{v\in\mathbb{C}^N, |v|\le 1} \|\operatorname{Op}(a)(\phi_k v) - \phi_k\, a(w_k)v\|_{L^2} \to 0.$$

Proof. By (6.7.1) the norm in question is $\|\operatorname{Op}(b_k)(u_0 v)\|$ with $b_k(w) = a(w+w_k) - a(w_k)$. All derivatives of b_k are bounded uniformly in k (as $\Gamma^0 \subset \mathcal{C}^\infty_{\mathrm{b}}$), and

$$|b_k(w)| \le |w| \sup_{t\in[0,1]} |\nabla a(w_k + tw)| \le C|w| \sup_{t\in[0,1]} \langle w_k + tw\rangle^{-1} \to 0$$

for each w. Now $\operatorname{Op}(b_k)u_0(x) = \int e^{ix\cdot\xi} b_k(x,\xi)\hat{u}_0(\xi)đ\xi$; integrating by parts, $x^\beta \operatorname{Op}(b_k)u_0 = \int e^{ix\cdot\xi}(-D_\xi)^\beta(b_k\hat{u}_0)đ\xi$, so $|\operatorname{Op}(b_k)u_0(x)| \le C\langle x\rangle^{-n-1}$ uniformly in k, while $\operatorname{Op}(b_k)u_0(x) \to 0$ pointwise by dominated convergence. Dominated convergence in L^2 gives $\|\operatorname{Op}(b_k)u_0\| \to 0$.

For the weak convergence it suffices, by density, to show $(\phi_k, f) \to 0$ for $f \in \mathcal{S}$, and for this it suffices to show that every subsequence has a further subsequence along which this holds. Write $w_k = (x_k,\xi_k)$. If $|x_k| \to \infty$ along a subsequence then $|(\phi_k, f)| \le \int u_0(x - x_k)|f(x)|\,dx \to 0$. Otherwise $x_k \to x_*$ along a subsequence and $|\xi_k| \to \infty$; then $(\phi_k, f) = \hat{g}_k(-\xi_k)$ with $g_k = u_0(\cdot - x_k)\bar{f} \to g_*$ in L^1, and $|\hat{g}_k(-\xi_k)| \le \|g_k - g_*\|_{L^1} + |\hat{g}_*(-\xi_k)| \to 0$ by the Riemann–Lebesgue lemma. □

Theorem 6.7.3. *If $A \in \Psi^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ has left symbol a then*

$$\inf_{K \text{ compact}} \|A+K\|_{\mathcal{B}(L^2)} = \inf_{E\in\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)} \|A+E\|_{\mathcal{B}(L^2)} = \limsup_{|w|\to\infty} \|a(w)\|. \tag{6.7.2}$$

If $A \in \Psi^0_{\mathrm{iso,ph}}$ the right side is $\sup_{p\in\mathbb{S}^{2n-1}}\|\sigma_0(A)(p)\|$.

Proof. Let μ be the right side of (6.7.2). Since residual operators are compact, the first infimum is at most the second.

Lower bound. Let K be compact. Choose $w_k \to \infty$ with $\|a(w_k)\| \to \mu$ and unit vectors v_k with $|a(w_k)v_k| = \|a(w_k)\|$; passing to a subsequence $v_k \to v$. With ϕ_k as in Lemma 6.7.2, $K(\phi_k v) \to 0$ since K is compact and $\phi_k \rightharpoonup 0$, so

$$\|A+K\| \ge \liminf_k \|(A+K)(\phi_k v)\| \ge \liminf_k |a(w_k)v| = \mu.$$

Upper bound. Let $C > \mu^2$ and choose R and $\delta' > 0$ such that $p = C - a^*a \geq 2\delta'$ for $|w| \geq R$. With $\chi \in \mathcal{C}^\infty_{\rm c}$ equal to 1 on $|w| \leq R$, $0 \leq \chi \leq 1$ and supported in $|w| \leq R+1$, the function $p' = (1-\chi)p + \chi C$ lies in Γ^0, satisfies $p' \geq 2\delta'$ everywhere and equals p for $|w| \geq R+1$. Since the square root is smooth on positive definite matrices, $q_0 = (p')^{1/2} \in \Gamma^0$ is hermitian, invertible, with $q_0^{-1} \in \Gamma^0$ (Lemma 6.1.2(ii)). Let $Q_0 = \mathrm{Op}(q_0)$. By Theorem 6.2.7, $G_1 = Q_0^*Q_0 - C + A^*A$ is self-adjoint and lies in $\Psi^{-2}_{\rm iso}$, since its symbol is $p' - p$ (compactly supported) modulo Γ^{-2}. Suppose $Q = Q_0 + R_1 + \cdots + R_{k-1}$ has been found with $R_j \in \Psi^{-2j}_{\rm iso}$ and $G_k = Q^*Q - C + A^*A \in \Psi^{-2k}_{\rm iso}$, and let $g_k = \sigma_L(G_k)$, so that $g_k - g_k^* \in \Gamma^{-2k-2}$. Set $R_k = \mathrm{Op}(r_k)$, $r_k = -\frac{1}{2}q_0^{-1}g_k$. Then $(Q+R_k)^*(Q+R_k) - C + A^*A = G_k + Q^*R_k + R_k^*Q + R_k^*R_k$, whose symbol is, modulo Γ^{-2k-2},

$$g_k + q_0 r_k + r_k^* q_0 = g_k - \tfrac{1}{2}g_k - \tfrac{1}{2}g_k^* \equiv 0.$$

Taking $Q \in \Psi^0_{\rm iso}$ with $\sigma_L(Q) \sim q_0 + \sum_k r_k$ we obtain $Q^*Q = C - A^*A + G$ with $G \in \Psi^{-\infty}_{\rm iso}$ self-adjoint. Hence for $u \in \mathcal{S}$

$$\|Au\|^2 = C\|u\|^2 - \|Qu\|^2 + (Gu, u) \leq C\|u\|^2 + (Gu, u).$$

Given $\delta > 0$, approximate the compact operator G within $\delta/2$ by a finite rank operator $\sum_{j \leq J} f_j \otimes \bar{g}_j$ and then replace the g_j by nearby $g_j' \in \mathcal{S}$, to get a finite-dimensional $W = \mathrm{span}\{g_j'\} \subset \mathcal{S}$ with $\|Gu\| \leq \delta\|u\|$ for $u \perp W$. Let $\Pi_W \in \Psi^{-\infty}_{\rm iso}$ be the orthogonal projection onto W. Then for $u \in \mathcal{S}$, $u' = (\mathrm{Id} - \Pi_W)u \perp W$ and $\|A(\mathrm{Id} - \Pi_W)u\|^2 \leq (C+\delta)\|u\|^2$. Since $A\Pi_W \in \Psi^{-\infty}_{\rm iso}$ and $C > \mu^2$, $\delta > 0$ are arbitrary, the second infimum is at most μ.

For polyhomogeneous A, $a - a_0 \to 0$ at infinity, a_0 being homogeneous of degree 0 outside the unit ball, so $\mu = \sup_{\mathbb{S}^{2n-1}} \|a_0\|$. □

Corollary 6.7.4. *An operator $A \in \Psi^0_{\rm iso}(\mathbb{R}^n; \mathbb{C}^N)$ is compact on $L^2(\mathbb{R}^n; \mathbb{C}^N)$ if and only if $\sigma_L(A)(w) \to 0$ as $|w| \to \infty$. In particular $A \in \Psi^0_{\rm iso,ph}(\mathbb{R}^n; \mathbb{C}^N)$ is compact if and only if $\sigma_0(A) = 0$, that is $A \in \Psi^{-1}_{\rm iso,ph}(\mathbb{R}^n; \mathbb{C}^N)$.*

Proof. If A is compact the left side of (6.7.2) vanishes. Conversely if the right side vanishes, A is a norm limit of residual, hence compact, operators. □

Corollary 6.7.5. *An operator $A \in \Psi^0_{\rm iso}(\mathbb{R}^n; \mathbb{C}^N)$ is Fredholm on $L^2(\mathbb{R}^n; \mathbb{C}^N)$ if and only if it is elliptic.*

Proof. If A is elliptic, (6.6.2) with $G \in \Psi^0_{\rm iso}$ bounded shows that A is Fredholm. Conversely, if A is Fredholm there is a bounded B with $BA = \mathrm{Id} - P$, P of finite rank. If A is not elliptic there are $w_k \to \infty$ and unit vectors v_k with $|a(w_k)v_k| \to 0$; passing to a subsequence $v_k \to v$. By Lemma 6.7.2, $\|A(\phi_k v)\| \to 0$, hence $\phi_k v = BA(\phi_k v) + P(\phi_k v) \to 0$, since P is compact and $\phi_k \rightharpoonup 0$. This contradicts $\|\phi_k v\| = 1$. □

In particular ellipticity is necessary in Proposition 6.6.3.

6.8 The harmonic oscillator

The *harmonic oscillator* is the differential operator

$$H = \sum_{j=1}^n (D_j^2 + x_j^2) = \Delta + |x|^2.$$

It is an elliptic element of $\Psi^2_{\rm iso,ph}(\mathbb{R}^n)$ with symbol $|w|^2$. Its spectral decomposition is computed explicitly using the *creation* and *annihilation* operators

$$C_j = D_j + ix_j, \qquad A_j = D_j - ix_j = C_j^*, \qquad j = 1, \ldots, n.$$

Lemma 6.8.1. *The following identities hold for $j,k=1,\dots,n$:*

$$[A_j,A_k]=[C_j,C_k]=0,\qquad [A_j,C_k]=2\delta_{jk},$$
$$H=\sum_j C_jA_j+n=\sum_j A_jC_j-n,\qquad [C_j,H]=-2C_j,\qquad [A_j,H]=2A_j.$$

Hence $HC_j=C_j(H+2)$ and $HA_j=A_j(H-2)$.

Proof. Since $[D_j,x_k]=-i\delta_{jk}$, $C_jA_j=D_j^2+x_j^2-i[D_j,x_j]=D_j^2+x_j^2-1$ and $A_jC_j=D_j^2+x_j^2+1$, which gives the formulæ for H and $[A_j,C_j]=2$; the other commutators of the A's and C's are clearly zero. Then $[C_j,H]=\sum_k C_k[C_j,A_k]=-2C_j$ and $[A_j,H]=\sum_k[A_j,C_k]A_k=2A_j$. □

Proposition 6.8.2. *For $u\in\mathcal{S}(\mathbb{R}^n)$,*

$$(Hu,u)=\sum_j\|A_ju\|^2+n\|u\|^2. \tag{6.8.1}$$

If $u\in\mathcal{S}'(\mathbb{R}^n)$ and $(H-\lambda)u=0$ for some $\lambda\in\mathbb{C}$, then $u\in\mathcal{S}(\mathbb{R}^n)$, and $u\neq 0$ implies $\lambda\in\mathbb{R}$, $\lambda\geq n$. The operator H is invertible on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, with inverse $H^{-1}\in\Psi^{-2}_{\mathrm{iso,ph}}(\mathbb{R}^n)$, which is a compact, self-adjoint, injective operator on $L^2(\mathbb{R}^n)$. More generally $H^p\in\Psi^{2p}_{\mathrm{iso,ph}}(\mathbb{R}^n)$ for all $p\in\mathbb{Z}$.

Proof. (6.8.1) follows from $H=\sum C_jA_j+n$ and $C_j^*=A_j$. Since $H-\lambda$ is elliptic, eigenfunctions in $\mathcal{S}'$ lie in $\mathcal{S}$ (Theorem 6.6.2); then $\lambda\|u\|^2=(Hu,u)$ is real and at least $n\|u\|^2$. Thus $\mathrm{Nul}(H)=\mathrm{Nul}(H^*)=\{0\}$ on $\mathcal{S}'$, and Proposition 6.6.3 gives $G\in\Psi^{-2}_{\mathrm{iso,ph}}$ with $HG=GH=\mathrm{Id}$ on $\mathcal{S}'$. By Proposition 6.7.1, G is compact on L^2; it is injective since $HG=\mathrm{Id}$, and self-adjoint since $(Gf,g)=(Gf,HGg)=(HGf,Gg)=(f,Gg)$ for $f,g\in\mathcal{S}$. □

Proposition 6.8.3. *The eigenvalues of H (on $\mathcal{S}'(\mathbb{R}^n)$) are*

$$\mathrm{spec}(H)=\{n,n+2,n+4,\dots\}=n+2\mathbb{N}_0.$$

The eigenvalue n has a one-dimensional eigenspace spanned by the ground state $u_0=\pi^{-n/4}\exp(-|x|^2/2)$.

Proof. By (6.8.1), $Hu=nu$ if and only if $A_ju=0$ for all j. Now $A_ju=-i(\partial_j+x_j)u=-ie^{-|x|^2/2}\partial_j(e^{|x|^2/2}u)$, so for a distribution u, $A_ju=0$ for all j means that $e^{|x|^2/2}u$ is constant, i.e. $u\in\mathbb{C}u_0$. Similarly $C_jv=-ie^{|x|^2/2}\partial_j(e^{-|x|^2/2}v)$, so the only tempered solution of $C_jv=0$ is $v=0$ (as $e^{x_j^2/2}$ is not tempered); thus each C_j is injective on $\mathcal{S}'$.

If $Hu=\lambda u$ with $0\neq u\in\mathcal{S}$, then by Lemma 6.8.1 $H(A_ju)=(\lambda-2)A_ju$ and $H(C_ju)=(\lambda+2)C_ju$, with $C_ju\neq 0$. So $\lambda+2\mathbb{N}_0$ consists of eigenvalues. Since all eigenvalues are $\geq n$, $A^\alpha u=0$ when $\lambda-2|\alpha|<n$. Let k be the largest integer such that $A^\alpha u\neq 0$ for some $|\alpha|=k$, and $v=A^\alpha u$ for such α. Then $A_jv=0$ for all j, so $v\in\mathbb{C}u_0$ and $\lambda-2k=n$. Combined with $Hu_0=nu_0$ and the ladder property, the eigenvalues are exactly $n+2\mathbb{N}_0$. □

Proposition 6.8.4. *For $k\in\mathbb{N}_0$ let E_k be the eigenspace of H with eigenvalue $n+2k$. Then the map*

$$\{\textit{polynomials on } \mathbb{R}^n \textit{ homogeneous of degree } k\}\ni p\longmapsto p(C)u_0\in E_k$$

is an isomorphism; thus $\{C^\alpha u_0;\ |\alpha|=k\}$ is a basis of E_k and $\dim E_k=\binom{n+k-1}{k}$.

Proof. The C_j commute, so $p(C)$ is well defined, and by Lemma 6.8.1, $HC^\alpha u_0 = C^\alpha(H + 2|\alpha|)u_0 = (n+2|\alpha|)C^\alpha u_0$. From $[A_j, C_k] = 2\delta_{jk}$ and $A_j u_0 = 0$ we get $A_j C^\alpha u_0 = 2\alpha_j C^{\alpha-e_j} u_0$ (zero if $\alpha_j = 0$), hence for $|\beta| = |\alpha|$

$$A^\beta C^\alpha u_0 = \begin{cases} 0, & \beta \neq \alpha, \\ 2^{|\alpha|}\alpha!\, u_0, & \beta = \alpha. \end{cases} \tag{6.8.2}$$

Applying A^β to $\sum_{|\alpha|=k} c_\alpha C^\alpha u_0 = 0$ gives $c_\beta = 0$, so the map is injective.

For surjectivity we show by induction on k: if $v \in E_k$ and $A^\beta v = 0$ for all $|\beta| = k$, then $v = 0$. For $k = 0$ this is trivial, as $A^0 = \mathrm{Id}$. For $k > 0$ and j fixed, $A_j v \in E_{k-1}$ (or is 0) and $A^\beta(A_j v) = A^{\beta+e_j} v = 0$ for $|\beta| = k - 1$, so $A_j v = 0$ by induction; then $v \in E_0 = \mathbb{C}u_0$ and $Hv = (n + 2k)v$ force $v = 0$. Now let $v \in E_k$ be orthogonal to all $C^\alpha u_0$, $|\alpha| = k$. Then $(A^\alpha v, u_0) = (v, C^\alpha u_0) = 0$, and $A^\alpha v \in E_0 = \mathbb{C}u_0$, so $A^\alpha v = 0$ for all $|\alpha| = k$, whence $v = 0$. Since $E_k \subset \mathcal{S}$ is finite-dimensional (Proposition 6.6.3 applied to $H - n - 2k$), E_k is spanned by the $C^\alpha u_0$. □

Eigenfunctions with different eigenvalues are orthogonal, H being symmetric on $\mathcal{S}$, and by (6.8.2), $(C^\alpha u_0, C^\beta u_0) = (A^\beta C^\alpha u_0, u_0) = \delta_{\alpha\beta} 2^{|\alpha|}\alpha!$ for $|\alpha| = |\beta|$, since $\|u_0\| = 1$. Hence the *Hermite functions*

$$u_\alpha = (2^{|\alpha|}\alpha!)^{-1/2} C^\alpha u_0, \qquad \alpha \in \mathbb{N}_0^n, \tag{6.8.3}$$

form an orthonormal set in $L^2(\mathbb{R}^n)$, with $Hu_\alpha = (n + 2|\alpha|)u_\alpha$. In one variable, $(\partial - x)f = e^{x^2/2}\partial(e^{-x^2/2} f)$ gives $C^k e^{-x^2/2} = i^k H_k(x) e^{-x^2/2}$, where $H_k(x) = (-1)^k e^{x^2} \frac{d^k}{dx^k} e^{-x^2}$ is the k-th Hermite polynomial, and in general $u_\alpha(x) = \prod_j u_{\alpha_j}(x_j)$. One also checks from (6.8.2) that

$$C_j u_\alpha = (2\alpha_j + 2)^{1/2} u_{\alpha+e_j}, \qquad A_j u_\alpha = (2\alpha_j)^{1/2} u_{\alpha-e_j}. \tag{6.8.4}$$

Lemma 6.8.5. *If K is a compact self-adjoint operator on a Hilbert space $V \neq \{0\}$ and $K \neq 0$, then K has an eigenvector with non-zero eigenvalue.*

Proof. Let $\tau = \|K\| > 0$ and choose unit vectors ϕ_j with $\|K\phi_j\| \to \tau$. Passing to a subsequence, $\phi_j \rightharpoonup \phi$ weakly with $\|\phi\| \leq 1$, and $K\phi_j \to K\phi$ by compactness. Thus $\|K\phi\| = \tau \leq \tau\|\phi\|$, so $\|\phi\| = 1$ and ϕ maximizes $\|K\psi\|^2 = (K^2\psi, \psi)$ on the unit sphere. For $\phi' \perp \phi$ the function $t \mapsto \|K(\phi + t\phi')\|^2 / \|\phi + t\phi'\|^2$, $t \in \mathbb{R}$, has a maximum at $t = 0$, so $\mathrm{Re}(K^2\phi, \phi') = 0$; replacing ϕ' by $i\phi'$, $(K^2\phi, \phi') = 0$. Hence $K^2\phi = \tau^2\phi$, i.e. $(K - \tau)(K + \tau)\phi = 0$. Either $(K + \tau)\phi \neq 0$ is an eigenvector with eigenvalue τ, or ϕ is one with eigenvalue $-\tau$. □

Theorem 6.8.6. *The Hermite functions u_α, $\alpha \in \mathbb{N}_0^n$, form an orthonormal basis of $L^2(\mathbb{R}^n)$; that is, for $u \in L^2(\mathbb{R}^n)$*

$$u = \sum_\alpha c_\alpha u_\alpha, \qquad c_\alpha = (u, u_\alpha), \tag{6.8.5}$$

with convergence in L^2.

Proof. Let W be the closed span of the u_α and $V = W^\perp$. The operator $G = H^{-1}$ of Proposition 6.8.2 satisfies $Gu_\alpha = (n + 2|\alpha|)^{-1} u_\alpha$, so $G(W) \subset W$ and, G being self-adjoint, $G(V) \subset V$. If $V \neq \{0\}$, the restriction of G to V is compact, self-adjoint and injective, so by Lemma 6.8.5 there is $0 \neq f \in V$ with $Gf = \mu f$, $\mu \neq 0$. Then $Hf = \mu^{-1} f$, so f is an eigenfunction of H, hence lies in some $E_k \subset W$ by Proposition 6.8.4, a contradiction. □

Proposition 6.8.7. *If $u \in \mathcal{S}(\mathbb{R}^n)$ then the coefficients in (6.8.5) are rapidly decreasing, $|c_\alpha| \leq C_N(1 + |\alpha|)^{-N}$ for all N, and the series converges absolutely in $\mathcal{S}(\mathbb{R}^n)$. Conversely, every such series with rapidly decreasing coefficients converges in $\mathcal{S}(\mathbb{R}^n)$.*

Proof. Since $H^N u \in \mathcal{S}$, $(n+2|\alpha|)^N |c_\alpha| = |(u, H^N u_\alpha)| = |(H^N u, u_\alpha)| \le \|H^N u\|$. By (6.8.4) and $2ix_j = C_j - A_j$, $2D_j = C_j + A_j$, each $x^\beta D^\gamma u_\alpha$ is a combination of at most $2^{|\beta|+|\gamma|}$ functions $u_{\alpha'}$ with coefficients bounded by $(2|\alpha| + 2|\beta| + 2|\gamma|)^{(|\beta|+|\gamma|)/2}$. By the Sobolev embedding $\sup |v| \le C \sum_{|\mu|\le n+1} \|D^\mu v\|_{L^2}$, the Schwartz seminorms of u_α are therefore bounded by $C_k(1+|\alpha|)^{M_k}$, and the series converges absolutely in $\mathcal{S}$ when the coefficients are rapidly decreasing; its sum is u since it converges to u in L^2. □

For $N \in \mathbb{N}_0$ let Π_N be the orthogonal projection onto $E_0 \oplus \cdots \oplus E_N$, i.e. $K_{\Pi_N} = \sum_{|\alpha|\le N} u_\alpha(x)\overline{u_\alpha(y)}$; it is a finite rank element of $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Corollary 6.8.8. *For every $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $\Pi_N A \Pi_N \to A$ in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ as $N \to \infty$. In particular the finite rank elements are dense in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, and also in $\mathcal{B}(L^2)$ norm.*

Proof. The functions $u_\alpha(x)\overline{u_\beta(y)}$, $(\alpha,\beta) \in \mathbb{N}_0^{2n}$, are, up to factors ± 1 (note $\overline{u_\beta} = (-1)^{|\beta|} u_\beta$), the Hermite functions of $\mathbb{R}^{2n}$. Hence by Proposition 6.8.7 in $\mathbb{R}^{2n}$ the expansion $K_A = \sum_{\alpha,\beta} c_{\alpha\beta} u_\alpha(x)\overline{u_\beta(y)}$ converges absolutely in $\mathcal{S}(\mathbb{R}^{2n})$; its partial sum over $|\alpha|, |\beta| \le N$ is the kernel of $\Pi_N A \Pi_N$. The last statement follows from Proposition 6.4.3. □

Remark 6.8.9. The same arguments show that for $p \in \mathbb{Z}$ the eigenvalues of H^p are $(n+2k)^p$, with the same eigenspaces E_k. Thus H^p is compact if $p < 0$ and $H^p = \sum_\alpha (n+2|\alpha|)^p u_\alpha \otimes \bar{u}_\alpha$ is of trace class if $p < -n$.

6.9 Isotropic Sobolev spaces

The space $L^2(\mathbb{R}^n)$ is associated with the inclusion $\mathcal{S}(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$, and operators of order 0 are bounded on it. The isotropic Sobolev space of order s consists of the distributions mapped into L^2 by all isotropic operators of order s. For each $s \in \mathbb{R}$ we fix the elliptic operator $\Lambda_s = \mathrm{Op}(\langle w \rangle^s) \in \Psi^s_{\mathrm{iso,ph}}(\mathbb{R}^n)$, and let P_s be the orthogonal projection onto its (finite-dimensional) null space and $G_s \in \Psi^{-s}_{\mathrm{iso,ph}}$ its generalized inverse (Proposition 6.6.3), $G_s \Lambda_s = \mathrm{Id} - P_s$.

Definition 6.9.1. For $s \in \mathbb{R}$,

$$H^s_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ Au \in L^2(\mathbb{R}^n)\ \forall\, A \in \Psi^s_{\mathrm{iso}}(\mathbb{R}^n)\},$$

with the norm $\|u\|_s^2 = \|\Lambda_s u\|_{L^2}^2 + \|P_s u\|_{L^2}^2$.

Proposition 6.9.2. (i) *If $A \in \Psi^s_{\mathrm{iso}}(\mathbb{R}^n)$ is elliptic then $H^s_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in \mathcal{S}';\ Au \in L^2\}$, and $\|Au\| + \|\Pi_0 u\|$, with Π_0 the projection onto $\mathrm{Nul}(A)$, is an equivalent norm.*

(ii) *$H^s_{\mathrm{iso}}(\mathbb{R}^n)$ is a Hilbert space in which $\mathcal{S}(\mathbb{R}^n)$ is dense; $H^0_{\mathrm{iso}} = L^2$.*

(iii) *For $k \in \mathbb{N}_0$, $H^k_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in L^2;\ x^\alpha D^\beta u \in L^2\ \forall\, |\alpha| + |\beta| \le k\}$ with equivalent norm $\big(\sum_{|\alpha|+|\beta|\le k} \|x^\alpha D^\beta u\|^2\big)^{1/2}$.*

(iv) *The L^2 pairing extends to a non-degenerate pairing $H^s_{\mathrm{iso}} \times H^{-s}_{\mathrm{iso}} \to \mathbb{C}$ identifying H^{-s}_{iso} with the dual of H^s_{iso} (via $v \mapsto (\cdot, v)$, an antilinear isomorphism).*

(v) *Each $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is bounded $H^s_{\mathrm{iso}} \to H^{s-m}_{\mathrm{iso}}$ for all s.*

(vi) *$\bigcap_s H^s_{\mathrm{iso}} = \mathcal{S}(\mathbb{R}^n)$, $\bigcup_s H^s_{\mathrm{iso}} = \mathcal{S}'(\mathbb{R}^n)$, and for $s > t$ the inclusion $H^s_{\mathrm{iso}} \to H^t_{\mathrm{iso}}$ is compact.*

Proof. (i) If A is elliptic of order s with parametrix B, any $A' \in \Psi^s_{\mathrm{iso}}$ can be written $A' = (A'B)A + A'(\mathrm{Id} - BA)$, with $A'B \in \Psi^0_{\mathrm{iso}}$ bounded on L^2 and $A'(\mathrm{Id} - BA)$ residual, mapping $\mathcal{S}'$ to $\mathcal{S}$. So $Au \in L^2$ implies $A'u \in L^2$. If G is the generalized inverse of A, $u = GAu + \Pi_0 u$, so $\|A'u\| \le \|A'G\|\|Au\| + \|A'\Pi_0\|\|\Pi_0 u\|$; by symmetry the norms are equivalent.

(ii) Let u_k be Cauchy. Then $\Lambda_s u_k \to f$ in L^2 and $P_s u_k \to g$. Since $\Pi_1 \Lambda_s u_k = 0$ (Π_1 the projection onto $\operatorname{Nul} \Lambda_s^*$), also $\Pi_1 f = 0$. As G_s is continuous on $\mathcal{S}'$, $u_k = G_s \Lambda_s u_k + P_s u_k \to u = G_s f + g$ in $\mathcal{S}'$, and $\Lambda_s u = (\mathrm{Id} - \Pi_1) f = f$, $P_s u = g$ (note $P_s G_s = 0$ by construction). Thus $u_k \to u$ in H^s_{iso}. For density, given $u \in H^s_{\mathrm{iso}}$ choose $f_k \in \mathcal{S}$ with $f_k \to \Lambda_s u$ in L^2; then $u_k = G_s f_k + P_s u \in \mathcal{S}$ and $u_k \to u$ in H^s_{iso}. That $H^0_{\mathrm{iso}} = L^2$ follows from (i) with $A = \mathrm{Id}$.

(iii) Since $x^\alpha D^\beta \in \Psi^{|\alpha|+|\beta|}_{\mathrm{iso}} \subset \Psi^k_{\mathrm{iso}}$ for $|\alpha|+|\beta| \le k$, elements of H^k_{iso} satisfy the conditions. Conversely the operator $Q_k = \sum_{|\alpha|+|\beta|\le k} (x^\alpha D^\beta)^* (x^\alpha D^\beta) \in \Psi^{2k}_{\mathrm{iso,ph}}$ has principal symbol $q_k = \sum_{|\alpha|+|\beta|=k} x^{2\alpha} \xi^{2\beta}$, which is positive on $\mathbb{S}^{2n-1}$ (if $w_i \neq 0$ the term w_i^{2k} occurs), so Q_k is elliptic. If B is a parametrix and $A \in \Psi^k_{\mathrm{iso}}$, then $Au = \sum (ABD^\beta x^\alpha)(x^\alpha D^\beta u) + A(\mathrm{Id} - BQ_k)u$ with $ABD^\beta x^\alpha \in \Psi^0_{\mathrm{iso}}$. The norm equivalence follows as in (i).

(iv) For $u \in \mathcal{S}$, $v \in H^{-s}_{\mathrm{iso}}$: $(u, v) = (G_s \Lambda_s u + P_s u, v) = (\Lambda_s u, G_s^* v) + (u, P_s v)$, and $G_s^* \in \Psi^{-s}_{\mathrm{iso}}$, so $|(u, v)| \le C \|u\|_s \|v\|_{-s}$; the pairing extends by density. Conversely a continuous linear functional ℓ on H^s_{iso} is, by the Riesz theorem, $\ell(u) = (u, w)_s$ for some $w \in H^s_{\mathrm{iso}}$, i.e. $\ell(u) = (u, v)$ with $v = \Lambda_s^* \Lambda_s w + P_s w$, and $v \in H^{-s}_{\mathrm{iso}}$ since $A\Lambda_s^* \in \Psi^0_{\mathrm{iso}}$ for $A \in \Psi^{-s}_{\mathrm{iso}}$. Non-degeneracy follows from the density of $\mathcal{S}$.

(v) If $A' \in \Psi^{s-m}_{\mathrm{iso}}$ then $A'A \in \Psi^s_{\mathrm{iso}}$, so A maps H^s_{iso} into H^{s-m}_{iso}, and boundedness follows from $\|\Lambda_{s-m} Au\| \le \|\Lambda_{s-m} A G_s\| \|\Lambda_s u\| + \|\Lambda_{s-m} A P_s\| \|P_s u\|$.

(vi) If $u \in H^k_{\mathrm{iso}}$ for all k then $x^\alpha D^\beta u \in L^2$ for all α, β, so $u \in \mathcal{S}$ by Sobolev embedding. A tempered distribution satisfies $|\langle u, \phi\rangle| \le C \sum_{|\alpha|+|\beta|\le k} \sup |x^\alpha D^\beta \phi| \le C' \|\phi\|_{k+n+1}$ (Sobolev embedding again), so by (iii), (iv) it lies in $H^{-k-n-1}_{\mathrm{iso}}$. For compactness, $\Lambda_t u = \Lambda_t G_s \Lambda_s u + \Lambda_t P_s u$ where $\Lambda_t G_s \in \Psi^{t-s}_{\mathrm{iso}}$ is compact on L^2 (Proposition 6.7.1) and $\Lambda_t P_s$ has finite rank. □

Proposition 6.9.3. *An operator $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is Fredholm as a map $H^s_{\mathrm{iso}}(\mathbb{R}^n) \to H^{s-m}_{\mathrm{iso}}(\mathbb{R}^n)$ if and only if it is elliptic; then its null space and index are those of Proposition 6.6.3, independent of s.*

Proof. If A is elliptic, (6.6.2) with G bounded $H^{s-m}_{\mathrm{iso}} \to H^s_{\mathrm{iso}}$ shows that A is Fredholm, with null space N_0 and range $\{f;\ \Pi_1 f = 0\}$. In particular $\Lambda_t : H^t_{\mathrm{iso}} \to L^2$ and $G_t : L^2 \to H^t_{\mathrm{iso}}$ are Fredholm. If $A : H^s_{\mathrm{iso}} \to H^{s-m}_{\mathrm{iso}}$ is Fredholm, then $T = \Lambda_{s-m} A G_s \in \Psi^0_{\mathrm{iso}}$ is Fredholm on L^2, so elliptic by Corollary 6.7.5. Since $\sigma_L(T) - \langle w\rangle^{s-m} a\, \sigma_L(G_s) \in \Gamma^{-2}$ and $\langle w\rangle^{s-m}$ and $\sigma_L(G_s)$ are elliptic of orders $s - m$ and $-s$, a is elliptic of order m. □

Remark 6.9.4. For the scattering calculus the natural spaces carry two orders, $H^{m,l}_{\mathrm{sc}}(\mathbb{R}^n) = \langle x\rangle^{-l} H^m(\mathbb{R}^n)$, and analogous results hold for $\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$ acting between them; we do not pursue this here.

6.10 The residual group

Definition 6.10.1. The *residual isotropic group* is

$$G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) = \{\mathrm{Id} + A;\ A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n),\ \mathrm{Id} + A \text{ invertible on } L^2(\mathbb{R}^n)\}.$$

By Corollary 6.4.8, $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ is a group, the inverse of $\mathrm{Id} + A$ being of the form $\mathrm{Id} + B$ with $B \in \Psi^{-\infty}_{\mathrm{iso}}$, and $\mathrm{Id} + A$ is in $G^{-\infty}_{\mathrm{iso}}$ as soon as it is injective (or surjective) on L^2. We identify $\mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ with $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^{2n})$ and regard $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ as an open subset of this Fréchet space.

Proposition 6.10.2. *$G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ is open in $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, and multiplication and inversion are smooth; the derivative of inversion at g is $v \mapsto -g^{-1}vg^{-1}$. The exponential map of Proposition 6.5.1 takes values in $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and restricts to a homeomorphism of a neighbourhood of 0 onto a neighbourhood of Id.*

Proof. Openness follows from the openness of the invertible operators in $\mathcal{B}(L^2)$ and the continuity of $\Psi^{-\infty}_{\mathrm{iso}} \to \mathcal{B}(L^2)$ (Proposition 6.4.3). Multiplication $(\mathrm{Id}+A)(\mathrm{Id}+B) = \mathrm{Id}+A+B+AB$ is continuous and bilinear in (A,B), hence smooth. If $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+B$, then as in the proof of Proposition 6.4.6, $B = -A + A^2 + ABA$. Since $A \mapsto B$ is continuous into $\mathcal{B}(L^2)$, (6.4.1) shows it is continuous into $\Psi^{-\infty}_{\mathrm{iso}}$. From $(g+tv)^{-1} - g^{-1} = -t(g+tv)^{-1}vg^{-1}$ the derivative of inversion is $-g^{-1}vg^{-1}$, continuous in (g,v), and by induction inversion is $\mathcal{C}^\infty$. The last statement is Proposition 6.5.1. □

6.10.1 Differential forms

Here a map $f : U \to F$ from an open subset U of a Fréchet space E into a Fréchet space F is $\mathcal{C}^1$ if the directional derivatives $Df(u;v) = \frac{d}{dt}f(u+tv)|_{t=0}$ exist and $(u,v) \mapsto Df(u;v)$ is continuous on $U \times E$; $\mathcal{C}^k$ and $\mathcal{C}^\infty$ are defined inductively. For $E = \mathcal{S}(\mathbb{R}^p)$, the continuous k-linear forms on E are identified, by the Schwartz kernel theorem, with $\mathcal{S}'(\mathbb{R}^{kp})$, and the alternating ones with

$$\Lambda^k\mathcal{S}(\mathbb{R}^p) = \{u \in \mathcal{S}'(\mathbb{R}^{kp});\ u(x_{e(1)},\dots,x_{e(k)}) = \mathrm{sgn}(e)\, u(x_1,\dots,x_k)\ \forall\, e\},$$

e ranging over the permutations of $\{1,\dots,k\}$ and $x_i \in \mathbb{R}^p$. A *k-form* on an open set $U \subset \mathcal{S}(\mathbb{R}^p)$ is a map $\alpha : U \times E^k \to \mathbb{C}$, alternating and k-linear in the last k arguments (so $\alpha(u) \in \Lambda^k\mathcal{S}(\mathbb{R}^p)$ for each u), which is smooth on $U \times E^k$. The exterior derivative is defined by the usual formula for constant vector fields,

$$d\alpha(u;v_0,\dots,v_k) = \sum_{i=0}^{k}(-1)^i D_u\alpha(u;v_0,\dots,\widehat{v_i},\dots,v_k)[v_i],$$

so that $df(u;v) = Df(u;v)$ for functions; the symmetry of second derivatives gives $d^2 = 0$, and $d(\alpha\wedge\beta) = d\alpha\wedge\beta + (-1)^{\deg\alpha}\alpha\wedge d\beta$. Applying this with $U = G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $E = \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^{2n})$, we obtain the de Rham complex of $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$; for example the Maurer–Cartan form $g^{-1}dg$, $(g;v) \mapsto g^{-1}v$, is a $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$-valued 1-form. These forms are used in Chapter 7.

6.11 Symplectic invariance and the Weyl quantization

The composition law of the isotropic calculus has a more invariant formulation, which exhibits its invariance under linear symplectic maps.

6.11.1 Symplectic vector spaces

A *symplectic vector space* is a real vector space W with an antisymmetric non-degenerate bilinear form ω:

$$\omega(w_1,w_2) + \omega(w_2,w_1) = 0, \qquad \omega(v,w) = 0\ \forall\, w \in W \implies v = 0.$$

A subspace V is *isotropic* if ω vanishes on $V \times V$, and *Lagrangian* if moreover $2\dim V = \dim W$.

Lemma 6.11.1. *Every symplectic vector space W has a Lagrangian subspace, and for every Lagrangian subspace U_1 there is a Lagrangian subspace U_2 with $W = U_1 \oplus U_2$.*

Proof. Every one-dimensional subspace is isotropic, by antisymmetry. Let V be an isotropic subspace of maximal dimension and U a complement of V. The pairing $\omega : V \times U \to \mathbb{R}$ is non-degenerate: if $u \in U$ and $\omega(v,u) = 0$ for all $v \in V$ then $V + \mathbb{R}u$ is isotropic, so $u = 0$ by maximality; if $v \in V$ and $\omega(v,u) = 0$ for all $u \in U$ then, since ω vanishes on V, $\omega(v,\cdot) = 0$ and $v = 0$. Hence $\dim U = \dim V$ and V is Lagrangian.

Now let U_1 be Lagrangian, $\dim W = 2n$, and let V be an isotropic subspace with $V \cap U_1 = \{0\}$, of maximal dimension among such subspaces. Suppose $\dim V < n$. The map $V \to U_1'$, $v \mapsto \omega(v,\cdot)|_{U_1}$, is injective, for if $\omega(v, U_1) = 0$ then $U_1 + \mathbb{R}v$ is isotropic, so $v \in U_1$ and $v = 0$. Hence $U_1 \to V'$, $u \mapsto \omega(u,\cdot)|_V$, is surjective. Choose $w \notin V \oplus U_1$, possible since $\dim V + \dim U_1 < 2n$, and $u_1 \in U_1$ with $\omega(u_1, v) = \omega(w, v)$ for all $v \in V$. Then $V \oplus \mathbb{R}(w - u_1)$ is isotropic and meets U_1 only in 0 (as $w \notin V + U_1$), contradicting maximality. Thus $\dim V = n$ and $U_2 = V$ is a Lagrangian complement. □

Given a Lagrangian decomposition $W = U_1 \oplus U_2$, the pairing $\omega : U_2 \times U_1 \to \mathbb{R}$ is non-degenerate. Choosing a basis $e_1, \dots, e_n$ of U_1 and the basis $f_1, \dots, f_n$ of U_2 with $\omega(f_i, e_j) = \delta_{ij}$, and writing $w = \sum x_i e_i + \sum \xi_i f_i$,

$$\omega(w, w') = \sum_{i=1}^{n} (\xi_i x_i' - x_i \xi_i'), \qquad \text{i.e.} \quad \omega = \sum_{i=1}^{n} d\xi_i \wedge dx_i. \tag{6.11.1}$$

These are *Darboux coordinates*. The *Liouville measure* $\nu = |\omega^n / n!|$ is, in Darboux coordinates, Lebesgue measure $dx\, d\xi$, and is invariant under the symplectic group $\mathrm{Sp}(W)$ of linear maps preserving ω.

6.11.2 Weyl quantization

For $a \in \Gamma^m(\mathbb{R}^{2n})$ the *Weyl quantization* of a is

$$\mathrm{Op^W}(a)u(x) = \iint e^{i(x-y)\cdot\xi} a(\tfrac{x+y}{2}, \xi) u(y)\, dy đ\xi,$$

i.e. $\mathrm{Op^W}(a) = \mathrm{Op}(a(\frac{x+y}{2}, \xi))$. Its kernel K and a are related by

$$K(s + \tfrac{t}{2}, s - \tfrac{t}{2}) = \int e^{it\cdot\xi} a(s,\xi) đ\xi, \qquad a(s,\xi) = \int e^{-it\cdot\xi} K(s + \tfrac{t}{2}, s - \tfrac{t}{2})\, dt. \tag{6.11.2}$$

Proposition 6.11.2. *For $a \in \Gamma^m(\mathbb{R}^{2n})$, $\mathrm{Op^W}(a) \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ with*

$$\sigma_L(\mathrm{Op^W}(a)) \sim \sum_{\alpha} \frac{(i/2)^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha a,$$

and $\mathrm{Op^W}(a)^* = \mathrm{Op^W}(\bar a)$. Conversely every $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is $\mathrm{Op^W}(a)$ for a unique $a \in \Gamma^m(\mathbb{R}^{2n})$, the Weyl symbol *of A; $A \in \Psi^m_{\mathrm{iso,ph}}$ if and only if $a \in \Gamma^m_{\mathrm{ph}}$, and $A \in \Psi^{-\infty}_{\mathrm{iso}}$ if and only if $a \in \mathcal{S}(\mathbb{R}^{2n})$.*

Proof. The first statement is Proposition 6.2.6 applied to the amplitude $a(\frac{x+y}{2}, \xi) \in \mathcal{A}^m$ (Lemma 6.2.5(ii) with $\lambda = \frac{1}{2}$), since $D_y^\alpha [a(\frac{x+y}{2}, \xi)]|_{y=x} = 2^{-|\alpha|} D_x^\alpha a(x,\xi)$. The adjoint formula is immediate from the kernel. By (6.11.2), a is determined by K, and $a \in \mathcal{S}(\mathbb{R}^{2n})$ if and only if $K \in \mathcal{S}(\mathbb{R}^{2n})$. Given A with left symbol b, choose $a' \in \Gamma^m$ with $a' \sim \sum_\alpha \frac{(-i/2)^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha b$; by the formal identity $e^{\frac{i}{2} D_x \cdot D_\xi} e^{-\frac{i}{2} D_x \cdot D_\xi} = 1$, $A - \mathrm{Op^W}(a') \in \Psi^{-\infty}_{\mathrm{iso}} = \mathrm{Op^W}(\mathcal{S})$, so $A = \mathrm{Op^W}(a)$ with $a - a' \in \mathcal{S}$. The expansions preserve polyhomogeneity. □

Proposition 6.11.3. *If $a, b \in \mathcal{S}(\mathbb{R}^{2n})$ then $\operatorname{Op}^{\mathrm{W}}(a)\operatorname{Op}^{\mathrm{W}}(b) = \operatorname{Op}^{\mathrm{W}}(a\#_{\mathrm{W}}b)$ where*

$$a\#_{\mathrm{W}}b(w) = \pi^{-2n}\iint_{\mathbb{R}^{2n}\times\mathbb{R}^{2n}} e^{-2i\omega(w_1,w_2)}a(w+w_1)b(w+w_2)\,dw_1\,dw_2, \tag{6.11.3}$$

ω being the standard symplectic form (6.11.1) on $\mathbb{R}^{2n}$.

Proof. Let $A = \operatorname{Op}^{\mathrm{W}}(a)$, $B = \operatorname{Op}^{\mathrm{W}}(b)$ with kernels K_A, K_B, and let c be the Weyl symbol of AB. By (6.11.2),

$$\begin{aligned} c(s,\zeta) &= \int e^{-it\cdot\zeta}\int K_A(s+\tfrac{t}{2},z)K_B(z,s-\tfrac{t}{2})\,dz\,dt \\ &= (2\pi)^{-2n}\int e^{i\Phi}a(\tfrac{s}{2}+\tfrac{t}{4}+\tfrac{z}{2},\xi)b(\tfrac{z}{2}+\tfrac{s}{2}-\tfrac{t}{4},\eta)\,dt\,dz\,d\xi\,d\eta, \\ \Phi &= -t\cdot\zeta + (s+\tfrac{t}{2}-z)\cdot\xi + (z-s+\tfrac{t}{2})\cdot\eta, \end{aligned}$$

all integrals converging absolutely. Change variables to $X = \frac{z}{2}+\frac{t}{4}-\frac{s}{2}$, $Y = \frac{z}{2}-\frac{t}{4}-\frac{s}{2}$, $\Xi = \xi-\zeta$, $\mathrm{H} = \eta-\zeta$. Then $z-s = X+Y$, $t = 2(X-Y)$, $dt\,dz = 4^n\,dX\,dY$ and $\Phi = 2(X\cdot\mathrm{H} - Y\cdot\Xi)$, so

$$c(s,\zeta) = \pi^{-2n}\int e^{2i(X\cdot\mathrm{H}-Y\cdot\Xi)}a(s+X,\zeta+\Xi)b(s+Y,\zeta+\mathrm{H})\,dX\,d\Xi\,dY\,d\mathrm{H}.$$

With $w = (s,\zeta)$, $w_1 = (X,\Xi)$, $w_2 = (Y,\mathrm{H})$ we have $\omega(w_1,w_2) = \Xi\cdot Y - X\cdot\mathrm{H}$, which gives (6.11.3). □

Theorem 6.11.4. *Let (W,ω) be a symplectic vector space of dimension $2n$ with Liouville measure ν. Then*

$$a\#b(w) = \pi^{-2n}\int_{W\times W} e^{-2i\omega(w_1,w_2)}a(w+w_1)b(w+w_2)\,d\nu(w_1)\,d\nu(w_2) \tag{6.11.4}$$

defines an associative product on $\mathcal{S}(W)$. For any Lagrangian decomposition $W = U_1\oplus U_2$ and Darboux coordinates as above, $(\mathcal{S}(W),\#)$ is isomorphic, via the Weyl quantization, to $(\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n),\circ)$. The product extends uniquely to continuous bilinear maps $\Gamma^m(W)\times\Gamma^{m'}(W)\to\Gamma^{m+m'}(W)$, which in any Darboux coordinates correspond to composition in $\Psi^{\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, and for every $F\in\mathrm{Sp}(W)$

$$F^*(a\#b) = (F^*a)\#(F^*b). \tag{6.11.5}$$

Proof. In Darboux coordinates (6.11.4) becomes (6.11.3), so the isomorphism (and hence associativity) follows from Propositions 6.11.2 and 6.11.3. For $F\in\mathrm{Sp}(W)$, substituting $v_i = Fw_i$ in (6.11.4), which preserves ω and ν, gives (6.11.5) for $a,b\in\mathcal{S}(W)$. In Darboux coordinates the Weyl product of symbols in Γ^m, $\Gamma^{m'}$ is continuous into $\Gamma^{m+m'}$: it is the composite of the continuous maps $a\mapsto\sigma_L(\operatorname{Op}^{\mathrm{W}}(a))$, its inverse (both continuous by the closed graph theorem, as in Theorem 6.2.7(iv)) and the product of Theorem 6.2.7(iv). Since $\mathcal{S}$ is dense in Γ^m for the topology of $\Gamma^{m+\epsilon}$ (Lemma 6.1.2(iv)) and $\epsilon>0$ is arbitrary, the extension is unique, and (6.11.5) extends by continuity because F^* is continuous on each $\Gamma^m(W)$. In particular, since two Darboux coordinate systems differ by an element of $\mathrm{Sp}(W)$, the extended product does not depend on the choice. □

Remark 6.11.5. The Fourier transform is an instance of this invariance. A computation with the operators $\operatorname{Op}^{\mathrm{W}}(e^{i(p\cdot x+q\cdot\xi)}) = e^{i(p\cdot x+q\cdot D)}$ (or a comparison with Proposition 6.2.10) shows that the Weyl symbol of $\mathcal{F}A\mathcal{F}^{-1}$ is $a\circ J$, where $J(x,\xi) = (-\xi,x)$ is symplectic. More generally each $F\in\mathrm{Sp}(\mathbb{R}^{2n})$ is implemented, in the sense that $\operatorname{Op}^{\mathrm{W}}(F^*a) = U_F^{-1}\operatorname{Op}^{\mathrm{W}}(a)U_F$, by a unitary operator U_F on $L^2(\mathbb{R}^n)$, determined up to a constant factor of modulus one; these operators generate the metaplectic group, see [8, Section 18.5].

6.12 Complex order

The identification $\Gamma^0_{\mathrm{ph}}(\mathbb{R}^{2n}) = \mathrm{RC}^*\,\mathcal{C}^\infty(\mathbb{S}^{2n}_+)$ of Proposition 6.1.8 allows us to define isotropic operators of complex order. For $z \in \mathbb{C}$ set

$$\Gamma^z_{\mathrm{ph}}(\mathbb{R}^N) = \langle w\rangle^z\,\Gamma^0_{\mathrm{ph}}(\mathbb{R}^N) = \mathrm{RC}^*\big(\rho^{-z}\mathcal{C}^\infty(\mathbb{S}^N_+)\big), \qquad \Psi^z_{\mathrm{iso}}(\mathbb{R}^n) = \mathrm{Op}\big(\Gamma^z_{\mathrm{ph}}(\mathbb{R}^{2n})\big).$$

For real z this agrees with Γ^z_{ph} as defined before, since $\langle w\rangle^z \in \Gamma^z_{\mathrm{ph}}$. Since $|\partial^\alpha\langle w\rangle^z| \le C_\alpha\langle w\rangle^{\operatorname{Re} z-|\alpha|}$, $\Gamma^z_{\mathrm{ph}} \subset \Gamma^{\operatorname{Re} z}$ and $\Psi^z_{\mathrm{iso}} \subset \Psi^{\operatorname{Re} z}_{\mathrm{iso}}$. The definition does not depend on the choice of boundary defining function: if $\rho' = f\rho$ with $0 < f \in \mathcal{C}^\infty(\mathbb{S}^N_+)$ then $f^z = e^{z\log f}$ is smooth, so $\rho'^{-z}\mathcal{C}^\infty = \rho^{-z}\mathcal{C}^\infty$.

Proposition 6.12.1. *For $z, z' \in \mathbb{C}$, $\Psi^z_{\mathrm{iso}}(\mathbb{R}^n) \circ \Psi^{z'}_{\mathrm{iso}}(\mathbb{R}^n) \subset \Psi^{z+z'}_{\mathrm{iso}}(\mathbb{R}^n)$ and $\Psi^z_{\mathrm{iso}}(\mathbb{R}^n)^* = \Psi^{\bar z}_{\mathrm{iso}}(\mathbb{R}^n)$.*

Proof. Differentiation maps $\langle w\rangle^z\Gamma^0_{\mathrm{ph}}$ into $\langle w\rangle^z\Gamma^{-1}_{\mathrm{ph}}$, since $\partial_{w_j}(\langle w\rangle^z f) = \langle w\rangle^z(z w_j\langle w\rangle^{-2} f + \partial_{w_j} f)$. Hence the terms of (6.2.4) satisfy $\frac{1}{\alpha!}\partial_\xi^\alpha a\, D_x^\alpha b \in \langle w\rangle^{z+z'}\Gamma^{-2|\alpha|}_{\mathrm{ph}}$. If $c = a\#b$ then $c \in \Gamma^\mu$, $\mu = \operatorname{Re}(z+z')$, and the remainders of the expansion lie in $\Gamma^{\mu-2N}$; multiplying by $\langle w\rangle^{-(z+z')} \in \Gamma^{-\mu}$, $\langle w\rangle^{-(z+z')}c$ has an asymptotic expansion with terms in $\Gamma^{-2j}_{\mathrm{ph}}$ and remainders in Γ^{-2N}, so it lies in Γ^0_{ph}. Adjoints are treated in the same way. □

Even more useful are holomorphic families. A map $\Omega \to \mathcal{C}^\infty(\mathbb{S}^N_+)$, with $\Omega \subset \mathbb{C}$ open, is *holomorphic* if it is given by an element of $\mathcal{C}^\infty(\Omega \times \mathbb{S}^N_+)$ satisfying the Cauchy–Riemann equation in the first variable. If $h : \Omega \to \mathbb{C}$ is holomorphic, a family $A(z) \in \Psi^{h(z)}_{\mathrm{iso}}(\mathbb{R}^n)$, $z \in \Omega$, is a *holomorphic family of order h* if $\sigma_L(A(z)) = \langle w\rangle^{h(z)}\,\mathrm{RC}^*\,F(z)$ with $F : \Omega \to \mathcal{C}^\infty(\mathbb{S}^{2n}_+)$ holomorphic.

Proposition 6.12.2. *If h, g are holomorphic on an open set $\Omega \subset \mathbb{C}$ and $A(z)$, $B(z)$ are holomorphic families of isotropic operators of orders $h(z)$ and $g(z)$, then $A(z) \circ B(z)$ is a holomorphic family of order $h(z) + g(z)$.*

Sketch of proof. It suffices to work on an open Ω' with compact closure in Ω, on which $\operatorname{Re} h, \operatorname{Re} g < M$ and $\operatorname{Re}(h+g) > -M$. The map $z \mapsto \langle w\rangle^{h(z)}$ is holomorphic into Γ^M (its derivative $h'(z)\log\langle w\rangle\,\langle w\rangle^{h(z)}$ lies in Γ^M, as $\operatorname{Re} h < M$ strictly), and $\mathrm{RC}^* : \mathcal{C}^\infty(\mathbb{S}^{2n}_+) \to \Gamma^0$ is continuous linear, so $a(z) = \sigma_L(A(z))$ and $b(z)$ are holomorphic Γ^M-valued. By Theorem 6.2.7(iv), $c(z) = a(z)\#b(z)$ is holomorphic into Γ^{2M} and

$$c(z) = \sum_{|\alpha|<N} \frac{1}{\alpha!}\partial_\xi^\alpha a(z)\, D_x^\alpha b(z) + R_N(z),$$

with R_N holomorphic into Γ^{2M-2N}. Using (6.1.1), $\partial_{w_j}(\langle w\rangle^h\,\mathrm{RC}^*\,F) = \langle w\rangle^{h-1}\,\mathrm{RC}^*\big(hX_jF + V_jF\big)$ with $X_j = w_j\langle w\rangle^{-1}$ and V_j smooth on $\mathbb{S}^{2n}_+$, so $\langle w\rangle^{-h-g}$ times each term of the sum is RC^* of a holomorphic family in $\mathcal{C}^\infty(\mathbb{S}^{2n}_+)$, while $\langle w\rangle^{-h-g}R_N(z)$ is holomorphic into Γ^{3M-2N}, hence, by Proposition 6.1.8(iii), into $\mathcal{C}^k(\mathbb{S}^{2n}_+)$ for $2N > 3M + k + 1$. Thus $\rho^{h+g}c(z)$ (pushed forward by RC) is holomorphic into $\mathcal{C}^k(\mathbb{S}^{2n}_+)$ for every k, which means that it is given by an element of $\mathcal{C}^\infty(\Omega' \times \mathbb{S}^{2n}_+)$ holomorphic in z. □

6.13 Exercises

Exercise 6.1. For $k \in \mathbb{N}$ define $H^{-k}_{\mathrm{iso}}(\mathbb{R}^n)$ as the space of $u \in \mathcal{S}'(\mathbb{R}^n)$ which can be written $u = \sum_{|\alpha|+|\beta|\le k} x^\alpha D^\beta u_{\alpha\beta}$ with $u_{\alpha\beta} \in L^2(\mathbb{R}^n)$. Show directly that each $A \in \Psi^p_{\mathrm{iso}}(\mathbb{R}^n)$, $p \in \mathbb{Z}$,

maps $H^k_{\rm iso}$ to $H^{k-p}_{\rm iso}$ for all $k \in \mathbb{Z}$ (with $H^k_{\rm iso}$, $k \geq 0$, as in Proposition 6.9.2(iii)), and that these definitions agree with Definition 6.9.1. Deduce that if $A \in \Psi^m_{\rm iso}(\mathbb{R}^n)$ is elliptic, $u \in \mathcal{S}'(\mathbb{R}^n)$ and $Au \in H^{m'}_{\rm iso}(\mathbb{R}^n)$, then $u \in H^{m'+m}_{\rm iso}(\mathbb{R}^n)$.

Exercise 6.2. Show that for $\epsilon > 0$, $H^\epsilon_{\rm iso}(\mathbb{R}^n) \subset \langle x\rangle^{-\epsilon} L^2(\mathbb{R}^n) \cap H^\epsilon(\mathbb{R}^n)$, and that the inclusion is strict. [Hint: compose with $\langle D\rangle^\epsilon$ and $\langle x\rangle^\epsilon$, using the scattering calculus.] Deduce, without using Proposition 6.7.1, that $H^\epsilon_{\rm iso}(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n)$ is compact, and hence that each element of $\Psi^{-\epsilon}_{\rm iso}(\mathbb{R}^n)$ is compact on $L^2(\mathbb{R}^n)$.

Exercise 6.3. Let $E \in \Psi^{-\infty}_{\rm iso}(\mathbb{R}^n)$. Show that there is an operator $F \in \Psi^{-\infty}_{\rm iso}(\mathbb{R}^n)$ such that $(\mathrm{Id}+E)(\mathrm{Id}+F) = \mathrm{Id}+G$ with $G \in \Psi^{-\infty}_{\rm iso}(\mathbb{R}^n)$ of finite rank. Deduce that an elliptic $A \in \Psi^m_{\rm iso}(\mathbb{R}^n)$ has a parametrix $B \in \Psi^{-m}_{\rm iso}(\mathbb{R}^n)$ up to finite rank errors, i.e. $AB - \mathrm{Id}$ and $BA - \mathrm{Id}$ are finite rank elements of $\Psi^{-\infty}_{\rm iso}(\mathbb{R}^n)$, and hence that $A : H^M_{\rm iso}(\mathbb{R}^n) \to H^{M-m}_{\rm iso}(\mathbb{R}^n)$ is Fredholm for every M. (Recall that a bounded operator between Hilbert spaces is Fredholm if it has finite-dimensional null space and closed range of finite codimension.)

Exercise 6.4. Suppose $A \in \Psi^m_{\rm iso}(\mathbb{R}^n)$ is elliptic and has index zero as an operator on $\mathcal{S}'(\mathbb{R}^n)$. Show that there is a finite rank $E \in \Psi^{-\infty}_{\rm iso}(\mathbb{R}^n)$ such that $A+E$ is an isomorphism of $\mathcal{S}'(\mathbb{R}^n)$, of $\mathcal{S}(\mathbb{R}^n)$ and of $H^s_{\rm iso}(\mathbb{R}^n) \to H^{s-m}_{\rm iso}(\mathbb{R}^n)$ for every s, with inverse in $\Psi^{-m}_{\rm iso}(\mathbb{R}^n)$.

Exercise 6.5. [Koszul complex] Let $\Lambda^k = \Lambda^k \mathbb{R}^n$ with basis dx^I, I strictly increasing k-tuples, declared orthonormal, and let $\mathcal{S}'(\mathbb{R}^n;\Lambda^k)$ be the space of $u = \sum_I u_I dx^I$, $u_I \in \mathcal{S}'(\mathbb{R}^n)$. Using the annihilation operators define

$$D : \mathcal{S}'(\mathbb{R}^n;\Lambda^k) \to \mathcal{S}'(\mathbb{R}^n;\Lambda^{k+1}), \qquad Du = \sum_{j=1}^n \sum_I A_j u_I \, dx_j \wedge dx^I.$$

(a) Show that $D^2 = 0$ and that the formal adjoint is $D^* u = \sum_j \sum_I C_j u_I \, \iota_j dx^I$, where ι_j, contraction with e_j, is the adjoint of $dx_j\wedge$.

(b) Show that $D + D^* \in \Psi^1_{\rm iso}(\mathbb{R}^n;\Lambda^*)$ is elliptic, and that $(D+D^*)^2 = DD^* + D^*D$ acts on $\mathcal{S}'(\mathbb{R}^n;\Lambda^k)$ as $H - n + 2k$.

(c) Deduce that the null space of $D + D^*$ on $\mathcal{S}'(\mathbb{R}^n;\Lambda^*)$ is $\mathbb{C}\exp(-|x|^2/2) \subset \mathcal{S}'(\mathbb{R}^n;\Lambda^0)$, that $\{u \in \mathcal{S}'(\mathbb{R}^n;\Lambda^0);\ Du = 0\} = \mathbb{C}\exp(-|x|^2/2)$, and that for $k \geq 1$ $\{u \in \mathcal{S}'(\mathbb{R}^n;\Lambda^k);\ Du = 0\} = D\,\mathcal{S}'(\mathbb{R}^n;\Lambda^{k-1})$.

(d) Conclude that $D + D^*$, as an element of $\Psi^1_{\rm iso}(\mathbb{R}^n;\Lambda^{\rm even},\Lambda^{\rm odd})$, is elliptic of index 1.

Exercise 6.6. [Isotropic operator wavefront set] For $a \in \Gamma^m(\mathbb{R}^{2n})$ define the *cone support* $\operatorname{cone\,supp}(a) \subset \mathbb{R}^{2n}\setminus\{0\}$ as the complement of the set of $\bar w \neq 0$ having a conic neighbourhood Γ such that $\sup_{w\in\Gamma,|w|\geq 1}\langle w\rangle^N |\partial^\alpha a(w)| < \infty$ for all N and α. For $A \in \Psi^m_{\rm iso}(\mathbb{R}^n)$ set $\mathrm{WF}'_{\rm iso}(A) = \operatorname{cone\,supp}(\sigma_L(A))$. Show that

$$\mathrm{WF}'_{\rm iso}(A+B) \subset \mathrm{WF}'_{\rm iso}(A) \cup \mathrm{WF}'_{\rm iso}(B), \quad \mathrm{WF}'_{\rm iso}(AB) \subset \mathrm{WF}'_{\rm iso}(A) \cap \mathrm{WF}'_{\rm iso}(B),$$

$\mathrm{WF}'_{\rm iso}(A^*) = \mathrm{WF}'_{\rm iso}(A)$, that $\mathrm{WF}'_{\rm iso}(A)$ is the same when defined using the Weyl symbol, and that $\mathrm{WF}'_{\rm iso}(A) = \emptyset$ if and only if $A \in \Psi^{-\infty}_{\rm iso}(\mathbb{R}^n)$.

Exercise 6.7. [Isotropic partition of unity] Let U_i, $i = 1,\dots,N$, be an open cover of $\mathbb{S}^{2n-1}$ and $\tilde U_i = \{w \neq 0;\ w/|w| \in U_i\}$. Show that there are $A_i \in \Psi^0_{\rm iso,ph}(\mathbb{R}^n)$ with $\mathrm{WF}'_{\rm iso}(A_i) \subset \tilde U_i$ and $\mathrm{Id} - \sum_i A_i \in \Psi^{-\infty}_{\rm iso}(\mathbb{R}^n)$.

Exercise 6.8. [Isotropic wavefront set] For $u \in \mathcal{S}'(\mathbb{R}^n)$ let

$$\mathrm{WF}_{\mathrm{iso}}(u) = \bigcap\{\mathrm{Char}_{\mathrm{iso}}(A);\ A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n),\ Au \in \mathcal{S}(\mathbb{R}^n)\},$$

where $\mathrm{Char}_{\mathrm{iso}}(A)$ is the cone over the zero set of $\sigma_0(A)$ in $\mathbb{S}^{2n-1}$. Show that $\mathrm{WF}_{\mathrm{iso}}(u)$ is a closed cone, that $\mathrm{WF}_{\mathrm{iso}}(u) = \emptyset$ if and only if $u \in \mathcal{S}(\mathbb{R}^n)$, that $\mathrm{WF}_{\mathrm{iso}}(Au) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cap \mathrm{WF}_{\mathrm{iso}}(u)$ for $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, and that $\mathrm{WF}_{\mathrm{iso}}(\hat{u}) = \{(\xi, -x);\ (x,\xi) \in \mathrm{WF}_{\mathrm{iso}}(u)\}$.

Exercise 6.9. Verify in detail that $\mathcal{F}\,\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)\,\mathcal{F}^{-1} = \Psi^{l,m}_{\mathrm{sc}}(\mathbb{R}^n)$, and compute the left symbol of $\mathcal{F}A\mathcal{F}^{-1}$ in terms of that of $A \in \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$ modulo $S^{l-1,m-1}$.

Notes

This chapter follows Sections 3.1–3.13 of the original notes, reorganised. The general (non-classical) isotropic class Γ^m is the symbol class introduced by Shubin [15], where much of the material here (in particular the harmonic oscillator and the isotropic Sobolev spaces) can be found; the Weyl calculus and its symplectic invariance are treated in [8, Section 18.5]. The scattering calculus on $\mathbb{R}^n$ is the model for Melrose's scattering calculus on asymptotically Euclidean manifolds (R. B. Melrose, *Spectral and scattering theory for the Laplacian on asymptotically Euclidean spaces*, 1994); see also the work of H. O. Cordes and C. Parenti on SG-calculi.

The main departures from the original are as follows. The reduction theorem is proved with a new class of amplitudes (Definition 6.2.4) which contains the products $a(x,\xi)b(y,\xi)$ and the Weyl amplitudes, and the isotropic decay of the remainder is obtained by intersecting weighted traditional symbol spaces; this replaces the original argument with a two-region decomposition of $\mathbb{R}^{3n}$. The generalized inverse of an elliptic operator is constructed on $\mathcal{S}'$ by a finite-dimensional argument, the essential norm estimate is proved with coherent states (which also give the necessity of ellipticity for the Fredholm property), the harmonic oscillator section uses the generalized inverse H^{-1} to prove completeness of the Hermite functions, and the Sobolev spaces of all real orders are treated uniformly. Problems on traces and determinants from the original list are in Chapter 7.

Chapter 7

Traces, determinants and the index

In this chapter we use the isotropic calculus of Chapter 6 as a laboratory for three closely related constructions: traces, determinants and the index. On the residual algebra $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, whose elements have Schwartz kernels, there is a unique trace, and it gives rise to the Fredholm determinant on $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$; the determinant leads directly to analytic Fredholm theory and so to the meromorphy of the resolvent of an elliptic operator. On the full algebra of polyhomogeneous operators of integral order the trace no longer extends, but complex powers allow us to define the *residue trace*, a genuine trace which only sees one term of the symbol expansion, and the *regularized trace*, which extends the trace but fails to be a trace; the failure is measured by the residue trace and an exterior derivation. These ingredients give a formula for the index of an elliptic operator, an obstruction to inverting families of elliptic operators by smoothing perturbations, and finally the determinant line bundle of such a family, together with a natural connection whose curvature is that obstruction.

7.1 Setup

We recall from Chapter 6 what we need, and fix notation. Write $Z=(x,\xi)\in\mathbb{R}^{2n}$. For $m\in\mathbb{R}$ the space of *isotropic symbols* of order m is

$$\Gamma^m(\mathbb{R}^{2n})=\big\{a\in\mathcal{C}^\infty(\mathbb{R}^{2n});\ |D_Z^\alpha a(Z)|\le C_\alpha\langle Z\rangle^{m-|\alpha|}\ \forall\,\alpha\big\},$$

a Fréchet space with the best constants C_α as seminorms, and $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)=\mathrm{Op}(\Gamma^m(\mathbb{R}^{2n}))$, with the left quantization of the Introduction. The polyhomogeneous subspace $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ consists of the operators whose left symbol has an expansion $a\sim\sum_{j\ge0}a_{m-j}$ with a_{m-j} homogeneous of degree $m-j$ in $|Z|\ge1$; equivalently, $\langle Z\rangle^{-m}a$ is smooth on the radial compactification $\overline{\mathbb{R}^{2n}}$, i.e. in polar coordinates $Z=\theta/t$, $t=|Z|^{-1}$, it is smooth up to $t=0$. The same definition with m replaced by $z\in\mathbb{C}$ gives the operators $\Psi^z_{\mathrm{iso,ph}}(\mathbb{R}^n)\subset\Psi^{\Re z}_{\mathrm{iso}}(\mathbb{R}^n)$ of complex order. We write $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n)=\bigcup_{m\in\mathbb{Z}}\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$. The *residual* operators are

$$\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)=\bigcap_m\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)=\bigcap_m\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n),$$

which are exactly the operators with Schwartz kernels in $\mathcal{S}(\mathbb{R}^{2n})$; we identify $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ with $\mathcal{S}(\mathbb{R}^{2n})$, including its topology.

All of this applies to operators on $\mathbb{C}^N$-valued functions: $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ consists of $N\times N$ matrices of elements of $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ (see Exercise 7.9). Everything below holds in this generality, with tr denoting the trace of matrices; to keep the notation light we often suppress $\mathbb{C}^N$.

We use the following results from Chapter 6.

(I1) *Algebra.* Composition maps $\Psi^m_{\mathrm{iso}} \times \Psi^{m'}_{\mathrm{iso}}$ to $\Psi^{m+m'}_{\mathrm{iso}}$ continuously for the symbol topologies, and $\Psi^z_{\mathrm{iso,ph}} \times \Psi^{z'}_{\mathrm{iso,ph}}$ to $\Psi^{z+z'}_{\mathrm{iso,ph}}$; the left symbol of a composite is given by $\sigma_L(AB) \sim \sum_\alpha \frac{1}{\alpha!}\partial_\xi^\alpha a\, D_x^\alpha b$. Adjoints and transposes of elements of Ψ^m_{iso} (resp. $\Psi^z_{\mathrm{iso,ph}}$) lie in the same space. In particular $\Psi^{-\infty}_{\mathrm{iso}}$ is a two-sided ideal. The principal symbol is multiplicative.

(I2) *Boundedness.* Every element of Ψ^m_{iso} is continuous on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$; elements of Ψ^0_{iso} are bounded on $L^2(\mathbb{R}^n)$, with norm bounded by a symbol seminorm; elements of negative order are compact on L^2, and residual operators map $\mathcal{S}'(\mathbb{R}^n)$ continuously into $\mathcal{S}(\mathbb{R}^n)$.

(I3) *Ellipticity.* An element $A \in \Psi^z_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$ is elliptic if its principal symbol is invertible on the sphere. It then has a parametrix $B \in \Psi^{-z}_{\mathrm{iso,ph}}$, with $AB - \mathrm{Id}$ and $BA - \mathrm{Id}$ residual; any two parametrices differ by a residual operator. If $Au \in \mathcal{S}$ for some $u \in \mathcal{S}'$, then $u \in \mathcal{S}$. There is a *generalized inverse* $B \in \Psi^{-z}_{\mathrm{iso,ph}}$ with

$$AB = \mathrm{Id} - \Pi_1, \quad BA = \mathrm{Id} - \Pi_0, \tag{7.1.1}$$

where Π_0, Π_1 are the L^2-orthogonal projections onto the finite dimensional spaces $\operatorname{Nul} A \subset \mathcal{S}$ and $\operatorname{Nul} A^* \subset \mathcal{S}$ (null spaces on $\mathcal{S}'$). Replacing B by $(\mathrm{Id} - \Pi_0)B(\mathrm{Id} - \Pi_1)$, which changes neither identity since $A\Pi_0 = 0$ and $\Pi_1 A = 0$, we may and do assume $\Pi_0 B = 0 = B\Pi_1$. In particular an elliptic A which is a bijection on $\mathcal{S}'$ (we then say A is *invertible*) has inverse in $\Psi^{-z}_{\mathrm{iso,ph}}$. For order 0, A is invertible in this sense if and only if it is invertible on L^2.

(I4) *The residual algebra.* If $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ then $\mathrm{Id} + E$ is Fredholm on L^2 of index zero; if it is invertible on L^2 its inverse is $\mathrm{Id} + F$ with $F \in \Psi^{-\infty}_{\mathrm{iso}}$. If $\operatorname{Nul}(\mathrm{Id} + E) \neq \{0\}$ there is a finite rank $F' \in \Psi^{-\infty}_{\mathrm{iso}}$, mapping $\operatorname{Nul}(\mathrm{Id} + E)$ isomorphically onto $\operatorname{Nul}(\mathrm{Id} + E^*)$ and vanishing on its orthocomplement, such that $\mathrm{Id} + E + sF'$ is invertible for all $s \neq 0$. (This is also the special case $A = \mathrm{Id} + E$ of Proposition 7.4.2 below.) The series $\exp E = \sum_k E^k/k!$ converges in $\mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}$.

(I5) *The harmonic oscillator.* The eigenfunctions $u_\alpha = c_\alpha C^\alpha e^{-|x|^2/2}$, $\alpha \in \mathbb{N}_0^n$, of $H = |D|^2 + |x|^2$ (with $C_j = D_j + ix_j$ and normalizing constants $c_\alpha > 0$) are real-valued and form an orthonormal basis of $L^2(\mathbb{R}^n)$; $Hu_\alpha = (n + 2|\alpha|)u_\alpha$; and each Schwartz seminorm of u_α is bounded by a polynomial in $|\alpha|$. See also Exercise 7.6.

Definition 7.1.1. Let Y be a manifold and $\Omega \subset \mathbb{C}$ open.

(a) A family $E_y \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $y \in Y$, is *smooth* if its kernel $E(y, x, x')$ is in $\mathcal{C}^\infty(Y \times \mathbb{R}^{2n})$ and, in local coordinates, $\langle (x, x')\rangle^k \partial_y^\gamma \partial^\alpha_{x,x'} E$ is bounded on $K \times \mathbb{R}^{2n}$ for every compact K and all k, γ, α. A family $E(z)$, $z \in \Omega$, is *holomorphic* if it is smooth and E is holomorphic in z.

(b) Let h be holomorphic on Ω. A family $A(z) = \mathrm{Op}(a(z))$, $z \in \Omega$, is a *holomorphic family of order* $h(z)$ if $a \in \mathcal{C}^\infty(\Omega \times \mathbb{R}^{2n})$ is holomorphic in z and

$$a(z; r\theta) = r^{h(z)}\, b(z; 1/r, \theta), \quad r \geq 1,\ \theta \in \mathbb{S}^{2n-1}, \tag{7.1.2}$$

with $b \in \mathcal{C}^\infty(\Omega \times [0, 1] \times \mathbb{S}^{2n-1})$ holomorphic in z. Then $A(z) \in \Psi^{h(z)}_{\mathrm{iso,ph}}$, and the term of homogeneity $h(z) - j$ in the expansion of $a(z)$ is $|Z|^{h(z)-j} b_j(z; Z/|Z|)$, with $b_j(z; \theta) = \frac{1}{j!}\partial_t^j b(z; 0, \theta)$. Families of fixed order m depending smoothly on $y \in Y$ are defined in the same way, with b smooth in (y, t, θ).

(I6) *Families.* Composites of holomorphic (or smooth) families are holomorphic (or smooth), with orders adding; this includes composites with residual families. If $A(z)$ is a holomorphic family of elliptic operators of order $h(z)$, then near each point of Ω there is a holomorphic family $B(z)$ of order $-h(z)$ such that $A(z)B(z) - \mathrm{Id}$ and $B(z)A(z) - \mathrm{Id}$ are holomorphic families in $\Psi^{-\infty}_{\mathrm{iso}}$ (the parametrix construction of Chapter 6, with Borel's lemma applied in the variable t uniformly in z); the same holds for smooth families. Finally

$$H(z) = \mathrm{Op}(\langle Z\rangle^z)\ \mathrm{Id}_N \tag{7.1.3}$$

is an entire holomorphic family of order z with $H(0) = \mathrm{Id}$.

We complete these facts with three elementary lemmas about residual operators. Fix seminorms on $\mathcal{S}(\mathbb{R}^{2n})$ by

$$\|K\|_{(k)} = \sum_{|\alpha|+|\beta|+|\gamma|+|\delta|\le k} \sup_{x,x'} |x^\alpha D_x^\beta x'^\gamma D_{x'}^\delta K(x,x')|.$$

For $E \in \Psi^{-\infty}_{\mathrm{iso}}$ we have $\|E\|_{\mathcal{B}(L^2)} \le \|E\|_{L^2(\mathbb{R}^{2n})} \le C\|E\|_{(2n+1)}$.

Lemma 7.1.2. *If $E_1, E_2 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and $T \in \mathcal{B}(L^2(\mathbb{R}^n))$, then $E_1 T E_2 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and for each k there are C and k' such that $\|E_1 T E_2\|_{(k)} \le C\|T\|_{\mathcal{B}(L^2)}\|E_1\|_{(k')}\|E_2\|_{(k')}$.*

Proof. The map $x' \mapsto E_2(\cdot, x')$ is smooth into $L^2(\mathbb{R}^n)$, so the kernel of E_1TE_2 is $K(x,x') = \int E_1(x,w)\,(TE_2(\cdot,x'))(w)\,dw$, and we may differentiate under T: $x^\alpha D_x^\beta x'^\gamma D_{x'}^\delta K(x,x') = \int x^\alpha D_x^\beta E_1(x,w)\,(T[x'^\gamma D_{x'}^\delta E_2(\cdot,x')])(w)\,dw$. By Cauchy–Schwarz this is bounded by $\|T\|$ times $\sup_x \|x^\alpha D_x^\beta E_1(x,\cdot)\|_{L^2} \sup_{x'} \|x'^\gamma D_{x'}^\delta E_2(\cdot,x')\|_{L^2}$, and each factor is bounded by a seminorm $\|\cdot\|_{(k')}$. □

Lemma 7.1.3 (Finite rank approximation)**.** *Let π_N be the orthogonal projection onto $L_N = \mathrm{span}\{u_\alpha; |\alpha| \le N\}$. For $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $\pi_N E \pi_N \to E$ in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ as $N \to \infty$. If E_y is a smooth family, the convergence holds locally uniformly in y together with all y-derivatives.*

Proof. The functions $u_\alpha \otimes u_\beta$ form an orthonormal basis of $L^2(\mathbb{R}^{2n})$ of eigenfunctions of the harmonic oscillator $H_{2n} = H_x + H_{x'}$ with eigenvalues $2n + 2|\alpha| + 2|\beta|$. Thus the coefficients $e_{\alpha\beta} = (E, u_\alpha \otimes u_\beta)$ satisfy $(2n + 2|\alpha| + 2|\beta|)^k |e_{\alpha\beta}| = |(H_{2n}^k E, u_\alpha \otimes u_\beta)| \le \|H_{2n}^k E\|_{L^2}$, a continuous seminorm of E. Since the seminorms of $u_\alpha \otimes u_\beta$ grow only polynomially, $\sum e_{\alpha\beta} u_\alpha \otimes u_\beta$ converges absolutely to E in $\mathcal{S}(\mathbb{R}^{2n})$, and its partial sum over $|\alpha|, |\beta| \le N$ is the kernel of $\pi_N E \pi_N$. For a smooth family the same estimates hold for $\partial_y^\gamma E_y$, locally uniformly. □

Lemma 7.1.4. *The set $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ of $\mathrm{Id} + E$, $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, which are invertible on L^2 is a group, open in $\mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. The inverse is $\mathrm{Id} + F$ with*

$$F = -E + E(\mathrm{Id}+E)^{-1}E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n), \tag{7.1.4}$$

and F depends continuously on E. If E_y is a smooth (holomorphic) family with $\mathrm{Id} + E_y$ invertible for all y, then F_y is a smooth (holomorphic) family.

Proof. From $(\mathrm{Id}+E)(\mathrm{Id}+F) = \mathrm{Id} = (\mathrm{Id}+F)(\mathrm{Id}+E)$ we get $F = -E - EF$ and $F = -E - FE$, so $F = -E + E^2 + EFE = -E + E(\mathrm{Id}+E)^{-1}E$, which is residual by Lemma 7.1.2. Openness holds because invertibility in $\mathcal{B}(L^2)$ is open and $\|E\|_{\mathcal{B}(L^2)} \le C\|E\|_{(2n+1)}$. Inversion is smooth (indeed analytic) on the invertible elements of the Banach algebra $\mathcal{B}(L^2)$, so $y \mapsto (\mathrm{Id} + E_y)^{-1}$ is smooth (holomorphic) in norm; the trilinear estimate of Lemma 7.1.2 and the product rule then show that F_y is smooth (holomorphic), and continuity follows in the same way. □

Lemma 7.1.5 (Local inversion)**.** *Let $A(z)$ be a holomorphic family of elliptic operators of order $h(z)$ on Ω, and suppose $A(p)$ is invertible. Then $A(z)$ is invertible for z near p and $A(z)^{-1}$ is a holomorphic family of order $-h(z)$ near p. The same holds for smooth families of fixed order.*

Proof. Let $B(z)$ be a holomorphic parametrix near p, $B(z)A(z) = \mathrm{Id} + E(z)$, $A(z)B(z) = \mathrm{Id} + E'(z)$. Set $B'(z) = B(z) - E(p)A(p)^{-1}$; then $B'(z)A(z) = \mathrm{Id} + E_1(z)$ with $E_1(z) = E(z) - E(p)A(p)^{-1}A(z)$ a holomorphic residual family vanishing at p. By Lemma 7.1.4, $\mathrm{Id} + E_1(z)$ is invertible near p with holomorphic inverse $\mathrm{Id} + S(z)$, so $L(z) = (\mathrm{Id} + S(z))B'(z)$ is a holomorphic family of order $-h(z)$ with $L(z)A(z) = \mathrm{Id}$. In the same way, with $B''(z) = B(z) - A(p)^{-1}E'(p)$, we find a right inverse $R(z)$. Then $L = LAR = R$ is a two-sided inverse. □

7.2 Traces on the residual algebra

Matrices

On the algebra $M(N;\mathbb{C})$ of $N \times N$ matrices, $\operatorname{tr} A = \sum_i A_{ii}$ satisfies $\operatorname{tr}(AB) = \sum_{i,k} A_{ik}B_{ki} = \operatorname{tr}(BA)$, that is,

$$\operatorname{tr}([A, B]) = 0 \quad \forall\, A, B \in M(N;\mathbb{C}),$$

and the normalization $\operatorname{tr} \mathrm{Id} = N$. These properties characterize it.

Lemma 7.2.1. *If $F : M(N;\mathbb{C}) \to \mathbb{C}$ is linear and $F([A, B]) = 0$ for all A, B, then $F = F(E_{11}) \operatorname{tr}$.*

Proof. Let E_{jk} be the matrix units, $E_{jk}E_{pq} = \delta_{kp}E_{jq}$. For $j \neq k$, $E_{jk} = [E_{jj}, E_{jk}]$, so $F(E_{jk}) = 0$; and $E_{jj} - E_{11} = [E_{j1}, E_{1j}]$, so $F(E_{jj}) = F(E_{11})$. Hence $F(A) = \sum_{j,k} A_{jk}F(E_{jk}) = F(E_{11}) \operatorname{tr} A$. □

The trace on residual operators

For $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ with kernel $A(x, x')$ we set

$$\operatorname{Tr}(A) = \int_{\mathbb{R}^n} \operatorname{tr} A(x, x)\, dx. \tag{7.2.1}$$

This is a continuous linear functional on $\Psi^{-\infty}_{\mathrm{iso}}$. It extends to operators of sufficiently negative order. If $a \in \Gamma^m(\mathbb{R}^{2n})$ with $m < -2n$, then $\int |a(x, \xi)|\, d\xi \leq C\langle x\rangle^{m+n}$, so the kernel $K(x, x') = \int e^{i(x-x')\cdot\xi} a(x, \xi) d\!\!{}^{-}\xi$ of $\mathrm{Op}(a)$ is continuous, $|K(x, x)| \leq C\langle x\rangle^{m+n}$ is integrable, and we define

$$\operatorname{Tr}(\mathrm{Op}(a)) = \int K(x, x)\, dx = (2\pi)^{-n} \int_{\mathbb{R}^{2n}} \operatorname{tr} a(x, \xi)\, dx\, d\xi, \qquad a \in \Gamma^m(\mathbb{R}^{2n}),\ m < -2n. \tag{7.2.2}$$

This is continuous on Ψ^m_{iso} for the symbol topology and agrees with (7.2.1) on residual operators. In particular Tr is defined on $\Psi^z_{\mathrm{iso,ph}}$ for $\Re z < -2n$.

Lemma 7.2.2. *If $A \in \Psi^p_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$, $B \in \Psi^q_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ and $p + q < -2n$, then $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$. In particular this holds if one of A, B is residual and the other is arbitrary.*

Proof. For residual A and B this is Fubini's theorem,

$$\operatorname{Tr}(AB) = \iint \operatorname{tr}(A(x, w)B(w, x))\, dw\, dx = \operatorname{Tr}(BA).$$

In general, let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^{2n})$ equal 1 near 0, and let $a_k(Z) = \chi(Z/k)a(Z)$, $b_k(Z) = \chi(Z/k)b(Z)$. Then $a - a_k$ is supported in $|Z| \geq k$, where $\langle Z\rangle^{-\epsilon} \leq k^{-\epsilon}$, and the derivatives of $\chi(Z/k)$ are bounded by $C\langle Z\rangle^{-|\beta|}$ uniformly in k; hence $a_k \to a$ in $\Gamma^{p+\epsilon}$ for every $\epsilon > 0$, and similarly $b_k \to b$ in $\Gamma^{q+\epsilon}$. Choose ϵ with $p + q + 2\epsilon < -2n$. The operators $A_k = \mathrm{Op}(a_k)$, $B_k = \mathrm{Op}(b_k)$ are residual, so $\mathrm{Tr}(A_kB_k) = \mathrm{Tr}(B_kA_k)$, and by the continuity of composition (see (I1)) and of the trace on $\Psi_{\mathrm{iso}}^{p+q+2\epsilon}$ we may let $k \to \infty$. □

Lemma 7.2.3. *Let $L \subset \mathcal{S}(\mathbb{R}^n;\mathbb{C}^N)$ be a finite dimensional subspace and Π the L^2-orthogonal projection onto L. Then $\Pi \in \Psi_{\mathrm{iso}}^{-\infty}$. If $T \in \Psi_{\mathrm{iso}}^{-\infty}$ satisfies $T = \Pi T\Pi$, then $\mathrm{Tr}(T) = \mathrm{tr}(T|_L)$, the trace of T as an endomorphism of L. More generally $\mathrm{Tr}(\Pi T) = \mathrm{Tr}(\Pi T\Pi) = \mathrm{tr}(\Pi T|_L)$ for all $T \in \Psi_{\mathrm{iso}}^m$.*

Proof. If $\varphi_1, \dots, \varphi_l$ is an orthonormal basis of L, the kernel of Π is $\sum_i \varphi_i(x)\overline{\varphi_i(x')}$ (a matrix-valued function when $N > 1$) and that of $T = \Pi T\Pi$ is $\sum_{i,j} t_{ij}\varphi_i(x)\overline{\varphi_j(x')}$, $t_{ij} = (T\varphi_j, \varphi_i)$. Thus $\mathrm{Tr}(T) = \sum_{i,j} t_{ij}(\varphi_i, \varphi_j) = \sum_i t_{ii}$. Finally $\mathrm{Tr}(\Pi T(\mathrm{Id} - \Pi)) = \mathrm{Tr}(T(\mathrm{Id} - \Pi)\Pi) = 0$ by Lemma 7.2.2. □

Taking $L = L_N$ in Lemma 7.2.3 and using Lemma 7.1.3, we see that $\mathrm{Tr}(A) = \lim_N \sum_{|\alpha|\leq N}(Au_\alpha, u_\alpha)$, so Tr coincides with the trace of A as a trace-class operator on L^2. The trace is essentially the only continuous trace on the residual algebra.

Proposition 7.2.4. *If $F : \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n) \to \mathbb{C}$ is continuous, linear and satisfies $F([A, B]) = 0$ for all $A, B \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$, then $F = c\,\mathrm{Tr}$ for some $c \in \mathbb{C}$. Consequently $F([A, B]) = 0$ also for all $A \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ and $B \in \Psi_{\mathrm{iso}}^m(\mathbb{R}^n)$, $m \in \mathbb{R}$.*

Proof. For a finite dimensional $L \subset \mathcal{S}(\mathbb{R}^n)$ with orthonormal basis $\varphi_1, \dots, \varphi_l$, the operators P_{ij} with kernels $\varphi_i(x)\overline{\varphi_j(x')}$ satisfy $P_{ij}P_{kl} = \delta_{jk}P_{il}$, so they span a subalgebra of $\Psi_{\mathrm{iso}}^{-\infty}$ isomorphic to $M(l;\mathbb{C})$, on which Tr is the matrix trace by Lemma 7.2.3. By Lemma 7.2.1, $F = F(P_{11})\,\mathrm{Tr}$ on this subalgebra. Applying this to $L = \mathrm{span}\{\varphi, \psi\}$ for two unit vectors $\varphi, \psi \in \mathcal{S}$ shows that $c = F(\varphi\otimes\overline{\varphi})$ is independent of φ. Every finite rank $A \in \Psi_{\mathrm{iso}}^{-\infty}$ has range, and range of A^*, in $\mathcal{S}$, and so lies in the subalgebra associated with $L = \mathrm{Ran}\,A + \mathrm{Ran}\,A^*$; thus $F = c\,\mathrm{Tr}$ on finite rank operators, and hence everywhere by Lemma 7.1.3 and continuity. The last statement then follows from Lemma 7.2.2. □

7.3 The Fredholm determinant

For matrices, $\det : M(N;\mathbb{C}) \to \mathbb{C}$ is multiplicative, $\det \mathrm{Id} = 1$, and $\mathrm{GL}(N;\mathbb{C}) = \{A; \det A \neq 0\}$. Multiplicativity and normalization do not determine det (consider $(\det)^k$), but they do together with the derivative condition $\frac{d}{ds}\det(\mathrm{Id} + sA)|_{s=0} = \mathrm{tr}\,A$, and for any smooth curve A_s in $\mathrm{GL}(N;\mathbb{C})$ one has Jacobi's formula

$$\frac{d}{ds}\det A_s = \det A_s\ \mathrm{tr}\Big(A_s^{-1}\frac{dA_s}{ds}\Big) \tag{7.3.1}$$

(Exercise 7.7). Locally along such a curve this reads $d\log\det A = \mathrm{tr}(A^{-1}dA)$; the right side is a closed 1-form on $\mathrm{GL}(N;\mathbb{C})$, independent of any choice of logarithm. We use the analogous 1-form

$$\alpha = \mathrm{Tr}(A^{-1}dA) \quad \text{on} \quad G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n),$$

where, at $A = \mathrm{Id} + E$, dA is the identification of tangent vectors with elements of $\Psi_{\mathrm{iso}}^{-\infty}$ (so $A^{-1}dA$ takes values in $\Psi_{\mathrm{iso}}^{-\infty}$ and Tr applies). To avoid the differential geometry of Fréchet manifolds we only integrate α along paths. A *path* in $G_{\mathrm{iso}}^{-\infty}$ is a continuous map

$\gamma : [s_0, s_1] \to G_{\mathrm{iso}}^{-\infty}$, $\gamma(s) = \mathrm{Id} + E(s)$, which is piecewise a smooth family in the sense of Definition 7.1.1, and we set

$$\int_\gamma \alpha = \int_{s_0}^{s_1} \mathrm{Tr}(\gamma(s)^{-1}\gamma'(s))\, ds.$$

That α is closed is the content of Exercise 7.8; we shall not need it, since the following lemma, which compares α with its finite dimensional analogue, gives path independence directly.

Lemma 7.3.1. *$G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ is path connected: if $\mathrm{Id} + E \in G_{\mathrm{iso}}^{-\infty}$ there is a path in $G_{\mathrm{iso}}^{-\infty}$ from Id to $\mathrm{Id} + E$.*

Proof. E is compact on L^2, so its non-zero spectrum consists of isolated eigenvalues accumulating only at 0. For $c \neq 0$, $\mathrm{Id} + cE$ is invertible unless $-1/c$ is an eigenvalue, so the set Λ of c with $|c| \leq 2$ and $\mathrm{Id} + cE$ not invertible is finite, and $0, 1 \notin \Lambda$. If $c : [0,1] \to \{|c| \leq 2\} \setminus \Lambda$ is a smooth path from 0 to 1, then $\mathrm{Id} + c(s)E$ is the required path. □

Lemma 7.3.2. *Let $\gamma(s) = \mathrm{Id} + E(s)$, $s \in [0,1]$, be a path in $G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ with $E(0) = 0$. Then*

$$\exp\Big(\int_\gamma \alpha\Big) = \lim_{N\to\infty} \det\big(\mathrm{Id} + \pi_N E(1)|_{L_N}\big),$$

where $\pi_N E(1)|_{L_N}$ is regarded as an endomorphism of the finite dimensional space L_N. In particular the left side depends only on $\gamma(1)$.

Proof. Set $E_N(s) = \pi_N E(s) \pi_N$ and $\gamma_N = \mathrm{Id} + E_N$. By Lemma 7.1.3, $E_N \to E$ and $E_N' \to E'$ in $\Psi_{\mathrm{iso}}^{-\infty}$, uniformly on each interval of smoothness. Let $M = \sup_s \|\gamma(s)^{-1}\|_{\mathcal{B}(L^2)} < \infty$. For N large, $\|E_N(s) - E(s)\|_{\mathcal{B}(L^2)} < 1/(2M)$ for all s, so $\gamma_N(s)$ is invertible with $\gamma_N(s)^{-1} \to \gamma(s)^{-1}$ uniformly in norm. By (7.1.4) and Lemma 7.1.2, $\gamma_N^{-1} - \mathrm{Id} \to \gamma^{-1} - \mathrm{Id}$ uniformly in $\Psi_{\mathrm{iso}}^{-\infty}$, hence $\gamma_N^{-1}\gamma_N' \to \gamma^{-1}\gamma'$ uniformly in $\Psi_{\mathrm{iso}}^{-\infty}$, and, Tr being continuous, $\int_{\gamma_N} \alpha \to \int_\gamma \alpha$.

Now $\gamma_N(s)$ preserves the decomposition $L^2 = L_N \oplus L_N^\perp$, acting as $g_N(s) = \mathrm{Id} + \pi_N E(s)|_{L_N}$ on L_N and as the identity on $L_N^\perp$. Hence $\gamma_N^{-1}\gamma_N' = \pi_N \gamma_N^{-1}\gamma_N' \pi_N$ acts as $g_N^{-1}g_N'$ on L_N, and by Lemma 7.2.3 and Jacobi's formula (7.3.1),

$$\int_{\gamma_N} \alpha = \int_0^1 \mathrm{tr}(g_N^{-1}g_N')\, ds, \qquad \exp\int_{\gamma_N} \alpha = \frac{\det g_N(1)}{\det g_N(0)} = \det g_N(1).$$

Letting $N \to \infty$ proves the lemma. □

Definition 7.3.3. The *Fredholm determinant* $\det_{\mathrm{Fr}} : \mathrm{Id} + \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n; \mathbb{C}^N) \to \mathbb{C}$ is

$$\det_{\mathrm{Fr}}(\mathrm{Id} + E) = \begin{cases} \exp\big(\int_\gamma \alpha\big) & \text{if } \mathrm{Id} + E \in G_{\mathrm{iso}}^{-\infty}, \\ 0 & \text{otherwise,} \end{cases}$$

where γ is any path in $G_{\mathrm{iso}}^{-\infty}$ from Id to $\mathrm{Id} + E$.

By Lemmas 7.3.1 and 7.3.2 this is well defined. Concatenating a path from Id to G_1 with a path δ in $G_{\mathrm{iso}}^{-\infty}$ from G_1 to G_2 gives

$$\det_{\mathrm{Fr}}(G_2) = \det_{\mathrm{Fr}}(G_1) \exp\Big(\int_\delta \alpha\Big). \tag{7.3.2}$$

In particular, if G_y is a smooth family in $G_{\mathrm{iso}}^{-\infty}$, then $\det_{\mathrm{Fr}}(G_y)$ is smooth and $d \log \det_{\mathrm{Fr}}(G_y) = \mathrm{Tr}(G_y^{-1} dG_y)$: applying (7.3.2) along curves shows that the directional derivatives are $\det_{\mathrm{Fr}}(G)\,\mathrm{Tr}(G^{-1}\partial G)$, which are continuous, and iterating gives smoothness. For a holomorphic family the Cauchy–Riemann equations follow in the same way, so $\det_{\mathrm{Fr}}(G(z))$ is holomorphic. Also $\det_{\mathrm{Fr}}$ is continuous on $G_{\mathrm{iso}}^{-\infty}$, since for G' close to G the segment from G to G' lies in $G_{\mathrm{iso}}^{-\infty}$.

Proposition 7.3.4. *The Fredholm determinant has the following properties.*

(i) $\det_{\mathrm{Fr}}(AB) = \det_{\mathrm{Fr}}(A)\det_{\mathrm{Fr}}(B)$ *for* $A, B \in \mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}$, *and* $\det_{\mathrm{Fr}}(\mathrm{Id}) = 1$.

(ii) $\det_{\mathrm{Fr}}(A) \neq 0$ *if and only if* A *is invertible.*

(iii) $\det_{\mathrm{Fr}}(\exp E) = \exp \mathrm{Tr}(E)$ *for all* $E \in \Psi^{-\infty}_{\mathrm{iso}}$; *if* $\|E\|_{\mathcal{B}(L^2)} < 1$, *then* $\log(\mathrm{Id}+E) = \sum_{k\geq 1}(-1)^{k+1}E^k/k$ *converges in* $\Psi^{-\infty}_{\mathrm{iso}}$ *and* $\det_{\mathrm{Fr}}(\mathrm{Id}+E) = \exp \mathrm{Tr}\log(\mathrm{Id}+E)$.

(iv) If $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ *is invertible (with inverse in* $\Psi^{-m}_{\mathrm{iso,ph}}$*) and* $B \in \mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}$, *then* $ABA^{-1} \in \mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}$ *and* $\det_{\mathrm{Fr}}(ABA^{-1}) = \det_{\mathrm{Fr}}(B)$.

Proof. (i) If A and B are invertible, let γ_1, γ_2 be paths from Id to A and B; then $\gamma = \gamma_1\gamma_2$ is a path from Id to AB and $\gamma^{-1}\gamma' = \gamma_2^{-1}(\gamma_1^{-1}\gamma_1')\gamma_2 + \gamma_2^{-1}\gamma_2'$. By Lemma 7.2.2, $\mathrm{Tr}(\gamma_2^{-1}X\gamma_2) = \mathrm{Tr}(X\gamma_2\gamma_2^{-1}) = \mathrm{Tr}(X)$ for residual X, so $\int_\gamma \alpha = \int_{\gamma_1}\alpha + \int_{\gamma_2}\alpha$. If AB is invertible then A is surjective and B is injective, so both are invertible since they have index zero (I4). Hence if one of A, B is not invertible, neither is AB, and both sides vanish.

(ii) is immediate from the definition.

(iii) The path $s \mapsto \exp(sE)$ has $\gamma^{-1}\gamma' = E$. If $\|E\| < 1$, the path $\mathrm{Id}+sE$, $s \in [0,1]$, lies in $G^{-\infty}_{\mathrm{iso}}$, and $(\mathrm{Id}+sE)^{-1}E = \sum_{k\geq 0}(-s)^k E^{k+1}$, the series converging in $\Psi^{-\infty}_{\mathrm{iso}}$ uniformly for $s \in [0,1]$ because, by Lemma 7.1.2, $\|E^{k+2}\|_{(j)} = \|E\,E^k E\|_{(j)} \leq C\|E\|^k_{\mathcal{B}(L^2)}\|E\|^2_{(j')}$. Integrating term by term gives $\int_\gamma \alpha = \sum_{k\geq 0}(-1)^k \mathrm{Tr}(E^{k+1})/(k+1) = \mathrm{Tr}\log(\mathrm{Id}+E)$.

(iv) $ABA^{-1} - \mathrm{Id} = A(B-\mathrm{Id})A^{-1}$ is residual and ABA^{-1} is invertible if and only if B is. If γ is a path from Id to B, then $A\gamma A^{-1}$ is a path from Id to ABA^{-1} and $\mathrm{Tr}(A\gamma^{-1}\gamma' A^{-1}) = \mathrm{Tr}(\gamma^{-1}\gamma')$ by Lemma 7.2.2. □

Lemma 7.3.5 (Finite dimensional reduction)**.** *Let* Π *be the orthogonal projection onto a finite dimensional subspace* $L \subset \mathcal{S}(\mathbb{R}^n;\mathbb{C}^N)$, *and let* $R \in \Psi^{-\infty}_{\mathrm{iso}}$ *satisfy* $R = R\Pi$. *Then* $\mathrm{Id}+R$ *is invertible if and only if* $\mathrm{Id}_L + \Pi R|_L$ *is invertible, and*

$$\det_{\mathrm{Fr}}(\mathrm{Id}+R) = \det\bigl(\mathrm{Id}_L + \Pi R|_L\bigr).$$

The same holds if $R = \Pi R$.

Proof. With respect to $L^2 = L \oplus L^\perp$ (columns corresponding to the domain), $\mathrm{Id}+cR = \begin{pmatrix} \mathrm{Id}+cr & 0 \\ c\rho & \mathrm{Id} \end{pmatrix}$ with $r = \Pi R|_L$ and $\rho = (\mathrm{Id}-\Pi)R|_L$. This is invertible if and only if $q(c) = \det(\mathrm{Id}_L + cr) \neq 0$, which proves the first statement for $c = 1$. If $q(1) = 0$ both sides vanish. Otherwise, since the polynomial q has $q(0) = 1$, it has finitely many zeros, and we may choose a smooth path $c(s)$ from 0 to 1 avoiding them; $\gamma(s) = \mathrm{Id} + c(s)R$ is then a path in $G^{-\infty}_{\mathrm{iso}}$. From the triangular form, $T = \gamma^{-1}\gamma' = \begin{pmatrix} (\mathrm{Id}+cr)^{-1}c'r & 0 \\ * & 0 \end{pmatrix}$, so $T = T\Pi$ and, by Lemma 7.2.3 and Lemma 7.2.2, $\mathrm{Tr}(T) = \mathrm{Tr}(\Pi T\Pi) + \mathrm{Tr}((\mathrm{Id}-\Pi)T\Pi) = \mathrm{tr}\bigl((\mathrm{Id}+cr)^{-1}c'r\bigr) + \mathrm{Tr}(\Pi(\mathrm{Id}-\Pi)T)$. The last term vanishes and by Jacobi's formula the first is $\frac{d}{ds}\log q(c(s))$, so $\exp\int_\gamma \alpha = q(1)/q(0) = \det(\mathrm{Id}_L + r)$. The case $R = \Pi R$ follows by taking adjoints, or by the same argument with upper triangular matrices. □

In particular, for finite rank residual operators the Fredholm determinant is an ordinary determinant, and $\det_{\mathrm{Fr}}(\mathrm{Id} + \pi_N E \pi_N) = \det(\mathrm{Id} + \pi_N E|_{L_N})$.

Theorem 7.3.6. *The Fredholm determinant is continuous on* $\mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$. *If* E_y *is a smooth (resp. holomorphic) family in* $\Psi^{-\infty}_{\mathrm{iso}}$, *then* $\det_{\mathrm{Fr}}(\mathrm{Id}+E_y)$ *is a smooth (resp. holomorphic) function. Moreover, for every* $E \in \Psi^{-\infty}_{\mathrm{iso}}$,

$$\det_{\mathrm{Fr}}(\mathrm{Id}+E) = \lim_{N\to\infty} \det\bigl(\mathrm{Id} + \pi_N E|_{L_N}\bigr). \tag{7.3.3}$$

Proof. Near invertible points this was shown above. Let $\mathrm{Id}+E_0$ be non-invertible, let Π_0 be the projection onto $\mathrm{Nul}(\mathrm{Id}+E_0)\subset\mathcal{S}$, and let F be as in (I4), so that $G_0=\mathrm{Id}+E_0+F$ is invertible and $F=F\Pi_0$. For E near E_0 the operator G_0+E-E_0 is invertible; let $K(E)=(G_0+E-E_0)^{-1}$, which lies in $G^{-\infty}_{\mathrm{iso}}$ and depends continuously on E (Lemma 7.1.4). Then

$$\mathrm{Id}+E=K(E)^{-1}\big(\mathrm{Id}-K(E)F\big),\qquad \det_{\mathrm{Fr}}(\mathrm{Id}+E)=\det_{\mathrm{Fr}}(K(E))^{-1}\det\big(\mathrm{Id}-\Pi_0K(E)F|_L\big),$$

using Proposition 7.3.4(i) and Lemma 7.3.5 with $L=\operatorname{Ran}\Pi_0$ and $R=-K(E)F=R\Pi_0$. The first factor is continuous and non-vanishing, and the second is a polynomial in the matrix entries $(K(E)Fe_j,e_i)$ for an orthonormal basis e_i of L, which are continuous in E. If $E=E_y$ is a smooth or holomorphic family, both factors are smooth or holomorphic. Finally (7.3.3) follows from $\det_{\mathrm{Fr}}(\mathrm{Id}+\pi_NE\pi_N)=\det(\mathrm{Id}+\pi_NE|_{L_N})$, the convergence $\pi_NE\pi_N\to E$ and continuity. □

Remark 7.3.7. Thus $\det_{\mathrm{Fr}}$ is an entire function on $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$: holomorphic along every holomorphic family, and in particular $c\mapsto\det_{\mathrm{Fr}}(\mathrm{Id}+cE)$ is entire. In view of Proposition 7.3.4, $G^{-\infty}_{\mathrm{iso}}$ is the complement of the (singular) hypersurface $\{\mathrm{Id}+E;\ \det_{\mathrm{Fr}}(\mathrm{Id}+E)=0\}$. The trace and the determinant extend further, to operators of order less than $-2n$; see Exercise 7.16.

7.4 Index and invertible perturbations

For an elliptic $A\in\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ the null spaces of A and A^* on $\mathcal{S}'$ are finite dimensional subspaces of $\mathcal{S}$, and the *index* is

$$\operatorname{ind}(A)=\dim\mathrm{Nul}(A)-\dim\mathrm{Nul}(A^*).$$

For $m=0$ this is the index of A as a Fredholm operator on L^2, since $\mathrm{Ran}(A)$ is closed with orthocomplement $\mathrm{Nul}(A^*)$ there.

Proposition 7.4.1. *If $A\in\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ is elliptic and B is any parametrix, then $[A,B]=(AB-\mathrm{Id})-(BA-\mathrm{Id})\in\Psi^{-\infty}_{\mathrm{iso}}$ and*

$$\operatorname{ind}(A)=\mathrm{Tr}([A,B]).\tag{7.4.1}$$

Consequently $\operatorname{ind}(A+E)=\operatorname{ind}(A)$ for $E\in\Psi^{-\infty}_{\mathrm{iso}}$, and if $A(z)$ is a holomorphic (or smooth) family of elliptic operators on a connected set, then $\operatorname{ind}A(z)$ is constant.

Proof. For the generalized inverse, (7.1.1) and Lemma 7.2.3 give $\operatorname{ind}(A)=\operatorname{Tr}\Pi_0-\operatorname{Tr}\Pi_1=\mathrm{Tr}(\mathrm{Id}-BA)-\mathrm{Tr}(\mathrm{Id}-AB)=\mathrm{Tr}([A,B])$. Any other parametrix is $B'=B+E'$ with E' residual, and $\mathrm{Tr}([A,B'])-\mathrm{Tr}([A,B])=\mathrm{Tr}([A,E'])=0$ by Lemma 7.2.2. A parametrix of A is also one of $A+E$, and $\mathrm{Tr}([E,B])=0$. For a family, choose locally a holomorphic parametrix $B(z)$ (I6); then $[A(z),B(z)]$ is a holomorphic residual family, so $\operatorname{ind}A(z)$ is continuous, hence locally constant, hence constant. □

Proposition 7.4.2. *Let $A\in\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ be elliptic with $\operatorname{ind}(A)=0$. Then there is a finite rank $F\in\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ such that $A+sF$ is invertible for every $s\neq0$; its inverse lies in $\Psi^{-m}_{\mathrm{iso,ph}}$.*

Proof. Let $e_1,\dots,e_p$ and $f_1,\dots,f_p$ be orthonormal bases of $\mathrm{Nul}(A)$ and $\mathrm{Nul}(A^*)$, both in $\mathcal{S}$. Let F and G be the operators with kernels $\sum_j f_j(x)\overline{e_j(x')}$ and $\sum_j e_j(x)\overline{f_j(x')}$, so

$F = \Pi_1 F = F\Pi_0$, $G = \Pi_0 G = G\Pi_1$, $FG = \Pi_1$, $GF = \Pi_0$. With B the generalized inverse, normalized as in (I3), and using $A\Pi_0 = 0$, $\Pi_1 A = 0$, $\Pi_0 B = 0$, $B\Pi_1 = 0$,

$$(A + sF)(B + s^{-1}G) = AB + s^{-1}A\Pi_0 G + sF\Pi_0 B + FG = \mathrm{Id} - \Pi_1 + \Pi_1 = \mathrm{Id},$$
$$(B + s^{-1}G)(A + sF) = BA + sB\Pi_1 F + s^{-1}G\Pi_1 A + GF = \mathrm{Id}\,.$$

Thus $A + sF$ is invertible with inverse $B + s^{-1}G \in \Psi^{-m}_{\mathrm{iso,ph}}$. □

Write $\mathrm{Ell}^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N) \subset \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ for the elliptic elements, $\mathrm{Ell}^0_{\mathrm{iso},k}$ for those of index k, and $G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ for the group of invertible elements (on L^2, or equivalently in $\Psi^0_{\mathrm{iso,ph}}$). Then $G^0_{\mathrm{iso}} \subset \mathrm{Ell}^0_{\mathrm{iso},0} \subset \mathrm{Ell}^0_{\mathrm{iso}} \subset \Psi^0_{\mathrm{iso,ph}}$.

Corollary 7.4.3. *$G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ is dense in $\mathrm{Ell}^0_{\mathrm{iso},0}(\mathbb{R}^n;\mathbb{C}^N)$ for the topology of $\Psi^0_{\mathrm{iso,ph}}$: indeed $A + sF \to A$ as $s \to 0$.*

By Proposition 7.4.1 the index is defined on the quotient. Let

$$\mathcal{A}^m = \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)/\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N);$$

then $\mathcal{A}^{\mathbb{Z}} = \bigcup_m \mathcal{A}^m$ is an algebra, A is elliptic if and only if its class $a = [A]$ is invertible in $\mathcal{A}^{\mathbb{Z}}$ (the inverse being the class of a parametrix), and

$$\mathrm{ind} : \{a \in \mathcal{A}^0 \text{ invertible}\} \to \mathbb{Z}, \qquad \mathrm{ind}(a) = \mathrm{ind}(A) \ \text{ if } a = [A],$$

is well defined. We return to it in Section 7.10.

7.5 Analytic Fredholm theory

A family $M(z)$ with values in $\Psi^{-\infty}_{\mathrm{iso}}$ is *meromorphic* near p if $M(z) = \sum_{k=1}^{K}(z-p)^{-k}M_{-k} + M_0(z)$ with $M_{-k} \in \Psi^{-\infty}_{\mathrm{iso}}$ and M_0 a holomorphic family; we say it has *finite rank singular part* if the M_{-k} have finite rank.

Lemma 7.5.1 (Fredholm alternative)**.** *Let $\Omega \subset \mathbb{C}$ be open and connected and let $A(z)$ be a holomorphic family in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$. Then either $\mathrm{Id} + A(z)$ is invertible for no $z \in \Omega$, or it is invertible outside a discrete set $D \subset \Omega$; in the latter case $(\mathrm{Id} + A(z))^{-1} = \mathrm{Id} + B(z)$ where B is meromorphic in $\Psi^{-\infty}_{\mathrm{iso}}$ with finite rank singular parts.*

Proof. By Theorem 7.3.6, $f(z) = \det_{\mathrm{Fr}}(\mathrm{Id} + A(z))$ is holomorphic on Ω, so either $f \equiv 0$ or its zero set D is discrete; and $\mathrm{Id} + A(z)$ is invertible exactly when $f(z) \neq 0$. Away from D, $B(z)$ is holomorphic by Lemma 7.1.4. Near $p \in D$ we proceed as in the proof of Theorem 7.3.6: with $L = \mathrm{Nul}(\mathrm{Id} + A(p))$, Π_0 the projection onto L and $G_0 = \mathrm{Id} + A(p) + F$ invertible, $F = F\Pi_0$ of finite rank, the family $K(z) = (G_0 + A(z) - A(p))^{-1}$ is holomorphic in $G^{-\infty}_{\mathrm{iso}}$ near p and $\mathrm{Id} + A(z) = K(z)^{-1}(\mathrm{Id} + R(z))$ with $R(z) = -K(z)F = R(z)\Pi_0$. The matrix $\mathrm{Id}_L + \Pi_0 R(z)|_L$ has determinant $\det_{\mathrm{Fr}}(\mathrm{Id} + R(z)) = f(z)\det_{\mathrm{Fr}}(K(z)) \not\equiv 0$, so its inverse $M(z)$ is a meromorphic matrix (Cramer's rule). A direct computation, using $R = R\Pi_0$ and $(\mathrm{Id}_L + \Pi_0 R\Pi_0)M = \mathrm{Id}_L$, shows $(\mathrm{Id} + R)(\mathrm{Id} - RM\Pi_0) = \mathrm{Id} + R - R\Pi_0 = \mathrm{Id}$, so

$$(\mathrm{Id} + A(z))^{-1} = K(z) - \sum_{i,j} m_{ij}(z)\, R(z) P_{ij} K(z),$$

where P_{ij} are the matrix units with respect to an orthonormal basis of L and m_{ij} the entries of M. Each $R(z)P_{ij}K(z)$ is a holomorphic family of residual operators of rank at most one, whose Taylor coefficients therefore have finite rank; multiplying by the meromorphic scalars m_{ij} gives a singular part of finite rank. □

This result for smoothing operators, which really follows from the corresponding result for matrices, implies the analogous statement for holomorphic families of elliptic operators.

Proposition 7.5.2. *Let $\Omega \subset \mathbb{C}$ be open and connected, h holomorphic on Ω, and $A(z)$ a holomorphic family of elliptic operators of order $h(z)$. If $A(z_0)$ is invertible for some $z_0 \in \Omega$, then $A(z)$ is invertible outside a discrete set $D \subset \Omega$, and near each $p \in \Omega$*

$$A(z)^{-1} = B(z) + H(z),$$

with $B(z)$ a holomorphic family of order $-h(z)$ and $H(z)$ meromorphic in $\Psi^{-\infty}_{\mathrm{iso}}$ with finite rank singular part ($H = 0$ if $p \notin D$).

Proof. By Proposition 7.4.1, $\operatorname{ind} A(z) = \operatorname{ind} A(z_0) = 0$ for all z. Fix $p \in \Omega$. By Proposition 7.4.2 there is a finite rank residual F with $A(p) + F$ invertible, so by Lemma 7.1.5 $A(z) + F$ is invertible on a disc U_p around p, with inverse $C(z)$ a holomorphic family of order $-h(z)$. Then $A(z) = (A(z) + F)(\mathrm{Id} - C(z)F)$, so on U_p, $A(z)$ is invertible exactly where $g_p(z) = \det_{\mathrm{Fr}}(\mathrm{Id} - C(z)F) \neq 0$.

Let W be the set of $p \in \Omega$ such that $A(z)$ is invertible at points arbitrarily close to p. If $p \in W$ then $g_p \not\equiv 0$ on the disc U_p, so its zeros are isolated, and every point of U_p is in W; thus W is open. Its complement is the interior of the set of non-invertible points, also open. Since $z_0 \in W$ and Ω is connected, $W = \Omega$, and the non-invertible points form a discrete set.

Near p, Lemma 7.5.1 gives $(\mathrm{Id} - C(z)F)^{-1} = \mathrm{Id} + M(z)$ with $M = \sum_{k=1}^{K} (z-p)^{-k} M_{-k} + M_0(z)$ as above, so $A(z)^{-1} = (\mathrm{Id} + M(z))C(z)$. Set $B = C + M_0 C$, a holomorphic family of order $-h(z)$, and $H = \sum_k (z-p)^{-k} M_{-k} C(z)$. Each $M_{-k}C(z)$ is a holomorphic residual family with range in the finite dimensional space $\operatorname{Ran} M_{-k}$, so the singular part of H has finite rank. □

7.6 Resolvent and spectrum

One direct application of analytic Fredholm theory is to the resolvent of an elliptic operator of positive order. For simplicity we take the order to be a positive integer m, so that $A - \lambda \in \Psi^m_{\mathrm{iso,ph}}$ for $\lambda \in \mathbb{C}$; the case of non-integral order needs only notational changes, replacing $\Psi^m_{\mathrm{iso,ph}}$ by the class of operators with symbols in Γ^m having an expansion in terms of homogeneous functions of degrees $m - j$ and $-j$.

Proposition 7.6.1. *Let $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$, $m \in \mathbb{N}$, be elliptic, and suppose that for some $\lambda' \in \mathbb{C}$ both $A - \lambda'$ and $A^* - \overline{\lambda'}$ have trivial null space. Then the set $\operatorname{Spec}(A)$ of $\lambda \in \mathbb{C}$ for which $A - \lambda$ is not invertible is discrete, and $(A - \lambda)^{-1}$ is a meromorphic family in $\Psi^{-m}_{\mathrm{iso,ph}}$ with finite rank residual singular parts. At $\tilde{\lambda} \in \operatorname{Spec}(A)$ the generalized eigenspace*

$$E_{\tilde{\lambda}} = \{u \in \mathcal{S}'(\mathbb{R}^n; \mathbb{C}^N);\ (A - \tilde{\lambda})^p u = 0 \text{ for some } p \in \mathbb{N}\}$$

is a finite dimensional subspace of $\mathcal{S}$, and

$$(A - \lambda)^{-1} = -\sum_{j=1}^{k} (\lambda - \tilde{\lambda})^{-j} (A - \tilde{\lambda})^{j-1} P + (\textit{holomorphic near } \tilde{\lambda}),$$

where P is a projection (not necessarily orthogonal) onto $E_{\tilde{\lambda}}$ commuting with A, and k is the least integer with $(A - \tilde{\lambda})^k P = 0$.

Proof. Since $m > 0$, $A - \lambda$ is, for every λ, elliptic with the same principal symbol as A, and $\lambda \mapsto A - \lambda$ is an entire family of order m. The hypothesis says $\operatorname{ind}(A - \lambda') = 0$ and $A - \lambda'$ is invertible, so Proposition 7.5.2 applies with $\Omega = \mathbb{C}$. Replacing A by $A - \tilde{\lambda}$ we may take $\tilde{\lambda} = 0$ and write $(A - \lambda)^{-1} = \sum_{j \geq -k} \lambda^j S_j$, with $S_{-1}, \dots, S_{-k}$ finite rank residual. Comparing coefficients in $(A - \lambda)(A - \lambda)^{-1} = \mathrm{Id} = (A - \lambda)^{-1}(A - \lambda)$ gives $AS_j - S_{j-1} = \delta_{j0}\,\mathrm{Id} = S_j A - S_{j-1}$, so with $R_j = S_{-j}$,

$$R_{j+1} = AR_j = R_j A \ (1 \leq j \leq k), \quad R_{k+1} = 0, \qquad \text{hence } R_j = A^{j-1}R_1,\ A^k R_1 = 0.$$

Thus $\operatorname{Ran} R_1 \subset E_0$ and R_1 commutes with A. Conversely, if $A^p u = 0$, then $(A - \lambda)(-\sum_{j=0}^{p-1} \lambda^{-j-1} A^j u) = u$, so $(A - \lambda)^{-1} u = -\sum_{j<p} \lambda^{-j-1} A^j u$ and comparing the coefficients of λ^{-1}, $R_1 u = -u$. Hence $P = -R_1$ is a projection with range exactly E_0, which is therefore finite dimensional and contained in $\mathcal{S}$. □

Note also that $\operatorname{Ran} R_k \subset \operatorname{Ran} R_{k-1} \subset \cdots \subset \operatorname{Ran} R_1$ and $\operatorname{Nul} R_k \supset \cdots \supset \operatorname{Nul} R_1$; precisely, $u \in \operatorname{Ran} R_p$, $p \geq 1$, if and only if $u = A^{p-1} u_1$ for some $u_1 \in \operatorname{Ran} R_1$.

7.7 The residue trace

We have shown that there is a unique trace on the residual algebra. Following ideas of Seeley [11], Guillemin [4, 5] and Wodzicki [13, 14], we now look for traces on the algebra $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n)$. For $A \in \Psi^{-\infty}_{\mathrm{iso}}$, by (7.2.2), $\operatorname{Tr}(A) = (2\pi)^{-n} \int a_L(x, \xi)\, dx\, d\xi$ is the integral of the left symbol; this is also true for any amplitude: if $A = \operatorname{Op}(a)$ with $a(x, x', \xi)$ an amplitude of order $-\infty$, then $\operatorname{Tr}(A) = (2\pi)^{-n} \int a(x, x, \xi)\, dx\, d\xi$. The integral extends to Ψ^m_{iso} for $m < -2n$, as we saw, but not beyond. To go further we regularize it by embedding operators in holomorphic families, which moves us from the symbol classes Γ^m to the polyhomogeneous operators.

Lemma 7.7.1. *Let $\Omega \subset \mathbb{C}$ be open and connected, $m \in \mathbb{C}$, and $A(z)$ a holomorphic family of order $m + z$ on Ω, with b, b_j as in (7.1.2). Suppose Ω contains points with $\Re z < -2n - \Re m$. Then $\operatorname{Tr}(A(z))$, defined where $\Re z < -2n - \Re m$, extends to a meromorphic function on Ω with at most simple poles, at the points $z_j = j - 2n - m \in \Omega$, $j \in \mathbb{N}_0$, and*

$$\operatorname{Res}_{z=z_j} \operatorname{Tr}(A(z)) = -(2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \operatorname{tr} b_j(z_j; \theta)\, d\theta. \tag{7.7.1}$$

If $A(z) = A_y(z)$ depends smoothly on a parameter y, so do the residues and the coefficients of the Laurent expansions.

Proof. For $\Re z < -2n - \Re m$ we have $\operatorname{Tr}(A(z)) = f_1(z) + f_2(z)$ with

$$f_1(z) = (2\pi)^{-n} \int_{|Z| \leq 1} \operatorname{tr} a(z; Z)\, dZ,$$
$$f_2(z) = (2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \int_1^\infty r^{m+z+2n-1} \operatorname{tr} b(z; 1/r, \theta)\, dr\, d\theta.$$

The first is holomorphic on Ω. Write $b(z; t, \theta) = \sum_{j<k} t^j b_j(z; \theta) + t^k b_{(k)}(z; t, \theta)$, with $b_{(k)}$ smooth and holomorphic in z (Taylor's formula with integral remainder). Since $\int_1^\infty r^{m+z+2n-1-j}\, dr = (j - 2n - m - z)^{-1}$ for $\Re z < j - 2n - \Re m$,

$$f_2(z) = (2\pi)^{-n} \sum_{j<k} \frac{1}{j - 2n - m - z} \int_{\mathbb{S}^{2n-1}} \operatorname{tr} b_j(z; \theta)\, d\theta + f_2^{(k)}(z),$$

where $f_2^{(k)}(z) = (2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \int_1^\infty r^{m+z+2n-1-k} \operatorname{tr} b_{(k)}(z; 1/r, \theta)\, dr\, d\theta$ is holomorphic in $\Re z < k - 2n - \Re m$, the integral converging locally uniformly there. This identity holds pointwise

for $\Re z < -2n - \Re m$, and its right side is meromorphic on $\Omega_k = \{z \in \Omega; \Re z < k - 2n - \Re m\}$; the right sides for k and $k+1$ agree on Ω_k (the difference is the $j = k$ term, computed by the same integral). Since Ω is connected, these define the meromorphic continuation, with the stated poles and residues. All the expressions depend smoothly on parameters if b does. □

We continue to write $\operatorname{Tr}(A(z))$ for the meromorphic continuation. By uniqueness of continuation, any identity between such functions which holds for $\Re z \ll 0$ holds on Ω. For a holomorphic family $A(z)$ of order $m + z$ we have $b_j(0;\theta)$ equal to the coefficient of the term of homogeneity $m - j$ in the symbol of $A(0)$. Taking $j = 2n + m$, the residue at $z = 0$ only depends on $A(0)$, and it vanishes unless $m \in \mathbb{Z}$, $m \geq -2n$.

Definition 7.7.2. For $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$, $m \in \mathbb{C}$, the *residue trace* is

$$\operatorname{Tr}_{\mathrm{R}}(A) = -\operatorname{Res}_{z=0} \operatorname{Tr}(A(z)),$$

where $A(z)$ is any holomorphic family of order $m + z$, on a connected neighbourhood of 0 containing points with $\Re z < -2n - \Re m$, with $A(0) = A$; for instance $A(z) = AH(z)$, with $H(z)$ from (7.1.3).

Proposition 7.7.3. *The residue trace is well defined, and*

$$\operatorname{Tr}_{\mathrm{R}}(A) = (2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \operatorname{tr} a_{-2n}(\theta)\, d\theta, \tag{7.7.2}$$

where a_{-2n} is the term of homogeneity $-2n$ in the expansion of the left symbol of A (so $\operatorname{Tr}_{\mathrm{R}}$ vanishes unless $m \in \mathbb{Z}$, $m \geq -2n$; in particular on $\Psi^{-2n-1}_{\mathrm{iso,ph}}$). It is a trace on $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$:

$$\operatorname{Tr}_{\mathrm{R}}([A, B]) = 0 \quad \forall\, A, B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N). \tag{7.7.3}$$

Proof. The formula is (7.7.1) with $j = 2n + m$, and shows that $\operatorname{Tr}_{\mathrm{R}}(A)$ is independent of the family. If $A \in \Psi^{m_1}_{\mathrm{iso,ph}}$, $B \in \Psi^{m_2}_{\mathrm{iso,ph}}$, then $C(z) = [AH(z), B]$ is an entire family of order $m_1 + m_2 + z$ with $C(0) = [A, B]$. For $\Re z < -2n - m_1 - m_2$, $\operatorname{Tr}(C(z)) = \operatorname{Tr}(AH(z)B) - \operatorname{Tr}(BAH(z)) = 0$ by Lemma 7.2.2, since $AH(z) \in \Psi^{m_1+\Re z}_{\mathrm{iso}}$. Hence the continuation vanishes identically and $\operatorname{Tr}_{\mathrm{R}}([A, B]) = 0$. □

Remark 7.7.4. The left symbol in (7.7.2) may be replaced by the right symbol (or the Weyl symbol). Indeed the right symbol of A has the expansion $a_R \sim \sum_\alpha \frac{1}{\alpha!}(-D_\xi)^\alpha \partial_x^\alpha a_L$, and each term with $\alpha \neq 0$ is a derivative $\partial_{Z_k} g$ of a polyhomogeneous function. If g is homogeneous of degree $1 - 2n$ then, integrating $\partial_{Z_k} g$ over $1 \leq |Z| \leq R$ and using the divergence theorem,

$$\log R \int_{\mathbb{S}^{2n-1}} \partial_{Z_k} g\, d\theta = \int_{\mathbb{S}^{2n-1}} (R^{1-2n} R^{2n-1} - 1) g(\theta) \theta_k\, d\theta = 0,$$

so the homogeneous term of degree $-2n$ of each such derivative integrates to zero over the sphere. The residue trace is, up to a constant multiple, the unique trace on $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n)$; this is the isotropic analogue of Wodzicki's theorem [13], and we shall neither prove nor use it.

7.8 Exterior derivation

Definition 7.8.1. A *regularizing family* is an entire holomorphic family $A(z)$ of order z, with values in $\Psi^z_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$, with scalar principal symbol and $A(0) = \mathrm{Id}$. Its derivative at 0 is denoted $\dot{A} = \frac{d}{dz} A(z)|_{z=0}$.

If $\sigma_L(A(z)) = \langle Z\rangle^z \tilde{a}(z;Z)$ with $\tilde{a}$ holomorphic in z with values in $\mathcal{C}^\infty(\overline{\mathbb{R}^{2n}})$, then $A(0) = \mathrm{Id}$ means $\tilde{a}(0) = 1$, so

$$\dot{A} = \mathrm{Op}(\log\langle Z\rangle + c), \qquad c = \partial_z \tilde{a}(0) \in S^0_{\mathrm{ph}}(\mathbb{R}^{2n}) \text{ with scalar leading term.}$$

Thus $\dot{A} \in \Psi^\epsilon_{\mathrm{iso}}$ for every $\epsilon > 0$, but $\dot{A} \notin \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$ because of the logarithm. For the standard family $H(z)$ of (7.1.3), $\dot{H} = \mathrm{Op}(\log\langle Z\rangle) = \mathrm{Op}(\frac{1}{2}\log(1+|x|^2+|\xi|^2))$.

Proposition 7.8.2. *For a regularizing family $A(z)$, the map*

$$D_A B = [\dot{A}, B]$$

is a derivation of $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ with $D_A : \Psi^m_{\mathrm{iso,ph}} \to \Psi^{m-1}_{\mathrm{iso,ph}}$ and $D_A(\Psi^{-\infty}_{\mathrm{iso}}) \subset \Psi^{-\infty}_{\mathrm{iso}}$. Equivalently, $D_A B = \frac{d}{dz}(A(z)BA(z)^{-1})\big|_{z=0}$. For two regularizing families, $D_{A_2} - D_{A_1} = [\dot{A}_2 - \dot{A}_1, \cdot\,]$ is an inner derivation, given by commutation with an element of $\Psi^0_{\mathrm{iso,ph}}$. Moreover

$$\mathrm{Tr_R}(D_A B) = 0 \quad \forall\, B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N). \tag{7.8.1}$$

Proof. A commutator is a derivation. For $B \in \Psi^m_{\mathrm{iso,ph}}$, $K(z) = [A(z), B]/z$ is an entire holomorphic family of order $m + z$ (the numerator is a holomorphic family vanishing at $z = 0$, and dividing its normalized symbol by z preserves holomorphy), with $K(0) = [\dot{A}, B]$; hence $D_A B \in \Psi^m_{\mathrm{iso,ph}}$. Since the principal symbol of $A(z)$ is scalar, $[A(z), B]$ has order $m + z - 1$, so in fact $D_A B \in \Psi^{m-1}_{\mathrm{iso,ph}}$. Residual operators form an ideal in $\bigcup_m \Psi^m_{\mathrm{iso}}$, so D_A preserves them. Near $z = 0$, $A(z)$ is invertible with holomorphic inverse by Lemma 7.1.5, and $\frac{d}{dz}(A(z)BA(z)^{-1})|_{z=0} = \dot{A}B - B\dot{A}$ by the product rule. The statement about two families follows since $\dot{A}_2 - \dot{A}_1 = \mathrm{Op}(c_2 - c_1) \in \Psi^0_{\mathrm{iso,ph}}$. Finally, $K(z)$ has value $D_A B$ at 0 and $\mathrm{Tr}(K(z)) = z^{-1}(\mathrm{Tr}(A(z)B) - \mathrm{Tr}(BA(z))) = 0$ for $\Re z \ll 0$ by Lemma 7.2.2, so the residue at 0 vanishes. □

Remark 7.8.3. D_H is not an inner derivation of $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$, which is why we call it an *exterior* derivation. Suppose $[\dot{H} - P, B] = 0$ for all $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$, for some $P \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$, and let c be the left symbol of $C = \dot{H} - P$. The composition formula is exact when one factor has polynomial symbol, and gives $[C, x_j] = \mathrm{Op}(-i\partial_{\xi_j} c)$ and $[C, D_j] = \mathrm{Op}(-D_{x_j} c)$; hence c is constant, which is impossible since $c - \log\langle Z\rangle$ is polyhomogeneous while $\log\langle Z\rangle$ is not.

7.9 The regularized trace

Let $A(z)$ be a regularizing family. For $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ the function $\mathrm{Tr}(BA(z))$ is meromorphic on $\mathbb{C}$ with at most a simple pole at 0, with residue $-\mathrm{Tr_R}(B)$. We define the *regularized trace* as the finite part at $z = 0$, the constant term of the Laurent expansion:

$$\overline{\mathrm{Tr}}_A(B) = \operatorname*{FP}_{z=0} \mathrm{Tr}(BA(z)) = \lim_{z\to 0}\Big(\mathrm{Tr}(BA(z)) + \frac{\mathrm{Tr_R}(B)}{z}\Big). \tag{7.9.1}$$

If B has order less than $-2n$ (in particular if B is residual), then $\mathrm{Tr}(BA(z))$ is holomorphic near 0 and $\overline{\mathrm{Tr}}_A(B) = \mathrm{Tr}(B)$; thus $\overline{\mathrm{Tr}}_A$ extends the trace. By Lemma 7.7.1, if B_y is a smooth family then $\overline{\mathrm{Tr}}_A(B_y)$ is smooth in y and $d\,\overline{\mathrm{Tr}}_A(B_y) = \overline{\mathrm{Tr}}_A(dB_y)$. In contrast with the residue trace, $\overline{\mathrm{Tr}}_A$ depends on the choice of regularizing family, though only through $\dot{A}$.

Lemma 7.9.1. *If $A_1(z)$, $A_2(z)$ are regularizing families, then for $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$,*

$$\overline{\mathrm{Tr}}_{A_2}(B) - \overline{\mathrm{Tr}}_{A_1}(B) = -\mathrm{Tr_R}(B(\dot{A}_2 - \dot{A}_1)).$$

Proof. $L(z) = (A_2(z) - A_1(z))/z$ is an entire family of order z with $L(0) = \dot{A}_2 - \dot{A}_1 \in \Psi^0_{\mathrm{iso,ph}}$, and $\operatorname{Tr}(BA_2(z)) - \operatorname{Tr}(BA_1(z)) = z \operatorname{Tr}(BL(z))$. The function $\operatorname{Tr}(BL(z))$ has at most a simple pole at 0 with residue $-\operatorname{Tr_R}(BL(0))$, which is therefore the value at 0 of $z \operatorname{Tr}(BL(z))$. □

In particular, if B has non-integral order, $\overline{\operatorname{Tr}}_A(B)$ is independent of A. The regularized trace is *not* a trace; its failure to be one is measured by the residue trace and the exterior derivation.

Proposition 7.9.2 (Trace defect formula)**.** *For a regularizing family $A(z)$ and $B_1, B_2 \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$,*

$$\overline{\operatorname{Tr}}_A([B_1, B_2]) = \operatorname{Tr_R}(B_1 D_A B_2) = -\operatorname{Tr_R}(B_2 D_A B_1). \tag{7.9.2}$$

Proof. For $\Re z \ll 0$, Lemma 7.2.2 gives $\operatorname{Tr}(B_2 B_1 A(z)) = \operatorname{Tr}(B_1 A(z) B_2)$, so

$$\operatorname{Tr}([B_1, B_2]A(z)) = \operatorname{Tr}(B_1[B_2, A(z)]) = -z \operatorname{Tr}(B_1 K(z)),$$

where $K(z) = [A(z), B_2]/z$ is the entire family of order $m_2 + z$ with $K(0) = D_A B_2$ used in the proof of Proposition 7.8.2. By continuation this holds for all z; since $\operatorname{Tr}(B_1 K(z))$ has at most a simple pole at 0 with residue $-\operatorname{Tr_R}(B_1 D_A B_2)$, the right side is regular at 0 with value $\operatorname{Tr_R}(B_1 D_A B_2)$. The second equality follows from (7.8.1) and the trace property: $0 = \operatorname{Tr_R}(D_A(B_1 B_2)) = \operatorname{Tr_R}(B_2 D_A B_1) + \operatorname{Tr_R}(B_1 D_A B_2)$. □

Corollary 7.9.3. *If $G \in G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n; \mathbb{C}^N)$ and $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$, then $\overline{\operatorname{Tr}}_A(G^{-1}BG) = \overline{\operatorname{Tr}}_A(B)$.*

Proof. $\overline{\operatorname{Tr}}_A(G^{-1}BG) - \overline{\operatorname{Tr}}_A(BGG^{-1}) = \overline{\operatorname{Tr}}_A([G^{-1}, BG]) = -\operatorname{Tr_R}(BG\, D_A(G^{-1}))$, which vanishes since $D_A(G^{-1}) = D_A(G^{-1} - \mathrm{Id})$ is residual and $\operatorname{Tr_R}$ vanishes on residual operators. □

7.10 The index formula

Let D_H be the exterior derivation of the standard family (7.1.3). It preserves $\Psi^{-\infty}_{\mathrm{iso}}$, and $\operatorname{Tr_R}$ vanishes there, so both descend to the quotient algebra $\mathcal{A}^{\mathbb{Z}} = \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}/\Psi^{-\infty}_{\mathrm{iso}}$: $D_H : \mathcal{A}^m \to \mathcal{A}^{m-1}$ and $\operatorname{Tr_R} : \mathcal{A}^{\mathbb{Z}} \to \mathbb{C}$.

Theorem 7.10.1. *If $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$, $m \in \mathbb{Z}$, is elliptic with parametrix B, then*

$$\operatorname{ind}(A) = \operatorname{Tr_R}(AD_H B) = -\operatorname{Tr_R}(BD_H A). \tag{7.10.1}$$

Thus, for invertible $a \in \mathcal{A}^{\mathbb{Z}}$, $\operatorname{ind}(a) = \operatorname{Tr_R}(a\, D_H(a^{-1}))$. The index is locally constant along smooth families of elliptic operators, and $\operatorname{ind}(a_1 a_2) = \operatorname{ind}(a_1) + \operatorname{ind}(a_2)$.

Proof. Since $[A, B]$ is residual, $\operatorname{ind}(A) = \operatorname{Tr}([A, B]) = \overline{\operatorname{Tr}}_H([A, B]) = \operatorname{Tr_R}(AD_H B)$ by (7.4.1) and (7.9.2). If A_t is a smooth family with smooth family of parametrices B_t (I6), the right side of (7.10.1) is an integral of finitely many terms of the symbol expansions, so it is continuous in t; being integer-valued it is locally constant. (Alternatively, $[A_t, B_t]$ is a smooth residual family.) For additivity,

$$a_1 a_2\, D_H(a_2^{-1} a_1^{-1}) = a_1 (a_2 D_H(a_2^{-1})) a_1^{-1} + a_1 D_H(a_1^{-1}),$$

and the residue trace of the first term is $\operatorname{Tr_R}(a_2 D_H(a_2^{-1}))$ by the trace property. □

Both properties hold for general Fredholm operators, but the point here is that they follow from a simple algebraic formula.

7.11 The variation 1-form

The index is the obstruction to perturbing a single elliptic operator, by a smoothing operator, to be invertible. We now derive a closely related obstruction for families. Let Y be a manifold and

$$Y \ni y \longmapsto A_y \in \mathrm{Ell}^0_{\mathrm{iso},0}(\mathbb{R}^n;\mathbb{C}^N) \tag{7.11.1}$$

a smooth family of elliptic operators of order 0 and index 0. The *families perturbative invertibility question* asks whether there is a smooth family

$$Y \ni y \longmapsto E_y \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N) \quad \text{with} \quad A_y + E_y \in G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)\ \forall\, y. \tag{7.11.2}$$

Since the index vanishes, such E_y exists for each fixed y (Proposition 7.4.2), and also locally: if $A_{y_0}+F$ is invertible then so is A_y+F for y near y_0, with inverse a smooth family in $\Psi^0_{\mathrm{iso,ph}}$ by Lemma 7.1.5. The issue is the global existence of a smooth (or even continuous) choice.

Operator-valued differential forms are handled in the obvious way: if $\omega = \sum \omega_j\, dy_j$ and $\eta = \sum \eta_k\, dy_k$ in local coordinates, $\omega\wedge\eta = \sum_{j,k}\omega_j\eta_k\, dy_j\wedge dy_k$, and Tr_{R} and $\overline{\mathrm{Tr}}_H$ are applied coefficientwise. Let $a_y = [A_y] \in \mathcal{A}^0$ and let B_y be a smooth family of parametrices, so $a_y^{-1} = [B_y]$. We set

$$\theta = a^{-1}da, \qquad \varphi = a^{-1}D_H a,$$

represented by $B_y\, dA_y$ and $B_y D_H A_y$, and define

$$\beta = \tfrac{1}{2}\,\mathrm{Tr}_{\mathrm{R}}(a^{-1}da\wedge a^{-1}da\; a^{-1}D_Ha) = \tfrac{1}{2}\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\,\varphi), \qquad \gamma = \tfrac{1}{2}\,\mathrm{Tr}_{\mathrm{R}}(\theta\varphi). \tag{7.11.3}$$

These are smooth forms on Y of degrees 2 and 1; they depend only on a_y, since Tr_{R} vanishes on residual operators.

Proposition 7.11.1. *The 2-form β is closed. If there is a smooth family E_y as in (7.11.2), then, with $\tilde A_y = A_y + E_y$ and*

$$\bar\alpha = \overline{\mathrm{Tr}}_H(\tilde A_y^{-1}d\tilde A_y), \tag{7.11.4}$$

we have $d\bar\alpha = -\beta + d\gamma$. In particular β is exact, so its de Rham class $[\beta]\in H^2(Y;\mathbb{C})$ is an obstruction to the existence of a smooth invertible perturbation.

Proof. Suppose first that $\tilde A = A+E$ is invertible on Y, and compute with $\theta = \tilde A^{-1}d\tilde A$ and $\varphi = \tilde A^{-1}D_H\tilde A$, which represent the classes above. Since $d(\tilde A^{-1}) = -\tilde A^{-1}d\tilde A\,\tilde A^{-1}$,

$$d\theta = -\theta\wedge\theta = -\tfrac{1}{2}\sum_{j,k}[\theta_j,\theta_k]\, dy_j\wedge dy_k.$$

Since d commutes with $\overline{\mathrm{Tr}}_H$, the trace defect formula (7.9.2) gives

$$d\bar\alpha = -\tfrac{1}{2}\sum_{j,k}\mathrm{Tr}_{\mathrm{R}}(\theta_j D_H\theta_k)\, dy_j\wedge dy_k = -\tfrac{1}{2}\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge D_H\theta).$$

As D_H is a derivation commuting with d (it is commutation with the fixed operator $\dot H$), $D_H(\tilde A^{-1}) = -\tilde A^{-1}(D_H\tilde A)\tilde A^{-1}$ and

$$D_H\theta = -\varphi\theta + \tilde A^{-1}d(D_H\tilde A) = d\varphi + \theta\varphi - \varphi\theta.$$

By the trace property, $\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\varphi\theta) = \sum \mathrm{Tr}_{\mathrm{R}}(\theta_k\theta_j\varphi)\, dy_j\wedge dy_k = -\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\varphi)$, so $\mathrm{Tr}_{\mathrm{R}}(\theta\wedge D_H\theta) = \mathrm{Tr}_{\mathrm{R}}(\theta\wedge d\varphi) + 2\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\varphi)$. On the other hand $d(\theta\varphi) = d\theta\,\varphi - \theta\wedge d\varphi$, so $2\,d\gamma = -\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\varphi) - \mathrm{Tr}_{\mathrm{R}}(\theta\wedge d\varphi)$. Combining,

$$\mathrm{Tr}_{\mathrm{R}}(\theta\wedge D_H\theta) = 2\beta - 2\,d\gamma, \qquad d\bar\alpha = -\beta + d\gamma.$$

Thus $\beta = d(\gamma-\bar\alpha)$ is exact. In general, invertible perturbations exist locally on Y, as noted above, so β is locally exact and hence closed. □

The form γ is defined on all of Y for any family (7.11.1), so $[\beta] = [\beta - d\gamma]$. On G^0_{iso}, $\bar\alpha$ extends the 1-form $d\log\det_{\mathrm{Fr}} = \mathrm{Tr}(A^{-1}dA)$ from $G^{-\infty}_{\mathrm{iso}}$, but it is not closed in general, because the regularized trace is not a trace. To explain the topological meaning of $[\beta]$ we now construct the determinant line bundle.

7.12 The determinant bundle

The determinant bundle was introduced by Quillen [10] for families of Cauchy–Riemann operators. Recall that $\det_{\mathrm{Fr}}$ is a character of $G^{-\infty}_{\mathrm{iso}}$: $\det_{\mathrm{Fr}}(AB) = \det_{\mathrm{Fr}}(A)\det_{\mathrm{Fr}}(B)$. Instead of trying to extend it multiplicatively to G^0_{iso} (see Remark 7.12.4), we consider all its possible values at once. Write $G^0 = G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ and $G^{-\infty} = G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$; the latter is a normal subgroup of G^0, and $\mathcal{G} = G^0/G^{-\infty}$ is the set of classes $a = [A] \in \mathcal{A}^0$ of invertible operators, which by Proposition 7.4.2 are exactly the invertible elements of $\mathcal{A}^0$ of index zero. On $G^0 \times \mathbb{C}$ define

$$(A,z) \sim (A',z') \iff A' = A(\mathrm{Id}+E),\ z' = \det_{\mathrm{Fr}}(\mathrm{Id}+E)\,z \ \text{ for some } \mathrm{Id}+E \in G^{-\infty}. \tag{7.12.1}$$

Lemma 7.12.1. (7.12.1) *is an equivalence relation. Let $\mathcal{D} = (G^0\times\mathbb{C})/\!\sim$ and $\pi : \mathcal{D} \to \mathcal{G}$, $\pi[(A,z)] = [A]$. Each fibre $\mathcal{D}_a = \pi^{-1}(a)$ is a one-dimensional complex vector space with the operations $[(A,z)] + [(A,w)] = [(A,z+w)]$, $\lambda[(A,z)] = [(A,\lambda z)]$.*

Proof. Reflexivity, symmetry and transitivity follow from $\det_{\mathrm{Fr}}(\mathrm{Id}) = 1$, $\det_{\mathrm{Fr}}((\mathrm{Id}+E)^{-1}) = \det_{\mathrm{Fr}}(\mathrm{Id}+E)^{-1}$ and multiplicativity. If $[A] = [A_0] = a$ then $A_0^{-1}A \in G^0 \cap (\mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}) = G^{-\infty}$, so every element of $\mathcal{D}_a$ has a representative (A_0, w), and w is unique since $(A_0,w) \sim (A_0,w')$ forces $\mathrm{Id}+E = \mathrm{Id}$. Thus $[(A_0,w)] \mapsto w$ is a bijection $\mathcal{D}_a \to \mathbb{C}$. The operations above are well defined because the relation is linear in z, and they correspond to those of $\mathbb{C}$ under this bijection. □

Now let $y \mapsto A_y$ be a smooth family (7.11.1) over a manifold Y and $a_y = [A_y]$. Set $\mathcal{D}_Y = \bigsqcup_{y\in Y} \mathcal{D}_{a_y}$. A *local invertible lift* on $U \subset Y$ open is a smooth family $\tilde A_y = A_y + E_y \in G^0$, $y \in U$, with E_y residual; it defines the section $s_{\tilde A}(y) = [(\tilde A_y, 1)]$ of $\mathcal{D}_Y$ over U. Local invertible lifts exist near every point of Y.

Proposition 7.12.2. *$\mathcal{D}_Y$ has a unique structure of smooth complex line bundle over Y for which the sections $s_{\tilde A}$ are smooth. If $\tilde A' = \tilde A(\mathrm{Id}+X)$ is another local invertible lift, then X_y is a smooth residual family and*

$$s_{\tilde A'} = \det_{\mathrm{Fr}}(\mathrm{Id}+X)^{-1} s_{\tilde A}. \tag{7.12.2}$$

If (7.11.2) has a solution, $\mathcal{D}_Y$ is trivial.

Proof. $\mathrm{Id}+X = \tilde A^{-1}\tilde A' = \mathrm{Id} + \tilde A^{-1}(E'-E)$, and $\tilde A^{-1}$ is a smooth family in $\Psi^0_{\mathrm{iso,ph}}$ by Lemma 7.1.5, so X is a smooth residual family. By (7.12.1), $(\tilde A, \det_{\mathrm{Fr}}(\mathrm{Id}+X)^{-1}) \sim (\tilde A(\mathrm{Id}+X), 1)$, which is (7.12.2). The transition functions $\det_{\mathrm{Fr}}(\mathrm{Id}+X_y)^{-1}$ are smooth and non-vanishing by Theorem 7.3.6, and the cocycle condition holds automatically since all sections live in the same space $\mathcal{D}_Y$. Hence the local trivializations $s_{\tilde A}$ define a smooth line bundle structure, clearly unique. A global invertible lift gives a global non-vanishing section. □

The projection to $\mathcal{G}$ and the quotient map fit into the commutative diagram

$$\begin{array}{ccc} G^0 \times \mathbb{C} & \xrightarrow{/\sim} & \mathcal{D} \\ \downarrow & & \downarrow \pi \\ G^0 & \xrightarrow{/G^{-\infty}} & \mathcal{G}, \end{array}$$

and the pull-back of $\mathcal{D}$ to G^0 is canonically trivial, with the section $A \mapsto [(A,1)]$. Over Y, the form $\bar\alpha$ of (7.11.4) defines a connection on $\mathcal{D}_Y$.

Theorem 7.12.3. *There is a unique connection ∇ on $\mathcal{D}_Y$ such that, for every local invertible lift $\tilde A$,*

$$\nabla s_{\tilde A} = -\overline{\mathrm{Tr}}_H(\tilde A^{-1} d\tilde A) \otimes s_{\tilde A}.$$

Its curvature is the globally defined 2-form $\beta - d\gamma$ of (7.11.3).

Proof. Write $\bar\alpha_{\tilde A} = \overline{\mathrm{Tr}}_H(\tilde A^{-1} d\tilde A)$. We must check that the local definitions agree. With $\tilde A' = \tilde A(\mathrm{Id} + X)$,

$$\tilde A'^{-1} d\tilde A' = (\mathrm{Id}+X)^{-1}(\tilde A^{-1} d\tilde A)(\mathrm{Id}+X) + (\mathrm{Id}+X)^{-1} dX,$$

so by Corollary 7.9.3, and since $\overline{\mathrm{Tr}}_H = \mathrm{Tr}$ on residual operators,

$$\bar\alpha_{\tilde A'} = \bar\alpha_{\tilde A} + \mathrm{Tr}((\mathrm{Id}+X)^{-1} dX) = \bar\alpha_{\tilde A} + d \log \det_{\mathrm{Fr}}(\mathrm{Id}+X).$$

With $f = \det_{\mathrm{Fr}}(\mathrm{Id}+X)^{-1}$, the Leibniz rule and (7.12.2) give

$$\nabla s_{\tilde A'} = \nabla(f s_{\tilde A}) = (d\log f - \bar\alpha_{\tilde A})\, s_{\tilde A'} = -\bar\alpha_{\tilde A'}\, s_{\tilde A'},$$

as required. The curvature is $-d\bar\alpha_{\tilde A} = \beta - d\gamma$ by Proposition 7.11.1. □

Thus $\beta - d\gamma$ is a curvature form, and by Chern–Weil theory the class $\frac{i}{2\pi}[\beta]$ represents the first Chern class of $\mathcal{D}_Y$ in $H^2(Y;\mathbb{C})$; in particular it is the image of an integral class. This is the topological origin of the obstruction in Proposition 7.11.1: the family A_y admits a smooth invertible perturbation only if $\mathcal{D}_Y$ is trivial.

Remark 7.12.4. If $f : G^0 \to \mathbb{C}^*$ were a multiplicative extension of $\det_{\mathrm{Fr}}$ which is smooth along smooth families, then the map $[(A,z)] \mapsto z/f(A)$ would be well defined on $\mathcal{D}$, since

$$\det_{\mathrm{Fr}}(\mathrm{Id}+E)\, z/f(A(\mathrm{Id}+E)) = z/f(A),$$

and would give a smooth non-vanishing linear functional on each $\mathcal{D}_Y$; hence every $\mathcal{D}_Y$ would be trivial and every $[\beta]$ would vanish. So any family with $[\beta] \neq 0$ shows that no such extension exists.

7.13 Exercises

These exercises cover both this chapter and Chapter 6.

Exercise 7.1 (Isotropic Sobolev spaces)**.** For $k \in \mathbb{N}_0$ define

$$H^k_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in L^2(\mathbb{R}^n);\ x^\alpha D_x^\beta u \in L^2(\mathbb{R}^n)\ \forall\, |\alpha| + |\beta| \le k\},$$
$$H^{-k}_{\mathrm{iso}}(\mathbb{R}^n) = \Big\{u \in \mathcal{S}'(\mathbb{R}^n);\ u = \sum_{|\alpha|+|\beta|\le k} x^\alpha D_x^\beta u_{\alpha\beta},\ u_{\alpha\beta} \in L^2(\mathbb{R}^n)\Big\}.$$

Show that if $A \in \Psi^p_{\mathrm{iso,ph}}(\mathbb{R}^n)$, $p \in \mathbb{Z}$, then $A : H^k_{\mathrm{iso}} \to H^{k-p}_{\mathrm{iso}}$ for all $k \in \mathbb{Z}$. Deduce, using elliptic isotropic operators, that the definition

$$H^s_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ Au \in L^2(\mathbb{R}^n)\ \forall\, A \in \Psi^{-s}_{\mathrm{iso,ph}}(\mathbb{R}^n)\}, \quad s \in \mathbb{R},$$

is consistent with the previous one, and that

$$A \in \Psi^M_{\mathrm{iso,ph}}(\mathbb{R}^n) \implies A : H^s_{\mathrm{iso}} \to H^{s-M}_{\mathrm{iso}}, \qquad \bigcap_s H^s_{\mathrm{iso}} = \mathcal{S}(\mathbb{R}^n), \quad \bigcup_s H^s_{\mathrm{iso}} = \mathcal{S}'(\mathbb{R}^n),$$
$$A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n) \text{ elliptic},\ u \in \mathcal{S}'(\mathbb{R}^n),\ Au \in H^s_{\mathrm{iso}} \implies u \in H^{s+m}_{\mathrm{iso}}.$$

Exercise 7.2. Show that if $\epsilon > 0$ then $H^\epsilon_{\mathrm{iso}}(\mathbb{R}^n) \subset \langle x\rangle^{-\epsilon} L^2(\mathbb{R}^n) \cap H^\epsilon(\mathbb{R}^n)$, and deduce that the inclusion $H^\epsilon_{\mathrm{iso}}(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n)$ is compact.

Exercise 7.3. Using Exercise 7.2, or otherwise, show that each element of $\Psi^{-\epsilon}_{\mathrm{iso,ph}}(\mathbb{R}^n)$, $\epsilon > 0$, is compact on $L^2(\mathbb{R}^n)$.

Exercise 7.4. Without using the Fredholm theory of (I4), show that if $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ there is $F \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ such that $(\mathrm{Id}+E)(\mathrm{Id}+F) = \mathrm{Id}+G$ with $G \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ of finite rank, i.e. $G \cdot \mathcal{S}(\mathbb{R}^n)$ is finite dimensional. (Hint: write $E = E_1 + E_2$ with E_1 of finite rank and $\|E_2\|_{\mathcal{B}(L^2)} < 1$, and use Lemma 7.1.4.)

Exercise 7.5. Using Exercise 7.4, show that an elliptic $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ has a parametrix $B \in \Psi^{-m}_{\mathrm{iso,ph}}(\mathbb{R}^n)$ up to finite rank error, i.e. $AB - \mathrm{Id}$ and $BA - \mathrm{Id}$ are finite rank elements of $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. Deduce that A is Fredholm as an operator $H^M_{\mathrm{iso}}(\mathbb{R}^n) \to H^{M-m}_{\mathrm{iso}}(\mathbb{R}^n)$ for every M. (An operator between Hilbert spaces is Fredholm if it is bounded, with finite dimensional null space and closed range of finite codimension.)

Exercise 7.6 (The harmonic oscillator)**.** Show that $H = |D|^2 + |x|^2 = \sum_j (D_j^2 + x_j^2)$ is an elliptic element of $\Psi^2_{\mathrm{iso,ph}}(\mathbb{R}^n)$. With the creation and annihilation operators $C_j = D_j + ix_j$ and $A_j = D_j - ix_j = C_j^*$, show that

$$H = \sum_{j=1}^n C_j A_j + n = \sum_{j=1}^n A_j C_j - n,$$
$$[A_j, H] = 2A_j, \quad [C_j, H] = -2C_j, \quad [C_l, C_j] = 0, \quad [A_l, A_j] = 0, \quad [A_l, C_j] = 2\delta_{lj}\,\mathrm{Id}\,.$$

Knowing that $(H-\lambda)u = 0$ with $u \in \mathcal{S}'(\mathbb{R}^n)$ implies $u \in \mathcal{S}(\mathbb{R}^n)$ (why?), show that

$$E_\lambda = \{u \in \mathcal{S}'(\mathbb{R}^n);\ (H-\lambda)u = 0\} \neq \{0\} \iff \lambda \in n + 2\mathbb{N}_0,$$
$$E_{n+2k} = \Big\{ \sum_{|\alpha|=k} c_\alpha C^\alpha e^{-|x|^2/2};\ c_\alpha \in \mathbb{C}\Big\}, \quad k \in \mathbb{N}_0.$$

Exercise 7.7 (Determinants of matrices)**.** Show that there is a unique function $\det : M(N;\mathbb{C}) \to \mathbb{C}$ which is linear in each column, vanishes when two columns are equal, and satisfies $\det \mathrm{Id} = 1$, and that $\det A = \sum_\sigma \mathrm{sgn}(\sigma) \prod_i A_{i\sigma(i)}$. Show that $\det(AB) = \det A \det B$ (consider $A \mapsto \det(AB)$ for fixed B), that A is invertible if and only if $\det A \neq 0$, that $\frac{d}{ds}\det(\mathrm{Id}+sA)|_{s=0} = \mathrm{tr}\, A$, and deduce Jacobi's formula (7.3.1). Show also that det is the unique multiplicative polynomial function on $M(N;\mathbb{C})$ with $\frac{d}{ds}\det(\mathrm{Id}+sA)|_{s=0} = \mathrm{tr}\, A$.

Exercise 7.8 (The 1-form α is closed)**.** Let $A(s,t)$ be a smooth two-parameter family in $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and $\alpha = \mathrm{Tr}(A^{-1}\partial_s A)\,ds + \mathrm{Tr}(A^{-1}\partial_t A)\,dt$ the pull-back of $\mathrm{Tr}(A^{-1}dA)$. Show that

$$d\alpha = \mathrm{Tr}([A^{-1}\partial_t A, A^{-1}\partial_s A])\,ds \wedge dt,$$

and hence that $d\alpha = 0$. Explain why this implies that $\mathrm{Tr}(A^{-1}dA)$ is closed on $G^{-\infty}_{\mathrm{iso}}$, and show directly, using Lemma 7.3.2, that its integral over any closed path lies in $2\pi i\mathbb{Z}$.

Exercise 7.9 (Bundle coefficients)**.** If V and W are finite dimensional complex vector spaces, show that the space $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; V, W)$ of operators from $\mathcal{S}(\mathbb{R}^n; V)$ to $\mathcal{S}'(\mathbb{R}^n; W)$ given by matrices with entries in $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$, with respect to bases of V and W, is independent of the choice of bases. Extend the results of Chapter 6 to these spaces.

Exercise 7.10 (Ellipticity is necessary for the Fredholm property)**.** Let $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$ be Fredholm as an operator $H^s_{\mathrm{iso}} \to H^{s-m}_{\mathrm{iso}}$ for some s. Show that A is elliptic, as follows.

1. Reduce to the case $m = s = 0$, using invertible elliptic operators of appropriate orders.
2. If the principal symbol of A is not invertible at $\theta \in \mathbb{S}^{2n-1}$, construct a sequence $u_k \in \mathcal{S}$ with $\|u_k\|_{L^2} = 1$, $u_k \to 0$ weakly and $Au_k \to 0$ in L^2 (consider $u_k = e^{ik^2 x\cdot\xi_0}\phi(x/k - kx_0)v$, suitably normalized, with $(x_0, \xi_0) = \theta$ and v in the null space of the principal symbol at θ).
3. Show that a Fredholm operator satisfies $\|u\| \leq C\|Au\| + C\|\Pi u\|$ with Π the projection onto its finite dimensional null space, and derive a contradiction.

Exercise 7.11 (The Koszul complex)**.** Let $\Lambda^k = \Lambda^k \mathbb{C}^n$ with basis $dx^\alpha = dx^{\alpha_1} \wedge \cdots \wedge dx^{\alpha_k}$, $\alpha_1 < \cdots < \alpha_k$, declared orthonormal, and let $\mathcal{S}'(\mathbb{R}^n; \Lambda^k)$ be the space of $u = \sum_\alpha u_\alpha \, dx^\alpha$, $u_\alpha \in \mathcal{S}'(\mathbb{R}^n)$. Using the annihilation operators of Exercise 7.6 define

$$D : \mathcal{S}'(\mathbb{R}^n; \Lambda^k) \to \mathcal{S}'(\mathbb{R}^n; \Lambda^{k+1}), \qquad Du = \sum_{j=1}^n \sum_\alpha A_j u_\alpha \, dx^j \wedge dx^\alpha.$$

Show that $D^2 = 0$, and that the formal adjoint of D with respect to the L^2 pairing is $D^* u = \sum_{j,\alpha} C_j u_\alpha \, \iota_j dx^\alpha$, where ι_j is contraction with e_j, the adjoint of $dx^j \wedge$. Show that $D + D^* \in \Psi^1_{\mathrm{iso,ph}}(\mathbb{R}^n; \Lambda^*)$ is elliptic and that

$$(D + D^*)^2 = DD^* + D^* D = H - n + 2\mathcal{N},$$

where $\mathcal{N} = k$ on Λ^k. Deduce that the null space of $D + D^*$ on $\mathcal{S}'(\mathbb{R}^n; \Lambda^*)$ is spanned by $e^{-|x|^2/2} \in \mathcal{S}(\mathbb{R}^n; \Lambda^0)$, and that

$$\{u \in \mathcal{S}'(\mathbb{R}^n);\ Du = 0\} = \mathbb{C} e^{-|x|^2/2},$$
$$\{u \in \mathcal{S}'(\mathbb{R}^n; \Lambda^k);\ Du = 0\} = D\,\mathcal{S}'(\mathbb{R}^n; \Lambda^{k-1}),\ k \geq 1.$$

Conclude that $D + D^*$, as an elliptic element of $\Psi^1_{\mathrm{iso,ph}}(\mathbb{R}^n; \Lambda^{\mathrm{even}}, \Lambda^{\mathrm{odd}})$, has index 1 (and as an operator from odd to even forms, index -1).

Exercise 7.12 (Isotropic essential support)**.** For $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ with left symbol a, define the isotropic essential support (operator wavefront set) $\mathrm{WF}'_{\mathrm{iso}}(A) \subset \mathbb{R}^{2n} \setminus \{0\}$ as the complement of the set of $Z_0 \neq 0$ having a conic neighbourhood Γ in which a is of order $-\infty$: $|D^\alpha a(Z)| \leq C_{\alpha,k}\langle Z\rangle^{-k}$ for $Z \in \Gamma$ and all α, k. Show that $\mathrm{WF}'_{\mathrm{iso}}(A)$ is closed and conic, that it is unchanged if the left symbol is replaced by the right or Weyl symbol, and that

$$\mathrm{WF}'_{\mathrm{iso}}(A + B) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cup \mathrm{WF}'_{\mathrm{iso}}(B), \qquad \mathrm{WF}'_{\mathrm{iso}}(AB) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cap \mathrm{WF}'_{\mathrm{iso}}(B),$$
$$\mathrm{WF}'_{\mathrm{iso}}(A) = \emptyset \iff A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n).$$

Exercise 7.13 (Isotropic partition of unity)**.** Let U_i be a finite open cover of $\mathbb{S}^{2n-1}$ and $\tilde{U}_i = \{Z \in \mathbb{R}^{2n} \setminus \{0\};\ Z/|Z| \in U_i\}$ the corresponding conic cover of $\mathbb{R}^{2n} \setminus \{0\}$. Show that there are $A_i \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n)$ with $\mathrm{WF}'_{\mathrm{iso}}(A_i) \subset \tilde{U}_i$ and $\mathrm{Id} - \sum_i A_i \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Exercise 7.14. Suppose $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ is elliptic and has index zero. Show that the operator $A + F$ of Proposition 7.4.2 is an isomorphism of $\mathcal{S}(\mathbb{R}^n)$, of $\mathcal{S}'(\mathbb{R}^n)$, and from $H^s_{\mathrm{iso}}(\mathbb{R}^n)$ to $H^{s-m}_{\mathrm{iso}}(\mathbb{R}^n)$ for every s.

Exercise 7.15 (Isotropic wavefront set)**.** For $A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n)$ let $\mathrm{Char}_{\mathrm{iso}}(A) = \{Z \neq 0;\ \sigma_0(A)(Z/|Z|) = 0\}$, where $\sigma_0(A)$ is the principal symbol. For $u \in \mathcal{S}'(\mathbb{R}^n)$ define

$$\mathrm{WF}_{\mathrm{iso}}(u) = \bigcap \{\mathrm{Char}_{\mathrm{iso}}(A);\ A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n),\ Au \in \mathcal{S}(\mathbb{R}^n)\}.$$

Show that $\mathrm{WF}_{\mathrm{iso}}(u)$ is a closed cone, that $\mathrm{WF}_{\mathrm{iso}}(u) = \emptyset$ if and only if $u \in \mathcal{S}(\mathbb{R}^n)$, and that $\mathrm{WF}_{\mathrm{iso}}(Bu) \subset \mathrm{WF}_{\mathrm{iso}}(u) \cap \mathrm{WF}'_{\mathrm{iso}}(B)$ for $B \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$. (Use Exercises 7.12 and 7.13.)

Exercise 7.16 (Traces and determinants of lower regularity)**.** Let $m < -2n$. Show that Tr is continuous on $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, that $\mathrm{Id} + E$ with $E \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ invertible on L^2 has inverse $\mathrm{Id} + F$ with $F \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, and that the definition of $\det_{\mathrm{Fr}}$ by integration of $\mathrm{Tr}(A^{-1}dA)$ along paths extends to $\mathrm{Id} + \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ with the properties of Proposition 7.3.4 and Theorem 7.3.6. (Hint: approximate E by $\mathrm{Op}(\chi(Z/k)e)$ as in the proof of Lemma 7.2.2, which converges in $\Psi^{m+\epsilon}_{\mathrm{iso}}$, and show that the corresponding determinants converge.)

Notes

The main departures from the original notes are as follows. The Fredholm determinant is constructed by integrating $\mathrm{Tr}(A^{-1}dA)$ along paths and comparing with finite dimensional determinants via the harmonic oscillator basis (Lemma 7.3.2); this gives path independence and the limit formula (7.3.3) at once, and the holomorphy across non-invertible operators is reduced, via Lemma 7.3.5, to a finite dimensional determinant. The residue trace is normalized as $-\mathrm{Res}_{z=0}$, so that (7.7.2) holds with a positive sign; this agrees with the usual convention for Wodzicki's residue [13], since z here corresponds to $-s$ in the zeta-function regularization $\mathrm{Tr}(AQ^{-s})$. The exterior derivation is defined directly as commutation with $\dot{A}$, which avoids inverting the regularizing family away from $z = 0$. The determinant line bundle is constructed over finite dimensional families, where it is an honest smooth line bundle, and we have added the connection of Theorem 7.12.3, whose curvature is the obstruction form of Proposition 7.11.1.

For the residue trace on compact manifolds see [13, 5]; for the isotropic (Shubin) calculus and its spectral theory see [15]; the determinant bundle originates in [10]. The Chern–Weil description of the first Chern class used at the end of Section 7.12 can be found, for instance, in Appendix C of J. Milnor and J. Stasheff, *Characteristic classes*.

Part III

Boundary problems and propagation

Chapter 8

Hamilton–Jacobi theory and transport equations

Elliptic operators have no singular solutions: if P is elliptic and $Pu \in \mathcal{C}^\infty$ then $u \in \mathcal{C}^\infty$ (Chapter 5). For non-elliptic operators, such as the wave operator, singular solutions do exist, and the simplest of them are conormal distributions. In this chapter we find out when, and how, one can solve $Pu \in \mathcal{C}^\infty$ with $u \in I^m(X, Y; \Omega^{\frac{1}{2}})$ conormal to a hypersurface Y. The principal symbol of Pu vanishes exactly when Y is *characteristic*; characteristic hypersurfaces are found by solving the eikonal equation $p(x, dh) = 0$, which we do by the Hamilton–Jacobi method, i.e. by symplectic geometry on T^*X. The next term in the symbol of Pu is then given by a first-order differential operator along the Hamilton vector field of p on N^*Y, and requiring it to vanish leads to the *transport equations*. Solving these iteratively produces solutions of $Pu \in \mathcal{C}^\infty$ with prescribed restriction to a transversal hypersurface ("travelling waves"); this is used in Chapter 9 to construct a parametrix for the wave equation.

We use the conormal distributions of Chapter 4 (whose notation and normalization of orders we follow) and the pseudodifferential operators on manifolds of Chapter 5. Throughout, X is a manifold of dimension n, $\pi : T^*X \to X$ is the projection, and operators act on half-densities.

8.1 Pseudodifferential operators on conormal distributions

Theorem 8.1.1. *Let $Y \subset X$ be a closed embedded submanifold and $A \in \Psi^{m'}(X; \Omega^{\frac{1}{2}})$ properly supported. Then*

$$A : I^m(X, Y; \Omega^{\frac{1}{2}}) \longrightarrow I^{m+m'}(X, Y; \Omega^{\frac{1}{2}}), \qquad \sigma_{m+m'}(Au) = \sigma_{m'}(A)|_{N^*Y}\, \sigma_m(u),$$

and A preserves the classical subspaces if it has a polyhomogeneous symbol. The product on the right makes sense since $\sigma_{m'}(A) \in S^{m'-[1]}(T^*X)$ restricts to a class in $S^{m'-[1]}(N^*Y)$.

Proof. By a partition of unity we may assume u has compact support in an adapted chart. Since A is properly supported and its kernel is smooth off the diagonal, ϕAu is smooth if $\phi \in \mathcal{C}^\infty$ vanishes near $\operatorname{supp} u$, and for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$ of the chart, $\phi A\chi$ (with $\chi = 1$ near $\operatorname{supp} u$) is, in the chart, $\operatorname{Op}(b)$ plus a smoothing operator, with $b \in S^{m'}(\mathbb{R}^n; \mathbb{R}^n)$ compactly supported in z (Chapter 5). The result therefore follows from the local computation of Chapter 4: if $u = (\mathcal{I}[a] + f)|dz|^{1/2}$ then $\operatorname{Op}(b)u$ has full symbol congruent to $b(0, y, \xi, 0)a(y, \xi)$ modulo $S^{M+m'-1}$, and $b(0, y, \xi, 0)$ represents $\sigma_{m'}(A)$ at $(0, y; \xi, 0) \in N^*Y$. Here and below $\mathcal{I}[a](x, y) = \int e^{ix\cdot\xi} a(y, \xi) \bar{d}\xi$ in adapted coordinates $(x, y) \in \mathbb{R}^k \times \mathbb{R}^d$, and $M = m + \frac{n}{4} - \frac{k}{2}$. □

In particular, if A is elliptic and $\sigma_m(u)$ is not in $S^{m+\frac{n}{4}-1}$, then $Au \notin \mathcal{C}^\infty$: there are no conormal solutions of $Au \in \mathcal{C}^\infty$. Suppose instead that $P \in \Psi^{m'}(X;\Omega^{\frac12})$ is not elliptic, with homogeneous principal symbol p, and that we look for $u \in I^m(X,Y;\Omega^{\frac12})$ which is genuinely singular, say with $\sigma_m(u)$ elliptic (bounded below by $c\langle\xi\rangle^{m+\frac{n}{4}}|dy\,d\xi|^{1/2}$ for large ξ), and $Pu \in \mathcal{C}^\infty$. By Theorem 8.1.1, $p|_{N^*Y}\sigma_m(u) = 0$ modulo lower order, so p must vanish on N^*Y.

Definition 8.1.2. Let $P \in \Psi^{m'}(X;\Omega^{\frac12})$ have principal symbol p, homogeneous of degree m' on $T^*X \setminus 0$, and characteristic variety $\Sigma(P) = \{p = 0\} \subset T^*X \setminus 0$. A closed embedded submanifold Y is *characteristic* for P if $N^*Y \setminus 0 \subset \Sigma(P)$, and characteristic on an open cone $\Gamma \subset N^*Y \setminus 0$ if $\Gamma \subset \Sigma(P)$.

Corollary 8.1.3. *If Y is characteristic for P on the open cone Γ, and $u \in I^m(X,Y;\Omega^{\frac12})$ has $\mathrm{WF}(u) \subset \Gamma$, then $Pu \in I^{m+m'-1}(X,Y;\Omega^{\frac12})$.*

Proof. By Chapter 4 (the characterization of the wavefront set of a conormal distribution by its full symbol), locally the full symbol a of u is of order $-\infty$ outside any conic neighbourhood of the closed cone $\mathrm{WF}(u)$; choosing a cut-off ψ homogeneous of degree 0, equal to one near $\mathrm{WF}(u)$ and supported in Γ, $a = \psi a$ modulo $S^{-\infty}$, and $p\psi = 0$ for large ξ. Hence $\sigma_{m+m'}(Pu) = p|_{N^*Y}\sigma_m(u) = 0$ and $Pu \in I^{m+m'-1}$ by the exact sequence of Chapter 4. □

So for characteristic Y we only gain one order; we return to this in Section 8.4. First we note that characteristic hypersurfaces and singular solutions do exist.

Example 8.1.4 (Plane waves)**.** Let $P = D_t^2 - \sum_{j=1}^n D_{x_j}^2 = D_t^2 - \Delta$ be the wave operator on $\mathbb{R}^{n+1} = \mathbb{R}_t \times \mathbb{R}^n_x$, acting on half-densities via the trivialization $|dt\,dx|^{1/2}$, so $p(t,x;\tau,\xi) = \tau^2 - |\xi|^2$. A hypersurface $H = \{h(t,x) = 0\}$ with $dh \neq 0$ on H has $N^*H = \{(t,x;\lambda\,dh(t,x));\ h(t,x) = 0,\ \lambda \neq 0\}$, so H is characteristic if and only if h satisfies the *eikonal equation*

$$\Big(\frac{\partial h}{\partial t}\Big)^2 = |d_x h|^2 \quad \text{on } \{h = 0\}.$$

Linear solutions are $h = at + b\cdot x$ with $a^2 = |b|^2$; up to scaling these are the planes $t = \omega\cdot x$, $\omega \in \mathbb{S}^{n-1}$. For $v \in I^{m_0}(\mathbb{R},\{0\})$ (a conormal distribution at a point, in one variable),

$$u = v(t-\omega\cdot x)\,|dt\,dx|^{1/2} \in I^{m_0-\frac{n}{4}}(\mathbb{R}^{n+1},H;\Omega^{\frac12}), \qquad Pu = (1-|\omega|^2)v''(t-\omega\cdot x)|dt\,dx|^{1/2} = 0.$$

(The order follows from the linear change of coordinates $s = t-\omega\cdot x$, $x' = x$, of determinant one, and the relation $M = m + \frac{n+1}{4} - \frac12$ between the order of u and that of its symbol, which is $m_0 - \frac14$.) For instance $\delta(t-\omega\cdot x)$ is a travelling wave. We shall find all characteristic hypersurfaces locally, by the Hamilton–Jacobi method.

8.2 Symplectic geometry

8.2.1 The canonical forms

The cotangent bundle carries a canonical one-form α, defined by

$$\alpha_{(x,\zeta)}(v) = \zeta(\pi_* v), \qquad v \in T_{(x,\zeta)}(T^*X).$$

If $x_1,\dots,x_n$ are local coordinates on X and $\zeta = \sum_j \xi_j dx_j$, then (x,ξ) are coordinates on T^*X and $\alpha = \sum_j \xi_j\,dx_j$. The *symplectic form* is

$$\omega = d\alpha = \sum_{j=1}^n d\xi_j \wedge dx_j.$$

A section $\theta : X \to T^*X$ is the same thing as a one-form on X, and the defining property of α gives $\theta^*\alpha = \theta$, hence $\theta^*\omega = d\theta$.

A *symplectic manifold* is a manifold S with a closed two-form ω which is non-degenerate at each point; then $\dim S = 2n$ is even and $\omega^{\wedge n} \neq 0$. A *symplectic vector space* is a real vector space E with a non-degenerate antisymmetric bilinear form ω; non-degeneracy means that $\tilde{\omega} : E \to E'$, $\tilde{\omega}(v)(w) = \omega(v, w)$, is an isomorphism. Each tangent space T_zS of a symplectic manifold is a symplectic vector space. For a subspace $W \subset E$ set $W^\omega = \{v \in E;\ \omega(v, w) = 0\ \forall\ w \in W\}$. A subspace is *Lagrangian* if $\omega|_{W\times W} = 0$ and $2 \dim W = \dim E$.

Lemma 8.2.1. *If W is a subspace of a symplectic vector space E then* $\dim W + \dim W^\omega = \dim E$*, and W is Lagrangian if and only if $W^\omega = W$.*

Proof. Let $W^0 \subset E'$ be the annihilator of W, of dimension $\dim E - \dim W$. By definition $v \in W^\omega$ if and only if $\tilde{\omega}(v) \in W^0$, so $\tilde{\omega}$ restricts to an isomorphism $W^\omega \to W^0$ and the dimension formula follows. If W is Lagrangian then $W \subset W^\omega$ and the two have the same dimension, so they are equal. Conversely if $W = W^\omega$ then ω vanishes on W, and $2 \dim W = \dim E$ by the dimension formula. □

A submanifold Λ of a symplectic manifold is *Lagrangian* if $T_\lambda\Lambda$ is a Lagrangian subspace of $T_\lambda S$ for every $\lambda \in \Lambda$, i.e. $\dim \Lambda = \frac{1}{2} \dim S$ and the pull-back of ω to Λ vanishes.

Lemma 8.2.2. *The fibres T^*_xX and, for a submanifold $Y \subset X$, the conormal bundle N^*Y are Lagrangian submanifolds of T^*X.*

Proof. On a fibre $\pi_*v = 0$ for tangent vectors, so α pulls back to zero, hence so does $\omega = d\alpha$. On N^*Y, if v is tangent at (y, ζ) then $\pi_*v \in T_yY$ and ζ annihilates T_yY, so again α pulls back to zero. Both have dimension n. □

Lemma 8.2.3. *Let $\Lambda \subset T^*X$ be a submanifold of dimension n such that $\pi|_\Lambda$ is a local diffeomorphism near $\bar{\lambda} \in \Lambda$. Then near $\bar{\lambda}$, $\Lambda = \{(x, dh(x))\}$ for some h defined near $\pi(\bar{\lambda})$, unique up to an additive constant, if and only if Λ is Lagrangian near $\bar{\lambda}$.*

Proof. Near $\bar{\lambda}$, Λ is the image of a section θ of T^*X over a neighbourhood U of $\pi(\bar{\lambda})$, and $\theta : U \to \Lambda$ is a diffeomorphism with $\theta^*\omega = d\theta$. So Λ is Lagrangian if and only if the one-form θ is closed, which on a ball is equivalent to $\theta = dh$. □

8.2.2 Hamilton vector fields

On a symplectic manifold the isomorphism $\tilde{\omega}$ associates to each function f its *Hamilton vector field* H_f, defined by

$$\omega(v, H_f) = df(v) \quad \forall\ v \in T_zS.$$

On T^*X, in canonical coordinates,

$$H_f = \sum_{j=1}^{n} \Big(\frac{\partial f}{\partial \xi_j} \frac{\partial}{\partial x_j} - \frac{\partial f}{\partial x_j} \frac{\partial}{\partial \xi_j} \Big). \tag{8.2.1}$$

By antisymmetry $H_f f = df(H_f) = \omega(H_f, H_f) = 0$: f is constant along the integral curves of H_f. The flow $\exp(sH_f)$ preserves ω, since by Cartan's formula $\mathcal{L}_{H_f}\omega = d\iota_{H_f}\omega + \iota_{H_f}d\omega = -d\,df = 0$. If f is real, H_f is a real vector field; if f is homogeneous of degree μ in the fibre variables then by (8.2.1) H_f is homogeneous of degree $\mu - 1$ (its ∂_x-components are

homogeneous of degree $\mu-1$ and its ∂_ξ-components of degree μ), and its flow commutes with the dilations of the fibres up to reparametrization. The condition $d_{\mathrm{fib}}f \neq 0$, i.e. $(\partial f/\partial\xi_1,\dots,\partial f/\partial\xi_n)\neq 0$, means precisely that $\pi_* H_f \neq 0$, i.e. H_f is transversal to the fibres of T^*X.

Lemma 8.2.4. *If $\Lambda\subset S$ is Lagrangian and $f\in\mathcal{C}^\infty(S)$ vanishes on Λ, then H_f is tangent to Λ.*

Proof. For $v\in T_\lambda\Lambda$, $\omega(v,H_f)=df(v)=0$; so $H_f(\lambda)\in(T_\lambda\Lambda)^\omega=T_\lambda\Lambda$ by Lemma 8.2.1. □

In particular, if Y is characteristic for P on $\Gamma\subset N^*Y$, then H_p is tangent to Γ: a characteristic conormal bundle is a union of integral curves of H_p (bicharacteristics). We use this to construct characteristic hypersurfaces.

8.3 Hamilton–Jacobi theory

Let $p\in\mathcal{C}^\infty(W)$ be a real-valued function on an open set $W\subset T^*X$. We seek solutions of the *eikonal equation*

$$p(x,dh(x))=0 \tag{8.3.1}$$

in a neighbourhood of a point $\bar x$, with h prescribed on a hypersurface through $\bar x$. Note that (8.3.1) is required on an open set, not only on $\{h=0\}$. By Lemma 8.2.3, solving it amounts to finding a Lagrangian submanifold $\Lambda\subset p^{-1}(0)$ which projects diffeomorphically to a neighbourhood of $\bar x$; by Lemma 8.2.4 such a Λ must be a union of integral curves of H_p.

Theorem 8.3.1 (Hamilton–Jacobi)**.** *Let $p\in\mathcal{C}^\infty(W)$ be real, $\bar z\in W$ with $p(\bar z)=0$, $\bar x=\pi(\bar z)$, and let $H=\{t=0\}$ be a hypersurface through $\bar x$, $t\in\mathcal{C}^\infty(X)$ real with $dt\neq 0$, such that*

$$H_p(\pi^*t)(\bar z)\neq 0. \tag{8.3.2}$$

Let $f\in\mathcal{C}^\infty(H)$ satisfy $df(\bar x)=\bar z|_{T_{\bar x}H}$. Then there are a neighbourhood O of $\bar x$ and $h\in\mathcal{C}^\infty(O)$ such that

$$p(x,dh(x))=0 \quad \text{in } O, \qquad h|_{H\cap O}=f|_{H\cap O}, \qquad dh(\bar x)=\bar z,$$

and any other such solution coincides with h near $\bar x$. Moreover the graph of dh is the union of the integral curves of H_p through the submanifold Λ_0 defined in (8.3.3) below.

Proof. Choose coordinates (t,x') near $\bar x=0$, $x'\in\mathbb{R}^{n-1}$, with dual coordinates (τ,ξ'); then $H=\{t=0\}$, $\bar z=(0,0;\bar\tau,\bar\xi')$ with $\bar\xi'=d_{x'}f(0)$, and (8.3.2) says $\partial_\tau p(\bar z)\neq 0$.

Initial manifold. By the implicit function theorem there is a unique smooth function $\tau(x')$, defined near 0, with $\tau(0)=\bar\tau$ and $p(0,x';\tau(x'),d_{x'}f(x'))=0$. Set

$$\Lambda_0=\{(0,x';\tau(x'),d_{x'}f(x'))\}, \tag{8.3.3}$$

an $(n-1)$-dimensional submanifold of $p^{-1}(0)$. It consists exactly of the points of $p^{-1}(0)$ near $\bar z$ lying over H whose restriction to TH is df. The pull-back of ω to Λ_0 is $d\tau\wedge dt+\sum d\xi'_j\wedge dx'_j$ restricted, which is $0+d(d_{x'}f)=0$ since $dt=0$ on Λ_0.

Flow-out. Let $\Phi_s=\exp(sH_p)$ and $\Psi(s,x')=\Phi_s(0,x';\tau(x'),d_{x'}f(x'))$ for $|s|<\epsilon$ and x' near 0. At $(0,0)$, $\partial_s\Psi=H_p(\bar z)$, whose projection to X has ∂_t-component $\partial_\tau p(\bar z)\neq 0$, while $\partial_{x'_j}\Psi$ projects to $\partial_{x'_j}$. Hence $\pi\circ\Psi$ has invertible differential at $(0,0)$, so, shrinking the domain, Ψ is an embedding onto an n-dimensional submanifold Λ and $\pi|_\Lambda$ is a diffeomorphism onto a neighbourhood O of $\bar x$. Since $H_pp=0$, $\Lambda\subset p^{-1}(0)$.

Λ is Lagrangian. At a point $z\in\Lambda_0$, $T_z\Lambda=T_z\Lambda_0\oplus\mathbb{R}H_p(z)$; ω vanishes on $T_z\Lambda_0\times T_z\Lambda_0$, $\omega(H_p,H_p)=0$, and for $v\in T_z\Lambda_0$, $\omega(v,H_p)=dp(v)=0$ because $\Lambda_0\subset p^{-1}(0)$. So $T_z\Lambda$ is

Lagrangian. Since $\Psi(s+s',x') = \Phi_s\Psi(s',x')$, the differential of Φ_s maps $T_z\Lambda$ onto $T_{\Phi_s(z)}\Lambda$, and Φ_s preserves ω; so every tangent space of Λ is Lagrangian.

Conclusion. By Lemma 8.2.3, Λ is the graph of dh for some $h \in \mathcal{C}^\infty(O)$, and $p(x,dh) = 0$. The points of Λ over H are those of Λ_0 (as $\pi|_\Lambda$ is injective and $\pi(\Lambda_0) = H \cap O$), so $d(h|_H) = d_{x'}f$; adjusting the constant, $h|_H = f$, and $dh(\bar{x}) = \bar{z}$ since $\bar{z} \in \Lambda_0$.

Uniqueness. If h' is another solution, its graph Λ' is Lagrangian and contained in $p^{-1}(0)$, so H_p is tangent to Λ' by Lemma 8.2.4. The points $(0,x';\partial_t h'(0,x'), d_{x'}f(x'))$ of Λ' over H satisfy $p = 0$ and are close to $\bar{z}$, so by the uniqueness in the implicit function theorem $\partial_t h'(0,x') = \tau(x')$, i.e. $\Lambda_0 \subset \Lambda'$. The integral curves of H_p starting on Λ_0 therefore stay in Λ', so $\Lambda \subset \Lambda'$ near $\bar{z}$. Both are graphs over a neighbourhood of $\bar{x}$, hence $dh = dh'$ there, and $h = h'$ since they agree on H. □

Note that (8.3.2) implies $d_{\mathrm{fib}}p(\bar{z}) \neq 0$; conversely, if $d_{\mathrm{fib}}p(\bar{z}) \neq 0$ then $\pi_* H_p(\bar{z}) \neq 0$ and hypersurfaces H satisfying (8.3.2) exist.

For homogeneous p we obtain characteristic hypersurfaces with prescribed intersection with H. If $G \subset H$ is a hypersurface in H and $\bar{x} \in G$, write $N^*_{\bar{x}}G \subset T^*_{\bar{x}}H$ for its conormal space in H; and for $\bar{z} \in T^*_{\bar{x}}X$ write $\bar{z}_H = \bar{z}|_{T_{\bar{x}}H} \in T^*_{\bar{x}}H$.

Corollary 8.3.2. *Let $p \in \mathcal{C}^\infty(T^*X \setminus 0)$ be real and homogeneous of degree m', $\bar{z} \in T^*X \setminus 0$ with $p(\bar{z}) = 0$, $H = \{t = 0\}$ a hypersurface through $\bar{x} = \pi(\bar{z})$ satisfying (8.3.2), and $G \subset H$ a hypersurface of H through $\bar{x}$ with $\bar{z}_H \in N^*_{\bar{x}}G \setminus 0$. Then there is a hypersurface $\tilde{G}$ of X, defined near $\bar{x}$, with $\tilde{G} \cap H = G$ near $\bar{x}$, $\bar{z} \in N^*\tilde{G}$, and*

$$\Gamma = \{(x, \lambda\, d\tilde{g}(x));\ x \in \tilde{G},\ \lambda > 0\} \subset \Sigma(p),$$

where $\tilde{g}$ is a defining function of $\tilde{G}$ with $d\tilde{g}(\bar{x}) = \bar{z}$; that is, $\tilde{G}$ is characteristic on the half Γ of $N^*\tilde{G} \setminus 0$ containing $\bar{z}$. Moreover $\tilde{G}$ is unique near $\bar{x}$.

Proof. Write $G = \{g_0 = 0\}$ near $\bar{x}$ with $g_0 \in \mathcal{C}^\infty(H)$, $dg_0 \neq 0$; since $\bar{z}_H$ is a nonzero multiple of $dg_0(\bar{x})$ we may rescale g_0 so that $dg_0(\bar{x}) = \bar{z}_H$. Theorem 8.3.1 with $f = g_0$ gives $\tilde{g}$ with $p(x,d\tilde{g}) = 0$, $\tilde{g}|_H = g_0$ and $d\tilde{g}(\bar{x}) = \bar{z} \neq 0$. Then $\tilde{G} = \{\tilde{g} = 0\}$ is a hypersurface near $\bar{x}$, $\tilde{G} \cap H = \{g_0 = 0\} = G$, and $p(x, \lambda d\tilde{g}(x)) = \lambda^{m'} p(x, d\tilde{g}(x)) = 0$ for $\lambda > 0$.

For uniqueness, let $\tilde{G}'$ be another such hypersurface, with half-conormal bundle $\Gamma' \subset \Sigma(p)$ containing $\bar{z}$. Over G, the points of Γ and of Γ' near $\bar{z}$ are covectors in $\Sigma(p)$ whose restrictions to TH lie in N^*G; by (8.3.2) and the implicit function theorem (as for Λ_0 above, now with the extra dilation parameter) these sets coincide near $\bar{z}$; call this $(n-1)$-dimensional manifold Γ_0. Since Γ, Γ' are Lagrangian (Lemma 8.2.2) and contained in $p^{-1}(0)$, H_p is tangent to both, so both contain the flow-out of Γ_0 by H_p. This flow-out is n-dimensional near $\bar{z}$: $H_p(\bar{z})$ is not tangent to Γ_0 since $\pi_* H_p(\bar{z}) \notin T_{\bar{x}}H \supset \pi_* T\Gamma_0$. So it is an open subset of both Γ and Γ', and $\tilde{G} = \pi(\Gamma) = \pi(\Gamma') = \tilde{G}'$ near $\bar{x}$. □

8.4 The transport equation

8.4.1 The symbol of Pu for characteristic Y

We now compute the next term in the symbol of Pu when Y is characteristic. We assume that $P \in \Psi^{m'}(X; \Omega^{\frac{1}{2}})$ is properly supported and has a polyhomogeneous symbol: in local coordinates, acting on coefficients with respect to $|dz|^{1/2}$, $P = \mathrm{Op}(b)$ modulo smoothing operators, with

$$b \sim b_{m'} + b_{m'-1} + \cdots, \qquad b_{m'-j} \text{ homogeneous of degree } m'-j \text{ in } |\zeta| \geq 1.$$

Differential operators are the main examples. The principal symbol is $p = b_{m'}$, and the *subprincipal symbol* is

$$p_s = b_{m'-1} - \frac{1}{2i}\sum_{j=1}^{n}\frac{\partial^2 p}{\partial z_j \partial \zeta_j}. \tag{8.4.1}$$

For operators on half-densities p_s is an invariantly defined function on $T^*X\setminus 0$, homogeneous of degree $m'-1$ [8, Theorem 18.1.33]; we shall only use its restriction to N^*Y, whose invariance follows from the next theorem. Recall that the Lie derivative acts on half-densities on a manifold with coordinates w by $\mathcal{L}_V(\beta|dw|^{1/2}) = (V\beta + \frac{1}{2}(\operatorname{div} V)\beta)|dw|^{1/2}$, $\operatorname{div} V = \sum \partial_{w_j} V_j$.

Theorem 8.4.1. *Let P be as above, $Y \subset X$ a closed embedded submanifold and $\Gamma \subset N^*Y\setminus 0$ an open cone on which $p = 0$, so that H_p is tangent to Γ. If $u \in I^m(X,Y;\Omega^{\frac12})$ and $\mathrm{WF}(u) \subset \Gamma$, then $Pu \in I^{m+m'-1}(X,Y;\Omega^{\frac12})$ and*

$$\sigma_{m+m'-1}(Pu) = \big(-i\mathcal{L}_{H_p} + p_s\big)\sigma_m(u). \tag{8.4.2}$$

Here $\sigma_m(u) \in S^{m+\frac{n}{4}}(N^*Y;\Omega^{\frac12})$ is, modulo $S^{-\infty}$, supported in a closed cone contained in Γ, and $\mathcal{L}_{H_p}$ (with H_p restricted to Γ) and multiplication by p_s map $S^r(\Gamma;\Omega^{\frac12})$ to $S^{r+m'-1}(\Gamma;\Omega^{\frac12})$, since H_p and p_s are homogeneous of degree $m'-1$; so the right side of (8.4.2) is well defined in $S^{m+\frac{n}{4}+m'-1-[1]}(N^*Y;\Omega^{\frac12})$.

Proof. That $Pu \in I^{m+m'-1}$ is Corollary 8.1.3. The formula is local; in adapted coordinates (x,y) with $u = (\mathcal{I}[a]+f)|dz|^{1/2}$, the full symbol a' of Pu is given by the expansion of Chapter 4,

$$a' \sim \sum_{\alpha,\gamma}\frac{1}{\alpha!\gamma!}(-D_\xi)^\alpha\partial_x^\alpha\Big(\partial_\eta^\gamma b(x,y,\xi,0)D_y^\gamma a(y,\xi)\Big)\Big|_{x=0}.$$

Each α or γ lowers the order by one, so modulo $S^{M+m'-2}$ only the terms with $|\alpha|+|\gamma|\le 1$ and the two leading terms of b contribute. Since $p(0,y,\xi,0) = 0$ where a is not of order $-\infty$,

$$\begin{aligned} a' &\equiv b_{m'-1}a + \sum_l \partial_{\eta_l}p\, D_{y_l}a - \sum_j D_{\xi_j}(\partial_{x_j}p\, a)\\ &= -i\Big(\sum_l \partial_{\eta_l}p\,\partial_{y_l}a - \sum_j \partial_{x_j}p\,\partial_{\xi_j}a\Big) + \Big(b_{m'-1} + i\sum_j \partial_{x_j}\partial_{\xi_j}p\Big)a, \end{aligned}$$

all functions of (y,ξ) evaluated at $x=0$, $\eta=0$. On $N^*Y = \{x=0,\ \eta=0\}$ we have $\partial_\xi p = 0$ and $\partial_y p = 0$ (as p vanishes there), so by (8.2.1)

$$H_p|_{N^*Y} = \sum_l \partial_{\eta_l}p\,\partial_{y_l} - \sum_j \partial_{x_j}p\,\partial_{\xi_j},$$

which is the first-order operator above. With respect to the half-density $|dy\,d\xi|^{1/2}$ its divergence is $\sum_l \partial_{y_l}\partial_{\eta_l}p - \sum_j \partial_{\xi_j}\partial_{x_j}p$, so

$$-iH_p a = -i\mathcal{L}_{H_p}a + \frac{i}{2}\Big(\sum_l \partial_{y_l}\partial_{\eta_l}p - \sum_j \partial_{x_j}\partial_{\xi_j}p\Big)a,$$

where $\mathcal{L}_{H_p}a$ denotes the coefficient of $\mathcal{L}_{H_p}(a|dy\,d\xi|^{1/2})$. Substituting,

$$a' \equiv -i\mathcal{L}_{H_p}a + \Big(b_{m'-1} + \frac{i}{2}\sum_{j=1}^{n}\frac{\partial^2 p}{\partial z_j\partial\zeta_j}\Big)a = -i\mathcal{L}_{H_p}a + p_s a$$

modulo $S^{M+m'-2}$, since $\frac{i}{2} = -\frac{1}{2i}$. This is (8.4.2). □

Thus, to obtain $Pu \in I^{m+m'-2}$ we must solve the *transport equation* $(-i\mathcal{L}_{H_p}+p_s)\sigma_m(u) = 0$, a first-order ordinary differential equation along the integral curves of H_p in Γ; the lower order terms are then removed by solving inhomogeneous transport equations. To set up the initial value problem we need restriction to a transversal hypersurface.

8.4.2 Restriction to a transversal hypersurface

Let $t \in \mathcal{C}^\infty(X)$ be real with $dt \neq 0$ on $H = \{t = 0\}$. Restriction of half-densities to H is defined using $|dt|^{1/2}$: in coordinates (t, w),

$$(\phi(t,w)|dt\,dw|^{1/2})\big|_H = \phi(0,w)|dw|^{1/2}; \tag{8.4.3}$$

this is independent of the coordinates w (Exercise 8.2). Let $\tilde{G} \subset X$ be a hypersurface transversal to H, i.e. $dt|_{T\tilde{G}} \neq 0$ at each point of $G = \tilde{G} \cap H$; then G is a hypersurface in H (Exercise 8.1). Near any point of G there are coordinates (x_1, t, y'), $y' \in \mathbb{R}^{n-2}$, with $\tilde{G} = \{x_1 = 0\}$ and $H = \{t = 0\}$. The restriction map $j : N^*\tilde{G}|_G \to N^*G$, $\zeta \mapsto \zeta|_{TH}$, is an isomorphism, in coordinates $(t = 0, y'; \xi_1) \mapsto (y'; \xi_1)$. The set $N^*\tilde{G}|_G = N^*\tilde{G} \cap \pi^{-1}(H)$ is the hypersurface $\{\pi^*t = 0\}$ of $N^*\tilde{G}$, so half-densities on $N^*\tilde{G}$ restrict to it as in (8.4.3), using $|d\pi^*t|^{1/2}$.

Proposition 8.4.2. *With H, $\tilde{G}$, G as above, restriction to H defines a map*

$$I^m(X, \tilde{G}; \Omega^{\frac12}) \ni u \longmapsto u|_H \in I^{m+\frac14}(H, G; \Omega^{\frac12}), \qquad \sigma_{m+\frac14}(u|_H) = (j^{-1})^*(\sigma_m(u)|_{\{\pi^*t=0\}}),$$

mapping I^{m-l} to $I^{m-l+\frac14}$, and $\mathrm{WF}(u|_H) \subset j(\mathrm{WF}(u) \cap \pi^{-1}(H))$.

Proof. Since $\mathrm{WF}(u) \subset N^*\tilde{G}$ and $N^*\tilde{G} \cap N^*H = 0$ by transversality, the restriction (pull-back by the inclusion) of u to H is well defined [7, Theorem 8.2.4]; for half-densities we combine it with (8.4.3). In coordinates as above, $u = (\mathcal{I}[a] + f)|dx_1\,dt\,dy'|^{1/2}$ with $a = a(t, y', \xi_1) \in S^M$, $M = m + \frac{n}{4} - \frac12$, and $\mathcal{I}[a]$ is a smooth function of t with values in distributions in (x_1, y'); so

$$u|_H = (\mathcal{I}[a(0,\cdot)] + f(\cdot, 0, \cdot))|dx_1\,dy'|^{1/2}.$$

Now H has dimension $n-1$ and G codimension one in it, so this is an element of $I^{m'}(H, G; \Omega^{\frac12})$ with $M = m' + \frac{n-1}{4} - \frac12$, i.e. $m' = m + \frac14$. Its symbol is $a(0, y', \xi_1)|dy'\,d\xi_1|^{1/2}$, which is the stated formula since $\sigma_m(u) = a|dt\,dy'\,d\xi_1|^{1/2}$. The wavefront set statement follows from the characterization of WF by the full symbol (Chapter 4). □

8.4.3 Travelling waves

We now construct conormal solutions of $Pu \in \mathcal{C}^\infty$ with prescribed restriction to a transversal hypersurface. We first solve the transport equations.

Lemma 8.4.3. *Under the hypotheses of Corollary 8.3.2, use coordinates (x_1, t, y') near $\bar{x}$ with $\tilde{G} = \{x_1 = 0\}$, $H = \{t = 0\}$, so that (t, y', ξ_1) are coordinates on $N^*\tilde{G}$ and $\Gamma = \{\xi_1 > 0\}$. Let q be a smooth function on Γ homogeneous of degree $m' - 1$. Then there are $\epsilon > 0$ and a neighbourhood U of $\bar{y}' = 0$ such that for every $c \in S^{r+m'-1}$ and $\beta_0 \in S^r$ (classes of half-densities on Γ, respectively on $\Gamma \cap \{t = 0\}$, of the stated orders, supported in $\xi_1 \geq 1$) the problem*

$$(-i\mathcal{L}_{H_p} + q)\beta = c \quad \text{in } \{|t| < \epsilon,\ y' \in U\}, \qquad \beta|_{t=0} = \beta_0$$

has a unique solution $\beta \in S^r$ there (restriction to $t = 0$ taken as in (8.4.3)). If β_0 and c are of order $-\infty$ outside closed cones K_0, K, then β is of order $-\infty$ outside the union of K and the H_p-flow-out of K_0.

Proof. Since H_p is tangent to Γ and homogeneous of degree $m'-1$, in the coordinates (t, y', ξ_1)

$$H_p = \xi_1^{m'-1}\Big(e_0(t,y')\partial_t + \sum_l e_l(t,y')\partial_{y'_l} + e_n(t,y')\xi_1\partial_{\xi_1}\Big),$$

and $e_0(0,0) \neq 0$ by (8.3.2). Writing $\beta = \tilde\beta |dt\, dy'\, d\xi_1|^{1/2}$ and dividing by $-i\xi_1^{m'-1}e_0$, the equation becomes

$$\Big(\partial_t + \sum_l \frac{e_l}{e_0}\partial_{y'_l} + \frac{e_n}{e_0}\xi_1\partial_{\xi_1}\Big)\tilde\beta + \kappa(t,y')\tilde\beta = \tilde c,$$

with κ smooth (the divergence of H_p and q both being $\xi_1^{m'-1}$ times functions of (t,y')) and $\tilde c = i\xi_1^{1-m'}e_0^{-1}c'$, c' the coefficient of c, of the same order as $\tilde\beta$. The vector field in parentheses has integral curves $s \mapsto (s, Y(s,y'_0), \xi_1 E(s,y'_0))$ with $E>0$ smooth, the ξ_1-component being linear in ξ_1 because its coefficient is. Hence the solution is given explicitly by

$$\tilde\beta(s, Y(s,y'_0), \xi_1 E(s,y'_0)) = e^{-k(s,y'_0)}\Big(\tilde\beta_0(y'_0,\xi_1) + \int_0^s e^{k(\sigma,y'_0)}\tilde c(\sigma, Y(\sigma,y'_0), \xi_1 E(\sigma,y'_0))\, d\sigma\Big),$$

$k(s,y'_0) = \int_0^s \kappa(\sigma, Y(\sigma,y'_0))\, d\sigma$, for $|s|<\epsilon$ and y'_0 near 0. The change of variables $(s, y'_0, \xi_1) \mapsto (s, Y, \xi_1 E)$ is a diffeomorphism near $\{s=0,\ y'_0=0\}$ which is linear and positive in the fibre variable, so it preserves symbol classes; hence β has order r, and the support statement is clear from the formula. □

Theorem 8.4.4 (Travelling waves)**.** *Let $P \in \Psi^{m'}(X;\Omega^{\frac12})$ be properly supported with polyhomogeneous symbol and real principal symbol p. Let $\bar z \in T^*X \setminus 0$ with $p(\bar z)=0$, $\bar x = \pi(\bar z)$, let $H=\{t=0\}$ be a hypersurface through $\bar x$ with $H_p(\pi^*t)(\bar z)\neq 0$, let $G\subset H$ be a hypersurface of H with $\bar z_H \in N^*_{\bar x}G\setminus 0$, and let $\tilde G$ be the characteristic hypersurface of Corollary 8.3.2. Then there are a neighbourhood O of $\bar x$ and an open conic neighbourhood γ of $\bar z_H$ in $T^*H\setminus 0$ such that for every $u_0 \in I^{m+\frac14}(H,G;\Omega^{\frac12})$ with $\mathrm{WF}(u_0)\subset\gamma$ there exists $u\in I^m(X,\tilde G;\Omega^{\frac12})$ with*

$$u|_{H\cap O} = u_0|_{H\cap O}, \qquad Pu \in \mathcal{C}^\infty(O;\Omega^{\frac12}),$$

and $\mathrm{WF}(u)\cap T^*O \subset \Gamma$. If P and u_0 are classical, u can be taken classical.

Proof. Use the coordinates of Lemma 8.4.3, and choose ϵ, U as there and γ so small that, in these coordinates, $j^{-1}(\gamma)$ lies over $\{y'\in U\}$ and within a cone whose H_p-flow-out for $|t|<\epsilon$ stays in Γ. Let $O=\{|x_1|<\epsilon', |t|<\epsilon,\ y'\in U'\}$ with $U'\Subset U$, and fix $\phi\in\mathcal{C}^\infty_c(\{|t|<\epsilon, y'\in U\})$ equal to one near $\bar O$. We construct $u_j \in I^{m-j}(X,\tilde G;\Omega^{\frac12})$, with wavefront sets in Γ and supports in $\operatorname{supp}\phi$, such that for $v_N = u'_0+\cdots+u_N$ (we write u'_0 for the first term to avoid confusion with the data u_0)

$$Pv_N \in I^{m+m'-N-2} \text{ in } O, \qquad u_0 - v_N|_H \in I^{m+\frac14-N-1}(H,G;\Omega^{\frac12}) \text{ in } H\cap O. \tag{8.4.4}$$

For $N=-1$ (with $v_{-1}=0$) the second condition holds and the first is vacuous with the convention $Pv_{-1}=0$. Given v_{N-1}, let $g=-Pv_{N-1}\in I^{m+m'-N-1}$ in O and $e = u_0 - v_{N-1}|_H \in I^{m+\frac14-N}(H,G;\Omega^{\frac12})$, both with wavefront sets in the appropriate cones. By Lemma 8.4.3 (with $q=p_s|_\Gamma$) solve

$$(-i\mathcal{L}_{H_p} + p_s)\beta_N = \sigma_{m+m'-N-1}(g), \qquad \beta_N|_{t=0} = (j)^*\sigma_{m+\frac14-N}(e),$$

with $\beta_N \in S^{m+\frac n4-N}$, of order $-\infty$ outside the flow-out cone, and let $u_N\in I^{m-N}$ be ϕ times a distribution with principal symbol β_N (Chapter 4). By Theorem 8.4.1, $\sigma_{m+m'-N-1}(Pu_N) =$

$\sigma_{m+m'-N-1}(g)$ in O (where $\phi = 1$ and, P being pseudolocal, $P(\phi w) - \phi P w$ is smooth), so $Pv_N = -g + Pu_N \in I^{m+m'-N-2}$ in O; and by Proposition 8.4.2, $\sigma_{m+\frac14-N}(u_N|_H) = \sigma_{m+\frac14-N}(e)$, so $e - u_N|_H \in I^{m+\frac14-N-1}$. This proves (8.4.4).

Let $\tilde u \sim \sum_N u_N$ be an asymptotic sum in $I^m(X, \tilde G; \Omega^{\frac12})$, with support in $\operatorname{supp}\phi$. Then $P\tilde u \in \bigcap_N I^{m+m'-N} = \mathcal{C}^\infty$ in O, and $u_0 - \tilde u|_H \in \bigcap_N I^{m+\frac14-N}(H, G) = \mathcal{C}^\infty$ in $H \cap O$, by Proposition 8.4.2. Finally choose $w \in \mathcal{C}^\infty(X; \Omega^{\frac12})$ with $w|_H = u_0 - \tilde u|_H$ near $H \cap \bar O$ (in coordinates, $w = \chi(t)\times$ the smooth error, extended constantly in t), and set $u = \tilde u + w$. Then $u|_{H\cap O} = u_0$, $Pu = P\tilde u + Pw \in \mathcal{C}^\infty(O)$, and $\mathrm{WF}(u) \cap T^*O \subset \Gamma$. In the classical case all symbols above are classical. □

The transport equation (8.4.2) for the leading symbol, and its inhomogeneous versions for the lower order terms, are thus solved by integration along the bicharacteristics (Duhamel's principle). In Chapter 9 this construction, applied to the wave operator with $H = \{t = 0\}$, gives a parametrix for the Cauchy problem.

8.5 Exercises

Exercise 8.1. Let $\tilde G \subset X$ be a hypersurface and $t \in \mathcal{C}^\infty(X)$ real with $dt|_{T_q\tilde G} \neq 0$ for all $q \in G = \tilde G \cap \{t = 0\}$. Show that $H = \{t = 0\}$ is a hypersurface near G and that $G = H \cap \tilde G$ is a submanifold of codimension one in both H and $\tilde G$.

Exercise 8.2. Assuming $dt \neq 0$ on $H = \{t = 0\}$, show that dt gives an isomorphism of line bundles $\Omega^{\frac12}(H) \cong \Omega^{\frac12}_H(X) \otimes |N^*H|^{-1/2}$, where $\Omega^{\frac12}_H(X)$ is the restriction to H of $\Omega^{\frac12}(X)$ and $|N^*H|^{-1/2}$ is trivialized by $|dt|^{-1/2}$, and hence that (8.4.3) defines a coordinate-independent restriction map $\mathcal{C}^\infty(X; \Omega^{\frac12}) \to \mathcal{C}^\infty(H; \Omega^{\frac12})$.

Exercise 8.3. Prove Proposition 8.4.2 directly from the coordinate formula: show that the restriction defined in adapted coordinates is independent of the choice of coordinates (x_1, t, y') with t fixed.

Exercise 8.4. For the wave operator $P = D_t^2 - \Delta$ on $\mathbb{R}^{n+1}$, $H = \{t = 0\}$ and $f(x) = \omega \cdot x$, $|\omega| = 1$, find the two solutions $h_\pm$ of the eikonal equation $p(x, dh) = 0$ with $h|_H = f$ given by Theorem 8.3.1 (one for each choice of $\bar z$ over 0). Do the same with $f(x) = |x|$ near a point $x \neq 0$, and describe the characteristic hypersurfaces through $\{t = 0,\ |x| = 1\}$. Show that the light cone $\{t^2 = |x|^2\}$ is characteristic away from its vertex.

Exercise 8.5. Show that the subprincipal symbol of $D_t^2 - \Delta$ on $\mathbb{R}^{n+1}$, acting on half-densities via $|dt\,dx|^{1/2}$, vanishes, and solve the transport equation for the plane waves of Example 8.1.4. For a Riemannian metric g on X compute the subprincipal symbol of $D_t^2 - \Delta_g$ acting on half-densities by $u \mapsto (D_t^2 - \Delta_g)(u\,|dg|^{-1/2})\,|dg|^{1/2}$, $|dg|$ the Riemannian density.

Notes

This chapter is a rewriting of the second half of the conormal/wave kernel chapter of the 1998 notes. The section on Riemann metrics and quantization there was only a heading and has been omitted. The Hamilton–Jacobi theorem is proved here in full (the original only sketched the strategy), the transport equation is derived for submanifolds of any codimension with the zeroth order term identified as the subprincipal symbol, the restriction

theorem (an exercise in the original) is proved, and the construction of travelling waves is made precise. For the general theory see [8, Sections 18.1, 25.2] and [6], where conormal distributions are replaced by Lagrangian distributions and the analogue of (8.4.2) holds on any Lagrangian submanifold contained in $\Sigma(P)$.

Chapter 9

The wave equation and Weyl asymptotics

In this chapter the theory of conormal distributions (Chapter 4) and the Hamilton–Jacobi and transport equations (Chapter 8) are applied to the wave operator $P = D_t^2 - \Delta$ on $\mathbb{R} \times X$, where X is a compact Riemannian manifold. We first solve the Cauchy problem and construct the forward fundamental solution, with energy estimates and finite speed of propagation. Then we construct, for short times, a parametrix for the wave kernel as a superposition of conormal distributions associated with characteristic hypersurfaces (equivalently, as an oscillatory integral whose phase solves the eikonal equation). Restricting this parametrix to the diagonal and integrating gives the singularity at $t = 0$ of the trace of the wave group, $T(t) = \sum_j \cos(t\lambda_j)$; a Tauberian argument then yields Weyl's asymptotic formula for the eigenvalues of Δ with remainder $O(\lambda^{n-1})$. Finally we explain how the propagation of singularities leads to the Poisson relation: the singularities of T lie at 0 and at the lengths of closed geodesics.

9.1 Setup

9.1.1 The Laplacian on a compact manifold

Throughout, X is a compact $\mathcal{C}^\infty$ manifold without boundary, of dimension $n \geq 1$, with a Riemannian metric g. In local coordinates $g = \sum g_{ij}(x)\, dx^i dx^j$, $(g^{ij}) = (g_{ij})^{-1}$, $|g| = \det(g_{ij})$, the Riemannian measure is $d\mathrm{vol} = |g|^{1/2} dx$ and the (non-negative) Laplacian is

$$\Delta u = -|g|^{-1/2} \sum_{i,j=1}^{n} \partial_i (|g|^{1/2} g^{ij} \partial_j u), \qquad \int_X \Delta u\, \bar{v}\, d\mathrm{vol} = \int_X \langle du, dv \rangle_g \, d\mathrm{vol}.$$

For a covector we write $|\xi|_{g(x)} = (\sum g^{ij}(x)\xi_i\xi_j)^{1/2}$ and set $q(x,\xi) = |\xi|_{g(x)}$. We use $d\mathrm{vol}$ to identify densities with functions: $\mathcal{C}^{-\infty}(X)$ is the dual of $\mathcal{C}^\infty(X)$ under $\langle u, v\rangle = \int_X uv\, d\mathrm{vol}$, and the Schwartz kernel K_A of an operator A is taken with respect to $d\mathrm{vol}$, $Au(x) = \int_X K_A(x,y)u(y)\, d\mathrm{vol}(y)$. In particular $K_{\mathrm{Id}} = |g(y)|^{-1/2}\delta(x-y)$ in local coordinates.

We use the following consequences of the elliptic theory on compact manifolds (Chapter 5; see also [15]).

(F1) There is an orthonormal basis $\{\phi_j\}_{j\geq 0}$ of $L^2(X)$ of real eigenfunctions $\phi_j \in \mathcal{C}^\infty(X)$, $\Delta\phi_j = \lambda_j^2 \phi_j$, with $0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$ (repeated with multiplicity).

(F2) For $s \in \mathbb{R}$, writing $u_j = \langle u, \phi_j\rangle$ and $\langle\lambda\rangle = (1+\lambda^2)^{1/2}$,

$$H^s(X) = \Big\{u \in \mathcal{C}^{-\infty}(X);\ \|u\|_s^2 = \sum_j \langle\lambda_j\rangle^{2s}|u_j|^2 < \infty\Big\}$$

with a norm equivalent to the usual one. Thus $\mathcal{C}^\infty(X) = \bigcap_s H^s(X)$, $\mathcal{C}^{-\infty}(X) = \bigcup_s H^s(X)$, every $u \in \mathcal{C}^{-\infty}(X)$ is the sum of its expansion $\sum_j u_j\phi_j$ (in some H^s), and for $u \in \mathcal{C}^\infty(X)$ the expansion converges in $\mathcal{C}^\infty(X)$. For $s > n/2 + k$ there is the Sobolev embedding $\|u\|_{\mathcal{C}^k} \le C\|u\|_s$.

For a bounded Borel function F on $[0,\infty)$ we set $F(\Lambda)u = \sum_j F(\lambda_j)u_j\phi_j$; here $\Lambda = \Delta^{1/2}$. Such an operator is bounded on every $H^s(X)$ with norm $\le \sup|F|$. We shall use $\cos(t\Lambda)$ and $\Lambda^{-1}\sin(t\Lambda)$, the latter defined by the function $\lambda \mapsto \sin(t\lambda)/\lambda$ (equal to t at $\lambda = 0$). The counting function is

$$N(\lambda) = \#\{j;\ \lambda_j \le \lambda\} = \text{number of eigenvalues of } \Delta \text{ in } [0,\lambda^2], \text{ with multiplicity.}$$

Lemma 9.1.1. *For every $s > n/2$ there is C_s with $\sum_j \langle\lambda_j\rangle^{-2s}|\phi_j(x)|^2 \le C_s^2$ for all $x \in X$; hence $\sum_j \langle\lambda_j\rangle^{-2s} \le C_s^2\,\mathrm{vol}(X)$ and $N(\lambda) \le C_s^2\,\mathrm{vol}(X)\langle\lambda\rangle^{2s}$.*

Proof. By the Sobolev embedding, $u \mapsto u(x)$ is a continuous linear functional on $H^s(X)$ of norm at most C_s, uniformly in x. In the spectral description (F2), $u(x) = \sum_j u_j\phi_j(x)$, and the norm of this functional is exactly $(\sum_j \langle\lambda_j\rangle^{-2s}|\phi_j(x)|^2)^{1/2}$. Integrating over X gives the second statement, and $N(\lambda)\langle\lambda\rangle^{-2s} \le \sum_{\lambda_j\le\lambda}\langle\lambda_j\rangle^{-2s}$ gives the third. □

In particular $\sum_j \langle\lambda_j\rangle^{-M} < \infty$ for $M > n$, and the sup norms of ϕ_j and their derivatives grow at most polynomially in λ_j.

9.1.2 The wave operator and its bicharacteristics

The wave operator is $P = D_t^2 - \Delta$ on $\mathbb{R}\times X$, $D_t = -i\partial_t$. In the canonical coordinates (t,x,τ,ξ) of $T^*(\mathbb{R}\times X) = T^*\mathbb{R}\times T^*X$ its principal symbol is

$$p(t,x,\tau,\xi) = \tau^2 - |\xi|^2_{g(x)},$$

and the characteristic set is $\Sigma(P) = \mathrm{Char}(P) = \{p = 0\}\setminus 0$. On $\Sigma(P)$ we have $\tau = \pm|\xi|_g$ with $\tau \ne 0$, $\xi \ne 0$, so $d_{(\tau,\xi)}p = (2\tau, -2g^{-1}\xi) \ne 0$: P is of real principal type and $\Sigma(P)$ is a smooth conic hypersurface. We use the Hamilton vector field $H_f = \sum_j(\partial_{\xi_j}f\,\partial_{x_j} - \partial_{x_j}f\,\partial_{\xi_j})$ (with the corresponding τ, t terms on $T^*(\mathbb{R}\times X)$); a *null bicharacteristic* of P is an integral curve of H_p in $\Sigma(P)$. Let

$$G_s = \exp(sH_q)\colon T^*X\setminus 0 \to T^*X\setminus 0, \qquad q = |\xi|_g,$$

be the (homogeneous) geodesic flow. It is complete since X is compact, commutes with dilations in ξ and preserves q, and the projection to X of $s \mapsto G_s(x,\xi)$ is the unit speed geodesic with initial velocity $g^{-1}\xi/|\xi|_g$: indeed $H_q = q^{-1}H_{q^2/2}$ on $T^*X\setminus 0$, and Hamilton's equations for $\frac12 q^2$ are the geodesic equations ($\dot x = g^{-1}\xi$ is the Legendre transform).

Lemma 9.1.2. *Along a null bicharacteristic, τ is constant and t is strictly monotone. Parametrised by t, the null bicharacteristic through $(\bar t,\bar x,\tau,\bar\xi)$ with $\tau = \mp|\bar\xi|_g$ is*

$$t \longmapsto (t, G_{\pm(t-\bar t)}(\bar x,\bar\xi),\tau), \qquad t\in\mathbb{R}.$$

In particular its projection to X is a unit speed geodesic.

Proof. $H_p = 2\tau\partial_t - H_{q^2} = 2\tau\partial_t - 2qH_q$, so $\dot\tau = 0$, $\dot t = 2\tau \ne 0$ and $d(x,\xi)/dt = -(q/\tau)H_q = \pm H_q$ when $\tau = \mp q$. □

9.1.3 Conormal distributions and oscillatory integrals

We recall from Chapter 4 the local description of conormal distributions associated with a hypersurface, which is all we need. Let M be a manifold of dimension N and $H \subset M$ a closed embedded hypersurface. Then $u \in I^m(M,H)$ if $u \in \mathcal{C}^{-\infty}(M)$ is smooth on $M \setminus H$ and near each point of H, in local coordinates $(s,z) \in \mathbb{R}\times\mathbb{R}^{N-1}$ with $H=\{s=0\}$,

$$u = \frac{1}{2\pi}\int_{\mathbb{R}} e^{is\rho}a(z,\rho)\,d\rho + \mathcal{C}^\infty, \qquad a \in S^{m+\frac{N}{4}-\frac12}(\mathbb{R}^{N-1}_z;\mathbb{R}_\rho). \tag{9.1.1}$$

(We work with functions rather than half-densities; in fixed coordinates the two differ by a smooth non-vanishing factor.) The class of a modulo symbols of one order lower is the local form of the principal symbol. Any defining function h of H (with $dh \neq 0$ on H) may be used in place of s, since it can be taken as a coordinate; and amplitudes $a(s,z,\rho)$ depending also on s give elements of the same space (Chapter 4; a special case is proved in Lemma 9.5.3 below). For $M=\mathbb{R}$, $H=\{0\}$: a compactly supported $u \in \mathcal{C}^{-\infty}(\mathbb{R})$ lies in $I^m(\mathbb{R},\{0\})$ iff $\hat u \in S^{m-\frac14}(\mathbb{R})$. We always have $\mathrm{WF}(u) \subset N^*H$.

A hypersurface $H=\{h=0\}$ is *characteristic* for P if $N^*H \subset \Sigma(P)$, i.e. $p(dh)=0$ on H; for P this is the eikonal equation $(\partial_t h)^2 = |d_x h|_g^2$. Characteristic hypersurfaces are obtained by Hamilton–Jacobi theory, and the symbols of conormal solutions of $Pu \in \mathcal{C}^\infty$ satisfy transport equations along H_p (Chapter 8). For P we carry this out explicitly in Section 9.4.

Finally we use standard facts about oscillatory integrals [7, Sections 8.1, 8.2]. Let $\Omega \subset \mathbb{R}^d$ be open and $\Phi \in \mathcal{C}^\infty(\Omega\times(\mathbb{R}^N\setminus 0))$ real, homogeneous of degree one in θ, with $d_{z,\theta}\Phi \neq 0$, and let $a \in S^\mu(\Omega;\mathbb{R}^N)$ vanish near $\theta=0$. Then $I(z)=\int e^{i\Phi(z,\theta)}a(z,\theta)\,d\theta$, defined as $\lim_{\epsilon\downarrow 0}\int e^{i\Phi}a(z,\theta)\chi_0(\epsilon\theta)\,d\theta$ for any $\chi_0 \in \mathcal{C}^\infty_{\mathrm c}(\mathbb{R}^N)$ with $\chi_0(0)=1$, is a distribution on Ω, and

$$\mathrm{WF}(I) \subset \{(z, d_z\Phi(z,\theta));\ d_\theta\Phi(z,\theta)=0,\ \theta \in \operatorname{cone\,supp} a\}. \tag{9.1.2}$$

In particular $I \in \mathcal{C}^\infty$ if $a \in S^{-\infty}$, or near points where $d_\theta\Phi \neq 0$ on the support of a. If some of the variables z are parameters t and $d_{z',\theta}\Phi \neq 0$ in the remaining variables, I is a smooth function of t with values in distributions.

9.2 The Cauchy problem

Theorem 9.2.1. *Let $s \in \mathbb{R}$, let $I \subset \mathbb{R}$ be an open interval containing 0, and let $u_0 \in H^s(X)$, $u_1 \in H^{s-1}(X)$, $f \in \mathcal{C}^0(I;H^{s-1}(X))$. There is a unique $u \in \mathcal{C}^0(I;H^s)\cap\mathcal{C}^1(I;H^{s-1})$ with*

$$Pu = f \quad \text{in } I\times X, \qquad u(0)=u_0, \qquad D_t u(0)=u_1, \tag{9.2.1}$$

namely

$$u(t) = \cos(t\Lambda)u_0 + i\Lambda^{-1}\sin(t\Lambda)u_1 - \int_0^t \Lambda^{-1}\sin((t-r)\Lambda)f(r)\,dr. \tag{9.2.2}$$

It satisfies, for $t \in I$, the energy estimate

$$\|u(t)\|_s + \|D_t u(t)\|_{s-1} \leq 4(1+|t|)\Big(\|u_0\|_s + \|u_1\|_{s-1} + \Big|\int_0^t \|f(r)\|_{s-1}dr\Big|\Big), \tag{9.2.3}$$

and if $f=0$ the energy $\|\partial_t u(t)\|^2_{L^2} + \|du(t)\|^2_{L^2}$ is independent of t (for $s \geq 1$). Uniqueness holds in the larger class $\mathcal{C}^2(I;H^{-M}(X))$, any M. If $u_0,u_1 \in \mathcal{C}^\infty(X)$ and $f \in \mathcal{C}^\infty(I\times X)$ then $u \in \mathcal{C}^\infty(I\times X)$.

Proof. Expanding in eigenfunctions, u solves (9.2.1) iff its coefficients $u_j(t) = \langle u(t), \phi_j\rangle$ solve $u_j'' + \lambda_j^2 u_j = -f_j$, $u_j(0) = u_{0,j}$, $u_j'(0) = iu_{1,j}$ (recall $D_t^2 = -\partial_t^2$ and $\langle \Delta u, \phi_j\rangle = \lambda_j^2 u_j$), whose unique solution is the j-th coefficient of (9.2.2). This proves uniqueness in $\mathcal{C}^2(I; H^{-M})$: if the data vanish, every u_j vanishes, hence $u = 0$ by (F2). For the estimate use $|\sin(t\lambda)/\lambda| \le \min(|t|, \lambda^{-1}) \le 2(1+|t|)\langle\lambda\rangle^{-1}$:

$$\langle\lambda_j\rangle^s |u_j(t)| \le \langle\lambda_j\rangle^s |u_{0,j}| + 2(1+|t|)\Big(\langle\lambda_j\rangle^{s-1}|u_{1,j}| + \Big|\int_0^t \langle\lambda_j\rangle^{s-1}|f_j(r)|\,dr\Big|\Big),$$

$$\langle\lambda_j\rangle^{s-1} |D_t u_j(t)| \le \langle\lambda_j\rangle^s |u_{0,j}| + \langle\lambda_j\rangle^{s-1}|u_{1,j}| + \Big|\int_0^t \langle\lambda_j\rangle^{s-1}|f_j(r)|\,dr\Big|,$$

the second because $D_t u_j = i\lambda_j \sin(t\lambda_j)u_{0,j} + \cos(t\lambda_j)u_{1,j} + i\int_0^t \cos((t-r)\lambda_j) f_j(r)\,dr$. Taking ℓ^2 norms in j (and Minkowski's inequality for the integral) gives (9.2.3); the same bounds and dominated convergence show that (9.2.2) lies in $\mathcal{C}^0(I; H^s) \cap \mathcal{C}^1(I; H^{s-1})$. If $f = 0$ then $|u_j'|^2 + \lambda_j^2 |u_j|^2$ is constant for each j, and summing gives the conservation of $\|\partial_t u\|^2 + \langle \Delta u, \bar u\rangle = \|\partial_t u\|^2 + \|du\|^2$. Smoothness is the case $Y = \{\mathrm{pt}\}$ of Proposition 9.2.2. □

The *wave group* is the map $U(t)\colon (u_0, u_1) \mapsto (u(t), D_t u(t))$ for $f = 0$; explicitly

$$U(t) = \begin{pmatrix} \cos t\Lambda & i\Lambda^{-1}\sin t\Lambda \\ i\Lambda \sin t\Lambda & \cos t\Lambda \end{pmatrix}.$$

By Theorem 9.2.1 it is a strongly continuous group of bounded operators on $H^s(X) \oplus H^{s-1}(X)$ for each s: $U(t+t') = U(t)U(t')$ by uniqueness. Its (distributional) trace is $2\sum_j \cos(t\lambda_j)$, see Section 9.6.

We need a version with parameters. Let Y be a compact manifold (possibly a point) and let Δ_X act on the first factor of $X \times Y$.

Proposition 9.2.2. *Let $I \ni 0$ be an open interval. If $w \in \mathcal{C}^2(I; H^{-M}(X \times Y))$ satisfies $(D_t^2 - \Delta_X)w = 0$ and $w(0) = D_t w(0) = 0$, then $w = 0$. If $F \in \mathcal{C}^\infty(I \times X \times Y)$ and $w_0, w_1 \in \mathcal{C}^\infty(X \times Y)$, the solution of $(D_t^2 - \Delta_X)w = F$, $w(0) = w_0$, $D_t w(0) = w_1$, given by (9.2.2) with $\Lambda = \Lambda_X$, lies in $\mathcal{C}^\infty(I \times X \times Y)$.*

Proof. Choose a metric on Y with Laplacian Δ_Y and eigenbasis ψ_l, $\Delta_Y \psi_l = \mu_l^2 \psi_l$. Then $\{\phi_j \otimes \psi_l\}$ is an orthonormal basis of $L^2(X \times Y)$ of eigenfunctions of the Laplacian $\Delta_X + \Delta_Y$ of the product metric, so by (F2) on $X \times Y$, $H^s(X \times Y)$ is characterised by $\sum_{j,l}(1 + \lambda_j^2 + \mu_l^2)^s |w_{jl}|^2 < \infty$. Uniqueness follows as before from the ODEs $w_{jl}'' + \lambda_j^2 w_{jl} = 0$. The operators $\cos(t\Lambda_X)$ and $\Lambda_X^{-1}\sin(t\Lambda_X)$ act on the coefficients by multiplication by functions bounded by $1 + |t|$, so they are bounded on every $H^s(X \times Y)$, locally uniformly and strongly continuously in t. Hence $w \in \mathcal{C}^0(I; H^s(X \times Y))$ for all s; the equation gives $\partial_t^2 w = -\Delta_X w - F \in \mathcal{C}^0(I; H^{s-2})$, and inductively $w \in \mathcal{C}^k(I; H^s(X \times Y))$ for all k, s. By the Sobolev embedding, $w \in \mathcal{C}^\infty(I \times X \times Y)$. □

Theorem 9.2.3 (Finite propagation speed)**.** *Let $K \subset X$ be closed, $K_r = \{x \in X;\ d(x, K) \le r\}$ for $r \ge 0$, where d is the Riemannian distance. If u is the solution of (9.2.1) with $f = 0$ and $\operatorname{supp} u_0 \cup \operatorname{supp} u_1 \subset K$ (any s), then $\operatorname{supp} u(t) \subset K_{|t|}$ for all t.*

Proof. First let $u_0, u_1 \in \mathcal{C}^\infty(X)$, so $u \in \mathcal{C}^\infty(\mathbb{R} \times X)$. Fix $x_0 \in X$ and $0 < R < \operatorname{inj}(X)$ (the injectivity radius, positive as X is compact) with $R < d(x_0, K)$. On the ball $B(x_0, \operatorname{inj} X)$ the function $d(x_0, \cdot)^2$ is smooth, so $\rho_\epsilon = (d(x_0,\cdot)^2 + \epsilon^2)^{1/2}$ is smooth there and $|d\rho_\epsilon|_g = d(x_0,\cdot)/\rho_\epsilon < 1$. Let $\theta_\delta \in \mathcal{C}^\infty(\mathbb{R})$ be non-decreasing, $\theta_\delta = 0$ on $(-\infty, 0]$, $\theta_\delta = 1$ on $[\delta, \infty)$, and $\psi(t, x) = \theta_\delta(R - t - \rho_\epsilon(x))$, a smooth function supported in $\{\rho_\epsilon \le R - t\} \subset B(x_0, R)$

for $t \geq 0$. With the energy density $e = |\partial_t u|^2 + |du|_g^2$ and $\partial_t^2 u = -\Delta u$ we have $\partial_t e = 2\,\mathrm{Re}(\overline{\partial_t u}\,\partial_t^2 u + \langle d\partial_t u, du\rangle_g)$ and, integrating by parts, the terms with two derivatives of u cancel:

$$\frac{d}{dt}\int_X \psi e\,d\mathrm{vol} = \int_X (\partial_t\psi)e\,d\mathrm{vol} - 2\,\mathrm{Re}\int_X \overline{\partial_t u}\,\langle d\psi, du\rangle_g\,d\mathrm{vol} \leq \int_X (\partial_t\psi + |d\psi|_g)e\,d\mathrm{vol} \leq 0,$$

because $\partial_t\psi = -\theta_\delta'$ and $|d\psi|_g = \theta_\delta'|d\rho_\epsilon|_g \leq \theta_\delta'$. At $t = 0$ the integral vanishes since u_0, u_1 vanish on $\bar{B}(x_0, R)$. Letting $\delta, \epsilon \to 0$ we get $e(t,x) = 0$ for $0 \leq t < R$, $d(x_0, x) < R - t$, hence $\partial_t u = 0$ there, and since $u(0) = 0$ on $B(x_0, R)$, $u = 0$ on this cone. Given $0 \leq t < \mathrm{inj}(X)$ and $x_0 \notin K_t$ we may choose $R \in (t, \min(\mathrm{inj}\,X, d(x_0, K)))$, so $u(t, x_0) = 0$: $\mathrm{supp}\,u(t) \subset K_t$ for $0 \leq t < \mathrm{inj}\,X$. Since $(K_a)_b \subset K_{a+b}$ by the triangle inequality, applying this repeatedly (to the solution with data at time t_1, using uniqueness and time-translation invariance) gives all $t \geq 0$; $t \leq 0$ follows by time reversal $t \mapsto -t$.

For distributional data use duality: $\cos(t\Lambda)$ and $\Lambda^{-1}\sin(t\Lambda)$ are symmetric for the bilinear pairing (the ϕ_j are real). If $\phi \in \mathcal{C}^\infty(X)$ has $\mathrm{supp}\,\phi \cap K_{|t|} = \emptyset$, then $d(\mathrm{supp}\,\phi, K) > |t|$, so by the smooth case $\cos(t\Lambda)\phi$ and $\Lambda^{-1}\sin(t\Lambda)\phi$ vanish near K and $\langle u(t), \phi\rangle = \langle u_0, \cos(t\Lambda)\phi\rangle + i\langle u_1, \Lambda^{-1}\sin(t\Lambda)\phi\rangle = 0$. □

9.3 The forward fundamental solution

For $\phi \in \mathcal{C}_c^\infty(\mathbb{R} \times X)$ set

$$E\phi(t) = -\int_{-\infty}^{t} \Lambda^{-1}\sin((t-r)\Lambda)\phi(r)\,dr, \qquad E^\dagger\phi(t) = -\int_t^{\infty} \Lambda^{-1}\sin((r-t)\Lambda)\phi(r)\,dr.$$

By Theorem 9.2.1 and Proposition 9.2.2 (with data at a time before, resp. after, $\mathrm{supp}\,\phi$) these are the unique solutions in $\mathcal{C}^\infty(\mathbb{R} \times X)$ of $Pu = \phi$ vanishing for $t \ll 0$, resp. $t \gg 0$, and they depend continuously on ϕ. The Schwartz kernel of E is

$$E(t, x; t', x') = -H(t - t')\,K_{\sin}(t - t', x, x'), \qquad K_{\sin}(t, x, y) = \sum_j \frac{\sin(t\lambda_j)}{\lambda_j}\phi_j(x)\phi_j(y), \tag{9.3.1}$$

(H the Heaviside function), and $E^\dagger$ is its transpose for the pairing $\langle u, v\rangle = \int uv\,dt\,d\mathrm{vol}$, since $K_{\sin}$ is symmetric in (x, y). For $A \subset \mathbb{R} \times X$ let $J^\pm(A) = \{(t, x);\ \exists (t', x') \in A,\ d(x, x') \leq \pm(t - t')\}$ be the causal future and past.

Theorem 9.3.1. *Let $f \in \mathcal{C}^{-\infty}(\mathbb{R} \times X)$ have $\mathrm{supp}\,f \subset [T, \infty) \times X$ for some T. Then there is a unique $u \in \mathcal{C}^{-\infty}(\mathbb{R} \times X)$ with support bounded below in t such that $Pu = f$. It satisfies $\mathrm{supp}\,u \subset J^+(\mathrm{supp}\,f) \subset [T, \infty) \times X$, it is given by $\langle u, \phi\rangle = \langle f, E^\dagger\phi\rangle$ for $\phi \in \mathcal{C}_c^\infty(\mathbb{R} \times X)$ (so $u = Ef$, extending E), and $u \in \mathcal{C}^\infty$ if $f \in \mathcal{C}^\infty$. In particular E is the unique forward fundamental solution of P: $PE = EP = \mathrm{Id}$ on distributions with support bounded below, and $\mathrm{supp}\,E \subset \{(t, x, t', x');\ d(x, x') \leq t - t'\}$.*

Proof. By Theorem 9.2.3 (applied with initial time $\sup\{t; (t, x) \in \mathrm{supp}\,\phi\}$), $\mathrm{supp}\,E^\dagger\phi \subset J^-(\mathrm{supp}\,\phi)$, so $\mathrm{supp}\,f \cap \mathrm{supp}\,E^\dagger\phi$ is compact and $\langle f, E^\dagger\phi\rangle$ is defined; it is continuous in ϕ, so it defines $u \in \mathcal{C}^{-\infty}(\mathbb{R} \times X)$. Since P is formally self-adjoint for the bilinear pairing ($P^t = P$), $\langle Pu, \phi\rangle = \langle f, E^\dagger P\phi\rangle = \langle f, \phi\rangle$, because $E^\dagger P\phi - \phi$ solves the homogeneous equation and vanishes for $t \gg 0$, hence is zero by Theorem 9.2.1. If $\mathrm{supp}\,\phi \cap J^+(\mathrm{supp}\,f) = \emptyset$ then $J^-(\mathrm{supp}\,\phi) \cap \mathrm{supp}\,f = \emptyset$, so $\langle u, \phi\rangle = 0$; this gives the support statement. For uniqueness let v have support bounded below and $Pv = 0$. For $\phi \in \mathcal{C}_c^\infty$, $\mathrm{supp}\,v \cap \mathrm{supp}\,E^\dagger\phi$ is compact, so $\langle v, \phi\rangle = \langle v, PE^\dagger\phi\rangle = \langle Pv, E^\dagger\phi\rangle = 0$. If $f \in \mathcal{C}^\infty$, the Duhamel integral

$-\int_{-\infty}^{t}\Lambda^{-1}\sin((t-r)\Lambda)f(r)\,dr$ is a smooth solution (Proposition 9.2.2) vanishing for $t<T$, hence equals u. The statements about E are a reformulation, the support of the kernel coming from Theorem 9.2.3. □

Remark 9.3.2. On non-compact manifolds (for instance $\mathbb{R}^n$ with a metric that is Euclidean outside a compact set) the spectral formula is not available, and one instead constructs E from a local parametrix E' with $PE'=\mathrm{Id}+F$, where F has a smooth kernel supported in $t\geq t'$; $\mathrm{Id}+F$ is then inverted by the Volterra (Neumann) series $\sum_k(-F)^{\circ k}$ for the t-convolution, which converges because the k-fold composition gains a factor $|t-t'|^{k-1}/(k-1)!$ (see Exercise 9.5).

9.4 The short-time structure of the wave kernel

We now describe the Schwartz kernels

$$K_{\cos}(t,x,y)=\sum_j\cos(t\lambda_j)\phi_j(x)\phi_j(y),\qquad K_{\sin}(t,x,y)=\sum_j\frac{\sin(t\lambda_j)}{\lambda_j}\phi_j(x)\phi_j(y)$$

of $\cos(t\Lambda)$ and $\Lambda^{-1}\sin(t\Lambda)$ for small $|t|$. For fixed y, $K_{\cos}$ solves $PK_{\cos}=0$ with $K_{\cos}(0)=\delta_y$, $D_tK_{\cos}(0)=0$, and $K_{\sin}$ solves $PK_{\sin}=0$ with $K_{\sin}(0)=0$, $\partial_tK_{\sin}(0)=\delta_y$. Both series converge in $\mathcal{C}^\infty(\mathbb{R};H^{-M}(X\times X))$ for M large, by Lemma 9.1.1.

9.4.1 Plane waves

The initial datum δ_y is conormal to the point $\{y\}$, but the only characteristic submanifolds for P are hypersurfaces: if $Z\subset\mathbb{R}\times X$ has codimension k and $N^*Z\subset\Sigma(P)\cup 0$, then each fibre N^*_zZ is a k-dimensional linear space on which the Lorentzian quadratic form p of signature $(1,n)$ vanishes, so $k\leq 1$. The way around this is to write δ as a superposition of distributions conormal to hyperplanes. In $\mathbb{R}^n$, using polar coordinates $\xi=r\omega$ and symmetrising $(r,\omega)\mapsto(-r,-\omega)$,

$$\delta(x)=(2\pi)^{-n}\int_{\mathbb{S}^{n-1}}\int_0^\infty e^{irx\cdot\omega}r^{n-1}\,dr\,d\omega=\frac{1}{2(2\pi)^{n-1}}\int_{\mathbb{S}^{n-1}}f_n(x\cdot\omega)\,d\omega,\tag{9.4.1}$$

where

$$f_n(s)=\frac{1}{2\pi}\int_{\mathbb{R}}e^{isr}|r|^{n-1}\,dr=|D_s|^{n-1}\delta(s).$$

For n odd, $f_n=D_s^{n-1}\delta$; for n even, $|D_s|^{n-1}$ is a pseudodifferential operator and f_n is not supported at 0.

Lemma 9.4.1. *$f_n\in I^{n-\frac34}(\mathbb{R},\{0\})$ (modulo a smooth function), and the pull-back $f_n(x\cdot\omega)$ under the submersion $\mathbb{R}^n\times\mathbb{S}^{n-1}\ni(x,\omega)\mapsto x\cdot\omega$ lies in $I^{\frac n2-\frac14}(\mathbb{R}^n\times\mathbb{S}^{n-1},R)$, $R=\{x\cdot\omega=0\}$.*

Proof. Let $\rho\in\mathcal{C}^\infty(\mathbb{R})$ vanish near 0 and equal 1 for $|r|\geq 1$. Then $f_n=\frac{1}{2\pi}\int e^{isr}\rho(r)|r|^{n-1}dr+\frac{1}{2\pi}\int e^{isr}(1-\rho(r))|r|^{n-1}dr$; the second term is smooth (the inverse Fourier transform of a compactly supported function) and the first is of the form (9.1.1) with symbol $\rho|r|^{n-1}\in S^{n-1}$, so of order m with $m+\frac14-\frac12=n-1$. The map $(x,\omega)\mapsto x\cdot\omega$ has differential $\omega\cdot dx+x\cdot d\omega\neq 0$, so it is a submersion, R is a smooth hypersurface and $x\cdot\omega$ is a defining function. The pull-back is $\frac{1}{2\pi}\int e^{i(x\cdot\omega)r}\rho(r)|r|^{n-1}dr+\mathcal{C}^\infty$, with the same symbol, now on a manifold of dimension $N=2n-1$; thus $m+\frac{2n-1}{4}-\frac12=n-1$, i.e. $m=\frac n2-\frac14$. □

Proposition 9.4.2 (Radon inversion)**.** *If $\pi_1\colon\mathbb{R}^n\times\mathbb{S}^{n-1}\to\mathbb{R}^n$ is the projection, then $(\pi_1)_*(f_n(x\cdot\omega)\,d\omega)=2(2\pi)^{n-1}\delta$; this is (9.4.1).*

Proof. For $\phi \in \mathcal{S}(\mathbb{R}^n)$ and fixed ω, $\int f_n(x\cdot\omega)\phi(x)\,dx = \langle f_n, \mathcal{R}_\omega\phi\rangle$, where $\mathcal{R}_\omega\phi(s) = \int_{x\cdot\omega=s}\phi$ satisfies $\widehat{\mathcal{R}_\omega\phi}(r) = \hat\phi(r\omega)$. Hence $\langle f_n,\mathcal{R}_\omega\phi\rangle = \frac{1}{2\pi}\int_{\mathbb{R}}|r|^{n-1}\hat\phi(-r\omega)\,dr$ and, by polar coordinates and Fourier inversion, $\int_{\mathbb{S}^{n-1}}\langle f_n,\mathcal{R}_\omega\phi\rangle\,d\omega = \frac{2}{2\pi}\int_{\mathbb{R}^n}\hat\phi(\xi)\,d\xi = 2(2\pi)^{n-1}\phi(0)$. □

Thus, to solve the Cauchy problem with datum δ_y we may solve it, for each ω, with data conormal to the hyperplane $\{(x-y)\cdot\omega=0\}$, using the two characteristic hypersurfaces through it, and then integrate over ω. It is convenient to keep the radial variable as well: this amounts to an oscillatory integral whose phase solves the eikonal equation.

9.4.2 The eikonal equation

Work in a coordinate chart $U\subset\mathbb{R}^n$ of X, and let $q(x,\eta) = |\eta|_{g(x)}$.

Lemma 9.4.3. *Let $W \Subset U$ be open. There exist $t_0>0$ and unique real $\varphi_\pm\in\mathcal{C}^\infty((-t_0,t_0)\times W\times(\mathbb{R}^n\setminus 0))$, homogeneous of degree one in η, with*

$$\partial_t\varphi_\pm = \pm q(x,d_x\varphi_\pm),\qquad \varphi_\pm(0,x,\eta) = x\cdot\eta. \tag{9.4.2}$$

Moreover $\varphi_-(t,x,\eta) = \varphi_+(-t,x,\eta)$, $d_x\varphi_\pm\neq 0$, and for $y\in W$:

$$d_\eta(\varphi_\pm(t,x,\eta) - y\cdot\eta) = 0 \iff (x,d_x\varphi_\pm(t,x,\eta)) = G_{\mp t}(y,\eta), \tag{9.4.3}$$

in which case $(x,d_x\varphi_\pm)$ lies on the geodesic flow-out and $d(x,y)\le|t|$.

Proof. This is the method of characteristics (the Hamilton–Jacobi theorem of Chapter 8) for $F_\pm = \tau\mp q(x,\xi)$ with initial surface $\{t=0\}$, which is non-characteristic since $\partial_\tau F_\pm = 1$. Explicitly, consider φ_+ and let $(x(t),\xi(t)) = G_{-t}(x_0,\eta)$, i.e. $\dot x = -\partial_\xi q$, $\dot\xi = \partial_x q$, which are the x,ξ components of the bicharacteristics of F_+. For $|t|<t_0$ small, uniformly for $|\eta|=1$ and x_0 near $\bar W$, the map $x_0\mapsto x(t;x_0,\eta)$ is a diffeomorphism onto a neighbourhood of $\bar W$, and φ_+ is defined by $\varphi_+(t,x(t;x_0,\eta),\eta) = x_0\cdot\eta$, extended homogeneously; the Hamilton–Jacobi theorem shows that it solves (9.4.2) and that $d_x\varphi_+(t,x(t;x_0,\eta),\eta) = \xi(t)\neq 0$ (consistently, $\frac{d}{dt}\varphi_+(t,x(t),\eta) = q - \xi\cdot\partial_\xi q = 0$ by Euler's relation). Uniqueness also follows from the characteristics. Since $\psi(t)=\varphi_+(-t)$ solves the equation with the sign $-$, $\varphi_-(t)=\varphi_+(-t)$.

For (9.4.3), differentiate $\varphi_+(t,x(t;x_0,\eta),\eta) = x_0\cdot\eta$ in η:

$$d_\eta\varphi_+(t,x(t),\eta) + \xi(t)\cdot\partial_\eta x(t;x_0,\eta) = x_0.$$

The second term vanishes: with $w=\partial_\eta x\cdot\delta\eta$, $v=\partial_\eta\xi\cdot\delta\eta$, $\frac{d}{dt}(\xi\cdot w) = \partial_x q\cdot w - \xi\cdot(\partial_x\partial_\xi q\, w + \partial_\xi^2 q\, v) = 0$, since $\xi\cdot\partial_\xi\partial_x q = \partial_x q$ and $\xi\cdot\partial_\xi^2 q = 0$ by homogeneity, and $w=0$ at $t=0$. (This is the invariance of the Liouville form $\xi\cdot dx$ under the homogeneous flow G_s.) Hence $d_\eta\varphi_+(t,x,\eta) = y$ iff $x = x(t;y,\eta)$, i.e. iff $(x,d_x\varphi_+(t,x,\eta)) = G_{-t}(y,\eta)$. The case of φ_- follows from $\varphi_-(t)=\varphi_+(-t)$. Since G_s moves base points along unit speed geodesics, $d(x,y)\le|t|$. □

9.4.3 Transport equations

For a real phase $\phi(t,x)$ and an amplitude $a(t,x)$, a direct computation (using $\Delta e^{i\phi} = e^{i\phi}(|d_x\phi|_g^2 + i\Delta\phi)$) gives

$$P(e^{i\phi}a) = e^{i\phi}\Big[((\partial_t\phi)^2 - |d_x\phi|_g^2)a - iL_\phi a + Pa\Big], \tag{9.4.4}$$

$$L_\phi a = 2(\partial_t\phi\,\partial_t a - \langle d_x\phi, d_x a\rangle_g) + (\partial_t^2\phi + \Delta\phi)a.$$

If ϕ solves the eikonal equation the first term vanishes, and L_ϕ is a first order operator whose principal part is the vector field $V_\phi = 2(\partial_t\phi\,\partial_t - \sum g^{ij}\partial_i\phi\,\partial_j)$, the projection to $\mathbb{R}\times X$ of H_p restricted to the Lagrangian manifold $\{(t,x,d\phi)\}$. For $\phi = \varphi_\pm(t,x,\eta) - y\cdot\eta$ the ∂_t-coefficient is $\pm 2q(x,d_x\varphi_\pm) \neq 0$, so V_ϕ is transversal to $\{t=0\}$ and its integral curves are (reparametrised) unit speed geodesics. If $a \sim \sum_{k\geq 0} a_k$ with a_k homogeneous of degree $\mu - k$ in η, then $P(e^{i\phi}a) = e^{i\phi}r$ with $r \in S^{-\infty}$ in η provided the *transport equations*

$$L_\phi a_0 = 0, \qquad L_\phi a_k = -iPa_{k-1} \quad (k \geq 1) \tag{9.4.5}$$

hold; both sides of the k-th equation are homogeneous of degree $\mu + 1 - k$. Each is a linear ODE along the integral curves of V_ϕ, uniquely solvable for small $|t|$ given $a_k(0,\cdot)$, and the solution is homogeneous of the right degree.

9.4.4 The parametrix

Fix $\chi \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $\chi(\eta) = 0$ for $|\eta| \leq \frac{1}{2}$ and $\chi(\eta) = 1$ for $|\eta| \geq 1$. Choose a finite cover of X by coordinate charts U_k, open sets $V_k \Subset U_k$ still covering X, a partition of unity χ_k with $\operatorname{supp}\chi_k \subset V_k$, and $\delta_0 > 0$ such that the $3\delta_0$-neighbourhood of $\bar V_k$ is relatively compact in U_k for all k. Let W_k be the $2\delta_0$-neighbourhood of $\bar V_k$ and $\zeta_k \in \mathcal{C}^\infty_{\mathrm{c}}(W_k)$ with $\zeta_k = 1$ on the δ_0-neighbourhood of $\bar V_k$.

Theorem 9.4.4. *There are $t_0 \in (0,\delta_0)$, phase functions $\varphi_{k,\pm}$ on $(-t_0,t_0)\times W_k\times(\mathbb{R}^n\setminus 0)$ as in Lemma 9.4.3, and polyhomogeneous symbols $a_{k,\pm} \in S^0_{\mathrm{ph}}$, $b_{k,\pm} \in S^{-1}_{\mathrm{ph}}$ in $\eta\in\mathbb{R}^n$, depending smoothly on $(t,x,y) \in (-t_0,t_0)\times W_k\times V_k$, such that*

$$K_{\cos}(t,x,y) \equiv \sum_{k,\pm} \zeta_k(x)\chi_k(y)(2\pi)^{-n}\int e^{i(\varphi_{k,\pm}(t,x,\eta)-y\cdot\eta)}a_{k,\pm}(t,x,y,\eta)\chi(\eta)\,d\eta,$$
$$K_{\sin}(t,x,y) \equiv \sum_{k,\pm} \zeta_k(x)\chi_k(y)(2\pi)^{-n}\int e^{i(\varphi_{k,\pm}(t,x,\eta)-y\cdot\eta)}b_{k,\pm}(t,x,y,\eta)\chi(\eta)\,d\eta,$$

modulo $\mathcal{C}^\infty((-t_0,t_0)\times X\times X)$. At $t=0$,

$$a_{k,\pm}(0,x,y,\eta) - \tfrac{1}{2}|g(y)|^{-1/2} \in S^{-1}, \qquad b_{k,\pm}(0,x,y,\eta) \mp \frac{|g(y)|^{-1/2}}{2i\,q(x,\eta)} \in S^{-2}. \tag{9.4.6}$$

Proof. Apply Lemma 9.4.3 on W_k (with $W_k \Subset U_k$), shrinking t_0 so that also all integral curves of the transport fields starting in W_k stay in the $3\delta_0$-neighbourhood for $|t| < t_0$. We construct the cosine term; write $\phi_\pm = \varphi_{k,\pm} - y\cdot\eta$ and $S = \sum_\pm (2\pi)^{-n}\int e^{i\phi_\pm}a_\pm\chi\,d\eta$.

Amplitudes. We seek $a_\pm \sim \sum_{j\geq 0} a_{\pm,j}$, $a_{\pm,j}$ homogeneous of degree $-j$, solving (9.4.5) for $\phi_\pm$, with initial values fixed by the requirements $S(0) \equiv |g(y)|^{-1/2}\delta(x-y)$, $D_tS(0)\equiv 0$. Since $\phi_\pm(0) = (x-y)\cdot\eta$ and $\partial_t\phi_\pm(0) = \pm q(x,\eta)$, these amount to

$$a_{+,0} + a_{-,0} = |g(y)|^{-1/2}, \qquad a_{+,j} + a_{-,j} = 0 \ (j\geq 1),$$
$$q(x,\eta)(a_{+,j} - a_{-,j}) + D_t a_{+,j-1} + D_t a_{-,j-1} = 0 \ (j \geq 0)$$

at $t=0$ (with $a_{\pm,-1} = 0$), the last being the part homogeneous of degree $1-j$ of $\sum_\pm(\partial_t\phi_\pm a_\pm + D_t a_\pm)|_{t=0}$. Inductively in j: the initial values of $a_{\pm,j}$ are determined by these equations, given $a_{\pm,j-1}$ for all t, and then $a_{\pm,j}$ is determined for $|t|<t_0$ by the transport equation. In particular $a_{\pm,0}(0) = \frac{1}{2}|g(y)|^{-1/2}$. Let $a_\pm$ be a Borel sum of the series, multiplied by χ. Then $PS = (2\pi)^{-n}\sum_\pm\int e^{i\phi_\pm}r_\pm\,d\eta$ with $r_\pm \in S^{-\infty}$, so $PS\in\mathcal{C}^\infty$; similarly $S(0) - |g(y)|^{-1/2}\delta(x-y)$ and $D_tS(0)$ are oscillatory integrals with amplitudes in $S^{-\infty}$, hence smooth. For the sine term the conditions are $\sum_\pm b_\pm(0) \in S^{-\infty}$ and

$\sum_{\pm}(i\partial_t\phi_\pm b_\pm + \partial_t b_\pm)(0) - |g(y)|^{-1/2} \in S^{-\infty}$, giving $b_{\pm,0}(0) = \pm|g(y)|^{-1/2}/(2iq(x,\eta))$ and then the same induction.

Singular support. The phase $\phi_\pm$ is non-degenerate in (y,η) ($d_y\phi_\pm = -\eta \neq 0$), so S is a smooth function of t with values in distributions on $W_k \times V_k$, and by (9.1.2) and (9.4.3) it is smooth on $\{d(x,y) > |t|\}$.

Comparison. Let $w = \chi_k(y)K_{\cos} - \zeta_k(x)\chi_k(y)S$ on $(-t_0,t_0)\times X\times X$. Then

$$Pw = -\zeta_k\chi_k PS - [P,\zeta_k]\chi_k S \in \mathcal{C}^\infty,$$

the commutator term because $d\zeta_k = 0$ on the δ_0-neighbourhood of $\operatorname{supp}\chi_k$, while S is smooth where $d(x,y) > |t|$ and $|t| < t_0 < \delta_0$. Also $w(0) = \chi_k(y)(K_{\mathrm{Id}} - \zeta_k(x)S(0))$ and $D_t w(0)$ are smooth, since $\zeta_k(x) = 1$ near the support of $\chi_k(y)\delta(x-y)$. Both $K_{\cos}$ and S lie in $\mathcal{C}^\infty((-t_0,t_0); H^{-M}(X\times X))$ for M large, so by Proposition 9.2.2 (with $Y = X$ the y-factor) $w \in \mathcal{C}^\infty$. Summing over k proves the claim for $K_{\cos}$, and the same argument applies to $K_{\sin}$. □

Remark 9.4.5 (Conormal superposition). In polar coordinates $\eta = r\omega$, homogeneity gives $\varphi_\pm(t,x,r\omega) - ry\cdot\omega = rh_\pm(t,x,y,\omega)$, so each term of the parametrix is

$$(2\pi)^{1-n}\int_{\mathbb{S}^{n-1}} u_\pm(t,x;y,\omega)\,d\omega, \qquad u_\pm = \frac{1}{2\pi}\int_0^\infty e^{irh_\pm(t,x,y,\omega)}a_\pm(t,x,y,r\omega)\chi(r\omega)r^{n-1}\,dr.$$

For fixed (y,ω), $u_\pm$ is a conormal distribution, of order $\frac{3n-3}{4}$, associated with the hypersurface $H_\pm(y,\omega) = \{h_\pm = 0\} \subset (-t_0,t_0)\times W_k$, which is characteristic for P ($(\partial_t h_\pm)^2 = |d_x h_\pm|_g^2$ by (9.4.2)) and meets $\{t=0\}$ in the hyperplane $\{(x-y)\cdot\omega = 0\}$; there are exactly two such characteristic hypersurfaces, corresponding to the two points $\tau = \pm|\omega|_g$ of $\Sigma(P)$ over the conormal $\omega\,dx$. The amplitude of $u_\pm$ solves the transport equation, and at $t=0$ the sum $u_+ + u_-$ reproduces the plane wave decomposition (9.4.1) of δ_y. As a function of all variables (t,x,y,ω), $u_\pm$ is conormal with respect to the hypersurface $\{h_\pm = 0\}$, smooth since $d_y h_\pm = -\omega \neq 0$, and the kernel is its push-forward along $\mathbb{S}^{n-1}$.

Corollary 9.4.6. *For $|t| < t_0$ the wavefront sets of $K_{\cos}$ and $K_{\sin}$ are contained in*

$$\Gamma = \{(t,x,y;\tau,\xi,-\eta);\ \eta\neq 0,\ \tau = \pm|\eta|_{g(y)},\ (x,\xi) = G_{\mp t}(y,\eta)\}.$$

In particular their singular supports are contained in $\{d(x,y) = |t|\}$ for $|t| < \min(t_0, \operatorname{inj} X)$.

Proof. By (9.1.2), the wavefront set of each term is contained in the set of points

$$(t,x,y;\partial_t\phi_\pm, d_x\phi_\pm, d_y\phi_\pm) \quad\text{with}\quad d_\eta\phi_\pm = 0, \qquad \phi_\pm = \varphi_{k,\pm}(t,x,\eta) - y\cdot\eta.$$

By (9.4.3) there $(x, d_x\varphi_\pm) = G_{\mp t}(y,\eta)$, while $d_y\phi_\pm = -\eta$ and $\partial_t\phi_\pm = \pm q(x,d_x\varphi_\pm) = \pm q(y,\eta)$ since G preserves q. Within the injectivity radius the geodesic of length $|t|$ is minimising, so $d(x,y) = |t|$. □

By Lemma 9.1.2, Γ is the union of the null bicharacteristics of $D_t^2 - \Delta_x$ (acting in (t,x), with (y,η) fixed) through the points $(0,y,y;\pm|\eta|,\eta,-\eta)$, i.e. the flow-out of the wavefront set of the initial datum $\delta(x-y)$.

9.5 Operations on conormal distributions

We record the behaviour of conormal distributions under push-forward along fibres and restriction to transversal submanifolds, and the reduction of amplitudes depending on the normal variable. Together these compute the trace of the wave group in Section 9.6.

Proposition 9.5.1 (Push-forward)**.** *Let M_1 be a manifold of dimension N_1, Z a compact manifold of dimension d with a smooth positive density ν, $\pi\colon M_1\times Z\to M_1$ the projection, and $H_1\subset M_1$ a hypersurface. If $u\in I^m(M_1\times Z, H_1\times Z)$ then $\pi_* u=\int_Z u\,\nu\in I^{m+\frac{d}{4}}(M_1,H_1)$; in local coordinates, with $u=\frac{1}{2\pi}\int e^{is\rho}a(z',w,\rho)\,d\rho$, the push-forward has amplitude $\int_Z a(z',w,\rho)\,\nu(w)$.*

Proof. π_* maps $\mathcal{C}^\infty$ to $\mathcal{C}^\infty$ and distributions smooth off $H_1\times Z$ to distributions smooth off H_1. Near H_1 choose coordinates (s,z') on M_1 with $H_1=\{s=0\}$; then $s\circ\pi$ defines $H_1\times Z$ and, using a partition of unity on Z and coordinates w there, u is locally as in (9.1.1) with $a(z',w,\rho)\in S^\mu$, $\mu=m+\frac{N_1+d}{4}-\frac12$. Integrating in w gives an amplitude in $S^\mu(\mathbb{R}^{N_1-1};\mathbb{R})$, and $\mu=m'+\frac{N_1}{4}-\frac12$ with $m'=m+\frac{d}{4}$. □

Proposition 9.5.2 (Restriction)**.** *Let $H\subset M$ be a hypersurface, $\dim M=N$, and $Y\subset M$ a submanifold of codimension k transversal to H ($T_zH+T_zY=T_zM$ for $z\in H\cap Y$). Then restriction extends from $\mathcal{C}^\infty$ to a map $I^m(M,H)\to I^{m+\frac{k}{4}}(Y,H\cap Y)$, and the local amplitude restricts.*

Proof. By transversality there are local coordinates (s,z',z'') with $H=\{s=0\}$ and $Y=\{z''=0\}$, $z''\in\mathbb{R}^k$. Then $u=\frac{1}{2\pi}\int e^{is\rho}a(z',z'',\rho)\,d\rho$ depends smoothly on z'' as a distribution in (s,z'), and $u|_Y=\frac{1}{2\pi}\int e^{is\rho}a(z',0,\rho)\,d\rho$, consistent with the restriction of smooth functions (and with $N^*Y\cap N^*H=0$). The symbol order $\mu=m+\frac{N}{4}-\frac12$ is unchanged, and $\mu=m'+\frac{N-k}{4}-\frac12$ gives $m'=m+\frac{k}{4}$. □

Lemma 9.5.3. *Let $B\in\mathcal{C}^\infty(\mathbb{R}\times\mathbb{R})$ satisfy $B(t,\rho)=0$ for $|t|\geq t_1$ and $|\partial_t^k\partial_\rho^l B(t,\rho)|\leq C_{kl}\langle\rho\rangle^{m-l}$. Then*

$$J_B(\lambda)=\int_{\mathbb{R}}\Big(\int_{\mathbb{R}} e^{it(\rho-\lambda)}B(t,\rho)\,dt\Big)d\rho$$

(an absolutely convergent iterated integral) lies in $S^m(\mathbb{R})$, and $J_B(\lambda)\sim 2\pi\sum_{k\geq0}\frac{i^k}{k!}\partial_t^k\partial_\rho^k B(0,\lambda)$; in particular $J_B(\lambda)-2\pi B(0,\lambda)\in S^{m-1}(\mathbb{R})$. If B_ϵ is bounded in this class for $\epsilon\in(0,1]$ and $B_\epsilon\to B$ pointwise, then $J_{B_\epsilon}\to J_B$ pointwise. Consequently $v=\frac{1}{2\pi}\int e^{it\rho}B(t,\rho)\,d\rho\in I^{m+\frac14}(\mathbb{R},\{0\})$, with $\hat v=J_B/(2\pi)$.

Proof. Let $\hat B(\sigma,\rho)=\int e^{-it\sigma}B(t,\rho)\,dt$; then $|\partial_\sigma^j\partial_\rho^l\hat B|\leq C_{Njl}\langle\sigma\rangle^{-N}\langle\rho\rangle^{m-l}$, and $J_B(\lambda)=\int\hat B(\lambda-\rho,\rho)\,d\rho=\int\hat B(\sigma,\lambda-\sigma)\,d\sigma$. Taylor's formula in the second variable gives $\hat B(\sigma,\lambda-\sigma)=\sum_{k<K}\frac{(-\sigma)^k}{k!}\partial_\rho^k\hat B(\sigma,\lambda)+R_K$ with, by Peetre's inequality $\langle\lambda-s\sigma\rangle^\mu\leq 2^{|\mu|}\langle\lambda\rangle^\mu\langle\sigma\rangle^{|\mu|}$, $|R_K|\leq C_N\langle\sigma\rangle^{K+|m-K|-N}\langle\lambda\rangle^{m-K}$, which integrates to $O(\langle\lambda\rangle^{m-K})$. Since $\sigma^k\hat B=\widehat{D_t^kB}$ and $\int\hat f\,d\sigma=2\pi f(0)$, the k-th term is $2\pi\frac{(-1)^k}{k!}D_t^k\partial_\rho^kB(0,\lambda)=2\pi\frac{i^k}{k!}\partial_t^k\partial_\rho^kB(0,\lambda)$. Finally $\partial_\lambda J_B=J_{\partial_\rho B}$ and $\partial_\rho B$ is of the same type with m replaced by $m-1$, so the bound $|J_B|\leq C\langle\lambda\rangle^m$ (case $K=0$) gives the symbol estimates. The convergence statement follows by dominated convergence from the uniform bound $|\hat B_\epsilon(\lambda-\rho,\rho)|\leq C\langle\lambda-\rho\rangle^{-N}\langle\rho\rangle^m$. For the last statement, $\hat v(\lambda)=\frac{1}{2\pi}J_B(\lambda)\in S^m$ and v has compact support, so $v\in I^{m+\frac14}(\mathbb{R},\{0\})$. □

In Lemma 9.5.3 the amplitude $B(t,\rho)$ is replaced by the amplitude $\hat v(\rho)$ independent of t; this is the one-dimensional case of the statement that amplitudes may depend on the normal variable.

9.6 The trace of the wave group

For $\psi\in\mathcal{S}(\mathbb{R})$ let $C_\psi=\int\psi(t)\cos(t\Lambda)\,dt=\tilde\psi(\Lambda)$, where $\tilde\psi(\lambda)=\int\psi(t)\cos(t\lambda)\,dt=\frac12(\hat\psi(\lambda)+\hat\psi(-\lambda))$ is rapidly decreasing. By Lemma 9.1.1 and the polynomial bounds on $\|\phi_j\|_{\mathcal{C}^k}$, the

kernel $K_{C_\psi}(x,y)=\sum_j \tilde\psi(\lambda_j)\phi_j(x)\phi_j(y)$ converges in $\mathcal{C}^\infty(X\times X)$; C_ψ is smoothing and of trace class, and

$$\operatorname{Tr} C_\psi=\sum_j \tilde\psi(\lambda_j)=\int_X K_{C_\psi}(x,x)\,d\mathrm{vol}(x).$$

Since $\sum_j\langle\lambda_j\rangle^{-M}<\infty$ for $M>n$, this is a continuous linear functional on $\mathcal{S}(\mathbb{R})$:

Definition 9.6.1. The *(even) wave trace* is the tempered distribution

$$T(t)=\operatorname{Tr}\cos(t\Lambda)=\sum_{j\ge 0}\cos(t\lambda_j),\qquad T(\psi)=\operatorname{Tr} C_\psi.$$

The trace of the wave group $U(t)$ is $2T$. With the measure $\mu=\sum_j\delta_{\lambda_j}$ on $[0,\infty)$ and its even part $\mu_e=\frac12\sum_j(\delta_{\lambda_j}+\delta_{-\lambda_j})$, $T(t)=\int e^{it\sigma}d\mu_e(\sigma)$, i.e. $\hat T=2\pi\mu_e$, and $N(\lambda)=\mu([0,\lambda])$. For $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$,

$$\widehat{\psi T}(\lambda)=T(\psi e^{-i\lambda\cdot})=\frac12\sum_j\big(\hat\psi(\lambda-\lambda_j)+\hat\psi(\lambda+\lambda_j)\big).\tag{9.6.1}$$

Let ω_n denote the volume of the unit ball in $\mathbb{R}^n$ (so $\mathrm{vol}(\mathbb{S}^{n-1})=n\omega_n$).

Theorem 9.6.2. *Let $t_0>0$ be as in Theorem 9.4.4. For every $\psi\in\mathcal{C}^\infty_{\mathrm{c}}((-t_0,t_0))$, $\psi T\in I^{n-\frac34}(\mathbb{R},\{0\})$, i.e. $\widehat{\psi T}\in S^{n-1}(\mathbb{R})$; more precisely, $\widehat{\psi T}$ is polyhomogeneous as $\lambda\to\pm\infty$ and*

$$\widehat{\psi T}(\lambda)=\frac{n\,\omega_n}{2(2\pi)^{n-1}}\,\psi(0)\,\mathrm{vol}(X)\,|\lambda|^{n-1}+O(|\lambda|^{n-2}),\qquad |\lambda|\to\infty.\tag{9.6.2}$$

Proof. Reduction to the parametrix. With $\psi_\lambda=\psi e^{-i\lambda\cdot}$, $\widehat{\psi T}(\lambda)=\int_X K_{C_{\psi_\lambda}}(x,x)\,d\mathrm{vol}$ (the definition of C_ψ extends to complex ψ), and $K_{C_{\psi_\lambda}}=\int\psi_\lambda(t)K_{\cos}(t)\,dt$. By Theorem 9.4.4, $K_{\cos}=\sum_{k,\pm}S_{k,\pm}+R$, with $S_{k,\pm}$ the terms of the parametrix, on $(-t_0,t_0)\times X\times X$ with R smooth; the contribution $\iint\psi(t)e^{-it\lambda}R(t,x,x)\,dt\,d\mathrm{vol}$ of R is $O(|\lambda|^{-\infty})$. Fix k and the sign $+$ (the sign $-$ is analogous) and write $\varphi=\varphi_{k,+}$, $a=a_{k,+}$. Then

$$\int\psi_\lambda(t)S_{k,+}(t,x,y)\,dt=(2\pi)^{-n}\int e^{-iy\cdot\eta}\alpha(x,y,\eta)\,d\eta,$$
$$\alpha=\zeta_k(x)\chi_k(y)\chi(\eta)\int e^{i(\varphi(t,x,\eta)-t\lambda)}\psi(t)a\,dt.$$

For fixed λ, $|\partial_t\varphi-\lambda|=|q(x,d_x\varphi)-\lambda|\ge c|\eta|$ for $|\eta|\ge C(1+|\lambda|)$, so repeated integration by parts with $(\partial_t\varphi-\lambda)^{-1}D_t$ shows that α and all its (x,y)-derivatives are rapidly decreasing in η. Hence this kernel is smooth, given by an absolutely convergent integral, and its value at $x=y$ is the absolutely convergent iterated integral $(2\pi)^{-n}\int(\int\cdots dt)\,d\eta$. Thus

$$I_{k,+}(\lambda)=(2\pi)^{-n}\int_X\int_{\mathbb{R}^n}\Big(\int_{\mathbb{R}}e^{i(\varphi(t,x,\eta)-x\cdot\eta-t\lambda)}b(t,x,\eta)\,dt\Big)d\eta\,d\mathrm{vol}(x),$$

with $b(t,x,\eta)=\psi(t)\chi_k(x)a(t,x,x,\eta)\chi(\eta)$ (using $\zeta_k\chi_k=\chi_k$). By dominated convergence we may insert $\chi_0(\epsilon\eta)$, $\chi_0\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, $\chi_0(0)=1$, and let $\epsilon\to0$; for $\epsilon>0$ the integral is absolutely convergent and Fubini's theorem allows the t-integration to be done last.

Restriction to the diagonal. Since $\varphi(0,x,\eta)=x\cdot\eta$,

$$\varphi(t,x,\eta)-x\cdot\eta=tQ(t,x,\eta),\qquad Q(t,x,\eta)=\int_0^1 q\big(x,d_x\varphi(st,x,\eta)\big)\,ds,$$

with Q smooth, positive, homogeneous of degree one in η and $Q(0,x,\eta)=|\eta|_{g(x)}$. (Thus the restriction of the characteristic hypersurface $\{h_+=0\}$ of Remark 9.4.5 to the diagonal

is exactly $\{t=0\}$, the transversal situation of Proposition 9.5.2.) In polar coordinates $\eta = r\omega$ and then $\rho = rQ(t,x,\omega)$ for fixed (t,x,ω), $d\eta = \rho^{n-1}Q(t,x,\omega)^{-n}d\rho\,d\omega$, and we obtain $I_{k,+}(\lambda) = \lim_{\epsilon\to 0}\iint e^{it(\rho-\lambda)}B_\epsilon(t,\rho)\,dt\,d\rho$ with

$$B_\epsilon(t,\rho) = (2\pi)^{-n}\int_X\int_{\mathbb{S}^{n-1}} b\Big(t,x,\frac{\rho\omega}{Q}\Big)\chi_0\Big(\frac{\epsilon\rho\omega}{Q}\Big)\frac{\rho^{n-1}}{Q(t,x,\omega)^n}\,d\omega\,\mathrm{dvol}(x)$$

for $\rho>0$, and $B_\epsilon = 0$ for $\rho\le 0$ (as $\chi = 0$ near 0). This is the push-forward of Proposition 9.5.1 along $X\times\mathbb{S}^{n-1}$. The B_ϵ are bounded in the class of Lemma 9.5.3 with $m = n-1$ and converge to $B = B_0$, whose amplitude is polyhomogeneous in ρ. By Lemma 9.5.3, $I_{k,+} = J_B \in S^{n-1}$, polyhomogeneous, and $I_{k,+}(\lambda) = 2\pi B(0,\lambda) + O(\langle\lambda\rangle^{n-2})$. In particular $I_{k,+}(\lambda) = O(|\lambda|^{-\infty})$ as $\lambda\to-\infty$.

Leading term. At $t=0$, $b(0,x,\eta) = \psi(0)\chi_k(x)a(0,x,x,\eta)$ for $|\eta|\ge 1$, and by (9.4.6) $a(0,x,x,\eta) = \frac12|g(x)|^{-1/2} + S^{-1}$. Hence for $\rho\to+\infty$

$$B(0,\rho) = \frac{\psi(0)}{2(2\pi)^n}\rho^{n-1}\int_X \chi_k(x)|g(x)|^{-1/2}\int_{\mathbb{S}^{n-1}}|\omega|^{-n}_{g(x)}\,d\omega\,\mathrm{dvol}(x) + O(\rho^{n-2}).$$

In polar coordinates the ellipsoid $\{\eta;\ |\eta|_{g(x)}\le 1\}$ has volume $\frac1n\int_{\mathbb{S}^{n-1}}|\omega|^{-n}_{g(x)}d\omega$, and it is the image of the unit ball under a linear map of determinant $\det(g^{ij})^{-1/2} = |g(x)|^{1/2}$; so the inner integral is $n\omega_n|g(x)|^{1/2}$ and $B(0,\rho) = \frac{n\omega_n}{2(2\pi)^n}\psi(0)\rho^{n-1}\int_X\chi_k\,\mathrm{dvol} + O(\rho^{n-2})$. The terms with the sign $-$ have phase $\varphi_{k,-} - x\cdot\eta = -tQ_-$ and give the same expression with λ replaced by $-\lambda$, rapidly decreasing as $\lambda\to+\infty$. Summing over k ($\sum\chi_k = 1$) and multiplying by 2π gives (9.6.2). □

Remark 9.6.3. For the circle $X = \mathbb{R}/L\mathbb{Z}$, $\lambda = 2\pi|k|/L$ for $k\in\mathbb{Z}$ and $T(t) = \sum_{k\in\mathbb{Z}}\cos(2\pi kt/L)$, while by Poisson summation $\sum_k\hat\psi(\lambda - 2\pi k/L) = L\sum_m\psi(mL)e^{-imL\lambda} = L\psi(0)$ if $\operatorname{supp}\psi\subset(-L,L)$. Thus $\widehat{\psi T} = L\psi(0)$, in agreement with (9.6.2) ($n=1$, $\omega_1 = 2$). The restriction on the support of ψ is necessary: T is singular at every $t\in L\mathbb{Z}$, the lengths of the closed geodesics (Section 9.8).

9.7 Weyl asymptotics

Theorem 9.7.1 (Weyl's law with remainder)**.** *Let X be a compact Riemannian manifold of dimension $n\ge 2$. Then*

$$N(\lambda) = (2\pi)^{-n}\omega_n\,\mathrm{vol}(X)\lambda^n + O(\lambda^{n-1}),\qquad \lambda\to\infty.$$

Proof. Write $C_W = (2\pi)^{-n}\omega_n\,\mathrm{vol}(X)$.

Choice of a test function. Let $\theta\in\mathcal{C}^\infty_{\mathrm{c}}((-t_0/2,t_0/2))$ be real, even, non-negative and not identically zero, normalised by $\int\theta^2 = 1$, and let $\psi = \theta*\theta$. Then $\psi\in\mathcal{C}^\infty_{\mathrm{c}}((-t_0,t_0))$ is even, $\psi(0) = 1$ and $\hat\psi = \hat\theta^2 = |\hat\theta|^2\ge 0$. Set $\rho = \hat\psi/(2\pi)$: $\rho\in\mathcal{S}(\mathbb{R})$ is even and non-negative, $\int\rho = \psi(0) = 1$, and $\rho(0) = (2\pi)^{-1}(\int\theta)^2 > 0$, so there are $a, c_0 > 0$ with $\rho\ge c_0$ on $[-a,a]$.

Smoothed density. By (9.6.1), $(\rho*\mu)(\lambda) = \sum_j\rho(\lambda-\lambda_j) = \pi^{-1}\widehat{\psi T}(\lambda) - \sum_j\rho(\lambda+\lambda_j)$. For $\lambda\ge 0$, $\sum_j\rho(\lambda+\lambda_j)\le C_M\sum_j(1+\lambda+\lambda_j)^{-M} = O(\lambda^{-\infty})$, by Lemma 9.1.1. Hence by Theorem 9.6.2

$$(\rho*\mu)(\lambda) = nC_W\lambda^{n-1} + O(\lambda^{n-2}),\quad \lambda\to+\infty;\qquad (\rho*\mu)(\lambda) = O(\langle\lambda\rangle^{-\infty}),\quad \lambda\to-\infty, \tag{9.7.1}$$

the second because $\lambda-\lambda_j\le\lambda<0$. Note $\pi^{-1}\frac{n\omega_n}{2(2\pi)^{n-1}} = n(2\pi)^{-n}\omega_n$.

Step 1: a priori bound. Since $\rho \geq 0$ and $\rho \geq c_0$ on $[-a,a]$, $\mu([\lambda-a,\lambda+a]) \leq c_0^{-1}(\rho*\mu)(\lambda) \leq C(1+|\lambda|)^{n-1}$ for all $\lambda \in \mathbb{R}$ (the function $\rho*\mu$ is continuous). Covering an interval of length $2r$ by at most $1+r/a$ intervals of length $2a$,

$$\mu([\lambda-r,\lambda+r]) \leq C'(1+r)^n(1+|\lambda|)^{n-1}, \qquad \lambda\in\mathbb{R},\ r\geq 0. \tag{9.7.2}$$

Step 2: the smoothed counting function. Let $N_\rho(\lambda) = \int N(\lambda-\sigma)\rho(\sigma)\,d\sigma = \sum_j R(\lambda-\lambda_j)$, $R(s) = \int_{-\infty}^{s}\rho$. Termwise differentiation (justified by the locally uniform convergence of $\sum_j \rho(\lambda-\lambda_j)$) gives $N_\rho' = \rho*\mu$, and $N_\rho(\lambda)\to 0$ as $\lambda\to-\infty$. Thus, by (9.7.1) and $n\geq 2$,

$$N_\rho(\lambda) = \int_{-\infty}^{1}(\rho*\mu)\,ds + \int_1^\lambda \left(nC_W s^{n-1} + O(s^{n-2})\right)ds = C_W\lambda^n + O(\lambda^{n-1}).$$

Step 3: removing the smoothing. Since $\int\rho = 1$, $\rho\geq 0$ and N is non-decreasing, $|N(\lambda)-N(\lambda-\sigma)| \leq \mu([\lambda-|\sigma|,\lambda+|\sigma|])$, so by (9.7.2)

$$|N(\lambda)-N_\rho(\lambda)| \leq \int |N(\lambda)-N(\lambda-\sigma)|\rho(\sigma)\,d\sigma \leq C'(1+|\lambda|)^{n-1}\int(1+|\sigma|)^n\rho(\sigma)\,d\sigma.$$

Combining with Step 2 proves the theorem. □

Remark 9.7.2. (i) The remainder cannot be improved in general: on the round sphere $\mathbb{S}^n$ the eigenvalues $k(k+n-1)$ of Δ have multiplicities of order k^{n-1}, so N has jumps of order λ^{n-1} (Exercise 9.8). The remainder is $o(\lambda^{n-1})$ if the set of periodic points of the geodesic flow on the unit cosphere bundle has measure zero (Duistermaat–Guillemin, Ivrii); this uses the structure of T at the non-zero periods. The weaker formula $N(\lambda)\sim C_W\lambda^n$ can also be obtained from the heat kernel or, as in [4], from residues of the zeta function of Δ.

(ii) For $n=1$ each component of X is a circle and everything is explicit (Remark 9.6.3); in that case $N(\lambda) = C_W\lambda + O(1)$, but the argument of Step 2 only gives $O(\log\lambda)$ unless one uses that the lower order terms in (9.6.2) vanish.

(iii) The proof only used: the trace formula (9.6.2) for *one* positive test function with small support, the positivity of μ, and a polynomial a priori bound. See [8, Section 17.5] for the same method based on the Hadamard parametrix.

9.8 Propagation of singularities and the Poisson relation

In this section we explain, with some steps only sketched, how the global behaviour of T is governed by closed geodesics. The full theory requires Fourier integral operators [6] and the propagation theorem for operators of real principal type; for the strictly hyperbolic case see [8, Chapter XXIII].

Let $C_+ \subset \Sigma(P)\times\Sigma(P)$ be the forward bicharacteristic relation: $(\zeta,\zeta')\in C_+$ if ζ lies on the null bicharacteristic through ζ' and $t(\zeta)\geq t(\zeta')$. Write Δ^* for the diagonal of $(T^*(\mathbb{R}\times X)\setminus 0)^2$ and $\mathrm{WF}'(E) = \{(\zeta,\zeta');\ (\zeta,-\zeta')\in\mathrm{WF}(E)\}$.

Lemma 9.8.1. *$\mathrm{WF}'(E)\cap\{|t-t'|<t_0\}\subset\Delta^*\cup C_+$, and $\mathrm{WF}(E)$ contains no points with zero covector in either factor.*

Sketch of proof. By (9.3.1), $E = -H(t-t')K_{\sin}(t-t',x,x')$. For $0<|t-t'|<t_0$, Corollary 9.4.6 gives $\mathrm{WF}(E)\subset$ pull-back of Γ, and by Lemma 9.1.2 a point of Γ at time $t-t'>0$ corresponds exactly to a pair in C_+ (the point (t',x',τ,η) is carried to

$(t, G_{\mp(t-t')}(x', \eta), \tau)$ for $\tau = \pm|\eta|$). At $t = t'$, $\mathrm{WF}(H(t-t'))$ consists of covectors $(\sigma, 0; -\sigma, 0)$ and $\mathrm{WF}(K_{\sin}(t-t', x, x'))$ of covectors $(\tau, \xi; -\tau, -\xi)$ with $x = x'$, $\tau = \pm|\xi|$; their sum is never zero, so the product is defined [7, Section 8.2], and all resulting points over $\{t = t', x = x'\}$ have the form $(\zeta, -\zeta)$ with $\zeta \neq 0$, i.e. lie in Δ^* in the primed notation. The last claim follows from the same description. □

Theorem 9.8.2 (Propagation of singularities)**.** *Let $u \in \mathcal{C}^{-\infty}(\mathbb{R} \times X)$. Then $\mathrm{WF}(u) \setminus \mathrm{WF}(Pu) \subset \Sigma(P)$, and if $\gamma\colon J \to \Sigma(P) \setminus \mathrm{WF}(Pu)$ is a null bicharacteristic on an interval J with $\gamma(s_0) \in \mathrm{WF}(u)$ for some $s_0 \in J$, then $\gamma(J) \subset \mathrm{WF}(u)$.*

Sketch of proof. The first statement is microlocal elliptic regularity. Let $z = \gamma(s_0) \in \mathrm{WF}(u) \setminus \mathrm{WF}(Pu)$, $\bar{t} = t(z)$, and parametrise γ by t (Lemma 9.1.2). As $\mathrm{WF}(Pu)$ is closed, there is $\delta \in (0, t_0/2)$ such that $\gamma(t) \notin \mathrm{WF}(Pu)$ for $|t - \bar{t}| \leq 2\delta$. Let $t_1 \in (\bar{t} - \delta, \bar{t})$ and suppose $\gamma(t_1) \notin \mathrm{WF}(u)$; then, $\mathrm{WF}(u)$ being closed, $\gamma(t) \notin \mathrm{WF}(u)$ for $|t - t_1| \leq \epsilon$. Choose $\chi \in \mathcal{C}^\infty(\mathbb{R})$ with $\chi = 0$ for $t < t_1 - \epsilon$, $\chi = 1$ for $t > t_1 + \epsilon$. Then χu has support bounded below and $P(\chi u) = g = \chi P u + [P, \chi]u$, so $\chi u = Eg$ by Theorem 9.3.1. Since $\chi u = u$ near $\bar{t}$ and only g on $\{t > \bar{t} - 2\delta\}$ influences Eg near $\bar{t}$ modulo $\mathcal{C}^\infty$ (smoothness of E off WF', finite speed), the wavefront calculus for operators with kernel E [7, Section 8.2] and Lemma 9.8.1 give $z \in \mathrm{WF}(g) \cup C_+ \circ \mathrm{WF}(g)$. Now $z \notin \mathrm{WF}(g)$, so some $\zeta' \in \mathrm{WF}(g)$ lies on the backward bicharacteristic segment of z within time 2δ. It is not in $\mathrm{WF}(Pu)$, hence $\zeta' \in \mathrm{WF}([P, \chi]u) \subset \mathrm{WF}(u) \cap \{|t - t_1| \leq \epsilon\}$, i.e. $\zeta' = \gamma(t)$ with $|t - t_1| \leq \epsilon$, a contradiction. Therefore $\gamma(t) \in \mathrm{WF}(u)$ for $t \in (\bar{t} - \delta, \bar{t}]$; applying this to $u(-t, x)$ (which reverses time and is handled by the backward fundamental solution) also for $t \in [\bar{t}, \bar{t} + \delta)$. So the set of $s \in J$ with $\gamma(s) \in \mathrm{WF}(u)$ is open, and it is closed in J since $\mathrm{WF}(u)$ is closed; J being connected, it is all of J. □

The same argument applies to $D_t^2 - \Delta_X$ on $\mathbb{R} \times X \times Y$ (fundamental solution $E \otimes \mathrm{Id}_Y$) at points where $(\tau, \xi) \neq 0$, for distributions whose wavefront set does not meet $\{\tau = 0,\ \xi = 0\}$; the bicharacteristics now leave the (y, η) components fixed.

Proposition 9.8.3. *For all $t \in \mathbb{R}$, $\mathrm{WF}(K_{\cos}) \subset \Gamma$, with Γ as in Corollary 9.4.6.*

Sketch of proof. $K_{\cos}$ is annihilated by $D_t^2 - \Delta_x$ and by $D_t^2 - \Delta_y$ (as $\cos(t\Lambda)$ commutes with Δ), so $\mathrm{WF}(K_{\cos}) \subset \{\tau^2 = |\xi|^2 = |\eta|^2\}$, which does not meet $\{\tau = 0, \xi = 0\}$. By the extension of Theorem 9.8.2, $\mathrm{WF}(K_{\cos})$ is a union of complete bicharacteristics of $D_t^2 - \Delta_x$; each of these meets $\{|t| < t_0\}$, where $\mathrm{WF}(K_{\cos}) \subset \Gamma$ by Corollary 9.4.6. Since Γ is a union of such bicharacteristics, $\mathrm{WF}(K_{\cos}) \subset \Gamma$. □

Let $\mathcal{L}$ be the set of lengths of closed (non-constant, unit speed) geodesics of X. It is closed, and $\inf \mathcal{L} \geq 2\,\mathrm{inj}(X) > 0$, since a geodesic loop at x cannot be shorter than $2\,\mathrm{inj}(X)$.

Theorem 9.8.4 (Poisson relation)**.** $\operatorname{sing\,supp} T \subset \{0\} \cup \mathcal{L} \cup (-\mathcal{L})$.

Sketch of proof. Let $\iota\colon \mathbb{R} \times X \to \mathbb{R} \times X \times X$, $(t, x) \mapsto (t, x, x)$, and $\pi\colon \mathbb{R} \times X \to \mathbb{R}$. The conormal bundle of $\iota(\mathbb{R} \times X)$ consists of covectors $(0, \xi, -\xi)$, which are not in Γ (there $\tau \neq 0$), so $\iota^* K_{\cos}$ is defined and $\mathrm{WF}(\iota^* K_{\cos}) \subset \{(t, x; \tau, \xi - \eta);\ (t, x, x; \tau, \xi, -\eta) \in \Gamma\}$ [7, Section 8.2]. Since integration over t against ψ commutes with ι^* and turns $K_{\cos}$ into the smooth kernel K_{C_ψ}, $T = \pi_* \iota^* K_{\cos}$. For the push-forward, $\mathrm{WF}(\pi_* v) \subset \{(t, \tau);\ \exists x,\ (t, x; \tau, 0) \in \mathrm{WF}(v)\}$. Hence if $(t, \tau) \in \mathrm{WF}(T)$ there are x and $\eta \neq 0$ with $(x, \eta) = G_{\mp t}(x, \eta)$, i.e. either $t = 0$, or the geodesic through x in the direction $g^{-1}\eta$ is closed of length dividing $|t|$ — in particular $|t| \in \mathcal{L}$ (a multiple of a closed geodesic is a closed geodesic). □

Corollary 9.8.5. *The conclusions of Theorem 9.6.2 hold for every $\psi \in \mathcal{C}_{\mathrm{c}}^\infty((-\inf \mathcal{L}, \inf \mathcal{L}))$.*

Proof. Take $\psi_0 \in \mathcal{C}^\infty_c((-t_0,t_0))$ with $\psi_0 = 1$ near 0. Then $(1-\psi_0)\psi T$ is smooth with compact support by Theorem 9.8.4, so its Fourier transform is rapidly decreasing, and Theorem 9.6.2 applies to $\psi_0\psi$, with $(\psi_0\psi)(0) = \psi(0)$. □

9.9 Exercises

Exercise 9.1. Let E be a real vector space with a non-degenerate antisymmetric bilinear form ω. Show that $\dim E = 2n$ is even and that there is a basis $v_1,\dots,v_n,w_1,\dots,w_n$ with dual basis v_j^*, w_j^* such that $\omega = \sum_j v_j^* \wedge w_j^*$. *Hint:* choose $v_1 \neq 0$, then w_1 with $\omega(v_1,w_1) = 1$; then v_2 in the ω-orthogonal complement of $\operatorname{span}(v_1,w_1)$ (why is this complement a complement, and why is ω non-degenerate on it?), and so on. Deduce that there is a linear isomorphism $L\colon E \to \mathbb{R}^{2n}$ with $\omega = L^*\omega_0$, where $\omega_0 = \sum_j d\xi_j \wedge dx_j$ is the standard symplectic form.

Exercise 9.2. Extend Exercise 9.1: if $V \subset E$ is a Lagrangian subspace ($\dim V = n$, $\omega|_V = 0$), then L can be chosen to map V onto $\{x=0\} \subset \mathbb{R}^{2n}$. *Hint:* construct the basis with all $v_j \in V$ (or all w_j, depending on the sign convention).

Exercise 9.3. Let (S,ω) be a symplectic manifold and H_f the Hamilton vector field of $f \in \mathcal{C}^\infty(S)$, defined by $\iota_{H_f}\omega = -df$ (or with the opposite sign). Show that the Poisson bracket $\{f,g\} = H_f g$ makes $\mathcal{C}^\infty(S)$ into a Lie algebra, and that $[H_f,H_g] = H_{\{f,g\}}$. *Hint:* the Jacobi identity is equivalent to $d\omega = 0$.

Exercise 9.4. Let $H_0 = \{(x-y)\cdot\omega = 0\} \subset \mathbb{R}^n$ and let $H_\pm$ be the two characteristic hypersurfaces of Remark 9.4.5 through $\{0\}\times H_0$. Show that for $v_0 \in I^m(\mathbb{R}^n,H_0)$ and $v_1 \in I^{m+1}(\mathbb{R}^n,H_0)$ there are conormal $u_\pm$, associated with $H_\pm$ and solving $Pu_\pm \in \mathcal{C}^\infty$ near $t=0$, with $u_+ + u_- = v_0$ and $D_t(u_+ + u_-) = v_1$ at $t=0$ modulo $\mathcal{C}^\infty$, and that their full symbols at $t=0$ are uniquely determined. (Write the conditions on the symbols as a triangular system, as in the proof of Theorem 9.4.4.)

Exercise 9.5. Let $F \in \mathcal{C}^0([0,t_1]\times X\times X)$ with $|F| \le M$, extended by 0 for $t<0$, and define the t-convolution $(F\circ G)(t,x,y) = \int_0^t \int_X F(t-s,x,z)G(s,z,y)\,d\mathrm{vol}(z)\,ds$. Show that the iterates $F^{(1)} = F$, $F^{(k)} = F^{(k-1)}\circ F$ satisfy $|F^{(k)}(t,x,y)| \le M^k \operatorname{vol}(X)^{k-1}t^{k-1}/(k-1)!$, and deduce that $\mathrm{Id}+F$ (with Id the identity for t-convolution, kernel $\delta(t)\delta_y(x)$) has inverse $\mathrm{Id}+\tilde F$, $\tilde F = \sum_{k\ge1}(-1)^k F^{(k)}$, with $\tilde F$ continuous. Show that if F is smooth and vanishes to infinite order at $t=0$ then so does $\tilde F$, and use this, together with a local parametrix, to construct a forward fundamental solution as in Remark 9.3.2.

Exercise 9.6. For $X = \mathbb{R}/L\mathbb{Z}$ compute $N(\lambda)$ exactly and verify $N(\lambda) = \frac{L}{\pi}\lambda + O(1)$. Show that $T = \sum_{k\in\mathbb{Z}} \cos(2\pi kt/L) = L\sum_{m\in\mathbb{Z}}\delta(t-mL)$ and compare with Theorem 9.8.4.

Exercise 9.7. For the flat torus $X = \mathbb{R}^n/2\pi\mathbb{Z}^n$ show that the eigenvalues of Δ are $|k|^2$, $k\in\mathbb{Z}^n$, and deduce from Theorem 9.7.1 the lattice point estimate $\#\{k\in\mathbb{Z}^n;\ |k|\le\lambda\} = \omega_n\lambda^n + O(\lambda^{n-1})$ $(n\ge2)$.

Exercise 9.8. On the round sphere $\mathbb{S}^2$ the eigenvalues of Δ are $k(k+1)$, $k\in\mathbb{N}_0$, with multiplicity $2k+1$. Show that $C_W = 1$ and that $N(\lambda)-\lambda^2$ takes values $\ge \frac12\lambda$ and $\le -\frac12\lambda$ for arbitrarily large λ, so the remainder in Theorem 9.7.1 is not $o(\lambda^{n-1})$ in general. Compute T for $\mathbb{S}^2$ and locate its singularities.

Exercise 9.9. Show that f_n is homogeneous of degree $-n$, that $f_n = D_s^{n-1}\delta$ for n odd, and that for n even $f_n(s) = c_n s^{-n}$ for $s\neq0$, with a constant $c_n \neq 0$ (compute it). Deduce from Proposition 9.4.2 that, for n odd, δ is a superposition of distributions supported on hyperplanes, whereas for n even it is not.

Notes

Compared with the original notes, we have restricted attention to compact X (dropping the case of $\mathbb{R}^n$ with a metric that is Euclidean at infinity), and solved the Cauchy problem and constructed the forward fundamental solution by spectral decomposition, keeping the Volterra-series construction as a remark and an exercise. The short-time parametrix is written as an oscillatory integral whose phase solves the eikonal equation; in polar coordinates it is exactly the superposition of conormal solutions associated with characteristic hypersurfaces through hyperplanes that the original uses (Remark 9.4.5). The order of $f_n(x \cdot \omega)$ in Lemma 9.4.1 is $\frac{n}{2} - \frac{1}{4}$ (not $\frac{n}{4} - \frac{1}{4}$), and Weyl's formula with the remainder $O(\lambda^{n-1})$ is what the Tauberian argument proves; the original stated $o(\lambda^{n-1})$, which is false in general (Exercise 9.8).

The remainder $O(\lambda^{n-1})$ is due to Avakumović, Levitan and Hörmander (L. Hörmander, *The spectral function of an elliptic operator*, Acta Math. 121 (1968)); see also [8, Section 17.5]. The Poisson relation and the improvement to $o(\lambda^{n-1})$ under a condition on periodic geodesics are due to Chazarain, Colin de Verdière and Duistermaat–Guillemin (J. J. Duistermaat and V. W. Guillemin, *The spectrum of positive elliptic operators and periodic bicharacteristics*, Invent. Math. 29 (1975)), and Ivrii. For Fourier integral operators and Lagrangian distributions, of which the wave kernel is the basic example, see [6], [2]; for a different proof of the leading term see [4].

Part IV

Index theory

Chapter 10

Homotopy invariance of the index and Toeplitz operators

This chapter opens the part of the book devoted to index theory. Let X be a compact manifold without boundary and E, F complex vector bundles over X. By Chapter 5 an elliptic operator $A \in \Psi^m(X; E, F)$ has a parametrix and is Fredholm, so its index

$$\operatorname{ind}(A) = \dim \operatorname{Nul}(A) - \dim\bigl(\mathcal{C}^\infty(X; F)/A\,\mathcal{C}^\infty(X; E)\bigr)$$

is defined. The first two sections show that $\operatorname{ind}(A)$ can be written as a difference of traces of smoothing operators, and deduce from this that the index is constant along smooth families of elliptic operators. It therefore depends only on the homotopy class of the principal symbol. The index theorem computes this integer from topological data. To set that up we take *odd K-theory* to be the homotopy classes of maps into the group $G^{-\infty}(Y; E)$ of invertible smoothing perturbations of the identity. We show that this is an abelian group and introduce its even counterpart, which is classified by a loop group $G^{-\infty}_{(1)}$ (Section 10.3). The bi-invariant *Chern forms* $\operatorname{Tr}((A^{-1}dA)^{2k-1})$ on $G^{-\infty}$ give de Rham classes on these K-groups. The simplest of them, $\operatorname{Tr}(A^{-1}dA)$, has integral periods, and this gives a direct construction of the Fredholm determinant (Section 10.4). The last three sections turn to the Toeplitz algebra on the circle. It is the model for the "symbol sequence" that leads to Bott periodicity in Chapter 11. We compute the index of Toeplitz operators. We show that families of elliptic operators with vanishing "families index" can be perturbed to be invertible, and that the matrix groups can be deformed away inside the invertible Toeplitz operators.

10.1 The index as a trace

10.1.1 What we use from earlier chapters

Throughout, X is a compact manifold without boundary, of positive dimension, and E, F are complex vector bundles over it. Operators in $\Psi^m(X; E, F)$ are classical (one-step polyhomogeneous). As in Chapters 4 and 5 their Schwartz kernels are conormal distributions at the diagonal. The principal symbol $\sigma_m(A)$ is a section of $\hom(\pi^*E, \pi^*F)$ over $T^*X \setminus 0$, homogeneous of degree m. After fixing a Riemannian metric we identify it with its restriction to the cosphere bundle S^*X. We use the following facts.

(i) The smoothing operators are $\Psi^{-\infty}(X; E, F) = \mathcal{C}^\infty(X^2; \operatorname{Hom}(E, F) \otimes \Omega_R)$, with $\operatorname{Hom}(E, F) = \pi_L^* F \otimes \pi_R^* E'$ and Ω_R the density bundle from the right factor. They

carry the $\mathcal{C}^\infty$ (Fréchet) topology. A smoothing operator maps $\mathcal{C}^{-\infty}(X;E)$ continuously into $\mathcal{C}^\infty(X;F)$. In particular it is bounded on L^2 for any choice of Hermitian metrics and of a smooth positive density, and the operator norm is a continuous seminorm on $\Psi^{-\infty}$.

(ii) If $A \in \Psi^m$ and $R \in \Psi^{-\infty}$ then AR and RA are smoothing (when defined), and depend continuously on R. The formal adjoint and the transpose of A lie in Ψ^m, and $\sigma_m(A^*) = \sigma_m(A)^*$.

(iii) $A \in \Psi^m(X;E,F)$ is bounded $H^s(X;E) \to H^{s-m}(X;F)$ for every s. If A is elliptic, i.e. $\sigma_m(A)$ is invertible on S^*X, it has a parametrix $B_0 \in \Psi^{-m}(X;F,E)$:

$$B_0 A = \mathrm{Id}_E - R, \quad AB_0 = \mathrm{Id}_F - R', \quad R \in \Psi^{-\infty}(X;E),\ R' \in \Psi^{-\infty}(X;F). \tag{10.1.1}$$

(iv) Finite rank smoothing operators are dense in $\Psi^{-\infty}(X;E,F)$. A smoothing operator of finite rank has kernel $\sum_{i=1}^N f_i(x)e_i(y)$ with $f_i \in \mathcal{C}^\infty(X;F)$ and $e_i \in \mathcal{C}^\infty(X;E'\otimes\Omega)$. The approximation can be made uniform for families: if $a \in \mathcal{C}^\infty(Z;\Psi^{-\infty}(X;E,F))$ with Z compact, the approximating kernels may be taken as $\sum_{i,j\le N} c_{ij}(z) f_i(x) e_j(y)$ with fixed f_i, e_j.

(v) The trace $\mathrm{Tr}(A) = \int_X \mathrm{tr}\, A(x,x)$ is a continuous linear functional on $\Psi^{-\infty}(X;E)$. It equals the matrix trace on finite rank operators, and $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ for $A \in \Psi^{-\infty}(X;E,F)$, $B \in \Psi^{-\infty}(X;F,E)$. The proof in Chapter 5 for $E = F$ applies verbatim.

(vi) If $A \in \Psi^{-\infty}(X;E)$ and $\mathrm{Id}+A$ is invertible on $L^2(X;E)$ (equivalently on $\mathcal{C}^\infty(X;E)$), then $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+B$ with $B \in \Psi^{-\infty}(X;E)$. The group

$$G^{-\infty}(X;E) = \{\mathrm{Id}+A;\ A \in \Psi^{-\infty}(X;E),\ \mathrm{Id}+A \text{ invertible}\}$$

is an open subset of $\mathrm{Id}+\Psi^{-\infty}(X;E)$, a topological group, and connected.

We identify $G^{-\infty}(X;E)$ with the open subset $\{A;\ \mathrm{Id}+A \text{ invertible}\} \subset \Psi^{-\infty}(X;E)$ whenever we speak of smooth maps into it. We also need two simple facts about finite rank operators.

Lemma 10.1.1. *If $\pi \in \Psi^{-\infty}(X;E)$ is a projection ($\pi^2 = \pi$) of finite rank, then* $\mathrm{Tr}(\pi) = \dim \mathrm{Ran}(\pi)$.

Proof. Let $f_1, \dots, f_N \in \mathcal{C}^\infty(X;E)$ be a basis of $\mathrm{Ran}(\pi)$. Then $\pi u = \sum_i \ell_i(u) f_i$, where the ℓ_i are continuous linear functionals given by smooth sections e_i of $E'\otimes\Omega$. Since $\pi f_j = f_j$ we get $\ell_i(f_j) = \delta_{ij}$, so $\mathrm{Tr}(\pi) = \sum_i \ell_i(f_i) = N$. □

Lemma 10.1.2. *If $A \in \Psi^m(X;E,F)$ and $C \in \Psi^{-\infty}(X;F,E)$ then* $\mathrm{Tr}(AC) = \mathrm{Tr}(CA)$, *the traces being taken on $\Psi^{-\infty}(X;F)$ and $\Psi^{-\infty}(X;E)$ respectively.*

Proof. Both sides are continuous in C by (ii) and (v). By (iv) it suffices to treat C of rank one, $Cf = \ell(f)w$ with $w \in \mathcal{C}^\infty(X;E)$ and $\ell(f) = \int_X v \cdot f$, $v \in \mathcal{C}^\infty(X;F'\otimes\Omega)$. Then AC is the rank one operator $f \mapsto \ell(f)\, Aw$ and CA is $g \mapsto \ell(Ag)\, w$. The functional $g \mapsto \ell(Ag)$ is given by the smooth section $A^t v$, where $A^t \in \Psi^m$ is the transpose. Thus

$$\mathrm{Tr}(AC) = \int_X v \cdot Aw = \int_X (A^t v) \cdot w = \mathrm{Tr}(CA). \qquad \square$$

10.1.2 The generalized inverse

The parametrix of an elliptic operator can be improved to an exact inverse up to finite rank projections.

Proposition 10.1.3. *Let $A \in \Psi^m(X;E,F)$ be elliptic. Choose Hermitian metrics on E, F and a smooth positive density on X, and let $A^* \in \Psi^m(X;F,E)$ be the formal adjoint. Then* $\mathrm{Nul}(A) \subset \mathcal{C}^\infty(X;E)$ *and* $\mathrm{Nul}(A^*) \subset \mathcal{C}^\infty(X;F)$ *are finite dimensional and*

$$\mathcal{C}^\infty(X;F) = A(\mathcal{C}^\infty(X;E)) \oplus \mathrm{Nul}(A^*), \tag{10.1.2}$$

an L^2-orthogonal decomposition. Let π and π' be the orthogonal projections onto $\mathrm{Nul}(A)$ *and* $\mathrm{Nul}(A^*)$*. There is a unique $B \in \Psi^{-m}(X;F,E)$, the* generalized inverse *of A, such that*

$$BA = \mathrm{Id}_E - \pi, \quad AB = \mathrm{Id}_F - \pi', \quad \pi B = 0, \quad B\pi' = 0. \tag{10.1.3}$$

The projections π, π' are self-adjoint finite rank smoothing operators, and

$$\mathrm{ind}(A) = \dim \mathrm{Nul}(A) - \dim \mathrm{Nul}(A^*) = \mathrm{Tr}(\pi) - \mathrm{Tr}(\pi').$$

Proof. Let B_0, R, R' be as in (10.1.1). As a bounded map $A : H^m(X;E) \to L^2(X;F)$, A satisfies $B_0A = \mathrm{Id} - R$ and $AB_0 = \mathrm{Id} - R'$, where R is compact on H^m and R' is compact on L^2. Indeed both map into $\mathcal{C}^\infty$ continuously from H^{-N} for every N, and $H^N \hookrightarrow H^m$ is compact for $N > m$ (Rellich). By Appendix A $A : H^m \to L^2$ is Fredholm. In particular its range is closed. A distribution $u \in H^m$ with $Au = 0$ satisfies $u = Ru \in \mathcal{C}^\infty$, so $\mathrm{Nul}(A)$ is the same on H^m and on $\mathcal{C}^\infty$, and is finite dimensional. The orthogonal complement of $A(H^m)$ in L^2 consists of the $v \in L^2$ with $A^*v = 0$ in the sense of distributions. Since $B_0^*A^* = \mathrm{Id} - (R')^*$, such v satisfy $v = (R')^*v \in \mathcal{C}^\infty$, so this complement is $\mathrm{Nul}(A^*)$, a finite dimensional subspace of $\mathcal{C}^\infty(X;F)$. If $f \in \mathcal{C}^\infty(X;F)$ is orthogonal to $\mathrm{Nul}(A^*)$ then $f = Au$ with $u \in H^m$, and $u = B_0 f + Ru \in \mathcal{C}^\infty$. This proves (10.1.2). The orthogonal projection onto a finite dimensional space $W \subset \mathcal{C}^\infty$ with orthonormal basis w_i has kernel $\sum_i w_i(x)\overline{w_i(y)}\nu(y)$, so π and π' are self-adjoint finite rank smoothing operators.

The map $A : \mathrm{Nul}(A)^\perp \cap H^m \to A(H^m)$ is a continuous bijection between Hilbert spaces. Let $B : L^2(X;F) \to H^m(X;E)$ be its inverse on $A(H^m)$, extended by zero on $\mathrm{Nul}(A^*)$. By the open mapping theorem B is bounded, and (10.1.3) holds on L^2. These four identities clearly determine B. Substituting $\mathrm{Id} = AB_0 + R'$ on the right and $\mathrm{Id} = B_0A + R$ on the left,

$$\begin{aligned} B &= B(AB_0 + R') = (\mathrm{Id} - \pi)B_0 + BR' = (\mathrm{Id} - \pi)B_0 + (B_0A + R)BR' \\ &= (\mathrm{Id} - \pi)B_0 + B_0(\mathrm{Id} - \pi')R' + RBR'. \end{aligned}$$

The first two terms lie in $\Psi^{-m}(X;F,E)$. The kernel of RBR' is smooth. The map $y \mapsto R'(\cdot, y)$ is smooth from X into $\mathcal{C}^\infty(X;F)$, hence into L^2. Then B is bounded on L^2, and $R : L^2 \to \mathcal{C}^\infty$ is continuous. So $y \mapsto RBR'(\cdot, y)$ is a smooth map into $\mathcal{C}^\infty(X;E)$. Thus $B \in \Psi^{-m}(X;F,E)$. Finally $\mathrm{Tr}(\pi) = \dim \mathrm{Nul}(A)$ and $\mathrm{Tr}(\pi') = \dim \mathrm{Nul}(A^*) = \dim \mathcal{C}^\infty(X;F)/A\mathcal{C}^\infty(X;E)$ by Lemma 10.1.1 and (10.1.2). □

Any two parametrices of A differ by a smoothing operator. Indeed, if $B_1A - \mathrm{Id}$ and $AB_2 - \mathrm{Id}$ are smoothing then $B_1 - B_2 = B_1(\mathrm{Id} - AB_2) - (\mathrm{Id} - B_1A)B_2 \in \Psi^{-\infty}$. In particular the generalized inverse is a parametrix in the sense of (10.1.1).

Theorem 10.1.4. *Let $A \in \Psi^m(X;E,F)$ be elliptic and let $B \in \Psi^{-m}(X;F,E)$ be any parametrix, so that* $\mathrm{Id}_E - BA$ *and* $\mathrm{Id}_F - AB$ *are smoothing. Then*

$$\mathrm{ind}(A) = \mathrm{Tr}(\mathrm{Id}_E - BA) - \mathrm{Tr}(\mathrm{Id}_F - AB). \tag{10.1.4}$$

Proof. For the generalized inverse $\tilde{B}$ of Proposition 10.1.3 this is the identity $\operatorname{ind}(A) = \operatorname{Tr}(\pi) - \operatorname{Tr}(\pi')$. For a general parametrix, $C = B - \tilde{B} \in \Psi^{-\infty}(X; F, E)$ and

$$(\operatorname{Tr}(\mathrm{Id} - BA) - \operatorname{Tr}(\mathrm{Id} - AB)) - (\operatorname{Tr}(\mathrm{Id} - \tilde{B}A) - \operatorname{Tr}(\mathrm{Id} - A\tilde{B})) = \operatorname{Tr}(AC) - \operatorname{Tr}(CA) = 0$$

by Lemma 10.1.2. □

Remark 10.1.5. The right side of (10.1.4) is $\operatorname{Tr}([A, B])$ in a formal sense: $\mathrm{Id} - BA$ and $\mathrm{Id} - AB$ are individually trace class, but AB and BA are not. The formula shows that the index measures how far the trace identity $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ fails when the operators are not smoothing.

10.2 Homotopy invariance of the index

10.2.1 Smooth families

Let Z be a manifold, possibly with boundary or corners (we mostly use $Z = [0, 1]$ and $Z = [0, 1] \times X'$ with X' compact). A *smooth family* $A \in \mathcal{C}^\infty(Z; \Psi^m(X; E, F))$ is a family $A_z \in \Psi^m(X; E, F)$ whose kernels have the following properties. Away from $Z \times \mathrm{Diag}$ they are smooth on $Z \times (X^2 \setminus \mathrm{Diag})$. In local coordinates and trivializations near the diagonal they are of the form

$$\int e^{i(x-y)\cdot\xi} a(z, x, \xi) đ\xi + K(z, x, y),$$

where K is smooth in all variables and a is a classical symbol of order m depending smoothly on z. That is, every z-derivative of a satisfies the symbol estimates locally uniformly in z, and the homogeneous terms of the expansion of a are smooth in z. For $m = -\infty$ this simply means $A \in \mathcal{C}^\infty(Z \times X^2; \operatorname{Hom}(E, F) \otimes \Omega_R)$.

All the constructions of Chapters 4 and 5 (composition, adjoints, quantization of symbols, asymptotic summation, parametrices) are given by formulas that are continuous in the symbol seminorms, so they apply to smooth families with the result again smooth. The only point needing a comment is asymptotic summation (Borel's lemma). The cut-offs $\chi(\epsilon_j \xi)$ can be chosen with ϵ_j decreasing so fast that the estimates hold uniformly on an exhaustion of Z by compact sets. We record the consequences we need.

Lemma 10.2.1. *Let $A \in \mathcal{C}^\infty(Z; \Psi^m(X; E, F))$ be a smooth family with A_z elliptic for every $z \in Z$.*

(a) There is a smooth family $B \in \mathcal{C}^\infty(Z; \Psi^{-m}(X; F, E))$ with $\mathrm{Id} - B_z A_z$ and $\mathrm{Id} - A_z B_z$ in $\mathcal{C}^\infty(Z; \Psi^{-\infty})$.

(b) If $m = 0$, $E = F$ and every A_z is invertible, then $z \mapsto A_z^{-1}$ is a smooth family in $\mathcal{C}^\infty(Z; \Psi^0(X; E))$.

Moreover, given $a \in \mathcal{C}^\infty(Z \times S^*X; \hom(\pi^*E, \pi^*F))$ there is a smooth family $Q(a) \in \mathcal{C}^\infty(Z; \Psi^m(X; E, F))$ with $\sigma_m(Q(a)_z) = a_z$, depending linearly on a.

Proof. The quantization Q is obtained by extending a homogeneously of degree m, cutting off near the zero section, quantizing in coordinate charts and summing with a partition of unity. Each step preserves smooth dependence on z. For (a), let $B' = Q(\sigma_m(A)^{-1})$ (with $-m$ in place of m). Then $\mathrm{Id} - B'A \in \mathcal{C}^\infty(Z; \Psi^{-1})$, and the left parametrix $B_L \sim \sum_{j \geq 0} (\mathrm{Id} - B'A)^j B'$ is an asymptotic sum of smooth families, so can be chosen smooth in z. Similarly for a right parametrix B_R, and $B_L - B_R = B_L(\mathrm{Id} - AB_R) - (\mathrm{Id} - B_L A)B_R$ is a smooth family of smoothing operators.

For (b), let $G_z = A_z^{-1}$. It lies in Ψ^0 by Proposition 10.1.3 with $\pi = \pi' = 0$. A smooth family in Ψ^0 is a smooth family of bounded operators on L^2, because the L^2 norm of an operator of order 0 is bounded by a symbol seminorm. So $z \mapsto G_z$ is smooth into the bounded operators, being the inverse of a smooth family of invertible bounded operators. With B from (a) and $R = \mathrm{Id} - BA$, $R' = \mathrm{Id} - AB$, the argument in the proof of Proposition 10.1.3 (with $\pi = \pi' = 0$) gives

$$G = G(AB + R') = B + GR' = B + (BA + R)GR' = B + BR' + RGR'.$$

Since R_z, R'_z are smooth families of smoothing operators and G_z is smooth in operator norm, the kernel of RGR' is smooth in all variables. Hence G is a smooth family. □

10.2.2 Invariance

Theorem 10.2.2. *If $A \in \mathcal{C}^\infty([0,1]; \Psi^m(X; E, F))$ is a smooth family with A_t elliptic for every $t \in [0,1]$, then $\mathrm{ind}(A_t)$ is independent of t.*

Proof. Choose a smooth family of parametrices B_t as in Lemma 10.2.1. By Theorem 10.1.4

$$\mathrm{ind}(A_t) = \mathrm{Tr}(\mathrm{Id} - B_t A_t) - \mathrm{Tr}(\mathrm{Id} - A_t B_t).$$

The arguments of the traces are continuous (indeed smooth) maps $[0,1] \to \Psi^{-\infty}$ and the trace is continuous, so $\mathrm{ind}(A_t)$ is a continuous integer-valued function on $[0,1]$. Hence it is constant. □

Remark 10.2.3. The same conclusion holds for families that are only continuous in t (in the topology of Ψ^m). One way to see this is to note that such a family is norm-continuous as a family of Fredholm operators $H^m \to L^2$. Its index there equals the index on $\mathcal{C}^\infty$, by the proof of Proposition 10.1.3. The index of Fredholm operators is locally constant in the norm topology (Appendix A).

Two elliptic operators $A, A' \in \Psi^m(X; E, F)$ with the same principal symbol are joined by the smooth family $(1-t)A + tA'$, which has constant principal symbol and so is elliptic. Hence $\mathrm{ind}(A) = \mathrm{ind}(A')$. More precisely, we have the following.

Corollary 10.2.4. *Let $A_0, A_1 \in \Psi^m(X; E, F)$ be elliptic. Suppose that*

$$a \in \mathcal{C}^\infty([0,1] \times S^*X; \hom(\pi^*E, \pi^*F))$$

is invertible at every point, with $a_0 = \sigma_m(A_0)$ and $a_1 = \sigma_m(A_1)$. Then $\mathrm{ind}(A_0) = \mathrm{ind}(A_1)$. Thus the index of an elliptic operator depends only on the homotopy class of its principal symbol in $\mathcal{C}^\infty(S^*X; \mathrm{Iso}(\pi^*E, \pi^*F))$.

Proof. With Q as in Lemma 10.2.1,

$$A_t = Q(a)_t + (1-t)(A_0 - Q(a)_0) + t(A_1 - Q(a)_1)$$

is a smooth family with $\sigma_m(A_t) = a_t$ invertible, and A_0, A_1 are its end points. Apply Theorem 10.2.2. □

Remark 10.2.5. (1) A continuous homotopy of invertible symbols between smooth end points can be approximated, with the same end points, by a smooth one, since the invertible elements form an open set. So continuous homotopies suffice in Corollary 10.2.4.

(2) The order is also irrelevant. For each $s \in \mathbb{R}$ there is an invertible elliptic $\Lambda_s \in \Psi^s(X; E)$. Take any elliptic $\Lambda \in \Psi^{s/2}(X; E)$ and set $\Lambda_s = \Lambda^*\Lambda + \pi$, with π the orthogonal

projection onto $\mathrm{Nul}(\Lambda)$. If $\Lambda_s u = 0$ then $\|\Lambda u\|^2 + \|\pi u\|^2 = 0$, so $u = 0$. Since Λ_s is self-adjoint, (10.1.2) shows it is also surjective. For $A \in \Psi^m(X; E, F)$ elliptic, $A\Lambda_{-m} \in \Psi^0(X; E, F)$ has the same index, since $\mathrm{Nul}(A\Lambda_{-m}) = \Lambda_{-m}^{-1}\,\mathrm{Nul}(A)$ and the ranges coincide. So the index is a function on the homotopy classes of $\mathcal{C}^\infty(S^*X; \mathrm{Iso}(\pi^*E, \pi^*F))$. Even this contains more information than the index sees. The index theorem eliminates the extra data by passing to K-theory, which we now introduce.

10.3 Smoothing groups and K-theory

10.3.1 Odd K-theory

Let Y be a compact manifold of positive dimension and E a vector bundle over Y. Let X be a compact manifold, possibly with boundary or corners. A *smooth map* $f : X \to G^{-\infty}(Y; E)$ is a map such that $f - \mathrm{Id} \in \mathcal{C}^\infty(X; \Psi^{-\infty}(Y; E)) = \mathcal{C}^\infty(X \times Y^2; \mathrm{Hom}(E) \otimes \Omega_R)$ and $f(x)$ is invertible for each x. Then $x \mapsto f(x)^{-1} - \mathrm{Id}$ is again smooth. If $f = \mathrm{Id} + a$ and $f^{-1} = \mathrm{Id} + b$, the identity $b = -a + a^2 + aba$ shows that b is smooth into $\Psi^{-\infty}$ once it is continuous in L^2 norm, and the latter holds because inversion is continuous on bounded operators. A smooth homotopy is a smooth map on $[0, 1] \times X$. As in Chapter 5 we set

$$K^{-1}(X) = [X; G^{-\infty}(Y; E)], \tag{10.3.1}$$

the set of smooth homotopy classes of smooth maps. We recall the result of Chapter 5: (10.3.1) is independent of (Y, E). The natural bijections are induced by finite rank approximation, which deforms any smooth f to a map of the form

$$f(x) = \mathrm{Id} + \sum_{i,j=1}^{N} (g_{ij}(x) - \delta_{ij})\phi_i \otimes \phi_j^*, \quad g \in \mathcal{C}^\infty(X; \mathrm{GL}(N, \mathbb{C})), \tag{10.3.2}$$

for an orthonormal family $\phi_1, \dots, \phi_N \in \mathcal{C}^\infty(Y; E)$, where $\phi_i \otimes \phi_j^*$ denotes $u \mapsto (u, \phi_j)\phi_i$. Such an f acts as the matrix $g(x)$ on $W_N = \mathrm{span}\{\phi_i\}$ and as the identity on $W_N^\perp$. It is invertible precisely when $g(x)$ is. To see that a smooth $f = \mathrm{Id} + a$ is homotopic to a map of this form, approximate a uniformly on X in $\mathcal{C}^\infty(X; \Psi^{-\infty})$ by $a^{(N)} = \sum_{i,j \le N} c_{ij}(x)\phi_i \otimes \phi_j^*$, as in (iv). Since $G^{-\infty}$ is open and X compact, $\mathrm{Id} + (1 - s)a + sa^{(N)}$ is invertible for $s \in [0, 1]$ once N is large.

Proposition 10.3.1. *Pointwise multiplication induces a product on $K^{-1}(X)$ which makes it an abelian group, with identity the class of the constant map* Id *and inverse $[f]^{-1} = [f^{-1}]$. The product is also induced by the "direct sum": if f and f' are of the form (10.3.2) with matrices g, g' acting on orthogonal spaces W, W', then $[f][f']$ is represented by the map acting as $g \oplus g'$ on $W \oplus W'$ and as the identity on $(W \oplus W')^\perp$.*

Proof. If f_s, f_s' are homotopies then so is $f_s f_s'$. Hence the product is well defined on homotopy classes. Associativity, the identity and the inverse are inherited from $G^{-\infty}$. For commutativity, deform f and f' to the form (10.3.2) with matrices g, g' with respect to the same orthonormal family $\phi_1, \dots, \phi_N$. Enlarge the family to $\phi_1, \dots, \phi_{2N}$ (possible since $\dim Y > 0$, so $\mathcal{C}^\infty(Y; E)$ is infinite dimensional). Let W_1 and W_2 be spanned by the first and last N elements. For $\theta \in [0, \pi/2]$ let R_θ be the unitary operator acting on $W_1 \oplus W_2 = \mathbb{C}^N \oplus \mathbb{C}^N$ by the block rotation

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

and as the identity on $(W_1 \oplus W_2)^\perp$. Then $R_\theta - \mathrm{Id}$ is a finite rank smoothing operator and $R_\theta f' R_\theta^{-1}$ is a smooth homotopy in $G^{-\infty}$. At $\theta = \pi/2$ it acts as $\mathrm{Id}_{W_1} \oplus g'$, with g' now acting on W_2; this is a direct computation with $R_{\pi/2} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Thus f' is homotopic to $f'' = \mathrm{Id}_{W_1} \oplus g'$, which commutes with $f = g \oplus \mathrm{Id}_{W_2}$, and

$$[f][f'] = [ff''] = [g \oplus g'] = [f''f] = [f'][f].$$

The middle term is the direct sum description. □

Remark 10.3.2. In terms of homotopy groups, the assertion that (10.3.1) is odd K-theory, i.e. that $G^{-\infty}$ is a classifying space for K^{-1}, amounts to Bott periodicity in the form

$$\pi_k(G^{-\infty}(Y;E)) = \begin{cases} 0 & k \text{ even}, \\ \mathbb{Z} & k \text{ odd}. \end{cases} \tag{10.3.3}$$

This is proved in Chapter 11. In the standard topological treatment $K^{-1}(X)$ is defined by stabilizing maps into $\mathrm{GL}(N,\mathbb{C})$ as $N \to \infty$ [17]. The group $G^{-\infty}$ makes the stabilization automatic: by (10.3.2) every map is stably a map into some $\mathrm{GL}(N,\mathbb{C})$, and the rotation argument above is the usual proof that the stabilized product is commutative.

10.3.2 Even K-theory

Let $\mathbb{S}^1 = \{e^{i\theta}\}$ with base point 1.

Definition 10.3.3. $K^{-2}(X)$ is the set of smooth homotopy classes of *pointed* smooth maps $f : X \times \mathbb{S}^1 \to G^{-\infty}(Y;E)$, meaning $f(x,1) = \mathrm{Id}$ for all $x \in X$, where homotopies are also required to be pointed at every time.

Pointwise product makes $K^{-2}(X)$ a group. To see that it is abelian, and how it is related to K^{-1}, we compare it with $K^{-1}(X \times \mathbb{S}^1)$.

Lemma 10.3.4. *The map $\Phi : K^{-1}(X) \times K^{-2}(X) \to K^{-1}(X \times \mathbb{S}^1)$, $\Phi([g],[h]) = [gh]$, where g is regarded as a map on $X \times \mathbb{S}^1$ constant in θ, is an isomorphism of groups. Its inverse is*

$$[f] \mapsto \big([f_1],[f_2]\big), \quad f_1(x) = f(x,1), \quad f_2(x,\theta) = f(x,1)^{-1} f(x,\theta). \tag{10.3.4}$$

In particular $K^{-2}(X)$ is abelian and $K^{-1}(X \times \mathbb{S}^1) \cong K^{-1}(X) \oplus K^{-2}(X)$.

Proof. Φ is well defined, since homotopies of g and h multiply to a homotopy of gh. The map (10.3.4) is also well defined. If F is a homotopy from f to f', then $F(\cdot,1)$ is a homotopy from f_1 to f_1' and $F(\cdot,1)^{-1}F$ is a pointed homotopy from f_2 to f_2'. The two maps are mutually inverse at the level of maps, since $f = f_1 f_2$ and $(gh)_1 = g$, $(gh)_2 = h$. Finally Φ is a homomorphism because $K^{-1}(X \times \mathbb{S}^1)$ is abelian by Proposition 10.3.1:

$$\Phi([g][g'],[h][h']) = [g][g'][h][h'] = [g][h][g'][h'] = \Phi([g],[h])\Phi([g'],[h']).$$

Since Φ restricted to $\{1\} \times K^{-2}(X)$ is an injective homomorphism into an abelian group, $K^{-2}(X)$ is abelian. □

10.3.3 The loop group

Pointed maps may be further normalized to be *flat* at the base point, i.e. to differ from the identity by a map vanishing to infinite order at $X \times \{1\}$. Such maps are naturally Schwartz functions on $X \times \mathbb{R}$.

Definition 10.3.5. The (smooth, flat, pointed) loop group of $G^{-\infty}(Y;E)$ is

$$G^{-\infty}_{(1)}(Y;E) = \{a \in \mathcal{S}(\mathbb{R};\Psi^{-\infty}(Y;E));\ \mathrm{Id}+a(t) \in G^{-\infty}(Y;E)\ \forall\, t \in \mathbb{R}\},$$

where $\mathcal{S}(\mathbb{R};\Psi^{-\infty}(Y;E)) = \mathcal{S}(\mathbb{R}\times Y^2;\mathrm{Hom}(E)\otimes\Omega_R)$ is the space of kernels that are Schwartz in t, uniformly with all derivatives in Y^2. The product is $(\mathrm{Id}+a)(\mathrm{Id}+b) = \mathrm{Id}+a+b+ab$, computed pointwise in t.

Lemma 10.3.6. *$G^{-\infty}_{(1)}(Y;E)$ is an open subset of $\mathcal{S}(\mathbb{R};\Psi^{-\infty}(Y;E))$ and a topological group.*

Proof. If $a \in G^{-\infty}_{(1)}$ then $\|a(t)\|_{L^2} \to 0$ as $|t| \to \infty$. So $\sup_t \|(\mathrm{Id}+a(t))^{-1}\| < \infty$ by continuity of inversion, and every a' with $\sup_t \|a'(t)-a(t)\|_{L^2}$ small also has $\mathrm{Id}+a'(t)$ invertible for all t. The supremum is bounded by a Schwartz seminorm, so $G^{-\infty}_{(1)}$ is open. Products of Schwartz families are Schwartz. For the inverse $\mathrm{Id}+b(t)$ we have $b = -a + a^2 + aba$ pointwise. Here $b(t)$ is bounded on L^2 uniformly in t, as are its t-derivatives, by repeated differentiation of $\partial_t b = -(\mathrm{Id}+b)(\partial_t a)(\mathrm{Id}+b)$. The kernel of aba is then Schwartz in t, because each factor a contributes Schwartz decay and $\mathcal{C}^\infty$ regularity in the outer variable. So $b \in G^{-\infty}_{(1)}$, and composition and inversion are continuous for the same reasons. □

Proposition 10.3.7. *Let $\gamma(t) = \exp\big(i\pi(1+t/\langle t\rangle)\big)$, a diffeomorphism of $\mathbb{R}$ onto $\mathbb{S}^1 \setminus \{1\}$. Pulling back by γ gives a bijection*

$$K^{-2}(X) \cong [X; G^{-\infty}_{(1)}(Y;E)]$$

onto the smooth homotopy classes of smooth maps $X \to G^{-\infty}_{(1)}(Y;E)$.

Proof. First, pulling back by γ identifies the functions on $\mathbb{S}^1$ vanishing to infinite order at 1 with $\mathcal{S}(\mathbb{R})$, with parameters. Indeed, near $t = \pm\infty$ the angle $\theta = \pi(1+t/\langle t\rangle)$ satisfies $\theta \sim \pi/(2t^2)$ and $2\pi - \theta \sim \pi/(2t^2)$. Moreover $d\theta/dt = \pi\langle t\rangle^{-3}$, so ∂_t and ∂_θ differ by factors that are polynomially bounded in t or in θ^{-1}. Infinite order vanishing in θ is therefore equivalent to rapid decay of all t-derivatives. Hence smooth maps $X \to G^{-\infty}_{(1)}$ correspond exactly to smooth pointed maps $X \times \mathbb{S}^1 \to G^{-\infty}$ that are flat at $X \times \{1\}$, and the same holds for homotopies.

It remains to show that every pointed map is homotopic, through pointed maps, to a flat one, and that two flat maps which are pointed homotopic are homotopic through flat maps. Let $f = \mathrm{Id}+F$ be pointed. Then $F(x,1) = 0$, so by compactness there is $\epsilon > 0$ with $\|F(x,e^{i\theta})\|_{L^2} < 1$ for $|\theta| \le 2\epsilon$. Choose $\rho \in \mathcal{C}^\infty(\mathbb{S}^1)$ with $0 \le \rho \le 1$, $\rho = 1$ for $|\theta| \le \epsilon$ and $\rho = 0$ for $|\theta| \ge 2\epsilon$. Then

$$f_s = \mathrm{Id}+(1-s\rho)F, \quad s \in [0,1],$$

is a pointed homotopy. Where $\rho \neq 0$ we have $\|(1-s\rho)F\|_{L^2} < 1$, so f_s is invertible. The end point f_1 equals Id near $X \times \{1\}$, so it is flat. If f is already flat, every f_s is flat. Applying the same construction to a pointed homotopy between flat maps, with X replaced by $[0,1] \times X$, gives a homotopy through flat maps from f_1 to f_1'. Combined with the flat homotopies f_s and f_s' at the ends, this connects f to f' through flat maps. □

Remark 10.3.8. Thus, essentially by definition, $G^{-\infty}(Y;E)$ and $G^{-\infty}_{(1)}(Y;E)$ are classifying spaces for odd and even K-theory respectively. The second is the space of based loops in the first, so $\pi_k(G^{-\infty}_{(1)}) = \pi_{k+1}(G^{-\infty})$, and (10.3.3) becomes $\pi_0(G^{-\infty}_{(1)}) = \mathbb{Z}$, $\pi_1(G^{-\infty}_{(1)}) = 0$ and so on. In Chapter 11 $G^{-\infty}_{(1)}$ is reinterpreted as the "symbol group" for elliptic Toeplitz operators on the circle with values in smoothing operators on Y. This leads to a *classifying sequence*

$$G^{-\infty}(\mathbb{S}^1 \times Y;E) \longrightarrow G \longrightarrow G^{-\infty}_{(1),0}(Y;E), \tag{10.3.5}$$

in which G is a weakly contractible group of invertible Toeplitz perturbations of the identity, and the subscript 0 denotes the identity component of the loop group, on which the index vanishes. The last three sections of this chapter prepare for this.

10.4 Chern forms and the Fredholm determinant

10.4.1 Forms with values in smoothing operators

Since $G^{-\infty} = G^{-\infty}(Y;E)$ is an open subset of $\mathrm{Id} + \Psi^{-\infty}(Y;E)$, we may treat it as a (Fréchet) manifold whose tangent space at every point is $\Psi^{-\infty}(Y;E)$. The tangent vector to a smooth curve $\mathrm{Id} + a_t$ at $A = \mathrm{Id} + a_0$ is $dA = \frac{d}{dt}a_t|_{t=0}$. The *Maurer–Cartan form* $A^{-1}dA$, the map $T_A G^{-\infty} = \Psi^{-\infty} \ni b \mapsto A^{-1}b$, is a $\Psi^{-\infty}$-valued 1-form. It is invariant under left multiplication: if $L_B(A) = BA$ then $L_B^*(A^{-1}dA) = A^{-1}B^{-1}B\,dA = A^{-1}dA$. Products of such forms are defined by composition of the values combined with antisymmetrization, and Tr is applied to the values. Explicitly, the *(unnormalized) odd Chern forms* are

$$\begin{aligned} u_{2k-1} &= \mathrm{Tr}\big((A^{-1}dA)^{2k-1}\big), \\ u_{2k-1}(b_1,\dots,b_{2k-1}) &= \sum_{\sigma\in\Sigma_{2k-1}} \mathrm{sgn}(\sigma)\,\mathrm{Tr}\big(A^{-1}b_{\sigma(1)}A^{-1}b_{\sigma(2)}\cdots A^{-1}b_{\sigma(2k-1)}\big), \end{aligned} \tag{10.4.1}$$

for $k = 1, 2, \dots$. They are smooth by the smoothness of composition, inversion and the trace.

To avoid differential calculus on Fréchet manifolds we work with the pull-backs of these forms under smooth maps $f : X \to G^{-\infty}(Y;E)$, where X is a finite dimensional manifold. Let

$$\Lambda^p(X;\Psi^{-\infty}(Y;E)) = \mathcal{C}^\infty\big(X \times Y^2; \Lambda^p T^*X \boxtimes (\mathrm{Hom}(E) \otimes \Omega_R)\big)$$

be the p-forms on X with values in smoothing operators on Y. We also allow 0-forms of the form $c\,\mathrm{Id} + a$. The wedge product of $\alpha \otimes a$ and $\beta \otimes b$ is $\alpha \wedge \beta \otimes ab$, extended by linearity and locally finite sums. The differential d acts in the X variables, and Tr acts on the values, producing ordinary forms. The following are immediate from the corresponding identities for scalar forms and from $\mathrm{Tr}(ab) = \mathrm{Tr}(ba)$:

$$\begin{aligned} d(\omega\wedge\eta) &= d\omega\wedge\eta + (-1)^p\omega\wedge d\eta, \qquad d\,\mathrm{Tr}(\omega) = \mathrm{Tr}(d\omega), \\ \mathrm{Tr}(\omega\wedge\eta) &= (-1)^{pq}\,\mathrm{Tr}(\eta\wedge\omega), \end{aligned} \tag{10.4.2}$$

for ω of degree p, η of degree q, and at least one of them smoothing-valued. For the last identity, write locally $\omega = \sum \alpha_I \otimes a_I$ and $\eta = \sum \beta_J \otimes b_J$. Then $\mathrm{Tr}(\omega\wedge\eta) = \sum \alpha_I \wedge \beta_J\,\mathrm{Tr}(a_I b_J) = (-1)^{pq}\sum \beta_J \wedge \alpha_I\,\mathrm{Tr}(b_J a_I)$. With the antisymmetrization convention of (10.4.1), the pull-back of u_{2k-1} by f is

$$f^*u_{2k-1} = \mathrm{Tr}\big(\theta^{2k-1}\big), \quad \theta = f^{-1}df \in \Lambda^1(X;\Psi^{-\infty}(Y;E)).$$

Lemma 10.4.1. *For every smooth $f : X \to G^{-\infty}(Y;E)$, with $\theta = f^{-1}df$:*

(a) $d\theta = -\theta \wedge \theta$*;*

(b) $\operatorname{Tr}(\theta^{2k}) = 0$ *for* $k \geq 1$*;*

(c) $f^{*}u_{2k-1} = \operatorname{Tr}(\theta^{2k-1})$ *is closed;*

(d) $f^{*}u_{2k-1}$ *is unchanged if* f *is replaced by* Bf *or* fB *with* $B \in G^{-\infty}(Y;E)$ *fixed. That is,* u_{2k-1} *is bi-invariant.*

Proof. Differentiating $f^{-1}f = \mathrm{Id}$ gives $d(f^{-1}) = -f^{-1}(df)f^{-1}$, so $d\theta = d(f^{-1}) \wedge df = -\theta \wedge \theta$. By (10.4.2) with $\omega = \theta$ and $\eta = \theta^{2k-1}$, $\operatorname{Tr}(\theta^{2k}) = (-1)^{2k-1}\operatorname{Tr}(\theta^{2k})$, which is (b). For (c), the Leibniz rule and (a) give

$$d(\theta^{n}) = \sum_{j=0}^{n-1} (-1)^{j}\theta^{j} \wedge d\theta \wedge \theta^{n-1-j} = -\Big(\sum_{j=0}^{n-1}(-1)^{j}\Big)\theta^{n+1}.$$

For $n = 2k-1$ odd the sum is 1, so $d\operatorname{Tr}(\theta^{2k-1}) = -\operatorname{Tr}(\theta^{2k}) = 0$ by (b). For (d), $(Bf)^{-1}d(Bf) = \theta$, while $(fB)^{-1}d(fB) = B^{-1}\theta B$ and $\operatorname{Tr}((B^{-1}\theta B)^{2k-1}) = \operatorname{Tr}(B^{-1}\theta^{2k-1}B) = \operatorname{Tr}(\theta^{2k-1})$. □

Lemma 10.4.2. *Let* $F : [0,1] \times X \to G^{-\infty}(Y;E)$ *be smooth and* $f_t = F(t,\cdot)$*. Then*

$$f_1^{*}u_{2k-1} - f_0^{*}u_{2k-1} = d\int_0^1 v_t\,dt, \quad v_t = (2k-1)\operatorname{Tr}(f_t^{-1}\partial_t f_t\,(f_t^{-1}d_X f_t)^{2k-2}).$$

Proof. On $[0,1]\times X$ write $\Theta = F^{-1}dF = dt\,\alpha + \theta_t$ with $\alpha = f_t^{-1}\partial_t f_t$ and $\theta_t = f_t^{-1}d_X f_t$. Expanding $\operatorname{Tr}(\Theta^{2k-1})$, the terms containing dt are

$$\sum_{j=0}^{2k-2}\operatorname{Tr}(\theta_t^{j} \wedge dt\,\alpha\,\theta_t^{2k-2-j}) = dt \wedge \sum_{j}(-1)^{j}\operatorname{Tr}(\theta_t^{j}\alpha\theta_t^{2k-2-j}) = dt \wedge v_t.$$

The last step uses (10.4.2) to move θ_t^{j} to the end at the cost of $(-1)^{j(2k-2-j)} = (-1)^{j}$. So $\operatorname{Tr}(\Theta^{2k-1}) = dt \wedge v_t + f_t^{*}u_{2k-1}$, and this is closed by Lemma 10.4.1. Its dt-component gives $\partial_t(f_t^{*}u_{2k-1}) - d_X v_t = 0$. Integrating in t proves the lemma. □

Proposition 10.4.3. *For each* $k \in \mathbb{N}$ *the de Rham class of* $f^{*}u_{2k-1}$ *defines a group homomorphism*

$$U_{2k-1} : K^{-1}(X) \longrightarrow H^{2k-1}_{\mathrm{dR}}(X;\mathbb{C}), \quad [f] \mapsto [f^{*}u_{2k-1}].$$

Proof. The class is well defined on homotopy classes by Lemma 10.4.2. For additivity, deform f and f' as in the proof of Proposition 10.3.1 so that $f = g \oplus \mathrm{Id}$ and $f' = \mathrm{Id} \oplus g'$ act on orthogonal finite dimensional spaces W_1, W_2. Then $\theta = f^{-1}df$ and $\theta' = (f')^{-1}df'$ satisfy $\theta \wedge \theta' = 0 = \theta' \wedge \theta$, and $(ff')^{-1}d(ff') = \theta + \theta'$. Hence $(\theta+\theta')^{2k-1} = \theta^{2k-1} + (\theta')^{2k-1}$ and $(ff')^{*}u_{2k-1} = f^{*}u_{2k-1} + (f')^{*}u_{2k-1}$. □

Remark 10.4.4. For a map of the form (10.3.2), $f^{*}u_{2k-1} = \operatorname{tr}((g^{-1}dg)^{2k-1})$ is the familiar form built from the matrix-valued map g, since the trace of a finite rank operator is the matrix trace. With suitable normalizing constants c_k the combination $\sum_k c_k U_{2k-1} : K^{-1}(X) \to H^{\mathrm{odd}}_{\mathrm{dR}}(X;\mathbb{C})$ is the *odd Chern character*; the constants are fixed in Chapter 13. It is a standard result of topology that its range spans $H^{\mathrm{odd}}(X;\mathbb{C})$ and that its null space is the torsion subgroup of $K^{-1}(X)$ (see [17]). We do not use this.

10.4.2 Even Chern forms

Even forms are defined in the same way on the loop group. A smooth map $f : X \to G^{-\infty}_{(1)}(Y;E)$ is a map $F : X \times \mathbb{R} \to G^{-\infty}(Y;E)$ with $F - \mathrm{Id}$ Schwartz in t, smooth in x. Set

$$f^*u_{2k} = \int_{\mathbb{R}} \mathrm{Tr}\big((F^{-1}d_XF)^{2k}\, F^{-1}\partial_tF\big)\, dt, \quad k = 0,1,2,\dots. \tag{10.4.3}$$

The integrand is Schwartz in t. By the computation in the proof of Lemma 10.4.2, with $(2k+1)$ in place of $(2k-1)$, we have $\mathrm{Tr}((F^{-1}dF)^{2k+1}) = dt \wedge v + w$ on $X \times \mathbb{R}$, where $v = (2k+1)\,\mathrm{Tr}(F^{-1}\partial_tF(F^{-1}d_XF)^{2k})$. Since $\mathrm{Tr}(\alpha\theta^{2k}) = \mathrm{Tr}(\theta^{2k}\alpha)$, this gives

$$(2k+1)f^*u_{2k} = \int_{\mathbb{R}} v\, dt.$$

This is the integral over the fibres of $X \times \mathbb{R} \to X$ of the closed form F^*u_{2k+1}. Because $\partial_t w = d_X v$ and $w \to 0$ rapidly as $|t| \to \infty$, we get $d_X \int_{\mathbb{R}} v\, dt = \int_{\mathbb{R}} \partial_t w\, dt = 0$. So f^*u_{2k} is closed. The argument of Lemma 10.4.2, applied on $[0,1] \times X \times \mathbb{R}$, shows that its class is a homotopy invariant. Exactly as in Proposition 10.4.3, using Proposition 10.3.7, we obtain homomorphisms

$$U_{2k} : K^{-2}(X) \longrightarrow H^{2k}_{\mathrm{dR}}(X;\mathbb{C}), \quad k \in \mathbb{N}_0.$$

A suitable combination of them is the (ordinary) Chern character.

The simplest and most fundamental cases are the first odd form and its even counterpart:

$$u_1 = \mathrm{Tr}(A^{-1}dA) \text{ on } G^{-\infty}, \qquad u_0(A) = \int_{\mathbb{R}} \mathrm{Tr}(A(t)^{-1}\partial_tA(t))\, dt,\ A \in G^{-\infty}_{(1)}. \tag{10.4.4}$$

Proposition 10.4.5. *For every piecewise smooth closed curve $\gamma : \mathbb{S}^1 \to G^{-\infty}(Y;E)$,*

$$\int_\gamma u_1 \in 2\pi i\mathbb{Z}.$$

Consequently $u_0/2\pi i$ is an integer-valued, locally constant function on $G^{-\infty}_{(1)}(Y;E)$.

Proof. The integral is unchanged by reparametrization, so we may assume γ smooth (round off corners by a reparametrization that is flat at them). By Lemma 10.4.2 with $X = \mathbb{S}^1$ and $k = 1$, homotopic loops have equal integrals. By finite rank approximation we may therefore replace γ by a loop of the form (10.3.2) with $g : \mathbb{S}^1 \to \mathrm{GL}(N,\mathbb{C})$. Then $\gamma^*u_1 = \mathrm{tr}(g^{-1}dg)$. By the matrix identity $d\det g = \det g\ \mathrm{tr}(g^{-1}dg)$,

$$\int_\gamma u_1 = \int_{\mathbb{S}^1} \frac{d\det g}{\det g} = 2\pi i\ \mathrm{wind}(\det g),$$

where wind is the winding number about 0. If $A \in G^{-\infty}_{(1)}$, then $u_0(A)$ is the integral of u_1 over the closed loop $A \circ \gamma^{-1}$ in the notation of Proposition 10.3.7, extended by Id at 1, so it lies in $2\pi i\mathbb{Z}$. It is continuous on $G^{-\infty}_{(1)}$, hence locally constant. □

Remark 10.4.6. We shall see in Chapter 11 that $u_0/2\pi i : \pi_0(G^{-\infty}_{(1)}) \to \mathbb{Z}$ is a bijection, so it labels the components of the loop group. It is the simplest case of the index formula: under the identification of Remark 10.3.8 it computes the index of the corresponding Toeplitz operator, up to sign.

10.4.3 The Fredholm determinant

Proposition 10.4.5 yields a direct construction of the Fredholm determinant on $G^{-\infty}(Y;E)$. Any two points of $G^{-\infty}$ are joined by a piecewise linear path. Indeed $G^{-\infty}$ is connected, and each point $A \in G^{-\infty}$ has a convex neighbourhood $\{A+b;\ \|b\|_{L^2} < \|A^{-1}\|^{-1}\}$ inside $G^{-\infty}$. After reparametrization such a path is smooth.

Theorem 10.4.7. *For $A \in G^{-\infty}(Y;E)$ let $\gamma : [0,1] \to G^{-\infty}(Y;E)$ be any piecewise smooth path with $\gamma(0) = \mathrm{Id}$ and $\gamma(1) = A$, and set*

$$\det(A) = \exp\Big(\int_\gamma u_1\Big) = \exp\Big(\int_0^1 \operatorname{Tr}\big(\gamma(t)^{-1}\gamma'(t)\big)\,dt\Big). \tag{10.4.5}$$

This is independent of γ. It defines a smooth homomorphism $\det : G^{-\infty}(Y;E) \to \mathbb{C}^*$ with $d\log\det = u_1$, i.e. $\frac{d}{ds}\det(A_s) = \det(A_s)\operatorname{Tr}(A_s^{-1}\frac{dA_s}{ds})$ along smooth curves. If $A = \mathrm{Id}+a$ with a of finite rank and range in a finite dimensional space $W \subset \mathcal{C}^\infty(Y;E)$, then $\det(A)$ is the determinant of the restriction $A|_W : W \to W$.

Proof. If γ_1, γ_2 are two such paths, then γ_1 followed by γ_2 reversed is a closed piecewise smooth loop. By Proposition 10.4.5 the two integrals differ by an element of $2\pi i\mathbb{Z}$. For multiplicativity, if γ_A and γ_B join Id to A and B, then $\gamma_A\gamma_B$ joins Id to AB and

$$(\gamma_A\gamma_B)^{-1}(\gamma_A\gamma_B)' = \gamma_B^{-1}(\gamma_A^{-1}\gamma_A')\gamma_B + \gamma_B^{-1}\gamma_B'.$$

Taking traces gives $\det(AB) = \det(A)\det(B)$. For smoothness near $A_0 \in G^{-\infty}$, choose the path to pass through A_0 and then follow the segment to A. This gives

$$\det(A) = \det(A_0)\exp\Big(\int_0^1 \operatorname{Tr}\big((A_0 + s(A-A_0))^{-1}(A-A_0)\big)\,ds\Big),$$

which is smooth in A near A_0. Differentiating along a curve gives $d\log\det = u_1$. Finally, suppose a has range in W. Then A preserves W and induces the identity on $\mathcal{C}^\infty(Y;E)/W$. In terms of a decomposition $\mathcal{C}^\infty = W \oplus W'$ it is upper triangular, $\left(\begin{smallmatrix} g & c \\ 0 & \mathrm{Id}\end{smallmatrix}\right)$ with $g = A|_W \in \mathrm{GL}(W)$. Since $\mathrm{GL}(W)$ is connected we can join A to Id by $\gamma(t) = \left(\begin{smallmatrix} g_t & tc \\ 0 & \mathrm{Id}\end{smallmatrix}\right)$ with g_t a path in $\mathrm{GL}(W)$ from Id to g. Then $\gamma^{-1}\gamma'$ has range in W, and its trace is $\operatorname{tr}(g_t^{-1}g_t')$. Integrating gives $\det(A) = \det(g)$. □

By Theorem 10.4.7 the determinant agrees with the matrix determinant under any finite rank embedding $\mathrm{GL}(N,\mathbb{C}) \hookrightarrow G^{-\infty}$. It is the same function as the Fredholm determinant constructed for the residual isotropic algebra in Chapter 7; the characterizing properties are identical. It extends to all smoothing perturbations of the identity, vanishing exactly on the non-invertible ones.

Theorem 10.4.8. *The function $a \mapsto \det(\mathrm{Id}+a)$, defined on $\{a \in \Psi^{-\infty}(Y;E);\ \mathrm{Id}+a \in G^{-\infty}\}$, extends uniquely to a continuous function on $\Psi^{-\infty}(Y;E)$. The extension is smooth, entire along complex lines (i.e. $z \mapsto \det(\mathrm{Id}+a+zb)$ is entire), and vanishes exactly where $\mathrm{Id}+a$ is not invertible. Moreover $G^{-\infty}$ is dense in $\mathrm{Id}+\Psi^{-\infty}(Y;E)$, and $\frac{d}{ds}\det(\mathrm{Id}+sa)|_{s=0} = \operatorname{Tr}(a)$.*

Proof. Fix $a_0 \in \Psi^{-\infty}(Y;E)$. Using (iv) write $a_0 = p_0 + r_0$ with p_0 of finite rank, with range in a finite dimensional $W \subset \mathcal{C}^\infty(Y;E)$, and $\|r_0\|_{L^2} < 1/2$. For a in the neighbourhood $\mathcal{U} = \{\|a-a_0\|_{L^2} < 1/2\}$ put $r = a - p_0$. Then $\|r\| < 1$, so $\mathrm{Id}+r \in G^{-\infty}$, and

$$\mathrm{Id}+a = \big(\mathrm{Id}+p_0(\mathrm{Id}+r)^{-1}\big)(\mathrm{Id}+r).$$

The first factor is Id plus an operator with range in W. As in the proof of Theorem 10.4.7 it is invertible exactly when its restriction $g(a)$ to W is. Define on $\mathcal{U}$

$$D(a) = \det(g(a))\det(\mathrm{Id}+r), \quad g(a) = \big(\mathrm{Id}+p_0(\mathrm{Id}+a-p_0)^{-1}\big)\big|_W.$$

This is smooth and holomorphic along complex lines, because $a \mapsto (\mathrm{Id}+a-p_0)^{-1}$ is. It vanishes exactly where $\mathrm{Id}+a$ is not invertible. Where $\mathrm{Id}+a \in G^{-\infty}$ it equals $\det(\mathrm{Id}+a)$ by Theorem 10.4.7, which gives both multiplicativity and the value on finite rank perturbations.

Next we show that $\{a \in \mathcal{U};\ \mathrm{Id}+a \in G^{-\infty}\}$ is dense in $\mathcal{U}$. Let π_W be the orthogonal projection onto W, a finite rank smoothing operator, and for $a \in \mathcal{U}$ set $a_\epsilon = a + \epsilon\pi_W(\mathrm{Id}+r)$, $r = a - p_0$. Then

$$\mathrm{Id}+a_\epsilon = \big(\mathrm{Id}+p_0(\mathrm{Id}+r)^{-1} + \epsilon\pi_W\big)(\mathrm{Id}+r),$$

and the first factor is Id plus an operator with range in W, whose restriction to W is $g(a) + \epsilon\,\mathrm{Id}_W$. This is invertible for all small $\epsilon \neq 0$, and $a_\epsilon \to a$. Two continuous functions agreeing on a dense set agree. So the local definitions agree on overlaps and define a global extension, which is unique for the same reason. Finally $\det(\mathrm{Id}+sa) = \exp\int_0^s \mathrm{Tr}((\mathrm{Id}+\sigma a)^{-1}a)\,d\sigma$ for small s, whose derivative at 0 is $\mathrm{Tr}(a)$. □

10.5 The Szegő projector and Toeplitz operators

10.5.1 Motivation

The next step towards the index theorem is a group-level version of the symbol sequence

$$\Psi^{-1}(Z;\mathbb{C}^N) \longrightarrow \Psi^0(Z;\mathbb{C}^N) \xrightarrow{\sigma_0} \mathcal{C}^\infty(S^*Z; M(N,\mathbb{C})).$$

Passing to invertible elements gives

$$G^{-1}(Z;\mathbb{C}^N) \longrightarrow G^0(Z;\mathbb{C}^N) \longrightarrow \mathcal{C}^\infty(S^*Z;\mathrm{GL}(N,\mathbb{C})), \tag{10.5.1}$$

where $G^0(Z;\mathbb{C}^N)$ is the group of invertible elliptic elements of $\Psi^0(Z;\mathbb{C}^N)$ (their inverses lie in Ψ^0 by Proposition 10.1.3) and $G^{-1}(Z;\mathbb{C}^N) = G^0(Z;\mathbb{C}^N) \cap (\mathrm{Id}+\Psi^{-1}(Z;\mathbb{C}^N))$. The second map is in general not surjective. An elliptic symbol is the symbol of an invertible operator if and only if the index of some (hence every) operator with that symbol vanishes. So the index is the obstruction to exactness of (10.5.1).

We would like to arrange that the middle group is contractible and the group on the right is essentially a loop group $G^{-\infty}_{(1)}$. This requires two things. First, S^*Z should be a circle. Second, $\mathbb{C}^N$ should be replaced by an infinite dimensional space, so that $\mathrm{GL}(N,\mathbb{C})$ becomes one of the groups $G^{-\infty}$ (this is done in Chapter 11). A compact manifold with $S^*Z = \mathbb{S}^1$ does not exist, since Z would be a compact 1-dimensional manifold and $S^*\mathbb{S}^1 = \mathbb{S}^1 \sqcup \mathbb{S}^1$. There are two ways round this:

(A) replace the circle by the line $\mathbb{R}$ and use the isotropic calculus of Chapter 6, for which the "cosphere" is the boundary circle of the radial compactification of $T^*\mathbb{R} = \mathbb{R}^2$;

(B) keep the circle but discard half of $S^*\mathbb{S}^1$, which leads to the Toeplitz algebra.

We follow (B). It is less transparent geometrically but has a long history. A third route uses the Heisenberg calculus [22].

10.5.2 The Hardy space and the Szegő projector

Expanding in Fourier series, $\mathcal{C}^\infty(\mathbb{S}^1)$ is the direct sum of

$$\mathcal{C}^\infty_+(\mathbb{S}^1) = \Big\{\sum_{k\geq 0} c_k e^{ik\theta}\Big\}, \qquad \mathcal{C}^\infty_-(\mathbb{S}^1) = \Big\{\sum_{k<0} c_k e^{ik\theta}\Big\},$$

the coefficients being rapidly decreasing. The *Szegő projector* is

$$S : \mathcal{C}^\infty(\mathbb{S}^1) \longrightarrow \mathcal{C}^\infty_+(\mathbb{S}^1), \quad S\Big(\sum_{k\in\mathbb{Z}} c_k e^{ik\theta}\Big) = \sum_{k\geq 0} c_k e^{ik\theta}. \tag{10.5.2}$$

It satisfies $S^2 = S = S^*$ (with respect to $d\theta$), its null space is $\mathcal{C}^\infty_-$ and its range is $\mathcal{C}^\infty_+$. The product of two elements of $\mathcal{C}^\infty_+$ is again in $\mathcal{C}^\infty_+$, so the Hardy space $\mathcal{C}^\infty_+(\mathbb{S}^1)$ is a ring. It consists of the boundary values of holomorphic functions on the unit disc $\mathbb{D} \subset \mathbb{C}$ (here $\mathbb{D}$ is the open disc and $\overline{\mathbb{D}}$ its closure), so it should be thought of as the functions on "half" of $\mathbb{S}^1$. The cotangent bundle is $T^*\mathbb{S}^1 = \mathbb{S}^1 \times \mathbb{R}_\xi$, and the Fourier variable k corresponds to ξ. This suggests that S is microlocally the identity on the positive half $S^*_+\mathbb{S}^1 = \{\xi > 0\}$ and zero on the negative half $S^*_-\mathbb{S}^1$.

Proposition 10.5.1. *Restriction to the boundary is an isomorphism*

$$\mathcal{C}^\infty_{\mathrm{hol}}(\overline{\mathbb{D}}) = \{u \in \mathcal{C}^\infty(\overline{\mathbb{D}});\ \bar\partial u = 0 \text{ in } \mathbb{D}\} \longrightarrow \mathcal{C}^\infty_+(\mathbb{S}^1).$$

For $f \in \mathcal{C}^\infty(\mathbb{S}^1)$ the Cauchy integral

$$\tilde S(f)(z) = \frac{1}{2\pi i}\int_{|\tau|=1} \frac{f(\tau)}{\tau - z}\, d\tau, \quad |z| < 1,$$

is the holomorphic extension of Sf, and $Sf(e^{i\theta}) = \lim_{r\uparrow 1} \tilde S f(re^{i\theta})$.

Proof. If $a = \sum_{k\geq 0} a_k e^{ik\theta} \in \mathcal{C}^\infty_+$ then the coefficients are rapidly decreasing. Hence $u_a(z) = \sum_{k\geq 0} a_k z^k$ converges on $\overline{\mathbb{D}}$ with all derivatives, is holomorphic in $\mathbb{D}$, and restricts to a. Conversely, if $u \in \mathcal{C}^\infty_{\mathrm{hol}}(\overline{\mathbb{D}})$ has the Taylor series $\sum_{k\geq 0} b_k z^k$ in $\mathbb{D}$, then for $r < 1$ and $k \in \mathbb{Z}$

$$\frac{1}{2\pi}\int_0^{2\pi} e^{-ik\theta} u(re^{i\theta})\, d\theta = \begin{cases} b_k r^k & k \geq 0, \\ 0 & k < 0. \end{cases}$$

Letting $r \uparrow 1$, the Fourier coefficients of $u|_{\mathbb{S}^1}$ are b_k for $k \geq 0$ and 0 for $k < 0$. So the boundary value lies in $\mathcal{C}^\infty_+$, and it vanishes only if $u = 0$. Thus restriction is an isomorphism. For the Cauchy integral, by rapid convergence of the Fourier series we can integrate term by term. For $|z| < 1$,

$$\frac{1}{2\pi i}\int_{|\tau|=1} \frac{\tau^k}{\tau - z}\, d\tau = \begin{cases} z^k & k \geq 0, \\ 0 & k < 0, \end{cases}$$

by the residue theorem. For $k < 0$ the integrand $\tau^k/(\tau - z)$ is holomorphic outside the disc and $O(|\tau|^{-2})$ at infinity, so we deform the contour to infinity. So $\tilde S(f) = u_{Sf}$, and the boundary values follow from the first part. □

Theorem 10.5.2. *$S \in \Psi^0(\mathbb{S}^1)$, and its principal symbol is*

$$\sigma_0(S) = \begin{cases} 1 & \text{on } S^*_+\mathbb{S}^1 = \{\xi > 0\}, \\ 0 & \text{on } S^*_-\mathbb{S}^1 = \{\xi < 0\}. \end{cases}$$

More precisely, near the diagonal the kernel of S is $\int e^{i(\theta-\theta')\xi}\chi(\xi)đ\xi$ modulo a smooth function, where $\chi \in \mathcal{C}^\infty(\mathbb{R})$, $\chi(\xi) = 1$ for $\xi \geq 1$ and $\chi(\xi) = 0$ for $\xi \leq -1$. Thus the amplitude vanishes identically in the negative direction.

Proof. The kernel of S with respect to $d\theta'$ is the distribution

$$K_S(\theta,\theta') = \frac{1}{2\pi}\sum_{k\geq 0} e^{ik(\theta-\theta')} = \lim_{r\uparrow 1}\frac{1}{2\pi}\frac{1}{1-re^{i\phi}}, \quad \phi=\theta-\theta'.$$

The limit is taken in distributions, since the Fourier coefficients $r^k \to 1$ with uniform polynomial bounds. For $\phi \notin 2\pi\mathbb{Z}$ the limit is the smooth function $(2\pi)^{-1}(1-e^{i\phi})^{-1}$, locally uniformly with all derivatives. So K_S is smooth off the diagonal. Near $\phi = 0$, write $r = e^{-\epsilon}$, so $re^{i\phi} = e^{i(\phi+i\epsilon)}$, and use the Laurent expansion

$$\frac{1}{1-e^{iz}} = \frac{i}{z} + h(z), \quad h \text{ holomorphic in } |z|<2\pi \quad (h(0)=\tfrac{1}{2}).$$

Hence for $|\phi|<\pi$

$$K_S = \frac{1}{2\pi}\Big(\frac{i}{\phi+i0}+h(\phi)\Big), \qquad \frac{i}{\phi+i0} = \lim_{\epsilon\downarrow 0}\frac{i}{\phi+i\epsilon} = \lim_{\epsilon\downarrow 0}\int_0^\infty e^{i\phi\xi-\epsilon\xi}\,d\xi.$$

So $(2\pi)^{-1}i/(\phi+i0) = \int e^{i\phi\xi}H(\xi)đ\xi$ with H the Heaviside function. Since $H-\chi$ has compact support, $\int e^{i\phi\xi}(H-\chi)đ\xi$ is entire in ϕ. Thus near the diagonal K_S is the kernel of $\mathrm{Op}(\chi)$ modulo a smooth function. As χ is a classical symbol of order 0 (it is homogeneous of degree 0 for $|\xi|\geq 1$), $S\in\Psi^0(\mathbb{S}^1)$ with principal symbol χ restricted to $\{\xi=\pm 1\}$. □

10.5.3 The Toeplitz algebra

On $\mathbb{C}^N$-valued functions S acts componentwise. Set

$$\Psi^m_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N) = \{A\in\Psi^m(\mathbb{S}^1;\mathbb{C}^N);\ A=SAS\}, \quad m\in\{0,-\infty\},$$

regarded as operators on $\mathcal{C}^\infty_+(\mathbb{S}^1;\mathbb{C}^N)$. This is a subalgebra with identity S: if $A_i = SA_iS$ then $S(A_1A_2)S = (SA_1S)(SA_2S) = A_1A_2$. Its elements map $\mathcal{C}^\infty_-$ to zero and $\mathcal{C}^\infty$ into $\mathcal{C}^\infty_+$, and $\Psi^{-\infty}_{\mathcal{T}} = S\Psi^{-\infty}S$. Since $\sigma_0(A) = \sigma_0(S)\sigma_0(A)\sigma_0(S)$, the principal symbol of $A\in\Psi^0_{\mathcal{T}}$ vanishes on $S^*_-\mathbb{S}^1$. We call A *elliptic* (as a Toeplitz operator) if $\sigma_0(A)|_{S^*_+\mathbb{S}^1}$, a smooth function on $\mathbb{S}^1$ with values in $M(N,\mathbb{C})$, is invertible. The basic examples are the *Toeplitz operators* $T_a = SaS$ with multiplier $a\in\mathcal{C}^\infty(\mathbb{S}^1;M(N,\mathbb{C}))$. They are elliptic exactly when a takes values in $\mathrm{GL}(N,\mathbb{C})$.

Proposition 10.5.3. *If $A\in\Psi^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N)$ is elliptic, then $A+(\mathrm{Id}-S)\in\Psi^0(\mathbb{S}^1;\mathbb{C}^N)$ is elliptic, and A is Fredholm on $\mathcal{C}^\infty_+(\mathbb{S}^1;\mathbb{C}^N)$ with*

$$\mathrm{ind}_{\mathcal{T}}(A) = \dim\mathrm{Nul}(A|_{\mathcal{C}^\infty_+}) - \dim(\mathcal{C}^\infty_+/A\mathcal{C}^\infty_+) = \mathrm{ind}(A+\mathrm{Id}-S).$$

There is $B\in\Psi^0_{\mathcal{T}}$ with $AB-S$ and $BA-S$ in $\Psi^{-\infty}_{\mathcal{T}}$. If A is invertible on $\mathcal{C}^\infty_+$, its inverse lies in $\Psi^0_{\mathcal{T}}$. The Toeplitz index is constant along smooth families of elliptic elements of $\Psi^0_{\mathcal{T}}$.

Proof. The symbol of $P = A+\mathrm{Id}-S$ is $\sigma_0(A)$ on S^*_+ and Id on S^*_-, so P is elliptic. With respect to $\mathcal{C}^\infty = \mathcal{C}^\infty_+\oplus\mathcal{C}^\infty_-$, P is diagonal, acting as A on $\mathcal{C}^\infty_+$ and as Id on $\mathcal{C}^\infty_-$. So the null space and cokernel of P on $\mathcal{C}^\infty$ are those of A on $\mathcal{C}^\infty_+$. If Q is a parametrix (or the inverse) of P, then $B = SQS$ satisfies $BA = SQSAS = SQPS = S - S(\mathrm{Id}-QP)S$. The other identity is similar. The last statement follows from Theorem 10.2.2 applied to the smooth family $A_t+\mathrm{Id}-S$. □

10.5.4 The index on the circle

The *shifts* on $\mathcal{C}^\infty_+(\mathbb{S}^1)$ are

$$U = Se^{i\theta}S : \sum_{k\geq 0} c_k e^{ik\theta} \mapsto \sum_{k\geq 0} c_k e^{i(k+1)\theta}, \quad L = Se^{-i\theta}S : \sum_{k\geq 0} c_k e^{ik\theta} \mapsto \sum_{k\geq 1} c_k e^{i(k-1)\theta}.$$

Both are elliptic elements of $\Psi^0_{\mathcal{T}}(\mathbb{S}^1)$, and

$$LU = \mathrm{Id}, \quad UL = \mathrm{Id} - \pi_0, \tag{10.5.3}$$

where Id means S, the identity on $\mathcal{C}^\infty_+$, and π_0 is the orthogonal projection onto the constants. Thus U is injective with range of codimension one (complemented by the constants), and L is surjective with null space the constants:

$$\mathrm{ind}_{\mathcal{T}}(U) = -1, \qquad \mathrm{ind}_{\mathcal{T}}(L) = 1.$$

More generally $U^k = T_{e^{ik\theta}}$ for $k \geq 0$ has index $-k$ and $L^k = T_{e^{-ik\theta}}$ has index k.

Theorem 10.5.4. *If $a \in \mathcal{C}^\infty(\mathbb{S}^1; \mathrm{GL}(N,\mathbb{C}))$ then the Toeplitz operator $T_a = SaS$ on $\mathcal{C}^\infty_+(\mathbb{S}^1;\mathbb{C}^N)$ is Fredholm and*

$$\mathrm{ind}_{\mathcal{T}}(T_a) = -\operatorname{wind}(\det a) = -\frac{1}{2\pi i}\int_{\mathbb{S}^1} \mathrm{tr}(a^{-1}da).$$

Proof. The second equality is the computation in the proof of Proposition 10.4.5. Let $k = \operatorname{wind}(\det a)$. We claim that a is smoothly homotopic in $\mathcal{C}^\infty(\mathbb{S}^1;\mathrm{GL}(N,\mathbb{C}))$ to $a_k = \mathrm{diag}(e^{ik\theta}, 1, \dots, 1)$. Write $a = \mathrm{diag}(\det a, 1, \dots, 1)\, b$ with b a loop in $\mathrm{SL}(N,\mathbb{C})$. The scalar loop $\det a$ is homotopic in $\mathbb{C}^*$ to $e^{ik\theta}$: write $\det a = e^{ik\theta}e^{\phi(\theta)}$ with ϕ a smooth function on the circle, and deform ϕ to 0. Since $\mathrm{SL}(N,\mathbb{C})$ is connected and simply connected (a standard fact; it retracts onto $\mathrm{SU}(N)$), b is smoothly homotopic to the constant loop Id. The family T_{a_t} along the homotopy is a smooth family of elliptic Toeplitz operators, so by Proposition 10.5.3 $\mathrm{ind}_{\mathcal{T}}(T_a) = \mathrm{ind}_{\mathcal{T}}(T_{a_k})$. Now $T_{a_k} = U^k \oplus \mathrm{Id} \oplus \cdots \oplus \mathrm{Id}$ if $k \geq 0$ and $L^{-k} \oplus \mathrm{Id} \oplus \cdots \oplus \mathrm{Id}$ if $k < 0$. Its index is $-k$. □

Remark 10.5.5. Via finite rank embeddings $\mathrm{GL}(N,\mathbb{C}) \hookrightarrow G^{-\infty}$, Theorem 10.5.4 says that the Toeplitz index is the homomorphism $-U_1/2\pi i : K^{-1}(\mathbb{S}^1) \to \mathbb{Z}$. It vanishes on the classes of constant maps, so by Lemma 10.3.4 it factors through $K^{-2}(\mathrm{pt})$. By Proposition 10.3.7 it is there given by $-u_0/2\pi i$ on $\pi_0(G^{-\infty}_{(1)})$. That it is an isomorphism is the case $k = 1$ of (10.3.3). This is the prototype of the index formula.

10.6 Smoothing perturbations of elliptic families

Throughout this section X is a compact manifold (the parameter space, possibly with boundary). We use the standard L^2 structure on $\mathbb{S}^1$ and the projections onto finite Fourier series

$$\pi_r\Big(\sum_k c_k e^{ik\theta}\Big) = \sum_{|k|\leq r} c_k e^{ik\theta}, \quad r \in \mathbb{N}_0,$$

acting componentwise on $\mathbb{C}^N$-valued functions. They are self-adjoint finite rank smoothing operators of rank $(2r+1)N$.

Lemma 10.6.1. *If $R \in \mathcal{C}^\infty(X; \Psi^{-\infty}(\mathbb{S}^1;\mathbb{C}^N))$ then $R\pi_r \to R$ and $\pi_r R \to R$ in $\mathcal{C}^\infty(X;\Psi^{-\infty})$, uniformly on X, as $r \to \infty$.*

Proof. The kernel of $R_x\pi_r$ is

$$\int R_x(\theta,\theta'')\pi_r(\theta'',\theta')\,d\theta'' = \sum_{|k|\le r} b_k(x,\theta)e^{-ik\theta'}, \quad b_k(x,\theta) = \frac{1}{2\pi}\int R_x(\theta,\theta'')e^{ik\theta''}\,d\theta'',$$

i.e. the partial sums of the Fourier series of $R_x(\theta,\theta')$ in the second variable. Integration by parts in θ'' shows that every (x,θ)-derivative of b_k is $O(\langle k\rangle^{-q})$ for all q, uniformly in (x,θ). So the series converges with all derivatives. The case of $\pi_r R$ follows by taking adjoints. □

Lemma 10.6.2. *Let $A \in \mathcal{C}^\infty(X;\Psi^0(\mathbb{S}^1;\mathbb{C}^N))$ be a smooth family of elliptic operators and $A'_r(x) = A(x)(\mathrm{Id}-\pi_r)$. There is r_0 such that for all $r \ge r_0$ and $x \in X$:*

(a) $\mathrm{Nul}(A'_r(x)) = \mathrm{Ran}(\pi_r)$, *of dimension* $(2r+1)N$;

(b) the range of $A'_r(x)$ is closed, with complement

$$F_r(x) = \mathrm{Nul}\big((\mathrm{Id}-\pi_r)A(x)^*\big) = \mathrm{Ran}(A'_r(x))^\perp$$

of dimension $(2r+1)N - \operatorname{ind} A(x)$;

(c) the $F_r(x)$ form a smooth vector bundle $F_r \to X$.

Proof. Let Q be a smooth family of parametrices, $Q(x)A(x) = \mathrm{Id}-R(x)$ with $R \in \mathcal{C}^\infty(X;\Psi^{-\infty})$ (Lemma 10.2.1). Then

$$Q(x)A'_r(x) = (\mathrm{Id}-R(x))(\mathrm{Id}-\pi_r) = \big(\mathrm{Id}-R(x)(\mathrm{Id}-\pi_r)\big)(\mathrm{Id}-\pi_r).$$

By Lemma 10.6.1, $R(x)(\mathrm{Id}-\pi_r) \to 0$ uniformly in $\mathcal{C}^\infty(X;\Psi^{-\infty})$, hence in L^2 norm. So for $r \ge r_0$ the operator $\mathrm{Id}-R(x)(\mathrm{Id}-\pi_r)$ is invertible, with inverse $\mathrm{Id}-S_r(x)$ depending smoothly on x in $\Psi^{-\infty}$ by (vi). With $Q'_r = (\mathrm{Id}-S_r)Q$ this gives

$$Q'_r(x)A'_r(x) = \mathrm{Id}-\pi_r. \tag{10.6.1}$$

So $\mathrm{Nul}(A'_r) \subset \mathrm{Ran}(\pi_r)$, and the reverse inclusion is clear, which proves (a). $A'_r(x)$ is elliptic and differs from $A(x)$ by a smoothing operator, so it has the same index. By Proposition 10.1.3 its range has the orthogonal complement $\mathrm{Nul}((A'_r)^*) = F_r(x)$, of dimension $(2r+1)N - \operatorname{ind} A(x)$. This proves (b).

For (c), the dimension is locally constant by Theorem 10.2.2. Let $T(x) = A'_r(x)^*A'_r(x) + \pi_r$, a smooth family of elliptic self-adjoint operators of order 0. It is injective, since $T(x)u = 0$ implies $\|A'_r u\|^2 + \|\pi_r u\|^2 = 0$, so $u \in \mathrm{Ran}(\pi_r) \cap \mathrm{Nul}(\pi_r) = \{0\}$. It is therefore invertible by (10.1.2), and $G(x) = T(x)^{-1}$ is a smooth family by Lemma 10.2.1(b). Set $P(x) = A'_r G A'^*_r$. Since $A'_r\pi_r = 0$ we have $T\pi_r = \pi_r$, hence $G\pi_r = \pi_r$ and $GA'^*_r A'_r = G(T-\pi_r) = \mathrm{Id}-\pi_r$. Therefore $P^2 = A'_r(\mathrm{Id}-\pi_r)GA'^*_r = P$, $P^* = P$ and $PA'_r = A'_r(\mathrm{Id}-\pi_r) = A'_r$. Since also $\mathrm{Ran}\,P \subset \mathrm{Ran}\,A'_r$, $P(x)$ is the orthogonal projection onto $\mathrm{Ran}(A'_r(x))$. Thus $\mathrm{Id}-P(x)$ is the orthogonal projection onto $F_r(x)$. It is a smooth family in Ψ^0 whose values are smoothing (finite rank with range in $\mathcal{C}^\infty$). Its full symbol is therefore residual for each x, which for a smooth family of classical symbols means the family lies in $\mathcal{C}^\infty(X;\Psi^{-\infty})$. If $v_1,\dots,v_d$ is a basis of $F_r(x_0)$, then $(\mathrm{Id}-P(x))v_i$ are smooth sections of F_r which remain a basis near x_0. So F_r is a smooth vector bundle. □

The formal difference "$\mathrm{Ran}(\pi_r) \ominus F_r$" of the trivial bundle of null spaces and the bundle F_r complementary to the range is the *families index* of A, an element of $K^0(X)$. It is discussed in Chapter 13.

Theorem 10.6.3. *Let $A \in \mathcal{C}^\infty(X; \Psi^0(\mathbb{S}^1; \mathbb{C}^N))$ be a smooth family of elliptic operators. There is a smooth family $B \in \mathcal{C}^\infty(X; \Psi^{-\infty}(\mathbb{S}^1; \mathbb{C}^N))$ such that $A(x) + B(x)$ is invertible for every $x \in X$ if and only if, for some (equivalently every) $r \geq r_0$, the bundle F_r of Lemma 10.6.2 is isomorphic to the trivial bundle $X \times \mathbb{C}^{(2r+1)N}$.*

Proof. Suppose F_r is trivial for some $r \geq r_0$. Since $\operatorname{Ran}(\pi_r)$ has dimension $(2r+1)N$, there is a smooth bundle isomorphism $\Phi(x) : \operatorname{Ran}(\pi_r) \to F_r(x)$. Then $B_r(x) = \Phi(x)\pi_r$ is a smooth family of finite rank smoothing operators: its kernel is $\sum_j \Phi(x)e_j \otimes e_j^*$ for an orthonormal basis e_j of $\operatorname{Ran}(\pi_r)$. The operator $A_r'(x) + B_r(x)$ maps $\operatorname{Ran}(\operatorname{Id} - \pi_r)$ injectively onto $\operatorname{Ran}(A_r'(x))$ and $\operatorname{Ran}(\pi_r)$ isomorphically onto $F_r(x)$. As $L^2 = \operatorname{Ran}(A_r') \oplus F_r$, it is invertible. Thus $B = B_r - A\pi_r$ has the required property, since $A + B = A_r' + B_r$.

Conversely, suppose $A + B$ is invertible on X. By Lemma 10.6.1, $B\pi_r \to B$ in L^2 norm, uniformly on X. Since the inverses $(A(x) + B(x))^{-1}$ are bounded uniformly on the compact set X, $A + B\pi_r$ is invertible for all large r. Write

$$A + B\pi_r = A_r' + B_r, \quad B_r = (A + B)\pi_r.$$

Invertibility of $A_r' + B_r$ means that $L^2 = A_r'(\operatorname{Ran}(\operatorname{Id} - \pi_r)) \oplus B_r(\operatorname{Ran}(\pi_r))$. So $V(x) = B_r(x)\operatorname{Ran}(\pi_r)$ is a smooth bundle complementary to $\operatorname{Ran}(A_r'(x))$, trivialized by B_r (which is injective on $\operatorname{Ran}(\pi_r)$). The orthogonal projection $\operatorname{Id} - P(x)$ onto $F_r(x)$ restricts to an isomorphism $V(x) \to F_r(x)$, since its null space is $\operatorname{Ran}(A_r'(x))$. Hence F_r is trivial for all large r. Together with the first part, triviality for one $r \geq r_0$ implies triviality for all large r. □

In particular a necessary condition is that $\operatorname{ind} A(x) = 0$, and when X is a point the theorem reduces to the statement of Chapter 5 that an elliptic operator has an invertible smoothing perturbation if and only if its index vanishes. The proofs extend to any compact manifold in place of the circle (and to fibrations by compact manifolds) once a sequence of finite rank smoothing projections with the property of Lemma 10.6.1 is available. They also pass unchanged to the Toeplitz algebra if π_r is replaced by the projection π_r^+ onto $\operatorname{span}\{e^{ik\theta};\ 0 \leq k \leq r\}$. This projection commutes with S, lies in $\Psi_{\mathcal{T}}^{-\infty}$, and has rank $(r+1)N$.

Corollary 10.6.4. *Let $A \in \mathcal{C}^\infty(X; \Psi_{\mathcal{T}}^0(\mathbb{S}^1; \mathbb{C}^N))$ be a smooth family of elliptic Toeplitz operators. There is $B \in \mathcal{C}^\infty(X; \Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}^1; \mathbb{C}^N))$ with $A(x) + B(x)$ invertible on $\mathcal{C}_+^\infty$ for all x if and only if, for large r, the bundle $F_r^+(x) = \operatorname{Nul}((S - \pi_r^+)A(x)^*) \cap \mathcal{C}_+^\infty$ is trivial of rank $(r+1)N$. The inverses then lie in $\Psi_{\mathcal{T}}^0$.*

Corollary 10.6.5. *Let $A \in \mathcal{C}^\infty([0,1] \times X; \Psi_{\mathcal{T}}^0(\mathbb{S}^1; \mathbb{C}^N))$ (or $\mathcal{C}^\infty([0,1] \times X; \Psi^0(\mathbb{S}^1; \mathbb{C}^N))$) be elliptic at each point, with $A_0(x)$ invertible for all $x \in X$. Then there is a smooth family $B \in \mathcal{C}^\infty([0,1] \times X; \Psi_{\mathcal{T}}^{-\infty})$ (respectively $\Psi^{-\infty}$) with $B_0 = 0$ and $A_t(x) + B_t(x)$ invertible for all (t, x).*

Proof. We give the proof in the Ψ^0 case; the Toeplitz case is identical. Let r be large. Applying the converse half of Theorem 10.6.3 on X with $B = 0$ shows that the restriction of F_r to $\{0\} \times X$ is trivial. Since $[0,1] \times X$ retracts onto $\{0\} \times X$, the bundle F_r over $[0,1] \times X$ is isomorphic to the pull-back of its restriction to $\{0\} \times X$ (homotopy invariance of vector bundles [17]), so it is trivial. The first half of the theorem gives a smooth family B' of smoothing operators with $C_t = A_t + B_t'$ invertible everywhere. To make the perturbation vanish at $t = 0$, set

$$D_t = C_t C_0^{-1} A_0 = A_t + B_t, \quad B_t = -A_t(A_0 + B_0')^{-1}B_0' + B_t'C_0^{-1}A_0.$$

Here we used $C_0^{-1}A_0 = \mathrm{Id} - C_0^{-1}B_0'$. D_t is invertible as a product of invertibles, $D_0 = A_0$, and B_t is a smooth family of smoothing operators, since $C_0^{-1} \in \Psi^0$ by Lemma 10.2.1(b). It vanishes at $t = 0$ because $D_0 = A_0$. □

This allows homotopies of elliptic families to be lifted to homotopies of invertible families. It is used in Chapter 11.

10.7 Deforming matrices away in the Toeplitz group

Let

$$\tilde{G}^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N) = \{A \in \Psi^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N);\ A \text{ is elliptic and invertible on } \mathcal{C}^\infty_+,\ A^{-1} \in \Psi^0_{\mathcal{T}}\}$$

(the tilde distinguishes it from a smaller group used in Chapter 11). By Proposition 10.5.3 the last condition is automatic. The smoothing elements $G^{-\infty}_{\mathcal{T}} = \tilde{G}^0_{\mathcal{T}} \cap (S + \Psi^{-\infty}_{\mathcal{T}})$ form a subgroup containing a copy of $\mathrm{GL}(N,\mathbb{C})$:

$$\mathrm{GL}(N,\mathbb{C}) \ni g \longmapsto \mathrm{Id} - \pi_0 + \pi_0 g \pi_0 \in G^{-\infty}_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N), \tag{10.7.1}$$

which acts as g on the constants and as the identity on higher modes; here $\mathrm{Id} = S$. We also use the standard stabilization $\mathrm{GL}(N,\mathbb{C}) \hookrightarrow \mathrm{GL}(2N,\mathbb{C})$, $g \mapsto \left(\begin{smallmatrix} g & 0 \\ 0 & \mathrm{Id}_N \end{smallmatrix}\right)$.

Proposition 10.7.1. *Embed* $\mathrm{GL}(N,\mathbb{C})$ *in* $\tilde{G}^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^{2N})$ *by combining the stabilization with* (10.7.1) *(for* $2N$ *in place of* N*). Then there is a continuous map* $M : [0,1] \times \mathrm{GL}(N,\mathbb{C}) \to \tilde{G}^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^{2N})$*, smooth in all variables, such that* $M(0,g)$ *is the image of* g *and* $M(1,g) = \mathrm{Id}$ *for all* g*.*

Proof. Write $\mathbb{C}^{2N} = \mathbb{C}^N \oplus \mathbb{C}^N$ and operators as 2×2 block matrices of $N \times N$ Toeplitz operators. A constant matrix g commutes with S, so it defines the Toeplitz operator gS, which we write as g. The image of g is

$$M_0 = \begin{pmatrix} \mathrm{Id} - \pi_0 + \pi_0 g \pi_0 & 0 \\ 0 & \mathrm{Id} \end{pmatrix}.$$

The homotopy has three stages, each depending smoothly on g.

Stage 1. For $\tau \in [0, \pi/2]$ let

$$M_\tau = \begin{pmatrix} \cos\tau(\mathrm{Id} - \pi_0) + \pi_0 g \pi_0 & \sin\tau\, gU \\ -\sin\tau\, g^{-1}L & \cos\tau\ \mathrm{Id} \end{pmatrix}.$$

Its symbol on S^*_+ is $\left(\begin{smallmatrix} \cos\tau & \sin\tau\, g e^{i\theta} \\ -\sin\tau\, g^{-1}e^{-i\theta} & \cos\tau \end{smallmatrix}\right)$, of determinant 1, so M_τ is elliptic. To see that it is invertible, write $u = \sum_{k\geq 0} u_k e^{ik\theta}$, $v = \sum_{k\geq 0} v_k e^{ik\theta}$ with $u_k, v_k \in \mathbb{C}^N$. Then M_τ maps $(u_0, 0)$ to $(gu_0, 0)$. For $k \geq 1$ it maps the pair of modes $(u_k e^{ik\theta}, v_{k-1}e^{i(k-1)\theta})$ to $(f_k e^{ik\theta}, h_{k-1}e^{i(k-1)\theta})$ with

$$\begin{pmatrix} f_k \\ h_{k-1} \end{pmatrix} = \begin{pmatrix} \cos\tau & \sin\tau\, g \\ -\sin\tau\, g^{-1} & \cos\tau \end{pmatrix} \begin{pmatrix} u_k \\ v_{k-1} \end{pmatrix},$$

and

$$\begin{pmatrix} \cos\tau & \sin\tau\, g \\ -\sin\tau\, g^{-1} & \cos\tau \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & g^{-1} \end{pmatrix} R_\tau \begin{pmatrix} 1 & 0 \\ 0 & g \end{pmatrix},$$

where $R_\tau = \left(\begin{smallmatrix} \cos\tau & \sin\tau \\ -\sin\tau & \cos\tau \end{smallmatrix}\right)$. These matrices are invertible, with inverses of the same form with τ replaced by $-\tau$. So M_τ is invertible, and its inverse is the Toeplitz operator obtained by replacing τ by $-\tau$ and g by g^{-1} in the $(1,1)$ entry. Thus $M_\tau \in \tilde{G}^0_{\mathcal{T}}$. At $\tau = \pi/2$,

$$M_{\pi/2} = \begin{pmatrix} \pi_0 g \pi_0 & gU \\ -g^{-1}L & 0 \end{pmatrix}.$$

Stage 2. For τ decreasing from $\pi/2$ to 0 let

$$M'_\tau = \begin{pmatrix} \cos\tau\, g(\mathrm{Id} - \pi_0) + \pi_0 g \pi_0 & \sin\tau\, gU \\ -\sin\tau\, g^{-1}L & \cos\tau\, g^{-1} \end{pmatrix},$$

which equals $M_{\pi/2}$ at $\tau = \pi/2$. Its symbol is $\mathrm{diag}(g, g^{-1}) \left(\begin{smallmatrix} \cos\tau & \sin\tau\, e^{i\theta} \\ -\sin\tau\, e^{-i\theta} & \cos\tau \end{smallmatrix}\right)$, which is invertible. On modes it acts as g on u_0 and by the matrix $\left(\begin{smallmatrix} \cos\tau\, g & \sin\tau\, g \\ -\sin\tau\, g^{-1} & \cos\tau\, g^{-1} \end{smallmatrix}\right) = \mathrm{diag}(g, g^{-1}) R_\tau$ on the pairs (u_k, v_{k-1}). So it is again in $\tilde{G}^0_{\mathcal{T}}$. At $\tau = 0$ we reach the constant matrix

$$M'_0 = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix}.$$

Stage 3. It remains to deform $\mathrm{diag}(g, g^{-1})$ to the identity in $\mathrm{GL}(2N, \mathbb{C})$, uniformly in g. First take $\left(\begin{smallmatrix} \cos\tau\, g & \sin\tau \\ -\sin\tau & \cos\tau\, g^{-1} \end{smallmatrix}\right)$, $\tau \in [0, \pi/2]$. For $\cos\tau \neq 0$ the Schur complement of the invertible $(1,1)$ entry is $\cos\tau\, g^{-1} + \sin^2\tau (\cos\tau)^{-1} g^{-1} = (\cos\tau)^{-1} g^{-1}$, so the matrix is invertible. At $\tau = \pi/2$ it is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Then follow the rotations R_τ from $\tau = \pi/2$ back to $\tau = 0$. Constant invertible matrices are elements of $\tilde{G}^0_{\mathcal{T}}$. After smoothing the corners by reparametrization, the concatenation of the three stages is the required homotopy. □

Remark 10.7.2. Proposition 10.7.1 shows that the "$G^{-\infty}_{\mathcal{T}}$ part" of the Toeplitz group can be (weakly) deformed away after stabilization. In Chapter 11 this is combined with Atiyah's linearization of loops in $\mathrm{GL}(N, \mathbb{C})$, which shows how invertible elliptic Toeplitz operators can be deformed into the smoothing subgroup. Together these give the weak contractibility of the middle group in (10.3.5).

10.8 Exercises

Exercise 10.1. Let $A_1 \in \Psi^{m_1}(X; F, G)$ and $A_2 \in \Psi^{m_2}(X; E, F)$ be elliptic. Using Theorem 10.1.4 with the parametrix $B_2 B_1$ of $A_1 A_2$, show that $\mathrm{ind}(A_1 A_2) = \mathrm{ind}(A_1) + \mathrm{ind}(A_2)$. (Write $\mathrm{Id} - B_2 B_1 A_1 A_2 = (\mathrm{Id} - B_2 A_2) + B_2 R_1 A_2$ with $R_1 = \mathrm{Id} - B_1 A_1$, and use Lemma 10.1.2.)

Exercise 10.2. Show that $\mathrm{ind}(A^*) = -\mathrm{ind}(A)$ for elliptic A, and that every elliptic self-adjoint $A \in \Psi^m(X; E)$ has index zero.

Exercise 10.3. Let $A \in \mathcal{C}^\infty(X'; \Psi^m(X; E, F))$ be a smooth family of elliptic operators parametrized by a connected compact manifold X'. Show directly from Theorem 10.1.4 and Lemma 10.2.1 that $x \mapsto \mathrm{ind}\, A(x)$ is constant.

Exercise 10.4. Show that for $a \in \Psi^{-\infty}(Y; E)$ the function $z \mapsto \det(\mathrm{Id} + za)$ is entire, and that its zeros are the points $z = -1/\lambda$ with $\lambda \neq 0$ an eigenvalue of a. Show further that the order of the zero at $-1/\lambda$ is the algebraic multiplicity of λ. (Use the local formula in the proof of Theorem 10.4.8 with W containing the generalized eigenspace.)

Exercise 10.5. Let $f : \mathbb{S}^1 \to G^{-\infty}(\mathbb{S}^1)$, $f(e^{is}) = \mathrm{Id} + (e^{is} - 1)\pi_0$, where π_0 is the projection onto the constants. Compute $\int_{\mathbb{S}^1} f^* u_1$ and $\det f(e^{is})$. Deduce that $U_1 : K^{-1}(\mathbb{S}^1) \to H^1(\mathbb{S}^1; \mathbb{C}) \cong \mathbb{C}$ has image $2\pi i\mathbb{Z}$.

Exercise 10.6. Show that the forms $\mathrm{Tr}((A^{-1}dA)^{2k})$ on $G^{-\infty}$ vanish identically, directly from the antisymmetrized formula (10.4.1) with $2k$ entries. Show also that u_{2k-1} is invariant under right multiplication, using $R_B^*(A^{-1}dA) = B^{-1}(A^{-1}dA)B$.

Exercise 10.7. Show that $[S, f] \in \Psi^{-\infty}(\mathbb{S}^1)$ for every $f \in \mathcal{C}^\infty(\mathbb{S}^1)$. Use the symbol χ of Theorem 10.5.2 and the composition formula, noting that all ξ-derivatives of χ have compact support. Deduce that $T_a T_b - T_{ab} \in \Psi_{\mathcal{T}}^{-\infty}$ for $a, b \in \mathcal{C}^\infty(\mathbb{S}^1; M(N, \mathbb{C}))$. Also show that, if a and b are trigonometric polynomials, $T_a T_b - T_{ab}$ has finite rank.

Exercise 10.8. A compact manifold Z is recovered from the ring $\mathcal{C}^\infty(Z)$ as the set of its multiplicative linear functionals ("valuations") p, with $p(fg) = p(f)p(g)$ and $p(1) = 1$. Show that the continuous multiplicative linear functionals on the ring $\mathcal{C}_+^\infty(\mathbb{S}^1)$ are exactly $f \mapsto u_f(z)$, $z \in \overline{\mathbb{D}}$, where u_f is the holomorphic extension of Proposition 10.5.1. So the "space" of the Hardy ring is the closed disc, not half of the circle.

Exercise 10.9. Using Theorem 10.5.4, show that $\mathrm{ind}_{\mathcal{T}}(T_a T_b) = \mathrm{ind}_{\mathcal{T}}(T_a) + \mathrm{ind}_{\mathcal{T}}(T_b)$ for $a, b \in \mathcal{C}^\infty(\mathbb{S}^1; \mathrm{GL}(N, \mathbb{C}))$. Check this against Exercise 10.1 applied to $T_a + \mathrm{Id} - S$ and $T_b + \mathrm{Id} - S$.

Notes

The source lectures stated the existence of smooth families of parametrices, the smoothness of the bundles F_r, and the extension of the Fredholm determinant to all of $\Psi^{-\infty}$ without proof. Here they are supplied in Lemma 10.2.1, Lemma 10.6.2 and Theorem 10.4.8. The second stage of the homotopy in Proposition 10.7.1 has been corrected so that it starts where the first stage ends. The parametrization of $\mathbb{S}^1 \setminus \{1\}$ in Proposition 10.3.7 has been adjusted to omit the base point. The classical index theorem for Toeplitz operators on the circle (Theorem 10.5.4) is due to F. Noether and to Gohberg and Krein; it has been added as the simplest instance of the index formula. Loop groups and their relation with Toeplitz operators are treated at length in [21]. The approach via the isotropic algebra on $\mathbb{R}$ is carried out in the setting of manifolds with boundary in [20].

The Szegő projector also exists in higher dimensions. For the boundary $\mathbb{S}^{2n-1}$ of the unit ball in $\mathbb{C}^n$ (or of a strictly pseudoconvex domain) the orthogonal projection onto boundary values of holomorphic functions is a Fourier integral operator of Hermite type. The index of the Toeplitz operator $T_a = SaS$ with $a \in \mathcal{C}^\infty(\mathbb{S}^{2n-1}; \mathrm{GL}(N, \mathbb{C}))$ is a universal multiple of $\int_{\mathbb{S}^{2n-1}} \mathrm{tr}((a^{-1}da)^{2n-1})$, i.e. of the pairing of the odd Chern form u_{2n-1} with the fundamental class. This is a theorem of L. Boutet de Monvel (*On the index of Toeplitz operators of several complex variables*, Invent. Math. 50 (1979)). Theorem 10.5.4 is the case $n = 1$.

Chapter 11

Linearization of symbols, K-theory and Bott periodicity

In Chapter 10 the index of an elliptic operator was shown to be a homotopy invariant, the group $G^{-\infty}(Y;E)$ of invertible perturbations of the identity by smoothing operators was introduced as a model for odd K-theory, and the Toeplitz algebra on the circle was set up. This chapter combines these ingredients. The central technical result is a normal form, due to Atiyah, for loops of invertible matrices: after stabilization and deformation every (pointed) loop can be brought to the form $\pi_- e^{-i\theta} + \pi_0 + \pi_+ e^{i\theta}$ with commuting idempotents $\pi_\pm, \pi_0$. Applied to the symbols of Toeplitz operators this *linearization* reduces index computations to an explicit model. As consequences we obtain:

- the index formula for Toeplitz and pseudodifferential operators on the circle;
- the families index of elliptic Toeplitz families, with values in K^0, and the fact that it gives an isomorphism $K^{-2}(X) \cong K^0(X)$ (Bott periodicity in its first form);
- the weak contractibility of the stabilized group $G_{\mathcal{T}}^0$ of invertible Toeplitz operators, hence a *classifying sequence* $G_{\mathcal{T}}^{-\infty} \to G_{\mathcal{T}}^0 \to G_{\mathcal{T}}^0/G_{\mathcal{T}}^{-\infty}$ for K-theory;
- Bott periodicity in its usual form, $K_{\mathrm{c}}^0(\mathbb{R}^2 \times X) \cong K^0(X)$, and the homotopy groups of $G^{-\infty}$: $\pi_k(G^{-\infty}) = \mathbb{Z}$ for k odd and 0 for k even;
- the analytic index map $K_{\mathrm{c}}^0(T^*Z) \to \mathbb{Z}$, the starting point of the index theorem.

K-theory is defined from scratch in Section 11.2; we follow [17] for the topological background. We use Toeplitz operators on the circle (an alternative based on the isotropic calculus of Chapter 6 is indicated in the Notes); compare also [21] for the relation between loop groups and Toeplitz operators.

11.1 Conventions and background

Throughout, X denotes a compact manifold, possibly with boundary (we need $[0,1] \times X$), and all maps and homotopies are smooth. The circle is $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$ with variable θ; we write $z = e^{i\theta}$, so that $\mathbb{S}$ is also the unit circle in $\mathbb{C}$, and we use the point $1 \in \mathbb{S}$ ($\theta = 0$) as base point. $M(N,\mathbb{C})$ is the algebra of $N \times N$ matrices. For maps from a product we use the obvious notion of smoothness; a smooth map into a Fréchet space F (such as $\Psi^0(\mathbb{S})$ with its standard topology, see Chapters 4 and 5) is one all of whose derivatives exist and are continuous.

Two smooth homotopies $h, h' \colon [0,1] \times X \to W$ with $h_1 = h'_0$ can be concatenated to a smooth homotopy after reparametrizing each of them by a smooth non-decreasing surjection $[0,1] \to [0,1]$ which is constant near 0 and 1. We use this tacitly whenever homotopies are composed.

11.1.1 Facts recalled from Chapter 10

Let Y be a compact manifold of positive dimension, $E \to Y$ a Hermitian vector bundle, and fix a smooth positive density on Y; write $L^2(Y;E)$ for the corresponding Hilbert space.

(B1) $\Psi^{-\infty}(Y;E)$, the smoothing operators (kernels in $\mathcal{C}^\infty(Y^2; \operatorname{Hom}(E) \otimes \Omega_R)$), is a Fréchet algebra and

$$G^{-\infty}(Y;E) = \{\mathrm{Id} + A;\ A \in \Psi^{-\infty}(Y;E),\ \mathrm{Id} + A \text{ invertible on } L^2(Y;E)\}.$$

The inverse of $\mathrm{Id} + A \in G^{-\infty}(Y;E)$ is of the form $\mathrm{Id} + B$ with $B \in \Psi^{-\infty}(Y;E)$; $G^{-\infty}(Y;E)$ is an open subset of $\mathrm{Id} + \Psi^{-\infty}(Y;E)$, a topological group with smooth multiplication and inversion, and it is connected.

(B2) (Finite rank approximation.) Given a finite dimensional subspace $V_0 \subset \mathcal{C}^\infty(Y;E)$ there is an increasing sequence of finite dimensional subspaces $V_0 \subset V_1 \subset V_2 \subset \cdots \subset \mathcal{C}^\infty(Y;E)$ (for instance V_0 plus the span of the first j eigensections of a Laplacian) whose orthogonal projections Π_j satisfy $\Pi_j A \Pi_j \to A$ in $\Psi^{-\infty}(Y;E)$, uniformly with all derivatives on smooth families A parametrized by a compact manifold. For a finite dimensional subspace V with orthogonal projection Π_V we identify $\mathrm{GL}(V)$, and so $\mathrm{GL}(N,\mathbb{C})$ after choosing an orthonormal basis, with the subgroup $\{\mathrm{Id} - \Pi_V + g\Pi_V;\ g \in \mathrm{GL}(V)\} \subset G^{-\infty}(Y;E)$. Enlarging V to $V' \supset V$ corresponds to the stabilization $g \mapsto g \oplus \mathrm{Id}$.

(B3) $K^{-1}(X) = [X; G^{-\infty}(Y;E)]$, the set of smooth homotopy classes of smooth maps, is an abelian group under pointwise multiplication, independent of the choice of (Y,E).

(B4) For a smooth loop $b \colon \mathbb{S} \to G^{-\infty}(Y;E)$ the number

$$\operatorname{wind}(b) = \frac{1}{2\pi i} \int_0^{2\pi} \operatorname{Tr}\left(b^{-1} \frac{db}{d\theta}\right) d\theta$$

is an integer, the winding number of the Fredholm determinant along b. For b with values in $\mathrm{GL}(N,\mathbb{C})$ it is the winding number of $\det b$, since $d \log \det b = \operatorname{Tr}(b^{-1} db)$. It is invariant under homotopy of loops.

(B5) The Szegő projector S, which removes the negative Fourier modes, is a self-adjoint projection in $\Psi^0(\mathbb{S})$ with principal symbol equal to 1 on $\mathbb{S}_+ = \{\xi > 0\}$ and 0 on $\mathbb{S}_- = \{\xi < 0\}$ in $S^*\mathbb{S} = \mathbb{S}_+ \sqcup \mathbb{S}_-$. Its range on $\mathcal{C}^\infty(\mathbb{S})$ is $\mathcal{C}^\infty_+(\mathbb{S})$, the smooth functions with only non-negative Fourier modes, and on L^2 it is $L^2_+(\mathbb{S})$. The Toeplitz algebra is

$$\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) = \{A \in \Psi^0(\mathbb{S};\mathbb{C}^N);\ A = SAS\},$$

acting on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$; $\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) = S\Psi^{-\infty}(\mathbb{S};\mathbb{C}^N)S$. The (Toeplitz) symbol $\sigma_0(A) \in \mathcal{C}^\infty(\mathbb{S}; M(N,\mathbb{C}))$ is the restriction of the principal symbol to $\mathbb{S}_+ \cong \mathbb{S}$; it is multiplicative and for $a \in \mathcal{C}^\infty(\mathbb{S}; M(N,\mathbb{C}))$ the operator $T_a = SaS$ (multiplication by a, then projection) lies in $\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ with $\sigma_0(T_a) = a$. A constant matrix commutes with S, so $T_c = c$. A is *elliptic* if $\sigma_0(A)$ is invertible; then A has a parametrix in $\Psi^0_{\mathcal{T}}$ modulo $\Psi^{-\infty}_{\mathcal{T}}$, is Fredholm on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ and on $L^2_+(\mathbb{S};\mathbb{C}^N)$ with null spaces in $\mathcal{C}^\infty_+$, and its index depends only on the homotopy class of $\sigma_0(A)$ in $\mathcal{C}^\infty(\mathbb{S}; \mathrm{GL}(N,\mathbb{C}))$. If an elliptic A is invertible, then $A^{-1} \in \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$.

(B6) The shifts $U = T_z$ and $L = T_{z^{-1}}$ act by $U(e^{ik\theta}) = e^{i(k+1)\theta}$, $L(e^{ik\theta}) = e^{i(k-1)\theta}$ for $k \geq 1$ and $L(1) = 0$. Thus $LU = \mathrm{Id}$, $UL = \mathrm{Id} - P_0$, $\operatorname{ind} U = -1$ and $\operatorname{ind} L = 1$. Here, and below, P_k is the orthogonal projection onto the span of $e^{ij\theta}$, $0 \leq j \leq k$ (acting componentwise on vector valued functions). For a smooth family K of elements of $\Psi_{\mathcal{T}}^{-\infty}$ parametrized by a compact manifold, $KP_k \to K$ and $P_kK \to K$ in $\Psi_{\mathcal{T}}^{-\infty}$ uniformly as $k \to \infty$.

(B7) (Lifting homotopies.) If $A\colon [0,1] \times X \to \Psi_{\mathcal{T}}^{0}(\mathbb{S}; \mathbb{C}^N)$ is a smooth family of elliptic operators and $A_0(x)$ is invertible for all $x \in X$, then there is a smooth family $B\colon [0,1] \times X \to \Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}; \mathbb{C}^N)$ with $B_0 = 0$ such that $A_t(x) + B_t(x)$ is invertible for all (t, x).

A smooth family of invertible elliptic Toeplitz operators has smoothly varying inverses; we prove this in Lemma 11.5.1 below.

11.1.2 Elementary deformations of loops

A *pointed loop* (over X) is a smooth map $a\colon X \times \mathbb{S} \to \mathrm{GL}(N, \mathbb{C})$ with $a(x, 1) = \mathrm{Id}$ for all x; a pointed homotopy is a smooth family of pointed loops. Loops are *stabilized* by $a \mapsto a \oplus \mathrm{Id}_{M-N}$, and a, b are *stably pointed homotopic*, $a \sim b$, if $a \oplus \mathrm{Id}$ and $b \oplus \mathrm{Id}$ are pointed homotopic in some $\mathrm{GL}(M, \mathbb{C})$. The following three lemmas are used constantly.

Lemma 11.1.1. *Let $h\colon [0,1] \times X \times \mathbb{S} \to \mathrm{GL}(N, \mathbb{C})$ be smooth and set*

$$\hat{h}_t(x, \theta) = h_t(x, 1)^{-1} h_t(x, \theta).$$

Then $\hat{h}$ is a pointed homotopy; $\hat{h}_t = h_t$ whenever h_t is pointed, and $\hat{h}_t = \mathrm{Id}$ whenever $h_t(x, \theta)$ is independent of θ.

Proof. Immediate from the definition. □

Lemma 11.1.2 (Rotation)**.** *For pointed loops a, b with values in $\mathrm{GL}(N, \mathbb{C})$, $\operatorname{diag}(ab, \mathrm{Id}_N)$ and $\operatorname{diag}(a, b)$ are pointed homotopic in $\mathrm{GL}(2N, \mathbb{C})$. Consequently $ab \sim \operatorname{diag}(a, b) \sim ba$, $\operatorname{diag}(a_1 \cdots a_r, \mathrm{Id}, \ldots, \mathrm{Id}) \sim \operatorname{diag}(a_1, \ldots, a_r)$, and block sums of pointed loops are commutative up to $\sim$.*

Proof. Let $R_\tau = \begin{pmatrix} \cos\tau\, \mathrm{Id}_N & -\sin\tau\, \mathrm{Id}_N \\ \sin\tau\, \mathrm{Id}_N & \cos\tau\, \mathrm{Id}_N \end{pmatrix}$ and $h_\tau = \operatorname{diag}(a, \mathrm{Id}) R_\tau \operatorname{diag}(b, \mathrm{Id}) R_\tau^{-1}$, $\tau \in [0, \pi/2]$. Then $h_0 = \operatorname{diag}(ab, \mathrm{Id})$, $h_{\pi/2} = \operatorname{diag}(a, \mathrm{Id}) \operatorname{diag}(\mathrm{Id}, b) = \operatorname{diag}(a, b)$, and $h_\tau(x, 1) = \mathrm{Id}$. The remaining statements follow by iteration and by applying the same argument with the roles of the blocks exchanged (the constant permutation matrix exchanging two blocks is joined to Id by R_τ). □

Lemma 11.1.3 (Conjugation)**.** *Let a be a pointed loop in $\mathrm{GL}(N, \mathbb{C})$ and $c\colon X \to \mathrm{GL}(N, \mathbb{C})$ smooth (independent of θ). Then $cac^{-1} \sim a$; if c is constant then cac^{-1} and a are pointed homotopic without stabilization.*

Proof. If c is constant, join it to Id by a path c_t in the connected group $\mathrm{GL}(N, \mathbb{C})$; then $c_t a c_t^{-1}$ is a pointed homotopy. In general, $h_\tau = \operatorname{diag}(c, \mathrm{Id}) R_\tau \operatorname{diag}(c^{-1}, \mathrm{Id}) R_\tau^{-1}$ is a θ-independent path from Id to $\operatorname{diag}(c, c^{-1})$, and $h_\tau \operatorname{diag}(a, \mathrm{Id}) h_\tau^{-1}$ is a pointed homotopy from $\operatorname{diag}(a, \mathrm{Id})$ to $\operatorname{diag}(c, c^{-1}) \operatorname{diag}(a, \mathrm{Id}) \operatorname{diag}(c^{-1}, c) = \operatorname{diag}(cac^{-1}, \mathrm{Id})$. □

11.2 Vector bundles and K-theory

All vector bundles are smooth complex vector bundles of finite rank; $\mathbb{C}^N$ also denotes the trivial bundle. For bundles E, F over the same base we write $E \cong F$ if there is a smooth bundle isomorphism, i.e. a smooth map of total spaces which is a linear isomorphism $E_x \to F_x$ on each fibre.

11.2.1 Bundles

Lemma 11.2.1. *Let X be compact.*

(i) Every vector bundle E over X is isomorphic to the range of a smooth family of orthogonal projections $p\colon X \to M(N,\mathbb{C})$; in particular there is a bundle F with $E \oplus F \cong \mathbb{C}^N$.

(ii) If $\pi\colon X \to M(N,\mathbb{C})$ is a smooth family of idempotents, $\pi^2 = \pi$, then $\operatorname{Ran}\pi = \{(x,v);\ v \in \operatorname{Ran}\pi(x)\}$ is a smooth vector bundle.

Proof. (ii) The rank of $\pi(x)$ equals $\operatorname{tr}\pi(x)$, so is locally constant. If $v_1,\dots,v_r$ is a basis of $\operatorname{Ran}\pi(x_0)$ then $\pi(x)v_1,\dots,\pi(x)v_r$ are linearly independent for x near x_0, hence a smooth local frame of $\operatorname{Ran}\pi$.

(i) Choose open sets $U_1,\dots,U_s$ covering X over which E has trivializations $\tau_i\colon E|_{U_i} \to U_i \times \mathbb{C}^n$, and smooth $\psi_i \geq 0$ with $\operatorname{supp}\psi_i \subset U_i$ and $\sum_i \psi_i > 0$. Then

$$\iota(v) = (\psi_1(x)\tau_1(v),\dots,\psi_s(x)\tau_s(v)), \quad v \in E_x,$$

is a smooth fibrewise injective bundle map $E \to \mathbb{C}^N$, $N = sn$. The orthogonal projection $p(x)$ onto $\iota(E_x)$ depends smoothly on x (locally it is given by the Gram–Schmidt process applied to a smooth frame), so $E \cong \operatorname{Ran}p$, and $F = \operatorname{Ran}(\mathrm{Id} - p)$ is a complement. □

Lemma 11.2.2 (Homotopy invariance of bundles)**.** *Let W be a manifold.*

(i) If E is a vector bundle over $[0,1]\times W$, then $E|_{\{0\}\times W} \cong E|_{\{1\}\times W}$ as bundles over W.

(ii) If $\varpi\colon V \to W$ is a real vector bundle (for instance $V = \mathbb{R}^n \times W$ or $V = T^*Z$) and E is a vector bundle over the total space V, then $E \cong \varpi^*(E|_W)$, where W is identified with the zero section.

Proof. Choose a connection on E (patching local trivial connections with a partition of unity). In (i), parallel transport along the curves $t \mapsto (t,w)$ gives linear isomorphisms $E_{(0,w)} \to E_{(1,w)}$ which depend smoothly on w by smooth dependence of solutions of linear ordinary differential equations on parameters. In (ii), parallel transport along $t \mapsto t\zeta$, $t \in [0,1]$, from the zero section to $\zeta \in V$ gives a smooth isomorphism $\varpi^*(E|_W) \to E$. □

11.2.2 The group K^0

Definition 11.2.3. For compact X, $K^0(X)$ is the set of equivalence classes of pairs (E_+, E_-) of vector bundles over X, where $(E_+,E_-) \sim (F_+,F_-)$ if there is a bundle H with

$$E_+ \oplus F_- \oplus H \cong F_+ \oplus E_- \oplus H.$$

The class of (E_+,E_-) is written $[E_+]-[E_-]$, and $[E] = [E]-[0]$. Addition is $([E_+]-[E_-])+([F_+]-[F_-]) = [E_+\oplus F_+]-[E_-\oplus F_-]$.

The relation is reflexive and symmetric; it is transitive because if $E_+ \oplus F_- \oplus H \cong F_+ \oplus E_- \oplus H$ and $F_+ \oplus G_- \oplus H' \cong G_+ \oplus F_- \oplus H'$ then $E_+ \oplus G_- \oplus (F_+ \oplus F_- \oplus H \oplus H') \cong G_+ \oplus E_- \oplus (F_+ \oplus F_- \oplus H \oplus H')$. Addition is well defined, associative and commutative; $[E]-[E]$ is the zero element and $[E_-]-[E_+]$ is the inverse of $[E_+]-[E_-]$. Thus $K^0(X)$ is an abelian group. By Lemma 11.2.1 every element can be written $[E]-[\mathbb{C}^N]$ (add a complement of E_- to both bundles), and $[E_+]-[E_-]=0$ if and only if $E_+ \oplus \mathbb{C}^L \cong E_- \oplus \mathbb{C}^L$ for some L (replace H by $H \oplus H^\perp \cong \mathbb{C}^L$). A smooth map $f\colon X' \to X$ induces a homomorphism $f^*\colon K^0(X) \to K^0(X')$ by pull-back of bundles. By Lemma 11.2.2, $K^0([0,1]\times X) \to K^0(X)$, restriction to $\{t\}\times X$, is independent of t. The difference of ranks at a point identifies $K^0(\mathrm{pt}) = \mathbb{Z}$; the *rank* of $[E_+]-[E_-]$ at x is $\dim E_{+,x} - \dim E_{-,x}$. If $x_0 \in X$ is a base point, $\tilde K^0(X) = \ker\big(K^0(X) \to K^0(\{x_0\})\big)$ is the *reduced* group, and $K^0(X) = \mathbb{Z} \oplus \tilde K^0(X)$, the splitting being given by pull-back along $X \to \{x_0\}$.

11.2.3 Odd K-theory and pointed loops

Odd K-theory is $K^{-1}(X) = [X; G^{-\infty}(Y;E)]$, see (B3). The loops in $G^{-\infty}$ give a second model of even K-theory.

Definition 11.2.4. For compact X, $K^{-2}(X)$ is the set of smooth homotopy classes of smooth maps $f\colon X \times \mathbb{S} \to G^{-\infty}(Y;E)$ with $f(x,1) = \mathrm{Id}$ for all x, where homotopies are required to satisfy the same condition. It is a group under pointwise multiplication.

Equivalently, $K^{-2}(X)$ is the set of homotopy classes of smooth maps from X into the pointed loop group

$$\Omega G^{-\infty}(Y;E) = \{b \in \mathcal{C}^\infty(\mathbb{S}; G^{-\infty}(Y;E));\ b(1) = \mathrm{Id}\}$$

(this is homotopy equivalent to the flat loop group $G^{-\infty}_{(1)}(Y;E)$ of Chapter 10, by the flattening argument used there). The number $\mathrm{wind}(f(x,\cdot))$ of (B4) is locally constant in x; we denote by $\Omega_0 G^{-\infty}(Y;E)$ the loops with wind $= 0$.

Lemma 11.2.5. *Let $X' \subset X$ be closed (e.g. $X' = X \times \{1\} \subset X \times \mathbb{S}$).*

(i) Every smooth $f\colon X \to G^{-\infty}(Y;E)$ with $f = \mathrm{Id}$ on X' is homotopic, through maps equal to Id *on X', to a map into a subgroup $\mathrm{GL}(V) \cong \mathrm{GL}(N,\mathbb{C})$ as in (B2).*

(ii) If two such maps into $\mathrm{GL}(V)$ are homotopic in $G^{-\infty}(Y;E)$ through maps equal to Id *on X', they are homotopic through such maps in $\mathrm{GL}(V')$ for some finite dimensional $V' \supset V$.*

In particular $K^{-2}(X)$ is the set of stable pointed homotopy classes of pointed loops in the groups $\mathrm{GL}(N,\mathbb{C})$; in it $[a]+[b] = [ab] = [\mathrm{diag}(a,b)]$, and $K^{-2}(X)$ is abelian. The same holds for $K^{-1}(X)$, with free maps $X \to \mathrm{GL}(N,\mathbb{C})$.

Proof. (i) Write $f = \mathrm{Id} + A$. With Π_j as in (B2), the maps $f_t = \mathrm{Id} + (1-t)A + t\Pi_j A \Pi_j$ converge to f uniformly on $[0,1]\times X$ as $j \to \infty$; since $f(X)$ is compact and $G^{-\infty}$ is open, $f_t \in G^{-\infty}$ for j large. Where $A = 0$ also $f_t = \mathrm{Id}$, and $f_1 = \mathrm{Id} - \Pi_j + (\mathrm{Id} + \Pi_j A \Pi_j)\Pi_j$ has values in $\mathrm{GL}(V_j)$. (ii) Apply (i) to the homotopy, a map on $[0,1]\times X$, with $V_0 = V$ and closed set $[0,1]\times X' \cup \{0,1\}\times X$; on $\{0,1\}\times X$ the operator A satisfies $A = \Pi_V A \Pi_V = \Pi_j A \Pi_j$, so the approximation does not change the endpoints. The final statements follow, using Lemma 11.1.2. □

11.3 Linearization of loops

The following normal form is due to Atiyah [17].

Theorem 11.3.1. *Let $a\colon X\times\mathbb{S}\to \mathrm{GL}(N,\mathbb{C})$ be a pointed loop, X compact. Then for some $M\geq N$ the stabilized loop $a\oplus \mathrm{Id}_{M-N}$ is pointed homotopic in $\mathrm{GL}(M,\mathbb{C})$ to a loop of the form*

$$\pi_-(x)z^{-1}+\pi_0(x)+\pi_+(x)z, \tag{11.3.1}$$

where $\pi_-,\pi_0,\pi_+\colon X\to M(M,\mathbb{C})$ are smooth families of idempotents which commute with each other and satisfy $\pi_-+\pi_0+\pi_+=\mathrm{Id}$.

Note that (11.3.1) is invertible, with inverse $\pi_- z+\pi_0+\pi_+ z^{-1}$, and is automatically pointed. We call it a loop in *normal form*. The proof occupies the rest of this section and proceeds through Laurent polynomials, polynomials and linear loops.

Lemma 11.3.2. *A pointed loop a is pointed homotopic to a pointed Laurent polynomial $\sum_{|j|\leq q} a'_j(x)z^j$ with $a'_j\in\mathcal{C}^\infty(X;M(N,\mathbb{C}))$.*

Proof. The Fourier coefficients $a_j(x)=\frac{1}{2\pi}\int_0^{2\pi} e^{-ij\theta}a(x,\theta)\,d\theta$ satisfy $\sup_X|a_j|\leq C_m(1+|j|)^{-m}$ for every m, by integration by parts in θ, so $a_{(q)}=\sum_{|j|\leq q}a_j z^j\to a$ uniformly on $X\times\mathbb{S}$. Let $\delta=\min_{X\times\mathbb{S}}\|a^{-1}\|^{-1}>0$ and choose q with $\sup\|a_{(q)}-a\|<\delta$. Then $h_t=(1-t)a+ta_{(q)}=a\big(\mathrm{Id}+ta^{-1}(a_{(q)}-a)\big)$ is invertible for $t\in[0,1]$, and by Lemma 11.1.1 $\hat{h}_t$ is a pointed homotopy from a to $a_{(q)}(x,1)^{-1}a_{(q)}$, which is a pointed Laurent polynomial. □

So we may assume $a=z^{-q}b$ where $b(x,z)=\sum_{j=0}^p b_j(x)z^j$, $p=2q$, is a polynomial in z which is invertible for $|z|=1$ and satisfies $b(x,1)=\mathrm{Id}$. Since z^{-q} is a scalar, Lemma 11.1.2 gives

$$\mathrm{diag}(a,\mathrm{Id}_N)=\mathrm{diag}(b\,z^{-q}\,\mathrm{Id}_N,\mathrm{Id}_N)\sim\mathrm{diag}(b,z^{-q}\,\mathrm{Id}_N),$$
$$\mathrm{diag}(z^{-q}\,\mathrm{Id}_N,\mathrm{Id}_{(q-1)N})\sim z^{-1}\,\mathrm{Id}_{qN},$$

the second by writing $z^{-q}=z^{-1}\cdots z^{-1}$. It therefore suffices to deal with the polynomial loop b.

Lemma 11.3.3. *Let $b=\sum_{j=0}^p b_j z^j$ be a pointed polynomial loop in $\mathrm{GL}(N,\mathbb{C})$. Then $\mathrm{diag}(b,\mathrm{Id}_{pN})$ is pointed homotopic in $\mathrm{GL}((p+1)N,\mathbb{C})$ to a loop $\mathrm{Id}+(z-1)A(x)$, with $A\in\mathcal{C}^\infty(X;M((p+1)N,\mathbb{C}))$.*

Proof. Set $c_j(x,z)=\sum_{l=j}^p b_l(x)z^{l-j}$ for $0\leq j\leq p$, so $c_0=b$, $c_p=b_p$ and $c_j-zc_{j+1}=b_j$. In $(p+1)\times(p+1)$ block form with $N\times N$ blocks consider

$$T(t)=\begin{pmatrix} b & tc_1 & \cdots & tc_p\\ 0 & \mathrm{Id} & & \\ \vdots & & \ddots & \\ 0 & & & \mathrm{Id}\end{pmatrix},\qquad \Lambda(s)=\begin{pmatrix} \mathrm{Id} & & & \\ -sz & \mathrm{Id} & & \\ & \ddots & \ddots & \\ & & -sz & \mathrm{Id}\end{pmatrix},\qquad t,s\in[0,1].$$

$T(t)$ is block upper triangular with invertible diagonal blocks when $|z|=1$, and $\Lambda(s)$ is block lower triangular with identity diagonal, so both are invertible. The homotopy $T(t)$, $t\in[0,1]$, followed by $T(1)\Lambda(s)$, $s\in[0,1]$, starts at $\mathrm{diag}(b,\mathrm{Id})$ and ends at

$$T(1)\Lambda(1)=\begin{pmatrix} b_0 & b_1 & \cdots & b_{p-1} & b_p\\ -z & \mathrm{Id} & & & \\ & -z & \mathrm{Id} & & \\ & & \ddots & \ddots & \\ & & & -z & \mathrm{Id}\end{pmatrix}=B_0(x)+zB_1(x),$$

since the $(1,j)$ entry of the product is $c_j - zc_{j+1} = b_j$ for $j < p$ and $c_p = b_p$ for $j = p$. Applying Lemma 11.1.1 gives a pointed homotopy (the initial loop is pointed) ending at $(B_0+B_1)^{-1}(B_0+zB_1) = \mathrm{Id} + (z-1)A$ with $A = (B_0+B_1)^{-1}B_1$. □

Lemma 11.3.4. *Let $A \in \mathcal{C}^\infty(X; M(M,\mathbb{C}))$. The loop $\ell(x,z) = \mathrm{Id} + (z-1)A(x)$ is invertible for all $|z|=1$ if and only if no eigenvalue of any $A(x)$ lies on the line $\{\Re\lambda = \frac12\}$. In that case ℓ is pointed homotopic, through loops of the same form, to $(\mathrm{Id} - \pi) + z\pi$ where $\pi(x)$ is the spectral projection of $A(x)$ onto the generalized eigenspaces with $\Re\lambda > \frac12$.*

Proof. For $z \neq 1$, $\ell(z) = (z-1)(A-w)$ with $w = (1-z)^{-1}$, and as $z = e^{i\theta}$ runs over $\mathbb{S}\setminus\{1\}$,

$$\frac{1}{1-e^{i\theta}} = \frac12 + \frac{i}{2}\cot\frac{\theta}{2}$$

runs over the whole line $\Re w = \frac12$; at $z=1$, $\ell = \mathrm{Id}$. This proves the first statement. Now the set $\{(x,\lambda);\ \det(\lambda - A(x)) = 0\}$ is compact and disjoint from $X \times \{\Re\lambda = \frac12\}$, so there is $\delta > 0$ such that all eigenvalues satisfy $|\Re\lambda - \frac12| \geq \delta$. Let Γ be the positively oriented boundary of the rectangle $\{\frac12 + \frac{\delta}{2} \leq \Re\lambda \leq R,\ |\Im\lambda| \leq R\}$ with $R > 1 + \sup_X \|A\|$. Then

$$\pi(x) = \frac{1}{2\pi i}\int_\Gamma (\lambda - A(x))^{-1}\, d\lambda$$

is a smooth family of idempotents commuting with $A(x)$, with range the sum of the generalized eigenspaces of $A(x)$ for eigenvalues with $\Re\lambda > \frac12$ and null space the sum of the others (the Riesz projection). Set $A_t = (1-t)A + t\pi$. On $\operatorname{Ran}\pi$, which is A-invariant, $A_t = (1-t)A + t$ has eigenvalues $(1-t)\lambda + t$ with $\Re\lambda > \frac12$, so real part $> \frac12$; on $\operatorname{Nul}\pi$, $A_t = (1-t)A$ has eigenvalues $(1-t)\lambda$ with $\Re\lambda < \frac12$, so real part $< \frac12$. By the first part $\mathrm{Id} + (z-1)A_t$ is a pointed homotopy, and it ends at $\mathrm{Id} + (z-1)\pi = (\mathrm{Id} - \pi) + z\pi$. □

Proof of Theorem 11.3.1. By Lemma 11.3.2 and the reduction following it, $a \oplus \mathrm{Id}$ is stably pointed homotopic to $\mathrm{diag}(b, z^{-1}\,\mathrm{Id}_{qN})$, where b is a pointed polynomial loop of degree at most $p = 2q$. By Lemmas 11.3.3 and 11.3.4, $\mathrm{diag}(b, \mathrm{Id}_{pN}) \sim (\mathrm{Id} - \pi) + z\pi$. Hence, with $M = (3q+1)N$, $a \oplus \mathrm{Id}_{M-N}$ is pointed homotopic (after a constant permutation of blocks, Lemma 11.1.3) to (11.3.1) with $\pi_+ = \pi \oplus 0$, $\pi_0 = (\mathrm{Id} - \pi) \oplus 0$, $\pi_- = 0 \oplus \mathrm{Id}_{qN}$. □

Remark 11.3.5. The proof gives more: π_- may be taken constant. We shall not use this. The spectral projection in Lemma 11.3.4 can also be written as a contour integral of the logarithmic derivative of the linear loop over $|z|=1$; the formulation above does not require the coefficient of z to be invertible.

The decisive point for K-theory is that a loop in normal form is trivial as soon as $\operatorname{Ran}\pi_-$ and $\operatorname{Ran}\pi_+$ are isomorphic.

Lemma 11.3.6. *Let π_-, π_0, π_+ be commuting smooth idempotents on $\mathbb{C}^M$ over X with sum* Id*, and suppose there is a smooth bundle isomorphism $F\colon \operatorname{Ran}\pi_- \to \operatorname{Ran}\pi_+$. Then $\pi_- z^{-1} + \pi_0 + \pi_+ z$ is pointed homotopic to* Id *in* $\mathrm{GL}(M,\mathbb{C})$.

Proof. The fibres split as $\mathbb{C}^M = V_- \oplus V_0 \oplus V_+$, $V_\bullet = \operatorname{Ran}\pi_\bullet$. Regard $F\pi_-$ and $F^{-1}\pi_+$ as smooth maps into $M(M,\mathbb{C})$ and set, for $\tau \in [0, \pi/2]$,

$$h_\tau = \pi_0 + \cos\tau\,(z^{-1}\pi_- + z\pi_+) + \sin\tau\,(F^{-1}\pi_+ - F\pi_-).$$

h_τ is the identity on V_0 and preserves $V_- \oplus V_+$; identifying V_+ with V_- by F^{-1} it acts there as $\begin{pmatrix} \cos\tau\, z^{-1} & \sin\tau \\ -\sin\tau & \cos\tau\, z \end{pmatrix} \otimes \mathrm{Id}_{V_-}$, which has determinant 1. So h_τ is a loop in $\mathrm{GL}(M,\mathbb{C})$, h_0 is the given loop, and $h_{\pi/2} = \pi_0 + F^{-1}\pi_+ - F\pi_-$ is independent of z. By Lemma 11.1.1, $\hat{h}_\tau$ is a pointed homotopy from h_0 to Id. □

11.4 The index of Toeplitz operators on the circle

For commuting smooth idempotents π_-, π_0, π_+ on $\mathbb{C}^M$ with sum Id consider the family of Toeplitz operators on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^M)$

$$Q(x) = \pi_-(x)L + \pi_0(x) + \pi_+(x)U, \qquad \sigma_0(Q) = \pi_- z^{-1} + \pi_0 + \pi_+ z. \tag{11.4.1}$$

The matrices act pointwise and commute with S, U, L and the P_k. For $c \in \mathbb{C}^M$ write $c^\bullet = \pi_\bullet c$.

Lemma 11.4.1. *For every $k \geq 0$ the null space of $Q(x)(\mathrm{Id} - P_k)$ is $\operatorname{Ran} P_k$, and*

$$W_k(x) = \operatorname{span}\{e^{ij\theta}\mathbb{C}^M\}_{0\leq j\leq k-1} \oplus e^{ik\theta}\operatorname{Ran}(\pi_0+\pi_+)(x) \oplus e^{i(k+1)\theta}\operatorname{Ran}\pi_+(x)$$

is a complement to the range of $Q(x)(\mathrm{Id} - P_k)$ in $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^M)$. In particular $\operatorname{Nul} Q(x) = \operatorname{Ran}\pi_-(x)$ (constant functions), $\operatorname{Nul} Q(x)^* = \operatorname{Ran}\pi_+(x)^*$ and $\operatorname{ind} Q(x) = \operatorname{rank}\pi_- - \operatorname{rank}\pi_+$.

Proof. For $u = \sum_{j\geq k+1} c_j e^{ij\theta}$ (rapidly decreasing coefficients),

$$Q(\mathrm{Id} - P_k)u = \sum_{j\geq k+1} (c_j^- e^{i(j-1)\theta} + c_j^0 e^{ij\theta} + c_j^+ e^{i(j+1)\theta}) = \sum_m d_m e^{im\theta}$$

with $d_m^- = c_{m+1}^-$, $d_m^0 = c_m^0$, $d_m^+ = c_{m-1}^+$. These relations determine u from the image, so $Q(\mathrm{Id} - P_k)$ is injective on $\operatorname{Ran}(\mathrm{Id} - P_k)$, and its range consists exactly of the $\sum d_m e^{im\theta} \in \mathcal{C}^\infty_+$ with $d_m^- = 0$ for $m < k$, $d_m^0 = 0$ for $m < k+1$ and $d_m^+ = 0$ for $m < k+2$. The complement W_k follows. For a constant c_0, $Qc_0 = c_0^0 + c_0^+ e^{i\theta}$, so $\operatorname{Nul} Q = \operatorname{Ran}\pi_-$. The index is $\dim \operatorname{Ran} P_0 - \dim W_0 = M - \operatorname{rank}\pi_0 - 2\operatorname{rank}\pi_+ = \operatorname{rank}\pi_- - \operatorname{rank}\pi_+$. Finally $Q^* = \pi_-^* U + \pi_0^* + \pi_+^* L$ is of the same form, with the commuting idempotents $\pi_\pm^*$ in exchanged roles, so $\operatorname{Nul} Q^* = \operatorname{Ran}\pi_+^*$. □

Corollary 11.4.2 (Index formula for Toeplitz operators)**.** *If $P \in \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ is elliptic, then*

$$\operatorname{ind} P = \dim\operatorname{Nul} P - \dim\operatorname{Nul} P^* = \frac{i}{2\pi}\int_0^{2\pi} \operatorname{Tr}\Big(\sigma_0(P)^{-1}\frac{d\sigma_0(P)}{d\theta}\Big)d\theta = -\operatorname{wind}\sigma_0(P). \tag{11.4.2}$$

Proof. The last two expressions agree by (B4). Let $c = \sigma_0(P)(1) \in \mathrm{GL}(N,\mathbb{C})$. Then $c^{-1}P$ is elliptic with pointed symbol $c^{-1}\sigma_0(P)$; it has the same index as P since c is an invertible operator, and the same winding number since $\det c$ is constant. So we may assume $\sigma_0(P)$ pointed. Both sides of (11.4.2) depend only on the stable pointed homotopy class of the symbol: the index by (B5), and because stabilizing P by Id adds nothing to either side; the winding number by (B4). By Theorem 11.3.1 (with X a point) we may thus assume $\sigma_0(P) = \pi_- z^{-1} + \pi_0 + \pi_+ z$, and then P may be replaced by Q of (11.4.1). Then $\operatorname{ind} Q = \operatorname{rank}\pi_- - \operatorname{rank}\pi_+$ by Lemma 11.4.1, while $\det\sigma_0(Q) = z^{\operatorname{rank}\pi_+ - \operatorname{rank}\pi_-}$, since $\sigma_0(Q)$ acts as z^{-1}, 1, z on the summands of $\mathbb{C}^M = \operatorname{Ran}\pi_- \oplus \operatorname{Ran}\pi_0 \oplus \operatorname{Ran}\pi_+$, so $\operatorname{wind}\sigma_0(Q) = \operatorname{rank}\pi_+ - \operatorname{rank}\pi_-$. □

The same holds for pseudodifferential operators on the circle, except that now both halves $\mathbb{S}_\pm$ of $S^*\mathbb{S}$ contribute. For $P \in \Psi^0(\mathbb{S};\mathbb{C}^N)$ write $\sigma_\pm(\theta) = \sigma_0(P)(\theta, \pm 1)$.

Proposition 11.4.3. *If $P \in \Psi^0(\mathbb{S};\mathbb{C}^N)$ is elliptic, then*

$$\operatorname{ind} P = \frac{i}{2\pi}\int_0^{2\pi} \operatorname{Tr}\left(\sigma_+^{-1}\partial_\theta\sigma_+\right)d\theta - \frac{i}{2\pi}\int_0^{2\pi} \operatorname{Tr}\left(\sigma_-^{-1}\partial_\theta\sigma_-\right)d\theta = \operatorname{wind}\sigma_- - \operatorname{wind}\sigma_+.$$

Equivalently $\operatorname{ind} P = \frac{i}{2\pi}\int_{S^*\mathbb{S}} \operatorname{Tr}(\sigma_0(P)^{-1}d\sigma_0(P))$ *with $\mathbb{S}_+$ oriented by $d\theta$ and $\mathbb{S}_-$ by $-d\theta$.*

Proof. Let $S' = \mathrm{Id} - S$, the projection onto $\mathcal{C}^\infty_-(\mathbb{S})$, the span of the negative Fourier modes. By (B5), SPS' and $S'PS$ have vanishing principal symbol, so $P_t = P - t(SPS' + S'PS)$ is a family of elliptic operators with constant principal symbol, and $\operatorname{ind} P = \operatorname{ind} P_1$. Now $P_1 = SPS + S'PS'$ preserves the decomposition $\mathcal{C}^\infty(\mathbb{S};\mathbb{C}^N) = \mathcal{C}^\infty_+ \oplus \mathcal{C}^\infty_-$, so $\operatorname{ind} P = \operatorname{ind}(SPS|_{\mathcal{C}^\infty_+}) + \operatorname{ind}(S'PS'|_{\mathcal{C}^\infty_-})$. The first is an elliptic Toeplitz operator with symbol σ_+, of index $-\operatorname{wind}\sigma_+$ by Corollary 11.4.2. For the second, let $Ru(\theta) = e^{-i\theta}u(-\theta)$. Then $R(e^{ik\theta}) = e^{-i(k+1)\theta}$, so R is an isomorphism of $\mathcal{C}^\infty(\mathbb{S};\mathbb{C}^N)$ exchanging $\mathcal{C}^\infty_-$ and $\mathcal{C}^\infty_+$, and $RS'R^{-1} = S$. Thus $R(S'PS')R^{-1} = STS$ with $T = RPR^{-1} \in \Psi^0(\mathbb{S};\mathbb{C}^N)$; as R is the composite of the pull-back by $\theta \mapsto -\theta$ and a multiplication, $\sigma_0(T)(\theta,\xi) = \sigma_0(P)(-\theta,-\xi)$. So $S'PS'|_{\mathcal{C}^\infty_-}$ is conjugate to the elliptic Toeplitz operator with symbol $\sigma_-(-\theta)$, whose index is $-\operatorname{wind}(\sigma_-(-\cdot)) = \operatorname{wind}\sigma_-$. □

Remark 11.4.4. The index of an elliptic *differential* operator $P \in \mathrm{Diff}^m(\mathbb{S};\mathbb{C}^N)$ vanishes. Indeed $\sigma_m(P) = p(\theta)\xi^m$ with p invertible, and $P(\mathrm{Id} + D_\theta^2)^{-m/2} \in \Psi^0$ has the same index as P (the second factor is invertible) and symbol $\sigma_\pm = (\pm 1)^m p$. Since $\operatorname{wind}((-1)^m p) = \operatorname{wind} p$, the two terms in Proposition 11.4.3 cancel.

11.5 The families index

We first record that inverses of smooth families are smooth. The same argument applies to the operators with values in smoothing operators of Section 11.7.

Lemma 11.5.1. *Let Z be a manifold and $A \in \mathcal{C}^\infty(Z;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ a smooth family of elliptic operators, each invertible on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ (equivalently on $L^2_+(\mathbb{S};\mathbb{C}^N)$). Then $A^{-1} \in \mathcal{C}^\infty(Z;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$.*

Proof. The symbolic parametrix construction (Chapter 5) works with smooth dependence on parameters: it uses only the multiplicativity of the symbol, the inversion of the symbol and asymptotic summation. This gives a smooth family B with $BA = \mathrm{Id} - R$, $AB = \mathrm{Id} - R'$, where $R, R' \in \mathcal{C}^\infty(Z;\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$. The inclusion of $\Psi^0_{\mathcal{T}}$ into the bounded operators on L^2_+ is continuous, so $z \mapsto A(z)$ is smooth in operator norm, and hence so is $V = A^{-1}$. From $V = B + RV$ and $V = B + VR'$ we get $V = B + RB + RVR'$. The bilinear map $T \mapsto RTR'$ is continuous from bounded operators to smoothing operators (its kernel is $(w,w') \mapsto (TR'(\cdot,w'), \overline{R(w,\cdot)})$ and all derivatives fall on R and R'), so RVR', and hence V, is a smooth family in $\Psi^0_{\mathcal{T}}$. □

Now let Z be a compact manifold (possibly with boundary) and $A \in \mathcal{C}^\infty(Z;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ a smooth family of elliptic Toeplitz operators. Since $A(z)$ is Fredholm, the range of $A(z)(\mathrm{Id} - P_k) = A(z) - A(z)P_k$ is closed and of finite codimension, equal to $(k+1)N - \operatorname{ind} A(z)$ if its null space is $\operatorname{Ran} P_k$; the same holds on $\mathcal{C}^\infty_+$ and on L^2_+.

Definition 11.5.2. An integer $k \geq 0$ is *admissible* for A if $\operatorname{Nul}(A(z)(\mathrm{Id} - P_k)) = \operatorname{Ran} P_k$ for all $z \in Z$. A *complement* for (A,k) is a smooth vector bundle W over Z whose fibres $W(z) \subset \mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ satisfy $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N) = A(z)(\mathrm{Id} - P_k)\mathcal{C}^\infty_+ \oplus W(z)$; smoothness means that W has local frames given by smooth maps into $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$.

If k is admissible then so is every $k' \geq k$, since $\operatorname{Ran}(\mathrm{Id} - P_{k'}) \subset \operatorname{Ran}(\mathrm{Id} - P_k)$.

Lemma 11.5.3. *Let A be a smooth elliptic family as above.*

(i) There is k_0 such that every $k \geq k_0$ is admissible.

(ii) If k is admissible, the orthocomplements $F_k(z)$ of the ranges of $A'(z) = A(z)(\mathrm{Id} - P_k)$ in L^2_+ form a complement for (A, k), and the orthogonal projections onto $F_k(z)$ form a smooth family. Every complement W for (A, k) is isomorphic to F_k.

(iii) If k is admissible with complement W, then $W \oplus A(e^{i(k+1)\theta}\mathbb{C}^N) \cong W \oplus \mathbb{C}^N$ is a complement for $(A, k+1)$.

Consequently the class

$$\operatorname{ind}(A) = [\operatorname{Ran} P_k] - [W] = [\mathbb{C}^{(k+1)N}] - [W] \in K^0(Z)$$

is independent of the admissible k and of the complement W for (A, k).

Proof. (i) Choose a smooth family of parametrices B, so $BA = \mathrm{Id} - R$ with $R \in \mathcal{C}^\infty(Z; \Psi^{-\infty}_{\mathcal{T}})$. Then $BA(\mathrm{Id} - P_k) = (\mathrm{Id} - R(\mathrm{Id} - P_k))(\mathrm{Id} - P_k)$. By (B6), $R(\mathrm{Id} - P_k) \to 0$ in $\Psi^{-\infty}_{\mathcal{T}}$, hence in L^2 operator norm, uniformly on Z, so for $k \geq k_0$ the operators $\mathrm{Id} - R(\mathrm{Id} - P_k)$ are invertible and $(\mathrm{Id} - P_k)(\mathrm{Id} - R(\mathrm{Id} - P_k))^{-1} BA(\mathrm{Id} - P_k) = \mathrm{Id} - P_k$. Hence $\operatorname{Nul} A(\mathrm{Id} - P_k) = \operatorname{Ran} P_k$.

(ii) The family $C = A'^* A' + P_k$ is elliptic (its symbol is $\sigma_0(A)^*\sigma_0(A)$) and injective: $(Cu, u) = \|A'u\|^2 + \|P_k u\|^2 = 0$ forces $P_k u = 0$ and $A'u = 0$, so $u \in \operatorname{Ran}(\mathrm{Id} - P_k) \cap \operatorname{Ran} P_k = \{0\}$. Being self-adjoint and Fredholm it is invertible, and C^{-1} is a smooth family by Lemma 11.5.1. Since $CP_k = P_k$ and $C(\mathrm{Id} - P_k) = A'^* A'$, the family $G = C^{-1} A'^*$ satisfies $GA' = \mathrm{Id} - P_k$. Then $A'G$ is the orthogonal projection onto $\operatorname{Ran} A'$ (it is idempotent, self-adjoint as $A'G = A'C^{-1}A'^*$, and has range $\operatorname{Ran} A'$), so $\Pi = \mathrm{Id} - A'G$ is a smooth family of orthogonal projections onto F_k. Its range consists of smooth functions (it is the null space of the elliptic operator A'^*) and its rank $(k+1)N - \operatorname{ind} A(z)$ is locally constant. If $v_1, \dots, v_r$ is a basis of $F_k(z_0)$, then $\Pi(z)v_1, \dots, \Pi(z)v_r$ are smooth maps into $\mathcal{C}^\infty_+$, linearly independent near z_0, so form a local frame: F_k is a smooth bundle. As $\operatorname{Ran} A'$ is closed and $A'(\mathcal{C}^\infty_+) = \operatorname{Ran} A' \cap \mathcal{C}^\infty_+$ (if $A'v \in \mathcal{C}^\infty_+$ then $GA'v = (\mathrm{Id} - P_k)v \in \mathcal{C}^\infty_+$), F_k is a complement. Finally, if W is a complement then $\Pi|_W \colon W \to F_k$ is a smooth bundle map; it is injective because $\operatorname{Nul}\Pi \cap \mathcal{C}^\infty_+ = A'(\mathcal{C}^\infty_+)$ meets W only in 0, and $\dim W = \dim F_k$, both being the codimension of $A'(\mathcal{C}^\infty_+)$. So $W \cong F_k$.

(iii) Since A is injective on $\operatorname{Ran}(\mathrm{Id} - P_k)$, we have $A(\mathrm{Id} - P_k)\mathcal{C}^\infty_+ = A(\mathrm{Id} - P_{k+1})\mathcal{C}^\infty_+ \oplus A(e^{i(k+1)\theta}\mathbb{C}^N)$. Hence $W \oplus A(e^{i(k+1)\theta}\mathbb{C}^N)$ is a complement for $(A, k+1)$, and $v \mapsto A(e^{i(k+1)\theta}v)$ trivializes the second summand.

For the consequence: given admissible $k \leq k'$ with complements W, W', (iii) applied $k' - k$ times gives a complement $W \oplus \mathbb{C}^{(k'-k)N}$ for k', isomorphic to W' by (ii); since $\operatorname{Ran} P_{k'} = \operatorname{Ran} P_k \oplus \mathbb{C}^{(k'-k)N}$ the classes agree. □

Definition 11.5.4. The class $\operatorname{ind}(A) \in K^0(Z)$ of Lemma 11.5.3 is the *families index* of the elliptic family A.

For Z a point this is the numerical index, since $(k+1)N - \dim W = \operatorname{ind} A$. In the notation of Chapter 10, $F_k = \operatorname{Nul}((\mathrm{Id} - P_k)A^*)$, and Lemma 11.5.3 shows that the bundle considered there represents, together with the trivial bundle $\operatorname{Ran} P_k$, a well-defined element of $K^0(Z)$. The families index has the expected properties.

Proposition 11.5.5. *Let A, A' be smooth elliptic families of Toeplitz operators over Z.*

(i) (Homotopy) If A_t is an elliptic family over $[0,1] \times Z$, then $\operatorname{ind}(A_0) = \operatorname{ind}(A_1)$. In particular $\operatorname{ind}(A)$ depends only on the homotopy class of $\sigma_0(A)$ in $\mathcal{C}^\infty(Z \times \mathbb{S}; \mathrm{GL}(N, \mathbb{C}))$.

(ii) (Additivity) $\operatorname{ind}(A \oplus A') = \operatorname{ind}(A) + \operatorname{ind}(A')$; $\operatorname{ind}(A \oplus \mathrm{Id}) = \operatorname{ind}(A)$.

(iii) (Naturality) $\operatorname{ind}(f^* A) = f^* \operatorname{ind}(A)$ for smooth $f \colon Z' \to Z$.

(iv) If each $A(z)$ is invertible then $\operatorname{ind}(A) = 0$. *If* $b, c\colon Z \to \mathrm{GL}(N,\mathbb{C})$ *then* $\operatorname{ind}(bAc) = \operatorname{ind}(A)$.

(v) (Normal form) For Q as in (11.4.1), $\operatorname{ind}(Q) = [\operatorname{Ran}\pi_-] - [\operatorname{Ran}\pi_+]$.

Proof. (i) Choose k admissible for the family over $[0,1]\times Z$; the restrictions of its complement F_k to $t = 0, 1$ are complements for A_0, A_1 and are isomorphic by Lemma 11.2.2. Two families with homotopic symbols are joined by an elliptic family: first the linear homotopy between A and $T_{\sigma_0(A)}$, which has constant symbol, then T_{a_t} for a homotopy a_t of symbols. (ii) and (iii) are clear from the definition, using P_k diagonally; a complement for (Id, k) is $\operatorname{Ran} P_k$. (iv) If A is invertible every k is admissible and $A(\operatorname{Ran} P_k) \cong \mathbb{C}^{(k+1)N}$ is a complement. Multiplication by $b(z)$ and $c(z)$ commutes with P_k, so $(bAc)(\mathrm{Id} - P_k)$ has null space $c^{-1}\operatorname{Ran} P_k = \operatorname{Ran} P_k$, range $bA(\mathrm{Id} - P_k)\mathcal{C}^\infty_+$ and complement $bW \cong W$. (v) By Lemma 11.4.1 every k is admissible, with complement $W_k \cong \mathbb{C}^{kM} \oplus \operatorname{Ran}(\pi_0+\pi_+) \oplus \operatorname{Ran}\pi_+$ (a smooth bundle by Lemma 11.2.1(ii)); since $\mathbb{C}^M = \operatorname{Ran}\pi_- \oplus \operatorname{Ran}(\pi_0 + \pi_+)$, $[\mathbb{C}^{(k+1)M}] - [W_k] = [\operatorname{Ran}\pi_-] - [\operatorname{Ran}\pi_+]$. □

The families index is the only obstruction to invertibility after perturbation; this is the criterion of Chapter 10 (there phrased as triviality of the bundle F_k), up to stabilization.

Proposition 11.5.6. *Let A be a smooth elliptic family of Toeplitz operators on $\mathbb{C}^N$ over compact Z. Then* $\operatorname{ind}(A) = 0$ *in $K^0(Z)$ if and only if for some L there is a smooth family $B\colon Z \to \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^{N+L})$ such that $A \oplus \mathrm{Id}_L + B$ is invertible for all $z \in Z$.*

Proof. If $A \oplus \mathrm{Id}_L + B$ is invertible, then $\operatorname{ind}(A) = \operatorname{ind}(A \oplus \mathrm{Id}_L + B) = 0$ by Proposition 11.5.5. Conversely, let k be admissible with complement W. If $\operatorname{ind}(A) = 0$ then $W \oplus \mathbb{C}^J \cong \mathbb{C}^{(k+1)N+J}$ for some J. Choose L with $(k+1)L \geq J$; then $\tilde{A} = A \oplus \mathrm{Id}_L$ has complement $\tilde{W} = W \oplus \mathbb{C}^{(k+1)L} \cong \mathbb{C}^{(k+1)(N+L)} = \operatorname{Ran} P_k$, so there is a smooth bundle isomorphism $\Theta\colon \operatorname{Ran} P_k \to \tilde{W}$. Put $B = \Theta P_k - \tilde{A}P_k$; this is a smooth family of finite rank operators with smooth kernels, so $B \in \Psi^{-\infty}_{\mathcal{T}}$, and $\tilde{A} + B = \tilde{A}(\mathrm{Id} - P_k) + \Theta P_k$ maps $\operatorname{Ran}(\mathrm{Id} - P_k)$ injectively onto $\tilde{A}(\mathrm{Id} - P_k)\mathcal{C}^\infty_+$ and $\operatorname{Ran} P_k$ onto the complement $\tilde{W}$. So it is a bijection of $\mathcal{C}^\infty_+$, hence invertible, and the inverses form a smooth family by Lemma 11.5.1. □

11.6 The index isomorphism and Bott periodicity, first form

For a pointed loop $a\colon X \times \mathbb{S} \to \mathrm{GL}(N,\mathbb{C})$, $T_a(x) = Sa(x,\cdot)S$ is a smooth elliptic family of Toeplitz operators with symbol a; set

$$\operatorname{ind}(a) = \operatorname{ind}(T_a) \in K^0(X).$$

By Proposition 11.5.5(i) any elliptic family with symbol a has the same index. By Proposition 11.5.5(i),(ii), $\operatorname{ind}(a)$ depends only on the stable pointed homotopy class of a, so by Lemma 11.2.5 it defines a map

$$\operatorname{ind}\colon K^{-2}(X) \longrightarrow K^0(X),$$

which is a homomorphism since $[a] + [b] = [\operatorname{diag}(a,b)]$ and $T_{\operatorname{diag}(a,b)} = T_a \oplus T_b$. For X a point, $\operatorname{ind}(a) = -\operatorname{wind}(a)$ by Corollary 11.4.2; in general the rank of $\operatorname{ind}(a)$ at x is $-\operatorname{wind} a(x,\cdot)$.

Proposition 11.6.1. *If a is a pointed loop over X with* $\operatorname{ind}(a) = 0$ *in $K^0(X)$, then a is stably pointed homotopic to* Id.

Proof. By Theorem 11.3.1 and homotopy invariance we may assume $a = \pi_- z^{-1} + \pi_0 + \pi_+ z$, and then $[\operatorname{Ran}\pi_-] - [\operatorname{Ran}\pi_+] = \operatorname{ind}(a) = 0$ by Proposition 11.5.5(v), i.e. $\operatorname{Ran}\pi_- \oplus H \cong \operatorname{Ran}\pi_+ \oplus H$ for some bundle H. By Lemma 11.2.1 we may take $H = \operatorname{Ran} q$ for a smooth family of idempotents q on some $\mathbb{C}^K$. The loop

$$\beta_q = \operatorname{diag}\left(qz^{-1} + (\operatorname{Id} - q),\ (\operatorname{Id} - q) + qz\right)$$

on $\mathbb{C}^{2K}$ is in normal form with $\operatorname{Ran}\pi_-^\beta = \operatorname{Ran} q \oplus 0 \cong 0 \oplus \operatorname{Ran} q = \operatorname{Ran}\pi_+^\beta$, so β_q is pointed homotopic to Id by Lemma 11.3.6. Hence $a \sim \operatorname{diag}(a, \beta_q)$, and $\operatorname{diag}(a, \beta_q)$ is in normal form with idempotents $\pi_- \oplus q \oplus 0$, $\pi_0 \oplus (\operatorname{Id} - q) \oplus (\operatorname{Id} - q)$, $\pi_+ \oplus 0 \oplus q$, whose ranges satisfy $\operatorname{Ran}\pi_- \oplus H \cong \operatorname{Ran}\pi_+ \oplus H$. By Lemma 11.3.6 again, $\operatorname{diag}(a, \beta_q)$ is pointed homotopic to Id. □

Theorem 11.6.2 (Bott periodicity, first form)**.** *For every compact manifold X the index map* $\operatorname{ind}\colon K^{-2}(X) \to K^0(X)$ *is an isomorphism of abelian groups, natural with respect to pull-back. Its inverse, the* Bott map β*, is given as follows: if $E_\pm \cong \operatorname{Ran} p_\pm$ with $p_\pm\colon X \to M(N_\pm, \mathbb{C})$ smooth families of projections, then*

$$\beta([E_+] - [E_-]) = \left[\left(p_+ z^{-1} + (\operatorname{Id} - p_+)\right) \oplus \left((\operatorname{Id} - p_-) + p_- z\right)\right]. \tag{11.6.1}$$

Proof. Injectivity is Proposition 11.6.1, and naturality follows from Proposition 11.5.5(iii). The loop in (11.6.1) is in normal form with $\pi_- = p_+ \oplus 0$ and $\pi_+ = 0 \oplus p_-$, so by Proposition 11.5.5(v) its index is $[\operatorname{Ran} p_+] - [\operatorname{Ran} p_-] = [E_+] - [E_-]$. As every element of $K^0(X)$ is of the form $[E_+] - [E_-]$ and every bundle is the range of a family of projections (Lemma 11.2.1), ind is surjective, and β is its inverse. □

Concretely, for the operator with symbol (11.6.1), namely $(p_+ L + \operatorname{Id} - p_+) \oplus (\operatorname{Id} - p_- + p_- U)$, the null space is the bundle E_+ of constant sections and the null space of the adjoint is E_-. The Toeplitz operator L, of index $1 = [\mathbb{C}] \in K^0(\mathrm{pt})$, is the *Bott element*: $\beta[\mathbb{C}] = [z^{-1}]$.

Remark 11.6.3. The signs matter here: the loop $\pi_- z^{-1} + \pi_0 + \pi_+ z$ with $\operatorname{Ran}\pi_\pm \cong E_\pm$ represents $-\beta([E_+] - [E_-])$, because $\operatorname{ind} U = -1$.

Corollary 11.6.4. *If $x_0 \in X$ is a base point and $\tilde{K}^{-2}(X)$ denotes the kernel of restriction $K^{-2}(X) \to K^{-2}(\{x_0\})$, then* ind *restricts to an isomorphism $\tilde{K}^{-2}(X) \cong \tilde{K}^0(X)$. Moreover $K^{-2}(\mathrm{pt}) \cong \mathbb{Z}$, i.e. pointed loops in $G^{-\infty}(Y; E)$ (equivalently in $\mathrm{GL}(N, \mathbb{C})$, stably) are classified up to pointed homotopy by their winding number, and every winding number occurs.*

Proof. Restriction and pull-back along $X \to \{x_0\}$ commute with ind, and ind is an isomorphism for X and for $\{x_0\}$, so it maps the kernel of one restriction bijectively onto the kernel of the other. The second statement is the case $X = \mathrm{pt}$, where $\operatorname{ind}(a) = -\operatorname{wind}(a)$. □

11.7 The stabilized Toeplitz group and the classifying sequence

We now pass from matrices to operators with values in the smoothing operators on Y, to obtain groups which contain all the $\mathrm{GL}(N, \mathbb{C})$ at once.

11.7.1 Toeplitz operators with values in smoothing operators

Let Y, E be as in Section 11.1. On $\Psi^0(\mathbb{S})$ we use the Fréchet topology of Chapters 4 and 5: an element is a conormal distribution at the diagonal, i.e. the sum of a smooth kernel and, near the diagonal, the inverse Fourier transform of a classical symbol, and the seminorms are those of uniform convergence of all derivatives of these two pieces. Composition is continuous, and so is the inclusion into the bounded operators on L^2.

Definition 11.7.1. $\Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S},Y;E)$ is the space of smooth sections $A\in\mathcal{C}^\infty(Y^2;\Psi_{\mathcal{T}}^0(\mathbb{S})\otimes \operatorname{Hom}(E)\otimes\Omega_R)$, i.e. smooth maps $(y,y')\mapsto A(y,y')$ with values in Toeplitz operators tensored with $\operatorname{Hom}(E_{y'},E_y)$ and a density in y'. It acts on $\mathcal{C}^\infty(\mathbb{S}\times Y;E)$, and on $\mathcal{C}^\infty_+(\mathbb{S}\times Y;E)=S\mathcal{C}^\infty(\mathbb{S}\times Y;E)$, by $(Au)(\cdot,y)=\int_Y A(y,y')u(\cdot,y')$, and is an algebra with $(AB)(y,y')=\int_Y A(y,y'')\circ B(y'',y')$. The symbol

$$\sigma_0\colon \Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S},Y;E)\longrightarrow \mathcal{C}^\infty(\mathbb{S};\Psi^{-\infty}(Y;E)),\quad \sigma_0(A)(\theta)(y,y')=\sigma_0(A(y,y'))(\theta),$$

is multiplicative. The elements with $A(y,y')\in\Psi_{\mathcal{T}}^{-\infty}(\mathbb{S})$ form the ideal $\Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)=S\Psi^{-\infty}(\mathbb{S}\times Y;E)S$, the Toeplitz smoothing operators on $\mathbb{S}\times Y$.

If $V\subset\mathcal{C}^\infty(Y;E)$ has orthonormal basis $\phi_1,\dots,\phi_N$, then $(A_{lm})\mapsto\sum_{l,m}A_{lm}\,\phi_l(y)\phi_m(y')^*$ embeds $\Psi_{\mathcal{T}}^0(\mathbb{S};\mathbb{C}^N)$ as a subalgebra (without unit); correspondingly an operator A on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ is identified with $(\mathrm{Id}-\Pi_V)+A$ on $\mathcal{C}^\infty_+(\mathbb{S}\times Y;E)$. Different choices of V give embeddings which are conjugate by unitaries joined to the identity, and enlarging V corresponds to stabilization $A\mapsto A\oplus\mathrm{Id}$.

Definition 11.7.2. The *stabilized normalized Toeplitz group* is

$$G_{\mathcal{T}}^0(\mathbb{S},Y;E)=\{\mathrm{Id}+A;\ A\in\Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S},Y;E),\ \sigma_0(A)(1)=0,\\ (\mathrm{Id}+A)^{-1}=\mathrm{Id}+B\text{ with }B\in\Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S},Y;E)\},$$

and $G_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$ is its subgroup of elements with $A\in\Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$. Similarly $G_{\mathcal{T}}^0(\mathbb{S};\mathbb{C}^N)$ denotes the group of invertible $A\in\Psi_{\mathcal{T}}^0(\mathbb{S};\mathbb{C}^N)$ with $\sigma_0(A)(1)=\mathrm{Id}$, and $G_{\mathcal{T}}^{-\infty}(\mathbb{S};\mathbb{C}^N)$ the subgroup with $A-\mathrm{Id}\in\Psi_{\mathcal{T}}^{-\infty}$.

Elements of $G_{\mathcal{T}}^0$ are elliptic: $\mathrm{Id}+\sigma_0(A)(\theta)$ has inverse $\mathrm{Id}+\sigma_0(B)(\theta)$, so $\mathrm{Id}+\sigma_0(A)\in\Omega G^{-\infty}(Y;E)$. The normalization $\sigma_0(A)(1)=0$ removes the evaluation map $G_{\mathcal{T}}^0\to G^{-\infty}(Y;E)$ at $1\in\mathbb{S}$, which would otherwise carry odd K-theory. A map $f\colon X\to G_{\mathcal{T}}^0(\mathbb{S},Y;E)$ is smooth if $f-\mathrm{Id}$ is smooth into the Fréchet space $\Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S},Y;E)$; by Lemma 11.7.3 below, f^{-1} is then smooth too, and products of smooth maps are smooth.

Lemma 11.7.3. *Let $A\in\mathcal{C}^\infty(Z;\Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S},Y;E))$ be a smooth family such that, for all $z\in Z$ and $\theta\in\mathbb{S}$, $\mathrm{Id}+\sigma_0(A(z))(\theta)\in G^{-\infty}(Y;E)$ and $\mathrm{Id}+A(z)$ is invertible on $L^2_+(\mathbb{S}\times Y;E)$. Then $(\mathrm{Id}+A)^{-1}=\mathrm{Id}+B$ with $B\in\mathcal{C}^\infty(Z;\Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S},Y;E))$.*

Proof. The proof of Lemma 11.5.1 applies verbatim. The parametrix construction goes through for operators with values in $\Psi^{-\infty}(Y;E)$; the symbol of a parametrix is $(\mathrm{Id}+\sigma_0(A))^{-1}-\mathrm{Id}\in\mathcal{C}^\infty(\mathbb{S};\Psi^{-\infty}(Y;E))$, inversion in $G^{-\infty}(Y;E)$ being smooth, and the lower order corrections are obtained by the same asymptotic summation. This gives $\mathrm{Id}+B_0$ with errors R,R' in $\Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$, and then

$$V=(\mathrm{Id}+A)^{-1}=(\mathrm{Id}+B_0)+R(\mathrm{Id}+B_0)+RVR'.$$

□

Lemma 11.7.4. *Let $A \in \mathcal{C}^\infty(X; \Psi_{\mathcal{T}}^{0,-\infty}(\mathbb{S}, Y; E))$ with $\mathrm{Id} + A(x) \in G_{\mathcal{T}}^0$ for all x, X compact. If $A' \in \mathcal{C}^\infty(X; \Psi_{\mathcal{T}}^{0,-\infty})$ satisfies $\sigma_0(A')(1) = 0$ and is sufficiently close to A uniformly on X (in a suitable seminorm), then $\mathrm{Id} + A'(x) \in G_{\mathcal{T}}^0$ for all x. The same holds for $G_{\mathcal{T}}^0(\mathbb{S}; \mathbb{C}^N)$ and for the groups $G_{\mathcal{T}}^{-\infty}$.*

Proof. The symbols $\mathrm{Id} + \sigma_0(A')(\theta)$ are uniformly close to the compact set of values of $\mathrm{Id} + \sigma_0(A)$ in the open set $G^{-\infty}(Y; E)$, so $\mathrm{Id} + A'$ is elliptic; the operators are close in L^2 operator norm to invertible ones, so are invertible on $L^2_+(\mathbb{S} \times Y; E)$. By Lemma 11.7.3 (or Lemma 11.5.1) the inverses lie in the algebra. □

11.7.2 Weak contractibility

Proposition 11.7.5. *Let $g\colon X \to G_{\mathcal{T}}^0(\mathbb{S}; \mathbb{C}^N)$ be smooth. Then for some $M \geq N$, $g \oplus \mathrm{Id}_{M-N}$ is smoothly homotopic in $G_{\mathcal{T}}^0(\mathbb{S}; \mathbb{C}^M)$ to a map into $G_{\mathcal{T}}^{-\infty}(\mathbb{S}; \mathbb{C}^M)$.*

Proof. The symbol $a = \sigma_0(g)$ is a pointed loop, and $\mathrm{ind}(a) = \mathrm{ind}(g) = 0$ since g is invertible (Proposition 11.5.5(iv)). By Proposition 11.6.1 there is a pointed homotopy a_t, $t \in [0, 1]$, in some $\mathrm{GL}(M, \mathbb{C})$ from $a \oplus \mathrm{Id}$ to Id. The family $A_t = g \oplus \mathrm{Id} + T_{a_t - a_0}$ is elliptic with symbol a_t, and $A_0 = g \oplus \mathrm{Id}$ is invertible. By (B7) there is a smooth family B_t of Toeplitz smoothing operators with $B_0 = 0$ such that $A_t + B_t$ is invertible; its inverse is smooth by Lemma 11.5.1, and its symbol a_t is pointed. So $A_t + B_t$ is a homotopy in $G_{\mathcal{T}}^0(\mathbb{S}; \mathbb{C}^M)$, and $A_1 + B_1$ has symbol Id, i.e. lies in $G_{\mathcal{T}}^{-\infty}$. □

Theorem 11.7.6. *The group $G_{\mathcal{T}}^0(\mathbb{S}, Y; E)$ is weakly contractible: every smooth map $f\colon X \to G_{\mathcal{T}}^0(\mathbb{S}, Y; E)$ from a compact manifold is smoothly homotopic in $G_{\mathcal{T}}^0(\mathbb{S}, Y; E)$ to the constant map* Id.

Proof. Step 1: reduction to matrices. Write $f = \mathrm{Id} + A$. Let Π_j be the projections of (B2). The expansion of the smooth map $(x, y, y') \mapsto A(x)(y, y')$ with values in the Fréchet space $\Psi_{\mathcal{T}}^0(\mathbb{S})$ (tensored with $\mathrm{Hom}(E)$) shows, exactly as for smoothing operators, that $\Pi_j A \Pi_j \to A$ uniformly on X in $\Psi_{\mathcal{T}}^{0,-\infty}$. Since $\sigma_0(\Pi_j A \Pi_j)(1) = \Pi_j \sigma_0(A)(1) \Pi_j = 0$, Lemma 11.7.4 shows that $\mathrm{Id} + (1-t)A + t\Pi_j A \Pi_j$, $t \in [0, 1]$, is a homotopy in $G_{\mathcal{T}}^0$ for j large. Its end point is $(\mathrm{Id} - \Pi_j) + g$, with $g\colon X \to G_{\mathcal{T}}^0(\mathbb{S}; \mathbb{C}^N)$, $N = \dim V_j$.

Step 2: reduction to smoothing operators. By Proposition 11.7.5, after enlarging V_j (which does not change the element of $G_{\mathcal{T}}^0(\mathbb{S}, Y; E)$), g is homotopic in $G_{\mathcal{T}}^0(\mathbb{S}; \mathbb{C}^M)$ to a map $g' = \mathrm{Id} + K$, $K \in \mathcal{C}^\infty(X; \Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}; \mathbb{C}^M))$.

Step 3: reduction to the zero Fourier mode. By (B6), $P_k K P_k \to K$ uniformly, so by Lemma 11.7.4 $\mathrm{Id} + (1-t)K + tP_k K P_k$ is a homotopy in $G_{\mathcal{T}}^{-\infty}(\mathbb{S}; \mathbb{C}^M)$ for k large. Its end point acts on $\mathrm{Ran}\, P_k$, spanned by $e^{ij\theta} e_l$, $0 \leq j \leq k$, $1 \leq l \leq M$, and is the identity on $\mathrm{Ran}(\mathrm{Id} - P_k)$. Stabilize once more, to $\mathbb{C}^{(k+1)M}$ with orthonormal basis $e_{l,j}$, $1 \leq l \leq M$, $0 \leq j \leq k$, where $e_{l,0} = e_l$. For $1 \leq j \leq k$ the pairs $(e^{ij\theta} e_{l,0}, e_{l,j})$, the second being a constant function, are orthonormal and span mutually orthogonal planes. Let $\mathcal{R}_\tau$ be the unitary rotating each of these planes by the angle τ and fixing their orthocomplement. Then $\mathcal{R}_\tau - \mathrm{Id}$ has finite rank and smooth kernel, so $\mathcal{R}_\tau \in G_{\mathcal{T}}^{-\infty}(\mathbb{S}; \mathbb{C}^{(k+1)M})$, and conjugation by $\mathcal{R}_\tau$, $\tau \in [0, \pi/2]$, is a homotopy in $G_{\mathcal{T}}^{-\infty}$ ending at an operator of the form

$$g'' = \mathrm{Id} - P_0 + P_0 h P_0, \qquad h\colon X \to \mathrm{GL}((k+1)M, \mathbb{C}),$$

since $\mathcal{R}_{\pi/2}$ maps $\mathrm{Ran}\, P_k \otimes e_{\cdot,0}$ onto the constants $\mathrm{Ran}\, P_0$.

Step 4: rotating away the matrix. For $h \in \mathrm{GL}(N', \mathbb{C})$ consider on $\mathcal{C}^\infty_+(\mathbb{S}; \mathbb{C}^{N'})^2$ the two curves of Toeplitz operators, $c = \cos\tau$, $s = \sin\tau$,

$$M_\tau = \begin{pmatrix} c(\mathrm{Id} - P_0) + P_0 h P_0 & s\, hU \\ -s\, h^{-1} L & c \end{pmatrix}, \quad M'_\tau = \begin{pmatrix} c\, h(\mathrm{Id} - P_0) + P_0 h P_0 & s\, hU \\ -s\, h^{-1} L & c\, h^{-1} \end{pmatrix},$$

$\tau \in [0,\pi/2]$; they agree at $\tau = \pi/2$, $M_0 = \operatorname{diag}(g'', \mathrm{Id})$ and $M_0' = \operatorname{diag}(h, h^{-1})$. Both are invertible: on $(u_k e^{ik\theta}, v_k e^{i(k-1)\theta})$, $k \geq 1$, they act by the matrices $\begin{pmatrix} c & sh \\ -sh^{-1} & c \end{pmatrix} = \operatorname{diag}(h,1) R_\tau' \operatorname{diag}(h^{-1},1)$ and $\operatorname{diag}(h,h^{-1}) R_\tau'$, $R_\tau' = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$, and they map $(u_0, 0)$ to $(hu_0, 0)$. Let $C_\tau = \sigma_0(M_\tau)(1)$, a constant invertible matrix (it commutes with S). Then $C_\tau^{-1} M_\tau$ followed by $(C_\tau')^{-1} M_\tau'$ (run from $\tau = \pi/2$ to 0, $C_\tau' = \sigma_0(M_\tau')(1)$) is a homotopy in $G^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^{2N'})$, smooth in $h = h(x)$, from $\operatorname{diag}(g'', \mathrm{Id})$ to $\operatorname{diag}(h,h^{-1})^{-1} \operatorname{diag}(h,h^{-1}) = \mathrm{Id}$. (This is the deformation of Chapter 10, normalized at $1 \in \mathbb{S}$.)

Composing the homotopies of Steps 1–4, all of which take place in $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ via the embeddings above, gives a homotopy from f to Id. □

Remark 11.7.7. Continuous maps from compact manifolds into $G^0_{\mathcal{T}}$ can be approximated by smooth ones, and nearby maps are homotopic (linear homotopy and Lemma 11.7.4), so all homotopy groups of $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ vanish. Whether $G^0_{\mathcal{T}}$ is actually contractible is discussed in the Notes.

11.7.3 The classifying sequence

To describe the quotient $G^0_{\mathcal{T}} / G^{-\infty}_{\mathcal{T}}$ we use full symbols. On the circle, with the normal fibration of the diagonal given by $\theta - \theta'$, an operator $A \in \Psi^0(\mathbb{S})$ has, modulo $\Psi^{-\infty}$, a kernel given near the diagonal by the inverse Fourier transform of a classical symbol $a(\theta, \xi)$; its Taylor series at $\xi = +\infty$ in $\rho = 1/\xi$ is the *full symbol* at $\mathbb{S}_+$, $\Phi(A) = \sum_{j \geq 0} \rho^j a_j(\theta)$, with $a_0 = \sigma_0(A)|_{\mathbb{S}_+}$. For Toeplitz operators the full symbol at $\mathbb{S}_-$ vanishes, so by Borel's lemma (asymptotic completeness, Chapter 4) there is a short exact sequence

$$\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}) \longrightarrow \Psi^0_{\mathcal{T}}(\mathbb{S}) \xrightarrow{\Phi} \mathcal{C}^\infty(\mathbb{S})[[\rho]],$$

and by the composition formula $\Phi(AB) = \Phi(A) * \Phi(B)$ for a bilinear product $*$ on formal series whose leading term is the pointwise product. Applying Φ to the kernels $A(y, y')$ gives, for $\Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E)$, the exact sequence

$$\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S} \times Y; E) \longrightarrow \Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E) \xrightarrow{\Phi} \mathcal{C}^\infty(\mathbb{S}; \Psi^{-\infty}(Y; E))[[\rho]]. \tag{11.7.1}$$

Lemma 11.7.8. *If $A \in \Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E)$ and $\mathrm{Id} + \sigma_0(A)$ is a loop in $G^{-\infty}(Y; E)$, then $\mathrm{Id} + A$ is Fredholm on $L^2_+(\mathbb{S} \times Y; E)$ and $\operatorname{ind}(\mathrm{Id} + A) = -\operatorname{wind}(\mathrm{Id} + \sigma_0(A))$. If this index vanishes there is $K \in \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S} \times Y; E)$ of finite rank with $\mathrm{Id} + A + K$ invertible; if moreover $\sigma_0(A)(1) = 0$ then $\mathrm{Id} + A + K \in G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$.*

Proof. A parametrix (Lemma 11.7.3) inverts $\mathrm{Id} + A$ modulo compact operators, so it is Fredholm, with null space and orthocomplement of the range (the null space of the adjoint, which is of the same type) consisting of smooth sections. For the index, the family $\mathrm{Id} + (1-t)A + t\Pi_j A \Pi_j$ consists of elliptic, hence Fredholm, operators for j large and is continuous in norm, so the index is constant; the winding number of the symbol is constant as well. At $t = 1$ the operator is $\mathrm{Id} - \Pi_j$ plus an elliptic Toeplitz operator on $\mathbb{C}^{\dim V_j}$, and Corollary 11.4.2 applies. If the index vanishes, let K map an orthonormal basis of the null space to one of the orthocomplement of the range and vanish on the orthocomplement of the null space; it has finite rank and smooth kernel, $\mathrm{Id} + A + K$ is bijective on L^2_+, and Lemma 11.7.3 applies. □

Theorem 11.7.9 (Classifying sequence)**.** *For Y, E as above there is a short exact sequence of groups*

$$G_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)\longrightarrow G_{\mathcal{T}}^{0}(\mathbb{S},Y;E)\xrightarrow{\Phi}\mathcal{Q}, \tag{11.7.2}$$

where $\mathcal{Q}$ is the set of formal series $\sum_{j\geq 0}\rho^j b_j$ with $b_j\in\mathcal{C}^\infty(\mathbb{S};\Psi^{-\infty}(Y;E))$ and $\mathrm{Id}+b_0\in\Omega_0G^{-\infty}(Y;E)$ (pointed loops of winding number zero), with the group structure induced by $*$. Moreover, for every compact manifold X:

(i) $[X;G_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)]\cong K^{-1}(X)$*;*

(ii) $G_{\mathcal{T}}^{0}(\mathbb{S},Y;E)$ *is weakly contractible;*

(iii) if X is connected, $[X;\mathcal{Q}]\cong[X;\Omega_0G^{-\infty}(Y;E)]\cong\tilde{K}^0(X)$, where $\mathcal{Q}$ has the product topology of the coefficients.

Thus $G_{\mathcal{T}}^{-\infty}$ classifies odd K-theory, $G_{\mathcal{T}}^{0}$ is weakly contractible, and $\mathcal{Q}$ classifies reduced even K-theory.

Proof. By (11.7.1) the kernel of Φ on $G_{\mathcal{T}}^{0}$ is $G_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$ (an element of $G_{\mathcal{T}}^{0}$ with A smoothing has inverse $\mathrm{Id}+B$ with $B=-A-AB\in\Psi_{\mathcal{T}}^{-\infty}$). If $\mathrm{Id}+A\in G_{\mathcal{T}}^{0}$, then $\mathrm{Id}+\sigma_0(A)$ is a pointed loop in $G^{-\infty}(Y;E)$, and its winding number is $-\operatorname{ind}(\mathrm{Id}+A)=0$ by Lemma 11.7.8; so $\Phi(G_{\mathcal{T}}^{0})\subset\mathcal{Q}$. Conversely, given $\sum\rho^j b_j\in\mathcal{Q}$, Borel's lemma gives A with $\Phi(A)=\sum\rho^j b_j$, and by Lemma 11.7.8 there is $K\in\Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$ with $\mathrm{Id}+A+K\in G_{\mathcal{T}}^{0}$; since $\Phi(K)=0$, Φ is surjective. This proves exactness, and Φ is a homomorphism for the product $*$.

(i) $G_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$ is the group of invertible perturbations of the identity by smoothing operators on the Hilbert space $L^2_+(\mathbb{S}\times Y;E)$. The orthonormal system $e^{il\theta}\phi_m$, $l\geq 0$, with ϕ_m the eigensections of (B2), gives finite rank approximations exactly as in (B2), so the proof of Lemma 11.2.5 identifies $[X;G_{\mathcal{T}}^{-\infty}]$ with the stable homotopy classes of maps $X\to\mathrm{GL}(N,\mathbb{C})$, which is $K^{-1}(X)$.

(ii) is Theorem 11.7.6.

(iii) As a space, $\mathcal{Q}=\Omega_0G^{-\infty}(Y;E)\times\prod_{j\geq 1}\mathcal{C}^\infty(\mathbb{S};\Psi^{-\infty}(Y;E))$, and the second factor is a vector space; scaling it by $t\in[0,1]$ shows that projection onto b_0 induces a bijection $[X;\mathcal{Q}]\to[X;\Omega_0G^{-\infty}]$. A smooth map $X\to\Omega G^{-\infty}$ is the same as a pointed map $X\times\mathbb{S}\to G^{-\infty}$; for connected X its winding number is constant, and it is zero exactly when the rank of the index class is zero. By Theorem 11.6.2 the maps into $\Omega_0G^{-\infty}$ thus correspond to $\{\xi\in K^0(X);\ \operatorname{rank}\xi=0\}=\tilde{K}^0(X)$. □

Remark 11.7.10. If (11.7.2) were known to be a fibration, the long exact sequence of homotopy groups and (ii) would give $\pi_k(\mathcal{Q})\cong\pi_{k-1}(G_{\mathcal{T}}^{-\infty})$. This is consistent with the results above: by (iii) and Corollary 11.8.9 below, $\pi_k(\mathcal{Q})=\pi_k(\Omega_0G^{-\infty})=\pi_{k+1}(G^{-\infty})$, which by Bott periodicity equals $\pi_{k-1}(G^{-\infty})=\pi_{k-1}(G_{\mathcal{T}}^{-\infty})$.

11.8 Compactly supported K-theory and Bott periodicity

11.8.1 Definitions

Let M be a manifold, not necessarily compact. We write $K\Subset M$ for a compact subset.

Definition 11.8.1. (i) $K_{\mathrm{c}}^{-1}(M)$ is the set of smooth maps $f\colon M\to G^{-\infty}(Y;E)$ with $f=\mathrm{Id}$ on $M\setminus K$ for some $K\Subset M$, modulo homotopies f_t with $f_t=\mathrm{Id}$ on $M\setminus K$ for one $K\Subset M$ and all t.

(ii) $K_{\mathrm{c}}^{-2}(M)$ is defined in the same way, using maps $f\colon M\times\mathbb{S}\to G^{-\infty}(Y;E)$ with $f=\mathrm{Id}$ on $(M\setminus K)\times\mathbb{S}\cup M\times\{1\}$.

(iii) $K^0_{\mathrm{c}}(M)$ is the set of equivalence classes of triples (E_+, E_-, a), where $E_\pm$ are vector bundles over M and $a\colon E_+ \to E_-$ is a smooth bundle isomorphism over $M \setminus K$ for some $K \Subset M$. The equivalence relation is generated by:

(a) $(E_+, E_-, a) \sim (E'_+, E'_-, \phi_- a \phi_+^{-1})$ for bundle isomorphisms $\phi_\pm\colon E_\pm \to E'_\pm$ over M;

(b) $(E_+, E_-, a) \sim (E_+, E_-, a')$ if $a = a'$ outside a compact set;

(c) $(E_+, E_-, a_0) \sim (E_+, E_-, a_1)$ if a_t, $t \in [0,1]$, is a smooth family of isomorphisms over $M \setminus K$ for one $K \Subset M$;

(d) $(E_+, E_-, a) \sim (E_+ \oplus H, E_- \oplus H, a \oplus \mathrm{Id}_H)$ for any bundle H.

The sum is $[(E_+, E_-, a)] + [(F_+, F_-, b)] = [(E_+ \oplus F_+, E_- \oplus F_-, a \oplus b)]$.

The groups K^{-1}_{c}, K^{-2}_{c} are abelian groups by pointwise multiplication (as in Lemma 11.2.5). $K^0_{\mathrm{c}}(M)$ is an abelian monoid with zero $[(0,0,\cdot)]$; a triple (F, F, c) with c an isomorphism over all of M represents 0 (use (a) with $\phi_- = c^{-1}$ and then (d)). It is a group: $(E_+ \oplus E_-, E_- \oplus E_+, a \oplus a^{-1})$ is equivalent by (a), composing with the exchange of the summands of $E_- \oplus E_+$, to (F, F, b) with $F = E_+ \oplus E_-$ and $b = \begin{pmatrix} 0 & a^{-1} \\ a & 0 \end{pmatrix}$, and $b_\tau = \begin{pmatrix} \cos\tau & \sin\tau\, a^{-1} \\ \sin\tau\, a & -\cos\tau \end{pmatrix}$ satisfies $b_\tau^2 = \mathrm{Id}$ and joins b to the globally defined isomorphism $\mathrm{diag}(\mathrm{Id}, -\mathrm{Id})$.

For compact M the conditions at infinity are void, and $K^{-1}_{\mathrm{c}}(M) = K^{-1}(M)$, $K^{-2}_{\mathrm{c}}(M) = K^{-2}(M)$; also $K^0_{\mathrm{c}}(M) = K^0(M)$, since (E_+, E_-, a) may be taken with a defined on the empty set and then (a), (d) are exactly the relation of Definition 11.2.3.

11.8.2 Flattening and suspension

Lemma 11.8.2 (Flattening)**.** *Let $W_1 \subset M_1$ and $W_2 \subset M_2$ be points (or empty) in compact manifolds and $W = W_1 \times M_2 \cup M_1 \times W_2$. Every smooth $f\colon M_1 \times M_2 \to G^{-\infty}$ with $f = \mathrm{Id}$ on W is homotopic, through such maps, to one which is* Id *in a neighbourhood of W; and two maps which are* Id *near W and homotopic through maps equal to* Id *on W are homotopic through maps equal to* Id *near W.*

Proof. In a coordinate ball centred at $W_1 = \{w\}$, let $\psi_t(y) = ((1-t) + t\chi(|y|))y$, where $\chi \in \mathcal{C}^\infty([0,\infty))$ vanishes on $[0,\delta]$ and equals 1 on $[2\delta, \infty)$, and extend ψ_t by the identity. Then ψ_t fixes w, $\psi_0 = \mathrm{Id}$ and ψ_1 maps a neighbourhood of w to w. With the analogous map on M_2, $\Psi_t = \psi_t \times \psi_t$ maps W into W for all t, and $f \circ \Psi_t$ is the required homotopy. For the second statement apply Ψ_1 to the homotopy; $h \circ \Psi_t$ joins a map h which is Id near W to $h \circ \Psi_1$ through maps equal to Id near W, provided δ is small (by compactness Ψ_t maps a small neighbourhood of W into the set where $h = \mathrm{Id}$). □

Applied with $M_1 = \mathrm{pt}$ or with the stereographic identification $\mathbb{S}^k \setminus \{e\} \cong \mathbb{R}^k$ this shows that the homotopy groups $\pi_k(G^{-\infty}) = [\mathbb{S}^k, e; G^{-\infty}, \mathrm{Id}]$ (pointed smooth homotopy classes) satisfy $\pi_k(G^{-\infty}) \cong K^{-1}_{\mathrm{c}}(\mathbb{R}^k)$. Similarly, identifying $\mathbb{S} \setminus \{1\}$ with $\mathbb{R}$:

Lemma 11.8.3. *For every manifold M, $K^{-2}_{\mathrm{c}}(M) \cong K^{-1}_{\mathrm{c}}(\mathbb{R} \times M)$.*

Proof. By (the proof of) Lemma 11.8.2 in the circle factor, representatives and homotopies in $K^{-2}_{\mathrm{c}}(M)$ may be taken equal to Id near $M \times \{1\}$; restricted to $M \times (\mathbb{S} \setminus \{1\}) \cong M \times \mathbb{R}$ they are exactly the representatives and homotopies of $K^{-1}_{\mathrm{c}}(\mathbb{R} \times M)$, and conversely such maps extend by Id. □

11.8.3 Clutching

For $n \geq 1$ let $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ be the unit sphere with base point e, and write $K^{-1}(\mathbb{S}^{n-1} \times X, X)$ for the set of homotopy classes of maps $g\colon \mathbb{S}^{n-1} \times X \to G^{-\infty}(Y; E)$ with $g(e, x) = \mathrm{Id}$, through such maps. Thus $K^{-1}(\mathbb{S}^0 \times X, X) = K^{-1}(X)$ and $K^{-1}(\mathbb{S} \times X, X) = K^{-2}(X)$.

Lemma 11.8.4 (Clutching)**.** *For compact X and $n \geq 1$ there is a natural isomorphism*

$$K^0_{\mathrm{c}}(\mathbb{R}^n \times X) \cong K^{-1}(\mathbb{S}^{n-1} \times X, X),$$

whose inverse sends the class of a pointed g with values in $\mathrm{GL}(N, \mathbb{C})$ to the class of $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$, $\tilde{g}(r\omega, x) = g(\omega, x)$ for $r \geq 1$, $\omega \in \mathbb{S}^{n-1}$.

Proof. Let $\varpi\colon \mathbb{R}^n \times X \to X$. By Lemma 11.2.2(ii) every bundle on $\mathbb{R}^n \times X$ is isomorphic to a pull-back, so by (a) every class is represented by $(\varpi^* F_+, \varpi^* F_-, a)$ with a an isomorphism on $\{|v| \geq R\}$. Choose a bundle G and an isomorphism $\tau\colon F_- \oplus G \to \mathbb{C}^N$, and set

$$g(\omega, x) = \tau\,(a_R \oplus \mathrm{Id}_G)(\omega, x)\,(a_R \oplus \mathrm{Id}_G)(e, x)^{-1}\tau^{-1}, \qquad a_R(\omega, x) = a(R\omega, x),$$

where $a_R(\omega, x)\colon F_{+,x} \to F_{-,x}$. This is a pointed map into $\mathrm{GL}(N, \mathbb{C})$; let $\kappa[(E_+, E_-, a)]$ be its class.

Well-definedness. Changing R, or a by a homotopy (c), changes g by a pointed homotopy, and (b) is covered by changing R. Under (d), or a change of (G, τ), g is changed by stabilization and conjugation by a map $X \to \mathrm{GL}$ independent of ω, which does not change the class by Lemma 11.1.3. Under (a) with isomorphisms $\phi_\pm$ of the pull-back bundles, $\phi_\pm(sR\omega, x)$, $s \in [0, 1]$, joins $\phi_\pm|_{\{|v|=R\}}$ to $\phi_\pm(0, x)$, which is independent of ω; along this homotopy g is changed by a pointed homotopy, and at $s = 0$ it is replaced by its conjugate by $\tau(\phi_-(0, \cdot) \oplus \mathrm{Id})\tau^{-1}$ and again the class does not change. κ is additive, since direct sums of triples give block sums of the g's.

Surjectivity. For pointed g with values in $\mathrm{GL}(N, \mathbb{C})$ (Lemma 11.2.5) the triple $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$ of the statement is mapped to $[g]$ (with $G = 0$, $\tau = \mathrm{Id}$, $R = 1$).

Injectivity. Using (d) with $\varpi^* G$ and then (a) with τ we may assume $F_- = \mathbb{C}^N$; since $a_R(e, x)\colon F_+ \to \mathbb{C}^N$ is an isomorphism, (a) allows us to assume $F_+ = \mathbb{C}^N$ and $a_R(e, x) = \mathrm{Id}$. The homotopy $a_s(r\omega, x) = a\big(((1 - s)r + sR)\omega, x\big)$, $r \geq R$, shows that the triple is equivalent to $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$ with $\tilde{g}(r\omega, x) = g(\omega, x)$ for $r \geq R$. If $[g] = 0$, then by Lemma 11.2.5 $g \oplus \mathrm{Id}$ is joined to Id by a pointed homotopy g_t in some $\mathrm{GL}(N', \mathbb{C})$, and (d), (c) give $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g}) \sim (\mathbb{C}^{N'}, \mathbb{C}^{N'}, \tilde{g}_1) = (\mathbb{C}^{N'}, \mathbb{C}^{N'}, \mathrm{Id})$, which represents 0. □

11.8.4 Bott periodicity

Theorem 11.8.5 (Bott periodicity)**.** *For every compact manifold X there is a natural isomorphism*

$$K^0(X) \longrightarrow K^0_{\mathrm{c}}(\mathbb{R}^2 \times X).$$

Identifying $\mathbb{R}^2 = \mathbb{C}$ with variable w, it maps $[E]$ to the class of $(\varpi^* E, \varpi^* E, \bar{w}/|w|)$; that is, it is multiplication by the Bott class $b = [(\mathbb{C}, \mathbb{C}, \bar{w}/|w|)] \in K^0_{\mathrm{c}}(\mathbb{R}^2)$.

Proof. The composite of the inverse of the index isomorphism (Theorem 11.6.2) and of the inverse of the clutching isomorphism (Lemma 11.8.4, $n = 2$), $K^0(X) \to K^{-2}(X) = K^{-1}(\mathbb{S} \times X, X) \to K^0_{\mathrm{c}}(\mathbb{R}^2 \times X)$, is an isomorphism. By (11.6.1), if $E = \mathrm{Ran}\, p$, $p\colon X \to M(N, \mathbb{C})$, then $\beta[E]$ is represented by $pz^{-1} + (\mathrm{Id} - p)$, and the clutching construction gives $(\mathbb{C}^N, \mathbb{C}^N, p\,\bar{w}/|w| + \mathrm{Id} - p)$ for $|w| \geq 1$, since $z^{-1} = \bar{w}/|w|$ on the circle $\omega = w/|w|$. Writing $\mathbb{C}^N = E \oplus \mathrm{Ran}(\mathrm{Id} - p)$ and using (d) this is equivalent to $(\varpi^* E, \varpi^* E, \bar{w}/|w|)$. □

Remark 11.8.6. Periodicity can also be deduced from the chain $K_c^0(\mathbb{R}^2\times X)\cong K_c^{-1}(\mathbb{R}\times X)\cong K_c^{-2}(X)\cong K_c^0(X)$, which uses the clutching isomorphism for the non-compact manifold $\mathbb{R}\times X$ and the index isomorphism for non-compact X (Exercise 11.1). For compact X the clutching lemma with $n=1$ gives $K_c^0(\mathbb{R}\times X)\cong K^{-1}(X)$, and Lemma 11.8.3 gives $K^{-2}(X)\cong K_c^{-1}(\mathbb{R}\times X)$, so the chain is $K_c^0(\mathbb{R}^2\times X)\cong K^{-2}(X)\cong K_c^{-1}(\mathbb{R}\times X)$ and $K_c^0(\mathbb{R}\times X)\cong K^{-1}(X)$.

11.8.5 The homotopy groups of $G^{-\infty}$

Spheres need one more form of the clutching construction.

Lemma 11.8.7. *For $m\geq 1$, $\tilde K^0(\mathbb{S}^m)\cong\pi_{m-1}(G^{-\infty}(Y;E))$.*

Proof. Let $U_\pm=\{\pm x_{m+1}>-\epsilon\}\subset\mathbb{S}^m\subset\mathbb{R}^{m+1}$, so that $U_\pm$ are diffeomorphic to $\mathbb{R}^m$ and $U_+\cap U_-$ retracts, by a map r, onto the equator $\mathbb{S}^{m-1}$, which carries the base point e. For pointed $g\colon\mathbb{S}^{m-1}\to\mathrm{GL}(N,\mathbb{C})$ let E_g be the bundle obtained from $U_+\times\mathbb{C}^N$ and $U_-\times\mathbb{C}^N$ by identifying $(y,v)_-$ with $(y,g(r(y))v)_+$ over $U_+\cap U_-$, and set $\Psi(g)=[E_g]-[\mathbb{C}^N]$, an element of $\tilde K^0(\mathbb{S}^m)$. A pointed homotopy g_t defines a bundle over $[0,1]\times\mathbb{S}^m$, so $E_{g_0}\cong E_{g_1}$ by Lemma 11.2.2; $E_{g\oplus\mathrm{Id}}=E_g\oplus\mathbb{C}$ and $E_{\mathrm{diag}(g,h)}=E_g\oplus E_h$. By Lemma 11.2.5 and Lemma 11.1.2, Ψ induces a homomorphism $\pi_{m-1}(G^{-\infty})\to\tilde K^0(\mathbb{S}^m)$.

It is surjective: an element of $\tilde K^0(\mathbb{S}^m)$ has the form $[E]-[\mathbb{C}^N]$ with $\operatorname{rank}E=N$ ($\mathbb{S}^m$ is connected). E is trivial over $U_\pm$ (Lemma 11.2.2(ii)); for trivializations $\tau_\pm$ the transition map $\tilde g=\tau_+\tau_-^{-1}$ on $U_+\cap U_-$ is homotopic to $\tilde g\circ r$ (deform along the retraction), so $E\cong E_g$ with $g=\tilde g|_{\mathbb{S}^{m-1}}$; replacing τ_- by $g(e)\tau_-$ makes g pointed.

It is injective: if $[E_g]=[\mathbb{C}^N]$, then $E_g\oplus\mathbb{C}^L\cong\mathbb{C}^{N+L}$, i.e. $E_{g'}$, $g'=g\oplus\mathrm{Id}_L$, has a global trivialization. In the local trivializations over $U_\pm$ it is given by maps $\tau_\pm\colon U_\pm\to\mathrm{GL}(N+L,\mathbb{C})$ with $\tau_-=\tau_+g'$, i.e. $g'=\tau_+^{-1}\tau_-$, on the equator. As $U_\pm$ are star-shaped in suitable coordinates, $\tau_\pm$ are homotopic to constants, so g' is homotopic to a constant, and then (as GL is connected) to Id; normalizing by the value at e (Lemma 11.1.1) makes the homotopy pointed. So $[g]=0$. □

Lemma 11.8.8. *For $m\geq 0$, with base point $e\in\mathbb{S}^m$, $\tilde K^{-2}(\mathbb{S}^m)\cong\pi_{m+1}(G^{-\infty}(Y;E))$.*

Proof. An element of $\tilde K^{-2}(\mathbb{S}^m)$ is represented by $f\colon\mathbb{S}^m\times\mathbb{S}\to G^{-\infty}$ with $f=\mathrm{Id}$ on $\mathbb{S}^m\times\{1\}$ and such that the loop $f(e,\cdot)$ is null-homotopic through pointed loops h_s, $h_0=f(e,\cdot)$, $h_1=\mathrm{Id}$. Choose $\chi\in\mathcal{C}^\infty(\mathbb{S}^m)$ with values in $[0,1]$ and $\chi(e)=1$. Then $F_s(x,\theta)=f(x,\theta)f(e,\theta)^{-1}h_{s\chi(x)}(\theta)$ is a homotopy in K^{-2} from f to a map equal to Id on $W=\mathbb{S}^m\times\{1\}\cup\{e\}\times\mathbb{S}$. Similarly a homotopy H_t between two maps equal to Id on W is replaced by $H_t(x,\theta)H_t(e,\theta)^{-1}$, which is Id on W and has the same ends. Hence $\tilde K^{-2}(\mathbb{S}^m)$ is the set of homotopy classes, relative to W, of maps equal to Id on W. By Lemma 11.8.2 these are the classes of maps which are Id near W, and since $(\mathbb{S}^m\setminus\{e\})\times(\mathbb{S}\setminus\{1\})\cong\mathbb{R}^{m+1}$ this is $K_c^{-1}(\mathbb{R}^{m+1})\cong\pi_{m+1}(G^{-\infty})$. (For $m=0$, $\mathbb{S}^0=\{e,-e\}$ and the statement is $K^{-2}(\{-e\})=\pi_1(G^{-\infty})$.) □

Corollary 11.8.9. *For every compact Y of positive dimension and every bundle E,*

$$\pi_k(G^{-\infty}(Y;E))\cong\begin{cases}\mathbb{Z} & k\text{ odd},\\ 0 & k\text{ even}.\end{cases}$$

The isomorphism $\pi_1(G^{-\infty})\cong\mathbb{Z}$ is given by the winding number. Consequently also $K^{-1}(\mathbb{S}^k)=[\mathbb{S}^k;G^{-\infty}]=\pi_k(G^{-\infty})$, and $\tilde K^0(\mathbb{S}^k)$ is $\mathbb{Z}$ for k even and 0 for k odd, $k\geq 1$.

Proof. $\pi_0(G^{-\infty}) = 0$ since $G^{-\infty}$ is connected. For $m \geq 0$, $\pi_{m+1}(G^{-\infty}) \cong \tilde{K}^{-2}(\mathbb{S}^m) \cong \tilde{K}^0(\mathbb{S}^m)$ by Lemma 11.8.8 and Corollary 11.6.4. For $m = 0$, $\tilde{K}^0(\mathbb{S}^0) = K^0(\mathrm{pt}) = \mathbb{Z}$, so $\pi_1(G^{-\infty}) = \mathbb{Z}$, the isomorphism being minus the index, i.e. the winding number. For $m \geq 1$, $\tilde{K}^0(\mathbb{S}^m) \cong \pi_{m-1}(G^{-\infty})$ by Lemma 11.8.7, so $\pi_{m+1}(G^{-\infty}) \cong \pi_{m-1}(G^{-\infty})$ and the result follows by induction. Free and pointed homotopy classes of maps $\mathbb{S}^k \to G^{-\infty}$ agree, since $f \mapsto f(e)^{-1}f$ is a homotopy inverse to the inclusion of pointed maps ($G^{-\infty}$ is a connected group). □

This justifies calling $G^{-\infty}$ a classifying space for odd K-theory: its homotopy groups are those of the stable unitary group (Bott's original theorem).

11.9 The analytic index map

Let Z be a compact manifold and $\varpi\colon T^*Z \to Z$. An elliptic operator $P \in \Psi^0(Z; E_+, E_-)$ is Fredholm, and its index

$$\operatorname{ind} P = \dim \operatorname{Nul} P - \dim \left(\mathcal{C}^\infty(Z; E_-)/P\mathcal{C}^\infty(Z; E_+)\right)$$

is a homotopy invariant (Chapter 10), hence depends only on $E_\pm$ and the homotopy class of $\sigma_0(P) \in \mathcal{C}^\infty(S^*Z; \mathrm{Iso}(E_+, E_-))$. The principal symbol, extended homogeneously of degree 0 to $T^*Z \setminus 0$, defines a triple $(\varpi^*E_+, \varpi^*E_-, \sigma_0(P))$ and so a class in $K^0_{\mathrm{c}}(T^*Z)$.

Proposition 11.9.1. *There is a unique homomorphism* $\operatorname{ind}_a\colon K^0_{\mathrm{c}}(T^*Z) \to \mathbb{Z}$*, the* analytic index*, with* $\operatorname{ind} P = \operatorname{ind}_a[(\varpi^*E_+, \varpi^*E_-, \sigma_0(P))]$ *for every elliptic* $P \in \Psi^0(Z; E_+, E_-)$*.*

Proof. Every class is of this form. By Lemma 11.2.2(ii) and (a) a class is represented by $(\varpi^*E_+, \varpi^*E_-, a)$ with $E_\pm$ bundles over Z and a an isomorphism for $|\zeta| \geq R$ (with respect to some metric). The homotopy (c) $a_s(\zeta) = a(((1-s)|\zeta| + sR)\zeta/|\zeta|)$, $|\zeta| \geq R$, ends at a map homogeneous of degree 0, i.e. at the symbol σ of some elliptic $P \in \Psi^0(Z; E_+, E_-)$. Define ind_a of the class to be $\operatorname{ind} P$.

This does not depend on the choices: by homotopy invariance $\operatorname{ind} P$ depends only on the homotopy class of σ, which takes care of R, of (b) and of (c). Under (d), P is replaced by $P \oplus \mathrm{Id}_H$, with the same index. Under (a), bundle isomorphisms $\phi_\pm$ of $\varpi^*E_\pm$ are homotopic, through $\phi_\pm(t\zeta)$, to the pull-backs of their restrictions $\phi^0_\pm$ to the zero section; so σ is replaced by a symbol homotopic to $\phi^0_-\sigma(\phi^0_+)^{-1}$, which is the symbol of $\phi^0_- P(\phi^0_+)^{-1}$, an operator with the same index since $\phi^0_\pm$ are invertible. Additivity follows from $\operatorname{ind}(P \oplus P') = \operatorname{ind} P + \operatorname{ind} P'$, and uniqueness from the fact that every class arises from an elliptic operator. □

The problem posed by Gelfand around 1960 was to compute this map topologically. The answer of Atiyah and Singer is to define a second, *topological*, index map by embedding Z into a Euclidean space (or a sphere), constructing on the larger space an operator with the same index and with symbol derived from that of P, and using the fact, essentially Bott periodicity, that for suitable operators on Euclidean space the index map is an isomorphism; the two maps are then shown to coincide. This is taken up in Chapter 13.

11.10 Exercises

Exercise 11.1. Show that the index isomorphism of Theorem 11.6.2 carries over to non-compact manifolds: for every manifold M there is an isomorphism $K^{-2}_{\mathrm{c}}(M) \cong K^0_{\mathrm{c}}(M)$. (Hint: represent elements of $K^{-2}_{\mathrm{c}}(M)$ by loops equal to Id outside $K \times \mathbb{S}$, choose a compact manifold with boundary $M' \Subset M$ containing a neighbourhood of K, and apply the families

index over M', observing that the index class comes with a canonical trivialization over $M \setminus K$.)

Exercise 11.2. Show directly that the Toeplitz operator T_{z^k} has index $-k$. For a smooth family of projections p on $\mathbb{C}^N$ over X show that $pL + (\mathrm{Id} - p)$ is surjective with null space the constant sections of $\operatorname{Ran} p$, and compute $\operatorname{ind}(pL + \mathrm{Id} - p) = [\operatorname{Ran} p]$ directly from Definition 11.5.4 with $k = 0$.

Exercise 11.3. Show that for pointed loops a, b over X the families index satisfies $\operatorname{ind}(T_a T_b) = \operatorname{ind}(T_{ab}) = \operatorname{ind}(a) + \operatorname{ind}(b)$. (Use Lemma 11.1.2 and the fact that $T_a T_b - T_{ab}$ has vanishing symbol.)

Exercise 11.4. Let $a(\theta) = \operatorname{diag}(e^{2i\theta}, e^{-i\theta})$. Carry out the steps of the proof of Theorem 11.3.1 for a and find the resulting normal form; check that its index is -1 in accordance with Corollary 11.4.2.

Exercise 11.5. Show that $K^0_{\mathrm{c}}(\mathbb{R}) = 0$ and $K^0_{\mathrm{c}}(\mathbb{R}^2) \cong \mathbb{Z}$, generated by the Bott class $[(\mathbb{C}, \mathbb{C}, \bar{w}/|w|)]$.

Exercise 11.6. Let $H \to \mathbb{S}^2 = \mathbb{C} \cup \{\infty\}$ be the line bundle obtained by clutching the trivial bundles over the two hemispheres with the map $z \mapsto z$ on the equator (this is the tautological bundle over $\mathbb{CP}^1$, up to conjugation). Show that $[H] - [\mathbb{C}]$ generates $\tilde{K}^0(\mathbb{S}^2) \cong \mathbb{Z}$, and prove that $(H \otimes H) \oplus \mathbb{C} \cong H \oplus H$ by comparing clutching functions and using Lemma 11.1.2.

Exercise 11.7. Show that if $A \colon X \to \Psi^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N)$ is a smooth family of elliptic Toeplitz operators whose null spaces have constant dimension, then $\operatorname{Nul} A$ and $\operatorname{Nul} A^*$ are smooth vector bundles and $\operatorname{ind}(A) = [\operatorname{Nul} A] - [\operatorname{Nul} A^*]$.

Notes

The linearization of loops (Section 11.3) is the heart of Atiyah's proof of Bott periodicity via clutching functions [17]; the use of the families index of Toeplitz operators to invert the Bott map follows the source lectures and is closely related to the treatment of Bott periodicity through the Grassmannian of a polarized Hilbert space in [21]. Compared to the source we have replaced the residue argument for linear loops by the Riesz spectral projection (Lemma 11.3.4), proved that a normal form loop is trivial when the ranges of $\pi_\pm$ are isomorphic in a single lemma (Lemma 11.3.6), defined the families index through arbitrary smooth complements of the range, corrected the sign of the Bott map (which, with $\operatorname{ind} U = -1$, sends $[E_+] - [E_-]$ to the class with $p_+ z^{-1}$), corrected the second curve in the deformation of $\mathrm{Id} - P_0 + P_0 h P_0$ and normalized it at $1 \in \mathbb{S}$, and derived the homotopy groups of $G^{-\infty}$ from the index isomorphism for compact spheres together with the clutching construction, instead of from the (unproved) non-compact index isomorphism. The source also suggests, as an alternative to Toeplitz operators, the isotropic calculus on $\mathbb{R}$ of Chapter 6: there the 'cosphere bundle' of $\mathbb{R}$ is the circle at infinity of $\mathbb{R}^2$, the annihilation operator $x + \partial_x$ of the harmonic oscillator is Fredholm of index 1 (its null space is spanned by $e^{-x^2/2}$, that of its adjoint is trivial), with symbol $x + i\xi$ of degree one on the circle at infinity, and it plays the role of L; the arguments of this chapter carry over with P_k replaced by the spectral projections of the harmonic oscillator.

The source conjectures that $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ is actually contractible. Since it is an open subset of an affine Fréchet space (Lemma 11.7.4), it has the homotopy type of a CW complex by a theorem of R. Palais (Homotopy theory of infinite dimensional manifolds, Topology 5 (1966)), and a weakly contractible CW complex is contractible by Whitehead's theorem; we have not checked the details of this argument in the present setting.

Chapter 12

Fibrations and product-type operators

The families version of the Atiyah–Singer index theorem concerns elliptic operators on the fibres of a fibration $\phi : M \longrightarrow B$; rather than a family $A_b \in \Psi^m(Z; E, F)$ simply parametrized by $b \in B$, the manifold on which A_b acts is allowed to be 'twisted' by diffeomorphisms of the fibre Z as b varies. This chapter sets up the analytic machinery needed for that theorem and for the index formulae of Chapter 13 and Chapter 14.

In the first three sections we define fibrations, the space $\Psi^m(M/B; E, F)$ of pseudodifferential operators acting on the fibres, and the index bundle $\mathrm{ind}_a(A) \in K^0(B)$ of an elliptic family. The rest of the chapter constructs a larger algebra, the *product-type* pseudodifferential operators $\Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$, which contains both the ordinary pseudodifferential operators on the total space M and the fibrewise families. Their kernels are conormal, in a 'product' sense, to the nested pair of submanifolds $\mathrm{Diag}_M \subset M^2_\phi \subset M^2$, where M^2_ϕ is the fibre product. Following the pattern of Chapter 4, these kernels are defined as inverse Fourier transforms of symbols, now smooth (up to weights) on a compactification of a vector space W relative to a subspace V; this *relative compactification* turns out to be a product of two radial compactifications. A product-type operator has two symbols: the usual principal symbol, defined on a blown-up cosphere bundle, and a *base symbol*, which is a family of pseudodifferential operators on the fibres parametrized by $T^*B \setminus 0$. If both are invertible the operator is *fully elliptic*, has a parametrix modulo smoothing operators, and is Fredholm.

We use freely the radial compactification of a vector space (Chapter 1), conormal distributions and their symbols (Chapter 4), pseudodifferential operators on compact manifolds and their Fredholm properties (Chapter 5), and the elementary K-theory of Chapter 10 and Chapter 11. All manifolds are without boundary unless otherwise stated, and all vector bundles are complex.

12.1 Fibrations

Definition 12.1.1. A *fibration* (of compact manifolds) is a smooth surjective map $\phi : M \longrightarrow B$ between compact manifolds which is a *submersion*, i.e. $\phi_* : T_mM \longrightarrow T_{\phi(m)}B$ is surjective for every $m \in M$. The *fibre* over $b \in B$ is $Z_b = \phi^{-1}(b)$. We indicate a fibration by the diagram $Z — M \xrightarrow{\phi} B$, where Z is the typical fibre (see below).

If M is compact, B is connected and ϕ is a submersion, then ϕ is automatically surjective (if $M \neq \emptyset$): its range is open, since ϕ is a submersion, and closed, since M is compact.

Theorem 12.1.2 (Local triviality)**.** *Let $\phi : M \longrightarrow B$ be a fibration.*

(1) *Each fibre Z_b is a compact embedded submanifold of M of dimension* $\dim M - \dim B$*; if B is connected all fibres are diffeomorphic to a fixed compact manifold Z.*

(2) *Each $b \in B$ has an open neighbourhood U_b with a diffeomorphism $f_b : \phi^{-1}(U_b) \longrightarrow Z \times U_b$ such that $\pi_U \circ f_b = \phi$, where $\pi_U : Z \times U_b \longrightarrow U_b$ is the projection.*

(3) *For two such trivializations, $f_{b'} \circ f_b^{-1}$ is a diffeomorphism of $Z \times (U_b \cap U_{b'})$ of the form $(z,c) \longmapsto (f_{b'b}(c)(z), c)$ where $f_{b'b} : U_b \cap U_{b'} \longrightarrow \mathrm{Dfm}(Z)$ is a smooth family of diffeomorphisms of Z (meaning that $(z,c) \mapsto f_{b'b}(c)(z)$ is smooth), and the cocycle condition $f_{b''b'} f_{b'b} = f_{b''b}$ holds on $U_b \cap U_{b'} \cap U_{b''}$.*

Proof. If $t_1, \dots, t_p$ are local coordinates on B centred at b, the functions $\phi^* t_i$ have linearly independent differentials at each point of Z_b because ϕ is a submersion, so by the implicit function theorem $Z_b = \{\phi^* t_1 = \cdots = \phi^* t_p = 0\}$ is an embedded submanifold of codimension $p = \dim B$; it is compact, being closed in M.

Choose a Riemannian metric on M and let $H_m \subset T_m M$ be the orthogonal complement of $\ker \phi_* = T_m Z_{\phi(m)}$. Then $\phi_* : H_m \longrightarrow T_{\phi(m)} B$ is an isomorphism and depends smoothly on m, so each vector field v on an open set $U \subset B$ has a unique smooth 'horizontal lift' $\tilde{v}$ to $\phi^{-1}(U)$, with $\tilde{v}(m) \in H_m$ and $\phi_* \tilde{v} = v$. Let U_b be a coordinate ball, $\{|t| < r\}$, centred at b. For $c \in U_b$ with coordinates $t(c)$ consider the constant-coefficient vector field $v_c = \sum_i t_i(c) \partial_{t_i}$ and its horizontal lift $\tilde{v}_c$. The integral curve of v_c starting at b is $s \mapsto st(c)$, which stays in the compact set $\{|t| \le |t(c)|\} \subset U_b$ for $s \in [0,1]$; hence the integral curves of $\tilde{v}_c$ starting on Z_b stay in the compact set $\phi^{-1}(\{|t| \le |t(c)|\})$ and exist up to time 1. Let $F(z,c) = \exp(\tilde{v}_c)(z)$ be the time-one flow. By smooth dependence of solutions of ordinary differential equations on parameters, $F : Z_b \times U_b \longrightarrow \phi^{-1}(U_b)$ is smooth, and $\phi(F(z,c)) = c$. It is a bijection with smooth inverse $m \longmapsto (\exp(-\tilde{v}_{\phi(m)})(m), \phi(m))$, so $f_b = F^{-1}$ is a diffeomorphism as in (2), with Z replaced by Z_b. In particular the fibres over U_b are all diffeomorphic to Z_b; since B is connected, the set of $c \in B$ with Z_c diffeomorphic to a given fibre is open and closed, which proves (1) and (2) with a fixed Z. Part (3) is immediate from (2): $f_{b'} \circ f_b^{-1}$ commutes with the projections to $U_b \cap U_{b'}$, so has the stated form, and the cocycle condition is just associativity of composition. □

Conversely, M is recovered from the trivializations and the transition maps $f_{b'b}$; thus a fibration is a fibre bundle with fibre Z and structure group $\mathrm{Dfm}(Z)$. If the transition maps can be chosen to take values in a subgroup $G \subset \mathrm{Dfm}(Z)$, the structure group is said to be *reduced* to G.

Proposition 12.1.3 (Pull-back)**.** *Let $\phi : M \longrightarrow B$ be a fibration and $F : \tilde{B} \longrightarrow B$ a smooth map from a compact manifold. Then $\tilde{M} = \{(m, \tilde{b}) \in M \times \tilde{B}; \phi(m) = F(\tilde{b})\}$ is an embedded submanifold of $M \times \tilde{B}$ and $\tilde{\phi} : \tilde{M} \longrightarrow \tilde{B}$, $\tilde{\phi}(m, \tilde{b}) = \tilde{b}$, is a fibration with fibre $\tilde{\phi}^{-1}(\tilde{b}) = Z_{F(\tilde{b})}$.*

Proof. $\tilde{M}$ is the preimage of Diag_B under $\Phi = \phi \times F : M \times \tilde{B} \longrightarrow B \times B$. Since ϕ is a submersion, the range of Φ_* contains $TB \times \{0\}$, and $TB \times \{0\}$ together with $T\,\mathrm{Diag}_B$ spans $T(B \times B)$; so Φ is transversal to Diag_B and $\tilde{M}$ is an embedded submanifold of codimension $\dim B$, compact since it is closed. Its tangent space is $\{(v,w); \phi_* v = F_* w\}$; given $w \in T_{\tilde{b}} \tilde{B}$ we can solve $\phi_* v = F_* w$, so $\tilde{\phi}$ is a submersion, and it is clearly surjective. □

Proposition 12.1.4 (Composites and products)**.** *If $\phi' : M' \longrightarrow M$ and $\phi : M \longrightarrow B$ are fibrations then $\phi \circ \phi' : M' \longrightarrow B$ is a fibration; its fibre over b is $(\phi')^{-1}(Z_b)$, which is itself the total space of the fibration $\phi'| : (\phi')^{-1}(Z_b) \longrightarrow Z_b$ obtained by restriction. If $\phi_i : M_i \longrightarrow B_i$, $i = 1, 2$, are fibrations then so is $\phi_1 \times \phi_2 : M_1 \times M_2 \longrightarrow B_1 \times B_2$.*

Proof. Composites and products of surjective submersions are surjective submersions. The restriction of ϕ' to $(\phi')^{-1}(Z_b)$ is the pull-back of ϕ' under the inclusion $Z_b \hookrightarrow M$, so is a fibration by Proposition 12.1.3. □

Note that the fibre of $\phi \circ \phi'$ need not be diffeomorphic to the product $Z \times Z'$ of the fibres; it is in general a non-trivial fibration over Z.

12.1.1 Vertical bundles and connections

The *vertical tangent bundle* $T(M/B) \subset TM$ is the null bundle of ϕ_*, so $T_m(M/B) = T_m Z_{\phi(m)}$, and there are short exact sequences of vector bundles over M

$$0 \longrightarrow T(M/B) \longrightarrow TM \xrightarrow{\phi_*} \phi^*TB \longrightarrow 0, \qquad 0 \longrightarrow \phi^*T^*B \xrightarrow{\phi^*} T^*M \longrightarrow T^*(M/B) \longrightarrow 0, \tag{12.1.1}$$

where $T^*(M/B)$ is the dual of $T(M/B)$. The second sequence identifies ϕ^*T^*B with the subbundle of covectors annihilating the fibres. We denote by $S^*(M/B)$ and ϕ^*S^*B the corresponding sphere bundles (quotients of the complements of the zero sections by $\mathbb{R}_+$).

Definition 12.1.5. A *connection* on the fibration ϕ is a choice of a smooth subbundle $H \subset TM$ complementary to $T(M/B)$. Then $\phi_* : H \longrightarrow \phi^*TB$ is an isomorphism, so a connection is the same as a splitting of the first sequence in (12.1.1), and dually of the second.

The proof of Theorem 12.1.2 used the connection given by the orthogonal complement for a Riemannian metric, so connections always exist.

The *fibre density bundle* $\Omega(M/B) = \Omega(T(M/B))$ is the bundle of densities on the fibres. From (12.1.1) there is a canonical isomorphism $\Omega(M) \cong \Omega(M/B) \otimes \phi^*\Omega(B)$; in particular a smooth positive section of $\Omega(M/B)$ restricts to a smooth positive density on each fibre.

12.1.2 The fibre product

Applying Proposition 12.1.3 with $F = \phi$ gives the *fibre product*

$$M^2_\phi = M \times_\phi M = \{(m, m') \in M^2; \phi(m) = \phi(m')\},$$

an embedded submanifold of M^2 of codimension $b = \dim B$. It fibres over M (by either projection) with fibre Z, and over B with fibre Z^2. It contains the diagonal, so we have a nested pair of embedded submanifolds

$$\mathrm{Diag}_M \subset M^2_\phi \subset M^2. \tag{12.1.2}$$

If (z, y) are local coordinates on M in which $\phi(z, y) = y$ (*fibred coordinates*), and (z', y') is a second copy, then near a point of M^2_ϕ

$$\mathrm{Diag}_M = \{z = z',\ y = y'\} \subset M^2_\phi = \{y = y'\} \subset M^2. \tag{12.1.3}$$

We write $\pi_L, \pi_R : M^2 \longrightarrow M$ for the projections and also for their restrictions to M^2_ϕ, and set $\Omega_R = \pi_R^*\Omega(M/B)$ over M^2_ϕ; for bundles E, F over M we write $\mathrm{Hom}(E, F) = \pi_L^*F \otimes \pi_R^*E^*$ over M^2 or M^2_ϕ.

Lemma 12.1.6. *For a fibration $\phi : M \longrightarrow B$:*

(i) *the conormal bundle of the fibre diagonal is* $N^*_{(m,m')}M^2_\phi = \{(\phi^*\eta, -\phi^*\eta); \eta \in T^*_{\phi(m)}B\}$, *so* $N^*M^2_\phi \cong (\phi\circ\pi_L)^*T^*B$ *and its sphere bundle is the fibre product* $SN^*M^2_\phi \cong M^2_\phi \times_B S^*B$, *which is the fibre product of the pulled-back fibration* $\pi^*M = M\times_B S^*B \longrightarrow S^*B$, $\pi : S^*B \longrightarrow B$;

(ii) $N^*\operatorname{Diag}_M \cong T^*M$, *by* $(\xi, -\xi) \longmapsto \xi$;

(iii) *under* (ii), *the restriction* $N^*M^2_\phi|_{\operatorname{Diag}_M}$ *is identified with* $\phi^*T^*B \subset T^*M$, *and the quotient*

$$N^*\operatorname{Diag}_M / N^*M^2_\phi|_{\operatorname{Diag}_M}$$

with $T^*(M/B)$, *the conormal bundle of* Diag_M *as a submanifold of* M^2_ϕ.

Proof. M^2_ϕ is the preimage of Diag_B under $\phi\times\phi$, a submersion, so its conormal space at (m,m') is the pull-back of $N^*\operatorname{Diag}_B = \{(\eta,-\eta)\}$. This gives (i); the description of the sphere bundle follows. Part (ii) is standard, and (iii) follows by comparing (i) and (ii) with (12.1.1); in the coordinates (12.1.3) it says that $N^*M^2_\phi$ is spanned by the $dy_j - dy'_j$. □

12.2 Families of pseudodifferential operators

Throughout, $\phi : M \longrightarrow B$ is a fibration with typical fibre Z, and we set $f = \dim Z$, $b = \dim B$. One could define operators on the fibres by patching together, using Theorem 12.1.2, smooth families of operators on Z, relying on the invariance of pseudodifferential operators under diffeomorphisms. It is more direct to use conormal distributions on the fibre product. Recall from Chapter 4 and Chapter 5 that $\Psi^m(X;E,F) = I^m(X^2, \operatorname{Diag}; \operatorname{Hom}(E,F)\otimes\Omega_R)$, with the order normalization of Chapter 4 (for which a conormal distribution of order m at a submanifold of codimension c in a manifold of dimension n has a symbol of order $m + \frac{n}{4} - \frac{c}{2}$).

Definition 12.2.1. The space of *pseudodifferential operators on the fibres* of ϕ, acting from sections of E to sections of F, is

$$\Psi^m(M/B;E,F) = I^{m-\frac{b}{4}}(M^2_\phi, \operatorname{Diag}_M; \operatorname{Hom}(E,F)\otimes\Omega_R).$$

The shift in order is such that the symbols have order m: $\dim M^2_\phi = 2f + b$ and Diag_M has codimension f in M^2_ϕ, so the symbol order is $m - \frac{b}{4} + \frac{2f+b}{4} - \frac{f}{2} = m$.

Lemma 12.2.2. *Let* $U \subset B$ *be an open ball over which there is a trivialization* $\phi^{-1}(U) \cong Z\times U$ *as in Theorem 12.1.2, and bundle isomorphisms* $E|_{\phi^{-1}(U)} \cong \pi_Z^*E_0$, $F|_{\phi^{-1}(U)} \cong \pi_Z^*F_0$ *for bundles* E_0, F_0 *over* Z *(these exist since* U *is contractible). Then restriction of kernels gives an identification*

$$\Psi^m(M/B;E,F)\big|_U \cong \mathcal{C}^\infty(U; \Psi^m(Z;E_0,F_0)),$$

where $\Psi^m(Z;E_0,F_0)$ *carries its Fréchet topology and the right side consists of the smooth maps into it.*

Proof. Over U the trivialization identifies M^2_ϕ with $Z^2\times U$, Diag_M with $\operatorname{Diag}_Z \times U$ and Ω_R with the right density bundle of Z. The submanifold $\operatorname{Diag}_Z \times U$ is a product, the variables in U being variables along it, so by the definition of conormal distributions in local coordinates (Chapter 4) an element of $I^{m''}(Z^2\times U, \operatorname{Diag}_Z\times U)$ is locally the inverse Fourier transform in the normal variables of a symbol depending smoothly on the variables in Z and U; that is, it is a smooth family, parametrized by U, of elements of $I^{m''+\frac{b}{4}}(Z^2, \operatorname{Diag}_Z)$ (the shift by $\frac{b}{4}$ comes from the change of dimension). With $m'' = m - \frac{b}{4}$ this is $\mathcal{C}^\infty(U;\Psi^m(Z;E_0,F_0))$. □

In particular $\Psi^m(M/B;E,F)$ does not depend on choices, and its elements are the families which are, locally in B, smooth families of pseudodifferential operators on a fixed fibre, glued by the transition maps of Theorem 12.1.2(3).

Note that the elements of $\Psi^m(M/B;E,F)$, as operators on M, have Schwartz kernels supported in M^2_ϕ, with a δ-singularity in the normal directions, whereas elements of $\Psi^m(M;E,F)$ have kernels which are smooth off the diagonal. For $\dim B > 0$ the only operators in both spaces are fibrewise differential operators (Exercise 12.2).

Proposition 12.2.3. *Let $\phi : M \longrightarrow B$ be a fibration and E, F, G vector bundles over M.*

(1) (Action) *Each $A \in \Psi^m(M/B;E,F)$ defines a continuous linear map $A : \mathcal{C}^\infty(M;E) \longrightarrow \mathcal{C}^\infty(M;F)$. It acts on the fibres: if $u \in \mathcal{C}^\infty(M;E)$ and $u|_{Z_b} = 0$ then $Au|_{Z_b} = 0$, so*

$$A_b v = (Au)|_{Z_b}, \quad u \in \mathcal{C}^\infty(M;E),\ u|_{Z_b} = v,$$

defines $A_b \in \Psi^m(Z_b;E_b,F_b)$, where $E_b = E|_{Z_b}$. Moreover $A(\phi^*\chi\, u) = \phi^*\chi\, Au$ for all $\chi \in \mathcal{C}^\infty(B)$.

(2) (Smoothing families) $\Psi^{-\infty}(M/B;E,F) = \bigcap_m \Psi^m(M/B;E,F) = \mathcal{C}^\infty(M^2_\phi;\mathrm{Hom}(E,F)\otimes \Omega_R)$.

(3) (Symbol) *There is a principal symbol map giving a short exact sequence*

$$\Psi^{m-1}(M/B;E,F) \hookrightarrow \Psi^m(M/B;E,F) \xrightarrow{\sigma_m} \mathcal{C}^\infty_{[m]}(T^*(M/B)\setminus 0;\hom(E,F)),$$

where $\mathcal{C}^\infty_{[m]}$ denotes smooth sections, homogeneous of degree m, of the pull-back of $\hom(E,F)$ to $T^*(M/B)\setminus 0$; $\sigma_m(A)$ restricts on $T^*Z_b \setminus 0$ to $\sigma_m(A_b)$.

(4) (Composition) *If $A \in \Psi^m(M/B;F,G)$ and $B' \in \Psi^{m'}(M/B;E,F)$ then $AB' \in \Psi^{m+m'}(M/B;E,G)$, $(AB')_b = A_bB'_b$ and $\sigma_{m+m'}(AB') = \sigma_m(A)\sigma_{m'}(B')$.*

(5) (Adjoints) *Given Hermitian metrics on E, F and a positive fibre density, the fibrewise adjoints A_b^* form an element $A^* \in \Psi^m(M/B;F,E)$, with $\sigma_m(A^*) = \sigma_m(A)^*$.*

(6) (Asymptotic completeness) *If $A_j \in \Psi^{m-j}(M/B;E,F)$ for $j \in \mathbb{N}_0$, then there exists an element $A \in \Psi^m(M/B;E,F)$, unique modulo $\Psi^{-\infty}(M/B;E,F)$, such that $A - \sum_{j<N} A_j \in \Psi^{m-N}(M/B;E,F)$ for all N.*

Proof. All statements are local in B, since the kernels are supported in M^2_ϕ. Choose a finite cover of B by balls U_i as in Lemma 12.2.2 and a partition of unity $\chi_i \in \mathcal{C}^\infty_c(U_i)$. Multiplication by $\phi^*\chi_i$ commutes with A because the kernel of A is supported in M^2_ϕ, on which $\phi^*\chi_i(m) = \phi^*\chi_i(m')$. So $A = \sum_i \phi^*\chi_i\, A$ and each term is, by Lemma 12.2.2, a smooth family over U_i of pseudodifferential operators on Z, with compact support in U_i.

Every construction of the calculus on the compact manifold Z is continuous in the Fréchet topologies involved: the action $\Psi^m(Z)\times\mathcal{C}^\infty(Z)\longrightarrow\mathcal{C}^\infty(Z)$, composition, the symbol map, adjoints, and the quantization of symbols. Hence each of them maps smooth families to smooth families, and (1), (3), (4), (5) follow from the corresponding statements for a single compact manifold in Chapter 5. For (1) note also that a smooth family $u \in \mathcal{C}^\infty(U\times Z;E_0)$ is mapped to a smooth family, i.e. to an element of $\mathcal{C}^\infty(U\times Z;F_0)$. In (3) the homogeneous symbols of the local families patch together because the symbol is invariant under the diffeomorphisms $f_{b'b}(c)$, and its restriction to T^*Z_b is the symbol of A_b; the null space of σ_m is $\Psi^{m-1}(M/B)$ by the corresponding property of conormal distributions, and surjectivity follows by quantizing $\chi_i\sigma$ in each U_i and summing. Part (2) is the definition of conormal distributions of order $-\infty$, and (6) follows from asymptotic completeness of conormal distributions on M^2_ϕ (or locally from the parameter version of Borel's lemma) together with the partition of unity. □

Definition 12.2.4. $A \in \Psi^m(M/B; E, F)$ is *elliptic* if $\sigma_m(A)$ is invertible at each point of $T^*(M/B) \setminus 0$, i.e. if each A_b is elliptic.

Proposition 12.2.5. *If $A \in \Psi^m(M/B; E, F)$ is elliptic there exists $Q \in \Psi^{-m}(M/B; F, E)$ with*

$$QA - \mathrm{Id}_E \in \Psi^{-\infty}(M/B; E), \qquad AQ - \mathrm{Id}_F \in \Psi^{-\infty}(M/B; F).$$

Proof. Over each U_i as above, A is a smooth family of elliptic operators on Z. The construction of a parametrix in Chapter 5 — quantize the inverse of the principal symbol to get Q_0 with $Q_0 A = \mathrm{Id} - R$, R of order -1, then take $Q \sim \sum_j R^j Q_0$ — uses only operations continuous in the Fréchet topologies together with asymptotic summation, which can be carried out with smooth dependence on parameters. So there are $Q_i \in \mathcal{C}^\infty(U_i; \Psi^{-m}(Z))$ with $Q_i A - \mathrm{Id}$ and $A Q_i - \mathrm{Id}$ smoothing families over U_i. Set $Q = \sum_i \phi^* \chi_i \, Q_i \in \Psi^{-m}(M/B; F, E)$. Since multiplication by $\phi^*\chi_i$ commutes with fibre operators,

$$QA = \sum_i \phi^*\chi_i \, Q_i A = \mathrm{Id}_E - \sum_i \phi^*\chi_i (\mathrm{Id} - Q_i A),$$
$$AQ = \sum_i \phi^*\chi_i \, A Q_i = \mathrm{Id}_F - \sum_i \phi^*\chi_i (\mathrm{Id} - A Q_i),$$

and the error terms are smoothing families. □

Unlike the case of a single operator, the partition of unity on B introduces no error at all, because it commutes with the operators. The errors could also be improved directly: if $Q_0 A = \mathrm{Id} - R$ with $R \in \Psi^{-1}(M/B; E)$, the Neumann series $\sum_j R^j Q_0$ is asymptotically summable by Proposition 12.2.3(6).

12.3 The index bundle

We now associate to an elliptic family $A \in \Psi^m(M/B; E, F)$ an element of $K^0(B)$ generalizing the index of a single elliptic operator. The null spaces $\mathrm{Nul}(A_b)$ need not have constant dimension, so need not form a vector bundle; following [18] we first stabilize the family by a finite rank perturbation so that it becomes surjective.

For each $b \in B$, A_b is an elliptic operator on the compact manifold Z_b, so by Chapter 5 it is Fredholm as a map $H^{s+m}(Z_b; E_b) \longrightarrow H^s(Z_b; F_b)$ for every s, its null space is a finite-dimensional subspace of $\mathcal{C}^\infty(Z_b; E_b)$ independent of s, its range is closed and $H^s(Z_b; F_b) = \mathrm{Ran}(A_b) \oplus \mathrm{Nul}(A_b^*)$ (orthogonal decomposition when $s = 0$; in general $\mathrm{Nul}(A_b^*) \subset \mathcal{C}^\infty$ is a complement); the index does not depend on s.

Lemma 12.3.1. *In a local trivialization as in Lemma 12.2.2, if $A \in \mathcal{C}^\infty(U; \Psi^m(Z; E_0, F_0))$ then for each s the map $U \ni c \longmapsto A(c) \in \mathcal{B}(H^{s+m}(Z; E_0), H^s(Z; F_0))$ is smooth in the norm topology.*

Proof. The inclusion $\Psi^m(Z; E_0, F_0) \longrightarrow \mathcal{B}(H^{s+m}, H^s)$ is linear and has closed graph (convergence in either space implies convergence of the Schwartz kernels as distributions), so is continuous by the closed graph theorem. The composite of a smooth map into a Fréchet space with a continuous linear map into a Banach space is smooth. □

Lemma 12.3.2 (Stabilization)**.** *If $A \in \Psi^m(M/B; E, F)$ is elliptic there are $N \in \mathbb{N}_0$ and sections $s_1, \dots, s_N \in \mathcal{C}^\infty(M; F)$ such that for every $b \in B$ the map*

$$\tilde{A}_b : \mathcal{C}^\infty(Z_b; E_b) \oplus \mathbb{C}^N \longrightarrow \mathcal{C}^\infty(Z_b; F_b), \quad (u, c) \longmapsto A_b u + \sum_{j=1}^N c_j s_j|_{Z_b} \tag{12.3.1}$$

is surjective.

Proof. Fix $b_0 \in B$. Since $\mathrm{Ran}(A_{b_0})$ has finite codimension in $\mathcal{C}^\infty(Z_{b_0}; F_{b_0})$ we may choose $v_1, \dots, v_p \in \mathcal{C}^\infty(Z_{b_0}; F_{b_0})$ spanning a complement, and extend them to sections $s_j \in \mathcal{C}^\infty(M; F)$. In a trivialization over a ball $U \ni b_0$ consider

$$T(c) : H^m(Z; E_0) \oplus \mathbb{C}^p \longrightarrow L^2(Z; F_0), \quad T(c)(u, a) = A(c)u + \sum_j a_j s_j|_{Z_c}.$$

The v_j also span a complement to the range of A_{b_0} on H^m: this range is $\mathrm{Nul}(A_{b_0}^*)^\perp$, of codimension $\dim \mathrm{Nul}(A_{b_0}^*) = p$, and if $\sum_j c_j v_j = A_{b_0} u$ with $u \in H^m$ then $u \in \mathcal{C}^\infty$ by elliptic regularity, so $c = 0$. Thus $T(b_0)$ is surjective. By Lemma 12.3.1, $T(c)$ depends continuously on c in norm, and surjectivity is an open condition (a surjective bounded operator has a bounded right inverse, and small perturbations of an operator with a bounded right inverse still have one), so $T(c)$ is surjective for c in a neighbourhood U' of b_0. If $g \in \mathcal{C}^\infty(Z_c; F_c)$ and $g = T(c)(u, a)$ with $u \in H^m$, then $A_c u = g - \sum a_j s_j \in \mathcal{C}^\infty$, so $u \in \mathcal{C}^\infty$ by elliptic regularity; hence (12.3.1), with these p sections, is surjective for $c \in U'$. Covering B by finitely many such neighbourhoods and taking all the sections so obtained gives the lemma, since adding sections preserves surjectivity. □

Proposition 12.3.3. *If $\tilde{A}_b$ in (12.3.1) is surjective for all b then the null spaces $\mathrm{Nul}(\tilde{A}_b) \subset \mathcal{C}^\infty(Z_b; E_b) \oplus \mathbb{C}^N$ form a smooth vector bundle $\mathrm{Nul}(\tilde{A})$ over B, of rank $N + \mathrm{ind}(A_b)$, in the sense that each point has a neighbourhood U over which there are $e_1, \dots, e_r \in \mathcal{C}^\infty(\phi^{-1}(U); E) \oplus \mathcal{C}^\infty(U; \mathbb{C}^N)$ restricting to a basis of $\mathrm{Nul}(\tilde{A}_c)$ for each $c \in U$.*

Proof. As a map $H^{s+m} \oplus \mathbb{C}^N \longrightarrow H^s$, $\tilde{A}_c$ is Fredholm with index $\mathrm{ind}(A_c) + N$, and surjective (its range is closed and contains the dense subspace $\mathcal{C}^\infty$); its null space lies in $\mathcal{C}^\infty \oplus \mathbb{C}^N$ by elliptic regularity. So $\dim \mathrm{Nul}(\tilde{A}_c) = \mathrm{ind}(A_c) + N$, which is locally constant since the index is (Chapter 10).

Fix b_0 and work in a trivialization over a ball $U \ni b_0$. Let $K_0 = \mathrm{Nul}(\tilde{A}(b_0))$ and let P be the orthogonal projection onto K_0 in $H^m \oplus \mathbb{C}^N$ (for some choice of inner product). Then

$$T(c) = (\tilde{A}(c), P) : H^{s+m}(Z; E_0) \oplus \mathbb{C}^N \longrightarrow H^s(Z; F_0) \oplus K_0$$

is invertible for $c = b_0$ and every $s \geq 0$. By Lemma 12.3.1 it is invertible, with inverse depending smoothly on c, for c in a neighbourhood U_0 of b_0 when $s = 0$. For $s > 0$ and $c \in U_0$, $T(c)$ is Fredholm of index 0 on the H^{s+m} level and injective, since its null space is contained in that on the H^m level; so it is invertible for all $s \geq 0$ and all $c \in U_0$, and the inverse is smooth in c in each of these norms. If $\kappa_1, \dots, \kappa_r$ is a basis of K_0, the elements $e_i(c) = T(c)^{-1}(0, \kappa_i)$ lie in $\mathrm{Nul}(\tilde{A}(c))$ and are linearly independent, hence form a basis by the dimension count. Since $c \mapsto e_i(c)$ is smooth into $H^{s+m}(Z; E_0) \oplus \mathbb{C}^N$ for every s, Sobolev embedding shows that $e_i \in \mathcal{C}^\infty(U_0 \times Z; E_0) \oplus \mathcal{C}^\infty(U_0; \mathbb{C}^N)$. □

Definition 12.3.4. The *(analytic) index* of an elliptic family $A \in \Psi^m(M/B; E, F)$ is

$$\mathrm{ind}_a(A) = [\mathrm{Nul}(\tilde{A})] - [\mathbb{C}^N] \in K^0(B),$$

where $\tilde{A}$ is any stabilization as in Lemma 12.3.2.

Theorem 12.3.5. *Let $A \in \Psi^m(M/B; E, F)$ be elliptic.*

(i) $\mathrm{ind}_a(A)$ *is independent of the choice of stabilization.*

(ii) *If A_t, $t \in [0, 1]$, is a smooth family of elliptic elements of $\Psi^m(M/B; E, F)$ then $\mathrm{ind}_a(A_0) = \mathrm{ind}_a(A_1)$. Consequently $\mathrm{ind}_a(A)$ depends only on the homotopy class of $\sigma_m(A)$ through elliptic symbols.*

(iii) *If* $\dim \mathrm{Nul}(A_b)$ *is independent of* b*, then* $\mathrm{Nul}(A)$ *and* $\mathrm{Nul}(A^*)$ *are smooth vector bundles and* $\mathrm{ind}_a(A) = [\mathrm{Nul}(A)] - [\mathrm{Nul}(A^*)]$*. In particular for* B *a point,* $\mathrm{ind}_a(A) = \mathrm{ind}(A) \in \mathbb{Z} = K^0(\mathrm{pt})$.

(iv) $\mathrm{ind}_a(A_1 \oplus A_2) = \mathrm{ind}_a(A_1) + \mathrm{ind}_a(A_2)$.

Proof. (i) Let S and S' be stabilizations, by N and N' sections. Then the combination $S'' = (S, S')$, by all $N + N'$ sections, is again a stabilization. The map $\mathrm{Nul}(\tilde{A}'') \ni (u, c, c') \longmapsto c' \in \mathbb{C}^{N'}$ is a smooth bundle map, surjective because $\tilde{A}$ is (given c' solve $A_b u + S_b c = -S'_b c'$), and its null space is $\{(u, c, 0); A_b u + S_b c = 0\} = \mathrm{Nul}(\tilde{A})$. So $0 \to \mathrm{Nul}(\tilde{A}) \to \mathrm{Nul}(\tilde{A}'') \to \mathbb{C}^{N'} \to 0$ is a short exact sequence of vector bundles, which splits, and $[\mathrm{Nul}(\tilde{A}'')] - [\mathbb{C}^{N+N'}] = [\mathrm{Nul}(\tilde{A})] - [\mathbb{C}^N]$. By symmetry the same holds with S' in place of S.

(ii) A smooth family A_t is an elliptic family for the fibration $M \times [0,1] \longrightarrow B \times [0,1]$, and the proofs above apply verbatim over the compact base $B \times [0,1]$ (manifold with boundary); thus a single stabilization works for all t and $\mathrm{Nul}(\tilde{A})$ is a vector bundle over $B \times [0,1]$. Its restrictions to $B \times \{0\}$ and $B \times \{1\}$ are isomorphic [17]. If σ_t is a smooth homotopy of elliptic symbols, quantizing it with smooth dependence on t gives a family A'_t with $\sigma_m(A'_t) = \sigma_t$; if $\sigma_m(A) = \sigma_m(A')$ then $(1-t)A + tA'$ is elliptic for all t. Together these show the second statement.

(iii) Suppose $\dim \mathrm{Nul}(A_b) = r$ for all b. Fix b_0, work in a trivialization near b_0, let P be a projection of $H^m(Z; E_0)$ onto $K_0 = \mathrm{Nul}(A(b_0))$ and Q the projection of $L^2(Z; F_0)$ onto $\mathrm{Ran}(A(b_0))$ along $\mathrm{Nul}(A(b_0)^*)$. Then $T(c) = (QA(c), P) : H^m \longrightarrow \mathrm{Ran}(A(b_0)) \oplus K_0$ is invertible at b_0, hence for c near b_0, and $\mathrm{Nul}(QA(c)) = T(c)^{-1}(\{0\} \oplus K_0)$ has dimension r. Since $\mathrm{Nul}(A(c)) \subset \mathrm{Nul}(QA(c))$ and both have dimension r they are equal, so as in the proof of Proposition 12.3.3 $\mathrm{Nul}(A)$ is a smooth bundle. The same applies to A^*, since $\dim \mathrm{Nul}(A_b^*) = r - \mathrm{ind}(A_b)$ is also constant. Now choose a bundle G with $\mathrm{Nul}(A^*) \oplus G \cong \mathbb{C}^N$ and let $S_b : \mathbb{C}^N \longrightarrow \mathcal{C}^\infty(Z_b; F_b)$ be the inclusion on $\mathrm{Nul}(A_b^*)$ and zero on G_b; this is given by N smooth sections of F. Since $\mathcal{C}^\infty(Z_b; F_b) = \mathrm{Ran}(A_b) \oplus \mathrm{Nul}(A_b^*)$, $\tilde{A}$ is surjective, and $\mathrm{Nul}(\tilde{A}) = \mathrm{Nul}(A) \oplus G$. Hence $\mathrm{ind}_a(A) = [\mathrm{Nul}\, A] + [G] - [\mathrm{Nul}(A^*) \oplus G] = [\mathrm{Nul}\, A] - [\mathrm{Nul}\, A^*]$.

(iv) If $\tilde{A}_1$, $\tilde{A}_2$ are stabilizations then $\tilde{A}_1 \oplus \tilde{A}_2$ is a stabilization of $A_1 \oplus A_2$ with null bundle $\mathrm{Nul}(\tilde{A}_1) \oplus \mathrm{Nul}(\tilde{A}_2)$. □

Remark 12.3.6. The principal symbol of an elliptic family, $\sigma_m(A)$, is an isomorphism $\pi^*E \longrightarrow \pi^*F$ off the zero section of $\pi : T^*(M/B) \longrightarrow M$, so defines a class $[\sigma_m(A)] \in K_c^0(T^*(M/B))$. By Theorem 12.3.5(ii) the index depends only on the homotopy class of the symbol; arguing exactly as for a single operator in Chapter 10 (stabilization by adding trivial bundles, invariance under bundle isomorphisms, and the fact that every compactly supported class is represented by the symbol of an elliptic family of order 0) one obtains a well-defined homomorphism

$$\mathrm{ind}_a : K_c^0(T^*(M/B)) \longrightarrow K^0(B),$$

the *analytic index map* of the fibration; see [18]. Its identification with a topological index is the families index theorem of Chapter 13.

12.4 The relative compactification

We now turn to the product-type calculus. Its kernels will be conormal to a nested pair of submanifolds $Z \subset Y \subset X$; at a point of Z the symbols are functions on the conormal

space $W = N_p^* Z$, which contains the distinguished subspace $V = N_p^* Y$, and the relevant compactification of W is one which 'remembers' V.

Recall from Chapter 1 that the radial compactification of a real vector space U with a Euclidean norm is the closed half-sphere

$$\overline{U} = \{(t, u') \in \mathbb{R} \times U;\ t \geq 0,\ t^2 + |u'|^2 = 1\}, \quad U \ni u \longmapsto \Big(\frac{1}{\langle u \rangle}, \frac{u}{\langle u \rangle}\Big) \in \overline{U},$$

which embeds U as the interior of a compact manifold with boundary $\partial\overline{U} = SU$, the unit sphere; $\rho_U = \langle u \rangle^{-1}$ extends to a boundary defining function, near the boundary $1/|u|$ and $u/|u|$ give smooth coordinates, and $\rho_U^{-\mu}\mathcal{C}^\infty(\overline{U})$ is the space $S^\mu_{\mathrm{ph}}(U)$ of classical symbols of order μ. We shall use the following parameter version of the invariance of $\overline{U}$ under affine maps.

Lemma 12.4.1. *Let P be a manifold, possibly with corners, and $L \in \mathcal{C}^\infty(P; \mathrm{GL}(U))$, $c \in \mathcal{C}^\infty(P; U)$. Then $P \times U \ni (p, u) \longmapsto (p, L(p)u + c(p))$ extends to a diffeomorphism of $P \times \overline{U}$.*

Proof. Away from the boundary there is nothing to prove. Near a boundary point use the coordinates $x = 1/|u|$, $\theta = u/|u|$. Then $Lu + c = x^{-1}(L\theta + xc)$ and $L\theta \neq 0$, so for x small

$$\frac{1}{|Lu + c|} = \frac{x}{|L\theta + xc|}, \qquad \frac{Lu + c}{|Lu + c|} = \frac{L\theta + xc}{|L\theta + xc|}$$

are smooth functions of (p, x, θ) up to $x = 0$. The inverse map is of the same form, with L^{-1} and $-L^{-1}c$. □

Now let W be a real vector space of dimension $k+l$ and $V \subset W$ a subspace of dimension l; we assume $k, l > 0$ (the degenerate cases are discussed in Exercise 12.3). Choose a complement U, so $W = U \oplus V$, and Euclidean norms on U and V; we write $w = (\zeta, \eta)$, $\zeta \in U$, $\eta \in V$ and $\langle \zeta, \eta \rangle = (1 + |\zeta|^2 + |\eta|^2)^{1/2}$. Note that U is identified with W/V, so the variable ζ is the class of w in W/V.

Definition 12.4.2. The *relative compactification* of W with respect to V is the compact manifold with corners

$${}^V\overline{W} = \overline{U} \times \overline{V}, \quad \text{with} \quad \iota_V : W \ni (\zeta, \eta) \longmapsto \Big(\zeta, \frac{\eta}{\langle \zeta \rangle}\Big) \in U \times V \subset \overline{U} \times \overline{V}.$$

Its two boundary hypersurfaces are

$$H_W = \partial\overline{U} \times \overline{V} = SU \times \overline{V} \quad (\text{the } \textit{main face}), \qquad H_V = \overline{U} \times \partial\overline{V} = \overline{U} \times SV \quad (\text{the } \textit{front face}),$$

meeting in the corner $H_W \cap H_V = SU \times SV$. The functions

$$\rho_W = \frac{1}{\langle \zeta \rangle}, \qquad \rho_V = \frac{1}{\langle \eta/\langle \zeta \rangle \rangle} = \frac{\langle \zeta \rangle}{\langle \zeta, \eta \rangle}$$

on W extend to boundary defining functions of H_W and H_V respectively, and $\rho_W \rho_V = \langle \zeta, \eta \rangle^{-1}$.

Clearly ι_V is a diffeomorphism of W onto the interior, with inverse $(\zeta, q) \mapsto (\zeta, \langle \zeta \rangle q)$. Composing with the embeddings of $\overline{U}$ and $\overline{V}$ as half-spheres gives Melrose's original description: ${}^V\overline{W}$ is the closure of the image of

$$W \ni (\zeta, \eta) \longmapsto (t, s, \zeta', \eta') = \Big(\frac{1}{\langle \zeta \rangle}, \frac{\langle \zeta \rangle}{\langle \zeta, \eta \rangle}, \frac{\zeta}{\langle \zeta \rangle}, \frac{\eta}{\langle \zeta, \eta \rangle}\Big) \in \mathbb{R}^2 \times W,$$

which is $\{t, s \geq 0,\ t^2 + |\zeta'|^2 = 1 = s^2 + |\eta'|^2\}$, the product of two half-spheres. The point is that the identification of the interior with W is *not* the obvious product one: the second factor is compactified in the rescaled variable $q = \eta/\langle\zeta\rangle$.

Proposition 12.4.3. *Smooth coordinates (more precisely, smooth functions among which a coordinate system can be chosen) near the boundary of ${}^{V}\overline{W}$ are given by*

(i) *near interior points of H_W:* $1/|\zeta|$, $\zeta/|\zeta|$ *and* $\eta/|\zeta|$;

(ii) *near interior points of H_V:* ζ, $1/|\eta|$ *and* $\eta/|\eta|$;

(iii) *near the corner:* $1/|\zeta|$, $|\zeta|/|\eta|$, $\zeta/|\zeta|$ *and* $\eta/|\eta|$,

where the first one, respectively the first two in (iii)*, are local boundary defining functions.*

Proof. Near H_W, $|\zeta|$ is large and $\rho_W = \langle\zeta\rangle^{-1}$ is a smooth function of $1/|\zeta|$ (since $x(1+x^2)^{-1/2}$ is a smooth function of x near 0 with nonvanishing derivative), while $\zeta/\langle\zeta\rangle = (\zeta/|\zeta|)\,|\zeta|/\langle\zeta\rangle$. In the interior of H_W, $q = \eta/\langle\zeta\rangle$ is a smooth coordinate on V and $\eta/|\zeta| = q\langle\zeta\rangle/|\zeta|$. This gives (i). In (ii), ζ is a coordinate on U and $q = \eta/\langle\zeta\rangle$ is large, so $1/|q| = \langle\zeta\rangle/|\eta|$ and $q/|q| = \eta/|\eta|$ are coordinates on $\overline{V}$ near its boundary; multiplying by the positive smooth function $1/\langle\zeta\rangle$ gives (ii). Near the corner both $|\zeta|$ and $|q|$ are large and $1/|q| = \langle\zeta\rangle/|\eta|$ is a smooth positive multiple of $|\zeta|/|\eta|$, giving (iii). □

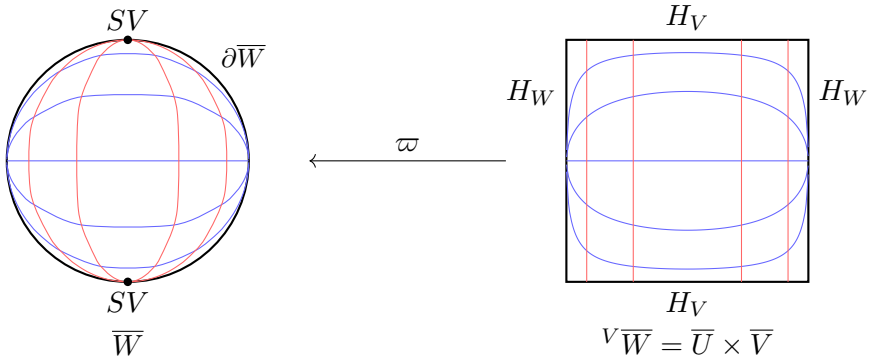


Figure 12.1: The relative compactification for $U = \mathbb{R}_\zeta$ (horizontal) and $V = \mathbb{R}_\eta$ (vertical). The lines $\eta = \text{const}$ (blue) all end at the two points $\zeta = \pm\infty$, $\eta/\langle\zeta\rangle = 0$ of H_W; the lines $\zeta = \text{const}$ (red), which in $\overline{W}$ all end at the two points of SV, end at distinct points of the front faces H_V. The blow-down map ϖ collapses each component of H_V to a point of SV.

Proposition 12.4.4 (Invariance)**.** *Let* $\mathrm{GL}(W, V) = \{L \in \mathrm{GL}(W); L(V) = V\}$. *The action of* $\mathrm{GL}(W, V)$ *on W and the action of W on itself by translations extend to smooth actions*

$$\mathrm{GL}(W, V) \times {}^{V}\overline{W} \longrightarrow {}^{V}\overline{W}, \qquad W \times {}^{V}\overline{W} \longrightarrow {}^{V}\overline{W}$$

by diffeomorphisms preserving H_W and H_V. Consequently ${}^{V}\overline{W}$, with the embedding ι_V, is independent of the choice of U and of the norms: two choices give compactifications related by a unique diffeomorphism extending the identity of W.

Proof. In terms of $W = U \oplus V$ an element $L \in \mathrm{GL}(W, V)$ is block lower triangular, $L(\zeta, \eta) = (A\zeta, C\zeta + D\eta)$ with $A \in \mathrm{GL}(U)$, $D \in \mathrm{GL}(V)$, $C \in \mathrm{Hom}(U, V)$. In the coordinates (ζ, q) of the interior, L becomes

$$(\zeta, q) \longmapsto \Big(A\zeta,\ g(\zeta)\Big(Dq + C\frac{\zeta}{\langle\zeta\rangle}\Big)\Big), \qquad g(\zeta) = \frac{\langle\zeta\rangle}{\langle A\zeta\rangle}. \tag{12.4.1}$$

Here $\zeta \mapsto A\zeta$ extends to a diffeomorphism of $\overline{U}$ by Lemma 12.4.1, and g and $\zeta/\langle\zeta\rangle$ are smooth on $\overline{U}$ (they are classical symbols of order 0), with $g > 0$. So for fixed $\zeta \in \overline{U}$ the second component is an invertible affine map of V whose coefficients are smooth on $\overline{U}$, and Lemma 12.4.1 with $P = \overline{U}$ shows that (12.4.1) extends to a diffeomorphism of $\overline{U} \times \overline{V}$; the same argument with $P = \mathrm{GL}(W,V) \times \overline{U}$ gives joint smoothness of the action. For a translation by (a,c) the lifted map is $(\zeta, q) \mapsto (\zeta + a,\ \tilde{g}(\zeta)(q + c/\langle\zeta\rangle))$ with $\tilde{g} = \langle\zeta\rangle/\langle\zeta + a\rangle$, and the same argument applies. Finally, changing the complement U and the norms amounts to composing ι_V with an element of $\mathrm{GL}(W,V)$. □

From (12.4.1) one sees that the boundary faces have some natural structure. As $|q| \to \infty$ the direction of the second component tends to the direction of Dq, so on H_V the lifted action is $(\zeta, \theta) \mapsto (A\zeta, D\theta/|D\theta|)$; thus there is a natural product decomposition

$$H_V = \overline{W/V} \times SV. \tag{12.4.2}$$

On the other hand on H_W the action is $(\omega, q) \mapsto (A\omega/|A\omega|, |\omega|_A^{-1}(Dq + C\omega))$, with $|\omega|_A = |A\omega|$; it covers the action of $\mathrm{GL}(W/V)$ on $S(W/V)$, but the off-diagonal part C acts on the fibres by translations. Thus H_W is naturally a fibration

$$\overline{V} \text{—} H_W \longrightarrow S(W/V), \tag{12.4.3}$$

trivial but not naturally a product, whose fibres are radial compactifications of affine spaces modelled on V.

Proposition 12.4.5. (i) *The identity of W extends to a smooth surjective map $\varpi : {}^V\overline{W} \longrightarrow \overline{W}$, the* blow-down map*, with $\varpi^*\rho = \rho_W\rho_V$ for $\rho = \langle\zeta,\eta\rangle^{-1}$. It restricts to a diffeomorphism of $H_W \setminus H_V$ onto $SW \setminus SV$ and maps H_V onto $SV = \partial\overline{V} \subset \partial\overline{W}$, by the projection $\overline{W/V} \times SV \longrightarrow SV$.*

(ii) *The projection $W \longrightarrow W/V$ extends to the fibration ${}^V\overline{W} \longrightarrow \overline{W/V}$, with fibre $\overline{V}$, given in the product description by projection onto the first factor; the inclusion $V \hookrightarrow W$ extends to the embedding $\overline{V} \ni \eta \longmapsto (0,\eta)$ as the fibre over 0.*

Proof. In the half-sphere description of $\overline{W}$,

$$\varpi(\zeta, q) = \Big(\frac{1}{\langle\zeta,\eta\rangle}, \frac{\zeta}{\langle\zeta,\eta\rangle}, \frac{\eta}{\langle\zeta,\eta\rangle}\Big) = \Big(\rho_W\rho_V,\ \rho_V\frac{\zeta}{\langle\zeta\rangle},\ \frac{q}{\langle q\rangle}\Big),$$

which is smooth on $\overline{U} \times \overline{V}$. On H_V ($\rho_V = 0$) the image is $(0,0,\theta)$, $\theta = \lim q/|q| \in SV$. On $H_W \setminus H_V$ ($\rho_W = 0$, $\rho_V > 0$) the image is $(0, \rho_V\omega, q/\langle q\rangle)$ with $\omega = \zeta/|\zeta|$, $\rho_V = \langle q\rangle^{-1}$, a point of SW with non-zero component in U; conversely a point $(\hat\zeta, \hat\eta) \in SW$ with $\hat\zeta \neq 0$ is the image of $\omega = \hat\zeta/|\hat\zeta|$, $q = \hat\eta/|\hat\zeta|$, and this inverse is smooth. Part (ii) is immediate from the definition. □

In fact ϖ identifies ${}^V\overline{W}$ with the manifold obtained from $\overline{W}$ by introducing polar coordinates around the submanifold SV of its boundary, the *blow-up* $[\overline{W}; SV]$ (Exercise 12.5); H_V is the 'front face' produced by the blow-up and H_W is the lift of the original boundary.

Remark 12.4.6. By Proposition 12.4.4 the construction applies fibrewise to a real vector bundle $W \longrightarrow X$ with a subbundle V: the transition maps of local trivializations in which V is trivialized as a subbundle take values in $\mathrm{GL}(W,V)$, so the fibrewise relative compactifications form a compact (if X is compact) manifold with corners ${}^V\overline{W}$, fibred over X, with boundary faces H_W and H_V and defining functions ρ_W, ρ_V well defined up to smooth positive factors. Moreover $H_V = \overline{W/V} \times_X SV$, and H_W fibres over $S(W/V)$.

12.5 Product-type symbols

We keep the notation $W = U \oplus V$, $w = (\zeta, \eta)$, $q = \eta/\langle\zeta\rangle$ of the previous section, and let $\Omega \subset \mathbb{R}^p$ be an open set of parameters. Estimates are required to hold uniformly for x in compact subsets of Ω. For $M, M' \in \mathbb{R}$ define the weight

$$w_{M,M'}(\zeta,\eta) = \langle q\rangle^{M}\langle\zeta\rangle^{M'} = \langle\zeta,\eta\rangle^{M}\langle\zeta\rangle^{M'-M} = \rho_V^{-M}\rho_W^{-M'}.$$

The first order M is the order at the front face H_V and the second, M', the order at the main face H_W.

Definition 12.5.1. The space $S^{M,M'}(\Omega; W, V)$ of *product-type symbols* consists of the $a \in \mathcal{C}^\infty(\Omega \times W)$ such that for all multi-indices

$$|\partial_x^\gamma\partial_\zeta^\alpha\partial_\eta^\beta a(x,\zeta,\eta)| \le C\langle\zeta\rangle^{M'-M-|\alpha|}\langle\zeta,\eta\rangle^{M-|\beta|} = C\, w_{M,M'}(\zeta,\eta)\langle\zeta\rangle^{-|\alpha|}\langle\zeta,\eta\rangle^{-|\beta|}. \quad (12.5.1)$$

The subspace of *classical* (polyhomogeneous) product-type symbols is

$$S^{M,M'}_{\mathrm{ph}}(\Omega; W, V) = \{a;\ a = \rho_V^{-M}\rho_W^{-M'}b \text{ on } \Omega\times W,\ b \in \mathcal{C}^\infty(\Omega \times {}^V\overline{W})\}.$$

By Proposition 12.4.4, $S^{M,M'}_{\mathrm{ph}}$ does not depend on the choices made, since the defining functions change by smooth positive factors; we shall see in Proposition 12.5.3(vii) that the same is true of $S^{M,M'}$. Thus $\langle\zeta,\eta\rangle^\mu \in S^{\mu,\mu}_{\mathrm{ph}}$, while $\langle\zeta\rangle^\mu \in S^{0,\mu}_{\mathrm{ph}}$ is constant along V. Each ∂_ζ-derivative improves the estimate by $\langle\zeta\rangle^{-1}$, each ∂_η-derivative by $\langle\zeta,\eta\rangle^{-1}$.

Lemma 12.5.2 (Peetre's inequality)**.** *For all M, M' there is C such that for all $w, \theta \in W$*

$$w_{M,M'}(w+\theta) \le C\, w_{M,M'}(w)\langle\theta\rangle^{|M|+|M'-M|}.$$

Proof. Apply Peetre's inequality $\langle x+y\rangle^s \le 2^{|s|/2}\langle x\rangle^s\langle y\rangle^{|s|}$ to each factor of $\langle\zeta,\eta\rangle^M\langle\zeta\rangle^{M'-M}$. □

Proposition 12.5.3. (i) *$a \in S^{M,M'}(\Omega; W, V)$ if and only if $c(x,\zeta,q) = a(x,\zeta,\langle\zeta\rangle q)$ satisfies the product estimates*

$$|\partial_x^\gamma\partial_\zeta^\alpha\partial_q^\beta c| \le C\langle\zeta\rangle^{M'-|\alpha|}\langle q\rangle^{M-|\beta|}.$$

(ii) *$S^{M,M'}_{\mathrm{ph}} \subset S^{M,M'}$.*

(iii) *$S^{M_1,M_1'} \cdot S^{M_2,M_2'} \subset S^{M_1+M_2,M_1'+M_2'}$, and the same for classical symbols.*

(iv) *$\partial_\zeta : S^{M,M'} \longrightarrow S^{M,M'-1}$ and $\partial_\eta : S^{M,M'} \longrightarrow S^{M-1,M'-1}$, and the same for classical symbols. Moreover $S^{M,M'} \subset S^{M+1,M'} \cap S^{M,M'+1}$ and $S^{M-1,M'-1} \subset S^{M,M'-1}$.*

(v) *The standard symbols on W are included: $S^\mu(\Omega; W) \subset S^{\mu,\mu}(\Omega; W, V)$ and $S^\mu_{\mathrm{ph}}(\Omega; W) \subset S^{\mu,\mu}_{\mathrm{ph}}(\Omega; W, V)$; more precisely $S^\mu_{\mathrm{ph}}(\Omega; W) = \varpi^*\big(\rho^{-\mu}\mathcal{C}^\infty(\Omega \times \overline{W})\big)$.*

(vi) *Symbols on W/V lift: if $a \in S^\mu(\Omega; W/V)$ then $a(x,\zeta) \in S^{0,\mu}(\Omega; W, V)$, and similarly for classical symbols.*

(vii) *If $L \in \mathcal{C}^\infty(\Omega; \mathrm{GL}(W, V))$ and $a \in S^{M,M'}$ then $a(x, L(x)w) \in S^{M,M'}$, and the same for classical symbols.*

(viii) *The residual spaces are*

$$S^{M,-\infty} = \bigcap_{M'} S^{M,M'} = \mathcal{S}(U; S^M(\Omega; V)), \quad S^{M,-\infty}_{\mathrm{ph}} = \bigcap_{M'} S^{M,M'}_{\mathrm{ph}} = \mathcal{S}(U; S^M_{\mathrm{ph}}(\Omega; V)),$$
$$S^{-\infty,-\infty} = \bigcap_{M,M'} S^{M,M'} = \bigcap_{M,M'} S^{M,M'}_{\mathrm{ph}} = \mathcal{C}^\infty(\Omega; \mathcal{S}(W)),$$

where $\mathcal{S}(U; S^M(\Omega; V))$ consists of the a with $|\partial_x^\gamma \partial_\zeta^\alpha \partial_\eta^\beta a| \leq C_N \langle\zeta\rangle^{-N} \langle\eta\rangle^{M-|\beta|}$ for all N, and the classical version requires in addition that $a(x, \zeta, \cdot)$ be classical, with all terms of the expansion rapidly decreasing in ζ.

Proof. (i) Consider the vector fields $L_j = \langle\zeta\rangle\partial_{\zeta_j}$ and $Q_i = \langle\zeta, \eta\rangle\partial_{\eta_i}$ in the coordinates (ζ, η), and $L'_j = \langle\zeta\rangle\partial_{\zeta_j}$, $Q'_i = \langle q\rangle\partial_{q_i}$ in the coordinates (ζ, q). Since $\partial_q = \langle\zeta\rangle\partial_\eta$ and $\langle q\rangle\langle\zeta\rangle = \langle\zeta, \eta\rangle$,

$$Q'_i = Q_i, \qquad L'_j = L_j + \frac{\zeta_j}{\langle\zeta\rangle} \sum_i \frac{q_i}{\langle q\rangle} Q_i.$$

The functions $\zeta_j/\langle\zeta\rangle$ and $q_i/\langle q\rangle$ are annihilated by Q', respectively L', and any number of applications of L', respectively Q', to them gives bounded functions of the same type. Hence the products of L's and Q's, and the products of L''s and Q''s, are linear combinations of each other with bounded coefficients all of whose L', Q' derivatives are bounded. By the usual induction, the estimates (12.5.1) are equivalent to $|Pa| \leq C w_{M,M'}$ for all products P of the L_j, Q_i and ∂_x (using $\partial_\zeta^\alpha = \langle\zeta\rangle^{-|\alpha|} L^\alpha +$ lower order terms with coefficients which are bounded symbols of negative order, and similarly for η), and the product estimates are equivalent to $|P'c| \leq C w_{M,M'}$ for all products P' of the L'_j, Q'_i and ∂_x. This proves (i).

(ii) If $b \in \mathcal{C}^\infty(\Omega \times \overline{U} \times \overline{V})$ then L'_j and Q'_i lift to smooth vector fields on $\overline{U} \times \overline{V}$ (they are tangent to the boundary), so $P'b$ is bounded for every product P'. The weight $\langle\zeta\rangle^{M'} \langle q\rangle^M$ is itself a product-type symbol in the sense of (i), so (ii) follows from (i) and Leibniz' formula.

(iii) and (iv) follow directly from (12.5.1) and Leibniz' formula, and for classical symbols from the fact that L'_j, Q'_i, and hence $\rho_W^{-1}\partial_{\zeta_j} = L_j$ and $(\rho_W \rho_V)^{-1}\partial_{\eta_i} = Q_i$ (in the original coordinates), are smooth vector fields on ${}^V\overline{W}$ tangent to the boundary; the inclusions in (iv) hold because $\langle q\rangle \geq 1$ and $\langle\zeta\rangle \geq 1$.

(v) For $a \in S^\mu(W)$, $|\partial_\zeta^\alpha \partial_\eta^\beta a| \leq C\langle\zeta, \eta\rangle^{\mu-|\alpha|-|\beta|} \leq C\langle\zeta\rangle^{-|\alpha|}\langle\zeta, \eta\rangle^{\mu-|\beta|}$, which is (12.5.1) with $M = M' = \mu$. The classical statement follows from Proposition 12.4.5(i) since $\varpi^*\rho = \rho_V \rho_W$.

(vi) is clear from (i), since c is independent of q; for classical symbols use Proposition 12.4.5(ii).

(vii) If $L(x)(\zeta, \eta) = (A\zeta, C\zeta + D\eta)$ then by the chain rule $\partial_\eta(a \circ L) = D^t(\partial_\eta a) \circ L$ and $\partial_\zeta(a \circ L) = A^t(\partial_\zeta a) \circ L + C^t(\partial_\eta a) \circ L$, and x-derivatives produce similar terms with coefficients $\partial_x A$, etc. Using (iv) and the fact that $w_{M,M'} \circ L$ is bounded above and below by multiples of $w_{M,M'}$ (since $\langle A\zeta\rangle \simeq \langle\zeta\rangle$ and $\langle L(\zeta, \eta)\rangle \simeq \langle\zeta, \eta\rangle$, locally uniformly in x), the estimates follow by induction. The classical case follows from the smoothness of the action in Proposition 12.4.4.

(viii) Since $\langle\zeta\rangle^{-N-M}\langle\zeta, \eta\rangle^{M-|\beta|}$ lies between $\langle\zeta\rangle^{-N-M\mp|M-|\beta||}\langle\eta\rangle^{M-|\beta|}$ (by Peetre's inequality in the form $\langle\eta\rangle^s\langle\zeta\rangle^{-|s|} \leq \langle\zeta, \eta\rangle^s \leq \langle\eta\rangle^s\langle\zeta\rangle^{|s|}$), the first identity is clear. For the classical case, $a \in S^{M,M'}_{\mathrm{ph}}$ for all M' means that $a = \langle q\rangle^M b(\zeta, q)$ where b is smooth on $\overline{U} \times \overline{V}$ and vanishes to infinite order at $H_W = SU \times \overline{V}$, i.e. b is a Schwartz function of ζ with values in the Fréchet space $\mathcal{C}^\infty(\overline{V})$. For fixed ζ, $\eta \mapsto a(\zeta, \eta)$ is obtained from $\langle q\rangle^M b(\zeta, q) \in \rho_V^{-M}\mathcal{C}^\infty(\overline{V}) = S^M_{\mathrm{ph}}(V)$ by the dilation $q = \eta/\langle\zeta\rangle$. Dilations by $\lambda \in (0, 1]$ act on $S^M_{\mathrm{ph}}(V)$ with each seminorm bounded by $C\lambda^{-K}$ times a seminorm, so

$a \in \mathcal{S}(U; S^M_{\mathrm{ph}}(V))$; the converse is proved in the same way using the inverse dilation. Finally, taking $M = M' = -N$ gives $w_{M,M'} = \langle \zeta, \eta \rangle^{-N}$, so $S^{-\infty,-\infty}$ consists of Schwartz functions; conversely $w_{M,M'} \geq \langle \zeta, \eta \rangle^{-|M|-|M'-M|}$, so Schwartz functions lie in every $S^{M,M'}$, and they are classical, being $\rho^\infty \mathcal{C}^\infty(\overline{W})$ lifted by ϖ. □

Lemma 12.5.4 (Translations)**.** *For $\theta \in W$ the translation $a \longmapsto a(\cdot+\theta)$ maps $S^{M,M'}_{\mathrm{ph}}(\Omega; W, V)$ and $S^{M,M'}(\Omega; W, V)$ into themselves; each seminorm of $a(\cdot + \theta)$ is bounded by $C\langle\theta\rangle^K$ times a seminorm of a.*

Proof. For $S^{M,M'}$ this is Lemma 12.5.2 applied to (12.5.1). For classical symbols, the translation lifts to a diffeomorphism T_θ of ${}^V\overline{W}$ depending smoothly on θ (Proposition 12.4.4) and preserving the boundary faces, so $T_\theta^*(\rho_V^{-M}\rho_W^{-M'})/(\rho_V^{-M}\rho_W^{-M'})$ is smooth and positive. From the explicit form $(\zeta, q) \mapsto (\zeta + \theta_1, \langle\zeta\rangle\langle\zeta+\theta_1\rangle^{-1}(q + \theta_2/\langle\zeta\rangle))$ one checks, using the coordinates of Proposition 12.4.3, that all derivatives of T_θ and of this quotient are bounded by powers of $\langle\theta\rangle$. □

Proposition 12.5.5 (Asymptotic completeness)**.** (i) *If $a_j \in S^{M,M'-j}_{\mathrm{ph}}(\Omega; W, V)$ for $j \in \mathbb{N}_0$, there is $a \in S^{M,M'}_{\mathrm{ph}}$ with $a - \sum_{j<N} a_j \in S^{M,M'-N}_{\mathrm{ph}}$ for all N; a is unique modulo $S^{M,-\infty}_{\mathrm{ph}}$.*

(ii) *If $a_j \in S^{M-j,M'}_{\mathrm{ph}}$, there is $a \in S^{M,M'}_{\mathrm{ph}}$ with $a - \sum_{j<N} a_j \in S^{M-N,M'}_{\mathrm{ph}}$ for all N; a is unique modulo $S^{-\infty,M'}_{\mathrm{ph}} = \bigcap_j S^{M-j,M'}_{\mathrm{ph}}$.*

(iii) *If $a_j \in S^{M-j,M'-j}_{\mathrm{ph}}$, there is $a \in S^{M,M'}_{\mathrm{ph}}$ with $a - \sum_{j<N} a_j \in S^{M-N,M'-N}_{\mathrm{ph}}$ for all N; a is unique modulo $S^{-\infty,-\infty}$.*

The same statements hold for the spaces $S^{M,M'}$.

Proof. Write $a_j = \rho_V^{-M}\rho_W^{-M'}\rho_W^j b_j$ in case (i). Near H_W the factor $\overline{U}$ has a collar $[0,1)_{\rho_W} \times SU$, so ${}^V\overline{W}$ has a collar $[0,1) \times H_W$, and Borel's lemma (with parameters in $\Omega \times H_W$) gives $b \in \mathcal{C}^\infty(\Omega \times {}^V\overline{W})$ with $b - \sum_{j<N} \rho_W^j b_j \in \rho_W^N \mathcal{C}^\infty$ for all N: take $b = \sum_j \chi(\rho_W/\epsilon_j)\rho_W^j b_j$ with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0,1))$ equal to 1 near 0 and $\epsilon_j \downarrow 0$ so fast that the j-th term has C^j norm on the j-th set of an exhaustion of Ω at most 2^{-j}. Then $a = \rho_V^{-M}\rho_W^{-M'}b$. Uniqueness holds since the difference of two choices vanishes to infinite order at H_W. Case (ii) is the same with ρ_V in place of ρ_W, and case (iii) the same with $\rho = \rho_V\rho_W$: since $|\partial^\alpha(\chi(\rho/\epsilon))| \leq C_\alpha \epsilon^{-|\alpha|}$ on $\{\rho \leq \epsilon\}$, the C^k norm of $\rho^j\chi(\rho/\epsilon)b_j$ is $O(\epsilon^{j-k})$, and the construction goes through. For non-classical symbols one uses instead the usual cut-offs $1 - \chi(\langle\zeta\rangle/R_j)$, $1 - \chi(\langle q\rangle/R_j)$, respectively $1 - \chi(\langle\zeta,\eta\rangle/R_j)$, with $R_j \to \infty$ rapidly, exactly as for standard symbols. □

12.5.1 Symbol maps in the model case

For $a \in S^{M,M'}_{\mathrm{ph}}(\Omega; W, V)$, $a = \rho_V^{-M}\rho_W^{-M'}b$, the restrictions of b to the two boundary faces are given by limits. For $w = (\zeta, \eta)$ with $\zeta \neq 0$ (that is, $w \notin V$) and for $\eta \in V \setminus 0$, put

$$\sigma_W(a)(x, w) = \lim_{t\to\infty} t^{-M'} a(x, tw), \qquad \sigma_V(a)(x, \zeta, \eta) = \lim_{t\to\infty} t^{-M} a(x, \zeta, t\eta). \tag{12.5.2}$$

Using $\langle tw\rangle \simeq t|w|$ and $tq \to \infty$ one finds

$$\sigma_W(a)(x, w) = |\zeta|^{M'-M}|w|^M b(x, \tfrac{\zeta}{|\zeta|}, \tfrac{\eta}{|\zeta|}), \qquad \sigma_V(a)(x, \zeta, \eta) = \langle\zeta\rangle^{M'-M}|\eta|^M b(x, \zeta, \tfrac{\eta}{|\eta|}),$$

where $(\zeta/|\zeta|, \eta/|\zeta|) \in SU \times V$ is a point of the interior of H_W and $(\zeta, \eta/|\eta|)$ a point of $H_V = \overline{U} \times SV$, and $|w|$ is the Euclidean norm on $W = U \oplus V$. Thus $\sigma_W(a)$ is homogeneous

of degree M' on $W \setminus V$, while for each $\eta \neq 0$, $\sigma_V(a)(x,\cdot,\eta)$ is a classical symbol of order $M'-M$ on $U = W/V$, and $\sigma_V(a)$ is homogeneous of degree M in η. Let $\Sigma_W^{M,M'}$ and $\Sigma_V^{M,M'}$ denote the spaces of such functions, i.e. of the functions s of this form with b a smooth function on $\Omega \times H_W$, respectively $\Omega \times H_V$.

Lemma 12.5.6. *The maps σ_W and σ_V give short exact sequences*

$$S^{M,M'-1}_{\mathrm{ph}} \hookrightarrow S^{M,M'}_{\mathrm{ph}} \xrightarrow{\sigma_W} \Sigma_W^{M,M'}, \qquad S^{M-1,M'}_{\mathrm{ph}} \hookrightarrow S^{M,M'}_{\mathrm{ph}} \xrightarrow{\sigma_V} \Sigma_V^{M,M'},$$
$$S^{M-1,M'-1}_{\mathrm{ph}} \hookrightarrow S^{M,M'}_{\mathrm{ph}} \xrightarrow{(\sigma_W,\sigma_V)} \{(s_W,s_V) \in \Sigma_W^{M,M'} \times \Sigma_V^{M,M'};\ (12.5.3)\ \textit{holds}\},$$

where the compatibility condition, which expresses the equality of the restrictions of $b|_{H_W}$ and $b|_{H_V}$ to the corner, is

$$\lim_{t\to\infty} t^{-M} s_W(x,\zeta+t\eta) = \lim_{r\to\infty} r^{M-M'} s_V(x,r\zeta,\eta), \qquad \zeta \in U\setminus 0,\ \eta \in V\setminus 0; \tag{12.5.3}$$

that is, the leading homogeneous term of s_W at V is the principal symbol of the classical symbol $s_V(x,\cdot,\eta)$ on U.

Proof. $b|_{H_W} = 0$ if and only if $b \in \rho_W \mathcal{C}^\infty$, which gives the null space of σ_W, and similarly for σ_V. If b vanishes on both faces then $b = \rho_V b'$ with b' vanishing on $H_W \setminus H_V$, hence by continuity on H_W, so $b = \rho_V\rho_W b''$. For surjectivity use the product structure: if $c_V \in \mathcal{C}^\infty(\Omega \times \overline{U} \times SV)$ and $c_W \in \mathcal{C}^\infty(\Omega \times SU \times \overline{V})$ agree on $SU \times SV$, let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0,1))$ be 1 near 0, set $\tilde{c}_V(x,\zeta,q) = \chi(\rho_V)c_V(x,\zeta,q/|q|)$ and

$$b = \tilde{c}_V + \chi(\rho_W)(c_W - \tilde{c}_V|_{H_W})(x,\zeta/|\zeta|,q).$$

Then $b|_{H_V} = c_V$, since $c_W - \tilde{c}_V|_{H_W}$ vanishes at the corner, and $b|_{H_W} = c_W$. Taking $c_W = 0$ or $c_V = 0$ (and dropping the compatibility) gives surjectivity of the separate maps. The computation of the limits above shows that the compatibility of $b|_{H_W}$ and $b|_{H_V}$ at the corner is (12.5.3). □

12.6 Product-type conormal distributions

Let $Z \subset Y \subset X$ be closed embedded submanifolds of a manifold X of dimension n, with $\dim Z = d$, $\dim Y = d+k$, so that Z has codimension k in Y and Y has codimension $l = n-d-k$ in X. (In this section Z denotes a submanifold, not the fibre of a fibration.) We define spaces $I^{m,m'}(X,Y,Z;E)$ of distributions which are conormal to Y of order m and, in a product sense, conormal to Z of order m'. Thus the *first* index refers to the larger submanifold Y.

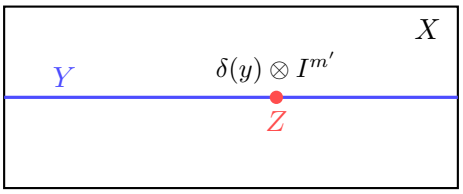


Figure 12.2: The nested pair $Z \subset Y \subset X$. A product-type conormal distribution is conormal to Y everywhere (for instance a δ-function in the directions normal to Y) and has an additional conormal singularity along Z. For the fibrewise and product-type operators, $X = M^2$, $Y = M^2_\phi$ and $Z = \mathrm{Diag}_M$.

12.6.1 The model case

Let $X = \mathbb{R}^n = \mathbb{R}^d_w \times \mathbb{R}^k_z \times \mathbb{R}^l_y$, $Y = \{y = 0\}$, $Z = \{z = 0, y = 0\}$. The conormal space to Z at each point is $W = \mathbb{R}^k_\zeta \times \mathbb{R}^l_\eta$, the dual of the normal variables (z, y), and the conormal space to Y is its subspace $V = \{\zeta = 0\}$, the annihilator of the z-directions; the quotient $W/V \cong \mathbb{R}^k_\zeta$ is the conormal space of Z in Y. For a compactly supported distribution u we write $\hat{u}(w, \zeta, \eta) = \int e^{-i(z\cdot\zeta + y\cdot\eta)} u(w, z, y)\, dz\, dy$ for the partial Fourier transform in the normal variables. Set

$$M = m + \frac{n}{4} - \frac{l}{2}, \qquad M' = m' + \frac{n}{4} - \frac{k+l}{2}; \tag{12.6.1}$$

these are the symbol orders corresponding, with the normalization of Chapter 4, to conormal order m at Y and m' at Z. Recall from Chapter 4 that, in these coordinates, $u \in I^m(\Omega, Y)$ if and only if for each $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ the partial Fourier transform of χu in y alone is a classical symbol of order M in η, depending smoothly on (w, z), and $u \in I^{m'}(\Omega, Z)$ if and only if each $\widehat{\chi u}$ lies in $S^{M'}_{\mathrm{ph}}(\mathbb{R}^d; W)$.

Definition 12.6.1. For $\Omega \subset \mathbb{R}^n$ open, $I^{m,m'}(\Omega, Y, Z)$ is the space of $u \in \mathcal{C}^{-\infty}(\Omega)$ such that for every $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, $\widehat{\chi u} \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^d; W, V)$, with M, M' given by (12.6.1).

Thus, for $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, $\chi u = \int e^{i(z\cdot\zeta + y\cdot\eta)} a(w, \zeta, \eta) đ\zeta đ\eta$ (an oscillatory integral) with $a = \widehat{\chi u}$ a product-type symbol, compactly supported in w. The definition is local: if $u \in I^{m,m'}(\Omega, Y, Z)$ and $\psi \in \mathcal{C}^\infty(\Omega)$ then $\psi u \in I^{m,m'}(\Omega, Y, Z)$, and membership can be checked using χ's from a partition of unity. The basic tool for handling these distributions is the following reduction of amplitudes.

Lemma 12.6.2 (Amplitudes)**.** *Let $r \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^n; W, V)$ have support in $K \times W$ for a compact $K \subset \mathbb{R}^n$, and let*

$$v(x) = \int e^{i(z\cdot\zeta + y\cdot\eta)} r(x, \zeta, \eta) đ\zeta đ\eta, \quad x = (w, z, y),$$

as an oscillatory integral. Then $\hat{v} \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^d; W, V)$ and, with $\xi = (\zeta, \eta)$,

$$\hat{v}(w, \xi) \sim \sum_\gamma \frac{1}{\gamma!} (-D_\xi)^\gamma \partial^\gamma_{(z,y)} r(w, 0, 0, \xi), \tag{12.6.2}$$

in the sense that the difference between $\hat{v}$ and the sum over $|\gamma| < N$ is in $S^{M,M'-N}$ for every N. The terms with $\gamma = (\alpha, 0)$ lie in $S^{M,M'-|\alpha|}_{\mathrm{ph}}$, those with $\gamma = (\alpha, \beta)$, $\beta \neq 0$, in $S^{M-1,M'-|\gamma|}_{\mathrm{ph}}$. In particular $v \in I^{m,m'}(\mathbb{R}^n, Y, Z)$.

Proof. Let $\tilde{r}(w, \theta; \xi) = \int e^{-i(z,y)\cdot\theta} r(w, z, y, \xi)\, dz\, dy$ be the Fourier transform of r in (z, y); since r has compact support in x and satisfies the estimates (12.5.1) with all x-derivatives, integration by parts shows that for each L the family $\langle\theta\rangle^L \tilde{r}(\cdot, \theta; \cdot)$, $\theta \in \mathbb{R}^{k+l}$, is bounded in $S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^d; W, V)$; that is, $\tilde{r}$ is rapidly decreasing in θ with values in $S^{M,M'}_{\mathrm{ph}}$. Inserting the definition of v and integrating first in (z, y) (which is justified by inserting a cut-off $\chi(\epsilon\xi)$ and letting $\epsilon \to 0$),

$$\hat{v}(w, \xi) = \int \tilde{r}(w, \xi - \xi'; \xi') đ\xi' = \int \tilde{r}(w, \theta; \xi - \theta) đ\theta.$$

By Lemma 12.5.4 the integrand is, for each θ, in $S^{M,M'}_{\mathrm{ph}}$ with seminorms bounded by $C_L\langle\theta\rangle^{K-L}$, so the integral converges in $S^{M,M'}_{\mathrm{ph}}$ and $\hat{v}$ is a classical product-type symbol. For the expansion, apply Taylor's formula in the last variable:

$$\tilde{r}(w,\theta;\xi-\theta)=\sum_{|\gamma|<N}\frac{(-\theta)^\gamma}{\gamma!}\partial_\xi^\gamma\tilde{r}(w,\theta;\xi)+\sum_{|\gamma|=N}\frac{N(-\theta)^\gamma}{\gamma!}\int_0^1(1-s)^{N-1}\partial_\xi^\gamma\tilde{r}(w,\theta;\xi-s\theta)\,ds.$$

Since $\int\theta^\gamma\tilde{r}(w,\theta;\xi)d\!\!{}^-\theta=D^\gamma_{(z,y)}r(w,0,0,\xi)$, the terms with $|\gamma|<N$ give the sum in (12.6.2), noting $(-\partial_\xi)^\gamma D^\gamma=(-D_\xi)^\gamma\partial^\gamma$. By Proposition 12.5.3(iv), $\partial_\xi^\gamma r\in S^{M,M'-N}$ when $|\gamma|=N$, so by Lemma 12.5.2 the remainder is bounded by $C\int\langle\theta\rangle^{-L}w_{M,M'-N}(\xi-s\theta)\,d\!\!{}^-\theta\le C'w_{M,M'-N}(\xi)$, and the same argument applies to its derivatives. The orders of the individual terms follow from Proposition 12.5.3(iv), since $(-D_\xi)^\gamma$ contains at least one η-derivative if $\beta\neq0$. □

Proposition 12.6.3. *Let $\Omega\subset\mathbb{R}^n$ be open.*

(i) $I^m(\Omega,Y)\subset I^{m,-\infty}(\Omega,Y,Z)=\bigcap_{m'}I^{m,m'}(\Omega,Y,Z)$, *with equality, and* $I^{m'}(\Omega,Z)\subset I^{m'-k/2,m'}(\Omega,Y,Z)$.

(ii) $\bigcap_j I^{m-j,m'-j}(\Omega,Y,Z)=\mathcal{C}^\infty(\Omega)$.

(iii) *If* $u\in I^{m,m'}(\Omega,Y,Z)$ *then* $u|_{\Omega\setminus Z}\in I^m(\Omega\setminus Z,Y\setminus Z)$; *in particular u is smooth off Y.*

(iv) *Multiplication by y_j maps $I^{m,m'}$ into $I^{m-1,m'-1}$, multiplication by z_j maps it into $I^{m,m'-1}$, and smooth vector fields tangent to both Y and Z map $I^{m,m'}$ into itself.*

Proof. (i) If $u\in I^m(\Omega,Y)$ and $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, the partial Fourier transform of χu in y is a classical symbol of order M in η which is smooth and compactly supported in (w,z); its Fourier transform in z is therefore in $\mathcal{S}(\mathbb{R}^k_\zeta;S^M_{\mathrm{ph}}(\mathbb{R}^d;\mathbb{R}^l))=S^{M,-\infty}_{\mathrm{ph}}$ by Proposition 12.5.3(viii). Conversely, if $\widehat{\chi u}\in S^{M,-\infty}_{\mathrm{ph}}$ then its inverse Fourier transform in ζ is smooth in (w,z) and a classical symbol in η, so $\chi u\in I^m(\Omega,Y)$. The second inclusion is Proposition 12.5.3(v): the symbol order M' of $I^{m'}(\Omega,Z)$ is the second order in (12.6.1), and the first order $M=M'$ corresponds to $m=m'-k/2$.

(ii) By Proposition 12.5.3(viii), $\widehat{\chi u}$ is Schwartz in ξ, so χu is smooth.

(iii) Points of $\Omega\setminus Z$ either lie off Y or have $z\neq0$. Let $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ and $a=\widehat{\chi u}$. Multiplication by y^β corresponds to applying $(i\partial_\eta)^\beta$ to a, which by Proposition 12.5.3(iv) gives an element of $S^{M-|\beta|,M'-|\beta|}$; since $w_{M-N,M'-N}\le\langle\xi\rangle^{-N+|M|+|M'-M|}$, $y^\beta\chi u\in C^j$ when $|\beta|$ is large compared to j, so χu is smooth where $y\neq0$.

Now consider $\{z\neq0\}$. The partial inverse Fourier transform $A(w,z,\eta)=\int e^{iz\cdot\zeta}a(w,\zeta,\eta)d\!\!{}^-\zeta$ satisfies, for $z\neq0$ and any N, $A=|z|^{-2N}\int e^{iz\cdot\zeta}(-\Delta_\zeta)^Na\,d\!\!{}^-\zeta$, and $(-\Delta_\zeta)^Na\in S^{M,M'-2N}$ satisfies $|\partial_\eta^\beta\partial_\zeta^\alpha\Delta_\zeta^Na|\le C\langle\zeta\rangle^{M'-M-2N-|\alpha|+|M-|\beta||}\langle\eta\rangle^{M-|\beta|}$ by Peetre's inequality. Taking N large, it follows that on $|z|>\epsilon$, A is smooth in (w,z) and satisfies symbol estimates of order M in η. To see that it is classical, write $a=\langle\zeta\rangle^{M'}\langle q\rangle^Mb$ and expand b at H_V: $b=\sum_{j<J}\rho_V^jb_j(w,\zeta,\eta/|\eta|)+\rho_V^Jb'_J$. Up to a term in $S^{M-J,M'}$, whose contribution to A on $|z|>\epsilon$ is $O(\langle\eta\rangle^{M-J})$ with all derivatives by the estimate just used, a is a sum of terms $(1-\chi(q))|\eta|^{M-j}\langle\zeta\rangle^{M'-M+j}b_j(w,\zeta,\hat\eta)$, $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(V)$ equal to 1 near 0 (absorbing smooth factors into the b_j). The cut-off $\chi(q)=\chi(\eta/\langle\zeta\rangle)$ is supported where $\langle\eta\rangle\le C\langle\zeta\rangle$, so after integrating by parts in ζ its contribution on $|z|>\epsilon$ is $O(\langle\eta\rangle^{-\infty})$; the remaining term contributes $|\eta|^{M-j}g_j(w,z,\hat\eta)$ with $g_j=\int e^{iz\cdot\zeta}\langle\zeta\rangle^{M'-M+j}b_j(w,\zeta,\hat\eta)d\!\!{}^-\zeta$, which is smooth for $z\neq0$. Hence A is a classical symbol of order M in η on $|z|>\epsilon$, and $\chi u=\int e^{iy\cdot\eta}A\,d\!\!{}^-\eta$ is conormal to Y there.

(iv) Multiplication by y_j and z_j corresponds to $i\partial_{\eta_j}$ and $i\partial_{\zeta_j}$ on the symbol, so this follows from Proposition 12.5.3(iv). The vector fields tangent to Y and Z are spanned over $\mathcal{C}^\infty$ by ∂_w, $z_i\partial_{z_j}$, $y_i\partial_{z_j}$ and $y_i\partial_{y_j}$, which correspond to ∂_w and to $-\partial_{\zeta_i}\zeta_j$, $-\partial_{\eta_i}\zeta_j$ and $-\partial_{\eta_i}\eta_j$; by Proposition 12.5.3 these preserve $S^{M,M'}_{\mathrm{ph}}$ ($\zeta_j \in S^{0,1}$, $\eta_j \in S^{1,1}$). Multiplication by smooth functions preserves the class by the definition. □

Proposition 12.6.4 (Invariance)**.** *Let $F : \Omega' \longrightarrow \Omega$ be a diffeomorphism between open subsets of $\mathbb{R}^n$ with $F(Y\cap\Omega') = Y\cap\Omega$ and $F(Z\cap\Omega') = Z\cap\Omega$. Then $F^* : I^{m,m'}(\Omega, Y, Z) \longrightarrow I^{m,m'}(\Omega', Y, Z)$. Moreover, near Z the symbol of F^*u is given, modulo $S^{M,M'-1}_{\mathrm{ph}}$, by*

$$\widehat{\chi F^*u}(w,\xi) \equiv \chi(w,0,0)\, a\big(F_w(w), (\Phi(w)^t)^{-1}\xi\big)\,|\det\Phi(w)|^{-1}, \tag{12.6.3}$$

where $a = \widehat{\psi u}$ for $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equal to 1 on $F(\operatorname{supp}\chi)$, $F_w(w)$ is the w-component of $F(w,0,0)$, and $\Phi(w)$ is the differential of F at $(w,0,0)$ acting on the normal variables (z,y).

Proof. By the localizability of the definition and Proposition 12.6.3(iii) together with the invariance of $I^m(\Omega, Y)$ under diffeomorphisms preserving Y (Chapter 4) and Proposition 12.6.3(i), it suffices to consider χF^*u where $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ has support in a small neighbourhood of a point of Z.

Write $F = (F_w, F_z, F_y)$. Since F_y vanishes on $Y = \{y = 0\}$ and F_z vanishes on $Z = \{z = y = 0\}$, Hadamard's lemma gives

$$F_z(x) = A(x)z + B(x)y, \qquad F_y(x) = C(x)y,$$

with A, B, C smooth matrices. At points of Z the differential of F is block triangular with diagonal blocks $\partial_w F_w$, A, C, so A and C are invertible near Z. Set $\Phi(x) = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$, so that $(F_z, F_y) = \Phi(x)(z,y)$. Then, with $a = \widehat{\psi u}$,

$$\begin{aligned}\chi F^*u(x) &= \chi(x)\int e^{i(F_z(x)\cdot\zeta + F_y(x)\cdot\eta)} a(F_w(x),\zeta,\eta)\,đ\zeta\,đ\eta \\ &= \int e^{i(z\cdot\zeta' + y\cdot\eta')}\chi(x)\, a\big(F_w(x), L(x)(\zeta',\eta')\big)\,|\det\Phi(x)|^{-1}đ\zeta'đ\eta',\end{aligned}$$

where $(\zeta',\eta') = \Phi(x)^t(\zeta,\eta) = (A^t\zeta, B^t\zeta + C^t\eta)$ and $L(x) = (\Phi(x)^t)^{-1}$; the change of variables is justified by inserting a cut-off $\chi(\epsilon\xi)$. The crucial point is that Φ^t, and hence $L(x)$, preserves $V = \{\zeta = 0\}$, so $L \in \mathcal{C}^\infty(\operatorname{supp}\chi; \mathrm{GL}(W,V))$. By Proposition 12.5.3(vii) the new amplitude $r(x,\xi') = \chi(x)a(F_w(x), L(x)\xi')|\det\Phi(x)|^{-1}$ lies in $S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^n; W, V)$ with compact support in x, so Lemma 12.6.2 shows that $\chi F^*u \in I^{m,m'}(\Omega', Y, Z)$; (12.6.3) is the leading term of (12.6.2). □

12.6.2 The general case

An *adapted chart* for $Z \subset Y \subset X$ near $p \in Z$ is a coordinate chart (w,z,y) in which $Y = \{y = 0\}$ and $Z = \{z = y = 0\}$; such charts exist near every point of Z (take coordinates on Y adapted to Z and extend them using a tubular neighbourhood of Y).

Definition 12.6.5. For a vector bundle E over X, $I^{m,m'}(X, Y, Z; E)$ is the space of $u \in \mathcal{C}^{-\infty}(X; E)$ such that $u|_{X\setminus Z} \in I^m(X\setminus Z, Y\setminus Z; E)$ and such that each point of Z has an adapted chart Ω, over which E is trivial, with $u|_\Omega \in I^{m,m'}(\Omega, Y, Z)\otimes\mathbb{C}^{\operatorname{rank} E}$.

By Proposition 12.6.4, if $u \in I^{m,m'}(X,Y,Z;E)$ then $u|_\Omega$ is of model form in *every* adapted chart; by Proposition 12.6.3(iii) the first condition is only a condition away from a neighbourhood of Z.

Proposition 12.6.6. *Let $Z \subset Y \subset X$ be as above, $k = \dim Y - \dim Z$, $l = \operatorname{codim} Y$.*

(i) *$I^{m,m'}(X,Y,Z;E)$ is a $\mathcal{C}^\infty(X)$-module, invariant under diffeomorphisms preserving Y and Z, and under differential operators generated by vector fields tangent to Y and Z; also $I^{m,m'} \subset I^{m+1,m'} \cap I^{m,m'+1}$.*

(ii) *$I^m(X,Y;E) = \bigcap_{m'} I^{m,m'}(X,Y,Z;E)$ and $I^{m'}(X,Z;E) \subset I^{m'-k/2,m'}(X,Y,Z;E)$.*

(iii) *$\bigcap_j I^{m-j,m'-j}(X,Y,Z;E) = \mathcal{C}^\infty(X;E)$.*

(iv) *Let $\Omega(N^*Y)$ be the density bundle of the conormal bundle. If $v \in I^{m''}(Y,Z;E|_Y \otimes \Omega(N^*Y))$ then $v\,\delta_Y$, defined by $\langle v\delta_Y, \psi\rangle = \langle v, \psi|_Y\rangle$ for test densities ψ on X, lies in $I^{(l-d-k)/4,\,m''+l/4}(X,Y,Z;E)$.*

(v) *Asymptotic completeness holds in the three forms of Proposition 12.5.5, modulo $I^{m,-\infty} = I^m(X,Y)$, modulo $I^{-\infty,m'} = \bigcap_j I^{m-j,m'}$, and modulo $\mathcal{C}^\infty(X;E)$ respectively.*

Proof. Parts (i)–(iii) follow from Proposition 12.6.3, Proposition 12.6.4 and the corresponding properties of $I^m(X,Y)$ away from Z. For (iv), in an adapted chart in which $\psi|_Y = \psi(w,z,0)|dw\,dz|$ (absorbing $|dy|$ into the trivialization of $\Omega(N^*Y)$), $v\delta_Y = v(w,z)\delta(y)$, so $\widehat{\chi v\delta_Y}(w,\zeta,\eta) = \widehat{\chi v}(w,\zeta)$ is independent of η and is a classical symbol of order $\mu = m'' + \frac{d+k}{4} - \frac{k}{2}$ in ζ. By Proposition 12.5.3(vi) it is in $S^{0,\mu}_{\mathrm{ph}}$; the orders in (12.6.1) with $M = 0$, $M' = \mu$ are $m = \frac{l}{2} - \frac{n}{4} = \frac{l-d-k}{4}$ and $m' = m'' + \frac{l}{4}$. Part (v) follows from Proposition 12.5.5 in adapted charts, patched with a partition of unity, together with asymptotic completeness of $I^m(X,Y)$. □

12.6.3 Symbol maps

At a point $p \in Z$ the symbol of $u \in I^{m,m'}(X,Y,Z;E)$ is a function on $W = N^*_pZ$ which is a product-type symbol with respect to $V = N^*_pY$. By Remark 12.4.6 the fibrewise relative compactifications form a manifold with corners ${}^{N^*Y}\overline{N^*Z}$ fibred over Z, with main face H_W, whose interior is identified by the blow-down map with $SN^*Z \setminus S(N^*Y|_Z)$ and which fibres over the sphere bundle of $N^*Z/N^*Y|_Z$, the conormal bundle of Z in Y, and with front face $H_V = \overline{N^*Z/N^*Y|_Z} \times_Z S(N^*Y|_Z)$.

Proposition 12.6.7 (Main symbol)**.** *For $u \in I^{m,m'}(X,Y,Z;E)$ the function $\sigma_W(\hat u)$ of (12.5.2), computed in adapted charts, defines a homogeneous function of degree M' on $N^*Z \setminus N^*Y|_Z$ with values in $E \otimes \Omega_{\mathrm{fib}}(N^*Z)$, smooth up to H_W after multiplication by the weight as in Lemma 12.5.6; this gives a short exact sequence*

$$I^{m,m'-1}(X,Y,Z;E) \hookrightarrow I^{m,m'}(X,Y,Z;E) \xrightarrow{\sigma_W} \Sigma^{M,M'}_W(N^*Z, N^*Y|_Z; E \otimes \Omega_{\mathrm{fib}}(N^*Z)).$$

Proof. By (12.6.3), under a change of adapted coordinates the symbol changes, modulo $S^{M,M'-1}_{\mathrm{ph}}$, by composition with the transpose inverse of the normal differential and multiplication by the inverse of its determinant. This is precisely the transformation law of a section of $\Omega_{\mathrm{fib}}(N^*Z)$ over N^*Z (as for the principal symbol of an element of $I^{m'}(X,Z)$ in Chapter 4), and the linear action extends smoothly to the relative compactification by Proposition 12.4.4. So $\sigma_W(u)$ is well defined. Exactness follows from Lemma 12.5.6 in adapted charts, using a partition of unity for surjectivity, and from the fact that away from Z the spaces $I^{m,m'-1}$ and $I^{m,m'}$ coincide. □

The other symbol is associated with Y. Away from Z, u is conormal to Y and has its usual principal symbol, a homogeneous function of degree M on $N^*(Y \setminus Z) \setminus 0$. At Z this symbol develops a conormal singularity: in an adapted chart, for $\eta \in \mathbb{R}^l \setminus 0$, set

$$\sigma_V(u)(w,z,\eta) = \int e^{iz\cdot\zeta}\sigma_V(\hat{u})(w,\zeta,\eta)đ\zeta. \tag{12.6.4}$$

Since $\sigma_V(\hat{u})(w,\cdot,\eta)$ is a classical symbol of order $M'-M$ on $\mathbb{R}^k$, $\sigma_V(u)$ is a distribution on $N^*Y \setminus 0$ (with coordinates (w,z,η)), homogeneous of degree M in η and conormal to $N^*Y|_Z$, with symbol of order $M'-M$ in the conormal variables ζ of Z in Y.

Proposition 12.6.8 (Symbol at Y)**.** *For $u \in I^{m,m'}(X,Y,Z;E)$, (12.6.4) defines invariantly a distribution $\sigma_V(u)$ on $N^*Y \setminus 0$ with values in $E \otimes \Omega_{\mathrm{fib}}(N^*Y)$, homogeneous of degree M and conormal to $N^*Y|_Z \setminus 0$ with symbol of order $M'-M$, which coincides with the usual symbol of u as a conormal distribution to Y over $Y \setminus Z$. There is a short exact sequence*

$$I^{m-1,m'}(X,Y,Z;E) \hookrightarrow I^{m,m'}(X,Y,Z;E) \xrightarrow{\sigma_V} \Sigma_V^{M,M'}(N^*Y, N^*Y|_Z; E \otimes \Omega_{\mathrm{fib}}(N^*Y)),$$

where the last space consists of all such homogeneous conormal distributions. The joint symbol (σ_W, σ_V) is surjective onto the pairs satisfying the compatibility condition (12.5.3) at the corner (the principal symbol of $\sigma_V(u)$ at $N^*Y|_Z$ is the leading part of $\sigma_W(u)$ at $N^*Y|_Z$), with null space $I^{m-1,m'-1}(X,Y,Z;E)$.

Proof. Away from Z, (12.6.4) is the usual principal symbol, by the computation in the proof of Proposition 12.6.3(iii) (the term $j=0$ there). Near Z consider a change of adapted coordinates as in the proof of Proposition 12.6.4. In the expansion (12.6.2) of the new symbol, the terms involving ∂_y lie in $S^{M-1,M'}$ and do not contribute to σ_V; the terms involving only ∂_z are exactly the terms of the expansion of Lemma 12.6.2 applied, on Y, to the amplitude $\lim_{t\to\infty} t^{-M} r(w,z,0,\zeta,t\eta)$. Now

$$L(\zeta, t\eta) = (A^{-t}\zeta,\ tC^{-t}\eta - C^{-t}B^tA^{-t}\zeta),$$

and the second term is a bounded translation in η for fixed ζ, which does not affect the limit in (12.5.2) (it is invisible at H_V). Hence

$$\sigma_V(F^*u)(\cdot,\cdot,\eta) = |\det C|^{-1}\,(F|_Y)^*\big[\sigma_V(u)(\cdot,\cdot,C^{-t}\eta)\big],$$

where the pull-back of the distribution in (w,z) by the diffeomorphism $F|_Y : (w,z) \mapsto (F_w(w,z,0), A(w,z,0)z)$ is computed, as in the standard invariance proof for $I^{\cdot}(Y,Z)$, by the same Kuranishi argument ($F_z|_Y = Az$, so $z\cdot\zeta$ becomes $z\cdot A^t\zeta$). This is the transformation law of a distribution on N^*Y with values in $\Omega_{\mathrm{fib}}(N^*Y)$ under the map induced by F, so $\sigma_V(u)$ is invariantly defined. The exact sequences and surjectivity of the joint symbol follow from Lemma 12.5.6, via a partition of unity, together with the usual symbol sequence for $I^m(X\setminus Z, Y\setminus Z)$; on the corner $H_V \cap H_W$ the restrictions of the two symbols correspond to the leading term of $\sigma_V(\hat{u})$ as a symbol in ζ, which is the principal symbol of the conormal distribution $\sigma_V(u)$. □

12.7 Product-type pseudodifferential operators

Let $\phi : M \longrightarrow B$ be a fibration with fibre Z, $f = \dim Z$, $b = \dim B$, and let E, F be vector bundles over M. We apply the theory of the previous section to the nested pair (12.1.2), $\mathrm{Diag}_M \subset M^2_\phi \subset M^2$, for which $d = f+b$, $k = f$ and $l = b$.

Definition 12.7.1. The space of *product-type pseudodifferential operators* associated with ϕ is

$$\Psi^{m,m'}_{\phi\text{-pt}}(M;E,F) = I^{m-\frac{f}{2},m'}(M^2, M^2_\phi, \mathrm{Diag}_M; \mathrm{Hom}(E,F)\otimes\Omega_R),$$

where $\Omega_R = \pi_R^*\Omega(M)$, identified with a space of operators through the Schwartz kernel theorem.

With this normalization the symbol orders (12.6.1) are exactly (m,m'): $M = m - \frac{f}{2} + \frac{f+b}{2} - \frac{b}{2} = m$ and $M' = m'$. The first order m is the order at the fibre diagonal M^2_ϕ; we call it the *base order*. The second, m', is the order at the diagonal, the *fibre order*. As we shall see, $\Psi^m(M) \subset \Psi^{m,m}_{\phi\text{-pt}}(M)$, $\Psi^{m'}(M/B) \subset \Psi^{0,m'}_{\phi\text{-pt}}(M)$, and compressions of operators of order m on the base lie in $\Psi^{m,-\infty}_{\phi\text{-pt}}(M)$.

12.7.1 Local structure

Let $x = (z,y)$ be fibred coordinates on an open set $\Omega \subset M$, so $\phi(z,y) = y$, and let $\xi = (\zeta,\eta)$ be the dual variables. Then $T^*_xM = \mathbb{R}^f_\zeta \times \mathbb{R}^b_\eta$ and $\phi^*T^*B = \{\zeta = 0\}$; we write $S^{m,m'}_{\mathrm{ph}}(\Omega)$ for $S^{m,m'}_{\mathrm{ph}}(\Omega; \mathbb{R}^{f+b}, \{\zeta = 0\})$. For a symbol $a(x,\xi)$, $\mathrm{Op}(a)$ is the left quantization.

Proposition 12.7.2. *If $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ and $\chi,\chi' \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ then, after trivializing E and F over Ω and $\Omega(M)$ by $|dx|$, $\chi A\chi' = \mathrm{Op}(a)$ with $a \in S^{m,m'}_{\mathrm{ph}}(\Omega)$, of compact support in x. Conversely $\mathrm{Op}(a)\chi' \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ for every such a. If $\operatorname{supp}\chi \cap \operatorname{supp}\chi' = \emptyset$ then the kernel of $\chi A\chi'$ is conormal to M^2_ϕ, of order $m - \frac{f}{2}$ there, and smooth off M^2_ϕ.*

Proof. The coordinates $(w,\tilde z,\tilde y) = (x, z-z', y-y')$ on $\Omega\times\Omega$ are adapted to (12.1.2), by (12.1.3). The partial Fourier transform of $\chi(x)K_A(x,x')\chi'(x')$ in $(\tilde z,\tilde y)$ is $a(x,\xi) = \int e^{-i\nu\cdot\xi}\chi(x)K_A(x,x-\nu)\chi'(x-\nu)\,d\nu$, and Fourier inversion gives $\chi K_A\chi' = \int e^{i(x-x')\cdot\xi}a(x,\xi)đ\xi$, i.e. $\chi A\chi' = \mathrm{Op}(a)$. By Definition 12.6.1 and the computation of the orders above, $a \in S^{m,m'}_{\mathrm{ph}}(\Omega)$. Conversely the kernel of $\mathrm{Op}(a)\chi'$ is of model form near Diag and, by Proposition 12.6.3(iii), conormal to M^2_ϕ away from it; it is smooth and rapidly decreasing away from $\{y = y'\}$, so extends by zero. The last statement is Proposition 12.6.3(iii). □

Away from the diagonal the kernels are conormal to M^2_ϕ but extend across the whole fibre, so it is natural to describe A over a trivializing open set of the base as an operator on B with values in operators on Z. Let $O \subset B$ be open, with coordinates y, a trivialization $\phi^{-1}(O) \cong Z\times O$ and bundle isomorphisms $E \cong \pi_Z^*E_0$, $F \cong \pi_Z^*F_0$ there.

Definition 12.7.3. $S^{m,m'}_{\mathrm{ph}}(O;Z;E_0,F_0)$ is the space of families $k(y,\eta)$, $(y,\eta)\in O\times\mathbb{R}^b$, of distributional kernels on Z^2 (sections of $\mathrm{Hom}(E_0,F_0)\otimes\Omega_R$) such that

(a) if $\chi \in \mathcal{C}^\infty(Z^2)$ vanishes near Diag_Z then χk is a classical symbol of order m in η with values in the Fréchet space $\mathcal{C}^\infty(Z^2;\mathrm{Hom}(E_0,F_0)\otimes\Omega_R)$;

(b) if $\psi,\psi' \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_Z)$ for a coordinate chart $\Omega_Z \subset Z$ then $\psi k(y,\eta)\psi' = \mathrm{Op}_z(a(\cdot,y,\cdot,\eta))$ with $a \in S^{m,m'}_{\mathrm{ph}}(\Omega_Z\times O)$, compactly supported in z.

For k with compact support in y we set $\mathrm{Op}_y(k)u(y) = \int e^{iy\cdot\eta}k(y,\eta)\hat u(\eta)đ\eta$, i.e. the kernel of $\mathrm{Op}_y(k)$ is $\int e^{i(y-y')\cdot\eta}k(y,\eta)đ\eta$. The non-classical spaces $S^{m,m'}(O;Z;E_0,F_0)$ are defined in the same way using non-classical symbols.

Proposition 12.7.4. *If $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ and $\chi,\chi' \in \mathcal{C}^\infty_{\mathrm{c}}(O)$ then $\phi^*\chi\, A\, \phi^*\chi' = \mathrm{Op}_y(k)$ with $k \in S^{m,m'}_{\mathrm{ph}}(O;Z;E_0,F_0)$ compactly supported in y; conversely $\mathrm{Op}_y(k)\phi^*\chi' \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ for each such k.*

Proof. Set $k(y,\eta) = \int e^{-i(y-y')\cdot\eta} K(\cdot,y,\cdot,y')\,dy'$, where K is the kernel of $\phi^*\chi A\phi^*\chi'$. Condition (a) holds since away from Diag_Z the kernel is conormal to $\{y = y'\}$ of symbol order m (the variables (z,z',y) are coordinates along M^2_ϕ, and the characterization of $I^\cdot(M^2,M^2_\phi)$ recalled in Section 12.6 applies, with $y-y'$ as normal variable). For (b) choose a partition of unity ψ_p on the chart with small supports. If $\operatorname{supp}\psi_p$ and $\operatorname{supp}\psi_q$ are close, then $\psi_p k \psi_q = \mathrm{Op}_z(a_{pq})$ with $a_{pq} \in S^{m,m'}_{\mathrm{ph}}$ by Proposition 12.7.2, since $\mathrm{Op}(a) = \mathrm{Op}_y(\mathrm{Op}_z(a))$. If they are disjoint then by (a) $\psi_p k\psi_q$ is a smooth kernel, compactly supported in the chart, which is a classical symbol in η; it equals $\mathrm{Op}_z(c)$ with $c(z,y,\zeta,\eta) = \int e^{-i(z-z')\cdot\zeta}(\psi_p k\psi_q)(z,z')\,dz'$, and c is Schwartz in ζ with values in classical symbols in η, so lies in $S^{m,-\infty}_{\mathrm{ph}} \subset S^{m,m'}_{\mathrm{ph}}$ by Proposition 12.5.3(viii). Summing gives (b). The converse follows by reversing these arguments. □

Lemma 12.7.5. *Let $k \in S^{m,m'}(O;Z;E_0,F_0)$ and $K \subset O$ compact.*

(i) *For all multi-indices γ, β the family*

$$\langle\eta\rangle^{-m+|\beta|}\partial_y^\gamma\partial_\eta^\beta k(y,\eta), \quad y \in K,\ \eta \in \mathbb{R}^b,$$

is bounded in $\Psi^{m'-m+|m-|\beta||}(Z;E_0,F_0)$.

(ii) *If k is classical, then for $\hat\eta \in \mathbb{S}^{b-1}$ the family $r^m k(y,\hat\eta/r)$ is smooth in $r \in [0,1)$ with values in $\Psi^\infty(Z;E_0,F_0)$, its j-th derivative in r being bounded in $\Psi^{m'-m+|m|+j}(Z;E_0,F_0)$. In particular*

$$\beta(k)(y,\eta) = \lim_{t\to\infty} t^{-m}k(y,t\eta) \tag{12.7.1}$$

exists in $\Psi^{m'-m+|m|+1}(Z;E_0,F_0)$, and lies in $\Psi^{m'-m}(Z;E_0,F_0)$; its local symbols are $\sigma_V(a)$ of (12.5.2).

Proof. (i) For the local symbols, $\partial^\beta_\eta a \in S^{m-|\beta|,m'-|\beta|}$, and by Peetre's inequality

$$\langle\zeta\rangle^{m'-m-|\alpha|}\langle\zeta,\eta\rangle^{m-|\beta|} \le C\langle\eta\rangle^{m-|\beta|}\langle\zeta\rangle^{m'-m+|m-|\beta||-|\alpha|};$$

the smooth parts are handled by (a). (ii) In terms of $a = \langle\zeta\rangle^{m'}\langle q\rangle^m b(\zeta,q)$ with $q = \hat\eta/(r\langle\zeta\rangle)$ we have $r^m a = \langle\zeta\rangle^{m'-m}(1+r^2\langle\zeta\rangle^2)^{m/2}b(\zeta,q)$. Near H_V, b is a smooth function of $\zeta \in \overline{U}$, $1/|q| = r\langle\zeta\rangle$ and $q/|q| = \hat\eta$, while where $|q|$ is bounded we have $r\langle\zeta\rangle \ge c > 0$ and $|\partial_r q| = |q|/r \le C\langle\zeta\rangle$. In either case each r-derivative produces at most a factor $\langle\zeta\rangle$ (also when it falls on $(1+r^2\langle\zeta\rangle^2)^{m/2}$), which gives the stated bounds; setting $r=0$ gives $\langle\zeta\rangle^{m'-m}b(\zeta,\infty\hat\eta)$, which is the symbol $\sigma_V(a)(\cdot,\hat\eta)$ of order $m'-m$. The smooth parts are classical in η by (a). □

Proposition 12.7.6. *Each $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ defines continuous linear maps $\mathcal{C}^\infty(M;E) \longrightarrow \mathcal{C}^\infty(M;F)$ and $\mathcal{C}^{-\infty}(M;E) \longrightarrow \mathcal{C}^{-\infty}(M;F)$. The transpose and, given Hermitian metrics and a positive density, the adjoint of A are product-type operators of the same orders.*

Proof. The transpose has kernel obtained by pulling back under the involution $(m,m') \mapsto (m',m)$ of M^2, which preserves M^2_ϕ and Diag_M; so it is in the class by Proposition 12.6.6(i), and similarly for the adjoint. The action on distributions follows by duality from that on smooth sections. Choose a partition of unity χ_i on B with supports so small that whenever

$\operatorname{supp}\chi_i \cap \operatorname{supp}\chi_j \neq \emptyset$, both lie in a trivializing open set. Then $A = \sum_{i,j} \phi^*\chi_i A \phi^*\chi_j$; if the supports are disjoint the kernel is supported off M_ϕ^2, so is smooth, and the term is smoothing. The other terms are of the form $\mathrm{Op}_y(k)$, and $\mathrm{Op}_y(k)u(\cdot,y) = \int e^{iy\cdot\eta} k(y,\eta)\hat{u}(\cdot,\eta)đ\eta$, where $\hat{u}(\cdot,\eta)$ is a Schwartz function of η with values in $\mathcal{C}^\infty(Z;E_0)$. By Lemma 12.7.5(i) and the continuity of the action $\Psi^\mu(Z)\times\mathcal{C}^\infty(Z)\longrightarrow\mathcal{C}^\infty(Z)$, the integrand and all its y-derivatives are bounded in $\mathcal{C}^\infty(Z;F_0)$ by $C\langle\eta\rangle^{-N}$, so the integral defines an element of $\mathcal{C}^\infty(O\times Z;F_0)$ depending continuously on u. □

12.7.2 Symbols

By Lemma 12.1.6, the conormal bundle $W = N^*\operatorname{Diag}_M$ is T^*M, with $V = N^*M_\phi^2|_{\mathrm{Diag}} = \phi^*T^*B$ and $W/V = T^*(M/B)$. The relative compactification (Remark 12.4.6) is a manifold with corners ${}^{\phi^*T^*B}\overline{T^*M}$ over M. Its main face

$$H_W = [S^*M;\phi^*S^*B]$$

has interior $S^*M\setminus\phi^*S^*B$ and fibres over $S^*(M/B)$ with fibre over a point above $b\in B$ a radially compactified affine space modelled on T_b^*B, by (12.4.3); its boundary, the corner, is $S^*(M/B)\times_M\phi^*S^*B$, which is the vertical cosphere bundle $S^*(\pi^*M/S^*B)$ of the pulled-back fibration $\pi^*M = M\times_B S^*B\longrightarrow S^*B$. The front face is $H_V = \overline{T^*(M/B)}\times_M\phi^*S^*B$. On the other hand $N^*M_\phi^2\setminus 0 = M_\phi^2\times_B(T^*B\setminus 0)$ is the fibre product for the fibration $\pi^*M\longrightarrow T^*B\setminus 0$, and its intersection with the conormal bundle of the diagonal is the diagonal of that fibre product. Since $\Omega(M)\cong\Omega(M/B)\otimes\phi^*\Omega(B)$ and $\Omega(B)\otimes\Omega(T_b^*B)$ is canonically trivial, the density factors in Proposition 12.6.7 and Proposition 12.6.8 are absorbed.

Accordingly we define the symbol spaces as follows. Choose a metric and let $|\zeta|$ denote the length of the image of $\xi\in T_x^*M$ in $T_x^*(M/B)$.

- $\Sigma_\phi^{m,m'}(M;E,F)$ is the space of $s\in\mathcal{C}^\infty(T^*M\setminus\phi^*T^*B;\hom(E,F))$, homogeneous of degree m', such that $|\zeta|^{m-m'}|\xi|^{-m}s$, restricted to $S^*M\setminus\phi^*S^*B$, extends to a smooth function on $[S^*M;\phi^*S^*B]$.
- $\mathcal{B}_\phi^{m,m'}(M;E,F)$ is the space of smooth families $P(b,\eta)\in\Psi^{m'-m}(Z_b;E_b,F_b)$, $(b,\eta)\in T^*B\setminus 0$, which are homogeneous of degree m, $P(b,t\eta)=t^mP(b,\eta)$ for $t>0$; that is, the homogeneous elements of $\Psi^{m'-m}(\pi^*M/(T^*B\setminus 0);E,F)$.

Theorem 12.7.7 (Symbol maps)**.** *There are two symbol maps on $\Psi_{\phi\text{-pt}}^{m,m'}(M;E,F)$:*

(i) *the* principal symbol $\sigma_{m,m'}:\Psi_{\phi\text{-pt}}^{m,m'}(M;E,F)\longrightarrow\Sigma_\phi^{m,m'}(M;E,F)$, *given in local coordinates by $\sigma_{m,m'}(A)(x,\xi)=\lim_{t\to\infty}t^{-m'}a(x,t\xi)$ for $\xi\notin\phi^*T^*B$;*

(ii) *the* base symbol $\beta_{m,m'}:\Psi_{\phi\text{-pt}}^{m,m'}(M;E,F)\longrightarrow\mathcal{B}_\phi^{m,m'}(M;E,F)$, *given in a local trivialization by* (12.7.1)*.*

They give short exact sequences

$$\Psi_{\phi\text{-pt}}^{m,m'-1}\hookrightarrow\Psi_{\phi\text{-pt}}^{m,m'}\xrightarrow{\sigma_{m,m'}}\Sigma_\phi^{m,m'},\qquad \Psi_{\phi\text{-pt}}^{m-1,m'}\hookrightarrow\Psi_{\phi\text{-pt}}^{m,m'}\xrightarrow{\beta_{m,m'}}\mathcal{B}_\phi^{m,m'},$$

and the joint symbol $(\sigma_{m,m'},\beta_{m,m'})$ has null space $\Psi_{\phi\text{-pt}}^{m-1,m'-1}(M;E,F)$ and range the compatible pairs, i.e. those satisfying

$$\lim_{t\to\infty}t^{-m}s(x,\tilde{\zeta}+t\,\phi^*\eta)=\sigma_{m'-m}(P(\phi(x),\eta))(x,\zeta)\tag{12.7.2}$$

for all $x \in M$, $\eta \in T^*_{\phi(x)}B \setminus 0$, $\zeta \in T^*_x(M/B) \setminus 0$ and any lift $\tilde\zeta \in T^*_xM$ of ζ; here $\sigma_{m'-m}(P(b,\eta))$ is the principal symbol of the operator $P(b,\eta)$ on Z_b.

Proof. The principal symbol is $\sigma_W(K_A)$ of Proposition 12.6.7 and the base symbol is $\sigma_V(K_A)$ of Proposition 12.6.8, under the identifications above; in particular both are invariantly defined. That $\sigma_V(K_A)$ is given by (12.7.1) is Lemma 12.7.5(ii) near the diagonal and the definition of the principal symbol of a conormal distribution away from it. The exact sequences and the range of the joint symbol are Proposition 12.6.7 and Proposition 12.6.8, and (12.7.2) is (12.5.3): indeed in local coordinates $\tilde\zeta + t\eta$ corresponds to $\zeta + t\eta$, and the right side of (12.5.3) is the principal symbol of the classical symbol $\sigma_V(a)(x,\cdot,\eta)$ of order $m'-m$, which is the local symbol of $\beta_{m,m'}(A)$. □

Concretely, the surjectivity of the base symbol amounts to the following: if $P \in \mathcal{B}^{m,m'}_\phi$ has local symbols $p(z,y,\zeta,\hat\eta)$ (of order $m'-m$ in ζ) near the fibre diagonal, then $a(z,y,\zeta,\eta) = (1-\chi(\eta/\langle\zeta\rangle))|\eta|^m p(z,y,\zeta,\eta/|\eta|)$, with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^b)$ equal to 1 near 0, lies in $S^{m,m'}_{\mathrm{ph}}$ and has $\sigma_V(a) = |\eta|^m p$. Note that the cut-off must be a function of $q = \eta/\langle\zeta\rangle$, not of η: $(1-\chi(\eta))|\eta|^m p$ is not smooth on the relative compactification at $H_W \cap \{q=0\}$.

12.7.3 Composition

Theorem 12.7.8. *For any fibration $\phi : M \longrightarrow B$ of compact manifolds and bundles E, F, G over M,*

$$\Psi^{m_1,m_1'}_{\phi\text{-pt}}(M;F,G) \circ \Psi^{m_2,m_2'}_{\phi\text{-pt}}(M;E,F) \subset \Psi^{m_1+m_2,m_1'+m_2'}_{\phi\text{-pt}}(M;E,G),$$

and both symbol maps are multiplicative:

$$\sigma_{m_1+m_2,m_1'+m_2'}(AA') = \sigma_{m_1,m_1'}(A)\,\sigma_{m_2,m_2'}(A'),$$
$$\beta_{m_1+m_2,m_1'+m_2'}(AA')(b,\eta) = \beta_{m_1,m_1'}(A)(b,\eta) \circ \beta_{m_2,m_2'}(A')(b,\eta),$$

the latter being composition of pseudodifferential operators on Z_b.

Proof. Step 1: reduction to a trivialization. Choose a finite cover of B by trivializing open sets, let $\delta > 0$ be a Lebesgue number for it (for some metric on B) and χ_i a partition of unity with $\operatorname{diam}\operatorname{supp}\chi_i < \delta/4$. Write $A = \sum A_{ij}$, $A_{ij} = \phi^*\chi_i A \phi^*\chi_j$, and similarly $A' = \sum A'_{kl}$. As in the proof of Proposition 12.7.6, the terms with $\operatorname{supp}\chi_i \cap \operatorname{supp}\chi_j = \emptyset$ are smoothing. If R is smoothing and A'' is any product-type operator then RA'' and $A''R$ are smoothing, since for instance the kernel of RA'' is $(x,x') \mapsto ((A'')^t K_R(x,\cdot))(x')$, which is smooth by Proposition 12.7.6 applied with the parameter x. Moreover $A_{ij}A'_{kl} = 0$ unless $\operatorname{supp}\chi_j \cap \operatorname{supp}\chi_k \neq \emptyset$. The remaining products involve four sets whose union has diameter less than δ, so lies in a trivializing set O. By Proposition 12.7.4 it therefore suffices to show that $\operatorname{Op}_y(k_1)\operatorname{Op}_y(k_2) = \operatorname{Op}_y(k)$ with $k \in S^{m_1+m_2,m_1'+m_2'}_{\mathrm{ph}}(O;Z;E_0,G_0)$ when $k_i \in S^{m_i,m_i'}_{\mathrm{ph}}(O;Z)$ have compact support in y.

Step 2: products on the fibre. We first show that the pointwise product $k_1(y,\eta)k_2(y,\eta)$ (composition of operators on Z) lies in $S^{m_1+m_2,m_1'+m_2'}_{\mathrm{ph}}(O;Z)$, with local symbol having leading part a_1a_2 modulo $S^{m_1+m_2,m_1'+m_2'-1}$. Decompose each k_i using a partition of unity ψ_p on Z whose supports are small compared to a Lebesgue number of a cover of Z by charts. A product of a piece $\psi_p k_1 \psi_q$ and a piece $\psi_r k_2 \psi_s$ vanishes unless q and r overlap. If both pieces are near the diagonal, all four supports lie in one chart and the product is $\operatorname{Op}_z(a_1)\operatorname{Op}_z(a_2) = \operatorname{Op}_z(c)$ with

$$c(z,y,\zeta,\eta) = \iint e^{-iz'\cdot\theta} a_1(z,y,\zeta+\theta,\eta) a_2(z+z',y,\zeta,\eta)\,dz'đ\theta.$$

This is the standard composition formula with (y,η) as parameters. The standard proof (Taylor expansion of a_1 in θ and integration by parts in z', Chapter 2) goes through verbatim with the weights $w_{M,M'}$ in place of $\langle\xi\rangle^\mu$, using Lemma 12.5.2 for translations in ζ alone and the fact that ∂_ζ lowers the second order; it gives

$$c \sim \sum_\alpha \frac{1}{\alpha!}\partial_\zeta^\alpha a_1\, D_z^\alpha a_2, \qquad \partial_\zeta^\alpha a_1\, D_z^\alpha a_2 \in S_{\mathrm{ph}}^{m_1+m_2,m_1'+m_2'-|\alpha|},$$

modulo $S^{m_1+m_2,-\infty}$. Moreover c is classical: after integrating by parts to make the integral absolutely convergent, the integrand is, for fixed (z',θ), a classical product-type symbol with seminorms $O(\langle z'\rangle^{-L}\langle\theta\rangle^{K-L})$ by Lemma 12.5.4. If one of the pieces, say $\psi_p k_1 \psi_q$, has a smooth kernel, then the product has kernel $(z,z') \mapsto (k_2(y,\eta)^t(\psi_p k_1\psi_q)(z,\cdot))(z')$, which by Lemma 12.7.5 is a classical symbol of order m_1+m_2 in η with values in $\mathcal{C}^\infty(Z^2)$: the factor $\langle\eta\rangle^{m_2}$ comes from (i), and classicality from (ii), since the smooth kernel absorbs the loss of orders in the r-derivatives. If both pieces are near the diagonal but $\operatorname{supp}\psi_p$ and $\operatorname{supp}\psi_s$ are far apart then the product vanishes by the choice of supports. Summing all pieces proves the claim.

Step 3: composition in the base. Exactly as for operators on $\mathbb{R}^b$,

$$\operatorname{Op}_y(k_1)\operatorname{Op}_y(k_2) = \operatorname{Op}_y(k), \quad k(y,\eta) = \iint e^{-iy'\cdot\theta} k_1(y,\eta+\theta)k_2(y+y',\eta)\,dy'đ\theta,$$

an oscillatory integral of products of operators on Z (the order of the factors matters). Taylor expansion of $k_1(y,\eta+\theta)$ in θ and integration by parts in y' give

$$k \sim \sum_\alpha \frac{1}{\alpha!}\partial_\eta^\alpha k_1 \cdot D_y^\alpha k_2, \qquad \partial_\eta^\alpha k_1 \cdot D_y^\alpha k_2 \in S_{\mathrm{ph}}^{m_1+m_2-|\alpha|,m_1'+m_2'-|\alpha|}(O;Z), \tag{12.7.3}$$

by Step 2 and Proposition 12.5.3(iv); the remainder after N terms is an integral of products $\partial_\eta^\alpha k_1(y,\eta+s\theta)\,\partial_y^\alpha k_2(y+y',\eta)$, $|\alpha|=N$, at different values of η. The argument of Step 2 applies to such products, and Peetre's inequality $w_{M,M'}(\zeta,\eta+s\theta) \le C w_{M,M'}(\zeta,\eta)\langle\theta\rangle^K$ shows, as in the scalar case, that the remainder lies in the non-classical space $S^{m_1+m_2-N,m_1'+m_2'-N}(O;Z)$. By Proposition 12.5.5(iii) there is a classical $\tilde k$ with the expansion (12.7.3); then $k-\tilde k \in \bigcap_N S^{-N,-N}(O;Z)$, and this space consists of Schwartz functions of η with values in smoothing operators on Z, which are classical. Hence $k \in S_{\mathrm{ph}}^{m_1+m_2,m_1'+m_2'}(O;Z)$.

Step 4: symbols. The local symbol of k is a_1a_2 modulo $S^{m_1+m_2,m_1'+m_2'-1}$, which gives multiplicativity of σ. All terms of (12.7.3) except the first have base order at most m_1+m_2-1, so $\beta(AA') = \lim t^{-m_1-m_2}k_1(y,t\eta)k_2(y,t\eta)$; by Lemma 12.7.5(ii) the factors converge in spaces of pseudodifferential operators of fixed order, and composition is continuous, so the limit is $\beta(A)\beta(A')$. □

Proposition 12.7.9. (i) $\bigcap_j \Psi_{\phi\text{-pt}}^{m-j,m'-j}(M;E,F) = \Psi^{-\infty}(M;E,F)$, *the smoothing operators.*

(ii) $\Psi_{\phi\text{-pt}}^{m,-\infty}(M;E,F) = \bigcap_{m'} \Psi_{\phi\text{-pt}}^{m,m'}(M;E,F)$ *consists of the operators whose kernels are conormal to* M_ϕ^2 *of order* $m-\frac{f}{2}$*; locally in* B *these are the* $\operatorname{Op}_y(k)$ *with* k *a classical symbol of order* m *with values in smoothing operators on* Z*.*

(iii) *Asymptotic completeness holds for* $\Psi_{\phi\text{-pt}}^{m,m'}$ *in the three forms of Proposition 12.5.5, modulo* $\Psi_{\phi\text{-pt}}^{m,-\infty}$*, modulo* $\Psi_{\phi\text{-pt}}^{-\infty,m'}$ *and modulo* $\Psi^{-\infty}(M)$ *respectively.*

Proof. This is Proposition 12.6.6(ii), (iii) and (v) in the present setting, together with Proposition 12.7.4. □

12.7.4 Subalgebras

Proposition 12.7.10. (i) (Fibrewise operators) $\Psi^{m'}(M/B;E,F) \subset \Psi^{0,m'}_{\phi\text{-pt}}(M;E,F)$; *for* $A \in \Psi^{m'}(M/B;E,F)$, $\sigma_{0,m'}(A) = \gamma^*\sigma_{m'}(A)$, *where* $\gamma : T^*M\backslash\phi^*T^*B \longrightarrow T^*(M/B)\backslash 0$ *is the quotient map, and* $\beta_{0,m'}(A)(b,\eta) = A_b$ *for all* $\eta \neq 0$.

(ii) (Ordinary operators) $\Psi^m(M;E,F) \subset \Psi^{m,m}_{\phi\text{-pt}}(M;E,F)$; $\sigma_{m,m}(A)$ *is the restriction of* $\sigma_m(A)$ *to* $T^*M \setminus \phi^*T^*B$, *and* $\beta_{m,m}(A)(b,\eta)$ *is the operator of multiplication on* Z_b *by* $x \longmapsto \sigma_m(A)(x,\phi^*\eta)$.

(iii) (Compressed base operators) *Let* $\pi_1 \in \Psi^{-\infty}(M/B;E)$ *and* $\pi_2 \in \Psi^{-\infty}(M/B;F)$ *be smooth families of projections. Their ranges are finite-dimensional and form vector bundles* $\tilde{E}$, $\tilde{F}$ *over* B, *subbundles of* $\mathcal{C}^\infty(M/B;E)$, $\mathcal{C}^\infty(M/B;F)$. *If* $P \in \Psi^m(B;\tilde{E},\tilde{F})$ *then* $\tilde{P} = \iota_2 P \pi_1$, *where* $\iota_2 : \mathcal{C}^\infty(B;\tilde{F}) \hookrightarrow \mathcal{C}^\infty(M;F)$, *lies in* $\Psi^{m,-\infty}_{\phi\text{-pt}}(M;E,F)$, *with* $\beta_{m,-\infty}(\tilde{P})(b,\eta) = \sigma_m(P)(b,\eta)\circ\pi_1(b)$ *(followed by the inclusion* $\tilde{F}_b \subset \mathcal{C}^\infty(Z_b;F_b)$*).*

Proof. (i) By Proposition 12.6.6(iv) with $Y = M^2_\phi$, $Z = \mathrm{Diag}_M$, $d = f+b$, $k = f$, $l = b$: an element of $I^{m'-b/4}(M^2_\phi, \mathrm{Diag})$ with values in $\mathrm{Hom}(E,F)\otimes\Omega_R(M/B)$ times $\delta_{M^2_\phi}$ (the factor $\Omega(N^*M^2_\phi) \cong \phi^*\Omega(T^*B)$ combining with $\Omega_R(M/B)$ to give $\Omega_R(M)$) lies in $I^{(b-2f-b)/4,\,m'} = I^{-f/2,m'}$, i.e. in $\Psi^{0,m'}_{\phi\text{-pt}}$. In local coordinates the symbol $a(z,y,\zeta)$ is independent of η, so $\sigma_W(a)$ is the pull-back of $\sigma_{m'}(A)$ and $\sigma_V(a) = a$; in the operator-valued description $k(y,\eta) = A_y$ is independent of η.

(ii) The local symbols of A lie in $S^m_{\mathrm{ph}} \subset S^{m,m}_{\mathrm{ph}}$ by Proposition 12.5.3(v), and the kernel is smooth off the diagonal; the limit (12.5.2) of $t^{-m}a(z,y,\zeta,t\eta)$ is the principal symbol at $(0,\eta)$, independent of ζ, i.e. a multiplication operator.

(iii) The rank of $\pi_1(b)$ is $\mathrm{Tr}\,\pi_1(b)$, a continuous integer-valued function, so locally constant, and near b_0 the sections $\pi_1(b)e_j$, for $e_j \in \mathcal{C}^\infty(M;E)$ restricting to a basis of $\mathrm{Ran}\,\pi_1(b_0)$, form a smooth local frame of $\tilde{E}$. Over a trivializing open set O choose smooth frames $e_1,\dots,e_r \in \mathcal{C}^\infty(\phi^{-1}(O);E)$ of $\tilde{E}$ and $f_1,\dots,f_s$ of $\tilde{F}$, and $e^*_j \in \mathcal{C}^\infty(\phi^{-1}(O);E^*\otimes\Omega(M/B))$ with $\pi_1 u = \sum_j e_j \int_{Z_b} e^*_j u$. If P has matrix (P_{ij}) in these frames, the kernel of $\tilde{P}$ over O is $\sum_{i,j} f_i(z,y)K_{P_{ij}}(y,y')e^*_j(z',y')$, i.e. $\tilde{P} = \mathrm{Op}_y(k)$ with $k(y,\eta) = \sum_{ij} f_i\, p_{ij}(y,\eta)\, e^*_j$, a classical symbol of order m with values in smoothing operators of finite rank on Z (up to a smoothing term coming from the left reduction in y). The base symbol is the limit (12.7.1). □

Remark 12.7.11. For a product $M = Z\times B$ one might expect the lift $\mathrm{Id}_Z \otimes P$ of $P \in \Psi^m(B)$ to be a product-type operator, of base order m and fibre order 0; this is not so, which is why the compression in (iii) is needed. Its local symbol $p(y,\eta)$ is independent of ζ, and near H_W, where $|\zeta| \to \infty$ with $q = \eta/\langle\zeta\rangle$ bounded, $p(y,\langle\zeta\rangle q)$ behaves like $\langle\zeta\rangle^m|q|^m p_m(y,q/|q|)$, which is not smooth at $q = 0$ unless p_m is a polynomial. In fact $\mathrm{Id}_Z\otimes P$ is product-type (and then lies in $\Psi^m(M) \subset \Psi^{m,m}_{\phi\text{-pt}}(M)$) only when P is a differential operator (Exercise 12.7).

12.7.5 Boundedness, ellipticity and the Fredholm property

Proposition 12.7.12. *Each* $A \in \Psi^{0,0}_{\phi\text{-pt}}(M;E,F)$ *extends to a bounded operator* $L^2(M;E) \longrightarrow L^2(M;F)$ *(for any choice of Hermitian metrics and smooth positive density).*

Proof. As in the proof of Proposition 12.7.6, up to smoothing terms (which are bounded) A is a finite sum of operators $\mathrm{Op}_y(k)$ with $k \in S^{0,0}_{\mathrm{ph}}(O;Z)$ compactly supported in $y \in O \subset \mathbb{R}^b$. By Lemma 12.7.5(i) with $\beta = 0$, the operators $\partial^\gamma_y k(y,\eta)$ are bounded in $\Psi^0(Z)$, hence

uniformly bounded on $L^2(Z;E_0)$ (the inclusion $\Psi^0(Z)\hookrightarrow\mathcal{B}(L^2(Z))$ is continuous). Write $k(y,\eta)=\int e^{iy\cdot\theta}\hat{k}(\theta,\eta)đ\theta$, where

$$\|\hat{k}(\theta,\eta)\|_{\mathcal{B}(L^2(Z))}\le C_N\langle\theta\rangle^{-2N}\sup_y\|(1-\Delta_y)^N k(y,\eta)\|\le C'_N\langle\theta\rangle^{-2N}.$$

Then

$$\mathrm{Op}_y(k)u=\int e^{iy\cdot\theta}\hat{k}(\theta,D_y)u đ\theta,$$

where $\hat{k}(\theta,D_y)$ is the Fourier multiplier, acting on the space $L^2(\mathbb{R}^b;L^2(Z;E_0))$, with operator-valued multiplier $\hat{k}(\theta,\eta)$; by Plancherel's theorem its norm is at most $\sup_\eta\|\hat{k}(\theta,\eta)\|$. Multiplication by $e^{iy\cdot\theta}$ is unitary, so

$$\|\mathrm{Op}_y(k)\|\le\int C'_N\langle\theta\rangle^{-2N}đ\theta<\infty.$$

□

Definition 12.7.13. $A\in\Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ is *fully elliptic* if

(a) $\sigma_{m,m'}(A)$ is invertible, in the sense that the normalized symbol $|\zeta|^{m-m'}|\xi|^{-m}\sigma_{m,m'}(A)$ is invertible at every point of $[S^*M;\phi^*S^*B]$, including its boundary, and

(b) $\beta_{m,m'}(A)(b,\eta):\mathcal{C}^\infty(Z_b;E_b)\longrightarrow\mathcal{C}^\infty(Z_b;F_b)$ is invertible for every $(b,\eta)\in T^*B\setminus 0$.

By (12.7.2), (a) implies that each $\beta(A)(b,\eta)$ is an elliptic operator of order $m'-m$ on Z_b; (b) then requires these elliptic operators to be invertible, which in particular forces their index to vanish.

Lemma 12.7.14. *If $P(p)\in\Psi^\mu(Z;E_0,F_0)$ is a smooth family of elliptic operators, depending on parameters p in a manifold, and each $P(p)$ is invertible on $\mathcal{C}^\infty$, then $P(p)^{-1}\in\Psi^{-\mu}(Z;F_0,E_0)$ is again a smooth family.*

Proof. Let $Q(p)$ be a smooth family of parametrices, $QP=\mathrm{Id}-R$, $PQ=\mathrm{Id}-R'$, with R, R' smooth families of smoothing operators. Then $P^{-1}=Q+RP^{-1}=Q+RQ+RP^{-1}R'$. The inverses $P(p)^{-1}:H^s\longrightarrow H^{s+\mu}$ depend smoothly on p in norm, by Lemma 12.3.1, for every s, and since R' maps H^{-N} to H^N and R maps H^{-N} to H^N smoothly in p for all N, $RP^{-1}R'$ is a smooth family of smoothing operators. □

Theorem 12.7.15. *If $A\in\Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ is fully elliptic there exists $Q\in\Psi^{-m,-m'}_{\phi\text{-pt}}(M;F,E)$ such that*

$$QA=\mathrm{Id}_E-R,\quad AQ=\mathrm{Id}_F-R',\qquad R\in\Psi^{-\infty}(M;E),\ R'\in\Psi^{-\infty}(M;F).$$

Conversely, if such a Q exists then A is fully elliptic.

Proof. The inverse of the normalized principal symbol is smooth on $[S^*M;\phi^*S^*B]$, so $\sigma_{m,m'}(A)^{-1}\in\Sigma_\phi^{-m,-m'}(M;F,E)$. By Lemma 12.7.14 applied to the fibration $\pi^*M\longrightarrow T^*B\setminus 0$, $\beta_{m,m'}(A)^{-1}\in\mathcal{B}_\phi^{-m,-m'}(M;F,E)$; it is homogeneous of degree $-m$, and its principal symbol is the inverse of that of $\beta(A)$, so by (12.7.2) the pair $(\sigma(A)^{-1},\beta(A)^{-1})$ is compatible. By Theorem 12.7.7 there is $Q_0\in\Psi^{-m,-m'}_{\phi\text{-pt}}(M;F,E)$ with $\sigma(Q_0)=\sigma(A)^{-1}$ and $\beta(Q_0)=\beta(A)^{-1}$. By Theorem 12.7.8, $Q_0A=\mathrm{Id}-R_0$ with $\sigma_{0,0}(R_0)=0$ and $\beta_{0,0}(R_0)=0$, so $R_0\in\Psi^{-1,-1}_{\phi\text{-pt}}(M;E)$. By Proposition 12.7.9(iii) the Neumann series $\sum_j R_0^j$ is asymptotically summable, to $S\in\Psi^{0,0}_{\phi\text{-pt}}$ with $S(\mathrm{Id}-R_0)-\mathrm{Id}\in\Psi^{-\infty}$, and $Q=SQ_0$ is a left parametrix.

A right parametrix Q' is constructed in the same way, and then $Q - Q' = Q(\mathrm{Id} - AQ') - (\mathrm{Id} - QA)Q'$ is smoothing, so Q is a two-sided parametrix. Conversely, if Q exists then by multiplicativity $\sigma(Q)\sigma(A) = \mathrm{Id}$ and $\beta(Q)\beta(A) = \mathrm{Id}$, and similarly on the other side, since smoothing operators have vanishing symbols. □

Corollary 12.7.16. *A fully elliptic $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$ is Fredholm as a map $\mathcal{C}^\infty(M; E) \longrightarrow \mathcal{C}^\infty(M; F)$, its null space and that of its adjoint consist of smooth sections, and*

$$\mathrm{ind}(A) = \mathrm{Tr}(\mathrm{Id}_E - QA) - \mathrm{Tr}(\mathrm{Id}_F - AQ)$$

for any parametrix Q as in Theorem 12.7.15. If A_t, $t \in [0,1]$, is a smooth family of fully elliptic operators then $\mathrm{ind}(A_t)$ *is constant. For $m = m' = 0$, A is also Fredholm on L^2, with the same index.*

Proof. With smoothing errors, the arguments of Chapter 5 and Chapter 10 for elliptic operators apply verbatim: $\mathrm{Nul}(A) \subset \mathrm{Nul}(\mathrm{Id} - R)$ is a finite-dimensional space of smooth sections, the range is closed of finite codimension, and the trace formula holds. The construction of the parametrix can be carried out with smooth dependence on t, so the right side of the trace formula is continuous in t, and integer valued. For $m = m' = 0$, A, Q are bounded on L^2 by Proposition 12.7.12 and R, R' are compact, so A is Fredholm on L^2; its null space, and that of its adjoint, are the same as on $\mathcal{C}^\infty$ by the parametrix identities. □

Proposition 12.7.17. *An operator $A \in \Psi^{0,0}_{\phi\text{-pt}}(M; E, F)$ is Fredholm on L^2 if and only if it is fully elliptic.*

Proof. Sufficiency is Corollary 12.7.16. Conversely suppose A is Fredholm on L^2; then there are a bounded Q and a compact K with $QA = \mathrm{Id} + K$, and the same holds for A^*. We sketch the construction of approximate null sequences. Suppose first that $(x_0, \xi_0) \in T^*M$, $\xi_0 \notin \phi^*T^*B$, and $e \in E_{x_0}$. In local coordinates let $u_t(x) = t^{n/4}\chi(t^{1/2}(x - x_0))e^{itx\cdot\xi_0}e$, $n = \dim M$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}$. Then $\|u_t\|$ is bounded below, $u_t \to 0$ weakly, and a direct computation with the local form $\mathrm{Op}(a)$ (Taylor expanding a at $(x_0, t\xi_0)$) shows that $Au_t - \sigma_{0,0}(A)(x_0, \xi_0)u_t \to 0$ in L^2. Since $Ku_t \to 0$, $\|u_t\| \leq \|Q\|\|Au_t\| + o(1)$, so $|\sigma_{0,0}(A)(x_0, \xi_0)e| \geq \|Q\|^{-1}|e|$. By continuity this lower bound extends to all of $[S^*M; \phi^*S^*B]$, and applying the same argument to A^* shows that $\sigma_{0,0}(A)$ is invertible. Next, for $(b_0, \eta_0) \in T^*B \backslash 0$ and $v \in \mathcal{C}^\infty(Z_{b_0}; E_{b_0})$, in a trivialization let $u_t(z, y) = t^{b/4}\chi(t^{1/2}(y - y_0))e^{ity\cdot\eta_0}v(z)$. The same computation, now using the operator-valued form $\mathrm{Op}_y(k)$ and Lemma 12.7.5, shows $Au_t - (\beta_{0,0}(A)(b_0, \eta_0)v) \otimes t^{b/4}\chi(t^{1/2}(y - y_0))e^{ity\cdot\eta_0} \to 0$, using the homogeneity of degree 0 of $\beta_{0,0}(A)$. Hence $\|\beta_{0,0}(A)(b_0, \eta_0)v\| \geq \|Q\|^{-1}\|v\|$; the same for A^*, with $\beta(A^*) = \beta(A)^*$, shows that $\beta_{0,0}(A)(b_0, \eta_0)$ is invertible on L^2, hence, being elliptic, on $\mathcal{C}^\infty$. □

12.7.6 The tensor product construction

The following construction, which is the starting point for the proof of the Atiyah–Singer index theorem in Chapter 13, shows how product-type operators connect the index of an operator on the base with operators on the total space of a fibration.

Proposition 12.7.18. *Let $\phi : M \longrightarrow B$ be a fibration, with Hermitian metrics and a fibre density chosen. Let $P \in \Psi^0(M/B; E_+, E_-)$ be an elliptic family such that each P_b is surjective and* $\mathrm{Nul}(P)$ *is a trivial line bundle, spanned by $e \in \mathcal{C}^\infty(M; E_+)$ with $\|e|_{Z_b}\| = 1$, and let $\pi_\nu = e \otimes e^* \in \Psi^{-\infty}(M/B; E_+)$ be the orthogonal projection onto* $\mathrm{Nul}(P)$. *Let $A \in \Psi^0(B; F_+, F_-)$ be elliptic, and let $A_\nu \in \Psi^{0,-\infty}_{\phi\text{-pt}}(M; E_+ \otimes F_+, E_+ \otimes F_-)$ be the*

compression $A_\nu u = e \otimes A(\int_{Z_b}(u,e)_{E_+})$ *of Proposition 12.7.10(iii) (here and below* $F_\pm$ *also denotes* $\phi^* F_\pm$*). Set*

$$H_+ = E_+ \otimes F_+ \oplus E_- \otimes F_-, \quad H_- = E_- \otimes F_+ \oplus E_+ \otimes F_-,$$
$$P_A = \begin{pmatrix} P & 0 \\ A_\nu & P^* \end{pmatrix} : \mathcal{C}^\infty(M; H_+) \longrightarrow \mathcal{C}^\infty(M; H_-),$$

where P *and* P^* *act on the first factors of the tensor products. Then* $P_A \in \Psi^{0,0}_{\phi\text{-pt}}(M; H_+, H_-)$ *is fully elliptic and* $\operatorname{ind}(P_A) = \operatorname{ind}(A)$.

Proof. By Proposition 12.7.10, P, $P^* \in \Psi^{0,0}_{\phi\text{-pt}}$ and $A_\nu \in \Psi^{0,-\infty}_{\phi\text{-pt}} \subset \Psi^{0,0}_{\phi\text{-pt}}$, with

$$\sigma_{0,0}(P_A) = \begin{pmatrix} \gamma^*\sigma_0(P) & 0 \\ 0 & \gamma^*\sigma_0(P)^* \end{pmatrix}, \qquad \beta_{0,0}(P_A)(b,\eta) = \begin{pmatrix} P_b & 0 \\ \pi_\nu a & P_b^* \end{pmatrix}, \quad a = \sigma_0(A)(b,\eta),$$

where $a : F_{+,b} \longrightarrow F_{-,b}$ is invertible. The principal symbol is invertible, P being elliptic. To see that $\beta(P_A)(b,\eta)$ is invertible, solve $P_b u = g$, $\pi_\nu a u + P_b^* v = h$. Since P_b is surjective with null space $\mathbb{C}e_b$, $u = u_0 + e_b \otimes c$ where u_0 is the solution orthogonal to e_b and $c \in F_{+,b}$ is free. Since P_b^* is injective with range $(\mathbb{C}e_b)^\perp$, the second equation splits into $\pi_\nu a u = \pi_\nu h$, i.e. $e_b \otimes ac = \pi_\nu h$, which determines c uniquely because a is invertible, and $P_b^* v = (\mathrm{Id} - \pi_\nu)h$, which determines v. So $\beta(P_A)(b,\eta)$ is bijective, and P_A is fully elliptic.

For the index, let $(u,v) \in \operatorname{Nul}(P_A)$. Then $Pu = 0$, so $u = e \otimes g$ with $g \in \mathcal{C}^\infty(B; F_+)$; and $A_\nu u + P^* v = e \otimes Ag + P^* v = 0$, where the first term lies in $\operatorname{Nul}(P) \otimes F_-$ and the second in its orthogonal complement, so $Ag = 0$ and $P^* v = 0$, i.e. $v = 0$. Conversely $(e \otimes g, 0) \in \operatorname{Nul}(P_A)$ if $Ag = 0$, so $\operatorname{Nul}(P_A) \cong \operatorname{Nul}(A)$. The adjoint is $P_A^* = \begin{pmatrix} P^* & A_\nu^* \\ 0 & P \end{pmatrix}$ with $A_\nu^* u = e \otimes A^* \int_{Z_b}(u,e)$, and the same argument gives $\operatorname{Nul}(P_A^*) \cong \operatorname{Nul}(A^*)$. Hence $\operatorname{ind}(P_A) = \operatorname{ind}(A)$. □

The operator P_A can be regarded as a 'Clifford' tensor product of P and A. In Proposition 12.7.18 its full ellipticity is guaranteed by the fibre operator P, while its index comes from the base operator A. In Chapter 13 P_A is deformed, through fully elliptic product-type operators (so without changing the index, by Corollary 12.7.16), to an ordinary pseudodifferential operator on M whose symbol, given a connection on ϕ, is of the form

$$\begin{pmatrix} \chi_1\sigma_0(P) & -\chi_2\sigma_0(A)^* \\ \chi_2\sigma_0(A) & \chi_1\sigma_0(P)^* \end{pmatrix},$$

with χ_1, χ_2 a partition of unity on S^*M subordinate to the complements of $\phi^* S^* B$ and of the vertical cosphere bundle. Applied to the fibration of the radially compactified normal bundle of an embedding $B \hookrightarrow \mathbb{S}^N$, this reduces the computation of the index of A to a problem on the sphere, which can then be solved using Bott periodicity (Chapter 11).

12.8 Exercises

Exercise 12.1. Let $\phi : M \longrightarrow B$ be a fibration. Show that there is an embedding $e : M \longrightarrow \mathbb{S}^N \times B$, with range in $(\mathbb{S}^N \setminus \{p\}) \times B$ for a fixed point p, such that $\pi_B \circ e = \phi$. *Hint:* embed M in a sphere avoiding one point (Whitney), and take e to be the product of this embedding with ϕ. Such an embedding identifies M with a fibration over B whose fibres are submanifolds of $\mathbb{R}^N$; it is used in the proof of the families index theorem.

Exercise 12.2. Let $\phi : M \longrightarrow B$ be a fibration with $\dim B > 0$. Show that $\Psi^m(M;E,F) \cap \Psi^m(M/B;E,F)$ is the space of fibrewise differential operators of order at most m (so it is $\{0\}$ for $m<0$). *Hint:* the kernel of an element of the intersection is smooth off Diag_M and supported in M^2_ϕ, which has positive codimension; so it is supported on the diagonal.

Exercise 12.3. Show that the definition of ${}^V\overline{W}$ extends to the degenerate cases, with ${}^{\{0\}}\overline{W} = \overline{W}$ and ${}^W\overline{W} = \overline{W}$, and that then $S^{M,M'}_{\mathrm{ph}}(W,\{0\}) = S^{M'}_{\mathrm{ph}}(W)$ and $S^{M,M'}_{\mathrm{ph}}(W,W) = S^{M}_{\mathrm{ph}}(W)$.

Exercise 12.4. In the notation of Section 12.4, show that the vector fields ∂_{ζ_j}, ∂_{η_i}, $\zeta_j\partial_{\zeta_{j'}}$, $\zeta_j\partial_{\eta_i}$ and $\eta_i\partial_{\eta_{i'}}$ on W lift to smooth vector fields on ${}^V\overline{W}$, tangent to its boundary, whereas $\eta_i\partial_{\zeta_j}$ does not. Relate this to Proposition 12.4.4 by observing that the first set spans the Lie algebra of the affine group of W preserving the direction V.

Exercise 12.5. Near a point of $SV \subset \partial\overline{W}$ use the coordinates $x = 1/|\eta|$, $v = \zeta/|\eta| \in \mathbb{R}^k$ and $\theta = \eta/|\eta|$ on $\overline{W}$, in which $SV = \{x = 0, v = 0\}$. Introduce polar coordinates $R = (x^2+|v|^2)^{1/2}$, $(x,v)/R$ in the half-space $x \geq 0$. Show that the map from a neighbourhood of H_V in ${}^V\overline{W}$ given in the coordinates of Proposition 12.4.3(ii),(iii) to $(R,(x,v)/R,\theta)$ is a diffeomorphism onto a neighbourhood of $\{R=0\}\times\mathbb{S}^k_+\times\mathbb{S}^{l-1}$. Conclude that ${}^V\overline{W}$ is the blow-up $[\overline{W};SV]$ of $\overline{W}$ along SV, with H_V the front face.

Exercise 12.6. Using (12.4.1), show that the identification (12.4.2) of H_V with $\overline{W/V}\times SV$ is natural, whereas H_W is only naturally a fibration over $S(W/V)$. In the setting of a nested pair $Z \subset Y \subset X$, show that H_W is naturally identified with the blow-up $[SN^*Z; S(N^*Y|_Z)]$, fibred over the cosphere bundle of Z in Y.

Exercise 12.7. Let $p \in S^m_{\mathrm{ph}}(\mathbb{R}^b)$ and regard $a(\zeta,\eta) = p(\eta)$ as a function on $\mathbb{R}^f\times\mathbb{R}^b$. Show that if $a \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^{f+b},\{\zeta=0\})$ for some M, M' then p is a polynomial. *Hint:* near the interior of H_W, ρ_V is bounded below and $a = p(q/\rho_W)$; differentiate $\rho_W^{M'}p(q/\rho_W)$ in q along the curves $q = \rho_W v$. Deduce the statement in Remark 12.7.11.

Exercise 12.8. For $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ show that $\sigma_{m,m'}(A^*) = \sigma_{m,m'}(A)^*$ and that $\beta_{m,m'}(A^*)(b,\eta)$ is the adjoint $\beta_{m,m'}(A)(b,\eta)^*$ (the adjoint on Z_b with respect to the induced metrics and fibre density).

Exercise 12.9. Let $P \in \Psi^{m'}(M/B;E,F)$ be an elliptic family, regarded as an element of $\Psi^{0,m'}_{\phi\text{-pt}}(M;E,F)$. Show that P is fully elliptic if and only if each P_b is invertible, and that then P is invertible with inverse in $\Psi^{-m'}(M/B;F,E)$. In particular, for $M = Z\times B$ with product metric, the fibre Laplacian $\Delta_Z + 1$ is a fully elliptic element of $\Psi^{0,2}_{\phi\text{-pt}}(M)$ which is not elliptic as an operator on M if $\dim B > 0$; while $\Delta_Z + \Delta_B$ and $\Delta_M + 1$ lie in $\Psi^{2,2}_{\phi\text{-pt}}(M)$ and are fully elliptic. Compute their base symbols.

Exercise 12.10. Let $A \in \Psi^m(M/B;E,F)$ be an elliptic family and $b \in B$. Show that the restriction of $\mathrm{ind}_a(A)$ to $\{b\}$, i.e. its image in $K^0(\{b\}) = \mathbb{Z}$, is $\mathrm{ind}(A_b)$, and that $\mathrm{ind}_a(A^*) = -\mathrm{ind}_a(A)$ when $m = 0$.

Notes

This chapter combines four lectures, with their (largely empty) addenda, of Melrose's 2005 course. The main departures from the original are as follows. The construction of the index bundle via finite rank projections π_N approximating the identity, which was stated without

proof in the original, has been replaced by the stabilization procedure of Atiyah and Singer [18], which the original adopted later in the course. The relative compactification has been presented as the product $\overline{U} \times \overline{V}$ in the rescaled variables $(\zeta, \eta/\langle\zeta\rangle)$; this makes its linear invariance and the structure of product-type symbols transparent, and the classical estimate characterization (12.5.1) is a by-product. The order conventions of the original are not consistent between lectures; here the first order always refers to the larger submanifold (for operators, the base order at M_ϕ^2) and the second to the smaller (the fibre order at the diagonal), so that $\Psi^m(M) \subset \Psi^{m,m}_{\phi\text{-pt}}(M)$, $\Psi^{m'}(M/B) \subset \Psi^{0,m'}_{\phi\text{-pt}}(M)$ and compressed base operators lie in $\Psi^{m,-\infty}_{\phi\text{-pt}}(M)$. The base symbol of an element of $\Psi^{m,m'}_{\phi\text{-pt}}(M)$ has fibre order $m' - m$ (not m'). The original asserted that operators on the base lift to $\Psi^{m,0}_{\phi\text{-pt}}(M)$; this is false unless they are differential operators (Remark 12.7.11), and the original's later arguments only use the compressed operators of Proposition 12.7.10(iii). The proofs of coordinate invariance, of the symbol maps and of the composition formula, which were left to unwritten addenda in the original, have been supplied, as has the proof that full ellipticity is equivalent to the Fredholm property.

Pseudodifferential operators with symbols of product type in two groups of variables go back to the 'bisingular' operators of Rodino; the calculus adapted to a fibration used here is that of Melrose's lectures, and is closely related to the calculi used for families of operators on manifolds with boundary in [20] and to the adiabatic limits of [19]. For the families index and its relation to K-theory see [18] and [17]; for the conormal distributions underlying the definitions see [8], Chapter 18.

Chapter 13

Multiplicativity, the Chern character and the families index theorem

In Chapter 12 the analytic index of a family of elliptic pseudodifferential operators on the fibres of a fibration $\phi : M \to B$ of compact manifolds was defined as a homomorphism $\mathrm{ind_a} : K_{\mathrm{c}}(T^*(M/B)) \to K^0(B)$, and the calculus of pseudodifferential operators of product type was set up. This chapter puts these together to prove the families index theorem of Atiyah and Singer in K-theory, and then passes to cohomology through the Chern character.

The proof rests on two properties of the analytic index. The first, *multiplicativity* (Section 13.2), says that for a tower of fibrations $\tilde{M} \to M \to B$ the index is unchanged when a family on M/B is "multiplied" by a family on the fibres of $\tilde{M} \to M$ whose index is the trivial line bundle. The product is first formed in the product-type calculus, where its index is computed directly, and is then deformed, through fully elliptic product-type families, into an ordinary family on $\tilde{M}/B$. The second property, *excision* (Section 13.3), says that the index of a symbol that is trivial outside an open set only depends on that open set. Embedding the fibration into a trivial one, $M \hookrightarrow \mathbb{R}^N \times B$, these two properties reduce the index to the case of the torus fibration $\mathbb{T}^N \times B \to B$. There the index is computed by multiplicativity and Bott periodicity. This gives the equality of the analytic and topological index maps (Theorem 13.4.4). One ingredient, the index of the Thom class of a non-trivial normal bundle, is not proved here; we say exactly what is used.

The second half of the chapter reviews Chern–Weil theory and defines the Chern character on K^0 (Section 13.5). It then defines the odd Chern character and computes the Chern character of the index bundle of a family of Toeplitz operators (Section 13.6). The relative Chern character of a compactly supported K-class is given by an explicit formula of Fedosov type (Section 13.7). Finally the cohomological form of the families index theorem is stated (Section 13.8).

13.1 Conventions and the index bundle

13.1.1 Superbundles and compactly supported K-theory

A *superbundle* over a manifold X is a pair $\mathbb{E} = (E_+, E_-)$ of complex vector bundles, always with Hermitian metrics. The *graded tensor product* of two superbundles is

$$(\mathbb{E} \otimes \mathbb{F})_+ = E_+ \otimes F_+ \oplus E_- \otimes F_-, \qquad (\mathbb{E} \otimes \mathbb{F})_- = E_- \otimes F_+ \oplus E_+ \otimes F_-. \tag{13.1.1}$$

Matrices of maps $(\mathbb{E}\otimes\mathbb{F})_+ \to (\mathbb{E}\otimes\mathbb{F})_-$ below are always written with respect to these ordered decompositions.

We recall from Chapter 11 (see also [17]) the compactly supported K-group of a manifold X. A *K_c-triple* (E_+, E_-, a) consists of complex vector bundles $E_\pm$ over X and $a \in \mathcal{C}^\infty(X \setminus K; \mathrm{Hom}(E_+, E_-))$ invertible, for some compact $K \subset X$. The group $K_c(X)$ is the set of equivalence classes of such triples under the equivalence relation generated by bundle isomorphisms, by changing a on a compact set, by homotopies of triples (triples on $[0,1]\times X$ with a invertible outside $[0,1]\times K$, K compact), and by stabilization $(E_+, E_-, a) \sim (E_+ \oplus H, E_- \oplus H, a \oplus \mathrm{Id}_H)$. The group operation is direct sum. We use the following standard facts.

(K1) If a is invertible on all of X the triple represents 0.

(K2) If X is compact, $[(E_+, E_-, a)] \mapsto [E_+] - [E_-]$ identifies $K_c(X)$ with $K^0(X)$.

(K3) An open embedding $j : U \hookrightarrow X$ induces a homomorphism $j_* : K_c(U) \to K_c(X)$, *extension by the identity*. Every class in $K_c(U)$ has a representative (E_+, E_-, a) with $E_\pm$ trivialized outside a compact set $C \subset U$ and $a = \mathrm{Id}$ outside C in these trivializations; j_* extends $E_\pm$ trivially and a by Id. This is independent of choices. ([17], §2.4; in the cases needed below the normalization is carried out in the proof of Theorem 13.3.1.)

(K4) A proper smooth map $f : X' \to X$ induces $f^* : K_c(X) \to K_c(X')$ by pulling back triples.

If $V \to Y$ is a real vector bundle over a compact manifold Y and SV its sphere bundle, a homogeneous map $a \in \mathcal{C}^\infty(V \setminus 0; \mathrm{Hom}(\pi^*E_+, \pi^*E_-))$ of degree 0 which is invertible defines the class $[(\pi^*E_+, \pi^*E_-, a)] \in K_c(V)$; here a is only used outside the compact set $\{|v| \le 1\}$. Every class in $K_c(V)$ arises in this way (Chapter 12): bundles over V are pulled back from Y, and a may be deformed to be homogeneous.

13.1.2 The Clifford product

The following elementary computation is used repeatedly. Let $p : E_+ \to E_-$ and $\alpha : F_+ \to F_-$ be linear maps between finite dimensional Hermitian spaces (or bounded operators between Hilbert spaces) and $c_1, c_2 \ge 0$. Set

$$m(c_1,c_2) = \begin{pmatrix} c_1\, p\otimes 1 & -c_2\, 1\otimes\alpha^* \\ c_2\, 1\otimes\alpha & c_1\, p^*\otimes 1 \end{pmatrix} : (\mathbb{E}\otimes\mathbb{F})_+ \longrightarrow (\mathbb{E}\otimes\mathbb{F})_-. \tag{13.1.2}$$

Lemma 13.1.1. *With the notation of* (13.1.2),

$$m^*m = \begin{pmatrix} c_1^2\, p^*p\otimes 1 + c_2^2\, 1\otimes\alpha^*\alpha & 0 \\ 0 & c_1^2\, pp^*\otimes 1 + c_2^2\, 1\otimes\alpha\alpha^* \end{pmatrix},$$

and mm^* is given by the same formula with (p,α) replaced by $(p^*,-\alpha)$ (and the factors in the other order). Hence m is invertible if either $c_1 > 0$ and p is invertible, or $c_2 > 0$ and α is invertible.

Proof. Since $m^* = \begin{pmatrix} c_1 p^*\otimes 1 & c_2\, 1\otimes\alpha^* \\ -c_2\, 1\otimes\alpha & c_1 p\otimes 1 \end{pmatrix}$, the off-diagonal entries of m^*m are $\pm c_1c_2(p^*\otimes\alpha^* - p^*\otimes\alpha^*) = 0$, because maps acting on different tensor factors commute. The diagonal entries are as stated. Note that m^* has the form (13.1.2) with (p,α) replaced by $(p^*,-\alpha)$

and the roles of the two summands interchanged, which gives mm^*. Under either hypothesis both m^*m and mm^* are sums of non-negative self-adjoint operators one of which is bounded below by a positive constant, so they are invertible, and then m is injective with closed range and surjective. □

Definition 13.1.2. Let $x_i = [(E_+^i, E_-^i, a_i)] \in K_{\mathrm{c}}(X_i)$, $i = 1, 2$, and extend a_i (after multiplying by a cut-off) to $a_i \in \mathcal{C}^\infty(X_i; \mathrm{Hom}(E_+^i, E_-^i))$, invertible outside a compact set K_i. The *exterior product* $x_1 \boxtimes x_2 \in K_{\mathrm{c}}(X_1 \times X_2)$ is the class of $\big((\mathbb{E}^1 \otimes \mathbb{E}^2)_+, (\mathbb{E}^1 \otimes \mathbb{E}^2)_-, a_1 \# a_2\big)$, $a_1 \# a_2 = m(1, 1)$ with $p = a_1$, $\alpha = a_2$. If V_1, V_2 are vector bundles over the same compact manifold Y and $x_i \in K_{\mathrm{c}}(V_i)$, the *product* $x_1 \cdot x_2 \in K_{\mathrm{c}}(V_1 \oplus V_2)$ is the pull-back of $x_1 \boxtimes x_2$ under the proper embedding $V_1 \oplus V_2 \hookrightarrow V_1 \times V_2$.

By Lemma 13.1.1, $a_1 \# a_2$ is invertible outside $K_1 \times K_2$. (Where it is invertible the ranks of the two bundles agree.) Changing the extension of a_1 inside K_1, the straight line homotopy between the two extensions gives a homotopy of triples, since $a_1 \# a_2$ stays invertible wherever a_2 is. Homotopies, bundle isomorphisms and direct sums pass through the construction. If x_2 is the stabilizing triple (H, H, Id) then $a_1 \#\, \mathrm{Id}$ is invertible everywhere, so the product vanishes by (K1). Thus the products are well defined and bilinear. This is the standard product in K-theory ([17], §2.6).

13.1.3 Fibrewise families and the analytic index

Let $\phi : M \to B$ be a fibration of compact manifolds with typical fibre Z, $Z_b = \phi^{-1}(b)$. From Chapter 12 we use:

(F1) The algebra $\Psi^m(M/B; E_+, E_-)$ of smooth families of pseudodifferential operators on the fibres, with principal symbol σ_m on $T^*(M/B) \setminus 0$, composition, adjoints (with respect to Hermitian metrics and a smooth positive fibre density), parametrices for elliptic families, and the fact that in a local trivialization $\phi^{-1}(U) \cong Z \times U$ an element of $\Psi^0(M/B)$ is a smooth family in $\Psi^0(Z)$, hence a norm-smooth family of bounded operators on $L^2(Z)$.

(F2) An elliptic $A \in \Psi^0(M/B; E_+, E_-)$ defines $[\sigma_0(A)] \in K_{\mathrm{c}}(T^*(M/B))$, and every class arises in this way.

(F3) The analytic index of A, defined by stabilizing with finite rank projections, depends only on $[\sigma_0(A)]$ and gives a homomorphism $\mathrm{ind_a} : K_{\mathrm{c}}(T^*(M/B)) \to K^0(B)$.

The degenerate case $M = B$, $\phi = \mathrm{Id}$, is allowed. Then $T^*(B/B) = B$, the "fibrewise operators" are bundle maps $a \in \mathcal{C}^\infty(B; \mathrm{Hom}(E_+, E_-))$, there is no ellipticity condition, and $K_{\mathrm{c}}(T^*(B/B)) = K^0(B)$.

13.1.4 Product-type families

Consider a tower of fibrations of compact manifolds

$$\begin{array}{ccc} Y & \longrightarrow & \tilde{M} \\ & & \big\downarrow{\scriptstyle \tilde{\phi}} \\ Z & \longrightarrow & M \\ & & \big\downarrow{\scriptstyle \phi} \\ & & B \end{array} \tag{13.1.3}$$

so $\tilde{M} \to B$ is a fibration with fibre $\tilde{M}_b = (\phi\tilde{\phi})^{-1}(b)$, itself fibred over $M_b = Z_b$ by $\tilde{\phi}$ with fibre Y. The space $\Psi^{m,m'}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; E_+, E_-)$ of *product-type families* consists of smooth families, parametrized by $b \in B$, of product-type operators on $\tilde{M}_b$ with respect to $\tilde{\phi} : \tilde{M}_b \to M_b$, as defined in Chapter 12; their kernels are product-type conormal distributions on the fibre product $\tilde{M} \times_B \tilde{M}$ with respect to $\tilde{M} \times_M \tilde{M} \supset \mathrm{Diag}$. Since all constructions in Chapter 12 are local in the base, they hold with parameters in B. Let

$$\Sigma = S(\tilde{\phi}^* T^*(M/B)) \subset S^*(\tilde{M}/B)$$

be the sphere bundle of the subbundle $\tilde{\phi}^* T^*(M/B) \subset T^*(\tilde{M}/B)$ (the annihilator of $T(\tilde{M}/M)$). We use the following from Chapter 12, in the families version and for order $(0,0)$.

(P1) There are two symbol maps. The *usual symbol* $\sigma(A)$ is a smooth function on the blow-up $[S^*(\tilde{M}/B); \Sigma]$. The *base symbol* $\beta(A)$ is a family, parametrized by $\eta \in S^*(M/B)$, of operators in $\Psi^0(Y_m)$, $Y_m = \tilde{\phi}^{-1}(m)$, m the base point of η. Both are multiplicative and compatible with adjoints. The two symbols determine A modulo $\Psi^{-\infty}(\tilde{M}/B)$, and any compatible pair arises.

(P2) A is *fully elliptic* if $\sigma(A)$ and $\beta(A)$ are invertible. A fully elliptic family has a parametrix modulo smoothing families, so it is Fredholm on $L^2(\tilde{M}_b)$ for each b; its null space consists of smooth sections. In a local trivialization of $\tilde{M} \to B$ it is a norm-smooth family of bounded operators.

(P3) A family $P \in \Psi^0(\tilde{M}/M; E_+, E_-)$ on the fibres of $\tilde{\phi}$ lies in $\Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B)$, with $\sigma(P) = \gamma^*\sigma_0(P)$ and $\beta(P)(\eta) = P_m$. Here $\gamma : [S^*(\tilde{M}/B); \Sigma] \to S^*(\tilde{M}/M)$ is the extension to the blow-up of the radial projection of the restriction map $T^*(\tilde{M}/B) \to T^*(\tilde{M}/M)$, which vanishes exactly on $\tilde{\phi}^* T^*(M/B)$.

(P4) $\Psi^0(\tilde{M}/B) \subset \Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B)$; $\sigma(A)$ is the pull-back of $\sigma_0(A)$ under the blow-down map, and $\beta(A)(\eta)$ is multiplication by the restriction of $\sigma_0(A)$ to Σ, i.e. to $\tilde{\phi}^* S^*(M/B)$, at η. For such A full ellipticity is ellipticity.

(P5) If $e \in \mathcal{C}^\infty(\tilde{M}; E_+)$ has L^2 norm one on each fibre Y_m, Π is the orthogonal projection onto $\mathbb{C}e$ on each fibre, and $A \in \Psi^0(M/B; F_+, F_-)$, then

$$A_\Pi : u \longmapsto e \otimes A\big(\langle u, e\rangle\big), \quad \langle u, e\rangle(m) = \int_{Y_m} (u, e)\, \nu,$$

lies in $\Psi^{-\infty,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; E_+ \otimes F_+, E_+ \otimes F_-)$, with $\sigma(A_\Pi) = 0$ and $\beta(A_\Pi)(\eta) = \Pi \otimes \sigma_0(A)(\eta)$.

Here ν is the fibre density on $\tilde{M}/M$ used to define adjoints, and the bundles $F_\pm$ over M are lifted to $\tilde{M}$ without change of notation. A fibrewise family P on $\tilde{M}/M$ acts on sections of $E_\pm \otimes F$ as $P \otimes \mathrm{Id}_F$, since F is constant on the fibres of $\tilde{\phi}$; we keep writing P.

13.1.5 Stabilization and the index bundle

The index bundle of a fibrewise elliptic family was constructed in Chapter 12 using finite rank projections. For the comparison of different families below, and for product-type families, an equivalent and more flexible construction is convenient: one stabilizes the *range* rather than the domain. In the next lemma, A is either an elliptic family in $\Psi^0(M/B; E_+, E_-)$ or a fully elliptic family in $\Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; E_+, E_-)$; to treat both cases at once we write

M for the total space and M_b for its fibres. Sections $v_1, \dots, v_N \in \mathcal{C}^\infty(M; E_-)$ define $S \in \mathcal{C}^\infty(M; \operatorname{Hom}(\mathbb{C}^N, E_-))$ and the operators

$$A_b \oplus S_b : L^2(M_b; E_+) \oplus \mathbb{C}^N \longrightarrow L^2(M_b; E_-), \quad (u, c) \longmapsto A_b u + \sum_j c_j v_j|_{M_b}.$$

Lemma 13.1.3 (Stabilization)**.** *(i) There exist N and S such that $A_b \oplus S_b$ is surjective for all $b \in B$.*

(ii) For such S, the null spaces $\operatorname{Nul}(A_b \oplus S_b) \subset \mathcal{C}^\infty(M_b; E_+) \oplus \mathbb{C}^N$ form a smooth vector bundle $\mathcal{N}(A, S)$ over B.

(iii) The class

$$\operatorname{ind}(A) = [\mathcal{N}(A, S)] - [\mathbb{C}^N] \in K^0(B)$$

is independent of S, and is unchanged under smooth homotopies of A through families of the same kind (elliptic, resp. fully elliptic).

(iv) For an elliptic $A \in \Psi^0(M/B; E_+, E_-)$, $\operatorname{ind}(A) = \operatorname{ind}_{\mathrm{a}}([\sigma_0(A)])$.

Proof. (i) Fix b_0 and a trivialization over a neighbourhood U of b_0. The operator A_{b_0} is Fredholm, so its range is closed of finite codimension. Since smooth sections are dense in L^2, there are smooth sections $w_1, \dots, w_r$ of E_- over M_{b_0} whose classes span $L^2/\operatorname{Ran}(A_{b_0})$. Extend them, using the trivialization and a cut-off on the base equal to 1 near b_0, to smooth sections over M supported in $\phi^{-1}(U)$. In the trivialization $b \mapsto A_b \oplus S_b$ is norm continuous and surjective bounded operators form an open set, so $A_b \oplus S_b$ is surjective for b near b_0. Cover B by finitely many such neighbourhoods and use all the sections so obtained; adding columns preserves surjectivity.

(ii) Let $\mathcal{N}_b = \operatorname{Nul}(A_b \oplus S_b)$. Fix b_0, let Π_0 be the orthogonal projection onto the finite dimensional space $\mathcal{N}_{b_0}$ in $L^2 \oplus \mathbb{C}^N$, and set

$$T_b(u, c) = \big(A_b u + S_b c,\ \Pi_0(u, c)\big) \in L^2(M_b; E_-) \oplus \mathcal{N}_{b_0}.$$

T_{b_0} is an isomorphism: it is injective since $\mathcal{N}_{b_0} \cap \mathcal{N}_{b_0}^\perp = 0$, and given (g, n) choose w with $(A_{b_0} \oplus S_{b_0})w = g$; then $w - \Pi_0 w + n$ is a preimage. Hence T_b is invertible for b near b_0, depending smoothly on b in the trivialization, and $n \mapsto T_b^{-1}(0, n)$ is an isomorphism $\mathcal{N}_{b_0} \to \mathcal{N}_b$ (if $w \in \mathcal{N}_b$ then $w = T_b^{-1}(0, \Pi_0 w)$). This gives local frames. The elements of $\mathcal{N}_b$ are smooth sections, by elliptic regularity: if Q is a parametrix, smooth in b, with $QA = \mathrm{Id} - R$, R smoothing, then $u = -QSc + Ru$. Applying this to the frame elements, which depend smoothly on b in L^2, shows that they are smooth on M. Smoothness of $\mathcal{N}$ does not depend on the trivialization, since its frames are smooth sections of E_+ over open subsets of M, plus vectors in $\mathbb{C}^N$.

(iii) Let S, S' (with N, N' columns) both satisfy (i) and let $S'' = (S, S')$. The sequence of bundles

$$0 \longrightarrow \mathcal{N}(A, S) \longrightarrow \mathcal{N}(A, S'') \longrightarrow \mathbb{C}^{N'} \longrightarrow 0, \quad (u, c) \mapsto (u, c, 0), \quad (u, c, c') \mapsto c',$$

is exact: the last map is surjective because, given c', we may solve $Au + Sc = -S'c'$. Exact sequences of vector bundles split, so $[\mathcal{N}(A, S'')] - [\mathbb{C}^{N+N'}] = [\mathcal{N}(A, S)] - [\mathbb{C}^N]$. By symmetry the same holds for S'. If A_t, $t \in [0, 1]$, is a smooth homotopy, apply (i) and (ii) on the compact manifold (with boundary) $B \times [0, 1]$. The resulting bundle over $B \times [0, 1]$ has isomorphic restrictions to $B \times \{0\}$ and $B \times \{1\}$ ([17], Lemma 1.4.3), and the restrictions of S stabilize A_0 and A_1. So $\operatorname{ind}(A_0) = \operatorname{ind}(A_1)$.

(iv) In Chapter 12 the index is $[\operatorname{Ran}\pi]-[W]$, where $\pi\in\Psi^{-\infty}(M/B;E_+)$ is a family of finite rank projections such that $\operatorname{Nul}(A(\operatorname{Id}-\pi))=\operatorname{Ran}\pi$ and $W=\operatorname{Nul}((\operatorname{Id}-\pi)A^*)$ are smooth bundles, W_b being the orthocomplement of the range of $A_b(\operatorname{Id}-\pi_b)$. Choose finitely many smooth sections $w_1,\dots,w_K$ of W (sections of E_- over M) spanning W_b for every b, and let S_W be the corresponding map from $\mathbb{C}^K$. Then $A(\operatorname{Id}-\pi)\oplus S_W$ is surjective. Its null space is $\operatorname{Ran}\pi\oplus L$, where L is the kernel of the surjective bundle map $\mathbb{C}^K\to W$, since the two terms of $A(\operatorname{Id}-\pi)u+S_Wc$ lie in complementary subspaces. As $L\oplus W\cong\mathbb{C}^K$, the stabilized class of $A(\operatorname{Id}-\pi)$ is $[\operatorname{Ran}\pi]+[L]-[\mathbb{C}^K]=[\operatorname{Ran}\pi]-[W]$. Finally $A(\operatorname{Id}-t\pi)$, $t\in[0,1]$, is a smooth homotopy of elliptic families, so by (iii) $\operatorname{ind}(A)=\operatorname{ind}(A(\operatorname{Id}-\pi))$. □

Remark 13.1.4. (a) If B is a point, $\dim\mathcal{N}(A,S)=\dim\operatorname{Nul}A+N-\operatorname{codim}\operatorname{Ran}A$, so $\operatorname{ind}(A)$ is the numerical index. (b) In the degenerate case $M=B$, for a bundle map $a:E_+\to E_-$ the sequence $0\to\mathcal{N}(a,S)\to E_+\oplus\mathbb{C}^N\to E_-\to 0$ is exact, so $\operatorname{ind}(a)=[E_+]-[E_-]$; that is, $\operatorname{ind}_{\mathrm{a}}$ is the identification (K2). (c) The index bundle is often written as $[\operatorname{Nul}(A+A')]-[\operatorname{Nul}((A+A')^*)]$ for a smoothing perturbation A' after which the null spaces form bundles; this is the construction of Chapter 12 in another guise. (d) The construction of (i)–(ii) is compatible with pull-back: for a smooth map $f:B'\to B$ and the pulled-back fibration, $\operatorname{ind}(f^*A)=f^*\operatorname{ind}(A)$, since f^*S stabilizes f^*A and $\mathcal{N}(f^*A,f^*S)=f^*\mathcal{N}(A,S)$.

By Lemma 13.1.3, the index of a fully elliptic product-type family is a well-defined element of $K^0(B)$, invariant under homotopies through fully elliptic families. For ordinary families it agrees with $\operatorname{ind}_{\mathrm{a}}$.

13.2 Multiplicativity

Throughout this section we consider the tower (13.1.3), a superbundle $\mathbb{E}=(E_+,E_-)$ over $\tilde M$, a superbundle $\mathbb{F}=(F_+,F_-)$ over M (also lifted to $\tilde M$), and $\mathbb{G}=\mathbb{E}\otimes\mathbb{F}$. We fix a fibre density ν on $\tilde M/M$ and Hermitian metrics.

13.2.1 The splitting of the symbol space

The vertical cotangent bundle $T^*(\tilde M/B)$ contains the subbundle $\tilde\phi^*T^*(M/B)$, the annihilator of $T(\tilde M/M)$, with quotient $T^*(\tilde M/M)$. A connection for $\tilde\phi$, i.e. a choice of complement H to $T(\tilde M/M)$ in $T(\tilde M/B)$, splits this sequence:

$$T^*(\tilde M/B)\cong T^*(\tilde M/M)\oplus\tilde\phi^*T^*(M/B),\tag{13.2.1}$$

where $T^*(\tilde M/M)$ is identified with the annihilator H° of H. Any two connections are homotopic. Let

$$\Sigma_1=S(\tilde\phi^*T^*(M/B)),\qquad \Sigma_2=S(H^\circ),$$

two disjoint closed subsets of $S^*(\tilde M/B)$. For elliptic families $P\in\Psi^0(\tilde M/M;E_+,E_-)$ and $A\in\Psi^0(M/B;F_+,F_-)$ write

$$p=\sigma_0(P),\qquad \alpha=\sigma_0(A),$$

homogeneous of degree 0 and invertible. The function p, composed with the restriction map $T^*(\tilde M/B)\to T^*(\tilde M/M)$, is defined on $S^*(\tilde M/B)\setminus\Sigma_1$, and α, composed with the projection onto the second summand in (13.2.1), is defined on $S^*(\tilde M/B)\setminus\Sigma_2$. Choose $\chi_1,\chi_2\in\mathcal{C}^\infty(S^*(\tilde M/B))$ with

$$\chi_1,\chi_2\ge 0,\quad \chi_1+\chi_2=1,\quad \chi_1=0\text{ near }\Sigma_1,\quad \chi_2=0\text{ near }\Sigma_2,\tag{13.2.2}$$

and define, with m as in (13.1.2),

$$a_P = m(\chi_1, \chi_2) = \begin{pmatrix} \chi_1\, p & -\chi_2\, \alpha^* \\ \chi_2\, \alpha & \chi_1\, p^* \end{pmatrix} \in \mathcal{C}^\infty(S^*(\tilde{M}/B); \operatorname{Hom}(G_+, G_-)). \tag{13.2.3}$$

It is smooth since each coefficient vanishes near the set where its factor is undefined. By Lemma 13.1.1 it is invertible at every point, since $\chi_1 > 0$ or $\chi_2 > 0$ everywhere. So a_P, extended as a homogeneous function of degree 0, is an elliptic symbol on $\tilde{M}/B$.

Lemma 13.2.1. *The class of a_P in $K_{\mathrm{c}}(T^*(\tilde{M}/B))$ is the product $[p] \cdot \tilde{\phi}^*[\alpha]$ of Definition 13.1.2, for the splitting (13.2.1). In particular it is independent of the choice of χ_1, χ_2 and of the connection.*

Proof. Write $\xi = (\xi_1, \xi_2)$ according to (13.2.1), and let p', α' be smooth extensions to all of $T^*(\tilde{M}/M)$, resp. $\tilde{\phi}^*T^*(M/B)$, equal to p, α for $|\xi_1| \geq 1$, resp. $|\xi_2| \geq 1$. These represent $[p]$ and $\tilde{\phi}^*[\alpha]$, and $m(1,1)$ formed from them represents the product. By compactness there is $\delta \in (0,1)$ with $|\xi_1| \geq \delta|\xi|$ on $\operatorname{supp}\chi_1$ and $|\xi_2| \geq \delta|\xi|$ on $\operatorname{supp}\chi_2$ (both extended homogeneously). Consider, for $t \in [0,1]$,

$$m_t = m((1-t)\chi_1 + t, (1-t)\chi_2 + t) \quad \text{formed with } p',\ \alpha'.$$

For $|\xi| \geq 2/\delta$ this is invertible by Lemma 13.1.1. If $t > 0$ both coefficients are positive, and $|\xi_1| \geq 1$ or $|\xi_2| \geq 1$. If $t = 0$, at a point where $\chi_1 > 0$ we have $|\xi_1| \geq 2$, and where $\chi_2 > 0$ we have $|\xi_2| \geq 2$. So m_t is a homotopy of triples. Moreover m_0 coincides with a_P for $|\xi| \geq 2/\delta$, and $m_1 = m(1,1)$. Different choices of cut-offs or connections are joined by homotopies, through which the same argument applies. □

13.2.2 The Clifford tensor product of families

We now make the following assumption on P:

$$\begin{gathered} P \in \Psi^0(\tilde{M}/M; E_+, E_-) \text{ is elliptic, each } P_m \text{ is surjective, and } \operatorname{Nul}(P_m) = \mathbb{C}e(m), \\ e \in \mathcal{C}^\infty(\tilde{M}; E_+),\ \textstyle\int_{Y_m} |e|^2 \nu = 1 \text{ for all } m \in M. \end{gathered} \tag{13.2.4}$$

So $\operatorname{ind}(P) = 1$ is the trivial line bundle, and the null bundle itself is trivialized. Let Π be the orthogonal projection onto $\mathbb{C}e$. The next lemma shows that any P with $\operatorname{ind}(P) = 1 \in K^0(M)$ can be brought to this form without changing its symbol class.

Lemma 13.2.2. *Let $P \in \Psi^0(\tilde{M}/M; E_+, E_-)$ be elliptic with $\operatorname{ind}(P) = [\mathbb{C}] \in K^0(M)$. Then for some N there is $P' \in \Psi^0(\tilde{M}/M; E_+ \oplus \mathbb{C}^N, E_- \oplus \mathbb{C}^N)$ with $P' - P \oplus \mathrm{Id} \in \Psi^{-\infty}$ satisfying (13.2.4). In particular $[\sigma_0(P')] = [\sigma_0(P)]$.*

Proof. Choose S with N columns as in Lemma 13.1.3 and let $L = \mathcal{N}(P, S)$, a bundle over M with $[L] - [\mathbb{C}^N] = [\mathbb{C}]$. Thus $L \oplus \mathbb{C}^k \cong \mathbb{C}^{N+1+k}$ for some k; adding k zero columns to S replaces L by $L \oplus \mathbb{C}^k$, so we may assume that $L \cong \mathbb{C}^{N+1}$. Choose a smooth frame $\ell_1, \dots, \ell_{N+1}$ of L and let $q : L \to \mathbb{C}^N$ be the first N coordinates, extended to $\mathcal{C}^\infty(Y_m; E_+) \oplus \mathbb{C}^N$ by composing with the orthogonal projection onto L. Normalize ν to have mass one on each fibre Y_m and let $\Pi_0 w = \int_{Y_m} w\, \nu$ be the projection of $\mathcal{C}^\infty(Y_m; \mathbb{C}^N)$ onto constants. Define

$$P'(u, w) = (Pu + S\Pi_0 w,\ (\mathrm{Id} - \Pi_0)w + q(u, \Pi_0 w)),$$

with $q(\cdot) \in \mathbb{C}^N$ regarded as a constant function. All the terms except P and Id have smooth kernels, so $P' - P \oplus \mathrm{Id}$ is smoothing. If $P'(u, w) = 0$ then the mean-zero part $(\mathrm{Id} - \Pi_0)w$ and

the constant $q(u, \Pi_0 w)$ both vanish, so $w = c$ is constant, $Pu + Sc = 0$ and $q(u, c) = 0$; thus $(u, c) \in \mathbb{C}\ell_{N+1}$. Conversely every multiple of ℓ_{N+1} gives an element of the null space. To see surjectivity, given (g, h) write $h = h_0 + h_1$ with $h_0 = \Pi_0 h$. Take $w = h_1 + c$, and choose (u, c) with $Pu + Sc = g$ and $q(u, c) = h_0$; this is possible since solutions of $Pu + Sc = g$ form an affine space modelled on L and $q : L \to \mathbb{C}^N$ is onto. Normalizing ℓ_{N+1} gives e. □

Under (13.2.4), and for any elliptic $A \in \Psi^0(M/B; F_+, F_-)$, define the *Clifford tensor product*

$$P \otimes A = \begin{pmatrix} P & 0 \\ A_\Pi & P^* \end{pmatrix} \in \Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; G_+, G_-), \tag{13.2.5}$$

acting from $G_+ = E_+ \otimes F_+ \oplus E_- \otimes F_-$ to $G_- = E_- \otimes F_+ \oplus E_+ \otimes F_-$, with A_Π as in (P5).

Proposition 13.2.3. *Under* (13.2.4), $P \otimes A$ *is fully elliptic and* $\operatorname{ind}(P \otimes A) = \operatorname{ind}(A)$ *in* $K^0(B)$.

Proof. By (P3) and (P5),

$$\sigma(P \otimes A) = \begin{pmatrix} p & 0 \\ 0 & p^* \end{pmatrix}, \qquad \beta(P \otimes A)(\eta) = \begin{pmatrix} P_m & 0 \\ \Pi \otimes \alpha(\eta) & P_m^* \end{pmatrix}.$$

The first is invertible. For the second, recall that P_m is surjective, so P_m^* is injective with closed range $(\mathbb{C}e)^\perp = \operatorname{Ran}(\mathrm{Id} - \Pi)$, while $\Pi \otimes \alpha$ takes values in $\operatorname{Ran}\Pi$. If $P_m u = 0$ and $(\Pi \otimes \alpha)u + P_m^* v = 0$ then $u = e \otimes f$, both terms vanish separately, $v = 0$, and $\alpha f = 0$ forces $f = 0$. Given (g, h), choose u_0 with $P_m u_0 = g$, write $h - (\Pi \otimes \alpha)u_0 = e \otimes k + P_m^* v$, and set $u = u_0 + e \otimes \alpha^{-1} k$; then (u, v) is a preimage. So the base symbol is invertible (its inverse is again a pseudodifferential operator on Y_m), and $P \otimes A$ is fully elliptic.

Choose $S \in \mathcal{C}^\infty(M; \operatorname{Hom}(\mathbb{C}^N, F_-))$ with $A \oplus S$ surjective, and let $\tilde{S} = e \otimes S$, mapping $\mathbb{C}^N$ into the second summand $E_+ \otimes F_-$ of G_-. We claim that $(P \otimes A) \oplus \tilde{S}$ is surjective on each fibre, with null space

$$\operatorname{Nul}\big((P \otimes A) \oplus \tilde{S}\big) = \{(e \otimes f, 0, c);\ (f, c) \in \operatorname{Nul}(A \oplus S)\}. \tag{13.2.6}$$

Indeed, $(P \otimes A)(u, v) + \tilde{S}c = (Pu,\ e \otimes (A\langle u, e\rangle + Sc) + P^* v)$. The second component is the sum of a term in $\operatorname{Ran}\Pi$ and a term in $\operatorname{Ran}(\mathrm{Id} - \Pi)$. If the whole vanishes then $u = e \otimes f$, $v = 0$ and $Af + Sc = 0$. Conversely, given (g, h) choose u_0 with $Pu_0 = g$, decompose $h - e \otimes A\langle u_0, e\rangle = e \otimes k + P^* v$, solve $Af + Sc = k$ and put $u = u_0 + e \otimes f$. The map $(f, c) \mapsto (e \otimes f, 0, c)$ is a smooth bundle isomorphism $\mathcal{N}(A, S) \to \mathcal{N}(P \otimes A, \tilde{S})$, so the two stabilized indices coincide. □

13.2.3 Deformation to an ordinary family

Now choose families

$$\tilde{A}_\pm \in \Psi^0(\tilde{M}/B; E_\pm \otimes F_+, E_\pm \otimes F_-), \quad \tilde{P}_\pm \in \Psi^0(\tilde{M}/B; E_+ \otimes F_\pm, E_- \otimes F_\pm),$$

with principal symbols $\sigma_0(\tilde{A}_\pm) = \chi_2\ \mathrm{Id}_{E_\pm} \otimes \alpha$ and $\sigma_0(\tilde{P}_\pm) = \chi_1\, p \otimes \mathrm{Id}_{F_\pm}$, extended homogeneously.

Theorem 13.2.4 (Multiplicativity)**.** *Assume* (13.2.4) *and let* $A \in \Psi^0(M/B; F_+, F_-)$ *be elliptic. The family*

$$\tilde{A}_P = \begin{pmatrix} \tilde{P}_+ & -\tilde{A}_-^* \\ \tilde{A}_+ & \tilde{P}_-^* \end{pmatrix} \in \Psi^0(\tilde{M}/B; G_+, G_-)$$

is elliptic with principal symbol a_P, *and*

$$\operatorname{ind}(\tilde{A}_P) = \operatorname{ind}(A) \in K^0(B).$$

Proof. That $\sigma_0(\tilde{A}_P) = a_P$ is immediate, so $\tilde{A}_P$ is elliptic. We connect $P \otimes A$ to $\tilde{A}_P$ by two homotopies of fully elliptic product-type families; by Proposition 13.2.3 and Lemma 13.1.3(iii) this proves the theorem.

First homotopy. For $t \in [0,1]$ let

$$Q_t = \begin{pmatrix} P & -t\tilde{A}_-^* \\ (1-t)A_\Pi + t\tilde{A}_+ & P^* \end{pmatrix} \in \Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; G_+, G_-),$$

so $Q_0 = P \otimes A$. On the blow-up of $\Sigma = \Sigma_1$ the functions $\gamma^* p$ and $\chi_2\alpha$ are smooth, and by (P3)–(P5)

$$\sigma(Q_t) = \begin{pmatrix} \gamma^* p & -t\chi_2\alpha^* \\ t\chi_2\alpha & \gamma^* p^* \end{pmatrix} = m(1, t\chi_2),$$

invertible by Lemma 13.1.1 since $\gamma^* p$ is invertible. For the base symbol note that $\chi_2 = 1$ near Σ_1, and on Σ_1 the projection onto the second summand of (13.2.1) is the identity, so $\beta(\tilde{A}_\pm)(\eta)$ is multiplication by $\mathrm{Id}_{E_\pm} \otimes \alpha(\eta)$. Hence, writing $\alpha = \alpha(\eta)$,

$$\beta(Q_t)(\eta) = \begin{pmatrix} P_m & -t\alpha^* \\ (1-t)\Pi\alpha + t\alpha & P_m^* \end{pmatrix}.$$

Here α acts on the F factor and commutes with P_m, P_m^* and Π. Decompose the domain as $\operatorname{Ran}\Pi \otimes F_+ \oplus \operatorname{Ran}(\mathrm{Id} - \Pi) \otimes F_+ \oplus E_- \otimes F_-$ and the range as $E_- \otimes F_+ \oplus \operatorname{Ran}\Pi \otimes F_- \oplus \operatorname{Ran}(\mathrm{Id} - \Pi) \otimes F_-$. Then $\beta(Q_t)(\eta)$ becomes

$$\begin{pmatrix} 0 & P_1 & -t\alpha^* \\ \alpha & 0 & 0 \\ 0 & t\alpha & P_1^* \end{pmatrix},$$

where $P_1 : \operatorname{Ran}(\mathrm{Id} - \Pi) \to L^2(Y_m; E_-)$ is the restriction of P_m, an isomorphism. The middle row is invertible on the first summand, and the remaining 2×2 block is $m(1,t)$ formed with P_1 and α, which is invertible by Lemma 13.1.1. So Q_t is fully elliptic for all $t \in [0,1]$, and $Q_1 = \left(\begin{smallmatrix} P & -\tilde{A}_-^* \\ \tilde{A}_+ & P^* \end{smallmatrix}\right)$.

Second homotopy. For $t \in [0,1]$ let

$$R_t = \begin{pmatrix} (1-t)P + t\tilde{P}_+ & -\tilde{A}_-^* \\ \tilde{A}_+ & (1-t)P^* + t\tilde{P}_-^* \end{pmatrix},$$

so $R_0 = Q_1$ and $R_1 = \tilde{A}_P$. Its usual symbol is $m(1 - t\chi_2, \chi_2)$ formed with $\gamma^* p$ and $\chi_2\alpha$ (recall $1 - t + t\chi_1 = 1 - t\chi_2$). Where $\chi_2 < 1$ the first coefficient is positive and p is invertible; where $\chi_2 = 1$ the second coefficient is positive and α is invertible. So $\sigma(R_t)$ is invertible. Since $\chi_1 = 0$ near Σ_1, $\beta(\tilde{P}_\pm) = 0$, and

$$\beta(R_t)(\eta) = \begin{pmatrix} (1-t)P_m & -\alpha^* \\ \alpha & (1-t)P_m^* \end{pmatrix} = m(1-t, 1),$$

which is invertible by Lemma 13.1.1 because α is invertible. Thus R_t is fully elliptic for all t. By (P4) the index of R_1 as a product-type family is its index as an ordinary family. □

Corollary 13.2.5. *For the tower* (13.1.3)*, let* $b \in K_{\mathrm{c}}(T^*(\tilde{M}/M))$ *satisfy* $\mathrm{ind}_{\mathrm{a}}(b) = 1 \in K^0(M)$*. Then for every* $x \in K_{\mathrm{c}}(T^*(M/B))$*,*

$$\mathrm{ind}_{\mathrm{a}}(b \cdot \tilde{\phi}^* x) = \mathrm{ind}_{\mathrm{a}}(x) \in K^0(B),$$

where the product is formed using (13.2.1)*. That is, the diagram*

$$\begin{array}{ccc} K_{\mathrm{c}}(T^*(M/B)) & \xrightarrow{b\cdot\tilde{\phi}^*(\cdot)} & K_{\mathrm{c}}(T^*(\tilde{M}/B)) \\ & {\scriptstyle \mathrm{ind_a}}\searrow \quad \swarrow{\scriptstyle \mathrm{ind_a}} & \\ & K^0(B) & \end{array}$$

commutes.

Proof. By (F2) write $b = [\sigma_0(P)]$ and $x = [\sigma_0(A)]$. By Lemma 13.2.2 we may assume (13.2.4). Then $b \cdot \tilde{\phi}^* x = [a_P] = [\sigma_0(\tilde{A}_P)]$ by Lemma 13.2.1, and Theorem 13.2.4 gives the result. □

Remark 13.2.6. The degenerate case $M = B$ is included: then A is a bundle map a over B, $\Sigma_1 = \emptyset$, and the corollary says $\mathrm{ind_a}(b \cdot x) = x$ for $x \in K^0(B) = K_{\mathrm{c}}(B)$.

The product is also natural with respect to open embeddings, in the following sense.

Lemma 13.2.7. *Let V_1, V_2 be vector bundles over a compact manifold Y, $O \subset Y$ open and $k : V_1|_O \hookrightarrow V_1$, $k' : (V_1 \oplus V_2)|_O \hookrightarrow V_1 \oplus V_2$ the inclusions. For $x_1 \in K_{\mathrm{c}}(V_1|_O)$ and $x_2 \in K_{\mathrm{c}}(V_2)$,*

$$k_*(x_1) \cdot x_2 = k'_*(x_1 \cdot x_2|_O) \in K_{\mathrm{c}}(V_1 \oplus V_2).$$

Proof. The product $x_1 \cdot x_2|_O$ is compactly supported since x_1 is supported over a compact subset of O. By (K3) represent x_1 by (E_+, E_-, a_1) with $E_\pm$ trivialized and $a_1 = \mathrm{Id}$ outside a compact set $C \subset V_1|_O$, and let a_2 (extended) represent x_2. Then $k_* x_1 \cdot x_2$ is represented by $m(1,1)$ formed with the extension of a_1 by Id. Choose $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(V_1|_O)$, $0 \le \psi \le 1$, equal to 1 near C, pulled back to $V_1 \oplus V_2$, and consider $m(1, (1-t) + t\psi)$. Where a_1 is not invertible, $\psi = 1$ and this equals $m(1,1)$, invertible where a_2 is; elsewhere the first coefficient is 1 and a_1 invertible. So it is a homotopy of triples. At $t = 1$, outside $\operatorname{supp}\psi$ the map is $\left(\begin{smallmatrix} \mathrm{Id} & 0 \\ 0 & \mathrm{Id} \end{smallmatrix}\right)$ in the trivializations, so the class is the extension by the identity of the triple $m(1,\psi)$ over $(V_1 \oplus V_2)|_O$, which in turn is homotopic there to $m(1,1)$ by the same homotopy. □

13.3 Excision

Let $\phi_j : M_j \to B$, $j = 1, 2$, be fibrations of compact manifolds and U a manifold with open embeddings $i_j : U \hookrightarrow M_j$ such that

$$\begin{array}{ccccc} M_1 & \xleftarrow{i_1} & U & \xrightarrow{i_2} & M_2 \\ & {\scriptstyle \phi_1}\searrow & \downarrow{\scriptstyle \phi_U} & \swarrow{\scriptstyle \phi_2} & \\ & & B & & \end{array} \tag{13.3.1}$$

commutes. Then ϕ_U is a submersion (with non-compact fibres), $T^*(U/B)$ is well defined, and i_j induce open embeddings $T^*(U/B) \hookrightarrow T^*(M_j/B)$, hence extension maps $(i_j)_*$ by (K3).

Theorem 13.3.1 (Excision)**.** *In the situation* (13.3.1)*,*

$$\mathrm{ind_a} \circ (i_1)_* = \mathrm{ind_a} \circ (i_2)_* : K_{\mathrm{c}}(T^*(U/B)) \longrightarrow K^0(B).$$

Proof. Step 1: normal form. Let $\varpi : T^*(U/B) \to U$ be the projection. Let $x \in K_{\mathrm{c}}(T^*(U/B))$ be represented by (E_+, E_-, a) with a invertible outside a compact set $K \subset T^*(U/B)$. Since $T^*(U/B)$ retracts onto U we may assume $E_\pm = \varpi^* E_\pm$ for bundles $E_\pm$ over U. The image $K' = \varpi(K)$ is compact, and over $U \setminus K'$ the map a is defined and invertible on the whole fibres of $T^*(U/B)$. Choose $\rho \in \mathcal{C}^\infty_{\mathrm{c}}(U)$, $0 \le \rho \le 1$, with $\rho = 1$ near K', and set $a_s(u,\xi) = a(u, (1 - s + s\rho(u))\xi)$. Where $\rho(u) = 1$, $a_s = a$; elsewhere $u \notin K'$ and a_s is invertible. So a_s is a homotopy of triples, and $a_1 = \varpi^* h$ over $U \setminus \operatorname{supp}\rho$, where $h(u) = a(u,0)$ is a bundle isomorphism $E_+ \to E_-$ there. Using h to identify E_+ with E_- outside a compact set, and after stabilization and the construction of (K3), we arrive at a representative of the following form: bundles $E_\pm$ over U with an isomorphism $\tau : E_+ \to E_-$ over $U \setminus C$, $C \subset U$ compact, both trivialized there, and a symbol a, homogeneous of degree 0 for $|\xi| \ge 1$, with $a = \tau$ over $U \setminus C$. Then $(i_j)_* x$ is represented by the bundles $\tilde{E}^j_\pm$ over M_j obtained by extending $E_\pm$ trivially, with the symbol $\tilde{a}^j$ extended by $\tau = \mathrm{Id}$.

Step 2: operators that agree. Choose compact sets $C \subset K_1 \subset \operatorname{int} K_2 \subset K_2 \subset \operatorname{int} K_3 \subset K_3 \subset U$ and $\psi, \psi' \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ with $\psi = 1$ near C, $\operatorname{supp}\psi \subset \operatorname{int} K_1$, $\psi' = 1$ on K_1, $\operatorname{supp}\psi' \subset \operatorname{int} K_2$. The bundle map $J = (1-\psi)\tau$ is defined on all of M_j (as τ outside $i_j(U)$). The symbol $\tilde{a}^j - J$ vanishes outside $\varpi^{-1}(\operatorname{supp}\psi)$. Choose $T_U \in \Psi^0(U/B; E_+, E_-)$ with $\sigma_0(T_U) = a - (1-\psi)\tau$ and replace it by $\psi' T_U \psi'$, which has the same symbol and Schwartz kernel supported in $K_2 \times_B K_2$. Set

$$P_j = J + (i_j)_* T_U \in \Psi^0(M_j/B; \tilde{E}^j_+, \tilde{E}^j_-),$$

where $(i_j)_* T_U$ is the operator whose kernel is the push-forward of that of T_U. Then $\sigma_0(P_j) = \tilde{a}^j$, so $[\sigma_0(P_j)] = (i_j)_* x$, and P_1, P_2 have the following locality properties:

(a) for every v and every point $y \notin i_j(K_2)$, $(P_j v)(y) = \tau(y) v(y)$; if moreover v vanishes near $i_j(K_2)$ then $P_j v = \tau v$ (both sides vanishing on $i_j(K_2)$);

(b) if v is supported in $i_j(L)$, $L \subset U$ compact, then $P_j v$ is supported in $i_j(L \cup K_2)$, and $i_1^*(P_1 v_1) = i_2^*(P_2 v_2)$ whenever v_1, v_2 are supported in $i_1(L)$, $i_2(L)$ and $i_1^* v_1 = i_2^* v_2$.

(In (b) only the restriction of the bundles to U, where they are the same, is involved.)

Step 3: common stabilization. Fix b. Sections supported in $i_j(\operatorname{int} K_3)$ span $L^2(M_{j,b}; \tilde{E}_-)$ modulo $\operatorname{Ran} P_{j,b}$: given g, choose $\psi_3 \in \mathcal{C}^\infty_{\mathrm{c}}(\operatorname{int} K_3)$ equal to 1 near K_2; then by (a) $P_j(\tau^{-1}(1-\psi_3)g) = (1-\psi_3)g$. Since the cokernel is finite dimensional and smooth sections are dense, it is spanned by finitely many smooth sections supported in $i_j(\operatorname{int} K_3)$. Running the proof of Lemma 13.1.3(i) for P_1 with this choice (and cut-offs in the base) gives sections $v_1, \dots, v_N$ of E_- supported in the interior of a compact subset of U, which we use for both j, so that $P_1 \oplus S$ is surjective.

Then $P_2 \oplus S$ is surjective too. Given g on $M_{2,b}$, write $g = \psi_3 g + (1-\psi_3)g$; the second term is $P_2(\tau^{-1}(1-\psi_3)g)$. For the first, transplant it to M_1 and solve $P_1 u + Sc = \psi_3 g$. Outside $i_1(K_2 \cup \operatorname{supp}\psi_3 \cup \operatorname{supp} S)$, (a) gives $\tau u = 0$ pointwise, so u is supported in $i_1(U)$; by (b) its transplant u_2 to M_2 satisfies $P_2 u_2 + Sc = \psi_3 g$. The same argument shows that the elements of $\operatorname{Nul}(P_j \oplus S)$ are supported in a fixed compact subset of $i_j(U)$, and that transplanting gives an isomorphism of bundles $\mathcal{N}(P_1, S) \cong \mathcal{N}(P_2, S)$ over B. Hence $\operatorname{ind_a}((i_1)_* x) = \operatorname{ind}(P_1) = \operatorname{ind}(P_2) = \operatorname{ind_a}((i_2)_* x)$ by Lemma 13.1.3. □

13.4 The families index theorem in K-theory

13.4.1 Embedding the fibration

Lemma 13.4.1. *Let $\phi : M \to B$ be a fibration of compact manifolds with fibres of dimension n. For N large there is an embedding $e : M \to \mathbb{R}^N \times B$ with $\pi_B \circ e = \phi$. If $\mathcal{V} \to M$ is*

the normal bundle, of rank $k = N - n$, there is a diffeomorphism Θ from the total space of $\mathcal{V}$ onto an open neighbourhood Ω of $e(M)$, with $\Theta(0_m) = e(m)$ and $\pi_B \circ \Theta = \phi \circ \varpi$, $\varpi : \mathcal{V} \to M$ the projection. Moreover $e(M)$ and Ω may be taken inside $(0, 2\pi)^N \times B$.

Proof. Let $\iota : M \to \mathbb{R}^N$ be an embedding (Whitney) and $e = (\iota, \phi)$; it is an injective immersion of a compact manifold, hence an embedding, and $\pi_B \circ e = \phi$. Define $\mathcal{V}_m = (d\iota(T_m(M/B)))^\perp \subset \mathbb{R}^N$ and $\Theta(m, v) = (\iota(m) + v, \phi(m))$. At the zero section $d\Theta(w, v) = (d\iota(w) + v, d\phi(w))$; if this vanishes then w is vertical, and $d\iota(w) + v = 0$ with $v \perp d\iota(w)$ forces $w = 0$, $v = 0$. Since $\dim M + k = N + \dim B$, Θ is a local diffeomorphism near the zero section, injective on it, hence (by the usual compactness argument) a diffeomorphism from $\{|v| < \delta\}$ onto an open set. Composing with $v \mapsto \delta v/(1 + |v|^2)^{1/2}$ gives Θ. Translating and scaling ι puts everything inside $(0, 2\pi)^N \times B$. □

Let $I = (0, 2\pi)$. Both Θ and the inclusion $\Omega \subset I^N \times B$ commute with the projections to B, so they induce maps of vertical cotangent bundles: $\Theta_* : K_{\mathrm{c}}(T^*(\mathcal{V}/B)) \to K_{\mathrm{c}}(T^*(\Omega/B))$ and the extension $j_* : K_{\mathrm{c}}(T^*(\Omega/B)) \to K_{\mathrm{c}}(T^*(I^N \times B/B)) = K_{\mathrm{c}}(T^*I^N \times B)$; note $T^*I^N \cong \mathbb{R}^{2N}$.

13.4.2 The Thom class

Consider $\mathcal{V}$ as a fibration over M. Its vertical cotangent bundle is $T^*(\mathcal{V}/M) = \varpi^*\mathcal{V}^*$, whose total space is $\mathcal{V} \oplus \mathcal{V}^* \cong \mathcal{V} \otimes \mathbb{C}$ via $(v, \eta) \mapsto v + i\eta^\sharp$. On the complex vector bundle $W = \varpi^*(\mathcal{V} \otimes \mathbb{C})$ over this total space, with the Hermitian metric induced by a metric on $\mathcal{V}$, Clifford multiplication $c(w) = w\wedge\cdot - \iota_w$ maps $\Lambda^{\mathrm{ev}}W \to \Lambda^{\mathrm{odd}}W$ and satisfies $c(w)^*c(w) = |w|^2$. The *Thom class* is

$$\lambda_{\mathcal{V}} = \left[(\Lambda^{\mathrm{ev}}W, \Lambda^{\mathrm{odd}}W, c)\right] \in K_{\mathrm{c}}(T^*(\mathcal{V}/M)). \tag{13.4.1}$$

By (13.2.1) for the tower $\mathcal{V} \to M \to B$ (with non-compact fibres, which does not affect the linear algebra), $T^*(\mathcal{V}/B) \cong T^*(\mathcal{V}/M) \oplus \varpi^*T^*(M/B)$, and we define the *Thom homomorphism*

$$\tau : K_{\mathrm{c}}(T^*(M/B)) \longrightarrow K_{\mathrm{c}}(T^*(\mathcal{V}/B)), \quad \tau(x) = \lambda_{\mathcal{V}} \cdot \varpi^*x.$$

(The product is compactly supported because $\lambda_{\mathcal{V}}$ is supported in the zero section, i.e. over a compact set.)

13.4.3 The Bott element and the trivial fibration

Let $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$, so that $I = (0, 2\pi) \subset \mathbb{S}$. Let $\Pi_>$ be the orthogonal projection of $L^2(\mathbb{S})$ onto $\overline{\mathrm{span}}\{e^{ik\theta}; k \geq 1\}$; it lies in $\Psi^0(\mathbb{S})$ with symbol 1 for $\xi > 0$ and 0 for $\xi < 0$ (it differs from the Szegő projector of Chapter 10 by the projection onto constants). Define

$$P_1 = e^{-i\theta}\Pi_> + (\mathrm{Id} - \Pi_>) \in \Psi^0(\mathbb{S}), \quad \sigma_0(P_1)(\theta, \xi) = \begin{cases} e^{-i\theta} & \xi > 0, \\ 1 & \xi < 0. \end{cases} \tag{13.4.2}$$

On Fourier coefficients, $(P_1u)_j = u_{j+1}$ for $j \geq 1$, $(P_1u)_0 = u_0 + u_1$ and $(P_1u)_j = u_j$ for $j < 0$. So P_1 is surjective and $\mathrm{Nul}(P_1) = \mathbb{C}(e^{i\theta} - 1)$; in particular $\mathrm{ind}(P_1) = 1$, and the constant family P_1 satisfies (13.2.4) for any fibration with fibre $\mathbb{S}$ of product form.

Let $f \in \mathcal{C}^\infty([0, 2\pi])$ be non-decreasing with $f = 0$ near 0, $f = 2\pi$ near 2π. The symbols with $e^{-i\theta}$ replaced by $e^{-i((1-s)\theta + sf(\theta))}$ are a homotopy of elliptic symbols, and at $s = 1$ the symbol is 1 for θ near 0. Hence there is a class

$$b_1 \in K_{\mathrm{c}}(T^*I) = K_{\mathrm{c}}(\mathbb{R}^2), \qquad (k_1)_*b_1 = [\sigma_0(P_1)] \in K_{\mathrm{c}}(T^*\mathbb{S}),$$

where $k_1 : T^*I \hookrightarrow T^*\mathbb{S}$; b_1 is the *Bott element*. Let $\mathbb{T}^N = \mathbb{S}^N$ and $k_N : T^*I^N \times B \hookrightarrow T^*(\mathbb{T}^N \times B/B)$. Set

$$\beta_N : K^0(B) \longrightarrow K_{\mathrm{c}}(T^*I^N \times B), \quad \beta_N(y) = b_1 \boxtimes \cdots \boxtimes b_1 \boxtimes y.$$

We use Bott periodicity in the following form.

(BP) For every manifold X, the map $K_{\mathrm{c}}(X) \ni x \mapsto b_1 \boxtimes x \in K_{\mathrm{c}}(T^*I \times X)$ is an isomorphism. Consequently each β_N is an isomorphism.

This is proved in Chapter 11 through the Toeplitz index; in the stated form (exterior multiplication by the generator) it is [17], Theorem 2.4.9. For compact X the two forms are equivalent by the following proposition, which shows that the index on the circle is a left inverse of $b_1 \boxtimes \cdot$.

Proposition 13.4.2 (Normalization)**.** *For every $N \geq 0$,*

$$\mathrm{ind}_{\mathrm{a}} \circ (k_N)_* \circ \beta_N = \mathrm{Id}_{K^0(B)}\,.$$

Hence, by (BP), $\mathrm{ind}_{\mathrm{a}} \circ (k_N)_* = \beta_N^{-1} : K_{\mathrm{c}}(T^*I^N \times B) \to K^0(B)$.

Proof. Induction on N; for $N = 0$ this is Remark 13.1.4(b). Consider the tower $\mathbb{T}^N \times B \to \mathbb{T}^{N-1} \times B \to B$, the first map forgetting the last circle factor, and the constant family P_1 on its fibres, which satisfies (13.2.4). Here the splitting (13.2.1) is the product decomposition. For $y \in K^0(B)$, Lemma 13.2.7 (applied in both factors) gives

$$[\sigma_0(P_1)] \cdot (k_{N-1})_*\beta_{N-1}(y) = (k_1)_*b_1 \cdot (k_{N-1})_*\beta_{N-1}(y) = (k_N)_*\beta_N(y).$$

So Corollary 13.2.5 and the inductive hypothesis give

$$\mathrm{ind}_{\mathrm{a}}\big((k_N)_*\beta_N(y)\big) = \mathrm{ind}_{\mathrm{a}}\big((k_{N-1})_*\beta_{N-1}(y)\big) = y.$$

□

Remark 13.4.3. The Thom class of the trivial line bundle, $\lambda_{\mathbb{R}} \in K_{\mathrm{c}}(T^*\mathbb{R})$, is represented by the symbol $x + i\xi$ on $\Lambda^0 \to \Lambda^1$, which has winding number 1 around the circle $|(x,\xi)| = R$ traversed counterclockwise in the (x,ξ)-plane. The symbol of b_1 has the same winding number (it is 1 on the lower edge $\xi = -R$ and on the vertical edges of a large square, and $e^{-if(\theta)}$ on the upper edge, traversed in the direction of decreasing θ). Since classes in $K_{\mathrm{c}}(\mathbb{R}^2)$ are determined by this winding number (Chapter 11), $\lambda_{\mathbb{R}} = b_1$ under $I \cong \mathbb{R}$. For a trivial bundle $\mathcal{V} = M \times \mathbb{R}^k$ it follows that $\lambda_{\mathcal{V}}$ is the pull-back of $b_1 \boxtimes \cdots \boxtimes b_1$ (k factors), since $\Lambda(W_1 \oplus W_2) = \Lambda W_1 \otimes \Lambda W_2$ as graded bundles and Clifford multiplication is the Clifford product (13.1.2).

13.4.4 The theorem

Using Lemma 13.4.1 and (BP) we define the *topological index*

$$\mathrm{ind}_{\mathrm{t}} = \beta_N^{-1} \circ j_* \circ \Theta_* \circ \tau : K_{\mathrm{c}}(T^*(M/B)) \longrightarrow K^0(B). \tag{13.4.3}$$

The definition involves the choice of the embedding (and of the tubular neighbourhood and the metrics); the following theorem shows a posteriori that $\mathrm{ind}_{\mathrm{t}}$ does not depend on these choices.

Theorem 13.4.4 (Atiyah–Singer, K-theory form)**.** *For every fibration $\phi : M \to B$ of compact manifolds,* $\mathrm{ind_a} = \mathrm{ind_t}$*, i.e. the diagram*

$$\begin{array}{ccccc} K_{\mathrm{c}}(T^*(M/B)) & \xrightarrow{\tau} & K_{\mathrm{c}}(T^*(\mathcal{V}/B)) & \xrightarrow{j_*\Theta_*} & K_{\mathrm{c}}(T^*I^N \times B) \\ & \searrow^{\mathrm{ind_a}} & & & \downarrow{\scriptstyle \beta_N^{-1}} \\ & & & & K^0(B) \end{array}$$

commutes. In particular $\mathrm{ind_t}$ *is independent of the embedding.*

Besides what was recalled in Section 13.1, the proof uses one result which we do not prove.

(TH) Let $\tilde{M} = S(\mathcal{V} \oplus \mathbb{R}) \to M$ be the fibrewise one-point compactification of $\mathcal{V}$ (the unit sphere bundle of $\mathcal{V} \oplus \mathbb{R}$), with fibre $\mathbb{S}^k$, and $k_{\mathcal{V}} : \mathcal{V} \hookrightarrow \tilde{M}$ the inverse stereographic projection from the section at infinity. Then

$$\mathrm{ind_a}\big((k_{\mathcal{V}})_*\lambda_{\mathcal{V}}\big) = 1 \in K^0(M).$$

Restricted to any point of M (using Remark 13.1.4(d)) this holds by Remark 13.4.3, Proposition 13.4.2 (with B a point) and excision (comparing the sphere $\mathbb{S}^k$ with the torus $\mathbb{T}^k$, both of which contain $\mathbb{R}^k$). By the same argument it holds in general when $\mathcal{V}$ is trivial. In general, however, a class in $K^0(M)$ is not determined by its restrictions to points. The standard proof shows that the index of the Bott operator on $\mathbb{S}^k$ is the trivial one-dimensional representation of $O(k)$, and then constructs the family over M from the orthonormal frame bundle of $\mathcal{V}$; see the Notes and [18].

Proof of Theorem 13.4.4. Let $x \in K_{\mathrm{c}}(T^*(M/B))$. Consider the tower $\tilde{M} \xrightarrow{\tilde{\phi}} M \xrightarrow{\phi} B$ with $\tilde{M} = S(\mathcal{V} \oplus \mathbb{R})$ as in (TH), and let $\hat{k} : T^*(\mathcal{V}/B) \hookrightarrow T^*(\tilde{M}/B)$ be the open embedding induced by $k_{\mathcal{V}}$. By (TH) the class $b = (k_{\mathcal{V}})_*\lambda_{\mathcal{V}}$ has $\mathrm{ind_a}(b) = 1$, so by Corollary 13.2.5 and Lemma 13.2.7 (with $Y = \tilde{M}$, $O = \mathcal{V}$; a splitting of $T^*(\tilde{M}/B)$ restricting to the one used for τ over $\mathcal{V}$ is chosen)

$$\mathrm{ind_a}(x) = \mathrm{ind_a}\big(b \cdot \tilde{\phi}^* x\big) = \mathrm{ind_a}\big(\hat{k}_*\tau(x)\big).$$

Now apply Theorem 13.3.1 with $U = \mathcal{V}$, $M_1 = \tilde{M}$, $i_1 = k_{\mathcal{V}}$, and $M_2 = \mathbb{T}^N \times B$, $i_2 = \Theta$ followed by the inclusion $\Omega \subset I^N \times B \subset \mathbb{T}^N \times B$ (all commuting with the maps to B):

$$\mathrm{ind_a}\big(\hat{k}_*\tau(x)\big) = \mathrm{ind_a}\big((k_N)_* j_*\Theta_*\tau(x)\big).$$

Finally Proposition 13.4.2 identifies the right side with $\beta_N^{-1}(j_*\Theta_*\tau(x)) = \mathrm{ind_t}(x)$. □

Remark 13.4.5. To summarize the logical structure: multiplicativity (Theorem 13.2.4), excision (Theorem 13.3.1) and the normalization (Proposition 13.4.2) are proved in this chapter from the calculus of Chapter 12; Bott periodicity (BP) comes from Chapter 11; the index of the Thom class (TH) is imported, except when the normal bundle is trivial. Traditionally (13.4.3) is the *definition* of the topological index. Read the other way, Theorem 13.4.4 is an effective method for computing the analytic index: everything has been reduced to the index on the trivial fibration, which is given by Bott periodicity, i.e. by repeated use of the Toeplitz index isomorphism of Chapter 11. In practice one may replace $\mathbb{R}^N \times B$ by $\mathbb{S}^N \times B$, as excision shows.

13.5 Chern–Weil theory and the Chern character

To pass from K-theory to cohomology we use the Chern character. We work with de Rham cohomology with complex coefficients; $H^{\mathrm{ev}}(X;\mathbb{C})$ denotes the cohomology in even degrees.

13.5.1 Connections and curvature

Let $E \to X$ be a complex vector bundle over a manifold. A *connection* on E is a differential operator $\nabla \in \mathrm{Diff}^1(X; E, \Lambda^1 \otimes E)$ with

$$\nabla(fu) = df \otimes u + f\nabla u, \quad f \in \mathcal{C}^\infty(X),\ u \in \mathcal{C}^\infty(X;E).$$

If $E = X \times \mathbb{C}^N$ then d, acting on the components, is a connection, and every connection is then $d + \gamma$ with γ a 1-form with values in $N \times N$ matrices. If ∇_i are connections on E over the sets of an open cover and ρ_i is a subordinate partition of unity then $\sum_i \rho_i \nabla_i$ is a connection. Hence every vector bundle admits connections, and the connections form an affine space over $\mathcal{C}^\infty(X; \Lambda^1 \otimes \mathrm{End}\, E)$. If $a : E \to \tilde{E}$ is a bundle isomorphism then $a\nabla a^{-1}$ is a connection on $\tilde{E}$.

A connection extends uniquely to $\nabla : \mathcal{C}^\infty(X; \Lambda^k \otimes E) \to \mathcal{C}^\infty(X; \Lambda^{k+1} \otimes E)$ for all k, so that

$$\nabla(\alpha \wedge u) = d\alpha \wedge u + (-1)^j \alpha \wedge \nabla u, \quad \alpha \in \mathcal{C}^\infty(X; \Lambda^j),\ u \in \mathcal{C}^\infty(X; \Lambda^* \otimes E).$$

(Locally $\nabla = d + \gamma$ with γ acting by exterior multiplication; the sign records the passage of ∇ past j factors.) The *curvature* $\omega_\nabla = \nabla^2$ commutes with multiplication by functions:

$$\nabla^2(fu) = \nabla(df \wedge u + f\nabla u) = -df \wedge \nabla u + df \wedge \nabla u + f\nabla^2 u = f\nabla^2 u,$$

so $\omega_\nabla \in \mathcal{C}^\infty(X; \Lambda^2 \otimes \mathrm{End}\, E)$; locally $\omega = d\gamma + \gamma \wedge \gamma$. The bundle $\Lambda^* \otimes \mathrm{End}\, E$ is a bundle of $\mathbb{Z}_2$-graded algebras, with product $(\alpha \otimes a)(\beta \otimes b) = \alpha \wedge \beta \otimes ab$, and $\mathrm{End}(E)$-valued forms act on $\Lambda^* \otimes E$. For $a \in \mathcal{C}^\infty(X; \Lambda^k \otimes \mathrm{End}\, E)$ set

$$\nabla a = [\nabla, a] = \nabla \circ a - (-1)^k a \circ \nabla \in \mathcal{C}^\infty(X; \Lambda^{k+1} \otimes \mathrm{End}\, E),$$

locally $da + \gamma \wedge a - (-1)^k a \wedge \gamma$. It satisfies $\nabla(ab) = (\nabla a)b + (-1)^k a \nabla b$, and the *Bianchi identity*

$$\nabla \omega_\nabla = [\nabla, \nabla^2] = 0.$$

The usual constructions are compatible with curvature. A connection on E induces one on E^* by $d(u^*(u)) = (\nabla^* u^*)(u) + u^*(\nabla u)$, with curvature $-\omega^t$. On $E \oplus F$, $\nabla_E \oplus \nabla_F$ has curvature $\omega_E \oplus \omega_F$. On $E \otimes F$, $\nabla(u \otimes v) = \nabla_E u \otimes v + u \otimes \nabla_F v$ has curvature

$$\omega_{E\otimes F} = \omega_E \otimes \mathrm{Id}_F + \mathrm{Id}_E \otimes \omega_F.$$

This gives connections on $\mathrm{Hom}(E,F) = F \otimes E^*$; the induced action on $a \in \mathcal{C}^\infty(X; \mathrm{Hom}(E,F))$ is $\nabla a = \nabla_F \circ a - a \circ \nabla_E$. For a smooth map $f : X' \to X$ the pull-back $f^*\nabla$ has curvature $f^*\omega_\nabla$. If $a : E \to \tilde{E}$ is an isomorphism, $a\nabla a^{-1}$ has curvature $a\omega_\nabla a^{-1}$.

13.5.2 The Chern character form

The fibre trace extends to $\mathrm{tr} : \mathcal{C}^\infty(X; \Lambda^j \otimes \mathrm{End}\, E) \to \mathcal{C}^\infty(X; \Lambda^j)$, $\mathrm{tr}(\alpha \otimes a) = \alpha\, \mathrm{tr}(a)$, and is graded-cyclic:

$$\mathrm{tr}(ab) = (-1)^{jk}\, \mathrm{tr}(ba), \quad a \in \Lambda^j \otimes \mathrm{End}\, E,\ b \in \Lambda^k \otimes \mathrm{End}\, E. \tag{13.5.1}$$

Lemma 13.5.1. *For any connection ∇ on E and $a \in \mathcal{C}^\infty(X; \Lambda^k \otimes \operatorname{End} E)$, $d \operatorname{tr}(a) = \operatorname{tr}(\nabla a)$.*

Proof. Both sides are local. In a trivialization $\nabla a = da + [\gamma, a]$, the bracket being graded, and $\operatorname{tr}[\gamma, a] = 0$ by (13.5.1); also $\operatorname{tr}(da) = d \operatorname{tr}(a)$. □

Definition 13.5.2. The *Chern character form* of (E, ∇) is

$$\operatorname{Ch}(\nabla) = \operatorname{tr} \exp\Big(\frac{i}{2\pi}\omega_\nabla\Big) = \operatorname{rank} E + \sum_{k \geq 1} \frac{i^k}{(2\pi)^k k!} \operatorname{tr}(\omega_\nabla^k) \in \mathcal{C}^\infty(X; \Lambda^{\mathrm{ev}}).$$

The sum is finite since $\omega^k = 0$ for $2k > \dim X$. Even forms commute, so exp has its usual properties. The normalization $i/2\pi$ makes the classes rational, as recalled below.

Proposition 13.5.3. *(i) $\operatorname{Ch}(\nabla)$ is closed.*

(ii) If $\nabla_t = \nabla + t\theta$, $\theta = \nabla' - \nabla$, then

$$\operatorname{Ch}(\nabla') - \operatorname{Ch}(\nabla) = d \operatorname{cs}(\nabla, \nabla'), \quad \operatorname{cs}(\nabla, \nabla') = \frac{i}{2\pi} \int_0^1 \operatorname{tr}\Big(\theta \exp\Big(\frac{i}{2\pi}\omega_{\nabla_t}\Big)\Big) dt.$$

So the class $\operatorname{Ch}(E) \in H^{\mathrm{ev}}(X; \mathbb{C})$ of $\operatorname{Ch}(\nabla)$ is independent of ∇.

(iii) $\operatorname{Ch}(E \oplus F) = \operatorname{Ch}(E) + \operatorname{Ch}(F)$, $\operatorname{Ch}(E \otimes F) = \operatorname{Ch}(E) \wedge \operatorname{Ch}(F)$, $\operatorname{Ch}(f^*E) = f^* \operatorname{Ch}(E)$, and $\operatorname{Ch}(E) = \operatorname{Ch}(\tilde{E})$ if $E \cong \tilde{E}$.

Proof. Write $c = i/2\pi$. (i) By Lemma 13.5.1, $d \operatorname{Ch}(\nabla) = \operatorname{tr}(\nabla e^{c\omega})$. Since ω is even, $\nabla(\omega^k) = \sum_j \omega^j (\nabla\omega) \omega^{k-1-j} = 0$ by Bianchi.

(ii) The curvature of ∇_t is $\omega_t = \omega + t\nabla\theta + t^2\theta^2$, so $\partial_t \omega_t = \nabla\theta + 2t\theta^2 = \nabla_t \theta$ (graded commutator, θ odd). By cyclicity, since ω_t is even,

$$\partial_t \operatorname{tr} e^{c\omega_t} = c \operatorname{tr}\big((\nabla_t \theta) e^{c\omega_t}\big) = c \operatorname{tr}\big(\nabla_t (\theta e^{c\omega_t})\big) = c\, d \operatorname{tr}(\theta e^{c\omega_t}),$$

using $\nabla_t e^{c\omega_t} = 0$ and Lemma 13.5.1. Integrate in t. (This is the transgression; equivalently, apply (i) to the connection $(1-t)\nabla + t\nabla' + dt\,\partial_t$ on the pull-back of E to $[0,1] \times X$ and integrate the dt-component.)

(iii) The curvature of a direct sum is block diagonal, so the traces add. For the tensor product, $\omega_E \otimes \mathrm{Id}$ and $\mathrm{Id} \otimes \omega_F$ commute, so $e^{c\omega_{E\otimes F}} = e^{c\omega_E} \otimes e^{c\omega_F}$ and $\operatorname{tr}_{E\otimes F} = \operatorname{tr}_E \otimes \operatorname{tr}_F$. Naturality follows from $\omega_{f^*\nabla} = f^*\omega_\nabla$, and invariance under isomorphism from $\operatorname{tr}(a\omega^k a^{-1}) = \operatorname{tr}(\omega^k)$. □

Definition 13.5.4. The *Chern character* $\operatorname{Ch} : K^0(X) \to H^{\mathrm{ev}}(X; \mathbb{C})$ is $\operatorname{Ch}([E_+] - [E_-]) = \operatorname{Ch}(E_+) - \operatorname{Ch}(E_-)$.

By Proposition 13.5.3(iii) this is well defined (it is unchanged by adding the same bundle to E_+ and E_- and by isomorphisms) and additive. For superbundles, with the graded tensor product (13.1.1), $\operatorname{Ch}(G_+) - \operatorname{Ch}(G_-) = (\operatorname{Ch}(E_+) - \operatorname{Ch}(E_-)) \wedge (\operatorname{Ch}(F_+) - \operatorname{Ch}(F_-))$. The product on $K^0(X)$ is induced by the graded tensor product, so Ch is a ring homomorphism,

$$\operatorname{Ch}(x \cdot y) = \operatorname{Ch}(x) \wedge \operatorname{Ch}(y), \quad x, y \in K^0(X).$$

With more care one shows that Ch takes values in rational cohomology. We will not prove, nor use, the theorem of Atiyah and Hirzebruch that $\operatorname{Ch} : K^0(X) \otimes \mathbb{C} \to H^{\mathrm{ev}}(X; \mathbb{C})$ is an isomorphism for compact X.

Example 13.5.5. Let $\pi \in \mathcal{C}^\infty(X; M(N,\mathbb{C}))$ be a smooth family of projections and $E = \pi\mathbb{C}^N$. The *Grassmann connection* is $\nabla u = \pi\, du$. From $\pi^2 = \pi$ we get $\pi\, d\pi + d\pi\, \pi = d\pi$, hence $\pi\, d\pi\, \pi = 0$ and $\pi(d\pi)^2 = (d\pi)^2\pi$. For $u = \pi u$,

$$\nabla^2 u = \pi\, d(\pi\, du) = \pi\, d\pi \wedge du = \pi\, d\pi \wedge (d\pi\, u + \pi\, du) = \pi(d\pi)^2 u,$$

so $\omega = \pi(d\pi)^2\pi$ and

$$\mathrm{Ch}(\nabla) = \sum_{k\geq 0} \frac{1}{k!}\Big(\frac{i}{2\pi}\Big)^k \operatorname{tr}(\pi(d\pi)^{2k}). \tag{13.5.2}$$

For the tautological line bundle over $\mathbb{C}P^1$, near $w = 0$ in the affine chart, $\pi = vv^*/|v|^2$ with $v = (1, w)$, and at $w = 0$ one finds $\operatorname{tr}(\pi(d\pi)^2) = d\bar w \wedge dw = 2i\, dx \wedge dy$. So $\frac{i}{2\pi}\operatorname{tr}(\pi(d\pi)^2)$ is negative at $w = 0$ (and, by $U(2)$-invariance, at every point), and the degree two part of Ch integrates to -1 (Exercise 13.2).

13.6 The odd Chern character and Toeplitz families

13.6.1 The odd Chern character

For a smooth map $a : X \to \mathrm{GL}(N,\mathbb{C})$ set $\theta = a^{-1}da$. In Chapter 10 it was shown that $\operatorname{tr}(\theta^{2k+1})$ is closed, that its class depends only on the homotopy class of a, and that it is additive. We fix the normalization:

$$\mathrm{Ch}_{\mathrm{odd}}(a) = \sum_{k\geq 0} c_k \operatorname{tr}((a^{-1}da)^{2k+1}), \qquad c_k = \frac{i}{2\pi}\Big(\frac{-i}{2\pi}\Big)^k \frac{k!}{(2k+1)!}. \tag{13.6.1}$$

The class of $\mathrm{Ch}_{\mathrm{odd}}(a)$ in $H^{\mathrm{odd}}(X;\mathbb{C})$ depends only on the class of a in $K^{-1}(X)$, since stabilization by Id does not change the forms, and by Chapter 10 the forms extend to maps into $G^{-\infty}$ (with Tr in place of tr), every class having a finite rank representative. For later use we recall the transgression formula from Chapter 10: if a_t is a smooth family and $\varphi = a_t^{-1}\partial_t a_t$ then

$$\partial_t \operatorname{tr}(\theta_t^{2k+1}) = (2k+1)\, d\operatorname{tr}(\varphi\, \theta_t^{2k}). \tag{13.6.2}$$

The normalization (13.6.1) is chosen so that the Chern character of the index of a Toeplitz family is the push-forward of the odd Chern character of its symbol, as we now show.

13.6.2 The Chern character of the Toeplitz index

Recall from Chapters 10 and 11 that a smooth family of elliptic Toeplitz operators $A \in \mathcal{C}^\infty(B; \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$, acting on the Hardy space $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$, has symbol $a = \sigma(A) \in \mathcal{C}^\infty(B \times \mathbb{S}; \mathrm{GL}(N,\mathbb{C}))$, and that the families index gives an isomorphism

$$\mathrm{ind}_{\mathcal{T}} : K^{-2}(B) \longrightarrow K^0(B)$$

on homotopy classes of such symbols normalized by $a(b,0) = \mathrm{Id}$. Our sign convention is that of Chapter 10: the lowering operator $L = T_{e^{-i\theta}}$ has index 1. For a form $\mu = d\theta \wedge \mu_1 + \mu_2$ on $B \times \mathbb{S}$, with μ_1, μ_2 not containing $d\theta$, set $\int_{\mathbb{S}} \mu = \int_0^{2\pi} \mu_1\, d\theta$; this commutes with d.

Proposition 13.6.1. *For any $a \in \mathcal{C}^\infty(B \times \mathbb{S}; \mathrm{GL}(N,\mathbb{C}))$ with $a(\cdot, 0) = \mathrm{Id}$,*

$$\mathrm{Ch}(\mathrm{ind}_{\mathcal{T}}[a]) = \int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(a) = \sum_{k\geq 0} \frac{(-1)^k i^{k+1} k!}{(2\pi)^{k+1}(2k+1)!} \int_{\mathbb{S}} \operatorname{tr}((a^{-1}da)^{2k+1}) \in H^{\mathrm{ev}}(B;\mathbb{C}).$$

Proof. Both sides depend only on $[a] \in K^{-2}(B)$ and are additive: the right side by (13.6.2) and since $\int_{\mathbb{S}}$ commutes with d, the left side because $\mathrm{ind}_{\mathcal{T}}$ is a homomorphism and Ch is additive. Since $\mathrm{ind}_{\mathcal{T}}$ is injective (Chapter 11) and every class in $K^0(B)$ has the form $[E]-[F]$ with $E=\pi_E\mathbb{C}^{N_1}$, $F=\pi_F\mathbb{C}^{N_2}$ for families of projections, it suffices to treat

$$a_E = \pi_E e^{-i\theta} + (\mathrm{Id}-\pi_E), \qquad a_F = \pi_F e^{i\theta} + (\mathrm{Id}-\pi_F),$$

whose Toeplitz families have null bundle E (constants), resp. cokernel bundle F, so $\mathrm{ind}_{\mathcal{T}}[a_E]=[E]$, $\mathrm{ind}_{\mathcal{T}}[a_F]=-[F]$.

For $a=a_E$, write $\pi=\pi_E$, $z=e^{i\theta}$ and $Q=\mathrm{Id}+(z-1)\pi$, so that $a^{-1}=Q$ and

$$a^{-1}da = -i\pi\, d\theta + \beta, \qquad \beta = (\bar z - 1)Q\, d\pi.$$

The form β is off-diagonal with respect to $\mathbb{C}^{N_1}=\operatorname{Ran}\pi\oplus\operatorname{Ran}(\mathrm{Id}-\pi)$, because $\pi\, d\pi\,\pi = 0 = (\mathrm{Id}-\pi)d\pi(\mathrm{Id}-\pi)$, and $\pi Q = z\pi$, $Q(\mathrm{Id}-\pi)=\mathrm{Id}-\pi$. Hence

$$\pi\beta^2\pi = (\bar z-1)^2 z\,\pi(d\pi)^2\pi = -(2-z-\bar z)\,\pi(d\pi)^2\pi, \qquad \pi\beta^{2k} = (-1)^k(2-2\cos\theta)^k\pi(d\pi)^{2k}.$$

In $\mathrm{tr}((a^{-1}da)^{2k+1})$ only terms with exactly one factor $-i\pi\,d\theta$ contribute to $\int_{\mathbb{S}}$. By (13.5.1) each of the $2k+1$ such terms can be cycled to the front without change of sign (the sign is $(-1)^{j(2k+1-j)}=1$), so the relevant part is

$$-i(2k+1)\,d\theta\wedge\mathrm{tr}(\pi\beta^{2k}) = -i(2k+1)(-1)^k(2-2\cos\theta)^k\,d\theta\wedge\mathrm{tr}\big(\pi(d\pi)^{2k}\big).$$

Since $\int_0^{2\pi}(2-2\cos\theta)^k d\theta = 4^k\int_0^{2\pi}\sin^{2k}(\theta/2)\,d\theta = 2\pi\frac{(2k)!}{(k!)^2}$,

$$c_k\int_{\mathbb{S}}\mathrm{tr}((a^{-1}da)^{2k+1}) = c_k(-i)(-1)^k 2\pi\frac{(2k+1)!}{(k!)^2}\,\mathrm{tr}(\pi(d\pi)^{2k}) = \frac{1}{k!}\Big(\frac{i}{2\pi}\Big)^k\,\mathrm{tr}(\pi(d\pi)^{2k}),$$

which is the degree $2k$ part of $\mathrm{Ch}(E)$ by (13.5.2). Finally $a_F(\theta)=a_{E'}(-\theta)$ with $E'=F$, and the reflection $\theta\mapsto-\theta$ reverses the orientation of $\mathbb{S}$, so $\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(a_F) = -\mathrm{Ch}(F)$. □

Corollary 13.6.2. *If $A\in\mathcal{C}^\infty(B;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ is an elliptic family of Toeplitz operators with symbol a, the Chern character of its index bundle is $\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(a)$.*

Proof. Let $g(b)=a(b,0)^{-1}$, independent of θ. Multiplying A by g, an invertible bundle map over B, does not change the index bundle, and ga is normalized. By the additivity of the odd Chern forms under pointwise products (Chapter 10; see also Exercise 13.4), $\mathrm{Ch}_{\mathrm{odd}}(ga)-\mathrm{Ch}_{\mathrm{odd}}(g)-\mathrm{Ch}_{\mathrm{odd}}(a)$ is exact, and $\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(g)=0$ because $\mathrm{Ch}_{\mathrm{odd}}(g)$ contains no $d\theta$. Since the families index depends only on the symbol, Proposition 13.6.1 applies to ga. □

Remark 13.6.3. For $k=0$ the formula reads $\operatorname{rank}\mathrm{ind}_{\mathcal{T}}(a)=\frac{i}{2\pi}\int_{\mathbb{S}}\mathrm{tr}(a^{-1}da) = -\frac{1}{2\pi i}\int\mathrm{tr}(a^{-1}da)$, which is Chapter 10's formula for the index of a single Toeplitz operator.

13.7 The relative Chern character

The cohomological index formula involves the Chern character of the symbol, a compactly supported class on $T^*(M/B)$. We define the Chern character on $K_{\mathrm{c}}(\operatorname{int}X)$, for X a compact manifold with boundary, by explicit forms. This covers $T^*(M/B)$, the interior of its radial compactification, and $\mathbb{R}^{2N}\times B$.

13.7.1 Relative de Rham cohomology

Let $\iota : \partial X \hookrightarrow X$ be the inclusion. The compactly supported de Rham cohomology $H^*_c(\operatorname{int} X)$ is isomorphic to the relative cohomology $H^*(X, \partial X)$. The latter is the cohomology of the complex of pairs

$$(\mu, \nu) \in \mathcal{C}^\infty(X; \Lambda^k) \oplus \mathcal{C}^\infty(\partial X; \Lambda^{k-1}), \quad D(\mu, \nu) = (d\mu, \iota^*\mu - d\nu),$$

and the isomorphism is induced by $\mu \mapsto (\mu, 0)$ for compactly supported μ. A closed pair is (μ, ν) with $d\mu = 0$ and $\iota^*\mu = d\nu$. The pairing with absolute classes,

$$\langle [(\mu, \nu)], [\varphi] \rangle = \int_X \mu \wedge \varphi - \int_{\partial X} \nu \wedge \iota^*\varphi, \quad d\varphi = 0, \tag{13.7.1}$$

is well defined (by Stokes' theorem, ∂X carrying the boundary orientation), and for $\nu = 0$ it is $\int_X \mu \wedge \varphi$.

13.7.2 The Chern form of a triple

Let (E_+, E_-, a) be a K_c-triple on $\operatorname{int} X$. As in the proof of Theorem 13.3.1 we may assume that $E_\pm$ and a extend smoothly to X and that a is invertible on a neighbourhood of ∂X. Choose connections $\nabla_\pm$ on $E_\pm$, with curvatures $\omega_\pm$, and let

$$\nabla a = \nabla_- \circ a - a \circ \nabla_+, \qquad \gamma = a^{-1}\nabla a,$$

an $\operatorname{End}(E_+)$-valued 1-form near ∂X. Choose $\rho \in \mathcal{C}^\infty(X)$ with $1 - \rho$ supported in the set where a is invertible and $\rho = 0$ near ∂X. Since $a^{-1}\nabla_- a = \nabla_+ + \gamma$,

$$\nabla^\rho = \rho\nabla_+ + (1-\rho)a^{-1}\nabla_- a = \nabla_+ + (1-\rho)\gamma \tag{13.7.2}$$

is a connection on E_+ which equals $a^{-1}\nabla_- a$ near ∂X. Set $c = i/2\pi$ and

$$\lambda_\rho = \operatorname{tr} e^{c\,\omega_{\nabla^\rho}} - \operatorname{tr} e^{c\,\omega_-}. \tag{13.7.3}$$

It is closed, and vanishes near ∂X since there $\omega_{\nabla^\rho} = a^{-1}\omega_- a$.

Proposition 13.7.1. *The class* $\operatorname{Ch}(E_+, E_-, a) = [\lambda_\rho] \in H^{\mathrm{ev}}_c(\operatorname{int} X)$ *depends only on the class of the triple in* $K_c(\operatorname{int} X)$*, and defines a homomorphism*

$$\operatorname{Ch} : K_c(\operatorname{int} X) \longrightarrow H^{\mathrm{ev}}_c(\operatorname{int} X) = H^{\mathrm{ev}}(X, \partial X).$$

If X is closed this is the Chern character of Definition 13.5.4.

Proof. All variations are handled by Proposition 13.5.3(ii). Changing ρ (or ∇_+ with ∇_-, a fixed) changes the connection on E_+ by a θ supported away from ∂X, so the transgression form is compactly supported. Changing ∇_- to ∇^t_- changes λ_ρ by $d(\operatorname{cs}(\nabla^\rho, \nabla^\rho_t) - \operatorname{cs}(\nabla_-, \nabla^t_-))$; near ∂X the two Chern–Simons forms are equal because the connections on E_+ there are $a^{-1}\nabla^t_- a$ and the trace is conjugation invariant. For a homotopy a_t the transgression form of ∇^ρ_t near ∂X is $c\int \operatorname{tr}(\theta_t e^{c\omega_t})dt$ with $\theta_t = \partial_t(a_t^{-1}\nabla_- a_t) = \nabla^\rho_t(\varphi_t)$, $\varphi_t = a_t^{-1}\partial_t a_t$; so there it equals $d(c\int \operatorname{tr}(\varphi_t e^{c\omega_t})dt)$, and subtracting d of χ times this, with $\chi = 1$ near ∂X and supported where $\rho = 0$, gives a compactly supported transgression. Bundle isomorphisms and stabilization by (H, H, Id) obviously do not change the class, and direct sums add. If $\partial X = \emptyset$ we may take $\rho = 1$. □

13.7.3 An explicit representative

The class can be represented without the cut-off ρ. On E_+ consider the linear path of connections from ∇_+ to $a^{-1}\nabla_- a$ near ∂X:

$$\nabla_+ + s\gamma, \quad s \in [0,1], \qquad \text{with curvature} \qquad \Omega(s) = (1-s)\omega_+ + s\, a^{-1}\omega_- a - s(1-s)\gamma^2.$$

To check the curvature, $(\nabla_+ + s\gamma)^2 = \omega_+ + s\nabla_+\gamma + s^2\gamma^2$, and at $s=1$ this is $a^{-1}\omega_- a$, so $\nabla_+\gamma = a^{-1}\omega_- a - \omega_+ - \gamma^2$.

Theorem 13.7.2 (Fedosov-type formula)**.** *The relative class* $\mathrm{Ch}(E_+, E_-, a) \in H^{\mathrm{ev}}(X, \partial X)$ *is represented by the closed pair*

$$\Big(\operatorname{tr} e^{c\omega_+} - \operatorname{tr} e^{c\omega_-},\ -c\,\iota^* \int_0^1 \operatorname{tr}\big(\gamma\, e^{c\Omega(s)}\big) ds\Big), \qquad c = \frac{i}{2\pi}.$$

Equivalently, for every closed form φ *on* X*,*

$$\int_X \lambda_\rho \wedge \varphi = \int_X (\operatorname{tr} e^{c\omega_+} - \operatorname{tr} e^{c\omega_-}) \wedge \varphi + c \int_{\partial X} \iota^*\Big(\int_0^1 \operatorname{tr}\big(a^{-1}(\nabla a) e^{c\Omega(s)}\big) ds \wedge \varphi \Big).$$

Proof. Let $\mu = c\int_0^1 \operatorname{tr}\big((1-\rho)\gamma\, e^{c\omega_t}\big) dt$, where ω_t is the curvature of $\nabla_+ + t(1-\rho)\gamma$. By Proposition 13.5.3(ii), $d\mu = \operatorname{tr} e^{c\omega_{\nabla^\rho}} - \operatorname{tr} e^{c\omega_+}$, which is λ_ρ minus the first entry of the pair. Near ∂X, $\rho = 0$ and $\omega_t = \Omega(t)$, so $\iota^*\mu$ is minus the second entry. Hence

$$(\lambda_\rho, 0) - \Big(\operatorname{tr} e^{c\omega_+} - \operatorname{tr} e^{c\omega_-}, -\iota^*\mu\Big) = (d\mu, \iota^*\mu) = D(\mu, 0).$$

Thus the pair is closed and cohomologous to $(\lambda_\rho, 0)$. The second statement is (13.7.1). □

Remark 13.7.3. The formula can be obtained by letting ρ tend to the characteristic function of X. Let x be a boundary defining function and $\rho = \chi(x/\epsilon)$ with $\chi \in \mathcal{C}^\infty(\mathbb{R})$, $\chi = 0$ on $(-\infty, \frac{1}{2}]$, $\chi = 1$ on $[1, \infty)$. From (13.7.2),

$$\omega_{\nabla^\rho} = \Omega(1-\rho) - d\rho \wedge \gamma,$$

and since $(d\rho)^2 = 0$, $\operatorname{tr} e^{c\omega_{\nabla^\rho}} = \operatorname{tr} e^{c\Omega(1-\rho)} - c\, d\rho \wedge \operatorname{tr}(\gamma e^{c\Omega(1-\rho)})$. As $\epsilon \to 0$ the first term tends to $\operatorname{tr} e^{c\omega_+}$ boundedly almost everywhere. In the second, $d\rho = \epsilon^{-1}\chi'(x/\epsilon)dx$; replacing the coefficients by their boundary values (with an error that tends to zero) and substituting $s = 1 - \chi(x/\epsilon)$ shows that it converges, as a current, to a multiple of $\delta(x)dx$ times $\iota^*\int_0^1 \operatorname{tr}(\gamma e^{c\Omega(s)})ds$. The limit is the "conormal" representative $\mu + \delta(x)dx \wedge \nu$ of the relative class; such a current is closed (on forms smooth up to the boundary) exactly when $d\mu = 0$ and $\iota^*\mu = \pm d\nu$, which is the closedness of the pair in Theorem 13.7.2. The proof given above avoids distributions. Note the analogy with Proposition 13.6.1, where the index class is also given by an integral of the odd forms of the symbol over a boundary circle.

For an elliptic family $A \in \Psi^0(M/B; E_+, E_-)$ we apply this with X the radial compactification of $T^*(M/B)$ (a ball bundle), $\partial X = S^*(M/B)$, the lifted bundles with pulled-back connections, and $a = \sigma_0(A)$ homogeneous. This defines

$$\mathrm{Ch}(\sigma_0(A)) \in H_{\mathrm{c}}^{\mathrm{ev}}(T^*(M/B); \mathbb{C}),$$

represented by the pair of Theorem 13.7.2, whose second entry lives on $S^*(M/B)$ and is built from $\sigma_0(A)^{-1}\nabla\sigma_0(A)$.

13.8 The cohomological form of the index theorem

Orient the fibres $T^*_b(Z)$ of $\pi : T^*(M/B) \to B$, where π is the composite of the bundle projection and ϕ, by $dz_1 \wedge d\zeta_1 \wedge \cdots \wedge dz_n \wedge d\zeta_n$ in canonical coordinates (z coordinates on the fibre Z_b, ζ the dual coordinates). This is the n-th power of the canonical symplectic form (up to a universal sign), so it is well defined even if Z is not orientable. Integration over the fibres is then a map $\pi_* : H^*_c(T^*(M/B)) \to H^{*-2n}(B)$. Let Td denote the Todd class, the multiplicative characteristic class with $\mathrm{Td}(L) = c_1(L)/(1 - e^{-c_1(L)})$ for a line bundle L.

Theorem 13.8.1 (Atiyah–Singer, cohomological form)**.** *For every $x \in K_c(T^*(M/B))$, with n the dimension of the fibres of ϕ,*

$$\mathrm{Ch}(\mathrm{ind_a}(x)) = (-1)^n \pi_*(\mathrm{Ch}(x) \wedge \pi_0^* \,\mathrm{Td}(T(M/B) \otimes \mathbb{C})) \in H^{\mathrm{ev}}(B; \mathbb{C}),$$

$\pi_0 : T^*(M/B) \to M$ the projection. In particular, for an elliptic family A, $\mathrm{Ch}(\mathrm{ind}\, A) = (-1)^n \int_{T^*(M/B)/B} \mathrm{Ch}(\sigma_0(A))\,\mathrm{Td}(T(M/B) \otimes \mathbb{C})$.

This is [18], Theorem 5.1. Since the sign depends on the orientation convention, it is often absorbed into it, as in the form $\mathrm{Ch}(\mathrm{ind}\, A) = \int \mathrm{Td} \cdot \mathrm{Ch}(\sigma(A))$. We explain how the theorem follows from Theorem 13.4.4; the topological inputs (C1) and (C4) below are not proved here.

Derivation from Theorem 13.4.4. We use the following properties of the Chern character of Proposition 13.7.1.

(C1) Ch is multiplicative for the products of Definition 13.1.2: $\mathrm{Ch}(x_1 \cdot x_2) = \mathrm{Ch}(x_1) \wedge \mathrm{Ch}(x_2)$. (This is standard; on a compact space it is Proposition 13.5.3(iii), and the compactly supported case follows by passing to one-point compactifications.)

(C2) Ch commutes with extensions j_* for open embeddings (extension of compactly supported forms by zero) and with pull-backs, since the forms λ_ρ are compactly supported and natural.

(C3) $\int_{T^*I} \mathrm{Ch}(b_1) = -1$ for the orientation $d\theta \wedge d\xi$ (Exercise 13.3). Hence by (C1), (C2), $\mathrm{Ch}(\beta_N^{-1} z) = (-1)^N \pi_* \,\mathrm{Ch}(z)$ for $z \in K_c(T^*I^N \times B)$, with the orientation above on T^*I^N.

(C4) For a real vector bundle $\mathcal{V}$ of rank k over M, integration over the fibres of $T^*(\mathcal{V}/M) \cong \mathcal{V} \otimes \mathbb{C}$ gives $\varpi_* \,\mathrm{Ch}(\lambda_{\mathcal{V}}) = (-1)^k \,\mathrm{Td}(\mathcal{V} \otimes \mathbb{C})^{-1}$. For trivial $\mathcal{V}$ this is (C3) with $B = M$ and Remark 13.4.3; in general it is the classical computation of Atiyah and Hirzebruch (see the Notes).

(C5) Td is multiplicative and $T(M/B) \oplus \mathcal{V}$ is trivial, so $\mathrm{Td}(\mathcal{V} \otimes \mathbb{C})^{-1} = \mathrm{Td}(T(M/B) \otimes \mathbb{C})$.

Then, with $x \in K_c(T^*(M/B))$ and the orientations multiplying as they should (all of them are products of oriented two-planes),

$$\begin{aligned} \mathrm{Ch}(\mathrm{ind_a}\, x) &= \mathrm{Ch}(\beta_N^{-1} j_* \Theta_* \tau(x)) = (-1)^N \pi_* \,\mathrm{Ch}(\lambda_{\mathcal{V}} \cdot \varpi^* x) \\ &= (-1)^N \pi_*(\varpi_* \,\mathrm{Ch}(\lambda_{\mathcal{V}}) \wedge \mathrm{Ch}(x)) = (-1)^{N+k} \pi_*(\mathrm{Td}(T(M/B) \otimes \mathbb{C}) \wedge \mathrm{Ch}(x)), \end{aligned}$$

and $(-1)^{N+k} = (-1)^n$ since $N = n + k$. □

Remark 13.8.2. Even without (C4) the derivation shows that $\mathrm{Ch}(\mathrm{ind_a}\, x) = \pi_*(\mathrm{Ch}(x) \wedge \pi_0^* \mathcal{T})$ for a class $\mathcal{T} \in H^{\mathrm{ev}}(M; \mathbb{C})$ which depends only on the fibration (through its normal bundle), not on x. The content of (C4) is the identification of $\mathcal{T}$ with $(-1)^n \,\mathrm{Td}(T(M/B) \otimes \mathbb{C})$. For B a point and $n = 1$, $M = \mathbb{S}$, the theorem reduces to $\mathrm{ind}(P_1) = 1 = -\int_{T^*\mathbb{S}} \mathrm{Ch}(\sigma_0(P_1))$, which is Exercise 13.3.

13.9 Exercises

Exercise 13.1. Show directly that the exterior product of Definition 13.1.2 is well defined on K_{c}-classes and bilinear, and that $x \boxtimes y$ corresponds to $y \boxtimes x$ under the flip $X_1 \times X_2 \cong X_2 \times X_1$. (For the latter, find a homotopy between $a_1 \# a_2$ and the conjugate of $a_2 \# a_1$ by the obvious bundle isomorphisms.)

Exercise 13.2. Complete Example 13.5.5: show that for the tautological line bundle over $\mathbb{C}P^1$, $\frac{i}{2\pi}\operatorname{tr}(\pi(d\pi)^2) = -\frac{1}{\pi}(1+|w|^2)^{-2} dx \wedge dy$ in the affine coordinate $w = x + iy$, so that its integral is -1.

Exercise 13.3. Let $X \subset T^*I$ be a large square $[\epsilon, 2\pi - \epsilon] \times [-R, R]$ in the coordinates (θ, ξ), oriented by $d\theta \wedge d\xi$, and represent b_1 by the trivial bundles with the trivial connection and the symbol of (13.4.2) modified as in the definition of b_1. Use Theorem 13.7.2 with $\varphi = 1$ to show that $\int_{T^*I} \operatorname{Ch}(b_1) = \frac{i}{2\pi} \int_{\partial X} a^{-1} da = -1$.

Exercise 13.4. Show that for maps $a, a' : X \to \mathrm{GL}(N, \mathbb{C})$ the forms $\operatorname{Ch}_{\mathrm{odd}}(aa')$ and $\operatorname{Ch}_{\mathrm{odd}}(a) + \operatorname{Ch}_{\mathrm{odd}}(a')$ differ by an exact form. (Use the homotopy $\left(\begin{smallmatrix} a & 0 \\ 0 & 1 \end{smallmatrix}\right) R_t \left(\begin{smallmatrix} a' & 0 \\ 0 & 1 \end{smallmatrix}\right) R_t^{-1}$ with R_t a rotation from Id to $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and (13.6.2).)

Exercise 13.5. Prove the base change property Remark 13.1.4(d) in detail, and deduce that if $\operatorname{ind}_{\mathrm{a}}(b) = 1$ in $K^0(M)$ for $b \in K_{\mathrm{c}}(T^*(\tilde{M}/M))$ then the restriction of b to the fibre over any $m \in M$ has numerical index 1.

Exercise 13.6. Show that $K_{\mathrm{c}}(\mathbb{R}^2) \to \mathbb{Z}$, taking the class of a triple $(\mathbb{C}^r, \mathbb{C}^r, a)$ with a invertible outside a disc to the winding number of $\det a$ on a large circle, is well defined, and that it maps b_1 to 1. Deduce from Proposition 13.4.2 with $N = 1$ and B a point that the index of a symbol in $K_{\mathrm{c}}(T^*I)$, extended to the circle, equals this winding number.

Notes

This chapter follows Lectures 19–21 of the 2005 course, with several changes. The proofs of multiplicativity and excision were completed: the invertibility of the symbols along the deformation is now deduced from the single identity of Lemma 13.1.1; the families index of product-type families and the comparison of the stabilization by sections with the construction of Chapter 12 (left to the addenda in the original) are in Lemma 13.1.3; the hypothesis of a trivialized null line bundle is reduced to $\operatorname{ind}_{\mathrm{a}}(b) = 1$ in Lemma 13.2.2. The normalization on the trivial fibration is proved via multiplicativity on the torus. In the original the curvature of the Grassmann connection carries a sign error, which propagated to the constants in the formula for the Chern character of a Toeplitz index; the corrected constants are in (13.6.1) (we checked the case $k = 1$ also numerically). The original section on the Bott element was empty; here its role is played by P_1, the normalization proposition and the imported statement (TH).

The families index theorem is due to Atiyah and Singer [18]; the K-theoretic method of proof, by embedding, excision and multiplicativity, goes back to M. F. Atiyah and I. M. Singer, *The index of elliptic operators I*, Ann. of Math. **87** (1968), 484–530, where the equivariant index of the Bott operator needed for (TH) is computed. For K-theory, products, the Thom isomorphism and Bott periodicity see [17]. The Riemann–Roch identity (C4) and the theorem that the Chern character is a rational isomorphism are due to M. F. Atiyah and F. Hirzebruch, *Vector bundles and homogeneous spaces*, Proc. Sympos. Pure Math. **3** (1961), 7–38. The explicit formula of Theorem 13.7.2 is of the type used by B. V. Fedosov in his work on index formulas.

Chapter 14

Eta forms and the index of product-type families

The index formula for product-type operators involves, besides the familiar Atiyah–Singer integrand, a 'regularized odd Chern character' of the base family, which we call an eta form. In this chapter we first construct the necessary regularization of the trace on a compact manifold, using holomorphic families of pseudodifferential operators: Seeley's theorem on the meromorphic continuation of traces, the residue trace, the regularized trace and the trace-defect formula (these are the analogues for compact manifolds of the constructions for the isotropic algebra in Chapter 7). We then specialize to the circle and define eta forms on the group of invertible Toeplitz operators, compute their exterior derivative and relate them to the odd Chern forms and the Fredholm determinant of Chapter 10. The main result, Theorem 14.8.1, is an index formula for fully elliptic operators of Toeplitz product type on $M \times \mathbb{S}$; the proof uses the product-type calculus of Chapter 12, the weak contractibility of the Toeplitz group from Chapter 11 and the Atiyah–Singer theorem of Chapter 13. A version for the full (non-Toeplitz) product-type calculus follows. The last section discusses, more briefly and partly conjecturally, the general product-type index problem, a corresponding cohomology theory and the determinant line bundle. The eta forms here are close relatives of the eta invariant of Atiyah–Patodi–Singer [16] and of the eta forms of Bismut and Cheeger [19]; see also [20].

Throughout, Z denotes a compact manifold without boundary of dimension $d \geq 1$ and $E \longrightarrow Z$ a complex vector bundle. We fix a smooth positive density on Z and a Hermitian structure on E, so that operators act on $L^2(Z;E)$; the trace does not depend on these choices.

14.1 The trace on operators of low order

14.1.1 Operators of complex order

Recall (Chapters 1 and 4) that the classical symbols of order m on $\mathbb{R}^d$ are exactly the functions of the form $\rho^{-m}b$ with $b \in \mathcal{C}^\infty(\overline{\mathbb{R}^d})$, where $\overline{\mathbb{R}^d}$ is the radial compactification and ρ is a boundary defining function, for instance $\rho = \langle\xi\rangle^{-1}$. This description makes sense for any $m \in \mathbb{C}$: we say that $a \in \mathcal{C}^\infty(U \times \mathbb{R}^d; M(N,\mathbb{C}))$ is a *classical symbol of order* $m \in \mathbb{C}$ if $a = \rho^{-m}b$ with $b \in \mathcal{C}^\infty(U \times \overline{\mathbb{R}^d}; M(N,\mathbb{C}))$. Such a symbol has an asymptotic expansion $a \sim \sum_{j\geq 0} a_{m-j}$ with a_{m-j} homogeneous of degree $m-j$ in $|\xi| \geq 1$, obtained from the Taylor series of b at the boundary. Correspondingly $\Psi^m(Z;E)$, $m \in \mathbb{C}$, denotes the operators whose kernels are smooth away from the diagonal and which in any coordinate chart U over

which E is trivial, after cut-off, are of the form $\mathrm{Op}(a)$ with a a classical symbol of order m (left quantization), modulo smoothing operators. The composition theorem and the symbol calculus of Chapter 5 hold without change for complex orders: $\Psi^m \circ \Psi^{m'} \subset \Psi^{m+m'}$, principal symbols multiply and $\Psi^m \subset \Psi^{m'}$ if $\Re m' - \Re m \in \mathbb{N}_0$ and $m' - m \in \mathbb{N}_0$. We write

$$\Psi^{\mathbb{Z}}(Z;E) = \bigcup_{m\in\mathbb{Z}} \Psi^m(Z;E)$$

for the algebra of classical operators of integral order.

Lemma 14.1.1. *If $A \in \Psi^m(Z;E)$ with $\Re m < -d$ then A is of trace class on $L^2(Z;E)$, its kernel K_A is continuous and*

$$\mathrm{Tr}(A) = \int_Z \mathrm{tr}_E\, K_A(z,z).$$

If U is a coordinate chart over which E is trivialized, $\phi,\psi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ and $A = \phi\,\mathrm{Op}(a)\psi$ with a a symbol of order m, then

$$\mathrm{Tr}(A) = (2\pi)^{-d} \int_U \int_{\mathbb{R}^d} \phi(z)\psi(z)\,\mathrm{tr}\, a(z,\xi)\, d\xi\, dz. \tag{14.1.1}$$

Proof. Using a partition of unity $\{\phi_i\}$ subordinate to coordinate charts and $\psi_i \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to 1 near $\mathrm{supp}\,\phi_i$, write $A = \sum_i \phi_i A \psi_i + R$ with $R = \sum_i \phi_i A(1-\psi_i)$ smoothing. Smoothing operators on a compact manifold are of trace class with trace given by the integral of the kernel over the diagonal, so it suffices to consider $A = \phi\,\mathrm{Op}(a)\psi$ in a chart. Write $a = a'\langle\xi\rangle^{m/2}$ with $a' = a\langle\xi\rangle^{-m/2}$ of order $m/2$. Since the second factor is a Fourier multiplier, $\mathrm{Op}(a')\,\mathrm{Op}(\langle\xi\rangle^{m/2}) = \mathrm{Op}(a)$ exactly in the left quantization, so $A = T_1T_2$ with $T_1 = \phi\,\mathrm{Op}(a')$, $T_2 = \mathrm{Op}(\langle\xi\rangle^{m/2})\psi$. The kernel of T_2 is $k(z-z')\psi(z')$ with $\hat{k} = \langle\xi\rangle^{m/2} \in L^2(\mathbb{R}^d)$ since $\Re m < -d$; hence $\int\int |k(z-z')\psi(z')|^2 dz\, dz' = \|k\|^2\|\psi\|^2$ and T_2 is Hilbert–Schmidt. The kernel of T_1 is $\phi(z)\int e^{i(z-z')\cdot\xi} a'(z,\xi)\,d\!\!\!^-\xi$ which, for each z, is in L^2 in z' with norm $\|a'(z,\cdot)\|_{L^2}$, uniformly bounded; as ϕ has compact support T_1 is Hilbert–Schmidt. Thus A is of trace class and $\mathrm{Tr}(A) = \int\int K_{T_1}(z,z')K_{T_2}(z',z)\,dz'\,dz$. Since a is integrable in ξ, locally uniformly in z, the kernel $K_A(z,z') = \phi(z)\int e^{i(z-z')\cdot\xi} a(z,\xi)\,d\!\!\!^-\xi\,\psi(z')$ is continuous and the double integral equals $\int K_A(z,z)\,dz$, which is (14.1.1). □

Lemma 14.1.2. *If $A \in \Psi^{m_1}(Z;E)$, $B \in \Psi^{m_2}(Z;E)$ and $\Re(m_1+m_2) < -d$ then $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.*

Proof. If A is smoothing then AB and BA are smoothing and both traces are the pairing of $K_A(z,z')$ with $K_B(z',z)$, so they agree. In general choose $\delta > 0$ with $\Re(m_1+m_2)+\delta < -d$ and smoothing operators $A_\epsilon \to A$ in $\Psi^{m_1+\delta}$ as $\epsilon \to 0$ (for instance by cutting off the local symbols with $\chi(\epsilon\xi)$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^d)$ equal to 1 near 0). Composition is continuous and, by (14.1.1), the trace is continuous on Ψ^m for $\Re m < -d$, so $\mathrm{Tr}(AB) = \lim \mathrm{Tr}(A_\epsilon B) = \lim \mathrm{Tr}(BA_\epsilon) = \mathrm{Tr}(BA)$. □

14.2 Holomorphic families

The space $\Psi^m(Z;E)$ of fixed order has a natural Fréchet topology and there is no difficulty in speaking of holomorphic maps $\Omega \longrightarrow \Psi^m(Z;E)$ for $\Omega \subset \mathbb{C}$ open. We need families in which the order varies holomorphically as well; these are not maps into a fixed space.

Definition 14.2.1. Let $\Omega \subset \mathbb{C}$ be open and $\mu : \Omega \longrightarrow \mathbb{C}$ holomorphic. A family $A(s) \in \Psi^{\mu(s)}(Z;E)$, $s \in \Omega$, is a *holomorphic family of order μ* if

(i) for every $\chi \in \mathcal{C}^\infty(Z^2)$ vanishing near the diagonal, $\chi K_{A(s)}$ is a holomorphic map from Ω to $\mathcal{C}^\infty(Z^2; \operatorname{Hom}(E) \otimes \Omega_R)$, and

(ii) for every coordinate chart U over which E is trivial and all $\phi, \psi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$,

$$\phi A(s) \psi = \operatorname{Op}(\rho^{-\mu(s)} b(s)) + R(s),$$

where $b \in \mathcal{C}^\infty(\Omega \times U \times \overline{\mathbb{R}^d}; M(N, \mathbb{C}))$ is holomorphic in s with support in a fixed compact subset of U in z, and $R(s)$ is a holomorphic family of smoothing operators.

Changing the boundary defining function ρ to $\rho' = f\rho$, $f > 0$ smooth, replaces $b(s)$ by $f^{\mu(s)} b(s)$, which is again smooth and holomorphic in s; so the definition does not depend on ρ. Its independence of the coordinates follows, exactly as for a single operator in Chapter 5, from the fact that the coordinate-change formula for symbols is an asymptotic sum whose terms are obtained from the symbol by applying differential operators in (z, ξ) and multiplying by smooth functions; these operations preserve $\rho^{-\mu(s)} \mathcal{C}^\infty(\Omega \times U \times \overline{\mathbb{R}^d})$, with holomorphic dependence, and the asymptotic summation can be carried out holomorphically (Borel's lemma with cut-offs chosen independently of s, uniformly on compact subsets of Ω). For μ constant this is the usual notion of a holomorphic family. In the same way one obtains:

Lemma 14.2.2. *If $A(s) \in \Psi^{\mu(s)}(Z; F, G)$ and $B(s) \in \Psi^{\nu(s)}(Z; E, F)$ are holomorphic families on Ω then $A(s)B(s)$ is a holomorphic family of order $\mu + \nu$.*

Sketch. Off the diagonal this follows from the corresponding property of compositions with smoothing operators. Near the diagonal, in local coordinates, the symbol of the composite has the expansion $\sum_\alpha \frac{1}{\alpha!} \partial_\xi^\alpha a(s) D_z^\alpha b(s)$, each term lying in $\rho^{-\mu(s)-\nu(s)} \mathcal{C}^\infty$ holomorphically, and the remainder after J terms satisfies symbol estimates of order $\Re(\mu + \nu) - J$ locally uniformly in s; by Cauchy's formula the same holds for its s-derivatives, which gives holomorphy of the composite symbol. □

We also need a division property.

Lemma 14.2.3. *If $F(s)$ is a holomorphic family of order μ on Ω and $F(s_0) = 0$ for some $s_0 \in \Omega$, then $F(s) = (s - s_0) F_1(s)$ with F_1 a holomorphic family of order μ.*

Proof. Off the diagonal $\chi K_{F(s)}$ is holomorphic and vanishes at s_0, so is divisible by $s - s_0$. Near the diagonal $\phi F(s) \psi = \operatorname{Op}(\rho^{-\mu(s)} b(s)) + R(s)$, and $\rho^{-\mu(s_0)} b(s_0)$ is a symbol of order $-\infty$, so $b(s_0)$ vanishes to infinite order at the boundary; subtracting $\rho^{-\mu(s)} \rho^{\mu(s_0)} b(s_0)$ (which changes R by a holomorphic smoothing family vanishing at s_0) we may assume $b(s_0) = 0$ and then $b(s) = (s - s_0) b_1(s)$ with b_1 smooth and holomorphic. Similarly $R(s) = (s - s_0) R_1(s)$. □

Proposition 14.2.4. *There is an entire family $E(s) \in \Psi^s(Z; E)$, $s \in \mathbb{C}$, with*

$$E(0) = \mathrm{Id}$$

and principal symbol $|\xi|_g^s \operatorname{Id}_E$ for a Riemannian metric g; in particular $E(s)$ is elliptic for every s.

Proof. Choose a finite cover by coordinate charts U_i over which E is trivialized, a subordinate partition of unity ϕ_i, functions $\psi_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$ equal to 1 near $\operatorname{supp} \phi_i$, and a metric g. Set

$$E(s) = \sum_i \phi_i \operatorname{Op}\big((1 + |\xi|^2_{g(z)})^{s/2} \operatorname{Id}\big) \psi_i.$$

This is an entire family of order s with principal symbol $|\xi|_g^s \operatorname{Id}$, and since $\operatorname{Op}(1) = \mathrm{Id}$, $E(0) = \sum_i \phi_i \psi_i = \sum_i \phi_i = \mathrm{Id}$. □

Using Seeley's complex powers [11] one can also take $E(s) = (\Delta + 1)^{s/2}$, with Δ a Laplacian, which is in addition invertible for all s; we shall not need this. We call an entire family $E(s)$ of order s with $E(0) = \mathrm{Id}$ *admissible*.

Theorem 14.2.5 (Seeley)**.** *Let $A(s)$ be a holomorphic family of order $m + s$, $m \in \mathbb{C}$, on a connected open set $\Omega \subset \mathbb{C}$ which meets the half-plane $\{\Re s < -d - \Re m\}$. Then the function $\mathrm{Tr}(A(s))$, holomorphic on $\Omega \cap \{\Re s < -d - \Re m\}$, extends to a meromorphic function on Ω with at most simple poles, located in $-d - m + \mathbb{N}_0$. The operator $A(s_j)$ at $s_j = j - d - m$ has integral order $j - d$, and the residue there is*

$$\operatorname*{Res}_{s=s_j} \mathrm{Tr}(A(s)) = -\operatorname{res}(A(s_j)), \tag{14.2.1}$$

where for $B \in \Psi^k(Z; E)$, $k \in \mathbb{Z}$,

$$\operatorname{res}(B) = (2\pi)^{-d} \sum_i \int_{U_i} \phi_i(z) \int_{|\xi|=1} \operatorname{tr} b^{(i)}_{-d}(z, \xi)\, dS(\xi)\, dz,$$

$b^{(i)}_{-d}$ being the term of homogeneity $-d$ in the symbol of B in the chart U_i (and 0 if $k < -d$); here $\{\phi_i\}$ is a partition of unity subordinate to charts U_i over which E is trivial.

Proof. As in the proof of Lemma 14.1.1, $A(s) = \sum_i \phi_i A(s)\psi_i + R(s)$ where $R(s)$ is a holomorphic family of smoothing operators, so $\mathrm{Tr}(R(s))$ is entire. By the definition, $\phi_i A(s)\psi_i = \mathrm{Op}(\rho^{-m-s} b_i(s))$ modulo holomorphic smoothing families, and we may choose $\rho = |\xi|^{-1}$ in $|\xi| \geq 1$. By (14.1.1), for $\Re s < -d - \Re m$,

$$\begin{aligned}\mathrm{Tr}(\phi_i A(s)\psi_i) = (2\pi)^{-d} \int_U \int_{|\xi|\leq 1} \operatorname{tr}(\cdots)\, d\xi\, dz \\ + (2\pi)^{-d} \int_U \int_{\mathbb{S}^{d-1}} \int_0^1 r^{-m-s-d-1} \operatorname{tr} b_i(s, z, r, \omega)\, dr\, d\omega\, dz,\end{aligned}$$

where we have used polar coordinates $\xi = r^{-1}\omega$ in $|\xi| \geq 1$, so that $d\xi = r^{-d-1} dr\, d\omega$, and b_i is smooth in $r \in [0, 1]$ (this is the meaning of smoothness on $\overline{\mathbb{R}^d}$). The first term is entire. Taylor expanding, $b_i(s, z, r, \omega) = \sum_{j<J} b_{i,j}(s, z, \omega) r^j + r^J b_{i,J}(s, z, r, \omega)$ with all coefficients smooth and holomorphic in s, so

$$\int_0^1 r^{-m-s-d-1} b_i\, dr = \sum_{j<J} \frac{b_{i,j}(s, z, \omega)}{j - d - m - s} + \int_0^1 r^{J-m-s-d-1} b_{i,J}\, dr,$$

and the last integral is holomorphic in $\Re s < J - d - \Re m$. Since J is arbitrary this gives the meromorphic continuation, with at most simple poles at the points s_j, and residue $-(2\pi)^{-d} \sum_i \int \int \operatorname{tr} b_{i,j}(s_j, z, \omega)\, d\omega\, dz$ there. Now $r^{-m-s_j+j} b_{i,j}(s_j, z, \omega) = |\xi|^{-d} b_{i,j}(s_j, z, \omega)$ is precisely the term of homogeneity $-d$ in the symbol of $\phi_i A(s_j)\psi_i$. Finally the full symbol of $\phi_i B \psi_i$ is ϕ_i times the local symbol of B modulo $S^{-\infty}$, because $\psi_i \equiv 1$ near $\operatorname{supp} \phi_i$; this gives (14.2.1). □

14.3 The residue trace and the regularized trace

Proposition 14.3.1. *For $A \in \Psi^m(Z; E)$, $m \in \mathbb{Z}$, and any holomorphic family $A(s)$ of order $m + s$ on a connected open set containing 0 and meeting $\{\Re s < -d - m\}$, with $A(0) = A$,*

$$\operatorname*{Res}_{s=0} \mathrm{Tr}(A(s)) = -\operatorname{res}(A).$$

The residue trace $\operatorname{Tr}_{\mathrm{R}}(A) := \operatorname{Res}_{s=0}\operatorname{Tr}(AE(s))$, *$E$ admissible, therefore does not depend on E, and*

$$\operatorname{Tr}_{\mathrm{R}}(A) = -\operatorname{res}(A) = -(2\pi)^{-d}\int_{S^*Z}\operatorname{tr}\sigma_{-d}(A), \tag{14.3.1}$$

where $\sigma_{-d}(A)$ is the homogeneous term of degree $-d$ in the symbol of A and the integral means, in local coordinates, $\int\int_{|\xi|=1}\operatorname{tr}\sigma_{-d}(A)\,dS(\xi)\,dz$; the integrand is a well-defined density on S^*Z although $\sigma_{-d}(A)$ itself depends on the coordinates. Moreover $\operatorname{Tr}_{\mathrm{R}}$ vanishes on $\Psi^m(Z;E)$ for $m < -d$ and on $\Psi^{-\infty}(Z;E)$, and

$$\operatorname{Tr}_{\mathrm{R}}([A,B]) = 0 \quad \forall\ A, B \in \Psi^{\mathbb{Z}}(Z;E).$$

Proof. The first statement is (14.2.1) at $s_j = 0$, since the expression for res only involves $A(0) = A$. By Lemma 14.2.2, $AE(s)$ is such a family. The formula (14.2.1) is a priori written with a partition of unity; applying it to an operator supported in the intersection of two charts shows that the density $\big(\int_{|\xi|=1}\operatorname{tr} a_{-d}(z,\xi)dS(\xi)\big)dz$ is independent of the coordinates (one can also check this directly, since for a function h homogeneous of degree $-d$ the form $\iota_R(h\,d\xi)$, R the radial vector field, is closed on $\mathbb{R}^d\setminus 0$). If $m < -d$, $AE(s)$ has order $m+s$ with $\Re(m+s) < -d$ near $s = 0$, so there is no pole.

For the trace property, $\operatorname{Tr}(ABE(s)) = \operatorname{Tr}(BE(s)A)$ for $\Re s$ very negative by Lemma 14.1.2, hence for all s by meromorphic continuation, so

$$\operatorname{Tr}([A,B]E(s)) = \operatorname{Tr}\big(B[E(s),A]\big).$$

The family $[E(s),A]$ is holomorphic of order m_A+s and vanishes at $s = 0$, so by Lemma 14.2.3 $[E(s),A] = sC(s)$ with C holomorphic. Since $\operatorname{Tr}(BC(s))$ has at most a simple pole at 0, $s\operatorname{Tr}(BC(s))$ is regular there, so $\operatorname{Tr}_{\mathrm{R}}([A,B]) = 0$. □

Thus, with our convention that $E(s)$ has order $+s$, the residue trace is minus the Wodzicki–Guillemin residue [13, 5].

Definition 14.3.2. For an admissible family E the *regularized trace* of $A \in \Psi^{\mathbb{Z}}(Z;E)$ is the finite part

$$\overline{\operatorname{Tr}}_E(A) = \operatorname*{FP}_{s=0}\operatorname{Tr}(AE(s)) = \lim_{s\to 0}\Big(\operatorname{Tr}(AE(s)) - \frac{1}{s}\operatorname{Tr}_{\mathrm{R}}(A)\Big).$$

Clearly $\overline{\operatorname{Tr}}_E$ is linear and $\overline{\operatorname{Tr}}_E(A) = \operatorname{Tr}(A)$ if $A \in \Psi^m$, $m < -d$, in particular on smoothing operators. From the proof of Theorem 14.2.5 the Laurent coefficients of $\operatorname{Tr}(AE(s))$ at $s = 0$ are given by convergent integrals of the (Taylor coefficients of the) symbol of $AE(s)$, so $\overline{\operatorname{Tr}}_E$ is continuous on each $\Psi^m(Z;E)$. However $\overline{\operatorname{Tr}}_E$ is not a trace, and it depends on E.

Lemma 14.3.3. *If E and E' are admissible then $E'(s) = E(s) + sF(s)$ with F an entire family of order s; setting $D(E,E') = F(0) \in \Psi^0(Z;E)$,*

$$\overline{\operatorname{Tr}}_{E'}(A) = \overline{\operatorname{Tr}}_E(A) + \operatorname{Tr}_{\mathrm{R}}\big(AD(E,E')\big) \quad \forall\ A \in \Psi^{\mathbb{Z}}(Z;E).$$

Proof. The first statement is Lemma 14.2.3. Then $\operatorname{Tr}(AE'(s)) = \operatorname{Tr}(AE(s)) + s\operatorname{Tr}(AF(s))$; here $AF(s)$ is holomorphic of order $m+s$ with value $AD(E,E')$ at 0, so $s\operatorname{Tr}(AF(s))$ is regular at 0 with value $\operatorname{Tr}_{\mathrm{R}}(AD(E,E'))$, while the residues of $\operatorname{Tr}(AE'(s))$ and $\operatorname{Tr}(AE(s))$ agree. □

Proposition 14.3.4. *The regularized traces $\overline{\operatorname{Tr}}_E$, as E ranges over admissible families, form an affine space modelled on $\Psi^0(Z;E)/\Psi^{-\infty}(Z;E)$: the map $E' \longmapsto [D(E,E')]$ identifies $\{\overline{\operatorname{Tr}}_{E'}\}$ with $\Psi^0/\Psi^{-\infty}$.*

Proof. Every $D \in \Psi^0$ occurs: $E'(s) = E(s) + sDE(s)$ is admissible with $D(E, E') = D$. By Lemma 14.3.3 it remains to see that the functional $A \mapsto \mathrm{Tr}_{\mathrm{R}}(AD)$ on $\Psi^{\mathbb{Z}}$ vanishes if and only if $D \in \Psi^{-\infty}$. If D is smoothing so is AD. Conversely, if $D \notin \Psi^{-\infty}$ let $k \in \mathbb{N}_0$ be such that $D \in \Psi^{-k}$ has non-vanishing principal symbol $\sigma_{-k}(D)$. For $A \in \Psi^{k-d}$ the term of homogeneity $-d$ in the symbol of AD is $\sigma_{k-d}(A)\sigma_{-k}(D)$, so by (14.3.1)

$$\mathrm{Tr}_{\mathrm{R}}(AD) = -(2\pi)^{-d} \int_{S^*Z} \mathrm{tr}(\sigma_{k-d}(A)\sigma_{-k}(D)).$$

Choosing $\sigma_{k-d}(A) = \chi\, \sigma_{-k}(D)^*$ (extended homogeneously) with $0 \le \chi$, $\chi > 0$ somewhere where $\sigma_{-k}(D) \neq 0$, makes this non-zero. □

14.3.1 The trace-defect formula

The failure of $\overline{\mathrm{Tr}}_E$ to be a trace is measured by a derivation.

Definition 14.3.5. For an admissible family E and $A \in \Psi^{\mathbb{Z}}(Z; E)$ let

$$D_E A = \frac{d}{ds}[E(s), A]\Big|_{s=0},$$

so that $[E(s), A] = sC(s)$ with C holomorphic of order $m + s$ and $C(0) = D_E A$.

Formally, if $E(s) = Q^s$ then $D_E A = [\log Q, A]$; $\log Q$ is not in the algebra, but the commutator is.

Proposition 14.3.6. *D_E is a derivation of $\Psi^{\mathbb{Z}}(Z; E)$ mapping Ψ^m into Ψ^m; if the principal symbol of $E(s)$ is $|\xi|_g^s \,\mathrm{Id}$ (as in Proposition 14.2.4) then $D_E(\Psi^m) \subset \Psi^{m-1}$ and*

$$\sigma_{m-1}(D_E A) = -i\{\log|\xi|_g, \sigma_m(A)\}. \tag{14.3.2}$$

For two admissible families, $D_{E'}A = D_E A + [D(E, E'), A]$. Moreover

$$\mathrm{Tr}_{\mathrm{R}}(D_E A) = 0 \quad \forall\ A \in \Psi^{\mathbb{Z}}(Z; E).$$

Proof. Differentiating $[E(s), AB] = [E(s), A]B + A[E(s), B]$ at $s = 0$ gives the derivation property, and the difference formula by differentiating $[E(s) + sF(s), A]$. If the principal symbol e_s of $E(s)$ is scalar, the principal symbol of $[E(s), A]$ of order $m + s$ vanishes, and the term of order $m + s - 1$ is

$$-i\{e_s, \sigma_m(A)\} = -is|\xi|_g^s\{\log|\xi|_g, \sigma_m(A)\};$$

dividing by s and setting $s = 0$ gives (14.3.2). Finally, the family $C(s)$ has order $m + s$ and value $D_E A$ at 0, so $\mathrm{Tr}_{\mathrm{R}}(D_E A)$ is the residue of $\mathrm{Tr}(C(s)) = s^{-1}\,\mathrm{Tr}([E(s), A])$; but $\mathrm{Tr}([E(s), A]) = 0$ for $\Re s$ very negative by Lemma 14.1.2, hence identically. □

One can show that D_E is not an inner derivation (see Exercise 14.2 for the circle).

Lemma 14.3.7 (Trace-defect formula)**.** *For all $A, B \in \Psi^{\mathbb{Z}}(Z; E)$,*

$$\overline{\mathrm{Tr}}_E([A, B]) = \mathrm{Tr}_{\mathrm{R}}\big((D_E A)B\big) = \mathrm{Tr}_{\mathrm{R}}\big(BD_E A\big).$$

Proof. As in the proof of Proposition 14.3.1, $\mathrm{Tr}([A, B]E(s)) = \mathrm{Tr}(B[E(s), A]) = s\,\mathrm{Tr}(BC(s))$, which is regular at $s = 0$ with value $\mathrm{Res}_{s=0}\,\mathrm{Tr}(BC(s)) = \mathrm{Tr}_{\mathrm{R}}(BD_E A)$. Since $\mathrm{Tr}_{\mathrm{R}}([A, B]) = 0$, this value is $\overline{\mathrm{Tr}}_E([A, B])$; the second equality is the trace property of Tr_{R}. □

We shall apply this to operator-valued differential forms. If X is a manifold and $\alpha = \sum_I A_I\,dx^I$, $\beta = \sum_J B_J\,dx^J$ are smooth forms of degrees p and q with values in $\Psi^{\mathbb{Z}}(Z;E)$, products are taken in the algebra and in the exterior algebra, $\overline{\mathrm{Tr}}_E$, Tr_{R} and D_E act on coefficients, and the graded commutator is $[\alpha,\beta] = \alpha\beta - (-1)^{pq}\beta\alpha = \sum_{I,J}[A_I,B_J]\,dx^I\wedge dx^J$. Hence:

Corollary 14.3.8. *For operator-valued forms α, β as above, $\overline{\mathrm{Tr}}_E([\alpha,\beta]) = \mathrm{Tr}_{\mathrm{R}}((D_E\alpha)\wedge\beta)$, and Tr_{R} vanishes on graded commutators. Moreover $d\,\overline{\mathrm{Tr}}_E(\alpha) = \overline{\mathrm{Tr}}_E(d\alpha)$, by continuity and linearity of $\overline{\mathrm{Tr}}_E$.*

14.4 The circle

On $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$, with variable θ and dual variable ξ, we use the admissible family

$$E(s)e^{ik\theta} = \lambda_k^s e^{ik\theta}, \qquad \lambda_k = (1+k^2)^{1/2}, \tag{14.4.1}$$

acting diagonally on $\mathcal{C}^\infty(\mathbb{S};\mathbb{C}^N)$. It is the Fourier multiplier with symbol $\langle\xi\rangle^s$, an entire family of invertible elliptic operators of order s with $E(0) = \mathrm{Id}$. From now on $\overline{\mathrm{Tr}} = \overline{\mathrm{Tr}}_E$ and D_E refer to this family. Here $S^*\mathbb{S} = \mathbb{S}\times\{\pm1\}$.

Lemma 14.4.1. *(i) For $A\in\Psi^{-1}(\mathbb{S};\mathbb{C}^N)$ with symbol term a_{-1} of homogeneity -1,*

$$\mathrm{Tr}_{\mathrm{R}}(A) = -\frac{1}{2\pi}\int_0^{2\pi}\mathrm{tr}\big(a_{-1}(\theta,1)+a_{-1}(\theta,-1)\big)\,d\theta.$$

(ii) For $A\in\Psi^m(\mathbb{S};\mathbb{C}^N)$ with principal symbol a, $D_EA\in\Psi^{m-1}$ has principal symbol

$$\sigma_{m-1}(D_EA)(\theta,\xi) = -i\,\xi^{-1}\partial_\theta a(\theta,\xi);$$

in fact D_EA has full symbol $\sum_{k\ge1}\frac{1}{k!}\partial_\xi^k(\log\langle\xi\rangle)D_\theta^k a$.

(iii) If $f\in\mathcal{C}^\infty(\mathbb{S};M(N,\mathbb{C}))$ acts by multiplication and Π is the Szegő projector then

$$\overline{\mathrm{Tr}}(f) = 0, \qquad \overline{\mathrm{Tr}}(\Pi f\Pi) = \frac{1}{4\pi}\int_0^{2\pi}\mathrm{tr}\,f\,d\theta;$$

in particular $\overline{\mathrm{Tr}}(\Pi) = \frac12$, and $\mathrm{Tr}_{\mathrm{R}}(f) = \mathrm{Tr}_{\mathrm{R}}(\Pi f\Pi) = 0$.

Proof. (i) is (14.3.1) for $d=1$. For (ii), since the symbol $e_s = \langle\xi\rangle^s$ depends only on ξ, $\mathrm{Op}(a)\,\mathrm{Op}(e_s) = \mathrm{Op}(ae_s)$ while $\mathrm{Op}(e_s)\,\mathrm{Op}(a)$ has symbol $\sum_k\frac{1}{k!}\partial_\xi^k e_s D_\theta^k a$; every term with $k\ge1$ is divisible by s, with $\frac{d}{ds}\partial_\xi^k e_s|_{s=0} = \partial_\xi^k\log\langle\xi\rangle$. The leading term is $\partial_\xi\log\langle\xi\rangle\,D_\theta a = \xi\langle\xi\rangle^{-2}(-i\partial_\theta a)$. For (iii), the multiplication operator has diagonal matrix entries $\langle fe_k,e_k\rangle = \frac{1}{2\pi}\int f\,d\theta$ for every $k\in\mathbb{Z}$ (in $\Pi f\Pi$ only for $k\ge0$), so for $\Re s<-1$

$$\mathrm{Tr}(fE(s)) = \frac{1}{2\pi}\int\mathrm{tr}\,f\,d\theta\sum_{k\in\mathbb{Z}}\lambda_k^s, \quad \mathrm{Tr}(\Pi f\Pi E(s)) = \frac{1}{2\pi}\int\mathrm{tr}\,f\,d\theta\sum_{k\ge0}\lambda_k^s.$$

Now $\sum_{k\ge0}\lambda_k^s = 1+\zeta(-s)+h(s)$ with $h(s) = \sum_{k\ge1}(\lambda_k^s - k^s)$; since $\lambda_k^s - k^s = O(|s|k^{\Re s-2})$, h is holomorphic in $\Re s<1$ with $h(0)=0$. The Riemann zeta function is regular at 0 with $\zeta(0) = -\frac12$, so the finite part (in fact the value) at $s=0$ is $\frac12$, and $\sum_{k\in\mathbb{Z}}\lambda_k^s = 2\sum_{k\ge0}\lambda_k^s - 1$ has value 0. □

Example 14.4.2. Let A be multiplication by $e^{i\theta}$ and $B = e^{-i\theta}\Pi$. Then $AB = \Pi$ and $BA = e^{-i\theta}\Pi e^{i\theta}$ is the projection onto the span of the $e^{ik\theta}$, $k\ge-1$, so $[A,B] = -\pi_{-1}$, minus the projection onto $e^{-i\theta}$, and $\overline{\mathrm{Tr}}([A,B]) = -1$. On the other hand, by Lemma 14.4.1(ii), $(D_EA)B$ has symbol ξ^{-1} in $\xi>0$ and is rapidly decreasing in $\xi<0$, so $\mathrm{Tr}_{\mathrm{R}}((D_EA)B) = -\frac{1}{2\pi}\int_0^{2\pi}1\,d\theta = -1$, as it must be by Lemma 14.3.7. This confirms the signs in the trace-defect formula.

14.5 Odd Chern forms and the Toeplitz index

14.5.1 Toeplitz operators

We recall from Chapter 10. The Szegő projector $\Pi \in \Psi^0(\mathbb{S})$ projects onto the span $\mathcal{H}$ of the $e^{ik\theta}$, $k \geq 0$; its symbol is 1 on $\xi > 0$ and its full symbol is rapidly decreasing as $\xi \to -\infty$. The Toeplitz algebra

$$\Psi^m_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) = \Pi\Psi^m(\mathbb{S};\mathbb{C}^N)\Pi$$

acts on $\mathcal{H}\otimes\mathbb{C}^N$, with identity Π; the full symbols of its elements are rapidly decreasing as $\xi \to -\infty$, and we write $\sigma_0(A)(\theta)$ for the principal symbol at $(\theta,+1)$. An element of $\Psi^0_{\mathcal{T}}$ is Fredholm on $\mathcal{H}\otimes\mathbb{C}^N$ if and only if $\sigma_0(A)$ is invertible, and then

$$\operatorname{ind}(A) = \frac{i}{2\pi}\int_0^{2\pi} \operatorname{tr}\big(\sigma_0(A)^{-1}\partial_\theta\sigma_0(A)\big)d\theta = -\operatorname{wind}\det\sigma_0(A). \tag{14.5.1}$$

For instance the shift $U = \Pi e^{i\theta}\Pi$ has index -1. Let

$$G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) = \{B \in \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N);\ B \text{ invertible on } \mathcal{H}\otimes\mathbb{C}^N,\ B^{-1} \in \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)\}$$

and $G^{-\infty}_{\mathcal{T}} = G_{\mathcal{T}} \cap (\Pi + \Psi^{-\infty}_{\mathcal{T}})$. Symbols of elements of $G_{\mathcal{T}}$ are loops in $\mathrm{GL}(N,\mathbb{C})$ with winding number 0. Products of operators on $\mathcal{H}\otimes\mathbb{C}^N$ are products in $\Psi^{\mathbb{Z}}(\mathbb{S};\mathbb{C}^N)$, so $\overline{\mathrm{Tr}}$ and Tr_{R} apply to elements of $\Psi^{\mathbb{Z}}_{\mathcal{T}}$.

14.5.2 Odd Chern forms

For a manifold W and $a \in \mathcal{C}^\infty(W;\mathrm{GL}(N,\mathbb{C}))$ set

$$\mathrm{Ch}_{\mathrm{odd}}(a) = \sum_{k\geq 0} c_k \operatorname{tr}\big((a^{-1}da)^{2k+1}\big), \qquad c_k = -\frac{k!}{(2\pi i)^{k+1}(2k+1)!}. \tag{14.5.2}$$

The same formula, with tr replaced by Tr, defines $\mathrm{Ch}_{\mathrm{odd}}$ for maps into the group $G^{-\infty}$ of Chapter 10. The forms are closed: with $\nu = a^{-1}da$ we have $d\nu = -\nu^2$, hence

$$d(\nu^{2k+1}) = \sum_{j=0}^{2k}(-1)^j\nu^j d\nu\,\nu^{2k-j} = -\nu^{2k+2}, \tag{14.5.3}$$

and $\operatorname{tr}(\nu^{2k+2}) = 0$ because $\operatorname{tr}(\nu\wedge\nu^{2k+1}) = -\operatorname{tr}(\nu^{2k+1}\wedge\nu)$. The constants c_k are fixed by Proposition 14.5.2 below; note $c_0 = i/2\pi$, consistent with (14.5.1).

Conventions. On a product $W\times\mathbb{S}$ every form can be written uniquely as $\omega = \omega_0 + \omega_1\wedge d\theta$ with ω_0,ω_1 not containing $d\theta$, and we set $\int_{\mathbb{S}}\omega = \int_0^{2\pi}\omega_1\,d\theta$, a form on W. With this ('fibre last') convention $d\int_{\mathbb{S}} = \int_{\mathbb{S}} d$, $\int_{\mathbb{S}}(\pi^*\alpha\wedge\omega) = \alpha\wedge\int_{\mathbb{S}}\omega$, and $\int_{W\times\mathbb{S}} = \int_W\int_{\mathbb{S}}$ when $W\times\mathbb{S}$ carries the product orientation (first W, then $d\theta$).

Lemma 14.5.1. *Let $a_t \in \mathcal{C}^\infty(W;\mathrm{GL}(N,\mathbb{C}))$ depend smoothly on $t\in[0,1]$ and regard a as a map on $W\times[0,1]$. Then*

$$\mathrm{Ch}_{\mathrm{odd}}(a) = \mathrm{Ch}_{\mathrm{odd}}(a_t) + \Theta(a_t)\wedge dt, \qquad \Theta(a_t) = \sum_{k\geq 0}(2k+1)c_k\operatorname{tr}\big((a_t^{-1}da_t)^{2k}a_t^{-1}\partial_t a_t\big),$$

and $\partial_t\,\mathrm{Ch}_{\mathrm{odd}}(a_t) = d\Theta(a_t)$.

Proof. Write $a^{-1}Da = \nu + \lambda\,dt$, $D = d + dt\,\partial_t$, $\nu = a_t^{-1}da_t$, $\lambda = a_t^{-1}\partial_t a_t$. The dt-component of $(\nu+\lambda dt)^{2k+1}$ is

$$\sum_j \nu^j \lambda\, dt\, \nu^{2k-j} = \sum_j (-1)^{2k-j}\nu^j\lambda\nu^{2k-j}\wedge dt,$$

and $\operatorname{tr}(\nu^j\lambda\nu^{2k-j}) = (-1)^{j(2k-j)}\operatorname{tr}(\lambda\nu^{2k}) = (-1)^j\operatorname{tr}(\lambda\nu^{2k})$, giving $(2k+1)\operatorname{tr}(\nu^{2k}\lambda)\wedge dt$. The second identity is the dt-component of $D\,\mathrm{Ch}_{\mathrm{odd}}(a) = 0$, using $dt\wedge\partial_t\omega = -\partial_t\omega\wedge dt$ for ω of odd degree. □

Applying this to the rotation homotopy from $g\oplus h$ to $gh\oplus \mathrm{Id}$ shows that $\mathrm{Ch}_{\mathrm{odd}}(gh) - \mathrm{Ch}_{\mathrm{odd}}(g) - \mathrm{Ch}_{\mathrm{odd}}(h)$ is exact (Exercise 14.4).

Proposition 14.5.2. *Let W be compact, $b\in\mathcal{C}^\infty(W\times\mathbb{S};\mathrm{GL}(N,\mathbb{C}))$ and let $T(w)\in\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ be a smooth family with symbol $b(w,\cdot)$. Then the Chern character of the index bundle $\operatorname{ind}(T)\in K^0(W)$ is*

$$\mathrm{Ch}(\operatorname{ind}T) = \Big[\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(b)\Big]\in H^{\mathrm{even}}(W;\mathbb{C}), \tag{14.5.4}$$

where Ch *of a bundle with curvature Ω is represented by* $\operatorname{tr}\exp(i\Omega/2\pi)$.

Proof. This is the families index theorem for Toeplitz operators of Chapter 13; we recall the structure of the argument and verify the constants. Both sides are additive under direct sums, unchanged by stabilization and homotopy invariant (the right side by Lemma 14.5.1). Multiplying T by the θ-independent invertible matrix $b(w,1)^{-1}$ changes neither side (for the right side use Exercise 14.4 and the fact that $\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(g)=0$ for θ-independent g). By the homotopy results of Chapter 11, after these reductions and stabilization b may be deformed to $\pi_- e^{-i\theta}+\pi_0+\pi_+e^{i\theta}$, with $\pi_\pm,\pi_0$ smooth families of commuting projections on some $\mathbb{C}^M$ summing to Id. This is the product of the commuting symbols $\pi_+e^{i\theta}+(\mathrm{Id}-\pi_+)$ and $\pi_-e^{-i\theta}+(\mathrm{Id}-\pi_-)$; the index is additive under composition (the Toeplitz operator of a product differs from the product of the Toeplitz operators by a compact family) and $\mathrm{Ch}_{\mathrm{odd}}$ is additive up to exact forms (Exercise 14.4), so it suffices to consider the two factors. The second is obtained from a symbol of the first type by $\theta\mapsto-\theta$, which reverses the sign of both sides. So it suffices to consider $b=\pi e^{i\theta}+\pi'$, with Toeplitz operator $\pi U+\pi'$: it is injective with cokernel $\pi\mathbb{C}^M$ (in the constants), so $\operatorname{ind}(T) = -[V]$, $V=\operatorname{Ran}\pi$. The connection πd on V has curvature $\Omega = \pi\,d\pi\wedge d\pi\,\pi$, so the left side of (14.5.4) is $-\sum_k \frac{1}{k!}(\frac{i}{2\pi})^k\operatorname{tr}\Omega^k$.

On the other hand, writing d' for the differential on $W\times\mathbb{S}$,

$$b^{-1}d'b = i\pi\,d\theta+\nu,\quad \nu = f\,\pi\,d\pi\,\pi' + g\,\pi' d\pi\,\pi,\quad f = 1-e^{-i\theta},\ g=e^{i\theta}-1,$$

using $\pi\,d\pi\,\pi = 0$ and $\pi' d\pi\,\pi' = 0$. The $d\theta$-component of $(b^{-1}d'b)^{2k+1}$ has trace $(2k+1)i\operatorname{tr}(\pi\nu^{2k}\pi)\wedge d\theta$ (by cyclicity, as in Lemma 14.5.1) and $\pi\nu^{2k}\pi = (fg)^k(\pi\,d\pi\,\pi' d\pi\,\pi)^k = (-1)^k(2-2\cos\theta)^k\Omega^k$. Since $\int_0^{2\pi}(2-2\cos\theta)^k d\theta = 2\pi\frac{(2k)!}{(k!)^2}$,

$$\int_{\mathbb{S}}\operatorname{tr}\big((b^{-1}d'b)^{2k+1}\big) = 2\pi i(-1)^k\frac{(2k+1)!}{(k!)^2}\operatorname{tr}\Omega^k,$$

and multiplying by c_k gives exactly $-\frac{1}{k!}(\frac{i}{2\pi})^k\operatorname{tr}\Omega^k$. □

14.5.3 The Fredholm determinant

Recall from Chapter 10 that on $G^{-\infty}$ the Fredholm determinant satisfies $d\log\det A = \operatorname{Tr}(A^{-1}dA)$, so that the degree-one part $c_0\operatorname{Tr}(A^{-1}dA)$ of $\mathrm{Ch}_{\mathrm{odd}}$ is $\frac{i}{2\pi}d\log\det A$, and that $\frac{1}{2\pi i}\int_\gamma\operatorname{Tr}(A^{-1}dA)\in\mathbb{Z}$ for every loop γ; this integer is the case $W=\{\mathrm{pt}\}$ of (14.5.1)–(14.5.4).

14.6 Eta forms

Let X be a manifold and $B \in \mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ a smooth family of invertible Toeplitz operators. Then

$$\alpha = B^{-1}dB \in \mathcal{C}^\infty(X; \Lambda^1 \otimes \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$$

and the symbol $b = \sigma_0(B) \in \mathcal{C}^\infty(X \times \mathbb{S}; \mathrm{GL}(N,\mathbb{C}))$ satisfies $\operatorname{wind}\det b(x,\cdot) = 0$. We write

$$\nu = b^{-1}d_X b, \qquad \mu = b^{-1}\partial_\theta b,$$

for the components of $b^{-1}d'b = \nu + \mu\, d\theta$ on $X \times \mathbb{S}$.

Definition 14.6.1. The *eta forms* of B are

$$\eta_{2k+1}(B) = \overline{\mathrm{Tr}}((B^{-1}dB)^{2k+1}), \qquad \eta(B) = \sum_{k\geq 0} c_k \eta_{2k+1}(B),$$

and the associated *symbolic correction forms* are

$$\gamma_{2k+1}(b) = \frac{i}{4\pi}\int_0^{2\pi} \mathrm{tr}(\nu^{2k+1}\mu)d\theta, \qquad \gamma(b) = \sum_{k\geq 0} c_k \gamma_{2k+1}(b).$$

The γ-forms are local in the symbol b; they vanish if b is independent of θ. If $B \in \mathcal{C}^\infty(X; G^{-\infty}_{\mathcal{T}})$ then $B^{-1}dB$ is smoothing, so $\overline{\mathrm{Tr}} = \mathrm{Tr}$ and $\eta(B) = \mathrm{Ch}_{\mathrm{odd}}(B)$, while $b = \mathrm{Id}$ and $\gamma(b) = 0$. Thus η *extends the odd Chern character from the normal subgroup* $G^{-\infty}_{\mathcal{T}}$ *to the whole group* $G_{\mathcal{T}}$, at the expense of no longer being closed.

Theorem 14.6.2. *For every smooth family* $B \in \mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ *with symbol* b,

$$d(\eta_{2k+1}(B) + \gamma_{2k+1}(b)) = \frac{i}{4\pi(2k+3)}\int_{\mathbb{S}} \mathrm{tr}((b^{-1}d'b)^{2k+3}) \tag{14.6.1}$$

and consequently

$$d(\eta(B) + \gamma(b)) = -\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b). \tag{14.6.2}$$

Proof. By Corollary 14.3.8 and (14.5.3) applied to α,

$$d\eta_{2k+1}(B) = \overline{\mathrm{Tr}}(d(\alpha^{2k+1})) = -\overline{\mathrm{Tr}}(\alpha^{2k+2}) = -\frac{1}{2}\overline{\mathrm{Tr}}([\alpha,\alpha^{2k+1}]) = -\frac{1}{2}\,\mathrm{Tr_R}((D_E\alpha)\wedge\alpha^{2k+1}),$$

since the graded commutator is $[\alpha,\alpha^{2k+1}] = 2\alpha^{2k+2}$. The operator $(D_E\alpha)\wedge\alpha^{2k+1}$ has order -1, and its full symbol is rapidly decreasing as $\xi \to -\infty$ because α is Toeplitz. By Lemma 14.4.1(i),(ii) only the principal symbol at $\xi = +1$, namely $-i(\partial_\theta\nu)\nu^{2k+1}$, contributes:

$$d\eta_{2k+1}(B) = -\frac{1}{2}\cdot\Big(-\frac{1}{2\pi}\Big)\int_0^{2\pi}\mathrm{tr}(-i\,\partial_\theta\nu\wedge\nu^{2k+1})d\theta = -\frac{i}{4\pi}\int_0^{2\pi}\mathrm{tr}(\partial_\theta\nu\wedge\nu^{2k+1})d\theta.$$

Differentiating $\nu = b^{-1}d_Xb$ and $\mu = b^{-1}\partial_\theta b$ gives the Maurer–Cartan identity $\partial_\theta\nu = d_X\mu + \nu\mu - \mu\nu$. By cyclicity $\mathrm{tr}(\nu\mu\nu^{2k+1}) = -\mathrm{tr}(\mu\nu^{2k+2})$, and by (14.5.3) $\mathrm{tr}(d\mu\wedge\nu^{2k+1}) = d\,\mathrm{tr}(\mu\nu^{2k+1}) + \mathrm{tr}(\mu\nu^{2k+2})$. Hence

$$\mathrm{tr}(\partial_\theta\nu\wedge\nu^{2k+1}) = d\,\mathrm{tr}(\mu\nu^{2k+1}) - \mathrm{tr}(\mu\nu^{2k+2}),$$

and, μ being of degree 0,

$$d\eta_{2k+1}(B) = -d\gamma_{2k+1}(b) + \frac{i}{4\pi}\int_0^{2\pi}\mathrm{tr}(\nu^{2k+2}\mu)\,d\theta.$$

Finally, the $d\theta$-component of $(\nu + \mu\, d\theta)^{2k+3}$ is $\sum_{j=0}^{2k+2}(-1)^j \nu^j \mu \nu^{2k+2-j} \wedge d\theta$, whose trace is $(2k+3)\operatorname{tr}(\nu^{2k+2}\mu) \wedge d\theta$ since $\operatorname{tr}(\nu^j \mu \nu^{2k+2-j}) = (-1)^j \operatorname{tr}(\mu \nu^{2k+2})$. This proves (14.6.1). For (14.6.2) note that the constants satisfy

$$c_k \cdot \frac{i}{4\pi(2k+3)} = -c_{k+1},$$

so the sum over k of (14.6.1) gives $-\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b)$ except for the term $c_0 \int_{\mathbb{S}} \operatorname{tr}(b^{-1}d'b) = c_0 \int_0^{2\pi} \operatorname{tr}\mu\, d\theta$ of degree 0. By (14.5.1) this is $\operatorname{ind}(B(x)) = 0$. □

Thus $-(\eta(B) + \gamma(b))$ is an explicit primitive of the form $\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b)$, which by Proposition 14.5.2 represents the Chern character of the index bundle of any Toeplitz family with symbol b, a class that vanishes here because the family B is invertible. Note that the variation of the eta form depends only on the leading part of the symbol, not on the lower order terms or on the smoothing part of B. The situation can be summarized in the diagram

$$\mathrm{Ch}_{\mathrm{odd}} \xleftarrow{\text{restrict}} \eta \xrightarrow{d(\cdot+\gamma)} -\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}$$

$$G_{\mathcal{T}}^{-\infty}(\mathbb{S};\mathbb{C}^N) \longrightarrow G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) \xrightarrow{\sigma_0} \mathcal{C}^\infty(\mathbb{S};\mathrm{GL}(N,\mathbb{C}))_0$$

where the right-hand group consists of loops of winding number 0. In the stabilized setting of Chapter 11 (where $\mathbb{C}^N$ is replaced by $\Psi^{-\infty}(Y;F)$ and tr by the trace on Y; all the arguments above go through verbatim) the bottom row is the classifying sequence for K-theory, with weakly contractible middle group, and the eta form is a transgression form for the even Chern character of the quotient. That the transgression only involves the principal symbol is a higher analogue of the 'miracle of the loop group' of Pressley and Segal [21].

Proposition 14.6.3. *Let $B \in \mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ and $g \in \mathcal{C}^\infty(X; G_{\mathcal{T}}^{-\infty})$. Then*

(i) $\eta_1(g) = \operatorname{Tr}(g^{-1}dg) = d\log\det g$;

(ii) $\eta_1(Bg) = \eta_1(B) + d\log\det g$ *and* $\eta_1(gB) = \eta_1(B) + d\log\det g$;

(iii) $d(\eta_1(B) + \gamma_1(b)) = \dfrac{i}{12\pi}\displaystyle\int_{\mathbb{S}} \operatorname{tr}((b^{-1}d'b)^3)$.

Proof. (i) was noted above and (iii) is (14.6.1) for $k = 0$. For (ii), $(Bg)^{-1}d(Bg) = g^{-1}\alpha g + g^{-1}dg$, and $g^{-1}\alpha g - \alpha = [g^{-1}\alpha, g]$, so by Lemma 14.3.7

$$\overline{\operatorname{Tr}}(g^{-1}\alpha g) - \overline{\operatorname{Tr}}(\alpha) = \operatorname{Tr}_{\mathrm{R}}(D_E(g^{-1}\alpha)\, g).$$

Since $g - \Pi$ is smoothing and $\sigma_0(g^{-1}\alpha) = \nu$, by Lemma 14.4.1 the right side is

$$\frac{i}{2\pi}\int_0^{2\pi} \partial_\theta \operatorname{tr}\nu\, d\theta = 0.$$

Also $\overline{\operatorname{Tr}}(g^{-1}dg) = \operatorname{Tr}(g^{-1}dg)$. For gB, we have $(gB)^{-1}d(gB) = B^{-1}(g^{-1}dg)B + \alpha$ and $\operatorname{Tr}(B^{-1}(g^{-1}dg)B) = \operatorname{Tr}(g^{-1}dg)$ because $g^{-1}dg$ is of trace class. □

Remark 14.6.4. Consider the principal bundle $G_{\mathcal{T}} \longrightarrow G_{\mathcal{T}}/G_{\mathcal{T}}^{-\infty}$ and the line bundle L associated to it by the character $\det : G_{\mathcal{T}}^{-\infty} \longrightarrow \mathbb{C}^*$. By (ii), and since $\gamma_1(\sigma_0(Bg)) = \gamma_1(\sigma_0(B))$, the 1-form $\eta_1 + \gamma_1$ on $G_{\mathcal{T}}$ is a connection form for L (its restriction to the fibres is $d\log\det$ and it transforms correctly); by (iii) its curvature is the 2-form $\frac{i}{12\pi}\int_{\mathbb{S}} \operatorname{tr}((b^{-1}d'b)^3)$,

which depends only on the symbol and so descends. The first Chern form $\frac{i}{2\pi}d(\eta_1+\gamma_1)$ of L is therefore $-\frac{1}{24\pi^2}\int_{\mathbb{S}}\operatorname{tr}((b^{-1}d'b)^3)$, which is minus the degree-two part of $\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(b)$. This is consistent with the identification of L with the determinant line bundle $\Lambda^{\max}(\mathrm{Nul})^*\otimes\Lambda^{\max}(\mathrm{coker})$ of the index bundle, whose first Chern class is $-c_1(\mathrm{ind})$. Thus η_1 is a regularized form of $d\log\det$ on the Toeplitz group; see also Section 14.10.4.

14.7 Product-type Toeplitz operators

14.7.1 The product-type calculus on $M\times\mathbb{S}$

Let M be a compact manifold without boundary of dimension $n\geq 1$ and consider the trivial fibration $\pi: M\times\mathbb{S}\longrightarrow M$. We use the following facts from Chapter 12 about the product-type operators $\Psi^{0,0}_{\mathrm{pt}}(M\times\mathbb{S};\mathbb{C}^N)$ associated to this fibration.

(P1) They form an algebra, acting on $\mathcal{C}^\infty(M\times\mathbb{S};\mathbb{C}^N)$ and boundedly on L^2, closed under adjoints. There are two multiplicative symbol maps. The *principal symbol* $\sigma(A)$ is a smooth function on the blow-up of $S^*(M\times\mathbb{S})$ along $S^*M\times\mathbb{S}$ (the covectors annihilating the fibres, $\tau=0$ where τ is dual to θ); this blow-up is the disjoint union of the two closed half-sphere bundles

$$S^*_\pm=\{\pm\tau\geq 0\}\subset S^*(M\times\mathbb{S}),\qquad \partial S^*_\pm=S^*M\times\mathbb{S}.$$

The *base family* $\beta(A)\in\mathcal{C}^\infty(S^*M;\Psi^0(\mathbb{S};\mathbb{C}^N))$ is a family of operators on the fibre parametrized by S^*M.

(P2) The symbols are compatible: the principal symbol of $\beta(A)(x)$ on $S^*\mathbb{S}=\mathbb{S}\times\{\pm1\}$ is the restriction of $\sigma(A)$ to $\{x\}\times\mathbb{S}\subset\partial S^*_\pm$. Conversely any compatible pair, depending smoothly on parameters, is the joint symbol of a smooth family of operators.

(P3) If $\sigma(A)=0$ and $\beta(A)=0$ then A is compact on L^2.

(P4) If A is *fully elliptic*, i.e. $\sigma(A)$ and all $\beta(A)(x)$ are invertible, then A is Fredholm on L^2 and on $\mathcal{C}^\infty$, with the same index; the index depends only on the joint symbol and is constant along smooth families of fully elliptic operators.

(P5) The ordinary operators $\Psi^0(M\times\mathbb{S};\mathbb{C}^N)$ are product-type, with $\sigma(P)$ the lift of the usual principal symbol and $\beta(P)(x)$ the multiplication operator by $\sigma(P)|_{\{x\}\times\mathbb{S}}$. Fibrewise families $Q\in\mathcal{C}^\infty(M;\Psi^0(\mathbb{S};\mathbb{C}^N))$ are product-type, with $\sigma(Q)$ the lift of the fibre symbol and $\beta(Q)(m,\xi)=Q(m)$; in particular multiplication by $h\in\mathcal{C}^\infty(M\times\mathbb{S};M(N,\mathbb{C}))$ has symbol h and base family the multiplication by $h(m,\cdot)$.

Since the Szegő projector Π, acting in θ, is a fibrewise operator, it is in $\Psi^{0,0}_{\mathrm{pt}}$ with $\sigma(\Pi)=1$ on S^*_+, $\sigma(\Pi)=0$ on S^*_-, and $\beta(\Pi)=\Pi$. The *product-type Toeplitz algebra* is

$$\Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M\times\mathbb{S};\mathbb{C}^N)=\Pi\,\Psi^{0,0}_{\mathrm{pt}}(M\times\mathbb{S};\mathbb{C}^N)\,\Pi,$$

acting on $\mathcal{H}(M\times\mathbb{S};\mathbb{C}^N)=\Pi\mathcal{C}^\infty(M\times\mathbb{S};\mathbb{C}^N)$ (or its L^2 closure), with identity Π. For A in this algebra the principal symbol vanishes on S^*_-; we set

$$a=\sigma(A)|_{S^*_+}\in\mathcal{C}^\infty(S^*_+;M(N,\mathbb{C})),$$
$$\beta=\beta(A)\in\mathcal{C}^\infty(S^*M;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)),\quad b=a|_{\partial S^*_+}=\sigma_0(\beta).$$

A is *fully elliptic* if a takes values in $\mathrm{GL}(N,\mathbb{C})$ and $\beta\in\mathcal{C}^\infty(S^*M;G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$; then A is Fredholm on $\mathcal{H}$ (apply (P4) to $A+\mathrm{Id}-\Pi$) and its index has the properties in (P4). We write $\mathrm{ind}(A)$ for the index on $\mathcal{H}$.

14.7.2 Geometry and orientations

Fix a metric on M and use the product metric on $M \times \mathbb{S}$. In each fibre of $S^*_+ \to M \times \mathbb{S}$, stereographic projection $(\xi, \tau) \mapsto \xi/(1+\tau)$ identifies the closed half-sphere $\{|\xi|^2 + \tau^2 = 1,\ \tau \geq 0\}$ diffeomorphically with the closed unit ball in $T^*_m M$; thus $S^*_+ \cong \overline{B^*M} \times \mathbb{S}$ is a ball bundle over $M \times \mathbb{S}$, with 'centre' the section $(\xi, \tau) = (0, 1)$, i.e. $d\theta$.

We orient cotangent bundles by the Liouville form: if $\omega = \sum_j d\xi_j \wedge dx_j$ then T^*M is oriented by $\omega^n/n!$. The ball bundle $\overline{B^*M}$ inherits this orientation, and $S^*M = \partial\overline{B^*M}$ is given the boundary orientation (outward normal first); we call this the *Liouville orientation* of S^*M, and similarly for $S^*(M \times \mathbb{S})$, whose open subset $\{\tau > 0\}$ orients S^*_+. For $n = 1$ the two components $\{\xi = \pm 1\}$ of $S^*M \cong M \times \{\pm 1\}$ receive opposite orientations ($\pm dx$ if M is oriented by dx).

Lemma 14.7.1. *With these orientations, $\partial S^*_+ = S^*M \times \mathbb{S}$ with the product orientation, S^*M first and $d\theta$ last.*

Proof. The Liouville form of $T^*(M \times \mathbb{S})$ is $\omega_M + d\tau \wedge d\theta$, and $(\omega_M + d\tau \wedge d\theta)^{n+1}/(n+1)! = \omega_M^n/n! \wedge d\tau \wedge d\theta$. At a point with $\tau = 0$ the outward radial vector R is tangent to the ξ directions, so $S^*(M \times \mathbb{S})$ is oriented there by $\iota_R(\omega_M^n/n!) \wedge d\tau \wedge d\theta = \sigma \wedge d\tau \wedge d\theta$, where σ is the Liouville orientation of S^*M. The outward normal of S^*_+ at $\tau = 0$ is $-\partial_\tau$, and $\iota_{-\partial_\tau}(\sigma \wedge d\tau \wedge d\theta) = (-1)^{2n}\sigma \wedge d\theta$ since σ has degree $2n - 1$. □

We use the Atiyah–Singer theorem in the following form, with $\mathrm{Td}(M)$ a closed form representing the Todd class of $TM \otimes \mathbb{C}$ (by Stokes' theorem the right side does not depend on the choice of form).

Theorem 14.7.2 (Atiyah–Singer, cosphere form)**.** *If W is a compact manifold without boundary and $P \in \Psi^0(W; \mathbb{C}^N)$ is elliptic with principal symbol $p \in \mathcal{C}^\infty(S^*W; \mathrm{GL}(N, \mathbb{C}))$ then*

$$\mathrm{ind}(P) = \int_{S^*W} \mathrm{Td}(W) \wedge \mathrm{Ch}_{\mathrm{odd}}(p),$$

where S^*W carries the Liouville orientation and only the component of degree $2 \dim W - 1$ is integrated.

This is the case of a point base of the families index theorem of Chapter 13, once the relative Chern character of the symbol class in $K_c(T^*W)$ is transgressed to the cosphere bundle; the normalization of $\mathrm{Ch}_{\mathrm{odd}}$ is the one forced by Bott periodicity, Proposition 14.5.2. For $W = \mathbb{S}$ it is the elementary formula $\mathrm{ind}\, P = -\mathrm{wind}\det p(\cdot, 1) + \mathrm{wind}\det p(\cdot, -1)$, which follows from (14.5.1) by splitting P into its positive and negative Toeplitz parts (see Chapter 11); with the Liouville orientations ($d\theta$ on $\xi = 1$, $-d\theta$ on $\xi = -1$) this agrees with the theorem, which checks the orientation convention.

14.8 The index formula

For a fully elliptic $A \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M \times \mathbb{S}; \mathbb{C}^N)$ with symbols (a, β), $b = \sigma_0(\beta)$, set

$$I(A) = \int_{S^*M} \mathrm{Td}(M) \wedge (\eta(\beta) + \gamma(b)) + \int_{S^*_+} \mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(a), \tag{14.8.1}$$

where $\mathrm{Td}(M)$ is pulled back to S^*M and to S^*_+, the orientations are those of the previous section, and only components of top degree are integrated. Note that $I(A)$ depends only on the joint symbol of A.

Theorem 14.8.1. *For every fully elliptic* $A \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M \times \mathbb{S}; \mathbb{C}^N)$,

$$\operatorname{ind}(A) = \int_{S^*M} \operatorname{Td}(M) \wedge (\eta(\beta) + \gamma(b)) + \int_{S^*_+(M\times\mathbb{S})} \operatorname{Td}(M) \wedge \operatorname{Ch}_{\mathrm{odd}}(a).$$

For M a point (formally $n = 0$) the formula reduces to (14.5.1), S^*_+ being the circle oriented by $d\theta$. The proof occupies the rest of this section. We write $\Delta(A) = \operatorname{ind}(A) - I(A)$.

Proposition 14.8.2. *If* (a_t, β_t), $t \in [0,1]$, *is a smooth family of compatible pairs of fully elliptic symbols then* I *is constant in* t. *Consequently* $\Delta(A)$ *depends only on the homotopy class of the joint symbol of* A *through fully elliptic symbols.*

Proof. Write $T = \operatorname{Td}(M)$, a closed even form, $X = S^*M$, and $I(t) = I_1(t) + I_2(t)$ for the two terms. By Lemma 14.5.1, $\partial_t \operatorname{Ch}_{\mathrm{odd}}(a_t) = d\Theta(a_t)$, so by Stokes' theorem and Lemma 14.7.1

$$I_2'(t) = \int_{S^*_+} T \wedge d\Theta(a_t) = \int_{X\times\mathbb{S}} T \wedge \Theta(b_t) = \int_X T \wedge \int_{\mathbb{S}} \Theta(b_t),$$

where $\Theta(b_t)$ is the restriction of $\Theta(a_t)$ to the boundary, given by the same formula in terms of b_t. Now apply Theorem 14.6.2 to the family β on $X \times [0,1]$, with total differential $D = d + dt\,\partial_t$. Writing $\eta(\beta) + \gamma(b) = \zeta(t) + \zeta'(t) \wedge dt$ and, by Lemma 14.5.1 applied on $X \times \mathbb{S} \times [0,1]$,

$$\int_{\mathbb{S}} \operatorname{Ch}_{\mathrm{odd}}(b) = \int_{\mathbb{S}} \operatorname{Ch}_{\mathrm{odd}}(b_t) + \int_{\mathbb{S}} (\Theta(b_t) \wedge dt) = \int_{\mathbb{S}} \operatorname{Ch}_{\mathrm{odd}}(b_t) - \Big(\int_{\mathbb{S}} \Theta(b_t)\Big) \wedge dt$$

(moving dt past $d\theta$), the dt-component of (14.6.2) reads $-\partial_t\zeta + d_X\zeta' = \int_{\mathbb{S}} \Theta(b_t)$, using $dt \wedge \partial_t\zeta = -\partial_t\zeta \wedge dt$ (ζ is odd). Since T is closed and X is compact without boundary,

$$I_1'(t) = \int_X T \wedge \partial_t\zeta = -\int_X T \wedge \int_{\mathbb{S}} \Theta(b_t) = -I_2'(t).$$

The second statement follows from (P2) and (P4): a homotopy of joint symbols is realized by a smooth family of fully elliptic operators, along which the index is constant. □

We use direct sums and stabilization freely: if A acts on $\mathbb{C}^N$-valued functions, $A \oplus \Pi$ acting on $\mathbb{C}^{N+K}$-valued functions has the same index and the same I, and both ind and I are additive under direct sums (all the forms involved are traces of block-diagonal expressions).

Lemma 14.8.3. *If* $A, A' \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M\times\mathbb{S}; \mathbb{C}^N)$ *are fully elliptic then* $\Delta(AA') = \Delta(A) + \Delta(A')$.

Proof. The index is additive under composition of Fredholm operators. Let R_t be the constant unitary $\left(\begin{smallmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{smallmatrix}\right) \otimes \mathrm{Id}_N$ on $\mathbb{C}^{2N}$; it commutes with Π and is an invertible element of the algebra. Then

$$C_t = R_t(A \oplus \Pi)R_t^{-1}(\Pi \oplus A'), \qquad 0 \le t \le \pi/2,$$

is a smooth family of fully elliptic operators with $C_0 = A \oplus A'$ and $C_{\pi/2} = \Pi \oplus AA'$. By Proposition 14.8.2 and additivity, $I(A) + I(A') = I(\Pi \oplus AA') = I(AA')$, since $I(\Pi) = 0$ ($\beta(\Pi)$ is constant and $a = \mathrm{Id}$). □

Lemma 14.8.4. *If* $h \in \mathcal{C}^\infty(M; \mathrm{GL}(N,\mathbb{C}))$ *and* H *denotes multiplication by* h *on* $\mathcal{H}(M \times \mathbb{S}; \mathbb{C}^N)$, *then* H *is fully elliptic and* $\operatorname{ind}(H) = I(H) = 0$.

Proof. H is invertible, so $\operatorname{ind}(H) = 0$. Its symbols are $a = h$ (pulled back to S^*_+) and $\beta = \Pi h \Pi$, constant in ξ and θ. Hence $\gamma(b) = 0$ and, by Lemma 14.4.1(iii), $\eta(\beta) = \frac{1}{2}\operatorname{Ch}_{\mathrm{odd}}(h)$ pulled back from M. The form $\operatorname{Td}(M) \wedge \operatorname{Ch}_{\mathrm{odd}}(h)$ is pulled back from M, which has dimension n, whereas $\dim S^*_+ = 2n + 1 > n$ and $\dim S^*M = 2n - 1$; so both integrals in (14.8.1) vanish when $n \geq 2$. If $n = 1$ the second still vanishes, and the first vanishes because $S^*M \cong M \times \{\pm 1\}$ with opposite orientations on the two components. □

Lemma 14.8.5. *Let $P \in \Psi^0(M \times \mathbb{S}; \mathbb{C}^N)$ be elliptic with principal symbol p equal to* Id *outside a compact subset of $\{\tau > 0\} \subset S^*(M \times \mathbb{S})$. Then $\Pi P \Pi \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}$ is fully elliptic with $a = p|_{S^*_+}$, $\beta = \Pi$, and*

$$\operatorname{ind}(\Pi P \Pi) = \operatorname{ind}(P) = I(\Pi P \Pi).$$

Proof. By (P5), $\beta(P)$ is multiplication by $p|_{\tau=0} = \mathrm{Id}$, so $\beta(\Pi P \Pi) = \Pi$ and $\sigma(\Pi P \Pi) = p$ on S^*_+. With respect to $L^2 = \mathcal{H} \oplus \mathcal{H}^\perp$ write $P = \Pi P \Pi + (\mathrm{Id} - \Pi) P (\mathrm{Id} - \Pi) + K$, $K = \Pi P (\mathrm{Id} - \Pi) + (\mathrm{Id} - \Pi) P \Pi$. Both symbols of K vanish ($\sigma(\Pi)\sigma(\mathrm{Id} - \Pi) = 0$ and $\Pi\,\mathrm{Id}\,(\mathrm{Id} - \Pi) = 0$), and both symbols of $(\mathrm{Id} - \Pi) P (\mathrm{Id} - \Pi) - (\mathrm{Id} - \Pi)$ vanish (on S^*_-, $p = \mathrm{Id}$). By (P3) these are compact, so $\operatorname{ind}(P) = \operatorname{ind}(\Pi P \Pi) + \operatorname{ind}((\mathrm{Id} - \Pi) + \text{compact}) = \operatorname{ind}(\Pi P \Pi)$. On the other hand $\eta(\Pi) = 0$ and $\gamma(\mathrm{Id}) = 0$, so $I(\Pi P \Pi) = \int_{S^*_+} \operatorname{Td}(M) \wedge \operatorname{Ch}_{\mathrm{odd}}(p)$. Since $p = \mathrm{Id}$ on S^*_-, $\operatorname{Ch}_{\mathrm{odd}}(p) = 0$ there, and $\operatorname{Td}(M \times \mathbb{S}) = \operatorname{Td}(M)$ (as $T(M \times \mathbb{S}) \otimes \mathbb{C} = \pi^*(TM \otimes \mathbb{C}) \oplus \mathbb{C}$); so Theorem 14.7.2 on $M \times \mathbb{S}$ gives $\operatorname{ind}(P) = I(\Pi P \Pi)$. □

We also need the weak contractibility of the normalized Toeplitz group from Chapter 11, in the following form. If $\beta' \in \mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N))$, X compact, satisfies the normalization $\sigma_0(\beta')(x, 1) = \mathrm{Id}$ (at $\theta = 0$), then after stabilization $\mathbb{C}^N \subset \mathbb{C}^K$, $\beta' \oplus \Pi$ is smoothly homotopic through $\mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^K))$ to the constant family Π. Indeed Chapter 11 gives such a homotopy in the group stabilized by the smoothing operators on an auxiliary manifold Y, with $\mathbb{C}^N$ embedded as the span of the first N eigenfunctions of a Laplacian on Y; compressing the homotopy by the spectral projection onto the first K eigenfunctions gives, for K large, a homotopy of the required kind, since the compressions converge in the $\mathcal{C}^\infty$ topology and invertibility is an open condition, uniformly on the compact parameter space.

Proof of Theorem 14.8.1. Let A be fully elliptic with symbols (a, β), $X = S^*M$. We show $\Delta(A) = 0$ in four steps; each step replaces A by an operator with the same Δ.

Homotopies with prescribed boundary values. Choose a collar $\partial S^*_+ \times [0, \epsilon)_r$ of the boundary of S^*_+ and $\phi \in \mathcal{C}^\infty([0, \epsilon))$ with $\phi = 0$ near 0 and $\phi = 1$ on $[\epsilon/2, \epsilon)$. If b_t is a smooth homotopy of $b_0 = a|_{\partial S^*_+}$ through maps into $\mathrm{GL}(N, \mathbb{C})$, then

$$a_t(q, r) = b_{t(1-\phi(r))}(q)\, b_0(q)^{-1} a(q, r) \text{ in the collar}, \qquad a_t = a \text{ elsewhere},$$

is a smooth homotopy of a through invertible symbols with $a_t|_{\partial S^*_+} = b_t$. So any smooth homotopy β_t of β through $\mathcal{C}^\infty(X; G_{\mathcal{T}})$ extends to a homotopy of fully elliptic compatible pairs (a_t, β_t), which does not change Δ (Proposition 14.8.2).

Step 1: β constant in θ. Let $g(x) = b(x, 1) \in \mathrm{GL}(N, \mathbb{C})$ and $\beta' = g^{-1}\beta$ (the matrix $g(x)^{-1}$ acting on $\mathcal{H} \otimes \mathbb{C}^N$, which commutes with Π). Then β' is normalized, so after stabilization there is a homotopy β'_t from β' to Π, and $\beta_t = g\beta'_t$ is a homotopy from β to $\Pi g \Pi$. After extending it as above we may assume that $\beta(x) = \Pi g(x) \Pi$ with $g \in \mathcal{C}^\infty(X; \mathrm{GL}(N, \mathbb{C}))$; then $b(x, \theta) = g(x)$.

Step 2: β pulled back from M. The submanifold $S^*_+ \cap \{\theta = 0\}$ is a closed ball bundle over M with boundary X, and a restricted to it extends g. Contracting the balls linearly to their centres (in the stereographic picture, $v \mapsto (1 - t)v$) gives a smooth homotopy g_t,

$g_0 = g$, from g to $h \circ \pi_M$, where $h(m)$ is the value of a at the centre over $(m,0)$. The family $\beta_t = \Pi g_t \Pi$ consists of invertible operators; extending as above we may assume $\beta = \Pi h \Pi$ with $h \in \mathcal{C}^\infty(M; \mathrm{GL}(N,\mathbb{C}))$.

Step 3: $\beta = \Pi$. Let H be multiplication by h and $A' = H^{-1}A$. Then $\beta(A') = \Pi$ and $a' = h^{-1}a$ satisfies $a'|_{\partial S^*_+} = \mathrm{Id}$. By Lemmas 14.8.3 and 14.8.4, $\Delta(A) = \Delta(H) + \Delta(A') = \Delta(A')$.

Step 4: reduction to an ordinary operator. Let $\kappa \in \mathcal{C}^\infty([0,\epsilon))$ be non-decreasing with $\kappa = 0$ on $[0,\epsilon/2]$ and $\kappa(r) = r$ near ϵ, and let $\Phi_t(q,r) = (q, (1-t)r + t\kappa(r))$ in the collar, $\Phi_t = \mathrm{Id}$ elsewhere. Then $a' \circ \Phi_t$ is a homotopy of invertible symbols with boundary value Id, and $a'' = a' \circ \Phi_1$ equals Id near ∂S^*_+. Keeping $\beta = \Pi$, this is a homotopy of fully elliptic pairs. Extend a'' by Id to a smooth function p on $S^*(M \times \mathbb{S})$ and choose $P \in \Psi^0(M \times \mathbb{S}; \mathbb{C}^N)$ with principal symbol p. Then $\Pi P \Pi$ has the joint symbol (a'', Π), so by Lemma 14.8.5 and Proposition 14.8.2, $\Delta(A') = \Delta(\Pi P \Pi) = 0$. □

Remark 14.8.6. The theorem contains the Atiyah–Singer formula on M. Let $\pi_{\le k}$ be the projection onto the span of $e^{ij\theta}$, $0 \le j \le k$, and let $Q \in \Psi^0(M; \mathbb{C}^N \otimes \mathbb{C}^{k+1})$ be elliptic, $\mathbb{C}^{k+1} \cong \operatorname{Ran} \pi_{\le k}$. The operator $A = Q\pi_{\le k} + (\Pi - \pi_{\le k})$ is in $\Psi^{0,0}_{\mathrm{pt},\mathcal{T}}$ (compare the compressions in Chapter 12) with $a = \mathrm{Id}$ and $\beta = \sigma(Q)\pi_{\le k} + \Pi - \pi_{\le k}$, a finite-rank perturbation of Π; so $b = \mathrm{Id}$, $\gamma(b) = 0$ and $\eta(\beta) = \mathrm{Ch}_{\mathrm{odd}}(\sigma(Q))$. Since $\mathrm{ind}(A) = \mathrm{ind}(Q)$, Theorem 14.8.1 reduces to Theorem 14.7.2 on M with the Liouville orientation of S^*M, consistently with Lemma 14.7.1. Conversely, one might try to prove the theorem by deforming every A to this form; but a fully elliptic A need not be deformable to one with $a = \mathrm{Id}$, because the obstruction, the index bundle over M of the Toeplitz family with symbol a restricted to the centre section, need not vanish (it only vanishes after pull-back to S^*M). This is why the proof above deforms in the opposite direction and uses the Atiyah–Singer theorem on $M \times \mathbb{S}$.

Remark 14.8.7. The symbolic term $\gamma(b)$ cannot be omitted. Take $n = 1$, $M = \mathbb{S}_x$, $N = 1$, and symbols $a_t = \exp(it\chi \sin(x+\theta))$ on S^*_+, where χ is a function on the fibres of S^*_+ (half-circles) equal to 1 at the end $\xi = +1$ and to 0 at the end $\xi = -1$. Thus $b_t = \exp(it \sin(x+\theta))$ on the component $\xi = +1$ of S^*M and $b_t = 1$ on $\xi = -1$. Since (a_t) starts at $a_0 = 1$, invertible base families β_t with symbols b_t and $\beta_0 = \Pi$ exist (lift the homotopy as in Chapter 10). In degree one, with $\mathrm{Td} = 1$ there,

$$\int_{S^*M} \gamma(b_t) = c_0 \int_0^{2\pi} \frac{i}{4\pi} \int_0^{2\pi} (it\cos(x+\theta))^2 \, d\theta \, dx = \frac{i}{2\pi} \cdot \frac{i}{4\pi} \cdot (-t^2) \cdot 2\pi^2 = \frac{t^2}{4},$$

which depends on t. Since I is constant along the family by Proposition 14.8.2, the remaining terms of I vary by $-t^2/4$, so the formula without γ cannot hold for all t.

14.9 Non-Toeplitz product-type operators

The formula extends to the full product-type calculus, where both halves $S^*_\pm$ of the blown-up cosphere bundle carry symbols. For a fully elliptic $P \in \Psi^{0,0}_{\mathrm{pt}}(M \times \mathbb{S}; \mathbb{C}^N)$ write $a_\pm = \sigma(P)|_{S^*_\pm}$, $\beta = \beta(P) \in \mathcal{C}^\infty(S^*M; \Psi^0(\mathbb{S}; \mathbb{C}^N))$ and $b_\pm = a_\pm|_{\partial S^*_\pm}$, regarded as functions on $S^*M \times \mathbb{S}$. The eta form $\eta(\beta)$ is defined exactly as before, with $\overline{\mathrm{Tr}}$ on $\Psi^0(\mathbb{S}; \mathbb{C}^N)$. For a map $c \in \mathcal{C}^\infty(S^*M \times \mathbb{S}; \mathrm{GL}(N,\mathbb{C}))$ let $\langle \mathrm{Ch}_{\mathrm{odd}}(c) \rangle = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{Ch}_{\mathrm{odd}}(c(\cdot,\theta))\, d\theta$ be the θ-average of the odd Chern forms of the maps $c(\cdot,\theta)$ on S^*M.

Proposition 14.9.1. *For every fully elliptic $P \in \Psi^{0,0}_{\mathrm{pt}}(M \times \mathbb{S}; \mathbb{C}^N)$,*

$$\operatorname{ind}(P) = \int_{S^*M} \operatorname{Td}(M) \wedge \big(\eta(\beta) + \langle \operatorname{Ch}_{\mathrm{odd}}(b_-)\rangle + \gamma(b_+) - \gamma(b_-)\big) + \int_{S^*_+} \operatorname{Td}(M) \wedge \operatorname{Ch}_{\mathrm{odd}}(a_+) + \int_{S^*_-} \operatorname{Td}(M) \wedge \operatorname{Ch}_{\mathrm{odd}}(a_-),$$

with $S^*_\pm$ oriented as subsets of $S^*(M \times \mathbb{S})$ and S^*M Liouville oriented.

Proof. Let $J : \mathcal{H}^\perp \longrightarrow \mathcal{H}$ be the unitary $(Ju)(\theta) = e^{-i\theta}u(-\theta)$, mapping $e^{-i(k+1)\theta}$ to $e^{ik\theta}$, and

$$\mathcal{U} : L^2(M \times \mathbb{S}; \mathbb{C}^N) \longrightarrow \mathcal{H}(M \times \mathbb{S}; \mathbb{C}^{2N}), \quad \mathcal{U}u = (\Pi u, J(\mathrm{Id} - \Pi)u).$$

The reflection $\theta \mapsto -\theta$ is a fibre-preserving diffeomorphism, so it preserves the product-type calculus, and the off-diagonal blocks $\Pi P(\mathrm{Id} - \Pi)$, $(\mathrm{Id} - \Pi)P\Pi$ have vanishing principal symbol. Hence $P^\sharp = \mathcal{U}P\mathcal{U}^{-1}$ is a fully elliptic element of $\Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M \times \mathbb{S}; \mathbb{C}^{2N})$ with

$$a^\sharp = a_+ \oplus R^* a_-, \qquad \beta^\sharp = \mathcal{U}\beta\mathcal{U}^{-1},$$

where $R : S^*_+ \longrightarrow S^*_-$ is induced by $(\theta, \tau) \mapsto (-\theta, -\tau)$, and $\operatorname{ind}(P) = \operatorname{ind}(P^\sharp)$. We evaluate $I(P^\sharp)$.

The map R preserves the Liouville form $d\tau \wedge d\theta$ and the radial structure, so it preserves orientations and $\int_{S^*_+} \operatorname{Td} \wedge \operatorname{Ch}_{\mathrm{odd}}(R^* a_-) = \int_{S^*_-} \operatorname{Td} \wedge \operatorname{Ch}_{\mathrm{odd}}(a_-)$. The boundary symbol of $P^\sharp$ is $b_+ \oplus b^R_-$ with $b^R_-(x, \theta) = b_-(x, -\theta)$; the substitution $\theta \mapsto -\theta$ changes the sign of μ and not of ν, so $\gamma(b^R_-) = -\gamma(b_-)$. Finally $\overline{\operatorname{Tr}}(\mathcal{U}Q\mathcal{U}^{-1}) = \overline{\operatorname{Tr}}_{E_1}(Q)$ where $E_1 = \mathcal{U}^{-1}E\mathcal{U}$ multiplies $e^{ij\theta}$ by λ^s_j for $j \geq 0$ and by $\lambda^s_{|j|-1}$ for $j < 0$. This is an admissible family and, by Lemma 14.3.3, $\overline{\operatorname{Tr}}_{E_1}(Q) = \overline{\operatorname{Tr}}(Q) + \operatorname{Tr}_{\mathrm{R}}(QD)$ where $D = \frac{d}{ds}(E_1 - E)|_{s=0}$ is the Fourier multiplier by $\log \lambda_{|j|-1} - \log \lambda_{|j|}$ for $j < 0$ and 0 for $j \geq 0$, with symbol ξ^{-1} (modulo lower order) in $\xi < 0$ and rapidly decreasing in $\xi > 0$. With $Q = (\beta^{-1}d\beta)^{2k+1}$, Lemma 14.4.1(i) gives

$$\operatorname{Tr}_{\mathrm{R}}(QD) = -\frac{1}{2\pi}\int_0^{2\pi} \operatorname{tr}(\nu_-^{2k+1} \cdot (-1))d\theta = \frac{1}{2\pi}\int_0^{2\pi} \operatorname{tr}\big((b_-^{-1}d_X b_-)^{2k+1}\big)d\theta,$$

so $\eta(\beta^\sharp) = \eta(\beta) + \langle \operatorname{Ch}_{\mathrm{odd}}(b_-)\rangle$. Now apply Theorem 14.8.1 to $P^\sharp$. □

As a check, let $P \in \Psi^0(M \times \mathbb{S}; \mathbb{C}^N)$ be an ordinary elliptic operator. Then β is multiplication by $b = b_+ = b_-$, so $\eta(\beta) = 0$ by Lemma 14.4.1(iii) and the γ-terms cancel. For fixed θ the map $b(\cdot, \theta)$ on S^*M extends over the ball bundle $S^*_+ \cap \{\theta\}$, hence is homotopic to a map pulled back from M, so $\int_{S^*M} \operatorname{Td} \wedge \operatorname{Ch}_{\mathrm{odd}}(b(\cdot, \theta)) = 0$ as in Lemma 14.8.4. The formula therefore reduces to Theorem 14.7.2 on $M \times \mathbb{S}$. The asymmetric term $\langle \operatorname{Ch}_{\mathrm{odd}}(b_-)\rangle$ reflects the fact that the regularization (14.4.1) is not invariant under J.

14.10 Towards general product-type index theorems

In this final section we indicate how the results above should fit into a general index theory for product-type families. Parts of it are conjectural, and are labelled as such.

14.10.1 Thom isomorphisms and the Todd class

For an oriented real vector bundle $V \longrightarrow M$ of rank r over a compact manifold, integration over the fibres gives the Thom isomorphism $H_c^*(V) \longrightarrow H^{*-r}(M)$. For a complex vector bundle there is also a Thom isomorphism $K_c(V) \longrightarrow K(M)$ in K-theory, extending Bott periodicity (the case of trivial V). Both sides carry Chern characters, giving a square

$$\begin{array}{ccc} K_c(V) & \xrightarrow{\text{Thom}} & K(M) \\ {\scriptstyle \text{Ch}}\downarrow & & \downarrow{\scriptstyle \text{Ch}} \\ H_c^{\text{even}}(V) & \xrightarrow[\text{Thom}]{} & H^{\text{even}}(M) \end{array}$$

which does *not* commute: the two composites differ by multiplication by a characteristic class of V, which (with appropriate conventions) is the Todd class. This is the origin of the factor $\mathrm{Td}(M) = \mathrm{Td}(TM \otimes \mathbb{C})$ in the index formula. A direct computation of the Chern character of the index bundle in the families index theorem of Chapter 13 produces the Todd class in the same way.

14.10.2 Product-type K-theory

Recall that for a fibration $\phi : M \longrightarrow B$ of compact manifolds the families index theorem is the equality of two maps

$$\mathrm{ind}_a = \mathrm{ind}_t : K_c^0(T^*(M/B)) \longrightarrow K^0(B),$$

the analytic index being defined by representing a class by the symbol of an elliptic family $A \in \Psi^0(M/B; \mathbb{E})$ and taking the stabilized index bundle, the topological index by embedding the fibration in a product $\mathbb{S}^N \times B$.

Consider now a compound fibration of compact manifolds

$$M \xrightarrow{\psi} Y \xrightarrow{\Phi} B, \qquad \phi = \Phi \circ \psi,$$

and the families $A \in \Psi^{0,0}_{\mathrm{pt}}(M/B; \mathbb{E})$ of operators on the fibres of ϕ which are of product type with respect to the fibration (over $\Phi^{-1}(b)$) induced by ψ. Such a family has two symbols,

$$\sigma(A) \in \mathcal{C}^\infty\big([S^*(M/B), \psi^*S^*(Y/B)]; \hom(\mathbb{E})\big), \qquad \beta(A) \in \Psi^0\big(\pi^*M/S^*(Y/B); \mathbb{E}\big),$$

the second being a family of operators on the fibres of ψ parametrized by $S^*(Y/B)$, and full ellipticity means invertibility of both. Then the analytic index $\mathrm{ind}_a(A) \in K^0(B)$ is well defined by stabilization and depends only on $(\sigma(A), \beta(A))$. Considering all compatible invertible pairs modulo bundle isomorphism, stabilization and homotopy one expects an abelian group $K^0_{c,\psi\text{-pt}}(T^*(M/B))$ (and an odd version $K^{-1}_{c,\psi\text{-pt}}$) on which the analytic index is defined. Forgetting β gives a map to $K_c^0(T^*(M/Y))$ (evaluation of σ in the vertical directions of ψ), and the families index of ψ, pulled back to $S^*(Y/B)$, gives a map $K_c^0(T^*(M/Y)) \longrightarrow K^0(S^*(Y/B))$ which vanishes on the image, expressing the invertibility of β. Identifying $K^{-1}(S^*(Y/B))$ with homotopy classes of maps into $G^{-\infty}$, i.e. with invertible base families with symbol the identity, one is led to:

Remark 14.10.1 (Conjectural six-term sequence). There should be an exact sequence

$$\begin{array}{ccccc} K^{-1}(S^*(Y/B)) & \longrightarrow & K^0_{c,\psi\text{-pt}}(T^*(M/B)) & \xrightarrow{\sigma} & K_c^0(T^*(M/Y)) \\ {\scriptstyle \text{ind}}\uparrow & & & & \downarrow{\scriptstyle \text{ind}} \\ K_c^{-1}(T^*(M/Y)) & \xleftarrow{\sigma} & K^{-1}_{c,\psi\text{-pt}}(T^*(M/B)) & \longleftarrow & K^0(S^*(Y/B)) \end{array}$$

and the analytic index should extend to an equality $\mathrm{ind}_a = \mathrm{ind}_t : K^0_{c,\psi\text{-pt}}(T^*(M/B)) \longrightarrow K^0(B)$ with a topological index defined by embedding. A natural problem is to find a more traditional realization of the group $K^0_{c,\psi\text{-pt}}(T^*(M/B))$, for which the six-term sequence should serve as a guide. Groups of this type were subsequently constructed by Melrose and Rochon; see [20] for the closely related families index for manifolds with boundary.

14.10.3 Product-type cohomology

The cohomological form of the index theorem expresses the Chern character of the index in terms of a Chern character of the symbol. Theorem 14.8.1 suggests what the latter should be in the product-type case: a pair consisting of the Chern form of σ and the eta form of β, related by Theorem 14.6.2. We formulate the corresponding cohomology in an abstract setting (correcting the degree conventions of the original notes).

Let Z be a compact manifold with boundary and $\psi : \partial Z \longrightarrow Y$ a fibration over a compact manifold Y without boundary, with compact oriented fibres of dimension f, and let ψ_* denote integration over the fibres with the 'fibre last' convention, so that $\psi_* d = d\psi_*$ and $\int_{\partial Z} = \int_Y \psi_*$ when ∂Z is oriented by Y followed by the fibres. Let $\iota : \partial Z \hookrightarrow Z$. Set

$$C^k(Z,\psi) = \mathcal{C}^\infty(Z;\Lambda^k) \oplus \mathcal{C}^\infty(Y;\Lambda^{k-f-1}), \qquad \mathcal{D}(u,\tau) = (du,\ \psi_*\iota^* u - d\tau).$$

Then $\mathcal{D}^2 = 0$; a pair is closed if $du = 0$ and $d\tau = \psi_*\iota^* u$, and exact if it is of the form $(dv, \psi_*\iota^* v - d\varsigma)$. Denote the cohomology by $H^k(Z,\psi)$.

Proposition 14.10.2. *(i) There is a long exact sequence*

$$\cdots \longrightarrow H^{k-f-1}(Y) \longrightarrow H^k(Z,\psi) \longrightarrow H^k(Z) \xrightarrow{\psi_*\iota^*} H^{k-f}(Y) \longrightarrow H^{k+1}(Z,\psi) \longrightarrow \cdots .$$

(ii) If $\dim Z = n$, *the pairing* $\langle (u,\tau), [Z,\psi]\rangle = \int_Z u - \int_Y \tau$ *on* $C^n(Z,\psi)$ *vanishes on exact pairs, so defines a linear form on* $H^n(Z,\psi)$.

Proof. (i) The sequence of complexes $0 \longrightarrow \mathcal{C}^\infty(Y;\Lambda^{\bullet-f-1}) \longrightarrow C^\bullet(Z,\psi) \longrightarrow \mathcal{C}^\infty(Z;\Lambda^\bullet) \longrightarrow 0$, $\tau \mapsto (0,\tau)$ and $(u,\tau) \mapsto u$, is exact and compatible with the differentials (up to the sign of d on the first complex, which does not affect cohomology). The connecting map sends $[u]$ to the class of $\psi_*\iota^* u$, since $\mathcal{D}(u,0) = (0, \psi_*\iota^* u)$. (ii) $\int_Z dv - \int_Y (\psi_*\iota^* v - d\varsigma) = \int_{\partial Z} v - \int_{\partial Z} v = 0$ by Stokes' theorem. □

When ψ is the identity of ∂Z ($f = 0$), $C^\bullet(Z,\psi)$ is the mapping cone of ι^*, and $H^k(Z,\psi)$ is the relative cohomology $H^k(Z,\partial Z)$, i.e. the compactly supported cohomology of the interior (Exercise 14.7).

In the setting of Theorem 14.8.1, with $Z = S^*_+$, $Y = S^*M$ and ψ the projection $S^*M \times \mathbb{S} \longrightarrow S^*M$ ($f = 1$), Theorem 14.6.2 says precisely that

$$\mathrm{Ch}_{\mathrm{pt}}(A) = \big(\mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(a),\ -\mathrm{Td}(M) \wedge (\eta(\beta) + \gamma(b))\big)$$

is closed, and Theorem 14.8.1 becomes

$$\mathrm{ind}(A) = \big\langle \mathrm{Ch}_{\mathrm{pt}}(A), [S^*_+, \psi]\big\rangle.$$

The proof of Proposition 14.8.2 amounts to showing that the class of $\mathrm{Ch}_{\mathrm{pt}}(A)$ is homotopy invariant. In general one expects a Chern character $K^0_{c,\psi\text{-pt}}(T^*(M/B)) \longrightarrow H^{\mathrm{even}}_{c,\psi\text{-pt}}(T^*(M/B))$, given by the pair (relative Chern form of σ, eta form of β) and defined using connections, mapping the K-theory six-term sequence of Remark 14.10.1 to the corresponding cohomological one, together with a push-forward $H^{\mathrm{even}}_{c,\psi\text{-pt}}(T^*(M/B)) \longrightarrow H^{\mathrm{even}}(B)$ when the fibres are oriented, so that $\mathrm{Ch}(\mathrm{ind}_a(A))$ is the push-forward of $\mathrm{Td} \cdot \mathrm{Ch}_{\mathrm{pt}}(A)$.

14.10.4 The determinant line bundle

The numerical index is the degree-zero part of the Chern character of the index bundle. In the odd case the degree-one part may be thought of as the 'spectral flow' of the phase of a determinant. In the even case the degree-two part corresponds to the determinant line bundle, which we now construct; the degree-three part in the odd case corresponds to the curvature of a gerbe, and the degree-four part in the even case to a 2-gerbe.

Let $\phi : M \longrightarrow B$ be a fibration with compact fibres Z_b and let $P \in \Psi^0(M/B; E_+, E_-)$ be an elliptic family of numerical index 0. Then each P_b can be perturbed by a smoothing operator to be invertible, so the sets

$$\mathcal{P}_b = \{P_b + Q_b;\ Q_b \in \Psi^{-\infty}(Z_b; E_+, E_-),\ (P_b + Q_b)^{-1} \in \Psi^0(Z_b; E_-, E_+)\}$$

are non-empty. Each $\mathcal{P}_b$ is a principal homogeneous space for the group $G^{-\infty}(Z_b; E_-)$ acting on the left: if $A, A' \in \mathcal{P}_b$ then $A' = (\mathrm{Id} + R)A$ with $\mathrm{Id} + R = A'A^{-1}$, $R = (A' - A)A^{-1}$ smoothing. Thus $\mathcal{P} = \bigcup_b \mathcal{P}_b \longrightarrow B$ behaves like a principal bundle, although the structure groups $G^{-\infty}(Z_b; E_-)$ form a bundle of groups rather than a fixed group. (Using exhausting families of smoothing projections, as in the stabilization of the index bundle in Chapter 10, one sees that the components of the space of sections of this bundle of groups are canonically $K^{-1}(B)$, so it is 'weakly trivial'.) The Fredholm determinant is a character of each $G^{-\infty}(Z_b; E_-)$, so we may form the associated line bundle.

Proposition 14.10.3. *Let* $\mathrm{Det}_b = (\mathcal{P}_b \times \mathbb{C})/\sim$, *where* $(A, z) \sim ((\mathrm{Id} + R)A, \det(\mathrm{Id} + R)^{-1} z)$ *for* $\mathrm{Id} + R \in G^{-\infty}(Z_b; E_-)$. *Then* $\mathrm{Det} = \bigcup_b \mathrm{Det}_b$ *is a smooth line bundle over* B, *and*

$$s(b) = [A, \det(A^{-1} P_b)], \qquad A \in \mathcal{P}_b,$$

is a well-defined smooth section of Det *which vanishes exactly at the points* b *where* P_b *is not invertible.*

Proof. The relation is an equivalence relation because det is multiplicative. For fixed $A_0 \in \mathcal{P}_b$, every element of $\mathcal{P}_b$ is uniquely $(\mathrm{Id} + R)A_0$, so $z \mapsto [A_0, z]$ is a bijection $\mathbb{C} \longrightarrow \mathrm{Det}_b$, and the linear structures induced by different choices of A_0 agree. Local triviality: if $A_0 = P_{b_0} + Q_0 \in \mathcal{P}_{b_0}$, choose a local trivialization of the fibration near b_0 and transport Q_0 to a smooth family Q_b; since invertibility is open, $P_b + Q_b \in \mathcal{P}_b$ for b near b_0, giving a local smooth section of $\mathcal{P}$, hence a local trivialization, and the transition functions are det of smooth families in $G^{-\infty}$, hence smooth. For s: $A^{-1}P_b = \mathrm{Id} - A^{-1}Q_b \in \mathrm{Id} + \Psi^{-\infty}$ has a Fredholm determinant, which is smooth in b for a smooth local section A. Replacing A by $(\mathrm{Id} + R)A$ gives $\det(A^{-1}(\mathrm{Id} + R)^{-1}P_b) = \det(\mathrm{Id} + R)^{-1} \det(A^{-1}P_b)$, using the multiplicativity of det and its invariance under conjugation by A; so s is well defined. Finally $\det(A^{-1}P_b) \neq 0$ if and only if $A^{-1}P_b$, equivalently P_b, is invertible. □

When P_b is invertible, $s(b) = [P_b, 1]$. The section s is the determinant of the family. The curvature of natural connections on Det (Quillen, Bismut–Freed) represents the degree-two part of $-\mathrm{Ch}(\mathrm{ind}\, P)$ (compare Remark 14.6.4); with the opposite convention for the equivalence relation one obtains the dual line bundle.

14.11 Exercises

Exercise 14.1. Let $U = \Pi e^{i\theta}\Pi$ and $L = \Pi e^{-i\theta}\Pi$ be the shifts on $\mathcal{H}$. Show that $[L, U]$ is the projection onto the constants, and verify the trace-defect formula $\overline{\mathrm{Tr}}([L, U]) = \mathrm{Tr}_{\mathrm{R}}((D_E L)U) = 1$ directly from Lemma 14.4.1.

Exercise 14.2. Show that D_E is not an inner derivation of $\Psi^{\mathbb{Z}}(\mathbb{S})$. (Hint: if $D_E = [C, \cdot]$ with $C \in \Psi^k(\mathbb{S})$, compare the principal symbols of $D_E f = -i\xi^{-1} f' + \dots$ and $[C, f]$, $f \in \mathcal{C}^\infty(\mathbb{S})$; show that $k \le 0$ is forced and that $k = 0$ would require $\partial_\xi \sigma_0(C) = \xi^{-1}$ with $\sigma_0(C)$ homogeneous of degree 0.)

Exercise 14.3. Let E' be another admissible family on $\mathbb{S}$ with $D = D(E, E') \in \Psi^{-1}(\mathbb{S})$ of principal symbol $\delta(\theta, \xi)$. Show that the eta forms defined using E' are

$$\eta'(B) = \eta(B) - \frac{1}{2\pi} \sum_k c_k \int_0^{2\pi} \operatorname{tr}(\nu^{2k+1})\, \delta(\theta, 1)\, d\theta,$$

so that they differ from η by a term local in the symbol of B. Deduce that Theorem 14.8.1 holds with η' in place of η provided γ is replaced by γ minus this local term.

Exercise 14.4. Let $g, h \in \mathcal{C}^\infty(W; \mathrm{GL}(N, \mathbb{C}))$. Using the homotopy $R_t(g \oplus \mathrm{Id}) R_t^{-1} (\mathrm{Id} \oplus h)$ and Lemma 14.5.1, show that $\mathrm{Ch}_{\mathrm{odd}}(gh) - \mathrm{Ch}_{\mathrm{odd}}(g) - \mathrm{Ch}_{\mathrm{odd}}(h)$ is exact, and that its degree-one part vanishes.

Exercise 14.5. Show that for elliptic $P \in \Psi^0(\mathbb{S}; \mathbb{C}^N)$, $\operatorname{ind}(P) = \int_{S^*\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(\sigma_0(P))$ with the Liouville orientation, by writing P, up to compact operators, as $\Pi P_+ \Pi + (\mathrm{Id} - \Pi) P_- (\mathrm{Id} - \Pi)$ and applying (14.5.1) to both parts (for the negative part use J from the proof of Proposition 14.9.1).

Exercise 14.6. Verify directly in local coordinates that for $n = 1$ the Liouville orientation of $S^*M \cong M \times \{\pm 1\}$ is $\pm dx$, and that the orientation of ∂S^*_+ is $\pm dx \wedge d\theta$ on the two components.

Exercise 14.7. Check that when $\psi : \partial Z \longrightarrow \partial Z$ is the identity, $H^k(Z, \psi)$ is naturally isomorphic to $H^k(Z, \partial Z) = H^k_c(Z \setminus \partial Z)$. (Hint: given a closed pair (u, τ), use a collar and a cut-off χ equal to 1 near ∂Z to replace it by the cohomologous pair $(u - d(\chi\tau), 0)$, with $u - d(\chi\tau)$ vanishing near ∂Z.)

Exercise 14.8. Give the details of Proposition 14.10.2(i), and show that the connecting map $H^k(Z) \longrightarrow H^{k-f}(Y)$ is induced by $\psi_* \iota^*$.

Exercise 14.9. In the setting of Proposition 14.10.3, suppose $P_b = P_0 + K_b$ where P_0 is a fixed invertible family (in a local trivialization of the fibration) and $K_b \in \Psi^{-\infty}$. Show that in the trivialization $z \mapsto [P_0, z]$ of Det the determinant section is $s(b) = \det(\mathrm{Id} + P_0^{-1} K_b)$, and that Det is trivial if every P_b is invertible.

Notes

Regularized traces of this kind were studied by Kontsevich and Vishik, and the trace-defect formula appears in the treatment of the b-trace in [16]; the residue trace is due to Wodzicki and Guillemin [13, 5], and the meromorphic continuation to Seeley [11]. In the APS index theorem the eta invariant arises in the same way, as a regularized trace of the indicial family [16]; the eta forms here are the corresponding objects for the odd Chern character, analogous to the eta forms of Bismut and Cheeger [19], and appear in the families index theorem of Melrose and Rochon [20].

The main departures from the original lecture notes are as follows. We have fixed all normalizations: with $E(s)$ of order $+s$ the residue trace is minus the Wodzicki residue, the trace-defect formula reads $\overline{\mathrm{Tr}}_E([A, B]) = \mathrm{Tr}_{\mathrm{R}}(B D_E A)$ with $D_E A = \frac{d}{ds}[E(s), A]|_{s=0}$, and

the constants of the odd Chern character are determined by the Toeplitz index (Proposition 14.5.2). The variation formula for the eta forms, Theorem 14.6.2, has been corrected; it contains a symbolic term γ which must also appear in the index formula (Remark 14.8.7), and with these normalizations the eta form enters with coefficient $+1$ rather than $-\frac{1}{2}$. The original proof of the index formula reduced to the Atiyah–Singer theorem on M by composing with the inverse of a bundle isomorphism; in the Toeplitz algebra such an inverse need not exist (Remark 14.8.6), and we have instead given a proof based on the weak contractibility of the Toeplitz group and the Atiyah–Singer theorem on $M \times \mathbb{S}$. The non-Toeplitz case is deduced by a unitary doubling. In the product-type cohomology the compatibility condition has been changed to $d\tau = \psi_* \iota^* u$ (push-forward along the fibres), which is what the variation formula requires; the suggested identification with the cohomology of the space obtained by collapsing the fibres of ψ refers to the pull-back condition and has been dropped. The original also asserts that D_E is the unique continuous outer derivation up to inner derivations and constant multiples; we have not proved this. For the Todd class in the comparison of Thom isomorphisms see Atiyah and Singer, *The index of elliptic operators III*, and for determinant line bundles Quillen and Bismut–Freed.

Part V

Algebraic structure

Chapter 15

Hochschild homology

The algebra $\Psi^{\mathbb{Z}}(X)$ of classical pseudodifferential operators of integral order on a compact manifold X, studied in Chapter 5, is a non-commutative deformation of a commutative algebra: modulo $\Psi^{-\infty}(X)$ it is, via the full symbol, the algebra of formal sums of homogeneous functions on $T^*X \setminus 0$, with a product whose leading term is pointwise multiplication and whose first correction is the Poisson bracket. Hochschild homology is the basic homological invariant of an associative algebra; in degree zero it is the dual of the space of traces. The aim of this chapter is to compute the Hochschild homology of the algebra $\Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)$ of formal symbols, following Brylinski–Getzler [1] and Wodzicki [13, 14]:

$$\mathrm{HH}_k\big(\Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)\big) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X), \qquad n = \dim X.$$

The computation proceeds in three stages. First (Sections 15.1 and 15.2) we set up the Hochschild complex and compute it for polynomial algebras by means of the Koszul resolution. Secondly (Sections 15.3 and 15.4) we prove the theorem of Hochschild–Kostant–Rosenberg and Connes that the Hochschild homology of $\mathcal{C}^\infty(X)$ is the space of differential forms, and deduce the analogous statement for the commutative algebra of homogeneous symbols. Finally (Sections 15.5, 15.6, 15.8 and 15.9) we filter the Hochschild complex of the symbol algebra by order; the resulting spectral sequence has E_1 term the homogeneous differential forms on $T^*X \setminus 0$, its first differential is the Poisson (Koszul–Brylinski) differential, a symplectic Hodge star identifies E_2 with de Rham cohomology, and a homogeneity argument shows that the spectral sequence degenerates and converges.

Throughout, vector spaces are complex, $\otimes$ without decoration is the algebraic tensor product over $\mathbb{C}$, and X is a smooth manifold of dimension n.

15.1 The Hochschild complex

15.1.1 Chains and differentials

Let $\mathcal{A}$ be an associative algebra over $\mathbb{C}$, not necessarily with identity. The space of (algebraic) Hochschild k-chains, $k \in \mathbb{N}_0$, is

$$C_k(\mathcal{A}) = \mathcal{A}^{\otimes(k+1)} = \mathcal{A} \otimes \mathcal{A} \otimes \cdots \otimes \mathcal{A}.$$

For $k \geq 1$ define the *face maps* $d_i : C_k(\mathcal{A}) \longrightarrow C_{k-1}(\mathcal{A})$, $0 \leq i \leq k$, by

$$\begin{aligned} d_i(a_0 \otimes \cdots \otimes a_k) &= a_0 \otimes \cdots \otimes a_{i-1} \otimes a_i a_{i+1} \otimes a_{i+2} \otimes \cdots \otimes a_k, \quad 0 \leq i \leq k-1, \\ d_k(a_0 \otimes \cdots \otimes a_k) &= a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}, \end{aligned} \tag{15.1.1}$$

and set

$$b' = \sum_{i=0}^{k-1}(-1)^i d_i, \qquad b = b' + (-1)^k d_k = \sum_{i=0}^{k}(-1)^i d_i \quad \text{on } C_k(\mathcal{A}), \tag{15.1.2}$$

with $b = b' = 0$ on $C_0(\mathcal{A}) = \mathcal{A}$.

Lemma 15.1.1. *For $k \geq 2$ and $0 \leq i < j \leq k$ the face maps satisfy $d_i d_j = d_{j-1} d_i$ as maps $C_k(\mathcal{A}) \longrightarrow C_{k-2}(\mathcal{A})$. Consequently $(b')^2 = 0$ and $b^2 = 0$.*

Proof. If $j \leq k-1$ both sides only multiply neighbouring factors and the identity is the associativity of the product (for $j = i+1$ both sides replace $a_i \otimes a_{i+1} \otimes a_{i+2}$ by $a_i a_{i+1} a_{i+2}$; for $j > i+1$ the two multiplications commute). If $j = k$ and $i = 0$, both sides give $a_k a_0 a_1 \otimes a_2 \otimes \cdots \otimes a_{k-1}$; if $j = k$ and $0 < i < k-1$, both give $a_k a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{k-1}$; and if $j = k$, $i = k-1$, both give $a_{k-1} a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-2}$.

Now $b^2 = \sum_{i=0}^{k-1} \sum_{j=0}^{k} (-1)^{i+j} d_i d_j$ on $C_k(\mathcal{A})$. The map $(i,j) \longmapsto (j-1, i)$ is a bijection from the pairs with $i < j \leq k$ onto the pairs (i', j') with $j' \leq i' \leq k-1$, and by the identity the corresponding terms are equal but carry the opposite signs $(-1)^{i+j}$ and $(-1)^{i+j-1}$; hence $b^2 = 0$. The same argument with k replaced by $k-1$ in the range of j gives $(b')^2 = 0$. □

Definition 15.1.2. The *Hochschild homology* of $\mathcal{A}$ is the homology of the complex $(C_\bullet(\mathcal{A}), b)$:

$$\mathrm{HH}_k(\mathcal{A}) = \frac{\mathrm{Nul}\big(b : C_k(\mathcal{A}) \to C_{k-1}(\mathcal{A})\big)}{\mathrm{Ran}\big(b : C_{k+1}(\mathcal{A}) \to C_k(\mathcal{A})\big)}, \qquad k \in \mathbb{N}_0.$$

In degree zero $b(a_0 \otimes a_1) = a_0 a_1 - a_1 a_0$, so $\mathrm{HH}_0(\mathcal{A}) = \mathcal{A}/[\mathcal{A}, \mathcal{A}]$, where $[\mathcal{A}, \mathcal{A}]$ is the span of the commutators. Thus the linear functionals on $\mathrm{HH}_0(\mathcal{A})$ are precisely the *traces* on $\mathcal{A}$, the linear maps τ with $\tau(ab) = \tau(ba)$. If $\mathcal{A}$ is commutative then $b = 0$ on $C_1(\mathcal{A})$ and $\mathrm{HH}_0(\mathcal{A}) = \mathcal{A}$.

15.1.2 The bar complex

The complex $(C_\bullet(\mathcal{A}), b')$ is the *bar complex*. It is, in contrast to the Hochschild complex, trivial for algebras with identity.

Lemma 15.1.3. *If $\mathcal{A}$ has an identity 1, then $s(a_0 \otimes \cdots \otimes a_k) = 1 \otimes a_0 \otimes \cdots \otimes a_k$ defines maps $s : C_k(\mathcal{A}) \to C_{k+1}(\mathcal{A})$ with*

$$b's + sb' = \mathrm{Id} \quad \text{on } C_k(\mathcal{A}),\ k \geq 0;$$

in particular $(C_\bullet(\mathcal{A}), b')$ is acyclic.

Proof. On $C_{k+1}(\mathcal{A})$, $d_0(1 \otimes a) = a$ for $a \in C_k(\mathcal{A})$, while for $1 \leq i \leq k$, $d_i(1 \otimes a) = 1 \otimes d_{i-1} a$. Hence $b'sa = a + \sum_{i=1}^{k} (-1)^i\, 1 \otimes d_{i-1} a = a - sb'a$. □

Definition 15.1.4. An associative algebra is *H-unital* if its bar complex $(C_\bullet(\mathcal{A}), b')$ is acyclic.

So every unital algebra is H-unital. For H-unital algebras Hochschild homology has good excision properties (Wodzicki); we shall not need these.

15.1.3 Hochschild homology as a derived functor

Suppose $\mathcal{A}$ has an identity and let $\mathcal{A}^e = \mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}$, so that $\mathcal{A}$-bimodules are the same as left $\mathcal{A}^e$-modules; $\mathcal{A}$ itself is an $\mathcal{A}^e$-module by $(a \otimes a') \cdot c = aca'$. Set

$$P_k = \mathcal{A} \otimes \mathcal{A}^{\otimes k} \otimes \mathcal{A}\ (= \mathcal{A}^{\otimes (k+2)}), \qquad (a \otimes a') \cdot (a_0 \otimes \cdots \otimes a_{k+1}) = aa_0 \otimes a_1 \otimes \cdots \otimes a_{k+1}a'.$$

Each $P_k \cong \mathcal{A}^e \otimes \mathcal{A}^{\otimes k}$ is a free $\mathcal{A}^e$-module, the bar differential b' of (15.1.2) (on $\mathcal{A}^{\otimes (k+2)}$, i.e. $b' = \sum_{i=0}^{k} (-1)^i d_i$) is $\mathcal{A}^e$-linear, and by Lemma 15.1.3 the complex

$$\ldots \xrightarrow{b'} P_2 \xrightarrow{b'} P_1 \xrightarrow{b'} P_0 \xrightarrow{m} \mathcal{A} \longrightarrow 0, \qquad m(a_0 \otimes a_1) = a_0 a_1, \tag{15.1.3}$$

is exact: it is a free resolution of $\mathcal{A}$ as an $\mathcal{A}^e$-module. Applying $\mathcal{A} \otimes_{\mathcal{A}^e} (\cdot)$ and using the isomorphism

$$\mathcal{A} \otimes_{\mathcal{A}^e} P_k \longrightarrow C_k(\mathcal{A}), \quad c \otimes (a_0 \otimes a_1 \otimes \cdots \otimes a_k \otimes a_{k+1}) \longmapsto a_{k+1} c a_0 \otimes a_1 \otimes \cdots \otimes a_k,$$

the differential $\mathrm{Id} \otimes b'$ becomes b: the term d_0 of b' gives d_0, and the last term $(-1)^k(\cdots \otimes a_k a_{k+1})$ of b' on P_k (with $a_{k+1} = 1$) gives $(-1)^k a_k c a_0 \otimes \cdots = (-1)^k d_k$. Thus

$$\mathrm{HH}_k(\mathcal{A}) = \mathrm{Tor}_k^{\mathcal{A}^e}(\mathcal{A}, \mathcal{A}).$$

The practical content of this is the following standard fact (the *comparison theorem* of homological algebra): if $Q_\bullet \to \mathcal{A} \to 0$ is another resolution of $\mathcal{A}$ by free (or projective) $\mathcal{A}^e$-modules, then there are $\mathcal{A}^e$-linear chain maps $\Phi : Q_\bullet \to P_\bullet$ and $\Psi : P_\bullet \to Q_\bullet$ inducing the identity on $\mathcal{A}$, and $\Psi\Phi$, $\Phi\Psi$ are $\mathcal{A}^e$-linearly chain homotopic to the identity. The maps are constructed degree by degree: on a basis element e of Q_k one chooses $\Phi(e)$ in P_k with $b'\Phi(e) = \Phi(\partial e)$, which is possible because $\Phi(\partial e)$ is a cycle (for $k = 0$, lies over the right element of $\mathcal{A}$) and $P_\bullet$ is exact; the homotopies are constructed in the same way. Consequently $\mathrm{HH}_\bullet(\mathcal{A})$ is also the homology of $\mathcal{A} \otimes_{\mathcal{A}^e} Q_\bullet$.

15.1.4 Topological algebras

For the 'large' algebras of analysis the algebraic tensor product is inappropriate and is replaced by a completion. If $\mathcal{A}$ is a Fréchet algebra (multiplication jointly continuous) one uses the completed projective tensor product $\mathcal{A}^{\hat{\otimes}(k+1)}$; the face maps extend by continuity and Lemmas 15.1.1 and 15.1.3 continue to hold, by density of the algebraic tensor product. The resulting homology is the *continuous* Hochschild homology.

For $\mathcal{A} = \mathcal{C}^\infty(X)$, X compact, a theorem of Grothendieck identifies $\mathcal{C}^\infty(X)\hat{\otimes}\mathcal{C}^\infty(Y)$ with $\mathcal{C}^\infty(X \times Y)$ (this rests on the nuclearity of $\mathcal{C}^\infty(X)$; the algebraic tensor product is in any case dense in $\mathcal{C}^\infty(X \times Y)$). We shall simply *define* the chain spaces for $\mathcal{C}^\infty(X)$ to be $\mathcal{C}^\infty(X^{k+1})$, see Section 15.3; for the symbol algebras the completion used is specified in Definitions 15.4.1 and 15.5.1.

15.2 Polynomial algebras

Let $\mathcal{A} = \mathbb{C}[x] = \mathbb{C}[x_1, \ldots, x_n]$, regarded as the algebra of polynomial functions on $\mathbb{R}^n$. Then $C_k(\mathcal{A}) = \mathbb{C}[z_0, z_1, \ldots, z_k]$, $z_j \in \mathbb{R}^n$, is the space of polynomials on $(\mathbb{R}^n)^{k+1}$, the element $a_0 \otimes \cdots \otimes a_k$ corresponding to $a_0(z_0)a_1(z_1)\cdots a_k(z_k)$. In these terms the face maps (15.1.1) are restrictions to diagonals:

$$\begin{aligned} (d_i u)(z_0, \ldots, z_{k-1}) &= u(z_0, \ldots, z_{i-1}, z_i, z_i, z_{i+1}, \ldots, z_{k-1}), \quad 0 \le i \le k-1, \\ (d_k u)(z_0, \ldots, z_{k-1}) &= u(z_0, z_1, \ldots, z_{k-1}, z_0). \end{aligned} \tag{15.2.1}$$

Let $\Omega^k_{\mathrm{pol}} = \mathbb{C}[x] \otimes \Lambda^k(\mathbb{C}^n)$ be the space of k-forms on $\mathbb{R}^n$ with polynomial coefficients, dx^j corresponding to the j-th basis vector e_j of $\mathbb{C}^n$.

15.2.1 The Hochschild–Kostant–Rosenberg map

For a function u of $(z_0, \dots, z_k)$ set

$$\varepsilon(u)(x) = \frac{1}{k!} \sum_{j_1,\dots,j_k=1}^{n} \frac{\partial^k u}{\partial z_1^{j_1} \cdots \partial z_k^{j_k}}(x, x, \dots, x)\, dx^{j_1} \wedge \cdots \wedge dx^{j_k}, \tag{15.2.2}$$

where z_l^j is the j-th coordinate of z_l. Thus $\varepsilon(a_0 \otimes a_1 \otimes \cdots \otimes a_k) = \frac{1}{k!} a_0\, da_1 \wedge \cdots \wedge da_k$. For $a_0, \dots, a_k \in \mathcal{A}$ we also set

$$\varphi(a_0; a_1, \dots, a_k) = \sum_{\sigma \in \mathfrak{S}_k} \operatorname{sgn}(\sigma)\, a_0 \otimes a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(k)} \in C_k(\mathcal{A}). \tag{15.2.3}$$

Lemma 15.2.1. *For any commutative algebra $\mathcal{A}$, define $\varepsilon : C_k(\mathcal{A}) \to \Omega^k_{\mathcal{A}}$ by $\varepsilon(a_0 \otimes \cdots \otimes a_k) = \frac{1}{k!} a_0\, da_1 \wedge \cdots \wedge da_k$, where $\Omega^\bullet_{\mathcal{A}}$ is any graded-commutative algebra containing $\mathcal{A}$ in degree zero, with a derivation d (for instance the de Rham complex of a manifold, with $\mathcal{A}$ a subalgebra of the smooth functions). Then $\varepsilon \circ b = 0$, $b\,\varphi(a_0; a_1, \dots, a_k) = 0$ and $\varepsilon(\varphi(a_0; a_1, \dots, a_k)) = a_0\, da_1 \wedge \cdots \wedge da_k$.*

Proof. Write $T_i = a_0 a_i\, da_1 \wedge \cdots \wedge \widehat{da_i} \wedge \cdots \wedge da_k$. By the Leibniz rule $d(a_i a_{i+1}) = a_{i+1} da_i + a_i da_{i+1}$, so $(k-1)!\,\varepsilon(b(a_0 \otimes \cdots \otimes a_k))$ equals

$$T_1 + \sum_{i=1}^{k-1} (-1)^i (T_{i+1} + T_i) + (-1)^k T_k = 0.$$

In $b\varphi$, the terms d_i with $1 \le i \le k-1$ cancel in pairs, since σ and $\sigma \circ (i\ i{+}1)$ have opposite signs and $\mathcal{A}$ is commutative. The term d_0 for σ is $\operatorname{sgn}(\sigma) a_0 a_{\sigma(1)} \otimes a_{\sigma(2)} \otimes \dots$, and the term $(-1)^k d_k$ for σ', where $\sigma'(j) = \sigma(j+1)$ for $j < k$ and $\sigma'(k) = \sigma(1)$, is $(-1)^k \operatorname{sgn}(\sigma') a_{\sigma(1)} a_0 \otimes a_{\sigma(2)} \otimes \dots$; since $\operatorname{sgn}(\sigma') = (-1)^{k-1} \operatorname{sgn}(\sigma)$ these cancel. Finally $\varepsilon\varphi = \frac{1}{k!} \sum_\sigma \operatorname{sgn}(\sigma) a_0\, da_{\sigma(1)} \wedge \cdots \wedge da_{\sigma(k)} = a_0\, da_1 \wedge \cdots \wedge da_k$. □

15.2.2 The Koszul resolution

We identify $\mathcal{A}^e = \mathbb{C}[x] \otimes \mathbb{C}[x]$ with $\mathbb{C}[x, y]$, polynomials in $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n$, via $a \otimes a' \mapsto a(x)a'(y)$; then $\mathcal{A}$ is the $\mathcal{A}^e$-module $\mathbb{C}[x, y]/(x_1 - y_1, \dots, x_n - y_n)$. The *Koszul complex* is

$$K_k = \mathbb{C}[x, y] \otimes \Lambda^k(\mathbb{C}^n), \quad \partial(f\, e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{l=1}^{k} (-1)^{l-1} (x_{i_l} - y_{i_l}) f\, e_{i_1} \wedge \cdots \wedge \widehat{e_{i_l}} \wedge \cdots \wedge e_{i_k},$$

with augmentation $K_0 = \mathbb{C}[x, y] \to \mathcal{A}$, $f \mapsto f(x, x)$.

Lemma 15.2.2. *The augmented Koszul complex $\cdots \to K_1 \to K_0 \to \mathcal{A} \to 0$ is exact, so $K_\bullet$ is a free resolution of $\mathcal{A}$ as an $\mathcal{A}^e$-module.*

Proof. In the variables $u = x - y$, $v = y$ identify K_k with the k-forms in u with coefficients in $\mathbb{C}[u, v]$, $e_i \leftrightarrow du_i$. Then $\partial = \iota_E$ is contraction with the Euler field $E = \sum_i u_i \partial_{u_i}$ and the augmentation is $f \mapsto f|_{u=0}$. Let d_u be the exterior derivative in u (with v as a parameter). By Cartan's formula $d_u \iota_E + \iota_E d_u = \mathcal{L}_E$, and $\mathcal{L}_E$ multiplies $u^\alpha du_I$ (with coefficients in $\mathbb{C}[v]$) by $|\alpha| + |I|$. Hence if $\omega \in K_k$, $k \ge 1$, is homogeneous of total degree $N = |\alpha| + |I| \ge 1$ in u and $\partial\omega = 0$, then $\omega = \partial(d_u\omega/N)$; splitting a general cycle into homogeneous parts, $K_\bullet$ is exact in degrees $k \ge 1$. In degree 0 the kernel of the augmentation consists of the f with no term of degree 0 in u, and such an f equals $\partial(\sum_N d_u f_N / N)$, f_N its homogeneous parts. Finally the augmentation is surjective. □

Theorem 15.2.3 (Hochschild–Kostant–Rosenberg)**.** *Let* $\mathcal{A} = \mathbb{C}[x_1, \dots, x_n]$. *The map* (15.2.2) *induces an isomorphism*

$$\varepsilon_* : \mathrm{HH}_k(\mathbb{C}[x]) \longrightarrow \Omega^k_{\mathrm{pol}} = \mathbb{C}[x] \otimes \Lambda^k(\mathbb{C}^n), \qquad k \in \mathbb{N}_0,$$

with inverse induced by $a_0\, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \longmapsto \varphi(a_0; x_{i_1}, \dots, x_{i_k})$. *In particular* $\mathrm{HH}_k(\mathbb{C}[x]) = 0$ *for* $k > n$.

Proof. Let $P_\bullet$ be the bar resolution (15.1.3) and define the $\mathcal{A}^e$-linear maps $\Phi : K_k \to P_k$ by

$$\Phi(e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{\sigma \in \mathfrak{S}_k} \mathrm{sgn}(\sigma)\, 1 \otimes x_{i_{\sigma(1)}} \otimes \cdots \otimes x_{i_{\sigma(k)}} \otimes 1.$$

This is a chain map. Indeed, applying b', the inner terms (multiplying two adjacent x's) cancel in pairs as in Lemma 15.2.1. The first term is $x_{i_{\sigma(1)}} \cdot (1 \otimes x_{i_{\sigma(2)}} \otimes \cdots \otimes 1)$ (left action, i.e. multiplication by the variable x) and, grouping the σ with $\sigma(1) = l$, contributes $\sum_l (-1)^{l-1} x_{i_l} \Phi(e_{i_1} \wedge \dots \widehat{e_{i_l}} \cdots \wedge e_{i_k})$; the last term is $(-1)^k (1 \otimes \cdots \otimes 1) \cdot x_{i_{\sigma(k)}}$ (right action, multiplication by y) and, grouping the σ with $\sigma(k) = l$ (sign $(-1)^{k-l}$), contributes $-\sum_l (-1)^{l-1} y_{i_l} \Phi(\dots \widehat{e_{i_l}} \dots)$. Together this is $\Phi(\partial e_{i_1} \wedge \cdots \wedge e_{i_k})$. In degree 0, Φ is the identity of $\mathcal{A}^e$, so Φ lies over the identity of $\mathcal{A}$.

By Lemma 15.2.2 both complexes are free resolutions of $\mathcal{A}$, so by the comparison theorem (Section 15.1.3) $\mathrm{Id} \otimes \Phi$ induces an isomorphism from the homology of $\mathcal{A} \otimes_{\mathcal{A}^e} K_\bullet$ to that of $\mathcal{A} \otimes_{\mathcal{A}^e} P_\bullet$, i.e. to $\mathrm{HH}_\bullet(\mathcal{A})$. Now $\mathcal{A} \otimes_{\mathcal{A}^e} K_k = \mathcal{A} \otimes \Lambda^k(\mathbb{C}^n) = \Omega^k_{\mathrm{pol}}$ and the induced differential vanishes, since $x_i - y_i$ acts on $\mathcal{A}$ as 0. Under the identification $\mathcal{A} \otimes_{\mathcal{A}^e} P_k = C_k(\mathcal{A})$, $\mathrm{Id} \otimes \Phi$ becomes $a_0\, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \mapsto \varphi(a_0; x_{i_1}, \dots, x_{i_k})$, which therefore induces an isomorphism $\varphi_* : \Omega^k_{\mathrm{pol}} \to \mathrm{HH}_k(\mathcal{A})$. By Lemma 15.2.1, ε descends to homology and $\varepsilon_* \varphi_* = \mathrm{Id}$, so $\varepsilon_* = \varphi_*^{-1}$. □

Remark 15.2.4. The same argument applies to the algebra $\mathbb{C}[r, r^{-1}]$ of Laurent polynomials in one variable. Here $\mathcal{A}^e = \mathbb{C}[r^{\pm 1}, s^{\pm 1}]$ and the Koszul resolution is $0 \to \mathcal{A}^e \xrightarrow{r-s} \mathcal{A}^e \to \mathcal{A} \to 0$; it is exact because $r - s$ is not a zero divisor and $f(r,s) - f(r,r)$ is divisible by $r - s$ in $\mathcal{A}^e$ (write $f = (rs)^{-N} g$ with g a polynomial). Hence $\mathrm{HH}_0(\mathbb{C}[r^{\pm 1}]) = \mathbb{C}[r^{\pm 1}]$, $\mathrm{HH}_1(\mathbb{C}[r^{\pm 1}]) = \mathbb{C}[r^{\pm 1}]\, dr$ and $\mathrm{HH}_k(\mathbb{C}[r^{\pm 1}]) = 0$ for $k \geq 2$, the isomorphisms again being induced by ε.

The original direct argument for Theorem 15.2.3 proceeds instead by successively restricting to chains vanishing on diagonals. This reduction is very useful for smooth functions, where no finite free resolution is at hand, and we now describe it.

15.3 Smooth functions

15.3.1 The complex

Let X be a compact manifold. We take as the space of Hochschild k-chains of $\mathcal{C}^\infty(X)$ the space $C_k = \mathcal{C}^\infty(X^{k+1})$, a completion of $\mathcal{C}^\infty(X)^{\otimes(k+1)}$ (see Section 15.1.4), with the face maps (15.2.1); these are the continuous extensions of (15.1.1), and $b = \sum_i (-1)^i d_i$ as before. The map $\varepsilon : C_k \to \Omega^k(X) = \mathcal{C}^\infty(X; \Lambda^k T^* X)$ is defined by (15.2.2) in local coordinates. This is invariant: the mixed derivative $\partial^k u / \partial z_1^{j_1} \cdots \partial z_k^{j_k}$, in which each variable is differentiated once, restricted to the diagonal, transforms as a section of $(T^*X)^{\otimes k}$ under change of coordinates, and $\varepsilon(u)$ is $\frac{1}{k!}$ times its image under $v_1 \otimes \cdots \otimes v_k \mapsto v_1 \wedge \cdots \wedge v_k$.

Lemma 15.3.1. $\varepsilon \circ b = 0$ *on* C_{k+1}, *and* $\varepsilon_* : \mathrm{HH}_k(\mathcal{C}^\infty(X)) \to \Omega^k(X)$ *is surjective.*

Proof. The value of $\varepsilon(bu)$ at $x \in X$ only involves derivatives of u of order at most $k+1$ at $(x, \dots, x)$. Replacing u, in local coordinates, by its Taylor polynomial of that order, which is a sum of products of functions of the individual variables, the first statement follows from Lemma 15.2.1. Embedding X in some $\mathbb{R}^N$, every k-form is a finite sum of forms $a_0\, da_1 \wedge \cdots \wedge da_k$ with $a_j \in \mathcal{C}^\infty(X)$ (take $a_1, \dots, a_k$ among the restricted coordinate functions). By Lemma 15.2.1, $\varphi(a_0; a_1, \dots, a_k)$ is a cycle with ε-image $a_0\, da_1 \wedge \cdots \wedge da_k$. □

The main result of this section is the corresponding injectivity.

Theorem 15.3.2 (Connes)**.** *For a compact manifold X, ε induces an isomorphism $\varepsilon_* : \mathrm{HH}_k(\mathcal{C}^\infty(X)) \longrightarrow \Omega^k(X)$ for each $k \in \mathbb{N}_0$.*

The bar complex $(C_\bullet, b')$ is acyclic by the continuous version of Lemma 15.1.3, with $(su)(z_0, \dots, z_{k+1}) = u(z_1, \dots, z_{k+1})$; so $\mathcal{C}^\infty(X)$ is (continuously) H-unital.

15.3.2 Reduction to normalized chains

The following discussion applies verbatim to $\mathcal{C}^\infty(X)$ and to the polynomial algebra (with $X = \mathbb{R}^n$ and C_k the polynomials on $(\mathbb{R}^n)^{k+1}$). For $u \in C_k$ consider the diagonals

$$D_i = \{z \in X^{k+1}; z_i = z_{i+1}\},\ 0 \le i \le k-1, \qquad D_k = \{z \in X^{k+1}; z_k = z_0\},$$

so that $d_i u$ is essentially the restriction of u to D_i, and set

$$N_k = \{u \in C_k;\ u|_{D_i} = 0,\ 0 \le i \le k-1\}, \qquad M_k = \{u \in N_k;\ u|_{D_k} = 0\},$$
$$B_k = d_{k+1}(N_{k+1}) = \{v(z_0, \dots, z_k, z_0);\ v \in N_{k+1}\}.$$

Note $N_0 = C_0$, $M_0 = C_0$, and $B_k \subset M_k$: a $v \in N_{k+1}$ vanishes on $\{z_k = z_{k+1}\}$, so $v(z_0, \dots, z_k, z_0)$ vanishes when $z_k = z_0$. If $u \in N_k$ then $d_i u = 0$ for $i < k$, so $bu = (-1)^k d_k u$; and $d_k u$ vanishes on the diagonals $\{z_i = z_{i+1}\}$, $i \le k-2$, of X^k. Thus $(N_\bullet, b)$ is a subcomplex of $(C_\bullet, b)$, and

$$H_k(N_\bullet, b) = M_k / B_k. \tag{15.3.1}$$

Proposition 15.3.3. *The inclusion $N_\bullet \hookrightarrow C_\bullet$ induces an isomorphism in homology. Hence $\mathrm{HH}_k = M_k/B_k$; in particular a chain in M_k which is a b-boundary in $C_\bullet$ lies in B_k.*

Proof. For $j \in \mathbb{N}_0$ let $J^j_m = \{u \in C_m;\ u|_{D_i} = 0 \text{ for } 0 \le i < \min(j, m)\}$. Thus $J^0 = C$, and $J^j_m = N_m$ for $m \le j$. Each $J^j_\bullet$ is a subcomplex: if $u \in J^j_m$ then $d_i u = 0$ for $i < \min(j, m)$, and for the other i the restriction $d_i u$ still vanishes on $\{z_l = z_{l+1}\}$ for $l < \min(j, m-1)$ (for $i \le m-1$ because $i \ge l+1$; for $i = m$ trivially).

Let $V^j_m = \{u \in J^j_m;\ u \text{ independent of } z_{j+1}\}$ for $m \ge j+1$, and $V^j_m = 0$ for $m \le j$. For $m \ge j+1$ every $u \in J^j_m$ decomposes uniquely as

$$u = u' + u'', \qquad u'(z) = u(z_0, \dots, z_j, z_j, z_{j+2}, \dots, z_m) \in V^j_m, \quad u'' = u - u' \in J^{j+1}_m,$$

(u'' vanishes on $\{z_j = z_{j+1}\}$; uniqueness because a function independent of z_{j+1} vanishing at $z_{j+1} = z_j$ vanishes). So $J^j_\bullet = J^{j+1}_\bullet \oplus V^j_\bullet$ as graded spaces. We show that $V^j_\bullet$ is a contractible subcomplex; then this is a direct sum of complexes and $J^{j+1} \hookrightarrow J^j$ is a quasi-isomorphism.

Let $u \in V^j_m$. In bu the terms $d_i u$, $i < j$, vanish, and $d_j u = d_{j+1} u$ (both delete the variable z_{j+1} on which u does not depend; if $j+1 = m$, d_{j+1} is the cyclic face d_m and the same holds), so these cancel. Hence

$$bu = \sum_{i=j+2}^{m} (-1)^i d_i u,$$

which is independent of z_{j+1} and lies in J^j_{m-1}; so $bu \in V^j_{m-1}$ (and $bu = 0$ if $m = j+1$). Define $H : V^j_m \to V^j_{m+1}$ by inserting a dummy variable in position $j+2$:

$$(Hu)(z_0, \dots, z_{m+1}) = u(z_0, \dots, z_{j+1}, z_{j+3}, \dots, z_{m+1}).$$

Then $d_{j+2}Hu = u$ (if $m = j+1$ this is the cyclic face, which also just deletes the dummy variable), $d_iHu = Hd_{i-1}u$ for $j+3 \le i \le m+1$ (for $i = m+1$ both sides equal $u(z_0, \dots, z_{j+1}, z_{j+3}, \dots, z_m, z_0)$ after renaming), hence

$$bHu = (-1)^j u + \sum_{i=j+3}^{m+1} (-1)^i Hd_{i-1}u = (-1)^j u - Hbu,$$

i.e. $bH + Hb = (-1)^j \,\mathrm{Id}$ on $V^j_\bullet$. Thus $V^j_\bullet$ is contractible.

Consequently $H_k(C) = H_k(J^0) \cong H_k(J^1) \cong \dots \cong H_k(J^{k+1})$, and J^{k+1} coincides with N in degrees $\le k+1$, so $H_k(J^{k+1}) = H_k(N)$. All isomorphisms are induced by inclusions. The last statement follows from (15.3.1). □

15.3.3 Structure of the normalized cycles

Lemma 15.3.4. (i) *B_k is a $\mathcal{C}^\infty(X^{k+1})$-submodule of M_k.*

(ii) *If $f \in \mathcal{C}^\infty(X)$ and $g \in M_k$, then $(f(z_j) - f(z_{j+1}))g \in B_k$ for $0 \le j \le k-1$, and $(f(z_k) - f(z_0))g \in B_k$.*

Proof. (i) If $\psi \in \mathcal{C}^\infty(X^{k+1})$ and $v \in N_{k+1}$, then $\tilde\psi v \in N_{k+1}$, where $\tilde\psi(z_0, \dots, z_{k+1}) = \psi(z_0, \dots, z_k)$, and $d_{k+1}(\tilde\psi v) = \psi\, d_{k+1}v$.

(ii) For $0 \le j \le k-1$ set

$$w(z_0, \dots, z_{k+1}) = (-1)^{j+1}(f(z_j) - f(z_{j+1}))\, g(z_0, \dots, z_j, z_{j+2}, \dots, z_{k+1}) \in C_{k+1}.$$

We compute bw. For $i < j$, $d_iw = 0$ since g vanishes on $\{z_i = z_{i+1}\}$; $d_jw = 0$ because of the first factor; for $j+2 \le i \le k$, $d_iw = 0$ because g vanishes on its diagonal D_{i-1}; and $d_{k+1}w = 0$ because g vanishes on its diagonal D_k (here $j \le k-1$ is used, so that z_{k+1} is a variable of g). Finally $d_{j+1}w$ identifies z_{j+1} with z_{j+2} and gives $(-1)^{j+1}(f(z_j) - f(z_{j+1}))g(z_0, \dots, z_k)$. Hence $bw = (f(z_j) - f(z_{j+1}))g$. This lies in M_k and is a boundary in $C_\bullet$, so it lies in B_k by Proposition 15.3.3. The last case follows from $f(z_k) - f(z_0) = -\sum_{j=0}^{k-1}(f(z_j) - f(z_{j+1}))$. □

Near the full diagonal $\Delta = \{z_0 = z_1 = \dots = z_k\}$ we use the following coordinates. Let $x_0 \in X$, choose coordinates near x_0 identifying a neighbourhood with an open set in $\mathbb{R}^n$, and on a neighbourhood $W = \{|x - x_0| < \delta,\ |y_j| < \delta\}$ of $(x_0, \dots, x_0)$ use

$$x = z_0, \qquad y_j = z_j - z_{j-1},\ j = 1, \dots, k. \tag{15.3.2}$$

Then $D_{j-1} = \{y_j = 0\}$ for $1 \le j \le k$, $D_k = \{y_1 + \dots + y_k = 0\}$ and $\Delta = \{y = 0\}$.

Lemma 15.3.5. *Let q be a complex-valued k-linear form on $\mathbb{R}^n$. Then $q(y_1, \dots, y_{k-1}, -(y_1 + \dots + y_{k-1})) \equiv 0$ if and only if q is alternating.*

Proof. If q is alternating the expression is $-\sum_{j<k} q(y_1, \dots, y_{k-1}, y_j) = 0$. Conversely each term $q(y_1, \dots, y_{k-1}, y_j)$, $j < k$, is of degree 2 in y_j and of degree 1 in the other y_i, $i < k$; these multidegrees are distinct, so each term vanishes separately. Thus q vanishes whenever the j-th and k-th arguments coincide, i.e. q is alternating in these two arguments. The transpositions $(j\ k)$ generate $\mathfrak{S}_k$, so q is alternating. □

Lemma 15.3.6. *Let $u \in N_k$. In the coordinates (15.3.2), $u = \sum_I y_{1,i_1} \cdots y_{k,i_k} w_I(x,y)$ on W, with w_I smooth, where $I = (i_1, \dots, i_k) \in \{1, \dots, n\}^k$ and $y_{j,i}$ is the i-th component of y_j. Hence the Taylor series of u at $y = 0$ (with coefficients smooth in x) starts with the k-linear term $u_{(k)}(x;y) = \sum_I w_I(x,0) y_{1,i_1} \cdots y_{k,i_k}$, and*

$$\varepsilon(u)(x) = \frac{1}{k!} \sum_I w_I(x,0)\, dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

If $u \in M_k$, then $u_{(k)}(x;\cdot)$ is alternating and $\varepsilon(u)(x) = 0$ if and only if $u_{(k)}(x;\cdot) = 0$, i.e. if and only if u vanishes to order $k+1$ at $(x, \dots, x)$.

Proof. By Hadamard's lemma in y_1, $u = \sum_i y_{1,i} u_i$ with $u_i = \int_0^1 (\partial_{y_{1,i}} u)(x, ty_1, y_2, \dots, y_k)\, dt$; the u_i still vanish on $\{y_j = 0\}$, $j \geq 2$. Repeating in $y_2, \dots, y_k$ gives the representation. Since $z_l = x + y_1 + \cdots + y_l$, the derivative ∂_{z_l} equals $\partial_{y_l} - \partial_{y_{l+1}}$ (∂_{y_l} for $l = k$). In $\partial_{z_1^{i_1}} \cdots \partial_{z_k^{i_k}} u$ at $y = 0$ only terms in which every factor $y_{l,\cdot}$ is differentiated survive; since ∂_{z_k} can only hit y_k, then $\partial_{z_{k-1}}$ only y_{k-1}, and so on, the result is $w_I(x,0)$, which gives the formula for $\varepsilon(u)$. If $u \in M_k$, the homogeneous term $u_{(k)}$ vanishes on $\{y_1 + \cdots + y_k = 0\}$, so it is alternating by Lemma 15.3.5. An alternating tensor is determined by its image in Λ^k, whence the last statement. □

Lemma 15.3.7. *Let $k \geq 1$ and $m \geq k+1$. Let $\mathcal{M}_m$ be the space of homogeneous polynomials q of degree m in $y = (y_1, \dots, y_k) \in (\mathbb{R}^n)^k$ which vanish on each of the subspaces $\{y_j = 0\}$, $1 \leq j \leq k$, and $\{y_1 + \cdots + y_k = 0\}$, and let $\mathcal{V}_m$ be the space of homogeneous polynomials of degree m in $(y_1, \dots, y_{k+1}) \in (\mathbb{R}^n)^{k+1}$ vanishing on each $\{y_j = 0\}$, $1 \leq j \leq k+1$. Then*

$$R : \mathcal{V}_m \longrightarrow \mathcal{M}_m, \quad (RV)(y_1, \dots, y_k) = V(y_1, \dots, y_k, -(y_1 + \cdots + y_k)),$$

is surjective.

Proof. Given $q \in \mathcal{M}_m$, the polynomial chain $u(z_0, \dots, z_k) = q(z_1 - z_0, \dots, z_k - z_{k-1})$ for $\mathbb{C}[x]$ lies in M_k, and $\varepsilon(u) = 0$ since u vanishes to order $m > k$ on Δ. By Theorem 15.2.3 and Proposition 15.3.3 (for polynomials) $u \in B_k$, i.e. $u = d_{k+1} v$ with $v \in N_{k+1}$. Writing v in the coordinates $x = z_0$, $y_j = z_j - z_{j-1}$, $1 \leq j \leq k+1$, as $V(x; y_1, \dots, y_{k+1})$, V vanishes on the subspaces $\{y_j = 0\}$, and $d_{k+1} v = V(x; y_1, \dots, y_k, -(y_1 + \cdots + y_k))$ because $z_{k+1} = z_0$ means $y_{k+1} = -(y_1 + \cdots + y_k)$. The component of V of degree 0 in x and degree m in y lies in $\mathcal{V}_m$ and is mapped by R to the corresponding component of u, which is q. □

Lemma 15.3.8. *Let $u \in M_k$, $k \geq 1$, satisfy $\varepsilon(u) = 0$. Then every point $p \in X^{k+1}$ has a neighbourhood on which u coincides with an element of B_k.*

Proof. First let $p \notin \Delta$. Then $p_j \neq p_{j+1}$ for some $0 \leq j \leq k-1$ (else all p_j coincide), and we choose $f \in \mathcal{C}^\infty(X)$ with $f(p_j) \neq f(p_{j+1})$ and $\psi \in \mathcal{C}^\infty(X^{k+1})$ supported where $f(z_j) \neq f(z_{j+1})$, equal to 1 near p. Then $g = \psi u / (f(z_j) - f(z_{j+1})) \in M_k$ and $u = (f(z_j) - f(z_{j+1}))g$ near p; this lies in B_k by Lemma 15.3.4.

Now let $p = (x_0, \dots, x_0) \in \Delta$ and use the coordinates (15.3.2). The Taylor series of u in y at $y = 0$ is $\sum_m u_{(m)}(x;y)$, where $u_{(m)}(x;\cdot)$ is homogeneous of degree m and vanishes on the same linear subspaces as u, so $u_{(m)} \in \mathcal{C}^\infty(B) \otimes \mathcal{M}_m$, $B = \{|x - x_0| < \delta\}$. By Lemma 15.3.6, $u_{(m)} = 0$ for $m \leq k$. For $m \geq k+1$, choosing a linear right inverse of R in Lemma 15.3.7, we find $V_{(m)} \in \mathcal{C}^\infty(B) \otimes \mathcal{V}_m$ with $RV_{(m)} = u_{(m)}$. By Borel's lemma (see [7, Section 1.2]), shrinking δ, there is a smooth $V(x; y_1, \dots, y_{k+1}) = \sum_m \chi(\lambda_m |y|) V_{(m)}(x;y)$, with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ equal to 1 near 0 and $\lambda_m \to \infty$ fast enough, having Taylor series $\sum_m V_{(m)}$ at $y = 0$; each term, hence V, vanishes on the subspaces $\{y_j = 0\}$. Let $\Theta \in \mathcal{C}^\infty_{\mathrm{c}}$ be a

cut-off near $(x_0,\dots,x_0)\in X^{k+2}$ and let $v=\Theta V\in N_{k+1}$, extended by zero (written in the variables $x=z_0$, $y_j=z_j-z_{j-1}$, $j\le k+1$). Then $s=d_{k+1}v\in B_k$ has the same Taylor series as u at Δ near p. So $g=u-s\in M_k$ vanishes to infinite order at Δ near p. Near p,

$$g=\sum_{j=1}^{k}\sum_{i=1}^{n} y_{j,i}\,g_{j,i},\qquad g_{j,i}=\frac{y_{j,i}\,g}{|y|^2},$$

where $g_{j,i}$, extended by 0 on Δ, is smooth since g is flat there, and vanishes wherever g does. Taking $f_i\in\mathcal{C}^\infty(X)$ equal to the i-th coordinate near x_0, and a cut-off ψ equal to 1 near p, we have $y_{j,i}=-(f_i(z_{j-1})-f_i(z_j))$ on the support of ψ, so near p, $g=-\sum_{j,i}(f_i(z_{j-1})-f_i(z_j))\psi g_{j,i}\in B_k$ by Lemma 15.3.4. Hence $u=s+g$ coincides near p with an element of B_k. □

Proof of Theorem 15.3.2. For $k=0$, $(bu)(z_0)=u(z_0,z_0)-u(z_0,z_0)=0$ for $u\in C_1$, so $\mathrm{HH}_0=\mathcal{C}^\infty(X)=\Omega^0(X)$. Let $k\ge1$. By Lemma 15.3.1 it remains to show injectivity. By Proposition 15.3.3 every class in HH_k is represented by some $u\in M_k$, and u is a boundary if (and only if) $u\in B_k$. Suppose $\varepsilon(u)=0$. By Lemma 15.3.8 and compactness there is a finite partition of unity $\sum_\beta\psi_\beta=1$ on X^{k+1} and $s_\beta\in B_k$ with $u=s_\beta$ on $\operatorname{supp}\psi_\beta$. Then $u=\sum_\beta\psi_\beta s_\beta\in B_k$ by Lemma 15.3.4(i). □

Remark 15.3.9. Compactness was only used to make the partitions of unity finite. With some care about supports (using locally finite partitions of unity and cut-offs in all variables) the same argument applies to non-compact X.

15.4 The commutative symbol algebra

From now on X is compact, $n=\dim X\ge1$, and

$$Y=T^*X\setminus 0,\qquad \rho_t(x,\xi)=(x,t\xi),\ t>0,\qquad S^*X=Y/\mathbb{R}^+.$$

A function a on Y is homogeneous of degree $m\in\mathbb{Z}$ if $\rho_t^*a=t^m a$ for all $t>0$, and similarly for differential forms; we write $\mathcal{C}^\infty_{(m)}(Y)$ and $\Omega^k_{(m)}(Y)$ for the smooth homogeneous functions and k-forms of degree m. (Equivalently, $\mathcal{C}^\infty_{(m)}(Y)$ is the space of sections of a line bundle over S^*X.) Choosing a Riemannian metric on X and setting $R=|\xi|$ identifies S^*X with $\{R=1\}$ and

$$Y\cong S^*X\times(0,\infty),\qquad (x,\xi)\longmapsto((x,\xi/|\xi|),|\xi|);\tag{15.4.1}$$

we write $s\in S^*X$ and $r=R$ for the two factors.

The *commutative algebra of formal symbols* is

$$\mathcal{A}_0=\Big\{a=\sum_{m\le M}a_m;\ a_m\in\mathcal{C}^\infty_{(m)}(Y),\ M\in\mathbb{Z}\Big\},\qquad (ab)_m=\sum_{l\in\mathbb{Z}}a_{m-l}b_l,$$

the sum being finite. It contains the algebra $\mathcal{A}_{\mathrm{fin}}=\bigoplus_m\mathcal{C}^\infty_{(m)}(Y)$ of finite sums.

Definition 15.4.1. For a multi-index $I=(I_0,\dots,I_k)\in\mathbb{Z}^{k+1}$ let $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ be the space of smooth functions on Y^{k+1} homogeneous of degree I_j in the j-th factor, and for $m\in\mathbb{Z}$ let $C_{m,k}=\bigoplus_{|I|=m}\mathcal{C}^\infty_{(I)}(Y^{k+1})$ (finite sums), $|I|=I_0+\cdots+I_k$. The space of Hochschild k-chains of the symbol algebra is

$$\mathcal{B}^{(k)}=\Big\{u=\sum_{m\le M}u_m;\ u_m\in C_{m,k},\ M\in\mathbb{Z}\Big\}.$$

Each $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ is the completed projective tensor product of the spaces $\mathcal{C}^\infty_{(I_j)}(Y)$, and $\mathcal{A}_0^{\otimes(k+1)} \subset \mathcal{B}^{(k)}$: a finite sum of products $a^0(\zeta_0)\cdots a^k(\zeta_k)$ of formal sums has, in each total degree m, only finitely many non-zero multihomogeneous components, because each a^j has degrees bounded above. The face maps (15.2.1), with $\zeta_j \in Y$ in place of z_j, map $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ to $\mathcal{C}^\infty_{(I')}(Y^k)$ where I' is obtained by adding two adjacent (or the last and the first) entries; they preserve the total degree and extend the face maps of $\mathcal{A}_0$. So $(\mathcal{B}^{(\bullet)}, b)$ is a complex, the direct product over m (with degrees bounded above) of the complexes $(C_{m,\bullet}, b)$. Homology commutes with such products, so

$$\mathrm{HH}_k(\mathcal{A}_0) = \Big\{ \sum_{m\le M} h_m;\ h_m \in H_k(C_{m,\bullet}, b) \Big\}. \tag{15.4.2}$$

Theorem 15.4.2. *The map ε of (15.2.2) (applied on Y) induces isomorphisms $H_k(C_{m,\bullet}, b) \cong \Omega^k_{(m)}(Y)$ for all $m \in \mathbb{Z}$, $k \in \mathbb{N}_0$. Hence*

$$\mathrm{HH}_k(\mathcal{A}_0) \cong \Big\{ \alpha = \sum_{m\le M} \alpha_m;\ \alpha_m \in \Omega^k_{(m)}(Y) \Big\},$$

the space of formal sums of homogeneous k-forms on $T^*X \setminus 0$.

Proof. Under (15.4.1), an element of the space $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ is of the form $r_0^{I_0}\cdots r_k^{I_k} v(s_0,\dots,s_k)$ with $v \in \mathcal{C}^\infty((S^*X)^{k+1})$, so

$$\bigoplus_m C_{m,k} = \mathcal{C}^\infty((S^*X)^{k+1}) \otimes \mathbb{C}[r_0^{\pm1},\dots,r_k^{\pm1}].$$

The face maps of the left side are the tensor products of the face maps of the Hochschild complexes of $\mathcal{C}^\infty(S^*X)$ (Section 15.3) and of $\mathbb{C}[r^{\pm1}]$. With the degeneracies s_i (insertion of the identity in position $i+1$) the Hochschild complex of a unital algebra is a simplicial vector space, and the complex above is the diagonal of the tensor product of two simplicial vector spaces. By the Eilenberg–Zilber theorem, the shuffle map sh from the tensor product complex $C_\bullet(\mathcal{C}^\infty(S^*X)) \otimes C_\bullet(\mathbb{C}[r^{\pm1}])$ to the diagonal complex is a quasi-isomorphism, and by the Künneth formula over $\mathbb{C}$ the homology of the tensor product complex is $\mathrm{HH}_\bullet(\mathcal{C}^\infty(S^*X)) \otimes \mathrm{HH}_\bullet(\mathbb{C}[r^{\pm1}])$. A direct computation from the definition of the shuffle map shows $\varepsilon(\mathrm{sh}(\alpha\otimes\beta)) = \varepsilon(\alpha)\wedge\varepsilon(\beta)$ (the $\binom{p+q}{p}$ shuffles each contribute the same wedge product, and $\binom{p+q}{p}/(p+q)! = 1/(p!q!)$). By Theorem 15.3.2 and Remark 15.2.4 it follows that ε induces an isomorphism of $H_k(\bigoplus_m C_{m,\bullet})$ onto

$$\Omega^k(S^*X)\otimes\mathbb{C}[r^{\pm1}] \ \oplus\ \Omega^{k-1}(S^*X)\otimes\mathbb{C}[r^{\pm1}]\,dr,$$

which by (15.4.1) is the space of finite sums of homogeneous k-forms on Y: a k-form homogeneous of degree m is exactly $r^m(\alpha' + \alpha''\wedge dr/r)$ with α', α'' forms on S^*X. Since ε maps $C_{m,k}$ to $\Omega^k_{(m)}(Y)$, the isomorphism respects the grading by m; (15.4.2) gives the last statement. □

15.5 The algebra of formal symbols

15.5.1 Full symbols and the star product

Let $\Psi^m(X)$ denote the classical pseudodifferential operators of order m on X (Chapter 5), $\Psi^{\mathbb{Z}}(X) = \bigcup_{m\in\mathbb{Z}} \Psi^m(X)$ and $\Psi^{-\infty}(X) = \bigcap_m \Psi^m(X)$. Then $\Psi^{-\infty}(X)$ is an ideal and we consider the quotient algebra

$$\mathcal{A} = \Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X).$$

The principal symbol gives exact sequences $0 \to \Psi^{m-1}(X) \to \Psi^m(X) \xrightarrow{\sigma_m} \mathcal{C}^\infty_{(m)}(Y) \to 0$, with $\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B)$ and

$$\sigma_{m+m'-1}([A,B]) = -i\{\sigma_m(A), \sigma_{m'}(B)\}, \qquad \{a,b\} = \sum_{j=1}^{n}\Big(\frac{\partial a}{\partial \xi_j}\frac{\partial b}{\partial x_j} - \frac{\partial a}{\partial x_j}\frac{\partial b}{\partial \xi_j}\Big) = H_a b,$$

the Poisson bracket, H_a being the Hamilton vector field of a (in canonical coordinates (x,ξ) on T^*X).

To compute Hochschild homology we need the full symbol. We use the following standard facts on global symbol calculus (see [1] and the references there). After the choice of a Riemannian metric, and hence of a global quantization map, there is a linear bijection

$$\sigma : \mathcal{A} \longrightarrow \mathcal{A}_0, \qquad \sigma\big(\Psi^m(X)\big) = \{a \in \mathcal{A}_0;\ a_j = 0 \text{ for } j > m\},$$

whose top-degree component on $\Psi^m(X)$ is the principal symbol, and there are bidifferential operators P_l, $l \in \mathbb{N}_0$, on Y with $P_l(\mathcal{C}^\infty_{(p)}(Y), \mathcal{C}^\infty_{(q)}(Y)) \subset \mathcal{C}^\infty_{(p+q-l)}(Y)$, such that the product transported to $\mathcal{A}_0$ is

$$\begin{aligned} a \star b &= \sigma\big(\sigma^{-1}(a)\sigma^{-1}(b)\big) = \sum_{l\ge 0} P_l(a,b), \\ P_0(a,b) = ab, \qquad & P_1(a,b) - P_1(b,a) = -i\{a,b\}. \end{aligned} \tag{15.5.1}$$

The sum is meaningful in $\mathcal{A}_0$: the component of degree m of $a \star b$ is the finite sum $\sum_{l\ge 0}\sum_p P_l(a_p, b_{m+l-p})$. In local coordinates, with the left quantization, the composition formula of Chapter 2 gives $P_l(a,b) = \sum_{|\alpha|=l}(\alpha!)^{-1}\partial_\xi^\alpha a\, D_x^\alpha b$, and then indeed

$$P_1(a,b) - P_1(b,a) = -i\sum_j (\partial_{\xi_j} a\, \partial_{x_j} b - \partial_{\xi_j} b\, \partial_{x_j} a).$$

The relations (15.5.1) for P_0 and the antisymmetric part of P_1 are forced by the principal symbol calculus; these, together with associativity of $\star$, are all that will be used.

15.5.2 Hochschild chains

Definition 15.5.1. The Hochschild complex of $\mathcal{A} \cong (\mathcal{A}_0, \star)$ has the chain spaces $\mathcal{B}^{(k)}$ of Definition 15.4.1, with face maps $d_i^\star = \sum_{l\ge 0} d_i^{(l)}$, where, for $u \in \mathcal{C}^\infty_{(I)}(Y^{k+1})$, $d_i^{(l)}u$ is obtained by applying the bidifferential operator P_l in the variables (ζ_i, ζ_{i+1}) (in (ζ_k, ζ_0) for $i = k$, the first argument being ζ_k) and then restricting as in (15.2.1). We write

$$b = \sum_{i=0}^{k}(-1)^i d_i^\star = \sum_{l\ge 0} b_l, \qquad b_l = \sum_{i=0}^{k}(-1)^i d_i^{(l)} : C_{m,k} \longrightarrow C_{m-l,k-1}.$$

On $a_0 \otimes \cdots \otimes a_k$ the map $d_i^{(l)}$ replaces $a_i \otimes a_{i+1}$ by $P_l(a_i, a_{i+1})$ (and $d_k^{(l)}$ gives $P_l(a_k, a_0) \otimes a_1 \otimes \cdots \otimes a_{k-1}$), so b extends the Hochschild differential of $(\mathcal{A}_0, \star)$. For $u = \sum_{m\le M} u_m$ the component of degree m of bu is the finite sum $\sum_{l\ge 0} b_l u_{m+l}$, so b is well defined on $\mathcal{B}^{(k)}$. Moreover $b^2 = 0$: on the algebraic tensor product of copies of $\mathcal{A}_{\mathrm{fin}}$ this is Lemma 15.1.1 (which only uses associativity), each component of b^2 is a finite sum of operators continuous in the $\mathcal{C}^\infty$ topology, and finite sums of products are dense in each $\mathcal{C}^\infty_{(I)}(Y^{k+1})$. The differential b_0 is the Hochschild differential of the commutative algebra $\mathcal{A}_0$ of Section 15.4. We define $\mathrm{HH}_k(\mathcal{A})$ to be the homology of $(\mathcal{B}^{(\bullet)}, b)$.

Remark 15.5.2 (Semiclassical interpretation). For $h > 0$ let $L_h : \mathcal{A}_0 \to \mathcal{A}_0$, $L_h(\sum_j a_j) = \sum_j h^j a_j$, and define $a \star_h b = L_h^{-1}(L_h a \star L_h b)$. Since $P_l(h^p a_p, h^q b_q) = h^{p+q} P_l(a_p, b_q)$ has degree $p + q - l$,

$$a \star_h b = \sum_{l \geq 0} h^l P_l(a, b),$$

a family of associative products, all isomorphic to $\star = \star_1$ via L_h, which at $h = 0$ reduces to the commutative product. The Hochschild differential of $\star_h$ is $b_h = \sum_l h^l b_l$. Thus the symbol algebra is a deformation of $\mathcal{A}_0$ in the direction of the Poisson bracket, and its Hochschild homology can be computed by expanding in powers of h, the 'semiclassical limit'. Expanding chains in powers of h is equivalent to filtering them by homogeneity, which is how we proceed: the spectral sequence of the next section is the one associated to the Taylor expansion of h-dependent chains at $h = 0$.

15.6 The filtration by order and its spectral sequence

15.6.1 The first terms

The complex $(\mathcal{B}^{(\bullet)}, b)$ is filtered by the subcomplexes $F_p\mathcal{B}^{(k)} = \{u;\ u_m = 0 \text{ for } m > p\}$ of chains of order at most p; $b(F_p) \subset F_p$ since each b_l lowers degrees by $l \geq 0$. The filtration is exhaustive, $\bigcap_p F_p = 0$, and complete: a series $\sum_j e_j$ with $e_j \in F_{p_j}$, $p_j \to -\infty$ and $\sup p_j < \infty$ converges in the sense that each component is a finite sum, and b commutes with such sums.

Comparing components of degree $m - l$ in $b^2 = 0$ gives $\sum_{i+j=l} b_i b_j = 0$ for every l, in particular

$$b_0^2 = 0, \qquad b_0 b_1 + b_1 b_0 = 0, \qquad b_1^2 + b_0 b_2 + b_2 b_0 = 0. \tag{15.6.1}$$

Define

$$E^1_{p,k} = H_k(C_{p,\bullet}, b_0),$$

the homology of the associated graded complex. By (15.6.1), b_1 maps b_0-cycles to b_0-cycles and b_0-boundaries to b_0-boundaries, so induces $d^1 : E^1_{p,k} \to E^1_{p-1,k-1}$, and $(d^1)^2 = 0$. Set

$$E^2_{p,k} = \frac{\mathrm{Nul}(d^1 : E^1_{p,k} \to E^1_{p-1,k-1})}{\mathrm{Ran}(d^1 : E^1_{p+1,k+1} \to E^1_{p,k})}.$$

These are the first two terms of the spectral sequence of the filtered complex; the higher terms E^r, with differentials $d^r : E^r_{p,k} \to E^r_{p-r,k-1}$, will not be needed explicitly. By Theorem 15.4.2,

$$E^1_{p,k} \cong \Omega^k_{(p)}(Y), \tag{15.6.2}$$

the isomorphism being induced by ε.

15.6.2 Degeneration and convergence

The following purely algebraic statement reduces the computation of $\mathrm{HH}_\bullet(\mathcal{A})$ to that of E^2. It applies to any complex of the form $C_k = \{\sum_{p \leq P} c_p;\ c_p \in C_{p,k}\}$ with $b = \sum_{l \geq 0} b_l$, $b_l : C_{p,k} \to C_{p-l,k-1}$, $b^2 = 0$, and E^1, E^2 defined as above.

Proposition 15.6.1. *Suppose there is an integer n such that $E^2_{p,k} = 0$ whenever $p \neq k - n$. Then every class in $H_k(C, b)$ has a representative $c \in F_{k-n}C_k$, and*

$$\Theta : H_k(C, b) \longrightarrow E^2_{k-n,k}, \qquad [c] \longmapsto [c_{k-n}],$$

is a well-defined isomorphism. Moreover all differentials d^r, $r \geq 2$, of the spectral sequence vanish.

Note that if $c \in F_{k-n}$ and $bc = 0$, then $b_0 c_{k-n} = (bc)_{k-n} = 0$ and $b_1 c_{k-n} = -b_0 c_{k-n-1}$, so c_{k-n} defines an element of $E^2_{k-n,k}$.

Proof. We use the following step.

(P) Let $c \in F_p C_k$ satisfy $b_0 c_p = 0$ and $b_1 c_p + b_0 c_{p-1} = 0$ (e.g. $bc = 0$, or more generally $bc \in F_{p-2}$), and suppose that the class of c_p in $E^2_{p,k}$ vanishes. Then there are $x \in C_{p+1,k+1}$ with $b_0 x = 0$ and $y \in C_{p,k+1}$ such that $c - b(x + y) \in F_{p-1}$.

Indeed, $[c_p] \in E^1_{p,k}$ is a d^1-cycle, and as it vanishes in E^2 it equals $d^1[x] = [b_1 x]$ for some b_0-cycle $x \in C_{p+1,k+1}$; so $c_p - b_1 x = b_0 y$ for some $y \in C_{p,k+1}$. The components of $b(x + y)$ of degree $p + 1$ and p are $b_0 x = 0$ and $b_1 x + b_0 y = c_p$, which proves (P).

Representatives. If $c \in F_P C_k$ is a cycle with $P > k - n$, then $E^2_{P,k} = 0$ and (P) replaces c by a homologous cycle in F_{P-1}. After finitely many steps $c \in F_{k-n}$.

Well-definedness. It suffices to show that if $c = be \in F_{k-n} C_k$, $e \in F_q C_{k+1}$, then $[c_{k-n}] = 0$ in E^2. If $q > k + 1 - n$, then $be \in F_{q-2}$ and $E^2_{q,k+1} = 0$, so by (P) applied to e we may replace e by $e - b(x + y) \in F_{q-1}$ without changing be. Repeating, we may assume $q \le k + 1 - n$. Then $b_0 e_{k+1-n} = (be)_{k+1-n} = 0$ and $c_{k-n} = b_1 e_{k+1-n} + b_0 e_{k-n}$, so the class of c_{k-n} in E^1 is $d^1[e_{k+1-n}]$, which vanishes in E^2.

Injectivity. Let $c \in F_{k-n} C_k$ be a cycle with $[c_{k-n}] = 0$ in E^2. By (P), $c - b(x_{k-n} + y_{k-n}) = c^{(1)} \in F_{k-n-1}$. Since $E^2_{p,k} = 0$ for $p < k - n$, (P) applies again to $c^{(1)}$, and so on: we obtain $e_p = x_p + y_p \in F_{p+1} C_{k+1}$, $p = k - n, k - n - 1, \dots$, with $c - b(\sum_{p \ge P} e_p) \in F_{P-1}$ for every P. By completeness $e = \sum_p e_p$ exists, and $c = be$.

Surjectivity. Let $x \in C_{k-n,k}$ with $b_0 x = 0$ represent a class in $E^2_{k-n,k}$. Then $[b_1 x] = d^1[x] = 0$ in E^1, i.e. $b_1 x = -b_0 y$ for some $y \in C_{k-n-1,k}$, and $c^{(1)} = x + y$ satisfies $bc^{(1)} \in F_{k-n-2}$. Suppose $c^{(r)} \in F_{k-n}$, $r \ge 1$, has $c^{(r)}_{k-n} = x$ and $z = bc^{(r)} \in F_p C_{k-1}$, $p = k - n - r - 1$. Then z is a cycle and $p \ne (k-1) - n$, so (P) gives $x' \in C_{p+1,k}$, $y' \in C_{p,k}$ with $z - b(x' + y') \in F_{p-1}$. Set $c^{(r+1)} = c^{(r)} - x' - y'$; since $p + 1 < k - n$, its component of degree $k - n$ is still x, and $bc^{(r+1)} \in F_{p-1}$. The corrections have degrees tending to $-\infty$, so $c = \lim_r c^{(r)}$ exists, $bc = 0$ and $\Theta[c] = [x]$.

Degeneration. Each $E^r_{p,k}$, $r \ge 2$, is a subquotient of $E^2_{p,k}$, so vanishes unless $p = k - n$. The differential d^r maps $E^r_{k-n,k}$ to $E^r_{k-n-r,k-1}$, which vanishes unless $k - n - r = k - 1 - n$, i.e. $r = 1$. Hence $d^r = 0$ for $r \ge 2$. □

15.7 The first differential: the Poisson complex

The canonical symplectic form on T^*X is $\omega = \sum_j d\xi_j \wedge dx_j$ in canonical coordinates. Its inverse is the Poisson bivector Π; we regard it as the antisymmetric bilinear form on T^*Y with $\Pi(da, db) = \{a, b\}$, i.e. $\Pi(\theta, \eta) = \sum_j (\theta(\partial_{\xi_j})\eta(\partial_{x_j}) - \theta(\partial_{x_j})\eta(\partial_{\xi_j}))$. Let $\iota_\Pi : \Omega^k(Y) \to \Omega^{k-2}(Y)$ be contraction with Π:

$$\iota_\Pi(\theta_1 \wedge \cdots \wedge \theta_k) = \sum_{j<l} (-1)^{j+l+1} \Pi(\theta_j, \theta_l)\, \theta_1 \wedge \cdots \wedge \widehat{\theta_j} \wedge \cdots \wedge \widehat{\theta_l} \wedge \cdots \wedge \theta_k;$$

in canonical coordinates $\iota_\Pi = \sum_j \iota_{\partial_{x_j}} \iota_{\partial_{\xi_j}}$, as one checks by expanding the right side on decomposable forms. Following Koszul and Brylinski, set

$$\delta = \iota_\Pi d - d\iota_\Pi : \Omega^k(Y) \longrightarrow \Omega^{k-1}(Y).$$

Since Π lowers homogeneity by one, δ maps $\Omega^k_{(p)}(Y)$ to $\Omega^{k-1}_{(p-1)}(Y)$.

Lemma 15.7.1. *For functions $a_0, \dots, a_k$,*

$$\delta(a_0\, da_1 \wedge \cdots \wedge da_k) = \sum_{j=1}^{k} (-1)^{j+1} \{a_0, a_j\}\, da_1 \wedge \cdots \wedge \widehat{da_j} \wedge \cdots \wedge da_k + \sum_{1 \le j < l \le k} (-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge da_1 \wedge \cdots \wedge \widehat{da_j} \wedge \cdots \wedge \widehat{da_l} \wedge \cdots \wedge da_k.$$

Moreover, if f is a function and β a form with constant coefficients in canonical coordinates, then $\delta(f\beta) = \iota_{H_f}\beta$.

Proof. Write D_{jl} for $da_1 \wedge \cdots \wedge da_k$ with da_j and da_l omitted (D_{0l}: only da_l omitted). Applying ι_Π to $da_0 \wedge da_1 \wedge \cdots \wedge da_k$, the pairs $(0, l)$ give $\sum_l (-1)^{l+1}\{a_0, a_l\} D_{0l}$ and the pairs $1 \le j < l$ give $\sum (-1)^{j+l+1}\{a_j, a_l\} da_0 \wedge D_{jl}$. On the other hand $d\iota_\Pi(a_0\, da_1 \wedge \cdots \wedge da_k) = \sum_{j<l} (-1)^{j+l+1}(\{a_j, a_l\} da_0 + a_0\, d\{a_j, a_l\}) \wedge D_{jl}$. Subtracting gives the formula. For the second statement, $d\beta = 0$ and $d\iota_\Pi\beta = 0$, so $\delta(f\beta) = \iota_\Pi(df \wedge \beta) - df \wedge \iota_\Pi\beta$. For vector fields X, Z and a 1-form θ one has $\iota_X\iota_Z(\theta \wedge \beta) - \theta \wedge \iota_X\iota_Z\beta = \theta(Z)\iota_X\beta - \theta(X)\iota_Z\beta$; with $X = \partial_{x_j}$, $Z = \partial_{\xi_j}$, $\theta = df$, summed over j, this is $\iota_V\beta$ with $V = \sum_j (\partial_{\xi_j} f\, \partial_{x_j} - \partial_{x_j} f\, \partial_{\xi_j}) = H_f$. □

Proposition 15.7.2. *Under the identification* (15.6.2), *$d^1 = -i\,\delta$.*

Proof. Homogeneous forms of degree p are finite sums of forms $\alpha = a_0\, da_1 \wedge \cdots \wedge da_k$ with a_j homogeneous of total degree p (use (15.4.1) and an embedding of S^*X in some $\mathbb{R}^N$). By Lemma 15.2.1 the chain $F = \varphi(a_0; a_1, \dots, a_k) \in C_{p,k}$ is a b_0-cycle with $\varepsilon(F) = \alpha$, so $d^1\alpha = \varepsilon(b_1 F)$.

Decompose $P_1 = S + A$ with $S(a,b) = \frac{1}{2}(P_1(a,b) + P_1(b,a))$ symmetric and $A(a,b) = -\frac{i}{2}\{a,b\}$ by (15.5.1). Replacing P_1 by S in b_1F, the inner terms cancel in pairs (as σ and $\sigma \circ (i\ i{+}1)$ have opposite signs) and the first and last terms cancel as in the proof of Lemma 15.2.1, now using $S(a_0, a_{\sigma(1)}) = S(a_{\sigma(1)}, a_0)$. Thus b_1F may be computed with P_1 replaced by $c\{\cdot,\cdot\}$, $c = -\frac{i}{2}$. Then $(k-1)!\,\varepsilon(b_1F)$ is the sum of the following, with the notation D_{jl} of the proof of Lemma 15.7.1:

(a) the first terms $c\sum_\sigma \operatorname{sgn}(\sigma)\{a_0, a_{\sigma(1)}\} da_{\sigma(2)} \wedge \cdots \wedge da_{\sigma(k)}$; grouping according to $\sigma(1) = j$ (sign $(-1)^{j-1}$) this is $c(k-1)!\sum_j (-1)^{j-1}\{a_0, a_j\} D_{0j}$;

(b) the last terms $(-1)^k c\sum_\sigma \operatorname{sgn}(\sigma)\{a_{\sigma(k)}, a_0\} da_{\sigma(1)} \wedge \cdots \wedge da_{\sigma(k-1)}$; grouping according to $\sigma(k) = j$ (sign $(-1)^{k-j}$) this equals the expression in (a);

(c) the inner terms $c\sum_\sigma \operatorname{sgn}(\sigma) \sum_{i=1}^{k-1} (-1)^i a_0\, da_{\sigma(1)} \wedge \cdots \wedge d\{a_{\sigma(i)}, a_{\sigma(i+1)}\} \wedge \cdots \wedge da_{\sigma(k)}$. Moving $d\{\cdot,\cdot\}$ to the front costs $(-1)^{i-1}$; for fixed i and $(\sigma(i), \sigma(i+1)) = (j, l)$, $j < l$, the sign of σ is $(-1)^{j+l+1}$ times the sign of the induced ordering of the remaining indices, and the pair (l, j) gives the same contribution since both $\{a_l, a_j\}$ and the sign change. Each i thus contributes $-2c(k-2)!\sum_{j<l}(-1)^{j+l+1} a_0\, d\{a_j, a_l\} \wedge D_{jl}$, and there are $k-1$ values of i.

Dividing by $(k-1)!$,

$$\varepsilon(b_1F) = 2c\Big(\sum_j (-1)^{j+1}\{a_0, a_j\} D_{0j} + \sum_{j<l} (-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge D_{jl}\Big) = -i\,\delta\alpha$$

by Lemma 15.7.1. □

In particular $\delta^2 = 0$, which can also be checked directly. The complex $(\Omega^\bullet(Y), \delta)$ is the *canonical* (or Koszul–Brylinski) complex of the Poisson manifold Y.

15.8 The E^2 term

15.8.1 The symplectic star operator

Fix $\zeta \in Y$ and let $V = T_\zeta Y$, of dimension $2n$. The bilinear form Π on V^* is non-degenerate; let $\Pi^\sharp : V^* \to V$ be defined by $\eta(\Pi^\sharp\theta) = \Pi(\theta, \eta)$, so that $\Pi^\sharp(df) = H_f$. On $\Lambda^k V^*$ define the bilinear form

$$\Lambda(\theta_1 \wedge \cdots \wedge \theta_k, \eta_1 \wedge \cdots \wedge \eta_k) = \det\big(\Pi(\theta_i, \eta_j)\big)_{i,j=1}^k,$$

which is non-degenerate since $\Lambda(\alpha, \beta) = (-1)^k\langle\alpha, \Lambda^k\Pi^\sharp\,\beta\rangle$ and $\Pi^\sharp$ is invertible. Let $\mathrm{vol} = \omega^n/n!$. The *symplectic star operator* $* : \Lambda^k V^* \to \Lambda^{2n-k}V^*$ is defined by

$$\alpha \wedge *\beta = \Lambda(\alpha, \beta)\,\mathrm{vol} \quad \text{for all } \alpha \in \Lambda^k V^*;$$

it is an isomorphism because the wedge pairing $\Lambda^k \times \Lambda^{2n-k} \to \Lambda^{2n}$ is non-degenerate. Applied pointwise it gives a bundle isomorphism, hence isomorphisms $* : \Omega^k(Y) \to \Omega^{2n-k}(Y)$, and in canonical coordinates it maps forms with constant coefficients to forms with constant coefficients (as Π and vol have constant coefficients). This is the operator W_ω of the original notes, which describes it in the basis $dx_i, d\xi_i$.

Lemma 15.8.1. *For $\theta \in V^*$ and $\beta \in \Lambda^k V^*$, $k \geq 1$,*

$$*^{-1}(\theta \wedge *\beta) = (-1)^{k-1}\iota_{\Pi^\sharp\theta}\beta.$$

Proof. Expanding the determinant along the first row, $\Lambda(\theta \wedge \alpha, \beta) = \Lambda(\alpha, \iota_{\Pi^\sharp\theta}\beta)$ for $\alpha \in \Lambda^{k-1}V^*$, since $\iota_v(\eta_1 \wedge \cdots \wedge \eta_k) = \sum_j (-1)^{j+1}\eta_j(v)\,\eta_1 \wedge \ldots \widehat{\eta_j} \cdots \wedge \eta_k$ and $\eta_j(\Pi^\sharp\theta) = \Pi(\theta, \eta_j)$. Let $\mu = *^{-1}(\theta \wedge *\beta) \in \Lambda^{k-1}V^*$. For all $\alpha \in \Lambda^{k-1}V^*$,

$$\Lambda(\alpha, \mu)\mathrm{vol} = \alpha \wedge *\mu = \alpha \wedge \theta \wedge *\beta = (-1)^{k-1}\theta \wedge \alpha \wedge *\beta = (-1)^{k-1}\Lambda(\alpha, \iota_{\Pi^\sharp\theta}\beta)\mathrm{vol},$$

and Λ is non-degenerate. □

Proposition 15.8.2 (Brylinski)**.** *On k-forms, $\delta = (-1)^{k-1} *^{-1} d *$. Moreover $*$ maps $\Omega^k_{(p)}(Y)$ onto $\Omega^{2n-k}_{(p+n-k)}(Y)$.*

Proof. Both sides are first-order differential operators, so it suffices to compare them on $f\beta$ with f a function and β a form with constant coefficients in canonical coordinates. Then $*\beta$ has constant coefficients, $d(f*\beta) = df \wedge *\beta$ and, by Lemma 15.8.1, $*^{-1}d*(f\beta) = (-1)^{k-1}\iota_{H_f}\beta$, which is $(-1)^{k-1}\delta(f\beta)$ by Lemma 15.7.1.

Since $\{a, b\}$ is homogeneous of degree $p + q - 1$ if a, b are homogeneous of degrees p, q, the function $\Lambda(\alpha, \beta)$ is homogeneous of degree $p + q - k$ for $\alpha \in \Omega^k_{(p)}$, $\beta \in \Omega^k_{(q)}$, and vol is homogeneous of degree n. Hence for such α, β,

$$\alpha \wedge \rho_t^*(*\beta) = \rho_t^*\big(t^{-p}\alpha \wedge *\beta\big) = t^{-p}\rho_t^*\big(\Lambda(\alpha, \beta)\mathrm{vol}\big) = t^{q+n-k}\alpha \wedge *\beta.$$

As the values of the forms $\alpha \in \Omega^k_{(p)}(Y)$ at any point span Λ^k, $\rho_t^*(*\beta) = t^{q+n-k} * \beta$. The same argument applies to $*^{-1}$. □

15.8.2 Homogeneous de Rham cohomology

Proposition 15.8.3. *Let $c \in \mathbb{Z}$. The cohomology of the complex $(\Omega^\bullet_{(c)}(Y), d)$ of forms homogeneous of degree c vanishes if $c \neq 0$, while for $c = 0$*

$$H^j\big(\Omega^\bullet_{(0)}(Y), d\big) \cong H^j(S^*X) \oplus H^{j-1}(S^*X).$$

Proof. Let E be the generator of the dilations ρ_t (the radial vector field $\sum_j \xi_j \partial_{\xi_j}$). Since ρ_t commutes with the flow of E, the contraction ι_E preserves homogeneity, and $\mathcal{L}_E \alpha = c\alpha$ for $\alpha \in \Omega^\bullet_{(c)}(Y)$. By Cartan's formula $d\iota_E + \iota_E d = \mathcal{L}_E$, so if $c \neq 0$ and $d\alpha = 0$ then $\alpha = d(\iota_E \alpha / c)$.

For $c = 0$ use (15.4.1). Every form on $S^*X \times (0,\infty)$ can be written $\alpha' + \alpha'' \wedge dr/r$ with α', α'' not containing dr; since $\rho_t^*(dr/r) = dr/r$, the form is homogeneous of degree 0 if and only if α' and α'' are (pull-backs of) forms on S^*X. As $d(\alpha' + \alpha'' \wedge dr/r) = d\alpha' + d\alpha'' \wedge dr/r$, the complex $\Omega^\bullet_{(0)}(Y)$ is the direct sum of the de Rham complex of S^*X and of the same complex shifted by one, which gives the result. □

The splitting depends on the choice of the radial function R; replacing R by aR, $a \in \mathcal{C}^\infty(S^*X)$ positive, changes α' by $\alpha'' \wedge d \log a$, which does not change the induced map on cohomology from the second summand, nor the subspace given by the first.

Theorem 15.8.4. *For the spectral sequence of Section 15.6,*

$$E^2_{p,k} \cong \begin{cases} H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X), & p = k-n, \\ 0, & p \neq k-n. \end{cases}$$

Proof. By (15.6.2) and Proposition 15.7.2, $E^2_{p,k}$ is the homology at $\Omega^k_{(p)}(Y)$ of $\Omega^{k+1}_{(p+1)} \xrightarrow{\delta} \Omega^k_{(p)} \xrightarrow{\delta} \Omega^{k-1}_{(p-1)}$. By Proposition 15.8.2, $*$ maps this sequence isomorphically onto $\Omega^{2n-k-1}_{(c)} \to \Omega^{2n-k}_{(c)} \to \Omega^{2n-k+1}_{(c)}$, $c = p+n-k$, intertwining δ with $\pm d$ (the sign depending only on the degree). So $E^2_{p,k} \cong H^{2n-k}(\Omega^\bullet_{(p+n-k)}(Y), d)$, and Proposition 15.8.3 gives the result. □

15.9 The Hochschild homology of the symbol algebra

Theorem 15.9.1 (Brylinski–Getzler, Wodzicki)**.** *Let X be a compact manifold of dimension $n \geq 1$. The Hochschild homology of the algebra $\mathcal{A} = \Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)$ of formal symbols, with the chain spaces of Definition 15.5.1, is*

$$\mathrm{HH}_k(\mathcal{A}) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X), \qquad k \in \mathbb{N}_0.$$

More precisely, every class in $\mathrm{HH}_k(\mathcal{A})$ is represented by a cycle of order at most $k-n$, and the map assigning to such a cycle the class in $E^2_{k-n,k}$ of the homogeneous k-form $\varepsilon(c_{k-n})$ of degree $k-n$ is an isomorphism; composing with Theorem 15.8.4 gives the displayed isomorphism. In particular $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k > 2n$ and $\mathrm{HH}_{2n}(\mathcal{A}) \cong H^0(S^*X)$.

Proof. By Theorem 15.8.4 the hypothesis of Proposition 15.6.1 holds, with $n = \dim X$. □

The degeneration of the spectral sequence at E^2, part of Proposition 15.6.1, thus comes down to the observation that the E^2 term is concentrated on the line $p = k-n$, where the order of a chain is tied to its degree; differentials d^r, $r \geq 2$, would have to connect different such lines.

Remark 15.9.2 (The residue trace). In degree 0, $\mathrm{HH}_0(\mathcal{A}) = \mathcal{A}/b(\mathcal{B}^{(1)})$, the quotient by (the closure of) the commutators, and its dual is the space of traces on $\mathcal{A}$. By Theorem 15.9.1, $\mathrm{HH}_0(\mathcal{A}) \cong H^{2n-1}(S^*X)$, and the isomorphism is computed as follows. Every $A \in \Psi^{\mathbb{Z}}(X)$ is congruent modulo $b(\mathcal{B}^{(1)})$ to some $A' \in \Psi^{-n}(X)$. The class of the function $a = \sigma_{-n}(A') \in \Omega^0_{(-n)}(Y)$ corresponds under $*$ to $a\,\mathrm{vol} \in \Omega^{2n}_{(0)}(Y)$, and writing $a\,\mathrm{vol} = \alpha'' \wedge dr/r$ as in Proposition 15.8.3, $\alpha'' = -\iota_E(a\,\mathrm{vol})|_{S^*X}$. Hence, if X is connected and $n \geq 2$, so that

S^*X is connected (and it is always orientable) and $H^{2n-1}(S^*X) \cong \mathbb{C}$, the isomorphism $\mathrm{HH}_0(\mathcal{A}) \cong \mathbb{C}$ is, up to a constant factor,

$$A \longmapsto \int_{S^*X} \sigma_{-n}(A')\, \iota_E\Big(\frac{\omega^n}{n!}\Big).$$

This is the noncommutative residue of Wodzicki and Guillemin [13, 5]: up to normalization it is $\int_{S^*X} \sigma_{-n}(A)\, \iota_E(\omega^n/n!)$, computed for general A in local coordinates from the term of homogeneity $-n$ of the full symbol. Thus, for $n \geq 2$ and X connected, the residue is the unique trace on $\Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)$ up to a multiple. For $n = 1$, S^*X has two components over each component of X, and there are correspondingly two independent residue traces, one for each direction of ξ.

15.10 Exercises

Exercise 15.1. Let $\mathcal{A} = \mathbb{C}^N$ with componentwise multiplication. Show that $\mathcal{A}$ is a projective $\mathcal{A}^e$-module (the idempotent $\sum_j e_j \otimes e_j \in \mathcal{A}^e$ splits the multiplication map $\mathcal{A}^e \to \mathcal{A}$) and deduce $\mathrm{HH}_0(\mathcal{A}) = \mathbb{C}^N$, $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k \geq 1$.

Exercise 15.2. Let $\mathcal{A} = M_N(\mathbb{C})$. Show that $\mathcal{A}$ is a projective $\mathcal{A}^e$-module, that $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k \geq 1$, and that the trace induces an isomorphism $\mathrm{HH}_0(\mathcal{A}) \cong \mathbb{C}$.

Exercise 15.3. Using Lemma 15.7.1, compute δ on 1-forms and on 2-forms on Y, and verify directly that $\delta^2 = 0$ on 2-forms. (Hint: use the Jacobi identity for the Poisson bracket.)

Exercise 15.4. Let $X = \mathbb{S}^1$. Show that S^*X is the disjoint union of two circles and deduce from Theorem 15.9.1 that $\mathrm{HH}_0(\mathcal{A}) \cong \mathbb{C}^2$, $\mathrm{HH}_1(\mathcal{A}) \cong \mathbb{C}^4$, $\mathrm{HH}_2(\mathcal{A}) \cong \mathbb{C}^2$ and $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k \geq 3$. Find explicit representatives of order -1 for the classes in $\mathrm{HH}_0(\mathcal{A})$, and the two traces.

Exercise 15.5. Show directly, using Theorem 15.8.4 and Proposition 15.6.1, that a class in $\mathrm{HH}_{2n}(\mathcal{A})$ is represented by a cycle of order at most n whose leading part corresponds to a locally constant multiple of ω^n on Y.

Notes

The polynomial case of the Hochschild–Kostant–Rosenberg theorem is proved here by the Koszul resolution instead of the direct reduction used in the original notes, which is kept, in a sharpened form (Proposition 15.3.3), for the proof of Connes' theorem. That proof was incomplete in the original (the key exactness statement was left open); the argument above reduces it, via the module property of the boundaries (Lemma 15.3.4) and Borel's lemma, to the polynomial case. Connes' original proof (A. Connes, *Non-commutative differential geometry*, Publ. Math. IHÉS **62**, 1985) uses a topological Koszul resolution. The Eilenberg–Zilber and Künneth theorems used in Theorem 15.4.2 are standard; see J.-L. Loday, *Cyclic homology*, Springer, 1992, Chapters 1 and 4, which also covers H-unitality. The identification $\mathcal{C}^\infty(X)\hat{\otimes}\mathcal{C}^\infty(Y) = \mathcal{C}^\infty(X \times Y)$ is in F. Trèves, *Topological vector spaces, distributions and kernels*, Academic Press, 1967.

The semiclassical family of the original has been replaced by the equivalent filtration by order, and the degeneration and convergence of the spectral sequence are proved in Proposition 15.6.1 by the homogeneity argument. The complex $(\Omega^\bullet, \delta)$ and the symplectic star operator are due to Koszul and to J.-L. Brylinski (*A differential complex for Poisson*

manifolds, J. Differential Geom. **28**, 1988). The main theorem is due to Brylinski–Getzler [1] and Wodzicki [13, 14]; these papers also treat cyclic homology and the algebras $\Psi^{-\infty}(X)$ and $\Psi^{\mathbb{Z}}(X)$, and they use a global symbol calculus (due to Widom) for which the product has the form (15.5.1).

Appendix A

Bounded operators on Hilbert space

This appendix collects the properties of bounded operators on complex Hilbert spaces which are used throughout the text: elementary facts about the algebra of bounded operators, compact operators, Fredholm operators and their index, and the Hilbert–Schmidt and trace-class ideals. Proofs are short and standard. Hilbert spaces H, H_1, $H_2, \dots$ are complex; where orthonormal bases are used they are assumed separable, so that bases are countable.

A.1 Bounded operators

Proposition A.1.1. *A linear map $A : H_1 \to H_2$ is continuous if and only if it is bounded, i.e. $\|A\| = \sup_{\|x\|\le 1} \|Ax\| < \infty$. The space $\mathcal{B}(H_1, H_2)$ of bounded operators is a Banach space with this norm, and $\|AB\| \le \|A\|\,\|B\|$ whenever the composite is defined. In particular $\mathcal{B}(H) = \mathcal{B}(H, H)$ is a Banach algebra.*

Proof. If A is continuous at 0 there is $\delta > 0$ with $\|Ax\| \le 1$ for $\|x\| \le \delta$, so $\|A\| \le \delta^{-1}$; conversely $\|Ax - Ay\| \le \|A\|\|x - y\|$. If (A_k) is Cauchy in norm then $A_k x$ is Cauchy for each x, and the limit Ax is linear with $\|A - A_k\| \to 0$. Finally $\|ABx\| \le \|A\|\|Bx\| \le \|A\|\|B\|\|x\|$. □

By the Riesz representation theorem each $A \in \mathcal{B}(H_1, H_2)$ has an adjoint $A^* \in \mathcal{B}(H_2, H_1)$, $(Ax, y) = (x, A^*y)$, with $(AB)^* = B^*A^*$, $A^{**} = A$ and $\|A^*\| = \|A\|$ (since $\|A\| = \sup_{\|x\|,\|y\|\le 1} |(Ax, y)|$). We also use the open mapping theorem: a bounded bijection between Banach spaces has bounded inverse.

Proposition A.1.2 (C^*-identity)**.** *For $A \in \mathcal{B}(H_1, H_2)$, $\|A^*A\| = \|AA^*\| = \|A\|^2$.*

Proof. $\|Ax\|^2 = (A^*Ax, x) \le \|A^*A\|\|x\|^2$, so $\|A\|^2 \le \|A^*A\| \le \|A^*\|\|A\| = \|A\|^2$. Apply this to A^* for the other identity. □

Proposition A.1.3 (Neumann series)**.** *If $A \in \mathcal{B}(H)$ and $\|A\| < 1$ then $\mathrm{Id} - A$ is invertible, with $(\mathrm{Id} - A)^{-1} = \sum_{k\ge 0} A^k$ and $\|(\mathrm{Id} - A)^{-1}\| \le (1 - \|A\|)^{-1}$. Consequently:*

(a) *$A - \lambda$ is invertible for $|\lambda| > \|A\|$, so the spectrum of A lies in $\{|\lambda| \le \|A\|\}$;*

(b) *the invertible elements of $\mathcal{B}(H_1, H_2)$ form an open set: if A is invertible and $\|E\| < \|A^{-1}\|^{-1}$ then $A + E = A(\mathrm{Id} + A^{-1}E)$ is invertible; and inversion is continuous.*

Proof. The series converges absolutely in the Banach space $\mathcal{B}(H)$ since $\|A^k\| \le \|A\|^k$, and $(\mathrm{Id}-A)\sum_{k<N} A^k = \sum_{k<N} A^k(\mathrm{Id}-A) = \mathrm{Id}-A^N \to \mathrm{Id}$. For (a) write $A-\lambda = -\lambda(\mathrm{Id}-A/\lambda)$. In (b) the inverse is $(A+E)^{-1} = \sum_k (-A^{-1}E)^k A^{-1}$, which depends continuously on E. □

The condition in (a) cannot be weakened to $|\lambda| \ge \|A\|$: for $A = \mathrm{Id}$, $\lambda = 1$.

Lemma A.1.4. *If $T \in \mathcal{B}(H)$ is self-adjoint then $\|T\| = \sup_{\|x\|=1} |(Tx, x)|$.*

Proof. Let M be the supremum; clearly $M \le \|T\|$. By polarization, for $\|x\| = \|y\| = 1$, $4\,\mathrm{Re}(Tx, y) = (T(x+y), x+y) - (T(x-y), x-y) \le M(\|x+y\|^2 + \|x-y\|^2) = 4M$. Taking $y = Tx/\|Tx\|$ gives $\|Tx\| \le M$. □

A.2 Compact operators

Definition A.2.1. An operator $K \in \mathcal{B}(H_1, H_2)$ is *compact* if the closure of the image of the unit ball is compact; equivalently, if every bounded sequence (x_n) has a subsequence along which Kx_n converges. The compact operators form a subspace $\mathcal{K}(H_1, H_2)$, and $\mathcal{K}(H) = \mathcal{K}(H, H)$.

Proposition A.2.2. (a) *If K is compact and A, B are bounded, then AKB is compact (when defined); $\mathcal{K}(H)$ is a two-sided ideal in $\mathcal{B}(H)$.*

(b) *K is compact if and only if $x_n \rightharpoonup x$ weakly implies $Kx_n \to Kx$ in norm.*

(c) *$\mathcal{K}(H_1, H_2)$ is the norm closure of the finite-rank operators.*

(d) *K is compact if and only if K^* is compact, if and only if K^*K is compact.*

(e) (Riesz) *If the closed unit ball of H is compact, $\dim H < \infty$. In particular if K is compact, $\mathrm{Nul}(\mathrm{Id}-K)$ is finite-dimensional.*

Proof. (a) Bounded operators map bounded sets to bounded sets and convergent sequences to convergent sequences.

(b) Suppose K compact and $x_n \rightharpoonup x$. Then (x_n) is bounded and $Kx_n \rightharpoonup Kx$, since $(Kx_n, z) = (x_n, K^*z)$. Every subsequence of (Kx_n) has a norm-convergent subsequence, whose limit must be Kx; hence $Kx_n \to Kx$. Conversely, every bounded sequence has a weakly convergent subsequence, which K maps to a convergent sequence.

(c) Finite-rank operators are compact, and a norm limit K of compact operators K_j is compact: given $\epsilon > 0$ choose j with $\|K - K_j\| < \epsilon$; then $K(B)$ lies within ϵ of the totally bounded set $K_j(B)$, so is totally bounded. Conversely, let K be compact and $\epsilon > 0$. Cover the compact set $\overline{K(B)}$ by finitely many balls $B(y_i, \epsilon)$, and let P be the orthogonal projection onto the span of the y_i. For $\|x\| \le 1$ choose i with $\|Kx - y_i\| < \epsilon$; then $\|Kx - PKx\| \le \|Kx - y_i\| + \|P(y_i - Kx)\| < 2\epsilon$. So $\|K - PK\| \le 2\epsilon$, and PK has finite rank.

(d) The adjoint of a finite-rank operator has finite rank and $\|K^* - F^*\| = \|K - F\|$, so the first equivalence follows from (c). If K is compact so is K^*K, by (a). Conversely, suppose K^*K is compact and $x_n \rightharpoonup x$, $\|x_n\| \le M$. Then $\|K(x_n - x)\|^2 = (K^*K(x_n - x), x_n - x) \le 2M\|K^*K(x_n - x)\| \to 0$ by (b), so K is compact by (b).

(e) If $\dim H = \infty$ an orthonormal sequence (e_n) has $\|e_n - e_m\| = \sqrt{2}$ for $n \ne m$, so has no convergent subsequence. On $\mathrm{Nul}(\mathrm{Id}-K)$, $x = Kx$, so the unit ball of this closed subspace is compact. □

Theorem A.2.3 (Spectral theorem)**.** *Let $T \in \mathcal{K}(H)$ be self-adjoint. Then there is an orthonormal sequence (e_k) (finite or infinite) of eigenvectors, $Te_k = \lambda_k e_k$, with real eigenvalues $|\lambda_1| \geq |\lambda_2| \geq \cdots > 0$ tending to zero if infinite, such that $H = \mathrm{Nul}(T) \oplus \overline{\mathrm{span}}\{e_k\}$ and $Tx = \sum_k \lambda_k (x, e_k) e_k$. Each non-zero eigenvalue has finite multiplicity.*

Proof. If $T \neq 0$, by Lemma A.1.4 there are unit vectors x_n with $(Tx_n, x_n) \to \lambda$, $|\lambda| = \|T\|$, λ real. Then $\|Tx_n - \lambda x_n\|^2 = \|Tx_n\|^2 - 2\lambda(Tx_n, x_n) + \lambda^2 \leq 2\lambda^2 - 2\lambda(Tx_n, x_n) \to 0$. Passing to a subsequence, $Tx_n \to y$, so $x_n \to y/\lambda$, and $e = y/\|y\|$ satisfies $Te = \lambda e$. The orthocomplement of e is invariant under T; repeating the argument there gives the sequence (e_k) with $|\lambda_k|$ non-increasing, the restriction of T to the orthocomplement of $e_1, \ldots, e_k$ having norm $|\lambda_{k+1}|$. If infinitely many $|\lambda_k| \geq \epsilon > 0$, then $\|Te_k - Te_j\|^2 = \lambda_k^2 + \lambda_j^2 \geq 2\epsilon^2$, contradicting compactness; so $\lambda_k \to 0$, and each eigenvalue occurs finitely often. On the orthocomplement W of all the e_k, which is invariant, $\|T|_W\| \leq |\lambda_k|$ for all k (or the process would have stopped), so $T|_W = 0$ and $W \subset \mathrm{Nul}(T)$; the reverse inclusion holds since $(x, e_k) = \lambda_k^{-1}(x, Te_k) = \lambda_k^{-1}(Tx, e_k)$. □

A.3 Fredholm operators

Definition A.3.1. An operator $A \in \mathcal{B}(H_1, H_2)$ is *Fredholm* if $\mathrm{Nul}(A)$ is finite-dimensional, $\mathrm{Ran}(A)$ is closed, and $\mathrm{Ran}(A)^\perp$ is finite-dimensional. Its *index* is $\mathrm{ind}(A) = \dim \mathrm{Nul}(A) - \dim \mathrm{Ran}(A)^\perp = \dim \mathrm{Nul}(A) - \dim \mathrm{Nul}(A^*)$.

Here $\mathrm{Ran}(A)^\perp = \mathrm{Nul}(A^*)$ for any bounded A, and if $\mathrm{Ran}(A)$ is closed it is the orthocomplement of $\mathrm{Nul}(A^*)$. Thus A is Fredholm if and only if A^* is, and $\mathrm{ind}(A^*) = -\mathrm{ind}(A)$.

Proposition A.3.2 (Generalized inverse)**.** *If $A \in \mathcal{B}(H_1, H_2)$ is Fredholm there is a unique $G \in \mathcal{B}(H_2, H_1)$ with*

$$GA = \mathrm{Id} - \pi_N, \qquad AG = \mathrm{Id} - \pi_C, \qquad G\pi_C = 0,$$

where π_N and π_C are the orthogonal projections onto $\mathrm{Nul}(A)$ and $\mathrm{Nul}(A^*)$. G is Fredholm with $\mathrm{ind}(G) = -\mathrm{ind}(A)$.

Proof. A restricts to a bounded bijection $\mathrm{Nul}(A)^\perp \to \mathrm{Ran}(A)$ between Hilbert spaces, which has bounded inverse A_0^{-1} by the open mapping theorem; set $G = A_0^{-1}(\mathrm{Id} - \pi_C)$. The identities are immediate and determine G. $\mathrm{Nul}(G) = \mathrm{Nul}(A^*)$ and $\mathrm{Ran}(G) = \mathrm{Nul}(A)^\perp$, so $\mathrm{ind}(G) = \dim \mathrm{Nul}(A^*) - \dim \mathrm{Nul}(A)$. □

Theorem A.3.3 (Atkinson)**.** *$A \in \mathcal{B}(H_1, H_2)$ is Fredholm if and only if there exists $B \in \mathcal{B}(H_2, H_1)$ such that $BA - \mathrm{Id}$ and $AB - \mathrm{Id}$ are compact. One may then take B such that both are of finite rank.*

Proof. If A is Fredholm, the generalized inverse is such a B, with finite-rank errors. Conversely suppose $BA = \mathrm{Id} - K$ and $AB = \mathrm{Id} - K'$ with K, K' compact. Then $\mathrm{Nul}(A) \subset \mathrm{Nul}(\mathrm{Id} - K)$, which is finite-dimensional by Proposition A.2.2(e). Let $L = \mathrm{Id} - K'$. On $\mathrm{Nul}(L)^\perp$ there is $c > 0$ with $\|Lx\| \geq c\|x\|$: otherwise there are unit vectors $x_n \in \mathrm{Nul}(L)^\perp$ with $Lx_n \to 0$; passing to a subsequence $K'x_n \to y$, so $x_n = Lx_n + K'x_n \to y$, with $\|y\| = 1$, $y \in \mathrm{Nul}(L)^\perp$ and $Ly = 0$, a contradiction. Hence $\mathrm{Ran}(L)$ is closed, and $\mathrm{Ran}(L)^\perp = \mathrm{Nul}(\mathrm{Id} - K'^*)$ is finite-dimensional. Since $\mathrm{Ran}(A) \supset \mathrm{Ran}(AB) = \mathrm{Ran}(L)$, $\mathrm{Ran}(A)$ is the sum of the closed subspace $\mathrm{Ran}(L)$ and a finite-dimensional space, hence closed, and $\mathrm{Ran}(A)^\perp \subset \mathrm{Ran}(L)^\perp$ is finite-dimensional. □

Lemma A.3.4. *If $F \in \mathcal{B}(H)$ has finite rank then $\mathrm{Id} - F$ is Fredholm of index zero.*

Proof. Let $V = \mathrm{Ran}(F) + \mathrm{Ran}(F^*)$, which is finite-dimensional. Then F maps V into itself and vanishes on $V^\perp \subset \mathrm{Nul}(F)$, so $\mathrm{Id} - F = (\mathrm{Id}_V - F|_V) \oplus \mathrm{Id}_{V^\perp}$, and the index is that of an endomorphism of a finite-dimensional space, which is zero. □

Proposition A.3.5. *If $B \in \mathcal{B}(H_1, H_2)$ and $A \in \mathcal{B}(H_2, H_3)$ are Fredholm, then AB is Fredholm and* $\mathrm{ind}(AB) = \mathrm{ind}(A) + \mathrm{ind}(B)$.

Proof. If B', A' are parametrices as in Theorem A.3.3 then $B'A'$ is one for AB. Consider the sequence of finite-dimensional spaces and natural maps

$$0 \to \mathrm{Nul}\, B \hookrightarrow \mathrm{Nul}\, AB \xrightarrow{B} \mathrm{Nul}\, A \to H_2/\mathrm{Ran}\, B \xrightarrow{A} H_3/\mathrm{Ran}\, AB \to H_3/\mathrm{Ran}\, A \to 0,$$

where the third map is inclusion followed by the quotient map and the last is the quotient map ($\mathrm{Ran}\, AB \subset \mathrm{Ran}\, A$). It is exact: the kernel of B on $\mathrm{Nul}\, AB$ is $\mathrm{Nul}\, B$; the image of $\mathrm{Nul}\, AB$ under B is $\mathrm{Nul}\, A \cap \mathrm{Ran}\, B$, the kernel of the third map; the kernel of the map induced by A consists of the classes of v with $Av \in \mathrm{Ran}\, AB$, i.e. $v \in \mathrm{Nul}\, A + \mathrm{Ran}\, B$; and the kernel of the last map is $\mathrm{Ran}\, A/\mathrm{Ran}\, AB$. The alternating sum of the dimensions in an exact sequence vanishes, and $\dim H_2/\mathrm{Ran}\, B = \dim \mathrm{Nul}\, B^*$ etc. since the ranges are closed; this gives the formula. □

Theorem A.3.6 (Stability of the index)**.** *Let $A \in \mathcal{B}(H_1, H_2)$ be Fredholm.*

(a) *If G is a generalized inverse of A and $\|E\| < \|G\|^{-1}$, then $A + E$ is Fredholm and* $\mathrm{ind}(A + E) = \mathrm{ind}(A)$. *Thus the Fredholm operators form an open subset of $\mathcal{B}(H_1, H_2)$ on which the index is locally constant, hence constant along continuous paths.*

(b) *If $K \in \mathcal{K}(H_1, H_2)$ then $A + K$ is Fredholm and* $\mathrm{ind}(A + K) = \mathrm{ind}(A)$.

Proof. (a) Since $\|GE\| < 1$ and $\|EG\| < 1$, $\mathrm{Id} + GE$ and $\mathrm{Id} + EG$ are invertible, and

$$(\mathrm{Id} + GE)^{-1} G(A + E) = \mathrm{Id} - (\mathrm{Id} + GE)^{-1} \pi_N,$$
$$(A + E) G (\mathrm{Id} + EG)^{-1} = \mathrm{Id} - \pi_C (\mathrm{Id} + EG)^{-1},$$

using $GA = \mathrm{Id} - \pi_N$ and $AG = \mathrm{Id} - \pi_C$. By Theorem A.3.3, $A + E$ is Fredholm. The first identity, Lemma A.3.4 and Proposition A.3.5 (invertible operators having index zero) give $\mathrm{ind}(G) + \mathrm{ind}(A + E) = 0$, and $\mathrm{ind}(G) = -\mathrm{ind}(A)$.

(b) A parametrix for A is a parametrix for $A + tK$ modulo compact operators, so $A + tK$ is Fredholm for $t \in [0, 1]$, and by (a) the index is locally constant, hence constant, in t. □

Theorem A.3.7 (Riesz theory)**.** *Let $K \in \mathcal{K}(H)$ and $\mu \in \mathbb{C} \setminus \{0\}$. Then $\mu - K$ is Fredholm of index zero, so $\mu \in \mathrm{Spec}(K)$ if and only if μ is an eigenvalue. For each $\epsilon > 0$ there are only finitely many eigenvalues with $|\mu| \geq \epsilon$; thus* $\mathrm{Spec}(K) \setminus \{0\}$ *is countable and discrete in $\mathbb{C} \setminus \{0\}$. For each eigenvalue $\mu \neq 0$ the generalized eigenspace $\bigcup_N \mathrm{Nul}((\mu - K)^N)$ is finite-dimensional.*

Proof. $\mu - K = \mu(\mathrm{Id} - K/\mu)$ is Fredholm of index zero by Theorem A.3.6(b); if it is injective it is therefore surjective, hence invertible. Suppose there were infinitely many distinct eigenvalues μ_k with $|\mu_k| \geq \epsilon$ and eigenvectors x_k; these are linearly independent. Let $M_k = \mathrm{span}\{x_1, \ldots, x_k\}$ and choose unit vectors $y_k \in M_k \ominus M_{k-1}$. Since $(K - \mu_k)M_k \subset M_{k-1}$, for $j < k$

$$Ky_k - Ky_j = \mu_k y_k + ((K - \mu_k)y_k - Ky_j), \quad (K - \mu_k)y_k - Ky_j \in M_{k-1},$$

so $\|Ky_k - Ky_j\| \geq |\mu_k| \geq \epsilon$, contradicting compactness. Finally the spaces $N_j = \mathrm{Nul}((\mu - K)^j)$ increase; if $N_{j-1} \neq N_j$ for all j, choose unit $u_j \in N_j \ominus N_{j-1}$. For $i < j$, $Ku_j - Ku_i = \mu u_j - \big((\mu - K)u_j + Ku_i\big)$ with the bracket in N_{j-1}, so $\|Ku_j - Ku_i\| \geq |\mu|$, again a contradiction. So $N_j = N_{j_0}$ for $j \geq j_0$, and $N_{j_0} = \mathrm{Nul}((\mu - K)^{j_0})$ is finite-dimensional because $(\mu - K)^{j_0} = \mu^{j_0}(\mathrm{Id} - K')$ with K' compact. □

A.4 Hilbert–Schmidt and trace-class operators

Definition A.4.1. An operator $A \in \mathcal{B}(H_1, H_2)$ is *Hilbert–Schmidt* if $\|A\|_2^2 = \sum_j \|Ae_j\|^2 < \infty$ for an orthonormal basis (e_j) of H_1.

For orthonormal bases (e_j) of H_1 and (f_k) of H_2, $\sum_j \|Ae_j\|^2 = \sum_{j,k} |(Ae_j, f_k)|^2 = \sum_k \|A^* f_k\|^2$, so $\|A\|_2$ does not depend on the basis and $\|A^*\|_2 = \|A\|_2$. The Hilbert–Schmidt operators form a Hilbert space with inner product $\langle A, B\rangle_2 = \sum_j (Ae_j, Be_j)$, also independent of the basis, and:

- $\|A\| \leq \|A\|_2$ (expand x in a basis and use Cauchy–Schwarz);
- $\|SAT\|_2 \leq \|S\|\,\|A\|_2\,\|T\|$ for bounded S, T;
- Hilbert–Schmidt operators are compact: if P_N is the projection onto $\mathrm{span}\{e_1, \dots, e_N\}$, $\|A - AP_N\|_2^2 = \sum_{j>N} \|Ae_j\|^2 \to 0$.

Proposition A.4.2. *Let (M_i, μ_i), $i = 1, 2$, be σ-finite measure spaces with $L^2(M_i)$ separable. An operator $A : L^2(M_1) \to L^2(M_2)$ is Hilbert–Schmidt if and only if it is given by a kernel $k \in L^2(M_2 \times M_1)$, $Au(x) = \int k(x,y)u(y)\,d\mu_1(y)$, and then $\|A\|_2 = \|k\|_{L^2}$. The same holds for vector-valued functions.*

Proof. If (e_j) and (f_k) are orthonormal bases of $L^2(M_1)$ and $L^2(M_2)$, the products $f_k(x)\overline{e_j(y)}$ form an orthonormal basis of $L^2(M_2 \times M_1)$. If A has kernel $k \in L^2$, then $(Ae_j, f_k) = \langle k, f_k \otimes \bar{e}_j\rangle$, so $\|A\|_2^2 = \sum_{j,k} |(Ae_j, f_k)|^2 = \|k\|_{L^2}^2$. Conversely, if A is Hilbert–Schmidt then $k = \sum_{j,k} (Ae_j, f_k) f_k \otimes \bar{e}_j \in L^2$ is a kernel for A, as one checks on the e_j. □

Definition A.4.3. An operator $A \in \mathcal{B}(H)$ is of *trace class* if it is a finite sum $A = \sum_k B_k C_k$ of products of Hilbert–Schmidt operators $C_k : H \to H'_k$, $B_k : H'_k \to H$. The trace norm $\|A\|_1$ is the infimum of $\sum_k \|B_k\|_2 \|C_k\|_2$ over such representations.

The trace-class operators form a two-sided ideal in $\mathcal{B}(H)$, contained in the compact operators, with $\|SAT\|_1 \leq \|S\|\,\|A\|_1\|T\|$ and $\|A\| \leq \|A\|_1$; $\|\cdot\|_1$ is a norm, and the ideal is complete with respect to it. Every finite-rank operator is of trace class. (Using the polar decomposition $A = U|A|$ one can show that $\|A\|_1 = \sum_j (|A|e_j, e_j)$, the sum of the singular values of A.)

Proposition A.4.4. *If $A \in \mathcal{B}(H)$ is of trace class then $\mathrm{Tr}(A) = \sum_j (Ae_j, e_j)$ converges absolutely, is independent of the orthonormal basis (e_j), and $|\mathrm{Tr}(A)| \leq \|A\|_1$. Moreover:*

(a) *$\mathrm{Tr}(XY) = \mathrm{Tr}(YX)$ if $X \in \mathcal{B}(H', H)$ and $Y \in \mathcal{B}(H, H')$ are Hilbert–Schmidt;*

(b) *$\mathrm{Tr}(ST) = \mathrm{Tr}(TS)$ if S is of trace class and T is bounded;*

(c) *for finite-rank A, $\mathrm{Tr}(A)$ is the trace of the matrix of A restricted to any finite-dimensional space containing $\mathrm{Ran}(A)$ and $\mathrm{Ran}(A^*)$;*

(d) *if $A = BC$ on $L^2(M)$, with B, C Hilbert–Schmidt with kernels b, c, then $\mathrm{Tr}(A) = \iint b(x,y)c(y,x)\,d\mu(y)\,d\mu(x)$.*

Proof. For $A = BC$, $\sum_j (BCe_j, e_j) = \sum_j (Ce_j, B^* e_j) = \langle C, B^* \rangle_2$, which converges absolutely by Cauchy–Schwarz with modulus at most $\|B\|_2 \|C\|_2$, and is independent of the basis. Summing over the terms of a representation gives the first statements.

(a) For orthonormal bases (e_j) of H and (f_k) of H',

$$\operatorname{Tr}(XY) = \sum_j (Ye_j, X^* e_j) = \sum_{j,k} (Ye_j, f_k)(f_k, X^* e_j) = \sum_{j,k} (Xf_k, e_j)(e_j, Y^* f_k) = \operatorname{Tr}(YX),$$

the double sum converging absolutely by Cauchy–Schwarz.

(b) It suffices to take $S = BC$; by (a), $\operatorname{Tr}(BCT) = \operatorname{Tr}(CTB) = \operatorname{Tr}(TBC)$.

(c) Take a basis containing an orthonormal basis of that space; A vanishes on its orthocomplement.

(d) $\operatorname{Tr}(BC) = \langle C, B^* \rangle_2$, and the kernel of B^* is $\overline{b(y,x)}$, so by Proposition A.4.2 this is $\iint c(y,x) b(x,y)$. □

Remark A.4.5 (Schatten classes). For a compact operator A let $s_1 \geq s_2 \geq \cdots \geq 0$ be its singular values, the eigenvalues of $(A^*A)^{1/2}$. For $1 \leq p < \infty$ the Schatten class $\mathcal{S}_p$ consists of those A with $\|A\|_p = (\sum_j s_j^p)^{1/p} < \infty$. These are two-sided ideals in $\mathcal{B}(H)$, Banach spaces under $\|\cdot\|_p$, with $\mathcal{S}_1$ the trace class, $\mathcal{S}_2$ the Hilbert–Schmidt operators, $\mathcal{S}_p \subset \mathcal{S}_q$ for $p \leq q$, and Hölder's inequality $\|AB\|_r \leq \|A\|_p \|B\|_q$ for $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$. By Lidskii's theorem the trace of an operator of trace class is the sum of its eigenvalues, counted with algebraic multiplicity.

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